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IP multimedia call control protocol based on
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and Session Description Protocol (SDP);
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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

- x the first digit:
  - 1 presented to TSG for information;
  - 2 presented to TSG for approval;
  - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

should indicates a recommendation to do something

should not indicates a recommendation not to do something

may indicates permission to do something

need not indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document defines a call control protocol for use in the IP Multimedia (IM) Core Network (CN) subsystem based on the Session Initiation Protocol (SIP), and the associated Session Description Protocol (SDP).

The present document is applicable to:

- the interface between the User Equipment (UE) and the Call Session Control Function (CSCF);
- the interface between the CSCF and any other CSCF;
- the interface between the CSCF and an Application Server (AS);
- the interface between the CSCF and an ISC gateway function;
- the interface between the ISC gateway function and an Application Server (AS);
- the interface between the CSCF and the Media Gateway Control Function (MGCF);
- the interface between the S-CSCF and the Multimedia Resource Function Controller (MRFC);
- the interface between the Application Server (AS) and the Multimedia Resource Function Controller (MRFC);
- the interface between the S-CSCF and the Media Resource Broker (MRB);
- the interface between the AS and the MRB;
- the interface between the MRB and the MRFC;
- the interface between the CSCF and the Breakout Gateway Control Function (BGCF);
- the interface between the BGCF and the MGCF;
- the interface between the CSCF and an IBCF;
- the interface between the IBCF and AS, MRFC or MRB;
- the interface between the E-CSCF and the Location Retrieval Function (LRF);
- the interface between the BGCF and any other BGCF;
- the interface between the CSCF and an external Multimedia IP network;
- the interface between the E-CSCF and the EATF;
- the interface between the E-CSCF and the terminating IMS network;
- the interface between the P-CSCF and the ATCF;
- the interface between the ATCF and the I-CSCF;
- the interface between the ATCF and the IBCF; and
- the interface between the transit function and the AS.

Where possible the present document specifies the requirements for this protocol by reference to specifications produced by the IETF within the scope of SIP and SDP. Where this is not possible, extensions to SIP and SDP are defined within the present document. The document has therefore been structured in order to allow both forms of specification.

As the IM CN subsystem is designed to interwork with different IP-Connectivity Access Networks (IP-CANs), the IP-CAN independent aspects of the IM CN subsystem are described in the main body and annex A of this specification. Aspects for connecting a UE to the IM CN subsystem through specific types of IP-CANs are documented separately in the annexes or in separate documents.

The document also specificies:

- HTTP for use by an AS and by an MRB in support of the provision of media resources; and
- HTTP for use by an IBCF and by an AS in support of the invocation of attestation and verification functions.

The document also specifies media-related requirements for the NAT traversal mechanisms defined in this specification.

NOTE: The present document covers only the usage of SIP and SDP to communicate with the enitities of the IM CN subsystem. It is possible, and not precluded, to use the capabilities of IP-CAN to allow a terminal containing a SIP UA to communicate with SIP servers or SIP UAs outside the IM CN subsystem, and therefore utilise the services provided by those SIP servers. The usage of SIP and SDP for communicating with SIP servers or SIP UAs outside the IM CN subsystem is outside the scope of the present document.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[1A] 3GPP TS 22.101: "Service aspects; Service principles".

[1B] 3GPP TS 22.003: "Circuit Teleservices supported by a Public Land Mobile Network (PLMN)".

[1C] 3GPP TS 22.011: "Service accessibility".

[2] 3GPP TS 23.002: "Network architecture".

[3] 3GPP TS 23.003: "Numbering, addressing and identification".

[4] 3GPP TS 23.060: "General Packet Radio Service (GPRS); Service description; Stage 2".

[4A] 3GPP TS 23.107: "Quality of Service (QoS) concept and architecture".

[4B] 3GPP TS 23.167: "IP Multimedia Subsystem (IMS) emergency sessions".

[4C] 3GPP TS 23.122: "Non-Access-Stratum (NAS) functions related to Mobile Station (MS) in idle mode".

[4D] 3GPP TS 23.140 Release 6: "Multimedia Messaging Service (MMS); Functional description; Stage 2".

[5] 3GPP TS 23.218: "IP Multimedia (IM) Session Handling; IM call model".

[6] 3GPP TS 23.221: "Architectural requirements".

[7] 3GPP TS 23.228: "IP multimedia subsystem; Stage 2".

[7A] 3GPP TS 23.234: "3GPP system to Wireless Local Area Network (WLAN) interworking; System description".

[7B] 3GPP TS 23.401: "GPRS enhancements for E-UTRAN access".

[7C] 3GPP TS 23.292: "IP Multimedia Subsystem (IMS) Centralized Services; Stage 2".

[7D] 3GPP TS 23.380: "IMS Restoration Procedures".

[7E] 3GPP TS 23.402: "Architecture enhancements for non-3GPP accesses".

[7F] 3GPP TS 23.334: "IMS Application Level Gateway (IMS-ALG) – IMS Access Gateway (IMS-AGW) interface".

[7G] 3GPP TS 24.103: "Telepresence using the IP Multimedia (IM) Core Network (CN) Subsystem (IMS); Stage 3".

[8] 3GPP TS 24.008: "Mobile radio interface layer 3 specification; Core Network protocols; Stage 3".

[8A] 3GPP TS 24.141: "Presence service using the IP Multimedia (IM) Core Network (CN) subsystem; Stage 3".

[8B] 3GPP TS 24.147: "Conferencing using the IP Multimedia (IM) Core Network (CN) subsystem; Stage 3".

[8C] 3GPP TS 24.234: "3GPP System to Wireless Local Area Network (WLAN) interworking; WLAN User Equipment (WLAN UE) to network protocols; Stage 3".

[8D] Void.

[8E] 3GPP TS 24.279: "Combining Circuit Switched (CS) and IP Multimedia Subsystem (IMS) services, stage 3, Release 7".

[8F] 3GPP TS 24.247: "Messaging service using the IP Multimedia (IM) Core Network (CN) subsystem; Stage 3".

[8G] 3GPP TS 24.167: "3GPP IMS Management Object (MO); Stage 3".

[8H] 3GPP TS 24.173: "IMS Multimedia telephony communication service and supplementary services; Stage 3".

[8I] 3GPP TS 24.606: "Message Waiting Indication (MWI) using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification".

[8J] 3GPP TS 24.301: "Non-Access-Stratum (NAS) protocol for Evolved Packet System (EPS); Stage 3".

[8K] 3GPP TS 24.323: "3GPP IMS service level tracing management object (MO)".

[8L] 3GPP TS 24.341: "Support of SMS over IP networks; Stage 3".

[8M] 3GPP TS 24.237: "IP Multimedia Subsystem (IMS) Service Continuity; Stage 3".

[8N] 3GPP TS 24.647: "Advice Of Charge (AOC) using IP Multimedia (IM) Core Network (CN) subsystem".

[8O] 3GPP TS 24.292: "IP Multimedia (IM) Core Network (CN) subsystem Centralized Services (ICS); Stage 3".

[8P] 3GPP TS 24.623: "Extensible Markup Language (XML) Configuration Access Protocol (XCAP) over the Ut interface for Manipulating Supplementary Services".

[8Q] 3GPP TS 24.182: "IP Multimedia Subsystem (IMS) Customized Alerting Tones (CAT); Protocol specification".

[8R] 3GPP TS 24.183: "IP Multimedia Subsystem (IMS) Customized Ringing Signal (CRS); Protocol specification".

[8S] 3GPP TS 24.616: "Malicious Communication Identification (MCID) using IP Multimedia (IM) Core Network (CN) subsystem".

[8T] 3GPP TS 24.305: "Selective Disabling of 3GPP User Equipment Capabilities (SDoUE) Management Object (MO)".

[8U] 3GPP TS 24.302: "Access to the Evolved Packet Core (EPC) via non-3GPP access networks; Stage 3".

[8V] 3GPP TS 24.303: "Mobility management based on Dual-Stack Mobile IPv6".

[8W] 3GPP TS 24.390: "Unstructured Supplementary Service Data (USSD) using IP Multimedia (IM) Core Network (CN) subsystem IMS".

[8X] 3GPP TS 24.139: "3GPP System-Fixed Broadband Access Network Interworking; Stage 3".

[8Y] 3GPP TS 24.322: "UE access to IMS services via restrictive access networks - stage 3".

[8Z] 3GPP TS 24.371: "Web Real Time Communication (WebRTC) Access to IMS".

[8ZA] 3GPP TS 24.525: "Business trunking; Architecture and functional description".

[8ZB] 3GPP TS 24.244: "Wireless LAN control plane protocol for trusted WLAN access to EPC; Stage 3".

[8ZC] 3GPP TS 24.337: "IP Multimedia (IM) Core Network (CN) subsystem IP Multimedia Subsystem (IMS) inter-UE transfer; Stage 3".

[8ZD] 3GPP TS 24.334: "Proximity-services (ProSe) User Equipment (UE) to Proximity-services (ProSe) Function Protocol aspects; Stage 3".

[8ZE] 3GPP TS 24.379: "Mission Critical Push To Talk (MCPTT) call control; Stage 3".

[8ZF] 3GPP TS 24.628: "Common Basic Communication procedures using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification".

[8ZG] 3GPP TS 24.604: "Communication Diversion (CDIV) using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification".

[8ZH] 3GPP TS 24.174: "Support of multi-device and multi-identity in the IP Multimedia Subsystem (IMS); Stage 3".

[8ZI] 3GPP TS 24.554: "Proximity-service (ProSe) in 5G System (5GS) protocol aspects; Stage 3"

[9] 3GPP TS 25.304: "User Equipment (UE) procedures in idle mode and procedures for cell reselection in connected mode".

[9A] 3GPP TS 25.331: "Radio Resource Control (RRC); Protocol Specification".

[9B] 3GPP TS 26.114: "IP Multimedia Subsystem (IMS); Multimedia Telephony; Media handling and interaction".

[9C] 3GPP TS 26.267: "eCall Data Transfer; In-band modem solution; General description".

[10] Void.

[10A] 3GPP TS 27.060: "Mobile Station (MS) supporting Packet Switched Services".

[11] 3GPP TS 29.061: "Interworking between the Public Land Mobile Network (PLMN) supporting Packet Based Services and Packet Data Networks (PDN)".

[11A] 3GPP TS 29.162: "Interworking between the IM CN subsystem and IP networks".

[11B] 3GPP TS 29.163: "Interworking between the IP Multimedia (IM) Core Network (CN) subsystem and Circuit Switched (CS) networks".

[11C] 3GPP TS 29.161: "Interworking between the Public Land Mobile Network (PLMN) supporting Packet Based Services with Wireless Local Access and Packet Data Networks (PDN)"

[11D] 3GPP TS 29.079: "Optimal Media Routeing within the IP Multimedia Subsystem".

[12] 3GPP TS 29.207 Release 6: "Policy control over Go interface".

[12A] 3GPP TS 29.273: "Evolved Packet System (EPS); 3GPP EPS AAA interfaces".

[13] Void.

[13A] 3GPP TS 29.209 Release 6: "Policy control over Gq interface".

[13B] 3GPP TS 29.212: "Policy and Charging Control (PCC); Reference points".

[13C] 3GPP TS 29.213: "Policy and charging control signalling flows and Quality of Service (QoS) parameter mapping".

[13D] 3GPP TS 29.214: "Policy and Charging Control over Rx reference point".

[14] 3GPP TS 29.228: "IP Multimedia (IM) Subsystem Cx and Dx Interfaces; Signalling flows and message contents".

[15] 3GPP TS 29.229: "Cx and Dx Interfaces based on the Diameter protocol, Protocol details".

[15A] 3GPP TS 29.311: "Service Level Interworking for Messaging Services".

[15B] 3GPP TS 31.103: "Characteristics of the IP multimedia services identity module (ISIM) application".

[15C] 3GPP TS 31.102: "Characteristics of the Universal Subscriber Identity Module (USIM) application".

[15D] 3GPP TS 31.111: "Universal Subscriber Identity Module (USIM) Application Toolkit (USAT)".

[16] 3GPP TS 32.240: "Telecommunication management; Charging management; Charging architecture and principles".

[17] 3GPP TS 32.260: "Telecommunication management; Charging management; IP Multimedia Subsystem (IMS) charging".

[17A] 3GPP TS 32.422: "Telecommunication management; Subscriber and equipment trace; Trace control and configuration management".

[18] 3GPP TS 33.102: "3G Security; Security architecture".

[19] 3GPP TS 33.203: "Access security for IP based services".

[19A] 3GPP TS 33.210: "3G security; Network Domain Security (NDS); IP network layer security".

[19B] 3GPP TS 36.304: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) procedures in idle mode".

[19C] 3GPP TS 33.328: "IP Multimedia Subsystem (IMS) media plane security".

[19D] 3GPP TS 33.310: "Network Domain Security (NDS); Authentication Framework (AF)".

[19E] 3GPP TS 36.413: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); S1 Application Protocol (S1AP)".

[19F] 3GPP TS 36.331: "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification".

[19G] 3GPP TS 38.331: " NR; Radio Resource Control (RRC); Protocol specification".

[20] 3GPP TS 44.018: "Mobile radio interface layer 3 specification; Radio Resource Control (RRC) protocol".

[20A] IETF RFC 2401 (November 1998): "Security Architecture for the Internet Protocol".

[20B] IETF RFC 1594 (March 1994): "FYI on Questions and Answers to Commonly asked "New Internet User" Questions".

[20C] Void.

[20D] Void.

[20E] IETF RFC 2462 (November 1998): "IPv6 Stateless Address Autoconfiguration".

[20F] IETF RFC 2132 (March 1997): "DHCP Options and BOOTP Vendor Extensions".

[20G] IETF RFC 2234 (November 1997): "Augmented BNF for Syntax Specification: ABNF".

[21] Void.

[22] IETF RFC 3966 (December 2004): "The tel URI for Telephone Numbers".

[23] IETF RFC 4733 (December 2006): "RTP Payload for DTMF Digits, Telephony Tones and Telephony Signals".

[24] IETF RFC 6116 (March 2011): "The E.164 to Uniform Resource Identifiers (URI) Dynamic Delegation Discovery System (DDDS) Application (ENUM)".

[25] IETF RFC 6086 (October 2009): "Session Initiation Protocol (SIP) INFO Method and Package Framework".

[25A] Void.

[26] IETF RFC 3261 (June 2002): "SIP: Session Initiation Protocol".

[27] IETF RFC 3262 (June 2002): "Reliability of provisional responses in Session Initiation Protocol (SIP)".

[27A] IETF RFC 3263 (June 2002): "Session Initiation Protocol (SIP): Locating SIP Servers".

[27B] IETF RFC 3264 (June 2002): "An Offer/Answer Model with Session Description Protocol (SDP)".

[28] IETF RFC 6665 (July 2012): "SIP Specific Event Notification".

[28A] Void.

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[270] 3GPP TS 38.300: "NR; NR and NG-RAN Overall Description; Stage 2".

[271] 3GPP TS 38.321: "NR; Medium Access Control (MAC) protocol specification".

[272] 3GPP TS 23.221: "Architectural requirements".

[273] 3GPP TS 29.514: "5G System; Policy Authorization Service; Stage 3".

[274] 3GPP TS 29.562: "Home Subscriber Server (HSS) Services for Interworking with the IP Multimedia Subsystem (IMS); Stage 3".

[275] 3GPP TS 23.502: "Procedures for the 5G System; Stage 2".

[276] 3GPP TS 26.238: "Uplink Streaming".

[277] IETF RFC 4574 (August 2006): "The Session Description Protocol (SDP) Label Attribute".

[278] IETF RFC 9027 (June 2021): "Assertion Values for Resource Priority Header and SIP Priority Header Claims in Support of Emergency Services Networks".

[279] IETF RFC 8443 (August 2018): "Personal Assertion Token (PASSporT) Extension for Resource Priority Authorization".

[280] IETF RFC 7230 (June 2014): "Hypertext Transfer Protocol (HTTP/1.1): Message Syntax and Routing".

[281] IETF RFC 7231 (June 2014): "Hypertext Transfer Protocol (HTTP/1.1): Semantics and Content".

[282] IETF RFC 7232 (June 2014): "Hypertext Transfer Protocol (HTTP/1.1): Conditional Requests".

[283] IETF RFC 7233 (June 2014): "Hypertext Transfer Protocol (HTTP/1.1): Range Requests".

[284] IETF RFC 7234 (June 2014): "Hypertext Transfer Protocol (HTTP/1.1): Caching".

[285] IETF RFC 7235 (June 2014): "Hypertext Transfer Protocol (HTTP/1.1): Authentication".

[286] IETF RFC 7616 (September 2015): "HTTP Digest Access Authentication".

[287] IETF RFC 8760 (March 2020): "The Session Initiation Protocol (SIP) Digest Access Authentication Scheme".

[288] 3GPP TS 29.510: "5G System; Network function repository services; Stage 3".

[289] IETF RFC 8445 (July 2018): "Interactive Connectivity Establishment (ICE): A Protocol for Network Address Translator (NAT) Traversal".

[290] IETF RFC 8839 (January 2021): "Session Description Protocol (SDP) Offer/Answer Procedures for Interactive Connectivity Establishment (ICE)".

[291] IETF RFC 8489 (February 2020): "Session Traversal Utilities for NAT (STUN)".

[292] IETF RFC 8656 (February 2020): "Traversal Using Relays around NAT (TURN): Relay Extensions to Session Traversal Utilities for NAT (STUN)".

[293] IETF RFC 8981 (February 2021): "Temporary Address Extensions for Stateless Address Autoconfiguration in IPv6".

[294] IETF RFC 9410 (July 2023): "Handling of Identity Header Errors for Secure Telephone Identity Revisited (STIR)".

[295] 3GPP TS 38.413: "NG-RAN; NG Application Protocol (NGAP)".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply.

3GPP PS data off status: indicates state of usage of the 3GPP PS data off. 3GPP PS data off status at the UE can be either "active" or "inactive".

Country: For the purposes of emergency service URNs in the present document, i.e. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69], an ISO 3166-1 alpha-2 code as specified in ISO 3166-1 [207] is used to identify a region or a country.

Entry point: In the case that "border control concepts", as specified in 3GPP TS 23.228 [7], are to be applied in an IM CN subsystem, then these are to be provided by capabilities within the IBCF, and the IBCF acts as an entry point for this network (instead of the I-CSCF). In this case the IBCF and the I-CSCF can be co-located as a single physical node. If "border control concepts" are not applied, then the I-CSCF is considered as an entry point of a network. If the P-CSCF is in the home network, then the I-CSCF is considered as an entry point for this document. Similary, in case that

"border control concepts", as specified in 3GPP TS 23.218 [5], are to be applied in an ISC interface, then these are to be provided by capabilities within the ISC gateway function, and the ISC gateway function acts as an entry point for this network.

Exit point: If operator preference requires the application of "border control concepts" as specified in 3GPP TS 23.228 [7], then these are to be provided by capabilities within the IBCF, and requests sent towards another network are routed via a local network exit point (IBCF), which will then forward the request to the other network (discovering the entry point if necessary). Similary, in case that "border control concepts", as specified in 3GPP TS 23.218 [5], are to be applied in an ISC interface, then these are to be provided by capabilities within the ISC gateway function, and requests sent towards another network are routed via a local network exit point (ISC gateway function).

Geo-local number: Either a geo-local service number as specified in 3GPP TS 23.228 [7] or a number in non-international format according to an addressing plan used at the current physical location of the user.

Home-local number: Either a home local service number as specified in 3GPP TS 23.228 [7] or a number in non-international format according to an addressing plan used in the home network of the user.

Main URI: In the case that the UE supports RFC 6140 [191] and performs the functions of an external attached network, the main URI is the URI which is used for the registration procedures in the To header of the REGISTER request as specified in RFC 6140 [191]; it represents the public user identities associated to that UE.

Newly established set of security associations: Two pairs of IPsec security associations that have been created at the UE and/or the P-CSCF after the 200 (OK) response to a REGISTER request was received.

Old set of security associations: Two pairs of IPsec security associations still in existence after another set of security associations has been established due to a successful authentication procedure.

Temporary set of security associations: Two pairs of IPsec security associations that have been created at the UE and/or the P-CSCF, after an authentication challenge within a 401 (Unauthorized) response to a REGISTER request was received. The SIP level lifetime of such created security associations will be equal to the value of reg-await-auth timer.

Integrity protected: See 3GPP TS 33.203 [19]. Where a requirement exists to send information "integrity-protected" the mechanisms specified in 3GPP TS 33.203 [19] are used for sending the information. Where a requirement exists to check that information was received "integrity-protected", then the information received is checked for compliance with the procedures as specified in 3GPP TS 33.203 [19].

Instance ID: An URN generated by the device that uniquely identifies a specific device amongst all other devices, and does not contain any information pertaining to the user (e.g., in GPRS instance ID applies to the Mobile Equipment rather than the UICC). The public user identity together with the instance ID uniquely identifies a specific UA instance. If the device has an IMEI available, it generates an instance ID based on its IMEI as defined in 3GPP TS 23.003 [3] clause 13. If the device has an MEID as defined in 3GPP2 S.R0048-A [86F] available, it generates an instance ID based on its MEID as defined in RFC 8464 [187]. If the device does not have an IMEI available and does not have an MEID available, the instance ID is generated as a string representation of a UUID as a URN as defined in RFC 4122 [154].

Resource reservation: Mechanism for reserving bearer resources that is required for certain access technologies.

Local preconditions: The indication of segmented status preconditions for the local reservation of resources as specified in RFC 3312 [30].

Alias URI, Alias SIP URI: A URI is an alias of another URI if the treatment of both URIs is identical, i.e. both URIs belong to the same set of implicitly registered public user identities, and are linked to the same service profile, and are considered to have the exact same service configuration for each and every service.

NOTE: The S-CSCF recognizes that a given URI is an alias of another URI using the grouping sent from the HSS (see 3GPP TS 29.228 [14]).

Globally Routeable SIP URI: a SIP URI of which the hostname part can be resolved to the IP address of the entry entity of the network reponsible for the identity represented by the userpart.

Initial registration: The registration procedure for a public user identity initiated by the UE in the absence of any valid registration.

Registration expiration interval: An indication on how long a registration is valid, indicated using the Expires header field, or the "expires" header field parameter within the Contact header field, according to the procedures specified in RFC 3261 [26].

Re-registration: The registration procedure initiated by the UE to refresh or update an already existing registration for a public user identity.

Registration of an additional public user identity: The registration procedure initiated by the UE to explicitly register an additional public user identity during the life time of the registration of another registered public user identity, where both public user identities have the same contact address and P-CSCF.

Emergency registration: A special registration that relates to binding of a public user identity to a contact address used for emergency service.

Initial emergency registration: An emergency registration that is also an initial registration.

Emergency reregistration: An emergency registration that is also a reregistration.

Back-to-Back User Agent (B2BUA): As given in RFC 3261 [26]. In addition, for the usage in the IM CN subsystem, a SIP element being able to handle a collection of "n" User Agents (behaving each one as UAC and UAS, according to SIP rules), which are linked by some application logic that is fully independent of the SIP rules.

UE private IP address: It is assumed that the NAT device performs network address translation between a private and a public network with the UE located in the private network and the IM CN subsystem in the public network. The UE is assumed to be configured with a private IP address. This address will be denoted as UE private IP address.

UE public IP address: The NAT device is assumed to be configured with one (or perhaps more) public address(es). When the UE sends a request towards the public network, the NAT replaces the source address in the IP header of the packet, which contains the UE private IP address, with a public IP addressed assigned to the NAT. This address will be denoted as UE public IP address.

Encapsulating UDP header: For the purpose of performing UDP encapsulation according to RFC 3948 [63A] each IPsec ESP packet is wrapped into an additional UDP header. This header is denoted as Encapsulating UDP header.

Port_Uenc: In most residential scenarios, when the NAT device performs address translation, it also performs translation of the source port found in the transport layer (TCP/UDP) headers. Following RFC 3948 [63A], the UE will use port 4500 as source port in the encapsulating UDP header when sending a packet. This port is translated by the NAT into an arbitrarily chosen port number which is denoted as port_Uenc.

Multiple registrations: An additional capability of the UE, P-CSCF and S-CSCF, such that the UE (as identified by the private user identity and instance-id), can create multiple simultaneous registration bindings (flows), associated with one or more contact addresses, to any public user identity, Without this capability, a new registration from the UE for a public user identity replaces the existing registration binding, rather than merely creating an additional binding.

IMS flow set: An IMS flow set is a set of flows as defined in RFC 5626 [92]. The flows in an IMS flow set are determined by a combination of transport protocol, IP addresses, and ports. An IMS flow set is established by a successful IMS registration procedure.

NOTE: For IPsec, the ports associated with the flow set include protected client ports and protected server ports as defined in 3GPP TS 33.203 [19] and an IMS flow set is made up of the following four flows:

- Flow 1: (IP address UE, port_uc) <--> (IP address P-CSCF, port_ps) over TCP;
- Flow 2: (IP address UE, port_uc) <--> (IP address P-CSCF, port_ps) over UDP;
- Flow 3: (IP address UE, port_us) <--> (IP address P-CSCF, port_pc) over TCP; and
- Flow 4: (IP address UE, port_us) <--> (IP address P-CSCF, port_pc) over UDP.

NOTE 3: For IPsec, according to 3GPP TS 33.203 [19], the P-CSCF can only select among flows 3 or 4 when forwarding requests towards the UE. According to 3GPP TS 33.203 [19], flow 2 is only used for UE generated requests and responses. The P-CSCF uses flow 2 to identify the correct IMS flow set.

NOTE 4: An IMS flow set can be considered as a realisation of a logical flow as used in RFC 5626 [92]. But this definition does not depend on any particular definition of a logical flow.

NOTE 5: For TLS, the ports associated with the flow set include a protected client port and a protected server port and an IMS flow set is made up of the following flow:

- (IP address UE, port) <--> (IP address P-CSCF, port) over TCP.

NOTE 6: For SIP digest without TLS, an IMS flow set is as defined in RFC 5626 [92].

IMS flow token: A IMS flow token is uniquely associated with a IMS flow set. When forwarding a request destined towards the UE, the P-CSCF selects the flow from the IMS flow set denoted by the IMS flow token as appropriate according to 3GPP TS 33.203 [19] and RFC 3261 [26].

IMS registration related signalling: Signalling carrying the SIP messages sent by the UE and the network during the IMS registration procedure, i.e., SIP REGISTER request, SIP SUBSCRIBE request, and their responses.

IP Association: A mapping at the P-CSCF of a UE's packet source IP address, the "sent-by" parameter in the Via header field, and, conditionally, the port with the identities of the UE. This association corresponds to the IP address check table specified in 3GPP TS 33.203 [19].

Authorised Resource-Priority header field: a Resource-Priority header field that is either received from another entity in the trust domain relating to the Resource-Priority header field, or which has been identified as generated by a subscriber known to have such priority privileges for the resource priority namespace and level of priority used within that namespace.

Temporarily authorised Resource-Priority header field: a Resource Priority header field that has been temporarily approved by the P-CSCF, the S-CSCF, or an IBCF. Temporarily authorised Resource-Priority heaer field appears in an INVITE request only, and is applied only in the direction P-CSCF to S-CSCF to AS, S-CSCF to AS, or IBCF to S-CSCF to AS, for the request, and the reverse direction for 1xx responses to that request. Subsequent requests in the same dialog will require an authorised Resource-Priority header field in order to obtain priority privileges. It is only valid when all entities are in the same trust domain for the Resource-Priority header field.

Network-initiated resource reservation: A mechanism of resource reservation where the IP-CAN on the behalf of network initiates the resources to the UE.

Trace depth: When SIP signalling is logged for debugging purposes, trace depth is the level of detail of what is logged.

P-CSCF restoration procedures: the procedures for the IP-CAN and the UE to handle P-CSCF service interruption scenarios (see 3GPP TS 23.380 [7D]).

HSS based P-CSCF restoration procedures: the procedures for the IP-CAN, the IM CN subsystem, the HSS and the UE to handle P-CSCF service interruption scenarios (see 3GPP TS 23.380 [7D]). In 5GS the procedure is called UDM/HSS based P-CSCF restoration (see 3GPP TS 23.380 [7D]) since the UDM participates in the procedure.

PCRF based P-CSCF restoration procedures: the procedures for the IP-CAN, the IM CN subsystem, the PCRF and the UE to handle P-CSCF service interruption scenarios (see 3GPP TS 23.380 [7D]). In 5GS the procedure is called PCF based P-CSCF restoration (see 3GPP TS 23.380 [7D]) since the PCF takes the role of the PCRF.

Public network traffic: traffic sent to the IM CN subsystem for processing according to normal rules of the NGN. This type of traffic is known as public network traffic.

Private network traffic: traffic sent to the IM CN subsystem for processing according to an agreed set of rules specific to an enterprise. This type of traffic is known as private network traffic. Private network traffic is normally within a single enterprise, but private network traffic can also exist between two different enterprises if not precluded for regulatory reasons.

NOTE 7: An IP-PBX or application functionality within the IM CN subsystem can change private network traffic to public network traffic and vice versa, by functionality known as "breakout" or "breakin" to the private network. As such a SIP transaction can be variously private network traffic and public network traffic on different hops across a SIP network.

Privileged sender: A privileged sender is allowed to send SIP messages where the identities in P-Asserted-Identity will be passed on in the P-CSCF and are not subject to further processing in the P-CSCF.

S-CSCF restoration procedures: the procedures for the IM CN subsystem and the UE to handle S-CSCF service interruption scenarios (see 3GPP TS 23.380 [7D]).

Loopback routeing: A method of routeing a SIP request back to the visited network for local breakout according to the roaming architecture for voice over IMS with local breakout as specified in 3GPP TS 23.228 [7].

UE performing the functions of an external attached network: an independent network connected to an IMS network over the Gm interface, through a single point and which is seen by the IMS network as a specific UE; e.g. an IP-PBX.

Static Mode of Operation: a mode of operation where the UE performing the functions of an external attached network does not initiate any IMS level registration procedures towards the operator IMS.

Canonical form of a SIP URI: Canoncial form of a SIP URI takes the form "sip:username@domain" as specified in RFC 3261 [26] subclause 10.3. SIP URI comparisons are performed as defined in RFC 3261 [26] subclause 19.1.4.

Originating home network: the home network of a user originating a transaction, and if applicable, the associated dialog.

Originating visited network: the visited network of a user originating a transaction, and if applicable, the associated dialog.

Terminating home network: the home network of a user terminating a transaction, and if applicable, the associated dialog.

Terminating visited network: the visited network of a user terminating a transaction, and if applicable, the associated dialog.

Type of emergency service: The type of emergency service is either an emergency call type standardized by 3GPP (see 3GPP TS 22.101 [8] subclause 10.1) or a similar capability not standardised by 3GPP and defined by national regulatory requirements. The generic (sos) service, identified by urn:service:sos, does not have a type of emergency service (even though usage of the generic (sos) service in the emergency call is defined).

Resource sharing: one dedicated EPS bearer is sharing resources among several ongoing sessions such that the highest GBR (and optionally MBR) to be shared for the set of PCC/QoS rules bound to the same bearer is used as input for the calculation of the GBR (and optionally MBR) of that bearer among the sessions sharing the resources.

Fully-Qualified Domain Name (FQDN): the syntax of the FQDN used in this specification is defined in RFC 3261 [26] subclause 25.1.

Trusted WLAN: A trusted non-3GPP access, where the non-3GPP access is a WLAN IP access.

Untrusted WLAN: An untrusted non-3GPP access, where the non-3GPP access is a WLAN IP access.

Calling number verification status determination: A feature which enables the terminating UE to determine whether number has been verified by the network as specified in RFC 8224 [252].

Calling number verification using signature verification and attestation information: A feature which enables a calling identity validation as specified in RFC 8224 [252] and uses an attestation information to vouch for the accuracy of the source of origin of the call. Attestation information consists of an attestation level and an origination identifier and may be included in the Identity header field as defined in RFC 8588 [261] and in the Attestation-Info and Origination-Id header fields as defined in subclauses 7.2.18 and 7.2.19.

Priority verification using assertion of priority information: A feature which enables validation of a priority level provided in the Resource-Priority header field as specified in RFC 8443 [279] and, by extension for emergency sessions, the header field value "psap-callback" provided in the Priority header field as specified in RFC 9027 [278]. As specified in RFC 8443 [279] the Identity header field is used for the purpose of authentication of the Resource-Priority header field and, by extension for emergency sessions, the Priority header field value "psap-callback".

For the purposes of the present document, the following terms and definitions given in RFC 3261 [26] apply (unless otherwise specified see clause 6).

Client
Dialog
Final response
Header
Header field
Loose routeing

Method
Option-tag (see RFC 3261 [26] subclause 19.2)
Provisional response
Proxy, proxy server
Recursion
Redirect server
Registrar
Request
Response
Server
Session
(SIP) transaction
Stateful proxy
Stateless proxy
Status-code (see RFC 3261 [26] subclause 7.2)
Tag (see RFC 3261 [26] subclause 19.3)
Target Refresh Request
User agent client (UAC)
User agent server (UAS)
User agent (UA)

For the purposes of the present document, the following terms and definitions given in 3GPP TS 23.002 [2] subclause 4.1.1.1 and subclause 4a.7 apply:

3GPP AAA proxy
3GPP AAA server
Breakout Gateway Control Function (BGCF)
Call Session Control Function (CSCF)
Home Subscriber Server (HSS)
Location Retrieval Function (LRF)
Media Gateway Control Function (MGCF)
MSC Server enhanced for IMS centralized services
Multimedia Resource Function Processor (MRFP)
Packet Data Gateway (PDG)
Subscription Locator Function (SLF)
WLAN UE

For the purposes of the present document, the following terms and definitions given in 3GPP TS 23.122 [4C] apply:

Equivalent Home PLMN (EHPLMN)
Home PLMN (HPLMN)
Visited PLMN (VPLMN)

For the purposes of the present document, the following terms and definitions given in 3GPP TS 23.218 [5] subclauses 3.1, 8 and 13 apply:

Filter criteria
Initial filter criteria
Initial request
ISC gateway function
Media Resource Broker (MRB)
Multimedia Resource Function Controller (MRFC)
Standalone transaction
Subsequent request

For the purposes of the present document, the following terms and definitions given in 3GPP TS 23.228 [7] subclauses 3.1, 4.3.3.1, 4.3.6, 4.6, 4.13, 4.15a, 5.2, 5.4.12.1, 5.10, annex U, and annex W apply:

Border control concepts
Geo-local service number
Home local service number
Implicit registration set
Interconnection Border Control Function (IBCF)

Interrogating-CSCF (I-CSCF)
IMS Application Level Gateway (IMS-ALG)
IMS application reference
IMS Application Reference Identifier (IARI)
IMS communication service
IMS Communication Service Identifier (ICSI)
IMS Services for roaming users in deployments without IMS-level roaming interfaces
Local service number
IP-Connectivity Access Network (IP-CAN)
P-CSCF enhanced for WebRTC (eP-CSCF)
Policy and Charging Rule Function (PCRF)
Private user identity
Proxy-CSCF (P-CSCF)
Public Service Identity (PSI)
Public user identity
Roaming Architecture for Voice over IMS with Local Breakout
Serving-CSCF (S-CSCF)
Statically pre-configured PSI
WebRTC IMS Client (WIC)

For the purposes of the present document, the following terms and definitions given in 3GPP TS 23.292 [7C] apply:

ICS UE
SCC AS

For the purposes of the present document, the following terms and definitions given in 3GPP TS 23.167 [4B] apply:

eCall over IMS
Emergency-CSCF (E-CSCF)
Geographical location information
Location identifier
Location information

For the purposes of the present document, the following terms and definitions given in 3GPP TR 33.203 [19] apply:

GPRS-IMS-Bundled Authentication (GIBA)
Port_pc
Port_ps
Port_uc
Port_us
Protected server port
Protected client port
spi_uc
spi_us

For the purposes of the present document, the following terms and definitions given in 3GPP TR 21.905 [1] apply:

IMS Credentials (IMC)
International Mobile Equipment Identity (IMEI)
IMS SIM (ISIM)
Serial NumbeR (SNR)
Type Approval Code (TAC)
Universal Integrated Circuit Card (UICC)
Universal Subscriber Identity Module (USIM)
User Equipment (UE)

For the purposes of the present document, the following terms and definitions given in RFC 2401 [20A] Appendix A apply:

Security association

A number of different security associations exist within the IM CN subsystem and within the underlying access transport. Within this document this term specifically applies to either:

i) the security association that exists between the UE and the P-CSCF. For this usage of the term, the term "security association" only applies to IPsec. This is the only security association that has direct impact on SIP; or

ii) the security association that exists between the WLAN UE and the PDG. This is the security association that is relevant to the discussion of Interworking WLAN as the underlying IP-CAN.

For the purposes of the present document, the following terms and definitions given in 3GPP TS 23.234 [7A] apply.

Interworking WLAN

For the purposes of the present document, the following terms and definitions given in ITU-T E.164 [57] apply:

International public telecommunication number

For the purposes of the present document, the following terms and definitions given in RFC 5012 [91] apply:

Emergency service identifier
Emergency service URN
Public Safety Answering Point (PSAP)
PSAP URI

For the purposes of the present document, the following terms and definitions given in RFC 5627 [93] apply:

Globally Routable User Agent URI (GRUU)

For the purposes of the present document, the following terms and definitions given in RFC 5626 [92] apply:

Flow

For the purposes of the present document, the following terms and definitions given in 3GPP TS 33.310 [19D] annex E and documents referenced therein:

TLS session

For the purposes of the present document, the following terms and definitions given in 3GPP TS 24.292 [8O] apply:

CS media

For the purposes of the present document, the following terms and definitions given in 3GPP TS 24.301 [8J] apply:

IMS Voice over PS Session (IMSVoPS) indicator
Persistent EPS bearer context

For the purposes of the present document, the following terms and definitions given in 3GPP TS 33.328 [19C] apply:

End-to-access edge security

For the purposes of the present document, the following terms and definitions given in 3GPP2 S.R0048-A v4.0 [86F] apply:

Mobile Equipment Identity (MEID)
Manufacturer code
Serial number

For the purposes of the present document, the following terms and definitions given in 3GPP TS 24.302 [8U] apply:

Restrictive non-3GPP access network
S2a
S2b
S2c
Trusted non-3GPP access
Untrusted non-3GPP access
Unauthenticated IMSI
Firewall traversal tunnel

For the purposes of the present document, the following terms and definitions given in 3GPP TS 32.240 [16] apply:

Charging Data Function (CDF);
Charging Data Record (CDR)
Online Charging Function (OCF)

For the purposes of the present document, the following terms and definitions given in 3GPP TS 32.260 [17] apply:

IM CN subsystem Charging Identifier (ICID)

For the purposes of the present document, the following terms and definitions given in RFC 8119 [230] apply:

Service access number

For the purposes of the present document, the following terms and definitions given in 3GPP TS 22.101 [1A] apply:

eCall
Minimum Set of Data (MSD)

For the purposes of the present document, the following terms and definitions given in 3GPP TS 22.011 [1C] apply:

3GPP PS data off
3GPP PS data off exempt services

For the purposes of the present document, the following terms and definitions given in 3GPP TS 23.402 [7E] apply.

TWAN

For the purposes of the present document, the following terms and definitions given in 3GPP TS 24.604 [8ZG] apply.

Diverting user
Diverted-to party

For the purposes of the present document, the following terms and definitions given in 3GPP TS 23.221 [272] apply:

Restricted Local Operator Services

For the purposes of the present document, the following terms and definitions given in 3GPP TS 23.501 [257] apply:

NR RedCap
Stand-alone Non-Public Network

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

1xx	A status-code in the range 101 through 199, and excluding 100
18x	A status-code in the range 180 through 189
2xx	A status-code in the range 200 through 299
3xx	A status-code in the range 300 through 399
4xx	A status-code in the range 400 through 499
5GC	5G Core Network
5GS	5G System
5G-AN	5G Access Network
5xx	A status-code in the range 500 through 599
6xx	A status-code in the range 600 through 699
AAA	Authentication, Authorization and Accounting
ANBR	Access Network Bitrate Recommendation
APN	Access Point
APN	Access Point Name
AS	Application Server
ATCF	Access Transfer Control Function

AUTN	Authentication TokeN
AVP	Attribute-Value Pair
B2BUA	Back-to-Back User Agent
BFCP	Binary Floor Control Protocol
BGCF	Breakout Gateway Control Function
c	conditional
BRAS	Broadband Remote Access Server
BSSID	Basic Service Set Identifier
CCF	Charging Collection Function
CDF	Charging Data Function
CDR	Charging Data Record
CH	Credentials Holder
CK	Ciphering Key
CN	Core Network
CPC	Calling Party's Category
CLF	Connectivity session Location and repository Function
CSCF	Call Session Control Function
DHCP	Dynamic Host Configuration Protocol
DNN	Data Network Name
DNS	Domain Name System
DOCSIS	Data Over Cable Service Interface Specification
DRVCC	Dual Radio Voice Call Continuity
DTD	Document Type Definition
DTLS	Datagram Transport Layer Security
DTMF	Dual Tone Multi Frequency
DVB	Digital Video Broadcast
DVB-RCS2	Second Generation DVB Interactive Satellite System
e2ae-security	End-to-access edge security
EATF	Emergency Access Transfer Function
EC	Emergency Centre
ECF	Event Charging Function
ECI	E-UTRAN Cell Identity
ECN	Explicit Congestion Notification
E-CSCF	Emergency CSCF
EF	Elementary File
eP-CSCF	P-CSCF enhanced for WebRTC
ePDG	Evolved Packet Data Gateway
EPS	Evolved Packet System
FAP	cdma2000® 1x Femtocell Access Point
FQDN	Fully Qualified Domain Name
GBA	Generic Bootstrapping Architecture
GBR	Guaranteed Bit Rate
GCID	GPRS Charging Identifier
GGSN	Gateway GPRS Support Node
GPON	Gigabit-capable Passive Optical Networks
GPRS	General Packet Radio Service
GRUU	Globally Routable User agent URI
GSTN	General Switched Telephone Network
HPLMN	Home PLMN
HSS	Home Subscriber Server
HTTP	HyperText Transfer Protocol
i	irrelevant
IARI	IMS Application Reference Identifier
IBCF	Interconnection Border Control Function
ICE	Interactive Connectivity Establishment
I-CSCF	Interrogating CSCF
ICS	Implementation Conformance Statement
ICID	IM CN subsystem Charging Identifier
ICSI	IMS Communication Service Identifier
ID	Identifier
IK	Integrity Key
IKEv2	Internet Key Exchange Protocol Version 2

IM	IP Multimedia
IMC	IMS Credentials
IMEI	International Mobile Equipment Identity
IMS	IP Multimedia core network Subsystem
IMS-AGW	IMS Access Gateway
IMS-ALG	IMS Application Level Gateway
IMSI	International Mobile Subscriber Identity
IMSVoPS	IMS Voice over PS Session
IOI	Inter Operator Identifier
IP	Internet Protocol
IP-CAN	IP-Connectivity Access Network
IPsec	IP security
IPv4	Internet Protocol version 4
IPv6	Internet Protocol version 6
ISC	IP Multimedia Subsystem Service Control
ISIM	IM Subscriber Identity Module
I-WLAN	Interworking – WLAN
IWF	Interworking Function
KMS	Key Management Service
LRF	Location Retrieval Function
m	mandatory
MAC	Message Authentication Code
MBR	Maximum guaranteed Bit Rate
MCC	Mobile Country Code
MCPTT	Mission Critical Push To Talk
MEID	Mobile Equipment IDentity
MGCF	Media Gateway Control Function
MGW	Media Gateway
MNC	Mobile Network Code
MRB	Media Resource Broker
MRFC	Multimedia Resource Function Controller
MRFP	Multimedia Resource Function Processor
MSC	Mobile-services Switching Centre
MSD	Minimum Set of emergency related Data
MSRP	Message Session Relay Protocol
n/a	not applicable
NAI	Network Access Identifier
NA(P)T	Network Address (and Port) Translation
NASS	Network Attachment Subsystem
NAT	Network Address Translation
NCC	Network Control Center
NCC_ID	Network Control Center Identifier
NID	Network Identifier
NP	Number Portability
o	optional
OCF	Online Charging Function
OLI	Originating Line Information
OMR	Optimal Media Routeing
PCC	Policy and Charging Control
PCF	Policy Control Function
PCO	Protocol Configuration Options
PCRF	Policy and Charging Rules Function
P-CSCF	Proxy CSCF
PDG	Packet Data Gateway
PDN	Packet Data Network
PDP	Packet Data Protocol
PDU	Protocol Data Unit
P-GW	PDN Gateway
PICS	Protocol Implementation Conformance Statement
PIDF-LO	Presence Information Data Format Location Object
PLMN	Public Land Mobile Network
PSAP	Public Safety Answering Point

PSI	Public Service Identity
PSTN	Public Switched Telephone Network
QCI	QoS Class Identifier
QoS	Quality of Service
RAND	RANDom challenge
RCS	Return Channel via Satellite
RCST	Return Channel via Satellite Terminal
RES	RESponse
RLOS	Restricted Local Operator Services
RTCP	Real-time Transport Control Protocol
RTP	Real-time Transport Protocol
SAC	Service Area Code
SAI	Service Area Identifier
SBA	Service Based Architecture
SBI	Service Based Interface
S-CSCF	Serving CSCF
SCTP	Stream Control Transmission Protocol
SDES	Session Description Protocol Security Descriptions for Media Streams
SDP	Session Description Protocol
SDU	Service Data Unit
SIP	Session Initiation Protocol
SLF	Subscription Locator Function
SNPN	Stand-alone Non-Public Network
SNR	Serial Number
SQN	SeQuence Number
SRVCC	Single Radio Voice Call Continuity
STUN	Session Traversal Utilities for NAT
SVN	Satellite Virtual Network
SVN-MAC	SVN Medium Access Control label
TAC	Type Approval Code
TFT	Traffic Flow Template
TP	Telepresence
TLS	Transport Layer Security
TRF	Transit and Roaming Function
TURN	Traversal Using Relay NAT
TWAG	Trusted WLAN Access Gateway
TWAN	Trusted WLAN
UA	User Agent
UAC	User Agent Client
UAS	User Agent Server
UDM	Unified Data Management
UDPTL	UDP Transport Layer
UDVM	Universal Decompressor Virtual Machine
UE	User Equipment
UICC	Universal Integrated Circuit Card
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
URN	Uniform Resource Name
USAT	Universal Subscriber Identity Module Application Toolkit
USIM	Universal Subscriber Identity Module
VPLMN	Visited PLMN
WebRTC	Web Real-Time Communication
WIC	WebRTC IMS Client
WLAN	Wireless Local Area Network
x	prohibited
xDSL	Digital Subscriber Line (all types)
XGPON1	10 Gigabit-capable Passive Optical Networks
XMAC	expected MAC
XML	eXtensible Markup Language

3A Interoperability with different IP-CAN

The IM CN subsystem can be accessed by UEs resident in different types of IP-CAN. The main body of this document, and annex A, are general to UEs and IM CN subsystems that are accessed using any type of IP-CAN. Requirements that are dependent on the type of IP-CAN are covered in annexes B, E, H, L, M, O, Q, R, S, U and W.

At any given time, for a given SIP transaction or dialog, the UE sees only one type of IP-CAN, as reported to it by the lower layers. The UE follows the procedures of the IP-CAN specific annex related to the last type of IP-CAN reported, even if it is different to one used previously. In particular, handover at the radio layers between two different access technologies can result in such a change while the dialog or transaction proceeds.

At any given time, for a given SIP transaction or dialog, the P-CSCF sees only one type of IP-CAN, as determined by interface to a particular resource architecture, e.g. policy and charging control, and by the access technology reported to it over that interface, or in the absence of this, by preconfiguration in the system. The P-CSCF follows the procedures of the IP-CAN specific annex related to the last type of IP-CAN determined, even if it is different to one used previously. In particular, handover at the radio layers between two different access technologies can result in such a change while the dialog or transaction proceeds.

It is the responsibility of the IP-CAN to ensure that usage of different bearer resources are synchronised on the handover from one IP-CAN to another, e.g. so that a signalling bearer provided by one IP-CAN is a signalling bearer (if provided by that IP-CAN) after handover, and that the appropriate QoS and resource reservation exists after handover. There is no SIP signalling associated with handover at the IP-CAN, and therefore no change in SIP state at one entity is signalled to the peer SIP entity when handover occurs.

In particular the following constraints exist that can have an impact on P-CSCF usage:

1) some IP-CANs can explicitly label a bearer as a signalling bearer, while others provide a bearer that has appropriate QoS, but no explicit labelling. Therefore if handover occurs from an IP-CAN with explicit labelling, to an IP-CAN with no explicit labelling, and then back to an IP-CAN with explicit labelling, the signalling will then be on a bearer that is not explicitly labelled; and

2) some IP-CANs support signalling of grouping of media within particular bearers, while others do not. Therefore if handover occurs from an IP-CAN with grouping, to an IP-CAN with no grouping, and then back to an IP-CAN with grouping, the signalled grouping can have been lost.

When a UE supports multiple IP-CANs, but does not support handover between those IP-CANs, the annex specific to that IP-CAN applies unmodified.

Where handover between IP-CANs occurs without a reregistration in the IM CN subsystem, the same identies and security credentials for access to the IM CN subsystem are used before and after the handover.

At the P-CSCF, the access technology can variously use the PCRF or PCF or NASS in support of both signalling and media bearer provision (or indeed use neither). How to determine which applies is up to network dependent rules, but can be specific to the access technology used by each different UE. Not all access technologies are defined for use with NASS, and not all access technologies are defined for use with the PCRF or PCF.

4 General

4.1 Conformance of IM CN subsystem entities to SIP, SDP and other protocols

SIP defines a number of roles which entities can implement in order to support capabilities. These roles are defined in annex A.

Each IM CN subsystem functional entity using an interface at the Gm reference point, the Ma reference point, the Mg reference point, the Mi reference point, the Mj reference point, the Mk reference point, the Ml reference point, the Mm reference point, the Mr reference point, the Mr' reference point, the Cr reference point, the Mw reference point, the I2 reference point, the I4 reference point and the Ici reference point, and also using the IP multimedia Subsystem Service

Control (ISC) Interface, shall implement SIP, as defined by the referenced specifications in Annex A, and in accordance with the constraints and provisions specified in annex A, according to the following roles.

Each IM CN subsystem entity using an interface at the Rc reference point and the Ms reference point shall implement HTTP as defined in RFC 7230 [280], RFC 7231 [281], RFC 7232 [282], RFC 7233 [283], RFC 7234 [284] and RFC 7235 [285].

Each IM CN subsystem entity using an interface at the W2 reference point may implement SIP as an option. The detailed procedures of W2 interface are defined in 3GPP TS 24.371 [8Z].

The Gm reference point, the W2 reference point, the Ma reference point, the Mg reference point, the Mi reference point, the Mj reference point, the Mk reference point, the Ml reference point, the Mm reference point, the Mr reference point, the Mw reference point, the Cr reference point, the I2 reference point, the I4 reference point and the ISC reference point are defined in 3GPP TS 23.002 [2]. The Ici reference point and the Ms reference point are defined in 3GPP TS 23.228 [7]. The Mr' reference point and the Rc reference point are defined in 3GPP TS 23.218 [5].

For SIP:

- The User Equipment (UE) shall provide the User Agent (UA) role, with the exceptions and additional capabilities to SIP as described in subclause 5.1, with the exceptions and additional capabilities to SDP as described in subclause 6.1, and with the exceptions and additional capabilities to SigComp as described in subclause 8.1. The UE shall also provide the access technology specific procedures described in the appropriate access technology specific annex (see subclause 3A and subclause 9.2.2). The UE may include one or several interconnected SIP elements registered as a single logical entity when the UE performs the functions of an external attached network (e.g. an enterprise network). This specification does not place any constraint on the SIP role played by each of the elements as long as the compound entity appears to the IM CM subsystem as a SIP UA with the aforementioned exceptions and additional capabilities except for the modifications defined by the UE performing the functions of an external attached network modifying role in annex A.

NOTE 1: When the UE performs the functions of an external attached network (e.g. an enterprise network), the internal structure of this UE is outside the scope of this specification. It is expected that in the most common case, several SIP elements will be connected to an additional element directly attached to the IM CN subsystem.

- The P-CSCF shall provide the proxy role, with the exceptions and additional capabilities to SIP as described in subclause 5.2, with the exceptions and additional capabilities to SDP as described in subclause 6.2, and with the exceptions and additional capabilities to SigComp as described in subclause 8.2. Under certain circumstances, if the P-CSCF provides an application level gateway functionality (IMS-ALG), the P-CSCF shall provide the UA role with the additional capabilities, as follows:

  a) when acting as a subscriber to or the recipient of event information (see subclause 5.2);

  b) when performing P-CSCF initiated dialog-release, even when acting as a proxy for the remainder of the dialog (see subclause 5.2);

  c) when performing NAT traversal procedures (see subclause 6.7.2);

  d) when performing media plane security procedures (see subclause 5.2); and

  e) when providing in-call access update procedures (see subclause 5.2.14).

  The P-CSCF shall also provide the access technology specific procedures described in the appropriate access technology specific annex (see subclause 3A and subclause 9.2.2).

- The I-CSCF shall provide the proxy role, with the exceptions and additional capabilities as described in subclause 5.3.

- The S-CSCF shall provide the proxy role, with the exceptions and additional capabilities as described in subclause 5.4, and with the exceptions and additional capabilities to SDP as described in subclause 6.3. Under certain circumstances as described in subclause 5.4, the S-CSCF shall provide the UA role with the additional capabilities, as follows:

  a) the S-CSCF shall also act as a registrar. When acting as a registrar, or for the purposes of executing a third-party registration, the S-CSCF shall provide the UA role;

b) as the notifier of event information the S-CSCF shall provide the UA role;

c) when providing a messaging mechanism by sending the MESSAGE method, the S-CSCF shall provide the UA role; and

d) when performing S-CSCF initiated dialog release the S-CSCF shall provide the UA role, even when acting as a proxy for the remainder of the dialog.

- The MGCF shall provide the UA role, with the exceptions and additional capabilities as described in subclause 5.5, and with the exceptions and additional capabilities to SDP as described in subclause 6.4.

- The BGCF shall provide the proxy role, with the exceptions and additional capabilities as described in subclause 5.6.

- The AS, acting as terminating UA, or redirect server (as defined in 3GPP TS 23.218 [5] subclause 9.1.1.1), shall provide the UA role, with the exceptions and additional capabilities as described in subclause 5.7.2, and with the exceptions and additional capabilities to SDP as described in subclause 6.6.

- The AS, acting as originating UA (as defined in 3GPP TS 23.218 [5] subclause 9.1.1.2), shall provide the UA role, with the exceptions and additional capabilities as described in subclause 5.7.3, and with the exceptions and additional capabilities to SDP as described in subclause 6.6.

- The AS, acting as a SIP proxy (as defined in 3GPP TS 23.218 [5] subclause 9.1.1.3), shall provide the proxy role, with the exceptions and additional capabilities as described in subclause 5.7.4.

- The AS, performing 3rd party call control (as defined in 3GPP TS 23.218 [5] subclause 9.1.1.4), shall provide the UA role, with the exceptions and additional capabilities as described in subclause 5.7.5, and with the exceptions and additional capabilities to SDP as described in subclause 6.6. An AS performing media control of an MRFC shall also support the procedures and methods described in subclause 10.2.

NOTE 2: Subclause 5.7 and its subclauses define only the requirements on the AS that relate to SIP. Other requirements are defined in 3GPP TS 23.218 [5].

- The AS, receiving third-party registration requests, shall provide the UA role, with the exceptions and additional capabilities as described in subclause 5.7.

- The MRFC shall provide the UA role, with the exceptions and additional capabilities as described in subclause 5.8, and with the exceptions and additional capabilities to SDP as described in subclause 6.5. The MRFC shall also support the procedures and methods described in subclause 10.3 for media control.

- In inline aware mode, the MRB shall provide the UA role, with the exceptions and additional capabilities as described in subclause 5.8A. In inline unaware mode, the MRB shall provide the proxy role, with the exceptions and additional capabilities as described in subclause 5.8A. The MRB shall also support the procedures and methods described in subclause 10.4 for media control.

- The IBCF shall provide the proxy role, with the exceptions and additional capabilities to SIP as described in subclause 5.10. If the IBCF provides an application level gateway functionality (IMS-ALG), then the IBCF shall provide the UA role, with the exceptions and additional capabilities to SIP as described in subclause 5.10, and with the exceptions and additional capabilities to SDP as described in subclause 6.7. If the IBCF provides screening functionality, then the IBCF may provide the UA role, with the exceptions and additional capabilities to SIP as described in subclause 5.10.

- The E-CSCF shall provide the proxy role, with the exceptions and additional capabilities as described in subclause 5.11. Under certain circumstances as described in subclause 5.11, the E-CSCF shall provide the UA role in accordance with RFC 3323 [33], with the additional capabilities, as follows:

a) when operator policy (e.g. determined by national regulatory requirements applicable to emergency services) allows user requests for suppression of public user identifiers and location information, then the E-CSCF shall provide the UA role, with the exceptions and additional capabilities to SIP as described in subclause 5.11;

b) when performing E-CSCF initiated dialog release the E-CSCF shall provide the UA role, even when acting as a proxy for the remainder of the dialog, e.g. for any of the reasons specified in RFC 6442 [89] or RFC 3323 [33];

c) when acting as a notifier for the dialog event package the E-CSCF shall provide the UA role; and

d) if operator policy allows any LRF to provide a location by value using the mechanism defined in subclause 5.11.3. the E-CSCF shall provide the UA role.

- The LRF shall provide the UA role.

- The ISC gateway function shall provide the proxy role, with the exceptions and additional capabilities to SIP as described in subclause 5.13. If the ISC gateway function provides an application level gateway functionality (IMS-ALG), then the ISC gateway function shall provide the UA role, with the exceptions and additional capabilities to SIP as described in subclause 5.13, and with the exceptions and additional capabilities to SDP as described in subclause 6.7.

- The MSC Server enhanced for ICS shall provide the UA role, with the exceptions and additional capabilities as described in 3GPP TS 24.292 [8O].

- The MSC server enhanced for SRVCC using SIP interface shall provide the UA role, with the exceptions and additional capabilities as described in 3GPP TS 24.237 [8M].

- The MSC server enhanced for DRVCC using SIP interface shall provide the UA role, with the exceptions and additional capabilities as described in 3GPP TS 24.237 [8M].

- The EATF shall provide the UA role, with the exceptions and additional capabilities as described in 3GPP TS 24.237 [8M].

- The ATCF shall:

  a) provide the proxy role, with the exceptions and additional capabilities as described in 3GPP TS 24.237 [8M]; and

  b) provide the UA role, with the exceptions and additional capabilities as described in 3GPP TS 24.237 [8M].

- Where access to the IM CN subsystem is provided using Web Real-Time Communication (WebRTC) in accordance with 3GPP TS 24.371 [8Z], the eP-CSCF shall act as the P-CSCF in regard to the Mw reference point. For SIP, conformance of the eP-CSCF and WIC (or whatever functionality is downloaded to the WIC) is not specified by this document unless 3GPP TS 24.371 [8Z] specifies that these entities act as specified for the interface Gm reference point, in which case existing P-CSCF and UE procedures apply, with the exceptions and additional capabilities as described in 3GPP TS 24.371 [8Z]. For SDP, these entities act as specified for the interface Gm reference point, in which case existing P-CSCF and UE procedures apply, with the exceptions and additional capabilities as described in 3GPP TS 24.371 [8Z].

In addition to the roles specified above, the P-CSCF, the I-CSCF, the IBCF, the S-CSCF, the BGCF, the E-CSCF and the ISC gateway function can act as a UA when providing server functionality to return a final response for any of the reasons specified in RFC 3261 [26].

In addition to the roles specified above the S-CSCF, AS and an entity hosting the additional routeing capabilities as specified in subclause I.3 can act as a UA when providing either client or server functionality when the event package associated with overload control is deployed.

NOTE 3: Annex A can change the status of requirements in referenced specifications. Particular attention is drawn to table A.4 and table A.162 for capabilities within referenced SIP specifications, and to table A.317 and table A.328 for capabilities within referenced SDP specifications. The remaining tables build on these initial tables.

NOTE 4: The allocated roles defined in this clause are the starting point of the requirements from the IETF SIP specifications, and are then the basis for the description of further requirements. Some of these extra requirements formally change the proxy role into a B2BUA. In all other respects other than those more completely described in subclause 5.2 the P-CSCF implements proxy requirements. Despite being a B2BUA a P-CSCF does not implement UA requirements from the IETF RFCs, except as indicated in this specification, e.g., relating to registration event subscription.

NOTE 5: Except as specified in clause 5 or otherwise permitted in RFC 3261, the functional entities providing the proxy role are intended to be transparent to data within received requests and responses. Therefore these entities do not modify message bodies. If local policy applies to restrict such data being passed on, the functional entity has to assume the UA role and reject a request, or if in a response and where such procedures apply, to pass the response on and then clear the session using the BYE method.

All the above entities are functional entities that could be implemented in a number of different physical platforms coexisting with a number of other functional entities. The implementation shall give priority to transactions at one functional entity, e.g. that of the E-CSCF, over non-emergency transactions at other entities on the same physical implementation. Such priority is similar to the priority within the functional entities themselves specified elsewhere in this document.

Additional routeing functionality can be provided to support the ability for the IM CN subsystem to provide transit functionality as specified in Annex I. The additional routeing functionality shall assume the proxy role.

4.2 URI and address assignments

In order for SIP and SDP to operate, the following prerequisite conditions apply:

1) I-CSCFs used in registration are allocated SIP URIs. Other IM CN subsystem entities may be allocated SIP URIs. For example sip:pcscf.home1.net and sip:<impl-specific-info>@pcscf.home1.net are valid SIP URIs. If the user part exists, it is an essential part of the address and shall not be omitted when copying or moving the address. How these addresses are assigned to the logical entities is up to the network operator. For example, a single SIP URI may be assigned to all I-CSCFs, and the load shared between various physical boxes by underlying IP capabilities, or separate SIP URIs may be assigned to each I-CSCF, and the load shared between various physical boxes using DNS SRV capabilities.

2) All IM CN subsystem entities are allocated IP addresses. Any IM CN subsystem entities can be allocated IPv4 only, IPv6 only or both IPv4 and IPv6 addresses. For systems providing access to IM CN subsystem using a GPRS IP-CAN or an EPS IP-CAN this is specified in 3GPP TS 23.221 [6] subclause 5.1. For systems providing access to IM CN subsystem using a cdma2000® packet data subsystem IP-CAN this is specified in subclause M.2.2.1. For systems providing access to IM CN subsystem using a 5GS IP-CAN this is specified in 3GPP TS 23.501 [257], subclause 5.8.2.2.

3) The subscriber is allocated a private user identity by the home network operator. This private user identity is available to the SIP application within the UE. Depending on the network operator, various arrangements exist within the UE for retaining this information:

   a) where an ISIM is present, within the ISIM, see subclause 5.1.1.1A;

   b) where no ISIM is present but USIM is present, the private user identity is derived (see subclause 5.1.1.1A);

   c) neither ISIM nor USIM is present, but IMC is present, within IMC (see subclause 5.1.1.1B.1);

   d) when neither ISIM nor USIM nor IMC is present, the private user identity is available to the UE via other means (see subclause 5.1.1.1B.2).

NOTE 1: 3GPP TS 33.203 [19] specifies that a UE attached to a 3GPP network has an ISIM or a USIM.

NOTE 2: The SIP URIs can be resolved by using any of public DNSs, private DNSs, or peer-to-peer agreements.

4) The subscriber is allocated one or more public user identities by the home network operator. The public user identity shall take the form of SIP URI as specified in RFC 3261 [26] or tel URI as specified in RFC 3966 [22]. At least one of the public user identities is a SIP URI. All registered public user identities are available to the SIP application within the UE, after registration. Depending on the network operator, various arrangements exist within the UE for retaining this information:

   a) where an ISIM is present, at least one public user identity, which is a SIP URI, within the ISIM, see subclause 5.1.1.1A;

   b) where no ISIM is present but USIM is present, a temporary public user identity is derived (see subclause 5.1.1.1A);

   c) neither ISIM nor USIM is present, but IMC is present, within IMC (see subclause 5.1.1.1B.1);

d) when neither ISIM nor USIM nor IMC is present, the public user identities are available to the UE via other means (see subclause 5.1.1.1B.2).

NOTE 3: 3GPP TS 33.203 [19] specifies that a UE attached to a 3GPP network has an ISIM or a USIM.

5) If the UE supports GRUU (see table A.4, item A.4/53) or multiple registrations, then it shall have an Instance ID, in conformance with the mandatory requirements for Instance IDs specified in RFC 5627 [93] and RFC 5626 [92].

6) For each tel URI, there is at least one alias SIP URI in the set of implicitly registered public user identities that is used to implicitly register the associated tel URI.

NOTE 4: For each tel URI, there always exists a SIP URI that has identical user part as the tel URI and the "user" SIP URI parameter equals "phone" (see RFC 3261 [26] subclause 19.1.6), that represents the same public user identity. If a tel URI identifies a subscriber served by the IM CN subsystem, then the hostport parameter of the respective SIP URI contains the home network domain name of the IM CN subsystem to which the subscriber belongs.

6A) Identification of the UE to a PSAP with point of presence in the CS domain is not possible if a tel URI is not included in the set of implicitly registered public user identities. If the included tel URI is associated either with the first entry in the list of public user identities provisioned in the UE or with the temporary public user identity, then a PSAP can uniquely identify the UE if emergency registration is performed.

NOTE 5: The tel URI uniquely identifies the UE by not sharing any of the implicit registered public user identities in the implicit registration set that contains this tel URI.

NOTE 6: Emergency registration is not always needed or supported.

7) The public user identities may be shared across multiple UEs. A particular public user identity may be simultaneously registered from multiple UEs that use different private user identities and different contact addresses. When reregistering and deregistering a given public user identity and associated contact address, the UE will use the same private user identity that it had used during the initial registration of the respective public user identity and associated contact address. If the tel URI is a shared public user identity, then the associated alias SIP URI is also a shared public user identity. Likewise, if the alias SIP URI is a shared public user identity, then the associated tel URI is also a shared public user identity.

8) For the purpose of access to the IM CN subsystem, UEs can be allocated IPv4 only, IPv6 only or both IPv4 and IPv6 addresses. For systems providing access to IM CN subsystem using a GPRS IP-CAN or an EPS IP-CAN this is specified in 3GPP TS 23.221 [6] subclause 5.1 (see subclause 9.2.1 for the assignment procedures). For systems providing access to IM CN subsystem using a cdma2000® network this is specified in subclause M.2.2.1. For systems providing access to IM CN subsystem using a 5GS IP-CAN this is specified in 3GPP TS 23.501 [257], subclause 5.8.2.2.

9) For the purpose of indicating an IMS communication service to the network, UEs are assigned ICSI values appropriate to the IMS communication services supported by the UE, coded as URNs as specified in subclause 7.2A.8.2.

NOTE 7: cdma2000® is a registered trademark of the Telecommunications Industry Association (TIA-USA).

10) E-CSCFs are allocated multiple SIP URIs. The SIP URI configured in the P-CSCF, AS or IBCF to reach the E-CSCF is distinct from the one given by the E-CSCF to the EATF such that EATF can reach the E-CSCF.

11) If the UE supports RFC 6140 [191] and performs the functions of an external attached network, the subscriber is allocated one or usually more public user identities by the home network operator. The public user identity(s) shall be allocated as global numbers in the international format.

4.2A Transport mechanisms

This document makes no requirement on the transport protocol used to transfer signalling information over and above that specified in RFC 3261 [26] clause 18, unless such requirement is defined in the access technology specific annex for the current access technology (see subclause 3A). However, the UE and IM CN subsystem entities shall transport SIP messages longer than 1300 bytes according to the procedures of RFC 3261 [26] subclause 18.1.1, even if a mechanism exists of discovering a maximum transmission unit size longer than 1500 bytes.

NOTE 1: Support of SCTP as specified in RFC 4168 [96] is optional for IM CN subsystem entities implementing the role of a UA or proxy. SCTP transport between the UE and P-CSCF is not supported in the present document. Support of the SCTP transport is currently not described in 3GPP TS 33.203 [19].

For initial REGISTER requests, the UE and the P-CSCF shall apply port handling according to subclause 5.1.1.2 and subclause 5.2.2.

The UE and the P-CSCF shall send and receive request and responses other than initial REGISTER requests on the protected ports as described in 3GPP TS 33.203 [19].

In case of an emergency session if the UE does not have sufficient credentials to authenticate with the IM CN subsystem and regulations allow, the UE and P-CSCF shall send request and responses other than initial REGISTER requests on non protected ports.

NOTE 2: When TCP is used to carry SIP signalling between the UE and the P-CSCF, it is known that there is no NAT between the UE and the P-CSCF and neither TLS nor the multiple registration mechanism is used, then both the UE and the P-CSCF can decide to close an existing TCP connection subject to the conditions described in RFC 3261 [26].

4.2B Security mechanisms

4.2B.1 Signalling security

3GPP TS 33.203 [19] defines the security features and mechanisms for secure access to the IM CN subsystem. This document defines a number of access security mechanisms, as summarised in table 4-1.

Table 4-1: Summary of access security mechanisms to the IM CN subsystem

Mechanism	Authentication	Integrity protection	Use of security agreement in accordance with RFC 3329 [48]	Support (as defined in 3GPP TS 33.203 [19])
IMS AKA plus IPsec ESP (see 3GPP TS 33.203 [19] clause 6) (NOTE 8)	IMS AKA	IPsec ESP	Yes	Mandatory for all UEs containing a UICC, else optional. Mandatory for all P-CSCF, I-CSCF, S-CSCF.
IMS AKA using HTTP Digest AKAv2 without IPSec security association (see 3GPP TS 33.203 [19] annex X)	IMS AKA	TLS session (NOTE 7)	No	Mandatory for all UEs containing a WIC able to access to UICC. Mandatory for all eP-CSCF. Optional for S-CSCF.
SIP digest plus check of IP association (see 3GPP TS 33.203 [19] annex N) (NOTE 2)	SIP digest	None (NOTE 3)	No	Optional for UEs. Optional for P-CSCF, I-CSCF, S-CSCF.
SIP digest plus Proxy Authentication (see 3GPP TS 33.203 [19] annex N) (NOTE 2)	SIP digest	None (NOTE 3)	No	Optional for UEs Optional for P-CSCF, I-CSCF, S-CSCF
SIP digest with TLS (see 3GPP TS 33.203 [19] annex N and annex O)	SIP digest	TLS session	Yes	Optional for UEs. Optional for P-CSCF, I-CSCF, S-CSCF.
NASS-IMS bundled authentication (see 3GPP TS 33.203 [19] annex R) (NOTE 4, NOTE 5)	not applicable (NOTE 1)	None (NOTE 3)	No	No UE support required. Optional for P-CSCF, I-CSCF, S-CSCF.
GPRS-IMS-Bundled authentication (see 3GPP TS 33.203 [19] annex S) (NOTE 5)	not applicable (NOTE 1)	None (NOTE 3)	No	Optional for UEs. Optional for P-CSCF, I-CSCF, S-CSCF.
Trusted node authentication (see 3GPP TS 33.203 [19] annex U)	not applicable (NOTE 6)	None (NOTE 3)	No	No UE support required. Optional for I-CSCF, S-CSCF.
SIP over TLS with client certificate authentication (see 3GPP TS 33.203 [19] annex O)	TLS client certificate	TLS session	No	Mandatory for a UE performing the functions of an external attached network operating in static mode. Optional for IBCF and P-CSCF.

NOTE 1: Authentication is not provided as part of the IM CN subsystem signalling.
NOTE 2: The term "SIP digest without TLS" is used in this specification to refer to both "SIP digest plus check of IP association" and "SIP digest plus Proxy Authentication".
NOTE 3: This security mechanism does not allow SIP requests to be protected using an IPsec security association because it does not perform a key agreement procedure.
NOTE 4: A P-Access-Network-Info aware P-CSCF is required in order to provide NASS-IMS bundled authentication.
NOTE 5: The P-CSCF is restricted to the home network when performing this security mechanism.
NOTE 6: Trusted node authentication. For example the MSC server enhanced for IMS centralized services has authenticated the UE and as a consequence S-CSCF will skip authentication.
NOTE 7: SIP requests received at the eP-CSCF are protected by a TLS session established prior registration (see 3GPP TS 33.203 [19] annex X).
NOTE 8: IMS AKA and IPsec ESP mechanism includes support of "AKAv2-SHA-256" and "AKAv1-MD5" digest algorithms, but "AKAv1-MD5" algorithm is only supported for backward compatibility.

Specification of the mechanisms identified within table 4-1 within this document are provided in clause 5. Subclauses where security procedures are required consist of a general subclause applicable whichever security mechanisms are in use, and a separate subclause for each security mechanism identified by a row within table 4-1.

For access to the IM CN subsystem different than WebRTC TLS is optional to implement and is used only in combination with SIP digest authentication. For WebRTC based access to the IM CN subsystem TLS can be used in combination with IMS AKA using HTTP Digest AKAv2 without IPSec security association. Authentication associated with registration to the IM CN subsystem is applicable to IMS AKA and SIP digest and is covered in subclause 5.1.1 for the UE, subclause 5.2.2 for the P-CSCF and subclause 5.4.1 for the S-CSCF. Additionally, SIP digest allows for authentication to also occur on an initial request for a dialog or a request for a standalone transaction, this additional capability is covered in subclause 5.1.2A and subclause 5.4.3.2.

If a UE that implements SIP digest is configured not to use TLS, then the UE does not establish a TLS session toward the P-CSCF. If a UE supports TLS, then the UE supports TLS as described in 3GPP TS 33.203 [19].

For SIP digest authentication, the P-CSCF can be configured to have TLS required or disabled:

- if TLS is required, the P-CSCF requires the establishment of a TLS session from all SIP digest UEs, in order to access IMS subsequent to registration; or

- if TLS is disabled, the P-CSCF does not allow the establishment of a TLS session from any UE.

NOTE: The mechanism to configure the P-CSCF to have TLS required or disabled is outside the scope of this specification.

SIP digest cannot be used in conjunction with the procedures of Annex F.

For emergency calls, 3GPP TS 33.203 [19] specifies some relaxations, which are further described in the subclauses of this document relating to emergency calls.

3GPP TS 33.210 [19A] defines the security architecture for network domain IP based control planes.
3GPP TS 33.210 [19A] applies for security mechanisms between entities in the IM CN subsystem.

4.2B.2 Media security

3GPP TS 33.328 [19C] defines mechanisms for support of security on the media plane.

This document defines the required elements for signalling the support of media security.

The media security mechanisms are summarised as shown in table 4-2.

Table 4-2: Summary of media security mechanisms to the IM CN subsystem

Mechanism	Applicable to media	Support required by UE	Support required by IM CN subsystem entities	Network support outside IM CN subsystem entities
End-to-access-edge media security using SDES.	RTP based media only.	Support RFC 3329 additions specified in subclause 7.2A.7 and SDP extensions specified in table A.317, items A.317/34, A.317/36 and A.317/37.	P-CSCF (IMS-ALG) is required. P-CSCF support of RFC 3329 additions specified in subclause 7.2A.7 and SDP extensions specified in table A.317, items A.317/34, A.317/36 and A.317/37. (NOTE)	Not applicable.
End-to-access-edge media security using DTLS-SRTP.	RTP based media only.	Support RFC 3329 additions specified in subclause 7.2A.7 and SDP extensions specified in table A.317, items A.317/51 and A.317/55.	P-CSCF (IMS-ALG) is required. P-CSCF support of RFC 3329 additions specified in subclause 7.2A.7 and SDP extensions specified in table A.317, items A.317/51 and A.317/55. (NOTE)	Not applicable.
End-to-access-edge media security for MSRP using TLS and certificate fingerprints.	MSRP based media only.	Support RFC 3329 additions specified in subclause 7.2A.7 and SDP extensions specified in table A.317, items A.317/40, A.317/40A, A.317/51 and A.317/37A.	P-CSCF (IMS-ALG) is required. P-CSCF support of RFC 3329 additions specified in subclause 7.2A.7 and SDP extensions specified in table A.317, items A.317/40, A.317/40A, A.317/51 and A.317/37A. (NOTE)	Not applicable.
End-to-access-edge media security for BFCP using TLS and certificate fingerprints.	BFCP based media only.	Support RFC 3329 additions specified in subclause 7.2A.7 and SDP extensions specified in table A.317, items A.317/28, A.317/51 and A.317/37B.	P-CSCF (IMS-ALG) is required. P-CSCF support of RFC 3329 additions specified in subclause 7.2A.7 and SDP extensions specified in table A.317, items A.317/28, A.317/51 and A.317/37B. (NOTE)	Not applicable.
End-to-access-edge media security for UDPTL using DTLS and certificate fingerprints.	UDPTL based media only.	Support RFC 3329 additions specified in subclause 7.2A.7 and SDP extensions specified in table A.317, items A.317/52, A.317/51 and A.317/37C.	P-CSCF (IMS-ALG) is required. P-CSCF support of RFC 3329 additions specified in subclause 7.2A.7 and SDP extensions specified in table A.317, items A.317/52, A.317/51 and A.317/37C. (NOTE)	Not applicable.

End-to-end media security using SDES.	RTP based media only.	Support SDP extensions specified in table A.317, items A.317/34 and A.317/36.	Not applicable.	Not applicable.
End-to-end media security using KMS.	RTP based media only.	Support SDP extensions specified in table A.317, items A.317/34 and A.317/35.	Not applicable.	GBA and KMS support required.
End-to-end media security for MSRP using TLS and KMS.	MSRP based media only.	Support SDP extensions specified in table A.317, items A.317/40, A.317/40A and A.317/35, and support RFC 4279 [218].	Not applicable.	GBA and KMS support required.
NOTE: Support of end-to-access-edge media security is determined entirely by the network operator of the P-CSCF, which need not be the same network operator as that of the S-CSCF.				

For RTP media security using SDES, the UE supports the SDES key management protocol and optionally the KMS key management protocol as defined in 3GPP TS 33.328 [19C] and SRTP as defined in RFC 3711 [169] for secure transport of media.

For end-to-access-edge media security of RTP media using DTLS-SRTP, the UE supports DTLS-SRTP as defined in RFC 5763 [222] and RFC 5764 [223] with certificate fingerprints as defined in 3GPP TS 33.328 [19C].

For end-to-access-edge media security for MSRP using TLS and certificate fingerprints, the UE supports MSRP over TLS as defined in RFC 4975 [178] and RFC 6714 [214] with certificate fingerprints as defined in 3GPP TS 33.328 [19C].

For end-to-access-edge media security for BFCP using TLS and certificate fingerprints, the UE supports BFCP over TLS as defined in RFC 4583 [108] with certificate fingerprints as defined in 3GPP TS 33.328 [19C].

For end-to-access-edge media security for UDPTL using DTLS and certificate fingerprints, the UE supports UDPTL over DTLS as defined in RFC 7345 [217] and RFC 8842 [240], with certificate fingerprints as defined in 3GPP TS 33.328 [19C].

For end-to-end media security for MSRP using TLS and KMS, the UE supports MSRP over TLS as defined in RFC 4975 [178] and RFC 6714 [214] with pre-shared key ciphersuites as defined in RFC 4279 [218] and the KMS key management protocol as defined in 3GPP TS 33.328 [19C]. The certificate fingerprints are not indicated.

There is no support for media security in the MGCF, because there would be no end-to-end media security support on calls interworked with the CS domain and the CS user. In this release of this document, there is no support for media security in the MRF. End-to-access-edge media security is not impacted by this absence of support.

For emergency calls, it is not expected that PSAPs would support end-to-end media security and therefore the procedures of this document do not allow the UE to establish such sessions with end-to-end media security. End-to-access-edge media security is not impacted and can be used on emergency calls.

When the UE performs the functions of an external attached network (e.g. an enterprise network):

- where end-to-access-edge media security is used, the UE functionality is expected to be in the gateway of the external attached network, and support for further media security is outside the scope of this document; and

- where end-to-end media security is used, the UE functionality is expected to be supported by the endpoints in the attached network.

4.3 Routeing principles of IM CN subsystem entities

Each IM CN subsystem functional entity shall apply loose routeing policy as described in RFC 3261 [26], when processing a SIP request. In cases where the I-CSCF, IBCF, S-CSCF and the E-CSCF may interact with strict routers in non IM CN subsystem networks, the I-CSCF, IBCF, S-CSCF and E-CSCF shall use the routeing procedures defined in RFC 3261 [26] to ensure interoperability with strict routers.

4.4 Trust domain

4.4.1 General

A trust domain can apply for specific header fields, tel URI parameters and SIP URI parameters within the IM CN subsystem.

For the IM CN subsystem, this trust domain consists of the functional entities that belong to the same operator's network (P-CSCF, the eP-CSCF, the E-CSCF, the I-CSCF, the IBCF, the S-CSCF, the BGCF. the MGCF, the MRFC, the MRB, the EATF, the ATCF, the ISC gateway function, and all ASs that are included in the trust domain). Additionally, other nodes within the IM CN subsystem that are not part of the same operator's domain may or may not be part of the trust domain, depending on whether an interconnect agreement exists with the remote network. SIP functional entities that belong to a network for which there is an interconnect agreement are part of the trust domain. ASs outside the operator's network can also belong to the trust domain if they have a trusted relationship with the home network.

NOTE 1: Whether any peer functional entity is regarded as part of the same operator's domain, and therefore part of the same trust domain, is dependent on operator policy which is preconfigured into each functional entity.

NOTE 2: For the purpose of this document, the PSAP is typically regarded as being within the trust domain, except where indicated. National regulator policy applicable to emergency services determines the trust domain applicable to certain header fields. This means that e.g. the handling of the P-Access-Network-Info header field, P-Asserted-Identity header field and the History-Info header field can be as if the PSAP is within the trust domain, and trust domain issues will be resolved accordingly.

The trust domain can exist for a number of purposes:

a) for the protection of information specific to an operator;

b) to provide for privacy requirements of the end user; or

c) to ensure that information is only passed to another entity if certain responsibilities related to that information are met by the receiving entity, for example that the signalled requirements in the Privacy header field will be met (see subclause 4.4.2 and 4.4.4).

Within the IM CN subsystem trust domains will be applied to a number of header fields. These trust domains do not necessarily contain the same functional entities or cover the same operator domains. The procedures in this subclause apply to the functional entities in clause 5 in the case where a trust domain boundary for that header field, tel URI parameter, or SIP URI parameter, exists at that functional entity.

Where the IM CN subsystem supports business communication, different trust domains can apply to public network traffic, and to private network traffic belonging to each supported corporate network.

NOTE 3: Where an external attached network (e.g. an enterprise network) is in use, the edges of the trust domains need not necessarily lie at the P-CSCF. In this release of the specification, the means by which the P-CSCF learns of such attached devices, and therefore different trust domain requirements to apply, is not provided in the specification and is assumed to be by configuration or by a mechanism outside the scope of this release of the specification.

A trust domain applies for the purpose of the following header fields: P-Asserted-Identity, P-Access-Network-Info, History-Info, Resource-Priority, P-Asserted-Service, Reason (only in a response), P-Profile-Key, P-Private-Network-Indication, P-Served-User, P-Early-Media, Feature-Caps Restoration-Info, Relayed-Charge, Service-Interact-Info, Cellular-Network-Info, Response-Source, Attestation-Info, Origination-Id, Additional-Identity and Priority-Verstat. A trust domain applies for the purpose of the CPC and OLI tel URI parameters. A trust domain applies for the iotl SIP URI parameter. The trust domains of these header fields and parameters need not have the same boundaries. Clause 5 defines additional procedures concerning these header fields, tel URI parameters and SIP URI parameter.

4.4.2 P-Asserted-Identity

RFC 3325 [34] provides for the existence and trust of an asserted identity within a trust domain. A functional entity at the boundary of the trust domain will need to determine whether to remove the P-Asserted-Identity header field according to RFC 3325 [34] when SIP signalling crosses the boundary of the trust domain. The priv-value "id" shall not

be removed from the Privacy header field when SIP signalling crosses the boundary of the trust domain. Subclause 5.4 identifies additional cases for the removal of the P-Asserted-Identity header field.

4.4.3 P-Access-Network-Info

A functional entity at the boundary of the trust domain shall remove any P-Access-Network-Info header field according to RFC 7315 [52].

4.4.4 History-Info

A functional entity at the boundary of the trust domain will need to determine whether to remove the History-Info header field according to RFC 7044 [66] subclause 10.1.2 when SIP signalling crosses the boundary of the trust domain. Subclause 5.4 identifies additional cases for the removal of the History-Info header field.

4.4.5 P-Asserted-Service

A functional entity at the boundary of the trust domain will need to determine whether to remove the P-Asserted-Service header field according to RFC 6050 [121] when SIP signalling crosses the boundary of the trust domain.

4.4.6 Resource-Priority

If Priority verification using assertion of priority information features described in subclause 3.1 is supported then a functional entity at the boundary of the trust domain will need to determine, based on the operator policy, whether to remove a Resource-Priority header field.

Otherwise, if Priority verification using assertion of priority information features described in subclause 3.1 is not supported a functional entity shall only include a Resource-Priority header field in a request or response forwarded to another entity within the trust domain. If a request or response is forwarded to an entity outside the trust domain, the functional entity shall remove the Resource-Priority header field from the forwarded request or response. If a request or response is received from an untrusted entity (with the exception requests or responses received by the P-CSCF from the UE for which procedures are defined in subclause 5.2) that contains the Resource-Priority header field, the functional entity shall remove the Resource-Priority header field before forwarding the request or response within the trust domain.

NOTE: Alternate treatments can be applied when a non-trusted Resource-Priority header field is received over the boundary of trust domain. The exact treatment (e.g. removal, modification, or passing of the Resource-Priority header field) is left to national regulation and network configuration.

4.4.7 Reason (in a response)

A functional entity shall only include a Reason header field in a response forwarded to another entity within the trust domain (as specified in RFC 6432 [130]). If a response is forwarded to an entity outside the trust domain, the functional entity shall remove the Reason header field from the forwarded response.

NOTE: A Reason header field can be received in a response from outside the trust domain and will not be removed.

4.4.8 P-Profile-Key

A functional entity at the boundary of the trust domain will need to determine whether to remove the P-Profile-Key header field as defined in RFC 5002 [97] when SIP signalling crosses the boundary of the trust domain.

4.4.9 P-Served-User

A functional entity at the boundary of the trust domain will need to determine whether to remove the P-Served-User header field according to RFC 5502 [133] when SIP signalling crosses the boundary of the trust domain.

4.4.10 P-Private-Network-Indication

A functional entity shall only include a P-Private-Network-Indication header field in a request forwarded to another entity within the trust domain. If a request is forwarded to an entity outside the trust domain, the functional entity shall remove the P-Private-Network-Indication header field from the forwarded request. If a request is received from an untrusted entity that contains the P-Private-Network-Indication header field, the functional entity shall remove the P-Private-Network-Indication header field before forwarding the request within the trust domain.

NOTE 1: Other entities within the enterprise will frequently be part of this trust domain.

NOTE 2: The presence of the P-Private-Network-Indication header field is an indication that the request constitutes private network traffic. This can modify the trust domain behaviour for other header fields.

NOTE 3: If a trust domain boundary is encountered for this header field without appropriate business communication processing, then this can be an indication that misconfiguration has occurred in the IM CN subsystem. Removal of this header field changes the request from private network traffic to public network traffic.

4.4.11 P-Early-Media

A functional entity at the boundary of the trust domain will need to determine whether to remove the P-Early-Media header field as defined in RFC 5009 [109] when SIP signalling crosses the boundary of the trust domain.

4.4.12 CPC and OLI

Entities in the IM CN subsystem shall restrict "cpc" and "oli" URI parameters to specific domains that are trusted and support the "cpc" and "oli" URI parameters. Therefore for the purpose of the "cpc" and "oli" URI parameters within this specification, a trust domain also applies.

SIP functional entities within the trust domain shall remove the "cpc" and "oli" URI parameters when the SIP signalling crosses the boundary of the trust domain.

4.4.13 Feature-Caps

A functional entity at the boundary of the trust domain shall remove all Feature-Caps header fields received from UEs and external networks outside the trust domain.

NOTE: A UE that is a privileged sender is considered as part of the trust domain.

4.4.14 Priority

Based on local policy, a functional entity at the boundary of the trust domain shall remove all Priority header fields with a "psap-callback" header field value.

4.4.15 iotl

Entities in the IM CN subsystem shall restrict "iotl" URI parameter to specific domains that are trusted and support the "iotl" URI parameter. Support implies that the parameter is removed before the containing request is sent over an interface of a different type. Therefore for the purpose of the "iotl" URI parameter within this specification, a trust domain also applies.

SIP functional entities within the trust domain shall remove the "iotl" URI parameter when the SIP signalling crosses the boundary of the trust domain.

4.4.16 Restoration-Info

A functional entity at the boundary of the trust domain will need to determine whether to remove the Restoration-Info header field when SIP signalling crosses the boundary of the trust domain.

4.4.17 Relayed-Charge

Entities in the IM CN subsystem shall restrict the Relayed-Charge header field to specific domains that are trusted and support the Relayed-Charge header field. Trust implies that the sending domain intends the receiving domain to have the contents of this header field. Therefore for the purpose of the Relayed-Charge header field within this specification, a trust domain also applies.

SIP functional entities within the trust domain shall remove the Relayed-Charge header field when the SIP signalling crosses the boundary of the trust domain.

4.4.18 Service-Interact-Info

A functional entity at the boundary of the trust domain shall remove all Service-Interact-Info header fields defined in subclause 7.2.when SIP signalling crosses the boundary of the trust domain.

4.4.19 Cellular-Network-Info

A functional entity shall only include a Cellular-Network-Info header field in a request forwarded to another entity within the trust domain. If a request is forwarded to an entity outside the trust domain, the functional entity shall remove the Cellular-Network-Info header field from the forwarded request. If a request is received from an untrusted entity that contains the Cellular-Network-Info header field, the functional entity shall remove Cellular-Network-Info header field before forwarding the request within the trust domain.

4.4.20 Response-Source

A functional entity at the boundary of the trust domain will need to determine whether to remove the Response-Source header field according to subclause 7.2.17. when SIP signalling crosses the boundary of the trust domain.

4.4.21 Attestation-Info header field

A functional entity at the boundary of the trust domain will need to determine whether to remove the Attestation-Info header field according to subclause 7.2.18. when SIP signalling crosses the boundary of the trust domain.

4.4.22 Origination-Id header field

A functional entity at the boundary of the trust domain will need to determine whether to remove the Origination-Id header field according to subclause 7.2.19 when SIP signalling crosses the boundary of the trust domain.

4.4.23 Additional-Identity header field

A functional entity at the boundary of the trust domain will need to determine whether to remove the Additional-Identity header field according to subclause 7.2.20 when SIP signalling crosses the boundary of the trust domain.

4.4.24 Priority-Verstat header field

A functional entity at the boundary of the trust domain will need to determine whether to remove the Priority-Verstat header field according to subclause 7.2.21 when SIP signalling crosses the boundary of the trust domain.

4.5 Charging correlation principles for IM CN subsystems

4.5.1 Overview

This subclause describes charging correlation principles to aid with the readability of charging related procedures in clause 5. See 3GPP TS 32.240 [16] and 3GPP TS 32.260 [17] for further information on charging.

The IM CN subsystem generates and retrieves the following charging correlation information for later use with offline and online charging:

1. IM CN subsystem Charging Identifier (ICID);
2. Access network charging information;
3. Inter Operator Identifier (IOI);
4. Charging function addresses:
   a. Charging Data Function (CDF);
   b. Online Charging Function (OCF);
5. IM CN subsystem Functional Entity Identifier.

How to use and where to generate the parameters in IM CN subsystems are described further in the subclauses that follow. The charging correlation information is encoded in the P-Charging-Vector header field as defined in subclause 7.2A.5 and in RFC 7315 [52]. The P-Charging-Vector header field contains the following header field parameters: "icid-value", "icid-generated-at", "related-icid", "related-icid-generated-at", "access-network-charging-info", "orig-ioi", "term-ioi", "transit-ioi" and "fe-identifier".

The offline and online charging function addresses are encoded in the P-Charging-Function-Addresses as defined in RFC 7315 [52]. The P-Charging-Function-Addresses header field contains the following header field parameters: "ccf" for CDF and "ecf" for OCF.

NOTE: P-Charging-Function-Addresses parameters were defined using previous terminology.

4.5.2 IM CN subsystem charging identifier (ICID)

The ICID is the session level data shared among the IM CN subsystem entities including ASs in both the calling and called IM CN subsystems. The ICID is used also for session unrelated messages (e.g. SUBSCRIBE request, NOTIFY request, MESSAGE request) for the correlation with CDRs generated among the IM CN subsystem entities.

The first IM CN subsystem entity involved in a SIP transaction will generate the ICID and include it in the "icid-value" header field parameter of the P-Charging-Vector header field in the SIP request.

For a dialog relating to a session, the generation of the ICID will be performed only on the initial request. This ICID will be used for the initial request and any response to the initial request, and all subsequent SIP messages ina P-Charging-Vector header field.

For all other transactions, generation of the ICID will be performed on each SIP request. This ICID will be used for the SIP request and any response to the SIP request in a P-Charging-Vector header field.

The "icid-value" header field parameter is inserted in the IM CN subsystem, as summarised in table 4-2A.

NOTE: This summary is also applicable for SIP messages which are not specified in clause 5, although each procedure for the P-Charging-Vector header field in clause 5 is described only for specific SIP message(s) (e.g. only for a 200 (OK) response).

Table 4-2A: Summary of ICID insertion in the IM CN subsystem

Inserted in	For initial or standalone SIP message	For subsequent SIP message
Any request	The first IM CN subsystem entity receiving the request inserts the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17].	The first IM CN subsystem entity receiving the request inserts the "icid-value" header field parameter set to the value populated in the initial request for the dialog.
Any response to the request	The first IM CN subsystem entity receiving the response inserts the "icid-value" header field parameter set to the value populated in the initial request for the dialog or the standalone request.	The first IM CN subsystem entity receiving the response inserts the "icid-value" header field parameter set to the value populated in the initial request for the dialog.

See 3GPP TS 32.260 [17] for requirements on the format of ICID. The P-CSCF will generate an ICID for UE-originated calls. The I-CSCF will generate an ICID for UE-terminated calls if there is no ICID received in the initial request (e.g. the calling party network does not behave as an IM CN subsystem). The AS will generate an ICID when acting as an originating UA. The MGCF will generate an ICID for PSTN/PLMN originated calls. The MSC server will generate an ICID for ICS and SRVCC originated calls. Each entity that processes the SIP request will extract the ICID for possible later use in a CDR. The I-CSCF and S-CSCF are also allowed to generate a new ICID for UE-terminated calls received from another network.

There is also an ICID generated by the P-CSCF with a REGISTER request that is passed in a unique instance of P-Charging-Vector header field. The valid duration of the ICID is specified in 3GPP TS 32.260 [17].

The "icid-value" header field parameter is included in any request and response that includes the P-Charging-Vector header field. However, the P-Charging-Vector (and ICID) is not passed to the UE.

The ICID is also passed from the P-CSCF to the IP-CAN via PCRF. The interface supporting this operation is outside the scope of this document.

4.5.2A Related ICID

During the process of SRVCC access transfer the MSC server or the P-CSCF generates an ICID for the target access leg. For the purpose of charging correlation between the source access leg and the target access leg when the user is roaming the SCC AS and the ATCF includes the ICID used on the source access leg in the "related-icid" header field parameter of the P-Charging-Vector header field returned in 1xx and 2xx responses to the initial INVITE request.

During the process of dual radio access transfer the MSC server or the P-CSCF generates an ICID for the target access leg. For the purpose of charging correlation between the source access leg and the target access leg when the user is roaming the SCC AS includes the ICID used on the source access leg in the "related-icid" header field parameter of the P-Charging-Vector header field returned in 1xx and 2xx responses to the initial INVITE request.

4.5.3 Access network charging information

4.5.3.1 General

The access network charging information are the media flow level data shared among the IM CN subsystem entities for one side of the session (either the calling or called side). GPRS charging information (GGSN identifier and PDP context information) is an example of access network charging information.

4.5.3.2 Access network charging information

The IP-CAN provides the access network charging information to the IM CN subsystem. This information is used to correlate IP-CAN CDRs with IM CN subsystem CDRs, i.e. the access network charging information is used to correlate the bearer level with the session level.

The access network charging information is generated at the first opportunity after the resources are allocated at the IP-CAN. The access network charging information is passed from IP-CAN to P-CSCF via PCRF, over the Rx and Gx interfaces. Access network charging information will be updated with new information during the session as media flows are added or removed. The P-CSCF provides the access network charging information to the S-CSCF. The S-CSCF may also pass the information to an AS, which may be needed for online pre-pay applications. The access network charging information for the originating network is used only within that network, and similarly the access network charging information for the terminating network is used only within that network. Thus the access network charging information are not shared between the calling and called networks. The access network charging information is not passed towards the external ASs from its own network.

The access network charging information is populated in the P-Charging-Vector header field.

The access network charging information can be included in a P-Charging-Vector header field in dialog forming requests, mid-dialog requests, and responses. This is dependant on when updated information is avialable in the P-CSCF.

4.5.4 Inter operator identifier (IOI)

The Inter Operator Identifier (IOI) is a globally unique identifier to share between sending and receiving networks, service providers or content providers.

The sending network populates the "orig-ioi" header field parameter of the P-Charging-Vector header field in a request and thereby identifies the operator network from which the request was sent. The "term-ioi" header field parameter is left out of the P-Charging-Vector header field in this request. The sending network retrieves the "term-ioi" header field parameter from the P-Charging-Vector header field in a response to the request, which identifies the operator network from which the response was sent.

The receiving network retrieves the "orig-ioi" header field parameter from the P-Charging-Vector header field in the request, which identifies the operator network from which the request was sent. The receiving network populates the "term-ioi" header field parameter of the P-Charging-Vector header field in the response to the request, which identifies the operator network from which the response was sent.

The "orig-ioi" and "term-ioi" header field parameters are inserted in the IM CN subsystem, as summarised in table 4-2B.

NOTE: This summary is also applicable for SIP messages which are not specified in clause 5, although each procedure for the P-Charging-Vector header field in clause 5 is described only for specific SIP message(s) (e.g. only for a 200 (OK) response).

Table 4-2B: Summary of IOI insertion in the IM CN subsystem

Inserted in	For initial, standalone or subsequent SIP message
Any request	The IM CN subsystem entity in the sending network: 1) removes any received "orig-ioi" header field parameter, if present; 2) inserts the "orig-ioi" header field parameter to a value that identifies the sending network of the request; and 3) does not insert the "term-ioi" header field parameter.
Any response to the request	The IM CN subsystem entity in the receiving network: 1) removes any received "orig-ioi" and "term-ioi" header field parameters, if present; 2) inserts the "orig-ioi" header field parameter set to the previously received value of "orig-ioi" header field parameter, if received in the request; and 3) inserts the "term-ioi" header field parameter to a value that identifies the receiving network from which the response is sent.

There are three types of IOI:

a) Type 1 IOI, between the visited network and the home network. This includes the following cases:

- between the P-CSCF (possibly in the visited network) and the S-CSCF in the home network. This is exchanged in REGISTER requests and responses, and in all session-related and session-unrelated requests and responses;

- between the SCC AS in the home network and the ATCF (possible in the visited network). This is exchanged in MESSAGE requests and responses;

NOTE: For applications where the primary relationship is home and visited network, request and responses to the request will normally contain a type 1 IOI value.-between the MSC server (possibly in the visited network) and the S-CSCF in the home network. This is exchanged in REGISTER requests and responses, and in all session-related and session-unrelated requests and responses; and

- when using Roaming Architecture for Voice over IMS with Local Breakout and loopback routeing occurs, between the S-CSCF of the home network and the TRF of the visited network or between the BGCF of the home network and the TRF of the visited network. This is exchanged in all session-related requests and responses.

b) Type 2 IOI, between originating network and the terminating network. This includes the following cases:

- between the S-CSCF of the home originating network and the S-CSCF of the home terminating network or between the S-CSCF of the home originating network and the MGCF when a call/session is terminated at the PSTN/PLMN;

- between the MGCF and the S-CSCF of the home terminating network when a call/session is originated from the PSTN/PLMN or with a PSI AS when accessed across I-CSCF; and

- when using Roaming Architecture for Voice over IMS with Local Breakout and loopback routeing occurs, between the TRF of the visited network and the S-CSCF of the home terminating network.

This is exchanged in all session-related and session-unrelated requests and responses.

Additionally, for emergency transactions, a type 2 IOI is exchanged between the E-CSCF and the MGCF or IBCF where the request is routed to a PSAP. In scenarios where the E-CSCF receives emergency requests from an S-CSCF, a type 2 IOI is exchanged. This can also occur where the E-CSCF receives emergency requests from an IBCF.

c) Type 3 IOI, between the S-CSCF or I-CSCF of the home operator network and any AS. Type 3 IOI are also used between E-CSCF and LRF, between E-CSCF and EATF, and between transit function and AS. The type 3 IOI is exchanged in all session-related and session-unrelated requests and responses.

Each entity that processes the SIP request will extract the IOI for possible later use in a CDR. The valid duration of the IOI is specified in 3GPP TS 32.240 [16].

4.5.4A Transit inter operator identifier (Transit IOI)

The Transit Inter Operator Identifier (Transit IOI) is a globally unique identifier to share between sending, transit and receiving networks, service providers or content providers.

A network sending a request can retrieve the "transit-ioi" header field parameter value(s) from the P-Charging-Vector header field in the response to the sent request, which identify the operator network(s) which the response was transitting.

The transit network(s) populate(s) the "transit-ioi" header field parameter value(s) of the P-Charging-Vector header field in a request and thereby identify(ies) the operator network(s) which the request was transitting. The "transit-ioi" header field parameter is an indexed value that is incremented each time a value is added. When the index is calculated then "void" entries are taken into account.

The transit network(s) populate(s) the "transit-ioi" header field parameter value(s) of the P-Charging-Vector header field in the response to the request and thereby identify(ies) the operator network(s) which the response was transitting. The "transit-ioi" header field parameter is an indexed value that is incremented each time a value is added. When the index is calculated then "void" entries are taken into account.

A network receiving a request can retrieve the "transit-ioi" header field parameter value(s) from the P-Charging-Vector header field in the request, which identify the operator network(s) which the request was transitting.

EXAMPLE:

Transit-IOI in a request when arriving on the terminating side:

P-Charging-Vector: icid-value="AyretyU0dm+6O2IrT5tAFrbHLso=023551024"; orig-ioi=home1.net; transit-ioi="operatorA.1, void, operatorB.3"

Transit-IOI in the corresponding response when arriving on the originating side:

P-Charging-Vector: icid-value="AyretyU0dm+6O2IrT5tAFrbHLso=023551024"; orig-ioi=home1.net; term-ioi=home2.net; transit-ioi="operatorB.1, void, operatorA.3"

The Transit IOI is exchanged between functional entities, as summarised in table 4-3.

Table 4-3: Summary of Transit IOI exchange in the IM CN subsystem

Item	Entities adding Transt IOI (NOTE 1)	Entities deleting Transit IOI (NOTE 1)
1	additional routeing functions as defined in annex I or IBCF (NOTE 2) in the transit network located between the visited network and the home network	S-CSCF of the home network; P-CSCF of the visited network; or TRF of the visited originating network when using Roaming Architecture for Voice over IMS with Local Breakout and loopback routeing occurs
2	additional routeing functions as defined in annex I or IBCF (NOTE 2) in the transit network located between originating network and the terminating network	S-CSCF of the home terminating network; S-CSCF of the home originating network; MGCF when a call/session is terminated at the PSTN/PLMN; or TRF of the visited originating network when using Roaming Architecture for Voice over IMS with Local Breakout and loopback routeing occurs
3	additional routeing functions as defined in annex I colocated with MGCF when a call/session was transitting the PSTN/PLMN	S-CSCF of the home terminating network
NOTE 1: Transit IOIs can also be exchanged with non 3GPP networks, e.g. IPX networks. NOTE 2: In the transit network, the IBCF acting as an entry point adds the Transit IOI in requests and the IBCF acting as an exit point adds the Transit IOI in responses, as described in subclause 5.10.		

Each entity that processes the SIP requests and responses may extract the Transit IOI for charging purposes, as described in 3GPP TS 32.260 [17].

4.5.5 Charging function addresses

Charging function addresses are distributed to each of the IM CN subsystem entities in the home network for one side of the session (either the calling or called side) and provide a common location for each entity to send charging information. Charging Data Function (CDF) addresses are used for offline billing. Online Charging Function (OCF) addresses are used for online billing.

There may be multiple addresses for CDF and OCF addresses populated into the P-Charging-Function-Addresses header field of the SIP request or response. The header field parameters are "ccf" and "ecf" for CDF and OCF, respectively. At least one instance of either "ccf" or "ecf" header field paramter is required. If "ccf" header field parameter is included for offline charging, then a secondary "ccf" header field paramter may be included by each network for redundancy purposes, but the first instance of "ccf" header field parameter is the primary address. If ecf address is included for online charging, then a secondary instance may also be included for redundancy.

The CDF and/or OCF addresses are retrieved from an Home Subscriber Server (HSS) via the Cx interface and passed by the S-CSCF to subsequent entities. The charging function addresses are passed from the S-CSCF to the IM CN subsystem entities in its home network, but are not passed to the visited network or the UE. When the P-CSCF is allocated in the visited network, then the charging function addresses are obtained by means outside the scope of this document. The AS receives the charging function addresses from the S-CSCF via the ISC interface. CDF and/or OCF addresses may be allocated as locally preconfigured addresses. The AS can also retrieve the charging function address from the HSS via Sh interface.

4.5.6 Relayed charge parameters

Where there is a desire to pass charging information to an entity over an interface to which the charging information is not directly related, then the Relayed-Charge header field is used. This is used only in accordance with the capabilities specified in this document, which currently specify only the relay of a transit IOI to an associated AS.

4.5.7 Loopback-indication parameter

When there is a desire to send the information that loopback has been applied to an entity, then the loopback-indication parameter is used. This parameter can e.g. be evaluated when processing the P-CSCF CDR and possibly the ATCF CDR to know whether or not to attempt to locate a correlated TRF CDR even for the non-loopback scenario when no such CDR exists. It is used only in accordance with the capabilities specified in this document.

4.5.8 IM CN subsystem Functional Entity Identifier

4.5.8.1 General

Different rules for generating and processing of charging information apply. In order to inform the billing domain which IM CN subsystem functional entities have created charging information, the IM CN subsystem functional entities and ASs include an "fe-identifier" header field parameter as additional information in the P-Charging-Vector header field when generating charging information as specified in 3GPP TS 32.260 [17].

NOTE: Within the billing domain this information given within the "fe-identifier" header field in a final response allows the billing domain to compare the generated data records for specific IM CN subsystem functional entities with the recorded addresses/identifiers of the IM CN subsystem functionial entities themselves. Thus information can be compared and missing information can be identified.

4.5.8.2 Tracking of IM CN subsystem functional entities generating charging information

Each IM CN subsystem functional entity that generates charging events, includes its own address or specific IM CN subsystem functional entity identifier within the "fe-addr" element of the "fe-identifier" header field parameter of the P-Charging-Vector header field into the initial SIP request to be sent from own domain.

The last element of the operator domain stores the "fe-identifier" header field parameter in the P-Charging-Vector header field and removes the "fe-identifier" header field parameter from the P-Charging-Vector header field.

When receiving the final SIP response sent back to its own domain, the last element of the operator domain deletes, if received, the "fe-identifier" header field parameter from the P-Charging-Vector header field, adds the stored "fe-identifier" and adds its own "fe-addr" to the "fe-identifier".

4.5.8.3 Tracking of applications generating charging information

Each application for which the hosting AS is generating charging events, includes the address or specific identifier of the AS within the "as-addr" element and the application identifier within the "ap-id" element of the "fe-identifier" header field parameter of the P-Charging-Vector header field into the initial SIP request to be sent.

The final SIP response sent back by the last element of the operator domain supporting the "fe-identifier" header field contains the list of addresses and application identifiers received within the initial SIP request.

4.6 Support of local service numbers

For the IM CN subsystem, the support of local service numbers is provided by an AS in the subscriber's home network as described in subclause 5.7.1.7.

4.7 Emergency service

4.7.1 Introduction

The need for support of emergency calls in the IM CN subsystem is determined by national regulatory requirements.

4.7.2 Emergency calls generated by a UE

If the UE cannot detect the emergency call attempt, the UE initiates the request as per normal procedures as described in subclause 5.1.2A. Depending on network policies, for a non-roaming UE or for a roaming UE where the P-CSCF is in the same network where the UE is roaming an emergency call attempt can succeed even if the UE did not detect that an emergency session is being requested, otherwise the network rejects the request indicating to the UE that the attempt was for an emergency service.

The UE procedures for UE detectable emergency calls are defined in subclause 5.1.6.

The P-CSCF, S-CSCF, IBCF, and E-CSCF procedures for emergency service are described in subclause 5.2.10, 5.4.8, 5.10.3.2 and 5.11, respectively.

Access dependent aspects of emergency service (e.g. whether the access technology defines emergency bearers, emergency registration support and location provision) are defined in the access technology specific annexes for each access technology.

There are a number of variants within these procedures and which variant gets used depends on a number of issues. These conditions are defined more specifically in 3GPP TS 23.167 [4B] and, where appropriate, in the access technology specific annex, but are summarised as follows:

a) if the UE knows that it is in its own home network, then an existing registration is permitted to be used for signalling the emergency call, except where item c) applies. The access technology specific annexes define the mechanism by which home network determination is made;

b) if emergency calls are permitted without security credentials (or additionally where the authentication is not possible or has failed), then the emergency call is made directly without use of any security association created by a registration, and therefore without the registration; and

c) where the access technology defines emergency bearers for the support of emergency calls, a new emergency registration is required so that these emergency bearers can be used for both signalling and media, unless an existing emergency registration exists on those emergency bearers.

4.7.3 Emergency calls generated by an AS

In certain circumstances an AS can identify that a request is an emergency call. This may relate to a request received from a UE (or subscription-based business trunking), or may be a call generated by an AS on behalf of a UE as far as the IM CN subsystem operation is concerned. These applications are outside the scope of this document to define.

Procedures in support of an AS initiating emergency calls are provided in subclause 5.7.1.14.

4.7.4 Emergency calls received from an enterprise network

An IBCF can also route emergency calls received from an enterprise network (peering-based business trunking) to an E-CSCF.

4.7.5 Location in emergency calls

A number of mechanisms also exist for providing location in support of emergency calls, both for routeing to a PSAP, and for use by the PSAP itself, in the IM CN subsystem:

a) by the inclusion by the UE of the Geolocation header field containing a location by reference or by value (see RFC 6442 [89]);

b) by the inclusion by the UE of a P-Access-Network-Info header field, which contains a cell identifier or location identitifier, which is subsequently mapped, potentially by the recipient, into a real location;

c) by the inclusion by the P-CSCF of a P-Access-Network-Info header field based on information supplied by either the PCRF or the NASS, and which contains a cell identifier or location identitifier, which is subsequently mapped, potentially by the recipient, into a real location;

d) by the allocation of a location reference that relates to the call by the LRF. Location is then supplied to the recipient over the Le interface (see 3GPP TS 23.167 [4B] for a definition of the Le interface) along with other call information. The LRF can obtain the location from entities outside the IM CN subsystem, e.g. by the e2 interface from the NASS (see ETSI TS 283 035 [98] or from the Gateway Mobile Location Centre (GMLC); and

e) by the inclusion by the S-CSCF of a P-Access-Network-Info header field based on information supplied by HSS, and which contains a location identitifier, which is subsequently mapped, potentially by the recipient, into a real location.

Mechanisms also exist for providing emergency-related information to a PSAP, in requests subsequent to routeing an initial request to a PSAP, in the IM CN subsystem:

a) by the inclusion by the UE of the Geolocation header field containing a location by reference or by value (see RFC 6442 [89]);

b) by the inclusion by the UE of a P-Access-Network-Info header field, which contains a cell identifier or location identitifier, which is subsequently mapped, potentially by the recipient, into a real location;

c) by the inclusion by the P-CSCF of a P-Access-Network-Info header field based on information supplied by either the PCRF or the NASS, and which contains a cell identifier or location identitifier, which is subsequently mapped, potentially by the recipient, into a real location;

d) by the inclusion by the UE of the emergency-related information as specified in subclause 5.1.6.10;

e) by the inclusion by the S-CSCF of a P-Access-Network-Info header field based on information supplied by HSS, and which contains a location identitifier, which is subsequently mapped, potentially by the recipient, into a real location; and

f) by LRF requesting the location from the UE via E-CSCF as specified in subclause 5.12.3.2, subclause 5.11.4.3, subclause 5.11.4.4, subclause 5.11.5 and subclause 5.1.6.12.

The E-CSCF routes such a subsequent request to the PSAP using normal SIP procedures.

NOTE 1: Mechanisms independent from SIP for providing the emergency related information to a PSAP after session setup exist and are not listed. The use of such mechanisms is not precluded.

Which means of providing location is used depends on local regulatory and operator requirements. One or more mechanisms can be used. Location can be subject to privacy constraints.

NOTE 2: A similar variety of mechanisms also exists for normal calls, where location can be made use of by the recipient or by an intermediate AS, again subject to privacy constraints. The LRF is not involved in a normal call, but an AS can obtain location from the e2 interface from the NASS (see ETSI TS 283 035 [98] or from the Gateway Mobile Location Centre (GMLC).

4.7.6 eCall type of emergency service

A PSAP supporting eCall over IMS supports:

- receipt of the minimum set of emergency related data (MSD) in an INVITE or INFO request;

- the EmergencyCallData.eCall.MSD Info-Package and the ability to request an updated MSD by including an "application/EmergencyCallData.Control+xml" MIME body part containing a "request" element with an "action" attribute set to "send-data" and a "datatype" attribute set to "eCall.MSD" in an INFO request as defined in RFC 8147 [244];

- sending of an acknowledgement for an MSD received in an INVITE request by including, in the final response to the INVITE request, an "application/EmergencyCallData.Control+xml" MIME body part with an "ack" element containing a "received" attribute set to "true" or "false" and a "ref" attribute set to the Content-ID of the MIME body part containing the MSD sent by the UE, as defined in RFC 8147 [244];

- receipt of the MSD using audio media stream encoded as described in 3GPP TS 26.267 [9C];

- the ability to request an updated MSD using audio media stream encoded as described in 3GPP TS 26.267 [9C]; and

- the ability to request an updated MSD using audio media stream encoded as described in 3GPP TS 26.267 [9C], if the remote UA modifies an IMS emergency call of the eCall type of emergency service and stops supporting the EmergencyCallData.eCall.MSD Info-Package defined in RFC 8147 [244].

NOTE: The remote UA modifies an IMS emergency call of the eCall type of emergency service and stops supporting the EmergencyCallData.eCall.MSD Info-Package defined in RFC 8147 [244] when SRVCC access transfer takes place.

4.8 Tracing of signalling

4.8.1 General

IM CN subsystem entities can log SIP signalling, for debugging or tracing purposes, as described in 3GPP TS 32.422 [17A]. Debugging of SIP signalling is configured using the 3GPP IMS service level tracing management object (MO), specified in 3GPP TS 24.323 [8K]. This management object can provide configuration data to a UE or an S-CSCF, including in a visited network. Logging always begins with the initial request that creates a dialog and ends when a pre-configured stop trigger occurs or the dialog ends, whichever occurs first, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K].

4.8.2 Trace depth

The depth parameter in trace control and configuration indicates which SIP requests and responses are logged. If the trace depth is "maximum" then all requests and responses within a dialog or standalone transaction are logged. If the trace depth is "minimum" then all requests and responses except for non-reliable 1xx responses (including 100 (Trying) responses) and the ACK request are logged.

4.9 Overlap signalling

4.9.1 General

This subclause explains the overlap signalling impacts on the core entities of the IM CN subsystem.

The support of overlap signalling, and each of the overlap signalling method, within the IM CN subsystem, are optional and is dependent on the network policy.

Only one overlap signalling method shall be used within one IM CN subsystem.

NOTE: Interworking between the overlap signalling methods is not specified in this release.

4.9.2 Overlap signalling methods

4.9.2.1 In-dialog method

4.9.2.1.1 General

The in-dialog method uses INFO requests or INVITE requests in order to transport additional digits. Before an early dialog has been established, upon reception of a 404 (Not Found) or 484 (Address Incomplete) response to an earlier INVITE request, new INVITE requests will be sent to transfer additional digits (as specified in 3GPP TS 29.163 [11B]). Once an entity establishes an early dialog, by sending a provisional response to a INVITE request, INFO requests will be sent to carry additional digits on the early dialog.

The message body, and associated header values, which is used to carry additional digits in INFO requests is defined in 3GPP TS 29.163 [11B].

4.9.2.2 Multiple-INVITE method

4.9.2.2.1 General

The multiple-INVITE method uses INVITE requests with the same Call ID and From header in order to transport digits (as specified in 3GPP TS 29.163 [11B]).

4.9.3 Routeing impacts

4.9.3.1 General

If overlap dialing is supported, the IM CN subsystem needs to be configured in such a manner that erroneous routeing of INVITE requests with incomplete numbers towards others entities than the corresponding INVITE requests with full numbers is avoided, for instance towards a default destination for unknown numbers such as a PSTN. Possibly impacted nodes include the S-CSCF for the UE-originated case, the transit routeing function, the I-CSCF, and application servers.

A misrouteing can be avoided by configuring the entity sending overlap signalling in such a manner that it will send the first INVITE request with a sufficient number of digits to find a suitable entry in the translation database. If ENUM is used, the ENUM database in a typical deployment contains sufficient information about the first digits, as required to identify the destination IP domain. Therefore, ENUM is able to handle incomplete numbers in such deployments. As another alternative, the routeing entity can reject calls with unknown numbers with a 404 (Not Found) response, using entries in the routeing database to identify calls towards the PSTN. The S-CSCF for the UE-originated case could also forward calls with unknown numbers to the BGCF, if the BGCF is configured to reject calls to unknown destinations with a 404 (Not Found) response.

4.9.3.2 Deterministic routeing

If the multiple-INVITE method is used for overlap signalling, if an entity receives a INVITE request outside an existing dialog with the same Call ID and From header field as a previous INVITE request during a certain period of time, the entity shall route the new INVITE request to the same next hop as the previous INVITE request.

NOTE: INVITE requests with the same Call ID and From header fields received in sequence during a certain period of time belong to the same call. The routeing towards the same next hop could be achieved by an appropriately configured database or by the entity comparing the Call ID and From header fields of each INVITE request outside an existing dialog with Call IDs and "tag" From header field parameters of previous INVITE requests. If the entity compares the Call ID and From header field, it stores the information about received Call ID and From header fields at least for a time in the order of call setup times. If paths have been established at registration time, deterministic routeing will be automatic for entities on these paths.

4.9.3.3 Digit collection

Entities performing routeing decicisions may require additional digits for a decision where to route an INVITE request. These entities may interact with a routeing database to reach this decision.

If no suitable entry in a database is found for the digits received in a INVITE request, an entity can reject the INVITE request with a 404 (Not Found) or 484 (Address Incomplete) response. This method of digit collection can be performed by a SIP proxy and is suitable both for the in-dialog and multiple-INVITE overlap signalling methods. Replying with a 404 (Not Found) response avoids the need to keep apart uncomplete and unknown numbers. The 484 (Address Incomplete) response requires the recognition of incomplete numbers.

NOTE: An HSS does not support the recognition of incomplete numbers. A routeing database being queried by ENUM also does not support the recognition of incomplete numbers.

As an alternative for the in-dialogue method, the digit collection function described in annex N.2 may be invoked. It shall be performed by an entity acting as a B2BUA. The digit collection function requires the ability to recognise incomplete number.

4.10 Dialog correlation for IM CN subsystems

4.10.1 General

The Call-ID header field in combination with the tags in the From header field and in the To header field is the standard mechanism to identify SIP messages which belong to the same dialog. However the Call-ID header field is often changed by B2BUAs and other SIP intermediaries in the end-to-end message path.

To solve this problem, a Session-ID header field containing a globally unique session identifier, as defined in RFC 7989 [162], can be used to correlate SIP messages belonging to the same session. In the case of a concatenation of dialogs, the dialog correlation mechanism indicates that these dialogs belong to the same session.

The usage of the Session-ID header field is specified in annex A.

4.10.2 CONF usage

In case of the activation of a 3PTY conference, in the INVITE request to the CONF AS the Session-ID header field is added to the URIs in the URI list, in order to indicate the dialogs which are to be included to the 3PTY conference at the CONF AS, as described in 3GPP TS 24.147 [8B].

4.11 Priority mechanisms

In support of priority, the IM CN subsystem uses the mechanisms of RFC 4412 [116]. The request for prioritisation of a transaction / dialog may, for some deployments, be marked with the Resource-Priority header field by the UE. For other deployments, the request is not marked for priority by the UE, but the request is instead identified as a priority request and marked for priority (via a Resource-Priority header field) by a functional entity (e.g., P-CSCF) within the network. Subsequent to successful authorisation at an authorisation point (e.g. AS), request is considered to be authorised.

The characteristics of any priority scheme is defined by the namespace that is used. This determines how priority is applied to the SIP signalling, to the bearer carrying the SIP signalling, and to the bearers carrying any media. Different priority levels exist within each namespace. Priority levels in one namespace have no relationship to the priority levels in any other namespace, i.e. priority level "1" in namespace "A" may have an entirely different level and characteristic of priority treatment to an identically labelled priority level "1" in namespace "B".

A network can support multiple namespaces. It is up to the network operator (potentially based on regulatory or contractural obligations) to define the relationship between the priority mechanisms for each namespace, and indeed with calls that are not given any priority. It is normal that prioritised calls do not have access to 100% of any available resource and indeed are limited to a much lower figure. Priority is optional, and this document places no requirement on a conformant IM CN subsystem implementation to support priority, or indeed any namespace in a priority scheme. Regulators can however place their own requirements on an operator. Emergency transactions or dialogs (see subclause 4.7) can also have their own priority scheme.

RFC 4412 [116] specifies several resource priority namespaces. For example, certain national MPS implementations use resource priority namespaces of ETS (Emergency Telecommunications Service) and WPS (Wireless Priority Service).

Several ways of using priority exist, depending on the authorisation mechanism adopted. These are identified as follows. In each of these authorisation means authorisation to use the service, the namespace, and the priority level within that namespace:

1) Authorisation based on subscription in the IM CN subsystem only, priority requested by the UE using the Resource Priority header field. Whether the user is allowed to use priority or not, and the appropriate namespace and priority levels, is stored as part of the user profile in the HSS. As part of the reg event package subscription, this information is given to the P-CSCF when the contact information for any public user identity changes, and based on this information, the P-CSCF acts as the authorisation point for priority on individual requests. At the P-CSCF, when a Resource-Priority header field is received from the UE, if the requested priority equates to a value (namespace and priority level) that the P-CSCF knows is allowed for that public user identity, the priority is authorised.

2) Authorisation based on a database deployed by an AS; priority requested by the UE using a special dialstring. In this case the user requires no priority subscription information in the HSS. Specific dialstrings are configured in the P-CSCF. When a request is received from the UE by the P-CSCF, if the request contains a specific dialstring that is recognised by the P-CSCF as being eligible for priority treatment, the request is marked for temporary priority, subject to subsequent authorisation by an authorisation point (i.e., AS). And all such requests are routed to an AS. Final authorisation is granted by the AS, based on a PIN or password exchange with the UE. Subsequent requests or responses after authorisation are only given priority by the P-CSCF and S-CSCF if some backwards indication is received for that specific dialog. The definition of this backwards indication is outside the scope of this document (because non-standardised mechanisms have already been implemented in association with this approach).

3) Authorisation based on subscription in the IM CN subsystem and on a database deployed by an AS; priority requested by the UE using a special dialstring. Specific dialstrings are configured in the P-CSCF. When a request is received from the UE by the P-CSCF, if the request contains a specific dialstring that is recognised by the P-CSCF as being eligible for priority treatment, the request is marked for temporary priority, subject to subsequent authorisation by an authorisation point (i.e., AS). Based on iFC functionality that exists at the S-CSCF (from the users subscription in the HSS), such requests are routed to an AS. Final authorisation is granted by the AS, based on a PIN or password exchange with the UE or based on user profile. Subsequent requests or responses after authorisation are only given priority by the P-CSCF and S-CSCF if some backwards indication is received for that specific dialog. The definition of this backwards indication is outside the scope of this document (because non-standardised mechanisms have already been implemented in association with this approach).

Some administrations can require the use of multiple approaches in the same network.

For the cases of interworking with other networks, where the P-CSCF of the other network does not support priority, but it is intended or required to give users of that P-CSCF priority in the home network, provision is made for recognition of dialstrings by the IBCF and the S-CSCF. In such scenarios, when the IBCF or S-CSCF recognize that a request contains a dialstring as being eligible for priority treatment, the request is marked by the IBCF or S-CSCF for temporary priority, subject to subsequent authorisation by an authorisation point (i.e. AS). This mechanism does not have an impact on the network where the P-CSCF resides.

Where the network has a requirement to prioritise emergency calls, it can either perform this function by the use of the "esnet" namespace in the Resource-Priority header field (as defined in RFC 7135 [197]), or by recognition of the presence of the service URN relating to an emergency. Where the Resource-Priority header field is used for this purpose, it is inserted by the entity identifying the emergency call, i.e. the P-CSCF or the IBCF. There is no usage of this namespace from the UE, and when this namespace is used, the trust domain implementation is set to remove it if it occurs from the UE.

Where a network has requirements on attestation and signing of priority IMS sessions (e.g., MPS sessions) the Priority verification using assertion of priority information feature described in subclause 3.1 shall be supported and the Calling number verification using signature verification and attestation information feature described in subclause 3.1 may be supported.

Where the network has requirements on attestation and signing of originating calling identification information for emergency and emergency callback IMS sessions, and on authentication of a Resource-Priority header field and a header field value "psap-callback" of a Priority header field, Calling number verification using signature verification and attestation information and Priority verification using assertion of priority information features described in subclause 3.1 shall be supported.

4.12 Overload control

Usage of overload control is independent of the nature of any SIP using entity, i.e. there are no specific requirements for any particular IMS functional entity implementing SIP. The capability however is not extended to the UE except when performing the function of an externally attached network.

Two mechanisms are defined as follows:

- a feedback based mechanism defined in RFC 7339 [199], where the feedback is given in the Via header field of signalling messages supporting the traffic. RFC 7339 [199] also defines the default algorithm for usage of the feedback based mechanism in the IM CN subsystem (i.e. loss-based algorithm). Additional algorithms are either already defined, e.g. the rate-based scheme defined in RFC 7415 [200] or can also be defined in the future. As it is carried in the Via header fields the nature of the mechanism is hop by hop.

- an event package for distributing load filters defined in RFC 7200 [201], which can be either used in a hop-by-hop manner between adjacent entities in a similar manner to the feedback based mechanism, or can be used on a wider basis across the network, subject to the restrictions given in annex A. In this manner it can be used to address expected overload situations, e.g. for voting calls initiated by a specific television programme.

When the load filters based mechanism is used in the IMS, the default algorithm is loss-based (i.e. the filter specifies the relative percentage of incoming requests that can be accepted).

The S-CSCF, application servers and entities that implement the additional routeing capability can use both mechanisms in parallel on the same interfaces.

There are no specific reasons why one protocol mechanism should be specified over another, although some discussion is given in the documents specifying the mechanisms themselves. It is regarded as a deployment issue as to which mechanisms are supported, and which algorithms are supported within those mechanisms, beyond those that the mechanisms themselves identify as mandatory. An operator will need to take a network wide view to planning their overload control strategy, it cannot be performed on ad-hoc basis as nodes are deployed.

For the distribution of load filters mechanism, typical deployments might include an S-CSCF subscribing to the load control event package at an AS, an AS subscribing to the load control event package at an AS, and an entity hosting additional routeing capabilities as specified in subclause I.3 subscribing to the load-control event package at the AS.

Based on regional/national requirements and network operator policy, priority calls (e.g., multimedia priority service) are exempted from SIP overload controls up to the point where further exemption would cause network instability. Therefore, SIP messages related to priority calls have the highest priority, and are last to be dropped or rejected, when an IM CN subsystem functional entity decides it is necessary to apply traffic reduction. The interaction between SIP overload control and priority services is covered in RFC 7339 [199] and RFC 7200 [201].

Based on regional/national requirements and network operator policy, emergency calls are exempted from SIP overload controls up to a configured threshold. Therefore, when an IM CN subsystem functional entity decides it is necessary to apply traffic reduction due to overload control, SIP messages related to emergency calls are not dropped while the configured threshold regarding the amount of the ongoing emergency calls is not reached.

Mid-dialog SIP messages have higher priority with regard to initial SIP requests, and therefore are last to be dropped or rejected, when an IM CN subsystem functional entity decides it is necessary to apply traffic reduction due to overload control.

Operation between two network operators is supported. If two network operators wish to implement overload control, it is a matter for bilateral agreement as to what is supported.

Operation with enterprise networks is supported. The network operator and the enterprise operator will need to agree on the overload control options to be supported.

4.13 II-NNI traversal scenario

4.13.1 General

Within the IM CN subsystem, the signalling path between a calling user and a called user can be divided into one or more traffic legs, referred to as II-NNI traversal scenarios. Each II-NNI traversal scenario can span networks belonging to different operators and will have its own characteristics that can be different from other II-NNI traversal scenarios in the same call.

Dialog creating SIP requests and standalone requests can contain an "iotl" SIP URI parameter as specified in RFC 7549 [225] in a Request-URI or in one or more Route header fields. The "iotl" SIP URI parameter is appended to the URI representing the end of the II-NNI traversal scenario. The value of "iotl" SIP URI parameter can be used to identify the II-NNI traversal scenario.

If the "iotl" SIP URI parameter is not included in a dialog creating SIP requests or a standalone request, the II-NNI traversal scenario type can be determined by analysing the content of the SIP request or using a default II-NNI traversal scenario type.

NOTE: How the content of the SIP request can be used to determine the II-NNI traversal scenario is implementation dependent and outside the scope of this specification.

The "iotl" SIP URI parameter is included by the start of the II-NNI traversal scenario (e.g. P-CSCF, S-CSCF, BGCF or SCC AS) and removed by the end of the II-NNI traversal scenario (e.g. S-CSCF, TRF or P-CSCF).

4.13.2 Identifying the II-NNI traversal scenario

The "iotl" SIP URI parameter specified in RFC 7549 [225] containing traffic leg information can be used to identify the II-NNI traversal scenario type. The II-NNI traversal scenario type can be used by intermediary entities (e.g. IBCF and transit networks) to make policy decisions related to e.g. media anchoring, screening of SIP signalling, insertion of media functions (e.g. transcoder) and charging.

One example on how the "iotl" SIP URI parameter is included in the Route header field by the P-CSCF in a visited network when sending a request towards the home network is shown below.

EXAMPLE: Route: <sip:ibcf-vA1.visited-A.net;lr>,<sip:home-abc@scscf-hA1.home-A.net;lr:iotl=visitedA-homeA>

If neither the Request-URI nor any of the Route header fields included in the SIP request contains the "iotl" SIP URI parameter, the II-NNI traversal scenario type can be determined by analysing the content of the SIP request or using a default II-NNI traversal scenario type. The recommended II-NNI traversal scenario type default value is "homeA-homeB".

NOTE: How the content of the SIP request can be used to determine the II-NNI traversal scenario is implementation dependent and outside the scope of this specification.

4.13.3 Security aspects

When receiving a dialog creating SIP request or a standalone SIP request from outside the trust domain the IBCF acting as an entry point removes any "iotl" SIP URI parameter according to subclause 4.4.15 and assume the default II-NNI traversal scenario type or can use trusted elements of the SIP request to determine the II-NNI traversal scenario type.

NOTE: Examples of trusted elements are protocol elements within the trust domain and protocol elements manipulated, checked or added by a previous hop secured by network domain security.

4.14 Restoration procedures

4.14.1 General

The present document includes optional restoration procedures for failure of P-CSCF and S-CSCF. The general mechanism is to inform the UE that one of the entities along its registration path is not working, and hence the UE needs to perform an initial registration. For systems providing access to IM CN subsystem using a GPRS IP-CAN, EPS IP-CAN or a 5GS IP-CAN, the mechanism to trigger the UE can be to use the Protocol Configuration Options IE or extended Protocol Configuration Options IE specified in 3GPP TS 24.008 [8], include an 3GPP IM CN subsystem XML body in a 504 (Server Time-out) response, or disconnect the PDN connection.

4.14.2 P-CSCF restoration procedures

P-CSCF restoration procedures are implemented in the S-CSCF, the IBCF and the UE.

When the UE originates a session it can detect that a P-CSCF is not reachable based on no response from the P-CSCF in which case the UE selects another P-CSCF if possible and performs a new initial registration.

UDM/HSS or HSS based P-CSCF restoration applies to UE terminating requests where the SIP entity neighbouring the P-CSCF (S-CSCF, IBCF) can detect that a P-CSCF is not reachable. When the neighbouring entity is the S-CSCF, the S-CSCF can in this case initiate the P-CSCF restoration. The S-CSCF sends an indication to the HSS to initiate the restoration. If the terminating user is roaming, the neighbouring entity is an entry IBCF which uses the Restoration-Info header field to inform the S-CSCF about the failure in a 408 (Request Timeout) response for INVITE requests or a 504 (Server Time-out) response for non-INVITE requests and the S-CSCF can send an indication to the HSS to initiate the restoration.

PCF or PCRF based P-CSCF restoration applies to UE terminating INVITE requests. For PCF or PCRF based P-CSCF restoration the S-CSCF uses the Restoration-Info header field to send the IMSI in initial INVITE requests to an alternative P-CSCF. When the user is roaming, the IBCF selects an alternative P-CSCF and forwards the IMSI of the terminating user in a Restoration-Info header field.

Restoration can also be initiated when the P-CSCF has restarted, and lost all bindings for a particular user. In this case the P-CSCF rejects the incoming request with a 404 (Not Found) response. If the home network applies UDM/HSS or HSS based P-CSCF restoration the S-CSCF initiates the restoration procedure by sending an indication to the HSS. If PCF or PCRF based restoration is used, the S-CSCF initiate the PCF or PCRF based P-CSCF restoration procedure for the served user by including the IMSI in a Restoration-Header field included in an initial INVITE request.

NOTE: In the rest of the present document where the "PCRF based P-CSCF restoration" procedure is mentioned the "PCF based P-CSCF restoration" procedure also applies, and where the "HSS based P-CSCF restoration" procedure is mentioned the "UDM/HSS based P-CSCF restoration" procedure also applies.

4.14.3 S-CSCF restoration procedures

The P-CSCF can inform the UE about S-CSCF failures in a 504 (Server Time-out) response using the 3GPP IM CN subsystem XML body defined in subclause 7.6, in accordance with subclause 5.2.6.3.2A, when the P-CSCF is unable to forward a request to an S-CSCF.

When the S-CSCF receives a request initiated by the served user for which the S-CSCF does not have the user profile or does not trust the data that it has (e.g. due to restart) the S-CSCF can if it fails to retrieve the data from the HSS trigger a registration by sending a 504 (Server Time-out) response using the 3GPP IM CN subsystem XML body defined in subclause 7.6 to the UE, in accordance with subclause 5.4.3.2.

An I-CSCF can reselect S-CSCF if the previously selected S-CSCF is not available.

If an IBCF acting as an entry point in the originating home network cannot forward the request the IBCF can trigger the UE to perform initial registration by including the 3GPP IM CN subsystem XML body in a 504 (Server Time-out) response, in accordance with subclause 5.10.3.5.

4.15 Resource sharing

Resource sharing allows two or more sessions to use the same resources for one or more media streams in uplink, downlink or both uplink and downlink direction.

A P-CSCF that supports resource sharing can determine that there is a potential for resource sharing based on local configuration or defer the determination of potential resource sharing to an AS in the home network.

If the determination of potential resource sharing is deferred to an AS in the home network:

- the P-CSCF on the originating side indicates that resource sharing is supported in the initial REGISTER request in the Resource-Share header field defined in subclause 7.2.13. The Resource-Share header field is included in the third-party REGISTER request towards the AS; and

- if the "message/sip" MIME body in the third-party REGISTER request included the Resource-Share header field with the value "supported", the AS in the home network includes the Resource-Share header field containing the rules for resource sharing in responses and requests towards the P-CSCF.

If the rules for resource sharing are updated, the updated rule will be sent to P-CSCF in one of the sessions that share resources. The updated resource sharing rules will then be applied for all sessions that are sharing resources.

NOTE: In this release of the technical specification the UE cannot indicate support of resource sharing. However, the Resource-Share header field is not removed from requests and responses towards the UE and the UE can use the information in the header field to adapt its behaviour according to the information.

4.16 Priority sharing

Priority sharing allows two or more sessions with different priority to share the same bearer.

The determination of the use of priority sharing is deferred to an AS in the home network:

1) if P-CSCF supports priority sharing and if according to local policy, the P-CSCF indicate that priority sharing is supported by including the g.3gpp.priority-share feature-capability indicator defined in subclause 7.9A.10 in a Feature-Caps header field in the REGISTER request;

NOTE: The Feature-Caps header field with the g.3gpp.priority-share feature-capability indicator is included in the "message/sip" MIME body in the third-party REGISTER request sent over the ISC interface.

2) if the "message/sip" MIME body in the third-party REGISTER request included the g.3gpp.priority-share feature-capability indicator and:

a) if the AS determined to enable priority sharing, the AS includes the Priority-Share header field with a value "allowed" in a request or response sent towards the P-CSCF; or

b) if the AS determined to disable priority sharing, the AS includes the Priority-Share header field with a value "not-allowed" in a request or response sent towards the P-CSCF.

4.17 3GPP PS data off

The UE and the network can support the 3GPP PS data off.

When 3GPP PS data off is supported and active, IP packets that are associated with services that are not a 3GPP PS data off exempt service are prevented from transport over EPS IP-CAN, GPRS IP-CAN and 5GS IP-CAN as specified in 3GPP TS 23.228 [7]. The UE may be configured by the HPLMN, the EHPLMN or the subscribed SNPN with up to two indications whether a 3GPP IMS service is a 3GPP PS Data Off exempt service, one indication is valid when the UE is in the HPLMN, the EHPLMN, or a subscribed SNPN and the other indication is valid when the UE is in the VPLMN or, if the UE supports access to an SNPN using credentials from a CH, a non-subscribed SNPN. When the UE is only configured with the indication valid for the UE camping in HPLMN the EHPLMN or a subscribed SNPN, the UE shall use this indication also when the UE is in the VPLMN or the non-subscribed SNPN.

When 3GPP PS data off is supported and active and the UE is configured, either as specified in 3GPP TS 24.167 [8G] or in 3GPP TS 31.102 [15C], with services that are 3GPP PS data off exempt, then the UE will not send uplink IP packets related to any services that are not 3GPP PS data off exempt over EPS IP-CAN, GPRS IP-CAN and 5GS IP-CAN. The UE informs the network about its 3GPP PS data off status by including a g.3gpp.ps-data-off media feature tag specified in subclauce 7.9.8 in all REGISTER requests sent over GPRS IP-CAN, EPS IP-CAN or 5GS IP-CAN. The UE reregisters over EPS IP-CAN, GPRS IP-CAN and 5GS IP-CAN every time the 3GPP PS data off status is changed or the UE is provided by the network with a new list of 3GPP PS data off exempt services while the 3GPP PS data off status is "active".

An AS handling a service is configured with information whether the service is a 3GPP PS data off exempt service. If the 3GPP PS data off status is active and the service is not a 3GPP PS data off exempt service, the AS prevents downlink IP packets of the service from reaching the UE over EPS IP-CAN, GPRS IP-CAN and 5GS IP-CAN. The AS shall be configured with up to two indications whether a 3GPP IMS service is a 3GPP PS Data Off exempt service, one indication is valid for non-roaming users, and the other indication is valid for users roaming in the various VPLMNs with whom roaming agreements exist or users accessing non-subscribed SNPNs. When the AS is only configured with the indication valid for the UE camping in the HPLMN, the EHPLMN or the subscribed SNPN, the AS shall use this indication also when the UE is in the VPLMN or non-subscribed SNPN.

4.18 Dynamic Service Interaction

Dynamic Service Interaction allows that different ASs involved in the same IMS session (within an operator network or across networks) exchange information about executed services to avoid conflicting interactions between these services. Dynamic Service Interaction information is included in a SIP header field Service-Interact-Info defined in subclause 7.2.14.

If an AS which supports dynamic service interaction:

- provides one or more services:

  a) the AS inserts in a SIP message the Service-Interact-Info header field with the identities of the services which have been performed; and

  b) if the AS identified services which should be further avoided the AS adds the identities of those services in the Service-Interact-Info header field; and

- receives a SIP message containing the Service-Interact-Info header field, the AS takes the received Service-Interact-Info header field information into account as described in subclause 7.2.14.3.

4.19 Restricted Local Operator Services

The UE and the network can support Restricted Local Operator Services (RLOS).

RLOS services are operator defined services that are offered to UEs when using an EPS IP-CAN as specified in annex L in the following scenarios:

- UE is successfully registered using IMS AKA or GPRS-IMS bundled authentication; or
- UE has attempted to register and the registration is rejected from the network with a 403 (Forbidden) response.

RLOS services are offered only for the UE-originating case.

RLOS services can be offered to an operator's own subscribers and roaming subscribers.

5 Application usage of SIP

5.1 Procedures at the UE

5.1.0 General

The UE procedures for UE detectable emergency calls are defined in subclause 5.1.6. Exceptions to UE procedures for SIP that do not relate to emergency, are documented in subclause 5.1.6 and shall apply. These exceptions include handling of a response to a request not detected by the UE as relating to an emergency.

When sending a failure response to any received request, depending on operator policy, the UE may insert a Response-Source header field with an "fe" header field parameter constructed with the URN namespace "urn:3gpp:fe", the fe-id part of the URN set to "ue" and optionally an appropriate fe-param part of the URN set in accordance with subclause 7.2.17. A UE when sending a failure response will add in the URN the "side" header field parameter set to:

- "orig" for a UE-originating case; and
- "term" for a UE-terminating case.

5.1.1 Registration and authentication

5.1.1.1 General

The UE shall register public user identities (see table A.4/1 and dependencies on that major capability).

NOTE 1: The UE can use multiple Contact header field values simultaneously containing the same IP address and port number in the contact address.

In case a UE registers several public user identities at different points in time, the procedures to re-register, deregister and subscribe to the registration-state event package for these public user identities can remain uncoordinated in time.

The UE can register any one of its public user identities with any IP address acquired by the UE. The same public user identity can be bound to more than one IP address of the UE. While having valid registrations of previously registered public user identities, the UE can register any additional public user identity with any of its IP addresses. When binding any one of its public user identities to an additional contact address, the UE shall follow the procedures described in RFC 5626 [92].

If SIP digest without TLS is used, the UE shall not include signalling plane security mechanisms in the header fields defined in RFC 3329 [48] in any SIP messages.

NOTE 2: The UE determines if SIP digest is used with or without TLS based on device configuration. If SIP digest with TLS is used, then the UE includes the TLS signalling plane security mechanism in the header fields defined in RFC 3329 [48] as described in subclause 5.1.1.2.4.

SIP requests that indicate security mechanisms for both the signalling plane and the media plane can contain multiple instances or a single instance of the Security-Client, Security-Verify, or Security-Server header fields defined in RFC 3329 [48].

In case a device performing address and/or port number conversions is provided by a NA(P)T or NA(P)T-PT, the UE may need to modify the SIP contents according to the procedures described in either annex F or annex K.

NOTE 3: If UE populates the display-name of the Contact header field included in the REGISTER request with UE name, other UEs of the user can discover the UE name of the UE in the reg event package notification. The UE name is a text string chosen by the user allowing the user to distinguish individual UEs of the same user.

5.1.1.1A Parameters contained in the ISIM

This subclause applies when a UE contains either an ISIM or a USIM.

The ISIM shall always be used for authentication to the IM CN subsystem, if it is present, as described in 3GPP TS 33.203 [19].

The ISIM is preconfigured with all the necessary parameters to initiate the registration to the IM CN subsystem. These parameters include:

- the private user identity;
- one or more public user identities; and
- the home network domain name used to address the SIP REGISTER request

The first public user identity in the list stored in the ISIM is used in emergency registration requests.

In case the UE does not contain an ISIM, the UE shall:

- generate a private user identity;
- generate a temporary public user identity; and
- generate a home network domain name to address the SIP REGISTER request to;

in accordance with the procedures in clause C.2.

The temporary public user identity is only used in REGISTER requests, i.e. initial registration, re-registration, UE-initiated deregistration.

The UE shall not reveal to the user the temporary public user identity if the temporary public user identity is barred. The temporary public user identity is not barred if received by the UE in the P-Associated-URI header field.

If the UE is unable to derive the parameters in this subclause for any reason, then the UE shall not proceed with the request associated with the use of these parameters and will not be able to register to the IM CN subsystem.

5.1.1.1B Parameters provisioned to a UE without ISIM or USIM

5.1.1.1B.1 Parameters provisioned in the IMC

In case the UE contains neither an ISIM nor a USIM, but IMC is present the UE shall use preconfigured parameters in the IMC to initiate the registration to the IM CN subsystem and for authentication.

The following IMS parameters are assumed to be available to the UE:

- a private user identity;
- a public user identity; and
- a home network domain name to address the SIP REGISTER request to.

These parameters may not necessarily reside in a UICC.

The first public user identity in the list stored in the IMC is used in emergency registration requests.

5.1.1.1B.2 Parameters when UE does not contain ISIM, USIM or IMC

If the UE contains neither ISIM, nor USIM nor IMC, the UE shall generate a temporary public user identity, a private user identity and a home network domain name to address the SIP REGISTER request to, according 3GPP TS 23.003 [3].

5.1.1.2 Initial registration

5.1.1.2.1 General

The initial registration procedure consists of the UE sending an unprotected REGISTER request and, if challenged depending on the security mechanism supported for this UE, sending the integrity-protected REGISTER request or other appropriate response to the challenge. The UE can register a public user identity with any of its contact addresses at any time after it has acquired an IP address, discovered a P-CSCF, and established an IP-CAN bearer that can be used for SIP signalling. However, the UE shall only initiate a new registration procedure when it has received a final response from the registrar for the ongoing registration, or the previous REGISTER request has timed out.

When registering any public user identity belonging to the UE, the UE shall either use an already active pair of security associations or a TLS session to protect the REGISTER requests, or register the public user identity via a new initial registration procedure.

When binding any one of its public user identities to an additional contact address via a new initial registration procedure, the UE shall follow the procedures described in RFC 5626 [92]. The set of security associations or a TLS session resulting from this initial registration procedure will have no impact on the existing set of security associations or TLS sessions that have been established as a result of previous initial registration procedures. However, if the UE registers any one of its public user identities with a new contact address via a new initial registration procedure and does not employ the procedures described in RFC 5626 [92], then the new set of security associations or TLS session shall replace any existing set of security association or TLS session.

If the UE detects that the existing security associations or TLS sessions associated with a given contact address are no longer active (e.g., after receiving no response to several protected messages), the UE shall:

- consider all previously registered public user identities bound to this security associations or TLS session that are only associated with this contact address as deregistered; and

- stop processing all associated ongoing dialogs and transactions that were using the security associations or TLS session associated with this contact address, if any (i.e. no further SIP signalling will be sent by the UE on behalf of these transactions or dialogs).

The UE shall send the unprotected REGISTER requests to the port advertised to the UE during the P-CSCF discovery procedure. If the UE does not receive any specific port information during the P-CSCF discovery procedure, or if the UE was pre-configured with the P-CSCF's IP address or domain name and was unable to obtain specific port information, the UE shall send the unprotected REGISTER request to the SIP default port values as specified in RFC 3261 [26].

NOTE 1: The UE will only send further registration and subsequent SIP messages towards the same port of the P-CSCF for security mechanisms that do not require to use negotiated ports for exchanging protected messages.

The UE shall extract or derive a public user identity, the private user identity, and the domain name to be used in the Request-URI in the registration, according to the procedures described in subclause 5.1.1.1A or subclause 5.1.1.1B. A public user identity may be input by the end user.

On sending an unprotected REGISTER request, the UE shall populate the header fields as follows:

a) a From header field set to the SIP URI that contains:

   1) if the UE supports RFC 6140 [191] and performs the functions of an external attached network, the main URI of the UE; else

   2) the public user identity to be registered;

b) a To header field set to the SIP URI that contains:

1) if the UE supports RFC 6140 [191] and performs the functions of an external attached network, the main URI of the UE; else

2) the public user identity to be registered;

c) a Contact header field set to include SIP URI(s) containing the IP address or FQDN of the UE in the hostport parameter. If the UE:

1) supports GRUU (see table A.4, item A.4/53);

2) supports multiple registrations;

3) has an IMEI available; or

4) has an MEID available;

the UE shall include a "+sip.instance" header field parameter containing the instance ID. Only the IMEI shall be used for generating an instance ID for a multi-mode UE that supports both 3GPP and 3GPP2 defined radio access networks.

NOTE 2: The requirement placed on the UE to include an instance ID based on the IMEI or the MEID when the UE does not support GRUU and does not support multiple registrations does not imply any additional requirements on the network.

If the UE supports multiple registrations it shall include a "reg-id" header field parameter as described in RFC 5626 [92].

The UE shall include all supported ICSI values (coded as specified in subclause 7.2A.8.2) in a g.3gpp.icsi-ref media feature tag as defined in subclause 7.9.2 and RFC 3840 [62] for the IMS communication services it intends to use, and IARI values (coded as specified in subclause 7.2A.9.2), for the IMS applications it intends to use in a g.3gpp.iari-ref media feature tag as defined in subclause 7.9.3 and RFC 3840 [62].

The UE shall include the media feature tags defined in RFC 3840 [62] and RFC 5688 [120] for all supported streaming media types.

If the UE supports RFC 6140 [191] and performs the functions of an external attached network, for the registration of bulk number contacts the UE shall include a Contact URI without a user portion and containing the "bnc" URI parameter.

If the UE has no specific reason not to include a user part in the URI of the contact address (eg. some UE performing the functions of an external attached network), the UE should include a user part in the URI of the contact address such that the user part is globally unique and does not reveal any private information;

NOTE 3: A time-based UUID (Universal Unique Identifier) generated as per subclause 4.2 of RFC 4122 [154] is globally unique and does not reveal any private information.

d) a Via header field set to include the sent-by field containing the IP address or FQDN of the UE and the port number where the UE expects to receive the response to this request when UDP is used. For TCP, the response is received on the TCP connection on which the request was sent. For the UDP, the UE shall also include a "rport" header field parameter with no value in the Via header field. Unless the UE has been configured to not send keep-alives, and unless the UE is directly connected to an IP-CAN for which usage of NAT is not defined, it shall include a "keep" header field parameter with no value in the Via header field, in order to indicate support of sending keep-alives associated with the registration, as described in RFC 6223 [143];

NOTE 4: When sending the unprotected REGISTER request using UDP, the UE transmit the request from the same IP address and port on which it expects to receive the response to this request.

e) a registration expiration interval value of 600 000 seconds as the value desired for the duration of the registration;

NOTE 5: The registrar (S-CSCF) might decrease the duration of the registration in accordance with network policy. Registration attempts with a registration period of less than a predefined minimum value defined in the registrar will be rejected with a 423 (Interval Too Brief) response.

f) a Request-URI set to the SIP URI of the domain name of the home network used to address the REGISTER request;

g) the Supported header field containing the option-tag "path", and

1) if GRUU is supported, the option-tag "gruu"; and

2) if multiple registrations is supported, the option-tag "outbound".

h) if a security association or TLS session exists, and if available to the UE (as defined in the access technology specific annexes for each access technology), a P-Access-Network-Info header field set as specified for the access network technology (see subclause 7.2A.4);

i) a Security-Client header field to announce the media plane security mechanisms the UE supports, if any, labelled with the "mediasec" header field parameter specified in subclause 7.2A.7;

NOTE 6: The "mediasec" header field parameter indicates that security mechanisms are specific to the media plane.

j) if the UE supports RFC 6140 [191] and performs the functions of an external attached network, for the registration of bulk number contacts the UE shall include a Require header field containing the option-tag "gin"; and

k) if the UE supports RFC 6140 [191] and performs the functions of an external attached network, for the registration of bulk number contacts the UE shall include a Proxy-Require header field containing the option-tag "gin".

On receiving a 401 (Unauthorized) response to the REGISTER request, the UE shall:

a) if available, store the announcement of media plane security mechanisms the P-CSCF (IMS-ALG) supports labelled with the "mediasec" header field parameter specified in subclause 7.2A.7 and received in the Security-Server header field, if any. Once the UE chooses a media security mechanism from the list received in the Security-Server header field from the server, the UE may initiate that mechanism on a media level when it initiates new media in an existing session.

NOTE 7: The "mediasec" header field parameter indicates that security mechanisms are specific to the media plane.

On receiving the 200 (OK) response to the REGISTER request, the UE shall:

a) store the expiration time of the registration for the public user identities found in the To header field value and bind it either to the respective contact address of the UE or to the registration flow and the associated contact address (if the multiple registration mechanism is used);

NOTE 8: If the UE supports RFC 6140 [191] and performs the functions of an external attached network, the To header field will contain the main URI of the UE.

b) store as the default public user identity the first URI on the list of URIs present in the P-Associated-URI header field and bind it to the respective contact address of the UE and the associated set of security associations or TLS session;

NOTE 9: When using the respective contact address and associated set of security associations or TLS session, the UE can utilize additional URIs contained in the P-Associated-URI header field and bound it to the respective contact address of the UE and the associated set of security associations or TLS session, e.g. for application purposes.

c) treat the identity under registration as a barred public user identity, if it is not included in the P-Associated-URI header field;

d) store the list of service route values contained in the Service-Route header field and bind the list either to the contact address or to the registration flow and the associated contact address (if the multiple registration mechanism is used), and the associated set of security associations or TLS session over which the REGISTER request was sent;

NOTE 10: When multiple registration mechanism is not used, there will be only one list of service route values bound to a contact address. However, when multiple registration mechanism is used, there will be different list of service route values bound to each registration flow and the associated contact address.

NOTE 11: The UE will use the stored list of service route values to build a proper preloaded Route header field for new dialogs and standalone transactions (other than REGISTER method) when using either the respective contact address or the registration flow and the associated contact address (if the multiple registration mechanism is used), and the associated set of security associations or TLS session.

e) if the UE indicated support for GRUU in the Supported header field of the REGISTER request then:

- if the UE did not use the procedures specified in RFC 6140 [191] for registration, find the Contact header field within the response that matches the one included in the REGISTER request. If this contains a "pub-gruu" header field parameter or a "temp-gruu" header field parameter or both, then store the value of those parameters as the GRUUs for the UE in association with the public user identity and the contact address that was registered; and

- if the UE used the procedures specified in RFC 6140 [191] for registration then find the Contact header field within the response that matches the one included in the REGISTER request. If this contains a "pub-gruu" header field parameter then store the value of the "pub-gruu" header field parameter for use for generating public GRUUs for registering UAs as specified in RFC 6140 [191]. If this contains a "temp-gruu-cookie" header field parameter then store the value of the "temp-gruu-cookie" header field parameter for use for generating temporary GRUUs for registering UAs as specified in RFC 6140 [191];

NOTE 12: When allocating public GRUUs to registering UAs the functionality within the UE that performs the role of registrar will add an "sg" SIP URI parameter that uniquenly identifies that UA to the public GRUU it received in the "pub-gruu" header field parameter. The procedures for generating a temporary GRUU using the "temp-gruu-cookie" header field parameter are specified in subclause 7.1.2.2 of RFC 6140 [191].

f) if the REGISTER request contained the "reg-id" and "+sip.instance" Contact header field parameter and the "outbound" option tag in a Supported header field, the UE shall check whether the option-tag "outbound" is present in the Require header field:

- if no option-tag "outbound" is present, the UE shall conclude that the S-CSCF does not support the registration procedure as described in RFC 5626 [92], and the S-CSCF has followed the registration procedure as described in RFC 5627 [93] or RFC 3261 [26], i.e., if there is a previously registered contact address, the S-CSCF replaced the old contact address and associated information with the new contact address and associated information (see bullet e) above). Upon detecting that the S-CSCF does not support the registration procedure as defined in RFC 5626 [92], the UE shall refrain from registering any additional IMS flows for the same private identity as described in RFC 5626 [92]; or

NOTE 13: Upon replaces the old contact address with the new contact address, the S-CSCF performs the network initiated deregistration procedure for the previously registered public user identities and the associated old contact address as described in subclause 5.4.1.5. Hence, the UE will receive a NOTIFY request informing the UE about the deregistration of the old contact address.

- if an option-tag "outbound" is present, the UE may establish additional IMS flows for the same private identity, as defined in RFC 5626 [92];

g) if available, store the announcement of media plane security mechanisms the P-CSCF (IMS-ALG) supports labelled with the "mediasec" header field parameter specified in subclause 7.2A.7 and received in the Security-Server header field, if any. Once the UE chooses a media security mechanism from the list received in the Security-Server header field from the server, it may initiate that mechanism on a media level when it initiates new media in an existing session;

NOTE 14: The "mediasec" header field parameter indicates that security mechanisms are specific to the media plane.

h) if the Via header field contains a "keep" header field parameter with a value, unless the UE detects that it is not behind a NAT, start to send keep-alives associated with the registration towards the P-CSCF, as described in RFC 6223 [143];

i) if a Feature-Caps header field, as specified in RFC 6809 [190] is received, a UE supporting the Feature-Caps header field shall consider the ICSI values received in the Feature-Caps header field of 200 (OK) response as supported by the IM subsystem for the established registration or registration flow (if the multiple registration mechanism is used);

NOTE 15: The UE and related applications can use the ICSI values received in the Feature-Caps header field of 200 (OK) response to improve the user experience.

j) void; and

k) if the 200 (OK) response includes a Feature-Caps header field, as specified in RFC 6809 [190], with a "+g.3gpp.verstat" header field parameter and if the UE supports calling number verification status determination, determine that the home network supports calling number verification using signature verification and attestation information, as defined in subclause 3.1.

On receiving a 305 (Use Proxy) response to the unprotected REGISTER request, unless otherwise specified in access specific annexes (as described in annex B, annex L or annex U), the UE shall:

a) ignore the contents of the Contact header field if it is included in the received message;

NOTE 16: The 305 response is not expected to contain a Contact header field.

b) release all IP-CAN bearers used for the transport of media according to the procedures in subclause 9.2.2;

c) initiate either a new P-CSCF discovery procedure as described in subclause 9.2.1, or select a new P-CSCF, if the UE was pre-configured with more than one P-CSCF's IP addresses or domain names;

d) select a P-CSCF address, which is different from the previously used address, from the address list; and

e) perform the procedures for initial registration as described in subclause 5.1.1.2.

On receiving a 423 (Interval Too Brief) response to the REGISTER request, the UE shall:

- send another REGISTER request populating the registration expiration interval value with an expiration timer of at least the value received in the Min-Expires header field of the 423 (Interval Too Brief) response.

On receiving a 408 (Request Timeout) response or 500 (Server Internal Error) response or 504 (Server Time-Out) or 600 (Busy Everywhere) response or 403 (Forbidden) response for an initial registration, the UE may attempt to perform initial registration again.

When the timer F expires at the UE, the UE:

a) shall mark the currently used P-CSCF address as unavailable for the last duration of the retry delay time computed by the algorithm defined in subclause 4.5 of RFC 5626 [92] plus 5 minutes;

b) if there is a locally stored P-CSCF address as specified in subclause 5.1.9 which is different from the currently used P-CSCF address and which is not marked as unavailable, may initiate an initial registration as specified in subclause 5.1.1.2 using that P-CSCF; and

c) if there is no locally stored P-CSCF address as specified in subclause 5.1.9 which is different from the currently used P-CSCF address and which is not marked as unavailable, may get a new set of P-CSCF-addresses as described in subclause 9.2.1 unless otherwise specified in the access specific annexes (as described in annex B, annex L or annex U) and initiate an initial registration as specified in subclause 5.1.1.2.

NOTE 17: It is an implementation option whether these actions are also triggered by other means than expiration of timer F, e.g. based on ICMP messages.

On receiving a 4xx, 5xx (except 503) or 6xx response to the REGISTER request, whereby the response contains a Retry-After header field, the UE shall not automatically attempt an initial registration via the same IP-CAN and the same P-CSCF for the amount of time indicated in the Retry-After header field. If the UE is power cycled, the UE can attempt an initial registration. If no initial registration occurs within the time period indicated by the Retry-After header field, the counter of unsuccessful initial registration attempts is reset.

On receiving a 503 response with a Retry-After header field to the REGISTER request and the Retry-After header field indicates time bigger than the value for timer F as specified in table 7.7.1, the UE:

a) shall mark the currently used P-CSCF address as unavailable for the time indicated by the Retry-After header field;

b) if there is a locally stored P-CSCF address as specified in subclause 5.1.9 which is different from the currently used P-CSCF address and which is not marked as unavailable, may initiate an initial registration as specified in subclause 5.1.1.2 using that P-CSCF; and

c) if there is no locally stored P-CSCF address as specified in subclause 5.1.9 which is different from the currently used P-CSCF address and which is not marked as unavailable, may get a new set of P-CSCF addresses as described in subclause 9.2.1 unless otherwise specified in the access specific annexes (as described in annex B, annex L or annex U) and initiate an initial registration as specified in subclause 5.1.1.2.

NOTE 18: if the Retry-After header field indicates time smaller than the value for timer F as specified in table 7.7.1, the UE continues using the currently used P-CSCF address.

For each 4xx, 5xx or 6xx response received without a Retry-After header field to the REGISTER request, the UE shall:

a) use the mechanism defined in subclause 4.5 of RFC 5626 [92] for determination of the retry delay time before each new registration attempt;

b) mark the currently used P-CSCF address as unavailable for the last duration of the retry delay time computed by the algorithm defined in subclause 4.5 of RFC 5626 [92] plus 5 minutes; and

c) initiate an initial registration as specified in subclause 5.1.1.2 after the amount of time of the last retry delay time computed by the algorithm defined in subclause 4.5 of RFC 5626 [92]; and

- if there is a locally stored P-CSCF address as specified in subclause 5.1.9 which is different from the currently used P-CSCF address and which is not marked as unavailable, may initiate the initial registration using that P-CSCF; and

- if there is no locally stored P-CSCF address as specified in subclause 5.1.9 which is different from the currently used P-CSCF address and which is not marked as unavailable, may get a new set of P-CSCF addresses as described in subclause 9.2.1 unless otherwise specified in the access specific annexes (as described in annex B, annex L or annex U) and initiate the initial registration as specified in subclause 5.1.1.2.

The values of max-time and base-time (if all failed) may be provided by the network to the UE with the management objects specified in 3GPP TS 24.167 [8G]. If no values of the parameters max-time and base-time (if all failed) have been provided to the UE by the network, the default values defined in subclause 4.5 of RFC 5626 [92] shall be used. Other mechanisms may be used as well and are outside the scope of the present document.

NOTE 19: If the UE stops initiating initial registration, the UE can attempt to perform initial registration with the network again based on the local configuration and not wait till power-cycled. The local configuration is outside the scope of this specification.

5.1.1.2.2 Initial registration using IMS AKA

On sending a REGISTER request, as defined in subclause 5.1.1.2.1, the UE shall additionally populate the header fields as follows:

a) an Authorization header field, with:

- the "username" header field parameter, set to the value of the private user identity;

- the "realm" header field parameter, set to the domain name of the home network;

- the "uri" header field parameter, set to the SIP URI of the domain name of the home network;

- the "nonce" header field parameter, set to an empty value; and

- the "response" header field parameter, set to an empty value;

NOTE 1: If the UE specifies its FQDN in the hostport parameter in the Contact header field and in the sent-by field in the Via header field, then it has to ensure that the given FQDN will resolve (e.g., by reverse DNS lookup) to the IP address that is bound to the security association.

NOTE 2: The UE associates two ports, a protected client port and a protected server port, with each pair of security association. For details on the selection of the port values see 3GPP TS 33.203 [19].

b) additionally for the Contact header field, if the REGISTER request is protected by a security association, include the protected server port value in the hostport parameter;

c) additionally for the Via header field, for UDP, if the REGISTER request is protected by a security association, include the protected server port value in the sent-by field; and

d) a Security-Client header field set to specify the signalling plane security mechanism the UE supports, the IPsec layer algorithms the UE supports and the parameters needed for the security association setup. The UE shall support the setup of two pairs of security associations as defined in 3GPP TS 33.203 [19]. The syntax of the parameters needed for the security association setup is specified in annex H of 3GPP TS 33.203 [19]. The UE shall support the "ipsec-3gpp" security mechanism, as specified in RFC 3329 [48]. The UE shall support the IPsec layer algorithms for integrity and confidentiality protection as defined in 3GPP TS 33.203 [19], and shall announce support for them according to the procedures defined in RFC 3329 [48].

On receiving the 200 (OK) response to the REGISTER request defined in subclause 5.1.1.2.1, the UE shall additionally:

1) If the UE supports multiple registrations and the REGISTER request contained the "+sip.instance" header field parameter and the "reg-id" header field parameter in the Contact header field, and the "outbound" option-tag in the Supported header field, the UE shall check whether the option-tag "outbound" is present in the Require header field. If the option-tag "outbound" is present, then the UE shall use the bidirectional flow as defined in RFC 5626 [92] as follows:

   a) for UDP, the bidirectional flow consists of two unidirectional flows, i.e. the first unidirectional flow is identified with the UE's protected client port, the P-CSCF's protected server port, and the respective IP addresses. The UE uses this flow to send the requests and responses to the P-CSCF. The second unidirectional flow is identified with the P-CSCF's protected client port, the UE's protected server port and the IP addresses. The second unidirectional flow is used by the UE to receive the requests and responses from the P-CSCF; or

   b) for TCP, the bidirectional flow is the TCP connection between the UE and the P-CSCF. This TCP connection was established by the UE, i.e. from the UE's protected client port and the UE's IP address to the P-CSCF's protected server port and the P-CSCF's IP address. This TCP connection is used to exchange SIP messages between the UE and the P-CSCF; and

2) set the security association lifetime to the longest of either the previously existing security association lifetime (if available), or the lifetime of the just completed registration plus 30 seconds.

NOTE 3: If the UE receives Authentication-Info, it will proceed as described in RFC 3310 [49] when AKAv1 is used or as described in RFC 4169 [227] when AKAv2 is used.

When a 401 (Unauthorized) response to a REGISTER is received the UE shall behave as described in subclause 5.1.1.5.1.

5.1.1.2.3 Initial registration using SIP digest without TLS

On sending a REGISTER request, as defined in subclause 5.1.1.2.1, the UE shall additionally populate the header fields as follows:

a) an Authorization header field as defined in RFC 7616 [286] and RFC 8760 [287] unless otherwise specified in the access specific annexes, with:

   - the "username" header field parameter, set to the value of the private user identity;
   - the "realm" header field parameter, set to the domain name of the home network;
   - the "uri" header field directive, set to the SIP URI of the domain name of the home network;
   - the "nonce" header field parameter, set to an empty value; and
   - the "response" header field parameter, set to an empty value;

b) the hostport parameter in the Contact header field with the port value of an unprotected port where the UE expects to receive subsequent requests; and

c) the sent-by field in the Via header field with the port value of an unprotected port where the UE expects to receive responses to the request.

The UE shall use the locally available public user identity, the private user identity, and the domain name to be used in the Request-URI in the registration. The method whereby the public user identity and private user identity are made available to the UE is outside the scope of this document (e.g. a public user identity could be input by the end user).

When a 401 (Unauthorized) response to a REGISTER is received the UE shall behave as described in subclause 5.1.1.5.4.

5.1.1.2.4 Initial registration using SIP digest with TLS

On sending a REGISTER request, as defined in subclause 5.1.1.2.1, the UE shall additionally populate the header fields as follows:

a) an Authorization header field set in accordance with subclause 5.1.1.2.3 unless otherwise specified in the access specific annexes; and

b) a Security-Client header field set to specify the signalling plane security mechanism the UE supports. The UE shall support the setup of a TLS session as defined in 3GPP TS 33.203 [19]. The UE shall support the "tls" security mechanism, as specified in RFC 3329 [48]. The UE shall support TLS for integrity and confidentiality protection as defined in RFC 3261 [26], and shall announce support for them according to the procedures defined in RFC 3329 [48].

On receiving the 200 (OK) response to the REGISTER request defined in subclause 5.1.1.2.1, the UE shall additionally:

a) set the TLS session lifetime to the longest of either the previously existing TLS session lifetime (if available), or the lifetime of the just completed registration plus 30 seconds.

If a UE supports TLS, then the UE shall support TLS ciphersuites as described in 3GPP TS 33.203 [19]. TLS session lifetime is determined by local configuration of the UE.

For SIP digest with TLS, the UE associates a protected server port with the TLS session port on the UE.

When a 401 (Unauthorized) response to a REGISTER is received the UE shall behave as described in subclause 5.1.1.5.6.

5.1.1.2.5 Initial registration using NASS-IMS bundled authentication

On sending a REGISTER request, as defined in subclause 5.1.1.2.1, the UE shall additionally populate the header fields as follows:

a) optionally, an Authorization header field, with the "username" header field parameter, set to the value of the private user identity;

NOTE 1: In case the Authorization header field is absent, the mechanism only supports that one public user identity is associated with only one private user identity. The public user identity is set so that it is possible to derive the private user identity from the public user identity by removing SIP URI scheme and the following parts of the SIP URI if present: port number, URI parameters, and To header field parameters.

On receiving the 200 (OK) response to the REGISTER request defined in subclause 5.1.1.2.1, there are no additional requirements for the UE.

NOTE 2: When NASS-IMS bundled authentication is in use, a 401 (Unauthorized) response to the REGISTER request is not expected to be received.

5.1.1.2.6 Initial registration using GPRS-IMS-Bundled authentication

On sending a REGISTER request, as defined in subclause 5.1.1.2.1, the UE shall additionally populate the header fields as follows:

a) an Authorization header field as defined in RFC 7616 [286] and RFC 8760 [287] shall not be included, in order to indicate support for GPRS-IMS-Bundled authentication.

b) the Security-Client header field as defined in RFC 3329 [48] shall not contain signalling plane security mechanisms;

c) a From header field set to a temporary public user identity as defined in 3GPP TS 23.003 [3], as the public user identity to be registered;

d) a To header field set to a temporary public user identity as defined in 3GPP TS 23.003 [3], as the public user identity to be registered;

e) the Contact header field with the port value of an unprotected port where the UE expects to receive subsequent mid-dialog requests; and

f) the Via header field with the port value of an unprotected port where the UE expects to receive responses to the request.

NOTE 1: Since the private user identity is not included in the REGISTER requests when GPRS-IMS-Bundled authentication is used for registration, re-registration and de-registration procedures, all REGISTER requests from the UE use the temporary IMPU as the public user identity even when the implicitly registered IMPUs are available at the UE. The UE does not use the temporary public user identity in any non-registration SIP requests.

On receiving the 200 (OK) response to the REGISTER request defined in subclause 5.1.1.2.1, there are no additional requirements for the UE.

NOTE 2: When GPRS-IMS-Bundled authentication is in use, a 401 (Unauthorized) response to the REGISTER request is not expected to be received.

5.1.1.3 Subscription to the registration-state event package

Upon receipt of a 2xx response to the initial registration, the UE shall subscribe to the reg event package for the public user identity registered at the user's registrar (S-CSCF) as described in RFC 3680 [43] and RFC 6665 [28].

NOTE 1: If the UE supports RFC 6140 [191] and performs the functions of an external attached network, the subscription will be directed to the main URI, as described in RFC 6140 [191].

The UE shall subscribe to the reg event package upon registering a new contact address via an initial registration procedure. If the UE receives a NOTIFY request via the newly established subscription dialog and via the previously established subscription dialogs (there will be at least one), the UE may terminate the previously established subscription dialogs and keep only the newly established subscription dialog.

The UE shall use the default public user identity for subscription to the registration-state event package.

NOTE 2: The subscription information stored in the HSS ensures that the default public user identity is a SIP URI.

On sending a SUBSCRIBE request, the UE shall populate the header fields as follows:

a) a Request-URI set to the resource to which the UE wants to be subscribed to, i.e. to the SIP URI that is the default public user identity used for subscription;

b) a From header field set to the SIP URI that is the default public user identity used for subscription;

c) a To header field set to the SIP URI that is the default public user identity used for subscription;

d) an Event header field set to the "reg" event package;

e) an Expires header field set to 600 000 seconds as the value desired for the duration of the subscription;

f) void; and

g) void.

Upon receipt of a dialog establishing NOTIFY request, as specified in RFC 6665 [28], associated with the SUBSCRIBE request, the UE shall:

1) store the information for the established dialog;

2) store the expiration time as indicated in the "expires" header field parameter of the Subscription-State header field, if present, of the NOTIFY request. Otherwise the expiration time is retrieved from the Expires header field of the 2xx response to SUBSCRIBE request; and

3) follow the procedures specified in RFC 6665 [28].

If continued subscription is required, the UE shall automatically refresh the subscription to the reg event package, for a previously registered public user identity, either 600 seconds before the expiration time if the initial subscription was for greater than 1200 seconds, or when half of the time has expired if the initial subscription was for 1200 seconds or less. If a SUBSCRIBE request to refresh a subscription fails with a non-481 response, the UE shall still consider the original subscription valid for the duration of the most recently known "Expires" value according to RFC 6665 [28]. Otherwise, the UE shall consider the subscription invalid and start a new initial subscription according to RFC 6665 [28].

5.1.1.3A Void

5.1.1.4 User-initiated reregistration and registration of an additional public user identity

5.1.1.4.1 General

The UE can perform the reregistration of a previously registered public user identity bound to any one of its contact addresses and the associated set of security associations or TLS sessions at any time after the initial registration has been completed.

The UE can perform the reregistration of a previously registered public user identity over any existing set of security associations or TLS session that is associated with the related contact address.

The UE can perform the reregistration of a previously registered public user identity via an initial registration as specified in subclause 5.1.1.2, when binding the previously registered public user identity to new contact address or to the registration flow and the associated contact address (if the multiple registration mechanism is used).

The UE can perform registration of additional public user identities at any time after the initial registration has been completed. The UE shall perform the registration of additional public user identities either:

- over the existing set of security associations or TLS sessions, if appropriate to the security mechanism in use, that is associated with the related contact address; or

- via an initial registration as specified in subclause 5.1.1.2.

The UE can fetch bindings as defined in RFC 3261 [26] at any time after the initial registration has been completed. The procedure for fetching bindings is the same as for a reregistration except that the REGISTER request does not contain a Contact header field.

Unless either the user or the application within the UE has determined that a continued registration is not required the UE shall reregister an already registered public user identity either 600 seconds before the expiration time if the previous registration was for greater than 1200 seconds, or when half of the time has expired if the previous registration was for 1200 seconds or less, or when the UE intends to update its capabilities according to RFC 3840 [62] and RFC 5688 [120] or when the UE needs to modify the ICSI values that the UE intends to use in a g.3gpp.icsi-ref media feature tag or IARI values that the UE intends to use in the g.3gpp.iari-ref media feature tag.

When sending a protected REGISTER request, the UE shall use a security association or TLS session associated either with the contact address or with the registration flow and the associated contact address used to send the request, see 3GPP TS 33.203 [19], established as a result of an earlier initial registration.

The UE shall extract or derive a public user identity, the private user identity, and the domain name to be used in the Request-URI in the registration, according to the procedures described in subclause 5.1.1.1A or subclause 5.1.1.1B.

On sending a REGISTER request that does not contain a challenge response, the UE shall populate the header fields as follows:

a) a From header field set to the SIP URI that contains:

1) if the UE supports RFC 6140 [191] and performs the functions of an external attached network, the main URI of the UE; else

2) the public user identity to be registered;

b) a To header field set to the SIP URI that contains:

1) if the UE supports RFC 6140 [191] and performs the functions of an external attached network, the main URI of the UE; else

2) the public user identity to be registered;

c) a Contact header field set to include SIP URI(s) that contain(s) in the hostport parameter the IP address or FQDN of the UE, and containing the instance ID of the UE in the "+sip.instance" header field parameter, if the UE:

1) supports GRUU (see table A.4, item A.4/53);

2) supports multiple registrations;

3) has an IMEI available; or

4) has an MEID available.

Only the IMEI shall be used for generating an instance ID for a multi-mode UE that supports both 3GPP and 3GPP2 defined radio access networks.

NOTE 1: The requirement placed on the UE to include an instance ID based on the IMEI or the MEID when the UE does not support GRUU and does not support multiple registrations does not imply any additional requirements on the network.

If the UE support multiple registrations, it shall include "reg-id" header field as described in RFC 5626 [92]. The UE shall include all supported ICSI values (coded as specified in subclause 7.2A.8.2) in a g.3gpp.icsi-ref media feature tag as defined in subclause 7.9.2 and RFC 3840 [62] for the IMS communication it intends to use, and IARI values (coded as specified in subclause 7.2A.9.2), for the IMS applications it intends to use in a g.3gpp.iari-ref media feature tag as defined in subclause 7.9.3 and RFC 3840 [62].

If the UE supports RFC 6140 [191] and performs the functions of an external attached network, for the registration of bulk number contacts the UE shall include a Contact URI without a user portion and containing the "bnc" URI parameter.

If a user part has previously been included in an initial REGISTER request, the UE shall use the user part which was previously used to create the binding being refreshed or removed;

d) a Via header field set to include the IP address or FQDN of the UE in the sent-by field. For the TCP, the response is received on the TCP connection on which the request was sent. If the UE previously has previously negotiated sending of keep-alives associated with the registration, it shall include a "keep" header field parameter with no value in the Via header field, in order to indicate continuous support to send keep-alives, as described in RFC 6223 [143];

e) a registration expiration interval value, set to 600 000 seconds as the value desired for the duration of the registration;

NOTE 2: The registrar (S-CSCF) might decrease the duration of the registration in accordance with network policy. Registration attempts with a registration period of less than a predefined minimum value defined in the registrar will be rejected with a 423 (Interval Too Brief) response.

f) a Request-URI set to the SIP URI of the domain name of the home network used to address the REGISTER request;

g) the Supported header field containing the option-tag "path", and:

1) if GRUU is supported, the option-tag "gruu"; and

2) if multiple registrations is supported, the option-tag "outbound";

h) if available to the UE (as defined in the access technology specific annexes for each access technology), a P-Access-Network-Info header field set as specified for the access network technology (see subclause 7.2A.4);

i) a Security-Client header field to announce the media plane security mechanisms the UE supports, if any, labelled with the "mediasec" header field parameter specified in subclause 7.2A.7;

NOTE 3: The "mediasec" header field parameter indicates that security mechanisms are specific to the media plane.

j) if the UE supports RFC 6140 [191] and performs the functions of an external attached network, for the registration of bulk number contacts the UE shall include a Require header field containing the option-tag "gin"; and

k) if the UE supports RFC 6140 [191] and performs the functions of an external attached network, for the registration of bulk number contacts the UE shall include a Proxy-Require header field containing the option-tag "gin".

On receiving the 200 (OK) response to the REGISTER request, the UE shall:

a) bind the new expiration time of the registration for this public user identity found in the To header field value either to the contact address or to the registration flow and the associated contact address used in this registration;

NOTE 4: If the UE supports RFC 6140 [191] and performs the functions of an external attached network, the To header field will contain the main URI of the UE.

b) store the list of service route values contained in the Service-Route header field and bind the list either to the contact address or to the registration flow and the associated contact address (if the multiple registration mechanism is used);

NOTE 5: The stored list of service route values will be used to build a proper preloaded Route header field for new dialogs and standalone transactions (other than REGISTER method) when using either the respective contact address or the registration flow and the associated contact address (if the multiple registration mechanism is used).

NOTE 6: If the list of Service-Route headers saved from a previous registration and bound either to this contact address or to the registration flow and the associated contact address (if the multiple registration mechanism is used), and the associated set of security associations or TLS session already exist, then the received list of Service-Route headers replaces the old list.

NOTE 7: The UE can utilize additional URIs contained in the P-Associated-URI header field, e.g. for application purposes.

c) if the UE indicated support for GRUU in the Supported header field of the REGISTER request then:

- if the UE did not use the procedures specified in RFC 6140 [191] for registration find the Contact header field within the response that matches the one included in the REGISTER request. If this contains a "pub-gruu" header field parameter or a "temp-gruu" header field parameter or both, then store the value of those parameters as the GRUUs for the UE in association with the public user identity and the contact address that was registered; and

- if the UE used the procedures specified in RFC 6140 [191]for registration then find the Contact header field within the response that matches the one included in the REGISTER request. If this contains a "pub-gruu" header field parameter then store the value of the "pub-gruu" header field parameter for use for generating public GRUUs for registering UAs as specified in RFC 6140 [191]. If this contains a "temp-gruu-cookie" header field parameter then store the value of the "temp-gruu-cookie" header field parameter for use for generating temporary GRUUs for registering UAs as specified in RFC 6140 [191];

NOTE 8: When allocating public GRUUs to registering UAs the functionality within the UE that performs the role of registrar will add an "sg" SIP URI parameter that uniquenly identifies that UA to the public GRUU it received in the "pub-gruu" header field parameter. The procedures for generating a temporary GRUU using the "temp-gruu-cookie" header field parameter are specified in subclause 7.1.2.2 of RFC 6140 [191].

d) store the announcement of the media plane security mechanisms the P-CSCF (IMS-ALG) supports received in the Security-Server header field and labelled with the "mediasec" header field parameter specified in

subclause 7.2A.7, if any. Once the UE chooses a media security mechanism from the list received in the Security-Server header field from the server, it may initiate that mechanism on a media level when it initiates new media in an existing session;

NOTE 9: The "mediasec" header field parameter indicates that security mechanisms are specific to the media plane.

e) if the Via header field contains a "keep" header field parameter with a value, continue to send keep-alives as described in RFC 6223 [143], towards the P-CSCF;

f) if the 200 (OK) response contains the Authentication-Info header field including a nextnonce field, store the contained nonce as a nonce for authentication associated to the same registration or registration flow (if the multiple registration mechanism is used) and shall delete any other previously stored nonce value for authentication for this registration or registration flow (if the multiple registration mechanism is used); and

NOTE 10: The related registration flow or registration is identified by the couple instance-id and reg-id if the multiple registration mechanism is used or by contact address if not.

g) if a Feature-Caps header field is received, a UE supporting the Feature-Caps header field shall consider the ICSI values received in the Feature-Caps header field of 200 (OK) response as supported for the established registration or registration flow (if the multiple registration mechanism is used) according to RFC 6809 [190]; and

NOTE 11: The UE and related applications can use the ICSI values received in the Feature-Caps header field to improve the user experience.

h) void.

When a 401 (Unauthorized) response to a REGISTER is received the UE shall behave as described in subclause 5.1.1.5.1.

On receiving a 423 (Interval Too Brief) response to the REGISTER request, the UE shall:

- send another REGISTER request populating the registration expiration interval value with an expiration timer of at least the value received in the Min-Expires header field of the 423 (Interval Too Brief) response.

On receiving a 408 (Request Timeout) response or 500 (Server Internal Error) response or 504 (Server Time-Out) response or 403 (Forbidden) response for a reregistration, the UE shall perform the procedures for initial registration as described in subclause 5.1.1.2.

On receiving a 305 (Use Proxy) response to the REGISTER request, unless otherwise specified in the access specific annexes (as described in annex B, annex L or annex U), the UE shall:

a) ignore the contents of the Contact header field if it is included in the received message;

NOTE 12: The 305 response is not expected to contain a Contact header field.

b) release all IP-CAN bearers used for the transport of media according to the procedures in subclause 9.2.2;

c) initiate either a new P-CSCF discovery procedure as described in subclause 9.2.1, or select a new P-CSCF, if the UE was pre-configured with more than one P-CSCF's IP addresses or domain names;

d) select a P-CSCF address, which is different from the previously used address, from the address list; and

e) perform the procedures for initial registration as described in subclause 5.1.1.2.

When the timer F expires at the UE:

1) the UE shall stop processing of all ongoing dialogs and transactions associated with that flow, if any (i.e. no further SIP signalling will be sent by the UE on behalf of these transactions or dialogs); and

2) after releasing all IP-CAN bearers used for the transport of media according to the procedures in subclause 9.2.2:

   a) the UE may select a different P-CSCF address from the list of P-CSCF addresses discovered during the procedures described in subclause 9.2.1 or from its pre-configured list of P-CSCF's IP addresses or domain names;

b) if no response has been received when attempting to contact all P-CSCFs known by the UE, the UE may get a new set of P-CSCF-addresses as described in subclause 9.2.1 unless otherwise specified in the access specific annexes (as described in annex B, annex L or annex U);

c) the UE may perform the procedures for initial registration as described in subclause 5.1.1.2; and

d) the UE shall perform the procedures in RFC 5626 [92] to form a new flow to replace the failed one if it supports multiple registrations. If failed registration attempts occur in the process of creating a new flow, the UE shall implement the flow recovery procedures defined in subclause 4.5 of RFC 5626 [92] for determination of the retry delay time before each new registration attempt. The UE shall use the values of the parameters max-time and base-time, of the algorithm defined in subclause 4.5 of RFC 5626 [92]. If no values of the parameters max-time and base-time (if all failed) have been provided to the UE by the network, the default values defined in subclause 4.5 of RFC 5626 [92] shall be used.

NOTE 13: It is an implementation option whether these actions are also triggered by other means than expiration of timer F, e.g. based on ICMP messages.

5.1.1.4.2 IMS AKA as a security mechanism

On sending a REGISTER request, as defined in subclause 5.1.1.4.1, the UE shall additionally populate the header fields as follows:

a) an Authorization header field, with:

- the "username" header field parameter set to the value of the private user identity;
- the "realm" header field parameter directive, set to the value as received in the "realm" WWW-Authenticate header field parameter;
- the "uri" header field parameter, set to the SIP URI of the domain name of the home network;
- the "nonce" header field parameter, set to last received nonce value; and
- the "response" header field parameter, set to the last calculated response value;

NOTE 1: If the UE specifies its FQDN in the hostport parameter in the Contact header field and in the sent-by field in the Via header field, then it has to ensure that the given FQDN will resolve (e.g., by reverse DNS lookup) to the IP address that is bound to the security association.

NOTE 2: The UE associates two ports, a protected client port and a protected server port, with each pair of security associations. For details on the selection of the protected port value see 3GPP TS 33.203 [19].

NOTE 3: If the UE is setting up an additional registration using procedures specified in RFC 5626 [92] and the UE accesses the network through 3GPP or 3GPP2 systems without any NAT, the flow is considered to be "logical flow".

b) additionally for the Contact header field, include the protected server port value in the hostport parameter;

c) additionally for the Via header field, for UDP, if the REGISTER request is protected by a security association, include the protected server port value in the sent-by field;

d) a Security-Client header field, set to specify the signalling plane security mechanism it supports, the IPsec layer algorithms for security and confidentiality protection it supports and the new parameter values needed for the setup of two new pairs of security associations. For further details see 3GPP TS 33.203 [19] and RFC 3329 [48]; and

e) a Security-Verify header field that contains the content of the Security-Server header field received in the 401 (Unauthorized) response of the last successful authentication.

On receiving the 200 (OK) response to the REGISTER request, the UE shall additionally:

a) set the security association lifetime associated with either this contact address or the registration flow and the associated contact address (if the multiple registration mechanism is used), and the associated set of security associations to the longest of either the previously existing security association lifetime, or the lifetime of the just completed registration plus 30 seconds.

NOTE 4: If the UE receives Authentication-Info, it will proceed as described in RFC 3310 [49] when AKAv1 is used or as described in RFC 4169 [227] when AKAv2 is used.

5.1.1.4.3 SIP digest without TLS as a security mechanism

On sending a REGISTER request, as defined in subclause 5.1.1.4.1, the UE shall additionally populate the header fields as follows:

a) an Authorization header field as defined in RFC 7616 [286] and RFC 8760 [287], including:

- the "username" header field parameter, set to the value of the private user identity;
- the "realm" header field parameter, set to the domain name of the home network;
- the "uri" header field parameter, set to the SIP URI of the domain name of the home network;
- the "nonce" header field parameter, set to the stored nonce value for authentication for the related registration or registration flow (if the multiple registration mechanism is used); and

NOTE: The related registration flow or registration is identified by the couple instance-id and reg-id if the multiple registration mechanism is used or by contact address if not.

- the "response" header field parameter, set to the challenge response, constructed using the stored nonce value for authentication for the same registration or registration flow (if the multiple registration mechanism is used), along with "algorithm", "cnonce", "qop", and "nc" header field parameters as specified in RFC 7616 [286] and RFC 8760 [287];

b) the Contact header field with the port value of an unprotected port where the UE expects to receive subsequent requests; and

c) the Via header field with the port value of an unprotected port where the UE expects to receive responses to the request.

5.1.1.4.4 SIP digest with TLS as a security mechanism

On sending a REGISTER request, as defined in subclause 5.1.1.4.1, the UE shall additionally populate the header fields as follows:

a) an Authorization header field set in accordance with subclause 5.1.1.4.3;

b) the Security-Client header field set to specify the signalling plane security mechanism the UE supports. The UE shall support the setup of a TLS session as defined in 3GPP TS 33.203 [19]. The UE shall support the "tls" security mechanism, as specified in RFC 3329 [48]. The UE shall support TLS for integrity and confidentiality protection as defined in RFC 3261 [26], and shall announce support for them according to the procedures defined in RFC 3329 [48]; and

c) a Security-Verify header field that contains the content of the Security-Server header field received in the 401 (Unauthorized) response of the last successful authentication.

On receiving the 200 (OK) response to the REGISTER request defined in subclause 5.1.1.2.1, the UE shall additionally:

a) set the lifetime of the respective TLS session to the value configured.

5.1.1.4.5 NASS-IMS bundled authentication as a security mechanism

On sending a REGISTER request, as defined in subclause 5.1.1.4.1, the UE shall additionally populate the header fields as follows:

a) optionally, an Authorization header field, with the "username" header field parameter, set to the value of the private user identity;

NOTE 1: In case the Authorization header field is absent, the mechanism only supports that one public user identity is associated with only one private user identity.

On receiving the 200 (OK) response to the REGISTER request defined in subclause 5.1.1.2.1, there are no additional requirements for the UE.

NOTE 2: When NASS-IMS bundled authentication is in use, a 401 (Unauthorized) response to the REGISTER request is not expected to be received.

5.1.1.4.6 GPRS-IMS-Bundled authentication as a security mechanism

On sending a REGISTER request, as defined in subclause 5.1.1.4.1, the UE shall additionally populate the header fields as follows:

a) an Authorization header field as defined in RFC 7616 [286] and RFC 8760 [287] shall not be included, in order to indicate support GPRS-IMS-Bundled authentication.

b) security agreement header field values as required by RFC 3329 [48] shall not contain signalling plane security mechanisms;

c) a From header field set to a temporary public user identity derived from the IMSI, as defined in 3GPP TS 23.003 [3], as the public user identity to be registered;

d) a To header field set to a temporary public user identity derived from the IMSI, as defined in 3GPP TS 23.003 [3], as the public user identity to be registered;

e) the Contact header field with the port value of an unprotected port where the UE expects to receive subsequent mid-dialog requests; and

f) the Via header field with the port value of an unprotected port where the UE expects to receive responses to the request.

NOTE 1: Since the private user identity is not included in the REGISTER requests when GPRS-IMS-Bundled authentication is used for registration, re-registration and de-registration procedures, all REGISTER requests from the UE use the IMSI-derived IMPU as the public user identity even when the implicitly registered IMPUs are available at the UE. The UE does not use the temporary public user identity (IMSI-derived IMPU) in any non-registration SIP requests.

On receiving the 200 (OK) response to the REGISTER request defined in subclause 5.1.1.4.1, there are no additional requirements for the UE.

NOTE 2: When GPRS-IMS-Bundled authentication is in use, a 401 (Unauthorized) response to the REGISTER request is not expected to be received.

5.1.1.5 Authentication

5.1.1.5.1 IMS AKA - general

Authentication is performed during initial registration. A UE can be re-authenticated during subsequent reregistrations, deregistrations or registrations of additional public user identities. When the network requires authentication or re-authentication of the UE, the UE will receive a 401 (Unauthorized) response to the REGISTER request.

On receiving a 401 (Unauthorized) response to the REGISTER request containing one or more WWW-Authenticate header fields the UE shall select the topmost header field that it supports (i.e. where the "algorithm" WWW-Authenticate header field parameter is "AKAv2-SHA-256" or "AKAv1-MD5"), the UE shall:

NOTE 1: The "AKAv1-MD5" algorithm is only supported for backward compatibility.

1) extract the RAND and AUTN parameters;

2) check the validity of a received authentication challenge, as described in 3GPP TS 33.203 [19] i.e. the locally calculated XMAC must match the MAC parameter derived from the AUTN part of the challenge; and the SQN parameter derived from the AUTN part of the challenge must be within the correct range; and

3) check the existence of the Security-Server header field as described in RFC 3329 [48]. If the Security-Server header field is not present or it does not contain the parameters required for the setup of the set of security associations (see annex H of 3GPP TS 33.203 [19]), the UE shall abandon the authentication procedure and send a new REGISTER request with a new Call-ID.

In the case that the 401 (Unauthorized) response to the REGISTER request is deemed to be valid the UE shall:

1) calculate the RES parameter and derive the keys CK and IK from RAND as described in 3GPP TS 33.203 [19];

2) set up a temporary set of security associations for this registration based on the static list and parameters the UE received in the 401 (Unauthorized) response and its capabilities sent in the Security-Client header field in the REGISTER request. The UE sets up the temporary set of security associations using the most preferred mechanism and algorithm returned by the P-CSCF and supported by the UE and using IK and CK (only if encryption enabled) as the shared key. The UE shall use the parameters received in the Security-Server header field to setup the temporary set of security associations. The UE shall set a temporary SIP level lifetime for the temporary set of security associations to the value of reg-await-auth timer;

3) store the announcement of the media plane security mechanisms the P-CSCF (IMS-ALG) supports received in the Security-Server header field and labelled with the "mediasec" header field parameter specified in subclause 7.2A.7, if any

NOTE 2: The "mediasec" header field parameter indicates that security mechanisms are specific to the media plane.

4) send another REGISTER request towards the protected server port indicated in the response using the temporary set of security associations to protect the message. The header fields are populated as defined for the initial REGISTER request that was challenged with the received 401 (Unauthorized) response, with the addition that the UE shall include an Authorization header field containing:

   - the "realm" header field parameter set to the value as received in the "realm" WWW-Authenticate header field parameter;

   - the "username" header field parameter, set to the value of the private user identity;

   - the "response" header field parameter that contains the RES parameter, as described in RFC 3310 [49] when AKAv1 is used or as described in RFC 4169 [227] when AKAv2 is used;

   - the "uri" header field parameter, set to the SIP URI of the domain name of the home network;

   - the "algorithm" header field parameter, set to the value received in the 401 (Unauthorized) response; and

   - the "nonce" header field parameter, set to the value received in the 401 (Unauthorized) response.

   The UE shall also insert the Security-Client header field that is identical to the Security-Client header field that was included in the previous REGISTER request (i.e. the REGISTER request that was challenged with the received 401 (Unauthorized) response). The UE shall also insert the Security-Verify header field into the request, by mirroring in it the content of the Security-Server header field received in the 401 (Unauthorized) response. The UE shall set the Call-ID of the security association protected REGISTER request which carries the authentication challenge response to the same value as the Call-ID of the 401 (Unauthorized) response which carried the challenge.

NOTE 3: The Security-Client header field contains signalling plane security mechanism and if the UE supports media plane security, then media plane security mechanisms are contained, too.

On receiving the 200 (OK) response for the security association protected REGISTER request registering a public user identity with the associated contact address, the UE shall:

- change the temporary set of security associations to a newly established set of security associations, i.e. set its SIP level lifetime to the longest of either the previously existing set of security associations SIP level lifetime, or the lifetime of the just completed registration plus 30 seconds; and

- if this is the only set of security associations available toward the P-CSCF, use the newly established set of security associations for further messages sent towards the P-CSCF. If there are additional sets of security associations (e.g. due to registration of multiple contact addresses), the UE can either use them or use the newly established set of security associations for further messages sent towards the P-CSCF as appropriate.

NOTE 4: If the UE has registered multiple contact addresses, the UE can either send requests towards the P-CSCF over the newly established set of security associations, or use different UE's contact address and associated set of security associations when sending the requests towards the P-CSCF. Responses towards the P-CSCF that are sent via UDP will be sent over the same set of security associations that the related request was received on. Responses towards the P-CSCF that are sent via TCP will be sent over the same set of security associations that the related request was received on.

When the first request or response protected with the newly established set of security associations is received from the P-CSCF or when the lifetime of the old set of security associations expires, the UE shall delete the old set of security associations and related keys it may have with the P-CSCF after all SIP transactions that use the old set of security associations are completed.

NOTE 5: If the UE has registered multiple contact addresses, the S-CSCF can use different contact address when sending the requests destined for the UE. In this case the UE will not receive the subsequent requests over the newly established set of security associations.

Whenever the 200 (OK) response is not received before the temporary SIP level lifetime of the temporary set of security associations expires or a 403 (Forbidden) response is received, the UE shall consider the registration to have failed. The UE shall delete the temporary set of security associations it was trying to establish, and use the old set of security associations. The UE should send an unprotected REGISTER request according to the procedure specified in subclause 5.1.1.2 if the UE considers the old set of security associations to be no longer active at the P-CSCF.

In the case that the 401 (Unauthorized) response is deemed to be invalid then the UE shall behave as defined in subclause 5.1.1.5.3.

5.1.1.5.2 Void

5.1.1.5.3 IMS AKA abnormal cases

If, in a 401 (Unauthorized) response, either the MAC or SQN is incorrect the UE shall respond with a further REGISTER indicating to the S-CSCF that the challenge has been deemed invalid as follows:

- in the case where the UE deems the MAC parameter to be invalid the subsequent REGISTER request shall contain no "auts" Authorization header field parameter and an empty "response" Authorization header field parameter, i.e. no authentication challenge response;

- in the case where the UE deems the SQN to be out of range, the subsequent REGISTER request shall contain the "auts" Authorization header field parameter (see 3GPP TS 33.102 [18]).

NOTE: In the case of the SQN being out of range, a "response" Authorization header field parameter can be included by the UE, based on the procedures described in RFC 3310 [49].

Whenever the UE detects any of the above cases, the UE shall:

- send the REGISTER request using an existing set of security associations, if available (see 3GPP TS 33.203 [19]);

- populate a new Security-Client header field within the REGISTER request and associated contact address, set to specify the security mechanisms it supports, the IPsec layer algorithms for integrity and confidentiality protection it supports and the parameters needed for the new security association setup. These parameters shall contain new values for spi_uc, spi_us and port_uc; and

- not create a temporary set of security associations.

On receiving a 420 (Bad Extension) in which the Unsupported header field contains the value "sec-agree" and if the UE supports GPRS-IMS-Bundled authentication, the UE shall initiate a new authentication attempt with the GPRS-IMS-Bundled authentication procedures as specified in subclause 5.1.1.2.6.

5.1.1.5.4 SIP digest without TLS – general

On receiving a 401 (Unauthorized) response to the REGISTER request containing one or more WWW-Authenticate header fields the UE shall select the topmost header field that it supports (i.e. where the "algorithm" WWW-Authenticate header field parameter is "SHA2-256", "SHA2-512/256" or "MD5"), and the UE shall:

NOTE 1: The MD5 algorithm is only supported for backward compatibility.

1) extract the digest-challenge parameters as indicated in RFC 7616 [286] and RFC 8760 [287] from the WWW-Authenticate header field;

2) store the contained nonce value as the nonce for authentication associated to the same registration or registration flow (if the multiple registration mechanism is used) and delete any other previously stored nonce value for authentication for this registration or registration flow (if the multiple registration mechanism is used);

NOTE 2: The related registration flow or registration is identified by the couple instance-id and reg-id if the multiple registration mechanism is used or by contact address if not.

3) calculate digest-response parameters as indicated in RFC 7616 [286] and RFC 8760 [287];

4) send another REGISTER request containing an Authorization header field. The header fields are populated as defined in subclause 5.1.1.2.3, with the addition that the UE shall include an Authorization header field containing a challenge response, constructed using the stored nonce value for authentication for the same registration or registration flow (if the multiple registration mechanism is used) , "algorithm", "cnonce", "qop", and "nc" header field parameters as indicated in RFC 7616 [286] and RFC 8760 [287]. The UE shall set the Call-ID of the REGISTER request which carries the authentication challenge response to the same value as the Call-ID of the 401 (Unauthorized) response which carried the challenge. If SIP digest without TLS is used, the UE shall not include RFC 3329 [48] header fields with this REGISTER.

On receiving the 200 (OK) response for the REGISTER request, if the "algorithm" Authentication-Info header field parameter is "SHA2-256", "SHA2-512/256" or "MD5", the UE shall authenticate the S-CSCF using the "rspauth" Authentication-Info header field parameter as described in RFC 7616 [286] and RFC 8760 [287]. If the nextnonce field is present in the Authentication-Info header field the UE shall store the contained nonce value as the nonce for authentication associated to the same registration or registration flow (if the multiple registration mechanism is used) and shall delete any other previously stored nonce value for authentication for this registration or registration flow (if the multiple registration mechanism is used).

5.1.1.5.5 SIP digest without TLS – abnormal procedures

On receiving a 403 (Forbidden) response, the UE shall consider the registration to have failed.

5.1.1.5.6 SIP digest with TLS – general

On receiving a 401 (Unauthorized) response to the REGISTER request, the procedures in subclause 5.1.1.5.4 apply with the following differences:

- The UE shall check the existence of the Security-Server header field as described in RFC 3329 [48]. If the Security-Server header field is not present or the list of supported security mechanisms does not include "tls", the UE shall abandon the authentication procedure and send a new REGISTER request.

In the case that the 401 (Unauthorized) response to the REGISTER is deemed to be valid the UE shall:

- store the announcement of the media plane security mechanisms the P-CSCF (IMS-ALG) supports labelled with the "mediasec" header field parameter specified in subclause 7.2A.7 and received in the Security-Server header field, if any; and

NOTE 1: The "mediasec" header field parameter indicates that security mechanisms are specific to the media plane.

- send another REGISTER request using the TLS session to protect the message.

The header fields are populated as defined for the initial request, with the addition that the UE shall include an Authorization header field containing a challenge response, constructed using the stored nonce value for authentication for the same registration or registration flow (if the multiple registration mechanism is used), "algorithm", "cnonce", "qop", and "nc" header field parameters as indicated in RFC 7616 [286] and RFC 8760 [287]. The UE shall also insert

the Security-Client header field that is identical to the Security-Client header field that was included in the previous REGISTER request (i.e. the REGISTER request that was challenged with the received 401 (Unauthorized) response). The UE shall also insert the Security-Verify header field into the request, by mirroring in it the content of the Security-Server header field received in the 401 (Unauthorized) response. The UE shall set the Call-ID to the same value as the Call-ID of the 401 (Unauthorized) response which carried the challenge.

NOTE 2: The Security-Client header field contains signalling plane security mechanism and if the UE supports media plane security, then media plane security mechanisms are contained, too.

When SIP digest with TLS is used, and for the case where the 401 (Unauthorized) response to the REGISTER request is deemed to be valid, the UE shall establish the TLS session as described in 3GPP TS 33.203 [19]. The UE shall use this TLS session to send all further messages towards the P-CSCF towards the protected server port.

NOTE 3: The related registration flow or registration is identified by the couple instance-id and reg-id if the multiple registration mechanism is used or by contact address if not.

5.1.1.5.7 SIP digest with TLS – abnormal procedures

On receiving a 403 (Forbidden) response, the UE shall consider the registration to have failed. If performing SIP digest with TLS, the UE should send an initial REGISTER according to the procedure specified in subclause 5.1.1.2 if the UE considers the TLS session to be no longer active at the P-CSCF.

5.1.1.5.8 NASS-IMS bundled authentication – general

NASS-IMS bundled authentication is only applicable to UEs that contain neither USIM nor ISIM. Authentication is achieved via the registration and re-registration procedures as defined in subclause 5.1.1.2 and subclause 5.1.1.4. NASS-bundled authentication is granted by the network upon receipt by the UE of a 200 (OK) response to the initial REGISTER request.

There is no separate authentication procedure.

5.1.1.5.9 NASS-IMS bundled authentication – abnormal procedures

There is no separate authentication procedure, and therefore no abnormal procedures.

5.1.1.5.10 GPRS-IMS-Bundled authentication – general

Authentication is achieved via the registration and re-registration procedures as defined in subclause 5.1.1.2 and subclause 5.1.1.4. GPRS-IMS-Bundled authentication is granted by the network upon receipt by the UE of a 200 (OK) response to the initial REGISTER request.

5.1.1.5.11 GPRS-IMS-Bundled authentication – abnormal procedures

There is no separate authentication procedure and therefore no abnormal procedures.

5.1.1.5.12 Abnormal procedures for all security mechanisms

A UE shall only respond to two consecutive invalid challenges. The UE may attempt to register with the network again after an implementation specific time.

5.1.1.5A Network-initiated re-authentication

At any time, the UE can receive a NOTIFY request carrying information related to the reg event package (as described in subclause 5.1.1.3). If:

- the state attribute in any of the <registration> elements is set to "active";
- the value of the <uri> sub-element inside the <contact> sub-element is set to the Contact address that the UE registered; and
- the event attribute of that <contact> sub-element(s) is set to "shortened";

the UE shall:

1) use the expires attribute of the <contact> sub-element that the UE registered to adjust the expiration time for that public user identity; and

2) start the re-authentication procedures at the appropriate time (as a result of the S-CSCF procedure described in subclause 5.4.1.6) by initiating a reregistration as described in subclause 5.1.1.4, if required.

NOTE: When authenticating a given private user identity, the S-CSCF will only shorten the expiry time within the <contact> sub-element that the UE registered using its private user identity. The <contact> elements for the same public user identity, if registered by another UE using different private user identities remain unchanged. The UE will not initiate a reregistration procedure, if none of its <contact> sub-elements was modified.

5.1.1.5B Change of IPv6 address due to privacy

Stateless address autoconfiguration as described in RFC 2462 [20E] defines how an IPv6 prefix and an interface identifier is used by the UE to construct a complete IPv6 address.

If the UE receives an IPv6 prefix, the UE may change the interface identity of the IPv6 address as described in RFC 8981 [293] due to privacy but this can result in service discontinuity for services provided by the IM CN subsystem.

NOTE: When the UE constructs new IPv6 address by changing the interface identity, the UE can either transfer all established dialogs to new IPv6 address as specified in 3GPP TS 24.237 [8M] and subsequently relinquish the old IPv6 address, or terminate all established dialogs and transactions. While transferring the established dialogs to new IPv6 address, the UE will have double registration, i.e. one registration for the old IPv6 address and another for the new IPv6 address.

The procedure described below assumes that the UE will terminate all established dialogs and transactions and temporarily disconnect the UE from the IM CN subsystem until the new registration is performed. If the UE decides to change the IPv6 address due to privacy and terminate all established dialogs and transaction, associated with old IPv6 address, the UE shall:

1) terminate all ongoing dialogs (e.g., sessions) and transactions (e.g., subscription to the reg event) that were using the old IPv6 address;

2) deregister all registered public user identities that were using the old IPv6 address as described in subsclause 5.1.1.4;

3) construct a new IPv6 address according to the procedures specified in RFC 8981 [293];

4) register the public user identities that were deregistered in step 2 above with a new IPv6 address, as follows:

   a) by performing an initial registration as described in subsclause 5.1.1.2; and

   b) by performing a subscription to the reg event package as described in subsclause 5.1.1.3; and

5) subscribe to other event packages it was subscribed to before the change of IPv6 address procedure started.

To ensure a maximum degree of continuous service to the end user, the UE should transfer all established dialogs to the new IPv6 address as specified in 3GPP TS 24.237 [8M] rather than terminate all established dialogs and transactions and temporarily disconnect the UE from the IM CN subsystem as described above.

5.1.1.6 User-initiated deregistration

5.1.1.6.1 General

For any public user identity that the UE has previously registered, the UE can deregister via a single registration procedure:

- all contact addresses bound to the indicated public user identity;

- some contact addresses bound to the indicated public user identity;

- a particular contact address bound to the indicated public user identity; or

- when the UE supports multiple registrations (i.e. the "outbound" option tag is included in the Supported header field) one or more flows bound to the indicated public user identity.

The UE can deregister a public user identity that it has previously registered with its contact address at any time. The UE shall protect the REGISTER request using a security association or TLS session that is associated with contact address, see 3GPP TS 33.203 [19], established as a result of an earlier registration, if one is available.

The UE shall extract or derive a public user identity, the private user identity, and the domain name to be used in the Request-URI in the registration, according to the procedures described in subclause 5.1.1.1A or subclause 5.1.1.1B.

Prior to sending a REGISTER request for deregistration, the UE shall release all dialogs that were using the contact addresses or the flow that is going to be deregistered and related to the public user identity that is going to be deregistered or to one of the implicitly registered public user identities. However:

- if the dialog that was established by the UE subscribing to the reg event package used the public user identity that is going to be deregistered; and

- this dialog is the only remaining dialog used for subscription to reg event package of the user, i.e. there are no other contact addresses registered with associated subscription to the reg event package of the user;

then the UE shall not release this dialog.

On sending a REGISTER request that will remove the binding between the public user identity and one of its contact addresses or one of its flows, the UE shall populate the header fields as follows:

a) a From header field set to the SIP URI that contains:

   1) if the UE supports RFC 6140 [191] and performs the functions of an external attached network, the main URI of the UE; else

   2) the public user identity to be deregistered;

b) a To header field set to the SIP URI that contains:

   1) if the UE supports RFC 6140 [191] and performs the functions of an external attached network, the main URI of the UE; else

   2) the public user identity to be deregistered;

c) a Contact header field set to the SIP URI(s) that contain(s) in the hostport parameter the IP address of the UE or FQDN, and:

   1) if the UE is removing the binding between the public user identity indicated in the To header field, (together with the associated implicitly registered public user identities), and the contact address indicated in the Contact header field; and

      - if the UE supports GRUU, or multiple registrations (i.e. the "outbound" option tag is included in the Supported header field), or has an IMEI available, or has an MEID available, the Contact header field also contains the "+sip.instance" header field parameter. Only the IMEI shall be used for generating an instance ID for a multi-mode UE that supports both 3GPP and 3GPP2 defined radio access networks;

      - if the UE supports multiple registrations (i.e. the "outbound" option tag is included in the Supported header field), the Contact header field does not contain the "reg-id" header field parameter;

      - if the UE does not supports GRUU and does not support multiple registrations (i.e. the "outbound" option tag is not included in the Supported header field), and does not have an IMEI available, and does not have an MEID available, the Contact header field does not contain either the "+sip.instance" header field parameter or the "reg-id" header field parameter;

   NOTE 1: Since the contact address is deregistered, if there are any flows that were previously registered with the respective contact address, all flows terminating at the respective contact address are removed.

   2) if the UE is removing the binding between the public user identity indicated in the To header field, (together with the associated implicitly registered public user identities) and one of its flows, the Contact header field

contains the "+sip.instance" header field parameter and the "reg-id" header field parameter that identifies the flow; and

NOTE 2: The requirement placed on the UE to include an instance ID based on the IMEI when the UE does not support GRUU and does not support multiple registrations does not imply any additional requirements on the network.

3) if the UE supports RFC 6140 [191] and performs the functions of an external attached network, for the registration of bulk number contacts the UE shall include a Contact URI without a user portion and containing the "bnc" URI parameter;

d) a Via header field set to include the IP address or FQDN of the UE in the sent-by field;

e) a registration expiration interval value set to the value of zero, appropriate to the deregistration requirements of the user;

f) a Request-URI set to the SIP URI of the domain name of the home network used to address the REGISTER request;

g) if available to the UE (as defined in the access technology specific annexes for each access technology), a P-Access-Network-Info header field set as specified for the access network technology (see subclause 7.2A.4);

h) a Security-Client header field to announce the media plane security mechanisms the UE supports, if any;

NOTE 3: The "mediasec" header field parameter indicates that security mechanisms are specific to the media plane.

i) if the UE supports RFC 6140 [191] and performs the functions of an external attached network, for the registration of bulk number contacts the UE shall include a Require header field containing the option-tag "gin"; and

j) if the UE supports RFC 6140 [191] and performs the functions of an external attached network, for the registration of bulk number contacts the UE shall include a Proxy-Require header field containing the option-tag "gin".

For a public user identity that the UE has registered with multiple contact addresses or multiple flows (e.g. via different P-CSCFs), the UE shall also be able to deregister multiple contact addresses or multiple flows, bound to its public user identity, via single deregistration proceduere as specified in RFC 3261 [26]. The UE shall send a single REGISTER request, using one of its contact addresses and the associated set of security associations or TLS session, containing a list of Contact headers. Each Contact header field is populated as specifed above in bullets a) through i).

The UE can deregister all contact addresses bound to its public user identity and associated with its private user identity. The UE shall send a single REGISTER request, using one of its contact addresses and the associated set of security associations or TLS session, containing a public user identity that is being deregistered in the To header field, and a single Contact header field with value of "*" and the Expires header field with a value of "0". The UE shall not include the "instance-id" feature tag and the "reg-id" header field parameter in the Contact header field in the REGISTER request.

NOTE 4: All entities subscribed to the reg event package of the user will be informed via NOTIFY request which contact addresses bound to the public user identity have been deregistered.

When a 401 (Unauthorized) response to a REGISTER request is received the UE shall behave as described in subclause 5.1.1.5.1.

On receiving the 200 (OK) response to the REGISTER request, the UE shall:

- remove all registration details relating to this public user identity and the associated contact address.

- store the announcement of the media plane security mechanisms the P-CSCF (IMS-ALG) supports labelled with the "mediasec" header field parameter specified in subclause 7.2A.7 and received in the Security-Server header field, if any.

NOTE 5: The "mediasec" header field parameter indicates that security mechanisms are specific to the media plane.

If there are no more public user identities registered with this contact address, the UE shall delete any stored media plane security mechanisms and related keys and any security associations or TLS sessions and related keys it may have towards the IM CN subsystem.

If all public user identities are deregistered and all security association or TLS session is removed, then the UE shall consider subscription to the reg event package cancelled (i.e. as if the UE had sent a SUBSCRIBE request with an Expires header field containing a value of zero).

5.1.1.6.2 IMS AKA as a security mechanism

On sending a REGISTER request, as defined in subclause 5.1.1.6.1, the UE shall additionally populate the header fields as follows:

a) an Authorization header field, with:

- the "username" header field parameter, set to the value of the private user identity;
- the "realm" header field parameter, set to the value as received in the "realm" WWW-Authenticate header field parameter;
- the "uri" header field parameter, set to the SIP URI of the domain name of the home network;
- the "nonce" header field parameter, set to last received nonce value; and
- the response directive, set to the last calculated response value;

b) additionally for each Contact header field and associated contact address, include the associated protected server port value in the hostport parameter;

c) additionally for the Via header field, include the protected server port value bound to the security association in the sent-by field;

NOTE 1: If the UE specifies its FQDN in the hostport parameter in the Contact header field and in the sent-by field in the Via header field, then it has to ensure that the given FQDN will resolve (e.g., by reverse DNS lookup) to the IP address that is bound to the security association.

d) a Security-Client header field, set to specify the signalling plane security mechanisms it supports, the IPsec layer algorithms for integrity and confidentiality protection it supports and the new parameter values needed for the setup of two new pairs of security associations. For further details see 3GPP TS 33.203 [19] and RFC 3329 [48]; and

e) a Security-Verify header field that contains the content of the Security-Server header field received in the 401 (Unauthorized) response of the last successful authentication.

NOTE 2: When the UE has received the 200 (OK) response for the REGISTER request of the only public user identity currently registered with this contact address and its associated set of implicitly registered public user identities (i.e. no other public user identity is registered), the UE removes the security association (between the P-CSCF and the UE) that were using this contact address. Therefore further SIP signalling using this security association (e.g. the NOTIFY request containing the deregistration event) will not reach the UE.

5.1.1.6.3 SIP digest without TLS as a security mechanism

On sending a REGISTER request, as defined in subclause 5.1.1.6.1, the UE shall additionally populate the header fields as follows:

a) an Authorization header field as defined in RFC 7616 [286] and RFC 8760 [287], including:

- the "username" header field parameter, set to the value of the private user identity;
- the "realm" header field parameter, set to the domain name of the home network;
- the "uri" header field parameter, set to the SIP URI of the domain name of the home network;
- the "nonce" header field parameter, set to an empty value; and
- the "response" header field parameter, set to an empty value;

b) for each Contact header field and associated contact address include the associated unprotected port value (where the UE was expecting to receive mid-dialog requests); and

c) the Via header field with the port value of an unprotected port where the UE expects to receive responses to the request.

5.1.1.6.4 SIP digest with TLS as a security mechanism

On sending a REGISTER request, as defined in subclause 5.1.1.6.1, the UE shall additionally populate the header fields as follows:

a) an Authorization header field set in accordance with subclause 5.1.1.6.3; and

b) a Security-Client header field, set to specify the signalling plane security mechanism it supports. For further details see 3GPP TS 33.203 [19] and RFC 3329 [48]; and

c) a Security-Verify header field that contains the content of the Security-Server header field received in the 401 (Unauthorized) response of the last successful authentication.

5.1.1.6.5 NASS-IMS bundled authentication as a security mechanism

On sending a REGISTER request, as defined in subclause 5.1.1.6.1, the UE shall additionally populate the header fields as follows:

a) optionally, an Authorization header field, with the "username" header field parameter, set to the value of the private user identity;

NOTE 1: In case the Authorization header field is absent, the mechanism only supports that one public user identity is associated with only one private user identity.

On receiving the 200 (OK) response to the REGISTER request defined in subclause 5.1.1.6.1, there are no additional requirements for the UE.

NOTE 2: When NASS-IMS bundled authentication is in use, a 401 (Unauthorized) response to the REGISTER request is not expected to be received.

5.1.1.6.6 GPRS-IMS-Bundled authentication as a security mechanism

On sending a REGISTER request, as defined in subclause 5.1.1.6.1, the UE shall additionally populate the header fields as follows:

a) an Authorization header field as defined in RFC 7616 [286] and RFC 8760 [287] shall not be included, in order to indicate support GPRS-IMS-Bundled authentication.

b) the Security-Verify header field and the Security-Client header field values as defined by RFC 3329 [48] shall not contain signalling plane security mechanisms;

c) a From header field set to a temporary public user identity derived from the IMSI, as defined in 3GPP TS 23.003 [3], as the public user identity to be deregistered;

d) a To header field set to a temporary public user identity derived from the IMSI, as defined in 3GPP TS 23.003 [3], as the public user identity to be deregistered;

e) for each Contact header field and associated contact address include the associated unprotected port value (where the UE was expecting to receive mid-dialog requests); and

f) the Via header field with the port value of an unprotected port where the UE expects to receive responses to the request.

NOTE 1: Since the private user identity is not included in the REGISTER requests when GPRS-IMS-Bundled authentication is used for registration, re-registration and de-registration procedures, all REGISTER requests from the UE use the IMSI-derived IMPU as the public user identity even when the implicitly registered IMPUs are available at the UE. The UE does not use the temporary public user identity (IMSI-derived IMPU) in any non-registration SIP requests.

On receiving the 200 (OK) response to the REGISTER request defined in subclause 5.1.1.6.1, there are no additional requirements for the UE.

NOTE 2: When GPRS-IMS-Bundled authentication is in use, a 401 (Unauthorized) response to the REGISTER request is not expected to be received.

5.1.1.7 Network-initiated deregistration

Upon receipt of a NOTIFY request, on any dialog which was generated during the subscription to the reg event package as described in subclause 5.1.1.3, including one or more <registration> element(s) which were registered by this UE, with:

1) the state attribute within the <registration> element set to "terminated", and within each <contact> element belonging to this UE, the state attribute set to "terminated" and the event attribute set either to "unregistered", or "rejected", or "deactivated", the UE shall remove all registration details relating to the respective public user identity (i.e. consider the public user identity indicated in the aor attribute of the <registration> element as deregistered); or

2) the state attribute within the <registration> element set to "active", and within a given <contact> element belonging to this UE, the state attribute set to "terminated", and the associated event attribute set either to "unregistered", or "rejected" or "deactivated", the UE shall consider the binding between the public user identity and either the contact address or the registration flow and the associated contact address (if the multiple registration mechanism is used) indicated in the respective <contact> element as removed. The UE shall consider its public user identity as deregistered when all bindings between the respective public user identity and all contact addresses and all registration flow and the associated contact address (if the multiple registration mechanism is used) belonging to this UE are removed.

NOTE 1: When multiple registration mechanism is used to register a public user identity and bind it to a registration flow and the associated contact address, there will be one <contact> element for each registration flow and the associated contact address.

NOTE 2: If the state attribute within the <registration> element is set to "active" and the <contact> element belonging to this UE is set to "active", the UE will consider that the binding between the public user identity and either the respective contact address or the registration flow and the associated contact address as left unchanged.

In case of a "deactivated" event attribute, the UE shall start the initial registration procedure as described in subclause 5.1.1.2. In case of a "rejected" event attribute, the UE shall release all dialogs related to those public user identities.

Upon receipt of a NOTIFY request, the UE shall delete all security associations or TLS sessions towards the P-CSCF either:

- if all <registration> element(s) have their state attribute set to "terminated" (i.e. all public user identities are deregistered) and the Subscription-State header field contains the value of "terminated"; or

- if each <registration> element that was registered by this UE has either the state attribute set to "terminated", or the state attribute set to "active" and the state attribute within the <contact> element belonging to this UE set to "terminated".

When all UE's public user identities are registered via a single P-CSCF and the subscription dialog to the reg event package of the UE is set via the respective P-CSCF, the UE shall delete these security associations or TLS sessions towards the respective P-CSCF when all public user identities have been deregistered and after the server transaction (as defined in RFC 3261 [26]) pertaining to the received NOTIFY request terminates.

NOTE 3: Deleting a security association or TLS session is an internal procedure of the UE and does not involve any SIP procedures.

NOTE 4: If all the public user identities (i.e. <contact> elements) registered by this UE are deregistered and the security associations or TLS sessions have been removed, the UE considers the subscription to the reg event package terminated since the NOTIFY request was received with Subscription-State header field containing the value of "terminated".

5.1.2 Subscription and notification

5.1.2.1 Notification about multiple registered public user identities

Upon receipt of a NOTIFY request for the dialog associated with the subscription to the reg event package the UE shall perform the following actions:

- store the information for the established dialog;
- store the expiration time as indicated in the "expires" header field parameter of the Subscription-State header field, if present, of the NOTIFY request. Otherwise the expiration time is retrieved from the Expires header field of the 2xx response to SUBSCRIBE request;
- if a <registration> element with state attribute "active", i.e. registered, is received for one or more public user identities, the UE shall store the indicated public user identities as registered;
- if a <registration> element with state attribute "active" is received, and the UE supports GRUU (see table A.4, item A.4/53), then for each public user identity indicated in the notification that contains a <pub-gruu> element or a <temp-gruu> element or both (as defined in RFC 5628 [94]), the UE shall store the value of those elements in association with the public user identity;
- if a <registration> element with state attribute "terminated", i.e. deregistered, is received for one or more public user identities, the UE shall store the indicated public user identities as deregistered and shall remove any associated GRUUs; and

NOTE 1: There can be public user identities which are automatically registered within the registrar (S-CSCF) of the user upon registration of one public user identity or when S-CSCF receives a Push-Profile-Request (PPR) from the HSS (as described in 3GPP TS 29.228 [14]) changing the status of a public user identity associated with a registered implicit set from barred to non-barred. Usually these automatically or implicitly registered public user identities belong to the same service profile of the user and they might not be available within the UE. The implicitly registered public user identities can also belong to different service profiles. The here-described procedures provide a different mechanism (to the 200 (OK) response to the REGISTER request) to inform the UE about these automatically registered public user identities.

NOTE 2: RFC 5628 [94] provides guidance on the management of temporary GRUUs, utilizing information provided in the reg event notification.

- follow the procedures specified in RFC 6665 [28].

5.1.2.2 General SUBSCRIBE requirements

If the UE receives a 503 (Service Unavailable) response to an initial SUBSCRIBE request containing a Retry-After header field, then the UE shall not automatically reattempt the request until after the period indicated by the Retry-After header field contents.

5.1.2A Generic procedures applicable to all methods excluding the REGISTER method

5.1.2A.1 UE-originating case

5.1.2A.1.1 General

The procedures of this subclause are general to all requests and responses, except those for the REGISTER method.

When the UE re-uses a previously registered contact address, the UE shall remove any parameters dedicated to registration from the Contact header field (e.g. "expires").

When the UE sends any request, the UE shall use either a given contact address that has been previously registered or a registration flow and the associated contact address (if the multiple registration mechanism is used) and shall:

- if IMS AKA is in use as a security mechanism:

a) if the UE has not obtained a GRUU, populate the Contact header field of the request with the protected server port and the respective contact address; and

b) include the protected server port and the respective contact address in the Via header field entry relating to the UE;

- if SIP digest without TLS is in use as a security mechanism:

a) if the UE has not obtained a GRUU, populate the Contact header field of the request with the port value of an unprotected port and the contact address where the UE expects to receive subsequent mid-dialog requests;

b) populate the Via header field of the request with the port value of an unprotected port and the respective contact address where the UE expects to receive responses to the request; and

c) if a nonce value for proxy authentication is stored for the related registration or registration flow (if the multiple registration mechanism is used), insert a Proxy-Authorization header field containing a challenge response, constructed using the stored nonce value for proxy authentication for the same registration or registration flow (if the multiple registration mechanism is used), "algorithm", "cnonce", "qop", and "nc" header field parameters as specified in RFC 7616 [286] and RFC 8760 [287];

- if SIP digest with TLS is in use as a security mechanism:

a) if the UE has not obtained a GRUU, populate the Contact header field of the request with the protected server port;

b) include the protected server port in the Via header field entry relating to the UE; and

c) if a nonce value for proxy authentication is stored for the related registration or registration flow (if the multiple registration mechanism is used), insert a Proxy-Authorization header field containing a challenge response, constructed using the stored nonce value for proxy authentication for the same registration or registration flow (if the multiple registration mechanism is used), "algorithm", "cnonce", "qop", and "nc" header field parameters as specified in RFC 7616 [286] and RFC 8760 [287];

- if NASS-IMS bundled authentication is in use as a security mechanism, and therefore no port is provided for subsequent SIP messages by the P-CSCF during registration, the UE shall send any request to the same port used for the initial registration as described in subclause 5.1.1.2;

- if GPRS-IMS-Bundled authentication is in use as a security mechanism, and therefore no port is provided for subsequent SIP messages by the P-CSCF during registration, the UE shall send any request to the same port used for the initial registration as described in subclause 5.1.1.2.

If SIP digest without TLS is used, the UE shall not include RFC 3329 [48] header field s in any SIP messages.

When SIP digest is in use, upon receiving a 407 (Proxy Authentication Required) response to an initial request, the originating UE shall:

- extract the digest-challenge parameters as indicated in RFC 7616 [286] and RFC 8760 [287] from the Proxy-Authenticate header field;

- if the contained nonce value is associated to the realm used for the related REGISTER request authentication, store the contained nonce as a nonce value for proxy authentication associated to the same registration or registration flow (if the multiple registration mechanism is used) and shall delete any other previously stored nonce value for proxy authentication for this registration or registration flow;

- calculate the response as described in RFC 7616 [286] and RFC 8760 [287] using the stored nonce value for proxy authentication associated to the same registration or registration flow (if the multiple registration mechanism is used); and

- send a new request containing a Proxy-Authorization header field in which the header field parameters are populated as defined in RFC 7616 [286] and RFC 8760 [287] using the calculated response.

Where a security association or TLS session exists, the UE shall discard any SIP response that is not protected by the security association or TLS session and is received from the P-CSCF outside of the registration and authentication procedures. The requirements on the UE within the registration and authentication procedures are defined in subclause 5.1.1.

For a UE performing the functions of an external attached network operating in static mode, authentication can take place without a registration based on TLS client certificate. Before any originating or terminating procedures can take place between the UE performing the functions of an external attached operating in static mode and the P-CSCF or between the UE performing the functions of an external attached network operating in static mode and the IBCF of the IMS network, for security and authentication between the UE performing the functions of an external attached network operating in static mode and the IMS network, the UE performing the functions of an external attached network operating in static mode shall use the TLS procedures according to 3GPP TS 33.310 [19D] using certificates.

In accordance with RFC 3325 [34] the UE may insert a P-Preferred-Identity header field in any initial request for a dialog or request for a standalone transaction as a hint for creation of an asserted identity (contained in the P-Asserted-Identity header field) within the IM CN subsystem.

NOTE 1: Since the S-CSCF uses the P-Asserted-Identity header field when checking whether the UE originating request matches the initial filter criteria, the P-Preferred-Identity header field inserted by the UE determines which services and applications are invoked.

When sending any initial request for a dialog or request for a standalone transaction using either a given contact address that has been previously registered or a registration flow and the associated contact address (if the multiple registration mechanism is used), the UE may include any of the following in the P-Preferred-Identity header field:

- a public user identity which has been registered by the user with the respective contact address;

- an implicitly registered public user identity returned in a registration-state event package of a NOTIFY request whose <uri> sub-element inside the <contact> sub-element of the <registration> element is the same as the contact address being used for this request and was not subsequently deregistered or that has not expired; or

- any other public user identity which the user has assumed by mechanisms outside the scope of this specification to have a current registration.

NOTE 2: The temporary public user identity specified in subclause 5.1.1.1 is not a public user identity suitable for use in the P-Preferred-Identity header field.

NOTE 3: Procedures in the network require international public telecommunication numbers when telephone numbers are used in P-Preferred-Identity header field.

NOTE 4: A number of header fields can reveal information about the identity of the user. Where privacy is required, implementers should also give consideration to other header fields that can reveal identity information. RFC 3323 [33] subclause 4.1 gives considerations relating to a number of header fields.

Where privacy is required, in any initial request for a dialog or request for a standalone transaction, the UE shall set a display-name of the From header field to "Anonymous" as specified in RFC 3261 [26] and set an addr-spec of the From header field to Anonymous User Identity as specified in 3GPP TS 23.003 [3].

NOTE 5: The contents of the From header field are not necessarily modified by the network based on any privacy specified by the user either within the UE indication of privacy or by network subscription or network policy. Therefore the user should include the value "Anonymous" whenever privacy is explicitly required. As the user can well have privacy requirements, terminal manufacturers should not automatically derive and include values in this header field from the public user identity or other values stored in or derived from the UICC. Where the user has not expressed a preference in the configuration of the terminal implementation, the implementation should assume that privacy is required. Users that require to identify themselves, and are making calls to SIP destinations beyond the IM CN subsystem, where the destination does not implement RFC 3325 [34], will need to include a value in the From header field other than Anonymous.

The UE shall determine the public user identity to be used for this request as follows:

1) if a P-Preferred-Identity was included, then use that as the public user identity for this request; or

2) if no P-Preferred-Identity was included, then use the default public user identity for the security association or TLS session and the associated contact address as the public user identity for this request;

The UE shall not include its "+sip.instance" header field parameter in the Contact header field in its non-register requests and responses except when the request or response is guaranteed to be sent to a trusted intermediary that will remove the "+sip.instance" header field parameter prior to forwarding the request or response to the destination.

NOTE 6: Such trusted intermediaries include an AS that all such requests as part of an application or service traverse. In order to ensure that all requests or responses containing the "+sip.instance" header field parameter are forwarded via the trusted intermediary the UE needs to have first verified that the trusted intermediary is present (e.g. first contacted via a registration or configuration procedure). Including the "+sip.instance" header field parameter containing an IMEI URN does not violate RFC 7254 [153] even when the UE requests privacy using RFC 3323 [33].

If this is a request for a new dialog, the Contact header field is populated as follows:

1) a contact header value which is one of:

   - if a public GRUU value ("pub-gruu" header field parameter) has been saved associated with the public user identity to be used for this request, and the UE does not indicate privacy of the P-Asserted-Identity, then the UE should insert the public GRUU ("pub-gruu" header field parameter) value as specified in RFC 5627 [93]; or

   - if a temporary GRUU value ("temp-gruu" header field parameter) has been saved associated with the public user identity to be used for this request, and the UE does indicate privacy of the P-Asserted-Identity, then the UE should insert the temporary GRUU ("temp-gruu" header field parameter) value as specified in RFC 5627 [93];

   - otherwise, a SIP URI containing the contact address of the UE that has been previously registered without any contact parameters dedicated to registration procedure;

   NOTE 7: The above items are mutually exclusive.

2) include an "ob" SIP URI parameter, if the UE supports multiple registrations, and the UE wants all subsequent requests in the dialog to arrive over the same flow identified by the flow token as described in RFC 5626 [92];

3) if the request is related to an IMS communication service that requires the use of an ICSI then the UE shall include in a g.3gpp.icsi-ref media feature tag, as defined in subclause 7.9.2 and RFC 3841 [56B], the ICSI value (coded as specified in subclause 7.2A.8.2) for the IMS communication service. The UE may also include other ICSI values that the UE is prepared to use for all dialogs with the terminating UE(s); and

4) if the request is related to an IMS application that is supported by the UE, then the UE may include in a g.3gpp.iari-ref media feature tag, as defined in subclause 7.9.3 and RFC 3841 [56B], the IARI value (coded as specified in subclause 7.2A.9.2) that is related to the IMS application and that applies for the dialog.

If this is a request within an existing dialog, and the request includes a Contact header field, then the UE should insert the previously used Contact header field.

If the UE support multiple registrations as specified in RFC 5626 [92], the UE should include option-tag "outbound" in the Supported header field.

If this is a request for a new dialog or standalone transaction and the request is related to an IMS communication service that requires the use of an ICSI then the UE:

1) shall include the ICSI value (coded as specified in subclause 7.2A.8.2), for the IMS communication service that is related to the request in a P-Preferred-Service header field according to RFC 6050 [121]. If a list of network supported ICSI values was received as specified in 3GPP TS 24.167 [8G], the UE shall only include an ICSI value that is in the received list;

NOTE 8: The UE only receives those ICSI values corresponding to the IMS communication services that the network provides to the user.

2) may include an Accept-Contact header field containing an ICSI value (coded as specified in subclause 7.2A.8.2) that is related to the request in a g.3gpp.icsi-ref media feature tag as defined in subclause 7.9.2 if the ICSI for the IMS communication service is known. The UE may remove one or more subclasses from an ICSI when including it in an Accept-Contact header field provided that the included ICSI corresponds to an IMS communication service.

NOTE 9: If the UE includes the same ICSI values into the Accept-Contact header field and the P-Preferred-Service header field, there is a possibility that one of the involved S-CSCFs or an AS changes the ICSI value in the P-Asserted-Service header field, which results in the message including two different ICSI values (one in the P-Asserted-Service header field, changed in the network and one in the Accept-Contact header field).

If an IMS application indicates that an IARI is to be included in a request for a new dialog or standalone transaction, the UE shall include an Accept-Contact header field containing an IARI value (coded as specified in subclause 7.2A.9.2) that is related to the request in a g.3gpp.iari-ref media feature tag as defined in subclause 7.9.3 and RFC 3841 [56B].

NOTE 10: RFC 3841 [56B] allows multiple Accept-Contact header fields along with multiple Reject-Contact header fields in a SIP request, and within those header fields, expressions that include one or more logical operations based on combinations of media feature tags. Which registered UE will be contacted depends on the Accept-Contact header field and Reject-Contact header field combinations included that evaluate to a logical expression and the relative qvalues of the registered contacts for the targeted registered public user identity. There is therefore no guarantee that when multiple Accept-Contact header fields or additional Reject-Contact header field(s) along with the Accept-Contact header field containing the ICSI value or IARI value are included in a request that the request will be routed to a contact that registered the same ICSI value or IARI value. Charging and accounting is based upon the contents of the P-Asserted-Service header field and the actual media related contents of the SIP request and not the Accept-Contact header field contents or the contact reached.

NOTE 11: The UE only includes the header field parameters "require" and "explicit" in the Accept-Contact header field containing the ICSI value or IARI value if the IMS communication service absolutely requires that the terminating UE understand the IMS communication service in order to be able to accept the session. Including the header field parameters "require" and "explicit" in Accept-Contact header fields in requests which don't absolutely require that the terminating UE understand the IMS communication service in order to accept the session creates an interoperability problem for sessions which otherwise would interoperate and violates the interoperability requirements for the ICSI in 3GPP TS 23.228 [7].

After the dialog is established the UE may change the dialog capabilities (e.g. add a media or request a supplementary service) if defined for the IMS communication service as identified by the ICSI value using the same dialog. Otherwise, the UE shall initiate a new initial request to the other user.

The UE can indicate privacy of the P-Asserted-Identity that will be generated by the P-CSCF in accordance with RFC 3323 [33], and the additional requirements contained within RFC 3325 [34].

If resource priority in accordance with RFC 4412 [116] is required for a dialog, then the UE shall include the Resource-Priority header field in all requests associated with that dialog.

NOTE 12: The case where the UE is unaware of the requirement for resource priority because the user requested the capability as part of the dialstring falls outside the scope of this requirement. Such cases can exist and will need to be dealt with by an appropriate functional entity (e.g. P-CSCF) to process the dialstring. For certain national implementations, signalling of a Resource-Priority header field to or from a UE is not required.

If available to the UE (as defined in the access technology specific annexes for each access technology), the UE shall insert a P-Access-Network-Info header field into any request for a dialog, any subsequent request (except CANCEL requests) or response (except CANCEL responses) within a dialog or any request for a standalone method (see subclause 7.2A.4). Insertion of the P-Access-Network-Info header field into the ACK request is optional.

NOTE 13: During the dialog, the points of attachment to the IP-CAN of the UE can change (e.g. UE connects to different cells). The UE will populate the P-Access-Network-Info header field in any request or response within a dialog with the current point of attachment to the IP-CAN (e.g. the current cell information).

NOTE 14: The value of the P-Access-Network-Info header field could be stale if the point of attachment of the UE with the network changes before the message is received by the network.

The UE shall build a proper preloaded Route header field value for all new dialogs and standalone transactions. The UE shall build a list of Route header field values made out of the following, in this order:

a) the P-CSCF URI containing the IP address acquired at the time of the P-CSCF discovery procedures which was used in registration of the contact address (or registration flow); and

NOTE 15: If the UE is provisioned with or receives a FQDN at the time of the P-CSCF discovery procedures, the FQDN is resolved to an IP address at the time of the P-CSCF discovery procedures.

b) the P-CSCF port based on the security mechanism in use:

- if IMS AKA or SIP digest with TLS is in use as a security mechanism, the protected server port learnt during the registration procedure;

- if SIP digest without TLS, NASS-IMS bundled authentication or GPRS-IMS-Bundled authentication is in use as a security mechanism, the unprotected server port used during the registration procedure;

c) and the values received in the Service-Route header field saved from the 200 (OK) response to the last registration or re-registration of the public user identity with associated contact address.

NOTE 16: When the UE registers multiple contact addresses, there will be a list of Service-Route headers for each contact address. When sending a request using a given contact address and the associated security associations or TLS session, the UE will use the corresponding list of Service-Route headers to construct a list of Route headers.

The UE may indicate that proxies should not fork the request by including a "no-fork" directive within the Request-Disposition header field in the request as described in RFC 3841 [56B].

If a request is for a new dialog or standalone transaction, and the request matches a trigger for starting logging of SIP signalling, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K], the UE shall:

- start to log SIP signalling for this dialog; and

- in any requests or responses sent on this dialog, append a "logme" header field parameter to the SIP Session-ID header field.

If a request or response is sent on a dialog for which logging of signalling is in progress, the UE shall check whether a trigger for stopping logging of SIP signalling has occurred, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K].

a) If a stop trigger event has occurred, the UE shall stop logging of signalling; or

b) if a stop trigger event has not occurred, the UE shall:

- in any requests or responses sent on this dialog, append a "logme" header field parameter to the SIP Session-ID header field; and

- log the request.

If the UE receives a 1xx or 200 (OK) response to an initial request for a dialog, the response containing a P-Asserted-Identity header field set to an emergency number as specified in 3GPP TS 22.101 [1A], the UE procedures in subclause 5.1.6.10 apply.

If the UE receives a 3xx response containing a Contact header field:

1) if the 3xx response is a 380 (Alternative Service) response to an INVITE request the response containing a P-Asserted-Identity header field with a value equal to the value of the last entry of the Path header field value received during registration and the response contains a 3GPP IM CN subsystem XML body that includes an <ims-3gpp> element, including a version attribute, with an <alternative-service> child element with the <type> child element set to "emergency" (see table 7.6.2) then the UE shall select a domain in accordance with the conventions and rules specified in 3GPP TS 22.101 [1A] and 3GPP TS 23.167 [4B], and:

- if the CS domain is selected, the UE behavior is defined in subclause 7.1.2 of 3GPP TS 23.167 [4B] and, where appropriate, in the access technology specific annex;

- if the IM CN subsystem is selected, the UE shall apply the procedures in subclause 5.1.6 with the exception of selecting a domain for the emergency call attempt; and

2) if the response is:

- not a 380 (Alternative Service) response; or

- a 380 (Alternative Service) response, and the response:

  i. does not contain a 3GPP IM CN subsystem XML body that includes an <ims-3gpp> element, including a version attribute, with an <alternative-service> child element with the <type> child element set to "emergency" (see table 7.6.2); or

  ii. does contain a 3GPP IM CN subsystem XML body that includes an <ims-3gpp> element, including a version attribute, with an <alternative-service> child element with the <type> child element set to "emergency" (see table 7.6.2), and the response;

    I) does not contain a P-Asserted-Identity header field; or

    II) does contain a P-Asserted-Identity header field with a value not equal to the value of the last entry of the Path header field value received during registration;

the UE should not automatically recurse on the Contact header field without first indicating the identity of the user to which a request will be sent and obtaining authorisation of the served user.

NOTE 17: The last entry on the Path header field value received during registration is the value of the SIP URI of the P-CSCF. If there are multiple registration flows associated with the registration, then the UE has received from the P-CSCF during registration multiple sets of Path header field values. The last entry of the Path header field value corresponding to the flow on which the 380 (Alternative Service) response was received is checked.

NOTE 18: A UE can still automatically recurse on 3xx responses as part of a service if the nature of the service enables the UE to identify 3xx responses as having originated from the home network and networks trusted by the home network and the nature of the service ensures that the charging for the requests sent as a result of the 3xx response is correlated with the original request.

NOTE 19: Automatically recursing on untrusted 3xx responses opens up the UE to being redirected to premium rate URIs without the user's consent.

The UE performing the functions of an external attached network operating in static mode shall send all requests using the already established TLS session as described in this subclause.

A UE supporting RFC 4028 [58], when it receives a 422 (Session Interval Too Small) to an INVITE request where the response contains a Min-SE header field, shall retry the request in accordance with RFC 4028 [58] subclause 7.4.

5.1.2A.1.2 Structure of Request-URI

The UE may include a SIP URI complying with RFC 3261 [26], a tel URI complying with RFC 3966 [22], a pres URI complying with RFC 3859 [179], an im URI complying with RFC 3860 [180] or a mailto URI complying with RFC 2368 [181].

NOTE: This version of the document does not specify how the UE determines the host part of the SIP URI.

The UE may use non-international formats of E.164 numbers or non-E.164 numbers, including geo-local numbers and home-local numbers and other local numbers (e.g. private number), in the Request-URI.

The actual value of the URI depends on whether user equipment performs an analysis of the dial string input by the end user or not, see subclauses 5.1.2A.1.3 and 5.1.2A.1.4.

5.1.2A.1.3 UE without dial string processing capabilities

In this case the UE does not perform any analysis of the dial string. This requires that the dialling plan is designed so it enables the network to differentiate local numbers from other numbers.

The dial string is sent to the network, in the Request-URI of a initial request or a stand alone transaction, using one of the following formats:

1) a tel-URI, syntactically complying with RFC 3966 [22], with the dial string encoded as a local number followed by a "phone-context" tel URI parameter value;

EXAMPLE: tel:00447700900123;phone-context=example.com

2) a SIP URI, syntactically complying with RFC 3261 [26], with the user=phone parameter, embedding a tel-URI with a "phone-context" tel URI parameter value;

EXAMPLE: sip:00447700900123;
phone-context=example.com@example.com;user=phone

3) a SIP URI, complying with RFC 3261 [26] and RFC 4967 [103], with the user=dialstring parameter and with a "phone-context" tel-URI parameter value in the user part; or

EXAMPLE: sip:00447700900123;
phone-context=example.com@example.com;user=dialstring

4) a SIP URI syntactically complying with RFC 3261 [26], where the user part contains the dial string and the domain name is specific enough to enable to network to understand that the user part contains a dial string.

EXAMPLE: sip:00447700900123@dialstrings.entreprise.com

For cases 1), 2), and 3) the UE shall set the "phone-context" tel URI parameter in accordance with subclause 5.1.2A.1.5.

5.1.2A.1.4 UE with dial string processing capabilities

In this case the UE performs sufficient dial string analysis (or receives an explicit indication from the user) to identify the type of numbering that is used and processes the dial string accordingly before building the Request-URI.

If the UE detects that a local dialling plan is being used, where the UE is able to identify a global telephone number, the UE shall translate the number into E.164 international format after removing all dial string elements used for local numbering detection purposes (e.g. escape codes).

If the UE detects that a local (private or public) dialling plan is being used and the UE is not able to identify a global number, it may decide to send the dial string unchanged to the network as described in subclause 5.1.2A.1.3 or the UE may decide to alter it to comply with the local numbering plan (e.g. remove all dial string elements used for local numbering detection).

In the latter case the local numbering information is sent using one of the following formats:

1) a tel-URI, complying with RFC 3966 [22], with a local number followed by a "phone-context" tel-URI parameter value;

2) a SIP URI, complying with RFC 3261 [26], with the "user" SIP URI parameter set to "phone" and a user part embedding a local number with a phone-context parameter; and

3) if the UE intends to send information related to supplementary services, a SIP URI, complying with RFC 3261 [26] and RFC 4967 [103], with the "user" SIP URI parameter set to "dialstring" and with a "phone-context" tel URI parameter value in the user part.

The UE shall set the "phone-context" tel URI parameter in accordance with subclause 5.1.2A.1.5.

NOTE: The way how the UE process the dial-string and handles special characters (e.g. pause) in order to produce a conformant SIP URI or tel-URI according to RFC 3966 [22] is implementation specific.

As a general rule, recognition of special service numbers shall take priority over other dialling plan issues. If the dial string equates to a pre-configured service URN as specified in RFC 5031 [69]) then the service-urn should be sent.

5.1.2A.1.5 Setting the "phone-context" tel URI parameter

When the UE uses home-local number, the UE shall include in the "phone-context" tel URI parameter the home network domain name in accordance with RFC 3966 [22].

When the UE uses geo-local number, the UE shall:

- if access technology information available to the UE (i.e., the UE can insert P-Access-Network-Info header field into the request), include the access technology information in the "phone-context" tel URI parameter according to RFC 3966 [22] as defined in subclause 7.2A.10; and

- if access technology information is not available to the UE (i.e., the UE cannot insert P-Access-Network-Info header field into the request), include in the "phone-context" tel URI parameter the home network domain name prefixed by the "geo-local." string according to RFC 3966 [22] as defined in subclause 7.2A.10.

When the UE uses other local numbers, than geo-local number or home local numbers, e.g. private numbers that are different from home-local number or the UE is unable to determine the type of the dialled number, the UE shall include a "phone-context" tel URI parameter set according to RFC 3966 [22], e.g. if private numbers are used a domain name to which the private addressing plan is associated. The "phone-context" value used in the case of other local numbers shall be different from "phone-context" values used with geo-local numbers and home-local numbers.

NOTE 1: The "phone-context" tel URI parameter value can be entered or selected by the subscriber, or can be a "pre-configured" value (e.g. using OMA-DM with the management object specified in 3GPP TS 24.167 [8G]) inserted by the UE.

NOTE 2: The way how the UE determines whether numbers in a non-international format are geo-local, home-local or relating to another network in absence of matching UE configuration in subclause 5.1.2A.1.5A, is implementation specific.

NOTE 3: Home operator's local policy can define a prefix string(s) to enable subscribers to differentiate dialling a geo-local number and/or a home-local number.

5.1.2A.1.5A Policy on local numbers

Policy on local numbers consists of zero or more parts of policy on local numbers

A part of policy on local numbers indicates an IMS communication service (identified by an ICSI) in which the UE is to use a number in non-international format without associated "phone-context" value as:

1) a geo-local number; or

2) a home-local number.

The UE may support the policy on local numbers.

If the UE supports the policy on local numbers:

1) if:

   a) the upper layers provide a number in non-international format to be included in Request-URI of a SIP request;

   b) the upper layers do not provide a "phone-context" value associated with the number;

   c) the UE is not configured to associate a particular "phone-context" value with the number; and

   d) the SIP request is related to an IMS communication service (identified by an ICSI) indicated in a part of the policy on local numbers such that:

      i) the part of the policy on local numbers indicates to use the number as a geo-local number, then the UE shall use the number as a geo-local number in subclause 5.1.2A.1.5; and

      ii) the part of the policy on local numbers indicates to use the number as a home-local number, then the UE shall use the number as a home-local number in subclause 5.1.2A.1.5; and

2) the UE may support being configured with the policy on local numbers using one or more of the following methods:

   a) the Policy_on_local_numbers node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C];

   b) the Policy_on_local_numbers node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

   c) the Policy_on_local_numbers node of 3GPP TS 24.167 [8G].

   If the UE is configured with both the Policy_on_local_numbers node of 3GPP TS 24.167 [8G] and the Policy_on_local_numbers node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or the

Policy_on_local_numbers node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B], then the Policy_on_local_numbers node of the $EF_{IMSConfigData}$ file shall take precedence.

NOTE: Precedence for files configured on both the USIM and ISIM is defined in 3GPP TS 31.103 [15B].

5.1.2A.1.6 Abnormal cases

In the event the UE receives a 504 (Server Time-out) response containing:

1) a P-Asserted-Identity header field set to a value equal to a URI:

   a) from the Service-Route header field value received during registration; or

   b) from the Path header field value received during registration; and

NOTE 1: If there are multiple registration flows associated with the registration, then the UE has received from the P-CSCF during registration multiple sets of Path header field and Service-Route header field values. The Path header field value and Service-Route header field value corresponding to the flow on which the 504 (Server Time-out) response was received are checked.

2) a Content-Type header field set according to subclause 7.6 (i.e. "application/3gpp-ims+xml"), independent of the value or presence of the Content-Disposition header field, independent of the value or presence of Content-Disposition parameters,

then the following treatment is applied:

a) if the 504 (Server Time-out) response includes an IM CN subsystem XML body as described in subclause 7.6 with the <ims-3gpp> element, including a version attribute, with the <alternative-service> child element:

   A) with the <type> child element set to "restoration" (see table 7.6.2); and

   B) with the <action> child element set to "initial-registration" (see table 7.6.3);

   then the UE:

   - shall initiate S-CSCF restoration procedures by performing an initial registration as specified in subclause 5.1.1.2; and

   - may provide an indication to the user based on the text string contained in the <reason> child element of the <alternative-service> child element of the <ims-3gpp> element.

NOTE 2: If the UE has discovered multiple P-CSCF addresses and has information that the P-CSCF was unable to forward the request resulting in sending back the 504 (Server Time-out) response, when starting the initial registration it is appropriate for the UE to select a P-CSCF address different from the one used for the registration binding on which the 504 (Server Time-out) response was received.

When sending a request from a contact address that has been previously registered (or via a registration flow if the multiple registration mechanism is used) which is bound to a public user identity by registration which used a P-CSCF address, and:

- if timer F expires in the "Trying" state of non-INVITE client transaction as described in IETF RFC 3261 [26];

- if a fatal transport error is reported by the transport layer in the "Trying" state of non-INVITE client transaction as described in IETF RFC 3261 [26];

- if timer B expires in the "Calling" state of INVITE client transaction as described in IETF RFC 6026 [163]; or

- if a fatal transport error is reported by the transport layer in "Calling" state of INVITE client transaction as described in IETF RFC 6026 [163];

then the UE shall:

1) if the multiple registration mechanism is not used:

   A) consider the contact address as not bound to any public user identity;

B) mark the currently used P-CSCF address (i.e. P-CSCF address using which the contact address was registered) as unavailable;

C) if there is a locally stored P-CSCF addressas specified in subclause 5.1.9 which is different than thecurrently used P-CSCF address and which is not marked as unavailable, initiate an initial registration as specified in subclause 5.1.1.2 using that P-CSCF; and

D) if there is no locally stored P-CSCF address as specified in subclause 5.1.9 which is different than the currently used P-CSCF address and which is not marked as unavailable, get a new set of P-CSCF addresses as described in subclause 9.2.1 unless otherwise specified in the access specific annexes (as described in annex B, annex L or annex U) and initiate an initial registration as specified in subclause 5.1.1.2; and

2) if the multiple registration mechanism is used, declare the registration flow dead as defined in RFC 5626 [92] and mark the currently used P-CSCF address as unavailable.

NOTE 3: When a fatal transport error occurs, further steps might be necessary to restore the transport layer, possibly including re-establishment of an IP-CAN bearer.

When sending a request from a contact address that has been previously registered (or via a registration flow if the multiple registration mechanism is used) which is bound to a public user identity by registration which used a P-CSCF address, and if a 503 (Service Unavailable) response without a Retry-After header field is received for request as described in IETF RFC 3261 [26], the UE shall:

1) if the multiple registration mechanism is not used:

A) consider the contact address as not bound to any public user identity;

B) mark the currently used P-CSCF address (i.e. P-CSCF address using which the contact address was registered) as unavailable;

C) if there is a locally stored P-CSCF address as specified in subclause 5.1.9 which is different than the currently used P-CSCF address and which is not marked as unavailable, initiate an initial registration as specified in subclause 5.1.1.2 using that P-CSCF; and

D) if there is no locally stored P-CSCF address as specified in subclause 5.1.9 which is different than the currently used P-CSCF address and which is not marked as unavailable, get a new set of P-CSCF addresses as described in subclause 9.2.1 unless otherwise specified in the access specific annexes (as described in annex B, annex L or annex U) and initiate an initial registration as specified in subclause 5.1.1.2; and

2) if the multiple registration mechanism is used, declare the registration flow dead as defined in RFC 5626 [92] and mark the currently used P-CSCF address as unavailable.

When sending a request from a contact address that has been previously registered (or via a registration flow if the multiple registration mechanism is used) which is bound to a public user identity by registration which used a P-CSCF address, and if a 503 (Service Unavailable) response with a Retry-After header field is received for request as described in IETF RFC 3261 [26] and :

- if the request was a non-INVITE request, the Retry-After header field indicates a time bigger than value for timer F as specified in table 7.7.1; and

- if the request was an INVITE request, the Retry-After header field indicates a time bigger than value for timer B as specified in table 7.7.1;

the UE shall:

1) if the multiple registration mechanism is not used:

A) consider the contact address as not bound to any public user identity;

B) mark the currently used P-CSCF address (i.e. P-CSCF address using which the contact address was registered) as unavailable for the time indicated by the Retry-After header field;

C) if there is a locally stored P-CSCF address as specified in subclause 5.1.9 which is different than the currently used P-CSCF address and which is not marked as unavailable, initiate an initial registration as specified in subclause 5.1.1.2 using that P-CSCF; and

D) if there is no locally stored P-CSCF address as specified in subclause 5.1.9 which is different than the currently used P-CSCF address and which is not marked as unavailable, get a new set of P-CSCF addresses as described in subclause 9.2.1 unless otherwise specified in the access specific annexes (as described in annex B, annex L or annex U) and initiate an initial registration as specified in subclause 5.1.1.2; and

2) if the multiple registration mechanism is used, declare the registration flow dead as defined in RFC 5626 [92] and mark the currently used P-CSCF address as unavailable for the time indicated by the Retry-After header field.

NOTE 4: if the Retry-After header field indicates time smaller than the value for timer F or timer B as specified in table 7.7.1, the UE continues using the currently used P-CSCF address.

5.1.2A.2 UE-terminating case

The procedures of this subclause are general to all requests and responses, except those for the REGISTER method.

Where a security association or TLS session exists, the UE shall discard any SIP request that is not protected by the security association or TLS session and is received from the P-CSCF outside of the registration and authentication procedures. The requirements on the UE within the registration and authentication procedures are defined in subclause 5.1.1.

If an initial request contains an Accept-Contact header field containing the g.3gpp.icsi-ref media feature tag with an ICSI value, the UE should invoke the IMS application that is the best match for the ICSI value.

If an initial request contains an Accept-Contact header field containing the g.3gpp.iari-ref media feature tag with an IARI value the UE should invoke the IMS application that is the best match for the IARI value.

The UE can receive multiple ICSI values, IARI values or both in an Accept-Contact header field. In this case it is up to the implementation which of the multiple ICSI values or IARI values the UE takes action on.

NOTE 1: The application verifies that the contents of the request (e.g. SDP media capabilities, Content-Type header field) are consistent with the ICSI value in the g.3gpp.icsi-ref media feature tag and IARI value contained in the g.3gpp.iari-ref media feature tag.

If an initial request does not contain an Accept-Contact header field containing a g.3gpp.icsi-ref media feature tag or a g.3gpp.iari-ref media feature tag the UE shall invoke the application that is the best match based on the contents of the request (e.g. SDP media capabilities, Content-Type header field, media feature tag).

The UE can indicate privacy of the P-Asserted-Identity that will be generated by the P-CSCF in accordance with RFC 3323 [33], and the additional requirements contained within RFC 3325 [34].

NOTE 2: In the UE-terminating case, this version of the document makes no provision for the UE to provide a P-Preferred-Identity in the form of a hint.

NOTE 3: A number of header fields can reveal information about the identity of the user. Where, privacy is required, implementers should also give consideration to other header fields that can reveal identity information. RFC 3323 [33] subclause 4.1 gives considerations relating to a number of header fields.

The UE shall not include its "+sip.instance" header field parameter in the Contact header field in its non-register requests and responses except when the request or response is guaranteed to be sent to a trusted intermediary that will remove the "+sip.instance" header field parameter prior to forwarding the request or response to the destination.

NOTE 4: Such trusted intermediaries include an AS that all such requests as part of an application or service traverse. In order to ensure that all requests or responses containing the "+sip.instance" header field parameter are forwarded via the trusted intermediary the UE needs to have first verified that the trusted intermediary is present (e.g first contacted via a registration or configuration procedure). Including the "+sip.instance" header field parameter containing an IMEI URN does not violate RFC 7254 [153] even when the UE requests privacy using RFC 3323 [33].

If the response includes a Contact header field, and the response is sent within an existing dialog, and the Contact address previously used in the dialog was a GRUU, then the UE should insert the previously used GRUU value in the Contact header field as specified in RFC 5627 [93].

If the response includes a Contact header field, and the response is not sent within an existing dialog, the Contact header field is populated as follows:

1) if a public GRUU value ("pub-gruu" header field parameter) has been saved associated with the public user identity from the P-Called-Party-ID header field, and the UE does not indicate privacy of the contents of the P-Asserted-Identity header field, then the UE should insert the public GRUU ("pub-gruu" header field parameter) value as specified in RFC 5627 [93];

2) if a temporary GRUU value ("temp-gruu" header field parameter) has been saved associated with the public user identity from the P-Called-Party-ID header field, and the UE does indicate privacy of the P-Asserted-Identity, then should insert the temporary GRUU ("temp-gruu" header field parameter) value in the Contact header field as specified in RFC 5627 [93];

NOTE 5: The above items 1 and 2 are mutually exclusive.

3) if the request is related to an IMS communication service that requires the use of an ICSI then the UE shall include in a g.3gpp.icsi-ref media feature tag as defined in subclause 7.9.2 and RFC 3841 [56B] the ICSI value (coded as specified in subclause 7.2A.8.2), for the IMS communication service and then the UE may include the IARI value for any IMS application that applies for the dialog, (coded as specified in subclause 7.2A.9.2), that is related to the request in a g.3gpp.iari-ref media feature tag as defined in subclause 7.9.3 and RFC 3841 [56B]. The UE may also include other ICSI values that the UE is prepared to use for all dialogs with the originating UE(s) and other IARI values for the IMS application that is related to the IMS communication service; and

4) if the request is related to an IMS application that is supported by the UE when the use of an ICSI is not needed, then the UE may include the IARI value (coded as specified in subclause 7.2A.9.2), that is related to any IMS application and that applies for the dialog, in a g.3gpp.iari-ref media feature tag as defined in subclause 7.9.3 and RFC 3841 [56B].

After the dialog is established the UE may change the dialog capabilities (e.g. add a media or request a supplementary service) if defined for the IMS communication service as identified by the ICSI value using the same dialog. Otherwise, the UE shall initiate a new initial request to the other user.

If the UE did not insert a GRUU in the Contact header field then the UE shall include a contact address that has been previously registered with contact parameters used for registration removed and a port in the address in the Contact header field as follows:

- if IMS AKA or SIP digest with TLS is being used as a security mechanism, the protected server port value as in the initial registration; or

- if SIP digest without TLS is being used as a security mechanism, the port value of an unprotected port where the UE expects to receive subsequent mid-dialog requests. The UE shall set the unprotected port value to the port value used in the initial registration.

If the UE receives a Resource-Priority header field in accordance with RFC 4412 [16] in an initial request for a dialog, then the UE shall include the Resource-Priority header field in all requests associated with that dialog.

NOTE 6: For certain national implementations, signalling of a Resource-Priority header field to and from a UE is not required.

If available to the UE (as defined in the access technology specific annexes for each access technology), the UE shall insert a P-Access-Network-Info header field into any response to a request for a dialog, any subsequent request (except CANCEL requests) or response (except CANCEL responses) within a dialog or any response to a standalone method (see subclause 7.2A.4).

The terminating UE shall not include the P-Early-Media header field in any SIP messages, unless the terminating UE is a UE performing the functions of an external attached network that is allowed to send early media.

If a request or response is sent on a dialog for which logging of signalling is in progress, the UE shall check whether a trigger for stopping logging of SIP signalling has occurred, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K].

a) If a stop trigger event has occurred, the UE shall stop logging of signalling; or

b) if a stop trigger event has not occurred, the UE shall:

   - in any requests or responses sent on this dialog, append a "logme" header field parameter to the SIP Session-ID header field; and

- log the request or response.

The UE shall not support RFC 7090 [209] (see table A.4, item A.4/116) and, in this version of the specification, the UE shall not perform any specific procedures beyond those defined in RFC 3261 [26] for the Priority header field.

NOTE 7: The mechanism specified in RFC 7090 [209] is based on the presence of a trust domain for the Priority header field in the operator's network. The UE is not aware whether a trust domain for the Priority header field exists in the operator's network.

If the terminating UE needs to retrieve the last service access number when the AS applies a number translation as described in subclause 5.7.1.22; the terminating UE can find the requested service access number in the hi-entry within the History-Info header field having a hi-index that match the "mp" or "rc" header field parameter value of the last hi-entry containing a "cause" SIP URI parameter, defined in RFC 4458 [68], set to the value "380" defined in RFC 8119 [230]. If no "mp" or "rc" header field parameter is received in the concerned hi-entry, the service access number can be found in the hi-entry preceding the hi-entry with the "cause" SIP URI parameter set to "380".

If the terminating UE

a) supports calling number verification status determination;

b) during registration determined that the home network supports calling number verification using signature verification as and attestation information, as defined in subclause 3.1; and

c) receives initial INVITE request for a dialog or a MESSAGE request containing a P-Asserted-Identity header field or a From header field with a "verstat" tel URI parameter in a tel URI or a SIP URI with a user=phone parameter;

then the terminating UE shall:

a) determine the calling number verification status based on the "verstat" tel URI parameter value; and

b) if unable to determine the calling number verification status based on the "verstat" parameter value, discard the parameter and treat the P-Asserted-Identity header field and the From header field in the same way as if the parameter would not have been included.

5.1.3 Call initiation - UE-originating case

5.1.3.1 Initial INVITE request

Where multiple domains exist for initiating a call/session, before sending an initial INVITE request, the UE shall perform access domain selection in accordance with the appropriate specification for the IP-CAN in use, taking into account the media to be requested. Access domain selection allows the policy of the network operator to be taken into account before the initial INVITE request is sent. Access dependent aspects of access domain selection are defined in the access technology specific annexes for each access technology.

Upon generating an initial INVITE request, the UE shall include the Accept header field with "application/sdp", the MIME type associated with the 3GPP IM CN subsystem XML body (see subclause 7.6.1) and any other MIME type the UE is willing and capable to accept.

The "integration of resource management and SIP" extension is hereafter in this subclause referred to as "the precondition mechanism" and is defined in RFC 3312 [30] as updated by RFC 4032 [64].

The preconditions mechanism should be supported by the originating UE.

If the precondition mechanism is disabled as specified in subclause 5.1.5A, the UE shall not use the precondition mechanism.

The UE may initiate a session without the precondition mechanism if the originating UE does not require local resource reservation.

NOTE 1: The originating UE can decide if local resource reservation is required based on e.g. application requirements, current access network capabilities, local configuration, etc.

In order to allow the peer entity to reserve its required resources, if the precondition mechanism is enabled as specified in subclause 5.1.5A; the originating UE supporting the precondition mechanism should make use of the precondition mechanism, even if it does not require local resource reservation.

Upon generating an initial INVITE request using the precondition mechanism, the UE shall:

- indicate the support for reliable provisional responses and specify it using the Supported header field; and
- indicate the support for the preconditions mechanism and specify it using the Supported header field.

Upon generating an initial INVITE request using the precondition mechanism, the UE shall not indicate the requirement for the precondition mechanism by using the Require header field.

During the session initiation, if the originating UE indicated the support for the precondition mechanism in the initial INVITE request and:

a) the received response with an SDP body includes a Require header field with "precondition" option-tag, the originating UE shall include a Require header field with the "precondition" option-tag:
   - in subsequent requests that include an SDP body, that the originating UE sends in the same dialog as the response is received from; and
   - in responses with an SDP body to subsequent requests that include an SDP body and include "precondition" option-tag in Supported header field or Require header field received in-dialog; or

b) the received response with an SDP body does not include the "precondition" option-tag in the Require header field,
   - in subsequent requests that include an SDP body, the originating UE shall not include a Require or Supported header field with "precondition" option-tag in the same dialog;
   - in responses with an SDP body to subsequent requests with an SDP body but without "precondition" option-tag in the Require or Supported header field, the originating UE shall not include a Require or Supported header field with "precondition" option-tag in the same dialog; and
   - in responses with an SDP body to subsequent requests with an SDP body and with "precondition" option-tag in the Require or Supported header field, the originating UE shall include a Require header field with "precondition" option-tag in the same dialog.

NOTE 2: Table A.4 specifies that UE support of forking is required in accordance with RFC 3261 [26]. The UE can accept or reject any of the forked responses, for example, if the UE is capable of supporting a limited number of simultaneous transactions or early dialogs.

If the precondition mechanism is used, the UE shall, upon successful reservation of local resources, confirm the successful resource reservation (see subclause 6.1.2) within the next SIP request.

NOTE 3: The next SIP request will be either a PRACK request or an UPDATE request.

NOTE 4: The UE can receive a P-Early-Media header field authorizing an early-media flow while the required preconditions, if any, are not met and/or the flow direction is not enabled by the SDP direction parameter. According to RFC 5009 [109], an authorized early-media flow can be established only if the necessary conditions related to the SDP negotiation are met. These conditions can evolve during the session establishment.

NOTE 5: When the UE is confirming the successful resource reservation using an UPDATE request (or a PRACK request) and the UE receives a 180 (Ringing) response or a 200 (OK) response to the initial INVITE request before receiving a 200 (OK) response to the UPDATE request (or a 200 (OK) response to the PRACK request), the UE does not treat this as an error case and does not release the session.

NOTE 6: The UE procedures for rendering of the received early media and of the locally generated communication progress information are specified in 3GPP TS 24.628 [8ZF].

If the UE wishes to receive early media authorization indications, as described in RFC 5009 [109], the UE shall add the P-Early-Media header field with the "supported" parameter to the initial INVITE request.

A UE supporting the Session Timer extension as described in RFC 4028 [58] may support the extension being configured using Session_Timer_Support node specified in 3GPP TS 24.167 [8G].

If the UE supports the Session Timer extension, the UE shall include the option-tag "timer" in the Supported header field and should either insert a Session-Expires header field with the header field value set to the configured session timer interval value, or should not include the Session-Expires header field in the initial INVITE request. The header field value of the Session-Expires header field may be configured using local configuration or using the Session_Timer_Initial_Interval node specified in 3GPP 24.167 [8G]. If the UE is configured with both the local configuration and the Session_Timer_Initial_Interval node specified in 3GPP 24.167 [8G], then the local configuration shall take precedence.

If the UE inserts the Session-Expires header field in the initial INVITE request, the UE may also include the "refresher" parameter with the "refresher" parameter value set to "uac".

When a final answer is received for one of the early dialogs, the UE proceeds to set up the SIP session. The UE shall not progress any remaining early dialogs to established dialogs. Therefore, upon the reception of a subsequent final 200 (OK) response for an INVITE request (e.g., due to forking), the UE shall:

1) acknowledge the response with an ACK request; and

2) send a BYE request to this dialog in order to terminate it.

Upon receiving a 488 (Not Acceptable Here) response to an initial INVITE request, the originating UE should send a new INVITE request containing SDP according to the procedures defined in subclause 6.1.

NOTE 7: An example of where a new request would not be sent is where knowledge exists within the UE, or interaction occurs with the user, such that it is known that the resulting SDP would describe a session that did not meet the user requirements.

Upon receiving a 421 (Extension Required) response to an initial INVITE request in which the precondition mechanism was not used, including the "precondition" option-tag in the Require header field, if the UE supports the precondition mechanism and the precondition mechanism is enabled as specified in subclause 5.1.5A, the originating UE shall:

- send a new INVITE request using the precondition mechanism; and

- send an UPDATE request as soon as the necessary resources are available and a 200 (OK) response for the first PRACK request has been received.

Upon receiving a 503 (Service Unavailable) response to an initial INVITE request containing a Retry-After header field, then the originating UE shall not automatically reattempt the request via the same P-CSCF until after the period indicated by the Retry-After header field contents.

The UE may include a "cic" tel URI parameter in a tel URI, or in the userinfo part of a SIP URI with user=phone, in the Request-URI of an initial INVITE request if the UE wants to identify a user-dialed carrier, as described in RFC 4694 [112].

NOTE 8: The method whereby the UE determines when to include a "cic" tel-URI parameter and what value it should contain is outside the scope of this document (e.g. the UE could use a locally configured digit map to look for special prefix digits that indicate the user has dialled a carrier).

NOTE 9: The value of the "cic" tel-URI parameter reported by the UE is not dependent on UE location (e.g. the reported value is not affected by roaming scenarios).

In the event the UE receives a 380 (Alternative Service) response to an initial INVITE request the response containing a P-Asserted-Identity header field with a value equal to the value of the last entry of the Path header field value received during registration and the response containing a 3GPP IM CN subsystem XML body that includes an <ims-3gpp> element, including a version attribute, with an <alternative-service> child element with the <type> child element set to "emergency" (see table 7.6.2), the UE shall select a domain in accordance with the conventions and rules specified in 3GPP TS 22.101 [1A] and 3GPP TS 23.167 [4B], and:

- if the CS domain is selected, the UE behavior is defined in subclause 7.1.2 of 3GPP TS 23.167 [4B] and, where appropriate, in the access technology specific annex; and

- if the IM CN subsystem is selected, the UE shall apply the procedures in subclause 5.1.6 with the exception of selecting a domain for the emergency call attempt.

NOTE 10: The last entry on the Path header field value received during registration is the value of the SIP URI of the P-CSCF. If there are multiple registration flows associated with the registration, then the UE has received from the P-CSCF during registration multiple sets of Path header field values. The last entry of the Path header field value corresponding to the flow on which the 380 (Alternative Service) response was received is checked.

Upon receiving a 199 (Early Dialog Terminated) provisional response to an established early dialog the UE shall release resources specifically related to that early dialog.

The UE shall include the media feature tags defined in RFC 3840 [62] and RFC 5688 [120] for all supported streaming media types in the initial INVITE request.

If the UE sends a CANCEL request to cancel an initial INVITE request, the UE shall when applicable include in the CANCEL request a Reason header field with a protocol value set to "RELEASE_CAUSE" and a "cause" header field parameter as specified in subclause 7.2A.18.11.2. The UE may also include the "text" header field parameter with reason-text as specified in subclause 7.2A.18.11.2.

Upon receiving a 500 (Server Internal Error) response to an initial INVITE request including a Reason header field with a protocol value set to "FAILURE_CAUSE" and a cause header field parameter value set to "1" as specified in subclause 7.2A.18.12.2 and a Response-Source header field with a "fe" header field parameter set to "<urn:3gpp:fe:p-cscf.orig>", the UE can determine that the QoS or bearer resources in the originating IP-CAN is not available.

5.1.4 Call initiation - UE-terminating case

5.1.4.1 Initial INVITE request

The preconditions mechanism should be supported by the terminating UE.

The handling of incoming initial INVITE requests at the terminating UE is mainly dependent on the following conditions:

- the specific service requirements for "integration of resource management and SIP" extension (hereafter in this subclause known as the precondition mechanism and defined in RFC 3312 [30] as updated by RFC 4032 [64], and with the request for such a mechanism known as a precondition);
- the UEs configuration for the case when the specific service does not require the precondition mechanism; and
- the precondition disabling policy specified in subclause 5.1.5A, if supported by the UE.

If an initial INVITE request is received the terminating UE shall check whether the terminating UE requires local resource reservation.

NOTE 1: The terminating UE can decide if local resource reservation is required based on e.g. application requirements, current access network capabilities, local configuration, etc.

During the session initiation, if local resource reservation is required at the terminating UE and the terminating UE supports the precondition mechanism, and:

a) the received INVITE request includes the "precondition" option-tag in the Supported header field or Require header field and the precondition mechanism is enabled as specified in subclause 5.1.5A, the terminating UE shall use the precondition mechanism and shall include a Require header field with the "precondition" option-tag:
   - in responses to that INVITE request if those responses include an SDP body;
   - in responses to subsequent requests received in-dialog that include an SDP body and include "precondition" option-tag in Supported header field or Require header field; and
   - in subsequent requests that include an SDP body, that it sends towards the originating UE during the session initiation;

b) the received INVITE request includes the "precondition" option-tag in the Supported header field, and the precondition mechanism is disabled as specified in subclause 5.1.5A, the terminating UE shall not use the precondition mechanism:

c) the received INVITE request includes the "precondition" option-tag in the Require header field, and the precondition mechanism is disabled as specified in subclause 5.1.5A, the terminating UE shall reject the INVITE request with a 420 (Bad Extension) response; and

d) the received INVITE request does not include the "precondition" option-tag in the Supported header field or Require header field, the terminating UE shall not use the precondition mechanism.

During the session initiation, if local resource reservation is not required by the terminating UE and the terminating UE supports the precondition mechanism and:

a) the received INVITE request includes the "precondition" option-tag in the Supported header field and:

i) the required resources at the originating UE are not reserved, and the precondition mechanism is enabled as specified in subclause 5.1.5A, the terminating UE shall use the precondition mechanism and shall include a Require header field with the "precondition" option-tag:

- in responses to that INVITE request if those responses include an SDP body;

- in responses with an SDP body to subsequent requests received in-dialog that include an SDP body and include "precondition" option-tag in Supported header field or Require header field; and

- in subsequent requests that include an SDP body, that it sends towards the originating UE during the session initiation;

ii) the required local resources at the originating UE and the terminating UE are available and the precondition mechanism is enabled as specified in subclause 5.1.5A, the terminating UE may use the precondition mechanism; and

iii) the precondition mechanism is disabled as specified in subclause 5.1.5A, the terminating UE shall not use the precondition mechanism;

b) the received INVITE request does not include the "precondition" option-tag in the Supported header field or Require header field, the terminating UE shall not use the precondition mechanism;

c) the received INVITE request includes the "precondition" option-tag in the Require header field and the precondition mechanism is enabled as specified in subclause 5.1.5A, the terminating UE shall use the precondition mechanism; and

d) the received INVITE request includes the "precondition" option-tag in the Require header field, and the precondition mechanism is disabled as specified in subclause 5.1.5A, the terminating UE shall reject the INVITE request with a 420 (Bad Extension) response.

NOTE 2: Table A.4 specifies that UE support of forking is required in accordance with RFC 3261 [26].

NOTE 3: If the terminating UE does not support the precondition mechanism it will apply regular SIP session initiation procedures.

If the received INVITE request indicated support for reliable provisionable responses, but did not require their use and the terminating UE supports reliable provisional responses, and if:

a) the terminating UE requires a reliable alerting indication at the originating side;

b) the 18x response (other than 183 response) carries SDP or for other application related purposes that requires its reliable transport; or

c) the reliable 18x policy indicates (see subclause 5.1.4.2) the UE to do so;

the terminating UE shall send the 18x response (other than 183 response) reliably.

NOTE 4: If the terminating UE is configured by the home operator to send the 18x response (other than 183 response) reliably and the received INVITE request did not indicate support for reliable provisional responses, then the terminating UE sends the 18x response (other than 183 response) unreliably.

The terminating UE shall send the 18x responses (other than 183 response) unreliably if the reliable 18x policy (see subclause 5.1.4.2) indicates the UE to do so, unless the received INVITE request requires to use reliable provisional responses.

NOTE 5: Certain applications, services and operator policies might mandate the terminating UE to send a 199 (Early Dialog Terminated) provisional response (see RFC 6228 [142]) prior to sending a non-2xx final response to the INVITE request.

If the terminating UE uses the precondition mechanism, upon successful reservation of local resources:

- if the originating side requested confirmation for the result of the resource reservation (as defined in RFC 3312 [30]) at the terminating UE, the terminating UE shall confirm the successful resource reservation (see subclause 6.1.3) within an SIP UPDATE request; and

NOTE 6: Originating side requests confirmation for the result of the resource reservation at the terminating UE e.g. when an application server performs 3rd party call control. The request for confirmation for the result of the resource reservation at the terminating UE can be included e.g. in the SDP answer in the PRACK request.

- if the originating side did not request confirmation for the result of the resource reservation (as defined in RFC 3312 [30]) at the terminating UE, the terminating UE shall not confirm the successful resource reservation (see subclause 6.1.3) within an UPDATE request.

NOTE 7: The terminating UE can send an UPDATE request for reasons other than confirmation of the successful resource reservation.

NOTE 8: If the terminating UE supports the precondition mechanism and has indicated the resources are not available or sufficient during SDP negotiation, the UE sends 180 (Ringing) response after the resources on the terminating side are successfully reserved.

If the terminating UE included an SDP offer or an SDP answer in a reliable provisional response to the INVITE request and both the terminating UE and the originating UE support UPDATE method, then in order to remove one or more media streams negotiated in the session for which a final response to the INVITE request has not been sent yet, the terminating UE shall send an UPDATE request with a new SDP offer and delays sending of 200 (OK) response to the INVITE request till after reception of 200 (OK) response to the UPDATE request.

If the user does not accept a media stream accepted in the SDP answer and the terminating UE, the originating UE or both do not support the UPDATE method, then after reception of ACK request related to 200 (OK) response to the INVITE request, the UE shall modify the session.

The terminating UE shall include the media feature tags defined in RFC 3840 [62] and RFC 5688 [120] for all supported streaming media types in the SIP response other than the 100 (Trying) response to the SIP INVITE request.

If the received INVITE request was received over a registration for which the 200 (OK) contained a Feature-Caps header field including the "+sip.607" header field parameter the UE may send a 607 (Unwanted) response as specified in RFC 8197 [254].

NOTE 9: 607 (Unwanted) response is normally sent after user interaction.

If the terminating UE supports the Session Timer extension, as described in RFC 4028 [58], and if the received INVITE request includes the "timer" option tag in the Supported header field, then the terminating UE shall behave as described in RFC 4028 [58] with the following clarification:

- If the received INVITE request does not contain the Session-Expires header field, then the terminating UE shall include a Session-Expires header field with the header field value set to the greater of the configured session timer interval value or the value contained in the Min-SE header field (if present, in the received INVITE), and the "refresher" parameter set to the configured "refresher" parameter value in the 200 (OK) response to the INVITE request. The session timer interval value may be configured using local configuration or the Session_Timer_Initial_Interval node specified in 3GPP 24.167 [8G]. If the UE is configured with both the local configuration and the Session_Timer_Initial_Interval node, then the local configuration shall take precedence. The"refresher" parameter value may be configured using local configuration or using the Session_Timer_Initial_MT_Refresher node specified in 3GPP 24.167 [8G]. If the UE is configured with both the local configuration and the Session_Timer_Initial_MT_Refresher node, then the local configuration shall take precedence;

- If the received INVITE request includes the "timer" option tag in the Supported header field and contains the Session-Expires header field without "refresher" parameter, then the terminating UE shall include a Session-Expires header field with the "refresher" parameter set to the configured "refresher" parameter value in the 200 (OK) response to the INVITE request, and shall set the header field value of the Session-Expires header field of

the 200 (OK) response to the INVITE request to the value received in the INVITE request. The "refresher" parameter value may be configured using local configuration or using the Session_Timer_Initial_MT_Refresher node specified in 3GPP 24.167 [8G]. If the UE is configured with both the local configuration and the Session_Timer_Initial_MT_Refresher node specified in 3GPP 24.167 [8G], then the local configuration shall take precedence; or

- If the received INVITE request contains the Session-Expires header field with "refresher" parameter, then the terminating UE shall include a Session-Expires header field with the "refresher" parameter set to the received "refresher" parameter value in the 200 (OK) response to the INVITE request, and shall set the header field value of the Session-Expires header field of the 200 (OK) response to the INVITE request to the value received in the INVITE request.

5.1.4.2 Reliable 18x Policy

The reliable 18x policy consists of one or more reliable 18x policy parts.

The reliable 18x policy part consists of a mandatory send 18x reliably configuration and an optional ICSI condition.

A 18x response (other than 183 response) to an INVITE request is to be sent reliably according to the reliable 18x policy if:

1) an INVITE request indicates support for reliable provisional responses; and

2) the terminating UE supports reliable provisional responses;

and if the reliable 18x policy contains a reliable 18x policy part such that:

1) the send 18x reliably configuration indicates to send 18x responses reliably; and

2) the following is true:

   a) the corresponding INVITE request is subject to an IMS communication service identified in the ICSI condition of the reliable 18x policy part; or

   b) the reliable 18x policy part does not have the ICSI condition.

A 18x response (other than 183 response) to an INVITE request is to be sent unreliably according to the reliable 18x policy if the INVITE request does not require use of reliable provisional responses and the reliable 18x policy contains a reliable 18x policy part such that:

1) the send 18x reliably configuration indicates to send 18x responses unreliably; and

2) the following is true:

   a) the corresponding INVITE request is subject to an IMS communication service identified in the ICSI condition of the reliable 18x policy part; or

   b) the reliable 18x policy part does not have the ICSI condition.

If the INVITE request is subject to an IMS communication service which does not match the ICSI condition in any of the reliable 18x policy parts and if there is no reliable 18x policy part without ICSI, it is IMS communication service and/or implemention dependent whether to send the SIP 18x responses reliably.

NOTE 1: Some IMS communication services require that SIP 18x responses are not sent reliably. Mandating that the UE send all SIP 18x responses reliably could prevent those IMS communication services from operating correctly.

The UE may support the reliable 18x policy.

The UE may support being configured with the reliable 18x policy using one or more of the following methods:

a) the Reliable_18x_policy node of the $EF_{IMSConfigDat}$a file described in 3GPP TS 31.102 [15C];

b) the Reliable_18x_policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

c) the Reliable_18x_policy node of 3GPP TS 24.167 [8G].

If the UE is configured with both the Reliable_18x_policy node of 3GPP TS 24.167 [8G] and the Reliable_18x_policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or 3GPP TS 31.103 [15B], then the Reliable_18x_policy node of the $EF_{IMSConfigData}$ file shall take precedence.

NOTE 2: Precedence for files configured on both the USIM and ISIM is defined in 3GPP TS 31.103 [15B].

5.1.4A Session modification

5.1.4A.0 General

This subclause applies after the 2xx response to the initial INVITE request has been sent or received.

5.1.4A.1 Generating session modification request

If the precondition mechanism was used during the session establishment, as described in subclause 5.1.3.1 or 5.1.4.1, the UE shall indicate support of the precondition mechanism during a session modification. If the precondition mechanism was not used during the session establishment, the UE shall not indicate support of the precondition mechanism during a session modification.

In order to indicate support of the precondition mechanism during a session modification, upon generating a reINVITE request, an UPDATE request with an SDP body, or a PRACK request with an SDP body, the UE shall:

a) indicate the support for the precondition mechanism using the Supported header field;

b) not indicate the requirement for the precondition mechanism using the Require header field; and

c) if a re-INVITE request is being generated, indicate the support for reliable provisional responses using the Supported header field

and follow the SDP procedures in clause 6 for the precondition mechanism.

5.1.4A.2 Receiving session modification request

Upon receiving a reINVITE request, an UPDATE request, or a PRACK request that indicates support for the precondition mechanism by using the Supported header field or requires use of the precondition mechanism by using the Require header field, the UE shall:

a) if the precondition mechanism was used during the session establishment, as described in subclause 5.1.3.1 or 5.1.4.1, use the precondition mechanism for the session modification; and

b) if the precondition mechanism was not used during the session establishment, and:

   1) if the use of the precondition mechanism is required using the Require header field, reject the request by sending a 420 (Bad Extension) response; and

   2) if the support of the precondition mechanism is indicated using the Supported header field, not use the precondition mechanism for the session modification.

If the precondition mechanism is used for the session modification, the UE shall indicate support for the preconditions mechanism, using the Require header field, in responses that include an SDP body, to the session modification request.

5.1.5 Call release

If the UE sends a BYE request, the UE shall when applicable include in the BYE request a Reason header field with a protocol value set to "RELEASE_CAUSE" and a "cause" header field parameter as specified in subclause 7.2A.18.11.2. The UE may also include the "text" header field parameter with reason-text as specified in subclause 7.2A.18.11.2.

If the UE sends a BYE request, due to the call being unwanted, and the received INVITE request was received over a registration for which the 200 (OK) contained a Feature-Caps header field including the "+sip.607" header field parameter, the UE shall include in the BYE request a Reason header field with a protocol value set to "SIP" and a "cause" header field parameter set to "607" as specified in RFC 8197 [254]. The UE may also include the "text" header field parameter with reason-text as specified in RFC 8197 [254].

5.1.5A Precondition disabling policy

The precondition disabling policy indicates whether the UE is allowed to use the precondition mechanism or whether the UE is not allowed to use the precondition mechanism.

If the precondition disabling policy is not configured, the precondition disabling policy is assumed to indicate that the UE is allowed to use the precondition mechanism.

The UE may support the precondition disabling policy.

If the UE supports the precondition disabling policy, the UE may support being configured with the precondition disabling policy using one or more of the following methods:

a) the Precondition_disabling_policy node of the $EF_{IMSConfigDat}$a file described in 3GPP TS 31.102 [15C];

b) the Precondition_disabling_policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

c) the Precondition_disabling_policy node of 3GPP TS 24.167 [8G].

If the UE is configured with both the Precondition_disabling_policy node of 3GPP TS 24.167 [8G] and the Precondition_disabling_policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or 3GPP TS 31.103 [15B], then the Precondition_disabling_policy node of the $EF_{IMSConfigData}$ file shall take precedence.

NOTE: Precedence for files configured on both the USIM and ISIM is defined in 3GPP TS 31.103 [15B].

The precondition mechanims is disabled, if the UE supports the precondition disabling policy and the precondition disabling policy indicates that the UE is not allowed to use the precondition mechanism.

The precondition mechanism is enabled, if:

1) the UE does not support the precondition disabling policy; or

2) the UE supports the precondition disabling policy and the precondition disabling policy indicates that the UE is allowed to use the precondition mechanism.

5.1.6 Emergency service

5.1.6.1 General

A CS and IM CN subsystem capable UE shall follow the conventions and rules specified in 3GPP TS 22.101 [1A] and 3GPP TS 23.167 [4B] to select the domain for the emergency call attempt. If the CS domain is selected, the UE shall attempt an emergency call setup using appropriate access technology specific procedures.

NOTE 1: For CS systems based on 3GPP TS 24.008 [8], clause B.5 applies.

The UE shall determine, whether it is currently attached to its home operator's network (e.g. HPLMN or subscribed SNPN) or to a different network than its home operator's network (e.g. VPLMN or non-subscribed SNPN) by applying access technology specific procedures described in the access technology specific annexes.

If the IM CN subsystem is selected and the UE is currently attached to its home operator's network (e.g. HPLMN or subscribed SNPN) and the UE is currently registered and the IP-CAN does not define emergency bearers, the UE shall attempt an emergency call as described in subclause 5.1.6.8.4.

If the IM CN subsystem is selected and the UE is currently attached to its home operator's network (e.g. HPLMN or subscribed SNPN) and the UE is currently registered and the IP-CAN defines emergency bearers and the core network has indicated that it supports emergency bearers, the UE shall:

1) perform an initial emergency registration, as described in subclause 5.1.6.2; and

2) attempt an emergency call as described in subclause 5.1.6.8.3.

If the IM CN subsystem is selected and the UE is currently attached to its home operator's network (e.g. HPLMN or subscribed SNPN) and the UE is not currently registered, the UE shall:

1) perform an initial emergency registration, as described in subclause 5.1.6.2; and

2) attempt an emergency call as described in subclause 5.1.6.8.3.

If the IM CN subsystem is selected and the UE is attached to a different network than its home operator's network (e.g. VPLMN or non-subscribed SNPN), the UE shall:

1) perform an initial emergency registration, as described in subclause 5.1.6.2; and

2) attempt an emergency call as described in subclause 5.1.6.8.3.

If the UE supports the emerg-reg timer defined in table 7.8.1, the UE shall start the emerg-reg timer when the UE decides that an emergency call is to be established via the IM CN subsystem. The UE shall stop the timer when the UE determines that an initial emergency registration, as described in subclause 5.1.6.2, is not required or upon receipt of any final SIP response during the initial emergency registration. When the emerg-reg timer expires, the UE shall:

1) if the initial REGISTER request for the initial emergency registration has been sent, consider that the emergency registration attempt for this P-CSCF has failed. The UE may retry registration on a different P-CSCF and if the UE has no more available P-CSCFs the UE considers the emergency registration to have failed and applies the procedures related to emergency registration failure that are defined in 3GPP TS 23.167 [4B] subclause 6.1; and

2) if the initial REGISTER request for the initial emergency registration has not been sent:

   - if the UE has successfully established an IP-CAN bearer for an emergency session, consider that the emergency registration attempt for this P-CSCF has failed. The UE may retry registration on a different P-CSCF and if the UE has no more available P-CSCFs the UE considers the emergency registration to have failed and applies the procedures related to emergency registration failure that are defined in 3GPP TS 23.167 [4B] subclause 6.1; or

   - if the UE has not successfully established an IP-CAN bearer for an emergency session, consider that the attempt to set up the emergency call via the IM CN subsystem has failed, abort any ongoing IP-CAN procedures for the emergency registration, and apply the procedures for domain selection as defined in 3GPP TS 23.167 [4B] clause H.5.

The UE may support being pre-configured for the Emerg-reg timer using one or more of the following methods:

a) the Timer_Emerg-reg leaf of the $EF_{IMSConfigDat}$a file described in 3GPP TS 31.102 [15C];

b) the Timer_Emerg-reg leaf of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

c) the Timer_Emerg-reg leaf of 3GPP TS 24.167 [8G].

If the UE is configured with both the Timer_Emerg-reg leaf of 3GPP TS 24.167 [8G] and the Timer_Emerg-reg leaf of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or 3GPP TS 31.103 [15B], then the Timer_Emerg-reg leaf of the $EF_{IMSConfigData}$ file shall take precedence.

NOTE 2: Precedence for files configured on both the USIM and ISIM is defined in 3GPP TS 31.103 [15B].

If the IM CN subsystem is selected and the UE has no credentials the UE can make an emergency call without being registered. The UE shall attempt an emergency call as described in subclause 5.1.6.8.2.

An IP-CAN can, dependent on the IP-CAN capabilities, provide local emergency numbers (including information about emergency service categories or information about emergency service URNs) to the UE which has that capability, in order for the UE to recognize these numbers as emergency call.

5.1.6.2 Initial emergency registration

When the user initiates an emergency call, if emergency registration is needed (including cases described in subclause 5.1.6.2A), the UE shall perform an emergency registration prior to sending the SIP request related to the emergency call.

The UE shall have only one valid emergency registration at any given time. If the UE initiates a new emergency registration using different contact address, and the previous emergency registration has not expired, the UE shall consider the previous emergency registration as expired.

IP-CAN procedures for emergency registration are defined in 3GPP TS 23.167 [4B] and in each access technology specific annex.

When a UE performs an initial emergency registration the UE shall perform the actions as specified in subclause 5.1.1.2 with the following additions and modifications:

a) the UE shall include a "sos" SIP URI parameter in the Contact header field as described in subclause 7.2A.13, indicating that this is an emergency registration and that the associated contact address is allowed only for emergency service; and

b) the UE shall populate the From and To header fields of the REGISTER request with:

- the first entry in the list of public user identities provisioned in the UE;
- the default public user identity obtained during the normal registration, if the UE is not provisioned with a list of public user identites, but the UE is currently registered to the IM CN subsystem; and
- the derived temporary public user identity, in all other cases.

When the UE performs an initial emergency registration and whilst this emergency registration is active, the UE shall:

- handle the emergency registration independently from any other ongoing registration to the IM CN subsystem;
- handle any signalling or media related IP-CAN for the purpose of emergency calls independently from any other established IP-CAN for IM CN subsystem related signalling or media; and
- handle all SIP signalling and all media related to the emergency call independently from any other ongoing IM CN subsystem signalling and media.

If:

1) the UE receives a 420 (Bad Extension) response to the REGISTER request for initial emergency registration containing an "sos" SIP URI parameter in the Contact header field;
2) the UE does not support GPRS-IMS-Bundled authentication; and
3) the response contains a 3GPP IM CN subsystem XML body that includes an <ims-3gpp> element, including a version attribute, with an <alternative-service> child element with the <type> child element set to "emergency" (see table 7.6.2) and <action> child element set to "anonymous-emergencycall" (see table 7.6.3);

the UE shall attempt an emergency call as described in subclause 5.1.6.8.2.

If:

1) the UE receives a 403 (Forbidden) response to the REGISTER request for initial emergency registration containing an "sos" SIP URI parameter in the Contact header field; and
2) the response contains a 3GPP IM CN subsystem XML body that includes an <ims-3gpp> element, including a version attribute, with an <alternative-service> child element with the <type> child element set to "emergency" (see table 7.6.2) and <action> child element set to "anonymous-emergencycall" (see table 7.6.3);

the UE shall attempt an emergency call as described in subclause 5.1.6.8.2.

5.1.6.2A New initial emergency registration

The UE shall perform a new initial emergency registration, as specified in subclause 5.1.6.2, if the UE determines that:

- it has previously performed an emergency registration which has not yet expired; and
- it has obtained an IP address from the serving IP-CAN, as specified in subclause 9.2.1, different than the IP address used for the emergency registration.

5.1.6.3 Initial subscription to the registration-state event package

Upon receiving the 200 (OK) to the REGISTER request that completes the emergency registration, the UE shall not subscribe to the reg event package of the public user identity specified in the REGISTER request.

5.1.6.4 User-initiated emergency reregistration

The UE shall perform user-initiated emergency reregistration as specified in subclause 5.1.1.4 if half of the time for the emergency registration has expired and:

- the UE has emergency related ongoing dialog;
- standalone transactions exist; or
- the user initiates an emergency call.

The UE shall not perform user-initiated emergency reregistration in any other cases.

5.1.6.5 Authentication

When a UE performs authentication a UE shall perform the procedures as specified in subclause 5.1.1.5.

5.1.6.6 User-initiated emergency deregistration

Once the UE registers a public user identity and an associated contact address via emergency registration, the UE shall not perform user-initiated deregistration of the respective public user identity and the associated contact address.

NOTE: The UE will be deregistered when the emergency registration expires.

5.1.6.7 Network-initiated emergency deregistration

An emergency registration will not be deregistered by the network (see subclause 5.4.8.4).

5.1.6.8 Emergency session setup

5.1.6.8.1 General

The UE shall translate any user indicated emergency number as specified in 3GPP TS 22.101 [1A] to an emergency service URN, i.e. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69].

When an initial request for a dialog or a standalone transaction, or an unknown method transmitted as part of UE detected emergency call procedures as defined in subclause 5.1.6 is initiated:

- in event other than reception of a 380 (Alternative Service) response to an initial request for a dialog, or a standalone transaction, or an unknown method as defined in procedures in subclause 5.1.2A.1.1, subclause 5.1.3.1, subclause 5.1.6.8.1, subclause 5.1.6.8.3 and subclause 5.1.6.8.4; or
- upon reception of a 380 (Alternative Service) response to an initial request for a dialog, or a standalone transaction, or an unknown method as defined in procedures in subclause 5.1.2A.1.1, subclause 5.1.3.1, subclause 5.1.6.8.1, subclause 5.1.6.8.3 and subclause 5.1.6.8.4, and the 380 (Alternative Service) response does not contain a Contact header field containing a service URN with a top-level service type of "sos",

the Request-URI of the initial request for a dialog or the standalone transaction, or the unknown method transmitted as part of UE detected emergency call procedures as defined in subclause 5.1.6 shall include one of the following service URNs:

- "urn:service:sos", "urn:service:sos.ambulance", "urn:service:sos.police", "urn:service:sos.fire", "urn:service:sos.marine", "urn:service:sos.mountain", "urn:service:sos.ecall.manual", "urn:service:sos.ecall.automatic". If the UE can determine the type of emergency service the UE shall use an emergency service URN with a sub-service type corresponding to the type of emergency service.
- as derived from the information about emergency service URNs provided with local emergency numbers (see subclause 5.1.6.1).

NOTE 1: A service URN with a top-level service type of "sos" is used only when the user intends to establish an emergency call.

NOTE 2: In countries where a type of emergency service is required, due to national regulations, an emergency call request with emergency service URN "urn:service:sos" can fail.

When an initial request for a dialog or a standalone transaction, or an unknown method transmitted as part of UE detected emergency call procedures as defined in subclause 5.1.6 is initiated upon reception of 380 (Alternative Service) response to an initial request for a dialog, or a standalone transaction, or an unknown method as defined in procedures in subclause 5.1.2A.1.1, subclause 5.1.3.1, subclause 5.1.6.8.1, subclause 5.1.6.8.3 and subclause 5.1.6.8.4, and if the 380 (Alternative Service) response contains a Contact header field containing a service URN with a top-level service type of "sos", the UE shall set the Request-URI of the initial request for a dialog or the standalone transaction, or the unknown method transmitted as part of UE detected emergency call procedures as defined in subclause 5.1.6 to the service URN of the Contact header field of the 380 (Alternative Service) response.

In the event the UE receives a 380 (Alternative Service) response to an INVITE request the response including a 3GPP IM CN subsystem XML body as described in subclause 7.6 that includes an <ims-3gpp> element, including a version attribute, with an <alternative-service> child element with the <type> child element set to "emergency" (see table 7.6.2), the UE shall automatically send an ACK request to the P-CSCF as per normal SIP procedures and terminate the session. In addition, if the 380 (Alternative Service) response includes a P-Asserted-Identity header field with a value equal to the value of the last entry on the Path header field value received during registration:

- the UE may also provide an indication to the user based on the text string contained in the <reason> child element of the <alternative-service> child element of the <ims-3gpp> element; and

- one of subclause 5.1.6.8.3 or subclause 5.1.6.8.4 applies.

NOTE 3: Emergency numbers which the UE does not detect, will be treated as a normal call.

NOTE 4: The last entry on the Path header field value received during registration is the value of the SIP URI of the P-CSCF. If there are multiple registration flows associated with the registration, then the UE has received from the P-CSCF during registration multiple sets of Path header field values. The last entry of the Path header field value corresponding to the flow on which the 380 (Alternative Service) response was received is checked.

If the UE supports the emerg-request timer defined in Table 7.8.1, the UE shall start the emerg-request timer when sending the initial INVITE request for emergency service. The UE shall stop the timer upon receipt of any 18x provisional SIP response. When the emerg-request timer expires, the UE shall consider that the emergency service request has failed and apply the procedures related to emergency service request failure that are defined in 3GPP TS 23.167 [4B] subclause 7.3 with clarifications in clause H.5. The UE may support being configured for the emerg-request timer using one or more of the following methods:

a) the Timer_Emerg-request leaf of the $EF_{IMSConfigDat}$a file described in 3GPP TS 31.102 [15C];

b) the Timer_Emerg-request leaf of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

c) the Timer_Emerg-request leaf of 3GPP TS 24.167 [8G].

If the UE is configured with both the Timer_Emerg-request leaf of 3GPP TS 24.167 [8G] and the Timer_Emerg-request leaf of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or 3GPP TS 31.103 [15B], then the Timer_Emerg-request leaf of the $EF_{IMSConfigData}$ file shall take precedence.

NOTE 5: Precedence for files configured on both the USIM and ISIM is defined in 3GPP TS 31.103 [15B].

5.1.6.8.2 Emergency session set-up in case of no registration

When establishing an emergency session for an unregistered user, the UE is allowed to receive responses to emergency requests and requests inside an established emergency session on the unprotected ports. The UE shall reject or silently discard all other messages not arriving on a protected port. Additionally, the UE shall transmit signalling packets pertaining to the emergency session from the same IP address and unprotected port on which it expects to receive signalling packets containing the responses to emergency requests and the requests inside the established emergency session.

Prior to establishing an emergency session for an unregistered user, the UE shall acquire a local IP address, discover a P-CSCF, and establish an IP-CAN bearer that can be used for SIP signalling. The UE shall send only the initial INVITE requests to the port advertised to the UE during the P-CSCF discovery procedure. If the UE does not receive any

specific port information during the P-CSCF discovery procedure, the UE shall send the initial INVITE request to the SIP default port values as specified in RFC 3261 [26].

The UE shall apply the procedures as specified in subclause 5.1.2A.1 and subclause 5.1.3 with the following additions:

1) the UE shall set the From header field of the INVITE request to "Anonymous" as specified in RFC 3261 [26];

2) the UE shall include a service URN in the Request-URI of the initial INVITE request in accordance with subclause 5.1.6.8.1;

NOTE 1: Other specifications make provision for emergency service identifiers, which are not specifically the emergency service URN, to be recognised in the UE. Emergency service identifiers which the UE does not detect will be treated as a normal call by the UE.

3) the UE shall insert in the INVITE request, a To header field with the same emergency service URN as in the Request-URI;

4) if available to the UE (as defined in the access technology specific annexes for each access technology), the UE shall include in the P-Access-Network-Info header field in any request for a dialog, any subsequent request (except CANCEL requests) or response (except CANCEL responses) within a dialog or any request. Insertion of the P-Access-Network-Info header field into the ACK request is optional. The UE shall populate the P-Access-Network-Info header field with the current point of attachment to the IP-CAN as specified for the access network technology (see subclause 7.2A.4). The P-Access-Network-Info header field contains the location identifier such as the cell id, the line id or the identity of the WLAN access node, which is relevant for routeing the emergency call;

5) if defined by the access technology specific annex, the UE shall populate the P-Preferred-Identity header field in the INVITE request with an equipment identifier as a SIP URI. The special details of the equipment identifier to use depend on the IP-CAN;

6) a Contact header field set to include SIP URI that contains in the hostport parameter the IP address of the UE and an unprotected port where the UE will receive incoming requests belonging to this dialog. The UE shall also include a "sip.instance" media feature tag containing Instance ID as described in RFC 5626 [92]. The UE shall not include either the public or temporary GRUU in the Contact header field;

7) a Via header field set to include the IP address of the UE in the sent-by field and for the UDP the unprotected server port value where the UE will receive response to the emergency request, while for the TCP, the response is received on the TCP connection on which the emergency request was sent. For the UDP, the UE shall also include "rport" header field parameter with no value in the top Via header field. Unless the UE has been configured to not send keep-alives, and unless the UE is directly connected to an IP-CAN for which usage of NAT is not defined, it shall include a "keep" header field parameter with no value in the Via header field, in order to indicate support of sending keep-alives associated with, and during the lifetime of, the emergency session, as described in RFC 6223 [143];

NOTE 2: The UE inserts the same IP address and port number into the Contact header field and the Via header field, and sends all IP packets to the P-CSCF from this IP address and port number.

8) if the UE has its location information available, or a URI that points to the location information, the UE shall include a Geolocation header field in the INVITE request in the following way:

   - if the UE is aware of the URI that points to where the UE's location is stored, include the URI as the Geolocation header field value, as described in RFC 6442 [89]; or

   - if the UE is aware of its location information, include the location information in a PIDF location object, in accordance with RFC 4119 [90] and RFC 5491 [267], include the location object in a message body with the content type application/pidf+xml, and include a Content ID URL, referring to the message body, as the Geolocation header field value, as described RFC 6442 [89], and include a Content-Disposition header field with a disposition type "render" value and a "handling" header field parameter with an "optional" value, as described in RFC 3261 [26];

NOTE 3: If the location information is old or inaccurate, the UE does not consider location information to be available.

9) if the UE includes a Geolocation header field, the UE shall also include a Geolocation-Routing header field with a "yes" header field value, which indicates that the location of the UE can be used by other entities to make routing decisions, as described in RFC 6442 [89];

10) if the UE has neither geographical location information available, nor a URI that points to the location information, the UE shall not insert a Geolocation header field in the INVITE request; and

NOTE 4: It is suggested that UE's only use the option of providing a URI when the domain part belongs to the current P-CSCF or S-CSCF provider. This is an issue on which the network operator needs to provide guidance to the end user. A URI that is only resolvable to the UE which is making the emergency call is inapplicable in this area.

11) if support of the current location discovery during an emergency call is allowed in the IP-CAN specific annex and the UE supports the current location discovery during an emergency call, the UE shall include a Recv-Info header field as described in RFC 6086 [25], indicating the g.3gpp.current-location-discovery info package name and shall include an Accept header field indicating the "application/vnd.3gpp.current-location-discovery+xml" MIME type.

NOTE 5: During the dialog, the points of attachment to the IP-CAN of the UE can change (e.g. UE connects to different cells). The UE will populate the P-Access-Network-Info header field in any request or response within a dialog with the current point of attachment to the IP-CAN (e.g. the current cell information).

The UE shall build a proper preloaded Route header field value for all new dialogs. The UE shall build a Route header field value containing only the P-CSCF URI (containing the unprotected port number and the IP address acquired at the time of the P-CSCF discovery procedures which was used in registration of the contact address (or registration flow).

NOTE 6: If the UE is provisioned with or receives a FQDN at the time of the P-CSCF discovery procedures, the FQDN is resolved to an IP address at the time of the P-CSCF discovery procedures.

When a SIP transaction times out, i.e. timer B, timer F or timer H expires at the UE, the UE may behave as if timer F expired, as described in subclause 5.1.1.4.

NOTE 7: It is an implementation option whether these actions are also triggered by other means.

NOTE 8: A number of header fields can reveal information about the identity of the user. Where privacy is required, implementers should also give consideration to other header fields that can reveal identity information. RFC 3323 [33] subclause 4.1 gives considerations relating to a number of header fields.

NOTE 9: RFC 3261 [26] provides for the use of the Priority header field with a suggested value of "emergency". It is not precluded that emergency sessions contain this value, but such usage will have no impact on the processing within the IM CN subsystem.

If the response for the initial INVITE request indicates that the UE is behind NAT, and the INVITE request was sent over TCP connection, the UE shall keep the TCP connection during the entire duration of the emergency session. In this case the UE will receive all responses to the emergency requests and the requests inside the established emergency session over this TCP connection.

If the Via header field of any provisional response, or of the final 200 (OK) response, for the initial INVITE request contains a "keep" header field parameter with a value, unless the UE detects that it is not behind a NAT, the UE shall start to send keep-alives associated with the session towards the P-CSCF, as described in RFC 6223 [143].

5.1.6.8.3 Emergency session set-up within an emergency registration

After a successful initial emergency registration, the UE shall apply the procedures as specified in subclause 5.1.2A and 5.1.3 with the following additions:

1) the UE shall insert in the INVITE request, a From header field that includes the public user identity registered via emergency registration or the tel URI associated with the public user identity registered via emergency registration, as described in subclause 4.2;

2) the UE shall include a service URN in the Request-URI of the INVITE request in accordance with subclause 5.1.6.8.1;

3) the UE shall insert in the INVITE request, a To header field with the same emergency service URN as in the Request-URI;

4) if available to the UE, and if defined for the access type as specified in subclause 7.2A.4, the P-Access-Network-Info header field shall contain a location identifier such as the cell id, line id or the identity of the WLAN access node, which is relevant for routeing the IMS emergency call;

NOTE 1: The IMS emergency specification in 3GPP TS 23.167 [4B] describes several methods how the UE can get its location information from the access network or from a server. Such methods are not in the scope of this specification.

5) the UE shall insert in the INVITE request, one or two P-Preferred-Identity header field(s) that include the public user identity registered via emergency registration or the tel URI associated with the public user identity registered via emergency registration as described in subclause 4.2;

NOTE 2: Providing two P-Preferred-Identity header fields is usually supported by UE acting as enterprise network.

6) void;

7) if the UE has its location information available, or a URI that points to the location information, then the UE shall include a Geolocation header field in the INVITE request in the following way:

- if the UE is aware of the URI that points to where the UE's location is stored, include the URI as the Geolocation header field value, as described in RFC 6442 [89]; or

- if the UE is aware of its location information, include the location information in a PIDF location object, in accordance with RFC 4119 [90] and RFC 5491 [267], include the location object in a message body with the content type application/pidf+xml, and include a Content ID URL, referring to the message body, as the Geolocation header field value, as described RFC 6442 [89], and include a Content-Disposition header field with a disposition type "render" value and a "handling" header field parameter with an "optional" value, as described in RFC 3261 [26];

NOTE 3: If the location information is old or inaccurate, the UE does not consider location information to be available.

8) if the UE includes a Geolocation header field, the UE shall also include a Geolocation-Routing header field with a "yes" header field value, which indicates that the location of the UE can be used by other entities to make routing decisions, as described in RFC 6442 [89];

NOTE 4: It is suggested that UE's only use the option of providing a URI when the domain part belongs to the current P-CSCF or S-CSCF provider. This is an issue on which the network operator needs to provide guidance to the end user. A URI that is only resolvable to the UE which is making the emergency call is not desirable.

9) if the UE has neither geographical location information available, nor a URI that points to the location information, the UE shall not insert a Geolocation header field in the INVITE request; and

10) if support of the current location discovery during an emergency call is allowed in the IP-CAN specific annex and the UE supports the current location discovery during an emergency call, the UE shall include a Recv-Info header field as described in RFC 6086 [25], indicating the g.3gpp.current-location-discovery info package name and shall include an Accept header field indicating the "application/vnd.3gpp.current-location-discovery+xml" MIME type.

NOTE 5: RFC 3261 [26] provides for the use of the Priority header field with a suggested value of "emergency". It is not precluded that emergency sessions contain this value, but such usage will have no impact on the processing within the IM CN subsystem.

In the event the UE receives a 380 (Alternative Service) response with a P-Asserted-Identity header field with a value equal to the value of the last entry on the Path header field value received during registration, and the Content-Type header field set according to subclause 7.6 (i.e. "application/3gpp-ims+xml"), independent of the value or presence of the Content-Disposition header field, independent of the value or presence of Content-Disposition parameters, then the following treatment is applied:

1) if the 380 (Alternative Service) response includes a 3GPP IM CN subsystem XML body as described in subclause 7.6 the <ims-3gpp> element, including a version attribute, with the <alternative-service> child element with the <type> child element set to "emergency" (see table 7.6.2), then the UE shall:

a) if the CS domain is available to the UE, and no prior attempt using the CS domain for the current emergency call attempt has been made, attempt emergency call via CS domain using appropriate access technology specific procedures;

b) if the CS domain is not available to the UE or the emergency call has already been attempted using the CS domain, then perform one of the following actions:

- if the <action> child element of the <alternative-service> child element of the <ims-3gpp> element in the IM CN subsystem XML body as described in subclause 7.6 is set to "emergency-registration" (see table 7.6.3), perform an initial emergency registration using a different VPLMN or non-subscribed SNPN, if available, as described in subclause 5.1.6.2 and if the new emergency registration succeeded, attempt an emergency call as described in this subclause; or

- perform implementation specific actions to establish the emergency call; and

2) if the 380 (Alternative Service) response includes a 3GPP IM CN subsystem XML body as described in subclause 7.6 with the <ims-3gpp> element, including a version attribute, with the <alternative-service> child element with the <type> child element set to "emergency" (see table 7.6.2) then the UE may also provide an indication to the user based on the text string contained in the <reason> child element of the <alternative-service> child element of the <ims-3gpp> element.

NOTE 6: The last entry on the Path header field value received during registration is the value of the SIP URI of the P-CSCF. If there are multiple registration flows associated with the registration, then the UE has received from the P-CSCF during registration multiple sets of Path header field values. The last entry of the Path header field value corresponding to the flow on which the 380 (Alternative Service) response was received is checked.

5.1.6.8.4 Emergency session setup within a non-emergency registration

The UE shall apply the procedures as specified in subclauses 5.1.2A and 5.1.3 with the following additions:

1) the UE shall include a service URN in the Request-URI of the INVITE request in accordance with subclause 5.1.6.8.1;

2) the UE shall insert in the INVITE request, a To header field with the same emergency service URN as in the Request-URI;

3) the UE shall insert in the INVITE request, a From header field that includes the public user identity or the tel URI associated with the public user identity, as described in subclause 4.2;

4) if available to the UE, and if defined for the access type as specified in subclause 7.2A.4, the UE shall insert in the P-Access-Network-Info header field a location identifier such as the cell id, line id or the identity of the WLAN access node, which is relevant for routeing the IMS emergency call;

NOTE 1: 3GPP TS 23.167 [4B] describes several methods how the UE can get its location information from the access network or from a server. Such methods are not in the scope of this specification.

5) the UE shall insert in the INVITE request one or two P-Preferred-Identity header field(s) that include the public user identity or the tel URI associated with the public user identity as described in subclause 4.2;

NOTE 2: Providing two P-Preferred-Identity header fields is usually supported by UE acting as enterprise network.

6) if the UE has its location information available, or a URI that points to the location information, then the UE shall include a Geolocation header field in the INVITE request in the following way:

- if the UE is aware of the URI that points to where the UE's location is stored, include the URI as the Geolocation header field value, as described in RFC 6442 [89]; or

- if the UE is aware of its location information, include the location information in a PIDF location object, in accordance with RFC 4119 [90] and RFC 5491 [267], include the location object in a message body with the content type application/pidf+xml, and include a Content ID URL, referring to the message body, as the Geolocation header field value, as described RFC 6442 [89], and include a Content-Disposition header field with a disposition type "render" value and a "handling" header field parameter with an "optional" value, as described in RFC 3261 [26];

NOTE 3: If the location information is old or inaccurate, the UE does not consider location information to be available.

7) if the UE includes a Geolocation header field, the UE shall also include a Geolocation-Routing header field with a "yes" header field value, which indicates that the location of the UE can be used by other entities to make routing decisions, as described in RFC 6442 [89];

8) if the UE has neither geographical location information available, nor a URI that points to the location information, the UE shall not insert a Geolocation header field in the INVITE request;

NOTE 4: It is suggested that UE's only use the option of providing a URI when the domain part belongs to the current P-CSCF or S-CSCF provider. This is an issue on which the network operator needs to provide guidance to the end user. A URI that is only resolvable to the UE which is making the emergency call is not desirable.

9) if a public GRUU value ("pub-gruu" header field parameter) has been saved associated with the public user identity to be used for this request, then insert the public GRUU ("pub-gruu" header field parameter) value in the Contact header field as specified in RFC 5627 [93]. Otherwise the UE shall include the address in the Contact header field set to contain the IP address or FQDN of the UE, and the UE shall also include:

   - if IMS AKA or SIP digest with TLS is being used as a security mechanism, the protected server port value as in the initial registration; or

   - if SIP digest without TLS, NASS-IMS bundled authentication or GPRS-IMS-Bundled Authentication is being used as a security mechanism, the port value of an unprotected port where the UE expects to receive subsequent mid-dialog requests. The UE shall set the unprotected port value to the port value used in the initial registration; and

10) if support of the current location discovery during an emergency call is allowed in the IP-CAN specific annex and the UE supports the current location discovery during an emergency call, the UE shall include a Recv-Info header field as described in RFC 6086 [25], indicating the g.3gpp.current-location-discovery info package name and shall include an Accept header field indicating the "application/vnd.3gpp.current-location-discovery+xml" MIME type.

In the event the UE receives a 380 (Alternative Service) response with a P-Asserted-Identity header field with a value equal to the value of the SIP URI of the P-CSCF received in the Path header field during registration, and the Content-Type header field set according to subclause 7.6 (i.e. "application/3gpp-ims+xml"), independent of the value or presence of the Content-Disposition header field, independent of the value or presence of Content-Disposition parameters, then the following treatment is applied:

a) if the 380 (Alternative Service) response includes a 3GPP IM CN subsystem XML body as described in subclause 7.6 with an <ims-3gpp> element, including a version attribute, with an <alternative-service> child element with the <type> child element set to "emergency" (see table 7.6.2), then the UE shall:

   - if the <action> child element of the <alternative-service> child element of the <ims-3gpp> element in the IM CN subsystem XML body as described in subclause 7.6 is set to "emergency-registration" (see table 7.6.3), perform an initial emergency registration, as described in subclause 5.1.6.2 and attempt an emergency call as described in subclause 5.1.6.8.3;

   - attempt emergency call via CS domain using appropriate access technology specific procedures, if available and not already tried; or

   - perform implementation specific actions to establish the emergency call; and

b) if the 380 (Alternative Service) response includes a 3GPP IM CN subsystem XML body as described in subclause 7.6 with an <ims-3gpp> element, including a version attribute, with an <alternative-service> child element with the <type> child element set to "emergency" (see table 7.6.2) then the UE may also provide an indication to the user based on the text string contained in the <reason> child element of the <alternative-service> child element of the <ims-3gpp> element.

NOTE 5: RFC 3261 [26] provides for the use of the Priority header field with a suggested value of "emergency". It is not precluded that emergency sessions contain this value, but such usage will have no impact on the processing within the IM CN subsystem.

NOTE 6: The last entry on the Path header field value received during registration is the value of the SIP URI of the P-CSCF. If there are multiple registration flows associated with the registration, then the UE has received from the P-CSCF during registration multiple sets of Path header field values. The last entry of the Path header field value corresponding to the flow on which the 380 (Alternative Service) response was received is checked.

5.1.6.9 Emergency session release

Normal call release procedure shall apply, as specified in the subclause 5.1.5.

5.1.6.10 Successful or provisional response to a request not detected by the UE as relating to an emergency session

If the UE receives a 1xx or 200 (OK) response to an initial request for a dialog, the response containing a P-Asserted-Identity header field set to an emergency number as specified in 3GPP TS 22.101 [1A], and:

- if a public GRUU value (pub-gruu) has been saved associated with the public user identity, the public GRUU value has not been included in the Contact header field of the initial request for a dialog as specified in RFC 5627 [93];

- if a public GRUU value (pub-gruu) has not been saved and a protected server port was not included in the address in the Contact header field of the initial request for a dialog; or

- if the UE has its geographical location information available, or a URI that points to the location information, and the geographical location information or URI has not been included in the initial request for a dialog;

then the UE shall send an UPDATE request according to RFC 3311 [29]; and:

1) if available to the UE, and if defined for the access type as specified in subclause 7.2A.4, the UE shall include in the UPDATE request a P-Access-Network-Info header field and it shall contain a location identifier such as the cell id or the identity of the WLAN access node;

2) if the UE has its geographical location information available, or a URI that points to the location information, then the UE shall include it in the UPDATE request in the following way:

   I) if the UE is aware of the URI that points to where the UE's location is stored, include the URI as the Geolocation header field value, as described in RFC 6442 [89]; or

   II) if the UE is aware of its location information, include the location information in a PIDF location object, in accordance with RFC 4119 [90] and RFC 5491 [267], include the location object in a message body with the content type application/pidf+xml, and include a Content ID URL, referring to the message body, as the Geolocation header field value, as described RFC 6442 [89], and include a Content-Disposition header field with a disposition type "render" value and a "handling" header field parameter with an "optional" value, as described in RFC 3261 [26];

NOTE 1: If the location information is old or inaccurate, the UE does not consider location information to be available.

3) if the UE includes a Geolocation header field, the UE shall also include a Geolocation-Routing header field with a "yes" header field value, which indicates that the location of the UE can be used by other entities to make routing decisions, as described in RFC 6442 [89];

4) if the UE has neither geographical location information available, nor a URI that points to the location information, the UE shall not insert a Geolocation header field in the UPDATE request; and

5) if a public GRUU value ("pub-gruu" header field parameter) has been saved associated with the public user identity, then the UE shall insert the public GRUU ("pub-gruu" header field parameter) value in the Contact header field of the UPDATE request as specified in RFC 5627 [93]; otherwise the UE shall include the address in the Contact header field set in accordance with subclause 5.1.6.8.4, item 8.

NOTE 2: The IMS emergency specification in 3GPP TS 23.167 [4B] describes several methods how the UE can get its location information from the access network or from a server. Such methods are not in the scope of this specification.

NOTE 3: It is suggested that UEs only use the option of providing a URI when the domain part belongs to the current P-CSCF or S-CSCF provider. This is an issue on which the network operator needs to provide guidance to the end user. A URI that is only resolvable to the UE which is making the emergency call is not desirable.

NOTE 4: During the dialog, the points of attachment to the IP-CAN of the UE can change (e.g. UE connects to different cells). The UE will populate the P-Access-Network-Info header field in any request (except CANCEL requests) or response (except CANCEL responses) within a dialog with the current point of attachment to the IP-CAN (e.g. the current cell information).

NOTE 5: In this version of the specification, only requests creating a dialog can request emergency services.

If the UE receives a 1xx or 200 (OK) response to an initial request for a dialog the response containing a P-Asserted-Identity header field set to an emergency number as specified in 3GPP TS 22.101 [1A], then the UE may indicate the nature of the session to the user.

5.1.6.11 eCall type of emergency service

5.1.6.11.1 General

If the upper layers request establishment of an IMS emergency call of the manually initiated eCall type of emergency service, the service URN shall be "urn:service:sos.ecall.manual" as specified in RFC 8147 [244].

If the upper layers request establishment of an IMS emergency call of the automatically initiated eCall type of emergency service, the service URN shall be "urn:service:sos.ecall.automatic" as specified in RFC 8147 [244].

NOTE 1: The manually initiated eCall type of emergency service is used when the eCall IMS emergency session is invoked with user input. The automatically initiated eCall type of emergency service is used if the eCall IMS emergency session is invoked without user input.

NOTE 2: To ensure the identification and the related routing (e.g. to a test centre) of non-emergency calls initiated for eCall testing or eCall terminal reconfiguration purposes, "urn:service:sos.ecall.automatic" or "urn:service:sos.ecall.manual" are not used for such calls. Additionally the usage of any other service URN to establish non-emergency calls initiated for eCall testing or eCall terminal reconfiguration purposes needs to be in accordance with local operator policy.

5.1.6.11.2 Initial INVITE request

If the upper layers request establishment of an IMS emergency call of the automatically initiated eCall type of emergency service or of the manually initiated eCall type of emergency service and if allowed by IP-CAN specific annex, the UE shall send an INVITE request as specified in the procedures in subclause 5.1.6.8 with the following additions:

1) the UE shall set the Request-URI to "urn:service:sos.ecall.automatic" or "urn:service:sos.ecall.manual"; and

2) if the IP-CAN indicates the eCall support indication, the UE shall:

   a) insert a multipart/mixed body containing an "application/EmergencyCallData.eCall.MSD" MIME body part as defined in RFC 8147 [244], containing the MSD not exceeding 140 bytes and encoded in binary ASN.1 PER as specified in CEN EN 15722:2015 [245] and include a Content-Disposition header field with a "handling" header field parameter with an "optional" value, as described in RFC 3261 [26];

   b) insert an Accept header field indicating the UE is willing to accept an "application/EmergencyCallData.Control+xml" MIME type as defined in RFC 8147 [244]; and

   c) insert a Recv-Info header field set to "EmergencyCallData.eCall.MSD" as defined in RFC 8147 [244].

NOTE: Further content for the INVITE is as defined in RFC 8147 [244].

Then the UE shall proceed as follows:

1) if the UE receives a 200 (OK) response to the INVITE request not containing:

a) a multipart/mixed body containing an "application/EmergencyCallData.Control+xml" MIME body part as defined in RFC 8147 [244] with an "ack" element containing:

i) a "received" attribute set to "true"; and

ii) a "ref" attribute set to the Content-ID of the MIME body part containing the MSD sent by the UE;

then the UE shall send the MSD using audio media stream encoded as described in 3GPP TS 26.267 [9C];

2) if the UE receives a 200 (OK) response to the INVITE request containing:

a) a multipart/mixed body containing an "application/EmergencyCallData.Control+xml" MIME body part as defined in RFC 8147 [244] with an "ack" element containing:

i) a "received" attribute set to "true"; and

ii) a "ref" attribute set to the Content-ID of the MIME body part containing the MSD sent by the UE;

then the UE shall consider the initial MSD transmission as successful;

3) if the UE receives a 486 (Busy Here), 600 (Busy Everywhere) or 603 (Decline) response to the INVITE request containing:

a) a multipart/mixed body containing an "application/EmergencyCallData.Control+xml" MIME body part as defined in RFC 8147 [244] with an "ack" element containing:

i) a "received" attribute set to "true"; and

ii) a "ref" attribute set to the Content-ID of the MIME body part containing the MSD sent by the UE;

then the UE shall consider the initial MSD transmission as successful and shall perform domain selection to re-attempt the eCall as specified in 3GPP TS 23.167 [4B]; and

4) in all other cases, the UE shall perform domain selection to re-attempt the eCall as specified in 3GPP TS 23.167 [4B].

5.1.6.11.3 Transfer of an updated MSD

During an emergency session established for eCall type of emergency service as described in subclause 5.1.6.11.2, if the UE receives an INFO request with:

1) an Info-Package header field set to "EmergencyCallData.eCall.MSD" as defined in RFC 8147 [244];

2) a multipart/mixed body including:

a) an "application/EmergencyCallData.Control+xml" MIME body part as defined in RFC 8147 [244] containing a "request" element with an "action" attribute set to "send-data" and a "datatype" attribute set to "eCall.MSD"; and

b) a Content-Disposition header field set to "By-Reference" associated with the "application/EmergencyCallData.Control+xml" MIME body part; and

3) a Content-Disposition header field set to "Info-Package" associated with the multipart/mixed body;

the UE shall proceed as follows:

1) if the UE is able to provide an updated MSD, the UE shall send an INFO request containing:

a) an Info-Package header field set to "EmergencyCallData.eCall.MSD" as defined in RFC 8147 [244];

b) a multipart/mixed body including:

i) an "application/EmergencyCallData.eCall.MSD" MIME body part as defined in RFC 8147 [244] containing the MSD not exceeding 140 bytes and encoded in binary ASN.1 as specified in CEN EN 15722:2015 [245]; and

ii) a Content-Disposition header field set to "By-Reference" associated with the "application/EmergencyCallData.eCall.MSD" MIME body part; and

c) a Content-Disposition header field set to "Info-Package" associated with the multipart/mixed body; and

2) if the UE is not able to provide an updated MSD, the UE shall send an INFO request containing:

a) an Info-Package header field set to "EmergencyCallData.eCall.MSD" as defined in RFC 8147 [244];

b) a multipart/mixed body including:

i) an "application/EmergencyCallData.Control+xml" MIME body part as defined in RFC 8147 [244] with an "ack" element containing:

- a "ref" attribute set to the Content-ID of the "application/EmergencyCallData.Control+xml" MIME body part in the INFO request received by the UE; and

- an "actionResult" child element containing:

A) an "action" attribute set to "send-data";

B) a "success" attribute set to "false"; and

C) a "reason" attribute set to an appropriate value as defined in RFC 8147 [244]; and

ii) a Content-Disposition header field set to "By-Reference" associated with the "application/EmergencyCallData.Control+xml" MIME body part; and

c) a Content-Disposition header field set to "Info-Package" associated with the multipart/mixed body.

NOTE: Further content for the INFO request is as defined in RFC 8147 [244].

5.1.6.12 Current location discovery during an emergency call

5.1.6.12.1 General

The UE can be requested to provide the current location information during an established emergency call.

5.1.6.12.2 Current location information requested

If:

1) the UE indicated a Recv-Info header field with the g.3gpp.current-location-discovery info package name in the dialog of the emergency call;

2) the UE indicated an Accept header field indicating the "application/vnd.3gpp.current-location-discovery+xml" MIME type in the dialog of the emergency call; and

3) the dialog of the emergency call is a confirmed dialog;

then upon receiving an INFO request as described in RFC 6086 [25] as in-dialog request of the dialog of the emergency call:

1) with Info-Package header field as described in RFC 6086 [25], containing the g.3gpp.current-location-discovery info package name; and

2) with a request-for-current-location body as specified in subclause 7.12.2.2 in the MIME body of "application/vnd.3gpp.current-location-discovery+xml" MIME type;

the UE shall send a response to the INFO request according to RFC 6086 [25]. If a 200 (OK) response is sent to the INFO request, the UE shall perform the procedure in subclause 5.1.6.12.3.

5.1.6.12.3 Providing current location information

If the UE has its location information available, the UE shall provide the location information.

If the UE does not have its location information available, the UE may attempt to obtain the location information. If the location information becomes available at a subsequent time, the UE shall provide the location information. If the UE does not attempt to obtain its location information or when the UE stops attempting to obtain the location information, the UE shall inform the network about the location information not being available.

In order to provide the location information or to inform the network about the location information not being available, the UE shall apply the procedures as specified in subclauses 5.1.2A with the following additions.

The UE shall send a PUBLISH request as described in RFC 3903 [70] and RFC 3856 [74], as an in-dialog request of the dialog of the emergency call. In the PUBLISH request, the UE shall include an application/pidf+xml MIME body. If the UE has its location information available, the UE shall include the location information in a PIDF location object, in accordance with RFC 4119 [90] and RFC 5491 [267] in the application/pidf+xml MIME body.

NOTE: If the UE does not have its location information available, a PUBLISH request with an application/pidf+xml MIME body not containing a PIDF location object is sent.

5.1.7 Void

5.1.8 Void

5.1.9 P-CSCF addresses management

The UE shall locally store the P-CSCF addresses discovered during the procedures described in subclause 9.2.1.

Based on conditions identified in subclause 5.1.1.2.1 and subclause 5.1.2A.1.6, the UE marks the locally stored P-CSCF addresses as unavailable.

The UE shall not use a locally stored P-CSCF address which is marked as unavailable when performing initial registration.

After:

- a computed retry delay time determined by the algorithm defined in subclause 4.5 of RFC 5626 [92] plus 5 minutes (if no expiration time was identified in subclause 5.1.1.2.1 or subclause 5.1.2A.1.6); or

- a time indicated by the network for this P-CSCF (if expiration time was identified in in subclause 5.1.1.2.1 or subclause 5.1.2A.1.6)

after marking of a locally stored P-CSCF address as unavailable elapses, the UE clears the mark on the locally stored P-CSCF address.

If the UE performs a new P-CSCF discovery or is power-cycled, the UE shall clear all the availability marks on the locally stored P-CSCF addresses.

5.2 Procedures at the P-CSCF

5.2.1 General

Where the P CSCF provides emergency call support, the procedures of subclause 5.2.10 shall be applied first.

Subclause 5.2.2 through subclause 5.2.9 define P-CSCF procedures for SIP that do not relate to emergency. All SIP requests are first screened according to the procedures of subclause 5.2.10 to see if they do relate to an emergency.

For all SIP transactions identified:

- as relating to an emergency; or

- if priority is supported, as containing an authorised Resource-Priority header field or a temporarily authorised Resource-Priority header field, or, if such an option is supported, relating to a dialog which previously contained an authorised Resource-Priority header field;

the P-CSCF shall give priority over other transactions or dialogs. This allows special treatment of such transactions or dialogs. If the P-CSCF recognises the need for priority processing to a request or if the P-CSCF recognises the need to provide different priority processing than the one indicated by the originating UE, based on the information stored during registration, the P-CSCF may insert or modify Resource-Priority header in accordance with RFC 4412 [116].

NOTE 1: The special treatment can include filtering, higher priority processing, routeing, call gapping. The exact meaning of priority is not defined further in this document, but is left to national regulation and network configuration.

The P-CSCF shall support the Path and Service-Route header fields.

NOTE 2: The Path header field is only applicable to the REGISTER request and its 200 (OK) response. The Service-Route header field is only applicable to the 200 (OK) response of REGISTER request.

NOTE 3: In subsequent procedures, the P-CSCF can address the needs of individual users (e.g. in support of attached enterprise networks or in support of priority mechanisms, from information saved during registration. In this release of the specification, no information is specified in the registration procedures to perform this, and therefore this information has to either be associated with the user at time of registration from configured information, or by a mechanism outside the scope of this release of the specification.

When the P-CSCF sends any request or response to the UE, before sending the message the P-CSCF shall:

- remove the P-Charging-Function-Addresses and P-Charging-Vector header fields, if present.

When the P-CSCF receives any request or response from the UE, the P-CSCF:

1) shall remove the P-Charging-Function-Addresses and P-Charging-Vector header fields, if present. Also, the P-CSCF shall ignore any data received in the P-Charging-Function-Addresses and P-Charging-Vector header fields; and

2) may insert previously saved values into the P-Charging-Function-Addresses header field before forwarding the message;

NOTE 4: When the P-CSCF is located in the visited network, then it will not receive the P-Charging-Function-Addresses header field from the S-CSCF, IBCF, or I-CSCF. Instead, the P-CSCF discovers charging function addresses by other means not specified in this document.

3) shall remove the P-Access-Network-Info header field, if the request or the response include a P-Access-Network-Info header field with a "network-provided" parameter;

4) may insert a P-Access-Network-Info header field where:

   a) if no mechanism exists to support the access technology for this UE, the "network-provided" parameter is included, and the access-type field is set to a preconfigured value;

   b) if NASS is used to support the access technology for this UE, the "network-provided" parameter is included, and the access-type field is set:

      - when xDSL is the IP-CAN, to one of "ADSL", "ADSL2", "ADSL2+", "RADSL", "SDSL", "HDSL", "HDSL2", "G.SHDSL", "VDSL", "IDSL", or "xDSL", and the "dsl-location" parameter is set with the value received in the Location-Information header field in the User-Data Answer command as specified in ETSI ES 283 035 [98];

NOTE 5: xDSL is a general abbreviation for all types of Digital Subscriber Lines, and "xDSL" is a possible access-type value of the P-Access-Network-Info header field.

      - when Ethernet is the IP-CAN, to one of "IEEE-802.3", "IEEE-802.3a", "IEEE-802.3e", "IEEE-802.3i", "IEEE-802.3j", "IEEE-802.3u","IEEE-802.3ab"or "IEEE-802.3ae", "IEEE-802.3ak", "IEEE-802.3aq", "IEEE-802.3an", "IEEE-802.3y" or "IEEE-802.3z" and if NASS subsystem is used, and the "eth-location" parameter is set with the value received in the Location-Information header field in the User-Data Answer command as specified in ETSI ES 283 035 [98];

      - when Fiber is the IP-CAN, to one of "G-PON", "XGPON1" or "IEEE-802.3ah" and if NASS subsystem is used, and the "fiber-location" parameter is set with the value received in the Location-Information header field in the User-Data Answer command as specified in ETSI ES 283 035 [98];

c) if the PCRF is used to support the access technology for this UE and 3GPP-User-Location-Info as specified in 3GPP TS 29.214 [13D] is not available:

- if the IP-CAN-Type value provided by the PCRF is not "DVB-RCS2", then:

I) the access-type field or the access-class field is set to a value consistent with that received from the PCRF in the IP-CAN-Type, RAT-Type and AN-Trusted parameters using the procedures specified in 3GPP TS 29.214 [13D].

If the IP-CAN-Type parameter is set "Non-3GPP-EPS (6)" as specified in 3GPP TS 29.212 [13B], the RAT-Type parameter is set to "VIRTUAL (1)" as specified in 3GPP TS 29.212 [13B] and the AN-Trusted parameter is set to "UNTRUSTED (1)" as specified in 3GPP TS 29.273 [12A], the P-CSCF shall include the access-class field set to "untrusted-non-3GPP-VIRTUAL-EPC".

II) if a 3GPP-MS-TimeZone parameter is available from the PCRF, then the "local-time-zone" parameter and the "daylight-saving-time" parameter may also be added using this information;

III) the "network-provided" parameter is added;

IV) if a TWAN-Identifier as specified in 3GPP TS 29.214 [13D] is received from the PCRF, the received TWAN-Identifier contains the Circuit-ID and the associated "Relay Identity", the received TWAN-Identifier does not contain the "Civic Address Information" and the P-CSCF is able to deduce a Geographical Identifier from the Circuit-ID and the associated "Relay Identity", then, if required by local operator policy, the P-CSCF shall include an operator-specific-GI field. The P-CSCF can obtain a Geographical Identifier from the CLF by using the e2 interface (see ETSI ES 283 035 [98]);

NOTE 6: ETSI ES 283 035 [98] Release 3 enables querying a CLF using the User-Data-Request command in which the Global-Access-Id AVP contains the Fixed-Access-ID AVP set using the Circuit-ID value as the Logical-Access-ID and the "Relay Identity" as the Relay-Agent to get a corresponding Geographical Identifier. If multiple CLFs are deployed, the P-CSCF can dertermine which CLF to query based on the CGI or the SAI values or can use a DIAMETER proxy if deployed.

V) if WLAN Location Information as specified in 3GPP TS 23.402 [7E] is received from the PCRF, the received WLAN Location Information contains the location identifier and the P-CSCF is able to deduce a Geographical Identifier from the WLAN Location Information, then, if required by local operator policy, the P-CSCF shall include an operator-specific-GI field; and

VI) if:

A) the access-class field of the P-Access-Network-Info header field is set to "untrusted-non-3GPP-VIRTUAL-EPC"; or

B) the access-class field of the P-Access-Network-Info header field is set to "3GPP-WLAN" and the AN-Trusted parameter specified in 3GPP TS 29.273 [12A] is received from PCRF and is set to "UNTRUSTED (1)";

then:

A) if a UE-Local-IP-Address parameter specified in 3GPP TS 29.212 [13B] is received from the PCRF and if required by local operator policy, P-CSCF shall also include in the P-Access-Network-Info header field a UE-local-IP-address parameter set to the UE local IP address in the UE-Local-IP-Address parameter received from PCRF;

B) if a UDP-Source-Port parameter specified in 3GPP TS 29.212 [13B] is received from the PCRF and if required by local operator policy, the P-CSCF shall also include in the P-Access-Network-Info header field a UDP-source-port parameter set to the UDP port in the UDP-Source-Port parameter received from PCRF;

C) if a TCP-Source-Port parameter specified in 3GPP TS 29.212 [13B] is received from the PCRF and if required by local operator policy, the P-CSCF shall also include in the P-Access-Network-Info header field a TCP-source-port parameter set to the TCP port in the TCP-Source-Port parameter received from PCRF; and

D) if an AN-GW-Address parameter specified in 3GPP TS 29.212 [13B] is received from the PCRF and if required by local operator policy, the P-CSCF shall also include in the P-Access-Network-

Info header field an ePDG-IP-address parameter set to the ePDG IP address in the ePDG-IP-Address parameter received from PCRF; and

VII) if the P-CSCF supports reporting EPS fallback, then the "eps-fallback" parameter is included, if the information is available to the P-CSCF; and

- if the IP-CAN-Type value provided by the PCRF is "DVB-RCS2", then the "network-provided" parameter is included, the access-type field is set to "DVB-RCS2", and the "dvb-rcs2-node-id" parameter is set with the value provided by the IP-CAN provider;

d) if the PCRF is used to support the access technology for this UE and 3GPP-User-Location-Info as specified in 3GPP TS 29.214 [13D] is available;

I) the access-type field or the access-class field is set to a value consistent with that received from the PCRF in the IP-CAN-Type and RAT-Type parameters;

II) the access-info field is set to a value consistent with the information received from the PCRF in the 3GPP-User-Location-Info parameter;

III) if a 3GPP-MS-TimeZone parameter is available from the PCRF, then the "local-time-zone" parameter and the "daylight-saving-time" parameter may also be added using this information;

IV) the "network-provided" parameter is added;

V) if required by local operator policy and the P-CSCF is able to deduce a Geographical Identifier from the Cell Global Identity (CGI) or form the Service Area Identifier (SAI) received from the PCRF, the P-CSCF shall include an operator-specific-GI field. The P-CSCF can obtain a Geographical Identifier from the CLF by using the e2 interface (see ETSI ES 283 035 [98]); and

NOTE 7: ETSI ES 283 035 [98] Release 3 enables querying a CLF using the User-Data-Request command in which the Global-Access-Id AVP contains the 3GPP-User-Location-Info AVP with a CGI or a SAI value to get a corresponding Geographical Identifier. If multiple CLFs are deployed, the P-CSCF can determine which CLF to query based on the CGI or the SAI values or can use a DIAMETER proxy if deployed.

VI) if the P-CSCF supports reporting EPS fallback then the "eps-fallback" parameter is included, if the information is available to the P-CSCF; and

e) if DOCSIS is used, and proprietary means of obtaining a location are used, the access-type field is set to "DOCSIS" and the "network-provided" parameter is added; and

f) if none of NASS, PCRF and DOCSIS are used to support the access technology for the UE and the IP-CAN is not provided by the packet switched domain of the PLMN of the P-CSCF:

I) if the P-CSCF is unaware of the radio access technology used by the UE, the access-class field is set to "VIRTUAL-no-PS";

II) if the P-CSCF is aware that the radio access technology used by the UE is specified by IEEE Std 802.11 [248], the access-class field is set to "WLAN-no-PS"; and

III) the "network-provided" parameter is added;

5) shall remove all Feature-Caps header fields, if present, from a UE that is not considered as privileged sender;

6) may insert a P-Visited-Network-ID header field (except ACK, BYE, CANCEL, NOTIFY, PRACK, INFO and UPDATE) according to RFC 7976 [52A] with the value:

I) of a pre-provisioned string that identifies the network of the P-CSCF at the home network; or

II) if the UE is roaming in deployments without IMS-level roaming interfaces according to 3GPP TS 23.228 [7], a string that identifies the visited network of the UE including an indication that the P-CSCF is located in the home network;

7) may insert a P-Visited-Network-ID header field in 200 (OK) response to INVITE request and in 200 (OK) response to MESSAGE request according to draft-jesske-update-p-visited-network [52B]; and

8) if a Geolocation header field is received from the UE, shall remove any present loc-src parameter from the Geolocation header field.

When the P-CSCF receives any request or response containing the P-Media-Authorization header field, the P-CSCF shall remove the header field.

NOTE 8: Depending on the security mechanism in use, the P-CSCF can integrity protect all SIP messages sent to the UE outside of the registration and authentication procedures by using a security association or TLS session. The P-CSCF will discard any SIP message that is not protected by using a security association or TLS session and is received outside of the registration and authentication procedures. The integrity and confidentiality protection and checking requirements on the P-CSCF within the registration and authentication procedures are defined in subclause 5.2.2.

With the exception of 305 (Use Proxy) responses, the P-CSCF shall not recurse on 3xx responses.

NOTE 9: If the P-CSCF is connected to a PDF the requirements for this interconnection is specified in the Release 6 version of this specification.

The P-CSCF may add, remove, or modify, the P-Early-Media header field within forwarded SIP requests and responses according to procedures in RFC 5009 [109].

NOTE 10: The P-CSCF can use the P-Early-Media header field for the gate control procedures, as described in 3GPP TS 29.214 [13D]. In the presence of early media for multiple dialogs due to forking, if the P-CSCF is able to identify the media associated with a dialog, (i.e., if symmetric RTP is used by the UE and the P-CSCF can use the remote SDP information to determine the source of the media) the P-CSCF can selectively open the gate corresponding to an authorized early media flow for the selected media.

When SIP digest without TLS is used, the P-CSCF shall discard any SIP messages received outside of the registration and authentication procedures that do not map to an existing IP association as defined in subclause 5.2.3.

In case a device performing address and/or port number conversions is provided by a NA(P)T or NA(P)T-PT controlled by the P-CSCF, the P-CSCF may need to modify the SIP contents according to the procedures described in annex F. In case a device performing address and/or port number conversions is provided by a NA(P)T or NA(P)T-PT not controlled by the P-CSCF, the P-CSCF may need to modify the SIP contents according to the procedures described in annex K if both a "reg-id" and "+sip.instance" header field parameters are present in the received Contact header field as described in RFC 5626 [92].

The P-CSCF shall support the provision of the user-related policies (e.g. consideration of the user as a privileged sender):

- from the S-CSCF during registration; and
- by local configuration.

For the same policy, the precedence between the locally configured policy and a policy received during registration shall be based on local operator policy.

For UE performing the functions of an external attached networks using static mode of operation, the P-CSCF will receive requests to establish a TLS session that are not accompanied by the associated procedures of subclause 5.2.2. The P-CSCF shall permit the establishment of such TLS sessions, but subsequent operations without the reception of a REGISTER request shall only be permitted if the P-CSCF is configured for such a UE performing the functions of an external attached network using static mode of operation. Where a REGISTER request is received from a UE, the P-CSCF shall process the REGISTER request as defined in subclause 5.2.2, and shall not provide special procedures for a UE performing the functions of an external attached network using static mode of operation for the duration of the registration.

NOTE 11: For requests other than REGISTER received from UEs that are not configured in this manner, then the procedures of subclause 5.2.6.3.2A apply.

NOTE 12: The P-CSCF does not subscribe to the reg event package for a UE performing the functions of an external attached network using static mode of operation.

When sending a failure response to any received request, depending on operator policy, the P-CSCF may insert a Response-Source header field with an "fe" header field parameter constructed with the URN namespace "urn:3gpp:fe", the fe-id part of the URN set to "p-cscf" and optionally an appropriate fe-param part of the URN set in accordance with

subclause 7.2.17. A P-CSCF when sending a failure response will add in the URN the "side" header field parameter set to:

- "orig" for a UE-originating case; and
- "term" for a UE-terminating case.

5.2.2 Registration

5.2.2.1 General

The P-CSCF shall be prepared to receive the unprotected REGISTER requests on the SIP default port values as specified in RFC 3261 [26]. The P-CSCF shall also be prepared to receive the unprotected REGISTER requests on the port advertised to the UE during the P-CSCF discovery procedure.

NOTE 1: The P-CSCF will only accept further registration and subsequent SIP messages on the same ports for security mechanisms that do not require to use negotiated ports for exchanging protected messages.

The P-CSCF shall distinguish between security mechanisms through the use of the Security-Client header field and Authorization header field as follows:

1) if a REGISTER request from the UE contains a Security-Client header field and the Require and Proxy-Require header fields contain "sec-agree", then for an initial registration, the P-CSCF shall select the sec-mechanism and mode (as described in Annex H of 3GPP TS 33.203 [19]) from the corresponding parameters offered in the Security-Client header field according to its priorities, as follows:

   - if the P-CSCF selects the sec-mechanism "ipsec- 3gpp" then follow the procedures as described in subclause 5.2.2.2, in addition to the procedures described in this subclause;
   - if the P-CSCF selects the sec-mechanism "tls" then follow the procedures as described in subclause 5.2.2.4, in addition to the procedures described in this subclause.

NOTE 2: If the Security-Client header field contains only media plane security mechanisms then Require and Proxy-Require header fields will not contain "sec-agree". The P-CSCF will then continue as per the procedure in bullet 2), not select a signalling plane security mechanism and then distinguish signalling plane security based upon the Authorization header field as described in the steps below.

2) if:

   a) a REGISTER request from the UE does not contain a Security-Client header field;

   b) a REGISTER request from the UE contains a Security-Client header field containing only media plane security mechanisms and the Require and Proxy-Require header fields do not contain "sec-agree"; or

   c) the P-CSCF does not select any signalling plane security mechanism from the Security-Client header field;

   then the P-CSCF shall behave as follows, in addition to the procedures described in the remainder of this subclause:

   - if the REGISTER request does not contain an Authorization header field and was received over an access network defined in 3GPP specifications then follow the GPRS-IMS-Bundled authentication procedures as described in subclause 5.2.2.6; or
   - if the REGISTER request does not contain an Authorization header field and was received over a TISPAN NASS and the P-CSCF supports both SIP digest and NASS-IMS bundled authentication, then the P-CSCF shall perform the steps required for NASS-IMS bundled authentication, in subclause 5.2.2.5, as well as the steps required for SIP digest without TLS, in subclause 5.2.2.3, unless it is configured to behave differently or the P-CSCF only supports either SIP digest without TLS or NASS-IMS bundled authentication. If the NASS-IMS bundled authentication related query from the P-CSCF to the TISPAN NASS fails, then the P-CSCF shall only continue with the SIP digest related steps; or
   - if the REGISTER request does not contain an Authorization header field, and was received over an access other than defined in 3GPP specifications or TISPAN NASS, then follow the SIP digest without TLS procedures described in subclause 5.2.2.3; or

NOTE 3: How the P-CSCF recognizes over which access network a request was received is an implementation specific feature.

- if the REGISTER request contains an Authorization header field with an "algorithm" header field parameter set to "AKAv2-SHA-256" and the REGISTER request was received by eP-CSCF over TLS, then follow the IMS-AKA procedures for eP-CSCF defined in 3GPP TS 24.371 [8Z]; or

- if the REGISTER request contains an Authorization header field without an "algorithm" header field parameter set to "AKAv2-SHA-256" and was not received over a TISPAN NASS then follow the SIP digest without TLS procedures as described in subclause 5.2.2.3; or

- if the REGISTER request contains an Authorization header field and was received over a TISPAN NASS, and the P-CSCF supports both SIP digest and NASS-IMS bundled authentication, then the P-CSCF shall perform the steps required for NASS-IMS bundled authentication, in subclause 5.2.2.5, as well as the steps required for SIP digest without TLS, in subclause 5.2.2.3, unless it is configured to behave differently. If the NASS-IMS bundled authentication related query from the P-CSCF to the TISPAN NASS fails, then the P-CSCF shall only continue with the SIP digest related steps.

For subsequent registrations, the P-CSCF shall continue to use the selected mechanism.

NOTE 4: The steps required for SIP digest and for NASS-IMS bundled authentcation are not in contradiction. Rather, for NASS-IMS bundled authentication the P-CSCF needs to perform additional steps, namely an exchange with the TISPAN NASS and an inclusion of NASS location information in the REGISTER request, on top of the steps required for SIP digest.

NOTE 5: How the P-CSCF knows the access network type of a specific network interface is implementation-dependent (e.g. it can know the access network type from different UE IP address ranges or by using different network interfaces for different access network types).

When the P-CSCF receives a REGISTER request from the UE, the P-CSCF shall:

1) insert a Path header field in the request including an entry containing:

   - the SIP URI identifying the P-CSCF;

   - an indication that requests routed in this direction of the path (i.e. from the S-CSCF towards the P-CSCF) are expected to be treated as for the UE-terminating case;

   NOTE 6: This indication can e.g. be in a parameter in the URI, a character string in the user part of the URI or be a port number in the URI.

   - an IMS flow token in the user portion of the P-CSCF's SIP URI inserted into the Path header field, and the "ob" SIP URI parameter according to RFC 5626 [92]. The same SIP URI (user portion, hostport parameter and SIP URI parameters) shall be used for the initial registration, and the re-registrations, binding fetchings, and de-registration that refreshes of the respective registration;

   - the P-CSCF shall use a different IMS flow token for each registration. If the multiple registration mechanism is used, the P-CSCF shall also use a different IMS flow token for each registration flow associated with the registration;

   NOTE 7: The form of the IMS flow token is of local significance to the P-CSCF only and can thus be chosen freely by a P-CSCF implementation.

   NOTE 8: By inserting the "ob" SIP URI parameter in its SIP URI, the P-CSCF indicates that it supports multiple registrations as specified in RFC 5626 [92]. The presence of the "ob" SIP URI parameter is not an indication that the P-CSCF supports the keep-alive mechanism. When the P-CSCF detects that the UE is behind a NAT and the P-CSCF supports a keep-alive mechanism defined in RFC 5626 [92].

   - if

     a) the P-CSCF supports indicating the traffic leg associated with a URI as specified in RFC 7549 [225];

     b) the UE is roaming;

     c) the P-CSCF is not in the home network; and

d) required by local policy;

then the P-CSCF may append an "iotl" SIP URI parameter with a value set to "homeB-visitedB" to the SIP URI of the Path header field;

2) insert a Require header field containing the option-tag "path";

3) insert a P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17] and a type 1 "orig-ioi" header field parameter. The P-CSCF shall set the type 1 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The P-CSCF shall not include the type 1 "term-ioi" header field parameter;

4) insert a P-Visited-Network-ID header field, with the value:

- of a pre-provisioned string that identifies the network of the P-CSCF at the home network; or
- if the UE is roaming in deployments without IMS-level roaming interfaces according to 3GPP°TS°23.228°[7], a string that identifies the visited network of the UE including an indication that the P-CSCF is located in the home network.

EXAMPLE: A UE is roaming using a deployment without an IMS-level roaming interface and the P-CSCF receives via Rx interface a MCC with the value "111" and a MNC with the value "22" identifying the visited network. The domain name of the home network where the P-CSCF is located has the value "networkoperator". In this case, the P-CSCF can set up the P-Visited Network-ID header with a string which can look like: "s8hr.mnc22.mcc111.networkoperator".

NOTE 9: The information of the visited network of the UE is taken from Rx interface as defined in 3GPP°TS°29.214°[13D].

NOTE 10: If required, the P-CSCF can determine that a UE is roaming using deployment without IMS-level roaming interfaces, if the via Rx interface received network information of the roaming UE points to a different network as the P-CSCF belongs to.

4A) store the announcement of the media plane security mechanisms the UE supports labelled with the "mediasec" header field parameter specified in subclause 7.2A.7 and received in the Security-Client header field, if any. Also, if the Security-Client header field contains only media plane security mechanisms, remove the header field;

NOTE 11: The "mediasec" header field parameter indicates that security mechanisms are specific to the media plane.

4B) if the REGISTER request contains an Authorization header field, remove the "integrity-protected" header field parameter, if present;

4C) if the host portion of the sent-by field in the topmost Via header field contains a FQDN, or if it contains an IP address that differs from the source address of the IP packet, the P-CSCF shall add a "received" Via header field parameter in accordance with the procedure defined in RFC 3261 [26];

5) if the P-CSCF is located in the visited network, and local policy requires the application of IBCF capabilities in the visited network towards the home network:

a) if the request is not to be forwarded to an ATCF according to local policy select an exit point in visited network;

NOTE 12: The list of the exit points can be either obtained as specified in RFC 3263 [27A] or provisioned in the P-CSCF.

b) if the request is to be forwarded to an ATCF according to local policy:

i) insert a Route header field with the ATCF URI for originating requests; and

ii) forward the request; and

c) if the request is not to be forwarded to an ATCF according to local policy, then forward the request to the selected exit point.

If:

- no response is received to the REGISTER request and its retransmissions by the P-CSCF; or
- a 3xx response or 480 (Temporarily Unavailable) response to a REGISTER request is received;

the P-CSCF shall repeat the actions of this bullet with a different exit point or a different ATCF.

If the P-CSCF fails to forward the REGISTER request to any exit point or any ATCF, the P-CSCF shall send back a 504 (Server Time-Out) response to the user, in accordance with the procedures in RFC 3261 [26] unless local policy allows omitting the exit point;

NOTE 13: If the P-CSCF forwards the request to an IBCF in the visited network, the IBCF in the visited network can determine the entry point of the home network, as specified in RFC 3263 [27A] or the entry point of the home network can be provisioned in the IBCF in the visited network.

6) if the P-CSCF is located in the visited network and local policy does not require the application of IBCF capabilities in the visited network towards the home network:

   a) if the request is not to be forwarded to an ATCF according to local policy select an entry point of the home network;

NOTE 14: The list of the entry points can be either obtained as specified in RFC 3263 [27A] or provisioned in the P-CSCF.

   b) if the request is to be forwarded to an ATCF according to local policy:

      i) insert a Route header field with the ATCF URI for originating requests; and

      ii) forward the request; and

   c) if the request is not to be forwarded to an ATCF according to local policy, then forward the request to the selected entry point.

   If:

   - no response is received to the REGISTER request and its retransmissions by the P-CSCF; or
   - a 3xx response or 480 (Temporarily Unavailable) response to a REGISTER request is received;

   the P-CSCF shall repeat the actions of this bullet with a different entry point or a different ATCF.

   If the P-CSCF fails to forward the REGISTER request to any entry point or any ATCF, the P-CSCF shall send back a 504 (Server Time-Out) response to the user, in accordance with the procedures in RFC 3261 [26];

7) if the P-CSCF is located in the home network:

   a) if the request is not to be forwarded to an ATCF according to local policy select the I-CSCF of the home network;

NOTE 15: The list of the I-CSCFs can be either obtained as specified in RFC 3263 [27A] or provisioned in the P-CSCF.

   b) if the request is to be forwarded to an ATCF according to local policy:

      i) insert a Route header field with the ATCF URI for originating requests; and

      ii) forward the request; and

   c) if the request is not to be forwarded to an ATCF according to local policy, then forward the request to the selected I-CSCF.

   If:

   - no response is received to the REGISTER request and its retransmissions by the P-CSCF; or
   - a 3xx response or 480 (Temporarily Unavailable) response to a REGISTER request is received;

   the P-CSCF shall repeat the actions of this bullet with a different I-CSCF or a different ATCF.

If the P-CSCF fails to forward the REGISTER request to any I-CSCF or any ATCF, the P-CSCF shall send back a 504 (Server Time-Out) response to the user, in accordance with the procedures in RFC 3261 [26]; and

8) void.

When the P-CSCF receives a 200 (OK) response to a REGISTER request, the P-CSCF shall check the value of the registration expiration interval value. When the registration expiration interval value is different than zero, then the P-CSCF shall:

1) save the list of service route values in the Service-Route header fields preserving the order, and bind the list either to the contact address or to the registration flow and the associated contact address (if the multiple registration mechanism is used) and the associated security association or TLS session over which the REGISTER request was received. The P-CSCF shall store this list during the entire registration period of the respective public user identity and bind it either to the associated contact address or to the registration flow and the associated contact address (if the multiple registration mechanism is used). The P-CSCF shall use this list to validate the routeing information in the requests originated by the UE using either the respective contact address or the registration flow and the associated contact address, and received over the respective security association or a TLS session. If the list of Service-Route header fields already exists either for this contact address or the registration flow and the associated contact address (if the multiple registration mechanism is used), then the P-CSCF shall replace the already existing list of service route values with the list of Service-Route header fields received in the 200 (OK) response;

NOTE 16: When the UE registers multiple registration flows and the associated contact addresses, then the UE and the P-CSCF will have a list of Service-Route header fields for each registration flow and the associated contact address and the associated security association or TLS session. When sending a request using a given registration flow and the associated contact address and the associated security association or TLS session, the UE will use the corresponding list of Service-Route header fields, when building a list of Route header fields.

2) associate the list of service route values with the registered public user identity and either the associated contact address or the registration flow and the associated contact address (if the multiple registration mechanism is used) and the associated security association or TLS session;

3) store the public user identities, found in the P-Associated-URI header field value, including any associated display names, and any parameters associated with either the user or the identities of the user, and associate them to the registered public user identity, i.e. the registered public user identity and its associated set of implicitly registered public user identities are bound to the contact address and security association or TLS session over which the REGISTER request was received;

3A) if the user-related policies statically provisioned to the P-CSCF (see subclause 5.2.1) indicate that the URIs contained in the P-Associated-URI header field shall not be forwarded towards the UE, and the P-CSCF is located in the home operator network of the UE, then the P-CSCF shall remove all but the first URI contained in the P-Associated-URI header field of the 200 (OK) response;

NOTE 17: The URIs in the P-Associated-URI header field might need to be removed in case of the UE performs the functions of an external attached network (e.g an enterprise network).

4) store the default public user identity, including its associated display name, if provided, for use with procedures for the P-Asserted-Identity header field for requests received from the UE over the respective security association or TLS session. The default public user identity is the first on the list of URIs present in the P-Associated-URI header field;

NOTE 18: There can be more than one default public user identity stored in the P-CSCF, as the result of the multiple registrations of public user identities.

NOTE 19: For each contact address and the associated security association or TLS session the P-CSCF will maintain a list of registered public user identities and the associated default public user identities, that it will use when populating the P-Asserted Identity header.

5) store the values received in the P-Charging-Function-Addresses header field;

6) if a "term-ioi" header field parameter is received in the P-Charging-Vector header field, store the value of the received "term-ioi" header field parameter;

NOTE 20: Any received "term-ioi" header field parameter will contain a type 1 IOI. The type 1 IOI identifies the home network of the registered user.

7) if the P-CSCF included an IMS flow token and the "ob" SIP URI parameter in the Path header field of the REGISTER request, check for presence of the option-tag "outbound" in the Require header field of the a 200 (OK) response:

   - if the option-tag "outbound" is present, it indicates that the UE has successfully registered its public user identity with a new bidirectional flow as defined in RFC 5626 [92]. In this case the P-CSCF shall route the subsequent requests and responses destined for the UE as specified in RFC 5626 [92]; or

   - if the option-tag "outbound" is not present, it indicates that the public user identity has not been registered as specified in RFC 5626 [92]. In this case the P-CSCF shall route the subsequent requests and responses destined for the UE as specified in RFC 3261 [26];

8) if the P-CSCF detects that the UE is behind a NAT, and the UE's Via header field contains a "keep" header field parameter, the P-CSCF shall add a value to the parameter, to indicate that it is willing to receive keep-alives associated with the registration from the UE, as defined in RFC 6223 [143];

9) void; and

10) if the P-CSCF is located in the visited network, store the value of a "+g.3gpp.thig-path" Feature-Caps header field parameter, defined in subclause 7.9A.9, if included in the response. The P-CSCF shall remove the "+g.3gpp.thig-path" Feature-Caps header field parameter before forwarding the 200 (OK) response to the UE.

If the P-CSCF detects that the UE is behind a NAT, and the request was received over a TCP connection, the P-CSCF shall not close the TCP connection during the duration of the registration.

NOTE 21: The P-CSCF can conclude whether the UE is behind a NAT or not by comparing the IP address in the "received" header field parameter with the IP address in the sent-by parameter in the topmost Via header field. If the values do not match, the P-CSCF can conclude that the UE is behind a NAT.

5.2.2.2 IMS AKA as a security mechanism

When the P-CSCF receives a REGISTER request from the UE, as defined in subclause 5.2.2.1, the P-CSCF shall additionally:

1) insert the "integrity-protected" header field parameter (described in subclause 7.2A.2) with a value "yes" into the Authorization header field in case the REGISTER request was either received protected with the security association created during an ongoing authentication procedure and includes an authentication challenge response (i.e. RES parameter), or it was received on the security association created during the last successful authentication procedure, otherwise insert the parameter with the value "no";

1A) if the "reg-id" header field parameter was included in the Contact header field of the REGISTER request, insert in the Path header an IMS flow token and the "ob" URI parameter according to RFC 5626 [92]. The IMS flow token shall identify the flow from the P-CSCF toward the UE, as follows:

   a) for UDP, the IMS flow token identifies the unidirectional flow from the P-CSCF's protected client port and the P-CSCF's IP address to the UE's protected server port and the UE's IP address. This flow is used by the P-CSCF to send requests and responses to the UE. The P-CSCF shall receive the requests and responses from the UE on its protected server port; or

   b) for TCP, the IMS flow token identifies the existing TCP connection between the UE and the P-CSCF. This TCP connection is established by the P-CSCF, i.e. from the P-CSCF's protected client port and the P-CSCF's IP address to the UE's protected server port and the UE's IP address. This TCP connection is used to send SIP requests from the P-CSCF to the UE and receive SIP responses from the UE to the P-CSCF;

2) in case the REGISTER request was received without protection, on the default port or port advertised to UE for P-CSCF discovery:

   a) check the existence of the Security-Client header field. If the Security-Client header field is present, then remove and store it. If the Security-Client header field is not present, then the P-CSCF shall return a suitable 4xx response;

b) if the "rport" header field parameter is included in the Via header field, set the value of the "rport" header field parameter in the Via header field to the source port of the received REGISTER request;

c) insert the "received" header field parameter in the Via header field containing the source IP address that the request came from, as defined in RFC 3581 [56A]; and

NOTE 1: As defined in RFC 3581 [56A], the P-CSCF will insert a "received" header field parameter containing the source IP address that the request came from, even if it is identical to the value of the "sent-by" component.

NOTE 2: Upon receiving the unprotected REGISTER request the P-CSCF detects if the UE is behind a NAT.

3) in case the REGISTER request was received protected, then towards the port that was notified to the UE in the previous response:

a) check the security association which protected the request. If the security association is a temporary one, then the request is expected to contain a Security-Verify header field in addition to a Security-Client header field. If there are no such header fields, then the P-CSCF shall return a suitable 4xx response. If there are such header fields, then the P-CSCF shall compare the content of the Security-Verify header field with the content of the Security-Server header field sent earlier and the content of the Security-Client header field with the content of the Security-Client header field received in the challenged REGISTER request. If those do not match, then there is a potential man-in-the-middle attack. The request should be rejected by sending a suitable 4xx response. If the contents match, the P-CSCF shall remove the Security-Verify and the Security-Client header field;

b) if the security association the REGISTER request was received on, is an already established one, then:

- the P-CSCF shall remove the Security-Verify header field if it is present;

- a Security-Client header field containing new parameter values is expected. If the Security-Client header field or any required parameter is missing, then the P-CSCF shall return a suitable 4xx response; and

- the P-CSCF shall remove and store the Security-Client header field before forwarding the request to the S-CSCF;

c) check if the private user identity conveyed in the Authorization header field of the protected REGISTER request is the same as the private user identity which was previously challenged or authenticated. If the private user identities are different, the P-CSCF shall reject the REGISTER request by returning a 403 (Forbidden) response; and

d) ignore the "rport" Via header field parameter, if included.

NOTE 3: Once the IPsec security associations between the UE and the P-CSCF have been created, in case of UDP the P-CSCF sends the responses to a different UE's port then the one from which the request was received from the UE. For the TCP, the responses are sent on the TCP connection on which the request was received. Hence, the P-CSCF will ignore the "rport" Via header field parameter in all protected requests and responses, if received.

When the P-CSCF receives a 401 (Unauthorized) response to a REGISTER request, the P-CSCF shall:

1) delete any temporary set of security associations established towards the UE;

2) remove the "ck" and "ik" WWW-Authenticate header field parameters contained in the 401 (Unauthorized) response and bind the values to the proper private user identity and to the temporary set of security associations which will be setup as a result of this challenge. The P-CSCF shall forward the 401 (Unauthorized) response to the UE if and only if the "ck" and "ik" header field parameters have been removed;

3) insert a Security-Server header field in the response, containing the P-CSCF static signalling plane security list and the parameters needed for this security association setup, as specified in Annex H of 3GPP TS 33.203 [19]. The P-CSCF shall support the "ipsec-3gpp" security mechanism, as specified in RFC 3329 [48]. The P-CSCF shall support the IPsec layer algorithms for integrity and confidentiality protection as defined in 3GPP TS 33.203 [19] and shall announce support for them according to the procedures defined in RFC 3329 [48];

3A) insert a Security-Server header field to specify the media plane security mechanisms the P-CSCF (IMS-ALG) supports, if any, labelled with the "mediasec" header field parameter specified in subclause 7.2A.7;

NOTE 4 The "mediasec" header field parameter indicates that security mechanisms are specific to the media plane.

4) set up the temporary set of security associations for this registration with a temporary SIP level lifetime between the UE and the P-CSCF for the user identified with the private user identity. For further details see 3GPP TS 33.203 [19] and RFC 3329 [48]. The P-CSCF shall set the temporary SIP level lifetime for the temporary set of security associations to the value of reg-await-auth timer; and

5) send the 401 (Unauthorized) response to the UE using the security association with which the associated REGISTER request was protected, or unprotected in case the REGISTER request was received unprotected. If the 401 (Unauthorized) response to the unprotected REGISTER request is sent using UDP, the P-CSCF shall send the response to the IP address listed in the "received" Via header field parameter and the port in the "rport" Via header field parameter. In case of TCP, the P-CSCF shall send the response over the same TCP connection over which the request was received from the UE.

NOTE 5: The challenge in the 401 (Unauthorized) response sent back by the S-CSCF to the UE as a response to the REGISTER request is piggybacked by the P-CSCF to insert the Security-Server header field in it. The S-CSCF authenticates the UE, while the P-CSCF negotiates and sets up two pairs of security associations with the UE during the same registration procedure. For further details see 3GPP TS 33.203 [19].

When the P-CSCF receives a 200 (OK) response to a REGISTER request as defined in subclause 5.2.2.1, the P-CSCF shall additionally:

1) if an existing set of security association is available, set the SIP level lifetime of the security association to the longest of either the previously existing security association lifetime, or the lifetime of the just completed registration plus 30 seconds;

2) if a temporary set of security associations exists, change the temporary set of security associations to a newly established set of security associations, i.e. set its SIP level lifetime to the longest of either the previously existing set of security associations SIP level lifetime, or the lifetime of the just completed registration plus 30 seconds; and

3) protect the 200 (OK) response to the REGISTER request within the same security association to that in which the REGISTER request was protected.

If the P-CSCF receives a SIP message (including REGISTER requests) from the UE over the newly established set of security associations that have not yet been taken into use, the P-CSCF shall:

1) reduce the SIP level lifetime of the old set of security associations towards the same UE to 64*T1 (if currently longer than 64*T1); and

2) use the newly established set of security associations for further messages sent towards the UE as appropriate (i.e. take the newly established set of security associations into use).

NOTE 6: If the UE has registered other contact addresses and established security associations for these contact addresses, it can use them when sending subsequent SIP messages rather than using the newly established set of security associations. In this case the P-CSCF will not receive any SIP message over the newly established set of security associations.

NOTE 7: In this case, the P-CSCF will send requests (that specify the associated contact address in the Request-URI) towards the UE over the newly established set of security associations. Responses towards the UE that are sent via UDP will be sent over the newly established set of security associations. Responses towards the UE that are sent via TCP will be sent over the same set of security associations that the related request was received on.

NOTE 8: When receiving a SIP message (including REGISTER requests) from the UE over a set of security associations that is different from the newly established set of security associations, the P-CSCF will not take any action on any set of security associations.

When the SIP level lifetime of an old set of security associations is about to expire, i.e. their SIP level lifetime is shorter than 64*T1 and a newly established set of security associations has not been taken into use, the P-CSCF shall use the newly established set of security associations for further messages towards the UE as appropriate (see NOTE 2).

When sending the 200 (OK) response for a REGISTER request that concludes a re-authentication, the P-CSCF shall:

1) keep the set of security associations that was used for the REGISTER request that initiated the re-authentication;

2) keep the newly established set of security associations created during this authentication; and

3) go on using for further requests sent towards the UE the set of security associations and associated contact address that was used to protect the REGISTER request that initiated the re-authentication as appropriate (see NOTE 6).

When sending the 200 (OK) respone for a REGISTER request that concludes an initial authentication of the user registering its public user identity with a given contact address the associated security association, i.e. the REGISTER request that initiated the authentication was received unprotected, the P-CSCF shall:

1) keep the newly established set of security associations created during this authentication; and

2) use the kept newly established set of security associations and associated contact address for further messages sent towards the UE as appropriate (see NOTE 6).

NOTE 9: For each contact address or for each registration flow and the associated contact address and bound to a set of security associations the P-CSCF will maintain two Route header field lists. The first Route header field list (constructed form the Service-Route header fields received during the last registration procedure of either the respective contact address or a registration flow and the associated contact address) is used only to validate the routeing information in the initial requests for a dialog and stand alone transactions originating from the UE using either the respective contact address or a registration flow and the associated contact address and are the respective security association. This list is valid as long as there is at least one public user identity registered either with the associated contact address or a registration flow and the associated contact address. The second list is the list of Route header fields (constructed from the Record Route header fields in the initial INVITE request and associated response) is used during the duration of the call. Once the call is terminated, this list of Route header fields is discarded.

The P-CSCF shall delete any security association from the IPsec database when their SIP level lifetime expires.

The handling of the security associations at the P-CSCF is summarized in table 5.2.2-1.

Table 5.2.2-1: Handling of security associations at the P-CSCF

	Temporary set of security associations	Newly established set of security associations	Old set of security associations
SIP message received over newly established set of security associations that have not yet been taken into use	No action	Take into use	Reduce SIP level lifetime to 64*T1, if lifetime is larger than 64*T1
SIP message received over old set of security associations	No action	No action	No action
Old set of security associations currently in use will expire in 64*T1	No action	Take into use	No action
Sending an authorization challenge within a 401 (Unauthorized) response for a REGISTER request	Create Remove any previously existing temporary set of security associations	No action	No action
Sending 200 (OK) response for REGISTER request that concludes re-authentication	Change to a newly established set of security associations	Convert to and treat as old set of security associations (see next column)	Continue using the old set of security associations over which the REGISTER request, that initiated the re-authentication was received. Delete all other old sets of security associations immediately
Sending 200 (OK) response for REGISTER request that concludes initial authentication	Change to a newly established set of security associations and take into use immediately	Convert to old set of security associations, i.e. delete	Delete

5.2.2.3 SIP digest without TLS as a security mechanism

When the P-CSCF receives a REGISTER request from the UE, as defined in subclause 5.2.2.1, the P-CSCF shall additionally:

1) if the REGISTER request includes an Authorization header field update the "integrity-protected" header field parameter as follows:

   a) if the REGISTER request does not map to an existing IP association, and does not contain a challenge response, not include the "integrity-protected" header field parameter; or

   b) if the REGISTER request does not map to an existing IP association, and does contain a challenge response, include an "integrity-protected" header field parameter with the value set to "ip-assoc-pending"; or

   c) if the REGISTER request does map to an existing IP association, include an "integrity-protected" header field parameter with the value set to "ip-assoc-yes";

NOTE 1: The value of "ip-assoc-pending" for the "integrity-protected" header field parameter or the absence of an "integrity-protected" header field parameter in the Authorization header field is an indication to the I-CSCF and S-CSCF that this is an initial REGISTER request.

2) if the P-CSCF adds a "received" header field parameter and UDP is being used, also add an "rport" Via header field parameter with the IP source port of the received REGISTER request; and

3) if the REGISTER request does not contain an Authorization header field and the requests was received over a non 3GPP access network, insert a P-Access-Network-Info header field as described in subclause 5.2.1 step 4).

NOTE 2: How the P-CSCF recognizes over which access network a request was received is an implementation specific feature.

NOTE 3: Subclause 5.2.1 describes the encoding of the P-Access-Network-Info header field, the mandatory requirement is defined in the above bullet.

If the P-CSCF receives a 500 (Server Internal Error) or 504 (Server Time-Out) response to a REGISTER request, and if the REGISTER request is mapped to an existing IP association, then the P-CSCF shall delete the IP association.

NOTE 4: The P-CSCF deletes the IP association on receipt of 500 (Server Internal Error) or 504 (Server Time-Out) so that the next REGISTER request received from the UE will look like an initial REGISTER request.

When the P-CSCF receives a 401 (Unauthorized) response to a REGISTER request, the P-CSCF shall:

1) insert a Security-Server header field to specify the media plane security mechanisms the P-CSCF (IMS-ALG) supports, if any, labelled with the "mediasec" header field parameter specified in subclause 7.2A.7; and

2) send the 401 (Unauthorized) response to the UE unprotected as defined in clause 4 of RFC 3581 [56A].

NOTE 5: The P-CSCF does not include signalling plane security mechanisms because the Require and Proxy-Require header fields in the REGISTER request do not contain "sec-agree".

When the P-CSCF receives a 200 (OK) response to a REGISTER request as defined in subclause 5.2.2.1, and the registration expiration interval value is different than zero, the P-CSCF shall additionally:

a) create an IP association by storing and associating the UE's packet source IP address and port along with the "sent-by" parameter of the Via header field, cf. RFC 3261 [26], of the REGISTER request with the private user identity and all the successfully registered public user identities related to that private user identity;

b) overwrite any existing IP association which has the same pair of IP address and port, but a different private user identity; and

c) send the 200 (OK) response to the UE unprotected as defined in clause 4 of RFC 3581 [56A].

5.2.2.4 SIP digest with TLS as a security mechanism

TLS is optional to implement and is used only in combination with SIP digest authentication. If the P-CSCF supports TLS, then the P-CSCF shall support TLS as described in 3GPP TS 33.203 [19]. If the P-CSCF supports TLS, the P-CSCF shall support TLS ciphersuites as described in 3GPP TS 33.203 [19].

When the P-CSCF receives a REGISTER request from the UE, as defined in subclause 5.2.2.1, the P-CSCF shall additionally:

1) in case the REGISTER request was received without protection on the default port or port advertised to UE for P-CSCF discovery and with the Security-Client header field indicating "tls", then:

   a) remove and store the Security-Client header field;

   b) do not include the "integrity-protected" header field parameter in the Authorization header;

   c) if the "rport" header field parameter is included in the Via header field, set the value of the "rport" header field parameter in the Via header to the source port of the received REGISTER request; and

   d) insert the "received" header field parameter in the Via header containing the source IP address that the request came from, as defined in RFC 3581 [56A];

NOTE 1: The absence of an "integrity-protected" header field parameter in the Authorization header is an indication to the I-CSCF and S-CSCF that this is an initial REGISTER request.

NOTE 2: As defined in RFC 3581 [56A], the P-CSCF will insert a "received" header field parameter containing the source IP address that the request came from, even if it is identical to the value of the "sent-by" component.

NOTE 3: Upon receiving the unprotected REGISTER request the P-CSCF detects if the UE is behind a NAT.

2) if the REGISTER request was received protected with a TLS session, on the protected server port, created during an ongoing authentication procedure, where the Session ID for the TLS session is not yet bound to a private user identity, and contains an authentication challenge response (i.e. response parameter), then:

   a) check if the private user identity conveyed in the Authorization header of the protected REGISTER request is the same as the private user identity which was previously challenged. If the private user identities are different, the P-CSCF shall reject the REGISTER request by returning a 403 (Forbidden) response;

b) check the existence of the Security-Verify header field and the Security-Client header field. If there are no such headers, then the P CSCF shall return a suitable 4xx response. If there are such headers, then the P-CSCF shall compare the content of the Security-Verify header field with the content of the Security-Server header field sent earlier and the content of the Security-Client header field with the content of the Security-Client header field received in the challenged REGISTER request. If those do not match, then there is a potential man-in-the-middle attack. The P-CSCF should reject the request by sending a suitable 4xx response. If the contents match, the P-CSCF shall remove the Security-Verify and the Security-Client header fields;

c) include an "integrity-protected" header field parameter with the value set to "tls-pending"; and

d) if the hostport parameter in the Contact header field is in the form of a FQDN, the P-CSCF shall ensure that the given FQDN will resolve (e.g. by reverse DNS lookup) to the IP address bound to the TLS session; or

2A) if the REGISTER request was received, protected with a TLS session on the protected server port, created during an ongoing authentication procedure, where the Session ID for the TLS session is not yet bound to a private user identity, and does not contain an authentication challenge response (i.e. response parameter), reject the REGISTER request by returning a 403 (Forbidden) response;

3) if the REGISTER request was received on an existing TLS session created during a previous authentication procedure, then:

a) if the REGISTER request includes an Authorization header field, check if the private user identity conveyed in the Authorization header of the protected REGISTER request is the same as the private user identity which was previously authenticated, i.e. the private user identity previously associated with the Session ID for this TLS session. If the private user identities are different, the P-CSCF shall reject the REGISTER request by returning a 403 (Forbidden) response;

b) check the existence of the Security-Verify header field and Security-Client header field. If there are no such header fields, then the P-CSCF shall return a suitable 4xx response. If there are such headers, then the P-CSCF shall compare the content of the Security-Verify header field with the content of the Security-Server header field sent earlier and the content of the Security-Client header field with the content of the Security-Client header field received in the challenged REGISTER request. If those do not match, then there is a potential man-in-the-middle attack. The P-CSCF should reject the request by sending a suitable 4xx response;

c) the P-CSCF shall remove and store the Security-Client header field and remove the Security-Verify header field before forwarding the request to the S-CSCF; and

d) include an "integrity-protected" header field parameter with the value set to "tls-yes".

If the P-CSCF require security agreement, and the Security-Client header field is not present, then the P-CSCF shall return a suitable 4xx response.

When the P-CSCF receives a 401 (Unauthorized) response to a REGISTER request, the P-CSCF shall:

1) insert a Security-Server header field in the response, containing the P-CSCF selected signalling plane mechanism name, as specified in Annex H of 3GPP TS 33.203 [19]. The P-CSCF shall support and indicate the "tls" security mechanism, as specified in RFC 3329 [48]. The P-CSCF shall support the TLS ciphersuites as described in 3GPP TS 33.203 [19] and shall announce support for them according to the procedures defined in RFC 3329 [48]; and

1A) insert a Security-Server header field in the response, containing the P-CSCF static media plane security list, if any, labelled with the "mediasec" header field parameter specified in subclause 7.2A.7;

2) send the 401 (Unauthorized) response to the UE using the TLS session with which the associated REGISTER request was protected, or unprotected in case the REGISTER request was received unprotected. If the 401 (Unauthorized) response to the unprotected REGISTER request is sent using UDP, the P-CSCF shall send the response to the IP address listed in the "received" header field parameter and the port in the "rport" header field parameter. In case of TCP, the P-CSCF shall send the response over the same TCP connection over which the request was received from the UE.

NOTE 4: The challenge in the 401 (Unauthorized) response sent back by the S-CSCF to the UE as a response to the REGISTER request is piggybacked by the P-CSCF to insert the Security-Server header field in it.

When the P-CSCF receives a 200 (OK) response to a REGISTER request as defined in subclause 5.2.2.1, and the registration expiration interval value is different than zero, the P-CSCF shall additionally:

- create an association by storing and associating the UEs IP address and port of the TLS connection with the TLS Session ID, the private user identity and all the successfully registered public user identities related to that private user identity; and

- protect the 200 (OK) response to the REGISTER request within the same TLS session to that in which the request was protected.

5.2.2.5 NASS-IMS bundled authentication as a security mechanism

When the P-CSCF receives a REGISTER request from the UE, as defined in subclause 5.2.2.1, the P-CSCF shall additionally:

1) perform the NASS-IMS bundled authentication related query from the P-CSCF to the TISPAN NASS;

2) if the query in step 1) is successful, insert a P-Access-Network-Info header field as described in subclause 5.2.1 step 4); and

3) if the P-CSCF adds a "received" header field parameter and UDP is being used, the P-CSCF shall also add an "rport" Via header field parameter with the IP source port of the received REGISTER request.

When the P-CSCF receives a 200 (OK) response to a REGISTER request from the UE, as defined in subclause 5.2.2.1, the P-CSCF shall additionally:

1) store an association between the IP source address and port of the initial REGISTER request and the public user identities found in the P-Associated-URI header field value and associate them to the public user identity under registration;

2) store an association between the IP source address and port of the initial REGISTER request the default public user identity for use with procedures for the P-Asserted-Identity header field. The default public user identity is the first on the list of URIs present in the P-Associated-URI header field; and

3) insert a Security-Server header field to specify the media plane security mechanisms the P-CSCF (IMS-ALG) supports, if any, labelled with the "mediasec" header field parameter specified in subclause 7.2A.7.

NOTE 3: The P-CSCF does not include signalling plane security mechanisms because the Require and Proxy-Require header fields in the REGISTER request do not contain "sec-agree".

5.2.2.6 GPRS-IMS-Bundled authentication as a security mechanism

When the P-CSCF receives a SIP request from a GPRS-IMS-Bundled UE, the P-CSCF checks the IP address in the "sent-by" parameter of the Via header field provided by the UE as specified in RFC 3261 [6]. If the "sent-by" parameter contains a domain name, or if it contains an IP address that differs from the packet source IP address, the P-CSCF adds a "received" header field parameter to that Via header field value. This parameter contains the source IP address from which the packet was received.

When the P-CSCF receives a 200 (OK) response to a REGISTER request from the UE, as defined in subclause 5.2.2.1, the P-CSCF shall additionally:

1) store an association between the IP source address and port of the initial REGISTER request and the public user identities found in the P-Associated-URI header field value and associate them to the public user identity under registration;

2) store an association between the IP source address and port of the initial REGISTER request the default public user identity for use with procedures for the P-Asserted-Identity header field. The default public user identity is the first on the list of URIs present in the P-Associated-URI header field;

3) if the P-CSCF adds a "received" header field parameter and UDP is being used, the P-CSCF shall also add an "rport" Via header field parameter with the IP source port of the received REGISTER request; and

4) insert a Security-Server header field to specify the media plane security mechanisms the P-CSCF (IMS-ALG) supports, if any, labelled with the "mediasec" header field parameter specified in subclause 7.2A.7.

NOTE: The P-CSCF does not include signalling plane security mechanisms because the Require and Proxy-Require header fields in the REGISTER request do not contain "sec-agree".

5.2.2.7 P-CSCF reconfigured to not accept registrations

If the P-CSCF has been reconfigured to not accept initial registrations and reregistrations and to redirect incoming registrations to another P-CSCF, on recepetion of a REGISTER request the P-CSCF shall return a 305 (Use Proxy) response.

NOTE 1: As an example, this situation can happen in case the P-CSCF has been administratively configured to force the UEs to attempt initial registration with another P-CSCF before shutdown.

NOTE 2: The UE does not use the Contact header field of 305 (Use Proxy) response to determine the IP address of the new P-CSCF through which the UE will attempt a new initial registration.

5.2.3 Subscription to the user's registration-state event package

Upon receipt of a 200 (OK) response to the first initial REGISTER request (i.e. this was the first initial REGISTER request that the P-CSCF received from the user identified with its private user identity), the P-CSCF shall:

1) generate a SUBSCRIBE request in accordance with RFC 3680 [43] and RFC 6665 [28], with the following elements:

   a) a Request-URI set to the resource to which the P-CSCF wants to be subscribed to, i.e. to the SIP URI that is the default public user identity of the user;

   b) a From header field set to the P-CSCF's SIP URI;

   c) a To header field, set to the SIP URI that is the default public user identity of the user;

   d) an Event header field set to the "reg" event package;

   e) an Expires header field set to a value higher than the registration expiration interval value indicated in the 200 (OK) response to the REGISTER request;

   f) a P-Asserted-Identity header field:

      - if the received 200 (OK) response to the REGISTER request contained a "+g.3gpp.thig-path" Feature-Caps header field parameter, as defined in subclause 7.9A.9, and if the P-CSCF is located in the visited network, set to the value of the received "+g.3gpp.thig-path" Feature-Caps header field parameter; or

      - set to the SIP URI of the P-CSCF, which was inserted into the Path header field during the registration of the user to whose registration state the P-CSCF subscribes to; and

   g) a P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17] and a type 1 "orig-ioi" header field parameter. The P-CSCF shall set the type 1 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The P-CSCF shall not include the type 1 "term-ioi" header field parameter;

1A) if

   a) the P-CSCF supports indicating the traffic leg as specified in RFC 7549 [225];

   b) the SIP URI used in the Request-URI of the SUBSCRIBE request belongs to a UE that is roaming;

   c) the P-CSCF is not in the home network; and

   d) if required by local policy;

   then append the "iotl" SIP URI parameter set to "visitedA-homeA" to the Request-URI;

1B) if required by local policy, then insert a Route header field in the SUBSCRIBE request with the list of service route values saved from the Service-Route header field received in the 200 (OK) response to the last registration of the public user identity with associated contact address and skip steps 2 to 4;

2) if the P-CSCF is located in the visited network, and local policy requires the application of IBCF capabilities in the visited network towards the home network, then the P-CSCF shall forward the request to an IBCF in the visited network;

3) if the P-CSCF is located in the visited network and local policy does not require the application of IBCF capabilities in the visited network towards the home network, determine the entry point of the home network (e.g., by using DNS services) and send the SUBSCRIBE request to that entry point, according to the procedures of RFC 3261 [26]; and

4) if the P-CSCF is located in the home network, then the P-CSCF shall forward the request to an I-CSCF in the home network.

NOTE: The subscription to reg event package is done once per private user identity.

Upon receipt of a dialog establishing NOTIFY request, as specified in RFC 6665 [28], associated with the SUBSCRIBE request, the P-CSCF shall:

1) store the information for the so established dialog;

2) store the expiration time as indicated in the "expires" header field parameter of the Subscription-State header field, if present, of the received NOTIFY request. Otherwise the expiration time is retrieved from the Expires header field of the 2xx response to SUBSCRIBE request;

3) follow the procedures specified in RFC 6665 [28]; and

4) store the "icid-value" header field parameter and the "orig-ioi" header field parameter if present in the received P-Charging-Vector header field.

When sending a response to the NOTIFY request, the P-CSCF shall insert a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the NOTIFY request, a type 1 "term-ioi" header field parameter and the "icid-value" header field parameter. The P-CSCF shall set the type 1 "term-ioi" header field parameter to a value that identifies the sending network of the response, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the previously received value of "icid-value" header field parameter in the request.

If continued subscription is required the P-CSCF shall automatically refresh the subscription by the reg event package 600 seconds before the expiration time for a previously registered public user identity, either 600 seconds before the expiration time if the initial subscription was for greater than 1200 seconds, or when half of the time has expired if the initial subscription was for 1200 seconds or less. If a SUBSCRIBE request to refresh a subscription fails with a non-481 response, the P-CSCF shall still consider the original subscription valid for the duration of the most recently known "Expires" value according to RFC 6665 [28]. Otherwise, the P-CSCF shall consider the subscription invalid and start a new initial subscription according to RFC 6665 [28].

5.2.3A Void

5.2.3B SUBSCRIBE request

Upon receipt of a NOTIFY request with the Subscription-State header field set to "terminated", once the NOTIFY transaction is terminated, the P-CSCF can remove all the stored information related to the associated subscription.

5.2.4 Registration of multiple public user identities

Upon receipt of a NOTIFY request for the dialog associated with the subscription to the reg event package of the user, the P-CSCF shall:

- store the information for the established dialog;

- store the expiration time as indicated in the "expires" header field parameter of the Subscription-State header field, if present, of the SIP NOTIFY request. Otherwise the expiration time is retrieved from the Expires header field of the 2xx response to SIP SUBSCRIBE request;

- identify the public user identity as follows:

1) if no <wildcardedIdentity> subelement is included in the <registration> element, the public user identity is taken from the 'aor' attribute of the registration element; or

2) if a <wildcardedIdentity> sub element is included in the <registration> element, the wildcarded public user identity is taken from the <wildcardedIdentity> sub element. The wildcarded public user identity is treated as a public user identity in the procedures of this subclause;

3) for each public user identity whose state attribute in the <registration> element is set to "active", i.e. registered; and

   i) the state attribute within the <contact> sub-element is set to "active";

   ii) the value of the <uri> sub-element inside the <contact> sub-element is set to the contact address of the user's UE; and

   iii) the event attribute of that <contact> sub-element(s) is set to "registered" or "created";

   the P-CSCF shall:

   i) bind the indicated public user identity as registered to the contact address of the respective user, including any associated display names, and any parameters associated with either the user or the identities of the user;

   ii) add the public user identity to the list of the public user identities that are registered for the user saved against the contact address;

   iii) if the <actions> child element is included in the <registration> element, bind the policy received in the <actions> child element of the <registration> element to each contact address of the public user identity; and

   iv) if the <actions> child element is not included in the <registration> element, remove the policy bound to each contact address of the public user identity;

4) for each public user identity whose state attribute in the <registration> element is set to "active", i.e. registered: and

   i) the state attribute within the <contact> sub-element is set to "terminated";

   ii) the value of the <uri> sub-element inside the <contact> sub-element is set to the contact address of the user's UE; and

   iii) the event attribute of that <contact> sub-element(s) is set to "deactivated", "expired", "probation", "unregistered", or "rejected";

   the P-CSCF shall consider the binding between the indicated public user identity and the contact address and its related information as deregistered for this user, and shall release all stored information associated with the deregistered contact address and related information associated with this contact address; and

5) for each public user identity whose state attribute in the <registration> element is set to "terminated", i.e. deregistered; and for each <contact> sub-element, if

   i) the value of the <uri> sub-element inside each <contact> sub-element is set to the respective contact address of the user's UE; and

   ii) the event attribute of each <contact> sub-element(s) is set to "deactivated", "expired", "probation", "unregistered", or "rejected";

   the P-CSCF shall consider the indicated public user identity and all its contact addresses as deregistered for this UE, and shall release all stored information for these public user identity bound to the respective user and remove the public user identity from the list of the public user identities that are registered for the user;

- shall store the "orig-ioi" header field parameter if present in the received P-Charging-Vector header field; and

- follow the procedures specified in RFC 6665 [28].

When sending a response to the NOTIFY request, the P-CSCF shall insert a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the NOTIFY request, a type 1 "term-ioi" header field parameter and the "icid-value" header field parameter. The P-CSCF shall set the type 1 "term-ioi" header field parameter to a value that identifies the sending network of the response, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the value populated in the initial request for the dialog.

If the P-CSCF is informed that all contact addresses that are registered with this P-CSCF and belonging to the user using its private user identity have been deregistered, i.e. the state attribute within each <contact> sub-element is set to "terminated", the P-CSCF shall either unsubscribe to the reg event package or let the subscription expire.

NOTE 1: Since there can be other active registrations of the user via other P-CSCFs, the S-CSCF will not terminate the by sending a NOTIFY request that includes the Subscription-State header set to "terminated".

If all public user identities, that were registered by the user using its private user identity, have been deregistered, the P-CSCF, will receive from the S-CSCF a NOTIFY request that may include the Subscription-State header field set to "terminated", as described in subclause 5.4.2.1.2. If the Subscription-State header field was not set to "terminated", the P-CSCF may either unsubscribe to the reg event package of the user or let the subscription expire.

NOTE 2: Upon receipt of a NOTIFY request with the Subscription-State header field set to "terminated", the P-CSCF considers the subscription to the reg event package terminated (i.e. as if the P-CSCF had sent a SUBSCRIBE request with an Expires header field containing a value of zero).

NOTE 3: There can be public user identities which are implicitly registered within the registrar (S-CSCF) of the user upon registration of one public user identity. The procedures in this subclause provide a mechanism to inform the P-CSCF about these implicitly registered public user identities.

5.2.5 Deregistration

5.2.5.1 User-initiated deregistration

When the P-CSCF receives a 200 (OK) response to a REGISTER request (sent according to subclause 5.2.2) sent by this UE, then the P-CSCF shall check each Contact header field included in the response. If there is a Contact header field that contains the contact address registered via this P-CSCF via the respective security associations or TLS sessions, and the value of the registration expiration interval value equals zero, then the P-CSCF shall:

1) if the "reg-id" header field parameter is not included in the Contact header field, remove the binding between the public user identity found in the To header field (together with the implicitly registered public user identities) and the contact address indicated in the Contact header field, from the registered public user identities list belonging to this UE and all related stored information;

1A) if the "reg-id" header field parameter is included in the Contact header field, remove the public user identity found in the To header field (together with the implicitly registered public user identities) and the flow identified by the "reg-id" header field parameter and all its related stored information belonging to this UE;

2) if multiple registrations is not used, check if the UE has left any other registered public user identity. When all of the public user identities that were registered by this UE are deregistered, the P-CSCF shall delete any security associations, TLS sessions or IP associations towards the UE, after the server transaction (as defined in RFC 3261 [26]) pertaining to this deregistration terminates; and

2A) if multiple registrations is used, check if the UE has left any other registered public user identity that is bound to this flow. When all of the public user identities that were registered and are bound to this flow are deregistered, the P-CSCF shall delete any security associations or TLS sessions associated with this flow, after the server transaction (as defined in RFC 3261 [26]) pertaining to this deregistration terminates.

NOTE 1: Upon receipt of a NOTIFY request with all <registration> element(s) having their state attribute set to "terminated" (i.e. all public user identities are deregistered) and the Subscription-State header field set to "terminated", the P-CSCF considers the subscription to the reg event package terminated (i.e. as if the P-CSCF had sent a SUBSCRIBE request with an Expires header field containing a value of zero).

NOTE 2: There is no requirement to distinguish a REGISTER request relating to a registration from that relating to a deregistration. For administration reasons the P-CSCF may distinguish such requests, however this has no impact on the SIP procedures.

NOTE 3: When the P-CSCF has sent the 200 (OK) response for the REGISTER request of the only public user identity currently registered with its associated set of implicitly registered public user identities using the respective security association or TLS session (i.e. no other public user identity belonging to the user is registered with this contact address the associated security association or TLS session), the P-CSCF removes the security association or TLS session established between the P-CSCF and the UE. Therefore further SIP signalling sent over this security association or TLS session (e.g. the NOTIFY request containing the deregistration event) will not reach the UE.

5.2.5.2 Network-initiated deregistration

Upon receipt of a NOTIFY request on the dialog which was generated during subscription to the reg event package of the UE, as described in subclause 5.2.3, including one or more <registration> element(s) which were registered by the UE with either:

- the state attribute within the <registration> element set to "terminated"; or
- the state attribute within the <registration> element set to "active" and the state attribute within the <contact> sub-element belonging to this UE and registered via this P-CSCF set to "terminated", and the event attribute within the <contact> sub-element belonging to this UE set either to "unregistered", or "rejected" or "deactivated";

the P-CSCF shall remove all stored information for these public user identities for this UE and remove these public user identities from the list of the public user identities that are registered for the user.

NOTE 1: If all public user identities have been removed from the list of the public user identities registered via this P-CSCF, and the NOTIFY request indicates that the UE is still registered (e.g. via another P-CSCF), the P-CSCF can unsubscribe from the reg event package of the UE.

Upon receipt of a NOTIFY request with all <registration> element(s) having their state attribute set to "terminated" (i.e. all public user identities are deregistered) and the Subscription-State header field set to "terminated" or when all public user identities of the UE have been deregistered, the P-CSCF shall shorten any security associations or TLS sessions towards the UE.

NOTE 2: The security association between the P-CSCF and the UE is shortened to a value that will allow the NOTIFY request containing the deregistration event to reach the UE.

NOTE 3: When the P-CSCF receives the NOTIFY request with Subscription-State header field containing the value of "terminated", the P-CSCF considers the subscription to the reg event package terminated (i.e. as if the P-CSCF had sent a SUBSCRIBE request to the S-CSCF with an Expires header field containing a value of zero).

Upon receipt of a NOTIFY request for the dialog associated with the subscription to the reg event package of the user the P-CSCF shall store the "orig-ioi" header field parameter if present in the received P-Charging-Vector header field.

When sending a response to the NOTIFY request, the P-CSCF shall insert a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the NOTIFY request, a type 1 "term-ioi" header field parameter and the "icid-value" header field parameter. The P-CSCF shall set the type 1 "term-ioi" header field parameter to a value that identifies the sending network of the response, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the value populated in the initial request for the dialog.

5.2.6 General treatment for all dialogs and standalone transactions excluding the REGISTER method

5.2.6.1 Introduction

The procedures of subclause 5.2.6 and its subclauses are general to all requests and responses, except those for the REGISTER method.

5.2.6.2 Determination of UE-originated or UE-terminated case

Upon receipt of an initial request or a target refresh request or a stand-alone transaction, the P-CSCF shall:

- perform the procedures for the UE-terminating case as described in subclause 5.2.6.4 if the request makes use of the information for UE-terminating calls, which was added to the Path header field entry of the P-CSCF during registration (see subclause 5.2.2), e.g. the message is received at a certain port or the topmost Route header field contains a specific user part or parameter;

- perform the procedures for the UE-originating case as described in subclause 5.2.6.3 if this information is not used by the request.

5.2.6.3 Requests initiated by the UE

5.2.6.3.1 General for all requests

When the P-CSCF receives an initial request for a dialog or a request for a standalone transaction from a UE that is not considered as privileged sender, and:

- the request does not include any P-Preferred-Identity header field or none of the P-Preferred-Identity header fields included in the request matches any of the registered public user identities or any of the registered wildcarded public user identities, then the P-CSCF shall identify the originator and the served user of the request by the default public user identity;

- the request includes one or two P-Preferred-Identity header field(s) each of which matches one of the registered public user identity or a registered wildcarded public user identity, the P-CSCF shall identify the originator and the served user of the request by the public user identity from the first such P-Preferred-Identity header field; and

- the request includes two P-Preferred-Identity header fields, each of which matches a registered public user identity or a registered wildcarded public user identity, the P-CSCF shall identify the alternative identity of the originator of the request by the public user identity from the second such P-Preferred-Identity header field.

NOTE: When two identities are provided in the P-Preferred-Identity header fields, it is assumed that one is an alias of the other but P-CSCF does not have the information to verify this.

When the P-CSCF receives an initial request for a dialog or a request for a standalone transaction from a UE that is considered as privileged sender, and the request:

a) does not include any P-Preferred-Identity header field, then the P-CSCF shall identify the served user of the request by the default public user identity;

b) includes a P-Preferred-Identity header field that does not match one of the registered public user identities or wildcarded public user identities, then the P-CSCF shall identify the served user of the request by the default public user identity; or

c) includes a P-Preferred-Identity header field that matches one of the registered public user identities or wildcarded public user identities, then the P-CSCF shall identify the served user of the request by the public user identity from the P-Preferred-Identity header field.

If a P-Preferred-Identity header field value is a SIP URI with the user part starting with a "+", and a "user" SIP URI parameter with a "phone" value, then the P-CSCF shall translate the SIP URI to a tel URI with the user part of the SIP URI in the telephone-subscriber element in the tel URI when checking whether the header field value matches any of the registered public user identities. The P-CSCF shall not modify the header field value within the P-Preferred-Identity header field.

NOTE 1: The case above can occur when the UE receives a public user identity as a tel URI during registration, and then translates that into a SIP URI when sending an initial request for a dialog or a request for a standalone transaction.

In addition, if the request from a UE that is considered as privileged sender:

1) includes one or two P-Asserted-Identity header field(s) then the P-CSCF shall identify the originator of that request by the public user identity from the first P-Asserted-Identity header field; or

2) does not include a P-Asserted-Identity header field then the P-CSCF shall identify the originator of that request by the same identity that has been determined for the served user according to steps a), b), and c) above.

NOTE 2: If no security association was set-up during registration, the P-CSCF identifies the originator and served user of the request by using the IP association information stored during the registration for which it holds the list of registered public user identities.

NOTE 3: The contents of the From header field do not form any part of this decision process.

NOTE 4: The display-name portion of the P-Preferred-Identity header field and the registered public user identities is not included in the comparison to determine a match.

NOTE 5: The P-CSCF determines if the UE is considered as privileged sender using the user-related policies provisioned to the P-CSCF (see subclause 5.2.1).

When the P-CSCF receives from the UE an initial request for a dialog or a request for a standalone transaction, if the host portion of the sent-by field in the topmost Via header field contains a FQDN, or if it contains an IP address that differs from the source address of the IP packet, the P-CSCF shall add a "received" header field parameter in accordance with the procedure defined in RFC 3261 [26].

If the P-CSCF adds a "received" header field parameter and UDP is being used, the P-CSCF shall also add an "rport" header field parameter. If IMS AKA is used, the parameter value shall contain the UEs protected server port. Otherwise the parameter value shall contain the IP source of the request.

When the P-CSCF receives from the UE an initial request for a dialog or a request for a standalone transaction, and the request matches a trigger for starting logging of SIP signalling, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K], the P-CSCF shall treat the dialog as one for which logging of signalling is in progress and start to log SIP signalling for this dialog according to its trace configuration.

When the P-CSCF receives from the UE a request sent on a dialog for which logging of signalling is in progress, the P-CSCF shall check whether a trigger for stopping logging of SIP signalling has occurred, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K]. If a stop trigger event has occurred then the P-CSCF shall stop treating the dialog as one for which logging of signalling is in progress, else the P-CSCF shall append a "logme" header field parameter to the SIP Session-ID header field if the parameter is missing and determine, by checking its trace configuration, whether to log the request.

If:

- the P-CSCF supports indicating the traffic leg as specified in RFC 7459 [225];
- the UE is roaming;
- the P-CSCF is not in the home network; and
- required by local policy;

then the P-CSCF shall:

- if the bottommost Route header field does not contain the "tokenized-by" header field parameter and an "iotl" SIP URI parameter is not already included, append the "iotl" SIP URI parameter with a value set to "visitedA-homeA" to the URI of the bottommost Route header field; and
- if the bottommost Route header field contains the "tokenized-by" header field parameter and an "iotl" SIP URI parameter is not already included, append the "iotl" SIP URI parameter with a value set to "visitedA-homeA" to the URI of the second Route header field from the bottom.

NOTE 6: The bottommost Route header field contains the "iotl" SIP URI parameter if the S-CSCF added the "iotl" SIP URI parameter in the Service-Route header field during registration and if the home network does not apply topology hiding. The second Route header field from the bottom contains the "iotl" SIP URI parameter if the S-CSCF added the "iotl" SIP URI parameter in the Service-Route header field during registration and if the home network applies topology hiding.

If a P-CSCF supporting barring of premium numbers when roaming receives a request from a roaming UE and the Request-URI contains an E.164 number encoded as described in subclause 5.1.2A.1.2 which the P-CSCF is able to identify as a premium rate number in the country of the served network, the P-CSCF shall, based on local policy, add the "premium-rate" tel URI parameter specified in subclause 7.2A.17 set to a value "information" or "entertainment" as appropriate.

NOTE 7: The feature barring of premium numbers when roaming can be implemented in the P-CSCF or an IBCF of the visited network. Local policy ensures that the feature is only activiated in one of the two.

5.2.6.3.2 General for all responses

The P-CSCF shall log all SIP responses destined for the UE that contain a "logme" Session-ID header field parameter based on local policy.

5.2.6.3.2A Abnormal cases

When the P-CSCF is unable to forward the request to the next hop by the Route header fields, as determined by one of the following:

- there is no response to the service request and its retransmissions by the P-CSCF; or
- by unspecified means available to the P-CSCF;

and:

- the P-CSCF supports S-CSCF restoration procedures;

then the P-CSCF:

1) shall reject the request by returning a 504 (Server Time-out) response to the UE;
2) shall assume that the UE supports version 1 of the XML Schema for the 3GPP IM CN subsystem XML body if support for the 3GPP IM CN subsystem XML body as described in subclause 7.6 in the Accept header field is not indicated; and
3) shall include in the 504 (Server Time-out) response:
   - a Content-Type header field with the value set to associated MIME type of the 3GPP IM CN subsystem XML body as described in subclause 7.6.1;
   - a P-Asserted-Identity header field set to the value of the SIP URI of the P-CSCF included in the Path header field during the registration of the user whose UE sent the request causing this response (see subclause 5.2.2.1); and
   - a 3GPP IM CN subsystem XML body containing:
     a) an <ims-3gpp> element with the "version" attribute set to "1" and with an <alternative-service> child element, set to the parameters of the alternative service:
        i) a <type> child element, set to "restoration" (see table 7.6.2) to indicate that S-CSCF restoration procedures are supported;
        ii) a <reason> child element, set to an operator configurable reason; and
        iii) an <action> child element, set to "initial-registration" (see table 7.6.3).

NOTE 1: These procedures do not prevent the usage of unspecified reliability or recovery techniques above and beyond those specified in this subclause.

If the P-CSCF receives a SIP request different from REGISTER that does not map to an existing IP association and unless the P-CSCF detects that the request is related to an emergency communication that is to be handled according to subclause 5.2.10.2 and if the request is not received from a UE performing the functions of an external attached network using static mode of operation, the P-CSCF shall discard this SIP request, i.e. it shall not send back a 100 Trying SIP response to the UE and shall not try to forward the request to the S-CSCF.

NOTE 2: Reception of SIP requests not corresponding to a stored registration context may happen if the P-CSCF has lost the registration context. If the UE does not receive any SIP response to the sent SIP request (i.e. the SIP transaction timer B or F expires), the UE will perform a new initial registration procedure.

NOTE 3: The P-CSCF can identify that a request is received from a UE performing the functions of an external attached network using static mode of operation by evaluating the TLS session or by other means. Such operation requires preconfiguration of the capability for this attached network in the P-CSCF.

5.2.6.3.3 Initial request for a dialog

When the P-CSCF receives from the UE an initial request for a dialog, and a service route value list exists for the served user of the request, the P-CSCF shall:

1) remove its own SIP URI from the top of the list of Route header fields;

2) if the UE is performing the functions of an external attached network using static mode of operation:

   i) select an I-CSCF and insert a Route header field with the URI of the I-CSCF as the topmost Route header field; otherwise

NOTE 1: The list of the I-CSCFs can be either obtained as specified in RFC 3263 [27A] or be provisioned in the P-CSCF.

   ii) verify that the resulting list of Route header fields matches the list of URIs received in the Service-Route header field (during the last successful registration or re-registration). This verification is done on a per URI basis, not as a whole string. If the verification fails, then the P-CSCF shall either:

      a) return a 400 (Bad Request) response; the P-CSCF shall not forward the request, and shall not continue with the execution of steps 2 onwards; or

      b) replace the preloaded Route header field value in the request with the value of the Service-Route header field received during the last 200 (OK) response for the last successful registration or reregistration;

NOTE 2: The P-CSCF can identify that a request is received from a UE performing the functions of an external attached network using static mode of operation by evaluating the TLS session or by other means.

3) if the 200 (OK) response to the last REGISTER request, which created or refreshed the binding of the contact address from which the request is received, has not contained a Feature-Caps header field, specified in RFC 6809 [190] with a "+g.3gpp.atcf" header field parameter, then:

   a) if the P-CSCF is located in the visited network, and local policy requires the application of IBCF capabilities in the visited network towards the home network, select an IBCF in the visited network and add the URI of the selected IBCF to the topmost Route header field;

NOTE 3: It is implementation dependent as to how the P-CSCF obtains the address of the IBCF exit point.

3A) if the 200 (OK) response to the last REGISTER request, which created or refreshed the binding of the contact address from which the request is received, contained a Feature-Caps header field with a "+g.3gpp.atcf" header field parameter, then:

   a) add the ATCF URI for originating requests that the P-CSCF used to forward the last REGISTER request which created or refreshed the binding of the contact address from which the request is received, to the topmost Route header field;

4) add its own address to the Via header field. The P-CSCF Via header field entry is built in a format that contains the port number of the P-CSCF in accordance with the procedures of RFC 3261 [26], and either:

   a) the P-CSCF FQDN that resolves to the IP address, or

   b) the P-CSCF IP address;

5) when adding its own SIP URI to the Record-Route header field, build the P-CSCF SIP URI in a format that contains the port number of the P-CSCF where it awaits subsequent requests from the called party, and either:

   a) the P-CSCF FQDN that resolves to the IP address; or

   b) the P-CSCF IP address.

   If the Contact header field in the request contains an "ob" SIP URI parameter, the P-CSCF shall add a flow token and the "ob" SIP URI parameter to its SIP URI;

NOTE 4: The inclusion of these values in the Record-Route header field will ensure that all subsequent mid dialog requests destined for the UE are sent over the same IMS flow over which the initial dialog-forming request was received.

5A) if a P-Private-Network-Indication header field is included in the request, check whether the information saved during registration or from configuration allows the receipt of private network traffic from this source. If private network traffic is allowed, the P-CSCF shall check whether the received domain name in any included P-Private-Network-Indication header field in the request is the same as the domain name associated with that saved information. If private network traffic is not allowed, or the received domain name does not match, then the P-CSCF shall remove the P-Private-Network-Indication header field;

5B) if the served user of the request is understood from information saved during registration or from configuration to always send and receive private network traffic from this source, insert a P-Private-Network-Indication header field containing the domain name associated with that saved information;

5C) if the request is originated from a UE which the P-CSCF considers as privileged sender (including one which is also a UE performing the functions of an external attached network using static mode of operation), keep the P-Asserted-Identity header field unchanged if one was received, or include the originator of the request in the P-Asserted-Identity header field if no P-Asserted-Identity header field was received. In addition remove any P-Preferred-Identity header field, include the served user of the request in the P-Served-User header field as specified in RFC 5502 [133] and skip step 6) below;

NOTE 5: The P-CSCF determines if the UE is considered as privileged sender using the user-related policies provisioned to the P-CSCF (see subclause 5.2.1).

NOTE 6: The P-CSCF can retrieve the identity of the UE performing the functions of an external attached network from the subjectCommonName (CN) if it is not present in the subjectAltName in the certificates during the TLS session setup in accordance with the procedures of RFC 5280 [213] or by other means.

5D) if the request is originated from a UE performing the functions of an external attached network using static mode of operation and which the P-CSCF considers as is not a privileged sender, include the served user of the request in the P-Served-User header field as specified in RFC 5502 [133] and skip step 6) below;

6) remove any P-Preferred-Identity header field or P-Asserted-Identity header field, if present, and insert a P-Asserted-Identity header field with the value identifying the originator of the request and the value of the alternative identity of the originator of the request, if identified (see subclause 5.2.6.3.1), including the display name if previously stored during registration representing the served user of the request;

6A) if the identity of the served user of the request was taken from P-Preferred-Identity header field by matching a registered wildcarded public user identity, and the identity of the served user is not a distinct identity within the range of the wildcarded public user identity, include the wildcarded public user identity value in the P-Profile-Key header field as defined in RFC 5002 [97];

NOTE 7: The matching of distinct public user identities takes precedence over the matching of wildcarded public user identities.

7) add a P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17] and a type 1 "orig-ioi" header field parameter. The P-CSCF shall set the type 1 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The P-CSCF shall not include the type 1 "term-ioi" header field parameter. Based on local policy, the P-CSCF shall add an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier;

7A) if the request comes from a UE performing the functions of an external attached network using static mode of operation add the "orig" parameter to the dialog request to indicate that this is an originating request; and

8) if the request is an INVITE request, save the Contact, CSeq and Record-Route header field values received in the request such that the P-CSCF is able to release the session if needed. If the Contact header field in the INVITE request contains a GRUU, the P-CSCF shall save the GRUU received in the Contact header field of the request and associate that GRUU with the UE contact address used during registration or with the registration flow and the associated UE contact address used over which the INVITE request was received such that the P-CSCF is able to release the session if needed

before forwarding the request, based on the topmost Route header field, in accordance with the procedures of RFC 3261 [26].

NOTE 8: According to RFC 5626 [92], an approach such as having the Edge Proxy adds a Record-Route header field with a flow token is one way to ensure that mid-dialog requests are routed over the correct flow.

If the request comes from a UE performing the functions of an external attached network using static mode of operation:

- no response is received to the initial request for dialog and its retransmissions by the P-CSCF; or
- a 3xx response or 480 (Temporarily Unavailable) response is received,

the P-CSCF shall repeat the actions of the above bullets with a different I-CSCF.

If the P-CSCF fails to forward the initial request for dialog to any I-CSCF, the P-CSCF shall send back a 504 (Server Time-Out) response to the UE performing the functions of an external attached network, in accordance with the procedures in RFC 3261 [26].

5.2.6.3.4 Responses to an initial request for a dialog

When the P-CSCF receives any 1xx or 2xx response to the above request, the P-CSCF shall:

1) store the values received in the P-Charging-Function-Addresses header field;
2) store the list of Record-Route header fields from the received response;
3) store the dialog ID and associate it with the private user identity and public user identity involved in the session;
4) if a security association or TLS session exists, in the response rewrite its own Record Route entry to its own SIP URI that contains the protected server port number of the security association or TLS session established from the UE to the P-CSCF and either:
   a) the P-CSCF FQDN that resolves to the IP address of the security association or TLS session established from the UE to the P-CSCF; or
   b) the P-CSCF IP address of the security association or TLS session established from the UE to the P-CSCF;

   NOTE 1: The P-CSCF associates two ports, a protected client port and a protected server port, with each pair of security associations. For details on the selection of the protected port values see 3GPP TS 33.203 [19].
5) if SIP digest without TLS, NASS-IMS bundled authentication or GPRS-IMS-Bundled authentication is used, in the response rewrite its own Record-Route entry to its own SIP URI that contains an unprotected server port number where the P-CSCF expects subsequent requests from the UE; and
6) if the response corresponds to an INVITE request, save the Contact, From, To and Record-Route header field values received in the response such that the P-CSCF is able to release the session if needed;

before forwarding the response to the UE in accordance with the procedures of RFC 3261 [26].

NOTE 2: The P-CSCF can find the IMS communication service supported for the dialog, as determined by the originating home network, in the topmost occurance of the "+g.3gpp.icsi-ref" header field parameter of the Feature-Caps header field(s) of 18x or 2xx response. The information can be used for charging purpose or resource reservation purpose.

The P-CSCF shall forward the response to the UE using the mechanisms described in RFC 3261 [26] and RFC 3581 [56A], i.e. the P-CSCF shall send the response to the IP address indicated in the "received" header field parameter and, in case UDP is used, to the port indicated in the "rport" header field parameter (if present) of the Via header field associated with the UE. In case TCP is used, the P-CSCF shall use the TCP connection on which the REGISTER request was received for sending the response back to the UE.

5.2.6.3.5 Target refresh request for a dialog

When the P-CSCF receives from the UE a target refresh request for a dialog, the P-CSCF shall:

1) verify if the request relates to a dialog in which the originator of the request is involved:

   a) if the request does not relates to an existing dialog in which the originator is involved, then the P-CSCF shall answer the request by sending a 403 (Forbidden) response back to the originator. The P-CSCF will not forward the request. No other actions are required; or

   b) if the request relates to an existing dialog in which the originator is involved, then the P-CSCF shall continue with the following steps;

1A) remove its own SIP URI from the top of the list of Route header fields;

2) verify that the resulting list of Route header fields matches the list of Record-Route header fields constructed by inverting the order of the stored list of Record-Route header fields and removing its Record-Route header field value from the list. This verification is done on a per URI basis, not as a whole string. If the verification fails, then the P-CSCF shall either:

   a) return a 400 (Bad Request) response; the P-CSCF shall not forward the request, and shall not continue with the execution of steps 3 onwards; or

   b) replace the Route header field value in the request with the list of Record-Route header fields constructed by inverting the order of the stored list of Record-Route header fields and removing its Record-Route header value from the list;

3) add its own address to the Via header field. The P-CSCF Via header field entry is built in a format that contains the port number of the P-CSCF where it awaits the responses to come, and either:

   a) the P-CSCF FQDN that resolves to the IP address, or

   b) the P-CSCF IP address;

4) void

5) for INVITE dialogs (i.e. dialogs initiated by an INVITE request), replace the saved Contact and CSeq header field values received in the request such that the P-CSCF is able to release the session if needed. If the Contact header field in the INVITE request contains a GRUU, the P-CSCF shall save the GRUU received in the Contact header field of the request and associate that GRUU with the UE contact address used during session establishment or with the registration flow and the associated UE contact address used during session establishment such that the P-CSCF is able to release the session if needed;

NOTE: The replaced Contact header field value is valid only if a 1xx or 2xx response will be received for the request. In other cases the old value is still valid.

6) if the P-CSCF inserted the header field parameters into the Feature-Caps header field of the initial request for the dialog then when the target refresh request is forwarded in the same direction, the P-CSCF shall insert the header field parameters with the same parameter values in the Feature-Caps header field; and

7) add a P-Charging-Vector header field with the "icid-value" header field parameter set to the value populated in the initial request for the dialog and a type 1 "orig-ioi" header field parameter. The P-CSCF shall set the type 1 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The P-CSCF shall not include the type 1 "term-ioi" header field parameter;

before forwarding the request, based on the topmost Route header field, in accordance with the procedures of RFC 3261 [26].

5.2.6.3.6 Responses to a target refresh request for a dialog

When the P-CSCF receives a 1xx or 2xx response to the above request, the P-CSCF shall:

1) if a security association or TLS session exists, rewrite the address and port number of its own Record Route entry to the same value as for the response to the initial request for the dialog; and

2) replace the saved Contact header field value received in the response such that the P-CSCF is able to release the session if needed;

before forwarding the response to the UE in accordance with the procedures of RFC 3261 [26].

5.2.6.3.7 Request for a standalone transaction

When the P-CSCF receives from the UE the request for a standalone transaction, and a service route value list exists for the served user of the request, the P-CSCF shall:

1) remove its own SIP URI from the top of the list of Route header fields;

2) if the UE is performing the functions of an external attached network using static mode of operation:

   i) select an I-CSCF and insert a Route header field with the URI of the I-CSCF as the topmost Route header field; otherwise

NOTE 1: The list of the I-CSCFs can be either obtained as specified in RFC 3263 [27A] or be provisioned in the P-CSCF.

   ii) verify that the resulting list of Route header fields matches the list of URIs received in the Service-Route header field (during the last successful registration or re-registration). This verification is done on a per URI basis, not as a whole string. If the verification fails, then the P-CSCF shall either:

      a) return a 400 (Bad Request) response; the P-CSCF shall not forward the request, and shall not continue with the execution of steps 3 onwards; or

      b) replace the preloaded Route header field value in the request with the one received during the last registration in the Service-Route header field of the 200 (OK) response;

NOTE 2: The P-CSCF can identify that a request is received from a UE performing the functions of an external attached network using static mode of operation by evaluating the TLS session or by other means.

3) if the 200 (OK) response to the last REGISTER request, which created or refreshed the binding of the contact address from which the request is received, has not contained a Feature-Caps header field, specified in RFC 6809 [190] with a "+g.3gpp.atcf" header field parameter, then:

   a) if the P-CSCF is located in the visited network, and local policy requires the application of IBCF capabilities in the visited network towards the home network, select an IBCF in the visited network and add the URI of the selected IBCF to the topmost Route header field;

NOTE 3: It is implementation dependent as to how the P-CSCF obtains the address of the IBCF exit point.

3A) if the 200 (OK) response to the last REGISTER request, which created or refreshed the binding of the contact address from which the request is received, contained a Feature-Caps header field with a "+g.3gpp.atcf" header field parameter, then:

   a) add the ATCF URI for originating requests, that the P-CSCF used to forward the last REGISTER request which created or refreshed the binding of the contact address from which the request is received, to the topmost Route header field;

3B) if the request is originated from a UE which the P-CSCF considers as privileged sender (including one which is also a UE performing the functions of an external attached network using static mode of operation), keep the P-Asserted-Identity header field unchanged if one was received, or include the originator of the request in the P-Asserted-Identity header field if no P-Asserted-Identity header field was received. In addition remove any P-Preferred-Identity header field, include the served user of the request in the P-Served-User header field as specified in RFC 5502 [133] and skip step 4) below;

NOTE 4: The P-CSCF determines if the UE is considered as privileged sender using the user-related policies provisioned to the P-CSCF (see subclause 5.2.1).

NOTE 5: The P-CSCF can retrieve the identity of the UE performing the functions of an external attached network from the subjectCommonName (CN) if it is not present in the subjectAltName in the certificates during the TLS session setup in accordance with the procedures of RFC 5280 [213] or by other means.

3D) if the request is originated from a UE performing the functions of an external attached network using static mode of operation and which the P-CSCF considers as is not a privileged sender, include the served user of the request in the P-Served-User header field as specified in RFC 5502 [133] and skip step 4) below;

4) remove any P-Preferred-Identity header field or P-Asserted-Identity header field, if present, and insert P-Asserted-Identity header fields the value identifying the served user of the request and the value of the alternative identity of the originator of the request, if identified (see subclause 5.2.6.3.1), including the display name if previously stored during registration, representing the served user of the request;

4A) if the identity of the served user of the request was taken from P-Preferred-Identity header field by matching a registered wildcarded public user identity, and the identity of the served user is not a distinct identity within the range of the wildcarded public user identity, include the wildcarded public user identity value in the P-Profile-Key header field as defined in RFC 5002 [97];

NOTE 6: The matching of distinct public user identities takes precedence over the matching of wildcarded public user identities.

4B) if a P-Private-Network-Indication header field is included in the request, check whether the information saved during registration or from configuration allows the receipt of private network traffic from this source. If private network traffic is allowed, the P-CSCF shall check whether the received domain name in any included P-Private-Network-Indication header field in the request is the same as the domain name associated with that saved information. If private network traffic is not allowed, or the received domain name does not match, then the P-CSCF shall remove the P-Private-Network-Indication header field;

4C) if the served user of the request is understood from information saved during registration or from configuration to always send and receive private network traffic from this source, insert a P-Private-Network-Indication header field containing the domain name associated with that saved information; and

4D) if the request comes from a UE performing the functions of an external attached network using static mode of operation add the "orig" parameter to the dialog request to indicate that this is an originating request; and

5) add a P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17] and a type 1 "orig-ioi" header field parameter. The P-CSCF shall set the type 1 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The P-CSCF shall not include the type 1 "term-ioi" header field parameter. Based on local policy, the P-CSCF shall add an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier;

before forwarding the request, based on the topmost Route header field, in accordance with the procedures of RFC 3261 [26].

If the request comes from a UE performing the functions of an external attached network using static mode of operation:

- no response is received to the standalone SIP request and its retransmissions by the P-CSCF; or

- a 3xx response or 480 (Temporarily Unavailable) response is received,

the P-CSCF shall repeat the actions of the above bullets with a different I-CSCF.

If the P-CSCF fails to forward the standalone SIP request to any I-CSCF, the P-CSCF shall send back a 504 (Server Time-Out) response to the UE performing the functions of an external attached network, in accordance with the procedures in RFC 3261 [26].

5.2.6.3.8 Responses to a request for a standalone transaction

When the P-CSCF receives any response to the above request, the P-CSCF shall:

1) store the values received in the P-Charging-Function-Addresses header field;

before forwarding the response to the UE in accordance with the procedures of RFC 3261 [26].

NOTE: The P-CSCF can find the IMS communication service supported for the standalone transaction, as determined by the originating home network, in the topmost occurance of the "+g.3gpp.icsi-ref" header field parameter of the Feature-Caps header field(s) of 18x or 2xx response. The information can be used for charging purpose.

The P-CSCF shall forward the response to the UE using the mechanisms described in RFC 3261 [26] and RFC 3581 [56A], i.e. the P-CSCF shall send the response to the IP address indicated in the "received" header field parameter and, in case UDP is used, to the port indicated in the "rport" header field parameter (if present) of the Via header field associated with the UE. In case TCP is used, the P-CSCF shall use the TCP connection on which the REGISTER request was received for sending the response back to the UE.

5.2.6.3.9 Subsequent request other than a target refresh request

When the P-CSCF receives from the UE a subsequent request other than a target refresh request (including requests relating to an existing dialog where the method is unknown), the P-CSCF shall:

1) verify if the request relates to a dialog in which the originator of the request is involved:

   a) if the request does not relates to an existing dialog in which the originator is involved, then the P-CSCF shall answer the request by sending a 403 (Forbidden) response back to the originator. The P-CSCF will not forward the request. No other actions are required; or

   b) if the request relates to an existing dialog in which the originator is involved, then the P-CSCF shall continue with the following steps;

1A) remove its own SIP URI from the top of the list of Route header fields;

2) verify that the resulting list of Route header fields matches the list of Record-Route header fields constructed by inverting the order of the stored list of Record-Route header fields and removing its Record-Route header field from the list. This verification is done on a per URI basis, not as a whole string. If the verification fails, then the P-CSCF shall either:

   a) return a 400 (Bad Request) response; the P-CSCF shall not forward the request, and shall not continue with the execution of steps 3 onwards; or

   b) replace the Route header field value in the request with the list of Record-Route header fields constructed by inverting the order of the stored list of Record-Route header fields and removing its Record-Route header field from the list;

3) add a P-Charging-Vector header field with the "icid-value" header field parameter set to the value populated in the initial request for the dialog and a type 1 "orig-ioi" header field parameter. The P-CSCF shall set the type 1 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The P-CSCF shall not include the type 1 "term-ioi" header field parameter; and

4) for INVITE dialogs, replace the saved CSeq header field value received in the request such that the P-CSCF is able to release the session if needed;

before forwarding the request (based on the topmost Route header field), in accordance with the procedures of RFC 3261 [26].

5.2.6.3.10 Responses to a subsequent request other than a target refresh request

Void

5.2.6.3.11 Request for an unknown method that does not relate to an existing dialog

When the P-CSCF receives from the UE the request for an unknown method (that does not relate to an existing dialog), and a service route value list exists for the served user of the request, the P-CSCF shall:

1) if the UE performing the functions of an external attached network using static mode of operation:

   i) select an I-CSCF and insert a Route header field with the URI of the I-CSCF as the topmost Route header field; otherwise

NOTE 1: The list of the I-CSCFs can be either obtained as specified in RFC 3263 [27A] or be provisioned in the P-CSCF.

ii) verify that the list of URIs received in the Service-Route header field (during the last successful registration or re-registration) is included, preserving the same order, as a subset of the preloaded Route header fields in the received request. This verification is done on a per URI basis, not as a whole string. If the verification fails, then the P-CSCF shall either:

a) return a 400 (Bad Request) response; the P-CSCF shall not forward the request, and shall not continue with the execution of steps 2 onwards; or

b) replace the Route header field value in the request with the one received during the last registration in the Service-Route header field of the 200 (OK) response;

NOTE 2: The P-CSCF can identify that a request is received from a UE performing the functions of an external attached network using static mode of operation by evaluating the TLS session or by other means.

2) if the 200 (OK) response to the last REGISTER request, which created or refreshed the binding of the contact address from which the request is received, has not contained a Feature-Caps header field, specified in RFC 6809 [190] with a "+g.3gpp.atcf" header field parameter, then:

a) if the P-CSCF is located in the visited network, and local policy requires the application of IBCF capabilities in the visited network towards the home network, then the P-CSCF shall select an IBCF in the visited network and add the URI of the selected IBCF to the topmost Route header field;

NOTE 3: It is implementation dependent as to how the P-CSCF obtains the address of the IBCF exit point.

2A) if the 200 (OK) response to the last REGISTER request, which created or refreshed the binding of the contact address from which the request is received, contained a Feature-Caps header field with a "+g.3gpp.atcf" header field parameter, then:

a) add the ATCF URI for originating requests, that the P-CSCF used to forward the last REGISTER request which created or refreshed the binding of the contact address from which the request is received, to the topmost Route header field;

2B) if the request is originated from a UE which the P-CSCF considers as privileged sender (including one which is also a UE performing the functions of an external attached network using static mode of operation), keep the P-Asserted-Identity header field unchanged if one was received, or include the originator of the request in the P-Asserted-Identity header field if no P-Asserted-Identity header field was received. In addition remove any P-Preferred-Identity header field, include the served user of the request in the P-Served-User header field as specified in RFC 5502 [133] and skip step 3) below;

NOTE 4: The P-CSCF determines if the UE is considered privileged sender using based on the user-related policies provisioned to the P-CSCF (see subclause 5.2.1).

NOTE 5: The P-CSCF can retrieve the identity of the UE performing the functions of an external attached network from the subjectCommonName (CN) if it is not present in the subjectAltName in the certificates during the TLS session setup in accordance with the procedures of RFC 5280 [213] or by other means.

2C) if the request is originated from a UE performing the functions of an external attached network using static mode of operation and which the P-CSCF considers as is not a privileged sender, include the served user of the request in the P-Served-User header field as specified in RFC 5502 [133] and skip step 3) below;

3) remove any P-Preferred-Identity header field or P-Asserted-Identity header field, if present, and insert a P-Asserted-Identity header fields the value identifying the originator of the request and the value of the alternative identity of the originator of the request, if identified (see subclause 5.2.6.3.1), including the display name if previously stored during registration, representing the served user of the request;

3A) if the identity of the served user of the request was taken from P-Preferred-Identity header field by matching a registered wildcarded public user identity, and the identity of the served user is not a distinct identity within the range of the wildcarded public user identity, include the wildcarded public user identity value in the P-Profile-Key header field as defined in RFC 5002 [97]; and

3B) if the request comes from a UE performing the functions of an external attached network using static mode of operation add the "orig" parameter to the dialog request to indicate that this is an originating request; and

4) add a P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17] and a type 1 "orig-ioi" header field parameter. The P-CSCF shall set the type 1 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The P-CSCF shall not include the type 1 "term-ioi" header field parameter. Based on local policy, the P-CSCF shall add an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier;

NOTE 6: The matching of distinct public user identities takes precedence over the matching of wildcarded public user identities.

before forwarding the request, based on the topmost Route header field, in accordance with the procedures of RFC 3261 [26].

If the request comes from a UE performing the functions of an external attached network using static mode of operation:

- no response is received to the standalone SIP request and its retransmissions by the P-CSCF; or
- a 3xx response or 480 (Temporarily Unavailable) response is received,

the P-CSCF shall repeat the actions of the above bullets with a different I-CSCF.

If the P-CSCF fails to forward the unknown SIP request to any I-CSCF, the P-CSCF shall send back a 504 (Server Time-Out) response to the UE performing the functions of an external attached network using static mode of operation, in accordance with the procedures in RFC 3261 [26].

5.2.6.3.12 Responses to a request for an unknown method that does not relate to an existing dialog

NOTE: The P-CSCF can find the IMS communication service supported for the transaction, as determined by the originating home network, in the topmost occurance of the "+g.3gpp.icsi-ref" header field parameter of the Feature-Caps header field(s) of 18x or 2xx response. The information can be used for charging purpose.

5.2.6.4 Requests terminated by the UE

5.2.6.4.1 General for all requests

The P-CSCF shall log all SIP requests destined for the UE that contain a "logme" Session-ID header field parameter based on local policy.

If the serving network supports PCRF based P-CSCF restoration and the Restoration-Info header field is included in the incoming request, and the P-CSCF has no binding for the identity in the Request-URI, the P-CSCF shall:

- initiate the PCRF based P-CSCF restoration procedure as specified in 3GPP TS 23.380 [7D] using the IMSI value contained in the Restoration-Info header field; and
- reject the request with a 404 (Not Found) response.

If the P-CSCF supports PCRF based P-CSCF restoration procedures, the P-CSCF shall remove the Restoration-Info header field, if included in the incoming request.

If the serving network supports HSS based P-CSCF restoration procedures and the P-CSCF has no binding for the identity in the Request-URI, the P-CSCF shall reject the request with a 404 (Not Found) response.

If the user-related policies provisioned to the P-CSCF (see subclause 5.2.1) do not indicate that the served UE is authorized to send early media, the P-CSCF shall not allow media flows in forward and backward direction before the 200 (OK) response to the initial INVITE is received. Based on operator policy the P-CSCF shall either remove the P-Early-Media header field or replace the value of the P-Early-Media header field with "inactive", if received from the terminating UE.

If the user-related policies provisioned to the P-CSCF (see subclause 5.2.1) indicate that the served UE is authorized to send early media, the P-CSCF shall not remove or modify the P-Early-Media header field if received in an UPDATE request.

5.2.6.4.2 General for all responses

When the P-CSCF receives, destined for the UE, a response sent on a dialog for which logging of signalling is in progress, the P-CSCF shall check whether a trigger for stopping logging of SIP signalling has occurred, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K]. If a stop trigger event has occurred then the P-CSCF shall stop treating the dialog as one for which logging of signalling is in progress, else the P-CSCF shall append a "logme" header field parameter to the SIP Session-ID header field if the parameter is missing and determine, by checking its trace configuration, whether to log the request.

If the user-related policies provisioned to the P-CSCF (see subclause 5.2.1) do not indicate that the served UE is authorized to send early media, the P-CSCF shall not allow media flows in forward and backward direction before the 200 (OK) response to the initial INVITE is received. Based on operator policy the P-CSCF shall either remove the P-Early-Media header field or replace the value of the P-Early-Media header field with "inactive", if received from the terminating UE.

If the user-related policies provisioned to the P-CSCF (see subclause 5.2.1) indicate that the served UE is authorized to send early media, the P-CSCF shall not remove or modify the P-Early-Media header field if received in a 18x provisional response.

5.2.6.4.3 Initial request for a dialog

When the P-CSCF receives, destined for the UE, an initial request for a dialog, prior to forwarding the request, the P-CSCF shall:

1) if an indication has been received from the PCRF that the signalling bearer to the UE is lost, and has not recovered, reject the request by sending 500 (Server Internal Error) response;

NOTE 1: The signalling bearer can be considered as recovered by the P-CSCF when the registration timer expires in P-CSCF and the user is de-registered from the IM CN subsystem, a new REGISTER request from the UE is received providing an indication to the P-CSCF that the signalling bearer to that user has become available or a P-CSCF implementation dependent function which discovers that the signalling bearer is available to the UE.

NOTE 2: The Retry-After header field value is set based on operator policy.

2) convert the list of Record-Route header field values into a list of Route header field values and save this list of Route header fields;

3) if the request is an INVITE request, save a copy of the Contact, CSeq and Record-Route header field values received in the request such that the P-CSCF is able to release the session if needed;

4) if a security association or TLS session exists, when adding its own SIP URI to the top of the list of Record-Route header fields and save the list, build the P-CSCF SIP URI in a format that contains the protected server port number of the security association or TLS session established from the UE to the P-CSCF and either:

   a) the P-CSCF FQDN that resolves to the IP address of the security association or TLS session established from the UE to the P-CSCF; or

   b) the P-CSCF IP address of the security association or TLS session established from the UE to the P-CSCF;

5) if SIP digest without TLS, NASS-IMS bundled authentication or GPRS-IMS-Bundled authentication is used, when adding its own SIP URI to the top of the list of Record-Route header fields and saving the list, build the P-CSCF URI in a format that contains an unprotected server port number where the P-CSCF expects subsequent requests from the UE;

6) if a security association or TLS session exists, when adding its own address to the top of the received list of Via header fields and save the list, build the P-CSCF Via header field entry in a format that contains the protected server port number of the security association or TLS session established from the UE to the P-CSCF and either:

   a) the P-CSCF FQDN that resolves to the IP address of the security association or TLS session established from the UE to the P-CSCF; or

   b) the P-CSCF IP address of the security association or TLS session established from the UE to the P-CSCF;

NOTE 3: The P-CSCF associates two ports, a protected client port and a protected server port, with each pair of security associations or TLS session. For details of the usage of the two ports see 3GPP TS 33.203 [19].

7) if SIP digest without TLS, NASS-IMS bundled authentication or GPRS-IMS-Bundled athentication is used, when adding its own address to the top of the received list of Via header fields and saving the list, build the P-CSCF Via header field entry in a format that contains an unprotected server port number where the P-CSCF expects responses to the current request from the UE;

7A) if the recipient of the request is understood from information saved during registration or from configuration to always send and receive private network traffic from this source, remove the P-Private-Network-Indication header field containing the domain name associated with that saved information;

8) store the values received in the P-Charging-Function-Addresses header field;

9) store the "icid-value" header field parameter and if present, the "orig-ioi" header field parameter received in the P-Charging-Vector header field;

10) if the request contains an "fe-identifier" header field parameter, based on local policy, store the content of the "fe-identifier" header field parameter of the P-Charging-Vector header field; and

11) save a copy of the P-Called-Party-ID header field;

before forwarding the request to the UE either in accordance with the procedures of RFC 3261 [26] or as specified in RFC 5626 [92].

If no security association exists between the P-CSCF and the UE performing the functions of an external attached network operating in static mode, the P-CSCF shall initiate a TLS session towards the UE performing the functions of an external attached network operating in static mode before sending the initial request in accordance with 3GPP TS 33.310 [19D].

NOTE 4: The P-CSCF can identify that a call is directed to a UE performing the functions of an external attached network operating in static mode by evaluating the Route header field, the Request URI or other means.

Once the TLS session is set up (using the certificates) the P-CSCF shall send the initial request for dialog over the secure connection to the UE performing the functions of an external attached network operating in static mode.

5.2.6.4.4 Responses to an initial request for a dialog

When the P-CSCF receives any 1xx or 2xx response to the above request, the P-CSCF shall:

0A) if the response is originated from a UE which the P-CSCF considers as privileged sender, remove any P-Preferred-Identity header field, and skip step 1) below;

NOTE: The P-CSCF determines if the UE is considered privileged sender using based on the user-related policies provisioned to the P-CSCF (see subclause 5.2.1).

1) remove any P-Preferred-Identity header field or P-Asserted-Identity header field, if present, and insert a P-Asserted-Identity header field with the saved public user identity from the P-Called-Party-ID header field that was received in the request, plus the display name if previously stored during registration, representing the originator of the response;

2) verify that the list of Via header fields matches the saved list of Via header fields received in the request corresponding to the same dialog, including the P-CSCF Via header field value. This verification is done on a per Via header field value basis, not as a whole string. If the verification fails, then the P-CSCF shall either:

   a) discard the response; or

   b) replace the Via header field values with those received in the request;

3) verify that the list of URIs received in the Record-Route header field of the request corresponding to the same dialog is included, preserving the same order, as a subset of the Record-Route header field list of this response. This verification is done on a per URI basis, not as a whole string.

   If the verification fails, then the P-CSCF shall either:

   a) discard the response; or

b) replace the Record-Route header field values with those received in the request, and if a security association or TLS session exists, add its own Record-Route entry with its own SIP URI with the port number where it awaits subsequent requests from the calling party. The P-CSCF shall include in the Record-Route header field either:

- the P-CSCF FQDN that resolves to its IP address; or
- the P-CSCF IP address.

The P-CSCF shall remove the "comp" SIP URI parameter from the Record-Route header field.

If the verification is successful, the P-CSCF shall rewrite its own Record-Route entry to its SIP URI in a format that contains the port number where it awaits subsequent requests from the calling party. The P-CSCF shall include in the Record-Route header field either:

a) the P-CSCF FQDN that resolves to its IP address; or

b) the P-CSCF IP address.

The P-CSCF shall remove the "comp" SIP URI parameter from the Record-Route header field;

When adding its SIP URI to the Record-Route header field, the P-CSCF shall also copy the flow token and the "ob" SIP URI parameter from the Route header field of the initial dialog-forming request destined for the UE to its SIP URI, if the Route header field contained these values;

4) store the dialog ID and associate it with the private user identity and public user identity involved in the session;

5) if the response corresponds to an INVITE request, save the Contact, To, From and Record-Route header field value received in the response such that the P-CSCF is able to release the session if needed. If the Contact header field in the response to the INVITE request contains a GRUU, the P-CSCF shall save the GRUU received in the Contact header field of the response and associate that GRUU with the contact address which was used to send the INVITE request or with the registration flow and the associated UE contact address which was used to send on which the INVITE request such that the P-CSCF is able to release the session if needed; and

6) include in the P-Charging-Vector header field:

- an "icid-value" header field parameter set to the value received in the request;
- the "orig-ioi" header field parameter, if received in the request; and
- a type 1 "term-ioi" header field parameter that identifies the sending network;
- if the P-CSCF has stored an "fe-identifier" header field parameter of the P-Charging-Vector header field, based on local policy, the stored "fe-identifier" header field parameter and include its own address or identifier in the "fe-addr" element of the "fe-identifier" header field parameter of the P-Charging-Vector header field.

before forwarding the response in accordance with the procedures of RFC 3261 [26].

When the P-CSCF receives any other response to the above request, the P-CSCF shall:

1) verify that the list of Via header fields matches the saved list of Via header fields received in the request corresponding to the same dialog, including the P-CSCF Via header field value. This verification is done on a per Via header field value basis, not as a whole string. If these lists do not match, then the P-CSCF shall either:

a) discard the response; or

b) replace the Via header field values with those received in the request; and

2) include in the P-Charging-Vector header field:

a) an "icid-value" header field parameter set to the value received in the request;

b) the "orig-ioi" header field parameter, if received in the request; and

c) a type 1 "term-ioi" header field parameter that identifies the sending network;

before forwarding the response in accordance with the procedures of RFC 3261 [26].

5.2.6.4.5 Target refresh request for a dialog

When the P-CSCF receives, destined for the UE, a target refresh request for a dialog, prior to forwarding the request, the P-CSCF shall:

1) if a security association or TLS session exists, add its own address to the top of the received list of Via header fields and save the list. The P-CSCF Via header field entry is built in a format that contains the protected server port number of the security association or TLS session established from the UE to the P-CSCF and either:

   a) the P-CSCF FQDN that resolves to the IP address of the security association or TLS session established from the UE to the P-CSCF; or

   b) the P-CSCF IP address of the security association or TLS session established from the UE to the P-CSCF;

NOTE 1: The P-CSCF associates two ports, a protected client port and a protected server port, with each pair of security associations or TLS session. For details of the usage of the two ports see 3GPP TS 33.203 [19].

2) if SIP digest without TLS, NASS-IMS bundled authentication or GPRS-IMS-Bundled authentication is used, when adding its own address to the top of the received list of Via header fields and saving the list, build the P-CSCF Via header field entry in a format that contains an unprotected server port number where the P-CSCF expects responses to the current request from the UE;

3) if a security association or TLS session exists, when adding its own SIP URI to the top of the list of Record-Route header fields and save the list, build the P-CSCF SIP URI in a format that contains the protected server port number of the security association or TLS session established from the UE to the P-CSCF and either:

   a) the P-CSCF FQDN that resolves to the IP address of the security association or TLS session established from the UE to the P-CSCF; or

   b) the P-CSCF IP address of the security association or TLS session established from the UE to the P-CSCF;

4) if SIP digest without TLS, NASS-IMS bundled authentication or GPRS-IMS-Bundled authentication is used, when adding its own SIP URI to the top of the list of Record-Route header fields and saving the list, build the P-CSCF URI in a format that contains an unprotected server port number where the P-CSCF expects subsequent requests from the UE;

5) for INVITE dialogs, replace the saved Contact and CSeq header field values received in the request such that the P-CSCF is able to release the session if needed;

6) if the request is destined to a UE performing the functions of an external attached network operating in static mode, send the request using the already established TLS session as described in subclause 5.2.6.4.3; and

NOTE 2: The replaced Contact header field value is valid only if a 1xx or 2xx response will be received for the request. In other cases the old value is still valid.

7) store the "orig-ioi" header field parameter received in the P-Charging-Vector header field if present;

before forwarding the request to the UE in accordance with the procedures of either RFC 3261 [26] or RFC 5626 [92].

5.2.6.4.6 Responses to a target refresh request for a dialog

When the P-CSCF receives a 1xx or 2xx response to the above request, the P-CSCF shall:

1) verify that the list of Via header fields matches the saved list of Via header fields received in the request corresponding to the same dialog, including the P-CSCF Via header field value. This verification is done on a per Via header field value basis, not as a whole string. If the verification fails, then the P-CSCF shall either:

   a) discard the response; or

   b) replace the Via header field values with those received in the request;

2) if a security association or TLS session exists, rewrite its own Record-Route entry to the same value as for the response to the initial request for the dialog and remove the "comp" SIP URI parameter;

3) replace the saved Contact header field value received in the response such that the P-CSCF is able to release the session if needed. If the Contact header field in the response to the target refresh request for a dialog contains a GRUU, the P-CSCF shall save the GRUU received in the Contact header field of the response and associate that GRUU with the contact address which was used to send the target refresh request or with the registration flow and the associated UE contact address which was used to send the target refresh request such that the P-CSCF is able to release the session if needed;

4) if the P-CSCF inserted the header field parameters into the Feature-Caps header field of the initial request for the dialog then when the response is forwarded in the same direction, the P-CSCF shall insert the header field parameters with the same parameter values in the Feature-Caps header field; and

5) include in the P-Charging-Vector header field:

   - an "icid-value" header field parameter set to the value populated in the initial request for the dialog;

   - the "orig-ioi" header field parameter, if received in the request; and

   - a type 1 "term-ioi" header field parameter that identifies the sending network;

before forwarding the response in accordance with the procedures of RFC 3261 [26].

When the P-CSCF receives any other response to the above request, the P-CSCF shall:

1) verify that the list of Via header fields matches the saved list of Via header fields received in the request corresponding to the same dialog, including the P-CSCF Via header field value. This verification is done on a per Via header field value basis, not as a whole string. If the verification fails, then the P-CSCF shall either:

   a) discard the response; or

   b) replace the Via header field values with those received in the request; and

2) if a security association or TLS session exists, rewrite the IP address and the port number of its own Record-Route entry to the IP address and the port number where it awaits subsequent requests from the calling party and remove the "comp" SIP URI parameter;

3) include in the P-Charging-Vector header field:

   a) an "icid-value" header field parameter set to the value populated in the initial request for the dialog;

   b) the "orig-ioi" header field parameter, if received in the request; and

   c) a type 1 "term-ioi" header field parameter that identifies the sending network;

before forwarding the response in accordance with the procedures of RFC 3261 [26].

5.2.6.4.7 Request for a standalone transaction

When the P-CSCF receives, destined for the UE, a request for a standalone transaction, or a request for an unknown method (that does not relate to an existing dialog), prior to forwarding the request, the P-CSCF shall:

1) if an indication has been received from the PCRF that the signalling bearer to the UE is lost, and has not recovered, reject the request by sending 500 (Server Internal Error) response;

NOTE 1: The signalling bearer can be considered as recovered by the P-CSCF when the registration timer expires in P-CSCF and the user is de-registered from the IM CN subsystem, a new REGISTER request from the UE is received providing an indication to the P-CSCF that the signalling bearer to that user has become available or a P-CSCF implementation dependent function which discovers that the signalling bearer is available to the UE.

NOTE 2: The Retry-After header field value is set based on operator policy.

2) if a security association or TLS session exists, add its own address to the top of the received list of Via header fields and save the list. The P-CSCF Via header field entry is built in a format that contains the protected server port number of the security association or TLS session established from the UE to the P-CSCF and either:

a) the P-CSCF FQDN that resolves to the IP address of the security association or TLS session established from the UE to the P-CSCF; or

b) the P-CSCF IP address of the security association or TLS session established from the UE to the P-CSCF;

NOTE 3: The P-CSCF associates two ports, a protected client port and a protected server port, with each pair of security associations or TLS session. For details of the usage of the two ports see 3GPP TS 33.203 [19].

3) if SIP digest without TLS, NASS-IMS bundled authentication or GPRS-IMS-Bundled authentication is used, when adding its own address to the top of the received list of Via header fields and saving the list, build the P-CSCF Via header field entry in a format that contains an unprotected server port number where the P-CSCF expects responses to the current request from the UE;

3A) if the recipient of the request is understood from information saved during registration or from configuration to always send and receive private network traffic from this source, remove the P-Private-Network-Indication header field containing the domain name associated with that saved information;

4) store the values received in the P-Charging-Function-Addresses header field;

5) store the "icid-value" header field parameter and if present, the "orig-ioi" header field parameter received in the P-Charging-Vector header field;

6) if the request contains an "fe-identifier" header field parameter, based on local policy, store the content of the "fe-identifier" header field parameter of the P-Charging-Vector header field; and

7) save a copy of the P-Called-Party-ID header field;

before forwarding the request to the UE either in accordance with the procedures of RFC 3261 [26] or as specified in RFC 5626 [92].

If no security association exists between the P-CSCF and the UE performing the functions of an external attached network operating in static mode, the P-CSCF shall initiate a TLS session towards the UE performing the functions of an external attached network operating in static mode before sending the standalone SIP request in accordance with 3GPP TS 33.310 [19D].

NOTE 4: The P-CSCF can identify that a call is directed to a UE performing the functions of an external attached network operating in static mode by evaluating the Route header field, the Request URI or other means.

Once the TLS session is set up (using the certificates) the P-CSCF shall send the standalone SIP request over the secure connection to the UE performing the functions of an external attached network operating in static mode.

5.2.6.4.8 Responses to a request for a standalone transaction

When the P-CSCF receives any response to the above request, the P-CSCF shall:

1) verify that the list of Via header fields matches the saved list of Via header fields received in the request corresponding to the same dialog, including the P-CSCF Via header field value. This verification is done on a per Via header field value basis, not as a whole string. If these lists do not match, then the P-CSCF shall either:

a) discard the response; or

b) replace the Via header field values with those received in the request;

1A) if the response is originated from a UE which the P-CSCF considers as privileged sender, remove any P-Preferred-Identity header field, and skip step 2) below;

NOTE: The P-CSCF determines if the UE is considered privileged sender using based on the user-related policies provisioned to the P-CSCF (see subclause 5.2.1).

2) remove any P-Preferred-Identity header field or P-Asserted-Identity header field, if present, and insert an P-Asserted-Identity header field with the saved public user identity from the P-Called-Party-ID header field of the request, plus the display name if previously stored during registration, representing the originator of the response; and

3) include in the P-Charging-Vector header field:

- an "icid-value" header field parameter set to the value received in the request;
- the "orig-ioi" header field parameter, if received in the request;
- a type 1 "term-ioi" header field parameter that identifies the sending network; and
- if the P-CSCF has stored an "fe-identifier" header field parameter of the P-Charging-Vector header field, based on local policy, the stored "fe-identifier" header field parameter and include its own address or identifier in the "fe-addr" element of the "fe-identifier" header field parameter of the P-Charging-Vector header field.

before forwarding the response in accordance with the procedures of RFC 3261 [26].

5.2.6.4.9 Subsequent request other than a target refresh request

When the P-CSCF receives, destined for the UE, a subsequent request for a dialog that is not a target refresh request (including requests relating to an existing dialog where the method is unknown), prior to forwarding the request, the P-CSCF shall:

1) if a security association or TLS session exists, add its own address to the top of the received list of Via header fields and save the list The P-CSCF Via header field entry is built in a format that contains the protected server port number of the security association or TLS session established from the UE to the P-CSCF and either:

   a) the P-CSCF FQDN that resolves to the IP address of the security association or TLS session established from the UE to the P-CSCF; or

   b) the P-CSCF IP address of the security association or TLS session established from the UE to the P-CSCF;

   NOTE: The P-CSCF associates two ports, a protected client port and a protected server port, with each pair of security associations or TLS session. For details of the usage of the two ports see 3GPP TS 33.203 [19].

2) if SIP digest without TLS, NASS-IMS bundled authentication or GPRS-IMS-Bundled authentication is used, when adding its own address to the top of the received list of Via header fields and saving the list, build the P-CSCF Via header field entry in a format that contains an unprotected server port number where the P-CSCF expects responses to the current request from the UE;

3) void;

4) for INVITE dialogs, replace the saved CSeq header field value received in the request such that the P-CSCF is able to release the session if needed;

5) if the request is destined to a UE performing the functions of an external attached network operating in static mode, send the request using the already established TLS session as described in subclause 5.2.6.4.3; and

6) store the "orig-ioi" header field parameter received in the P-Charging-Vector header field if present;

before forwarding the request to the UE either in accordance with the procedures of RFC 3261 [26] or as specified in RFC 5626 [92].

5.2.6.4.10 Responses to a subsequent request other than a target refresh request

When the P-CSCF receives any response to the above request, the P-CSCF shall:

1) verify that the list of Via header fields matches the saved list of Via header fields received in the request corresponding to the same dialog, including the P-CSCF Via header field value. This verification is done on a per Via header field value basis, not as a whole string. If these lists do not match, then the P-CSCF shall either:

   a) discard the response; or

   b) replace the Via header field values with those received in the request; and

2) include in the P-Charging-Vector header field:

   a) an "icid-value" header field parameter set to the value received in the initial request for the dialog;

   b) the "orig-ioi" header field parameter, if received in the request; and

c) a type 1 "term-ioi" header field parameter that identifies the sending network;

before forwarding the response in accordance with the procedures of RFC 3261 [26].

5.2.6.4.11 Request for an unknown method that does not relate to an existing dialog

Void.

5.2.6.4.12 Responses to a request for an unknown method that does not relate to an existing dialog

Void.

5.2.7 Initial INVITE

5.2.7.1 Introduction

In addition to following the procedures for initial requests defined in subclause 5.2.6, initial INVITE requests also follow the procedures of this subclause.

5.2.7.2 UE-originating case

When the P-CSCF receives from the UE an INVITE request for which resource authorization procedure is required, if it receives from the IP-CAN (e.g. via PCRF) an indication that the requested resources for the multimedia session being established cannot be granted and this indication does not provide an acceptable bandwidth information:

- if the P-CSCF is unable to handle further requests from the UE (i.e. P-CSCF is overloaded by SIP requests), the P-CSCF shall return a 503 (Service Unavailable) response to the received INVITE request. Depending on local operator policy, the 503 (Service Unavailable) response may include a Retry-After header field; and

- if the P-CSCF is able to handle further requests from the UE (i.e. P-CSCF is not overloaded by SIP requests), the P-CSCF shall return a 500 (Server Internal Error) response to the received INVITE request. Depending on local operator policy, the 500 (Server Internal Error) response may include a Retry-After header field.

When the P-CSCF receives from the UE an INVITE request, the P-CSCF may require the periodic refreshment of the session to avoid hung states in the P-CSCF. If the P-CSCF requires the session to be refreshed, then the P-CSCF shall apply the procedures described in RFC 4028 [58] clause 8.

NOTE 1: Requesting the session to be refreshed requires support by at least one of the UEs. This functionality cannot automatically be granted, i.e. at least one of the involved UEs needs to support it.

The P-CSCF shall respond to all INVITE requests with a 100 (Trying) provisional response.

If received from the IP-CAN, the P-CSCF shall also include the access-network-charging-info parameter (e.g. received via the PCRF, over the Rx or Gx interfaces) in the P-Charging-Vector header field in the first request originated by the UE that traverses the P-CSCF, as soon as the charging information is available in the P-CSCF, e.g., after the local resource reservation is complete. Typically, this first request is an UPDATE request if the remote UA supports the "integration of resource management in SIP" extension or a re-INVITE request if the remote UA does not support the "integration of resource management in SIP" extension. See subclause 5.2.7.4 for further information on the access network charging information.

If:

- the UE is roaming;

- the P-CSCF is not in the home network; and

- an agreement exists with the home network operator (as identified by the bottom most URI in the list of URIs received in the Service-Route header field during the last successful registration or re-registration) to support Roaming Architecture for Voice over IMS with Local Breakout;

the P-CSCF may:

- insert into the request a Feature-Caps header field with the "+g.3gpp.trf" header field parameter as specified in RFC 6809 [190]. Based on local policy the P-CSCF shall insert the "+g.3gpp.trf" header field parameter with the parameter value set to the URI of the desired TRF; and

- if a TRF URI is included in the "+g.3gpp.trf" header field parameter and the P-CSCF supports indicating the traffic leg associated with a URI as specified in RFC 7549 [225] and if required by local policy, append an "iotl" SIP URI parameter with a value set to "homeA-visitedA" to the TRF URI.

If:

- the UE is roaming;

- the P-CSCF is not in the home network; and

- the visited network supports MRB functionality for the allocation of MRF resources and if an agreement exists with the home operator (identified by the bottom most URI in the list of URIs received in the Service-Route header field during the last successful registration or re-registration) to provide access to MRF resources from the visited network;

the P-CSCF may insert into the request a Feature-Caps header field with the "+g.3gpp.mrb" header field parameter, as specified in RFC 6809 [190]. Based on local policy the P-CSCF shall insert the "+g.3gpp.mrb" header field parameter with the parameter value set to the URI of the desired MRB.

The P-CSCF (IMS-ALG) shall transparently forward a received Contact header field towards the UE when the Contact header field contains a GRUU or a media feature tag indicating a capability for which the URI can be used.

NOTE 2: One example of such a media feature tag is the isfocus media feature tag where the URI in the Contact header field is used by conference services to transport the temporary conference identity that can be used when rejoining an ongoing conference.

NOTE 3: Various mechanisms can be applied to recognize the need for priority treatment (e.g., based on the dialled digits). The exact mechanisms are left to national regulation and network configuration.

Based on the alternative mechanism to recognize the need for priority treatment, the P-CSCF shall insert the temporarily authorised Resource-Priority header field with appropriate namespace and priority value in the INVITE request.

When the P-CSCF responds to the UE with a 500 (Server Internal Error) response after receiving an indication that radio/bearer resources are not available, then based on operator policy, the P-CSCF may include a Reason header field with a protocol value set to "FAILURE_CAUSE" and a "cause" header field parameter set to "1" as specified in subclause 7.2A.18.12.2 and a Response-Source header field with a "fe" header field parameter set to "<urn:3gpp:fe:p-cscf.orig>".

If the P-CSCF supports the in-call access update procedure and if a received a response to the INVITE request contains a Feature-Caps header field with a g.3gpp.in-call-access-update feature capability indicator, the P-CSCF shall store the value of this feature capability indicator.

5.2.7.3 UE-terminating case

When the P-CSCF receives an INVITE request destined for the UE the P-CSCF may require the periodic refreshment of the session to avoid hung states in the P-CSCF. If the P-CSCF requires the session to be refreshed, then the P-CSCF shall apply the procedures described in RFC 4028 [58] clause 8.

NOTE 1: Requesting the session to be refreshed requires support by at least one of the UEs. This functionality cannot automatically be granted, i.e. at least one of the involved UEs needs to support it in order to make it work.

When the P-CSCF receives an initial INVITE request destined for the UE, it will have a list of Record-Route header fields. Prior to forwarding the initial INVITE request, the P-CSCF shall respond to all INVITE requests with a 100 (Trying) provisional response.

If received from the IP-CAN, the P-CSCF shall also include the access-network-charging-info parameter (e.g. received via the PCRF, over the Rx or Gx interfaces) in the P-Charging-Vector header field in the first request or reliable response originated by the UE that traverses the P-CSCF, as soon as the charging information is available in the P-CSCF e.g., after the local resource reservation is complete. When the P-CSCF sends the response including P-Charging-

Vector header field, the P-CSCF shall set the "icid-value" header field parameter to the previously received value of "icid-value" header field parameter in the request. See subclause 5.2.7.4 for further information on the access network charging information.

The P-CSCF (IMS-ALG) shall transparently forward a received Contact header field towards the UE when the Contact header field contains a GRUU or a media feature tag indicating a capability for which the URI can be used.

NOTE 2: One example of such a media feature tag is the isfocus media feature tag where the URI in the Contact header field is used by conference services to transport the temporary conference identity that can be used when rejoining an ongoing conference.

If the P-CSCF supports the in-call access update procedure and if the received INVITE request contains a Feature-Caps header field with a g.3gpp.in-call-access-update feature capability indicator, the P-CSCF shall store the value of this feature capability indicator.

5.2.7.4 Access network charging information

The P-CSCF shall include the "access-network-charging-info" header field parameter within the P-Charging-Vector header field as described in subclause 7.2A.5.

5.2.8 Call release

5.2.8.1 P-CSCF-initiated call release

5.2.8.1.1 Cancellation of a session currently being established

Upon receipt of an indication that the signalling bearer is no longer available (e.g. an Rx interface message from PCRF), the P-CSCF shall cancel that dialog by applying the following steps:

1) if the P-CSCF serves the calling user of the session, send out a CANCEL request to cancel the INVITE request towards the terminating UE that includes:

   a) if a cause or error code was received from the entity controlling radio/bearer resources, a Reason header field, with an appropriate protocol value in the protocol field, and the "cause" header field parameter set to the received cause or error code; and

   b) if:

      - no cause or error code was received from the entity controlling radio/bearer resources; or

      - if the abort cause PS_TO_CS_HANDOVER was received over Rx from the entity controlling radio/bearer resources;

      a Reason header field containing a 503 (Service Unavailable) status code according to the procedures described in RFC 3261 [26] and RFC 3326 [34A]; and

2) if the P-CSCF serves the called user of the session, send out a 500 (Server Internal Error) response to the received INVITE request. If a cause or error code was received from the entity controlling radio/bearer resources the P-CSCF shall include a Reason header field, with a protocol value set to "FAILURE_CAUSE" in the "protocol" header field parameter as described in subclause 7.2A.18.12.2, and the "cause" header field parameter set to "2" as described in subclause 7.2A.18.12.2.

Upon receipt of an indication that QoS or bearer resources are no longer available for a media negotiated in a multimedia session currently being established (e.g. an Rx interface message from PCRF) and if no SIP message removing the media for which resources are no longer available is received within an operator defined time after the reception of the indication, the P-CSCF shall:

1) cancel that dialog by responding to the original INVITE request with a 500 (Server Internal Error) response. If a cause or error code was received from the entity controlling radio/bearer resources the P-CSCF shall include a Reason header field, with a protocol value set to "FAILURE_CAUSE" in the "protocol" header field parameter as described in subclause 7.2A.18.12.2, and the "cause" header field parameter set to "1" as described in subclause 7.2A.18.12.2; and

2) by sending out a CANCEL request to the INVITE request towards the terminating UE that includes a Reason header field containing a 503 (Service Unavailable) status code according to the procedures described in RFC 3261 [26] and RFC 3326 [34A].

5.2.8.1.2 Release of an existing session

Upon:

1) receipt of an indication that the radio/bearer resources are no longer available for a media negotiated in a session (e.g. an Rx interface message from PCRF) and if no SIP message removing the media for which resources are no longer available is received within an operator defined time after the reception of the indication;

2) receipt of an indication that the signalling bearer is no longer available (e.g. an Rx interface message from PCRF); or

3) detecting that the SDP offer conveyed in a SIP response contained parameters which are not allowed according to the local policy (as specified in the subclause 6.2);

the P-CSCF shall release the respective dialog by applying the following steps:

1) if the P-CSCF serves the calling user of the session, then the P-CSCF shall generate a BYE request destined for the called user based on the information saved for the related dialog, including:

   - a Request-URI, set to the stored Contact header field provided by the called user;
   - a To header field, set to the To header field value as received in the 200 (OK) response for the initial INVITE request;
   - a From header field, set to the From header field value as received in the initial INVITE request;
   - a Call-ID header field, set to the Call-Id header field value as received in the initial INVITE request;
   - a CSeq header field, set to the current CSeq value stored for the direction from the calling to the called user, incremented by one;
   - a Route header field, set to the routeing information towards the called user as stored for the dialog;
   - a Reason header field or Reason header fields that contains:

     a) if a cause or error code was received from the entity controlling radio/bearer resources, an appropriate protocol value in the protocol field, and the "cause" header field parameter set to the received cause or error code;

     b) if no cause or error code was received from the entity controlling radio/bearer resources, and if radio/bearer interface resources are no longer available, a 503 (Service Unavailable) response code;

     c) if no cause or error code was received from the entity controlling radio/bearer resources, and if the signalling bearer is no longer available, a 503 (Service Unavailable) response code;

     d) if no cause or error code was received from the entity controlling radio/bearer resources, and if a SDP offer conveyed in a SIP response contained parameters which are not allowed according to the local policy, a 488 (Not Acceptable Here) response code; and

     e) if the abort cause PS_TO_CS_HANDOVER was received over Rx from the entity controlling radio/bearer resources, a 503 (Service Unavailable) response code;

   - further header fields, based on local policy; and
   - send the generated BYE requests towards the called user;

2) if the P-CSCF serves the calling user of the session and upon detecting that the SDP offer conveyed in a SIP response contained parameters which are not allowed according to the local policy (as specified in the subclause 6.2), then the P-CSCF shall generate an additional BYE request destined for the calling user based on the information saved for the related dialog, including:

- a Request-URI, set to a contact address obtained from the stored Contact header field if provided by the calling user. If the stored Contact header field contains either a public or a temporary GRUU, the P-CSCF shall set the Request-URI either to:

  a) the stored UE IP address and the UE port associated with the respective GRUU, if the stored Contact header field contains either a public or a temporary GRUU and the bidirectional flow as defined in RFC 5626 [92] is not used for this session; or

  b) the UE IP address and UE port associated with the bidirectional flow that the P-CSCF uses to send the in-dialog requests toward the UE as defined in RFC 5626 [92];

- a To header field, set to the From header field value as received in the initial INVITE request;

- a From header field, set to the To header field value as received in the 200 (OK) response for the initial INVITE request;

- a Call-ID header field, set to the Call-Id header field value as received in the initial INVITE request;

- a CSeq header field, set to the current CSeq value stored for the direction from the called to the calling user, incremented by one;

- a Route header field, set to the routeing information towards the calling user as stored for the dialog;

- a Reason header field that contains a 488 (Not Acceptable Here) response code;

- further header fields, based on local policy; and

- send the BYE request either:

  a) to the contact address indicated in the Request-URI, if the dialog being released did not use the bidirectional flow to send the requests to the UE as defined in RFC 5626 [92]; or

  b) over the same flow that the P-CSCF uses to send the in-dialog requests toward the UE as defined in RFC 5626 [92];

3) If the P-CSCF serves the called user of the session, then the P-CSCF shall generate a BYE request destined for the calling user based on the information saved for the related dialog, including:

- a Request-URI, set to the stored Contact header field provided by the calling user;

- a To header field, set to the From header field value as received in the initial INVITE request;

- a From header field, set to the To header field value as received in the 200 (OK) response for the initial INVITE request;

- a Call-ID header field, set to the Call-Id header field value as received in the initial INVITE request;

- a CSeq header field, set to the current CSeq value stored for the direction from the called to the calling user, incremented by one;

- a Route header field, set to the routeing information towards the calling user as stored for the dialog;

- a Reason header field or Reason header fields that contains:

  a) if a cause or error code was received from the entity controlling radio/bearer resources, an appropriate protocol value in the protocol field, and the "cause" header field parameter set to the received cause or error code;

  b) if no cause or error code was received from the entity controlling radio/bearer resources, and if radio/bearer interface resources are no longer available, a 503 (Service Unavailable) response code;

  c) if no cause or error code was received from the entity controlling radio/bearer resources, and if a SDP offer conveyed in a SIP response contained parameters which are not allowed according to the local policy, a 488 (Not Acceptable Here) response code; and

  d) if the abort cause PS_TO_CS_HANDOVER was received over Rx from the entity controlling radio/bearer resources, a 503 (Service Unavailable) response code;

- further header fields, based on local policy; and
- send the generated BYE requests towards the calling user;

4) if the P-CSCF serves the called user of the session and upon detecting that the SDP offer conveyed in a SIP response contained parameters which are not allowed according to the local policy (as specified in the subclause 6.2), then the P-CSCF shall generate an additional BYE request destined for the called user based on the information saved for the related dialog, including:

- a Request-URI, set to a contact address obtained from the stored Contact header field if provided by the called user. If the stored Contact header field contains either a public or a temporary GRUU, the P-CSCF shall set the Request-URI either to:

  a) the stored UE IP address and the UE port associated with the respective GRUU, if the stored Contact header field contains either a public or a temporary GRUU and the bidirectional flow as defined in RFC 5626 [92] is not used for this session; or

  b) the UE IP address and the UE port associated with the bidirectional flow that the P-CSCF uses to send the in-dialog requests toward the UE as defined in RFC 5626 [92];

- a To header field, set to the To header field value as received in the 200 (OK) response for the initial INVITE request;
- a From header field, set to the From header field value as received in the initial INVITE request;
- a Call-ID header field, set to the Call-Id header field value as received in the initial INVITE request;
- a CSeq header field, set to the current CSeq value stored for the direction from the calling to the called user, incremented by one;
- a Route header field, set to the routeing information towards the called user as stored for the dialog;
- a Reason header field that contains a 488 (Not Acceptable Here) response code;
- further header fields, based on local policy; and
- send the BYE request either:

  a) to the contact address indicated in the Request-URI, if the dialog being released did not use t the bidirectional flow to send the requests to the UE as defined in RFC 5626 [92]; or

  b) over the same flow that the P-CSCF uses to send the in-dialog requests toward the UE as defined in RFC 5626 [92].

Upon receipt of the 2xx responses for the BYE requests, the P-CSCF shall delete all information related to the dialog and the related multimedia session.

5.2.8.1.3 Abnormal cases

Upon receipt of a request on a dialog for which the P-CSCF initiated session release, the P-CSCF shall terminate this received request and answer it with a 481 (Call/Transaction Does Not Exist) response.

5.2.8.1.4 Release of the existing dialogs due to registration expiration and deletion of the security association, IP association or TLS session

If there are still active dialogs associated with the user after the security associations, IP association or TLS sessions were deleted, the P-CSCF shall discard all information pertaining to these dialogs without performing any further SIP transactions with the peer entities of the P-CSCF.

NOTE: If the interface between the P-CSCF and the IP-CAN is supported, the P-CSCF will also indicate (e.g. via the Rx or Gx interface) that the session has been terminated.

5.2.8.2 Call release initiated by any other entity

When the P-CSCF receives a 2xx response for a BYE request matching an existing dialog, then the P-CSCF shall delete all the stored information related to the dialog.

5.2.8.3 Session expiration

If the P-CSCF requested the session to be refreshed periodically, and the P-CSCF got the indication that the session will be refreshed, when the session timer expires, the P-CSCF shall delete all the stored information related to the dialog.

NOTE: If the interface between the P-CSCF and the IP-CAN is supported, the P-CSCF will also indicate to the IP-CAN (e.g. via the Rx or Gx interface), that the session has terminated.

5.2.9 Subsequent requests

5.2.9.1 UE-originating case

The P-CSCF shall respond to all reINVITE requests with a 100 (Trying) provisional response.

For a reINVITE request or UPDATE request from the UE within the same dialog, the P-CSCF shall include the updated access-network-charging-info parameter from P-Charging-Vector header field when sending the SIP request to the S-CSCF. See subclause 5.2.7.4 for further information on the access network charging information.

For an ACK request from the UE sent on a dialog where a 200 (OK) has been received, the P-CSCF shall include the access-network-charging-info parameter from the P-Charging-Vector header field when updated access-network-charging-info is available when sending the ACK request to the S-CSCF. See subclause 5.2.7.4 for further information on the access network charging information.

If priority is supported, the P-CSCF shall adjust the priority treatment of transactions or dialogs according to the most recently received authorized Resource-Priority header field value.

NOTE: A UE-initiated priority upgrade request directly from the UE to the P-CSCF is not supported using a special dial string as described in clause 4.11.

5.2.9.2 UE-terminating case

The P-CSCF shall respond to all reINVITE requests with a 100 (Trying) provisional response.

For a reINVITE request or UPDATE request destined towards the UE within the same dialog, when the P-CSCF sends 200 (OK) response (to the INVITE request or UPDATE request) towards the S-CSCF, the P-CSCF shall include the updated access-network-charging-info parameter in the P-Charging-Vector header field. See subclause 5.2.7.4 for further information on the access network charging information.

If priority is supported, the P-CSCF shall adjust the priority treatment of transactions or dialogs according to the most recently received authorized Resource-Priority header field value and set the backwards indication accordingly.

5.2.10 Emergency service

5.2.10.1 General

If the P-CSCF belongs to a network where the registration is not required to obtain emergency service, the P-CSCF shall accept any unprotected request on the IP address and port advertised to the UE during the P-CSCF discovery procedure. The P-CSCF shall also accept any unprotected request on the same IP address and the default port as specified in RFC 3261 [26].

When the P-CSCF sends unprotected responses to the UE, it shall use the same IP address and port where the corresponding request was received.

The P-CSCF can handle emergency session and other requests from both a registered user as well as an unregistered user. Certain networks only allow emergency session from registered users.

NOTE 1: If only emergency setup from registered users is allowed, a request from an unregistered user is ignored since it is received outside of the security association, TLS session or IP association.

The P-CSCF can handle emergency session establishment within a non-emergency registration, i.e. one that did not contain the "sos" SIP URI parameter in the Contact header field of the 200 (OK) response.

If the network uses the Resource-Priority header field to control the priority of emergency calls, and the P-CSCF receives a REGISTER request containing an "sos" SIP URI parameter in the Contact header field, the P-CSCF shall, in addition to the normal handling of the REGISTER request, add a Resource-Priority header field containing a namespace of "esnet" as defined in RFC 7135 [197] to the REGISTER request.

Upon receiving the 200 (OK) response to the REGISTER request that completes the emergency registration, as identified by the presence of the "sos" SIP URI parameter in the Contact header field of the 200 (OK) response, the P-CSCF shall not subscribe to the registration event package for any emergency public user identity specified in the REGISTER request.

Upon reception of a REGISTER request containing an "sos" SIP URI parameter in the Contact header field and not containing an Authorization header field, if:

1) the network supports IMS Services for roaming users in deployments without IMS-level roaming interfaces;

2) the UE is roaming; and

3) there is no II-NNI to the HPLMN of the served user;

NOTE 2: The P-CSCF can determine whether the UE is roaming by analysing the home network domain name of the user received in the Request-URI in the REGISTER request.

NOTE 3: The P-CSCF can know if II-NNI to the HPLMN of the served user is supported by analysing the home network domain name of the user received in the Request-URI in the REGISTER request.

or:

1) if required by operator policy; and

2) the UE is not roaming;

the P-CSCF:

1) shall not forward the REGISTER request; and

2) if the PCRF is used to retrieve the EPS-level identities (i.e., IMSI, IMEI(SV)) as specified in 3GPP TS 29.214 [13D] and IMSI is retrieved:

   a) if the P-CSCF supports IMSI or IMEI verification upon reception of a REGISTER request without Authorization header;

      i) if the IMSI derived from public user identity conveyed in To header is different from the IMSI received from PCRF, shall reject the REGISTER request by returning a 403 (Forbidden) response and shall not perform the rest of steps; and

NOTE 4: The P-CSCF can also derive IMSI from derived private user identity. The private user identity can be derived from the public user identity being registered by removing URI scheme and the following parts of the URI if present: port number, URI parameters, and To header field parameters.

      ii) if the IMEI(SV) is retrieved from the PCRF and the TAC and SNR portions of the IMEI obtained from instance ID conveyed in Contact header field are different from the TAC and SNR portions of IMEI(SV) received from PCRF, reject the REGISTER request by returning a 403 (Forbidden) response and shall not perform the rest of steps;

   b) if MSISDN is retrieved:

      i) shall generate:

         - a SIP URI with user=phone for the retrieved MSISDN; and

         - a tel URI for the retrieved MSISDN;

and shall include the URIs in the associated set of implicitly registered public user identities bound to the contact address from which the REGISTER request was received; and

ii) shall treat the SIP URI with user=phone for the retrieved MSISDN as the default public user identity for requests received from the contact address from which the REGISTER request was received;

c) if MSISDN is not retrieved:

i) shall generate a temporary public user identity for the IMSI retrieved from the PCRF as specified in 3GPP TS 29.214 [13D] and shall include the temporary public user identity in the associated set of implicitly registered public user identities bound to the contact address from which the REGISTER request was received; and

ii) shall treat the temporary public user identity for the retrieved IMSI as the default public user identity for requests received from the contact address from which the REGISTER request was received; and

NOTE 5: In the case when MSISDN is not retrieved, if the temporary public user identity is not provisioned in the HSS or is provisioned in the HSS, but barred, then a PSAP callback is not possible.

d) shall send a 200 (OK) response for the REGISTER request. In the 200 (OK) response, the P-CSCF shall include a P-Associated-URI header field containing the list of the implicitly registered public user identities bound to the contact address from which the REGISTER request was received. The first URI in the list of public user identities will indicate the default public user identity.

The P-CSCF shall store a configurable list of local emergency service identifiers, i.e. emergency numbers (the emergency numbers that can be resolved in the network to which the P-CSCF belongs) and emergency service URNs (i.e. emergency service URNs identifying emergency services that can be resolved in the network to which the P-CSCF belongs). In addition to the configurable list of local emergency service identifiers, the P-CSCF shall store a configurable list of roaming partners' emergency service identifiers (i.e. the emergency service numbers or the emergency service URNs identifying emergency services, which can be resolved in the roaming partners' network). Each emergency number in a configurable list is mapped to an emergency service URN if the network is configured, for the emergency number, to:

- accept a received request including the emergency number; or

- reject, using a 380 (Alternative Service) response, a received request including the emergency number, and include in the response a Contact header field with the emergency service URN.

NOTE 6: The emergency service URN is common to all networks, although subtypes might either not necessarily be in use, or a different set of subtypes is in use in different networks.

Access technology specific procedures are described in each access technology specific annex to determine the originating network of the requests.

NOTE 7: Depending on local operator policy, the P-CSCF has the capability to reject requests relating to specific methods in accordance with RFC 3261 [26], as an alternative to the functionality described above.

5.2.10.2 General treatment for all dialogs and standalone transactions excluding the REGISTER method – requests from an unregistered user

If the P-CSCF receives an initial request for a dialog or standalone transaction, or an unknown method from an unregistered user on the IP address and the unprotected port advertised to the UE during the P-CSCF discovery or the SIP default port, the P-CSCF shall inspect the Request-URI independent of values of possible entries in the received Route header fields for emergency service identifiers. The P-CSCF shall consider the Request URI of the initial request as an emergency service identifier, if it is an emergency number or an emergency service URN in the list of local emergency service identifiers or in the list of roaming partners emergency service identifiers.

If the Request-URI is a service URN with a top-level service type of "sos" as specified in RFC 5031 [69] and the P-CSCF does not consider the Request URI of the initial request as an emergency service identifier, the P-CSCF may:

- remove the right most service identifier and re-inspect the Request-URI for emergency service identifiers; or

- set the Request-URI to an operator defined emergency service URN that matches one of the emergency service identifiers.

If the P-CSCF detects that the Request-URI of the initial request for a dialog or a standalone transaction, or an unknown method matches one of the emergency service identifiers, the P-CSCF:

1) shall include in the Request-URI an emergency service URN, i.e. a service URN with a top-level service type of "sos" in accordance with RFC 5031 [69]:

   a) if the received Request-URI matches an emergency service URN, as received in the Request-URI from the UE; and

   b) if the received Request-URI does not match an emergency service URN, as deduced from the Request-URI received from the UE;

NOTE 1: Bullet b) can happen if a request is received from a UE not following the procedures in the present document.

2) shall include a topmost Route header field set to the URI associated with an E-CSCF;

NOTE 2: How the list of E-CSCF is obtained by the P-CSCF is implementation dependent.

3) shall execute the procedure described in subclause 5.2.6.3.3, subclause 5.2.6.3.7, subclause 5.2.6.3.11 and subclause 5.2.7.2, as appropriate except for:

   - verifying the preloaded route against the received Service-Route header field;

   - routing to IBCF;

   - removing the P-Preferred-Identity header field;

   - inserting a P-Asserted-Identity header field; and

   - inserting a type 1 "orig-ioi" header field parameter in the P-Charging-Vector header field;

3A) void;

3B) where the network uses the Resource-Priority header field to control the priority of emergency calls, shall add a Resource-Priority header field containing a namespace of "esnet" as defined in RFC 7135 [197];

4) if the P-CSCF detects that the UE is behind a NAT, and the UE's Via header field contains a "keep" header field parameter, shall add a value to the parameter, to indicate that it is willing to receive keep-alives associated with the dialog from the UE, as defined in RFC 6223 [143]; and

5) if required by operator policy (e.g. when the network supports IMS services for roaming users in deployments without IMS-level roaming interfaces), and the P-CSCF supports including EPS-level identities (i.e. IMSI, IMEI(SV)) and MSISDN in a request from an unregistered user:

   a) shall attempt to retrieve from PCRF the EPS-level identities and MSISDN available for the IP-CAN session of the request:

   b) if a Subscription-Id AVP(s) as specified in 3GPP TS 29.214 [13D] with an MSISDN, an IMSI or both is(are) retrieved:

      i) shall remove from the request any P-Preferred-Identity header field;

      ii) if an MSISDN is retrieved, shall insert in the request a P-Asserted-Identity header field set to a tel URI carrying the MSISDN; and

      iii) if an IMSI is retrieved, shall insert in the request a P-Asserted-Identity header field set to a temporary public user identity derived from the IMSI, as defined in 3GPP TS 23.003 [3]; and

   c) if a User-Equipment-Info-Type AVP or a User-Equipment-Info-Extension AVP as specified in 3GPP TS 29.214 [13D] with an IMEI(SV) is retrieved where the TAC and SNR portions are different from the TAC and SNR portions of the IMEI received in the "+sip.instance" header field parameter of the Contact header field of the request and if according to operator policy, shall reject the request with 403 (Forbidden) response.

When the P-CSCF receives any 1xx or 2xx response to the above requests, the P-CSCF shall execute the appropriate procedure for the type of request described in subclause 5.2.6.3.4, subclause 5.2.6.3.8, and subclause 5.2.6.3.12, except

that the P-CSCF may rewrite the port number of its own Record-Route entry to an unprotected port where the P-CSCF wants to receive the subsequent incoming requests from the UE belonging to this dialog.

If the P-CSCF does not receive any response to the initial request for a dialog or standalone transaction or unknown method (including its retransmissions); or receives a 3xx response or 480 (Temporarily Unavailable) response to an initial request for a dialog or standalone transaction or an unknown method, the P-CSCF shall include a URI associated with a different E-CSCF in the topmost Route header field and forward the request.

When the P-CSCF received a subsequent request in the dialog from the UE, and the network uses the Resource-Priority header field to control the priority of emergency calls, the P-CSCF shall add a Resource-Priority header field containing a namespace of "esnet" as defined in RFC 7135 [197].

When the P-CSCF receives a target refresh request from the UE for a dialog, the P-CSCF shall execute the procedure described in subclause 5.2.6.3.5, except for inserting a type 1 "orig-ioi" header field parameter in the P-Charging-Vector header field.

When the P-CSCF receives from the UE subsequent requests other than a target refresh request (including requests relating to an existing dialog where the method is unknown), the P-CSCF shall execute the procedure described in subclause 5.2.6.3.9, except for inserting a type 1 "orig-ioi" header field parameter in the P-Charging-Vector header field.

When the P-CSCF receives any 1xx or 2xx response to the above requests, the P-CSCF shall execute the appropriate procedure for the type of request described in subclause 5.2.6.3.5 or subclause 5.2.6.3.9.

5.2.10.2A General treatment for all dialogs and standalone transactions excluding the REGISTER method – requests to an unregistered user

When the P-CSCF receives, destined for the UE, a target refresh request for a dialog, prior to forwarding the request, the P-CSCF shall execute the procedure described in step 5, the paragraph of subclause 5.2.6.4.5.

When the P-CSCF receives a 1xx or 2xx response to the above request the P-CSCF shall execute the procedure described in subclause 5.2.6.4.6, except for inserting type 1 "orig-ioi" and "term-ioi" header field parameters in the P-Charging-Vector header field.

When the P-CSCF receives any other response to the above request the P-CSCF shall execute the procedure described in steps in the paragraph of subclause 5.2.6.4.6 describing when the P-CSCF receives any other response to a target request, except for inserting type 1 "orig-ioi" and "term-ioi" header field parameters in the P-Charging-Vector header field.

When the P-CSCF receives, destined for the UE, a subsequent request for a dialog that is not a target refresh request (including requests relating to an existing dialog where the method is unknown), prior to forwarding the request, the P-CSCF shall execute the procedure described in steps 3 and 4 of subclause 5.2.6.4.9 describing when a P-CSCF receives a subsequent request.

When the P-CSCF receives any other response to the above request the P-CSCF shall execute the procedure described in steps in the paragraph of subclause 5.2.6.4.10 describing when the P-CSCF receives any other response to a subsequent request, except for inserting type 1 "orig-ioi" and "term-ioi" header field parameters in the P-Charging-Vector header field.

5.2.10.3 General treatment for all dialogs and standalone transactions excluding the REGISTER method after emergency registration

If the P-CSCF receives an initial request for a dialog, or a standalone transaction, or an unknown method, for a registered user over the security association, TLS session, or IP association that was created during the emergency registration, as identified by the presence of the "sos" SIP URI parameter in the Contact header field of the 200 (OK) response, the P-CSCF shall inspect the Request-URI independent of values of possible entries in the received Route header fields for emergency service identifiers. The P-CSCF shall consider the Request URI of the initial request as an emergency service identifier, if it is an emergency number or an emergency service URN from the configurable lists that are associated with:

- the country of the operator to which the P-CSCF belongs to; and

- for inbound roamers, the country from which the UE is roaming from. The P-CSCF determines the country to which the UE is belonging to based on the content of the P-Assserted-Identity header field which contains the home network domain name in a SIP URI belonging to the user.

If the P-CSCF detects that the Request-URI of the initial request for a dialog, or a standalone transaction, or an unknown method does not match any one of the emergency service identifiers in the associated lists, the P-CSCF shall either:

- reject the request by returning a 403 (Forbidden) response to the UE; or

- if the Request-URI is a service URN with a top-level service type of "sos" as specified in RFC 5031 [69]:

  1) the P-CSCF sets the Request-URI to an operator defined emergency service URN that matches one of the emergency service identifiers; or

  2) remove the right most service identifier and re-inspect the Request-URI for emergency service identifiers.

If the P-CSCF detects that the Request-URI of the initial request for a dialog, or a standalone transaction, or an unknown method matches one of the emergency service identifiers in the associated lists, the P-CSCF shall:

1) include in the Request-URI an emergency service URN, i.e. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69]:

   a) if the received Request-URI matches an emergency service URN, as received from the UE in the Request-URI; and

   b) if the received Request-URI does not match an emergency service URN, as deduced from the Request-URI received from the UE.

NOTE 1: Bullet b) can happen if a request is received from a UE not following the procedures in the present document.

1A) if the operator policy requires that emergency service requests are forwarded to the S-CSCF and the P-CSCF determines that the network to which the originating user is attached (see the IP-CAN specific annexes for the detailed procedure) is the network the P-CSCF is in and if the user is not roaming, then:

NOTE 2: The P-CSCF can know if the user is roaming by comparing the home network domain name of the user received in the Request-URI in the REGISTER request with its own domain name. If they are different the user is a roaming user.

   a) execute the procedure described in subclause 5.2.6.3.3, subclause 5.2.6.3.7, subclause 5.2.6.3.11 and subclause 5.2.7.2, as appropriate except for routing to IBCF;

   b) before the request is forwarded in the referenced procedures, include a bottom most Route header field set to the URI associated with an E-CSCF;

NOTE 3: It is implementation dependent as to how the P-CSCF obtains the list of E-CSCFs.

   c) afterwards upon receipt of a target refresh request or a subsequent request other than a target refresh request (including requests relating to an existing dialog where the method is unknown) for a dialog from the UE, execute the procedure described in subclause 5.2.6.3.5 and subclause 5.2.6.3.9; and

   d) afterwards upon receipt of any response from the UE to a target refresh request or a subsequent request other than a target refresh request (including requests relating to an existing dialog where the method is unknown) for a dialog, execute the procedure described in subclause 5.2.6.4.6 and subclause 5.2.6.4.10;

1B) if the condition for 1A) is not fulfilled then:

   a) execute the procedure described in subclause 5.2.6.3.3, subclause 5.2.6.3.7, subclause 5.2.6.3.11 and subclause 5.2.7.2, as appropriate except for:

      - verifying the preloaded route against the received Service-Route header field;

      - routing to IBCF; and

      - inserting a type 1 "orig-ioi" header field parameter in the P-Charging-Vector header field;

b) before the request is forwarded in the referenced procedures, remove all Route header fields and include a Route header field set to the URI associated with an E-CSCF;

NOTE 4: It is implementation dependent as to how the P-CSCF obtains the list of E-CSCFs.

c) afterwards upon receipt of a target refresh request or a subsequent request other than a target refresh request (including requests relating to an existing dialog where the method is unknown) for a dialog from the UE, execute the procedure described in subclause 5.2.6.3.5 and subclause 5.2.6.3.9, except for inserting a type 1 "orig-ioi" header field parameter in the P-Charging-Vector header field; and

d) afterwards upon receipt of any response from the UE to a target refresh request or a subsequent request other than a target refresh request (including requests relating to an existing dialog where the method is unknown) for a dialog, execute the procedure described in subclause 5.2.6.4.6 and subclause 5.2.6.4.10, except for inserting type 1 "orig-ioi" and "term-ioi" header field parameters in the P-Charging-Vector header field;

1C) if the request is from a UE that is not considered as privileged sender and if the alternative identity of the originator of the request was not identified (see subclause 5.2.6.3.1):

i) if the P-Asserted-Identity header field in the request to be sent contains a SIP URI and if a tel URI belongs to the set of implicitly registered public user identities that contains the SIP URI, add a second P-Asserted-Identity header field that contains the first tel URI of the implicitly registered public user identities; and

ii) if the P-Asserted-Identity header field in the request to be sent contains a tel URI, add a second P-Asserted-Identity header field that contains the first SIP URI of the implicitly registered public user identities that contains the tel URI;

2) if the request contains a Contact header field containing a GRUU the P-CSCF shall save the GRUU received in the Contact header field of the request and associate it with the UE IP address and UE port such that the P-CSCF is able to route target refresh request containing that GRUU in the Request-URI. The UE port used for the association is determined as follows:

- if IMS AKA or SIP digest with TLS is being used as a security mechanism, the UE protected server port for the security association on which the request was received; or

- if SIP digest without TLS, NASS-IMS bundled authentication or GPRS-IMS-Bundled Authentication is being used as a security mechanism, the UE unprotected port on which the request was received;

3) where the network uses the Resource-Priority header field to control the priority of emergency calls, add a Resource-Priority header field containing a namespace of "esnet" as defined in RFC 7135 [197]; and

4) if the P-CSCF supports calling number verification using signature verification and attestation information as specified in subclause 3.1 and if required by operator policy, the P-CSCF shall perform attestation of the user identity by inserting:

- a "verstat" tel URI parameter, specified in subclause 7.2A.20, to the tel URI or SIP URI with a user=phone parameter in the From header field or the P-Asserted-Identity header field;

- an Attestation-Info header field specified in subclause 7.2.18; and

- an Origination-Id header field, specified in subclause 7.2.19, set to a UUID identifying the P-CSCF which is configured based on local policy;

If the P-CSCF does not receive any response to an initial request for a dialog or standalone transaction or an unknown method sent to an E-CSCF (including its retransmissions); or receives a 480 (Temporarily Unavailable) response to an initial request for a dialog or standalone transaction or an unknown method sent to an E-CSCF, the P-CSCF shall include a URI, associated with a different E-CSCF, in the topmost Route header field and forward the request.

If the P-CSCF does not receive any response to an initial request for a dialog or standalone transaction or an unknown method sent to a S-CSCF (including its retransmissions); or receives a 480 (Temporarily Unavailable) response to an initial request for a dialog or standalone transaction or an unknown method sent to a S-CSCF, the P-CSCF shall include a URI, associated with a different E-CSCF, in the topmost Route header field of the initial request for a dialog or standalone transaction or an unknown method, and forward the request.

When the P-CSCF received a subsequent request in the dialog from the UE, and the network uses the Resource-Priority header field to control the priority of emergency calls, the P-CSCF shall add a Resource-Priority header field containing a namespace of "esnet" as defined in RFC 7135 [197].

When the P-CSCF receives a target refresh request for a dialog with the Request-URI containing a GRUU the P-CSCF shall:

- obtain the UE IP address and UE port associated to the GRUU contained in the Request-URI and rewrite the Request-URI with that UE IP address and UE port; and

- perform the steps in subclause 5.2.6.4.5 for when the P-CSCF receives, destined for the UE, a target refresh request for a dialog.

5.2.10.4 General treatment for all dialogs and standalone transactions excluding the REGISTER method - non-emergency registration

If the P-CSCF receives an initial request for a dialog, or a standalone transaction, or an unknown method, for a registered user, and the request is:

a) understood from saved or included information to relate to private network traffic (see subclause 5.2.6.3), and operator policy requires the P-CSCF to detect an emergency session request relating to private network traffic; or

b) not understood from saved or included information to relate to private network traffic (see subclause 5.2.6.3);

then the P-CSCF shall inspect the Request-URI independent of values of possible entries in the received Route header fields for emergency service identifiers. The P-CSCF shall consider the Request URI of the initial request as an emergency service identifier, if it is an emergency numbers or an emergency service URN from the configurable lists that are associated with:

- the country of the operator to which the P-CSCF belongs to;

- for inbound roamers, the country from which the UE is roaming from. The P-CSCF determines the country to which the UE is belonging to based on the content of the P-Assserted-Identity header field which contains the home network domain name in a SIP URI belonging to the user; and

- the country of roaming partners, if the request originates from a different country then the country of the network to which the P-CSCF belongs to. Access technology specific procedures are described in each access technology specific annex to determine from which country and roaming partner the request was originated. If the country from which the request originates can not be determined all lists are associated.

If the P-CSCF detects that the Request-URI of the initial request for a dialog, or a standalone transaction, or an unknown method matches one of the emergency service identifiers in the associated lists, the P-CSCF shall:

0A) determine the geographical location of the UE. Access technology specific procedures are described in each access technology specific annex:

   a) if the UE is roaming and the P-CSCF is in the home operator's network, or the SDP of the request describes CS media (see 3GPP TS 24.292 [8O]), then the P-CSCF:

      I) shall reject the request as specified in subclause 5.2.10.5;

   b) if the UE is roaming and the P-CSCF is in the same network where the UE is roaming, or the UE is not roaming, then the P-CSCF, depending on operator policies:

      I) may reject the request as specified in subclause 5.2.10.5; or

      II) may continue with the next steps;

NOTE 1: Roaming is when a UE is in a geographic area that is outside the serving geographic area of the home IM CN subsystem.

NOTE 2: Emergency service URN in the request-URI indicates for the network that the emergency call attempt is recognized by the UE.

1) include in the Request-URI an emergency service URN, i.e. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69], if necessary. If information on the type of emergency service is known include a sub-service type. The entry in the Request-URI that the P-CSCF includes shall be:

   - if the received Request-URI matches an emergency service URN, as received from the UE in the Request-URI; and

   - if the received Request-URI does not match an emergency service URN, as deduced from the Request-URI received from the UE;

1A) if operator policy requires that emergency service requests are forwarded to the S-CSCF and the P-CSCF determines that the network to which the originating user is attached (see the IP-CAN specific annexes for the detailed procedure) is the network the P-CSCF is in then:

   a) execute the procedure described in subclause 5.2.6.3.3, subclause 5.2.6.3.7, subclause 5.2.6.3.11 and subclause 5.2.7.2, as appropriate except for routing to IBCF;

   b) before the request is forwarded in the referenced procedures, include a bottom most Route header field set to the URI associated with an E-CSCF;

NOTE 3: It is implementation dependent as to how the P-CSCF obtains the list of E-CSCFs.

   c) afterwards upon receipt of a target refresh request or a subsequent request other than a target refresh request (including requests relating to an existing dialog where the method is unknown) for a dialog from the UE, execute the procedure described in subclause 5.2.6.3.5 and subclause 5.2.6.3.9; and

   d) afterwards upon receipt of any response from the UE to a target refresh request or a subsequent request other than a target refresh request (including requests relating to an existing dialog where the method is unknown) for a dialog, execute the procedure described in subclause 5.2.6.4.6 and subclause 5.2.6.4.10;

1B) if the condition for 1A) is not fulfilled then:

   a) execute the procedure described in subclause 5.2.6.3.3, subclause 5.2.6.3.7, subclause 5.2.6.3.11 and subclause 5.2.7.2, as appropriate except for:

      - verifying the preloaded route against the received Service-Route header field;

      - routing to IBCF; and

      - inserting a type 1 "orig-ioi" header field parameter in the P-Charging-Vector header field;

   b) before the request is forwarded in the referenced procedures, remove all Route header fields and include a Route header field set to the URI associated with an E-CSCF;

NOTE 4: It is implementation dependent as to how the P-CSCF obtains the list of E-CSCFs.

   c) afterwards upon receipt of a target refresh request or a subsequent request other than a target refresh request (including requests relating to an existing dialog where the method is unknown) for a dialog from the UE, execute the procedure described in subclause 5.2.6.3.5 and subclause 5.2.6.3.9, except for inserting a type 1 "orig-ioi" header field parameter in the P-Charging-Vector header field; and

   d) afterwards upon receipt of any response from the UE to a target refresh request or a subsequent request other than a target refresh request (including requests relating to an existing dialog where the method is unknown) for a dialog, execute the procedure described in subclause 5.2.6.4.6 and subclause 5.2.6.4.10, except for inserting type 1 "orig-ioi" and "term-ioi" header field parameters in the P-Charging-Vector header field;

1C) if the request is from a UE that is not considered as privileged sender and if the alternative identity of the originator of the request was not identified (see subclause 5.2.6.3.1):

   i) if the P-Asserted-Identity header field in the request to be sent contains a SIP URI and if a tel URI belongs to the set of implicitly registered public user identities that contains the SIP URI, add a second P-Asserted-Identity header field that contains the first tel URI of the implicitly registered public user identities; and

   ii) if the P-Asserted-Identity header field in the request to be sent contains a tel URI, add a second P-Asserted-Identity header field that contains the first SIP URI of the implicitly registered public user identities that contains the tel URI;

2) if the request contains a Contact header field containing a GRUU the P-CSCF shall save the GRUU received in the Contact header field of the request and associate it with the UE IP address and UE port such that the P-CSCF is able to route target refresh request containing that GRUU in the Request-URI. The UE port used for the association is determined as follows:

   - if IMS AKA or SIP digest with TLS is being used as a security mechanism, the UE protected server port for the security association on which the request was received; or

   - if SIP digest without TLS is being used as a security mechanism, the UE unprotected port on which the request was received; and

3) where the network uses the Resource-Priority header field to control the priority of emergency calls, add a Resource-Priority header field containing a namespace of "esnet" as defined in RFC 7135 [197].

If the P-CSCF does not receive any response to the initial request for a dialog or standalone transaction or an unknown method sent to an E-CSCF (including its retransmissions); or receives a 480 (Temporarily Unavailable) response to an initial request for a dialog or standalone transaction or an unknown method sent to an E-CSCF, the P-CSCF shall include a URI, associated with a different E-CSCF that has not been tried before for this initial request for the dialog or standalone transaction (including its retransmissions), in the topmost Route header field and forward the request.

If the P-CSCF does not receive any response to the initial request for a dialog or standalone transaction or an unknown method sent to a S-CSCF (including its retransmissions); or receives a 480 (Temporarily Unavailable) response to an initial request for a dialog or standalone transaction or an unknown method sent to a S-CSCF, the P-CSCF shall include a URI, associated with a different E-CSCF that has not been tried before for this initial request for the dialog or standalone transaction (including its retransmissions), in the topmost Route header field of the initial request for a dialog or standalone transaction or an unknown method, and forward the request.

If the P-CSCF:

- does not receive any response to this initial request for a dialog or standalone transaction or an unknown method (including its retransmissions); or receives a 3xx response or 480 (Temporarily Unavailable) response to an initial request for a dialog or standalone transaction or an unknown method, and if all E-CSCFs have been tried before for this initial request for the dialog or standalone transaction (including its retransmissions), the P-CSCF shall reject this request as specified in subclause 5.2.10.5;

- receives:

  1) any 4xx response other than a 480 (Temporarily Unavailable) response;

  2) any 5xx response;

  3) any 6xx response,

  and the entry in the Request-URI as received from the UE is not in accordance with RFC 5031 [69], then the P-CSCF shall reject this request as specified in subclause 5.2.10.5.

If the P-CSCF receives from the IP-CAN (e.g. via PCRF) an indication that the requested resources for the multimedia session being established cannot be granted and the entry in the Request-URI as received from the UE is not in accordance with RFC 5031 [69], then the P-CSCF shall:

- send a CANCEL request to cancel the request forwarded to the selected E-CSCF; and

- reject this request as specified in subclause 5.2.10.5.

When the P-CSCF received a subsequent request in the dialog from the UE, and the network uses the Resource-Priority header field to control the priority of emergency calls, the P-CSCF shall add a Resource-Priority header field containing a namespace of "esnet" as defined in RFC 7135 [197].

When the P-CSCF receives a target refresh request for a dialog with the Request-URI containing a GRUU the P-CSCF shall:

- obtain the UE IP address and UE port associated to the GRUU contained in the Request-URI and rewrite the Request-URI with that UE IP address and UE port; and

- perform the steps in subclause 5.2.6.4 for when the P-CSCF receives, destined for the UE, a target refresh request for a dialog.

5.2.10.5 Abnormal and rejection cases

If the IM CN subsystem to where the P-CSCF belongs to is not capable to handle emergency sessions or due to local policy does not handle emergency sessions or only handles certain type of emergency session request or does not support emergency sessions for either the geographical location of the UE is located or the IP-CAN to which the UE is attached, or the SDP of the request describes CS media (see 3GPP TS 24.292 [8O]), or for reasons described in subclause 5.2.10.4, the P-CSCF shall not forward the initial request for a dialog or standalone transaction or an unknown method. The P-CSCF:

I) shall reject the request by returning a 380 (Alternative Service) response;

II) if:

- support for the 3GPP IM CN subsystem XML body as described in subclause 7.6 in the Accept header field is not indicated, the P-CSCF shall assume that the UE supports version 1 of the 3GPP XML Schema for the IM CN subsystem XML; or

- if both the "sv" and "schemaversion" parameters are present, then the P-CSCF shall ignore the value of the "schemaversion" parameter;

III) shall include in the 380 (Alternative Service) response:

a) a Content-Type header field with the value set to associated MIME type of the 3GPP IM CN subsystem XML body as described in subclause 7.6.1;

b) a P-Asserted-Identity header field set to the value of the SIP URI of the P-CSCF included in the Path header field during the registration of the user whose UE sent the request causing this response (see subclause 5.2.2.1); and

c) if required by operator policy implementing national regulatory requirements, a Contact header field with an emergency service URN (i.e. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69]). If a type of emergency service can be deduced from the Request-URI received from the UE and if required by operator policy implementing national regulatory requirements, the P-CSCF shall include in the emergency service URN a sub-service type deduced from the Request-URI received from the UE; and

NOTE 1: If the Request-URI identifies an emergency service with a type of emergency service, and the 380 (Alternative Service) response does not contain a Contact header field with an emergency service URN or contains a Contact header field with an emergency service URN which does not include a sub-service type, and if, upon reception of the response, the UE performs the emergency call attempt in the IM CN subsystem, then the emergency call attempt in the IM CN subsystem can be misrouted.

IV) shall include an IM CN subsystem XML body with the following elements:

a) an <ims-3gpp> element with the "version" attribute set to "1" and with an <alternative-service> child element, set to the parameters of the alternative service;

i) a <type> child element, set to "emergency" (see table 7.6.2) to indicate that it was an emergency call;

ii) a <reason> child element, set to an operator configurable reason; and

iii) an <action> child element, set to "emergency-registration" (see table 7.6.3) if the P-CSCF is accordingly configured by the operator.

NOTE 2: Emergency service URN in the request-URI indicates for the network that the emergency call attempt is recognized by the UE.

NOTE 3: Some networks only allow session requests with a Request-URI containing an emergency service URN, i.e. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69].

Upon reception of a REGISTER request containing an "sos" SIP URI parameter in the Contact header field and containing an Authorization header field, if:

1) the network supports IMS Services for roaming users in deployments without IMS-level roaming interfaces;

2) required by operator policy;

3) the UE is roaming; and

4) there is no II-NNI to the HPLMN of the served user;

NOTE 4: The P-CSCF can determine whether the UE is roaming by analysing the home network domain name of the user received in the Request-URI in the REGISTER request.

NOTE 5: The P-CSCF can know if II-NNI to the HPLMN of the served user is supported by analysing the home network domain name of the user received in the Request-URI in the REGISTER request.

or:

1) if required by operator policy; and

2) the UE is not roaming;

the P-CSCF:

1) if the P-CSCF supports GPRS-IMS-Bundled authentication, shall reject the request by returning a 420 (Bad Extension) response in which the Unsupported header field contains the value "sec-agree"; and

2) if the P-CSCF does not support GPRS-IMS-Bundled authentication, shall reject the request by returning a 403 (Forbidden) response.

When the P-CSCF responds 420 (Bad Extension) or 403 (Forbidden) response, if required by operator policy implementing national regulatory requirements (i.e., the network support an emergency session for an unregistered user as described in subclause 5.2.10.2), the P-CSCF shall include:

1) a Content-Type header field with the value set to associated MIME type of the 3GPP IM CN subsystem XML body as described in subclause 7.6.1;

2) a Content-Disposition header field with a disposition type "render" value and a "handling" header field parameter with an "optional" value, as described in RFC 3261 [26];

3) a P-Asserted-Identity header field set to the value of the SIP URI of the P-CSCF; and

4) a 3GPP IM CN subsystem XML body containing:

   a) an <ims-3gpp> element with the "version" attribute set to "1" and with an <alternative-service> child element, set to the parameters of the alternative service:

      i) a <type> child element, set to "emergency" (see table 7.6.2) to indicate that it was an emergency call;

      ii) a <reason> child element, set to an operator configurable reason; and

      iii) an <action> child element, set to "anonymous-emergencycall" (see table 7.6.3) if the P-CSCF is accordingly configured by the operator.

5.2.11 Void

5.2.12 Resource sharing

The P-CSCF supporting resource sharing shall perform the actions defined in access technology specific annexes.

5.2.13 Priority sharing

The P-CSCF supporting priority sharing shall perform the actions defined in access technology specific annexes.

5.2.14 In-call access update

If the P-CSCF supports the in-call access update procedure, has received the g.3gpp.in-call-access-update feature-capability indicator during session set-up, and determines that the UE has changed location, then the P-CSCF shall generate a MESSAGE request populated as follows:

NOTE: This procedure can be used for both reporting change of IP-CAN and reporting change of PLMN.

1) the Request URI set to the stored value of the g.3gpp.in-call-access-update feature-capability indicator;

2) a From header field set to the FQDN of the P-CSCF sending the request;

3) a To header field, set to the same value as the Request-URI;

4) a P-Asserted-Identity header field set to the default public user identity of the served user;

5) a P-Charging-Vector header field with the "icid-value" header field parameter populated with the ICID value used for the dialog related to the access change and a type 1 "orig-ioi" header field parameter. The P-CSCF shall set the type 1 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The P-CSCF shall not include the type 1 "term-ioi" header field parameter;

6) a P-Access-Network-Info header field including network-provided location information;

7) a P-Visited-Network-ID header field; and

8) if the MESSAGE request is to be sent via the S-CSCF, a Route header field with the S-CSCF address as received in the Service-Route header field during registration.

5.3 Procedures at the I-CSCF

5.3.0 General

When sending a failure response to any received request, depending on operator policy, the I-CSCF may insert a Response-Source header field with an "fe" header field parameter constructed with the URN namespace "urn:3gpp:fe", the fe-id part of the URN set to "i-cscf" and optionally an appropriate fe-param part of the URN set in accordance with subclause 7.2.17.

5.3.1 Registration procedure

5.3.1.1 General

During the registration procedure the I-CSCF shall behave as a stateful proxy.

5.3.1.2 Normal procedures

When the I-CSCF receives a REGISTER request, the I-CSCF shall verify whether or not it has arrived from a trusted domain. If the request has not arrived from a trusted domain, the I-CSCF shall complete the processing of the request by responding with 403 (Forbidden) response. Otherwise, the I-CSCF starts the user registration status query procedure to the HSS as specified in 3GPP TS 29.228 [14].

NOTE 1: The I-CSCF can find out whether the request arrived from a trusted domain or not, from the procedures described in 3GPP TS 33.210 [19A].

NOTE 2: Different UEs, each with its own private user identity, can register the same shared public user identity. Registrations of all public user identities belonging to these UEs are directed to the same S-CSCF as described in 3GPP TS 29.228 [14].

If the REGISTER request does not include an Authorization header field and private user identity, the I-CSCF shall derive the private user identity from the public user identity being registered, contained in the To header field, by removing URI scheme and the following parts of the URI if present: port number, URI parameters, and To header field parameters.

Prior to performing the user registration query procedure to the HSS, the I-CSCF decides which HSS to query, possibly as a result of a query to the Subscription Locator Functional (SLF) entity as specified in 3GPP TS 29.228 [14]. As a result of the query the I-CSCF gets the Redirect-Host AVP.

If the user registration status query response from the HSS includes a valid SIP URI, the I-CSCF shall:

1) replace the Request-URI of the received REGISTER request with the SIP URI received from the HSS in the Server-Name AVP;

2) optionally include in the P-User-Database header field defined in RFC 4457 [82]:

   a) either the received Redirect-Host AVP value; or

   b) the HSS Group ID using the form "aaa://hss.5gc.gid.<GID>.invalid", if the HSS Group ID is received following procedures in clause X.3; and

3) forward the REGISTER request to the indicated S-CSCF.

NOTE 3: The P-User-Database header field can be included only if the I-CSCF can assume (e.g. based on local configuration) that the receiving S-CSCF will be able to process the header field.

If the user registration status query response from the HSS includes a list of capabilities, the I-CSCF shall:

1) select a S-CSCF that fulfils the indicated mandatory capabilities – if more than one S-CSCFs fulfils the indicated mandatory capabilities the S-CSCF which fulfils most of the possibly additionally indicated optional capabilities;

2) replace the Request-URI of the received REGISTER request with the URI of the S-CSCF;

3) optionally, include in the P-User-Database header field defined in RFC 4457 [82]:

   a) either the received Redirect-Host AVP value; or

   b) the HSS Group ID using the form "aaa://hss.5gc.gid.<GID>.invalid", if the HSS Group ID is received following procedures in clause X.3; and

4) forward the REGISTER request to the selected S-CSCF.

NOTE 4: The P-User-Database header field can be included only if the I-CSCF can assume (e.g. based on local configuration) that the receiving S-CSCF will be able to process the header field.

NOTE 5: It is important that the I-CSCF does not alter the Via header field for requests and responses sent in the direction from the UE to the S-CSCF in the case of GPRS-IMS-Bundled authentication

When the I-CSCF receives a 2xx response to a REGISTER request, the I-CSCF shall forward the 2xx response to the P-CSCF.

5.3.1.3 Abnormal cases

In the case of SLF query, if the SLF does not send HSS address to the I-CSCF, the I-CSCF shall send back a 403 (Forbidden) response to the UE.

If the HSS sends a negative response to the user registration status query request, the I-CSCF shall send back a 403 (Forbidden) response.

If the user registration status query procedure cannot be completed, e.g. due to time-out or incorrect information from the HSS, the I-CSCF shall send back a 480 (Temporarily Unavailable) response to the UE.

If a selected S-CSCF:

- does not respond to the REGISTER request and its retransmissions by the I-CSCF; or

- sends back a 3xx response or 480 (Temporarily Unavailable) response to a REGISTER request;

and:

- the REGISTER request did not include an "integrity-protected" header field parameter in the Authorization header field;

- the REGISTER request did include an "integrity-protected" header field parameter in the Authorization header field with a value set to "no" in the Authorization header field;

- the REGISTER request did include an "integrity-protected" header field parameter in the Authorization header field with a value set to other than "no" and the I-CSCF supports S-CSCF restoration procedures; or

- the REGISTER request did not include an Authorization header field and the I-CSCF supports S-CSCF restoration procedures;

then:

- if the I-CSCF has received the list of capabilities from the HSS, the I-CSCF shall select a new S-CSCF as described in subclause 5.3.1.2, based on the capabilities indicated from the HSS. The newly selected S-CSCF shall not be one of any S-CSCFs selected previously during this same registration procedure; or

- if the I-CSCF has received a valid SIP URI from the HSS because the S-CSCF is already assigned to other UEs sharing the same public user identity, it will request the list of capabilities from the HSS and, on receiving these capabilities, the I-CSCF shall select a new S-CSCF as described in subclause 5.3.1.2, based on the capabilities indicated from the HSS. The newly selected S-CSCF shall not be one of any S-CSCFs selected previously during this same registration procedure.

NOTE 1: Checking for the inclusion of the Authorization header field is necessary to prevent S-CSCF reselection in the case of GPRS-IMS-Bundled authentication or NASS-IMS bundled authentication when no Authorization header field is present in case I-CSCF does not support S-CSCF restoration procedures.

NOTE 2: In case the S-CSCF does not respond, the I-CSCF can apply a pre-configured timer based on local policy before re-selecting a new S-CSCF.

When forwarding the REGISTER request to the new S-CSCF, the I-CSCF includes the SIP URI parameter "scscf-reselection" to the Request-URI of the REGISTER request.

If a selected S-CSCF does not respond to a REGISTER request and its retransmissions by the I-CSCF and none of the conditions specified above in this case are fulfilled, the I-CSCF shall send back a or 504 (Server Time-Out) response to the user, in accordance with the procedures in RFC 3261 [26].

If the I-CSCF cannot select a S-CSCF which fulfils the mandatory capabilities indicated by the HSS, the I-CSCF shall send back a 600 (Busy Everywhere) response to the user.

5.3.2 Initial requests

5.3.2.1 Normal procedures

The I-CSCF may behave as a stateful proxy for initial requests.

Upon receipt of a request, the I-CSCF shall perform the originating procedures as described in subclause 5.3.2.1A if the topmost Route header field of the request contains the "orig" parameter. Otherwise, the I-CSCF shall continue with the rest of the procedures of this subclause.

When the I-CSCF receives a request, the I-CSCF shall verify whether it has arrived from a trusted domain or not. If the request has arrived from a non trusted domain, then the I-CSCF shall remove all P-Charging-Vector header fields and all P-Charging-Function-Addresses header fields the request may contain.

NOTE 1: The I-CSCF can find out whether the request arrived from a trusted domain or not, from the procedures described in 3GPP TS 33.210 [19A].

For all SIP transactions identified:

- if priority is supported, as containing an authorised Resource-Priority header field, or, if such an option is supported, relating to a dialog which previously contained an authorised Resource-Priority header field;

the I-CSCF shall give priority over other transactions or dialogs. This allows special treatment of such transactions or dialogs. If priority is supported, the I-CSCF shall adjust the priority treatment of transactions or dialogs according to the most recently received authorized Resource-Priority header field or backwards indication value.

NOTE 2: The special treatment can included filtering, higher priority processing, routeing, call gapping. The exact meaning of priority is not defined further in this document, but is left to national regulation and network configuration.

The I-CSCF shall discard the P-Profile-Key header field, if the I-CSCF receives the P-Profile-Key header field in a SIP request or response.

When the I-CSCF receives, destined for a served user or a PSI, an initial request for a dialog or standalone transaction the I-CSCF shall:

1) if the Request-URI includes:

   a) a pres: or an im: URI, then translate the pres: or im: URI to a public user identity and replace the Request-URI of the incoming request with that public user identity; or

   b) a SIP-URI that is not a GRUU and with the user part starting with a + and the "user" SIP URI parameter equals "phone" then replace the Request-URI with a tel-URI with the user part of the SIP-URI in the telephone-subscriber element in the tel-URI, and carry forward the tel-URI parameters that may be present in the Request-URI; or

   c) a SIP URI that is a GRUU, then obtain the public user identity or an identity of the UE that represents the functionality within the UE that performs the role of registrar from the Request-URI and use it for location query procedure to the HSS. When forwarding the request, the I-CSCF shall not modify the Request-URI of the incoming request;

NOTE 3: SRV records have to be advertised in DNS pointing to the I-CSCF for pres: and im: queries.

2) remove its own SIP URI from the topmost Route header field, if present; and

3) check if the domain name of the Request-URI matches with one of the PSI subdomains configured in the I-CSCF. If the match is successful, the I-CSCF resolves the Request-URI by an internal DNS mechanism into the IP address of the AS hosting the PSI and does not start the user location query procedure. Otherwise, the I-CSCF will start the user location query procedure to the HSS as specified in 3GPP TS 29.228 [14] for the called PSI or user, indicated in or derived from the Request-URI. Prior to performing the user location query procedure to the HSS, the I-CSCF decides which HSS to query, possibly as a result of a query to the Subscription Locator Functional (SLF) entity as specified in 3GPP TS 29.228 [14].

When the I-CSCF receives any response to such a request, the I-CSCF shall store the value of the "term-ioi" header field parameter received in the P-Charging-Vector header field, if present.

NOTE 4: A received "term-ioi" header field parameter will be a type 3 IOI if received from an AS hosting a PSI or a type 2 IOI if received from the S-CSCF of the served user. The type 3 IOI identifies the service provider from which the response was sent and the type 2 IOI identifies the network from which the response was sent.

When the I-CSCF receives an INVITE request, the I-CSCF may require the periodic refreshment of the session to avoid hung states in the I-CSCF. If the I-CSCF requires the session to be refreshed, then the I-CSCF shall apply the procedures described in RFC 4028 [58] clause 8.

NOTE 5: Requesting the session to be refreshed requires support by at least one of the UEs. This functionality cannot automatically be granted, i.e. at least one of the involved UEs needs to support it.

In case the I-CSCF is able to resolve the Request-URI into the IP address of the AS hosting the PSI, then the I-CSCF shall:

1) store the value of the "icid-value" header field parameter received in the P-Charging-Vector header field and retain the "icid-value" header field parameter in the P-Charging-Vector header field. If no P-Charging-Vector header field was found, then insert the P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17]. The I-CSCF shall insert a type 3 "orig-ioi" header field parameter in place of any received "orig-ioi" header field parameter. The I-CSCF shall set the type 3 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The I-CSCF shall not include the type 3 "term-ioi" header field parameter. Based on local policy, the I-CSCF shall add an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier; and

2) forward the request directly to the AS hosting the PSI.

Upon successful user location query, when the response contains the URI of the assigned S-CSCF, or the URI of an AS hosting the PSI, the I-CSCF shall:

1) insert the URI received from the HSS as the topmost Route header field;

2) store the value of the "icid-value" header field parameter received in the P-Charging-Vector header field and retain the P-Charging-Vector header field in the P-Charging-Vector header field. If no "icid-value" header field parameter was found, then insert the P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17];

2A) based on local policy, add an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier;

3) optionally, include in the P-User-Database header field defined in RFC 4457 [82]:

   a) either the received Redirect-Host AVP value; or

   b) the HSS Group ID using the form "aaa://hss.5gc.gid.<GID>.invalid", if the HSS Group ID is received following procedures in clause X.3;

3A) if the Wildcarded Identity value is received from the HSS in the Wildcarded-Identity AVP and the I-CSCF supports the SIP P-Profile-Key private header extension, include the wildcarded identity value in the P-Profile-Key header field defined in RFC 5002 [97]; and

4) forward the request based on the topmost Route header field.

NOTE 6: The P-User-Database header field can be included only if the I-CSCF can assume (e.g. based on local configuration) that the receiving S-CSCF will be able to process the header field.

Upon successful user location query, when the response contains information about the required S-CSCF capabilities, the I-CSCF shall:

1) if overlap signalling using the multiple-INVITEs method is supported as a network option, and if the I-CSCF receives an INVITE request outside an existing dialog with the same Call ID and From header as a previous INVITE request during a certain period of time, route the new INVITE to the same next hop as the previous INVITE request; otherwise

2) select a S-CSCF according to the method described in 3GPP TS 29.228 [14];

3) insert the URI of the selected S-CSCF as the topmost Route header field value;

4) execute the procedure described in step 2 and 3 in the above paragraph (upon successful user location query, when the response contains the URI of the assigned S-CSCF);

5) optionally, include in the P-User-Database header field defined in RFC 4457 [82]:

   a) either the received Redirect-Host AVP value; or

   b) the HSS Group ID using the form "aaa://hss.5gc.gid.<GID>.invalid", if the HSS Group ID is received following procedures in clause X.3;

6) if the Wildcarded Identity value is received from the HSS in the Wildcarded-Identity AVP and the I-CSCF supports the SIP P-Profile-Key private header extension, include the wildcarded identity value in the P-Profile-Key header field as defined in RFC 5002 [97]; and

NOTE 7: A Wildcarded Identity can be either a PSI or a public user identity.

7) forward the request to the selected S-CSCF.

NOTE 8: The P-User-Database header field can be included only if the I-CSCF can assume (e.g. based on local configuration) that the receiving S-CSCF will be able to process the header field.

Upon an unsuccessful user location query when the response from the HSS indicates that the user does not exist, and if the Request-URI is a tel URI containing a public telecommunications number as specified in RFC 3966 [22], the I-CSCF may support a local configuration option that indicates whether or not request routeing is to be attempted. If the local configuration option indicates that request routeing is to be attempted, then the I-CSCF shall perform one of the following procedures based on local operator policy:

1) forward the request to the transit functionality for subsequent routeing; or

2) invoke the portion of the transit functionality that translates the public telecommunications number contained in the Request-URI to a routeable SIP URI, and process the request based on the result, as follows:

   a) if the translation fails, the request may be forwarded to a BGCF or any other appropriate entity (e.g. a MRFC to play an announcement) in the home network, or the I-CSCF may send an appropriate SIP response to the originator, such as 404 (Not Found) or 604 (Does not exist anywhere). When forwarding the request to a BGCF or any other appropriate entity, the I-CSCF shall leave the original Request-URI containing the tel URI unmodified:

      i) if overlap signalling using the multiple-INVITEs method is supported as a network option, and if the I-CSCF receives an INVITE request outside an existing dialog with the same Call ID and From header as a previous INVITE request during a certain period of time, the I-CSCF shall route the new INVITE to the same next hop as the previous INVITE request; and

      ii) additional procedures apply if the I-CSCF supports NP capabilities and these capabilities are enabled by local policy, and the database used for translation from an international public telecommunications number to a SIP URI also provides NP data (for example, based on the PSTN Enumservice as defined by RFC 4769 [114] or other appropriate data bases). If the above translation from an international public telecommunications number to a SIP URI failed, but NP data was obtained from the database, then the I-CSCF shall replace the tel-URI in the Request-URI with the obtained NP data, prior to forwarding the request to the BGCF or other appropriate entity. The URI is updated by the I-CSCF by adding the NP parameters defined by RFC 4694 [112] to the tel-URI in the Request-URI: an "npdi" tel-URI parameter is added to indicate that NP data retrieval has been performed, and if the number is ported, an "rn" tel-URI parameter is added to identify the ported-to routeing number. The I-CSCF shall perform these procedures if the tel-URI in the received Request-URI does not contain an "npdi" tel-URI parameter. In addition, the I-CSCF may, based on local policy, perform these procedures when the tel-URI in the received Request-URI contains an "npdi" tel-URI parameter indicating that the NP data has been previously obtained; or

NOTE 9: The I-CSCF might need to replace NP data added by a previous network if the previous network's NP database did not contain the local ported data for the called number. When the I-CSCF replaces the tel URI in the Request-URI with the obtained NP data, all tel URI parameters in the received Request-URI will be replaced by the obtained NP data.

   b) if this translation succeeds, then replace the Request-URI with the routeable SIP URI and process the request as follows:

      - determine the destination address (e.g. DNS access) using the URI placed in the topmost Route header field if present, otherwise based on the Request-URI. If the destination requires interconnect functionalities (e.g. the destination address is of an IP address type other than the IP address type used in the IM CN subsystem), the I-CSCF shall:

         i) if the I-CSCF supports indicating the traffic leg as specified in RFC 7549 [225] and required by local policy, append the "iotl" SIP URI parameter set to "homeA-homeB" to the Request-URI; and

         ii) forward the request to the destination address via an IBCF in the same network;

      - if network hiding is needed due to local policy, put the address of the IBCF to the topmost Route header field;

      - route the request based on SIP routeing procedures; and

      - if overlap signalling using the multiple-INVITE method is supported as a network option, and if the I-CSCF receives an INVITE request outside an existing dialog with the same Call ID and From header as a previous INVITE request during a certain period of time, route the new INVITE to the same next hop as the previous INVITE request.

Upon an unsuccessful user location query when the response from the HSS indicates that the user does not exist, and if local operator policy does not indicate that request routeing is to be attempted, then, the I-CSCF shall return an appropriate unsuccessful SIP response. Upon an unsuccessful user location query when the response from the HSS indicates that the user does not exist, and if if the Request-URI is a SIP URI, the I-CSCF shall also return an appropriate unsuccessful SIP response. This response may be a 404 (Not found) or 604 (Does not exist anywhere) in the case the user is not a user of the home network.

Upon an unsuccessful user location query when the response from the HSS indicates that the user is not registered and no services are provided for such a user, the I-CSCF shall return an appropriate unsuccessful SIP response. This response may be a 480 (Temporarily unavailable) response if the user is recognized as a valid user, but is not registered at the moment and it does not have services for unregistered users.

When the I-CSCF receives an initial request for a dialog or standalone transaction, that contains a single Route header field pointing to itself, the I-CSCF shall determine from the entry in the Route header field whether it needs to do HSS query. In case HSS query not is needed, then the I-CSCF shall:

1) remove its own SIP URI from the topmost Route header field; and

2) route the request based on the Request-URI.

When the I-CSCF receives an initial request for a dialog or standalone transaction containing more than one Route header field, the I-CSCF shall:

1) remove its own SIP URI from the topmost Route header field; and

2) forward the request based on the topmost Route header field.

NOTE 10: In accordance with SIP the I-CSCF can add its own routeable SIP URI to the top of the Record-Route header field to any request, independently of whether it is an initial request. The P-CSCF will ignore any Record-Route header field that is not in the initial request of a dialog.

When the I-CSCF receives a response to an initial request (e.g. 183 (Session Progress) response or 2xx response), the I-CSCF shall store the values from the P-Charging-Function-Addresses header field, if present. If the next hop is outside of the current network, then the I-CSCF shall remove the P-Charging-Function-Addresses header field prior to forwarding the message.

When the I-CSCF receives any response to the initial request for a dialog or standalone transaction containing a "term-ioi" header field parameter in the P-Charging-Vector header field from the AS hosting the PSI, the I-CSCF shall:

1) remove all received "orig-ioi" and "term-ioi" header field parameters from the forwarded response;

2) insert the stored "orig-ioi" header field parameter if received in the request; and

3) insert a type 2 "term-ioi" header field parameter. The "term-ioi" header field parameter is set to a value that identifies the sending network of the response.

When the I-CSCF, upon sending an initial INVITE request to the S-CSCF, receives a 305 (Use Proxy) response from the S-CSCF, the I-CSCF shall forward the initial INVITE request to the SIP URI indicated in the Contact field of the 305 (Use Proxy) response, as specified in RFC 3261 [26].

5.3.2.1A Originating procedures for requests containing the "orig" parameter

The procedures of this subclause apply for requests received at the I-CSCF when the topmost Route header field of the request contains the "orig" parameter.

The I-CSCF shall verify for all requests whether they arrived from a trusted domain or not. If the request arrived from a non trusted domain, then the I-CSCF shall respond with 403 (Forbidden) response.

If the request arrived from a trusted domain, the I-CSCF shall perform the procedures below.

NOTE 1: The I-CSCF can find out whether the request arrived from a trusted domain or not, from the procedures described in 3GPP TS 33.210 [19A].

For all SIP transactions identified:

- if priority is supported, as containing an authorised Resource-Priority header field, or, if such an option is supported, relating to a dialog which previously contained an authorised Resource-Priority header field;

the I-CSCF shall give priority over other transactions or dialogs. This allows special treatment of such transactions or dialogs. If priority is supported, the I-CSCF shall adjust the priority treatment of transactions or dialogs according to the most recently received authorized Resource-Priority header field or backwards indication value.

NOTE 2 The special treatment can include filtering, higher priority processing, routeing, call gapping. The exact meaning of priority is not defined further in this document, but is left to national regulation and network configuration.

If the I-CSCF receives the P-Profile-Key header field in a SIP request or response the I-CSCF shall discard the P-Profile-Key header field.

When the I-CSCF receives an initial request for a dialog or standalone transaction the I-CSCF will start the user location query procedure to the HSS as specified in 3GPP TS 29.228 [14] for the calling user, indicated in either:

1) the P-Served-User header field, if included in the request; or

2) the P-Asserted-Identity header field, if the P-Served-User header field is not included in the request.

Prior to performing the user location query procedure to the HSS, the I-CSCF decides which HSS to query, possibly as a result of a query to the Subscription Locator Functional (SLF) entity as specified in 3GPP TS 29.228 [14].

When the I-CSCF receives an INVITE request, the I-CSCF may require the periodic refreshment of the session to avoid hung states in the I-CSCF. If the I-CSCF requires the session to be refreshed, the I-CSCF shall apply the procedures described in RFC 4028 [58] clause 8.

NOTE 3: Requesting the session to be refreshed requires support by at least one of the UEs. This functionality cannot automatically be granted, i.e. at least one of the involved UEs needs to support it.

When the response for user location query contains information about the required S-CSCF capabilities, the I-CSCF shall select a S-CSCF according to the method described in 3GPP TS 29.228 [14].

If the user location query was successful, the I-CSCF shall:

1) insert the URI of an AS hosting the PSI, or the URI of the S-CSCF - either received from the HSS, or selected by the I-CSCF based on capabilities - as the topmost Route header field appending the "orig" parameter to the URI of the S-CSCF;

2) store the value of the "icid-value" header field parameter received in the P-Charging-Vector header field and retain the "icid-value" header field parameter in the P-Charging-Vector header field. If no P-Charging-Vector header field was found, then insert the P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17];

2A) based on local policy, add an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier;

3) optionally, include in the P-User-Database header field defined in RFC 4457 [82]:

   a) either the received Redirect-Host AVP value; or

   b) the HSS Group ID using the form "aaa://hss.5gc.gid.<GID>.invalid", if the HSS Group ID is received following procedures in clause X.3;

4) if a wildcarded identity value is received from the HSS in the Wildcarded-Identity AVP and the I-CSCF supports the SIP P-Profile-Key private header extension, include the wildcarded public user identity value in the P-Profile-Key header field as defined in RFC 5002 [97]; and

5) forward the request based on the topmost Route header field.

NOTE 4: The P-User-Database header field can be included only if the I-CSCF can assume (e.g. based on local configuration) that the receiving S-CSCF will be able to process the header field.

Upon an unsuccessful user location query, the I-CSCF shall return an appropriate unsuccessful SIP response. This response may be a 404 (Not found) response or 604 (Does not exist anywhere) response in the case the user is not a user of the home network.

When the I-CSCF receives any response to the above request, and forwards it to AS, the I-CSCF shall:

- store the values from the P-Charging-Function-Addresses header field, if present. If the next hop is outside of the current network, then the I-CSCF shall remove the P-Charging-Function-Addresses header field prior to forwarding the message; and

- insert a P-Charging-Vector header field containing the type 3 "orig-ioi" header field parameter, if received in the request, and a type 3 "term-ioi" header field parameter in the response. The I-CSCF shall set the type 3 "term-ioi" header field parameter to a value that identifies the sending network of the response and the type 3 "orig-ioi" header field parameter is set to the previously received value of type 3 "orig-ioi" header field parameter.

5.3.2.2 Abnormal cases

In the case of SLF query, if the SLF does not send HSS address to the I-CSCF, the I-CSCF shall send back a 404 (Not Found) response to the UE.

Upon successful user location query, when the response contains the URI of the assigned S-CSCF, if the I-CSCF is unable to contact the assigned S-CSCF, as determined by one of the following:

- the S-CSCF does not respond to the service request and its retransmissions by the I-CSCF; or
- by unspecified means available to the I-CSCF;

and:

- the I-CSCF supports S-CSCF restoration procedures;

then:

- the I-CSCF shall explicitly request the list of capabilities from the HSS and, on receiving these capabilities, the I-CSCF shall select a new S-CSCF, based on the capabilities indicated from the HSS. The newly selected S-CSCF shall not be one of any S-CSCFs selected previously during this same terminating procedure. Re-selection shall be performed until SIP transaction timer expires as specified in RFC 3261 [26]. When forwarding the request to the new S-CSCF, the I-CSCF includes the SIP URI parameter "scscf-reselection" to the Request-URI of the request.

NOTE 1: These procedures do not prevent the usage of unspecified reliability or recovery techniques above and beyond those specified in this subclause.

Upon successful user location query, when the response contains information about the required S-CSCF capabilities, if the I-CSCF is unable to contact a selected S-CSCF, as determined by one of the following:

- the S-CSCF does not respond to the service request and its retransmissions by the I-CSCF; or
- by unspecified means available to the I-CSCF;

then:

- the I-CSCF shall select a new S-CSCF, based on the capabilities indicated from the HSS. The newly selected S-CSCF shall not be one of any S-CSCFs selected previously during this same terminating procedure. Re-selection shall be performed until SIP transaction timer expires as specified in RFC 3261 [26]. When forwarding the request to the new S-CSCF, the I-CSCF includes the SIP URI parameter "scscf-reselection" to the Request-URI of the request.

NOTE 2: These procedures do not prevent the usage of unspecified reliability or recovery techniques above and beyond those specified in this subclause.

If the I-CSCF receives a negative response to the user location query, the I-CSCF shall send back a 404 (Not Found) response.

If the I-CSCF receives a CANCEL request and if the I-CSCF finds an internal state indicating a pending Cx transaction with the HSS, the I-CSCF:

- shall answer the CANCEL request with a 200 (OK) response; and
- shall answer the original request with a 487 (Request Terminated) response.

NOTE 3: The I-CSCF will discard any later arriving (pending) Cx answer message from the HSS.

With the exception of 305 (Use Proxy) response, the I-CSCF may recurse on a 3xx response only when the domain part of the URI contained in the 3xx response is in the same domain as the I-CSCF. For the same cases, if the URI is an IP address, the I-CSCF shall only recurse if the IP address is known locally to be a address that represents the same domain as the I-CSCF.

5.3.3 Void

5.3.3.1 Void

5.3.3.2 Void

5.3.3.3 Void

5.3.4 Void

5.3.5 Subsequent requests

When the I-CSCF receives a subsequent request, the I-CSCF shall verify whether it has arrived from an entity within the trust domain or not. If the request has not arrived from an entity within the trust domain, then the I-CSCF shall remove all P-Charging-Vector header fields, if present.

If no P-Charging-Vector header field was found in the received subsequent request, then insert the P-Charging-Vector header field with the "icid-value" header field parameter set to the value populated in the initial request for the dialog.

When the I-CSCF receives any response to the subsequent request, the I-CSCF shall store the value of the "term-ioi" header field parameter received in the P-Charging-Vector header field, if present.

When the I-CSCF receives a subsequent request directly forwarded to the AS hosting the PSI, the I-CSCF shall insert a type 3 "orig-ioi" header field parameter in place of any received "orig-ioi" header field parameter, if the received request including a P-Charging-Vector header field. The I-CSCF shall set the type 3 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The I-CSCF shall not include the type 3 "term-ioi" header field parameter.

When the I-CSCF receives any response to the subsequent request from the AS hosting the PSI, if the received response containing a "term-ioi" header field parameter in the P-Charging-Vector header field, the I-CSCF shall:

1) remove all received "orig-ioi" and "term-ioi" header field parameters from the forwarded response;

2) insert the stored "orig-ioi" header field parameter if received in the request; and

3) insert a type 2 "term-ioi" header field parameter. The "term-ioi" header field parameter is set to a value that identifies the sending network of the response.

5.4 Procedures at the S-CSCF

5.4.0 General

Where the S-CSCF provides emergency call support, the procedures of subclause 5.4.8 shall be applied first.

Upon

1) a third-party registration due to initial registration on behalf of a served public user identity; or

2) a trigger to an AS for an unregistered public user identity and there is no IP address of that AS associated with that public user identity stored;

the S-CSCF shall store the IP address of the AS and associate the IP address with the public user identity and the AS SIP URI along with all URI parameters.

When sending a failure response to any received request, depending on operator policy, the S-CSCF may insert a Response-Source header field with an "fe" header field parameter constructed with the URN namespace "urn:3gpp:fe", the fe-id part of the URN set to "s-cscf" and optionally an appropriate fe-param part of the URN set in accordance with subclause 7.2.17. A S-CSCF when sending a failure response will add in the URN the "side" header field parameter set to:

- "orig" for a UE-originating case; and
- "term" for a UE-terminating case.

5.4.1 Registration and authentication

5.4.1.1 Introduction

The S-CSCF shall determine which authentication mechanism applies based on the contents of the REGISTER request and the authentication mechanism assigned in the HSS:

1) if the REGISTER request contains an Authorization header field with the "integrity-protected" header field parameter set to "no", the S-CSCF shall perform the initial registration procedures with IMS-AKA authentication described in subclauses 5.4.1.2.1 and 5.4.1.2.1A;

2) if the REGISTER request contains an Authorization header field with the "integrity-protected" header field parameter set to "yes", the S-CSCF shall perform the protected registration procedures with IMS-AKA as a security mechanism as described in subclause 5.4.1.2.2;

2A) if the REGISTER request contains an Authorization header field with the "integrity-protected" header field parameter set to "tls-connected" and with the "algorithm" header field parameter set to "AKAv2-SHA-256", and if the S-CSCF supports the IMS AKA using HTTP Digest AKAv2 without IPSec security association, the S-CSCF shall perform:

   a) if the REGISTER request does not contain an authentication challenge response, the initial registration procedures for IMS-AKA authentication described in subclauses 5.4.1.2.1 and 5.4.1.2.1A; or

   b) if the REGISTER request contains an authentication challenge response, the protected registration procedures with IMS-AKA as a security mechanism as described in subclause 5.4.1.2.2;

NOTE 1: 3GPP TS 33.203 [19] defines support of IMS AKA using http Digest AKAv2 without IPSec security association only for WebRTC.

3) if the REGISTER request does not contain an Authorization header field, then the S-CSCF shall identify the user by the public user identity as received in the To header field of the REGISTER request. The S-CSCF shall derive the private user identity from the public user identity being registered. The S-CSCF shall derive the private user identity by removing SIP URI scheme and the following parts of the SIP URI if present: port number, URI parameters, and To header field parameters or by alternative mechanisms to derive the private user identity if operator policy requires to do so. These alternative mechanisms are not defined in this version of the specification;

4) if the REGISTER request does not contain an Authorization header field and the access-type field in the P-Access-Network-Info header field indicated xDSL, Ethernet, or Fiber access, and containing the "network provided" header field parameter and the S-CSCF supports NASS-IMS-bundled authentication but does not support SIP digest, then the S-CSCF shall perform the initial registration procedures with NASS-IMS bundled authentication as a security mechanism as described in subclause 5.4.1.2.1D;

5) if the REGISTER request does not contain an Authorization header field and the access-type field in the P-Access-Network-Info header field indicates it is received from an IP-CAN different from 3GPP and containing the "network provided" header field parameter and the S-CSCF supports SIP digest but does not support NASS-IMS-bundled authentication, then the S-CSCF shall perform the initial registration procedures with SIP digest as a security mechanism as described in subclauses 5.4.1.2.1 and 5.4.1.2.1B;

6) if the REGISTER request does not contain an Authorization header field and there is no P-Access-Network-Info header field containing the "network provided" field or there is a P-Access-Network-Info header field indicating a 3GPP access network containing the "network provided", and the S-CSCF supports GPRS-IMS-Bundled

authentication, the S-CSCF shall perform the initial registration procedures with GPRS-IMS-Bundled authentication described in subclause 5.4.1.2.1E;

7) if the REGISTER request does not contain an Authorization header field, and the P-Access-Network-Info header field indicates it is received from an access network other than 3GPP, xDSL, Ethernet or Fiber and containing the "network provided" header field parameter, and the S-CSCF supports SIP digest and NASS-IMS bundled authentication, the S-CSCF shall perform the initial registration procedures with SIP digest as a security mechanism as described in subclauses 5.4.1.2.1 and 5.4.1.2.1B:

8) if the REGISTER request does not contain an Authorization header field, and the P-Access-Network-Info header field indicates it is received from a xDSL, Ethernet or Fiber access network, and containing the "network provided" header field parameter, and the S-CSCF supports SIP digest and NASS-IMS bundled authentication, the S-CSCF sends an authentication request for the user to the HSS indicating that the authentication scheme is unknown as described in 3GPP TS 29.228 [14]:

   - if the HSS responds with an authentication scheme of SIP digest, then the S-CSCF shall perform the initial registration procedures with SIP digest as a security mechanism as described in subclauses 5.4.1.2.1 and 5.4.1.2.1B; or

   - if the HSS responds with an authentication scheme of NASS-IMS bundled authentication and the request was received from a P-CSCF in the home network and the P-CSCF is "TISPAN-enabled", then the S-CSCF shall perform the initial registration procedures with NASS-IMS bundled authentication as a security mechanism as described in subclause 5.4.1.2.1D;

9) if the REGISTER request contains an Authorization header field without an "integrity-protected" header field parameter, the S-CSCF shall send an authentication request for the user to the HSS indicating that the authentication scheme is unknown as described in 3GPP TS 29.228 [14]:

   - if the HSS responds with an authentication scheme of NASS-IMS bundled authentication and the request was received from a P-CSCF is in the home network and the P-CSCF is "TISPAN-enabled", then the S-CSCF shall perform the initial registration procedures with NASS-IMS bundled authentication as a security mechanism as described in subclause 5.4.1.2.1D; or

   - if the HSS responds with an authentication scheme of SIP digest, then the S-CSCF shall perform the initial registration procedures with SIP digest as a security mechanism as described in subclauses 5.4.1.2.1 and 5.4.1.2.1B;

10) if the REGISTER request contains an Authorization header field with the "integrity-protected" header field parameter set to "tls-pending", "tls-yes", "ip-assoc-pending" or "ip-assoc-yes", the S-CSCF shall perform the protected registration procedures for SIP digest described in subclause 5.4.1.2.2A;

11) if the REGISTER request contains an Authorization header field with the "integrity-protected" header field parameter set to "auth-done", the S-CSCF shall perform the protected registration procedures described in subclause 5.4.1.2.2E; and

12) if the REGISTER request contains a JSON Web Token with the "3gpp-waf" JSON Web Token claim or with the "3gpp-wwsf" JSON Web Token claim, as defined in RFC 7519 [235], and if the S-CSCF supports WebRTC, and if the S-CSCF has received authorization information about WAF or WWSF entities from the HSS, or per configuration, then the S-CSCF shall check whether the WAF or WWSF is not barred, as specified in 3GPP TS 33.203 [9] annex X. If the WAF or the WWSF is barred, the S-CSCF shall send a 403 (Forbidden) response to the REGISTER request.

NOTE 2: The S-CSCF needs to be configured to know which P-CSCFs are "TISPAN-enabled" and uses the Via header field to determine which P-CSCF forwarded the registration request.

The S-CSCF shall act as the SIP registrar for all UEs belonging to the IM CN subsystem and with public user identities.

Subclause 5.4.1.2 through subclause 5.4.1.7 define S-CSCF procedures for SIP registration that do not relate to emergency. All registration requests are first screened according to the procedures of subclause 5.4.8.2 to see if they do relate to an emergency registration.

For all SIP registrations identified:

- as relating to an emergency; or

- if priority is supported, as containing an authorised Resource-Priority header field;

the S-CSCF shall give priority over other registrations. This allows special treatment of such registrations.

NOTE 3: The special treatment can include filtering, higher priority processing, routeing, call gapping. The exact meaning of priority is not defined further in this document, but is left to national regulation and network configuration.

The S-CSCF shall support the use of the Path and Service-Route header field. The S-CSCF shall also support the Require and Supported header fields. The Path header field is only applicable to the REGISTER request and its 200 (OK) response. The Service-Route header field is only applicable to the 200 (OK) response of REGISTER. The S-CSCF shall not act as a redirect server for REGISTER requests.

The network operator defines minimum and maximum times for each registration. These values are provided within the S-CSCF.

The procedures for notification concerning automatically registered public user identities of a user are described in subclause 5.4.2.1.2.

If the S-CSCF supports HSS based P-CSCF restoration procedures, and receives a REGISTER request from a P-CSCF that the S-CSCF considers is in a non-working state, the S-CSCF shall consider this P-CSCF as being in a working state.

If the S-CSCF supports PCRF based P-CSCF restoration procedures, and receives a REGISTER request from a P-CSCF that the S-CSCF considers is in a non-working state, the S-CSCF shall consider this P-CSCF as being in a working state.

In case a device performing address and/or port number conversions is provided by a NA(P)T or NA(P)T-PT, the S-CSCF may need to modify the SIP signalling according to the procedures described in annex K if both a "reg-id" and "+sip.instance" header field parameter are present in the received Contact header field as described in RFC 5626 [92].

5.4.1.2 Initial registration and user-initiated reregistration

5.4.1.2.1 Unprotected REGISTER

Any REGISTER request received unprotected by the S-CSCF without an Authorization header field, or with an Authorization header field having the "integrity-protected" header field parameter in the Authorization header field set to "no", or without an "integrity-protected" header field parameter is considered to be an initial registration. If such an initial registration contains a private user identity specifically reserved for IM CN subsystem registrations from an MSC Server enhanced for ICS as defined in 3GPP TS 23.003 [3], the S-CSCF shall respond with a 403 (Forbidden) response. The S-CSCF shall consider this registration attempt as failed.

NOTE 1: For NASS-IMS bundled authentication and GPRS-IMS-Bundled Authentication there is no distinction between a protected and an unprotected REGISTER. There is only an unprotected REGISTER to consider.

NOTE 2: If IMS AKA or SIP digest with TLS are used as a security mechanism, a 200 (OK) final response to an initial registration will only be sent back after the S-CSCF receives a correct authentication challenge response in a REGISTER request that is sent integrity protected.

NOTE 3: A REGISTER with the registration expiration interval value equal to zero will always be received protected. However, it is possible that in error conditions a REGISTER with the registration expiration interval value equal to zero can be received unprotected. In that instance the procedures below will be applied.

Upon receipt of a REGISTER request that is part of an initial registration as outlined above, for a public user identity for which the maximum number of allowed simultaneously registration flows for the used UE (i.e. linked to the same private user identity and instance ID) is reached, if the REGISTER is adding a new registration flow, then the S-CSCF shall reject the REGISTER by generating a 403 (Forbidden) response. If not, the S-CSCF shall continue with the rest of the procedures of this subclause.

Upon receipt of a REGISTER request that is part of an initial registration as outlined above, for a user identity linked to a private user identity and instance ID/reg-id if available, that has previously registered one or more public user identities, the S-CSCF shall:

1) perform the procedure below in this subclause for receipt of a REGISTER request for a public user identity which is not already registered, for the received public user identity;

2) if the multiple registrations is not used and if the authentication that in step 1) has been successful, and there are public user identities (including the public user identity being registered, if previously registered) that belong to this user that have been previously registered with the same private user identity, and with an old contact address different from the one received in the REGISTER request, and the previous registrations have not expired, perform the network initiated deregistration procedure (as described in subclause 5.4.1.5) for the previously registered public user identities belonging to this user including the public user identity being registered, if previously registered; and

3) if the multiple registrations is used (i.e., the "reg-id" header field parameter is included in the REGISTER request), and if the authentication that concludes the initial registration has been successful, and if the public user identity being registered has been previously registered with the same private user identity and the same "+sip.instance" and "reg-id" header field parameter values, and the previous registration has not expired:

   a) identify the registration flow being replaced;

   b) terminate any dialog, as specified in subclause 5.4.5.1.2, with a status code 480 (Temporarily Unavailable) in the Reason header field of the BYE request, associated with the registration flow being replaced; and

   c) send a NOTIFY request to the subscribers to the registration event package for the public user identity indicated in the REGISTER request, as described in subclause 5.4.2.1.2.

NOTE 4: The way the S-CSCF identifies the dialogs associated with the registration flow being replaced is implementation specific.

NOTE 5: The S-CSCF will inform the HSS that the previously registered public user identities, excluding the public user identity being registered, have been deregistered.

NOTE 6: Contact related to emergency registration is not affected. S-CSCF is not able deregister contact related to emergency registration and will not delete that.

When S-CSCF receives a REGISTER request with the "integrity-protected" header field parameter in the Authorization header field set to "no" and a non-empty "response" Authorization header field parameter, the S-CSCF shall ignore the value of the "response" header field parameter.

Upon receipt of a REGISTER request that is part of an initial registration as outlined above, for a public user identity which is not already registered linked to the same private user identity and the "+sip.instance" and "reg-id" header field parameters, if available, the S-CSCF shall:

1) identify the user by the public user identity as received in the To header field and if the REGISTER request includes an Authorization header field, identify the private user identity as received in the "username" Authorization header field parameter of the REGISTER request;

2) check if the P-Visited-Network-ID header field is included in the REGISTER request, and if it is included identify the visited network by the value of this header field;

3) select an authentication vector for the user. If no authentication vector for this user is available, after the S-CSCF has performed the Authentication procedure with the HSS, as described in 3GPP TS 29.228 [14], the S-CSCF shall select an authentication vector as described in 3GPP TS 33.203 [19].

   Prior to performing Authentication procedure with the HSS, the S-CSCF decides which HSS to query, possibly as a result of a query to the Subscription Locator Functional (SLF) entity as specified in 3GPP TS 29.228 [14] or use the value as received in the P-User-Database header field in the REGISTER request as defined in RFC 4457 [82];

NOTE 7: The HSS address received in the response to SLF query or as a value of P-User-Database header field can be used to address the HSS of the public user identity in further queries.

NOTE 8: At this point the S-CSCF informs the HSS, that the user currently registering will be served by the S-CSCF by passing its SIP URI to the HSS. This will be used by the HSS to direct all subsequent incoming initial requests for a dialog or standalone transactions destined for this user to this S-CSCF.

NOTE 9: When passing its SIP URI to the HSS, the S-CSCF may include in its SIP URI the transport protocol and the port number where it wants to be contacted.

4) store the "icid-value" header field parameter received in the P-Charging-Vector header field;

5) challenge the user by generating a 401 (Unauthorized) response for the received REGISTER request appropriate to the security mechanism in use;

6) send the so generated 401 (Unauthorized) response towards the UE, and if the URI in the first Path header field has an "ob" SIP URI parameter, include a Require header field with the option-tag "outbound" as described in RFC 5626 [92]; and

7) start timer reg-await-auth which guards the receipt of the next REGISTER request.

If the received REGISTER request indicates that the challenge sent previously by the S-CSCF to the UE was deemed to be invalid by the UE, the S-CSCF shall stop the timer reg-await-auth and proceed as described in the subclause 5.4.1.2.3.

5.4.1.2.1A Challenge with IMS AKA as security mechanism

On sending a 401 (Unauthorized) response to an unprotected REGISTER request, the S-CSCF shall populate the header fields as follows:

1) a WWW-Authenticate header field which transports:

   a) a globally unique name of the S-CSCF in the "realm" header field parameter;

   b) the RAND and AUTN parameters and optional server specific data for the UE in the "nonce" header field parameter;

   c) if the REGISTER request contains an Authorization header field with an "integrity-protected" header field parameter set to the value "no":

      - the security mechanism, in the "algorithm" header field parameter set to the value as received in the Authorization header field i.e. "AKAv2-SHA-256" or "AKAv1-MD5";

NOTE: The "AKAv1-MD5" algorithm is only supported for backward compatibility.

      - the IK (Integrity Key) parameter for the P-CSCF in the "ik" header field parameter (see subclause 7.2A.1); and

      - the CK (Cipher Key) parameter for the P-CSCF in the "ck" header field parameter (see subclause 7.2A.1); and

   d) if the REGISTER request does contain an Authorization header field with the "algorithm" header field parameter set to "AKAv2-SHA-256", and if the S-CSCF supports the IMS AKA using HTTP Digest AKAv2 without IPSec security association:

      - the security mechanism, which is "AKAv2-SHA-256" in the "algorithm" header field parameter.

The S-CSCF shall store the RAND parameter used in the 401 (Unathorized) response for future use in case of a resynchronisation. If a stored RAND already exists in the S-CSCF, the S-CSCF shall overwrite the stored RAND with the RAND used in the most recent 401 (Unauthorized) response.

5.4.1.2.1B Challenge with SIP digest as security mechanism

On sending a 401 (Unauthorized) response to an unprotected REGISTER request, the S-CSCF shall populate the header fields as follows:

1) for each supported digest algorithm the S-CSCF shall create a WWW-Authenticate header field as defined in RFC 7616 [286] and RFC 8760 [287], which transports:

   - a protection domain in the "realm" header field parameter;

   - a "nonce" header field parameter (generated by the S-CSCF);

- an "algorithm" header field parameter; and

- a "qop" header field parameter; if the qop value is not provided in the authentication vector, it shall contain the value "auth".

The S-CSCF shall include the WWW-Authenticate header fields to the 401 (Unauthorized) response in order of preference, starting with the most preferred algorithm.

5.4.1.2.1C Challenge with SIP digest with TLS as security mechanism

The procedures for subclause 5.4.1.2.1B apply.

NOTE: The S-CSCF is not able to distinguish between SIP Digest with TLS and SIP Digest without TLS for the case of an unprotected REGISTER request, therefore the procedures are the same for both.

5.4.1.2.1D Initial registration and user-initiated reregistration for NASS-IMS bundled authentication

Upon receipt of a REGISTER request that is determined to be NASS-IMS bundled authentication, for a user identity linked to a private user identity that has a registered public user identity but with a new contact address, the S-CSCF shall:

1) perform the procedure for receipt of a REGISTER request without the "integrity-protected" header field parameter in the Authorization header field or without the Authorization header field, for the received public user identity; and

2) if the Contact header field of the REGISTER request does not contain a "reg-id" header field parameter (i.e., the multiple registrations mechanism is not used), and the authentication has been successful, and there are public user identities (including the public user identity being registered, if previously registered) belonging to this user that have been previously registered with the same private user identity and with an old contact address different from the one received in the REGISTER request and if the previous registration have not expired:

   a) terminate all dialogs, if any, associated with the previously registered public user identities (including the public user identity being registered, if previously registered), with a status code 480 (Temporarily Unavailable) in the Reason header field of the BYE request, as specified in subclause 5.4.5.1.2;

   b) send a NOTIFY request, to the subscribers to the registration event package of the previously registered public user identities, that indicates that all previously registered public user identities (excluding the public user identity being registered) belonging to this user identified with its private user identity, have been deregistered, as described in subclause 5.4.2.1.2. For the public user identity being registered, the NOTIFY request contains the new contact information; and

NOTE 1: The last dialog to be terminated will be the dialog established by the UE subscribing to the reg event package. When sending the NOTIFY request to the UE over this dialog, the S-CSCF will terminate this dialog by setting in the NOTIFY request the Subscription-State header field to the value of "terminated".

   c) delete all information associated with the previously registered public user identities.

NOTE 2: Contact related to emergency registration is not affected. The S-CSCF is not able to deregister contact related to emergency registration and will not delete it.

Upon receipt of a REGISTER request that is determined to be NASS-IMS bundled authentication, for a public user identity for which the maximum number of allowed simultaneously registration flows is for the used UE (i.e. linked to the same private user identity and instance ID) is reached, if the REGISTER is adding a new registration flow, then the S-CSCF shall reject the REGISTER by generating a 403 (Forbidden) response. If not, the S-CSCF shall continue with the rest of the procedures of this subclause;

Upon receipt of a REGISTER request without the "integrity-protected" header field parameter in the Authorization header field or without an Authorization header field, which is not for an already registered public user identity linked to the same private user identity, the S-CSCF shall:

1) identify the user by the public user identity as received in the To header field of the REGISTER request and if the Authorization header field is present, the private user identity as received in the Authorization header field of the REGISTER request. If the Authorization header field is not present, the S-CSCF shall derive the private user

identity from the public user identity being registered by removing SIP URI scheme and the following parts of the SIP URI if present: port number, URI parameters, and To header field parameters;

2) check whether one or more Line-Identifiers previously received over the Cx interface, and stored as a result of a Authentication procedure with the HSS, are available for the user. If not, the S-CSCF performs the Authentication procedure with the HSS, as described in 3GPP TS 29.228 [14], in order to obtain these Line-Identifiers;

3) in the particular case where the S-CSCF received via the Cx interface one or more Line-Identifiers, compare each of Line-Identifiers with the "dsl-location", "eth-location" or "fiber-location" parameter of the P-Access-Network-Info header field (if present and if it includes the "network-provided" parameter):

   - if one of these match, the user is considered authenticated, behave as described in step 5) to 11) of subclause 5.4.1.2.2;

   - otherwise i.e. if these do not match, return a 403 (Forbidden) response to the REGISTER request; and

4) if no Line-Identifier is received over the Cx interface, send a 500 (Server Internal Error) response to the REGISTER request.

Upon receipt of a REGISTER request without the "integrity-protected" header field parameter in the Authorization header field or without an Authorization header field, for an already registered public user identity linked to the same private user identity, and for existing contact information, the S-CSCF shall behave as described in subclause 5.4.1.2.2F.

5.4.1.2.1E Initial registration and user-initiated reregistration for GPRS-IMS-Bundled authentication

Upon receipt of a REGISTER request without an Authorization header field, the S-CSCF shall:

1) identify the user by the public user identity as received in the To header field of the REGISTER request. The S-CSCF shall derive the private user identity from the public user identity being registered by removing URI scheme and the following parts of the URI if present: port number, URI parameters, and To header field parameters;

1A) if the maximum number of simultaneously registration flows allowed for the related public user identity for the used UE (i.e. linked to the same private user identity and instance ID) is reached, then the S-CSCF shall reject the REGISTER by generating a 403 (Forbidden) response. If not, the S-CSCF shall continue with the rest of the steps;

2) check if the P-Visited-Network-ID header field is included in the REGISTER request, and if it is included identify the visited network by the value of this header field;

3) check whether an IP address is stored for this UE. If no IP address (or prefix) is stored for the UE, query the HSS as described in 3GPP TS 29.228 [14] with the derived private user identity and the public user identity as input and store the received IP address (or prefix) of the UE; if the S-CSCF receives a prefix from the HSS, it will only check against prefixes otherwise it will check against the full IP address;

NOTE 1: At this point the S-CSCF informs the HSS, that the user currently registering will be served by the S-CSCF by passing its SIP URI to the HSS. This will be indicated by the HSS for all further incoming requests to this user, in order to direct all these requests directly to this S-CSCF.

4) check whether a "received" header field parameter exists in the Via header field provided by the UE. If a "received" header field parameter exists, the S-CSCF shall compare the IP address recorded in the "received" header field parameter against the UE's IP address stored during registration. In case of IPv6 stateless autoconfiguration, the S-CSCF shall compare the prefix of the IP address recorded in the "received" header field parameter against the UE's IP address prefix stored during registration. If no "received" header field parameter exists in the Via header field provided by the UE, then the S-CSCF shall compare IP address recorded in the "sent-by" parameter against the stored UE IP address. In case of IPv6 stateless autoconfiguration, S-CSCF shall compare the prefix of the IP address recorded in the "sent-by" parameter against the UE's IP address prefix stored during registration. In any case, if the stored IP address (or prefix) and the (prefix of the) IP address recorded in the Via header field provided by the UE do not match, the S-CSCF shall query the HSS as described in 3GPP TS 29.228 [14] with the derived private user identity and the public user identity as input and store the received IP address (or prefix) of the UE. If the stored IP address (or prefix) and the (prefix of the) IP address

recorded in the Via header field provided by the UE still do not match the S-CSCF shall reject the registration with a 403 (Forbidden) response and skip the following steps;

5) after performing the S-CSCF Registration/deregistration notification procedure with the HSS, as described in 3GPP TS 29.228 [14], store the following information in the local data:

   a) the list of public user identities, including the registered own public user identity and its associated set of implicitly registered public user identities and wildcarded public user identities due to the received REGISTER request. Each public user identity is identified as either barred or non-barred;

   b) all the service profile(s) corresponding to the public user identities being registered (explicitly or implicitly), including initial Filter Criteria (the initial Filter Criteria for the Registered and common parts is stored and the unregisterd part is retained for possible use later - in the case the S-CSCF is retained if the user becomes unregistered);

   c) if S-CSCF restoration procedures are supported, the restoration information if received as specified in 3GPP TS 29.228 [14]; and

   d) if PCRF based P-CSCF restoration procedures are supported, all the user profile(s) corresponding to the public user identities being registered (explicitly or implicitly), including the IMSI, if available;

NOTE 2: There might be more than one set of initial Filter Criteria received because some implicitly registered public user identities that are part of the same implicit registration set belong to different service profiles.

6) update registration bindings as follows:

   a) bind to each non-barred registered public user identity all registered contact information including all header field parameters contained in the Contact header field and all associated URI parameters with the exception of the "pub-gruu" and "temp-gruu" header field parameters as specified in RFC 5627 [93], and store information for future use; and

   b) if the Contact URI in the Contact header field does not contain a "bnc" URI parameter, then for each binding that contains a "+sip.instance" Contact header field parameter, assign a new temporary GRUU, as specified in subclause 5.4.7A.3;

NOTE 3: There might be more than one contact information available for one public user identity.

NOTE 4: The barred public user identities are not bound to the contact information.

7) check whether a Path header field was included in the REGISTER request and construct a list of preloaded Route header fields from the list of entries in the received Path header field. The S-CSCF shall preserve the order of the preloaded Route header fields and bind them either to the contact address of the UE or to the registration flow and the associated contact address (if the multiple registration mechanism is used) and the contact information that was received in the REGISTER request;

NOTE 5: If this registration is a reregistration or an initial registration (i.e., there are previously registered public user identities belonging to the user that have not been deregistered or expired), then a list of pre-loaded Route header fields will already exist. If multiple registration mechanism was not used, then the existing list of pre-loaded Route header fields is bound to a respective contact address of the UE. However, if multiple registration mechanism was used, then the existing list of pre-loaded Route header fields is bound to a registration flow and the associated contact address that was used to send the REGISTER request. In either case, the new list replaces the old list.

8) determine the duration of the registration by checking the registration expiration interval value in the received REGISTER request and bind it either to the respective contact address of the UE or to the registration flow and the associated contact address (if the multiple registration mechanism is used). Based on local policy, the S-CSCF may reduce the duration of the registration or send back a 423 (Interval Too Brief) response specifying the minimum allowed time for registration. The local policy can take into account specific criteria such as the used authentication mechanism to determine the allowed registration duration;

9) store the "icid-value" header field parameter received in the P-Charging-Vector header field;

9A) if an "orig-ioi" header field parameter is received in the P-Charging-Vector header field, store the value of the received "orig-ioi" header field parameter; and

NOTE 6: Any received "orig-ioi" header field parameter will be a type 1 IOI. The type 1 IOI identifies the network from which the request was sent.

10) create and send a 200 (OK) response for the REGISTER request as specified in subclause 5.4.1.2.2F.

When a user de-registers, or is de-registered by the HSS, the S-CSCF shall delete the IP address stored for the UE.

5.4.1.2.2 Protected REGISTER with IMS AKA as a security mechanism

Upon receipt of a REGISTER request with the "integrity-protected" header field parameter in the Authorization header field set to "yes" or to "tls-connected", the S-CSCF shall identify the user by the public user identity as received in the To header field and the private user identity as received in the Authorization header field of the REGISTER request, and:

If the maximum number of simultaneously registration flows allowed for the related public user identity for the used UE (i.e. linked to the same private user identity and instance ID) is reached, then the S-CSCF shall reject the REGISTER by generating a 403 (Forbidden) response. If not, the S-CSCF shall continue with rest of the procedures of this subclause;

In the case that there is no authentication currently ongoing for this user (i.e. no timer reg-await-auth is running):

1) check if the user needs to be reauthenticated.

   The S-CSCF may require authentication of the user for any REGISTER request, and shall always require authentication for REGISTER requests received without the "integrity-protected" header field parameter in the Authorization header field set to "yes" or "tls-connected".

   If the user needs to be reauthenticated, the S-CSCF shall proceed with the procedures as described for the unprotected REGISTER in subclause 5.4.1.2.1, beginning with step 3). If the user does not need to be reauthenticated, the S-CSCF shall proceed with the following steps in this paragraph; and

2) check whether a registration expiration interval value is included in the REGISTER request and its value. If the registration expiration interval value indicates a zero value, the S-CSCF shall perform the deregistration procedures as described in subclause 5.4.1.4. If the registration expiration interval value does not indicate zero, the S-CSCF:

   - if the REGISTER request does not contain a "reg-id" header field parameter and the contact address indicated in the Contact header field was not previously registered, send a 403 (Forbidden) response to the UE; and

NOTE 1: New contact address is always registered via an initial registration.

3) check whether the public user identity received in the To header field is already registered. If it is not registered, the S-CSCF shall proceed beginning with step 4B below. Otherwise, the S-CSCF shall:

   - send a 439 (First Hop Lacks Outbound Support) response to the UE, if the REGISTER request contains the "reg-id" Contact header field parameter and the "outbound" option tag in a Supported header field, but the first URI in the Path header field does not have an "ob" URI parameter; or

   - otherwise proceed beginning with step 6 below.

In the case that a timer reg-await-auth is running for this user the S-CSCF shall:

1) check if the Call-ID of the request matches with the Call-ID of the 401 (Unauthorized) response which carried the last challenge. The S-CSCF shall only proceed further if the Call-IDs match;

2) stop timer reg-await-auth;

3) check whether an Authorization header field is included, containing:

   a) the private user identity of the user in the "username" header field parameter;

   b) if the "integrity-protected" header field parameter is set to "yes", the "algorithm" header field parameter set to "AKAv2-SHA-256" or "AKAv1-MD5";

c) if the "integrity-protected" header field parameter is set to "tls-connected", the "algorithm" header field parameter set to "AKAv2-SHA-256" if the S-CSCF supports the IMS AKA using HTTP Digest AKAv2 without IPSec security association; and

d) the authentication challenge response needed for the authentication procedure in the "response" header field parameter.

The S-CSCF shall only proceed with the following steps in this paragraph if the authentication challenge response was included;

4) check whether the received authentication challenge response and the expected authentication challenge response (calculated by the S-CSCF using XRES and other parameters as described in RFC 3310 [49] when AKAv1 is used or as described in RFC 4169 [227] when AKAv2 is used) match. The XRES parameter was received from the HSS as part of the Authentication Vector. The S-CSCF shall only proceed with the following steps if the challenge response received from the UE and the expected response calculated by the S-CSCF match;

4A) if the Contact header field of the REGISTER request does not contain a "reg-id" header field parameter (i.e., the multiple registrations mechanism is not used), and there are public user identities (including the public user identity being registered, if previously registered) that belong to this user that have been previously registered with the same private user identity, and with an old contact address different from the one received in the REGISTER request and if the previous registrations have not expired:

a) terminate all dialogs, associated with the previously registered public user identities (including the public user identity being registered, if previously registered), with a status code 480 (Temporarily Unavailable) in the Reason header field of the BYE request, as specified in subclause 5.4.5.1.2;

b) send a NOTIFY request, to the subscribers to the registration event package of the previously registered public user identities, that indicates that all previously registered public user identities (excluding the public user identity being registered) belonging to this user identified with its private user identity, have been deregistered, as described in subclause 5.4.2.1.2. For the public user identity being registered, the NOTIFY request contains the new contact information; and

NOTE 2: The last dialog to be terminated will be the dialog established by the UE subscribing to the reg event package. When sending the NOTIFY request to the UE over this dialog, the S-CSCF will terminate this dialog by setting in the NOTIFY request the Subscription-State header field to the value of "terminated".

c) delete all information associated with the previously registered public user identities;

NOTE 3: Contact related to emergency registration is not affected. The S-CSCF is not able to deregister contact related to emergency registration and will not delete it.

4B) if the REGISTER request contains the "reg-id" Contact header field parameter and the "outbound" option tag in a Supported header field, but the first URI in the Path header field does not have an "ob" URI parameter, send a 439 (First Hop Lacks Outbound Support) response to the UE;

5) after performing the S-CSCF Registration/deregistration notification procedure with the HSS, as described in 3GPP TS 29.228 [14], store the following information in the local data:

a) the list of public user identities, including the registered own public user identity and its associated set of implicitly registered public user identities and wildcarded public user identities due to the received REGISTER request. Each public user identity is identified as either barred or non-barred;

b) all the service profile(s) corresponding to the public user identities being registered (explicitly or implicitly), including initial Filter Criteria(the initial Filter Criteria for the Registered and common parts is stored and the unregistered part is retained for possible use later - in the case of the S-CSCF is retained if the user becomes unregistered);

c) if S-CSCF restoration procedures are supported, the restoration information if received as specified in 3GPP TS 29.228 [14]; and

d) if PCRF based P-CSCF restoration procedures are supported, all the user profile(s) corresponding to the public user identities being registered (explicitly or implicitly), including the IMSI, if available;

NOTE 4: There might be more than one set of initial Filter Criteria received because some implicitly registered public user identities that are part of the same implicit registration set belong to different service profiles.

6) update registration bindings:

   a) if the Contact URI in the Contact header field does not contains a "bnc" URI parameter, then bind to each non-barred registered public user identity all registered contact information including all header field parameters contained in the Contact header field and all associated SIP URI parameters, with the exception of the "pub-gruu" and "temp-gruu" header field parameters as specified in RFC 5627 [93], and store information for future use;

   b) if the Contact URI in the Contact header field contains a "bnc" URI parameter, as a network option bind each non-barred registered public user identity to a contact address generated according to the procedures of RFC 6140 [191].

NOTE 5: It is assumed that when the Contact header field contains a "bnc" parameter, the associated public user identities obtained from the HSS are all of a form compatible with registration procedures as specified in RFC 6140 [191]; i.e. the set consists only of distinct public user identities contain global numbers in the international format or wildcarded public user identities representing multiple global numbers in the international format. The S-CSCF procedures for handling the error case where an associated public user identity is incompatible with RFC 6140 [191] is out of scope of this specification.

   c) if the Contact URI in the Contact header field does not contain a "bnc" URI parameter, then for each binding that contains a "+sip.instance" Contact header field parameter, assign a new temporary GRUU, as specified in subclause 5.4.7A.3;

   d) if the Contact header field of the REGISTER request contained a "+sip.instance" and a "reg-id" header field parameter, and the SIP URI in the Path header field inserted by the P-CSCF contained an "ob" SIP URI parameter header field, and:

      - if the public user identity has not previously been registered with the same "+sip.instance" and "reg-id" Contact header field parameter values, then create the registration flow in addition to any existing registration flow; or

      - if the public user identity has previously been registered with the same "+sip.instance" and "reg-id" header field parameter values, then determine whether the request refreshes or replaces an existing registration flow. If the request:

         i) refreshes an existing registration flow, then the S-CSCF shall leave the flow intact; or

         ii) replaces the existing registration flow with a new flow, then the S-CSCF shall:

            a) terminate any dialog, as specified in subclause 5.4.5.1.2, with a status code 480 (Temporarily Unavailable) in the Reason header field of the BYE request, associated with the registration flow being replaced; and

            b) send a NOTIFY request to the subscribers to the registration event package for the public user identity indicated in the REGISTER request, as described in subclause 5.4.2.1.2;

NOTE 6: The S-CSCF determines whether this REGISTER request replaces or refreshes an existing registration flow by examining the SIP URI in the Path header field inserted into the request by the P-CSCF (see subclause 5.2.2.1).

NOTE 7: The way the S-CSCF identifies the dialogs associated with the registration flow being replaced is implementation specific.

NOTE 8: There might be more than one contact information available for one public user identity.

NOTE 9: The barred public user identities are not bound to the contact information.

NOTE 10: Contact related to emergency registration is not affected. S-CSCF is not able deregister contact related to emergency registration and will not delete that.

7) check whether a Path header field was included in the REGISTER request and construct a list of preloaded Route header fields from the list of entries in the received Path header field. The S-CSCF shall preserve the order of the preloaded Route header fields and bind them either to the contact address of the UE or the registration flow and the associated contact address (if the multiple registration mechanism is used) and the contact information that was received in the REGISTER request;

NOTE 11: If this registration is a reregistration or an initial registration (i.e., there are previously registered public user identities belonging to the user that have not been deregistered or expired), then a list of pre-loaded Route header fields will already exist. If multiple registration mechanism was not used, then the existing list of pre-loaded Route header fields is bound to a respective contact address of the UE. However, if multiple registration mechanism was used, then the existing list of pre-loaded Route header fields is bound to a registration flow and the associated contact address that was used to send the REGISTER request. In either case, the new list replaces the old list.

8) determine the duration of the registration by checking the value of the registration expiration interval value in the received REGISTER request and bind it either to the respective contact address of the UE or to the registration flow and the associated contact address (if the multiple registration mechanism is used). Based on local policy, the S-CSCF may reduce the duration of the registration or send back a 423 (Interval Too Brief) response specifying the minimum allowed time for registration. The local policy can take into account specific criteria such as the used authentication mechanism to determine the allowed registration duration;

9) store the "icid-value" header field parameter received in the P-Charging-Vector header field;

10) if an "orig-ioi" header field parameter is received in the P-Charging-Vector header field, store the value of the received "orig-ioi" header field parameter; and

NOTE 12: Any received "orig-ioi" header field parameter will be a type 1 IOI. The type 1 IOI identifies the network from which the request was sent.

11) create and send a 200 (OK) response for the REGISTER request as specified in subclause 5.4.1.2.2F.

5.4.1.2.2A Protected REGISTER with SIP digest as a security mechanism

Upon receipt of a REGISTER request with the "integrity-protected" header field parameter in the Authorization header field set to "tls-pending", "tls-yes", "ip-assoc-pending", or "ip-assoc-yes", the S-CSCF shall identify the user by the public user identity as received in the To header field and the private user identity as received in the Authorization header field of the REGISTER request, and:

NOTE 1: Although the REGISTER request with the "integrity-protected" header field parameter set to "ip-assoc-pending" or "ip-assoc-yes" is handled as protected REGISTER request, the integrity of the request is actually not protected by SIP digest.

If the maximum number of simultaneously registration flows allowed for the related public user identity for the used UE (i.e. linked to the same private user identity and instance ID) is reached, then the S-CSCF shall reject the REGISTER by generating a 403 (Forbidden) response. If not, the S-CSCF shall continue with rest of the procedures of this subclause;

In the case that there is no authentication currently ongoing for this user (i.e. no timer reg-await-auth is running):

1) check if the user needs to be reauthenticated. The S-CSCF may require authentication of the user for any REGISTER request, and shall always require authentication for REGISTER requests received without the "integrity-protected" header field parameter in the Authorization header field set to "tls-yes".

   If the user needs to be reauthenticated and the REGISTER did not include an Authorization header field with a digest response, the S-CSCF shall proceed with the authentication procedures as described for the initial REGISTER in subclause 5.4.1.2.1 and subclause 5.4.1.2.1B.

   If the user needs to be reauthenticated and the REGISTER included an Authorization header field with a digest response, the S-CSCF shall proceed with the authentication procedures as described for the initial REGISTER in subclause 5.4.1.2.1 and subclause 5.4.1.2.1B and include the "stale" header field parameter with value "true" in the WWW-Authenticate header field.

In the case that a timer reg-await-auth is running for this user the S-CSCF shall:

1) check if the Call-ID of the request matches with the Call-ID of the 401 (Unauthorized) response which carried the last challenge. The S-CSCF shall only proceed further if the Call-IDs match;

2) stop timer reg-await-auth;

3) in the case the algorithm is "SHA2-256", "SHA2-512/256" or "MD5", check the following additional fields:

NOTE 2: The MD5 algorithm is only supported for backward compatibility.

- a "realm" header field parameter matching the "realm" header field parameter in the authentication challenge;
- an "algorithm" header field parameter which matches the "algorithm" header field parameter sent in the authentication challenge;
- "nonce" header field parameter matching the "nonce" header field parameter in the authentication challenge;
- a "cnonce" header field parameter; and
- a "nc" header field parameter.

The S-CSCF shall only proceed with the following steps in this paragraph if the authentication challenge response was included;

4) check whether the received authentication challenge response and the expected authentication challenge response match. The expected response is calculated by the S-CSCF as described in RFC 7616 [286] and RFC 8760 [287] using the H(A1) value provided by the HSS. If the received authentication challenge response and the expected authentication challenge response match, then the UE is considered authenticated. If the UE is considered authenticated, and if the "integrity-protected" header field parameter in the Authorization header field is set to the value "tls-pending" or "tls-yes", then the S-CSCF shall associate the registration with the local state of "tls-protected";

NOTE 3: The S-CSCF can have a local security policy to treat messages other than initial REGISTER requests, messages relating to emergency services, and error messages, differently depending on whether the registration is associated with the state "tls-protected".

4A) if the REGISTER request contains the "reg-id" Contact header field parameter and the "outbound" option tag in a Supported header field, but the first URI in the Path header does not have an "ob" URI parameter, send a 439 (First Hop Lacks Outbound Support) response to the UE;

5) after performing the S-CSCF Registration/deregistration notification procedure with the HSS, as described in 3GPP TS 29.228 [14], store the following information in the local data:

   a) the list of public user identities, including the registered own public user identity and its associated set of implicitly registered public user identities and wildcarded public user identities due to the received REGISTER request. Each public user identity is identified as either barred or non-barred;

   b) all the service profile(s) corresponding to the public user identities being registered (explicitly or implicitly), including initial Filter Criteria(the initial Filter Criteria for the Registered and common parts is stored and the unregistered part is retained for possible use later - in the case of the S-CSCF is retained if the user becomes unregistered);

   c) if S-CSCF restoration procedures are supported, the restoration information, if received, as specified in 3GPP TS 29.228 [14]; and

   d) if PCRF based P-CSCF restoration procedures are supported, all the user profile(s) corresponding to the public user identities being registered (explicitly or implicitly), including the IMSI, if available;

NOTE 4: There might be more than one set of initial Filter Criteria received because some implicitly registered public user identities that are part of the same implicit registration set belong to different service profiles.

6) update registration bindings:

   a) if the Contact URI in the Contact header field does not contains a "bnc" URI parameter, then bind to each non-barred registered public user identity all registered contact information including all header field parameters contained in the Contact header field and all associated URI parameters, with the exception of the "pub-gruu" and "temp-gruu" header field parameters as specified in RFC 5627 [93], and store information for future use;

   b) if the Contact URI in the Contact header field contains a "bnc" URI parameter, as a network option bind each non-barred registered public user identity to a contact address as specified in RFC 6140 [191].

NOTE 5: It is assumed that when the Contact header field contains a "bnc" parameter, the associated public user identities obtained from the HSS are all of a form compatible with registration procedures as specified in RFC 6140 [191]; i.e. the set consists only of distinct public user identities contain global numbers in the international format or wildcarded public user identities representing multiple global numbers in the international format. The S-CSCF procedures for handling the error case where an associated public user identity is incompatible with RFC 6140 [191] is out of scope of this specification.

c) if the Contact URI in the Contact header field does not contain a "bnc" URI parameter, then for each binding that contains a "+sip.instance" Contact header field parameter, assign a new temporary GRUU, as specified in subclause 5.4.7A.3;

d) if the Contact header field of the REGISTER request does not contain a "reg-id" header field parameter (i.e., the multiple registrations mechanism is not used), and there are public user identities (including the public user identity being registered, if previously registered) that belong to this user that have been previously registered with the same private user identity, and with an old contact address different from the one received in the REGISTER request and if the previous registrations have not expired:

- terminate all dialogs, associated with the previously registered public user identities (including the public user identity being registered, if previously registered), with a status code 480 (Temporarily Unavailable) in the Reason header field of the BYE request, as specified in subclause 5.4.5.1.2;

- send a NOTIFY request, to the subscribers to the registration event package of the previously registered public user identities, that indicates that all previously registered public user identities (excluding the public user identity being registered) belonging to this user identified with its private user identity, have been deregistered, as described in subclause 5.4.2.1.2. For the public user identity being registered, the NOTIFY request contains the new contact information; and

NOTE 6: The last dialog to be terminated will be the dialog established by the UE subscribing to the reg event package. When sending the NOTIFY request to the UE over this dialog, the S-CSCF will terminate this dialog by setting in the NOTIFY request the Subscription-State header field to the value of "terminated".

- delete all information associated with the previously registered public user identities;

NOTE 7: Contact related to emergency registration is not affected. The S-CSCF is not able to deregister contact related to emergency registration and will not delete it.

e) if the Contact header field of the REGISTER request contained a "+sip.instance" and a "reg-id" header field parameter, and the SIP URI in the Path header field inserted by the P-CSCF contained an "ob" SIP URI parameter header field, and:

- if the public user identity has not previously been registered with the same "+sip.instance" and "reg-id" Contact header field parameter values, then create the registration flow in addition to any existing registration flow; or

- if the public user identity has previously been registered with the same "+sip.instance" and "reg-id" header field parameter values, then determine whether the request refreshes or replaces an existing registration flow. If the request:

  i) refreshes an existing registration flow, then the S-CSCF shall leave the flow intact; or

  ii) replaces the existing registration flow with a new flow, then the S-CSCF shall:

    a) terminate any dialog, as specified in subclause 5.4.5.1.2, with a status code 480 (Temporarily Unavailable) in the Reason header field of the BYE request, associated with the registration flow being replaced; and

    b) send a NOTIFY request to the subscribers to the registration event package for the public user identity indicated in the REGISTER request, as described in subclause 5.4.2.1.2; and

f) store the used nonce as a valid nonce for this registration or registration flow (if multiple registration mechanism is used) for an operator configured duration.

NOTE 8: The S-CSCF determines whether this REGISTER request replaces or refreshes an existing registration flow by examining the SIP URI in the Path header field inserted into the request by the P-CSCF (see subclause 5.2.2.1).

NOTE 9: The way the S-CSCF identifies the dialogs associated with the registration flow being replaced is implementation specific.

NOTE 10: There might be more than one contact information available for one public user identity.

NOTE 11: The barred public user identities are not bound to the contact information.

NOTE 12: Contact related to emergency registration is not affected. S-CSCF is not able deregister contact related to emergency registration and will not delete that.

7) check whether a Path header field was included in the REGISTER request and construct a list of preloaded Route header fields from the list of entries in the received Path header field. The S-CSCF shall preserve the order of the preloaded Route header fields and bind them to either the contact address of the UE or the registration flow and the associated contact address (if the multiple registration mechanism is used) and contact information that was received in the REGISTER request;

NOTE 13: If this registration is a reregistration or an initial registration (i.e., there are previously registered public user identities belonging to the user that have not been deregistered or expired), then a list of pre-loaded Route header fields will already exist. If multiple registration mechanism was not used, then the existing list of pre-loaded Route header fields is bound to a respective contact address of the UE. However, if multiple registration mechanism was used, then the existing list of pre-loaded Route header fields is bound to a registration flow and the associated contact address that was used to send the REGISTER request. In either case, the new list replaces the old list.

8) determine the duration of the registration by checking the value of the registration expiration interval value in the received REGISTER request and bind it either to the respective contact address of the UE or to the registration flow and the associated contact address (if the multiple registration mechanism is used). Based on local policy, the S-CSCF may reduce the duration of the registration or send back a 423 (Interval Too Brief) response specifying the minimum allowed time for registration. The local policy can take into account specific criteria such as the used authentication mechanism to determine the allowed registration duration;

9) store the "icid-value" header field parameter received in the P-Charging-Vector header field;

10) if an "orig-ioi" header field parameter is received in the P-Charging-Vector header field, store the value of the received "orig-ioi" header field parameter; and

NOTE 14: Any received "orig-ioi" header field parameter will be a type 1 IOI. The type 1 IOI identifies the network from which the request was sent.

11) create and send a 200 (OK) response for the REGISTER request as specified in subclause 5.4.1.2.2F. The S-CSCF shall also store the nonce-count value in the received REGISTER request and include an Authentication-Info header field containing the fields described in RFC 7616 [286] and RFC 8760 [287] as follows:

- a "nextnonce" header field parameter if the S-CSCF requires a new nonce for subsequent authentication responses from the UE. In that case, the S-CSCF shall consider this nonce as a valid nonce for this registration or registration flow (if multiple registration mechanism is used) for an operator configured duration;

- a "qop" header field parameter matching the "qop" Authorization header field parameter sent by the UE;

- a "rspauth" header field parameter with a response-digest calculated as described in RFC 7616 [286] and RFC 8760 [287];

- a "cnonce" header field parameter matching the "cnonce" Authorization header field parameter sent by the UE; and

- a "nc" header field parameter matching the "nc" Authorization header field parameter sent by the UE.

5.4.1.2.2B Protected REGISTER with SIP digest with TLS as a security mechanism

The procedures for subclause 5.4.1.2.2A apply.

5.4.1.2.2C NASS-IMS bundled authentication as a security mechanism

There is no protected REGISTER when NASS-IMS bundled authentication is used as a security mechanism. The procedures of subclause 5.4.1.2.1D apply to all REGISTER requests.

5.4.1.2.2D GPRS-IMS-Bundled authentication as a security mechanism

There is no protected REGISTER when GPRS-IMS-Bundled authentication is used as a security mechanism. The procedures of subclause 5.4.1.2.1E apply to all REGISTER requests.

5.4.1.2.2E Protected REGISTER – Authentication already performed

The S-CSCF shall not perform authentication of the user for any REGISTER request with the "integrity-protected" header field parameter in the Authorization header set to "auth-done".

In this release of this document, when the registration procedure as specified in this subclause is performed, i.e., the REGISTER request contains the "integrity-protected" header field parameter in the Authorization header set to "auth-done", the S-CSCF shall not employ outbound registration as described in RFC 5626 [92].

Upon receipt of a REGISTER request with the "integrity-protected" header field parameter in the Authorization header set to "auth-done", the S-CSCF shall identify the user by the public user identity as received in the To header field and the private user identity as received in the Authorization header field of the REGISTER request.

In addition the S-CSCF shall check whether a registration expiration interval value is included in the REGISTER request and its value. If the registration expiration interval value indicates a zero value, the S-CSCF shall perform the deregistration procedures as described in subclause 5.4.1.4. If the registration expiration interval value does not indicate zero, the S-CSCF shall:

1) if the REGISTER request contains the "reg-id" header field parameter in the Contact header field, respond with a 403 (Forbidden) response to the REGISTER request; and

2) if there are public user identities (including the public user identity being registered, if previously registered) that belong to this user that have been previously registered with the same private user identity, and with an old contact address different from the one received in the REGISTER request and if the previous registrations have not expired:

   a) terminate all dialogs, associated with the previously registered public user identities (including the public user identity being registered, if previously registered), with a status code 480 (Temporarily Unavailable) in the Reason header field of the BYE request, as specified in subclause 5.4.5.1.2;

   b) send a NOTIFY request, to the subscribers to the registration event package of the previously registered public user identities, that indicates that all previously registered public user identities (excluding the public user identity being registered) belonging to this user identified with its private user identity, have been deregistered, as described in subclause 5.4.2.1.2. For the public user identity being registered, the NOTIFY request contains the new contact information; and

NOTE 1: The last dialog to be terminated will be the dialog established by the user (identified with its private user identity) subscribing to its own reg event package using the old contact address. When sending the NOTIFY request over this dialog, the S-CSCF will terminate this dialog by setting in the NOTIFY request the Subscription-State header field to the value of "terminated".

   c) delete all information associated with the previously registered public user identities;

Subsequently, the S-CSCF shall check whether the public user identity received in the To header field is already registered. If it is not registered, the S-CSCF shall proceed beginning with step 1 below. Otherwise, the S-CSCF shall proceed beginning with step 2 below.

1) after performing the S-CSCF Registration/deregistration notification procedure with the HSS, as described in 3GPP TS 29.228 [14], store the following information in the local data:

   a) the list of public user identities, including the registered own public user identity and its associated set of implicitly registered public user identities and wildcarded public user identities due to the received REGISTER request. Each public user identity is identified as either barred or non-barred;

b) all the service profile(s) corresponding to the public user identities being registered (explicitly or implicitly), including initial Filter Criteria(the initial Filter Criteria for the Registered and common parts is stored and the unregistered part is retained for possible use later - in the case of the S-CSCF is retained if the user becomes unregistered); and

c) if PCRF based P-CSCF restoration procedures are supported, all the user profile(s) corresponding to the public user identities being registered (explicitly or implicitly), including the IMSI, if available;

NOTE 2: There might be more than one set of initial Filter Criteria received because some implicitly registered public user identities that are part of the same implicit registration set belong to different service profiles.

2) update registration bindings:

a) if the Contact URI in the Contact header field does not contains a "bnc" URI parameter, then bind to each non-barred registered public user identity all registered contact information including all header parameters contained in the Contact header and all associated URI parameters, with the exception of the URI "pub-gruu" and "temp-gruu" parameters as specified in RFC 5627 [93], and store information for future use;

b) if the Contact URI in the Contact header field contains a "bnc" URI parameter, as a network option bind each non-barred registered public user identity to a contact address as specified in RFC 6140 [191].

NOTE 3: It is assumed that when the Contact header field contains a "bnc" parameter, the associated public user identities obtained from the HSS are all of a form compatible with registration procedures as specified in RFC 6140 [191]; i.e. the set consists only of distinct public user identities contain global numbers in the international format or wildcarded public user identities representing multiple global numbers in the international format. The S-CSCF procedures for handling the error case where an associated public user identity is incompatible with RFC 6140 [191] is out of scope of this specification.

c) if the Contact URI in the Contact header field does not contain a "bnc" URI parameter, then for each binding that contains a "+sip.instance" header field parameter, assign a new temporary GRUU, as specified in subclause 5.4.7A.3.

NOTE 4: There might be more than one contact information available for one public user identity.

NOTE 5: The barred public user identities are not bound to the contact information.

3) check whether a Path header was included in the REGISTER request and construct a list of preloaded Route headers from the list of entries in the received Path header field. The S-CSCF shall preserve the order of the preloaded Route header fields and bind them to the contact information that was received in the REGISTER request;

NOTE 6: If this registration is a reregistration or an initial registration (i.e., there are previously registered public user identities belonging to the user that have not been deregistered or expired), then a list of pre-loaded Route headers will already exist. The new list replaces the old list.

4) determine the duration of the registration by checking the value of the registration expiration interval value in the received REGISTER request. Based on local policy, the S-CSCF may reduce the duration of the registration or send back a 423 (Interval Too Brief) response specifying the minimum allowed time for registration. The local policy can take into account specific criteria such as the used authentication mechanism to determine the allowed registration duration;

5) store the "icid-value" header field parameter received in the P-Charging-Vector header;

6) if an "orig-ioi" header field parameter is received in the P-Charging-Vector header, store the value of the received "orig-ioi" header field parameter; and

NOTE 7: Any received "orig-ioi" header field parameter will be a type 1 IOI. The type 1 IOI identifies the network from which the request was sent.

7) create and send a 200 (OK) response for the REGISTER request as specified in subclause 5.4.1.2.2F.

5.4.1.2.2F Successful registration

If a 200 (OK) response is to be sent for a REGISTER request, the S-CSCF shall, in addition to any contents identified elsewhere in subclause 5.4.1.2, include:

a) the list of received Path header fields;

b) a P-Associated-URI header field containing the list of the registered distinct public user identity and its associated set of implicitly registered distinct public user identities. The first URI in the list of public user identities supplied by the HSS to the S-CSCF will indicate the default public user identity to be used by the S-CSCF. The public user identity indicated as the default public user identity must be a registered public user identity. The S-CSCF shall place the default public user identity as the first entry in the list of URIs present in the P-Associated-URI header field. The default public user identity will be used by the P-CSCF in conjunction with the procedures for the P-Asserted-Identity header field, as described in subclause 5.2.6.3. If the S-CSCF received a display name from the HSS for a public user identity, then the S-CSCF shall populate the P-Associated-URI header field entry for that public identity with the associated display name. The S-CSCF shall not add a barred public user identity to the list of URIs in the P-Associated-URI header field;

NOTE 1: The P-Associated-URI header field lists only the public user identity and its associated set of implicitly registered public user identities that have been registered, rather than the list of user's URIs that may be either registered or unregistered as specified in RFC 7315 [52]. If the registered public user identity which is not barred does not have any other associated public user identities or wildcarded public user identities, the P-Associated-URI header field lists only the registered public user identity itself. The P-Associated-URI header field does not list wildcarded public user identities.

c) a Service-Route header field containing:

A) the SIP URI identifying the S-CSCF containing an indication that subsequent requests routed via this service route (i.e. from the P-CSCF to the S-CSCF) was sent by the UE using either the contact address of the UE or the registration flow and the associated contact address (if the multiple registration mechanism is used) that has been registered and are treated as for the UE-originating case.

NOTE 2: This indication can e.g. be in a parameter in the URI, a character string in the user part of the URI or be a port number in the URI.

The S-CSCF shall use a different SIP URI for each registration. If the multiple registration mechanism is used, the S-CSCF shall also use a different SIP URI for each registration flow associated with the registration;

B) if network topology hiding is required a SIP URI identifying an IBCF as the topmost entry; and

NOTE 3: In accordance with the procedures described in RFC 3608 [38], an IBCF does not insert its own routable SIP URI to the Service-Route header field.

C) if

1) S-CSCF supports indicating the traffic leg associated with a URI as specified in RFC 7549 [225];

2) the UE is roaming;

3) the P-CSCF is not in the home network; and

4) required by local policy

then the S-CSCF may append an "iotl" SIP URI parameter with a value set to "visitedA-homeA" to the S-CSCF SIP URI in the Service-Route header field;

d) if the P-CSCF is in the same network as the S-CSCF a P-Charging-Function-Addresses header field containing the values received from the HSS. It can be determined if the P-CSCF is in the same network as the S-CSCF by the contents of the P-Visited-Network-ID header field included in the REGISTER request;

NOTE 4: The P-CSCF does not check the P-Charging-Function-Addresses header field, providing this header field to the visiting network could cause undefined charging behaviour.

e) a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the REGISTER request, a type 1 "term-ioi" header field parameter and the "icid-value" header field parameter. The S-CSCF shall set the type 1 "term-ioi" header field parameter to a value that identifies the sending network of the response, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the previously received value of "icid-value" header field parameter in the request;

f) a Contact header field listing all contact addresses for this public user identity, including all saved header field parameters and URI parameters (including all ICSI values and IARI values) received in the Contact header field of the REGISTER request,

g) GRUUs in the Contact header field. If the REGISTER request contained a Required or Supported header field containing the value "gruu" then for each contact address in the Contact header field that has a "+sip.instance" header field parameter:

   i) add "pub-gruu" header field parameter containing the public GRUU representing (as specified in subclause 5.4.7A.2) the association between the public user identity from the To header field in the REGISTER request and the instance ID contained in the "+sip.instance" header field parameter;

   ii) if the Contact URI in the Contact header field does not contain a "bnc" URI parameter, then add a "temp-gruu" header field parameters. containing the most recently assigned temporary GRUU representing (as specified in subclause 5.4.7A) the association between the public user identity from the To header field in the REGISTER request and the instance ID contained in the "+sip.instance" header field parameter; and

   iii) if the S-CSCF supports RFC 6140 [191] and the Contact URI in the Contact header field contains a "bnc" URI parameter, then add a "temp-gruu-cookie" header field parameter containing a value generated as specified in RFC 6140 [191];

h) if the received REGISTER request contained both a "reg-id" and "+sip.instance" header field parameters in the Contact header field, and the first URI within the Path header field contains the "ob" SIP URI parameter a Require header field with the "outbound" option-tag as described in RFC 5626 [92];

NOTE 5: There might be other contact addresses available, that this UE or other UEs have registered for the same public user identity.

i) void

j) optionally, a Feature-Caps header field including the ICSI values contained in the service profile of the served user except the ones that require explicit support indication of capabilities by intermediary entities and that have not been indicated as supported according to RFC 6809 [190] for the corresponding registration or registration flow (if multiple registration mechanism is used);

k) if the home network supports calling number verification using signature verification and attestation information, as defined in subclause 3.1,a Feature-Caps header field, as specified in RFC 6809 [190], including the "+g.3gpp.verstat" header field parameter; and

NOTE 6: If the network has indicated support for the calling number verification using signature verification and attestation information to a UE during registration, the network needs to perform calling number verification for all calls delivered to the registered contact address.

l) if the home network supports the response code 607 (Unwanted) as specified in RFC 8197 [254], a Feature-Caps header field including the "+sip.607" header field parameter.

and send the so created 200 (OK) response to the UE.

For all service profiles in the implicit registration set, the S-CSCF shall send a third-party REGISTER request, as described in subclause 5.4.1.7, to each AS that matches the Filter Criteria of the service profile from the HSS for the REGISTER event; and,

NOTE 7: If this registration is a reregistration, the Filter Criteria already exists in the local data.

NOTE 8: If the same AS matches the Filter Criteria of several service profiles for the event of REGISTER request, then the AS will receive several third-party REGISTER requests. Each of these requests will include a public user identity from the corresponding service profile.

The S-CSCF shall consider the public user identity being registered to be bound either to the contact address of the UE or to the registration flow and the associated contact address (if the multiple registration mechanism is used), as specified in the Contact header field, for the duration indicated in the registration expiration interval value.

5.4.1.2.3 Abnormal cases - general

In the case that the expiration timer from the UE is too short to be accepted by the S-CSCF, the S-CSCF shall:

- reject the REGISTER request with a 423 (Interval Too Brief) response, containing a Min-Expires header field with the minimum registration time the S-CSCF will accept.

On receiving a failure response to one of the third-party REGISTER requests, based on the information in the Filter Criteria the S-CSCF may:

- abort sending third-party REGISTER requests; and

- initiate network-initiated deregistration procedure.

If the Filter Criteria does not contain instruction to the S-CSCF regarding the failure of the contact to the AS, the S-CSCF shall not initiate network-initiated deregistration procedure.

In the case that the REGISTER request from the UE contains multiple SIP URIs which are different addresses as Contact header field entries, the S-CSCF shall store:

- the entry in the Contact header field with the highest value of the "q" header field parameter; or

- an entry decided by the S-CSCF based on local policy;

and include the stored entry in the 200 (OK) response.

In the case that the REGISTER request from the UE contains multiple SIP URIs which are the same addresses with the same value of the "q" Contact header field parameter, the S-CSCF shall not store multiple entries with the same "q" value but store one of the entries with the same "q" value based on local policy along with any entries that have different "q" values and include only the stored entries in the 200 (OK) response.

NOTE 1: The UE can register multiple SIP URIs in the Contact header field simultanously, provided they all contain the same IP address and port number. In this case the S-CSCF behaviour is as defined RFC 3261 [26] (i.e multiple Contact header field entries are bound to the public user identity in the To header field and are returned in the 200 (OK) response).

NOTE 2: If the timer reg-await-auth expires, the S-CSCF will consider the authentication to have failed. If the public user identity was already registered, the S-CSCF will leave it registered, as described in 3GPP TS 33.203 [19].

If the S-CSCF receives a new initial REGISTER request before the reg-await-auth timer expires, the S-CSCF shall:

1) stop the reg-await-auth timer; and

2) initiate the authentication procedures for initial registration as if there is no authentication currently ongoing for this user and send a 401 (Unauthorized) response containing a new challenge as described in subclause 5.4.1.

For any error response, the S-CSCF shall insert a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the REGISTER request, a type 1 "term-ioi" header field parameter and the "icid-value" header field parameter. The S-CSCF shall set the type 1 "term-ioi" header field parameter to a value that identifies the sending network of the response, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field and the "icid-value" header field parameter is set to the previously received value of "icid-value" header field parameter in the request.

NOTE 3: Any previously received "orig-ioi" header field parameter will be a type 1 IOI. The type 1 IOI identifies the visited network of the registered user.

If the Contact header field in the REGISTER request from the UE contains an invalid Contact URI as defined in RFC 6140 [191] (e.g., the Contact URI contains both a "bnc" and "user" URI parameter) then the S-CSCF shall reject the REGISTER request with a 400 (Bad Request) response.

5.4.1.2.3A Abnormal cases – IMS AKA as security mechanism

In the case that the REGISTER request, that contains the authentication challenge response from the UE does not match with the expected REGISTER request (e.g. wrong Call-Id or authentication challenge response) and the request has the "integrity-protected" header field parameter in the Authorization header field set to "yes", the S-CSCF shall:

- send a 403 (Forbidden) response to the UE. The S-CSCF shall consider this authentication attempt as failed. The S-CSCF shall not update the registration state of the subscriber.

NOTE 1: If the UE was registered before, it stays registered until the registration expiration time expires.

In the case that the REGISTER request from the UE containing an "auts" Authorization header field parameter, indicating that the SQN was deemed to be out of range by the UE), the S-CSCF will fetch new authentication vectors from the HSS. In order to indicate a resynchronisation, the S-CSCF shall include the value of the "auts" header field parameter received from the UE and the stored RAND, when fetching the new authentication vectors. On receipt of the new authentication vectors from the HSS, the S-CSCF shall either:

- send a 401 (Unauthorized) response to initiate a further authentication attempt, using these new vectors; or
- respond with a 403 (Forbidden) response if the authentication attempt is to be abandoned. The S-CSCF shall not update the registration state of the subscriber.

NOTE 2: If the UE was registered before, it stays registered until the registration expiration time expires.

NOTE 3: Since the UE responds only to two consecutive invalid challenges, the S-CSCF will send a 401 (Unauthorized) response that contains a new challenge only twice.

NOTE 4: In the case of an "auts" Authorization header field parameter being present in the REGISTER request, the "response" Authorization header field parameter in the same REGISTER request will not be taken into account by the S-CSCF.

In the case that the S-CSCF receives a REGISTER request with the "integrity-protected" header field parameter in the Authorization header field set to "yes", for which the public user identity received in the To header field and the private user identity received in the "username" Authorization header field parameter of the REGISTER request do not match to any registered user at this S-CSCF, if the S-CSCF supports S-CSCF restoration procedures, the S-CSCF shall behave as described in subclause 5.4.1.2.2, otherwise the S-CSCF shall:

- respond with a 500 (Server Internal Error) response to the UE.

NOTE 5: This error is not raised if there is a match on the private user identity, but no match on the public user identity.

5.4.1.2.3B Abnormal cases – SIP digest as security mechanism

In the case that the REGISTER request, that contains the authentication challenge response from the UE does not match with the expected REGISTER request (e.g. wrong Call-Id or authentication challenge response) and the request has the "integrity-protected" header field parameter in the Authorization header field set to either "tls-pending", "tls-yes", "ip-assoc-pending", or "ip-assoc-yes", the S-CSCF shall do one of the following:

- send a 403 (Forbidden) response to the UE. The S CSCF shall consider this authentication attempt as failed. The S-CSCF shall not update the registration state of the subscriber; or
- rechallenge the user by issuing a 401 (Unauthorized) response including a challenge as per the authentication procedures described in subclause 5.4.1.2.1B.

NOTE 1: If the UE was registered before, it stays registered until the registration expiration time expires.

In the case that the REGISTER request from the UE contains an invalid "nonce" Authorization header field parameter with a valid challenge response for that nonce (indicating that the client knows the correct username/password), or when the nonce-count value sent by the UE is not the expected value, the S-CSCF shall:

- send a 401 (Unauthorized) response to initiate a further authentication attempt with a fresh nonce and the "stale" header field parameter set to "true" in the WWW-Authenticate header field.

In the case that the S-CSCF receives a REGISTER request with the "integrity-protected" header field parameter in the Authorization header field set to "tls-pending", "tls-yes", "ip-assoc-pending", or "ip-assoc-yes", for which the public user identity received in the To header field and the private user identity received in the Authorization header field of the REGISTER request do not match to any registered or initial registration pending user at this S-CSCF, if the S-CSCF supports S-CSCF restoration procedures, the S-CSCF shall behave as described in subclause 5.4.1.2.2A, otherwise the S-CSCF shall:

- respond with a 500 (Server Internal Error) response to the UE.

NOTE 2: This error is not raised if there is a match on the private user identity, but no match on the public user identity.

5.4.1.2.3C Abnormal cases – SIP digest with TLS as security mechanism

The procedures for subclause 5.4.1.2.3B apply.

5.4.1.2.3D Abnormal cases – NASS-IMS bundled authentication as security mechanism

There are no abnormal cases for NASS-IMS bundled authentication.

5.4.1.2.3E Abnormal cases – GPRS-IMS-Bundled authentication as security mechanism

There are no abnormal cases for GPRS-IMS-Bundled authentication.

5.4.1.3 Authentication and reauthentication

Authentication and reauthentication is performed by the registration procedures as described in subclause 5.4.1.2.

5.4.1.4 User-initiated deregistration

5.4.1.4.1 Normal cases

When S-CSCF receives a REGISTER request with the registration expiration interval value containing the value zero, the S-CSCF shall:

1) verify that the REGISTER request is associated with an existing registered contact or an existing flow or, if the S-CSCF restoration procedures are supported by this S-CSCF, attempt to restore a contact or flow from HSS associated with the REGISTER request. If no associated contact or flow exists then the S-CSCF shall send a 481 (Call Leg/Transaction Does Not Exist) response to the UE and skip the remaining procedures in this subclause;

2) if IMS AKA is in use as the security mechanism, check whether the "integrity-protected" header field parameter in the Authorization header field set to "yes", indicating that the REGISTER request was received integrity protected. The S-CSCF shall only proceed with the following steps if the "integrity-protected" header field parameter is set to "yes";

3) if SIP digest without TLS or SIP digest with TLS is in use as a security mechanism, check whether the "integrity-protected" header field parameter in the Authorization header field set to "tls-yes" or "ip-assoc-yes", indicating that the REGISTER request was received from a previously registered user. If the "integrity-protected" header field parameter is set to "tls-pending", "ip-assoc-pending" or is not present the S-CSCF shall ensure authentication is performed as described in subclause 5.4.1.2.1 (and consequently subclause 5.4.1.2.1B or 5.4.1.2.1C) if local policy requires. The S-CSCF shall only proceed with the following steps if the "integrity-protected" header field parameter is set to "tls-yes", "ip-assoc-yes", or the required authentication is successfully performed if required by local policy;

4) if NASS-IMS bundled authentication is in use as a security mechanism, only proceed with the following steps if the "integrity-protected" header field parameter in the Authorization header field does not exist or without an Authorization header field, and one or more Line-Identifiers previously received over the Cx interface, stored as a result of an Authentication procedure with the HSS, as described in 3GPP TS 29.228 [14], are available for the user;

4A) if the security mechanism as described in subclause 5.4.1.2.2E is in use, check whether the "integrity-protected" header field parameter in the Authorization header field set to "auth-done". The S-CSCF shall only proceed with the following steps if the "integrity-protected" header field parameter is set to "auth-done";

5) release all INVITE dialogs that include this user's contact addresses or the flows that are being deregistered, and where these dialogs were initiated by or terminated towards these contact addresses and the same public user identity found that was To header field that was received REGISTER request or with one of the implicitly registered public user identities by applying the steps listed in subclause 5.4.5.1.2;

6) examine the Contact header field in the REGISTER request, and:

a) if the value "*" is not included in the Contact header field and:

i) if the "reg-id" header field parameter is not included in the Contact header field, then:

- remove the binding (i.e. deregister) between the public user identity found in the To header field together with the implicitly registered public user identities and the contact addresses specified in the REGISTER request. The S-CSCF shall only remove the contact addresses that were registered by this UE;

ii) if the "reg-id" header field parameter and "+sip.instance" header field parameter are included in the Contact header field, and the UE supports multiple registrations (i.e. the "outbound" option tag is included in the Supported header field), then:

- remove the binding (i.e. deregister) between the public user identity indicated in the To header field (together with the associated implicitly registered public user identities) and the flow identified by the "reg-id" header field parameter;

7) if the S-CSCF receives a REGISTER request with the value "*" in the Contact header field and the value zero in the Expires header field, remove all contact addresses that were bound to the public user identity found in the To header field and have been registered by this UE identified with its private user identity;

8) for all service profiles in the implicit registration set send a third-party REGISTER request, as described in subclause 5.4.1.7, to each AS that matches the Filter Criteria of the service profile from the HSS for the REGISTER event;

9) if this is a deregistration request for the only public user identity currently registered with its associated set of implicitly registered public user identities (i.e. no other is registered) and there are still active multimedia sessions that includes this user's registered contact address, where the session was initiated by or terminated towards the contact with the registered contact address for that public user identity which is currently registered or with one of the implicitly registered public user identities, release only each of these multimedia sessions associated with the registered contact address by applying the steps listed in subclause 5.4.5.1.2. The S-CSCF shall only release dialogs associated to the multimedia sessions originated or terminated towards the registered user's contact address; and

10) send a 200 (OK) response to a REGISTER request that contains a list of Contact header fields enumerating all contacts and flows that are currently registered, and all contacts that have been deregistered. For each contact address and the flow that has been deregistered, the Contact header field shall contain the contact address and the "reg-id" header field parameter that identifies the flow, if a flow was deregistered, and the associated information, and the registration expiration interval value shall be set to zero.

If all public user identities of the UE are deregistered, then the S-CSCF may consider the UE and P-CSCF subscriptions to the reg event package cancelled (i.e. as if the UE had sent a SUBSCRIBE request with an Expires header field containing a value of zero).

If the Authorization header field of the REGISTER request contained an "integrity-protected" header field parameter set to the value "no", the S-CSCF shall apply the procedures described in subclause 5.4.1.2.1.

On completion of the above procedures in this subclause and of the S-CSCF Registration/deregistration notification procedure with the HSS, as described in 3GPP TS 29.228 [14], for one or more public user identities, the S-CSCF shall:

1) update or remove those public user identities, their registration state and the associated service profiles from the local data; and

2) if all the contacts bound to the public user identities have been deregistered and there is no ongoing session due to the public user identities, then remove all the stored AS IP addresses which are associated with those public user identities.

Based on operators' policy the S-CSCF can request the HSS to either be kept or cleared as the S-CSCF allocated to this subscriber. If emergency contacts are still registered for this subscriber, the S-CSCF requests the HSS to be kept as the S-CSCF allocated to this subscriber.

5.4.1.4.2 Abnormal cases - IMS AKA as security mechanism

If case that the S-CSCF receives a REGISTER request with the "integrity-protected" header field parameter in the Authorization header set to "yes" or to "tls-connected", for which the public user identity received in the To header and the private user identity received in the Authorization header of the REGISTER request do not match to any registered user at this S-CSCF, if the S-CSCF supports S-CSCF restoration procedures as specified in 3GPP TS 23.380 [7D], the S-CSCF shall behave as described in subclause 5.4.1.4.1, otherwise the S-CSCF shall:

- respond with a 500 (Server Internal Error) response to the UE.

NOTE: This error is not raised if there is a match on the private user identity, but no match on the public user identity.

5.4.1.4.4 Abnormal cases – SIP digest with TLS as security mechanism

The procedures for subclause 5.4.1.4.2 apply.

5.4.1.4.5 Abnormal cases – NASS-IMS bundled authentication as security mechanism

There are no abnormal cases for NASS-IMS bundled authentication.

5.4.1.4.6 Abnormal cases – GPRS-IMS-Bundled authentication as security mechanism

There are no abnormal cases for GPRS-IMS-Bundled authentication.

5.4.1.5 Network-initiated deregistration

NOTE 1: A network-initiated deregistration event that occurs at the S-CSCF can be received from the HSS or can be an internal event in the S-CSCF.

For any registered public user identity, the S-CSCF can deregister:

- all contact addresses bound to the indicated public user identity (i.e. deregister the respective public user identity);
- some contact addresses bound to the indicated public user identity;
- a particular contact address bound to the indicated public user identity; or
- one or more registration flows and the associated contact address bound to the indicated public user identity, when the UE supports multiple registration procedure;

by sending a single NOTIFY request.

Prior to initiating the network-initiated deregistration for the only currently registered public user identity and its associated set of implicitly registered public user identities and wildcarded public user identities that have been registered either with the same contact address of the UE or the same registration flow and the associated contact address (if the multiple registration mechanism is used), i.e. there are no other public user identities registered either with this contact address or with this registration flow and the associated contact address (if the multiple registration mechanism is used), and there are still active multimedia sessions belonging either to this contact address or to this registration flow and the associated contact address (if the multiple registration mechanism is used), the S-CSCF shall release only multimedia sessions belonging to this contact address or to this registration flow and the associated contact address (if the multiple registration mechanism is used) as described in the following paragraph. The multimedia sessions for the same public user identity, if registered either with another contact address or another registration flow and the associated contact address (if the multiple registration mechanism is used) remain unchanged.

Prior to initiating the network-initiated deregistration while there are still active multimedia sessions that are associated with this user and contact, the S-CSCF shall release none, some or all of these multimedia sessions by applying the steps listed in subclause 5.4.5.1.2 under the following conditions:

- when the S-CSCF does not expect the UE to reregister a given public user identity and its associated set of implicitly registered public user identities that have been registered with respective contact address (i.e. S-CSCF will set the event attribute within the respective <contact> element to "rejected" for the NOTIFY request, as

described below), the S-CSCF shall release all sessions that are associated with the registered contact address for the public user identities using the contact address that is being deregistered, which includes the implicitly registered public user identities.

- when the S-CSCF expects the UE to reregister a given public user identity and its associated set of implicitly registered public user identities that have been registered with respective contact address (i.e. S-CSCF will set the event attribute within the respective <contact> element to "deactivated" for the NOTIFY request, as described below), the S-CSCF shall only release sessions that currently include the user's contact address, where the session was initiated by or terminated towards the user with the contact address registered to one of the public user identities using the contact address that is being deregistered, which includes the implicitly registered public user identities.

When a network-initiated deregistration event occurs for one or more public user identities that are bound either to one or more contact addresses or registration flows and the associated contact addresses (if the multiple registration mechanism is used), the S-CSCF shall send a NOTIFY request to all subscribers that have subscribed to the respective reg event package. For each NOTIFY request, the S-CSCF shall:

1) set the Request-URI and Route header field to the saved route information during subscription;

2) set the Event header field to the "reg" value;

3) in the body of the NOTIFY request, include as many <registration> elements as many public user identities the S-CSCF is aware of the user owns;

4) set the aor attribute within each <registration> element to one public user identity:

   a) set the <uri> sub-element inside each <contact> sub-element of each <registration> element to the respective contact address provided by the UE;

   b) if the public user identity:

      i) has been deregistered (i.e. all contact addresses and all registration flows and associated contact addresses bound to the indicated public user identity are removed) then:

         - set the state attribute within the <registration> element to "terminated";

         - set the state attribute within each <contact> element belonging to this UE to "terminated"; and

         - set the event attribute within each <contact> element belonging to this UE to either "unregistered", or "deactivated" if the S-CSCF expects the UE to reregister or "rejected" if the S-CSCF does not expect the UE to reregister; or

NOTE 2: If the multiple registration mechanism is used, then the reg-id header field parameter will be included as an <unknown-param> element within each respective <contact> element.

NOTE 3: The UE will consider its public user identity as deregistered when the binding between the respective public user identity and all contact addresses and all registration flows and associated contact addresses (if the multiple registration mechanism is used) belonging to the UE have been removed.

      ii) has been kept registered then:

         I) set the state attribute within the <registration> element to "active";

         II) set the state attribute within each <contact> element to:

            - for the binding between the public user identity and either the contact address or a registration flow and associated contact addresses (if the multiple registration mechanism is used) to be removed set the state attribute within the <contact> element to "terminated", and event attribute element to either "unregistered", or "deactivated" if the S-CSCF expects the UE to reregister or "rejected" if the S-CSCF does not expect the UE to reregister; or

            - for the binding between the public user identity and either the contact address or the registration flow and associated contact addresses (if the multiple registration mechanism is used) which remain unchanged, if any, leave the <contact> element unmodified, and if the contact has been assigned GRUUs and the Contact URI did not contain a "bnc" SIP URI parameter then set the <pub-gruu> and <temp-gruu> sub-elements of the <contact> element as specified in

RFC 5628 [94] and include the <unknown-param> sub-element within each <contact> to any additional header field parameters contained in the Contact header field of the REGISTER request according to RFC 3680 [43]; and

NOTE 4: There might be more than one contact information available for one public user identity. When deregistering this UE, the S-CSCF will only modify the <contact> elements that were originally registered by this UE using its private user identity. The <contact> elements of the same public user identity, if registered by another UE using different private user identities remain unchanged.

5) add a P-Charging-Vector header field with the "icid-value" header field parameter set to the value populated in the initial request for the dialog and a type 1 "orig-ioi" header field parameter. The S-CSCF shall set the type 1 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The S-CSCF shall not include the type 1 "term-ioi" header field parameter.

The S-CSCF shall only include the non-barred public user identities in the NOTIFY request.

When sending a final NOTIFY request with all <registration> element(s) having their state attribute set to "terminated" (i.e. all public user identities have been deregistered or expired), the S-CSCF shall also terminate the subscription to the registration event package by setting the Subscription-State header field to the value of "terminated".

Also, for all service profiles in the implicit registration set the S-CSCF shall send a third-party REGISTER request, as described in subclause 5.4.1.7, to each AS that matches the Filter Criteria of the service profile from the HSS as if a equivalent REGISTER request had been received from the user deregistering that public user identity, or combination of public user identities.

On completion of the above procedures for one or more public user identities linked to the same private user identity, the S-CSCF shall consider those public user identities and the associated implicitly registered public user identities which have no contact address or a registration flow and associated contact addresses (if the multiple registration mechanism is used) bound to them as deregistered. On completion of the S-CSCF Registration/deregistration notification procedure with the HSS, as described in 3GPP TS 29.228 [14], the S-CSCF shall:

1) update or remove those public user identities linked to the same private user identity, their registration state and the associated service profiles from the local data (based on operators' policy the S-CSCF can request of the HSS to either be kept or cleared as the S-CSCF allocated to this subscriber); and

2) if all the contacts bound to the public user identities have been deregistered and there is no ongoing session due to the public user identities, then remove all the stored AS IP addresses which are associated with those public user identities.

On the completion of the Network initiated de-registration by the HSS procedure, as described in 3GPP TS 29.228 [14], the S-CSCF shall remove:

1) those public user identities, their registration state and the associated service profiles from the local data; and

2) if all the contacts bound to the public user identities have been deregistered and there is no ongoing session due to the public user identities, then remove all the stored AS IP addresses which are associated with those public user identities.

5.4.1.6 Network-initiated reauthentication

The S-CSCF may request a subscriber to reauthenticate at any time, based on a number of possible operator settable triggers.

NOTE 1: Triggers for re-authentication include e.g. a current registration of the UE is set to expire at a predetermined time; one or more error conditions in the S-CSCF; the S-CSCF mistrusts the UE.

If the S-CSCF is informed that a private user identity needs to be re-authenticated, the S-CSCF shall generate a NOTIFY request on all dialogs which have been established due to subscription to the reg event package of that user. For each NOTIFY request the S-CSCF shall:

1) set the Request-URI and Route header field to the saved route information during subscription;

2) set the Event header field to the "reg" value;

3) in the body of the NOTIFY request, include as many <registration> elements as many public user identities the S-CSCF is aware of the user owns:

   a) set the <uri> sub-element inside the <contact> sub-element of each <registration> element to the contact address provided by the UE;

   b) set the aor attribute within each <registration> element to one public user identity;

   c) set the state attribute within each <registration> element to "active";

   d) set the state attribute within each <contact> element to "active";

   e) set the event attribute within each <contact> element that was registered by this UE to "shortened";

   f) set the expiry attribute within each <contact> element that was registered by this UE to an operator defined value; and

   g) if the Contact URI did not contain a "bnc" SIP URI parameter then set the <pub-gruu> and <temp-gruu> sub-elements within each <contact> element as specified in subclause 5.4.2.1.2; and

NOTE 2: There might be more than one contact information available for one public user identity. The S-CSCF will only modify the <contact> elements that were originally registered by this UE using its private user identity. The S-CSCF will not modify the <contact> elements for the same public user identitity, if registered by another UE using different private user identity.

4) set a P-Charging-Vector header field with the "icid-value" header field parameter set to the value populated in the initial request for the dialog and a type 1 "orig-ioi" header field parameter. The S-CSCF shall set the type 1 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The S-CSCF shall not include the type 1 "term-ioi" header field parameter.

Afterwards the S-CSCF shall wait for the user to reauthenticate (see subclause 5.4.1.2).

NOTE 3: Network initiated re-authentication can occur due to internal processing within the S-CSCF.

The S-CSCF shall only include the non-barred public user identities in the NOTIFY request.

When generating the NOTIFY request, the S-CSCF shall shorten the validity of all registration lifetimes associated with this private user identity to an operator defined value that will allow the user to be re-authenticated.

5.4.1.7 Notification of Application Servers about registration status

During registration, the S-CSCF shall include the P-Access-Network-Info header fields (as received in the REGISTER request from the UE and the P-CSCF) and a P-Visited-Network-ID header field (as received in the REGISTER request from the UE) in the third-party REGISTER request sent towards the ASs, if the AS is part of the trust domain. If the AS is not part of the trust domain, the S-CSCF shall not include any P-Access-Network-Info header field or P-Visited-Network-ID header field. The S-CSCF shall not include a P-Access-Network-Info header field in any responses to the REGISTER request.

If the registration procedure described in subclauses 5.4.1.2, 5.4.1.4 or 5.4.1.5 (as appropriate) was successful, the S-CSCF shall send a third-party REGISTER request to each AS with the following information:

a) the Request-URI, which shall contain the AS's SIP URI;

b) the From header field, which shall contain the S-CSCF's SIP URI;

c) the To header field, which shall contain a non-barred public user identity belonging to the service profile of the processed Filter Criteria. It may be either a public user identity as contained in the REGISTER request received from the UE or one of the implicitly registered public user identities in the service profile, as configured by the operator;

NOTE 1: For the whole implicit registration set only one public user identity per service profile appears in the third-party REGISTER requests. Thus, based on third-party REGISTER requests only, the ASs will not have complete information on the registration state of each public user identity in the implicit registration set. The only way to have a complete and continuously updated information (even upon administrative change in subscriber's profile) is to subscribe to the reg event package.

d) the Contact header field, which shall contain the S-CSCF's SIP URI;

e) for initial registration and user-initiated reregistration (subclause 5.4.1.2), the registration expiration interval value, which shall contain the same value that the S-CSCF returned in the 200 (OK) response for the REGISTER request received from the UE;

f) for user-initiated deregistration (subclause 5.4.1.4) and network-initiated deregistration (subclause 5.4.1.5), the registration expiration interval value, which shall contain the value zero;

NOTE 2: The user can have one or more contacts registered after a third-party deregister. If an AS needs more detailed knowledge of the user registration status, the AS can subscribe to the reg event package.

g) for initial registration and user-initiated reregistration (subclause 5.4.1.2), a message body, if there is Filter Criteria indicating the need to include HSS provided data for the REGISTER event (e.g. HSS may provide AS specific data to be included in the third-party REGISTER) or if there is Filter Criteria indicating the need to include the contents of the incoming REGISTER request or the contents of the 200 (OK) response to the incoming REGISTER request in the body of the third-party REGISTER. The S-CSCF shall format the MIME body and set the value of the Content-Type header field to include the MIME type specified in subclause 5.4.1.7A;

NOTE 3: When the AS is outside the trust domain for any header field that is permitted in the REGISTER request received from the UE or final response to the REGISTER request received from the UE, including an Include Register Request or Include Register Response indication in the initial Filter Criteria would cause the incoming REGISTER request or 200 (OK) response to the incoming REGISTER request contents to be delivered to the AS revealing information that AS is not trusted to obtain. Include Register Request and Include Register Response indication is therefore not included in the initial Filter Criteria for an AS that exists outside the trust domain for any such header field.

h) for initial registration and user-initiated reregistration, the P-Charging-Vector header field, which shall contain the same "icid-value" header field parameter that the S-CSCF received in the REGISTER request from the UE. The S-CSCF shall insert a type 3 orig-ioi parameter in place of any received "orig-ioi" header field parameter and shall set the type 3 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The S-CSCF shall not include the type 3 "term-ioi" header field parameter;

i) for initial registration and user-initiated reregistration, a P-Charging-Function-Addresses header field, which shall contain the values received from the HSS if the message is forwarded within the S-CSCF home network;

j) in case the received REGISTER request contained a P-User-Database header field and the AS belongs to the same operator as the S-CSCF, optionally a P-User-Database header field which shall contain the received value; and

k) void

l) if the S-CSCF supports using a token to identify the registration for initial registration and user initiated reregistration, a "+g.3gpp.registration-token" Contact header field parameter, as defined in subclause 7.9.7, set to a value identifying this registration among the set of registrations for the registered URI. The value shall be the same until the UE is deregistered.

NOTE 4: Setting the value of the registration-token to the same value as the S-CSCF will use for the "id" parameter identifying this contact in the "reg" event package allows the AS to retrieve the value using the "reg" event package.

For third-party REGISTER upon user-initiated reregistration, user-initiated deregistration or network-initiated deregistration, the S-CSCF shall send SIP REGISTER request towards the IP address associated with the corresponding registered public user identity stored as described in subclause 5.4.0.

When the S-CSCF receives any response to a third-party REGISTER request, the S-CSCF shall store the value of the "term-ioi" header field parameter received in the P-Charging-Vector header field, if present.

NOTE 5: Any received "term-ioi" header field parameter will be a type 3 IOI. The type 3 IOI identifies the service provider from which the response was sent.

If the S-CSCF fails to receive a SIP response or receives a 408 (Request Timeout) response or a 5xx response to a third-party REGISTER, the S-CSCF shall:

- if the default handling defined in the filter criteria indicates the value "SESSION_CONTINUED" as specified in 3GPP TS 29.228 [14] or no default handling is indicated, no further action is needed; and

- if the default handling defined in the filter criteria indicates the value "SESSION_TERMINATED" as specified in 3GPP TS 29.228 [14], initiate the network-initiated deregistration as described in subclause 5.4.1.5 for the currently registered public user identity and its associated set of implicitly registered non-barred public user identities bound to the contact(s) registered in the REGISTER request causing the third-party REGISTER request.

5.4.1.7A Including contents in the body of the third-party REGISTER request

If there is a service information XML element provided in the HSS Filter Criteria for an AS (see 3GPP TS 29.228 [14]), then in the third-party REGISTER request the S-CSCF shall:

- include in the message body the service information within the <service-info> XML which is a child XML element of an <ims-3gpp> element with the "version" attribute set to "1" element as described in subclause 7.6; and

- set the value of the content type to the MIME type specified in subclause 7.6.

If there is an Include Register Request XML element provided in the HSS Filter Criteria for an AS (see 3GPP TS 29.228 [14]), then in the third-party REGISTER request the S-CSCF shall:

- include in the message body the incoming SIP REGISTER request within a "message/sip" MIME body as defined in RFC 3261 [26]; and

- set the value of the content type to "message/sip".

If there is an Include Register Response XML element provided in the HSS Filter Criteria for an AS (see 3GPP TS 29.228 [14]), then in the third-party REGISTER request, the S-CSCF shall:

- include in the message body the 200 (OK) response to the incoming SIP REGISTER request within a "message/sip" MIME body as defined in RFC 3261 [26]; and

- set the value of the content type to "message/sip".

If there is more than one message body to be included in the third-party REGISTER request then in the third-party REGISTER request the S-CSCF shall:

- include a multipart message body and set the value of the Content-Type header field to "multipart/mixed" as specified in RFC 2046 [149] and RFC 5621 [150]; and

- set the Content-Type of the elements of the MIME body to the content type specified for the body.

If there is only one message body to be included in the third-party REGISTER request then the S-CSCF sets the Content-Type header field to the content type specified for the body.

5.4.1.8 Service profile updates

NOTE 1: The S-CSCF can receive an update of subscriber data notification on the Cx interface, from the HSS, which can affect the stored information about served public user identities. According to 3GPP TS 29.228 [14], the changes are guaranteed not to affect the default public user identity within the registration implicit set.

When receiving a Push-Profile-Request (PPR) from the HSS (as described in 3GPP TS 29.228 [14]), modifying the service profile of served public user identities, the S-CSCF shall:

1) if the modification consists in the addition of a new non-barred public user identity to an implicit set, or in the change of status from barred to non-barred for a public user identity already in the implicit set, add the public user identity to the list of registered, non-barred public user identities;

2) if the modification consists in the deletion of a public user identity while there are no active multimedia session belonging to this public user identity, or in the change of status from non-barred to barred of a public user identity in an implicit set, remove the public user identity from the list of registered, non-barred public user identities;

NOTE 2: As the S-CSCF checks the barring status of the public user identity on receipt of a initial request for a dialog, or a standalone transaction, the above procedures have no impact on transactions or dialogs already in progress and are effective only for new transactions and dialogs.

3) if the modification consists of deletion of a public user identity from an implicit registration set while there are active multimedia session belonging to this public user identity and contact, release these multimedia sessions as described in subclause 5.4.5.1.2 and after all multimedia sessions are released, remove the public user identity from the list of registered, non-barred public user identities; and

4) synchronize with the UE and IM CN entities, by either:

- performing the procedures for notification of the reg-event subscribers about registration state, as described in subclause 5.4.2.1.2; or

- triggering the UE to re-register, by:

a) depending on operator configuration, rejecting a request from the UE using a 504 (Server Time-out) response and indicating in the response that S-CSCF restoration procedures are supported, in accordance with subclause 5.4.3.2; or

b) shortening the life time of the current registration, as described in subclause 5.4.1.6, e.g. when a new trigger point of Register method is added in the iFCs.

NOTE 3: The UE procedure in response to receiving a rejection as described in item a) (see subclause 5.1.2A.1.6) is specified for UEs since 3GPP Rel-8.

5.4.2 Subscription and notification

5.4.2.1 Subscriptions to S-CSCF events

5.4.2.1.1 Subscription to the event providing registration state

When an incoming SUBSCRIBE request addressed to S-CSCF arrives containing the Event header field with the reg event package, the S-CSCF shall:

0) if the Request-URI of the SUBSCRIBE request contains a URI for which currently no binding exists, then send a 480 (Temporarily Unavailable) response indicating that the subscription was not successful and skip the remainder of the steps;

1) check if, based on the local policy, the request was generated by a subscriber who is authorised to subscribe to the registration state of this particular user. The authorized subscribers include:

- all public user identities this particular user owns, that the S-CSCF is aware of, and which are not-barred;

- all the entities identified by the Path header field (i.e. the P-CSCF to which this user is attached to or the IBCF which encrypted the Path header field); and

- all the ASs listed in the initial filter criteria that are part of the trust domain;

if the request is received from a subscriber which is not auhorized to subscribe to the registration state of this particular user, then send a 403 (Forbidden) response indicating that the subscription was not successful and skip the remainder of the steps;

NOTE 1: The S-CSCF finds the identity for authentication of the subscription in the P-Asserted-Identity header field received in the SUBSCRIBE request.

1A) if the Request-URI of the SUBSCRIBE request identifies a public user identity that was implicitly registered using the registration procedures defined in RFC 6140 [191] and performs the functions of an external attached network, and the registration is currently active, then skip the remainder of the procedure in this subclause and route the SUBSCRIBE request to the UE performing the functions of an externally attached network using the procedures defined in subclause 5.4.3.3;

1B) store the "icid-value" header field parameter received in the P-Charging-Vector header field;

2) store the value of the "orig-ioi" header field parameter received in the P-Charging-Vector header field if present;

NOTE 2: Any received "orig-ioi" header field parameter will be either a type 1 IOI or a type 3 IOI. The type 1 IOI identifies the sending network and the type 3 IOI identifies the service provider from which the request was sent.

3) generate a 200 (OK) response acknowledging the SUBSCRIBE request and indicating that the authorised subscription was successful as described in RFC 3680 [43] and RFC 6665 [28]. The S-CSCF shall populate the header fields as follows:

- an Expires header field, set to either the same or a decreased value as the Expires header field in SUBSCRIBE request;

- if the request originated from an ASs listed in the initial filter criteria, a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the SUBSCRIBE request, a type 3 "term-ioi" header field parameter and the "icid-value" header field parameter. The S-CSCF shall set the type 3 "term-ioi" header field parameter to a value that identifies the sending network of the response, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the previously received value of "icid-value" header field parameter in the request; and

- if the request originated from a public user identity this particular user owns, or any of the entities identified by the Path header field, a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the SUBSCRIBE request, a type 1 "term-ioi" header field parameter, and the "icid-value" header field parameter. The S-CSCF shall set the type 1 "term-ioi" header field parameter to a value that identifies the sending network of the response, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the previously received value of "icid-value" header field parameter in the request.

The S-CSCF may set the Contact header field to an identifier uniquely associated to the SUBSCRIBE request and generated within the S-CSCF, that may help the S-CSCF to correlate refreshes for the SUBSCRIBE dialog; and

NOTE 3: The S-CSCF could use such unique identifiers to distinguish between UEs, when two or more users, holding a shared subscription, register under the same public user identity.

4) determine the applicable private user identity as the private user identity included in a REGISTER request:

- which created (implicitly or explictly) a binding of the public user identity in the Request-URI of the SUBSCRIBE request to an contact address; and

- for which one of the following is true:

  a) the 200 (OK) response to the REGISTER request contained the Service-Route header field with the S-CSCF URI matching the URI in the top Route header field of the SUBSCRIBE request (i.e. the SUBSCRIBE request originated by a served UE); or

  b) the 200 (OK) response to the REGISTER request contained a Path header field with a URI matching the URI in the P-Asserted-Identity header field of the SUBSCRIBE request (i.e. the SUBSCRIBE request originated by a P-CSCF serving a UE).

NOTE 4: If the URI in the P-Asserted-Identity header field of the initial SUBSCRIBE request matches URIs of several Path header fields (e.g. the SUBSCRIBE request is originated by Rel-7 P-CSCF), the applicable private user identity is not determined.

Afterwards the S-CSCF shall perform the procedures for notification about registration state as described in subclause 5.4.2.1.2.

If the SUBSCRIBE request originated from an AS listed in the initial filter criteria, for any response that is not a 2xx response, the S-CSCF shall insert a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the SUBSCRIBE request, a type 3 "term-ioi" header field parameter and the "icid-value" header field parameter. The S-CSCF shall set the type 3 "term-ioi" header field parameter to a value that identifies the sending network of the response, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the previously received value of "icid-value" header field parameter in the request.

If the SUBSCRIBE request originated from a public user identity this particular user owns, or any of the entities identified by the Path header field, for any response that is not a 2xx response, the S-CSCF shall insert a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the SUBSCRIBE request, a type 1 "term-ioi" header field parameter and the "icid-value" header field parameter. The S-CSCF shall set the type 1 "term-ioi" header field parameter to a value that identifies the sending network of the response, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the previously received value of "icid-value" header field parameter in the request.

When the S-CSCF receives a subscription refresh request for a dialog that was established by the UE subscribing to the reg event package, the S-CSCF shall accept the request irrespective if the user's public user identity specified in the SUBSCRIBE request is either registered or has been deregistered.

5.4.2.1.2 Notification about registration state

The UE can bind any one of its public user identities either to its contact address or to a registration flow and the associated contact address (if the multiple registration mechanism is used) via a single registration procedure. When multiple registrations mechanism is used to register a public user identity and bind it to a registration flow and the associated contact address, the S-CSCF shall generate a NOTIFY request that includes one <contact> element for each binding between a public user identity and a registration flow and the associated contact address.

NOTE 1: If the UE binds a given public user identity to the same contact address but several registration flows and the associated contact address (via several registrations), then the NOTIFY request will contain one <contact> element for each registration flow and the associated contact address. Each respective <contact> elements will contain the same contact address in the <uri> sub-element, but different value in the "id" attribute and different "reg-id" value included in the respective <unknown-param> element.

For every successful registration that creates a new binding between a public user identity and either its contact address or the registration flow and the associated contact address (if the multiple registration mechanism is used, the NOTIFY request shall always include a new <contact> element containing new value in the "id" sub-element, the state attribute set to "active", and event attribute set to either "registered" or "created".

Any successful registration (that creates a new binding between a public user identity and either its contact address or a registration flow and associated contact address) may additionally replace or remove one or more existing bindings. In the NOTIFY request, for each replaced or removed binding, the <contact> element shall have the state attribute set to "terminated" and the event attribute set to "unregistered", "deactivated", or "rejected".

NOTE 2: When multiple registrations mechanism is not used, if the UE registers new contact address then all registrations, if any, using an old contact address are deregistered, i.e. the new registration replaces the old registrations. Hence, for each deregistered public user identity, the NOTIFY request will have the state attribute within the <registration> element set to "terminated" and the state attribute in the <contact> element set to "terminated" and the event attribute set to "unregistered", "deactivated", or "rejected".

NOTE 3: If the UE uses a multiple registrations mechanism to bind a public user identity to a new registration flow the registration flow and the associated contact address, and if the new registration flow replaces an existing registration flow, then for the registration flow and the associated contact address being replaced, the respective <contact> element in the NOTIFY request will have the state attribute set to "terminated" and the event attribute set to "unregistered", "deactivated", or "rejected".

The S-CSCF shall send a NOTIFY request:

- when an event pertaining to the user occurs. In this case the NOTIFY request is sent on all dialogs which have been established due to subscription to the reg event package of that user; and

- as specified in RFC 6665 [28].

When sending a NOTIFY request, the S-CSCF shall not use the default filtering policy as specified in RFC 3680 [43], i.e. the S-CSCF shall always include in every NOTIFY request the state information of all registered public user identities of the user (i.e. the full state information).

NOTE 4: Contact information related to emergency registration is not included.

When generating NOTIFY requests, the S-CSCF shall not preclude any valid reg event package parameters in accordance with RFC 3680 [43].

For each NOTIFY request triggered by an event and on all dialogs which have been established due to subscription to the reg event package of that user, the S-CSCF shall:

1) set the Request-URI and Route header field to the saved route information during subscription;

2) set the Event header field to the "reg" value;

3) in the body of the NOTIFY request, include one <registration> elements for each public user identity that the S-CSCF is aware the user owns.

   If the user shares one or more public user identities with other users, the S-CSCF shall include any contact addresses registered by other users of the shared public user identity in the NOTIFY request;

4) for each <registration> element:

   a) set the aor attribute to one public user identity or if the public user identity of this <registration> element is a wildcarded public user identity, then choose arbitrarily a public user identity that matches the wildcarded public user identity and the service profile of the wildcarded public user identity and set the aor attribute to this public user identity;

NOTE 5: If the public user identity of this <registration> element is a wildcarded public user identity, the value of the aor attribute will not be used by the receiver of the NOTIFY.

   b) set the <uri> sub-element inside each <contact> sub-element of the <registration> element to the contact address provided by the respective UE as follows:

      I) if the aor attribute of the <registration> element contains a SIP URI and if the Contact URI did not contain a "bnc" SIP URI parameter, then for each contact address that contains a "+sip.instance" Contact header field parameter, include <pub-gruu> and <temp-gruu> sub-elements within the corresponding <contact> element. The S-CSCF shall set the contents of these elements as specified in RFC 5628 [94]; or

      II) if the aor attribute of the <registration> element contains a tel-URI, determine its alias SIP URI and if the Contact URI did not contain a "bnc" SIP URI parameter then include a copy of the <pub-gruu> and <temp-gruu> sub-elements from that equivalent element;

   c) if the respective UE has provided a display-name in a Contact header field, set the <display-name> sub-element inside the respective <contact> sub-element of the <registration> element to the value provided by the UE according to RFC3680 [43];

   d) if the user owns a wildcarded public user identity then include a <wildcardedIdentity> sub-element as described in subclause 7.10.2;

   e) if the public user identity set in step a):

      I) has been deregistered either by the UE or the S-CSCF (i.e. upon the deregistration, there are no binding left between this public user identity and either a contact address or a registration flows and associated contact addresses that belong to this user) then:

         - set the state attribute within the <registration> element to "terminated";

         - set the state attribute within each <contact> element belonging to this user to "terminated"; and

         - set the event attribute within each <contact> element to "deactivated", "expired", "unregistered", "rejected" or "probation" according to RFC 3680 [43].

         If the public user identity has been deregistered for this user and this deregistration has already been indicated in the NOTIFY request, and no new registration for this user has occurred, its <registration> element shall not be included in the subsequent NOTIFY requests;

      II) has been registered by the UE (i.e. the public user identity has not been previously bound either to a contact address or to a registration flow and the associated contact address (if the multiple registration mechanism is used)) then:

         - set the <unknown-param> element to any additional header field parameters contained in the Contact header field of the REGISTER request according to RFC 3680 [43];

NOTE 6: If the multiple registration mechanism is used, then the reg-id header field parameter will be included as an <unknown-param> element.

- if the subscription contains, for the applicable private user identity (determined as described in subclause 5.4.2.1.1) and the public user identity, any of the policies described in subclause 7.10.3, then include the policy associated with the applicable private user identity and the public user identity using coding described in subclause 7.10.3;

- set the state attribute within the <registration> element to "active"; and:

- set the state attribute within the <contact> element belonging to this user to "active", include new value for the "id" attribute within the <contact> sub-element, and set the event attribute within this <contact> element to "registered";

NOTE 7: If this registration, that created new binding, additionally replaces or removes one or more existing registrations, then for the replaced or removed registrations the respective <registration> elements and <contact> elements will be modified accordingly.

III) has been re-registered (i.e. it has been previously registered) then:

- set the state attribute within the <registration> element to "active";

- if the subscription contains, for the applicable private user identity (determined as described in subclause 5.4.2.1.1) and the public user identity, any of the policies described in subclause 7.10.3, then include the policy associated with the applicable private user identity and the public user identity using coding described in subclause 7.10.3;

- set the <unknown-param> element to any additional header field parameters contained in the Contact header field of the REGISTER request according to RFC 3680 [43];

- for contact addresses to be registered: set the state attribute within the <contact> element to "active"; and set the event attribute within the <contact> element to "registered";

- for contact addresses to be re-registered, set the state attribute within the <contact> element to "active"; and set the event attribute within the <contact> element to "refreshed" or "shortened" according to RFC 3680 [43]; and

- for contact addresses that remain unchanged, if any, leave the <contact> element unmodified (i.e. the event attribute within the <contact> element includes the last event that caused the transition to the respective state);

IV) has been automatically registered or registered by the S-CSCF, and has not been previously automatically registered:

- set the <unknown-param> element to any additional header field parameters contained in the Contact header field of the REGISTER request according to RFC 3680 [43];

- set the state attribute within the <registration> element to "active";

- set the state attribute within the <contact> element to "active"; and

- set the event attribute within the <contact> element to "created"; or

V) is hosted (unregistered case) at the S-CSCF:

- set the state attribute within the <registration> element to "terminated";

- set the state attribute within each <contact> element to "terminated"; and

- set the event attribute within each <contact> element to "unregistered".

The S-CSCF shall also terminate the subscription to the registration event package by setting the Subscription-State header field to the value of "terminated"; and

NOTE 8: The value of "init" for the state attribute within the <registration> element is not used.

f) set the callid and cseq attributes for the <contact> as specified in RFC 3680 [43], and the first-cseq attribute as specified in RFC 5628 [94]; and

NOTE 9: Errata of RFC 5628 clarifies the usage of the first-cseq attribute of the <temp-gruu> element.

5) set the P-Charging-Vector header field with the "icid-value" header field parameter set to the value populated in the initial request for the dialog, and

- if the NOTIFY request is sent towards an AS listed in the initial filter criteria a type 3 "orig-ioi" header field parameter. The S-CSCF shall set the type 3 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The S-CSCF shall not include the type 3 "term-ioi" header field parameter; and

- if the NOTIFY request is sent towards a public user identity this particular user owns, or any of the entities identified by the Path header field, a type 1 "orig-ioi" header field parameter. The S-CSCF shall set the type 1 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The S-CSCF shall not include the type 1 "term-ioi" header field parameter.

NOTE 10: When sending a NOTIFY request to a subscriber subscribing or unsubscribing to the reg event package, or when the S-CSCF terminates the subscription, the event attribute within the <contact> element includes the last event that caused the transition to the respective state.

The S-CSCF shall only include the non-barred public user identities in the NOTIFY request.

EXAMPLE 1: If sip:user1_public1@home1.net is registered, the public user identity sip:user1_public2@home1.net can automatically be registered. Therefore the entries in the body of the NOTIFY request look like:

```
<?xml version="1.0"?>
<reginfo xmlns="urn:ietf:params:xml:ns:reginfo"
             xmlns:cp="urn:ietf:params:xml:ns:common-policy"
             xmlns:eri="urn:3gpp:ns:extRegInfo:1.0"
             version="0" state="full">
  <registration aor="sip:user1_public1@home1.net" id="as9"
                state="active">
    <contact id="76" state="active" event="registered">
           <uri>sip:[5555::aaa:bbb:ccc:ddd]</uri>
           <unknown-param name="audio"/>
    </contact>
  </registration>
  <registration aor="sip:user1_public2@home1.net" id="as10"
                state="active">
    <contact id="86" state="active" event="created">
           <uri>sip:[5555::aaa:bbb:ccc:ddd]</uri>
           <unknown-param name="audio"/>
    </contact>
    <cp:actions>
           <eri:rph ns="wps" val="1"/>
           <eri:privSender/>
    </cp:actions>
  </registration>
</reginfo>
```

EXAMPLE 2: If sip:user1_public1@home1.net is registered, the public user identity sip:ep_user1@home1.net can automatically be registered. sip:ep_user1@home1.net is a dedicated identity out of the related range indicated in the <wildcardedIdentity> element. Therefore the entries in the body of the NOTIFY request look like:

```
<?xml version="1.0"?>
<reginfo xmlns="urn:ietf:params:xmlns:reginfo"
             xmlns:ere="urn:3gpp:ns:extRegExp:1.0"
             version="0" state="full">
  <registration aor="sip:user1_public1@home1.net" id="as10"
                state="active">
    <contact id="86" state="active" event="created">
           <uri>sip:[5555::aaa:bbb:ccc:ddd]</uri>
    </contact>
   </registration>
   <registration aor="sip:ep_user1@home1.net" id="as11"
                 state="active">
    <contact id="86" state="active" event="created">
           <uri>sip:[5555::aaa:bbb:ccc:ddd]</uri>
```

```
      </contact>
      <ere:wildcardedIdentity>sip:ep_user!.*!@home1.net
      </ere:wildcardedIdentity>
    </registration>
</reginfo>
```

When sending a final NOTIFY request with all <registration> element(s) having their state attribute set to "terminated" (i.e. all public user identities have been deregistered, expired or are hosted (unregistered case) at the S-CSCF), the S-CSCF shall also terminate the subscription to the registration event package by setting the Subscription-State header field to the value of "terminated".

When all of a UE's contact addresses have been deregistered (i.e. there is no <contact> element set to "active" for this UE), the S-CSCF shall consider subscription to the reg event package belonging to the UE cancelled (i.e. as if the UE had sent a SUBSCRIBE request with an Expires header field containing a value of zero).

The S-CSCF shall only include the non-barred public user identities in the NOTIFY request.

When the S-CSCF receives any response to the NOTIFY request, the S-CSCF shall store the value of the "term-ioi" header field parameter received in the P-Charging-Vector header field, if present.

NOTE 11: Any received "term-ioi" header field parameter will be a type 3 IOI if received from an AS or a type 1 IOI if received from a public user identity this particular user owns, or any of the entities identified by the Path header field. The type 3 IOI identifies the service provider from which the response was sent and the type 1 IOI identifies the network from which the response was sent.

5.4.2.1.3 Void

5.4.2.1.4 Void

5.4.2.1A Outgoing subscriptions to load-control event

Based on operator policy, the S-CSCF may subscribe to the load-control event package with one or more target SIP entities. The list of target SIP entities is provisioned.

Subscription to the load-control event package is triggered by internal events (e.g. the physical device hosting the SIP entity is power-cycled) or through a management interface.

The S-CSCF shall perform subscriptions to the load-control event package to a target entity in accordance with RFC 6665 [28] and with RFC 7200 [201]. When subscribing to the load-control event, the S-CSCF shall:

1) Send a SUBSCRIBE request in accordance with RFC 6665 [28] and with RFC 7200 [201] to the target entity, with the following elements:

   - an Expires header field set to a network specific value;

2) If the target entity is located in a different network and local policy requires the application of IBCF capabilities, forward the request to an IBCF acting as an exit point.

The S-CSCF shall automatically refresh ongoing subscription to the load-control event package either 600 seconds before the expiration time if the initial subscription was for greater than 1200 seconds, or when half of the time has expired if the initial subscription was for 1200 seconds or less.

The S-CSCF can terminate a subscription according to RFC 6665 [28].

5.4.2.2 Other subscriptions

Upon receipt of a NOTIFY request with the Subscription-State header field set to "terminated" and the S-CSCF has retained the SIP dialog state information for the associated subscription, once the NOTIFY transaction is terminated, the S-CSCF can remove all the stored information related to the associated subscription.

5.4.3 General treatment for all dialogs and standalone transactions excluding requests terminated by the S-CSCF

5.4.3.1 Determination of UE-originated or UE-terminated case

Upon receipt of an initial request or a stand-alone transaction, the S-CSCF shall:

- perform the procedures for the UE-originating case as described in subclause 5.4.3.2 if the request makes use of the information for UE-originating calls, which was added to the Service-Route header field entry of the S-CSCF during registration (see subclause 5.4.1.2.2F), e.g. the message is received at a certain port or the topmost Route header field contains a specific user part or parameter; or,

- perform the procedures for the UE-originating case as described in subclause 5.4.3.2 if the topmost Route header field of the request contains the "orig" parameter. The S-CSCF shall remove the "orig" parameter from the topmost Route header field; or,

- perform the procedures for the UE-terminating case as described in subclause 5.4.3.3 if this information is not used by the request.

5.4.3.2 Requests initiated by the served user

For all SIP transactions identified:

- if priority is supported, as containing an authorised Resource-Priority header field or a temporarily authorised Resource-Priority header field, or, if such an option is supported, relating to a dialog which previously contained an authorised Resource-Priority header field;

the S-CSCF shall give priority over other transactions or dialogs. This allows special treatment of such transactions or dialogs. If priority is supported, the S-CSCF shall adjust the priority treatment of transactions or dialogs according to the most recently received authorized Resource-Priority header field or backwards indication value.

NOTE 1: The special treatment can include filtering, higher priority processing, routeing, call gapping. The exact meaning of priority is not defined further in this document, but is left to national regulation and network configuration.

When the S-CSCF receives from the UE an initial request for a dialog, which contains a GRUU and an "ob" SIP URI parameter in the Contact header field, and multiple contact addresses have been registered for the specific GRUU, then for all subsequent in-dialog requests sent toward the UE's, the S-CSCF shall populate the Request-URI with the registered contact address from which the UE sent the initial request for the dialog.

NOTE 2: When a given contact address is registered, the S-CSCF can use a dedicated value in its Service-Route header field entry to identify the given contact address. When the S-CSCF receives an initial request for a dialog, the S-CSCF can find out from which contact address the initial request was sent by looking at the preloaded Route header field (constructed from the Service-Route header field returned in the response for the REGISTER request) which contains the entry of the S-CSCF.

When performing SIP digest without TLS, when the S-CSCF receives from the served user an initial request for a dialog or a request for a standalone transaction, the S-CSCF may perform the steps in subclause 5.4.3.6 to challenge the request based on local policy.

NOTE 3: If the user registration is associated with the state "tls-protected", then the execution of Proxy-Authorization as described in subclause 5.4.3.6 is still possible, although it is unlikely this would add additional security provided the P-CSCF is trusted. Thus, in most cases the state "tls-protected" will be reason for the S-CSCF to not desire Proxy-Authentication for this user.

NOTE 4: The option for the S-CSCF to challenge the request does not apply to a request from an AS acting as an originating UA.

When performing GPRS-IMS-Bundled authentication, when the S-CSCF receives from the served user an initial request for a dialog or a request for a standalone transaction, the S-CSCF shall check whether a "received" header field parameter exists in the Via header field provided by the UE. If a "received" header field parameter exists, S-CSCF shall compare the (prefix of the) IP address received in the "received" header field parameter against the UE's IP address (or prefix) stored during registration. If no "received" header field parameter exists in the Via header field provided by the

UE, then S-CSCF shall compare the (prefix of the) IP address received in the "sent-by" parameter against the IP address (or prefix) stored during registration. If the stored IP address (or prefix) and the (prefix of the) IP address in the "received" Via header field parameter provided by the UE do not match, the S-CSCF shall reject the request with a 403 (Forbidden) response. In case the stored IP address (or prefix) and the (prefix of the) IP address in the "received" Via header field parameter provided by the UE do match, the S-CSCF shall proceed as described in the remainder of this subclause.

If the S-CSCF supports HSS based P-CSCF restoration, and receives a request from a P-CSCF that the S-CSCF considers is not reachable, the S-CSCF shall consider this P-CSCF as being reachable.

If the S-CSCF supports PCRF based P-CSCF restoration, and receives a request from a P-CSCF that the S-CSCF considers is in a not reachable, the S-CSCF shall consider this P-CSCF as being reachable.

When the S-CSCF receives from the served user or from a PSI an initial request for a dialog or a request for a standalone transaction, and the request is received either from a functional entity within the same trust domain or contains a valid original dialog identifier (see step 3) or the dialog identifier (From, To and Call-ID header fields) relates to an existing request processed by the S-CSCF, then prior to forwarding the request, the S-CSCF shall:

0) if the request is received from a P-CSCF that does not support the trust domain handling of the P-Served-User header field then remove any P-Served-User header fields;

1) determine the served user as follows:

   a) if the request contains a P-Served-User header field then

      i) determine the served user by taking the identity contained in a P-Served-User header field as defined in RFC 5502 [133]. Then check whether the determined served user is a barred public user identity. In case the said header field contains the served user identity is a barred public user identity for the user, then the S-CSCF shall reject the request by generating a 403 (Forbidden) response. Otherwise, the S-CSCF shall save the public user identity of the served user and continue with the rest of the steps;

NOTE 5: If the P-Served-User header field contains a barred public user identity, then the message has been received, either directly or indirectly, from a non-compliant entity which should have had generated the content with a non-barred public user identity.

   b) if the request does not contain a P-Served-User header field then

      i) determine the served user by taking the identity contained in one of the URI(s) of the P-Asserted-Identity header field. In case the determined served user is a barred public user identity, then the S-CSCF shall reject the request by generating a 403 (Forbidden) response. Otherwise, the S-CSCF shall save the public user identity of the served user and continue with the rest of the steps; and

      ii) if the P-Asserted-Identity header field contains two URIs and the URI other than the determined served user is not an alias of the determined served user or is barred then act based on local policy, e.g. reject the request by generating a 403 (Forbidden) response or remove the URI not identifying the determined served user from the P-Asserted-Identity header field;

NOTE 6: If the P-Asserted-Identity header field contains a barred public user identity, then the message has been received, either directly or indirectly, from a non-compliant entity which should have had generated the content with a non-barred public user identity.

1A) if the Contact is a GRUU, but is not valid as defined in subclause 5.4.7A.4, then return a 4xx response as specified in RFC 5627 [93];

2) store the value of the "orig-ioi" header field parameter received in the P-Charging-Vector header field if present, and remove it from any forwarded request;

NOTE 7: Any received "orig-ioi" header field parameter will be either a type 1 IOI or a type 3 IOI. The type 1 IOI identifies the network from which the request was sent and the type 3 IOI identifies the service provider from which the request was sent (AS initiating a session on behalf of a user or a PSI);

3) check if an original dialog identifier that the S-CSCF previously placed in a Route header field is present in the topmost Route header field of the incoming request.

- If not present, the S-CSCF shall build an ordered list of initial filter criteria based on the public user identity of the served user (as determined in step 1) of the received request as described in 3GPP TS 23.218 [5].

- If present, the request has been sent from an AS in response to a previously sent request, an ordered list of initial filter criteria already exists and the S-CSCF shall not change the ordered list of initial filter criteria even if the AS has changed the P-Served-User header field or the P-Asserted-Identity header field;

NOTE 8: An original dialog identifier is sent to each AS invoked due to iFC evaluation such that the S-CSCF can associate requests as part of the same sequence that trigger iFC evaluation in priority order (and not rely on SIP dialog information that can change due to B2BUA AS). If the same original dialog identifier is included in more than one request from a particular AS (based on service logic in the AS), then the S-CSCF will continue the iFC evaluation sequence rather than build a new ordered list of iFC;

4) remove its own SIP URI from the topmost Route header field;

4A) if a reference location was received from the HSS at registration as part of the user profile and the request does not contain a message body with the content type application/pidf+xml in accordance with RFC 6442 [89] and does not contain a P-Access-Network-Info header field containing the "network-provided" parameter, the S-CSCF shall insert a P-Access-Network-Info header field constructed according to the reference location received from the HSS and containing the "network-provided" parameter. The access type information received from the HSS shall be mapped into the corresponding access-type parameter of the P-Access-Network-Info header field and the location information shall be mapped into the location parameter corresponding to the access-type parameter, i.e. into "dsl-location" parameter, "fiber-location" parameter or "eth-location" parameter;

4B) if there was an original dialog identifier present in the topmost Route header field of the incoming request and the request is received from a functional entity within the same trust domain and contains a P-Asserted-Service header field, continue the procedure with step 5;

4C) if the request contains a P-Preferred-Service header field, check whether the ICSI value contained in the P-Preferred-Service header field is part of the set of the subscribed services for the served user and determine, using operator-configured data, whether the contents of the request match the ICSI for the subscribed service. The operator-configured data used to determine if there is a matching between the request and the ICSI value may be based on any information in the request (e.g. SDP media capabilities, Content-Type header field, request method). Then:

a) if there is no match between the request and the ICSI value, as an operator option, the S-CSCF may reject the request by generating a 403 (Forbidden) response. Otherwise remove the P-Preferred-Service header field and continue with the rest of the steps; and

b) if there is a match between the request and the ICSI value, then include a P-Asserted-Service header field in the request containing the ICSI value contained in the P-Preferred-Service header field, remove the P-Preferred-Service header field, and continue the procedure with step 5;

4D) if the request does not contain a P-Preferred-Service header field, check, using operator-configured data, whether the contents of the request match a subscribed service for each and any of the subscribed services for the served user:

a) if not, as an operator option, the S-CSCF may reject the request by generating a 403 (Forbidden) response; and

b) if so, and if the request is related to an IMS communication service and the IMS communication service requires the use of an ICSI value then select an ICSI value for the related IMS communication service and include a P-Asserted-Service header field in the request containing the selected ICSI value; and

NOTE 9: If more than one ICSI values match the contents of the request, the S-CSCF selects an ICSI value based on local policy.

c) if so, and if the request is related to an IMS communication service and the IMS communication service does not require the use of an ICSI value then continue without including an ICSI value; and

d) if so, and if the request does not relate to an IMS communication service (or if the S-CSCF is unable to unambiguously determine the service being requested but decides to allow the session to continue) then continue without including an ICSI value;

5) check whether the initial request matches any unexecuted initial filter criteria. If there is a match, then the S-CSCF shall select the first matching unexecuted initial filter criteria from the ordered list of initial filter criteria and the S-CSCF shall:

   a) insert the AS URI to be contacted into the Route header field as the topmost entry followed by its own URI populated as specified in the subclause 5.4.3.4;

NOTE 10: If the AS is accessed via an ISC gateway function, then the URI will be the address of the ISC gateway function.

   b) if the S-CSCF supports the P-Served-User extension as specified in RFC 5502 [133] and RFC 8498 [239] insert P-Served-User header field populated with the served user identity as determined in step 1. If required by operator policy, the S-CSCF shall:

      - if the associated session case is "Originating" as specified in 3GPP TS 29.228 [14], include the sescase header field parameter set to "orig" and the regstate header field parameter set to "reg";

      - if the associated session case is "Originating_Unregistered" as specified in 3GPP TS 29.228 [14], include the sescase header field parameter set to "orig" and the regstate header field parameter set to "unreg";

      - if the associated session case is "Originating_CDIV" as specified in 3GPP TS 29.228 [14], include the "orig-cdiv" header field parameter, defined in RFC 8498 [239]; and

   c) if the AS is located outside the trust domain then the S-CSCF shall remove the access-network-charging-info parameter in the P-Charging-Vector header field from the request that is forwarded to the AS; if the AS is located within the trust domain, then the S-CSCF shall retain the access-network-charging-info parameter in the P-Charging-Vector header field in the request that is forwarded to the AS;

   d) insert a type 3 "orig-ioi" header field parameter in place of any received "orig-ioi" header field parameters in the P-Charging-Vector header field. The S-CSCF shall set the type 3 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The S-CSCF shall not include the type 3 "term-ioi" header field parameter;

   e) remove the "transit-ioi" header field parameter, if received;

   f) based on operator policy insert in a Relayed-Charge header field the value of the received "transit-ioi" header field parameter in the P-Charging-Vector header field;

   g) based on local policy, the S-CSCF shall add an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier;

   h) if the S-CSCF supports using a token to identify the registration and if a registration exists, insert a "+g.3gpp.registration-token" Feature-Caps header field parameter, as defined in subclause 7.9A.8, set to the same value as included in the "+g.3gpp.registration-token" Contact header field parameter of the third party REGISTER request sent to the AS when the UE registered; and

   i) if an IP address associated with the served user and the AS SIP URI is stored as described in subclause 5.4.0 exists, then the S-CSCF forwards the SIP message to the IP address associated with the served user and the AS SIP URI;

NOTE 11: Depending on the result of processing the filter criteria the S-CSCF might contact one or more AS(s) before processing the outgoing Request-URI.

NOTE 12: An AS can activate or deactivate its own filter criteria via the Sh interface. As the S-CSCF checks initial filter criteria only on receipt of an initial request for a dialog, or a standalone transaction, a modified service profile will have no impact on transactions or dialogs already in progress and the modified profile will be effective only for new transactions and dialogs. If the S-CSCF receives a modification of the iFC during their execution, then it should not update the stored initial Filter Criteria until the iFC related to the initial request have been completely executed.

6) if there was no original dialog identifier present in the topmost Route header field of the incoming request store the value of the "icid-value" header field parameter received in the P-Charging-Vector header field and retain the "icid-value" header field parameter in the P-Charging-Vector header field. Optionally, the S-CSCF may generate a new, globally unique ICID and insert the new value in the "icid-value" header field parameter of the P-Charging-Vector header field when forwarding the message. If the S-CSCF creates a new ICID, then it is

responsible for maintaining the two ICID values in the subsequent messaging. Based on local policy, the S-CSCF shall add an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier if not already available;

7) in step 5, if the initial request did not match any unexecuted initial filter criteria (i.e. the request is not forwarded to an AS):

   a) remove the received "transit-ioi" from the P-Charging-Vector header field, if present;

   b) insert a type 2 "orig-ioi" header field parameter into the P-Charging-Vector header field. The S-CSCF shall set the type 2 "orig-ioi" header field parameter to a value that identifies the sending network. The S-CSCF shall not include the type 2 "term-ioi" header field parameter; and

   c) remove the Relayed-Charge header field, if present;

8) insert a P-Charging-Function-Addresses header field populated with values received from the HSS if the request does not contain a P-Charging-Function-Addresses header field and the message is forwarded within the S-CSCF home network, including towards AS;

9) if there was no original dialog identifier present in the topmost Route header field of the incoming request and if the served user is not considered a privileged sender then:

   a) if the P-Asserted-Identity header field contains only a SIP URI and if the S-CSCF has knowledge that the SIP URI contained in the received P-Asserted-Identity header field is an alias SIP URI for a tel URI, add a second P-Asserted-Identity header field containing this tel-URI, including the display name associated with the tel URI, if available; and

   b) if the P-Asserted-Identity header field contains only a tel URI, the S-CSCF shall add a second P-Asserted-Identity header field containing a SIP URI. The added SIP URI shall contain in the user part a "+" followed by the international public telecommunication number contained in tel URI, and user's home network domain name in the hostport part. The added SIP URI shall contain the same value in the display name as contained in the tel URI. The S-CSCF shall also add a "user" SIP URI parameter equals "phone" to the SIP URI;

NOTE 13: If tel URI is shared URI so is the alias SIP URI.

10) if the request is not forwarded to an AS and if the outgoing Request-URI is:

- a SIP URI with the user part starting with a + and the "user" SIP URI parameter equals "phone", and if configured per local operator policy, the S-CSCF shall perform the procedure described here. Local policy can dictate whether this procedure is performed for all domains of the SIP URI, only if the domain belongs to the home network, or not at all. If local policy indicates that the procedure is to be performed, then the S-CSCF shall translate the international public telecommunications number contained in the user part of the SIP URI (see RFC 3966 [22]) to a globally routeable SIP URI using either an ENUM/DNS translation mechanism with the format specified in RFC 6116 [24], or any other available database. Database aspects of ENUM are outside the scope of the present document. An S-CSCF that implements the additional routeing functionality described in annex I may forward the request without attempting translation. If an agreement exists between the home network and the visited network to support roaming architecture for voice over IMS with local breakout, the S-CSCF does not enable NP capabilities, and the S-CSCF decides to loopback the call to the visited network, the S-CSCF may forward the request without attempting translation. If a translation is in fact performed and it succeeds, the S-CSCF shall update the Request-URI with the globally routeable SIP URI either returned by ENUM/DNS or obtained from any other available database. If this translation fails, the request may be forwarded to a BGCF or any other appropriate entity (e.g. a MRFC to play an announcement) in the originator's home network or the S-CSCF may send an appropriate SIP response to the originator. When forwarding the request to a BGCF or any other appropriate entity, the S-CSCF shall leave the original Request-URI containing the SIP URI with "user" SIP URI parameter equals phone unmodified. If the request is forwarded, the S-CSCF shall remove the access-network-charging-info parameter from the P-Charging-Vector header field prior to forwarding the message;

- a SIP URI with a "user" SIP URI parameter equals "dialstring" and the domain name of the SIP URI belongs to the home network (i.e. the local number analysis and handling is either failed in the appropriate AS or the request has not been forwarded to AS for local number analysis and handling at all), either forward the request to any appropriate entity (e.g a MRFC to play an announcement) in the originator's home network or send an appropriate SIP response to the originator;

- a SIP URI with a local number (see RFC 3966 [22]) in the user part and a "user" SIP URI parameter equals "phone" and the domain name of the SIP URI belongs to the home network (i.e. the local number analysis and handling is either failed in the appropriate AS or the request has not been forwarded to AS for local number analysis and handling at all), either forward the request to to a BGCFor any appropriate entity (e.g a MRFC to play an announcement) in the originator's home network or send an appropriate SIP response to the originator;

- a tel URI containing a global number (see RFC 3966 [22]) in the international format, the S-CSCF shall translate the E.164 address to a globally routeable SIP URI using either an ENUM/DNS translation mechanism with the format specified in RFC 6116 [24], or any other available database. Database aspects of ENUM are outside the scope of the present document. An S-CSCF that implements the additional routeing functionality described in annex I may forward the request without attempting translation. If an agreement exists between the home network and the visited network to support roaming architecture for voice over IMS with local breakout, the S-CSCF does not enable NP capabilities, and the S-CSCF decides to loopback the call to the visited network, the S-CSCF may forward the request without attempting translation. If this translation is in fact performed and it succeeds, the S-CSCF shall update the Request-URI with the globally routeable SIP URI returned by ENUM/DNS or any other available database. If this translation fails, the request may be forwarded to a BGCF or any other appropriate entity (e.g a MRFC to play an announcement) in the originator's home network or the S-CSCF may send an appropriate SIP response to the originator. When forwarding the request to a BGCF or any other appropriate entity, the S-CSCF shall leave the original Request-URI containing the tel URI unmodified. If the request is forwarded, the S-CSCF shall remove the access-network-charging-info parameter from the P-Charging-Vector header field prior to forwarding the message;

- a tel URI containing a local number (see RFC 3966 [22]) (i.e. the local number analysis and handling is either failed in the appropriate AS or the request has not been forwarded to AS for local number analysis and handling at all), either forward the request to a BGCF or any other appropriate entity (e.g. a MRFC to play an announcement) in the originator's home network or send an appropriate SIP response to the originator;

- a pres URI or an im URI, the S-CSCF shall forward the request as specified in RFC 3861 [63]. In this case, the S-CSCF shall not modify the received Request-URI: and

- a service URN, e.g. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69]. In this case the S-CSCF shall not modify the received Request-URI.

NOTE 14: If there is no SIP-based transport found after applying the procedure specified in RFC 3861 [63], the S-CSCF can forward the request to a translating gateway.

Additional procedures apply if the S-CSCF supports NP capabilities and these capabilities are enabled by local policy, and the database used for translation from an international public telecommunications number to a SIP URI also provides NP data (for example, based on the PSTN Enumservice as defined by RFC 4769 [114] or other appropriate data bases). If the above translation from an international public telecommunications number to a SIP URI failed, but NP data was obtained from the database and there is no "npdi" parameter in the received request, then the S-CSCF shall, based on operator policy, replace the URI in the Request-URI with the obtained NP data, prior to forwarding the request to the BGCF or other appropriate entity. If the received request already contains a tel-URI "npdi" parameter, then the S-CSCF may update the URI with the obtained NP data. The URI is updated by the S-CSCF by adding NP parameters defined by RFC 4694 [112]. If the Request-URI is a tel-URI, then an "npdi" tel-URI parameter is added to indicate that NP data retrieval has been performed, and if the number is ported, an "rn" tel-URI parameter is added to identify the ported-to routeing number. If the Request-URI is in the form of a SIP URI user=phone, the "npdi" and "rn" tel-URI parameters are added as described above to the userinfo part of the SIP URI;

NOTE 15: When the S-CSCF replaces the tel-URI in the Request-URI with the obtained NP data, all tel URI parameters in the received Request-URI will be replaced by the obtained NP data.

10A) if the request is not forwarded to an AS and if local policy requires the application of additional routeing capabilities, as defined in annex I, the S-CSCF shall apply the additional routeing capabilities if they are locally available or forward the request to an entity that implements the additional routeing capabilities;

10B) if an agreement exists between the home network and the visited network to support Roaming Architecture for Voice over IMS with Local Breakout then continue with the following steps. Otherwise continue with step 11. If:

- the top most Route header contains an indication that this is the UE-originating case;

NOTE 16: This indication can e.g. be in a URI parameter, a character string in the user part of the URI or can be a port number in the URI added by the S-CSCF during the registration in the Service-Route header field.

- the UE is roaming (as identified by the P-Visited-Network-ID header field value in the original REGISTER request); and

- the request is an INVITE request;

determine if loopback routeing is applicable for this request using local policy, and save this decision for subsequent processing along with the following information:

a) any URI representing the TRF address preference received from the visited network; and

b) the ICID received in the request.

In addition, the S-CSCF shall also include in the request a Feature-Caps header field with the "+g.3gpp.home-visited" header field parameter according to RFC 6809 [190] with the "+g.3gpp.home-visited" header field parameter set to the identifier of the visited network received in the P-Visited-Network-ID header field in the original registration request;

10C) if the request is an INVITE request, then determine whether loopback is applied for this request. The information saved in step 10B, and the presence or absence of the Feature-Caps header field with the "+g.3gpp.home-visited" header field parameter in the received INVITE request are taken into account in making this decision:

a) if loopback routeing is not to be performed for this request remove any "+g.3gpp.trf" header field parameter or "+g.3gpp.home-visited" header field parameter from the Feature-Caps header field of the outgoing request;

b) if loopback routeing is applied for this request;

i) remove all entries in the Route header field;

ii) if a "+g.3gpp.trf" header field parameter with a parameter value containing a valid URI, is included in the Feature-Caps header field of the request, insert the URI in a Route header field;

iii) if a "+g.3gpp.trf" header field parameter, with a parameter value containing a valid URI is not included in the Feature-Caps header field of the request, insert a locally configured TRF address, associated with the visited network for this call, in the Route header field;

iv) remove any "+g.3gpp.home-visited" header field parameter from the Feature-Caps header field of the outgoing request;

v) insert the "+g.3gpp.loopback" header field parameter as specified in subclause 7.9A.4 in the Feature-Caps header field of the request, in accordance with the RFC 6809 [190]. If providing the identifier of the home network is supported by the S-CSCF and the visited network, the S-CSCF may based on operator agreement insert the "+g.3gpp.loopback" header field parameter set to the identifier of the home network;

vi) if included in the incoming request, remove the "+g.3gpp.trf" header field parameter from the Feature-Caps header field from the outgoing request;

vii) remove a type 2 "orig-ioi" header field parameter that was added in step 7 from the P-Charging-Vector header field and insert a type 1 "orig-ioi" header field parameter into the P-Charging-Vector header field. The S-CSCF shall set the type 1 "orig-ioi" header field parameter to a value that identifies the network in which the S-CSCF resides. The S-CSCF shall not include the "term-ioi" header field parameter; and

viii) if the S-CSCF supports indicating the traffic leg associated with a URI as specified in RFC 7549 [225] and if an "iotl" SIP URI parameter is not included in the TRF URI in the Route header field and if required by local policy, append an "iotl" URI parameter with a value set to "homeA-visitedA" to the URI in the Route header field; and

c) if the final decision on loopback routeing is deferred to a subsequent entity in the home network, the BGCF, then the S-CSCF includes, if absent, in the request a Feature-Caps header field with the "+g.3gpp.home-visited" header field parameter, with the parameter value set to the identifier of the visited network received in the P-Visited-Network-ID header field in the original registration request. The S-CSCF is expected to know by means of network configuration that such a subsequent entity exists; and

NOTE 17: The subsequent entity in the home network, the BGCF, will remove the "+g.3gpp.home-visited" header field parameter from the Feature-Caps header field when a final routeing decision is taken.

11) determine the destination address (e.g. DNS access) using the URI placed in the topmost Route header field if present, otherwise based on the Request-URI. If the destination requires interconnect functionalities (e.g. the destination address is of an IP address type other than the IP address type used in the IM CN subsystem), the S-CSCF shall forward the request to the destination address via an IBCF in the same network;

12) if network hiding is needed due to local policy, put the address of the IBCF to the topmost Route header field;

13) in case of an initial request for a dialog:

a) determine the need for GRUU processing. GRUU processing is required if:

- an original dialog identifier that the S-CSCF previously placed in a Route header field is not present in the topmost Route header field of the incoming request (this means the request is not returning after having been sent to an AS), and

- the contact address contains a GRUU that was either assigned by the S-CSCF that is valid as specified in subclause 5.4.7A.4 or a temporary GRUU self assigned by the UE based on the "temp-gruu-cookie" header parameter provided to the UE;

NOTE 18: The procedures for determining that a URI is a temporary GRUU assigned by the UE are specified in subclause 7.1.2.3 of RFC 6140 [191].

b) if GRUU processing is not required and the initial request originated from a served user, then determine the need to record-route for other reasons:

- if the request is routed to an AS which is part of the trust domain, the S-CSCF shall decide, based on operator policy, whether to record-route or not. The decision is configured in the S-CSCF using any information in the received request that may otherwise be used for the initial filter criteria. If the request is record-routed the S-CSCF shall create a Record-Route header field containing its own SIP URI;

- if the request is a SUBSCRIBE request and routed elsewhere, the S-CSCF shall decide, based on operator policy, whether to record-route or not. The decision is configured in the S-CSCF using any information in the received request (e.g. event package name). If the request is record-routed the S-CSCF shall create a Record-Route header field containing its own SIP URI; or

NOTE 19: Some subscriptions to event packages (e.g. presence) can result in virtually persistent subscriptions and if the S-CSCF Record-Routes this can prevent reassignment of the S-CSCF.

NOTE 20: If the S-CSCF does not Record-Route the initial SUBSCRIBE request, it will not be possible to perform SIP digest authentication of SIP requests sent inside the SIP dialog related to the associated subscription.

- if the request not a SUBSCRIBE request and is routed elsewhere, create a Record-Route header field containing its own SIP URI;

NOTE 21: For requests originated from a PSI the S-CSCF can decide whether to record-route or not based on operator policy.

c) if GRUU processing is required, the S-CSCF shall create a Record-Route header field containing its own SIP URI;

d) if GRUU processing is required, the S-CSCF shall save an indication that GRUU-routeing is to be performed for in-dialog requests that reach the S-CSCF because of the Record-route header field added in step c);

NOTE 22: The manner of representing the GRUU-routeing indication is a private matter for the S-CSCF. The indication is used during termination processing of in-dialog requests to cause the S-CSCF to replace a Request-URI containing a GRUU with the corresponding registered contact address. It can be saved using values in the Record-Route header field, or in dialog state.

14) based on the destination user (Request-URI), remove any P-Access-Network-Info header field and the access-network-charging-info parameter in the P-Charging-Vector header field prior to forwarding the message;

14A) if the request is not routed to an AS, to a BGCF or to an entity that implements the additional routeing functionality, remove the P-Served-User header field prior to forwarding the request;

14B) if the S-CSCF supports indicating the traffic leg as specified in RFC 7549 [225], the request is not routed to an AS, to a BGCF or to an entity that implements the additional routeing functionality, loopback routeing is not to be performed for this request, required by local policy and the Request-URI contains a SIP URI, append the "iotl" SIP URI parameter set to "homeA-homeB" to the Request-URI;

15) route the request based on SIP routeing procedures;

16) if the request is an INVITE request, save the Contact, CSeq and Record-Route header field values received in the request such that the S-CSCF is able to release the session if needed;

17) if the request contains a "logme"parameter in the Session-ID header field, treat this dialog as one for which logging is in progress and log SIP signalling for this dialog according to its trace configuration;

18) if the S-CSCF supports using a token to identify the registration and if the request is not forwarded to an AS, remove the "+g.3gpp.registration-token" Feature-Caps header field parameter, defined in subclause 7.9A.8, if received in the request; and

19) if the received request is an INVITE request or a MESSAGE request and the S-CSCF supports calling number verification using signature verification and attestation information as specified in subclause 3.1, the S-CSCF shall based on local policy perform attestation of the user identity by inserting:

- a "verstat" tel URI parameter, specified in subclause 7.2A.20, to the tel URI or SIP URI with a user=phone parameter in the From header field or the P-Asserted-Identity header field;

  an Origination-Id header field, specified in subclause 7.2.19, set to a UUID identifying the S-CSCF which is configured based on local policy and requirements from national regulation: and

- an Attestation-Info header field, specified in subclause 7.2.18, set to the value "A".

When the S-CSCF receives, an initial request for a dialog or a request for a standalone transaction, from an AS acting on behalf of an unregistered user, the S-CSCF shall:

1) execute the procedures described in the steps 1, 2, 3, 4, 4B, 4C, 4D, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 and 16 in the above paragraph (when the S-CSCF receives, from a registered served user, an initial request for a dialog or a request for a standalone transaction).

NOTE 23: When the S-CSCF does not have the user profile, before executing the actions as listed above, it initiates the S-CSCF Registration/deregistration notification procedure, as described in 3GPP TS 29.228 [14]; with the purpose of downloading the relevant user profile (i.e. for unregistered user) and informs the HSS that the user is unregistered. The S-CSCF will assess triggering of services for the unregistered user, as described in 3GPP TS 29.228 [14]. When requesting the user profile, and the request received by the S-CSCF contains a P-Profile-Key header field, the S-CSCF can include the header field value in S-CSCF Registration/deregistration notification. If the response from the HSS includes a Wildcarded Public Identity AVP, and if the request received by the S-CSCF did not include a P-Profile-Key header field, the S-CSCF uses the AVP value to set the P-Profile-Key header field before forwarding the request to an AS.

When the S-CSCF receives a request initiated by the served user for which the S-CSCF does not have the user profile or does not trust the data that it has (e.g. due to restart), the S-CSCF shall attempt to retrieve the user profile from the HSS. If the S-CSCF receives a Diameter result code of DIAMETER_UNABLE_TO_COMPLY as defined in 3GPP TS 29.228 [14], the S-CSCF supports S-CSCF restoration procedures, and the Request-URI of the request does not match an emergency service URN, i.e. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69], then the S-CSCF shall:

I) reject the request by returning a 504 (Server Time-out) response to the UE;

II) assume that the UE supports version 1 of the XML Schema for the 3GPP IM CN subsystem XML body if support for the 3GPP IM CN subsystem XML body as described in subclause 7.6 in the Accept header field is not indicated; and

III) include in the 504 (Server Time-out) response:

- a Content-Type header field with the value set to associated MIME type of the 3GPP IM CN subsystem XML body as described in subclause 7.6.1;

- a P-Asserted-Identity header field set to the value of the SIP URI of the S-CSCF included in the Service-Route header field (see subclause 5.4.1.2.2F) during the registration of the user whose UE sent the request causing this response; and

- a 3GPP IM CN subsystem XML body:

  a) an <ims-3gpp> element with the "version" attribute set to "1" and with an <alternative-service> child element, set to the parameters of the alternative service;

     i) a <type> child element, set to "restoration" (see table 7.6.2) to indicate that S-CSCF restoration procedures are supported;

     ii) a <reason> child element, set to an operator configurable reason; and

     iii) an <action> child element, set to "initial-registration" (see table 7.6.3).

NOTE 24: These procedures do not prevent the usage of unspecified reliability or recovery techniques above and beyond those specified in this subclause.

Depending on operator configuration (see subclause 5.4.1.8), when the S-CSCF receives a request with a Request-URI that does not match an emergency service URN, i.e. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69], the request initiated by the served user for which the S-CSCF has modified but not synchronized the service profile for the served user and the S-CSCF supports S-CSCF restoration procedures, then the S-CSCF shall reject the request as described in items I), II) and III).

If the S-CSCF:

a) fails to receive a SIP response within a configurable time; or

b) receives a 408 (Request Timeout) response or a 5xx response from the AS without previously receiving a 1xx response to the original SIP request, and without previously receiving a SIP request from the AS that contained the same original dialog identifier as the original request;

the S-CSCF shall:

- if the default handling defined in the filter criteria indicates the value "SESSION_CONTINUED" as specified in 3GPP TS 29.228 [14] or no default handling is indicated, execute the procedure from step 5; and

- if the default handling defined in the filter criteria indicates the value "SESSION_TERMINATED" as specified in 3GPP TS 29.228 [14], either forward the received response or, if the request is an initial INVITE request, send a 408 (Request Timeout) response or a 5xx response towards the served UE as appropriate (without verifying the matching of filter criteria of lower priority and without proceeding for further steps).

If the S-CSCF receives any final response from the AS, the S-CSCF shall forward the response towards the served UE (without verifying the matching of filter criteria of lower priority and without proceeding for further steps).

When the S-CSCF receives any response to the above request containing a Relayed-Charge header field, and the next hop is not an AS, the S-CSCF shall remove the Relayed-Charge header field from the forwarded response.

When the S-CSCF receives any response to the above request, the S-CSCF may:

1) apply any privacy required by RFC 3323 [33] and RFC 3325 [34] to the P-Asserted-Identity header field.

NOTE 25: The P-Asserted-Identity header field would normally only be expected in 1xx or 2xx responses.

NOTE 26: The optional procedure above is in addition to any procedure for the application of privacy at the edge of the trust domain specified by RFC 3325 [34].

When the S-CSCF receives any response to the above request, the S-CSCF shall:

1) If logging is in progress for this dialog, check whether a trigger for stopping logging of SIP signalling has occurred, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K]. If a stop trigger event has occurred then stop treating this as a dialog for which logging is in progress, else the S-CSCF shall append a "logme" header field parameter to the SIP Session-ID header field if the parameter is missing and determine, by checking its trace configuration, whether to log the response.

When the S-CSCF receives any response to the above request containing a "term-ioi" header field parameter in the P-Charging-Vector header field, the S-CSCF shall:

1) store the value of the received "term-ioi" header field parameter if present;

2) remove all received "orig-ioi", "term-ioi" and "transit-ioi" header field parameters from the forwarded response;

3) include the stored "orig-ioi" header field parameter if received in the request;

4) include a type 1 "term-ioi" header field parameter if next hop is not an AS, or a type 3 "term-ioi" header field parameter. The "term-ioi" header field parameter is set to a value that identifies the sending network of the response

NOTE 27: Any received "term-ioi" header field parameter will be a type 2 IOI, if received from an S-CSCF, or type 3 IOI, if received from an AS, or type 1 IOI if the S-CSCF performed loopback routeing for this request. A type 2 IOI identifies the sending network of the response, a type 3 IOI identifies the sending service provider of the response, and a type 1 IOI identifies the visited network of the served user.

5) based on operator policy include any received "transit-ioi" header field parameter, from the P-Charging-Vector header field, in a Relayed-Charge header field, if the next hop is an AS.

When the S-CSCF receives any 1xx or 2xx response to the initial request for a dialog, if the response corresponds to an INVITE request, the S-CSCF shall save the Contact and Record-Route header field values in the response in order to be able to release the session if needed.

When the S-CSCF receives any 1xx or 2xx response to an initial request for a dialog or a request for a standalone transaction, if the response is forwarded within the S-CSCF home network and not to an AS, the S-CSCF shall insert a P-Charging-Function-Addresses header field populated with values received from the HSS.

When the S-CSCF, upon sending an initial INVITE request that includes an IP address in the SDP offer (in "c=" parameter), receives an error response indicating that the IP address type is not supported, (e.g., the S-CSCF receives the 488 (Not Acceptable Here) with 301 Warning header field indicating "incompatible network address format"), the S-CSCF shall either:

- fork the initial INVITE request to the IBCF; or

- process the error response and forward it using the Via header field.

NOTE 28: If the S-CSCF knows that the originating UE supports both IPv6 and IPv4 addresses simultaneously, the S-CSCF will forward the error response to the UE using the Via header field. The present version of the specification does not specify how the S-CSCF determines whether the UE supports both IPv6 and IPv4 addressing simultaneously.

When the S-CSCF receives from the served user a target refresh request for a dialog, prior to forwarding the request the S-CSCF shall:

0A) if the dialog is related to an IMS communication service determine whether the contents of the request (e.g. SDP media capabilities, Content-Type header field) match the IMS communication service as received as the ICSI value in the P-Asserted-Service header field in the initial request. As an operator option, if the contents of the request do not match the IMS communication service the S-CSCF may reject the request by generating a status code reflecting which added contents are not matching. Otherwise, continue with the rest of the steps;

1) remove its own URI from the topmost Route header field;

2) create a Record-Route header field containing its own SIP URI;

3) for INVITE dialogs (i.e. dialogs initiated by an INVITE request), save the Contact and CSeq header field values received in the request such that the S-CSCF is able to release the session if needed;

4) in case the request is routed towards the destination user (Request-URI) or in case the request is routed to an AS located outside the trust domain, remove the access-network-charging-info parameter in the P-Charging-Vector header field;

5) route the request based on the topmost Route header field; and

6) if the request was sent on a dialog for which logging of signalling is in progress, check whether a trigger for stopping logging of SIP signalling has occurred, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K]. If a stop trigger event has occurred then stop logging of signalling, else determine, by checking its trace configuration, whether to log the response.

When the S-CSCF receives any response to the above request, the S-CSCF shall:

1) If logging is in progress for this dialog, check whether a trigger for stopping logging of SIP signalling has occurred, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K]. If a stop trigger event has occurred then stop logging of signalling, else determine, by checking its trace configuration, whether to log the response.

When the S-CSCF receives any 1xx or 2xx response to the target refresh request for an INVITE dialog, the S-CSCF shall replace the saved Contact header field values in the response such that the S-CSCF is able to release the session if needed.

If the S-CSCF inserted in the initial request for the dialog the header field parameters into the Feature-Caps header field then the S-CSCF shall include the header field parameters with the same parameter values into the Feature-Caps header field in any target refresh request for the dialog, and in each 1xx or 2xx response to target refresh request sent in the same direction.

When the S-CSCF receives from the served user a subsequent request other than a target refresh request for a dialog, prior to forwarding the request the S-CSCF shall:

1) remove its own URI from the topmost Route header field;

2) in case the request is routed towards the destination user (Request-URI) or in case the request is routed to an AS located outside the trust domain, remove the access-network-charging-info parameter in the P-Charging-Vector header field; and

3) route the request based on the topmost Route header field; and

4) if the request was sent on a dialog for which logging of signalling is in progress, check whether a trigger for stopping logging of SIP signalling has occurred, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K]. If a stop trigger event has occurred, stop logging of signalling, else determine, by checking its trace configuration, whether to log the request.

When the S-CSCF receives any response to the above request, the S-CSCF shall:

1) If logging is in progress for this dialog, check whether a trigger for stopping logging of SIP signalling has occurred, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K]. If a stop trigger event has occurred then stop logging of signalling, else determine, by checking its trace configuration, whether to log the response.

With the exception of 305 (Use Proxy) responses, the S-CSCF shall not recurse on 3xx responses.

5.4.3.3 Requests terminated at the served user

For all SIP transactions identified:

- if priority is supported, as containing an authorised Resource-Priority header field or a temporarily authorised Resource-Priority header field, or, if such an option is supported, relating to a dialog which previously contained an authorised Resource-Priority header field;

the S-CSCF shall give priority over other transactions or dialogs. This allows special treatment for such transactions or dialogs. If priority is supported, the S-CSCF shall adjust the priority treatment of transactions or dialogs according to the most recently received authorized Resource-Priority header field or backwards indication value.

NOTE 1: The special treatment can include filtering, higher priority processing, routeing, call gapping. The exact meaning of priority is not defined further in this document, but is left to national regulation and network configuration.

When the S-CSCF receives, destined for a registered served user, an initial request for a dialog or a request for a standalone transaction, and the request is received either from a functional entity within the same trust domain or

contains a valid original dialog identifier or the dialog identifier (From, To and Call-ID header fields) relates to an existing request processed by the S-CSCF, then prior to forwarding the request, the S-CSCF shall:

1) check if an original dialog identifier that the S-CSCF previously placed in a Route header field is present in the topmost Route header field of the incoming request.

   - If present, the request has been sent from an AS in response to a previously sent request.

   - If not present, it indicates that the request is visiting the S-CSCF for the first time and in this case the S-CSCF shall determine the served user by taking the identity contained in the Request-URI. If the Request-URI is a temporary GRUU assigned by the S-CSCF as defined in subclause 5.4.7A.3, then take the public user identity that is associated with the temporary GRUU to be the served user identity. Then check whether the determined served user identity is a barred public user identity. In case the served user identity is a barred public user identity for the user, then the S-CSCF shall reject the request by generating a 404 (Not Found) response. Otherwise, the S-CSCF shall save the Request-URI from the request, the served user identity and the public user identity of the served user and continue with the rest of the steps;

NOTE 2: An original dialog identifier is sent to each AS invoked due to iFC evaluation such that the S-CSCF can associate requests as part of the same sequence that trigger iFC evaluation in priority order (and not rely on SIP dialog information that can change due to B2BUA AS). If the same original dialog identifier is included in more than one request from a particular AS (based on service logic in the AS), then the S-CSCF will continue the iFC evaluation sequence rather than build a new ordered list of iFC;

2) remove its own URI from the topmost Route header field;

2A) if there was no original dialog identifier present in the topmost Route header field of the incoming request build an ordered list of initial filter criteria based on the public user identity in the Request-URI of the received request as described in 3GPP TS 23.218 [5].

3) if there was an original dialog identifier present in the topmost Route header field of the incoming request then check whether the Request-URI matches the saved Request-URI. The Request-URI and saved Request-URI are considered a match:

   a) if the canonical forms of the two Request-URI are equal to the saved value of the Request-URI;

   b) if the Request-URI is a GRUU (public or temporary) and the saved value of the Request-URI is a GRUU (public or temporary) and both GRUUs represent the same public user identity or represent public user identities that are alias SIP URIs of each other; or

   c) if the Request-URI is an alias SIP URI of the saved value of the Request-URI.

NOTE 3: The canonical form of the Request-URI is obtained by removing all URI parameters (including the user-param), and by converting any escaped characters into unescaped form. The alias SIP URI is defined in subclause 3.1.

If there is no match, then the S-CSCF shall decide whether to trigger the originating services to be executed after retargeting. The decision is configured in the S-CSCF and may use any information in the received request that is used for the initial filter criteria or an operator policy. The S-CSCF shall decide either to:

   a) stop evaluating current iFC. In that case, if the request is an INVITE request, save the Contact, CSeq and Record-Route header field values received in the request such that the S-CSCF is able to release the session if needed, forward the request based on the topmost Route header field or if not available forward the request based on the Request-URI (routeing based on Request-URI is specified in steps 2, 7 and 10 through 14a from subclause 5.4.3.2) and skip the following steps; or

   b) stop evaluating current iFC and build an ordered list of iFC with the originating services to be executed after retargeting as described in 3GPP TS 23.218 [5] criteria based on the public user identity of the served user and start the evaluation of that iFC as described in subclause 5.4.3.2 starting at step 4B of subclause 5.4.3.2;

NOTE 4: The S-CSCF assesses triggering of services for the originating services after retargeting means it evaluates IFCs with a SessionCase set to ORIGINATING_CDIV, as defined in 3GPP TS 29.228 [14]. If the P-Served-User extension specified in RFC 5502 [133] is supported, the S-CSCF uses the "orig-cdiv" header field parameter defined in RFC 8498 [239].

NOTE 5: The identity of the served user can be obtained from the History-Info header field (see RFC 7044 [66]) or the P-Served User header field as specified in RFC 5502 [133]. The served user can be a public user identity, a public GRUU, or a temporary GRUU. It needs to be ensure, that all ASs in the iFC can determine the served user correctly.

NOTE 6: The S-CSCF determines whether to apply a) or b) based on information in the initial Filter Criteria.

3A) if the Request-URI is a GRUU, but is not valid as defined in subclause 5.4.7A.4, then return a 4xx response as specified in RFC 5627 [93];

3B) if the Request-URI contains a public GRUU and the saved value of the Request-URI is a temporary GRUU, then replace the Request-URI with the saved value of the Request-URI;

3C) if the request contains a P-Asserted-Service header field check whether the IMS communication service identified by the ICSI value contained in the P-Asserted-Service header field is allowed by the subscribed services for the served user:

a) if so, continue from step 4; and

b) if not, as an operator option, the S-CSCF may reject the request by generating a 403 (Forbidden) response. Otherwise, remove the P-Asserted-Service header field and continue with the rest of the steps;

3D) if the request does not contain a P-Asserted-Service header field check if the contents of the request matches a subscribed service (e.g. SDP media capabilities, Content-Type header field) for each and any of the subscribed services for the served user:

a) if not, as an operator option, the S-CSCF may reject the request by generating a 403 (Forbidden) response. Otherwise, continue with the rest of the steps; and

b) if so, and if the request is related to an IMS communication service and the IMS communication service requires the use of an ICSI value then include a P-Asserted-Service header field in the request containing the ICSI value for the related IMS communication service, and use it as a header field in the initial request when matching initial filter criteria in step 4; and

c) if so, and if the request is related to an IMS communication service and the IMS communication service does not require the use of an ICSI value then continue without including an ICSI value; and

d) if so, and if the request does not relate to an IMS communication service (or if the S-CSCF is unable to unambiguously determine the service being requested but decides to allow the session to continue) then continue without inclding an ICSI value;

4) check whether the initial request matches any unexecuted initial filter criteria based on the public user identity of the served user in the priority order and apply the filter criteria on the SIP method as described in 3GPP TS 23.218 [5] subclause 6.5. If there is a match, then the S-CSCF shall select the first matching unexecuted initial filter criteria and:

a) if the Request-URI is a temporary GRUU as defined in subclause 5.4.7A.3, then replace the Request-URI with the public GRUU that is associated with the temporary GRUU (i.e. the public GRUU representing the same public user identity and instance ID as the temporary GRUU);

b) insert the AS URI to be contacted into the Route header field as the topmost entry followed by its own URI populated as specified in the subclause 5.4.3.4;

c) if the S-CSCF supports the P-Served-User extension as specified in RFC 5502 [133], insert the P-Served-User header field populated with the served user identity as determined in step 1. If required by operator policy, the S-CSCF shall:

- if the associated session case is "Terminating" as specified in 3GPP TS 29.228 [14], include the sescase header field parameter set to "term" and the regstate header field parameter set to "reg";

- if the associated session case is "Terminating_Unregistered" as specified in 3GPP TS 29.228 [14], include the sescase header field parameter set to "term" and the regstate header field parameter set to "unreg";

d) insert a type 3 "orig-ioi" header field parameter replacing any received "orig-ioi" header field parameter in the P-Charging-Vector header field. The type 3 "orig-ioi" header field parameter identifies the sending network of the request message. The S-CSCF shall not include the type 3 "term-ioi" header field parameter;

e) based on local policy, the S-CSCF shall add an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier if not already available;

f) remove the "transit-ioi" header field parameter, if received;

g) based on operator policy insert a Relayed-Charge header field containing the value of the received "transit-ioi" header field parameter in the P-Charging-Vector header field; and

h) if an IP address associated with the served user and the AS SIP URI is stored as described in subclause 5.4.0 exists, then the S-CSCF forwards the SIP message to the IP address associated with the served user and the AS SIP URI;

NOTE 7: Depending on the result of the previous process, the S-CSCF can contact one or more AS(s) before processing the outgoing Request-URI.

NOTE 8: If the Request-URI of the received terminating request contains a temporary GRUU, then step 4 replaces the Request-URI with the associated public GRUU before invoking the AS, and step 3B restores the original temporary GRUU when the request is returned from the AS.

NOTE 9: An AS can activate or deactivate its own filter criteria via the Sh interface. As the S-CSCF checks initial filter criteria only on receipt of an initial request for a dialog, or a standalone transaction, a modified service profile will have no impact on transactions or dialogs already in progress and the modified profile will be effective only for new transactions and dialogs. If the S-CSCF receives a modification of the iFC during their execution, then it should not update the stored initial Filter Criteria until the iFC related to the initial request have been completely executed.

5) if there was no original dialog identifier present in the topmost Route header field of the incoming request insert a P-Charging-Function-Addresses header field, if not present, populated with values received from the HSS if the message is forwarded within the S-CSCF home network, including towards AS;

6) if there was no original dialog identifier present in the topmost Route header field of the incoming request store the value of the "icid-value" header field parameter received in the P-Charging-Vector header field and retain the "icid-value" header field parameter in the P-Charging-Vector header field;

7) if there was no original dialog identifier present in the topmost Route header field of the incoming request:

- store the value of the "orig-ioi" header field parameter received in the P-Charging-Vector header field, if present;

- remove received "orig-ioi", "term-ioi" and "transit-ioi" header field parameters from the forwarded request if next hop is not an AS; and

- include a type 1 "orig-ioi" header field parameter if next hop is not an AS;

NOTE 10: Any received "orig-ioi" header field parameter will be a type 2 IOI. or type 3 IOI. A type 2 IOI identifies the sending network of the request message, a type 3 IOI identifies the sending service provider of the request message.

7A) if there was an original dialog identifier present in the topmost Route header field of the incoming request:

- store the value of the "orig-ioi" header field parameter received in the P-Charging-Vector header field, if present;

- remove the received "orig-ioi" header field parameter if next hop is not an AS;

- include a type 1 "orig-ioi" header field parameter if next hop is not an AS;

- based on local policy, the S-CSCF shall add an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or if not already available; and

- remove any received Relayed-Charge header field if next hop is not an AS;

NOTE 11: Any received "orig-ioi" header field parameter will be a type 3 IOI. A type 3 IOI identifies the sending service provider of the request message.

8) in the case:

i) there are no Route header fields in the request; and

ii) there are bindings saved during registration or re-registration as described in subclause 5.4.1.2 which are not marked as created by an emergency registration as described in subclause 5.4.8.2;

then, create a target set of potential routes from the list of preloaded routes associated with the bindings in item 8) ii), as follows:

a) if the Request-URI contains a valid GRUU assigned by the S-CSCF as defined in subclause 5.4.7A.4 that does not contain a "bnc" URI parameter, then the target set is determined by following the procedures for Request Targeting specified in RFC 5627 [93], using the public user identity and instance ID derived from the GRUU using the procedures of subclause 5.4.7A;

b) if the Request-URI contains a valid public GRUU assigned by the S-CSCF as defined in subclause 5.4.7A.4 that contains a "bnc" URI parameter then the target set is determined by following the procedures for routeing of public GRUUs specified in RFC 6140 [191].

NOTE 12: The procedures for Request Targeting for public GRUUs in subclause 7.1.1 of RFC 6140 [191] involve copying the "sg" SIP URI parameter from the Public GRUU into the Request-URI along with the bound registered Contact Address.

NOTE 13: In this release of the specification, use of preloaded routes saved during registration or re-registration which created or refreshed bindings marked as created by an emergency registration is out of scope.

c) if the Request-URI contains a temporary GRUU not assigned by the S-CSCF but that contains "temp-gruu-cookie" information provided by the S-CSCF to the UE in a "temp-gruu-cookie" header field parameter as specified in RFC 6140 [191] then the target set is determined by following the procedures for Request Targeting for temporary GRUUs specified in RFC 6140 [191]; or

NOTE 14: The procedures for obtaining the "temp-gruu-cookie" information from the temporary GRUU and for routeing of temporary GRUUs are specified in subclause 7.1.2.3 of RFC 6140 [191].

d) if the Request-URI contains a public user identity or a GRUU not assigned by the S-CSCF, then the target set is all the registered contacts saved for the destination public user identity;

9) if necessary perform the caller preferences to callee capabilities matching according to RFC 3841 [56B] to the target set;

NOTE 15: This might eliminate entries and reorder the target set.

NOTE 16: The S-CSCF performs caller preferences to callee capabilities matching also to select among multiple targets set to a single instance-id, when the UE has registered multiple registration flows.

10) in case there are no Route header fields in the request:

a) if there is more than one route in the target set determined in steps 8) and 9) above:

- if the fork directive in the Request-Disposition header field was set to "no-fork", use the contact with the highest qvalue parameter to build the target URI. In case no qvalue parameters were provided, the S-CSCF shall decide locally what contact address to be used to build the target URI;

- if the fork directive in the Request-Disposition header field was not set to "no-fork", fork the request or perform sequential search based on the relative preference indicated by the qvalue parameter of the Contact header field in the REGISTER request, as described in RFC 3261 [26]. In case no qvalue parameters were provided, then the S-CSCF determine the contact address to be used to build the target URI as directed by the Request-Disposition header field as described in RFC 3841 [56B]. If the Request-Disposition header field is not present, the S-CSCF shall decide locally whether to fork or perform sequential search among the contact addresses;

- in case that no route is chosen, return a 480 (Temporarily unavailable) response or another appropriate unsuccessful SIP response and terminate these procedures; and

- per the rules defined in RFC 5626 [92], the S-SCSF shall not populate the target set with more than one contact with the same public user identity and instance-id at a time. If a request for a particular public user identity and instance-id fails with a 430 response, the S-CSCF shall replace the failed branch with another target with the same public user identity and instance-id, but a different reg-id;

b) If no "Loose-Route Indication" indicating the HSS requires the loose-route mechanism as described in 3GPP TS 29.228 [14] has been received, in the service profile of the served public user identity, from the HSS during registration, build the Request-URI with the contents of the target URI determined in the previous step, otherwise the Request-URI is retained as received;

c) insert a P-Called-Party-ID SIP header field containing the contents of the Request-URI (if no "Loose-Route Indication" indicating the HSS requires the loose-route mechanism as described in 3GPP TS 29.228 [14] has been received, in the service profile of the served public user identity, from the HSS during registration, then exclude "rn" tel-URI parameter and "npdi" tel-URI parameter as defined in RFC 4694 [112]) received in the request unless the Request-URI contains a temporary GRUU in which case insert the public GRUU in the P-Called-Party-ID;

d) build the Route header field with the Path values from the chosen route and if "Loose-Route Indication" indicating the HSS requires the loose-route mechanism as described in 3GPP TS 29.228 [14] has been received, in the service profile of the served user identity, from the HSS during registration and the selected contact address was not registered as described in RFC 5626 [92], add the content of the target URI determined in step a), as last URI of the route. If the selected contact address was registered as described in RFC 5626 [92], the target URI determined in step a) is not added to the Route header field; and

e) save the Request-URI and the total number of Record-Route header fields as part of the dialog request state.

NOTE 17: For each initial dialog request terminated at a served user two pieces of state are maintained to assist in processing GRUUs: the chosen contact address to which the request is routed; and the position of an entry for the S-CSCF in the Record-Route header field that will be responsible for GRUU translation, if needed (the position is the number of entries in the list before the entry was added). The entry will be added in step 5) of the below procedures for handling S-CSCF receipt any 1xx or 2xx response to the initial request for a dialog. The S-CSCF can record-route multiple times, but only one of those (the last) will be responsible for gruu translation at the terminating end.

11) if the request is an INVITE request, save the Contact, CSeq and Record-Route header field values received in the request such that the S-CSCF is able to release the session if needed;

12) optionally, apply any privacy required by RFC 3323 [33] and RFC 3325 [34] to the P-Asserted-Identity header field and privacy required by RFC 7044 [66]. The S-CSCF shall not remove any priv-value from the Privacy header field;

NOTE 18: keeping the priv-value in the Privacy header field is necessary to indicate to the UE that the public user identity was not sent because of restriction. Although RFC 3323 [33] states that when a privacy service performs one of the functions corresponding to a privacy level listed in the Privacy header field, it SHOULD remove the corresponding priv-value from the Privacy header field, there is no harm that the S-CSCF does not remove the priv-values as there will be no other entity that would perform the privacy service after the S-CSCF.

NOTE 19: The optional procedure above is in addition to any procedure for the application of privacy at the edge of the trust domain specified by RFC 3325 [34].

13) in case of an initial request for a dialog, either:

- if the request is routed to an AS which is part of the trust domain, the S-CSCF shall decide, based on operator policy, whether to record-route or not. The decision is configured in the S-CSCF using any information in the received request that may otherwise be used for the initial filter criteria. If the request is record-routed the S-CSCF shall create a Record-Route header field containing its own SIP URI; or

- if the request is routed elsewhere, create a Record-Route header field containing its own SIP URI;

13A) if the request is routed towards the UE remove the P-User-Database header field and P-Served-User header field if present;

13B) void

13C) if the request was sent on a dialog for which logging of signalling is in progress, check whether a trigger for stopping logging of SIP signalling has occurred, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K]. If a stop trigger event has occurred, stop treating the dialog as one for which logging of signalling is in progress, else append a "logme" header field parameter to the SIP

Session-ID header field if the parameter is missing and determine, by checking its trace configuration, whether to log the request;

13D) if the request is routed towards the UE,

- the S-CSCF supports indicating the traffic leg as specified in RFC 7549 [225];
- the UE is roaming; and
- required by local policy;

then:

- if the bottommost Route header field does not contain the "tokenized-by" header field parameter and an "iotl" SIP URI parameter is not already included, append an "iotl" SIP URI parameter set to "homeB-visitedB" to the URI of the bottommost Route header field; and
- if the bottommost Route header field contains the "tokenized-by" header field parameter and an "iotl" SIP URI parameter is not already included, append an "iotl" SIP URI parameter set to "homeB-visitedB" to the URI of the second Route header field from the bottom;

NOTE 20: The bottommost Route header field contains an "iotl" SIP URI parameter if the P-CSCF added the "iotl" SIP URI parameter in the Path header field during registration and if the visited network does not apply topology hiding. The second Route header field from the bottom contains an "iotl" SIP URI parameter if the P-CSCF added the "iotl" SIP URI parameter in the Path header field during registration and if the visited network applied topology hiding.

13E) if the S-CSCF supports HSS based P-CSCF restoration and the S-CSCF considers the P-CSCF, identified by the bottommost Route header field, is not reachable:

- reject the request with a 480 (Temporarily Unavailable) response; and
- initiate the HSS based P-CSCF restoration procedure towards the served user as specified in 3GPP TS 23.380 [7D];

13F) if the S-CSCF supports PCRF based P-CSCF restoration procedures, insert a Restoration-Info header field including the IMSI value contained in the user profile of the registered served user as a quoted string defined in 3GPP TS 29.228 [14];

NOTE 21: If PCRF based P-CSCF restoration procedure is operated between the home network and the visited network, the operator policy depends on an agreement with the visited network operator.

13G) if the S-CSCF supports PCRF based P-CSCF restoration procedures,

- the request contains a topmost Route header field pointing to a P-CSCF, and
- the S-CSCF considers the P-CSCF is in a non-working state,

remove all entries in the Route header field and add a Route header field set to the URI associated with an alternative P-CSCF; and

NOTE 22: How the SIP URI of the alternative P-CSCF is obtained by the S-CSCF is implementation dependent. The S-CSCF can make sure that selected P-CSCF support the PCRF based P-CSCF restoration procedures based on local configuration.

NOTE 23: It is implementation dependent as to how the S-CSCF determines the P-CSCF is in non-working state.

14) forward the request based on the topmost Route header field.

If the S-CSCF receives any response to the above request, the S-CSCF shall:

1) If the response contains a "logme" header field parameter in the SIP Session-ID header field then log the response based on local policy.

If the S-CSCF supports HSS based P-CSCF restoration and

a) receives a 404 (Not Found) response;

b) fails to receive any SIP response from a P-CSCF serving a non-roaming user within a configurable time; or

NOTE 24: The configurable time needs to be less than timer B and timer F.

c) receives a 408 (Request Timeout) response or a 504 (Server Time-out) response:

- including a Restoration-Info header field defined in subclause 7.2.11 set to "noresponse"; and

- the "+g.3gpp.ics" Contact header field parameter with a value set to "server" was not included in the REGISTER request when the UE registered;

NOTE 25: If this Contact header field parameter is not included the S-CSCF can deduce that the P-CSCF did not respond to the request.

the S-CSCF shall:

- send a 480 (Temporarily Unavailable) response;

- initiate the HSS based P-CSCF restoration procedure towards the served user as specified in 3GPP TS 23.380 [7D]; and

- if b) or c) above applied consider the P-CSCF as not reachable.

If the S-CSCF supports PCRF based P-CSCF restoration and receives a 404 (Not Found) response, the S-CSCF shall consider the P-CSCF to be in a non-working state and shall initiate the PCRF based P-CSCF restoration procedure towards the served user as specified in 3GPP TS 23.380 [7D].

If the S-CSCF:

a) fails to receive a SIP response within a configurable time; or

b) receives a 408 (Request Timeout) response or a 5xx response from the AS without previously receiving a 1xx response to the original SIP request, and without previously receiving a SIP request from the AS that contained the same original dialog identifier as the original request;

the S-CSCF shall:

- if the default handling defined in the filter criteria indicates the value "SESSION_CONTINUED" as specified in 3GPP TS 29.228 [14] or no default handling is indicated, execute the procedure from step 4; and

- if the default handling defined in the filter criteria indicates the value "SESSION_TERMINATED" as specified in 3GPP TS 29.228 [14], either forward the received response or, if the request is an initial INVITE request, send a 408 (Request Timeout) response or a 5xx response towards the originating UE as appropriate (without verifying the matching of filter criteria of lower priority and without proceeding for further steps).

If the S-CSCF receives any final response from the AS, the S-CSCF shall forward the response towards the originating UE (without verifying the matching of filter criteria of lower priority and without proceeding for further steps).

When the S-CSCF receives any response to the above request and forwards it to an AS, the S-CSCF shall remove any "orig-ioi", "term-ioi" and "transit-ioi" header field parameter if received in a P-Charging-Vector header field, insert a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the request, a type 3 "term-ioi" header field parameter, and based on operator option insert a Relayed-Charge header field in the response. The S-CSCF shall set the type 3 "term-ioi" header field parameter to a value that identifies the sending network of the response, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter and include in the Relayed-Charge header field the received "transit-ioi" header field parameter from the P-Charging-Vector header field.

NOTE 26: Any received "term-ioi" header field parameter will be a type 1 IOI or a type 3 IOI. The type 1 IOI identifies the network from which the response was sent and the type 3 IOI identifies the service provider from which the response was sent.

When the S-CSCF receives, destined for an unregistered served user or a statically pre-configured PSI, an initial request for a dialog or a request for a standalone transaction, the S-CSCF shall:

1) Void.

2) execute the procedures described in 1, 2, 3, 3C, 3D, 4, 5, 6, 7, 11, 13, 13B, 13C and 14 in the above paragraph (when the S-CSCF receives, destined for the registered served user, an initial request for a dialog or a request for a standalone transaction).

3) In case that no more AS needs to be contacted, then S-CSCF shall return an appropriate unsuccessful SIP response. This response may be a 480 (Temporarily unavailable) and terminate these procedures.

NOTE 27: When the S-CSCF does not have the user profile, before executing the actions as listed above, it initiates the S-CSCF Registration/deregistration notification procedure, as described in 3GPP TS 29.228 [14]; with the purpose of downloading the relevant user profile (i.e. for unregistered user) and informs the HSS that the user is unregistered. The S-CSCF will assess triggering of services for the unregistered user, as described in 3GPP TS 29.228 [14]. When requesting the user profile the S-CSCF can include the information in the P-Profile-Key header field in S-CSCF Registration/deregistration notification When requesting the user profile, and the request received by the S-CSCF contains a P-Profile-Key header field, the S-CSCF can include the header field value in S-CSCF Registration/deregistration notification. If the response from the HSS includes a Wildcarded Public Identity AVP, and if the request received by the S-CSCF did not include a P-Profile-Key header field, the S-CSCF uses the AVP value to set the P-Profile-Key header field before forwarding the request to an AS.

Prior to performing S-CSCF Registration/Deregistration procedure with the HSS, the S-CSCF decides which HSS to query, possibly as a result of a query to the Subscription Locator Functional (SLF) entity as specified in 3GPP TS 29.228 [14] or use the value as received in the P-User-Database header field in the initial request for a dialog or a request for a standalone transaction as defined in RFC 4457 [82]. The HSS address received in the response to SLF query can be used to address the HSS of the public user identity with further queries.

If the HSS indicates to the S-CSCF that there is already another S-CSCF assigned for the user, the S-CSCF shall return a 305 (Use Proxy) response containing the SIP URI of the assigned S-CSCF received from the HSS in the Contact header field.

When the S-CSCF receives any response to the above request containing a Relayed-Charge header field, and the next hop is not an AS, the S-CSCF shall remove the Relayed-Charge header field.

When the S-CSCF receives any 1xx or 2xx response to the initial request for a dialog (whether the user is registered or not), the S-CSCF shall:

1) if the response corresponds to an INVITE request, save the Contact and Record-Route header field values in the response such that the S-CSCF is able to release the session if needed;

2) if the response is not forwarded to an AS (i.e. the response is related to a request that was matched to the first executed initial filter criteria):

   a) remove the received "transit-ioi" header field parameter if present and insert a type 2 "term-ioi" header field parameter in the P-Charging-Vector header field of the outgoing response. The type 2 "term-ioi" header field is set to a value that identifies the sending network of the response and the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter. Values of "orig-ioi" and "term-ioi" header field parameters in the received response are removed; and

   b) if the S-CSCF supports using a token to identify the registration, remove the "+g.3gpp.registration-token" Feature-Caps header field parameter, defined in subclause 7.9A.8, if received in the response;

3) in case the served user is not considered a privileged sender then:

   a) if the P-Asserted-Identity header field contains only a SIP URI and in the case where the S-CSCF has knowledge that the SIP URI contained in the received P-Asserted-Identity header field is an alias SIP URI for a tel URI, the S-CSCF shall add a second P-Asserted-Identity header field containing this tel URI, including the display name associated with the tel URI, if available; and

   b) if the P-Asserted-Identity header field contains only a tel URI, the S-CSCF shall add a second P-Asserted-Identity header field containing a SIP URI. The added SIP URI shall contain in the user part a "+" followed by the international public telecommunication number contained in tel URI, and user's home network domain name in the hostport part. The added SIP URI shall contain the same value in the display name as contained in the tel URI. The S-CSCF shall also add a "user" SIP URI parameter equals "phone" to the SIP URI;

4) in case the response is sent towards the originating user, the S-CSCF may retain the P-Access-Network-Info header field based on local policy rules and the destination user (Request-URI);

5) save an indication that GRUU routeing is to be performed for subsequent requests sent within this same dialog if:

   a) there is a record-route position saved as part of the initial dialog request state; and

   b) the contact address in the response is a valid GRUU assigned by the S-CSCF as specified in subclause 5.4.7A.4 or a temporary GRUU self assigned by the UE based on the "temp-gruu-cookie" header field parameter provided to the UE;

6) if the S-CSCF supports using a token to identify the registration and if a registration exists, add a "+g.3gpp.registration-token" Feature-Caps header field parameter, as defined in subclause 7.9A.8, set to the same value as included in the "+g.3gpp.registration-token" Contact header field parameter of the third party REGISTER request sent to the AS when the UE registered; and

NOTE 28: There could be several responses returned for a single request, and the decision to insert or modify the Record-Route needs to be applied to each. But a response might also return to the S-CSCF multiple times as it is routed back through AS. The S-CSCF will take this into account when carrying out step 5) to ensure that the information is stored only once.

7) if the response is forwarded within the S-CSCF home network and not to an AS, insert a P-Charging-Function-Addresses header field populated with values received from the HSS.

When the S-CSCF receives a response to a request for a standalone transaction (whether the user is registered or not), then:

1) in case the served user is not considered a privileged sender then:

   a) if the P-Asserted-Identity header field contains only a SIP URI and in the case where the S-CSCF has knowledge that the SIP URI contained in the received P-Asserted-Identity header field is an alias SIP URI for a tel URI, the S-CSCF shall add a second P-Asserted-Identity header field containing this tel URI, including the display name associated with the tel URI, if available; and

   b) if the P-Asserted-Identity header field contains only a tel URI, the S-CSCF shall add a second P-Asserted-Identity header field containing a SIP URI. The added SIP URI shall contain in the user part a "+" followed by the international public telecommunication number contained in tel URI, and user's home network domain name in the hostport part. The added SIP URI shall contain the same value in the display name as contained in the tel URI. The S-CSCF shall also add a "user" SIP URI parameter equals "phone" to the SIP URI; and

2) in case the response is forwarded to an AS that is located within the trust domain, the S-CSCF shall retain the access-network-charging-info parameter in the P-Charging-Vector header field; otherwise, the S-CSCF shall remove the access-network-charging-info parameter in the P-Charging-Vector header field.

When the S-CSCF receives the 200 (OK) response for a standalone transaction request, the S-CSCF shall:

1) insert a P-Charging-Function-Addresses header field populated with values received from the HSS if the message is forwarded within the S-CSCF home network, including towards an AS;

1A) if the S-CSCF supports using a token to identify the registration and if a registration exists, add a "+g.3gpp.registration-token" Feature-Caps header field parameter, as defined in subclause 7.9A.8, set to the same value as included in the "+g.3gpp.registration-token" Contact header field parameter of the third party REGISTER request sent to the AS when the UE registered;

1B) if the S-CSCF supports using a token to identify the registration in case the response is not forwarded to an AS the S-CSCF shall remove the "+g.3gpp.registration-token" Feature-Caps header field parameter, defined in subclause 7.9A.8, if received in the response; and

2) if the response is not forwarded to an AS (i.e. the response is related to a request that was matched to the first executed initial filter criteria), remove the received "orig-ioi", "term-ioi" and "transit-ioi" header field parameter if present and insert a type 2 "term-ioi" header field parameter in the P-Charging-Vector header field of the outgoing response. The type 2 "term-ioi" header field parameter is set to a value that identifies the sending network of the response and the type 2 "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter.

NOTE 29: If the S-CSCF forked the request of a stand alone transaction to multiple UEs and receives multiple 200 (OK) responses, the S-CSCF will select and return only one 200 (OK) response. The criteria that the S-CSCF employs when selecting the 200 (OK) response is based on the operator's policy (e.g. return the first 200 (OK) response that was received).

When the S-CSCF receives, destined for a served user, a target refresh request for a dialog, prior to forwarding the request, the S-CSCF shall:

0) if the dialog is related to an IMS communication service determine whether the contents of the request (e.g. SDP media capabilities, Content-Type header field) match the IMS communication service as received as the ICSI value in the P-Asserted-Service header field in the initial request. As an operator option, if the contents of the request do not match the IMS communication service the S-CSCF may reject the request by generating a status code reflecting which added contents are not matching. Otherwise, continue with the rest of the steps;

1) if the incoming request is received on a dialog for which GRUU routeing is to be performed and the Request-URI is not the GRUU for this dialog, then return a response of 400 (Bad Request).

2) if the incoming request is received on a dialog for which GRUU routeing is to be performed and the Request-URI contains the GRUU for this dialog then:

   i) if the Request-URI contains a valid GRUU assigned by the S-CSCF as defined in subclause 5.4.7A.4 that does not contain a "bnc" URI parameter, then perform the procedures for Request Targeting specified in RFC 5627 [93], using the public user identity and instance ID derived from the Request-URI, as specified in subclause 5.4.7A;

   ii) if the Request-URI contains a valid public GRUU assigned by the S-CSCF as defined in subclause 5.4.7A.4 that contains a "bnc" URI parameter then the target set is determined by following the procedures for routeing of public GRUUs specified in RFC 6140 [191]. or

NOTE 30: The procedures for Request Targeting for public GRUUs in subclause 7.1.1 of RFC 6140 [191] involve copying the "sg" SIP URI parameter from the Public GRUU into the Request-URI along with the bound registered Contact Address.

   iii) if the Request-URI contains a temporary GRUU not assigned by the S-CSCF but that contains "temp-gruu-cookie" information provided by the S-CSCF to the UE in a "temp-gruu-cookie" header field parameter as specified in RFC 6140 [191] then the target set is determined by following the procedures for routeing of temporary GRUUs specified in RFC 6140 [191];

NOTE 31: The procedures for obtaining the "temp-gruu-cookie" information from the temporary GRUU and for routeing of temporary GRUUs are specified in subclause 7.1.2.3 of RFC 6140 [191].

   iv) if no contact can be selected, return a response of 480 (Temporarily Unavailable);

3) remove its own URI from the topmost Route header field;

4) for INVITE dialogs (i.e. dialogs initiated by an INVITE request), save the Contact and CSeq header field values received in the request such that the S-CSCF is able to release the session if needed;

5) create a Record-Route header field containing its own SIP URI;

5A) void

5B) if the request was sent on a dialog for which logging of signalling is in progress, check whether a trigger for stopping logging of SIP signalling has occurred, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K]. If a stop trigger event has occurred, stop treating the dialog as one for which logging of signalling is in progress, else append a "logme" header field parameter to the SIP Session-ID header field if the parameter is missing and determine, by checking its trace configuration, whether to log the request; and

6) forward the request based on the topmost Route header field.

When the S-CSCF receives any response to the above request, the S-CSCF shall:

1) If the response contains a "logme" header field parameter in the SIP Session-ID header field then log the response based on local policy.

When the S-CSCF receives any 1xx or 2xx response to the target refresh request for a dialog (whether the user is registered or not), the S-CSCF shall:

1) for INVITE dialogs, replace the saved Contact header field values in the response such that the S-CSCF is able to release the session if needed; and

2) in case the response is forwarded to an AS that is located within the trust domain, the S-CSCF shall retain the access-network-charging-info parameter in the P-Charging-Vector header field; otherwise, the S-CSCF shall remove the access-network-charging-info parameter in the P-Charging-Vector header field.

When the S-CSCF receives, destined for the served user, a subsequent request other than target refresh request for a dialog, prior to forwarding the request, the S-CSCF shall:

1) if the incoming request is received on a dialog for which GRUU routeing is to be performed and the Request-URI is not the GRUU for this dialog, then return a response of 400 (Bad Request).

2) if the incoming request is received on a dialog for which GRUU routeing is to be performed and the Request-URI contains the GRUU for this dialog then:

   i) if the Request-URI contains a valid GRUU assigned by the S-CSCF as defined in subclause 5.4.7A.4 that does not contain a "bnc" URI parameter, then perform the procedures for Request Targeting specified in RFC 5627 [93], using the public user identity and instance ID derived from the Request-URI, as specified in subclause 5.4.7A;

   ii) if the Request-URI contains a valid public GRUU assigned by the S-CSCF as defined in subclause 5.4.7A.4 that contains a "bnc" URI parameter then the target set is determined by following the procedures for routeing of public GRUUs specified in RFC 6140 [191]; or

NOTE 32: The procedures for Request Targeting for public GRUUs in subclause 7.1.1 of RFC 6140 [191] involve copying the "sg" SIP URI parameter from the Public GRUU into the Request-URI along with the bound registered Contact Address.

   iii) if the Request-URI contains a temporary GRUU not assigned by the S-CSCF but that contains "temp-gruu-cookie" information provided by the S-CSCF to the UE in a "temp-gruu-cookie" header field parameter as specified in RFC 6140 [191] then the target set is determined by following the procedures for routeing of temporary GRUUs specified in RFC 6140 [191].

NOTE 33: The procedures for obtaining the "temp-gruu-cookie" information from the temporary GRUU and for routeing of temporary GRUUs are specified in subclause 7.1.2.3 of RFC 6140 [191].

   iv) if no contact can be selected, return a response of 480 (Temporarily Unavailable).

3) remove its own URI from the topmost Route header field;

3A) void

3B) if the request was sent on a dialog for which logging of signalling is in progress, check whether a trigger for stopping logging of SIP signalling has occurred, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K]. If a stop trigger event has occurred, stop treating the dialog as one for which logging of signalling is in progress, else append a "logme" header field parameter to the SIP Session-ID header field if the parameter is missing and determine, by checking its trace configuration, whether to log the request; and

4) forward the request based on the topmost Route header field.

When the S-CSCF receives any response to the above request, the S-CSCF shall:

1) If the response contains a "logme" header field parameter in the SIP Session-ID header field then log the response based on local policy.

When the S-CSCF receives a response to a a subsequent request other than target refresh request for a dialog, in case the response is forwarded to an AS that is located within the trust domain, the S-CSCF shall retain the access-network-charging-info parameter from the P-Charging-Vector header field; otherwise, the S-CSCF shall remove the access-network-charging-info parameter from the P-Charging-Vector header field.

With the exception of 305 (Use Proxy) responses, the S-CSCF shall not recurse on 3xx responses.

5.4.3.4 Original dialog identifier

The original dialog identifier is an implementation specific token that the S-CSCF encodes into the own S-CSCF URI in a Route header field, prior to forwarding the request to an AS. This is possible because the S-CSCF is the only entity that creates and consumes the value.

The token may identify the original dialog of the request, so in case an AS acting as a B2BUA changes the dialog, the S-CSCF is able to identify the original dialog when the request returns to the S-CSCF. In a case of a standalone transaction, the token indicates that the request has been sent to the S-CSCF from an AS in response to a previously sent request. The token can be encoded in different ways, such as e.g., a character string in the user-part of the S-CSCF URI, a parameter in the S-CSCF URI or port number in the S-CSCF URI.

The S-CSCF shall ensure that the value chosen is unique so that the S-CSCF may recognize the value when received in a subsequent message of one or more dialogs and make the proper association between related dialogs that pass through an AS.

An original dialog identifier is sent to each AS invoked due to iFC evaluation such that the S-CSCF can associate requests as part of the same sequence that trigger iFC evaluation in priority order (and not rely on SIP dialog information that may change due to B2BUA AS).

NOTE: If the same original dialog identifier is included in more than one request from a particular AS (based on service logic in the AS), then the S-CSCF will continue the iFC evaluation sequence. If the AS wants iFC evaluation to start from the beginning for a request, then AS should not include an original dialog identifier;

5.4.3.5 Void

5.4.3.6 SIP digest authentication procedures for all SIP request methods initiated by the UE excluding REGISTER

5.4.3.6.1 General

When the S-CSCF receives from the UE a request (excluding REGISTER), and SIP digest without TLS or SIP digest with TLS is supported and in use for this UE, the S-CSCF may perform the following steps if authentication of SIP request methods initiated by the UE excluding REGISTER is desired:

1) The S-CSCF shall identify the user by the public user identity as received in the P-Asserted-Identity header field;

2) If the public user identity does not match one of the registered public user identities, and the public user identity does not match one of the registered wildcarded public user identities, the S-CSCF may reject the request with a 400 (Bad Request) response or silently discard the request;

3) If the request does not contain a Proxy-Authorization header field or the Proxy-Authorization header field does not contain a digest response, the S-CSCF shall:

   a) challenge the user by generating a 407 (Proxy Authentication Required) response for the received request, including a Proxy-Authenticate header field as defined in RFC 7616 [286] and RFC 8760 [287], which includes:

      - a "realm" header field parameter;

      - a "nonce" header field parameter, with a newly generated value by the S-CSCF;

      - an "algorithm" header field parameter; and

      - a "qop" header field parameter; if the qop value is not provided in the authentication vector, it shall have the value "auth".

      The challenge parameters, with the exception of the "nonce" header field parameter, shall be the same as the ones used for the last successful registration.

NOTE 1: The usage of the same parameters for authentication of non-registration SIP requests requires the storage of these parameters during authentication of REGISTER requests, as retrieval of authentication vectors is only specified for REGISTER requests.

NOTE 2: If these parameters are not locally stored in the S-CSCF, i.e. when the S-CSCF has restarted, and the S-CSCF supports restoration as specified in 3GPP TS 23.380 [7D], subclause 4.4.2, the S-CSCF can fetch these parameters from the HSS.

b) send the so generated 407 (Proxy Authentication Required) response towards the UE;

c) retain the nonce and initialize the corresponding nonce count to a value of 1; and

d) start timer request-await-auth.

4) If the request contains a Proxy-Authorization header field, the S-CSCF shall:

a) check whether the Proxy-Authorization header field contains:

- the private user identity of the user in the "username" header field parameter;
- an "algorithm" header field parameter value which matches the "algorithm" header field parameter in the authentication challenge (i.e. "SHA2-256", "SHA2-512/256" or "MD5");
- a "response" header field parameter with the authentication challenge response;
- a "realm" header field parameter matching the "realm" header field parameter in the authentication challenge;
- "nonce" header field parameter matching a nonce that is deemed valid by the S-CSCF for the related registration or registration flow (i.e. a nonce that was set in a a Proxy-Authenticate header field of a 407 (Proxy Authentication Required) response to a non-REGISTER request for which the associated validity duration has not expired or in a WWW-Authenticate header field of a 401 (Unauthorized) response to a REGISTER request for which the associated validity duration has not expired, a nonce sent in a "nextnonce" header field parameter sent in a Authentication-Info header field of a 200 OK (OK) to REGISTER request) or if an authentication is ongoing for this request (i.e. a associated "req-await-auth" is running) matching the nonce that was sent in a Proxy-Authenticate header field of the 407 (Proxy Authentication Required) response to this request;

NOTE 3: The related registration flow or registration is identified by the couple instance-id and reg-id if the multiple registration mechanism is used or by contact address if not.

- a "uri" header field parameter matching the SIP Request-URI;
- a "cnonce" header field parameter; and
- a "nc" header field parameter with a value that equals the nonce-count expected by the S-CSCF. The S-CSCF may choose to accept a nonce-count which is greater than the expected nonce-count. If the S-CSCF uses this nonce-count and authentication is successful and the S-CSCF increments it for any subsequent authentication responses.

If any of the above checks do not succeed, the S-CSCF shall proceed as described in subclause 5.4.3.6.2, and skip the remainder of this procedure; and

b) check whether the received authentication challenge response and the expected authentication challenge response match. The S-CSCF shall compute the expected digest response as described in RFC 7616 [286] and RFC 8760 [287] using the H(A1) value contained within the authentication vector, and other digest parameters (i.e. nonce, cnonce, nc, qop).

In the case where the digest response does not match the expected digest response calculated by the S-CSCF, the S-CSCF shall consider the authentication attempt as failed and do one of the following:

1) rechallenge the user by issuing a 407 (Proxy Authentication Required) response including a challenge as per procedures described in this subclause; or

2) reject the request by issuing a 403 (Forbidden) response; or

3) reject the request without sending a response.

In the case where the digest response matches the expected digest response calculated by the S-CSCF, the S-CSCF shall:

1) consider the identity of the user verified and the request authenticated and continue with the procedures as described in subclause 5.4.3;

2) if the used nonce was not considered valid before the authentication succed (i.e a "req-await-auth" was running), add this nonce to the list of the valid nonces for the related registration or registration flow (if multiple registration mechanism is used) for an operator configured duration; and

3) stop the related "request-await-auth" running if any.

If the timer request-await-auth expires, the S-CSCF shall consider the authentication to have failed.

5.4.3.6.2 Abnormal cases

In the case that SIP digest is used and the request from the UE contains an invalid "nonce" Authorization header field parameter with a valid challenge response for that nonce (indicating that the client knows the correct username/password), or when the "nonce-count" Authorization header field parameter value sent by the UE is not the expected value, or when the Proxy-Authorization header field does not include the correct parameters, the S-CSCF shall:

- send a 407 (Proxy Authentication Required) response to initiate a further authentication attempt with a fresh nonce and the "stale" header field parameter set to "true" in the Proxy-Authenticate header field.

When the S-CSCF cannot forward an initial incoming request to an Application Server due to overload control mechanism, it shall either

- if the default handling defined in the filter criteria indicates the value "SESSION_CONTINUED" as specified in 3GPP TS 29.228 [14] or no default handling is indicated, execute the procedure from step 5 in subclause 5.4.3.2 or from step 4 in subclause 5.4.3.3 depending on the type of request; and

- if the default handling defined in the filter criteria indicates the value "SESSION_TERMINATED" as specified in 3GPP TS 29.228 [14], reject the request as specified in RFC 7339 [199] and RFC 7200 [201] (without verifying the matching of filter criteria of lower priority and without proceeding for further steps).

5.4.4 Call initiation

5.4.4.1 Initial INVITE

When the S-CSCF receives an INVITE request, either from the served user or destined to the served user, the S-CSCF may require the periodic refreshment of the session to avoid hung states in the S-CSCF. If the S-CSCF requires the session to be refreshed, the S-CSCF shall apply the procedures described in RFC 4028 [58] clause 8.

NOTE 1: Requesting the session to be refreshed requires support by at least one of the UEs. This functionality cannot automatically be granted, i.e. at least one of the involved UEs needs to support it.

For interworking with a visiting network, where the P-CSCF of the visiting network does not support priority but it is intended or required to give users of that P-CSCF priority in the home network, the S-CSCF in the home network shall recognize the need for priority treatment if such detection is not alternately provided via an IBCF in the home network.

NOTE 2: Various mechanisms can be applied to recognize the need for priority treatment (e.g., based on the dialled digits). The exact mechanisms are left to national regulation and network configuration.

When an S-CSCF interworks with a visiting network that does not support priority, and the S-CSCF recognizes the need for priority treatment, the S-CSCF shall insert the temporarily authorised Resource-Priority header field with appropriate namespace and priority value in the INVITE request.

When the S-CSCF receives an initial INVITE request destined for the served user, the S-CSCF shall either:

a) examine the SDP offer (the "c=" parameter) to detect if it contains an IP address type that is not supported by the IM CN subsystem; or

NOTE 3: The S-CSCF can, based on local policy, assume that a UE supports the IP address type of the SDP offer for media if it is identical to the address type of a contact that the UE has registered.

b) process the initial INVITE request without examining the SDP.

NOTE 4: If the S-CSCF knows that the terminating UE supports both IPv6 and IPv4 addressing simultaneously, the S-CSCF will forward the initial INVITE request to the UE without examining the SDP. The present version of the specification does not specify how the S-CSCF determines whether the UE supports both IPv6 and IPv4 addressing simultaneously.

NOTE 5: If the SDP offer contained an IP address type that is not supported by the IM CN subsystem, the S-CSCF will receive the 488 (Not Acceptable Here) response with 301 Warning header field indicating "incompatible network address format".

Subsequently, when the S-CSCF detects that the SDP offer contained an IP address type that is not supported by the IM CN subsystem (i.e., either case a) or b)), the S-CSCF shall either:

- return a 305 (Use Proxy) response to the I-CSCF with the Contact field containing the SIP URI of the IBCF, or
- forward the initial INVITE request to the IBCF. When forwarding the initial INVITE request, the S-CSCF shall not insert its SIP URI into the Record-Route header field.

If overlap signalling using the multiple-INVITE method is supported as a network option, several INVITE requests with the same Call ID and the same From header field (including "tag" header field parameter) can be received outside of an existing dialog. Such INVITE requests relate to the same call. If the S-CSCF receives an INVITE request from the served user outside an existing dialog with the same Call ID and From header field as a previous INVITE request during a certain period of time, it shall route the new INVITE request to the same next hop as the previous INVITE request.

5.4.4.2 Subsequent requests

5.4.4.2.1 UE-originating case

When the S-CSCF receives the request containing the access-network-charging-info parameter in the P-Charging-Vector, the S-CSCF shall store the access-network-charging-info parameter from the P-Charging-Vector header field. The S-CSCF shall retain access-network-charging-info parameter in the P-Charging-Vector header field when the request is forwarded to an AS. However, the S-CSCF shall not include the access-network-charging-info parameter in the P-Charging-Vector header field when the request is forwarded outside the home network of the S-CSCF.

When the S-CSCF receives any request or response (excluding CANCEL requests and responses) related to a UE-originated dialog or standalone transaction, the S-CSCF shall insert previously saved values into the P-Charging-Vector header field before forwarding the message within the S-CSCF home network, including towards AS.

When the S-CSCF receives any request or response (excluding ACK requests and CANCEL requests and responses) related to a UE-originated dialog or standalone transaction, the S-CSCF may insert previously saved values into the P-Charging-Function-Addresses header field before forwarding the message within the S-CSCF home network, including towards AS.

5.4.4.2.2 UE-terminating case

When the S-CSCF receives 180 (Ringing) or 200 (OK) (to INVITE) responses containing the access-network-charging-info parameter in the P-Charging-Vector, the S-CSCF shall store the access-network-charging-info parameter from the P-Charging-Vector header field. The S-CSCF shall retain the access-network-charging-info parameter in the P-Charging-Vector header field when the response is forwarded to an AS. However, the S-CSCF shall not include the access-network-charging-info parameter in the P-Charging-Vector header field when the response is forwarded outside the home network of the S-CSCF.

When the S-CSCF receives any request or response (excluding CANCEL requests and responses) related to a UE-terminated dialog or standalone transaction, the S-CSCF shall insert previously saved values into the P-Charging-Vector header field before forwarding the message within the S-CSCF home network, including towards AS.

When the S-CSCF receives any request or response (excluding ACK requests and CANCEL requests and responses) related to a UE-terminated dialog or standalone transaction, the S-CSCF may insert previously saved values into the P-

Charging-Function-Addresses header field before forwarding the message within the S-CSCF home network, including towards AS.

When the S-CSCF receives an error response (to INVITE) for an existing early dialog, and if the S-CSCF does not forward the response immediately (if the S-CSCF forked the INVITE request it may wait for additional final responses), the S-CSCF does not have knowledge of having received an 199 (Early Dialog Terminated) provisional response on the same early dialog, and the associated INVITE request included the "199" option-tag in the Supported header field, and the INVITE request did not include the "100rel" option tag in the Require header field, the S-CSCF shall trigger and send an unreliable 199 (Early Dialog Terminated) provisional response, using the same "tag" To header field parameter value as the error response, as specified in RFC 6228 [142].

When the S-CSCF has forked an initial INVITE request, and it has received:

- a 2xx response associated with one of the early dialogs, the S-CSCF shall in each CANCEL request it generates as specified in RFC 3261 [26] insert a Reason header field with a "SIP" protocol header field parameter value, a "200" cause header field parameter value, and a "Call completed elsewhere" text header field parameter value, as specified in RFC 3326 [34A]; or

- a 6xx response associated with one of the early dialogs, the S-CSCF shall, in each CANCEL request it generates as specified in RFC 3261 [26] insert a Reason header field with "SIP" protocol header field parameter value, a cause header field parameter value representing the response code (e.g. "603") in the received response, and a text header field parameter with a value associated with the response code (e.g. a "Declined" value in the case of a "603" response code), as specified in RFC 3326 [34A].

5.4.5 Call release

5.4.5.1 S-CSCF-initiated session release

5.4.5.1.1 Cancellation of a session currently being established

Upon receipt of a network internal indication to release a session which is currently being established, the S-CSCF shall:

1) cancel the related dialogs by sending the CANCEL request according to the procedures described in RFC 3261 [26]; and

2) send an appropriate response to the sender of the original INVITE request.

5.4.5.1.2 Release of an existing session

Upon receipt of a network internal indication to release an existing multimedia session, the S-CSCF shall:

1) if the S-CSCF serves the calling user of the session, generate a BYE request destined for the called user based on the information saved for the related dialog, including:

   - a Request-URI, set to the stored Contact header field provided by the called user;

   - a To header field, set to the To header field value as received in the 200 (OK) response for the initial INVITE request;

   - a From header field, set to the From header field value as received in the initial INVITE request;

   - a Call-ID header field, set to the Call-Id header field value as received in the initial INVITE request;

   - a CSeq header field, set to the CSeq value that was stored for the direction from the calling to the called user, incremented by one;

   - a Route header field, set to the routeing information towards the called user as stored for the dialog;

   - a Reason header field that contains proper SIP response code;

   - further header fields, based on local policy;

- treat the BYE request as if received directly from the calling user, i.e. the S-CSCF shall send the BYE request to the internal service control and based on the outcome further on towards the called user; and

2) if the S-CSCF serves the calling user of the session, generate an additional BYE request destined for the calling user based on the information saved for the related dialog, including:

   - a Request-URI, set to a contact address obtained from the stored Contact header field if provided by the calling user. If the stored Contact header field contained either a public or a temporary GRUU, the S-CSCF shall set the Request-URI either to:

     a) the contact address bound to the respective GRUU, if the stored Contact header field did not include an "ob" SIP URI parameter; or

     b) the contact address that the UE used to send the initial INVITE request, if the stored Contact header field included an "ob" SIP URI parameter;

NOTE 1: Since the same public GRUU can be bound to multiple contact addresses of the UE that were registered as specified in RFC 5626 [92], the S-CSCF selects the contact address that the UE used to send the initial INVITE request.

   - a To header field, set to the From header field value as received in the initial INVITE request;

   - a From header field, set to the To header field value as received in the 200 (OK) response for the initial INVITE request;

   - a Call-ID header field, set to the Call-Id header field value as received in the initial INVITE request;

   - a CSeq header field, set to the CSeq value that was stored for the direction from the called to the calling user, incremented by one – if no CSeq value was stored for that session the S-CSCF shall generate and apply a random number within the valid range for CSeqs;

   - a Route header field, set to the routeing information towards the calling user as stored for the dialog;

   - a Reason header field that contains proper SIP response code;

   - further header fields, based on local policy;

   - send the BYE request directly to the calling user.

3) if the S-CSCF serves the called user of the session, generate a BYE request destined for the called user based on the information saved for the related dialog, including:

   - a Request-URI, set to a contact address that the S-CSCF uses to send the in-dialog requests towards the called UE as defined in RFC 5626 [92] and RFC 5627 [93];

   - a To header, set to the To header value as received in the 200 (OK) response for the initial INVITE request;

   - a From header, set to the From header value as received in the initial INVITE request;

   - a Call-ID header, set to the Call-Id header value as received in the initial INVITE request;

   - a CSeq header, set to the CSeq value that was stored for the direction from the calling to the called user, incremented by one;

   - a Route header, set to the routeing information towards the called user as stored for the dialog;

   - a Reason header that contains proper SIP response code;

   - further headers, based on local policy;

   - send the BYE request directly to the called user; and

4) if the S-CSCF serves the called user of the session, generate an additional BYE request destined for the calling user based on the information saved for the related dialog, including:

   - a Request-URI, set to the stored Contact header field provided by the calling user;

- a To header, set to the From header field value as received in the initial INVITE request;
- a From header, set to the To header field value as received in the 200 (OK) response for the initial INVITE request;
- a Call-ID header, set to the Call-Id header field value as received in the initial INVITE request;
- a CSeq header, set to the CSeq value that was stored for the direction from the called to the calling user, incremented by one – if no CSeq value was stored for that session the BYE shall generate and apply a random number within the valid range for CSeqs;
- a Route header field, set to the routeing information towards the calling user as stored for the dialog;
- a Reason header field that contains proper SIP response code;
- further headers, based on local policy;
- treat the BYE request as if received directly from the called user, i.e. the S-CSCF shall send the BYE request to the internal service control and based on the outcome further on towards the calling user.

Upon receipt of the 2xx responses for both BYE requests, the S-CSCF shall release all information related to the dialog and the related multimedia session.

5.4.5.1.2A Release of the existing dialogs due to registration expiration

When:

1) the registration lifetime of the only public user identity currently registered with its associated set of implicitly registered public user identities (i.e. no other is registered) and bound either to the contact address of the UE or to the registration flow and the associated contact address (if the multiple registration mechanism is used) expires;
2) there are still active multimedia sessions that includes either this user's contact address or the registration flow and the associated contact address (if the multiple registration mechanism is used);
3) the session was initiated by or terminated towards the user using the public user identity currently registered or with one of the implicitly registered public used identities bound either to the contact addressof the UE or to the registration flow and the associated contact address (if the multiple registration mechanism is used);

then the S-CSCF shall:

- release each of these multimedia sessions by applying the steps listed in the subclause 5.4.5.1.2. The S-CSCF shall only release dialogs associated with the multi media sessions originated or terminated towards the registered user's contact address or the registration flow and the associated contact address (if the multiple registration mechanism is used).

5.4.5.1.3 Abnormal cases

Upon receipt of a request on a dialog for which the S-CSCF initiated session release, the S-CSCF shall terminate the received request and answer it with a 481 (Call/Transaction Does Not Exist) response.

5.4.5.2 Session release initiated by any other entity

Upon receipt of a 2xx response for a BYE request matching an existing dialog, the S-CSCF shall delete all the stored information related to the dialog.

5.4.5.3 Session expiration

If the S-CSCF requested the session to be refreshed periodically, and the S-CSCF got the indication that the session will be refreshed, when the session timer expires, the S-CSCF shall delete all the stored information related to the dialog.

5.4.6 Call-related requests

5.4.6.1 ReINVITE

5.4.6.1.1 Determination of served user

Void.

5.4.6.1.2 UE-originating case

For a reINVITE request or UPDATE request from the UE within the same dialog, the S-CSCF shall store the updated access-network-charging-info parameter from P-Charging-Vector header field in the received SIP request. The S-CSCF shall retain the access-network-charging-info parameter in the P-Charging-Vector header field when the request is forwarded to an AS. However, the S-CSCF shall not include the access-network-charging-info parameter in the P-Charging-Vector header field when the request is forwarded outside the home network of the S-CSCF.

For a reINVITE request from the UE, if the request is to be forwarded to an AS that is located within the trust domain, the S-CSCF shall retain the access-network-charging-info parameter from the P-Charging-Vector header field; otherwise, the S-CSCF shall remove the access-network-charging-info parameter from the P-Charging-Vector header field.

5.4.6.1.3 UE-terminating case

For a reINVITE request or UPDATE request destined towards the UE within the same dialog, when the S-CSCF receives the 200 (OK) response (to the INVITE request or UPDATE request), the S-CSCF shall store the updated access-network-charging-info parameter from the P-Charging-Vector header field. The S-CSCF shall retain the access-network-charging-info parameter in the P-Charging-Vector header field when the response is forwarded to the AS. However, the S-CSCF shall not include the access-network-charging-info parameter in the P-Charging-Vector header field when the 200 (OK) response is forwarded outside the home network of the S-CSCF.

For any SIP response to an INVITE request, if the response is to be forwarded to an AS that is located within the trust domain, the S-CSCF shall retain the access-network-charging-info parameter from the P-Charging-Vector header field; otherwise, the S-CSCF shall remove the access-network-charging-info parameter from the P-Charging-Vector header field.

5.4.7 Void

5.4.7A GRUU management

5.4.7A.1 Overview of GRUU operation

The S-CSCF provides a service of assigning and translating GRUUs for use by registered UEs, unless "Loose-Route Indication" indicating the HSS requires the loose-route mechanism as described in 3GPP TS 29.228 [14] has been provisioned in the service profile of the registered public user identity. This is conducted as specified in RFC 5627 [93] and RFC 5628 [94]. Two kinds of GRUUs are assigned: public GRUUs and temporary GRUUs.

NOTE: If the UE performs the functions of an external attached network (e.g an enterprise network) the UE could have self allocated its own GRUUs. In this version of the specification only UE self allocated public GRUUs are supported. Routeing to a specific UE self-allocated public GRUUs requires that "Loose-Route Indication" indicating the HSS requires the loose-route mechanism as described in 3GPP TS 29.228 [14] is provisioned in the service profile of the served public user identity. Use of UE self-allocated temporary GRUUs is not supported in this version of the specification and requests addressed to UE self allocated temporary GRUUs will fail to be routed to the UE.

Each assigned GRUU represents an association between a public user identity and an instance ID provided by a registering UE. It is used to address a particular UE that possesses the instance ID and registers with the public user identity. The GRUU also denotes a contact address registered with a public user identity when the contact address has a "+sip.instance" header field parameter containing the GRUU instance ID.

The S-CSCF issues GRUUs as part of the registration process, and also reports GRUUs as part of notifications for subscriptions to the "reg" event package. The S-CSCF always issues GRUUs in pairs – a public GRUU and a temporary GRUU. In case of implicit registration the S-CSCF assigns a unique public GRUU and a unique temporary GRUU for each public user identity.

The S-CSCF may also support the procedures for allocating public GRUUs and supporting the generation of temporary GRUUs by the functionality within the UE that performs the role of registrar as specified in RFC 6140 [191] as well as the procedures to route requests containing such GRUUs.

5.4.7A.2 Representation of public GRUUs

Each public GRUU shall conform to all requirements specified in RFC 5627 [93].

If the Contact URI in the Contact header field does not contain a "bnc" URI parameter, then the S-CSCF constructs a public GRUU by adding a "gr" SIP URI parameter to the canonical for m of the SIP URI which contains a public user identity.

If the Contact URI in the Contact header field contains a "bnc" URI parameter and if the S-CSCF supports RFC 6140 [191], then the S-CSCF constructs a public GRUU by adding both "bnc" and "gr" SIP URI parameters to the canonical form of the SIP URI from the To header field of the REGISTER request

The "gr" SIP URI parameter serves as an indicator that the URI is in fact a GRUU and if the "+sip.instance" header field parameter from the Contact address contains an IMEI URN or a MEID URN then it carries a value that encodes the IMEI based instance ID that is defined in 3GPP TS 23.003 [3] or the MEID based instance ID which is defined in RFC 8464 [187] otherwise it carries the value received in the "+sip.instance" header field parameter.

By default, the value of the "gr" SIP URI parameter is a copy of the value of the "+sip.instance" header field parameter from a Contact address registered with the S-CSCF, with escaping of special characters as specified in RFC3261 [26].

The public GRUU that is returned in the "pub-gruu" parameter in the 200 (OK) response to the REGISTER request is constructed using the canonical form of the SIP URI of the public user identity from the To header field of the REGISTER request provided that public user identity is not barred. If the public user identity from the To header field of the REGISTER request is barred then the public GRUU that is returned in the "pub-gruu" parameter in the 200 (OK) response to the REGISTER request is constructed using the canonical form of the SIP URI of the default public user identity.

NOTE 1: The default public user identity is always provisioned as a SIP URI.

If the "+sip.instance" header field parameter from the Contact address contains an IMEI URN, as specified in RFC 7254 [153] or an MEID URN, as specified in RFC 8464 [187], then the value of the "gr" SIP URI parameter is generated by the S-CSCF using the name-based UUID algorithm defined in RFC 4122 [154]. The following applies to the algorithm:

1) the "name space ID" shall be a UUID generated for use across the administrative domain and shall use the algorithm for creating a UUID from truly random numbers specified in RFC 4122 [154];

NOTE 2: If the generated UUID is changed, then newly created GRUUs will not match those that were created with the previous UUID. Therefore, the UUID needs to remain the same in order to create consistent GRUUs. This means that the namespace UUID needs to be the same for all S-CSCFs within the domain for which the public GRUU is hosted (it cannot be generated at run time by the S-CSCF as that would produce different values).

2) SHA-1 shall be used as the hash algorithm; and

3) the "name" is made up of a concatenation of the ASCII representation (see RFC 20 [212]) of:

   a) if IMEI, the TAC and SNR portions of the IMEI; or

   b) if MEID, the Manufacturer Code and the Serial Number portions of the MEID;

   from the "+sip.instance" header field parameter.

Only the IMEI shall be use for generating an instance ID for a multi-mode UE that supports both 3GPP and 3GPP2 defined radio access networks, and the S-CSCF shall follow the procedures for an IMEI as described above.

The S-CSCF shall store the "gr" parameter used in a public GRUU and the associated value received in a "+sip.instance" header field parameter.

The public GRUU for a particular association of public user identity and instance ID is persistent. The same public GRUU will be returned each time a registration is performed with a particular pair of public user identity and instance ID.

5.4.7A.3 Representation of temporary GRUUs

NOTE 1: For UEs performing the functions of an external attached network that support RFC 6140 [191] the S-CSCF does not allocate temporary GRUUs but assists the functionality within the UE that performs the role of registrar in allocating it's own temporary GRUUs by providing to the UE the "temp-gruu-cookie" header field parameter that uniquely identifies the registration. The functionality within the UE that performs the role of registrar then is able to allocate its own temporary GRUUs as per RFC 6140 [191] procedures.

Each temporary GRUU shall conform to all requirements specified in RFC 5627 [93].

Because of the limited lifetime of an temporary GRUU, only the S-CSCF that created a temporary GRUU is required to understand how to translate that GRUU to the corresponsing public user identity and instance ID.

The temporary GRUU that is returned in the "temp-gruu" parameter in the 200 (OK) response to the REGISTER request is mapped to the public user identity from the To header field of the REGISTER request provided that public user identity is not barred. If the public user identity from the To header field of the REGISTER request is barred then the temporary GRUU that is returned in the "temp-gruu" parameter in the 200 (OK) response to the REGISTER request is mapped to the default public user identity.

NOTE 2: The default public user identity is always provisioned as a SIP URI.

The specific representation of a temporary GRUU may be decided by each S-CSCF implementation. Temporary GRUUs must route to the assigning S-CSCF without requiring each assigned GRUU to be stored in the HSS.

The S-CSCF may choose a representation of temporary GRUUs that requires no extra state to be retained, such as that specified in RFC 5627 [93]. Alternatively, the S-CSCF may choose a stateful representation. This is an implementation choice.

NOTE 3: One possible implementation is for the S-CSCF to have a statically configured wildcard PSI that routes to it, with each temporary GRUU being encoded so that it matches the wildcard.

5.4.7A.4 GRUU recognition and validity

The S-CSCF shall recognize those GRUUs it has assigned, verify their validity, and extract the associated public user identity or stored identity of the UE that represents the functionality within the UE that performs the role of registrar and instance ID. This is true for both public GRUUs and temporary GRUUs.

NOTE 1: The S-CSCF only validates and extracts the associated public user identity and instance ID for GRUUs that it assigned.

GRUUs are distinguished from other URIs by the presence of a "gr" SIP URI parameter. Public GRUUs are distinguished from temporary GRUUs by the presence of a value for the "gr" SIP URI parameter.

The instance ID is obtained from a public GRUU by using the "gr" parameter to retrieve the stored associated instance ID. The public user identity or stored identity of the UE that represents the functionality within the UE that performs the role of registrar is extracted from a public GRUU by removing the "gr" SIP URI parameter.

The S-CSCF can recognize a public GRUU as valid if the "gr" parameter contains a value that was stored in the S-CSCF during generation of the public GRUU, and the derived public user identity compares equal, according to the comparison rules of RFC3261 [26], to a public user identity active within the S-CSCF or a stored identity of the UE that represents the functionality within the UE that performs the role of registrar from which a public GRUU was created. When validating public GRUUs the S-CSCF shall ignore the presence of any "sg" SIP URI parameter when determining if a public GRUU is one allocated by the S-CSCF.

NOTE 2: The UE that supports RFC 6140 [191] and performs the functions of an external attached network, adds a unique "sg" SIP URI parameter value to the public GRUU supplied by the S-CSCF when generating public GRUUs for its registering UAs.

The public user identity and instance ID are derived from a temporary GRUU via implementation specific means consistent with the way temporary GRUUs are constructed. The S-CSCF shall determine the validity of a temporary GRUU in conformance with RFC 5627 [93], and if the GRUU was allocated using RFC 6140 [191] procedures then in conformance with RFC 6140 [191] or using implementation specific means.

The S-CSCF regards a UE self-allocated public GRUU as valid if "Loose-Route Indication" indicating the HSS requires the loose-route mechanism as described in 3GPP TS 29.228 [14] is provisioned in the service profile of the served public user identity.

5.4.8 Emergency service

5.4.8.1 General

The S-CSCF shall handle the emergency registration as per the needs of the normal registration.

NOTE 1: Emergency specific procedures for the Cx interface are specified in annex G in 3GPP TS 29.228 [14].

NOTE 2: When receiving an emergency service request then the S-CSCF handles the emergency service request as per the procedures in subclause 5.4.3.2. The Route header field indicating the URI associated with the E-CSCF is included by a P-CSCF or an AS.

5.4.8.2 Initial emergency registration or user-initiated emergency reregistration

When the S-CSCF receives a REGISTER request; and the Contact header field includes a "sos" SIP URI parameter that indicates that this is an emergency registration, the S-CSCF shall perform the actions as specified in subclause 5.4.1.1 with the following additions:

1) when handling unprotected REGISTER request or protected REGISTER request, the S-CSCF:

   a) shall deregister only contacts that were registered as part of emergency registration; and

   b) shall not deregister contacts that were registered as part of non-emergency registration;

NOTE 1: other conditions triggering contact deregistration are described in subclause 5.4.1.

2) for the protected REGISTER request, when the S-CSCF receives a REGISTER request with the "integrity-protected" header field parameter in the Authorization header field set to "yes", "tls-yes" or "ip-assoc-yes", i.e. for the protected REGISTER request, and the Contact header field includes a "sos" SIP URI parameter that indicates that this is an emergency registration, the S-CSCF shall identify the user by the public user identity as received in the To header field and the private user identity as received in the Authorization header field of the REGISTER request;

3) if operator policy does not require that emergency service requests are forwarded to the S-CSCF, the S-CSCF shall not include a Service-Route header field in the 200 (OK) response to the REGISTER request;

4) the S-CSCF shall not include a temporary GRUU in the 200 (OK) response to the REGISTER request;

5) the S-CSCF shall in the Contact header field of the 200 (OK) response to the REGISTER request include only the URI that was successfully emergency registered and in this URI include the "sos" SIP URI parameter;

NOTE 2 Only including the emergency registered contact in the 200 (OK) response to the REGISTER request deviates from bullet 8 in section 10.3 of RFC 3261 [26].

NOTE 3: In the case where the S-CSCF returns a GRUU in the Contact header field of the 200 (OK) response to the REGISTER request, the "sos" SIP URI parameter is appended to the URI and not included as a Contact header field parameter. The public GRUU that is returned in the 200 (OK) response includes the "sos" SIP URI parameter as a parameter of the URI included in the "pub-gruu" Contact header field parameter.

6) store the Path header field and the contact information including all header field parameters contained in the Contact header field;

NOTE 4: The Path header field and contact information used for the emergency dialogs destined for the UE and obtained during the emergency registration can be different than the Path header field used for the non-emergency communication and obtained during the non-emergency registration.

NOTE 5: The S-CSCF will not perform the network initiated deregistration procedure for an emergency registration, but will let it expire. A new emergency registration will overwrite any previous emergency registration.

7) the S-CSCF shall not send any third-party REGISTER requests to any AS;

8) void

9) determine the duration of the registration by checking the value of the registration expiration interval value in the received REGISTER request and based on local policy; and

NOTE 6: The value of the emergency registration time is subject to national regulation and can be subject to roaming agreements.

10) for any bindings created by the emergency registration, mark those bindings as created by an emergency registration.

5.4.8.3 User-initiated emergency deregistration

When S-CSCF receives a REGISTER request with the registration expiration interval value containing zero and the Contact header field contains a contact address that has been registered for emergency service (i.e. the "sos" SIP URI parameter that indicates that this is an emergency registration is included in the Contact header field), the S-CSCF shall reject the REGISTER request by sending a 501 (Not Implemented) response.

NOTE: The UE cannot deregister its emergency public user identity.

5.4.8.4 Network-initiated emergency deregistration

The S-CSCF shall not perform a network-initiated emergency deregistration.

5.4.8.5 Network-initiated emergency reauthentication

If a given public user identity and the associated contact address have been registered via emergency registration, the S-CSCF shall not reauthenticate this public user identity.

5.4.8.6 Subscription to the event providing registration state

If a S-CSCF receives a SUBSCRIBE request addressed to S-CSCF containing the Event header field with the reg event package with the Contact header field that contains a contact address that has been registered for emergency service, the S-CSCF shall reject the SUBSCRIBE request for the reg-event package by sending a 489 (Bad Event) response.

5.4.8.7 Notification of the registration state

When the user performs an emergency registration or when the emergency registration expires, the S-CSCF shall not send a NOTIFY request to the subscribers to the reg event package of the respective user.

The contact address that has been registered for emergency service shall not be included in the NOTIFY requests sent to the subscribers to the reg event package of the user.

5.5 Procedures at the MGCF

5.5.1 General

The MGCF, although acting as a UA, does not initiate any registration of its associated addresses. These are assumed to be known by peer-to-peer arrangements within the IM CN subsystem. Therefore table A.4/1 and dependencies on that major capability shall not apply.

The use of the Path and Service-Route header fields shall not be supported by the MGCF.

For all SIP transactions identified:

- if priority is supported, as containing an authorised Resource-Priority header field, or, if such an option is supported, relating to a dialog which previously contained an authorised Resource-Priority header field;

the MGCF shall give priority over other transactions or dialogs. This allows special treatment of such transactions or dialogs. If priority is supported, the MGCF shall adjust the priority treatment of transactions or dialogs according to the most recently received authorized Resource-Priority header field or backwards indication value.

NOTE: The special treatment can include filtering, higher priority processing, routeing, call gapping. The exact meaning of priority is not defined further in this document, but is left to national regulation and network configuration.

When the MGCF sends any request or response related to a dialog, the MGCF may insert previously saved values into P-Charging-Vector and P-Charging-Function-Addresses header fields before sending the message.

The MGCF shall use a GRUU referring to itself (as specified in RFC 5627 [93]) when inserting a contact address in a dialog establishing or target refreshing SIP message. This specification does not define how GRUUs are created by the MGCF; they can be provisioned by the operator or obtained by any other mechanism. A GRUU used by the MGCF when establishing a dialog shall remain valid for the lifetime of the dialog. The GRUU used by the MGCF shall not reveal calling party related information.

The MGCF shall handle requests addressed to its currently valid GRUUs when received outside of the dialog in which the GRUU was provided.

EXAMPLE: Upon receipt of an INVITE request addressed to a GRUU assigned to a dialog it has active, and containing a Replaces header field referencing that dialog, the MGCF will be able to establish the new call replacing the old one.

The MGCF may support retrieval of NP data, subject to local policy. The interface used at the MGCF to retrieve the NP data is out of scope of this specification. Retrieval of NP data is relevant only if the Request-URI contains an international public telecommunications number. For requests from the IM CN subsystem network, if the Request-URI contains a tel-URI with an "npdi" tel-URI parameter, as defined in RFC 4694 [112], NP data has been obtained previously and NP data retrieval is not needed, but still may still be performed if required by local policy. If NP data is retrieved by the MGCF, and the request is routed to the IM CN subsystem, the MGCF shall add the tel-URI NP parameters to the Request-URI as defined in RFC 4694 [112]: an "npdi" tel-URI parameter is added to indicate that NP data retrieval has been performed, and if the number is ported, an "rn" tel-URI parameter is added to identify the ported-to routeing number.

The MGCF NP procedures also apply when the request contains a Request-URI in the form of a SIP URI user=phone, where the "npdi" and "rn" tel-URI parameters are contained in the userinfo part of the SIP URI.

The MGCF supports as a network option the inclusion of the XML MIME schema for PSTN. In cases where the XML MIME for PSTN is included the Content-Type header field is set to "application/vnd.etsi.pstn+xml" and the Content-Disposition to "signal" with the "handling" parameter set to "optional".

The MGCF shall log all SIP requests and responses that contain a "logme" header field parameter in the SIP Session-ID header field if required by local policy.

When sending a failure response to any received request, depending on operator policy, the MGCF may insert a Response-Source header field with an "fe" header field parameter constructed with the URN namespace "urn:3gpp:fe", the fe-id part of the URN set to "mgcf" and optionally an appropriate fe-param part of the URN set in accordance with subclause 7.2.17.

5.5.2 Subscription and notification

Void.

5.5.3 Call initiation

5.5.3.1 Initial INVITE

5.5.3.1.1 Calls originated from circuit-switched networks

When the MGCF receives an indication of an incoming call from a circuit-switched network, the MGCF shall:

1) generate an INVITE request:

   - set the Request-URI to the "tel" format using an E.164 address or to the "sip" format using an E164 address in the user portion and set user=phone in accordance with 3GPP TS 29.163 [11B];

NOTE 1: Details how to set the host portion are out of scope of the document. However, when a SIP URI is used the host portion needs to be part of the domain name space owned by the I-CSCF

   - include the "100rel" option tag in the Supported header field (as defined in RFC 3262 [27]);

   - include the "precondition" option tag in the Supported header field (as defined in RFC 3312 [30] as updated by RFC 4032 [64]) if the MGCF supports the SIP preconditions mechanism;

   - not indicate the requirement for the precondition mechanism by using the Require header field;

   - create a new, globally unique value for the "icid-value" header field parameter and insert it into the P-Charging-Vector header field;

   - insert a type 2 "orig-ioi" header field parameter into the P-Charging-Vector header field. The MGCF shall set the type 2 "orig-ioi" header field parameter to a value that identifies the sending network in which the MGCF resides and the type 2 "term-ioi" header field parameter shall not be included;

   - based on local policy, add an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier; and

   - if services that require knowledge of the adjacent network are provided within the network, based on operator policy, insert a Via "received-realm" header field parameter, as defined in RFC 8055 [208];

When the MGCF receives a 1xx or 2xx response to an initial request for a dialog, the MGCF shall store the value of the received "term-ioi" header field parameter received in the P-Charging-Vector header field, if present.

NOTE 2: Any received "term-ioi" header field parameter will be a type 2 IOI. The type 2 IOI identifies the sending network of the response message.

Upon receiving a 199 (Early Dialog Terminated) provisional response to an established early dialog the MGCF shall release resources specifically related to that early dialog.

Based upon local policy, the MGCF may support preferred circuit carrier access (RFC 4694 [112]). If such routeing is applicable for the call, the MGCF shall perform the interworking of the carrier identification code from the circuit switched signalling protocol as described in 3GPP TS 29.163 [11B].

If resource priority in accordance with RFC 4412 [116] is required for a dialog, then the MGCF shall include the Resource-Priority header field in all requests associated with that dialog.

If overlap signalling using the multiple-INVITE method is supported as a network option, several INVITE requests with the same Call ID and the same From header field (including "tag" header field parameter) that relate to the same call can be sent by the MGCF. The MGCF shall route those INVITE requests to the same next hop.

5.5.3.1.2 Calls terminating in circuit-switched networks

When the MGCF receives an initial INVITE request with Supported header field indicating "100rel", the MGCF shall:

1) based on local policy, store the "fe-identifier" header field parameter of the P-Charging-Vector header field, if present;

2) store the value of the "orig-ioi" header field parameter received in the P-Charging-Vector header field, if present;

NOTE 1: Any received "orig-ioi" header field parameter will be a type 2 IOI. The type 2 IOI identifies the sending network of the request message.

3) send a 100 (Trying) response;

4) after a matching codec is found or no codec is required at the MGW, send 183 "Session Progress" response:

- set the Require header field to the value of "100rel";
- store the values received in the P-Charging-Function-Addresses header field;
- store the value of the "icid-value" header field parameter received in the P-Charging-Vector header field; and
- insert a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the initial INVITE request, a type 2 "term-ioi" header field parameter and the "icid-value" header field parameter. The MGCF shall set the type 2 "term-ioi" header field parameter to a value that identifies the network in which the MGCF resides, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the previously received value of "icid-value" header field parameter in the request. Based on local policy, the MGCF shall include the stored "fe-identifier" header field parameter in the P-Charging-Vector header field.

If a codec is required and the MGCF does not find an available matching codec at the MGW for the received initial INVITE request, the MGCF shall:

- send 503 (Service Unavailable) response if the type of codec was acceptable but none were available and the MGCF is unable to handle further requests received from the same upstream entity on the transport address where the INVITE request was received (i.e. MGCF is overloaded by SIP requests);
- send 500 (Server Internal Error) response if the type of codec was acceptable but none were available and the MGCF is able to handle further requests received from the same upstream entity on the transport address where the INVITE request was received (i.e. MGCF is not overloaded by SIP requests); or
- send 488 (Not Acceptable Here) response if the type of codec was not supported, and may include SDP in the message body to indicate the codecs supported by the MGCF/MGW.

Based upon local policy, the MGCF may support preferred ciruit carrier access (RFC 4694 [112]), if such routeing is applicable for the call.

NOTE 2: Interworking of the "cic" tel-URI parameter, if present in a tel-URI or in the userinfo part of a SIP URI with user=phone Request-URI, to the circuit switched signalling protocol is described in 3GPP TS 29.163 [11B].

The MGCF may support resource priority in accordance with RFC 4412 [116] if required for a dialog. The MGCF shall use compatible namespace and priority levels to the capabilities supported in the CS network.

Based on local policy, the MGCF shall include the stored "fe-identifier" header field parameter in the P-Charging-Vector header field, add its own address or identifier as "fe-addr" element of the "fe-identifier" header field parameter of the P-Charging-Vector header field and send the P-Charging-Vector header field in the related final response.

5.5.3.2 Subsequent requests

5.5.3.2.1 Calls originating in circuit-switched networks

When the MGCF generate a subsequent request in accordance with 3GPP TS 29.163 [11B], the MGCF shall:

a) add a P-Charging-Vector header field with the "icid-value" header field parameter set to the value populated in the initial request for the dialog and a type 2 "orig-ioi" header field parameter. The MGCF shall set the type 2 "orig-ioi" header field parameter to a value that identifies the sending network in which the MGCF resides and shall not set the type 2 "term-ioi" header field parameter.

When the MGCF receives a 1xx or 2xx response to a subsequent request for a dialog, the MGCF shall store the value of the received "term-ioi" header field parameter in the P-Charging-Vector header field, if present.

When the MGCF receives 183 (Session Progress) response to an INVITE request, the MGCF shall:

- store the values received in the P-Charging-Function-Addresses header field.

The MGCF shall send an UPDATE request when the following conditions are fulfilled:

- conditions as specified in 3GPP TS 29.163 [11B]; and
- the MGCF receives 200 (OK) response to a PRACK request

NOTE: When the MGCF is confirming the successful resource reservation using an UPDATE request (or a PRACK request) and the MGCF receives a 180 (Ringing) response or a 200 (OK) response to the initial INVITE request before receiving a 200 (OK) response to the UPDATE request (or a 200 (OK) response to the PRACK request), the MGCF does not treat this as an error case and does not release the session.

5.5.3.2.2 Calls terminating in circuit-switched networks

When the MGCF receives a subsequent request, the MGCF shall:

1) store the value of the "orig-ioi" header field parameter received in the P-Charging-Vector header field, if present.

NOTE: Any received "orig-ioi" header field parameter will be a type 2 IOI. The type 2 IOI identifies the sending network of the request message.

When the MGCF generate a response to a subsequent request in accordance with 3GPP TS 29.163 [11B], the MGCF shall, insert a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the subsequent request, a type 2 "term-ioi" header field parameter and the "icid-value" header field parameter. The MGCF shall set:

1) the type 2 "term-ioi" header field parameter to a value that identifies the network in which the MGCF resides;
2) the "orig-ioi" header field parameter set to the previously received value of "orig-ioi" header field parameter in the subsequent request; and
3) the "icid-value" header field parameter set to the previously received value of "icid-value" header field parameter in the subsequent request.

When the MGCF receives an indication of a ringing for the called party of outgoing call to a circuit-switched network, the MGCF shall:

- send 180 (Ringing) response to the UE.

When the MGCF receives an indication of answer for the called party of outgoing call to a circuit-switched network, the MGCF shall:

- send 200 (OK) response to the UE. The 200 (OK) response shall include an P-Asserted-Identity header field if corresponding information is received from the circuit-switched network.

5.5.4 Call release

5.5.4.1 Call release initiated by a circuit-switched network

When the MGCF receives an indication of call release from a circuit-switched network, the MGCF shall:

- send a BYE request to the UE.

5.5.4.2 IM CN subsystem initiated call release

NOTE: The release of a call towards the circuit-switched network additionally requires signalling procedures other than SIP in the MGCF that are outside the scope of this document.

5.5.4.3 MGW-initiated call release

When the MGCF receives an indication from the MGW that the bearer was lost, the MGCF shall:

- send a BYE request towards the UE; and
- may include Error-Info header field with a pointer to additional information indicating that bearer was lost.

5.5.5 Call-related requests

5.5.5.1 Session modification

5.5.5.1.0 General

This subclause applies after the 2xx response to the initial INVITE request has been sent or received.

5.5.5.1.1 Session modifications originating from circuit-switched networks

If the precondition mechanism was used during the session establishment, as described in subclause 5.5.3.1.1 or 5.5.3.1.2, the MGCF shall indicate support of the precondition mechanism during a session modification. If the precondition mechanism was not used during the session establishment, the MGCF shall not indicate support of the precondition mechanism during a session modification.

In order to indicate support of the precondition mechanism during a session modification, upon generating a reINVITE request, an UPDATE request with an SDP body, or a PRACK request with an SDP body, the MGCF shall:

a) indicate the support for the precondition mechanism using the Supported header field;

b) not indicate the requirement for the precondition mechanism using the Require header field; and

c) if a reINVITE request is being generated, indicate the support for reliable provisional responses using the Supported header field,

and follow the SDP procedures in clause 6 for the precondition mechanism.

5.5.5.1.2 Session modifications terminating in circuit-switched networks

When the MGCF receives a reINVITE request for hold/resume operation, the MGCF shall:

- send a 100 (Trying) response;
- after performing interaction with MGW to hold/resume the media flow, send a 200 (OK) response.

Upon receiving a reINVITE request, an UPDATE request, or a PRACK request that indicates support for the precondition mechanism by using the Supported header field or requires use of the precondition mechanism by using the Require header field, the MGCF shall:

a) if the precondition mechanism was used during the session establishment, as described in subclause 5.5.3.1.1 or 5.5.3.1.2, use the precondition mechanism for the session modification; and

b) if the precondition mechanism was not used during the session establishment, and

   1) if use of the precondition mechanism is required using the Require header field, reject the request by sending a 420 (Bad Extension) response; and

   2) if the support of the precondition mechanism is indicated using the Supported header field, not use the precondition mechanism for the session modification.

If the precondition mechanism is used for the session modification, the MGCF shall indicate support for the preconditions mechanism, using the Require header field, in responses that include an SDP body, to the session modification request.

5.5.6 Further initial requests

When the MGCF responds to an OPTIONS request with a 200 (OK) response, the MGCF may include a message body with an indication of the DTMF capabilities and supported codecs of the MGCF/MGW.

NOTE: The detailed interface for requesting MGCF/MGW capabilities is not specified in this version of the document. Other solutions can be used in the interim.

5.6 Procedures at the BGCF

5.6.1 General

The use of the Path and Service-Route header fields shall not be supported by the BGCF.

For all SIP transactions identified:

- if priority is supported, as containing an authorised Resource-Priority header field, or, if such an option is supported, relating to a dialog which previously contained an authorised Resource-Priority header field;

the BGCF shall give priority over other transactions or dialogs. This allows special treatment of such transactions or dialogs. If priority is supported, the MGCF shall adjust the priority treatment of transactions or dialogs according to the most recently received authorized Resource-Priority header field or backwards indication value.

NOTE: The special treatment can include filtering, higher priority processing, routeing, call gapping. The exact meaning of priority is not defined further in this document, but is left to national regulation and network configuration.

When the BGCF receives any request or response related to a dialog or standalone transaction, the BGCF may insert previously saved values into a P-Charging-Vector header field before forwarding the message.

When the BGCF receives any request or response (excluding ACK requests and CANCEL requests and responses) related to a dialog or standalone transaction, the BGCF may insert previously saved values into a P-Charging-Function-Addresses header field before forwarding the message.

With the exception of 305 (Use Proxy) responses, the BGCF may recurse on a 3xx response only when the domain part of the URI contained in the 3xx response is in the same domain as the BGCF. For the same cases, if the URI is an IP address, the BGCF shall only recurse if the IP address is known locally to be a address that represents the same domain as the BGCF.

The BGCF shall log all SIP requests and responses that contain a "logme" header field parameter in the SIP Session-ID header field if required by local policy.

When sending a failure response to any received request, depending on operator policy, the BGCF may insert a Response-Source header field with an "fe" header field parameter constructed with the URN namespace "urn:3gpp:fe", the fe-id part of the URN set to "bgcf" and optionally an appropriate fe-param part of the URN set in accordance with subclause 7.2.17.

5.6.2 Common BGCF procedures

When determining where to route the received request, the originating BGCF may use the information obtained from other protocols or any other available databases.

The BGCF may support retrieval of NP data as part of the procedures to determine where to route the request. Retrieval of NP data by the BGCF is subject to local policy. Retrieval of NP data is relevant only if the Request-URI contains an international public telecommunications number. The interface used at the BGCF to retrieve the NP data is out of scope of this specification. If the Request-URI contains a tel-URI with an "npdi" tel-URI parameter, as defined in RFC 4694 [112], NP data has been obtained previously and NP data retrieval is only performed if required by local policy. If NP data is retrieved by the BGCF, the BGCF shall add the tel-URI NP parameters to the Request-URI as defined in RFC 4694 [112]: an "npdi" tel-URI parameter is added to indicate that NP data retrieval has been performed, and if the number is ported, an "rn" tel-URI parameter is added to identify the ported-to routeing number. The "rn" tel-URI parameter may be used by the BGCF for routeing the request.

The BGCF NP procedures also apply when the request contains a Request-URI in the form of a SIP URI user=phone, where the "npdi" and "rn" tel-URI parameters are contained in the userinfo part of the SIP URI.

When the BGCF receives a request, the BGCF shall forward the request:

- to an MGCF within its own network; or
- to another network containing a BGCF, or I-CSCF; or
- where the request is for another network, to an IBCF in its own network, if local policy requires IBCF capabilities towards another network; or
- where the Ici interface is used to interconnect two networks and the destination network is beyond such interface, to an IBCF in its own network.

When forwarding the request to the next hop, the BGCF may leave the received Request-URI unmodified.

If the request is not routed to a BGCF or to an entity that implements the additional routeing functionality, the BGCF shall remove the P-Served-User header field prior to forwarding the request.

When the BGCF receives a request and the Request-URI contains a tel URI in local number format or a SIP URI with the user part not starting with a + and the "user" SIP URI parameter equals "phone", the BGCF shall not forward the request to an entity in another network (e.g. BGCF, I-CSCF) unless the local policy (e.g. routeing of service numbers) requires forwarding the request outside the network. If local policy does not allow forwarding the request outside the network and additional routeing capabilities as defined in annex I are locally available, the BGCF shall attempt translation of the local number. If the translation fails, the BGCF shall send an appropriate SIP response to the originator. If local policy does not allow forwarding the request outside the network and additional routeing capabilities as defined in annex I are not locally available, the BGCF shall either:

- forward the request to any appropriate entity in its own network where additional routeing functionality are available; or
- send an appropriate SIP response to the originator.

The BGCF need not Record-Route the INVITE and the SUBSCRIBE requests. While the next entity may be a MGCF acting as a UA, the BGCF shall not apply the procedures of RFC 3323 [33] relating to privacy. The BGCF shall store the values received in the P-Charging-Function-Addresses header field. The BGCF shall store the value of the "icid-value" header field parameter received in the P-Charging-Vector header field and retain the "icid-value" header field parameter in the P-Charging-Vector header field.

NOTE 1: The means by which the decision is made to forward to an MGCF or to another network is outside the scope of the present document, but can be by means of a lookup to an external database, or can be by data held internally to the BGCF.

If the BGCF supports carrier routeing, then the BGCF shall support the following procedures, based on local policy:

a) if the BGCF is configured to populate an operator configured preassigned carrier into a tel-URI contained in the Request-URI, and a preassigned carrier is required for this call, then the BGCF shall include the "cic" tel-URI parameter in the Request-URI identifying the preassigned carrier (as described in RFC 4694 [112]); or

b) if the BGCF is configured to populate the freephone carrier ID, and a freephone carrier is required for this call, then the BGCF shall include the "cic" tel-URI parameter in the Request-URI identifying the freephone carrier (as described in RFC 4694 [112]).

The BGCF carrier routeing procedures also apply when the Request-URI is in the form of a SIP URI user=phone, where the "cic" tel-URI parameter is contained in the userinfo part of the SIP URI.

The BGCF shall not add the "cic" tel-URI parameter in the Request-URI if the parameter already exists in the tel-URI.

NOTE 2: Local policy should be able to control the interaction and precedence between routeing on "cic" parameter versus routeing based on "rn" parameter.

NOTE 3: The means to configure the BGCF with the pre-assigned carrier is outside the scope of this document.

If

a) the BGCF supports indicating the traffic leg as specified in RFC 7549 [225];

b) an "iotl" SIP URI parameter is not already included in the Request-URI; and

NOTE 4: If an "iotl" SIP URI parameter is included it contains the value "visitedA-homeB" inserted by the TRF in the roaming architecture for voice over IMS with local breakout scenario.

c) required by local policy;

the BGCF shall before forwarding the request:

a) if the Request-URI contains a SIP URI, append the "iotl" SIP URI parameter set to "homeA-homeB" to the Request-URI; and

b) if the Request-URI contains a tel URI that can be converted to a SIP URI by the BGCF:

- convert the tel URI in the Request-URI to the form of a SIP URI with user=phone; and
- append an "iotl" SIP URI parameter with a value set to "homeA-homeB" in the Request-URI.

NOTE 5: If the "iotl" SIP URI parameter can not be included by the above procedure, the upstream nodes have to determine the II-NNI traversal scenario by analysing the content of the SIP request (implementation dependent) or using the default II-NNI traversal scenario type.

5.6.3 Specific procedures for INVITE requests and responses

When the BGCF receives an INVITE request that contains a Feature-Caps header field with the "+g.3gpp.home-visited" header field parameter, the BGCF shall decide based on local policy whether to perform loopback routeing for this request. The BGCF shall:

a) if loopback routeing is not to be performed for this request remove any "+g.3gpp.trf" or "+g.3gpp.home-visited" header field parameter from the Feature-Caps header field of the outgoing request;

b) if loopback routeing is applied for this request:

i) remove all entries in the Route header field;

ii) if a "+g.3gpp.trf" header field parameter with a parameter value containing a valid URI, is included in the Feature-Caps header field of the request, insert the URI in a Route header field;

iii) if a "+g.3gpp.trf" header field parameter, with a parameter value containing a valid URI is not included in the Feature-Caps header field of the request, insert a locally configured TRF address, associated with the visited network for this call (as identified in the "+g.3gpp.home-visited" header field parameter), in the Route header field;

iv) remove any "+g.3gpp.home-visited" header field parameter from the Feature-Caps header field of the outgoing request;

v) if included in the incoming request, remove the "+g.3gpp.trf" header field parameter from the Feature-Caps header field from the outgoing request;

vi) insert the "+g.3gpp.loopback" header field parameter, as specified in subclause 7.9A.4 in the Feature-Caps header field of the request, in accordance with RFC 6809 [190]. If providing the identifier of the home network is supported by the BGCF and the visited network, the BGCF may based on operator agreement insert the "+g.3gpp.loopback" header field parameter set to the identifier of the home network;

vii) remove the "orig-ioi" header field parameter received in the P-Charging-Vector header field, if present. The BGCF shall insert a type 1 "orig-ioi" header field parameter into the P-Charging-Vector header field and shall set the type 1 "orig-ioi" header field parameter to a value that identifies the home network of the served user (i.e. the network in which the BGCF resides). The BGCF shall not include the "term-ioi" header field parameter; and

viii) if the BGCF supports indicating the traffic leg associated with a URI as specified in RFC 7549 [225] and if an "iotl" SIP URI parameter is not included in the TRF URI in the Route header field, the BGCF if

required by local policy, append an "iotl" URI parameter with a value set to "homeA-visitedA" to the URI in the Route header field; and

c) if the final decision on loopback routeing is deferred to a subsequent entity in the home network, a further BGCF, then, retain in the request a Feature-Caps header field with the "+g.3gpp.home-visited" header field parameter with the parameter value set to the identifier of the visited network. The BGCF is expected to know by means of network configuration that such a subsequent entity exists;

If the BGCF inserts its own Record-Route header field, the BGCF may require the periodic refreshment of the session to avoid hung states in the BGCF. If the BGCF requires the session to be refreshed, the BGCF shall apply the procedures described in RFC 4028 [58] clause 8.

NOTE: Requesting the session to be refreshed requires support by at least one of the UEs. This functionality cannot automatically be granted, i.e. at least one of the involved UEs needs to support it.

If overlap signalling using the multiple-INVITE method is supported as a network option, several INVITE requests with the same Call ID and the same From header field (including "tag" header field parameter) can be received outside of an existing dialog. Such INVITE requests relate to the same call and the BGCF shall route such INVITE request received during a certain period of time to the same next hop.

If the BGCF inserted in the initial request for the dialog the header field parameters into the Feature-Caps header field then the BGCF shall include the header field parameters with the same parameter values into the Feature-Caps header field in any target refresh request for the dialog, and in each 1xx or 2xx response to target refresh request sent in the same direction.

Based on local policy, the BGCF shall add an "fe-addr" element of the "fe-identifier" header field parameter of the P-Charging-Vector header field with its own address or identifier to an initial request.

5.6.4 Specific procedures for subsequent requests and responses

When the BGCF receives a subsequent request whose initial request applied loopback routeing, the BGCF shall:

a) remove the "orig-ioi" header field parameter received in the P-Charging-Vector header field, if present. The BGCF shall insert a type 1 "orig-ioi" header field parameter into the P-Charging-Vector header field and shall set the type 1 "orig-ioi" header field parameter to a value that identifies the home network of the served user (i.e. the network in which the BGCF resides). The BGCF shall not include the "term-ioi" header field parameter.

5.7 Procedures at the Application Server (AS)

5.7.1 Common Application Server (AS) procedures

5.7.1.0 General

When sending a failure response to any received request, depending on operator policy, the AS may insert a Response-Source header field with an "fe" header field parameter constructed with the URN namespace "urn:3gpp:fe", the fe-id part of the URN set to "as" and optionally an appropriate fe-param part of the URN set in accordance with subclause 7.2.17. An AS when sending a failure response will add in the URN the "role" header field parameter set to the corresponding AS role listed in subclause 7.2.17.

5.7.1.1 Notification about registration status

The AS may support the REGISTER method in order to discover the registration status of the user. If a REGISTER request arrives and the AS supports the REGISTER method, the AS shall store the registration expiration interval value from the request and generate a 200 (OK) response or an appropriate failure response. For the success case, the 200 (OK) response shall contain a registration expiration interval value equal to the value received in the REGISTER request. The AS shall store the values received in P-Charging-Function-Addresses header field. Also, the AS shall store the values of the "icid-value" header field parameter and "orig-ioi" header field parameter if present in the P-Charging-Vector header field from the REGISTER request.

NOTE 1: The user can have one or more contacts registered after a 3rd party REGISTER request with an Expires header field set to a value "0" has been received. If an AS needs more detailed knowledge of the user registration status, the AS can subscribe to the reg event package.

If a Contact header field is included in the REGISTER request including a "+g.3gpp.registration-token" header field parameter as defined in subclause 7.9.7, the AS supporting this feature shall store the value of the "+g.3gpp.registration-token" header field parameter.

NOTE 2: The S-CSCF can set this token to the same value as used in the "id" parameter identifying the contact in the "reg" event package, allowing the AS to retrieve the value from the "reg" event package. The AS can know by configuration or other means if the S-CSCF uses this value.

The AS shall insert a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the REGISTER request, a type 3 "term-ioi" header field parameter in the response to REGISTER and the "icid-value" header field parameter. The AS shall set the type 3 "term-ioi" header field parameter to a value that identifies the service provider from which the response is sent, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the previously received value of "icid-value" header field parameter in the request.

Upon receipt of a third-party REGISTER request, with the Content-Type header field or with a MIME body part's Content-Type header field set according to subclause 7.6 (i.e. "application/3gpp-ims+xml"), independent of the value or presence of the Content-Disposition header field or a MIME body part's Content-Type header field, independent of the value or presence of Content-Disposition parameters or MIME body part's Content-Disposition parameters, then the following treatment is applied:

- if the third-party REGISTER request includes an IM CN subsystem XML body with an <ims-3gpp> element, including a version attribute, with the <service-info> child element or a MIME body part containing an <ims-3gpp> element with a <service-info> XML child element as described in subclause 7.6, then the AS may retrieve the service information within the <service-info> XML child element of the <ims-3gpp> element.

Upon receipt of a third-party REGISTER request, with the Content-Type header field or with a body part's Content-Type header field set to "message/sip" and including a "message/sip" MIME body of the incoming REGISTER request, or the 200 (OK) response to the incoming REGISTER request then the AS may retrieve information from the "message/sip" MIME body or body part.

Upon receipt of a third-party REGISTER request, the AS may subscribe to the reg event package for the public user identity registered at the user's registrar (S-CSCF) as described in RFC 3680 [43] and RFC 6665 [28].

On sending a SUBSCRIBE request, the AS shall populate the header fields as follows:

a) a Request-URI set to the resource to which the AS wants to be subscribed to, i.e. to a SIP URI that contains the public user identity of the user that was received in the To header field of the third-party REGISTER request;

b) a From header field set to the AS's SIP URI;

c) a To header field, set to a SIP URI that contains the public user identity of the user that was received in the To header field of the third-party REGISTER request;

d) an Event header field set to the "reg" event package;

e) a P-Asserted-Identity header field set to the SIP URI of the AS; and

NOTE 3: The S-CSCF expects the SIP URI used in the P-Asserted-Identity header field to correspond to the SIP URI, which identified this AS in the initial filter criteria of the user to whose registration state the AS subscribes to.

f) a P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17] and a type 3 "orig-ioi" header field parameter. The type 3 "orig-ioi" header field parameter identifies the service provider from which the request is sent. The AS shall not include the type 3 "term-ioi" header field parameter.

Upon receipt of a dialog establishing NOTIFY request, as specified in RFC 6665 [28], associated with the SUBSCRIBE request, the AS shall:

1) store the information for the so established dialog;

2) store the expiration time as indicated in the "expires" header field parameter of the Subscription-State header field, if present, of the NOTIFY request. Otherwise the expiration time is retrieved from the Expires header field of the 2xx response to SUBSCRIBE request; and

3) follow the procedures specified in RFC 6665 [28].

Upon receipt of any response, the AS shall store the value of the "term-ioi" header field parameter received in the P-Charging-Vector header field if present.

NOTE 4: Any received term-ioi parameter will be a type 3 term-ioi. The type 3 term-ioi identifies the network operator from which the response was sent.

NOTE 5: Upon receipt of a NOTIFY request with all <registration> element(s) having their state attribute set to "terminated" (i.e. all public user identities are deregistered) and the Subscription-State header field set to "terminated", the AS considers the subscription to the reg event package terminated, i.e. as if the AS had sent a SUBSCRIBE request with an Expires header field containing a value of zero.

Upon receipt of a NOTIFY request for the dialog associated with the subscription to the reg event package, the AS shall:

- store the information for the established dialog;

- store the expiration time as indicated in the "expires" header field parameter of the Subscription-State header field, if present, of the NOTIFY request. Otherwise the expiration time is retrieved from the Expires header field of the 2xx response to SUBSCRIBE request;

- store the value of the "orig-ioi" header field parameters if present in the P-Charging-Vector header field. The AS shall insert a P-Charging-Vector header field in the response to the NOTIFY request containing the "orig-ioi" header field parameter, if received in the NOTIFY request, a type 3 "term-ioi" header field and the "icid-value" header field parameter;

- set the type 3 "term-ioi" header field parameter to a value that identifies the service provider from which the response is sent, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the previously received value of "icid-value" header field parameter in the request; and

- follow the procedures specified in RFC 6665 [28].

5.7.1.2 Extracting charging correlation information

When an AS receives an initial request for a dialog or a request for a standalone transaction, the AS shall store the values received in the P-Charging-Vector header field, e.g. "orig-ioi" header field parameter, if present, and "icid-value" header field parameter, and retain the P-Charging-Vector header field in the message. The AS shall store the values received in the P-Charging-Function-Addresses header field and retain the P-Charging-Function-Addresses header field in the message.

When an AS sends any request or response related to a dialog or standalone transaction, the AS shall insert previously saved values into the P-Charging-Vector header field and may insert previously saved values into the P-Charging-Function-Addresses header field before sending the message.

5.7.1.3 Access-Network-Info and Visited-Network-ID

The AS may receive information about the served user core network in REGISTER requests from S-CSCF. This information can be obtained either from the P-Access-Network-Info header field and P-Visited-Network-ID header field in the REGISTER request or can be obtained from those header fields in the body of the REGISTER request.

The AS may also receive information about the served user access network in other requests (excluding CANCEL requests and responses). This information can be obtained from the P-Access-Network-Info header field.

The AS can use the P-Access-Network-Info and P-Visited-Network-ID header fields to provide an appropriate service to the user.

5.7.1.3A Determination of the served user

5.7.1.3A.1 General

The determination of the served user is different per session:

- for an originating session, the procedure is described in subclause 5.7.1.3A.2; and
- for a terminating session the procedure is described in subclause 5.7.1.3A.3.

If the AS supports the P-Served-User header field as defined in RFC 5502 [133] and RFC 8498 [239] and the P-Served-User header field is included in the received request, the AS can determine the session case related to the request, as specified in 3GPP TS 29.228 [14], from the P-Served-User header field parameters if available.

5.7.1.3A.2 AS serving an originating user

If an AS receives a request on behalf of an originating user:

- and the AS supports the P-Served-User header field as defined in RFC 5502 [133], the AS shall determine the served user by taking the identity contained in the P-Served-User header field if available;
- otherwise, if the AS supports the History-Info header field as defined in RFC 7044 [66] the AS shall determine the served user from the content of the History-Info header field if available; and
- otherwise, the AS shall determine the served user by taking the identity contained in P-Asserted-Identity header field.

5.7.1.3A.3 AS serving a terminating user

If an AS receives a request on behalf of a terminating user:

- and the AS supports the P-Served-User header field as defined in RFC 5502 [133], the AS shall determine the served user by taking the identity contained in the P-Served-User header field if available;
- otherwise, if the AS supports the History-Info header field as defined in RFC 7044 [66] the AS shall determine the served user from the content of the History-Info header field if available; and
- otherwise, the AS shall determine the served user from the content of the Request-URI.

5.7.1.3B Determination of the used registration

A prerequisite for the procedure in this subclause is that a REGISTER request has been received including a Contact header field with a "+g.3gpp.registration-token" header field parameter and that the AS supports using this token to identify the registration.

When receiving an initial request for a dialog or a request for a standalone transaction, or a response to such request the AS shall if a "+g.3gpp.registration-token" header field parameter as defined in subclause 7.9A.8 is included in a Feature-Caps header field use this value to identify the registration used for this initial request for a dialog or this request for a standalone transaction or a response to such request by comparing it to the value of the "+g.3gpp.registration-token" Contact header field parameter stored when the user registered.

NOTE: The Include Register Request or Include Register Response indication in the initial Filter Criteria can be used to provide the incoming REGISTER request or 200 (OK) response to the incoming REGISTER request containing the instance ID to the AS. The AS can use the mechanism in this subclause to determine the instance ID for subsequent requests and responses.

If the AS routes the originating request to another entity than the S-CSCF, the AS shall remove the "+g.3gpp.registration-token" header field parameter from the Feature-Caps header field before forwarding the request.

5.7.1.4 User identity verification at the AS

The procedures at the AS to accomplish user identity verification are described with the help of figure 5-1. Procedures for user identity verification using the Identity header field are specified in subclause 5.7.1.25.

NOTE: Different means can be used to represent or transport the credentials. Such mechanisms are subject to operator policy and can e.g. include the P-Asserted-Identity header field, the Authorization header field or other mechanisms not specified by 3GPP TS 24.229.

When the AS receives a SIP initial or standalone request, excluding REGISTER request, that does not contain credentials, the AS shall:

a) if a Privacy header field is present in the initial or standalone request and the Privacy header field value is set to "id" or "user", then the user and the request are considered as anonymous, and no further actions are required. The AS shall consider the request as authenticated;

b) if there is no Privacy header field present in the initial or standalone request, or if the Privacy header field contains a value other than "id" or "user", then the AS shall check for the presence of a P-Asserted-Identity header field in the initial or standalone request. Two cases exist:

   i) the initial or standalone request contains a P-Asserted-Identity header field. This is typically the case when the user is located inside a trusted domain as defined by subclause 4.4. In this case, the AS is aware of the identity of the user and no extra actions are needed. The AS shall consider the request as authenticated; and

   ii) the initial or standalone request does not contain a P-Asserted-Identity header field. This is typically the case when the user is located outside a trusted domain as defined by subclause 4.4. In this case, the AS does not have a verified identity of the user. The AS shall check the From header field of the initial or standalone request. If the From header field value in the initial or standalone request is set to "Anonymous" as specified in RFC 3261 [26], then the user and the request are considered as anonymous and no further actions are required. If the From header field value does not indicate anonymity, then the AS shall challenge the user by issuing a 401 (Unauthorized) response including a challenge as per procedures described in RFC 3261 [26].

When the AS receives a SIP initial or standalone request that contains credentials but it does not contain a P-Asserted-Identity header field the AS shall check the correctness of the credentials as follows:

a) If the credentials are correct, then the AS shall consider the identity of the user verified, and the AS shall consider the request as authenticated;

b) If the credentials are not correct, the AS may either rechallenge the user by issuing a 401 (Unauthorized) response including a challenge as per procedures described in RFC 3261 [26] (up to a predetermined maximum number of times predefined in the AS configuration data), or consider the user as anonymous. If the user is considered anonymous, the AS shall consider the request as authenticated.

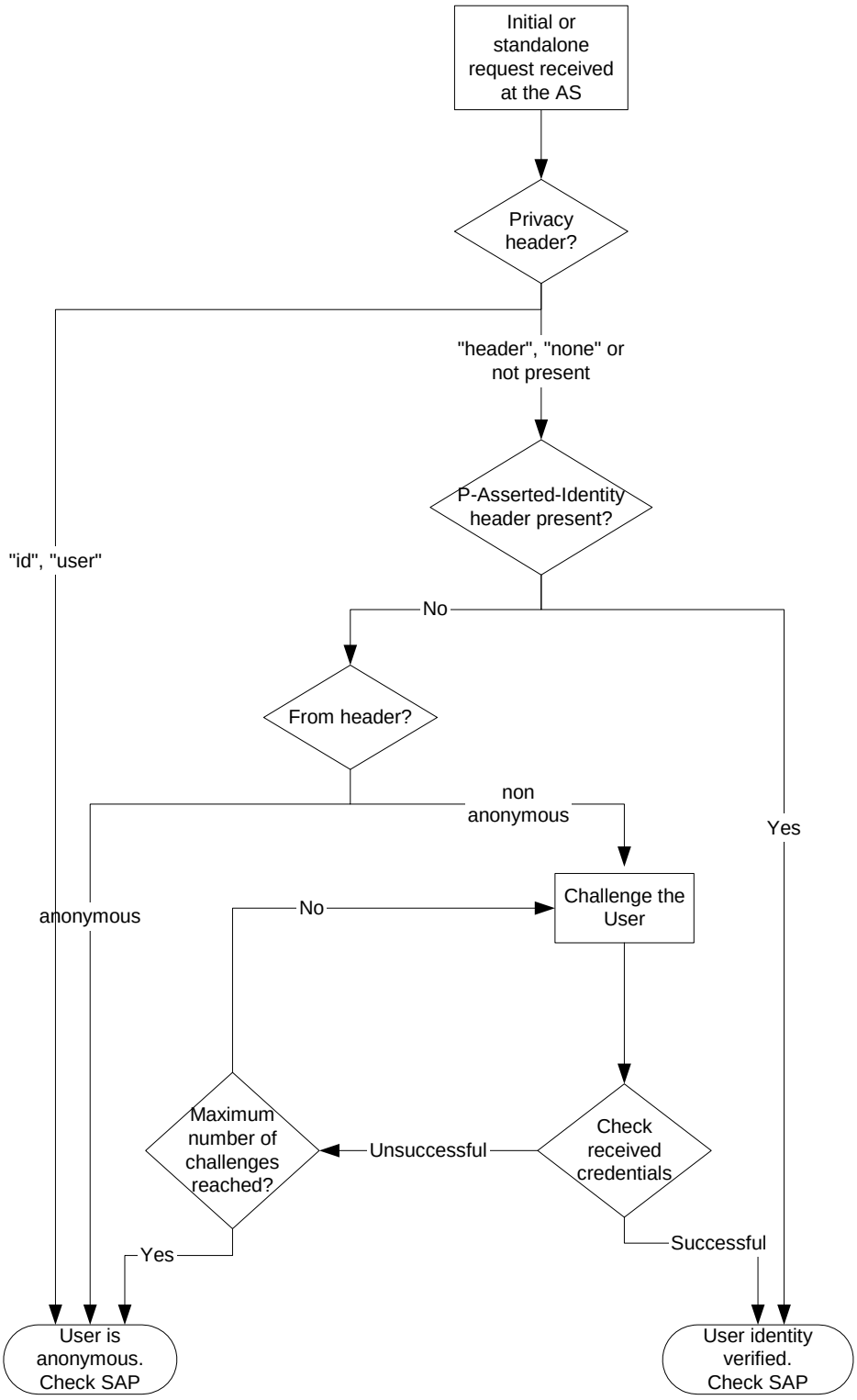


Figure 5-1: User identity verification flow at the AS

5.7.1.5 Request authorization

Once the AS have tried to verify the identity of the user, the AS either has a verified identity of the user or it considers the user as anonymous.

If the user is considered anonymous, the AS shall check whether the authorization policy defined for this request allows anonymous requests. If anonymous requests are allowed, then the AS can proceed with the requested functionality, otherwise, the AS shall not proceed with the requested functionality.

If the user is identified by an identity, the AS shall apply the authorization policy related to the requested functionality to detect whether the particular user is allowed to request the functionality. The authorization policy may require a verified identity of a user.

If the request is authorized then the AS shall continue with the procedures as defined for that request.

If the request is not authorized, the AS shall either:

- reject the request according to the procedures defined for that request e.g., by issuing a 403 (Forbidden) response; or
- send a 2xx final response if the authorization policy requires to deny the requested functionality, whilst appearing to the user as if the request has been granted.

5.7.1.6 Event notification throttling

If the AS has a local configuration information limiting the rate at which notification generation is allowed, then the AS shall take that information into account. Such local configuration information could be e.g. the shortest time period between issuing consecutive NOTIFY requests.

5.7.1.7 Local numbering

5.7.1.7.1 Interpretation of the numbers in a non-international format

If home operator's local policy defines a prefix string(s) to enable subscribers to differentiate dialling a geo-local number and/or a home-local number and if the phone number in a non-international format in the Request-URI includes such a prefix, the AS shall interpret the received number in a non-international format as a geo-local number or as a home-local number according to the prefix.

If the phone number in a non-international format in the Request-URI includes a "phone-context" tel URI parameter, the AS shall:

1) if the "phone-context" tel URI parameter contains access technology information or the home network domain name prefixed by the "geo-local." string, interpret it as a geo-local number;
2) if the "phone-context" tel URI parameter contains the home network domain name, interpret it as a home-local number; or
3) if the "phone-context" tel URI parameter contains any other value, apply general procedures for translation.

NOTE 1: If business communication services are provided to the calling user, and the "phone-context" tel URI parameter contains a value associated with a private numbers, it is expected that any needed translation of the number information is handled by the corresponding business communication AS.

If the phone number in a non-international format in the Request-URI includes both operator defined prefix and a "phone-context" tel URI parameter and those information are contradictory, the AS shall ignore either the prefix or the "phone-context" tel URI parameter according to operator policy.

If the phone number in a non-international format in the Request-URI does not include either a "phone-context" tel URI parameter or an operator defined prefix, the AS shall interpret the phone number in a non-international format either as a geo-local number or as a home-local number according to operator policy.

NOTE 2: Operator must ensure that service setting dialling strings do not reach local numbering AS by setting appropriately the precedences of the initial filter criteria.

5.7.1.7.2 Translation of the numbers in a non-international format

When an AS receives a request having a geo-local number in a non-international format in the Request-URI, the AS shall use the "phone-context" tel URI parameter to determine the visited access network, if the "phone-context" tel URI parameter in the Request-URI is available. If the "phone-context" tel URI parameter in the Request-URI is not available, the AS may determine the visited access network based on the available P-Access-Network-Info header fields containing the access-type field, if it is available in the received request, or by means outside the scope of this document.

If the visited access network is determined:

1) if the home network supports the roaming architecture for voice over IMS with local breakout, and an incoming INVITE request contains a Feature-Caps header field with the "+g.3gpp.home-visited" header field parameter, the AS may choose to not attempt translation of some geo-local numbers and defer the translation to the visited network after loopback has been performed; or

2) the AS shall attempt to determine whether the geo-local number:

   a) corresponds to a home local service number that can be used by a roaming user;

   b) is used to access a service in the visited network; or

   c) is used to access the local addressing plan of the visited network

- and translate the received geo-local number to a globally routeable SIP URI or an international tel URI:

NOTE 1: During the translation the AS can contact an entity in the visited access network for getting the needed information. The protocol and procedures for this is outside the scope of this specification.

NOTE 2: The AS can translate the tel URI to a SIP URI by including the 'telephone-subscriber' part of the received tel URI to the user part of the SIP URI and setting the domain name of the SIP URI to indicate the domain name of the network of the phone number based on the received "phone-context" tel URI parameter.

NOTE 3: In addition to the service numbers corresponding to a service in the visited network the home network can also support service numbers corresponding to services in the home network as geo-local service numbers.

When an AS receives a request having a home-local number in a non-international format in the Request-URI, the AS shall determine whether the home-local number is used to access a service or the local addressing plan and translate the received home-local number to a globally routeable SIP URI or an international tel URI:

When an AS receives a request having any other number in a non-international format in the Request-URI, the AS shall attempt to determine whether it is used to access a service in the third network or the local addressing plan of the third network and translate the received number in a non-international format to a globally routeable SIP URI or an international tel URI:

NOTE 4: The AS can translate the tel URI to a SIP URI by including the 'telephone-subscriber' part of the received tel URI to the user part of the SIP URI and setting the domain name of the SIP URI to indicate the domain name of the network of the phone number based on the received "phone-context" tel URI parameter;

NOTE 5: If business communication services are provided to the calling user, and the "phone-context" tel URI parameter contains a value associated with a private numbers, it is expected that any needed translation of the number information is handled by the corresponding business communication AS.

If the translation at the AS fails, the AS shall either send an appropriate SIP response or leave the Request-URI unmodified and route the request based on the topmost Route header field, based on local policy.

5.7.1.8 GRUU assignment and usage

It is possible for an AS to use a GRUU referring to itself when inserting a contact address in a dialog establishing or target refreshing SIP message.

This specification does not define how GRUUs are created by the AS; they can be provisioned by the operator or obtained by any other mechanism. The GRUU shall remain valid for the time period in which features addressed to it remain meaningful.

The AS shall handle requests addressed to its currently valid GRUUs when received outside of the dialog in which the GRUU was provided.

EXAMPLE: Upon receipt of an INVITE request addressed to a GRUU assigned to a dialog it has active, and containing a Replaces header field referencing that dialog, the AS will be able to establish the new call replacing the old one, if that is appropriate for the features being provided by the AS.

When an AS is acting as a routeing B2BUA (as defined in subclause 5.7.5) it may provide a contact address that is not a GRUU when the contact address in the incoming message that is being replaced is not a GRUU.

When an AS is acting as a routeing B2BUA forwards a SIP request it shall transparently forward a received Contact header field when the Contact header field contains a GRUU. When transparently forwarding a received Contact header field of a dialog-forming request, the AS shall include its own URI in a Record-Route header field in order to ensure that it is included on the route of subsequent requests.

When an AS acts as UA or Initiating B2BUA it shall use a GRUU as the contact address if the AS acts as a notifier per RFC 6665 [28] and RFC 7647 [231], otherwise the AS may provide a contact address that is not a GRUU in cases where it can ascertain that valid requests that could result from the use of that contact and follow the usage rules of RFC 5627 [93] will reach the element. In all other cases the AS shall use a GRUU.

An AS acting as a UA or an initiating B2BUA on behalf of a public user identity can provide a GRUU in the contact address referring to itself as described above. When the AS provides a GRUU on behalf of a user, subsequent dialog-initiating requests sent to that GRUU will be routed directly to the AS, thus bypassing terminating services assigned to the user. If the AS wishes to have terminating services applied for the user, the AS may generate a new terminating request addressed to a public GRUU associated with the public user identity of the user.

NOTE 1: If the AS wishes to have terminating services applied when the public user identity on whose behalf the AS is acting is unregistered, then the options available to the AS depend on whether or not the subscriber has ever previously registered with the IM CN subsystem. In the case where the public user identity had previously registered with the IM CN subsystem, then the AS can use the most recently allocated public GRUU if available. In the case where the user has never registered with the IM CN subsystem, then the AS can use the public user identity itself.

NOTE 2: Once terminating services have been applied, it is assumed that the terminating S-CSCF will route the request back to this AS via the initial filter criteria. In order for this to work, the initial filter criteria of the target user need to be configured so that the AS is invoked at the appropriate time relative to other terminating ASs (say, after the required terminating services have been applied). The mechanism to ensure that the AS is invoked by the initial filter criteria at the appropriate time is outside the scope of this specification (e.g. the user's filter criteria could be statically configured to invoke the AS at the correct time, or the AS could use the Dynamic Service Activation Information mechanism to activate the appropriate filter criteria).

When an AS acts as a UA or an initiating B2BUA, and is originating or terminating a request on behalf of a public user identity, and privacy is required, the AS shall ensure that any GRUU provided in the contact address in the request does not reveal the public user identity of the user.

5.7.1.9 Use of ICSI and IARI values

Based on service logic, an AS can validate an ICSI value received in an Accept-Contact header field or received in a P-Asserted-Service header field and reject the request if necessary.

A trusted AS may insert a P-Asserted-Service header field in a request for a new dialog or standalone transaction. An untrusted AS may insert a P-Preferred-Service header field in a request for a new dialog or standalone transaction. If the request is related to an IMS communication service that requires the use of an ICSI then the AS:

- shall include the ICSI value (coded as specified in subclause 7.2A.8.2), for the IMS communication service that is related to the request in either a P-Asserted-Service header field or a P-Preferred-Service header field depending whether the AS is trusted or not according to RFC 6050 [121].

When an AS that is acting as a UA or initiating B2BUA or routeing B2BUA sends an initial request for a dialog or a request for a standalone transaction, the AS may include an Accept-Contact header field containing:

- an ICSI value (coded as specified in subclause 7.2A.8.2) in a g.3gpp.icsi-ref media feature tag as defined in subclause 7.9.2 and RFC 3841 [56B]; and

- one or more IARI values (coded as specified in subclause 7.2A.9.2) that are related to the request in a g.3gpp.iari-ref media feature tag as defined in subclause 7.9.3 and RFC 3841 [56B];

if the ICSI or IARIs for the IMS communication service and IMS application are known.

The AS may:

- include the received ICSI and IARI values;

- replace or remove received ICSI and IARI values; or

- include new ICSI and IARI values.

When the AS acting as a UA or initiating B2BUA or routeing B2BUA sends a SIP request or a SIP response related to an IMS communication service, the AS may include in the Contact header field:

- in a g.3gpp.icsi-ref media feature tag as defined in subclause 7.9.2 one or more ICSI values (coded as specified in subclause 7.2A.8.2); and

- one or more IARI values (coded as specified in subclause 7.2A.9.2) in a g.3gpp.iari-ref media feature tag, for the IMS applications, that are related to the request as defined in subclause 7.9.2 and RFC 3840 [62];

if the ICSI or IARIs for the IMS communication service and IMS application are known. The AS may:

- include the received ICSI and IARI values;

- replace or remove received ICSI values; or

- include new ICSI and IARI values.

When sending a 18x or 2xx response to a request on behalf of an originating user, then the AS may insert a Feature-Caps header field with the "+g.3gpp.icsi-ref" header field parameter into the response, as specified in RFC 6809 [190]. The parameter value is set according to subclause 7.9A.2.

NOTE 3: The AS can insert the Feature-Caps header field e.g. based on operator policy, IMS communication service specification or depending whether other AS already inserted the Feature-Caps header field with the "+g.3gpp.icsi-ref" header field parameter in the response.

When sending a request on behalf of a terminating user, then the AS may insert a Feature-Caps header field with the "+g.3gpp.icsi-ref" header field parameter. The parameter value is set according to subclause 7.9A.2.

NOTE 4: The AS can insert the Feature-Caps header field e.g. based on operator policy, IMS communication service specification or depending whether other AS already inserted the Feature-Caps header field with the "+g.3gpp.icsi-ref" header field parameter in the request.

If the AS inserted the header field parameters in the Feature-Caps header field in the request or in 18x or 2xx response to request then the AS shall include the header field parameters with the same parameter values into the Feature-Caps header field in any target refresh request, and in each 1xx or 2xx response to target refresh request sent in the same direction.

5.7.1.10 Carrier selection

An AS may play a role in support of carrier selection as defined in RFC 4694 [112].

NOTE 1: In general, ASs do not need to support carrier selection, Rather a specific AS or a few ASs in a network will be used for carrier selection,

When an AS that supports carrier selection receives an initial request with a Request-URI in the form of a tel-URI that contains a "cic" tel-URI parameter inserted by the UE and if configured per operator policy, the AS may validate the value of the "cic" parameter. If an AS that supports carrier selection determines the "cic" parameter received in the

initial request to be valid, as configured per operator policy, the AS shall process the request accordingly. If an AS supports carrier selecton and determines the "cic" parameter received in the initial request to be invalid, then the AS shall remove the "cic" parameter and process the request as if no "cic" had been received from the UE.

When an AS that support carrier selection receives an initial request with a Request-URI in the form of a tel-URI, the AS may, based on operator policy, insert an appropriate value for the "cic" tel-URI parameter as defined in RFC 4694 [112].

NOTE 2: For example, the AS that supports preferred carrier could insert a "cic" tel-URI parameter that identifies the originating user's preassigned carrier, or the carrier assigned to a called freephone number.

When an AS that support carrier selection receives an initial request with a Request-URI in the form of a SIP URI user=dialstring (see RFC 4967 [103]), the AS may translate the SIP URI to a valid tel-URI or a valid SIP URI user=phone comprising a userinfo part containing the tel-URI and a domain matching the domain of the original SIP URI user=dialstring. If the received SIP URI user=dialstring is successfully converted, then the AS shall replace the Request-URI with the newly created tel-URI or SIP URI user=phone. The AS shall then process the request as if it had arrived from the UE containing this tel-URI or SIP URI user=phone in the Request-URI.

NOTE 3: This specification does not make any assumptions regarding how these procedures are mapped to ASs; whether all procedures are supported by a single AS or spread across multiple ASs. However, this specification does assume that the responsibility for ensuring that the UE complies with the carrier selection procedures defined in RFC 4694 [112] will be performed by a single AS (e.g. validate "cic"), and the filter criteria will be configured so that this AS is invoked before other ASs that have carrier selection responsibilities.

The AS carrier selection procedures also apply when the request contains a Request-URI in the form of a SIP URI user=phone, where the "cic" tel-URI parameter is contained in the userinfo part of the SIP URI.

5.7.1.11 Tracing

An AS can retrieve tracing configuration information from the HSS via the Sh reference point. The AS tracing configuration can use the parameters specified in 3GPP TS 24.323 [8K] but need not structure them as a management object.

5.7.1.12 Delivery of original destination identity

If the service the AS provides needs to deliver the original destination identity to the UE, the AS shall insert a new hi-entry to the last hi-entry of History-Info header field including "mp" header field parameter as specified in RFC 7044 [66].

NOTE: If the "mp" header field parameter is present in the hi-entries within the History-Info header field, the information of the original destination number or URI, e.g. the service number for freephone, will be found in the hi-entry whose index-val matches the value of the first hi-entry with "mp" header field parameter.

5.7.1.13 CPC and OLI

The AS may populate the "cpc" and "oli" URI parameters in each initial request for a dialog or a request for a standalone transaction in the tel URI or SIP URI representation of telephone numbers in the P-Asserted-Identity header field based on their origin source.

5.7.1.14 Emergency transactions

Identification of emergency transactions for termination in the public network by an AS are outside the scope of this document, and are dependent on many application specific considerations.

Where an AS decides to generate an emergency request on behalf of its served user, the AS shall meet the following conditions:

1) the UE is in the same network as the S-CSCF (i.e. that the UE is not roaming).

NOTE 1: How the above is determined is outside the scope of this document and will depend on the application supported. Possible mechanisms could be: 1) that the AS only receives requests that are from non-roaming UEs; 2) analysis of the P-Access-Network-Info header field in a received request from the UE.

The AS generate the request with the following contents:

1) include in the Request-URI an emergency service URN, i.e. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69]. An additional sub-service type can be added if information on the type of emergency service is known;

2) a Route header field with the topmost Route header field set to the URI associated with an E-CSCF;

3) if the AS is part of the trust domain of the network, a P-Asserted-Identity header field containing the identity of the UE served by the AS;

4) if the AS is not part of the trust domain of the network, a P-Preferred-Identity header field containing the identity of the UE served by the AS;

5) if a GRUU is available for the UE served by the AS, provide the GRUU as part of a Contact header field;

NOTE 2: If the AS is not already aware of the GRUU of the UE due to previously receiving it in a Contact header, and the UE is registered, the GRUU can be obtained using either the subscription to the reg events package or using the third-party registration procedure with the REGISTER request including a "message/sip" MIME body of the 200 (OK) response for the REGISTER request as described in subclause 5.7.1.1.

6) if a location is available at the AS in any form, include a Geolocation header field with that location;

7) a P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17];

8) if the AS supports calling number verification using signature verification and attestation information as specified in subclause 3.1 and if required by operator policy, the AS shall perform attestation of the user identity by inserting:

   - a "verstat" tel URI parameter, specified in subclause 7.2A.20, to the tel URI or SIP URI with a user=phone parameter in the From header field or the P-Asserted-Identity header field;

   - an Attestation-Info header field, specified in subclause 7.2.18; and

     an Origination-Id header field, specified in subclause 7.2.19, set to a UUID identifying the AS which is configured based on local policy and requirements from national regulation; and

9) if the AS supports priority verification using assertion of priority information as specified in subclause 3.1 and if required by operator policy, the AS shall add a Resource-Priority header field containing a namespace of "esnet" as defined in RFC 7135 [197].

If the AS does not receive any response to the INVITE request (including its retransmissions); or receives a 3xx response or 480 (Temporarily Unavailable) response to an INVITE request, the AS shall select a new E-CSCF and forward the INVITE request.

5.7.1.15 Protecting against attacks using 3xx responses

The AS upon receiving a 3xx response to a request from the served UE that contains a Contact header field may remove the Contact header field or modify the response code from 3xx to an appropriate non 2xx response code (e.g. 5xx response code) based on operator policy before sending the response towards the UE.

NOTE: Automatically recursing on untrusted 3xx responses opens up the UE to being redirected to premium rate URIs without the user's consent. An AS that protects against attacks using 3xx responses needs to ensure that it doesn't break services that depend on 3xx responses being passed to the UE. Subclause 5.1.2A.1.1 specifies how a UE protects itself against attacks using 3xx responses. For some services an AS could automatically recurse on a 3xx response.

5.7.1.16 Support of Roaming Architecture for Voice over IMS with Local Breakout

5.7.1.16.1 Preservation of parameters

An AS in a network supporting the roaming architecture for voice over IMS with local breakout shall not change the value of the "icid-value" header field parameter in the P-Charging-Vector header field received in the INVITE request when the AS sends any request or response related to an INVITE transaction.

NOTE: An AS that determines that loopback is not a viable option can change or remove any of the above parameters if required by the application logic performed by the AS.

5.7.1.16.2 Preference for loopback routeing not to occur

When the AS is accessed by a network supporting roaming architecture for voice over IMS with local breakout, and the incoming INVITE request contains a Feature-Caps header field with the "+g.3gpp.home-visited" header field parameter, the AS can indicate to the S-CSCF its preference for loopback routeing not to occur by removing the "+g.3gpp.home-visited" header field parameter from the Feature-Caps header field from any outgoing INVITE request back to the S-CSCF. Reasons for such an indication might include the need to terminate media streams at an MRF in the home network.

NOTE: If the original dialog identifier sent by the S-CSCF is not preserved by the AS in the outgoing requests, then loopback routeing will not occur.

5.7.1.17 Delivery of network provided location information

If the AS supports delivery of network provided location information, and the AS performs the retrieval of cell id and/or UE time zone via Sh interface, the AS shall insert the P-Access-Network-Info header field with the cell id, local-time-zone parameter and/or the "daylight-saving-time" parameter, including a "network-provided" parameter in the incoming request or response. Additionally, if required by local operator policy and the AS is able to deduce a Geographical Identifier from the Cell Global Identity (CGI) or form the Service Area Identifier (SAI), the AS shall include an operator-specific-GI header field parameter. The P-Access-Network-Info header field shall only be inserted if there is not already a network provided information.

When the AS receives, in a SIP request or response, a P-Access-Network-Info header field which does not contain the operator-specific-GI header field parameter, contains a Cell Global Identity (CGI) or a Service Area Identifier (SAI) information and contains the "network provided" header field parameter, if required by local operator policy and if the AS is able to deduce a Geographical Identifier from the contained Cell Global Identity (CGI) or form the contained Service Area Identifier (SAI), the AS shall instert an operator-specific-GI header field parameter in that received network provided P-Access-Network-Info header field.

The AS can obtain a Geographical Identifier from the CLF by using the e2 interface (see ETSI ES 283 035 [98]).

NOTE: ETSI ES 283 035 [98] Release 3 enables querying a CLF using the User-Data-Request command in which the Global-Access-Id AVP contains the 3GPP-User-Location-Info AVP with a CGI or a SAI value to get a corresponding Geographical Identifier. If multiple CLFs are deployed, the AS can dertermine which CLF to query based on the CGI or the SAI values or can use a DIAMETER proxy if deployed.

5.7.1.18 Delivery of MRB address information

A visited MRB address can be received during session establishment. If an AS receives the URI of an MRB in a "+g.3gpp.mrb" header field parameter included in a Feature-Caps header field of an INVITE request, it shall store this URI. If the AS requests allocation of MRF resources from an MRB in its own network, either in in-line or query mode, the AS shall forward the visited network MRB URI in the request.

5.7.1.19 Overload control

5.7.1.19.1 Outgoing subscriptions to load-control event

Based on operator policy, the AS may subscribe to the load-control event package with one ore more target SIP entities. The list of target SIP entities is provisioned.

Subscription to the load-control event package is triggered by internal events (e.g. the physical device hosting the SIP entity is power-cycled) or through a management interface.

The AS shall perform subscriptions to the load-control event package to a target entity in accordance with RFC 6665 [28] and with RFC 7200 [201]. When subscribing to the load-control event, the AS shall:

1) Send a SUBSCRIBE request in accordance with RFC 6665 [28] and with RFC 7200 [201] to the target entity, with the following elements:

   - an Expires header field set to a network specific value;

2) If the target entity is located in a different network and local policy requires the application of IBCF capabilities, forward the request to an IBCF acting as an exit point.

The AS shall automatically refresh ongoing subscriptions to the load-control event package either 600 seconds before the expiration time if the initial subscription was for greater than 1200 seconds, or when half of the time has expired if the initial subscription was for 1200 seconds or less.

The AS can terminate a subscription according to RFC 6665 [28].

5.7.1.19.2 Incoming subscriptions to load-control event

If subscriptions to load-control event package is supported, the AS shall handle incoming subscriptions to the load-control event package in accordance with RFC 6665 [28] and with RFC 7200 [201]. When the AS receives a SUBCRIBE request for the load-control event from an unauthorised or unexpected source, the AS shall generate a "403 forbidden" response to the SUBSCRIBE request.

If the AS receives a SUBSCRIBE request from an authorised source the AS shall:

2) Generate a "200 OK" response to the SUBSCRIBE request with the following settings:

   - an Expires header field, set to either the same or a decreased value as the Expires header field in SUBSCRIBE request; and

   - the Contact header field set to an identifier uniquely associated to the SUBSCRIBE request that may help correlating refreshes.

3) In case of an initial subscription, determine the list of load filters applicable to the subscriber, create an XML document to represent this information and send it as an attachement to a NOTIFY request towards the subscriber. If no applicable load filters are identified when the subscription request is received, an empty document is attached to the NOTIFY request.

Subsequent NOTIFY requests with updated or new filters may then be sent as the actual load of the target entity evolves.

5.7.1.20 Procedures in the AS for resource sharing

5.7.1.20.1 General

An AS supporting resource sharing shall use the "+g.3gpp.registration-token" header field parameter in the Contact header field of the incoming third-party REGISTER request and the "+g.3gpp.registration-token" header field parameter in the Feature-Caps header field in the initial INVITE request or provisional responses to the initial INVITE to identify the UE.

The AS supporting resource sharing shall only include the Resource-Share header field in requests and responses destined to the served user and in all other cases remove the header field.

5.7.1.20.2 UE-originating case

If the AS supporting resource sharing receives a response or request destined to the served user containing a Resource-Share header field, the AS shall remove that header field from the outgoing response or request.

Upon receiving an SDP answer in a provisional response or a 200 (OK) response to an initial INVITE request from a UE served by a P-CSCF supporting resource sharing, the AS supporting resource sharing shall determine whether resource sharing can be applied.

NOTE 1: An AS can learn that a P-CSCF supports resource sharing from the Resource-Share header field with the value "supported" in the initial REGISTER request that is contained in the "message/sip" MIME body of the third-party REGISTER request received when the UE registered.

NOTE 2: The conditions for resource sharing are outside the scope of the present document.

If the AS:

1) determines that at least one media stream in the initial SDP answer is subject to resource sharing, the AS shall in the outgoing response include a Resource-Share header field as described in subclause 7.2.13.4 with the following clarifications:

   a) the AS shall set the "origin" header field parameter to "session-initiator"; and

   b) if

      - a session exists that can share resources with the dialog created by the initial INVITE request involving the same UE, the AS shall include a new sharing key set to the value of the sharing key in the other session and use this sharing key to identify the resource sharing rule for this media stream in this session; and

      - no session exists that can share resources with the dialog created by the initial INVITE request, the AS shall include a new sharing key unique among all UEs registered by the user and use this sharing key to identify the resource sharing rule for this media stream in this session; and

2) determines that resource sharing can never be applied for any of the media streams in the SDP answer, the AS shall include a Resource-Share header field set to the value "no-media-sharing" in the outgoing response.

5.7.1.20.3 UE-terminating case

5.7.1.20.3.1 Determine resource sharing using the initial SDP offer

Upon receiving an initial INVITE request containing an initial SDP offer destined for the served user, the AS supporting resource sharing shall if at least one registered UE is served by a P-CSCF supporting resource sharingdetermine if resource sharing can be applied.

NOTE 1: An AS can learn that a P-CSCF supports resource sharing from the Resource-Share header field with the value "supported" in the initial REGISTER request that is contained in the "message/sip" MIME body of the third-party REGISTER request received when the UE registered.

NOTE 2: The condition for resource sharing is outside the scope of the present document.

If the AS:

1) determines that at least one media stream in the initial SDP answer is subject to resource sharing, the AS shall in the outgoing request include a Resource-Share header field as described in subclause 7.2.13.4 with the following clarifications:

   a) the AS shall set the "origin" header field parameter to "session-receiver";

   b) the AS shall include a new sharing key part that is determined as follows:

      A) if the AS is aware of only one registered contact:

         I) if a session exists where media in the existing session can be shared with media in the new SDP offer, the sharing key in the INVITE request shall be set to the value of the sharing key used in the existing session and the AS shall use this sharing key to identify the resource sharing rules for each media stream in the session; or

         II) if no session exists where media in the existing session can be shared with media in the new SDP offer, the AS shall create a new sharing key that is unique among all sessions that exist on the UE.

This new sharing key shall be included in the INVITE request and is used to identify the resource sharing rules for each media stream in this session; and

B) if the AS is aware of more than one registered contacts:

I) if the AS is using an Accept-Contact header field, Reject-Contact header field, and/or a GRUU in the request URI to target a specific registered UE and a session exists on the target UE where media in the existing session can be shared with media in the new SDP offer, the sharing key in the INVITE request shall be set to the value of the sharing key used in the existing session and the AS shall use this sharing key to identify the resource sharing rules for each media stream in the session; or

II) if no session exists on the target UE (or on the set of UEs when more than one UE is registered and the S-CSCF can fork the INVITE request to more than one UE) where media in the existing session can be shared with media in the new SDP offer, the AS shall create a new sharing key that is unique among all sessions that exist for all UEs registered for the server user. This new sharing key shall be included in the INVITE request and is used to identify the resource sharing rules for each media stream in this session; and

c) if the AS is aware of more than one registered contact and the AS is not using an Accept-Contact header field, Reject-Contact header field, and/or a GRUU in the request URI to target a specific registered UE (so that the S-CSCF is allowed to fork the INVITE request to one or more UEs) and sessions exist with any registered UE where media in an existing session can be shared with media in the new SDP offer, the existing-sharing-key-list part in the INVITE shall be set to the value of the sharing keys used in the existing sessions; and

2) determines that resource sharing can not be applied, the AS shall include a Resource-Share header field set to the value "no-media-sharing" in the outgoing request.

5.7.1.20.3.2 Determine resource sharing using the initial SDP answer

Upon receiving a PRACK request or an ACK request destined to a UE served by a P-CSCF supporting resource sharing that contains an initial SDP answer, the AS supporting resource sharing shall determine if resource sharing can be applied.

NOTE 1: An AS can learn that a P-CSCF supports resource sharing from the Resource-Share header field with the value "supported" in the initial REGISTER request when that is contained in the "message/sip" MIME body of the third-party REGISTER request received the UE registered.

NOTE 2: The condition for resource sharing is outside the scope of this technical specification.

If the AS:

1) determines that at least one media stream in the initial SDP answer is subject to resource sharing, the AS shall in the outgoing request include a Resource-Share header field as described in subclause 7.2.13.4 with the following clarifications:

a) the "origin" header field parameter shall be set to "session-receiver"; and

b) if

- a session exists that can share resources involving the same UE, the AS shall include a new sharing key set to the value of the sharing key in the other session and use this sharing key to identify the resource sharing rule for this media stream in this dialog; and

- no session exists that can share resources, the AS shall include a new sharing key unique among all UEs registered by the user and use this sharing key to identify the resource sharing rule for this media stream in this dialog; and

2) determines that resource sharing can never be applied for any of the media streams in the SDP answer, the AS shall include a Resource-Share header field with the value "no-media-sharing" as described in subclause 7.2.13.4 in the outgoing response.

5.7.1.20.4 Updating the resource sharing options

If the AS during the duration of the call determines that the resource sharing options needs to be changed (e.g. if media streams are added by the UE), then the AS shall include a Resource-Share header field with the updated resource sharing options as specified in subclause 7.2.13, in the outgoing request or response causing the reason for change.

NOTE: If more than one dialog exists and the update is sent before the session invitation is accepted by the terminating user, the resource sharing options are determined per individual dialog.

5.7.1.20.5 Abnormal cases

If the AS receives a request or response from a served user containing a Resource-Share field with the value "no-media-sharing", the AS shall no longer apply resource sharing with sessions involving the UE sending the request or response.

If the AS receives a request or response from a served user containg an SDP offer conflicting with an earlier decision to share resources, the AS shall include in the response carrying the SDP answer towards the served user a Resource-Share header field with the value "no-media-sharing" along with the "origin" header field parameter set to "session-initiator" or "session-receiver" as appropriate and no longer apply resource sharing with sessions involving the UE sending the request or response.

NOTE: A typical example when this can happen is the communication waiting use case. If the UE sends a 200 (OK) response to the INVITE request without putting the first call on hold, the UE's behaviour is then regarded as unpredictable and resource sharing cannot be used towards that UE.

5.7.1.21 Dynamic Service Interaction

If an AS supports dynamic service interaction, the AS should insert the Service-Interact-Info header field into the SIP messages before those messages are sent out. The Service-Interact-Info field is filled with the service identities of the services which have been executed. Additionally, the AS may insert into the Service-Interact-Info header field with the identities of the services which may have confliction with the executed services according to local policy.

NOTE: No service identifiers are defined in this release of the present document.

If the AS supports dynamic service interaction, it should take the information contained in any received Service-Interact-Info header field into account when executing service logic.

If the AS does not recognize the service identities contained in the Service-Interact-Info header field, they should be ignored.

5.7.1.22 Service access number translation

When the AS is accessed by a service access number (e.g. toll free, premium service) and performes a translation of this number into a routeable number, the AS shall include in the outgoing request:

1) the Request-URI set to the targeted SIP URI containing the "cause" SIP URI parameter, defined in RFC 4458 [68], set to the value "380" defined in RFC 8119 [230]; and

2) the History-Info header field constructed according to RFC 7044 [66] in which the hi-targeted-to-uri of the last hi-entry is set to the new Request-URI with the "cause" SIP URI parameter, defined in RFC 4458 [68], set to the value "380" defined in RFC 8119 [230].

A service access number does not constrain the number format. A local service number as defined in TS 23.228 [7] and described in subclause 5.7.1.7 can be a service access number.

5.7.1.23 Procedures in the AS for priority sharing

5.7.1.23.1 General

An AS supporting priority sharing and if according to local policy shall apply priority sharing as specified in the following subclauses.

NOTE: The MCPTT server is the only AS in the present document defined to use priority sharing (see 3GPP TS 24.379 [8ZE]).

An AS supporting priority sharing shall only include a Priority-Share header field in requests and responses destined to the served user and in all other cases remove the header field.

5.7.1.23.2 Session originating procedures

If the Feature-Caps header field with the g.3gpp.priority-share feature-capability indicator was included in the "message/sip" MIME body in the third-party REGISTER request and when the AS determines that priority sharing can be applied, the AS supporting priority sharing shall:

1) include the Priority-Share header field with the value "allowed" defined in subclause 7.2.16 in a 18x and 2xx response to the initial INVITE request; or

2) include the Priority-Share header field with the value "allowed" defined in subclause 7.2.16 in a subsequent request or a response to subsequent requests destined to the served user.

If priority sharing can not be applied any longer, the AS shall include the Priority-Share header field, defined in subclause 7.2.16, with the value "not-allowed" in a request or response destined to the served user.

NOTE: The AS can enable or disable priority sharing by including the Priority-Share header field with the value "allowed" or "not-allowed" as specified above until the session is released.

5.7.1.23.3 Session terminating procedures

If the incoming REGISTER request contained in the "message/sip" MIME body of a third-party REGISTER request the g.3gpp.priority-share feature-capability indicator in a Feature-Caps header field and when the AS determines that priority sharing can be applied, the AS supporting priority sharing shall:

1). include the Priority-Share header field with the value "allowed" defined in subclause 7.2.16 in the initial INVITE request destined to the served user; or

2) include Priority-Share header field with the value "allowed" defined subclause 7.2.16 in a subsequent request or responses to a subsequent request destined to the served user.

If priority sharing can not be applied any longer, the AS shall include the Priority-Share header field, defined in subclause 7.2.16, with the value "not-allowed" in a request or response destined to the served user.

NOTE: The AS can enable or disable priority sharing by including the Priority-Share header field with the value "allowed" or "not-allowed" as specified above until the session is released.

5.7.1.24 Handling re-INVITE request collisions

An AS shall handle re-INVITE request collisions as specified in RFC 3261 [26] with the clarification in this subclause.

When the AS receives a SIP 491 (Request Pending) response to a re-INVITE request initiated by the AS and sent towards the remote user, the AS shall:

1) if the AS is serving the user at the originating side of the call, act as the the owner of the Call-ID of the dialog ID and should select a randomly choosen time to be between 2.1 and 4 seconds in units of 10 ms; and

2) if the AS is serving the user at the terminating side of the call, act as not being the owner of the Call-ID of the dialog ID and should select a randomly choosen time to be between 0 and 2 seconds in units of 10 ms.

When the AS receives a SIP 491 (Request Pending) response to a re-INVITE request initiated by the AS and sent towards the served user, the AS shall:

1) if the AS is serving the user at the originating side of the call, act as not being the owner of the Call-ID of the dialog ID and should select a randomly choosen time to be between 0 and 2 seconds in units of 10 ms; and

2) if the AS is serving the user at the terminating side of the call, act as the owner of the Call-ID of the dialog ID and should select a randomly choosen time to be between 2.1 and 4 seconds in units of 10 ms.

5.7.1.25 Assertion verification using the Identity header field

5.7.1.25.1 General

RFC 8224 [252] describes a mechanism where an authentication service after verifying the calling number identity inserts a signature over selected header fields. A verification service can then use this signature to trust the correctness of the identity.

RFC 8946 [265] describes a mechanism where an authentication service after verifying the diverting number identity inserts a signature over selected header fields. A verification service can then use this signature to trust the correctness of the diverting identity.

RFC 8443 [279] describes a mechanism where an authentication service after verifying the Resource-Priority header field inserts a signature over the Resource-Priority header field. A verification service can then use this signature to trust the correctness of the Resource-Priority header field.

RFC 9027 [278] extends a mechanism defined in RFC 8443 [279] to enable authentication of the header field value "psap-callback" of the Priority header field and inserts a signature over the Resource-Priority header field and the header field value "psap-callback" provided in the Priority header field. A verification service can then use this signature to trust the correctness of the Resource-Priority header field and header field value "psap-callback" of the Priority header field.

5.7.1.25.2 Originating procedures

An originating AS supporting the calling number verification using signature verification and attestation information, as defined in subclause 3.1:

1) may based on local policy insert an Identity header field as specified in RFC 8224 [252] for all initial INVITE requests and MESSAGE requests and shall for this purpose use the identity in the P-Asserted-Identity header field or the From header field; or

NOTE 1: This option is kept from the original release-14 functionality. If the AS knows the IBCF supports invoking an AS for providing an Identity header field the below actions are more efficient.

2) may based on local policy perform attestation of the identity of the served user by:

   a) inserting a "verstat" tel URI parameter, specified in subclause 7.2A.20; in the From header field or the P-Asserted-Identity header field if not already present; and

   b) insert an Origination-Id header field as specified in subclause 7.2.19 and an Attestation-Info header field specified in subclause 7.2.18, if not alredy present.

If the AS performs originating services on behalf of a diverting user, the AS may assert the identity of the diverting user by inserting a "verstat" tel URI parameter, specified in subclause 7.2A.20, in the History-Info hi-entry representing the diverting user.

If the AS supports priority verification using assertion of priority information as specified in subclause 3.1 and if allowed by local operator policy, the AS may insert an Identity header field as described in RFC 8443 [279] and RFC 9027 [278], and shall for this purpose use the resource priority of the Resource-Priority header field in the received initial INVITE or re-INVITE request.

NOTE 2: For sessions terminating in another domain, only one of the following entities needs to be configured to provide an Identity header field for the resource priority: the IBCF or the AS. Which functional entity inserts the Identity header field is subject to network configuration and local policy.

NOTE 3: How the AS asserts the authenticity of the Resource-Priority header field value is out of scope of the present document.

5.7.1.25.3 Terminating procedures

Upon receiving an initial INVITE request or a MESSAGE request containing one or more Identity header fields, an AS supporting the calling number verification using signature verification and attestation information, as defined in subclause 3.1, shall if the network indicated support for the calling number verification during registration:

- if no "verstat" tel URI parameter is present for the identity to be verified in the From or P-Asserted-Identity header field, perform user identity verification of the originating user identity using the Identity header field containing a PASSporT SHAKEN JSON Web Token, specified in RFC 8588 [261] and based on local policy all Identity header fields containing a PASSporT div JSON Web Token, specified in RFC 8946 [265], in the received request. Based on the outcome of the verification insert a "verstat" tel URI parameter, specified in subclause 7.2A.20, with a value representing the outcome of the verification in the tel URI or SIP URI with the user=phone parameter of each P-Asserted-Identity header field or From header field where the URI contains the calling number that was tested for verification and based on local policy in all verified identities in the History-Info header field.

If no Identity header field is present in the received INVITE or MESSAGE request, but an Origination-Id header field along with an Attestation-Info header field set either to "B" or "C" is present, the AS shall set the verstat tel URI parameter to the value "No-TN-Validation".

If the AS supports priority verification using assertion of priority information as specified in subclause 3.1 and if allowed by local operator policy, the AS may verify that the Priority and Resource-Priority header field values are authorized. To do so, the AS:

- verifies the Identity header fields containing a PASSporT rph JSON Web Token as specified in RFC 8443 [279] and RFC 9027 [278] if included in the initial INVITE or re-INVITE request; and

- verifies that the Priority and Resource-Priority header field values are authorized by valid "rph" PASSporT claims.

The AS shall populate the Priority-Verstat header field associated with the Resource-Priority header field and include the Priority-Verstat header field in the forwarded SIP request.

The AS may report any verification failure of an Identity header field to the appropriate upstream signing service by populating a Reason header field in the next provisional or final response to the INVITE or MESSAGE request, where the Reason header field "protocol" value is set to "STIR", as specified in RFC 9410 [294], and the "cause" header field parameter contains the 4xx response code of the failing PASSporT, as defined in RFC 8224 [252]. Additionally, the AS may include the "ppi" header field parameter containing the failing PASSporT.

NOTE 1: For sessions originating in another domain, only one of the following entities needs to be configured to verify the Identity header field for the resource priority: the IBCF or the AS. Which functional entity inserts the Identity header field verification is subject to network configuration and local policy.

NOTE 2: Multiple Reason header fields with the protocol value set to "STIR" are not supported in the present document.

5.7.1.25.4 Procedures over the Ms reference point

If the AS receives a verificationRequest as specified in annex V, the AS verifies the request and when verified generates a verificationResponse, as specified in annex V, including a "verstat" claim for the verified identity.

If the AS receives a signingRequest, specified in annex V, the AS sends the signed information in a signingResponse as specified in annex V.

5.7.1.26 Procedures in the AS for 3GPP PS data off

An AS that supports 3GPP PS data off can receive in the message/SIP MIME body in a third party REGISTER request the REGISTER request sent by the UE. If this REGISTER request contains a "+g.3gpp.ps-data-off" Contact header field parameter the AS can determine that the UE supports 3GPP PS data off, and the value of the parameter indicates the 3GPP PS data off status. When the AS receives an initial request for a dialog or a standalone transaction destined to the served user, if:

- the latest "+g.3gpp.ps-data-off" Contact header field parameter, specified in subclauce 7.9.8, that was received in a third party REGISTER request, as specified above, was set to "active" in the UE; and

- the service the AS supports is not configured as a 3GPP PS data off exempt service to be used in the HPLMN,the EHPLMN or the subscribed SNPN, or the service the AS support is not configured as a 3GPP PS data off exempt service to be used in the VPLMN or the non-subscribed SNPN;

the AS shall not send the request to the UE via GPRS IP-CAN, EPS IP-CAN or 5GS IP-CAN.

5.7.1.27 AS support for in-call access update procedures

5.7.1.27.1 General

The AS can indicate that it supports in-call access update procedures by sending a feature-capability indicator including a public service identity indicating to where to send the updates.

When the AS receives a MESSAGE request with a P-Charging-Vector header field containing an "icid-value" header field parameter matching the "icid-value" header field parameter of an existing dialog, the AS can use the location information in the received MESSAGE request to update the location information for the dialog identified by the Call-Id.

5.7.1.27.2 Originating procedures

If the AS supports in-call access update procedures and the AS is interested in receiving updates of changes of IP-CAN or PLMN the AS shall include in responses to an initial INVITE request a g.3gpp.in-call-access-update feature-capability indicator with a value set to a public service identity resolving to the AS.

5.7.1.27.3 Terminating procedures

If the AS supports in-call access update procedures and the AS is interested in receiving updates of changes of IP-CAN or PLMN the AS shall include in the INVITE request a g.3gpp.in-call-access-update feature-capability indicator with a value set to a public service identity resolving to the AS.

5.7.2 Application Server (AS) acting as terminating UA, or redirect server

When acting as a terminating UA the AS shall behave as defined for a UE in subclause 5.1.4, with the exceptions identified in this subclause.

The AS, although acting as a UA, does not initiate any registration of its associated addresses. These are assumed to be known by peer-to-peer arrangements within the IM CN subsystem.

If the AS requires knowledge of the served user it shall determine the served user according to the applicable procedure in subclause 5.7.1.3A.

An AS acting as redirect server shall propagate any received IM CN subsystem XML message body in the redirected message.

When an AS acting as a terminating UA generates a subsequent request for a dialog, the AS shall insert a P-Charging-Vector header field with the "icid-value" header field parameter set to the value populated in the initial request for the dialog and a type 3 "orig-ioi" header field parameter. The AS shall set the type 3 "orig-ioi" header field parameter to a value that identifies the service provider from which the request is sent. The AS shall not include the type 3 "term-ioi" header field parameter.

When the AS acting as terminating UA receives a request, the AS shall store the value of the "orig-ioi" header field parameters received in the P-Charging-Vector header field if present.

NOTE 1: Any received orig-ioi parameter will be a type 3 orig-ioi. The orig-ioi identifies the network operator from which the request was sent.

When the AS acting as terminating UA generates a response to a request, the AS shall insert a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the request, a type 3 "term-ioi" header field parameter and the "icid-value" header field parameter. The AS shall set the type 3 "term-ioi" header field parameter to a value that identifies the service provider from which the response is sent, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the previously received value of "icid-value" header field parameter in the request.

The AS acting as terminating UA receiving an initial request with a P-Charging-Vector header field shall, based on local policy, store the "fe-identifier" header field parameter of the P-Charging-Vector header field.

The AS acting as terminating UA shall, based on local policy, include the stored "fe-identifier" header field parameter in the P-Charging-Vector header field, add its address or identifier and application id to the "as-addr" and "as-id"

elements of the "fe-identifier" header field parameter of the P-Charging-Vector header field and send the P-Charging-Vector header field in the related final response.

NOTE 2: An AS hosting multiple applications can add multiple pairs of "as-addr" and "as-id" header field parameters when executing these applications.

If resource priority in accordance with RFC 4412 [116] is required for a dialog, then the AS shall include the Resource-Priority header field in all requests associated with that dialog. If priority is supported, the AS shall take into account that subsequent received SIP requests or responses within the same dialog or transaction may have added or changed the Resource-Priority header field or backwards indication.

NOTE 3: An AS can modify the priority using the Resource-Priority header field in a dialog on behalf of a user. The mechanism for a user to invoke this AS action is out of scope of this release of the specification.

5.7.3 Application Server (AS) acting as originating UA

In order to support an AS acting as an originating UA, the AS has to be within the same trust domain as the S-CSCF to which requests will be sent.

When acting as an originating UA the AS shall behave as defined for a UE in subclause 5.1.3, with the exceptions identified in this subclause.

The AS, although acting as a UA, does not initiate any registration of its associated addresses and does not participate in any authentication procedures defined for a UE. These are assumed to be known by peer-to-peer arrangements within the IM CN subsystem.

When an AS acting as an originating UA generates an initial request for a dialog or a request for a standalone transaction, the AS shall insert a P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17] and a type 3 "orig-ioi" header field parameter. The AS shall set the type 3 "orig-ioi" header field parameter to a value that identifies the service provider from which the request is sent. The AS shall not include the type 3 "term-ioi" header field parameter.

NOTE 1: The AS can retrieve CDF and/or ODF addresses from HSS on Sh interface.

When the AS acting as an originating UA receives any response to a request, the AS shall store the value of the "term-ioi" header field parameter received in the P-Charging-Vector header field if present.

NOTE 2: Any received "term-ioi" header field parameter will be a type 3 IOI. The type 3 IOI identifies the network operator from which the response was sent.

When an AS acting as an originating UA generates a subsequent request for a dialog, the AS shall insert a P-Charging-Vector header field with the "icid-value" header field parameter set to the value populated in the initial request for the dialog and a type 3 "orig-ioi" header field parameter. The AS shall set the type 3 "orig-ioi" header field parameter to a value that identifies the service provider from which the request is sent. The AS shall not include the type 3 "term-ioi" header field parameter.

Based on local policy, the AS acting as an originating UA or application(s) hosted by the AS acting as originating UA shall add an "as-addr" and an "as-id" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier and application identifier to an initial request.

NOTE 3: An AS hosting multiple applications can add multiple pairs of "as-addr" and "as-id" header field parameters when executing these applications for an initial request.

The AS shall extract charging function addresses from any P-Charging-Function-Addresses header field that is received in any 1xx or 2xx responses to the requests.

The AS may also indicate that the proxies should not fork the request by including a "no-fork" directive within the Request-Disposition header field in the request as described in RFC 3841 [56B].

When sending any initial request, an identity is needed that will correlate with the service profile to be used at the S-CSCF. If the identity for that service profile corresponds to the value to be used to identify the caller to the destination user, include the identity in the P-Asserted-Identity header field. If the identity for that service profile does not correspond to the value to be used to identify the caller to the destination user, and the P-Served-User header field is supported by the S-CSCF, include the identity in the P-Served-User header field. This leaves the P-Asserted-Identity

header field for the identity to be used to identify the caller to the destination user. If the identity for that service profile matches a wildcarded identity and the P-Profile-Key header field is supported by the AS, include the wildcarded identity in the P-Profile-Key header field.

When sending an initial request on behalf of a PSI that is hosted by the AS, the AS shall:

- insert a Request-URI as determined by the service logic;

- insert a P-Asserted-Identity header field and possibly a P-Served-User header field containing the PSI as indicated earlier in this subclause;

- if the AS is not able to resolve the next hop address by itself or the operator policy does not allow it, insert a Route header field pointing either to the S-CSCF where the PSI is hosted, or to the entry point of the home network of the PSI or to the transit function. The AS shall append the "orig" parameter to the URI in the topmost Route header field; and

NOTE 4: The address of the S-CSCF hosting the PSI can be obtained by querying the HSS on the Sh interface.

NOTE 5: AS can only send the initial request to the entry point of the home network of the PSI only if the AS can assume (e.g. based on local configuration) that the receiving entry point will be able to process the request as an originating request.

- if the AS is able to resolve the next hop address by itself and the operator policy allows it, forward the originating request directly to the destination without involving any S-CSCF in the originating IM CN subsystem.

When sending an initial request on behalf of a public user identity, the AS shall:

- insert a Request-URI as determined by the service logic;

- insert a P-Asserted-Identity header field and possibly a P-Served-User header field containing the public user identity as indicated earlier in this subclause;

- if the AS intends to send the originating request to the home network of the public user identity or the operator policy requires it, insert a Route header field pointing to the S-CSCF where the public user identity on whose behalf the request is generated is registered or hosted (unregistered case) or to the entry point of the public user identity's network. The AS shall append the "orig" parameter to the URI in the topmost Route header field; and

NOTE 6: The address of the S-CSCF can be obtained either by querying the HSS on the Sh interface or during third-party registration.

NOTE 7: AS can send the initial request to the entry point of the public user identity's network or to the entry point of the home network of the PSI only if the AS can assume (e.g. based on local configuration) that the receiving entry point will be able to process the request as an originating request.

- if the AS intends to send the originating request directly to the terminating network and the operator policy allows it, forward the originating request directly to the destination without involving any S-CSCF in the originating IM CN subsystem.

When sending an initial request to a served public user identity, the AS shall insert:

- a Request-URI containing the served public user identity;

- a P-Asserted-Identity as determined by the service logic (e.g. the URI of the AS or the URI of the entity that triggered the SIP request, if the sending of the initial request is triggered by a non-SIP request); and

- a Route header field pointing to the S-CSCF where the public user identity to whom the request is generated is registered or hosted (unregistered case) or to the entry point of the public user identity's network. The AS shall not append the "orig" parameter to the URI in the topmost Route header field.

NOTE 8: The address of the S-CSCF can be obtained either by querying the HSS on the Sh interface or during third-party registration.

The AS can indicate privacy of the P-Asserted-Identity in accordance with RFC 3323 [33], and the additional requirements contained within RFC 3325 [34].

Where privacy is required, in any initial request for a dialog or request for a standalone transaction, the AS shall set a display-name of the From header field to "Anonymous" as specified in RFC 3261 [26] and set an addr-spec of the From header field to Anonymous User Identity as specified in 3GPP TS 23.003 [3].

NOTE 9: The contents of the From header field cannot be relied upon to be modified by the network based on any privacy specified by the user either within the AS indication of privacy or by network subscription or network policy. Therefore the AS includes the value "Anonymous" whenever privacy is explicitly required.

If resource priority in accordance with RFC 4412 [116] is required for a dialog, then the AS shall include the Resource-Priority header field in all requests associated with that dialog. If priority is supported, the AS shall take into account that subsequent received SIP requests or responses within the same dialog or transaction may have added or changed the Resource-Priority header field or backwards indication.

NOTE 10: An AS can modify the priority using the Resource-Priority header field in a dialog on behalf of a user. The mechanism for a user to invoke this AS action is out of scope of this release of the specification.

5.7.4 Application Server (AS) acting as a SIP proxy

When the AS acting as a SIP proxy receives a request from the S-CSCF, prior to forwarding the request, the AS shall:

- remove its own URI from the topmost Route header field;
- if the request contains a "logme" header field parameter in the SIP Session-ID header field then log the request if required by local policy; and
- after executing the required services, route the request based on the topmost Route header field.

When the AS acting as a SIP proxy receives any response to the above request, the AS shall:

- if the response contains a "logme" header field parameter in the SIP Session-ID header field then log the request if required by local policy.

The AS may modify the SIP requests based on service logic, prior to forwarding the request back to the S-CSCF.

The AS shall not fork the request if the fork-directive in the Request-Disposition header field is set to "no-fork" as described in RFC 3841 [56B].

If the AS requires knowledge of the served user it shall determine the served user according to the applicable procedure in subclause 5.7.1.3A.

An AS acting as a SIP proxy shall propagate any received IM CN subsystem XML message body in the forwarded message.

When the AS acting as a SIP proxy receives a request, the AS shall store the value of the "orig-ioi" header field parameter received in the P-Charging-Vector header field if present. The AS shall remove the "orig-ioi" header field parameter from the forwarded request and insert a type 3 "orig-ioi" header field parameter. The AS shall set the type 3 "orig-ioi" header field parameter to a value that identifies the service provider from which the request is sent. The AS shall not include the type 3 "term-ioi" header field parameter.

NOTE 1: A received orig-ioi parameter will be a type 3 orig-ioi. The orig-ioi identifies the network operator from which the request was sent.

When the AS acting as a SIP proxy forwards a response to a request, the AS shall remove any received "orig-ioi" and "term-ioi" header field parameters, and insert a P-Charging-Vector header field containing the previously stored "orig-ioi" header field parameter, if received in the request and a type 3 "term-ioi" header field parameter. The AS shall set the type 3 "term-ioi" header field parameter to a value that identifies the service provider from which the response is sent and the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter. Any values of "orig-ioi" or "term-ioi" header field parameters received in any response that is being forwarded are not used.

Based on local policy, the AS acting as a SIP proxy shall add an "as-addr" and an "as-id" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier and application identifier.

NOTE 2: An AS hosting multiple applications can add multiple pairs of "as-addr" and "as-id" header field parameters when executing these applications for an initial request.

5.7.5 Application Server (AS) performing 3rd party call control

5.7.5.1 General

The AS performing 3rd party call control acts as a B2BUA. There are two kinds of 3rd party call control:

- Routeing B2BUA: an AS receives a request, terminates it and generates a new request, which is based on the received request.

- Initiating B2BUA: an AS initiates two requests, which are logically connected together at the AS, or an AS receives a request and initiates a new request that is logically connected but unrelated to the incoming request from the originating user (e.g. the P-Asserted-Identity of the incoming request is changed by the AS). AS can initiate additional requests and associate them with a related incoming request.

When the AS acting as an initiating B2BUA receives a request and initiates a new request that is logically connected but unrelated to the incoming request from the originating user, the AS can include an original dialog identifier in the Route header field for the S-CSCF that it learned from the incoming request, per service logic needs.

NOTE 1: If the AS does not include the original dialog identifier in an initiated request, the S-CSCF can apply the default handling procedure relating to the incoming request if after a certain time no 1xx response is sent by the AS to the incoming request or if the AS forwards a 408 (Request Timeout) response or a 5xx response received from downstream as a response to the incoming request. To avoid the application of the default handling procedure by the S-CSCF when the AS is waiting for a SIP response for an initiated request, the AS can generate a SIP provisional response to the incoming request.

If the AS requires knowledge of the served user the AS shall determine the served user according to the applicable procedure in subclause 5.7.1.3A.

When the AS receives a terminated call and generates a new call, and dependent on whether the service allows the AS to change the P-Asserted-Identity for outgoing requests compared with the incoming request, the AS will select appropriate kind of 3rd party call control.

The B2BUA AS will internally map the message header fields between the two dialogs that it manages. It is responsible for correlating the dialog identifiers and will decide when to simply translate a message from one dialog to the other, or when to perform other functions. These decisions are specific to each AS and are outside the scope of the present document.

The AS, although acting as a UA, does not initiate any registration of its associated addresses. These are assumed to be known by peer-to-peer arrangements within the IM CN subsystem.

For standalone transactions, when the AS is acting as a Routeing B2BUA, the AS shall copy the remaining Route header field(s) unchanged from the received request for a standalone transaction to the new request for a standalone transaction.

When the AS receives a Replaces header field within an initial request for a dialog, the AS should check, whether the AS acts as a routeing B2BUA for the dialog identified in the Replaces header field. The AS should:

- if the AS acts as routeing B2BUA for the dialog indicated in the Replaces header field, include in the forwarded request a Replaces header field, indicating the dialog on the outgoing side that corresponds to the dialog identified in the received Replaces header field; or

- if the AS does not act as a routeing B2BUA for the dialog indicated in the Replaces header field, include in the forwarded request the Replaces header field as received in the incoming request.

When the AS receives a Target-Dialog header field within an initial request or a standalone transaction for a dialog, the AS shall:

- if the AS acts as routeing B2BUA for the dialog indicated in the Target-Dialog header field, include in the forwarded request a Target-Dialog header field, indicating the dialog on the outgoing side that corresponds to the dialog identified in the received Target-Dialog header field.

When the AS acting as a routeing B2BUA receives a request, the AS shall:

- store the value of the "orig-ioi" header field parameter received in the P-Charging-Vector header field if present; and

- remove the "orig-ioi" header field parameter from the forwarded request.

NOTE 2: Any received orig-ioi parameter will be a type 3 orig-ioi. The orig-ioi identifies the network operator from which the request was sent.

When an AS acts as a routeing B2BUA and the received Contact header field contains a media feature tag indicating a capability for which the Contact URI can be used by the remote party, the AS shall transparently forward the Contact header field. When transparently forwarding a received Contact header field of a dialog-forming request, the AS shall include its own URI in a Record-Route header field in order to ensure that it is included on the route of subsequent requests.

NOTE 3: One example of such a media feature tag is the isfocus media feature tag where the URI in the Contact header field is used by conference services to transport the temporary conference identity that can be used when rejoining an ongoing conference.

When the AS acting as a routeing B2BUA generates a response to a request, the AS shall insert a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the request, a type 3 "term-ioi" header field parameter and the "icid-value" header field parameter. The AS shall set the type 3 "term-ioi" header field parameter to a value that identifies the service provider from which the response is sent, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the previously received value of "icid-value" header field parameter in the request. Any values of "orig-ioi" or "term-ioi" header field parameter received in any response that is being forwarded are not used.

The AS shall transparently pass supported and unsupported signalling elements (e.g. SIP headers, SIP messages bodies), except signalling elements that are modified or deleted as part of the hosted service logic, or based on service provider policy.

If resource priority in accordance with RFC 4412 [116] is required for a dialog, then the AS shall include the Resource-Priority header field in all requests associated with that dialog. If priority is supported, the AS shall take into account that subsequent received SIP requests or responses within the same dialog or transaction may have added or changed the Resource-Priority header field or backwards indication.

The AS shall log all SIP requests and responses that contain a "logme" header field parameter in the SIP Session-ID header field if required by local policy.

5.7.5.2 Call initiation

5.7.5.2.1 Initial INVITE

When the AS acting as a Routeing B2BUA receives an initial INVITE request, the AS shall:

1) remove its own SIP URI from the topmost Route header field of the received INVITE request;

2) perform the AS specific functions. See 3GPP TS 23.218 [5];

3) if successful, generate and send a new INVITE request to establish a new dialog;

4) copy the remaining Route header field(s) unchanged from the received INVITE request to the new INVITE request;

5) copy the P-Asserted-Identity to the outgoing request;

6) if a Route header field is present, route the new INVITE request based on the topmost Route header field; and

NOTE 1: The topmost Route header field of the received INVITE request will contain the AS's SIP URI. The following Route header field will contain the SIP URI of the S-CSCF.

7) if no Route header field is present (e.g. the AS may be acting on behalf of a PSI):

a) insert a Route header field pointing either to the S-CSCF where the PSI is hosted or to the entry point of the home network of the PSI or to the transit function, if the AS is not able to resolve the next hop address by itself or the operator policy requires it; or

b) forward the originating request directly to the destination without involving any S-CSCF in the originating IM CN subsystem, if the AS is able to resolve the next hop address by itself, and the operator policy allows it.

NOTE 2: The address of the S-CSCF hosting the PSI can be obtained by querying the HSS on the Sh interface.

When the AS is acting as an Initiating B2BUA, the AS shall apply the procedures described in subclause 5.7.3 for any outgoing requests. The AS shall either set the "icid-value" header field parameter in the P-Charging-Vector header field to be the same as received or different.

NOTE 3: The AS can retrieve CDF and/or ODF adresses from HSS on Sh interface.

5.7.5.2.2 Subsequent requests

If the policy or service logic requires the AS to check whether the session is still alive, the AS shall send UPDATE requests periodically to the served UE. The UPDATE requests shall not contain SDP offer.

NOTE: The exact timing of sending the UPDATE requests is out of scope of this specification. Sending UPDATE requests too frequently can increase the load on the network and increase the probability of interactions delaying urgent requests (e.g., those related to session transfers). RFC 4028 [58] provides additional information on the problems caused by sending too frequest SIP "keep alives" and provides recommendations on suitable timer values to avoid such issues.

5.7.5.3 Call release

An AS may initiate a call release. See 3GPP TS 23.218 [5] for possible reasons. The AS shall simultaneously send the BYE request for both dialogs managed by the B2BUA.

5.7.5.4 Call-related requests

Void.

5.7.5.5 Further initial requests

When the AS is acting as an Initiating B2BUA the AS shall apply the procedures described in subclause 5.7.3 for the requests. The AS shall either set the "icid-value" header field parameter in the P-Charging-Vector header field to be the same as received or different. The AS may initiate any number of requests, per service logic needs.

5.7.5.6 Transcoding services invocation using third-party call control

An AS may invoke transcoding at an MRFC by the use of RFC 4117 [166], if the MRFC supports acting as the transcoding server described in RFC 4117 [166].

During the call setup, an AS may decide proactively to invoke transcoding when receiving an INVITE request, or reactively when the callee rejects the call setup using a 488 (Not Acceptable Here) response. To invoke transcoding using RFC 4117 [166], the AS shall act as a B2BUA between caller and callee and establish a third SIP dialogue towards the MRFC, supporting the transcoding as defined in subclause 6.6.

The SIP messages relating to the dialogue between AS and MRFC are sent either via the S-CSCF over the ISC and Mr interfaces, or directly over the Mr' interface.

5.7.6 Void

5.8 Procedures at the MRFC

5.8.1 General

Although the MRFC is acting as a UA, it is outside the scope of this specification how the MRFC associated addresses are made known to other entities.

For all SIP transactions identified:

- if priority is supported, as containing an authorised Resource-Priority header field, or, if such an option is supported, relating to a dialog which previously contained an authorised Resource-Priority header field;

the MRFC shall give priority over other transactions or dialogs. This allows special treatment of such transactions or dialogs. If priority is supported, the MGCF shall adjust the priority treatment of transactions or dialogs according to the most recently received authorized Resource-Priority header field or backwards indication value.

NOTE: This special treatment can include filtering, higher priority processing, routeing, call gapping. The exact meaning of priority is not defined further in this document, but is left to national regulation and network configuration.

When the MRFC sends any request or response (excluding CANCEL requests and responses) related to a dialog or standalone transaction, the MRFC may insert previously saved values into P-Charging-Vector header field before sending the message.

When the MRFC sends any request or response (excluding ACK requests and CANCEL requests and responses) related to a dialog or standalone transaction, the MRFC may insert previously saved values into P-Charging-Function-Addresses header fields before sending the message.

The MRFC shall use a GRUU referring to itself (as specified in RFC 5627 [93]) when inserting a contact address in a dialog establishing or target refreshing SIP message. This specification does not define how GRUUs are created by the MRFC; they can be provisioned by the operator or obtained by any other mechanism. A GRUU used by the MRFC when establishing a dialog shall remain valid for the lifetime of the dialog.

The MRFC shall handle requests addressed to its currently valid GRUUs when received outside of the dialog in which the GRUU was provided.

EXAMPLE: Upon receipt of an INVITE request addressed to a GRUU assigned to a dialog it has active, and containing a Replaces header field referencing that dialog, the MRFC will be able to establish the new call replacing the old one.

The MRFC shall log all SIP requests and responses that contain a "logme" header field parameter in the SIP Session-ID header field if required by local policy.

When sending a failure response to any received request, depending on operator policy, the MRFC may insert a Response-Source header field with an "fe" header field parameter constructed with the URN namespace "urn:3gpp:fe", the fe-id part of the URN set to "mrfc" and optionally an appropriate fe-param part of the URN set in accordance with subclause 7.2.17.

5.8.2 Call initiation

5.8.2.1 Initial INVITE

5.8.2.1.1 MRFC-terminating case

5.8.2.1.1.1 Introduction

The MRFC shall provide a P-Asserted-Identity header field in a response to the initial request for a dialog, or any response for a standalone transaction. It is a matter of network policy whether the MRFC expresses privacy according to RFC 3323 [33] with such responses.

When the MRFC receives an initial INVITE request, the MRFC shall store the values received in the P-Charging-Vector header field, e.g. "icid-value" header field parameter. The MRFC shall store the values received in the P-Charging-Function-Addresses header field.Based on local policy, the MRFC shall add an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier to an initial request.

If the MRFC receives an initial request with a P-Charging-Vector header field, the MRFC shall, based on local policy, store the "fe-identifier" header field parameter of the P-Charging-Vector header field.

When the MRFC receives a final response, the MRFC shall, based on local policy, include the stored "fe-identifier" header field parameter in the P-Charging-Vector header field and add its own address or identifier as an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header.

5.8.2.1.1.2 Tones and announcements

The MRFC can receive INVITE requests to set up a session to play tones and announcements. The MRFC acts as terminating UA in this case.

When the MRFC receives an INVITE request for a tone or announcement, the MRFC shall:

- send 100 (Trying) response.

5.8.2.1.1.3 Ad-hoc conferences

The MRFC can receive INVITE requests to set up an ad-hoc conferencing session (for example a multiparty call) or to add parties to the conference. The MRFC acts as terminating UA in this case.

When the MRFC receives an INVITE request for ad hoc conferencing, the MRFC shall:

- send 100 (Trying) response.

When the MRFC receives an INVITE request to add a party to an existing ad hoc conference (i.e. MRFC conference identifier), the MRFC shall:

- send 100 (Trying) response.

5.8.2.1.1.4 Transcoding

The MRFC may receive INVITE requests to set up transcoding between endpoints with incompatible codecs. The MRFC acts as terminating UA in this case.

When the MRFC receives an INVITE request for transcoding and a codec is supplied in SDP, the MRFC shall:

- send 100 (Trying) response.

When the MRFC receives an INVITE request with an indicator for transcoding but no SDP, the MRFC shall:

- send 183 (Session Progress) response with list of codecs supported by the MRFC/MRFP.

5.8.2.1.2 MRFC-originating case

The MRFC shall provide a P-Asserted-Identity header field in an initial request for a dialog, or any request for a standalone transaction. It is a matter of network policy whether the MRFC expresses privacy according to RFC 3323 [33] with such requests.

When an MRFC generates an initial request for a dialog or a request for a standalone transaction, the MRFC shall insert a P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17].

5.8.2.2 Subsequent requests

5.8.2.2.1 Tones and announcements

When the MRFC receives an ACK request for a session, this may be considered as an event to direct the MRFP to start the playing of a tone or announcement.

5.8.2.2.2 Transcoding

When the MRFC receives a PRACK request (in response to the 183 (Session Progress) response with an indicator for transcoding and codec supplied in SDP, the MRFC shall:

- after the MRFP indicates that the transcoding request is granted, send 200 (OK) response.

5.8.3 Call release

5.8.3.1 S-CSCF-initiated call release

5.8.3.1.1 Tones and announcements

When the MRFC receives a BYE request for a session, the MRFC directs the MRFP to stop the playing of a tone or announcement.

5.8.3.2 MRFC-initiated call release

5.8.3.2.1 Tones and announcements

When the MRFC has a timed session to play tones and announcements and the time expires, the MRFC shall:

- send a BYE request towards the UE.

When the MRFC is informed by the MRFP that tone or announcement resource has been released, the MRFC shall:

- send a BYE request towards the UE.

5.8.4 Call-related requests

5.8.4.1 ReINVITE

5.8.4.1.1 MRFC-terminating case

5.8.4.1.1.1 Ad-hoc conferences

The MRFC can receive reINVITE requests to modify an ad-hoc conferencing session (for example a multiparty call) for purposes of floor control and for parties to leave and rejoin the conference.

When the MRFC receives a reINVITE request, the MRFC shall:

- send 100 (Trying) response; and

- after the MRFP indicates that the conferencing request is granted, send 200 (OK) response. The MRFC may choose to send a 183 (Session Progress) response prior to the 200 (OK) response.

5.8.4.1.2 MRFC-originating case

Void.

5.8.4.2 REFER

5.8.4.2.1 MRFC-terminating case

Void.

5.8.4.2.2 MRFC-originating case

Void.

5.8.4.2.3 REFER initiating a new session

Void.

5.8.4.2.4 REFER replacing an existing session

Void.

5.8.4.3 INFO

Void.

5.8.5 Further initial requests

When the MRFC responds to an OPTIONS request with a 200 (OK) response, the MRFC may include a message body with an indication of the supported tones/announcement packages, DTMF capabilities, supported codecs and conferencing options of the MRFC/MRFP.

NOTE: As specified in RFC 6230 [146] an MRFC supporting the use of the control channel framework shall support the SYNC command to indicate the media control packages supported. Additionally each media control package should define an audit command for discovery of package capabilities (for example supported codecs and options).

5.8A Procedures at the MRB

For all SIP transactions identified:

- if priority is supported, as containing an authorised Resource-Priority header field, or, if such an option is supported, relating to a dialog which previously contained an authorised Resource-Priority header field;

the MRB shall give priority over other transactions or dialogs. This allows special treatment of such transactions or dialogs. If priority is supported, the MRB shall adjust the priority treatment of transactions or dialogs according to the most recently received authorized Resource-Priority header field or backwards indication value.

NOTE: This special treatment can include filtering, higher priority processing, routeing, call gapping. The exact meaning of priority is not defined further in this document, but is left to national regulation and network configuration.

The MRB shall log all SIP requests and responses that contain a "logme" header field parameter in the SIP Session-ID header field based on local policy.

5.9 Void

5.9.1 Void

5.10 Procedures at the IBCF

5.10.1 General

As specified in 3GPP TS 23.228 [7] border control functions may be applied between two IM CN subsystems or between an IM CN subsystem and other SIP-based multimedia networks based on operator preference. The IBCF may act both as an entry point and as an exit point for a network. If it processes a SIP request received from other network it functions as an entry point (see subclause 5.10.3) and it acts as an exit point whenever it processes a SIP request sent to other network (see subclause 5.10.2).

The functionalities of the IBCF are entry and exit point procedures as defined in subclause 5.10.2 and subclause 5.10.3 and additionally can include:

- network configuration hiding (as defined in subclause 5.10.4);
- application level gateway (as defined in subclause 5.10.5);
- transport plane control, i.e. QoS control (as defined in subclause 5.10.5);
- screening of SIP signalling (as defined in subclause 5.10.6);
- inclusion of an IWF if appropriate;
- media transcoding control (as defined in suclause 5.10.7);
- privacy protection (as defined in subclause 5.10.8);
- additional routeing functionality (as defined in Annex I); and
- invocation of an AS over the Ms reference point (as defined in subclause 5.10.10).

NOTE 1: The functionalities performed by the IBCF are configured by the operator, and it is network specific.

The IBCF shall log all SIP requests and responses that contain a "logme" header field parameter in the SIP Session-ID header field if required by local policy.

When an IBCF acting as an exit or an entry point receives a SIP request, the IBCF may reject the SIP request based on local policy by sending an appropriate SIP 4xx response.

NOTE 2: The local policy can take bilateral agreements between operators into consideration.

NOTE 3: Some SIP requests can be rejected by an AS instead of the IBCF according to local policy.

The IBCF, acting as B2BUA, which is located between visited network and home network shall preserve the dialog identifier, i.e. shall not change the Call-Id header field value, the "tag" header field parameter value of the From header field in any SIP INVITE request and any SIP response to the SIP INVITE request, and shall preserve the "tag" header field parameter value of the To header field, in any SIP response to the SIP INVITE request.

NOTE 4: The IBCF can identify whether it is located between visited network and home network based on local configuration or, if IBCF supports indicating traffic leg associated with a URI as specified in RFC 7549 [225], based on the value of the "iotl" SIP URI parameter.

If the IBCF has verified that an initial INVITE request is for a PSAP callback, then depending on local policy it may include a Priority header field with a "psap-callback" header field value in the INVITE request. If the IBCF included the Priority header field with a "psap-callback" header field value, if the IBCF supports priority verification using assertion of priority information as specified in subclause 3.1 and if required by operator policy, the IBCF shall add a Resource-Priority header field containing a namespace of "esnet" as defined in RFC 7135 [197] if not already present.

NOTE 5: The means for the IBCF to verify that a request is for a PSAP callback is outside the scope of this specification.

When receiving a dialog creating SIP request or a SIP stand-alone request and if an IBCF acting as an entry or exit point supports indicating the traffic leg as specified in RFC 7549 [225], the IBCF can identify the II-NNI traversal scenario as described in subclause 4.13 and make policy decisions based on the II-NNI traversal scenario type. If a received request contains more than one "iotl" SIP URI parameter the IBCF shall select one of the "iotl" SIP parameters in the received request in accordance with the RFC 7549 [225].

When sending a failure response to any received request, depending on operator policy, the IBCF may insert a Response-Source header field with an "fe" header field parameter constructed with the URN namespace "urn:3gpp:fe", the fe-id part of the URN set to "ibcf" and optionally an appropriate fe-param part of the URN set in accordance with subclause 7.2.17.

5.10.2 IBCF as an exit point

5.10.2.1 Registration

When IBCF receives a REGISTER request, the IBCF shall:

1) void;

2) if network topology hiding is required or IBCF is configured to perform application level gateway and/or transport plane control functionalities, then the IBCF shall add its own routeable SIP URI to the top of the Path header field; and

NOTE 1: The IBCF can include in the inserted SIP URI an indicator that identifies the direction of subsequent requests received by the IBCF i.e., from the S-CSCF towards the P-CSCF, to identify the UE-terminating case. The IBCF can encode this indicator in different ways, such as, e.g., a unique parameter in the URI, a character string in the username part of the URI, or a dedicated port number in the URI.

NOTE 2: Any subsequent request that includes the direction indicator (in the Route header field) or arrives at the dedicated port number, indicates that the request was sent by the S-CSCF towards the P-CSCF.

3) select an entry point of the home network and forward the request to that entry point.

   If the selected entry point:

   - does not respond to the REGISTER request and its retransmissions by the IBCF; or

   - sends back a 3xx response or 480 (Temporarily Unavailable) response to a REGISTER request;

   the IBCF shall select a new entry point and forward the REGISTER request to that entry point.

NOTE 3: The list of the entry points can be either obtained as specified in RFC 3263 [27A] or provisioned in the IBCF. The entry point can be an IBCF or an I-CSCF.

   If the IBCF fails to forward the REGISTER request to any entry point, the IBCF shall send back a 504 (Server Time-Out) response to the P-CSCF, in accordance with the procedures in RFC 3261 [26].

5.10.2.1A General

For all SIP transactions identified:

- if priority is supported, as containing an authorised Resource-Priority header field or a temporarily authorised Resource-Priority header field, or, if such an option is supported, relating to a dialog which previously contained an authorised Resource-Priority header field;

the IBCF shall give priority over other transactions or dialogs. This allows special treatment of such transactions or dialogs. If priority is supported, the IBCF shall adjust the priority treatment of transactions or dialogs according to the most recently received authorized Resource-Priority header field or backwards indication value.

NOTE 1: The special treatment can include filtering, higher priority processing, routeing, call gapping. The exact meaning of priority is not defined further in this document, but is left to national regulation and network configuration.

Based on local policy, the IBCF acting as an exit point shall add in responses in the P-Charging-Vector header field a "transit-ioi" header field parameter with an entry which identifies the operator network which the response is transitting or with a void entry.

Based on local policy the IBCF shall delete or void in responses in the P-Charging-Vector header field any received "transit-ioi" header field parameter value.

If an IBCF in the originating visited network, supporting barring of premium numbers when roaming, receives a request to be sent towards the originating home network and the request is originated from a roaming UE and the Request-URI contains an E.164 number encoded as described in subclause 5.1.2A.1.2 which the IBCF is able to identify as a premium rate number in the country of the served network, the IBCF shall, based on local policy, add the "premium-rate" tel URI parameter specified in subclause 7.2A.17 set to a value "information" or "entertainment" as appropriate.

NOTE 2: The feature barring of premium numbers when roaming can be implemented in the P-CSCF or an IBCF of the visited network. Local policy ensures that the feature is only activiated in one of the two.

NOTE 3: If the visited network supports indicating traffic leg as specified in RFC 7549 [225] the above request includes the "iotl" SIP URI parameter with the value "visitedA-homeA" in the bottommost Route header field.

5.10.2.2 Initial requests

Upon receipt of:

- an initial request for a dialog;
- a request for a standalone transaction, except the REGISTER method; or
- a request for an unknown method that does not relate to an existing dialog;

the IBCF shall:

1) if the request is an INVITE request, respond with a 100 (Trying) provisional response;

1A) remove its own SIP URI from the topmost Route header field;

2) if the request is an INVITE request and the IBCF is configured to perform application level gateway and/or transport plane control functionalities, save the Contact, CSeq and Record-Route header field values received in the request such that the IBCF is able to release the session if needed;

2A) If the request is a SUBSCRIBE and the IBCF does not need to act as B2BUA, based on operator policy, the IBCF shall determine whether or not to retain, for the related subscription, the SIP dialog state information and the duration information;

NOTE 1: The event package name can be taken into account to decide whether or not the SIP dialog state and the subscription duration information needs to be retained.

NOTE 2: The IBCF needs to insert its own URI in Record-Route of the initial SUBSCRIBE request and all subsequent NOTIFY requests if it decides to retain the SIP dialog state information.

2B) if the request is an initial request for a dialog and local policy requires the application of IBCF capabilities in subsequent requests, perform record route procedures as specified in RFC 3261 [26];

3) void;

4) void;

5) void;

5A) if the recipient of the request is understood from configured information to always send and receive private network traffic from this source, remove the P-Private-Network-Indication header field containing the domain name associated with that saved information;

6) store the values from the P-Charging-Function-Addresses header field, if present;

7) if the request is an initial request and "fe-identifier" header field parameter of P-Charging-Vector header field is applied in the operator domain;

   - store the "fe-identifier" header field parameter in the P-Charging-Vector header field; and

   - remove the "fe-identifier" header field parameter from the P-Charging-Vector header field;

8) remove some of the parameters from the P-Charging-Vector header field or the header field itself, depending on operator policy, if present;

9) remove the P-Charging-Function-Addresses header fields, if present; and

10) remove the Via "received-realm" header field parameter, as defined in RFC 8055 [208], if present, prior to forwarding the request;

and forward the request according to RFC 3261 [26].

NOTE 3: If IBCF processes a request without a pre-defined route (e.g. the subscription to reg event package originated by the P-CSCF), the next-hop address can be either obtained as specified in RFC 3263 [27A] or be provisioned in the IBCF.

When the IBCF receives an INVITE request, the IBCF may require the periodic refreshment of the session to avoid hung states in the IBCF. If the IBCF requires the session to be refreshed, the IBCF shall apply the procedures described in RFC 4028 [58] clause 8.

NOTE 4: Requesting the session to be refreshed requires support by at least one of the UEs. This functionality cannot automatically be granted, i.e. at least one of the involved UEs needs to support it.

When receiving a response to the initial request with a P-Charging-Vector header field, the IBCF acting as an exit point shall, if "fe-identifier" header field parameter of P-Charging-Vector header field is applied in the operator domain:

- remove any received "fe-identifier" header field parameter from the P-Charging-Vector header field; and

- add the "fe-identifier" header field parameter stored from the initial request to the P-Charging-Vector header field and add its own address or identifier as an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field.

With the exception of 305 (Use Proxy) responses, the IBCF shall not recurse on 3xx responses.

5.10.2.3 Subsequent requests

Upon receipt of a subsequent request, the IBCF shall:

1) if the request is an INVITE request, respond with a 100 (Trying) provisional response;

1A) if the request is a NOTIFY request with the Subscription-State header field set to "terminated" and the IBCF has retained the SIP dialog state information for the associated subscription, once the NOTIFY transaction is terminated, the IBCF can remove all the stored information related to the associated subscription;

2) if the request is a target refresh request and the IBCF is configured to perform application level gateway and/or transport plane control functionalities, save the Contact and CSeq header field values received in the request such that the IBCF is able to release the session if needed; and

3) if the subsequent request is other than a target refresh request (including requests relating to an existing dialog where the method is unknown) and the IBCF is configured to perform application level gateway and/or transport plane control functionalities, save the Contact and CSeq header field values received in the request such that the IBCF is able to release the session if needed;

and forwards the request, based on the topmost Route header field, in accordance with the procedures of RFC 3261 [26].

5.10.2.4 IBCF-initiated call release

If the IBCF provides transport plane control functionality and receives an indication of a transport plane related error the IBCF may:

1) generate a BYE request for the terminating side based on information saved for the related dialog; and

2) generate a BYE request for the originating side based on the information saved for the related dialog.

NOTE: Transport plane related errors can be indicated from e.g. TrGW, or PCRF. The protocol for indicating transport plane related errors to the IBCF is out of scope of this specification.

Upon receipt of the 2xx responses for both BYE requests, the IBCF shall release all information related to the dialog and the related multimedia session.

5.10.3 IBCF as an entry point

5.10.3.1 Registration

When IBCF receives a REGISTER request, the IBCF shall:

1) verify if it arrived from a trusted domain or not. If the request arrived from an untrusted domain, respond with 403 (Forbidden) response;

NOTE 1: The IBCF can find out whether the request arrived from a trusted domain or not, from the procedures described in 3GPP TS 33.210 [19A].

2) if network topology hiding, or screening of SIP signalling, is required or IBCF is configured to perform application level gateway and/or transport plane control functionalities, add its own routeable SIP URI to the top of the Path header field; and

NOTE 2: The IBCF can include in the inserted SIP URI an indicator that identifies the direction of subsequent requests received by the IBCF i.e., from the S-CSCF towards the P-CSCF, to identify the UE-terminating case. The IBCF can encode this indicator in different ways, such as, e.g., a unique parameter in the URI, a character string in the username part of the URI, or a dedicated port number in the URI.

NOTE 3: Any subsequent request that includes the direction indicator (in the Route header field) or arrives at the dedicated port number, indicates that the request was sent by the S-CSCF towards the P-CSCF.

3) If IBCF is colocated with an I-CSCF, or it has a preconfigured I-CSCF to be contacted, forward the request to that I-CSCF. Otherwise select an I-CSCF and forward the request to that I-CSCF.

NOTE 5: The selection of an I-CSCF can lead to additional delays.

If the selected I-CSCF:

- does not respond to the REGISTER request and its retransmissions by the IBCF; or

- sends back a 3xx response or 480 (Temporarily Unavailable) response to a REGISTER request;

the IBCF shall select a new I-CSCF and forward the REGISTER request to that I-CSCF.

NOTE 4: The list of the I-CSCFs can be either obtained as specified in RFC 3263 [27A] or provisioned in the IBCF.

If the IBCF fails to forward the REGISTER request to any I-CSCF, the IBCF shall send back a 504 (Server Time-Out) response towards the P-CSCF, in accordance with the procedures in RFC 3261 [26].

5.10.3.1A General

For all SIP transactions identified:

- if priority is supported (NOTE 1), as containing an authorised Resource-Priority header field or a temporarily authorised Resource-Priority header field, or, if such an option is supported, relating to a dialog which previously contained an authorised Resource-Priority header field;

the IBCF shall give priority over other transactions or dialogs. This allows special treatment of such transactions or dialogs. If priority is supported, the IBCF shall adjust the priority treatment of transactions or dialogs according to the most recently received authorized Resource-Priority header field or backwards indication value.

NOTE 1: For an INVITE request, various mechanisms can be applied to recognize the need for priority treatment (e.g., based on the dialled digits). The exact mechanisms are left to national regulation and network configuration.

Based on the alternative mechanism to recognize the need for priority treatment, the IBCF shall insert the temporarily authorised Resource-Priority header field with appropriate namespace and priority value in the INVITE request.

NOTE 2: The special treatment can include filtering, higher priority processing, routeing, call gapping. The exact meaning of priority is not defined further in this document, but is left to national regulation and network configuration.

Based on local policy, the IBCF acting as an entry point shall add in requests in the P-Charging-Vector header field a "transit-ioi" header field parameter with an entry which identifies the operator network which the request is transitting or with a void entry.

Based on local policy the IBCF shall delete or void in requests in the P-Charging-Vector header field any received "transit-ioi" header field parameter value.

NOTE 3: Only one "transit-ioi" header field parameter entry is added per transit network.

5.10.3.2 Initial requests

Upon receipt of:

- an initial request for a dialog;
- a request for a standalone transaction except the REGISTER request; or
- a request for an unknown method that does not relate to an existing dialog;

the IBCF shall verify whether the request is arrived from a trusted domain or not. If the request arrived from an untrusted domain, then the IBCF shall:

- if the topmost Route header field of the request contains the "orig" parameter, respond with 403 (Forbidden) response.

  Otherwise,

  - remove all P-Charging-Vector header fields and all P-Charging-Function-Addresses header fields the request may contain; and
  - remove all Feature-Caps header fields, if present.

Upon receipt of:

- an initial request for a dialog;
- a request for a standalone transaction except the REGISTER request; or
- a request for an unknown method that does not relate to an existing dialog;

the IBCF shall:

1) if the request is an INVITE request, then respond with a 100 (Trying) provisional response;

1A) if a P-Private-Network-Indication header field is included in the request, check whether the configured information allows the receipt of private network traffic from this source. If private network traffic is allowed, the IBCF shall check whether the received domain name in any included P-Private-Network-Indication header

field in the request is the same as the domain name associated with that configured information. If private network traffic is not allowed, or the received domain name does not match, then the IBCF shall remove the P-Private-Network-Indication header field;

1B) if the initiator of the request is understood from configured information to always send and receive private network traffic from this source, insert a P-Private-Network-Indication header field containing the domain name associated with that configured information;

1C) remove its own SIP URI from the topmost Route header field;

2) if the request is an INVITE request and the IBCF is configured to perform application level gateway and/or transport plane control functionalities, then the IBCF shall save the Contact, CSeq and Record-Route header field values received in the request such that the IBCF is able to release the session if needed;

2A) If the request is a SUBSCRIBE and the IBCF does not need to act as B2BUA, based on operator policy, the IBCF shall determine whether or not to retain, for the related subscription, the SIP dialog state information and the duration information;

NOTE 1: The event package name can be taken into account to decide whether or not the SIP dialog state and the subscription duration information needs to be retained.

NOTE 2: The IBCF needs to insert its own URI in Record-Route of the initial SUBSCRIBE request and all subsequent NOTIFY requests if it decides to retain the SIP dialog state information.

2B) if the request is an initial request for a dialog and local policy requires the application of IBCF capabilities in subsequent requests, perform record route procedures as specified in RFC 3261 [26];

2C) if

- the request is an initial request for a dialog, or a standalone request, and
- the Request-URI contains an emergency service URN, i.e. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69] and
- a P-Private-Network-Indication valid within the trust domain is not included, and
- based on local policy, no Route header field is remaining after step 1C) was executed,

then include a topmost Route header field set to the URI associated with an E-CSCF;

2D) if the network uses the Resource-Priority header field to control the priority of emergency calls, the IBCF shall add a Resource-Priority header field containing a namespace of "esnet" as defined in RFC 7135 [197];

3) void;

4) if IBCF receives an initial request for a dialog or standalone transaction, that contains a single Route header field pointing to itself, and it is co-located with an I-CSCF, or it has a preconfigured I-CSCF to be contacted, then forward the request to that I-CSCF. Otherwise select an I-CSCF and forward the request to that I-CSCF. If the single Route header field of the request contains the "orig" parameter, the IBCF shall insert the "orig" parameter to the URI of the I-CSCF;

NOTE 3: The selection of an I-CSCF can lead to additional delays.

5) if the request does not contain a Route header field or if it contains one or more Route header fields where the topmost Route header field does not contain the "orig" parameter, optionally – based on operator policy – append the "orig" parameter to the URI in the topmost Route header field of the next request sent from the IBCF to an entity of the IM CN subsystem for which it is an entry point;

NOTE 4: The appending of an "orig" parameter to the URI in the topmost Route header field enables an IM CN subsystem to perform originating services to the network that originated the initial request. The appending can be dependent on the network that originated the initial request as determined by e.g. origin IP address of the received request, etc.

6) if services that require knowledge of the adjacent network are provided within the network for which the IBCF is acting as an entry point, based on operator policy, insert a Via "received-realm" header field parameter, as defined in RFC 8055 [208];

6A) if the IBCF, acting as an entry point to a terminating visited network, PCRF based P-CSCF restoration procedures,

- the request contains a topmost Route header field pointing to a P-CSCF, and
- the IBCF considers the P-CSCF is in a non-working state,

remove all entries in the Route header field and add a Route header field set to the URI associated with an alternative P-CSCF;

NOTE 5: How the SIP URI of the alternative P-CSCF is obtained by the IBCF is implementation dependent. The IBCF can make sure that selected P-CSCF support the PCRF based P-CSCF restoration procedures based on local configuration.

NOTE 6: It is implementation dependent as to how the IBCF determines the P-CSCF is in non-working state.

7) if the initiator of the request is understood to always send and receive private network traffic:

NOTE 7: The IBCF can identify that a request is received from a source that always sends or receives private traffic by evaluating the TLS session or by other means.

a) add the identity of the initiator in a P-Served-User header field as defined in RFC 5502 [133] as a SIP URI identifying the initiator; and

NOTE 8: The IBCF can retrieve the identity of the initiator from the subjectCommonName (CN) if it is not present in the subjectAltName in the certificates during the TLS session setup in accordance with the procedures of RFC 5280 [213] or by other means.

b) if not already appended in 4) or 5) above, append the "orig" parameter to the URI in the topmost Route header field of the request sent from the IBCF to the entity of the IM CN subsystem for which it is an entry point;

8) if the request is an initial request and "fe-identifier" header field parameter of P-Charging-Vector header field is applied in the operator domain:

- remove any received "fe-identifier" header field parameter from the P-Charging-Vector header field; and
- add an "fe-addr" element in an "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier; and

9) if the IBCF supports calling number verification using signature verification and attestation information as described in subclause 3.1 and no Identity header field is received in an initial INVITE or MESSAGE request, based on local policy insert:

a) an Attestation-Info header field specified in subclause 7.2.18 set to a value "C", specified in RFC 8588 [261]; and

b) an Origination-Id header field specified in subclause 7.2.19 containing an "origid" claim as specified in RFC 8588 [261] set to a value identifying the source of the request;

and forward the request according to RFC 3261 [26].

When the IBCF receives an INVITE request, the IBCF may require the periodic refreshment of the session to avoid hung states in the IBCF. If the IBCF requires the session to be refreshed, the IBCF shall apply the procedures described in RFC 4028 [58] clause 8.

NOTE 9: Requesting the session to be refreshed requires support by at least one of the UEs. This functionality cannot automatically be granted, i.e. at least one of the involved UEs needs to support it.

If the serving network supports HSS based P-CSCF restoration as specified in 3GPP TS 23.380 [7D], the IBCF is acting as an entry point to a terminating visited network and the IBCF does not receive any response within a configured time:

NOTE 10:The configurable time needs to be less than timer B and timer F.

1) to an initial INVITE request, then if the Route header field contains only one entry the IBCF shall in the 408 (Request Timeout) response include a Restoration-Info header field specified in subclause 7.2.11 containing the value "noresponse"; and

2) to an initial non-INVITE request for a dialog, a standalone transaction or an unknown method that does not relate to an existing dialog, then if the Route header field contains only one entry the IBCF shall send a 504 (Server Time-out) response include a Restoration-Info header field specified in subclause 7.2.11 containing the value "noresponse".

NOTE 11: The IBCF determines if it is acting as an entry point to a terminating visited network based on configuration or other data in the incoming request, or the "iotl" SIP URI parameter specified in RFC 7549 [225].

NOTE 12: If there is only one entry in the Route header field it represents either an MSC server or a P-CSCF. The S-CSCF will use the g.3gpp.ics media feature tag to determine if it is the MSC server or the P-CSCF.

When the IBCF receives a response to an initial request (e.g. 183 or 2xx), the IBCF shall:

1) store the values from the P-Charging-Function-Addresses header field, if present;

2) remove the "fe-identifier" header field parameter from the P-Charging-Vector header field, if present; and

3) remove the P-Charging-Function-Addresses header field prior to forwarding the message;

With the exception of 305 (Use Proxy) responses, the IBCF shall not recurse on 3xx responses.

5.10.3.3 Subsequent requests

Upon receipt of a subsequent request, the IBCF shall:

1) if the request is an INVITE request, then respond with a 100 (Trying) provisional response;

1A) if the request is a NOTIFY request with the Subscription-State header field set to "terminated" and the IBCF has retained the SIP dialog state information for the associated subscription, once the NOTIFY transaction is terminated, the IBCF can remove all the stored information related to the associated subscription;

2) if the request is a target refresh request and the IBCF is configured to perform application level gateway and/or transport plane control functionalities, then the IBCF shall save the Contact and CSeq header field values received in the request such that the IBCF is able to release the session if needed;

3) if the subsequent request is other than a target refresh request (including requests relating to an existing dialog where the method is unknown) and the IBCF is configured to perform application level gateway and/or transport plane control functionalities, then the IBCF shall save the Contact and CSeq header field values received in the request such that the IBCF is able to release the session if needed;

4) void;

5) if the request is received from an untrusted domain, remove all Feature-Caps header fields, if present; and

6) if the subsequent request is received from an entity outside the trust domain, then the IBCF shall remove a P-Charging-Vector header field, if present;

and forwards the request, based on the topmost Route header field, in accordance with the procedures of RFC 3261 [26].

5.10.3.4 IBCF-initiated call release

If the IBCF provides transport plane control functionality and receives an indication of a transport plane related error the IBCF may:

1) generate a BYE request for the terminating side based on information saved for the related dialog; and

2) generate a BYE request for the originating side based on the information saved for the related dialog.

NOTE: Transport plane related errors can be indicated from e.g. TrGW or PCRF. The protocol for indicating transport plane related errors to the IBCF is out of scope of this specification.

Upon receipt of the 2xx responses for both BYE requests, the IBCF shall release all information related to the dialog and the related multimedia session.

5.10.3.5 Abnormal cases

When the IBCF acting as an entry point in the originating home network is unable to forward a SIP request, as determined by one of the following:

NOTE 1: If IBCF supports indicating traffic leg associated with a URI as specified in RFC 7549 [225], the IBCF can determine that IBCF is acting as an entry point in the originating home network by inspecting the value of the "iotl" SIP URI parameter, if an "iotl" SIP URI is included in the SIP request.

- there is no response to the SIP request and its retransmissions by the IBCF; or
- by unspecified means available to the IBCF;

and:

- the IBCF supports S-CSCF restoration procedures;

then the IBCF:

1) shall reject the request by returning a 504 (Server Time-out) response; and
2) shall include in the 504 (Server Time-out) response:
   - a Content-Type header field with the value set to associated MIME type of the 3GPP IM CN subsystem XML body as described in subclause 7.6.1;
   - a P-Asserted-Identity header field set to the value of the SIP URI of the IBCF included in the Path header field during the registration (see subclause 5.10.3.1); and
   - a 3GPP IM CN subsystem XML body containing:
     a) an <ims-3gpp> element with the "version" attribute set to "1" and with an <alternative-service> child element, set to the parameters of the alternative service:
        i) a <type> child element, set to "restoration" (see table 7.6.2) to indicate that restoration procedures are supported;
        ii) a <reason> child element, set to an operator configurable reason; and
        iii) an <action> child element, set to "initial-registration" (see table 7.6.3).

NOTE 2: These procedures do not prevent the usage of unspecified reliability or recovery techniques above and beyond those specified in this subclause.

5.10.4 THIG functionality in the IBCF

5.10.4.1 General

NOTE 1: THIG functionality is performed in I-CSCF in Release-5 and Release-6 and is compatible with the procedures specified in this subclause.

The following procedures shall only be applied if network topology hiding is required by the network. The network requiring network topology hiding is called the hiding network.

NOTE 2: Requests and responses are handled independently therefore no state information is needed for that purpose within an IBCF.

The IBCF shall apply network topology hiding to all header fields which reveal topology information, such as Via, Route, Record-Route, Service-Route, and Path.

Upon receiving an incoming REGISTER request for which network topology hiding has to be applied and which includes a Path header field, the IBCF shall add the routeable SIP URI of the IBCF to the top of the Path header field. The IBCF may:

1) include in the inserted SIP URI an indicator that identifies the direction of subsequent requests received by the IBCF i.e., from the S-CSCF towards the P-CSCF, to identify the UE-terminating case. The IBCF may encode this indicator in different ways, such as, e.g., a unique parameter in the URI, a character string in the username part of the URI, or a dedicated port number in the URI; and

2) if:

   a) IBCF supports indicating traffic leg associated with a URI as specified in RFC 7549 [225]; and

   b) if the SIP URI in the bottommost hidden Path header field contains an "iotl" SIP URI parameter;

   then append an "iotl" SIP URI parameter with the same value to its own SIP URI in the Path header field.

NOTE 3: Any subsequent request that includes the direction indicator (in the Route header field) or arrives at the dedicated port number, indicates that the request was sent by the S-CSCF towards the P-CSCF.

Upon receiving a 200 (OK) response to the REGISTER request and:

1. if the IBCF is located in the visited network; and

2. if the IBCF applied topology hiding on the Path header field contained in the REGISTER request;

the IBCF shall:

1. perform a decryption procedure, as described in subclause 5.10.4.3, on the received Path header field; and

2. insert a "+g.3gpp.thig-path" Feature-Caps header field parameter, as defined in subclause 7.9A.9, set to the same IBCF's SIP URI value as included in the Path header field of the REGISTER request sent to the home network.

NOTE 4: If a decryption of the Path header field contained in a 200 (OK) response on REGISTER request is not done then the UE will not perform restoration procedures if the P-CSCF rejects an initial request for a dialog or a request for a standalone transaction with a 504 (Server Time-out) response since there will be a mismatch between a SIP URI in the P-Asserted-Identity header field received in a valid 504 (Server Time-out) response and the SIP URIs the UE received in the Path header field.

Upon receiving an incoming initial request for which network topology hiding has to be applied and which includes a Record-Route header field, the IBCF shall add its own routeable SIP URI to the top of the Record-Route header field.

Upon receiving a 200 (OK) response to a REGISTER request for which network topology hiding has to be applied and which includes an URI identifying the IBCF in the topmost Service-Route header field and:

1) if IBCF supports indicating the traffic leg associated with a URI as specified in RFC 7549 [225]; and

2) if an "iotl" parameter is included in the bottommost SIP URI;

then append an "iotl" SIP URI parameter with the same value to its own SIP URI in the Service-Route header field.

When the home network IBCF receives a 504 (Server Time-out) response containing a P-Asserted-Identity header field set to the value of the S-CSCF's SIP URI for a roaming UE and if the home network is a hiding network then the IBCF shall replace the received P-Asserted-Identity header field with the P-Asserted-Identity header field set to the value of the own SIP URI.

NOTE 5: By provision or by obtaining from the corresponding request's Route header field, the IBCF deduces whether the received value of the P-Asserted-Identity header field in the 504 (Server Time-out) response is the value of S-CSCF's SIP URI.

5.10.4.2 Encryption for network topology hiding

Upon receiving an outgoing request/response from the hiding network the IBCF shall perform the encryption for network topology hiding purposes, i.e. the IBCF shall:

0) if applying encryption procedure on the Service-Route header field, exclude from the Service-Route header field the entry corresponding to its own SIP URI and use the remaining header field values which were added by one or more specific entity of the hiding network as input to encryption and skip item 1) below;

NOTE 1: In accordance with the procedures described in RFC 3608 [38], the IBCF does not insert its own routable SIP URI to the Service-Route header field i.e. the SIP URI identifying the IBCF in the topmost entry of the Service-Route header field is inserted by the S-CSCF. However this entry is excluded from encryption and will stay in the topmost entry of the Service-Route header field i.e. before the topmost encrypted entry.

1) use the whole header field values which were added by one or more specific entity of the hiding network as input to encryption, besides the UE entry;

2) not change the order of the header fields subject to encryption when performing encryption;

3) use for one encrypted string all received consecutive header field entries subject to encryption, regardless if they appear in separate consecutive header fields or if they are consecutive entries in a comma separated list in one header field;

4) construct a hostname that is the encrypted string in a way that allows to identify the encrypting network's name (i.e. the IBCF network);

NOTE 2: This is to allow the IBCF to identify that itself has encrypted the string when subsequently receiving the encrypted string. The details of encoding the encrypting networks's name are not specified as the IBCF is the creator and consumer of this value. This is needed because header field parameters (like "tokenized-by") are not required to be preserved when creating a route set.

5) append a "tokenized-by" header field parameter and set it to the value of the encrypting network's name, after the constructed hostname;

6) form one valid entry for the specific header field out of the resulting NAI, e.g. prepend "SIP/2.0/UDP" for Via header fields or "sip:" for Path, Service-Route, Route and Record-Route header fields;

7) if the IBCF encrypted an entry in the Route header field, then it also inserts its own URI before the topmost encrypted entry; and

8) if the IBCF encrypted an entry in the Via header field, then it also inserts its own URI before the topmost encrypted entry.

NOTE 3: Even if consecutive entries of the same network in a specific header field are encrypted, they will result in only one encrypted header field entry. For example:

```
Via: SIP/2.0/UDP ibcf1.home1.net;lr,
                 SIP/2.0/UDP Token(  SIP/2.0/UDP scscf1.home1.net;lr,
                         SIP/2.0/UDP pcscf1.home1.net;lr)@home1.net;
                             tokenized-by=home1.net,
                 SIP/2.0/UDP [5555::aaa:bbb:ccc:ddd]
```

NOTE 4: If multiple entries of the same network are within the same type of header fields, but they are not consecutive, then these entries will be tokenized to different strings. For example:

```
Record-Route:   sip:ibcf1.home1.net;lr,
    sip:Token(sip:scscf1.home1.net;lr)@home1.net;tokenized-by=home1.net,
    sip:as1.foreign.net;lr,
    sip:Token(sip:scscf1.home1.net;lr,
     sip:pcscf1.home1.net;lr)@home1.net;tokenized-by=home1.net
```

NOTE 5: If request will return to the hiding network (e.g. after visiting an AS), then the URI of IBCF is inserted. For example:

```
Route: sip:as1.foreign.net;lr,
    sip:ibcf1.home1.net;lr,
    sip:Token(sip:scscf1.home1.net;lr);tokenized-by=home1.net
```

5.10.4.3 Decryption for network topology hiding

Upon receiving and incoming requests/response to the hiding network the IBCF shall perform the decryption for network topology hiding purposes, i.e. the IBCF shall:

1) identify hostnames encrypted by the network this IBCF belongs to within all header fields of the incoming message;

2) use those hostnames that carry the identification of the hiding network as input to decryption;

3) use as encrypted string the hostname which follows the sent-protocol (for Via header fields, e.g. "SIP/2.0/UDP") or the URI scheme (for Path, Route and Record-Route header fields, e.g. "sip:");

4) replace all content of the received header field which carries encrypted information with the entries resulting from decryption.

EXAMPLE: An encrypted entry to a Via header field that looks like:

```
Via:SIP/2.0/UDP Token(SIP/2.0/UDP scscf1.home1.net;lr,
        SIP/2.0/UDP pcscf1.home1.net;lr);tokenized-by=home1.net
```

will be replaced with the following entries:

```
Via: SIP/2.0/UDP scscf1.home1.net;lr, SIP/2.0/UDP pcscf1.home1.net;lr
```

NOTE: Motivations for these decryption procedures are e.g. to allow the correct routeing of a response through the hiding network, to enable loop avoidance within the hiding network, or to allow the entities of the hiding network to change their entries within e.g. the Record-Route header field.

5.10.5 IMS-ALG functionality in the IBCF

The IBCF shall only apply the following procedures if application level gateway functionality is required by the network.

The IBCF acts as a B2BUA when it performs IMS-ALG functionality. As an IMS-ALG, the IBCF will internally map the message header fields between the two dialogs that it manages. It is responsible for correlating the dialog identifiers and will decide when to simply translate a message from one dialog to the other, or when to perform other functions. The IBCF, although acting as a UA, does not initiate any registration of its associated addresses. These are assumed to be known by peer-to-peer arrangements within the IM CN subsystem.

An IBCF may replace a contact address with a URI of its own when the contact address in the incoming message is not a GRUU. In all other cases the IBCF shall use a GRUU (e.g when the contact address is an IP address).

The IBCF shall transparently forward a received Contact header field when the Contact header field contains a GRUU or a media feature tag is included indicating a capability for which the Contact URI can be used by the remote party. When transparently forwarding a received Contact header field of a dialog-forming request, the IBCF shall include its own URI in a Record-Route header field in order to ensure that it is included on the route of subsequent requests.

NOTE: One example of such a media feature tag is the isfocus media feature tag used by conference services to transport the temporary conference identity that can be used when rejoining an ongoing conference.

The internal function of the IBCF as an IMS-ALG is defined in 3GPP TS 29.162 [11A].

If the IBCF receives a message with a body part for a UE from an S-CSCF, and:

- if the body part is the 3GPP IM CN subsystem XML body (as indicated by the Content-Type header field, see subclause 7.6) and the body part is not optional (as indicated by the (absence of the) Content-Disposition header field); or

- if a header field that describes the body is present and the header field's value is not understood (e.g. Content-Language header field or Content-Encoding header field);

then the IBCF shall transparently forward the message with the body part and the header field(s) that describe the body part.

5.10.6 Screening of SIP signalling

5.10.6.1 General

The IBCF may act as a B2BUA when it performs screening of SIP signalling functionality. In this case the B2BUA behaviour of the IBCF shall comply with the description given in subclause 5.10.5 for the IMS-ALG functionality.

NOTE: Many header fields are intended for end-to-end operation; removal of such header fields will impact the intended end-to-end operation between the end users. Additionally the IM CN subsystem does not preclude security mechanisms covering SIP header fields; any such removal can prevent validation of all header fields covered by the security mechanism.

5.10.6.2 IBCF procedures for SIP header fields

If specified by local policy rules, the IBCF may omit or modify any received SIP header fields prior to forwarding SIP messages, with the following exceptions.

As a result of any screening policy adopted, the IBCF should not modify at least the following header fields which would cause misoperation of the IM CN subsystem:

- Authorization; and
- WWW-Authenticate.

Where the IBCF appears in the path between the UE and the S-CSCF, some header fields are involved in the registration and authentication of the user. As a result of any screening policy adopted as part of normal operation, e.g. where the request or response is forwarded on, the IBCF should not modify as part of the registration procedure at least the following header fields:

- Path; and
- Service-Route.

NOTE 1: If the IBCF modifies SIP information elements (SIP header fields, SIP message bodies) other than as specified by SIP procedures (e.g., RFC 3261 [26]) caution needs to be taken that SIP functionality (e.g., routeing using Route, Record-Route and Via) is not impacted in a way that could create interoperability problems with networks that assume that this information is not modified.

NOTE 2: Where operator requirements can be achieved by configuration hiding, then these procedures can be used in preference to screening.

The IBCF may add, remove, or modify, the P-Early-Media header field within forwarded SIP requests and responses according to procedures in RFC 5009 [109].

NOTE 3: The IBCF can use the P-Early-Media header field for the gate control procedures,by through-connect control as described in 3GPP TS 29.162 [11A]. In the presence of early media for multiple dialogs due to forking, if the IBCF is able to identify the media associated with a dialog, (i.e., if symmetric RTP is used by the UE and the IBCF can use the remote SDP information to determine the source of the media) the IBCF can selectively open the gate corresponding to an authorized early media flow for the selected media.

The IBCF may add, or omit any P-Asserted-Identity header fields prior to forwarding SIP messages according to local policy.

NOTE 4: The IBCF can use the P-Asserted-Identy header field to trigger identity specific procedures in subsequent entities, e.g. for malicious call identification. As an example, a P-Asserted-Identity header field will be deleted and a new P-Asserted-Identity header field with operator specific content will be added to the outgoing request, if the request was received from a network which cannot support the deletion of INFO request which is needed for the support of the malicious call identification service.

When the IBCF, located in the home network, receives a SIP request from another entity within the same trust domain, the IBCF may police the ICSI value contained in the P-Asserted-Service header field.

5.10.6.3 IBCF procedures for SIP message bodies

If the IBCF acts as a B2BUA, and the IBCF receives a message with a body part for a UE from an S-CSCF, and:

- if the body part is the 3GPP IM CN subsystem XML body (as indicated by the Content-Type header field, see subclause 7.6) and the body part is not optional (as indicated by the (absence of the) Content-Disposition header field); or

- if a header field that describes the body is present and the header field's value is not understood (e.g. Content-Language header field or Content-Encoding header field),

then the IBCF shall transparently forward the message with the body part and the header field(s) that describe the body part.

If IP address translation (NA(P)T or IP version interworking) occurs on the user plane, the IBCF shall modify SDP according to subclause 6.7.1;

Additionally, the IBCF may take the followings action upon SIP message bodies:

1) examine the length of a SIP message body and if required by local policy, take an appropriate action (e.g. forward the message body transparently, reject the request, remove the body);

2) examine the characteristics of the SIP message body MIMEs (i.e. check the values of any Content-Type, Content-Disposition, and Content-Language header fields), take an appropriate action defined by local policy (e.g. forward the body unchanged, remove the SIP message body MIME, reject the call); and

3) examine the content of SIP message body MIMEs, and take appropriate action defined by local policy (e.g. forward the body unchanged, remove the SIP message body MIME, reject the call).

When the intended action of an IBCF, based on local policy, is to remove a message body MIME from a SIP message body, and a Content-Disposition header field with a "handling" parameter set to "required" is associated with the MIME, the IBCF shall reject the SIP request with the 415 (Unsupported Media Type) response code as specified in RFC 5621 [150].

5.10.7 Media transcoding control

The IBCF may perform the media transcoding control in order to allow establishing communication between IM CN subsystems using different media codecs based on the interworking agreement and session information. When performing media transcoding control the IBCF acts as a special case of an IMS-ALG compliant with the description given in subclause 5.10.5.

Upon receipt of any request containing an SDP offer, based on local policy and signalling inspection (e.g ICSI values, SDP), the IBCF may perform media transcoding control, as defined in subclause 6.7.1.3. Based on the local configuration determines the media which requires transcoding in the SDP offer.

5.10.8 Privacy protection at the trust domain boundary

In order to ensure privacy IBCF shall additionally to what is specified in subclause 4.4 and before sending the SIP requests or SIP responses outside the trust domain boundary perform the privacy protection as specified in RFC 3323 [33] and RFC 7044 [66] applicable to header fields with the clarifications in this subclause. If there are any conflicts between topology hiding specified in subclause 5.10.4 and the procedures in this subclause, the topology hiding takes precedence over privacy protection.

NOTE: The privacy protection for the History-Info header field is performed in accordance with RFC 7044 [66] subclause 10.1.2.

If a Privacy header field with a value different from "none" is received the IBCF shall:

1) if "header" privacy is requested as specified in RFC 3323 [33]:

   - remove all received Via header fields and then add a single Via header field with a URI of its own as described in RFC 3323 [33] subclause 5.1;

- if the Contact header field does not contain a GRUU or does not contain an isfocus media feature tag, replace the value of the URI of the Contact header field with a URI that does not dereference to the originator of the message as described in RFC 3323 [33] subclause 5.1; and

- remove any Record-Route header fields as described in RFC 3323 [33] subclause 5.1;

2) if "user" level privacy is requested as specified in RFC 3323 [33]:

- anonymize the From header field. The convention for configuring an anonymous From header field described in RFC 3323 [33] and RFC 3325 [34] should be followed; i.e. From: "Anonymous" <sip:anonymous@anonymous.invalid>;tag= xxxxxxx; and

3) if any modification of any dialog-matching headers for privacy protection reasons is done act as a transparent B2BUA as described in RFC 3323 [33] subclause 5.3.

If a Privacy header field is not received IBCF may based on local policy act as if "id", "user", "header" and "history" was received and perform privacy protection as specified in RFC 3325 [34], RFC 3323 [33] and RFC 7044 [66] with the clarifications above.

If a Privacy header field with the value "none" is received the IBCF should not protect the privacy of the identity information.

NOTE: A local policy can regard a Privacy header field with the value "none" the same as if no Privacy header field was received.

5.10.9 Roaming architecture for voice over IMS with local breakout

The IBCF shall apply OMR as specified in 3GPP TS 29.079 [11D] and in accordance with the roaming architecture for voice over IMS with local breakout when a session is identified as a roaming architecture for voice over IMS with local breakout session.

A session can be identified as a potential roaming architecture for voice over IMS with local breakout session when:

1) a received initial INVITE request contains a Feature-Caps header field with a "+g.3gpp.trf" header field parameter, a "+g.3gpp.loopback" header field parameter or any other implementation dependent indication; or

NOTE: An implementation dependent indication can e.g. be in a URI parameter, a character string in the user part of the URI or be a port number in the URI.

2) if indicating traffic leg as specified in RFC 7549 [225] is supported and used:

a) the "iotl" SIP URI parameter with the value "visitedA-homeA" in the bottommost Route header field; or

b) the "iotl" SIP URI parameter with the value "homeA-visitedA" in the bottommost Route header field.

5.10.10 HTTP procedures over the Ms reference point

5.10.10.1 General

General procedures over the Ms reference point are specified in clause V.2.

5.10.10.2 Procedures for an IBCF acting as an entry point

When receiving an initial INVITE, re-INVITE or MESSAGE request containing one or more SIP Identity header fields, the IBCF shall determine the information (originating identity, diverting identities, contents of the Resource-Priority and Priority header fields) to be verified by decoding the Identity header fields containing a PASSporT SHAKEN JSON Web Token and/or a PASSporT rph JSON Web Token with an optional PASSporT sph JSON Web Token. The IBCF uses the Identity header fields to:

1) build and send a verificationRequest, specified in annex V, to an AS for verification over the Ms reference point; and

2) shall upon receiving an HTTP 200 (OK) response to the above request, use:

- the verstat claim from this response to populate the "verstat" tel URI parameter associated with the originating identity and add this parameter to the verified identity in the SIP From header field or the SIP P-Asserted-Identity header field in the forwarded SIP request. Additionally, if the HTTP 200 (OK) response included verification results for the diverting identities, the IBCF shall based on local policy add the "verstat" tel URI parameter to the verified diverting identities in the History-Info header field if this field is available;

- the verstatPriority claim from this response to populate the Priority-Verstat header field associated with the Resource-Priority header field and with the header field value "psap-callback" of the Priority header field (if present) and include the Priority-Verstat header field in the forwarded SIP request; and

- the verifyResults from this response, if present, to store any of the PASSporT verification failure parameters shown in Table V.2.6.2-4.

Based on local policy, the IBCF may populate a Reason header field in the next provisional or final response of the INVITE or MESSAGE request, where the Reason header field protocol value is set to "STIR", as specified in RFC 9410 [294], and the "cause" header field parameter contains the stored "reasonCode" value. Additionally, the IBCF may include the "ppi" header field parameter containing the failing PASSporT.

NOTE 1: Multiple Reason header fields with the protocol value set to "STIR" are not supported in the present document.

Based on local policy, the IBCF may verify that the validated claims returned in the validClaims parameter of the verification response authorize the associated SIP header field values.

NOTE 2: For sessions originating in another domain, only one of the following entities needs to be configured to verify the Identity header field for the resource priority: the IBCF or the AS. Which functional entity inserts the Identity header field verification is subject to network configuration and local policy.

5.10.10.3 Procedures for an IBCF acting as an exit point

When receiving an initial INVITE or MESSAGE request containing:

NOTE 1: As part of the border control procedures the IBCF can apply privacy procedures and in these cases this procedure is not needed.

1) a "verstat" tel URI parameter in at least one of the SIP From header field or the SIP P-Asserted-Identity header field;

2) a SIP Attestation-Info header field as defined in subclause 7.2.18; and

3) a SIP Origination-Id header field as defined in subclause 7.2.19;

and if no Identity header field exists, the IBCF sends a signingRequest, specified in annex V, over the Ms reference point. When the HTTP 200 (OK) response to this request is received, the IBCF shall include value of the "identity" claim in an Identity header field in the forwarded SIP request.

When receiving an initial INVITE or MESSAGE request containing at least one Identity header field and a "verstat" tel URI parameter in a tel URI or a SIP URI with a user=phone parameter in one or more History-Info header field(s) or using other not specified means to determine that a diversion has occurred, then the IBCF sends a signingRequest, specified in annex V, over the Ms reference point for each of the identities to be signed. When the HTTP 200 (OK) response for any of these requests is received, the IBCF shall include the value of the "identity" claim in an Identity header field in the forwarded SIP request.

NOTE 2: As part of the border control procedures the IBCF can apply privacy procedures and in these cases this procedure is not needed.

When receiving an initial INVITE request containing the Resource-Priority header field and optionally the Priority header field with a "psap-callback" header field value or if the IBCF included the Priority header field with a "psap-callback" header field value and the Resource-Priority header field (as specified in subclause 5.10.1), the IBCF sends a signingRequest, over the Ms reference point, as specified in annex V, for the resource priority and optionally, the Priority header fields. When the HTTP 200 (OK) response to this request is received, the IBCF shall include the value of the "identity" claim in an Identity header field in the forwarded initial INVITE request.

When receiving a re-INVITE request containing the Resource-Priority header field, the IBCF sends a signingRequest, over the Ms reference point, as specified in annex V, for the resource priority. When the HTTP 200 (OK) response to

this request is received, the IBCF shall include the value of the "identity" claim in an Identity header field in the forwarded re-INVITE request.

NOTE 3: For sessions terminating in another domain, only one of the following entities needs to be configured to provide an Identity header field for the resource priority: the IBCF or the AS. Which functional entity inserts the Identity header field is subject to network configuration and local policy.

5.11 Procedures at the E-CSCF

5.11.1 General

The PSAP may either be directly connected to the IM CN subsystem or via the PSTN. Based on regional/national requirements and network operator policy, the PSAP may be connected to the IM CN subsystem of another network.

The E-CSCF can receive URIs for a domain for which the operator running the E-CSCF is not responsible. Where RFC 3261 [26] specifies a requirement that the SIP entity has to be responsible for the domain for particular functionality to occur, the E-CSCF may ignore this restriction.

NOTE 1: The E-CSCF would normally implement this override if the P-CSCF or S-CSCF is configured to pass on URIs (e.g. Request-URI) that are outside the responsible domain of the E-CSCF, otherwise emergency calls might not be routed to a PSAP. If the P-CSCF or S-CSCF does not do this, then the override need not be applied.

The E-CSCF retrieves a PSAP URI, based on the location of the UE and the requested type of emergency service. The PSAP URI can be retrieved from LRF (see subclause 5.11.3) or from local configuration. The PSAP address will either point to a PSAP connected to the IM CN subsystem or to a PSAP connected to the PSTN.

If operator policy determines that the E-CSCF selects the PSAP and if, based on the location information contained in the INVITE request, the E-CSCF fails to select the PSAP, the E-CSCF can interrogate an external server in order to retrieve location information.

NOTE 2: The protocol used between an E-CSCF and an external server is not specified in this version of the specification.

When the E-CSCF receives an emergency request for a dialog requesting privacy or a standalone emergency transaction requesting privacy or any request or response related to a UE-originated emergency dialog requesting privacy, and if operator policy (e.g. determined by national regulatory requirements applicable to emergency services) allows requests for suppression of public user identifiers and location information per 3GPP TS 22.101 [1A], the E-CSCF:

- shall provide the privacy service role according to RFC 3323 [33] and RFC 3325 [34];

NOTE 3: The procedure above is in addition to any procedure for the application of privacy at the edge of the trust domain specified by RFC 3325 [34] and subclause 4.4.

- shall remove any location object from the message's body with Content-Type header field containing the content type application/pidf+xml. If only one message body remains in the message's body then the E-CSCF sets the Content-Type header field to the content type specified for the body; and

- shall remove the Geolocation header field (if present) and the Geolocation-Routing header field (if present);

NOTE 4: Operator policy can require retention/removal of user location information from such request or response separately from user identity, based on the national regulatory requirements.

prior to forwarding any such request to a PSAP.

NOTE 5: If the routeing functions are supported by an LRF, this information is not removed before the request is sent to the LRF.

The E-CSCF shall log all SIP requests and responses that contain a "logme" header field parameter in the SIP Session-ID header field if required by local policy.

When sending a failure response to any received request, depending on operator policy, the E-CSCF may insert a Response-Source header field with an "fe" header field parameter constructed with the URN namespace "urn:3gpp:fe",

the fe-id part of the URN set to "e-cscf" and optionally an appropriate fe-param part of the URN set in accordance with subclause 7.2.17.

5.11.2 UE originating case

The E-CSCF may forward an emergency request to a PSAP in the IM CN subsystem, a PSAP attached to another network, or a PSAP in the PSTN. If the PSAP is attached to another network, the requrest can pass IBCF(s) before entering the other network. If the PSAP is located in the PSTN, the request will pass a BGCF and a MGCF before entering the PSTN.

Upon receipt of an initial request for a dialog, or a standalone transaction, or an unknown method including a Request-URI with an emergency service URN, i.e. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69], or an emergency number the E-CSCF shall:

1) if:

   a) the topmost Route header field of the received SIP INVITE request contains an E-CSCF URI inserted by a P-CSCF, an AS or an IBCF;

   NOTE 1: The E-CSCF is identified by two URIs, one preconfigured in the P-CSCF, AS or IBCF and one used to receive the request from EATF.

   b) the Contact header field includes an instance-id feature tag containing an IMEI URN as specified in RFC 7254 [153] or an MEID URN as specified in RFC 8464 [187]. Only the IMEI shall be used for generating an instance ID for a multi-mode UE that supports both 3GPP and 3GPP2 defined radio access networks; and

   c) required by the operator policy;

   then:

   a0) remove its own SIP URI from the topmost Route header field;

   a) insert URI of the EATF to be contacted into the Route header field as the topmost entry followed by own URI to be used to receive the request from EATF;

   b) insert a type 3 "orig-ioi" header field parameter in the P-Charging-Vector header field. The E-CSCF shall set the type 3 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The E-CSCF shall not include the type 3 "term-ioi" header field parameter;

   c) if required by national regulatory requirements applicable to emergency services, include:

      - a CPC with value "emergency"; and optionally

      - an OLI set to a value corresponding to the characteristics of the access used when the emergency request was initiated by the UE, i.e., an OLI that corresponds to a wireless access; and

   d) route the request based on SIP routeing procedures and do not continue with the rest of the steps;

1A) remove its own SIP URI from the topmost Route header field;

1B) if the received request does not contain a P-Charging-Vector header field, insert a P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17];

1C) if an "orig-ioi" header field parameter is received in the P-Charging-Vector header field, store the value of the received "orig-ioi" header field parameter;

NOTE 2: Any received "orig-ioi" header field parameter will be a type 2 IOI generated by an S-CSCF or passed on by an IBCF. The type 2 IOI identifies the network from which the request was sent.

1D) if operator policy determines that an LRF is to be used, forward the request to the LRF as indicated in subclause 5.11.3;

2) if the PSAP is the next hop, store the value of the "icid-value" header field parameter received in the P-Charging-Vector header field and remove the received information in the P-Charging-Vector header field, else keep the P-Charging-Vector if the next hop is an exit IBCF or a BGCF;

3) if the PSAP is the next hop remove the P-Charging-Function-Addresses header fields, if present, else keep the P-Charging-Function-Addresses header fields if the next hop is an exit IBCF or an BGCF;

4) if an IBCF or a BGCF is the next hop, delete any received "orig-ioi" header field parameter, and insert a type 2 "orig-ioi" header field parameter into the P-Charging-Vector header field. The E-CSCF shall set the type 2 "orig-ioi" header field parameter to a value that identifies the sending network. The E-CSCF shall not include the "term-ioi" header field parameter;

5) get location information as:

- geographical location information received in a PIDF location object as defined in RFC 4119 [90] and RFC 5491 [267], with the content type application/pidf+xml, as described RFC 6442 [89]; and

- location identifier as derived from the P-Access-Network-Info header field, if available.

NOTE 3: As an alternative to retrieve location information from the LRF the E-CSCF can also request location information from an external server. The address to the external server can be received in the Geolocation header field as specified in RFC 6442 [89]. The protocol used to retrieve the location information from the external server is not specified in this version of the specification.

5A) if the location is retrieved using information from the Geolocation header field, and if:

- the Geolocation-Routing header field is present, and includes a value not allowing routing of the request based on user location information;

- the Geolocation-Routing header field is present, and includes a value unknown to the E-CSCF; or

- the Geolocation-Routing header field is not present.

not use the location retrieved from the Geolocation header field when deciding where to forward the request.

6) select, based on location information and optionally type of emergency service:

a) a PSAP connected to the IM CN subsystem or another network, and add the PSAP URI to the topmost Route header field; or

NOTE 4: If the user did not request privacy or if national regulator policy applicable to emergency services does not require the user be allowed to request privacy, the E-CSCF conveys the Geolocation header field (if present), the Geolocation-Routing header field (if present), the location information in a PIDF location object (if present) and the P-Access-Network-Info header field containing the location identifier, if defined for the access type as specified in subclause 7.2A.4, to the PSAP.

b) a PSAP in the PSTN, add the BGCF URI to the topmost Route header field, add a PSAP URI in tel URI format to the Request-URI with an entry used in the PSTN/CS domain to address the PSAP and set the handling header field parameter value of the Content-Disposition header field associated with the application/pidf+xml message body (if present) to "optional";

NOTE 5: If the user did not request privacy or if national regulator policy applicable to emergency services does not require the user be allowed to request privacy, the E-CSCF conveys the Geolocation header field (if present), the Geolocation-Routing header field (if present), the location information in a PIDF location object (if present) and the P-Access-Network-Info header field containing the location identifier, if defined for the access type as specified in subclause 7.2A.4, towards the MGCF. The MGCF can translate the location information if included in INVITE (i.e. both the geographical location information in PIDF-LO and the location identifier in the P-Access-Network-Info header field) into ISUP signalling, see 3GPP TS 29.163 [11B].

NOTE 6: The way the E-CSCF determines the next hop address when the PSAP address is a tel URI is implementation dependent.

7) void;

8) if due to local policy or if the PSAP requires interconnect functionalities (e.g. PSAP address is of an IP address type other than the IP address type used in the IM CN subsystem, or the PSAP URI contains the domain name of another network), put the address of the IBCF to the topmost Route header field, in order to forward the request to the PSAP via an IBCF in the same network;

9) create a Record-Route header field containing its own SIP URI;

10) if the request is an INVITE request, save the Contact, CSeq and Record-Route header field values received in the request such that the E-CSCF is able to release the session if needed; and

11) if no P-Asserted-Identity header field is present and if required by operator policy governing the indication to PSAPs that a UE does not have sufficient credentials (e.g. determined by national regulatory requirements applicable to emergency services), insert a P-Asserted-Identity header field set to a non-dialable callback number (see ANSI/J-STD-036-B [176]);

NOTE 7: A P-Asserted-Identity header field that is present can contain a reference number used in the communication between the PSAP and LRF according to procedures in subclause 5.11.3. Such a P-Asserted-Identity header field would not be replaced with a P-Asserted-Identity header field set to a non-dialable callback number.

12) if required by national regulatory requirements applicable to emergency services, include:

- a CPC with value "emergency"; and optionally

- an OLI set to a value corresponding to the characteristics of the access used when the emergency request was initiated by the UE, i.e., an OLI that corresponds to a wireless access; and

13) route the request based on SIP routeing procedures.

NOTE 8: Depending on local operator policy, the E-CSCF has the capability to reject requests relating to specific methods in accordance with RFC 3261 [26], as an alternative to the functionality described above.

Upon receipt of an initial request for a dialog, a standalone transaction, or an unknown method, that does not include a Request-URI with an emergency service URN or an emergency number, the E-CSCF shall reject the request by sending a 403 (Forbidden) response.

When the E-CSCF receives the request containing the access-network-charging-info parameter in the P-Charging-Vector, the E-CSCF shall store the access-network-charging-info parameter from the P-Charging-Vector header field. The E-CSCF shall retain access-network-charging-info parameter in the P-Charging-Vector header field.

When the E-CSCF receives any request or response (excluding ACK requests and CANCEL requests and responses) related to a UE-originated dialog or standalone transaction, the E-CSCF may insert previously saved values into a P-Charging-Function-Addresses header field before forwarding the message.

When the E-CSCF receives any request or response related to a UE-originated dialog or standalone transaction, the E-CSCF may insert previously saved values into a P-Charging-Vector before forwarding the message. If the original request contained a P-Charging-Vector header field including an orig-IOI header field parameter, insert a type 2 "term-ioi" header field parameter in the P-Charging-Vector header field of the outgoing response. The type 2 "term-ioi" header field is set to a value that identifies the sending network of the response and the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter. Values of "orig-ioi" and "term-ioi" header field parameters in the received response are removed.

Based on local policy the E-CSCF shall add an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier to an initial request.

When the E-CSCF receives any 1xx or 2xx response related to a UE-originated dialog or standalone transaction, the E-CSCF shall remove any P-Preferred-Identity header field and P-Asserted-Identity header field, and insert a P-Asserted-Identity header field with the digits that can be recognized as a valid emergency number if dialled as a tel URI representing the number, before forwarding the message.

NOTE 9: Numbers that can be recognized as valid emergency numbers if dialled by the user are specified in 3GPP TS 22.101 [1A]. The emergency numbers 112 and 911 are stored on the ME, in accordance with 3GPP TS 22.101 [1A].

When the E-CSCF receives any response related to a UE-originated dialog or standalone transaction containing a "term-ioi" header field parameter, the E-CSCF shall store the value of the received "term-ioi" header field parameter received in the P-Charging-Vector header field, if present, and remove all received "orig-ioi" and "term-ioi" header field parameters.

NOTE 10: Any received "term-ioi" header field parameter will be a type 2 IOI. The IOI identifies the sending network of the response message.

When the E-CSCF receives an INVITE request from the UE, the E-CSCF may require the periodic refreshment of the session to avoid hung states in the E-CSCF. If the E-CSCF requires the session to be refreshed, the E-CSCF shall apply the procedures described in RFC 4028 [58] clause 8.

NOTE 11: Requesting the session to be refreshed requires support by at least the UE or the PSAP or MGCF. This functionality cannot automatically be granted, i.e. at least one of the involved UAs needs to support it in order to make it work.

When the E-CSCF receives a 2xx response related to a UE-originated dialog and if:

1) the E-CSCF supports the current location discovery during the emergency call;

2) the UE indicated a Recv-Info header field with the g.3gpp.current-location-discovery info package name in the dialog of the emergency call; and

3) the UE indicated an Accept header field indicating the "application/vnd.3gpp.current-location-discovery+xml" MIME type in the dialog of the emergency call;

the E-CSCF:

1) shall include an Allow header field indicating support of the PUBLISH method in the SIP 2xx response; and

2) shall include an Allow-Events header field indicating support of the presence event package in the SIP 2xx response;

before forwarding the message.

5.11.3 Use of an LRF

Where the network operator determines that an LRF is to be used, the E-CSCF shall route initial requests for a dialog and standalone requests that contain an emergency service URN, i.e. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69], or an emergency number, to the LRF in accordance with the procedures of RFC 3261 [26].

NOTE 1: The E-CSCF is by definition responsible for emergency service URNs and is therefore allowed to change the Request-URI of requests containing emergency service URNs when a 3xx or 416 response is received.

For the outgoing request, the E-CSCF shall:

1) insert a type 3 "orig-ioi" header field parameter in the P-Charging-Vector header field. The E-CSCF shall set the type 3 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The E-CSCF shall not include the type 3 "term-ioi" header field parameter; and

2) perform step 11 of subclause 5.11.2 before sending the INVITE request to the LRF.

When the E-CSCF receives any 3xx response to such a request, the E-CSCF shall select a Contact header field URI from the 3xx response according to RFC 3261 [26] and continue processing the steps given in subclause 5.11.2 with the following additions:

a) at step 6), if item a) applies, place the URI received in the selected Contact header field URI in the 3xx response in the topmost entry in the Route header field;

b) at step 6), if item b) applies, replace the original Request-URI with the URI received in the selected Contact header field URI in the 3xx response;

c) if the user did not request privacy or if national regulator policy applicable to emergency services does not require the user be allowed to request privacy, then if the selected Contact header field URI contains a P-Asserted-Identity header field encoded as a header field of the URI, replace all P-Asserted-Identity header fields in the original request with this value;

NOTE 2: Such a P-Asserted-Identity header field contains a reference number which is used in the communication between the PSAP and LRF.

d) if operator local policies allow insertion of UE location information and if the received 3xx response contains a message/external-body MIME type as specified in RFC 4483 [186] with "access-type" MIME type parameter containing "URL" and "URL" MIME type parameter containing an HTTP or HTTPS URI identifying a PIDF location object as defined in RFC 4119 [90] and RFC 5491 [267], then the E-CSCF shall insert a Geolocation header field containing this PIDF location object by reference (see RFC 6442 [89]);

e) if the location source parameter for the SIP Geolocation header field as defined in RFC 8787 [266] is supported, include a loc-src parameter in the Geolocation header field set to the domain name of the visited network; and

f) if operator policies allow forming requests from a URI and if 3xx response is received, then follow the procedures of RFC 3261 [26] subclause 19.1.5 with the following additions and clarifications:

- replacement or inclusion of any header field from the URI in the selected Contact header field is subject to operator policy; and

- if operator policy allows any LRF to provide a location by value, and the URI in the selected Contact header field contains the "Geolocation" header field, a "Geolocation-Routing" header field and a header field with hname "body" with a value, replace the entire message body with value of the header field with hname "body" in the URI in the selected Contact header field, otherwise do not perform this replacement.

If no 1xx or 2xx response to the request is received from the addressed PSAP within an operator settable timeout, or a 4xx – 5xx response is received, and additional URI values were included in the Contact header field of the response, the E-CSCF shall use these values according to RFC 3261 [26] in new requests that are otherwise generated according to the rules specified above.

If no 1xx or 2xx response to the request is received from the addressed PSAP within an operator settable timeout, or a 4xx – 5xx response is received, and all URI values included in the Contact header field of the 3xx response have been attempted, the E-CSCF shall use a default URI value configured in the E-CSCF in a new request that is otherwise generated according to the rules specified above.

If a 6xx response to the request is received, the E-CSCF acts in accordance with RFC 3261 [26].

When the E-CSCF receives any response related to the above request containing a "term-ioi" header field parameter, the E-CSCF shall store the value of the received "term-ioi" header field parameter received in the P-Charging-Vector header field, if present, and remove all received "orig-ioi" and "term-ioi" header field parameters from the forwarded response.

NOTE 3: Any received "term-ioi" header field parameter will be a type 3 IOI. The IOI identifies the sending network of the response message.

If no 3xx response to the request is received from the LRF within an operator settable timeout, the E-CSCF shall use a default URI value configured in the E-CSCF in a request that is otherwise generated according to the rules specified above.

5.11.4 Subscriptions to E-CSCF events

5.11.4.1 Subscription to the event providing dialog state

When an incoming SUBSCRIBE request addressed to the E-CSCF arrives containing the Event header field with the dialog event package, the E-CSCF shall:

1) based on the local policy, check if the request was generated by a subscriber who is authorised to subscribe to the dialog state of this particular user. The authorized subscribers include:

- all the LRFs that belong to the same network operator.

If the requester is not authorised, the E-CSCF shall reject the request with an appropriate 4xx – 6xx response;

2) store the "icid-value" header field parameter received in the P-Charging-Vector header field;

3) store the "orig-ioi" header field parameter received in the P-Charging-Vector header field if present; and

NOTE: Any received "orig-ioi" header field parameter will be a type 3 IOI. The type 3 IOI identifies the service provider from which the request was sent.

4) generate a 200 (OK) response acknowledging the SUBSCRIBE request and indicating that the authorised subscription was successful as described in RFC 4235 [171]. The E-CSCF shall populate the header fields as follows:

- an Expires header field, set to either the same or a decreased value as the Expires header field in the SUBSCRIBE request; and

- a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the SUBSCRIBE request, a type 3 "term-ioi" header field parameter and the "icid-value" header field parameter. The E-CSCF shall set the type 3 "term-ioi" header field parameter to a value that identifies the sending network of the response, the "orig-ioi" header field parameter is set to the previously received value of the "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the received value of the "icid-value" header field parameter in the request.

The E-CSCF may set the Contact header field to an identifier uniquely associated to the SUBSCRIBE request and generated within the E-CSCF, that may help the E-CSCF to correlate refreshes for the SUBSCRIBE.

Afterwards the E-CSCF shall perform the procedures for notification about dialog state as described in subclause 5.11.4.2.

When the E-CSCF receives a subscription refresh request for a dialog that was established by the UE subscribing to the dialog event package, the E-CSCF shall accept the request.

5.11.4.2 Notification about dialog state

The E-CSCF shall send a NOTIFY request when an event pertaining to the dialog or dialogs occurs, as specified in RFC 6665 [28].

When generating NOTIFY requests, the E-CSCF shall not preclude any valid dialog event package parameters in accordance with RFC 4235 [171]. Where RFC 4235 [171] expresses an option or only a recommendation as to the generation of a NOTIFY request, it is a matter of operator policy as to whether such requests are generated.

For each NOTIFY request triggered by an event and on all dialogs which have been established due to subscription to the dialog event package, and in addition to the requirements specified in RFC 4235 [171], the E-CSCF shall:

1) set the P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17], and a type 3 "orig-ioi" header field parameter. The E-CSCF shall set the type 3 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The E-CSCF shall not include the type 3 "term-ioi" header field parameter.

2) in the body of the NOTIFY request, include one <dialog> XML element for each dialog to be reported in accordance with the subscription; and

3) for each <dialog> XML element:

- if the subscription is for all dialogs, rather than a specific dialog, then include the call-id attribute.

If the subscription is to a specific dialog (or to a specific set of dialogs), when sending a final NOTIFY request with all dialogs set to a state of "terminated", the E-CSCF shall also terminate the subscription to the dialog event package by setting the Subscription-State header field to the value of "terminated".

When the E-CSCF receives any response to the NOTIFY request, the E-CSCF shall store the value of the "term-ioi" header field parameter received in the P-Charging-Vector header field, if present.

NOTE: Any received "term-ioi" header field parameter will be a type 3 IOI. The type 3 IOI identifies the service provider from which the response was sent.

5.11.4.3 Subscription to the presence event package

When an incoming SUBSCRIBE request addressed to the E-CSCF arrives containing the Event header field with the presence event package and a Target-Dialog header field:

1) based on the local policy, the E-CSCF shall check if the request was generated by a subscriber who is authorised to subscribe to the presence state of this particular user. The authorized subscribers include:

   - all the LRFs that belong to the same network operator.

   If the requester is not authorised, the E-CSCF shall reject the request with an appropriate 4xx – 6xx response;

2) the E-CSCF shall determine the dialog of the related emergency call, i.e. a confirmed dialog identified by:

   a) the call identifier in the callid portion of the Target-Dialog header field; and

   b) the "remote-tag" header field parameter of the Target-Dialog header field.

   If such dialog does not exist, the E-CSCF shall reject the request with an appropriate 4xx – 6xx response;

3) if :

   a) the UE did not indicate a Recv-Info header field with the g.3gpp.current-location-discovery info package name in the dialog of the related emergency call; or

   b) the UE did not indicate an Accept header field indicating the "application/vnd.3gpp.current-location-discovery+xml" MIME type in the dialog of the related emergency call;

   the E-CSCF shall reject the request with an appropriate 4xx – 6xx response;

4) the E-CSCF shall store the "icid-value" header field parameter received in the P-Charging-Vector header field;

5) the E-CSCF shall store the "orig-ioi" header field parameter received in the P-Charging-Vector header field if present; and

NOTE: Any received "orig-ioi" header field parameter will be a type 3 IOI. The type 3 IOI identifies the service provider from which the request was sent.

6) the E-CSCF shall generate a 200 (OK) response acknowledging the SUBSCRIBE request and indicating that the authorised subscription was successful as described in RFC 4235 [171]. The E-CSCF shall populate the header fields as follows:

   - an Expires header field, set to either the same or a decreased value as the Expires header field in the SUBSCRIBE request; and

   - a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the SUBSCRIBE request, a type 3 "term-ioi" header field parameter and the "icid-value" header field parameter. The E-CSCF shall set the type 3 "term-ioi" header field parameter to a value that identifies the sending network of the response, the "orig-ioi" header field parameter is set to the previously received value of the "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the received value of the "icid-value" header field parameter in the request;

7) the E-CSCF shall associate the dialog of the 200 (OK) response to the SUBSCRIBE request with the dialog of the related emergency call;

8) if the Expires header field of the SUBSCRIBE request is set to zero, the E-CSCF shall perform the procedure in subclause 5.11.5.2 in the dialog of the related emergency call and shall indicate that the receiving entity is requested to send the location information once; and

9) if the Expires header field of the SUBSCRIBE request is not set to zero, the E-CSCF shall perform the procedure in subclause 5.11.5.2 in the dialog of the related emergency call and shall indicate that the receiving entity is requested to start sending the location information.

The E-CSCF may set the Contact header field to an identifier uniquely associated to the SUBSCRIBE request and generated within the E-CSCF, that may help the E-CSCF to correlate refreshes for the SUBSCRIBE.

Afterwards the E-CSCF shall perform the procedures for notification about presence event as described in subclause 5.11.4.4.

When the E-CSCF receives a subscription refresh request for the subscription associated with the initial SUBSCRIBE request, the E-CSCF shall accept the request.

When the E-CSCF receives an unsubscription request for the subscription associated with the initial SUBSCRIBE request:

1) the E-CSCF shall accept the request; and

2) if the dialog of the related emergency call still exists, the E-CSCF shall perform the procedure in subclause 5.11.5.2 in the dialog of the related emergency call and shall indicate that the receiving entity is requested to stop sending the location information.

5.11.4.4 Notification about presence

Upon reception of a PUBLISH request in the dialog of the related emergency call as described in subclause 5.11.5.3, the E-CSCF shall send a NOTIFY request for the presence event package as specified in RFC 6665 [28]. The E-CSCF:

1) if the PUBLISH request contains a body of the "application/pidf+xml" MIME type, shall include in the NOTIFY request the body of the "application/pidf+xml" MIME type of the PUBLISH request;

2) if the PUBLISH request contains P-Access-Network-Info header field(s), shall include in the NOTIFY request the P-Access-Network-Info header field(s) of the PUBLISH request; and

3) shall set the P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17], and a type 3 "orig-ioi" header field parameter in the NOTIFY request. The E-CSCF shall set the type 3 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The E-CSCF shall not include the type 3 "term-ioi" header field parameter.

If the dialog of the related emergency call is terminated, the E-CSCF shall send a NOTIFY request for the presence event package indicating that the subscription is terminated by setting the Subscription-State header field to the "terminated" value. The E-CSCF shall set the P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17], and a type 3 "orig-ioi" header field parameter in the NOTIFY request. The E-CSCF shall set the type 3 "orig-ioi" header field parameter to a value that identifies the sending network of the request. The E-CSCF shall not include the type 3 "term-ioi" header field parameter.

When the E-CSCF receives any response to the NOTIFY request, the E-CSCF shall store the value of the "term-ioi" header field parameter received in the P-Charging-Vector header field, if present.

NOTE: Any received "term-ioi" header field parameter will be a type 3 IOI. The type 3 IOI identifies the service provider from which the response was sent.

5.11.5 Current location discovery during an emergency call

5.11.5.1 General

The UE can be requested to provide the current location information during an emergency call.

5.11.5.2 Requesting current location informaton

If:

1) the UE indicated a Recv-Info header field with the g.3gpp.current-location-discovery info package name in the dialog of the emergency call;

2) the UE indicated an Accept header field indicating the "application/vnd.3gpp.current-location-discovery+xml" MIME type in the dialog of the emergency call; and

3) the dialog of the emergency call is a confirmed dialog;

then in order to request providing of the location information, the E-CSCF shall send an INFO request as described in RFC 6086 [25], as an in-dialog request of the dialog of the emergency call towards the UE. In the INFO request:

1) the E-CSCF shall include an Info-Package header field as described in RFC 6086 [25], containing the g.3gpp.current-location-discovery info package name; and

2) the E-CSCF shall include an request-for-current-location body as specified in subclause 7.12.2.2 in the MIME body of "application/vnd.3gpp.current-location-discovery+xml" MIME type.

5.11.5.3 Receiving current location informaton

Upon receiving a PUBLISH request as described in RFC 3903 [70] as in-dialog request of the dialog of the emergency call, with Event header field containing the presence event package name, the E-CSCF shall perform the procedures described in subclause 5.11.4.4.

5.12 Location Retrieval Function (LRF)

5.12.1 General

The LRF can receive URIs for a domain for which the operator running the LRF is not responsible. Where RFC 3261 [26] specifies a requirement that the SIP entity has to be responsible for the domain for particular functionality to occur, the LRF may ignore this restriction.

NOTE: The LRF would normally implement this override if the P-CSCF is configured to pass on URIs (e.g. Request-URI) that are outside the responsible domain of the LRF, otherwise emergency calls might not be routed to a PSAP. If the P-CSCF does not do this, then the override need not be applied.

The LRF shall log all SIP requests and responses that contain a "logme" header field parameter in the SIP Session-ID header field if required by local policy.

When sending a failure response to any received request, depending on operator policy, the LRF may insert a Response-Source header field with an "fe" header field parameter constructed with the URN namespace "urn:3gpp:fe", the fe-id part of the URN set to "lrf" and optionally an appropriate fe-param part of the URN set in accordance with subclause 7.2.17.

5.12.2 Treatment of incoming initial requests for a dialog and standalone requests

The LRF shall respond to all received initial requests for a dialog, and to all standalone requests, as a redirect server as defined in subclause 8.3 of RFC 3261 [26] with the following additions:

1) the LRF shall generate a 300 (Multiple Choices) response to all such requests;

2) the LRF shall set the Contact header field of the response to a list (one or more) address(es) of PSAP(s), selected according to network operator policy;

NOTE 1: The mechanisms for selection of PSAP addresses are outside the scope of this specification, but can be based on a variety of input information including the value of the URN included in the Request-URI of the request, the value of the Geolocation header field and Geolocation-Routing header field received in the request, the value of the P-Access-Network-Info header field received in the request, any location known at the LRF for the requesting user as identified by the P-Access-Network-Info header field.

2A) if the location is retrieved using information from the Geolocation header field, and if:

- the Geolocation-Routing header field is present, and includes a value not allowing routing of the request based on user location information;

- the Geolocation-Routing header field is present, and includes a value unknown to the LRF; or

- the Geolocation-Routing header field is not present;

the LRF shall not use the location retrieved from the Geolocation header field when selecting PSAP(s);

3) the LRF shall insert a P-Charging-Vector header field containing the "orig-ioi" header field parameter, if received in the request, a type 3 "term-ioi" header field parameter and the "icid-value" header field parameter. The LRF shall set the type 3 "term-ioi" header field parameter to a value that identifies the service provider from which the response is sent, the "orig-ioi" header field parameter is set to the previously received value of "orig-ioi" header field parameter and the "icid-value" header field parameter is set to the previously received value of "icid-value" header field parameter in the request;

4) optionally, generate a reference identifier and set the P-Asserted-Identity header field encoded as a header field of the URI in the Contact header field to this value in each included Contact header field URI associated with a PSAP. The LRF shall maintain state for any generated reference identifier. If the LRF uses a SIP URI (or any other permitted URI scheme other than tel URI) as the reference identifier, the LRF has the responsibility of ensuring (e.g. by configuration) that the emergency request is being routed to an IP connected PSAP. Subclause 5.12.3.1 defines a means of maintaining the state of the reference identifier. If required by operator policy governing the indication to PSAPs that a UE does not have sufficient credentials (e.g. determined by national regulatory requirements applicable to emergency services), the reference identifier shall not be equal to a non-dialable callback number used to indicate the UE does not have credentials;

NOTE 2: The reference identifier is used to correlate information requested over the Le interface (see 3GPP TS 23.167 [4B]) and is not needed if the Le interface is not used. The protocol at the Le interface is not defined in this release.

NOTE 3: The reference identifier is managed by the RDF or the LRF. If the RDF manages the reference identifier, the LRF obtains the a reference identifier from the RDF. In some regional systems, this reference identifier is the ESQK.

5) if required by operator local policies, the LRF shall include a message/external-body MIME type as specified in RFC 4483 [186] with:

   a) "access-type" MIME type parameter containing "URL"; and

   b) "URL" MIME type parameter containing an HTTP or HTTPS URI identifying a PIDF location object as defined in RFC 4119 [90] and RFC 5491 [267]; and

6) if required by operator local policies, the LRF shall include geographical information, encoded as header fields of the URI in a Contact header field of the 300 (Multiple Choices) response, in the following way:

   a) if operator policy indicates location-by-reference is to be used:

      i. a Geolocation-Routing header field with value "yes"; and

      ii. a Geolocation header field that contains an HTTP URI or a HTTPS URI associated with a location-by-reference, as defined in RFC 6442 [89]; and

   b) if operator policy indicates location-by-value is to be used:

      i. a Geolocation-Routing header field with value "yes";

      ii. Geolocation header field with value associated with the location-by-value;

      iii. a header field with hname "body" and with a value that contains an escape encoded MIME body of multipart/mixed MIME type containing:

         - the MIME body from the received request; and

         - the geographical location information as PIDF location object in accordance with RFC 4119 [90] and RFC 5491 [267]; and

      iv. a Content-Type header field with multipart/mixed MIME type.

NOTE 4: The mechanisms for selection of PSAP addresses are outside the scope of this specification. See note 1.

NOTE 5: The body of the received request can include a PIDF location object and SDP.

5.12.3 Subscription and notification

5.12.3.1 Notification about dialog state

Based on operator policy, the LRF can either subscribe to all dialog information on an E-CSCF or individually subscribe to each dialog as it receives the requests.

NOTE 1: Subscription to dialog information is dependent on the use of Le interface as described in subclause 5.12.2.

In the case that the LRF is subscribing to all dialogs at the E-CSCF, the LRF shall generate a SUBSCRIBE request to the dialog state event package in accordance with RFC 6665 [28] and RFC 4235 [171]. The LRF shall include the following additional information in the SUBSCRIBE request:

a) the Request-URI set to an E-CSCF address;

NOTE 2: In this case, it is expected that the LRF will be configured with a set of E-CSCF addresses, and the LRF will subscribe to all of them.

b) no header field parameters in the Event header field;

c) an Expires header field set to 600 000 seconds; and

d) a P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17] and a type 3 "orig-ioi" header field parameter. The type 3 "orig-ioi" header field parameter identifies the service provider from which the request is sent. The LRF shall not include the type 3 "term-ioi" header field parameter.

Upon generation of a 300 response to an incoming dialog forming request that contains a reference identifier, and in the case that the LRF is subscribing to individual dialogs at the E-CSCF, the LRF shall generate a SUBSCRIBE request to the dialog state event package in accordance with RFC 6665 [28] and RFC 4235 [171]. The LRF shall include the following additional information in the SUBSCRIBE request:

a) the Request-URI set to the value of the P-Asserted-Identity in the original request to which the response was generated;

b) a Route header field that addresses the request to the E-CSCF. How such a value is determined depends on deployment;

NOTE 3: A number of mechanisms exist for identifying the required E-CSCF, however all suffer some restrictions. It is therefore a matter of configuration at deployment time to identify the solution that works for that particular deployment. Mechanisms that exist include:

i) if there is only one E-CSCF in the network, using the address of that E-CSCF preconfigured into the system;

ii) using the last entry in the Via header field of the original request to which the 3xx response was generated. If the deployment however includes some intermediate SIP proxy or B2BUA not otherwise included in the emergency call architecture this will not provide the desired result; or

iii) using the IP address from which the original request was received to which the 3xx response was generated. The request is sent to the same port number and IP address as the 3xx response was generated. If the deployment however includes some intermediate SIP proxy or B2BUA not otherwise included in the emergency call architecture this will not provide the desired result, and additionally, if the system is set up to use port numbers in a unidirectional manner, i.e. one port number for requests and another port number for responses, it will also not operate correctly.

c) the "call-id" and "to-tag" header field parameters in the Event header field set to the values in the original request to which the 3xx response was generated. No "from-tag" header field parameter can be included as it is not known by the LRF;

d) an Expires header field set to 86400 seconds; and

) a P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17] and a type 3 "orig-ioi" header field parameter. The type 3 "orig-ioi" header field parameter

identifies the service provider from which the request is sent. The LRF shall not include the type 3 "term-ioi" header field parameter.

In the case that the LRF is subscribing to individual dialogs at the E-CSCF, and a NOTIFY request is received indicating a state of "terminated", the LRF shall end the subscription to the dialog event package.

NOTE 4: Such NOTIFY requests will normally be accompanied by the Subscription-State header field set to the value of "terminated".

When, as a result of successful subscription to the dialog event package, the LRF receives a notification containing dialog updates, the LRF shall update its record for each dialog included in the event package information.

5.12.3.2 Notification about UE location

Based on operator policy, the LRF can subscribe to UE location as it receives the requests.

Upon generation of a 300 response to an incoming dialog forming request that contains a reference identifier, the LRF shall generate a SUBSCRIBE request to the presence event package in accordance with RFC 6665 [28] and RFC 3856 [74]. The LRF shall include the following additional information in the SUBSCRIBE request:

a) the Request-URI set to an E-CSCF address;

b) a Target-Dialog header field with the callid portion and the "remote-tag" header field parameter set to the values in the original request to which the 3xx response was generated. No "local-tag" header field parameter can be included as it is not known by the LRF;

c) an Expires header field set to 86400 seconds or to 0 seconds; and

) a P-Charging-Vector header field with the "icid-value" header field parameter populated as specified in 3GPP TS 32.260 [17] and a type 3 "orig-ioi" header field parameter. The type 3 "orig-ioi" header field parameter identifies the service provider from which the request is sent. The LRF shall not include the type 3 "term-ioi" header field parameter.

When, as a result of successful subscription to the presence event package, the LRF receives a notification containing the UE location, the LRF shall update its record for the dialog indicated in the Target-Dialog header field of the SUBSCRIBE request.

5.13 ISC gateway function

5.13.1 General

As specified in 3GPP TS 23.218 [5] border control functions may be applied between the IM CN subsystem and an application server based on operator preference. The ISC gateway function may act both as an entry point and as an exit point for a network. If it processes a SIP request received from another network it functions as an entry point (see subclause 5.13.3) and it acts as an exit point whenever it processes a SIP request sent to other network (see subclause 5.13.2).

In many cases, the ISC interface carries more than one hop of the session, e.g. the application server has applied a service to a SIP request and then returned the SIP request to the S-CSCF, or a AS acting as a third-party call controller generates multiple outgoing legs. In these cases all the requests relating to the session on all hops / legs should be configured to route through the same ISC gateway function.

NOTE 1: This is to provide for future requirements for the ISC gateway function that may need to provide correlation of the SIP transactions, and additional functionality based on that correlation.

This ISC gateway function exists on a one to one basis with its addressed AS, i.e. the URI used to address the ISC gateway function will always reach the same AS beyond the ISC gateway function.

The functionalities of the ISC gateway function are entry and exit point procedures as defined in subclause 5.13.2 and subclause 5.13.3 and additionally can include:

- network configuration hiding (as defined in subclause 5.13.4);

- application level gateway (as defined in subclause 5.13.5);
- transport plane control, i.e. QoS control (as defined in subclause 5.13.5); and
- screening of SIP signalling (as defined in subclause 5.13.6);

NOTE 2: The functionalities performed by the application level gateway are configured by the operator, and it is network specific.

The application level gateway shall log all SIP requests and responses that contain a "logme" header field parameter in the SIP Session-ID header field based on local policy.

5.13.2 ISC gateway function as an exit point

5.13.2.1 Registration

There are no specific requirements for the REGISTER method, i.e. the REGISTER method is treated as for other SIP methods.

5.13.2.2 General

This subclause applies for requests sent from the S-CSCF to the AS via the ISC gateway function.

For all SIP transactions identified:

- if priority is supported, as containing an authorised Resource-Priority header field or a temporarily authorised Resource-Priority header field, or, if such an option is supported, relating to a dialog which previously contained an authorised Resource-Priority header field, the ISC gateway function shall give priority over other transactions or dialogs. This allows special treatment of such transactions or dialogs. If priority is supported, the ISC gateway function shall adjust the priority treatment of transactions or dialogs according to the most recently received authorized Resource-Priority header field or backwards indication value.

NOTE: The special treatment can include filtering, higher priority processing, routeing, call gapping. The exact meaning of priority is not defined further in this document, but is left to national regulation and network configuration.

5.13.2.3 Initial requests

Upon receipt of:

- an initial request for a dialog;
- a request for a standalone transaction; or
- a request for an unknown method that does not relate to an existing dialog;

the ISC gateway function shall:

1) if the request is an INVITE request, respond with a 100 (Trying) provisional response;
2) remove the topmost entry from the Route header field in accordance with RFC 3261 [26] procedures for processing Route header fields, and then add as the topmost entry the URI of the application server associated with this ISC gateway function, followed by a next entry of a URI needed to reach this ISC gateway function from the application server;
3) if the request is an INVITE request and the ISC gateway function is configured to perform application level gateway and/or transport plane control functionalities, save the Contact, CSeq and Record-Route header field values received in the request such that the ISC gateway function is able to release the session if needed;
4) If the request is a SUBSCRIBE and the ISC gateway function does not need to act as B2BUA, based on operator policy, the ISC gateway function shall determine whether or not to retain, for the related subscription, the SIP dialog state information and the duration information;

NOTE 1: The event package name can be taken into account to decide whether or not the SIP dialog state and the subscription duration information needs to be retained.

NOTE 2: The ISC gateway function needs to insert its own URI in the Record-Route header field of the initial SUBSCRIBE request and all subsequent NOTIFY requests if it decides to retain the SIP dialog state information.

5) if the request is an initial request for a dialog and local policy requires the application of ISC gateway function capabilities in subsequent requests, perform record route procedures as specified in RFC 3261 [26];

6) if the recipient of the request is understood from configured information to always send and receive private network traffic from this source, remove the P-Private-Network-Indication header field containing the domain name associated with that saved information;

7) store the values from the P-Charging-Function-Addresses header field, if present;

8) if the request is an initial request and "fe-identifier" header field parameter of P-Charging-Vector header field is applied in the operator domain;

- store the "fe-identifier" header field parameter of the P-Charging-Vector header field; and

- remove the "fe-identifier" header field parameter from the P-Charging-Vector header field;

9) remove some of the parameters from the P-Charging-Vector header field or the header field itself, depending on operator policy, if present; and

NOTE 3: An example where an ISC-GW removes the P-Charging-Vector header field is where the request is forwarded to outside the trust domain.

10) remove the P-Charging-Function-Addresses header fields, if present, prior to forwarding the message;

and forwards the request according to RFC 3261 [26].

NOTE 4: If ISC gateway function processes a request without a pre-defined route (e.g. the subscription to reg event package originated by the AS), the next-hop address can be either obtained as specified in RFC 3263 [27A] or be provisioned in the ISC gateway function.

When the ISC gateway function receives an INVITE request, the ISC gateway function may require the periodic refreshment of the session to avoid hung states in the ISC gateway function. If the ISC gateway function requires the session to be refreshed, the ISC gateway function shall apply the procedures described in RFC 4028 [58] clause 8.

NOTE 5: Requesting the session to be refreshed requires support by at least one of the UEs. This functionality cannot automatically be granted, i.e. at least one of the involved UEs needs to support it.

When the ISC gateway function receives a response to any of the requests handled in this subclause, then the ISC gateway function shall:

1) in the P-Charging-Vector header field, subject to operator policy, reinsert any parameters that were removed and stored. In addition, where the operator policy requires it, include on behalf of the supported application server a type 3 "term-ioi" header field parameter. This IOI may represent either the network of the ISC gateway function or the network providing the AS.

In responses, if "fe-identifier" header field parameter of P-Charging-Vector header field is applied in the operator domain, the ISC gateway function acting as an exit point shall:

- delete in the P-Charging-Vector header field any received "fe-identifier" header field parameter; and

- add the stored"fe-identifier" to the P-Charging-Vector header field and include its own address or identifier as an "fe-addr" element of the "fe-identifier" header field parameter of the P-Charging-Vector header.

With the exception of 305 (Use Proxy) responses, the ISC gateway function shall not recurse on 3xx responses.

5.13.2.4 Subsequent requests

Upon receipt of a subsequent request, the ISC gateway function shall:

1) if the request is an INVITE request, respond with a 100 (Trying) provisional response;

2) if the request is a NOTIFY request with the Subscription-State header field set to "terminated" and the ISC gateway function has retained the SIP dialog state information for the associated subscription, once the NOTIFY transaction is terminated, the ISC gateway function can remove all the stored information related to the associated subscription;

3) if the request is a target refresh request and the ISC gateway function is configured to perform application level gateway and/or transport plane control functionalities, save the Contact and CSeq header field values received in the request such that the ISC gateway function is able to release the session if needed; and

4) if the subsequent request is other than a target refresh request (including requests relating to an existing dialog where the method is unknown) and the ISC gateway function is configured to perform application level gateway and/or transport plane control functionalities, save the Contact and CSeq header field values received in the request such that the ISC gateway function is able to release the session if needed;

and forwards the request, based on the topmost Route header field, in accordance with the procedures of RFC 3261 [26].

5.13.2.5 Call release initiated by ISC gateway function

If the ISC gateway function provides transport plane control functionality and receives an indication of a transport plane related error the ISC gateway function may:

1) generate a BYE request for the terminating side based on information saved for the related dialog; and

2) generate a BYE request for the originating side based on the information saved for the related dialog.

NOTE: Transport plane related errors can be indicated from e.g. TrGW. The protocol for indicating transport plane related errors to the ISC gateway function is out of scope of this specification.

Upon receipt of the 2xx responses for both BYE requests, the ISC gateway function shall release all information related to the dialog and the related multimedia session.

5.13.3 ISC gateway function as an entry point

5.13.3.1 Registration

There are no specific requirements for the REGISTER method, i.e. the REGISTER method is treated as for other SIP methods.

5.13.3.2 General

This subclause applies for requests sent from the AS to the S-CSCF via the ISC gateway function. Such requests come from the AS as a result of a request received from the S-CSCF and forwarded by the ISC gateway function.

For all SIP transactions identified:

- if priority is supported (NOTE), as containing an authorised Resource-Priority header field or a temporarily authorised Resource-Priority header field, or, if such an option is supported, relating to a dialog which previously contained an authorised Resource-Priority header field, the ISC gateway function shall give priority over other transactions or dialogs. This allows special treatment of such transactions or dialogs. If priority is supported, the ISC gateway function shall adjust the priority treatment of transactions or dialogs according to the most recently received authorized Resource-Priority header field or backwards indication value.

NOTE: The special treatment can include filtering, higher priority processing, routeing, call gapping. The exact meaning of priority is not defined further in this document, but is left to national regulation and network configuration.

5.13.3.3 Initial requests

Upon receipt of:

- an initial request for a dialog;
- a request for a standalone transaction; or
- a request for an unknown method that does not relate to an existing dialog;

the ISC gateway function shall verify whether the request is arrived from a trusted domain or not. If the request arrived from an untrusted domain, then the ISC gateway function shall:

- remove all P-Charging-Vector header fields and all P-Charging-Function-Addresses header fields the request may contain.

Upon receipt of:

- an initial request for a dialog;
- a request for a standalone transaction except the REGISTER request; or
- a request for an unknown method that does not relate to an existing dialog;

the ISC gateway function shall:

1) if the request is an INVITE request, respond with a 100 (Trying) provisional response;
2) remove the topmost entry from the Route header field in accordance with RFC 3261 [26] procedures for processing Route header fields;
3) if a P-Private-Network-Indication header field is included in the request, check whether the configured information allows the receipt of private network traffic from this source. If private network traffic is allowed, the ISC gateway function shall check whether the received domain name in any included P-Private-Network-Indication header field in the request is the same as the domain name associated with that configured information. If private network traffic is not allowed, or the received domain name does not match, then the ISC gateway function shall remove the P-Private-Network-Indication header field;
4) if the initiator of the request is understood from configured information to always send and receive private network traffic from this source, insert a P-Private-Network-Indication header field containing the domain name associated with that configured information;
5) if the request is an INVITE request and the ISC gateway function is configured to perform application level gateway and/or transport plane control functionalities, then the ISC gateway function shall save the Contact, CSeq and Record-Route header field values received in the request such that the ISC gateway function is able to release the session if needed;
6) If the request is a SUBSCRIBE and the ISC gateway function does not need to act as B2BUA, based on operator policy, the ISC gateway function shall determine whether or not to retain, for the related subscription, the SIP dialog state information and the duration information; and

NOTE 1: The event package name can be taken into account to decide whether or not the SIP dialog state and the subscription duration information needs to be retained.

NOTE 2: The ISC gateway function needs to insert its own URI in the Record-Route header field of the initial SUBSCRIBE request and all subsequent NOTIFY requests if it decides to retain the SIP dialog state information.

7) if the request is an initial request for a dialog and local policy requires the application of ISC gateway function capabilities in subsequent requests, perform record route procedures as specified in RFC 3261 [26];

When the ISC gateway function receives an INVITE request, the ISC gateway function may require the periodic refreshment of the session to avoid hung states in the ISC gateway function. If the ISC gateway function requires the session to be refreshed, the ISC gateway function shall apply the procedures described in RFC 4028 [58] clause 8.

NOTE 3: Requesting the session to be refreshed requires support by at least one of the UEs. This functionality cannot automatically be granted, i.e. at least one of the involved UEs needs to support it.

When receiving an initial request and "fe-identifier" header field parameter of P-Charging-Vector header field is applied in the operator domain, the ISC gateway function acting as an entry point shall:

- add an "fe-addr" element of the "fe-identifier" header field parameter to the P-Charging-Vector header field with its own address or identifier; and

- delete in the P-Charging-Vector header field any received "fe-identifier" header field parameter.

When the ISC gateway function receives a response to an initial request (e.g. 183 or 2xx), the ISC gateway function shall:

1) store the values from the P-Charging-Function-Addresses header field, if present;

2) remove the "fe-identifier" header field parameter from the P-Charging-Vector header field, if present; and

3) remove the P-Charging-Function-Addresses header field prior to forwarding the message.

With the exception of 305 (Use Proxy) responses, the ISC gateway function shall not recurse on 3xx responses.

5.13.3.4 Subsequent requests

Upon receipt of a subsequent request, the ISC gateway function shall:

1) if the request is an INVITE request, then respond with a 100 (Trying) provisional response;

2) if the request is a NOTIFY request with the Subscription-State header field set to "terminated" and the ISC gateway function has retained the SIP dialog state information for the associated subscription, once the NOTIFY transaction is terminated, the ISC gateway function can remove all the stored information related to the associated subscription;

3) if the request is a target refresh request and the ISC gateway function is configured to perform application level gateway and/or transport plane control functionalities, then the ISC gateway function shall save the Contact and CSeq header field values received in the request such that the ISC gateway function is able to release the session if needed;

4) if the subsequent request is other than a target refresh request (including requests relating to an existing dialog where the method is unknown) and the ISC gateway function is configured to perform application level gateway and/or transport plane control functionalities, then the ISC gateway function shall save the Contact and CSeq header field values received in the request such that the ISC gateway function is able to release the session if needed; and

5) if the subsequent request is received from outside the trust domain, then the ISC gateway function shall remove a P-Charging-Vector header field, if present;

and forwards the request, based on the topmost Route header field, in accordance with the procedures of RFC 3261 [26].

5.13.3.5 Call release initiated by the ISC gateway function

If the ISC gateway function provides transport plane control functionality and receives an indication of a transport plane related error the ISC gateway function may:

1) generate a BYE request for the terminating side based on information saved for the related dialog; and

2) generate a BYE request for the originating side based on the information saved for the related dialog.

NOTE: Transport plane related errors can be indicated from e.g. TrGW. The protocol for indicating transport plane related errors to the ISC gateway function is out of scope of this specification.

Upon receipt of the 2xx responses for both BYE requests, the ISC gateway function shall release all information related to the dialog and the related multimedia session.

5.13.4 THIG functionality in the ISC gateway function

The ISC gateway function shall act according to the procedures defined for the IBCF in subclause 5.10.4 with the following exceptions:

- there are no specific requirements for the REGISTER method, i.e. the REGISTER method is treated as for other SIP methods.

5.13.5 IMS-ALG functionality in the ISC gateway function

The ISC gateway function shall act according to the procedures defined for the IBCF in subclause 5.10.5.

5.13.6 Screening of SIP signalling

The ISC gateway function shall act according to the procedures defined for the IBCF in subclause 5.10.6.

NOTE 1: Subclause 5.10.6 identifies a number of header fields that should not be screened. It is not expected that the ISC gateway function will see these header fields.

NOTE 2: In identifying header fields to be screened, care is needed to ensure that header fields needed by application servers later in the filter criteria chain are not removed. The ordering of the applications in the filter criteria chain might be reordered if this is a constraint that cannot be met.

6 Application usage of SDP

6.1 Procedures at the UE

6.1.1 General

The "integration of resource management and SIP" extension is hereafter in this subclause referred to as "the precondition mechanism" and is defined in RFC 3312 [30] as updated by RFC 4032 [64].

In order to authorize the media streams, the P-CSCF and S-CSCF have to be able to inspect SDP message bodies. Hence, the UE shall not encrypt SDP message bodies.

During the session establishment procedure, and during session modification procedures, SIP messages shall only contain an SDP message body if that is intended to modify the session description, or when the SDP message body is included in the message because of SIP rules described in RFC 3261 [26].

NOTE 1: A codec can have multiple payload type numbers associated with it.

In order to support accurate bandwidth calculations, the UE may include the "a=ptime" attribute for all "audio" media lines as described in RFC 4566 [39]. If a UE receives an "audio" media line with "a=ptime" specified, the UE should transmit at the specified packetization rate. If a UE receives an "audio" media line which does not have "a=ptime" specified or the UE does not support the "a=ptime" attribute, the UE should transmit at the default codec packetization rate as defined in RFC 3551 [55A]. The UE will transmit consistent with the resources available from the network.

For "video" and "audio" media types that use the RTP/RTCP and where the port number is not zero, the UE shall specify the proposed bandwidth for each media stream using the "b=" media descriptor and the "AS" bandwidth modifier in the SDP.

NOTE 2: The above is the minimum requirement for all UEs. Additional requirements can be found in other specifications.

For "video" and "audio" media types that use the RTP/RTCP and where the port number is not zero, the UE may include for each RTP payload type "a=bw-info" SDP attribute(s) (defined in clause 19 of 3GPP TS 26.114 [9B]) to indicate the additional bandwidth information. The "a=bw-info" SDP attribute line(s) shall be specified in accordance with 3GPP TS 26.114 [9B]. The value of the "a=bw-info" SDP attribute(s) may affect the assigned QoS which is defined in 3GPP TS 29.213 [13C].

For "video" and "audio" media types that utilize the RTP/RTCP, in addition to the "b=AS" parameter, the UE may specify the "b=TIAS", and "a=maxprate" parameters in accordance with RFC 3890 [152]. The value of the parameter shall be determined as described in RFC 3890 [152]. The value or absence of the "b=" parameter(s) may affect the assigned QoS which is defined in 3GPP TS 29.213 [13C].

If a UE receives a media line which contains both a=ptime and a=maxprate, the UE should use the a=maxprate value, if this attribute is supported.

If multiple codecs are specified on the media line, "a=maxprate" (or "a=ptime" if "a=maxprate" is not available or not supported) should be used to derive the packetization time used for all codecs specified on the media line. Given that not all codecs support identical ranges of packetization, the UE should ensure that the packetization derived by "a=maxprate" (or "a=ptime" if "a=maxprate" is not available or not supported) is a valid packetization time for each codec specified in the list.

If the media line in the SDP message body indicates the usage of RTP/RTCP, and if the UE is configured to request an RTCP bandwidth level for the session is different than the default RTCP bandwidth as specified in RFC 3556 [56], then in addition to the "AS" bandwidth modifier in the media-level "b=" line, the UE shall include two media-level "b=" lines, one with the "RS" bandwidth modifier and the other with the "RR" bandwidth modifier as described in RFC 3556 [56] to specify the required bandwidth allocation for RTCP. The bandwidth-value in the b=RS: and b=RR: lines may include transport overhead as described in subclause 6.1 of RFC 3890 [152].

For other media streams the "b=" media descriptor may be included. The value or absence of the "b=" parameter will affect the assigned QoS which is defined in or 3GPP 29.213 [13C].

NOTE 3: In a two-party session where both participants are active, the RTCP receiver reports are not sent, therefore, the RR bandwidth modifier will typically get the value of zero.

If an in-band DTMF codec is supported by the application associated with an audio media stream, then the UE shall include, in addition to the payload type numbers associated with the audio codecs for the media stream, for each clock rate associated with the audio codecs for the media stream, a payload type number associated with the MIME subtype "telephone-event", to indicate support of in-band DTMF as described in RFC 4733 [23].

The UE shall inspect the SDP message body contained in any SIP request or response, looking for possible indications of grouping of media streams according to RFC 3524 [54] and perform the appropriate actions for IP-CAN bearer establishment for media according to IP-CAN specific procedures (see subclause B.2.2.5 for IP-CAN implemented using GPRS, subclause L.2.2.5 for IP-CAN implemented using EPS, and subclause U.2.2.5 for IP-CAN implemented using 5GS).

In case of UE initiated resource reservation and if the UE determines resource reservation is needed, the UE shall start reserving its local resources whenever it has sufficient information about the media streams, media authorization and used codecs available.

NOTE 4: Based on this resource reservation can, in certain cases, be initiated immediately after the sending or receiving of the initial SDP offer.

In order to fulfil the QoS requirements of one or more media streams, the UE may re-use previously reserved resources. In this case the UE shall indicate as met the local preconditions related to the media stream, for which resources are re-used.

If the SDP is affected due to a rejected IP-CAN bearer or a released IP-CAN bearer then the UE shall:

1) update the session according to RFC 3264 [27B] and set the ports of the media stream(s) for which IP-CAN resource was rejected or released to zero in the new SDP offer;

2) release the session according to RFC 3261 [26];

3) cancel the session setup or the session modification according to RFC 3261 [26]; or

4) reject the session setup or the session modification according to RFC 3261 [26].

If the SDP is affected due to a modified IP-CAN bearer, and the desired QoS resources for one or more media streams are no longer available at the UE due to the modification, then the UE shall:

1) update the session according to RFC 3264 [27B] and set the ports of the media stream(s) for which IP-CAN resource was modified to zero in the new SDP offer;

2) release the session according to RFC 3261 [26];

3) cancel the session setup or the session modification according to RFC 3261 [26]; or

4) reject the session setup or the session modification according to RFC 3261 [26].

NOTE 5: The UE can use one IP address for signalling (and specify it in the Contact header field) and different IP address(es) for media (and specify it in the "c=" parameter of the SDP).

If the UE wants to transport media streams with TCP and there are no specific alternative negotiation mechanisms defined for that particular application, then the UE shall support the procedures and the SDP rules specified in RFC 4145 [83].

The UE may support being configured with a media type restriction policy using one or more of the following methods:

a) the Media_type_restriction_policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C];

b) the Media_type_restriction_policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

c) the Media_type_restriction_policy node of 3GPP TS 24.167 [8G].

If the UE is configured with both the Media_type_restriction_policy node of 3GPP TS 24.167 [8G] and the Media_type_restriction_policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or 3GPP TS 31.103 [15B, then the Media_type_restriction_policy node of the $EF_{IMSConfigData}$ file shall take precedence.

NOTE 6: Precedence for files configured on both the USIM and ISIM is defined in 3GPP TS 31.103 [15B].

If the UE supports being configured with a media type restriction policy, the UE shall not include in a sent SDP message (SDP offer or SDP answer) a media stream with:

- non zero port number; and
- a media type which is restricted from inclusion in an SDP message according to the media type restriction policy.

NOTE 7: 488 (Not Acceptable Here) response is sent when all media types of all media streams of an SDP offer are restricted from inclusion in an SDP message according to the media type restriction policy.

6.1.2 Handling of SDP at the originating UE

An INVITE request generated by a UE shall contain a SDP offer and at least one media description. This SDP offer shall reflect the calling user's terminal capabilities and user preferences for the session.

If the desired QoS resources for one or more media streams have not been reserved at the UE when constructing the SDP offer, the UE:

- shall indicate the related local preconditions for QoS as not met, using the segmented status type, as defined in RFC 3312 [30] and RFC 4032 [64], as well as the strength-tag value "mandatory" for the local segment and the strength-tag value either "optional" or as specified in RFC 3312 [30] and RFC 4032 [64] for the remote segment, if the UE uses the precondition mechanism (see subclause 5.1.3.1); and
- if the UE uses the precondition mechanism (see subclause 5.1.3.1), shall not request confirmation for the result of the resource reservation (as defined in RFC 3312 [30]) at the terminating UE.

NOTE 1: Previous versions of this document mandated the use of the SDP inactive attribute. This document does not prohibit specific services from using direction attributes to implement their service-specific behaviours.

If the UE uses the precondition mechanism (see subclause 5.1.3.1), and the desired QoS resources for one or more media streams are available at the UE when the SDP offer is sent, the UE shall indicate the related local preconditions as met, using the segmented status type, as defined in RFC 3312 [30] and RFC 4032 [64], as well as the strength-tag value "mandatory" for the local segment and the strength-tag value either "optional" or as specified in RFC 3312 [30] and RFC 4032 [64] for the remote segment and shall not request confirmation for the result of the resource reservation (as defined in RFC 3312 [30]) at the terminating UE.

NOTE 2: If the originating UE does not use the precondition mechanism (see subclause 5.1.3.1), it will not include any precondition information in the SDP message body.

If the UE indicated support for end-to-access-edge media security using SDES during registration, and the P-CSCF indicated support for end-to-access-edge media security using SDES during registration, then upon generating an SDP

offer with an RTP based media, for each RTP based media except those for which the UE requests an end-to-end media security mechanism, the UE shall:

- offer SRTP transport protocol according to RFC 3711 [169] and the profile defined in 3GPP TS 33.328 [19C];
- include the SDP crypto attribute according to RFC 4568 [168] and the profile defined in 3GPP TS 33.328 [19C]; and
- include an SDP "a=3ge2ae:requested" attribute.

If the UE indicated support for the end-to-access-edge media security for RTP media using DTLS-SRTP and certificate fingerprints during registration, and the P-CSCF indicated support for the end-to-access-edge media security for RTP media using DTLS-SRTP and certificate fingerprints during registration, then upon generating an SDP offer with an RTP based media, for each RTP based media except those for which the UE requests an end-to-end media security mechanism, the UE shall:

- offer "UDP/TLS/RTP/SAVP" or "UDP/TLS/RTP/SAVPF" as the transport protocol according to RFC 5763 [222] and RFC 5764 [223] and the profile defined in 3GPP TS 33.328 [19C];
- include the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C];
- include the SDP "a=3ge2ae:requested" attribute; and
- include the SDP tls-id attribute according to RFC 8842 [240].

If the UE indicated support for the end-to-access-edge media security for MSRP using TLS and certificate fingerprints during registration, and the P-CSCF indicated support for the end-to-access-edge media security for MSRP using TLS and certificate fingerprints during registration, then upon generating an SDP offer with an MSRP based media, for each MSRP based media except those for which the UE requests an end-to-end security mechanism, the UE shall:

- offer MSRP over TLS transport protocol according to RFC 4975 [178], RFC 6714 [214] and the profile defined in 3GPP TS 33.328 [19C];
- include the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C]; and
- include the SDP "a=3ge2ae:requested" attribute.

NOTE 3: TLS client role and TLS server role are determined according to RFC 6135 [215] (referenced by RFC 6714 [214]). If the SDP answer contains the SDP setup attribute with "active" attribute value, the answerer performs the TLS client role. If the SDP answer contains the SDP setup attribute with "passive" attribute value, the offerer performs the TLS client role.

If the UE indicated support for the end-to-access-edge media security for BFCP using TLS and certificate fingerprints during registration, and the P-CSCF indicated support for the end-to-access-edge media security for BFCP using TLS and certificate fingerprints during registration, then upon generating an SDP offer with an BFCP based media, for each BFCP based media except those for which the UE requests an end-to-end security mechanism, the UE shall:

- offer BFCP over TLS transport protocol according to RFC 4583 [108] and the profile defined in 3GPP TS 33.328 [19C];
- include the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C]; and
- include the SDP "a=3ge2ae:requested" attribute.

Unless a new TLS session is negotiated, subsequent SDP offers and answers shall not impact the previously negotiated TLS roles.

NOTE 4: RFC 4583 [108] specifies that the SDP answerer will act as the TLS server but leaves the impact of SDP renegotiation on TLS unspecified.

If the UE indicated support for the end-to-access-edge media security for UDPTL using DTLS and certificate fingerprints during registration, and the P-CSCF indicated support for the end-to-access-edge media security for UDPTL using DTLS and certificate fingerprints during registration, then upon generating an SDP offer with an UDPTL

based media, for each UDPTL based media except those for which the UE requests an end-to-end security mechanism, the UE shall:

- offer UDPTL over DTLS transport protocol according to RFC 7345 [217], RFC 8842 [240] and the profile defined in 3GPP TS 33.328 [19C];
- include the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C];
- include the SDP "a=3ge2ae:requested" attribute; and
- include the SDP tls-id attribute according to RFC 8842 [240].

If the P-CSCF did not indicate support for end-to-access-edge media security using neither DTLS-SRTP nor SDES during registration, the UE shall not include an SDP "a=3ge2ae:requested" attribute in any RTP based media in any SDP offer.

If the P-CSCF did not indicate support for the end-to-access-edge media security for MSRP using TLS and certificate fingerprints during registration, the UE shall not include an SDP "a=3ge2ae:requested" attribute in any MSRP based media in any SDP offer.

If the P-CSCF did not indicate support for the end-to-access-edge media security for BFCP using TLS and certificate fingerprints during registration, the UE shall not include an SDP "a=3ge2ae:requested" attribute in any BFCP based media in any SDP offer.

If the P-CSCF did not indicate support for the end-to-access-edge media security for UDPTL using DTLS and certificate fingerprints during registration, the UE shall not include an SDP "a=3ge2ae:requested" attribute in any UDPTL based media in any SDP offer.

The UE shall not include an SDP "a=3ge2ae:requested" attribute in any media other than RTP based, MSRP based, BFCP based and UDPTL based in any SDP offer.

Upon generating an SDP offer with an MSRP based media protected by the end-to-end media security for MSRP using TLS and KMS, the UE shall:

- offer MSRP over TLS transport protocol according to RFC 4975 [178], RFC 6714 [214] and the profile defined in 3GPP TS 33.328 [19C]; and
- include the SDP key-mgmt attribute according to RFC 4567 [167] and the profile defined in 3GPP TS 33.328 [19C];

NOTE 5: SDP fingerprint attribute is not included.

Upon receiving an SDP answer to the SDP offer with the MSRP based media protected by the end-to-end media security for MSRP using TLS and KMS, and if the MSRP based media is accepted and associated with the SDP key-mgmt attribute as described in RFC 4567 [167] and the profile defined in 3GPP TS 33.328 [19C] in the SDP answer, then the UE indicate the pre-shared key ciphersuites according to RFC 4279 [218] and the profile defined in 3GPP TS 33.328 [19C] in TLS handshake of TLS connection transporting the MSRP based media.

When the UE detects that an emergency call is being made, the UE shall not include end-to-end media security on any media in the SDP offer.

Upon generating the SDP offer for an INVITE request generated after receiving a 488 (Not Acceptable Here) response, as described in subclause 5.1.3.1, the SDP offer shall contain a subset of the allowed media types, codecs and other parameters from the SDP message bodies of all 488 (Not Acceptable Here) responses so far received for the same session establishment attempt (i.e. a set of INVITE requests used for the same session establishment). For each media line, the UE shall order the codecs in the SDP offer according to the order of the codecs in the SDP message bodies of the 488 (Not Acceptable Here) responses.

NOTE 6: The UE can attempt a session establishment through multiple networks with different policies and potentially can need to send multiple INVITE requests and receive multiple 488 (Not Acceptable Here) responses from different CSCF nodes. The UE therefore takes into account the SDP message bodies of all the 488 (Not Acceptable Here) responses received related to the same session establishment when building a new INVITE request.

Upon confirming successful local resource reservation, the UE shall create an SDP offer in which the related local preconditions are set to met, using the segmented status type, as defined in RFC 3312 [30] and RFC 4032 [64].

Upon receiving an SDP answer, which includes more than one codec per media stream, excluding the in-band DTMF codec, as described in subclause 6.1.1, the UE shall:

- send an SDP offer at the first possible time, selecting only one codec per media stream; or
- if the UE is participant in a multi-stream multiparty multimedia conference session using simulcast (indicated by the presence of "a=simulcast" SDP attribute(s) in the SDP answer, as defined in RFC 8853 [249]), apply the procedures defined in 3GPP TS 26.114 [9B] annex S.

If the UE sends an initial INVITE request that includes only an IPv6 address in the SDP offer, and receives an error response (e.g., 488 (Not Acceptable Here) with 301 Warning header field) indicating "incompatible network address format", the UE shall send an ACK as per standard SIP procedures. Subsequently, the UE may acquire an IPv4 address or use an existing IPv4 address, and send a new initial INVITE request to the same destination containing only the IPv4 address in the SDP offer.

6.1.3 Handling of SDP at the terminating UE

Upon receipt of an initial SDP offer in which no precondition information is available, the terminating UE shall in the SDP answer:

- if, prior to sending the SDP answer the desired QoS resources have been reserved at the terminating UE, set the related media streams in the SDP answer to:
  - active mode, if the offered media streams were not listed as inactive; or
  - inactive mode, if the offered media streams were listed as inactive.

If the terminating UE had previously set one or more media streams to inactive mode and the QoS resources for those media streams are now ready, the UE shall set the media streams to active mode by applying the procedures described in RFC 4566 [39] with respect to setting the direction of media streams.

Upon sending a SDP answer to an SDP offer (which included one or more media lines which was offered with several codecs) the terminating UE shall:

- select exactly one codec per media line and indicate only the selected codec for the related media stream. In addition, the UE may indicate support of the in-band DTMF codec, as described in subclause 6.1.1; or
- if the UE is participant in a multi-stream multiparty multimedia conference session using simulcast (indicated by the presence of "a=simulcast" SDP attribute(s) in the SDP answer, as defined in RFC 8853 [249]), apply the procedures defined in 3GPP TS 26.114 [9B] annex S.

If the terminating UE does not support any of the offered codecs, or there are other parameters not acceptable to the UE, the UE shall send a 488 (Not Acceptable Here) response and shall in the response include an SDP in the message body containing the codecs and parameters supported by the UE.

Upon sending an SDP answer to an SDP offer, with the SDP answer including one or more media streams for which the originating side did indicate its local preconditions as not met, if the precondition mechanism is used by the terminating UE (see subclause 5.1.4.1), the terminating UE shall indicate its local preconditions and request the confirmation for the result of the resource reservation at the originating end point.

NOTE 1: If the terminating UE does not use the precondition mechanism (see subclause 5.1.4.1), it will ignore any precondition information received from the originating UE.

Upon receiving an initial INVITE request that includes the SDP offer containing an IP address type (in the "c=" parameter) that is not supported by the UE, the UE shall:

- if the UE is a UE performing the functions of an external attached network and
  1) if the received SDP offer contains an "altc" SDP attribute indicating an alternative and supported IP address; and
  2) the UE supports the "altc" SDP attribute;

select an IP address type in accordance with RFC 6947 [228]; or

- otherwise respond with a 488 (Not Acceptable Here) response including a 301 Warning header field indicating "incompatible network address format".

NOTE 2: Upon receiving an initial INVITE request that does not include an SDP offer, the UE can accept the request and include an SDP offer in the first reliable response. The SDP offer will reflect the called user's terminal capabilities and user preferences for the session.

If the UE receives an SDP offer that specifies different IP address type for media (i.e. specify it in the "c=" parameter of the SDP offer) that the UE is using for signalling, and if the UE supports both IPv4 and IPv6 addresses simultaneously, the UE shall accept the received SDP offer. Subsequently, the UE shall either acquire an IP address type or use an existing IP address type as specified in the SDP offer, and include it in the "c=" parameter in the SDP answer.

NOTE 3: Upon receiving an initial INVITE request, that includes an SDP offer containing connection addresses (in the "c=" parameter) equal to zero, the UE will select the media streams that is willing to accept for the session, reserve the QoS resources for accepted media streams, and include its valid connection address in the SDP answer.

If the UE supports the end-to-access-edge media security using SDES, upon receiving an SDP offer containing an RTP based media:

- transported using the SRTP transport protocol as defined in RFC 3711 [169];
- with an SDP crypto attribute as defined in RFC 4568 [168]; and
- with the SDP "a=3ge2ae:applied" attribute;

and if the UE accepts the RTP based media, then the UE shall generate the SDP answer with the related RTP based media:

- transported using the SRTP transport protocol according to RFC 3711 [169] and the profile defined in 3GPP TS 33.328 [19C]; and
- including an SDP crypto attribute according to RFC 4568 [168] and the profile defined in 3GPP TS 33.328 [19C].

If the UE supports the end-to-access-edge media security for RTP media using DTLS-SRTP and certificate fingerprints, upon receiving an SDP offer containing an RTP based media:

- transported using the "UDP/TLS/RTP/SAVP" or "UDP/TLS/RTP/SAVPF" as the transport protocol according to RFC 5763 [222] and RFC 5764 [223];
- with the SDP fingerprint attribute as defined in RFC 8122 [241]; and
- with the SDP "a=3ge2ae:applied" attribute;

and if the UE accepts the RTP based media, then the UE shall generate the SDP answer with the related RTP based media:

- transported using "UDP/TLS/RTP/SAVP" or "UDP/TLS/RTP/SAVPF" as the transport protocol according to RFC 5763 [222] and RFC 5764 [223] and the profile defined in 3GPP TS 33.328 [19C];
- including the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C]; and
- including the SDP tls-id attribute according to RFC 8842 [240].

If the UE supports the end-to-access-edge media security for MSRP using TLS and certificate fingerprints, upon receiving an SDP offer containing an MSRP based media:

- transported using the MSRP over TLS transport protocol as defined in RFC 4975 [178] and RFC 6714 [214];
- with the SDP fingerprint attribute as defined in RFC 8122 [241]; and
- with the SDP "a=3ge2ae:applied" attribute;

and if the UE accepts the MSRP based media, then the UE shall generate the SDP answer with the related MSRP based media:

- transported using the MSRP over TLS transport protocol according to RFC 4975 [178], RFC 6714 [214] and the profile defined in 3GPP TS 33.328 [19C]; and

- including the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C].

NOTE 4: TLS client role and TLS server role are determined according to RFC 6135 [215] (referenced by RFC 6714 [214]). If the SDP answer contains the SDP setup attribute with "active" attribute value, the answerer performs the TLS client role. If the SDP answer contains the SDP setup attribute with "passive" attribute value, the offerer performs the TLS client role.

If the UE supports the end-to-access-edge media security for BFCP using TLS and certificate fingerprints, upon receiving an SDP offer containing an BFCP based media:

- transported using the BFCP over TLS transport protocol as defined in RFC 4583 [108];

- with the SDP fingerprint attribute as defined in RFC 8122 [241]; and

- with the SDP "a=3ge2ae:applied" attribute;

and if the UE accepts the BFCP based media, then the UE shall generate the SDP answer with the related BFCP based media:

- transported using the BFCP over TLS transport protocol according to RFC 4583 [108] and the profile defined in 3GPP TS 33.328 [19C]; and

- including the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C].

Unless a new TLS session is negotiated, subsequent SDP offers and answers shall not impact the previously negotiated TLS roles.

NOTE 5: RFC 4583 [108] specifies that the SDP answerer will act as the TLS server but leaves the impact of SDP renegotiation on TLS unspecified.

If the UE supports the end-to-access-edge media security for UDPTL using DTLS and certificate fingerprints, upon receiving an SDP offer containing an UDPTL based media:

- transported using the UDPTL over DTLS transport protocol as defined in RFC 7345 [217] and RFC 8842 [240];

- with the SDP fingerprint attribute as defined in RFC 8122 [241]; and

- with the SDP "a=3ge2ae:applied" attribute;

and if the UE accepts the UDPTL based media, then the UE shall generate the SDP answer with the related UDPTL based media:

- transported using the UDPTL over DTLS transport protocol according to RFC 7345 [217], RFC 8842 [240] and the profile defined in 3GPP TS 33.328 [19C];

- including the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C]; and

- including the SDP tls-id attribute according to RFC 8842 [240].

Upon receiving an SDP offer containing an MSRP based media:

- transported using the MSRP over TLS transport protocol as defined in RFC 4975 [178] and RFC 6714 [214]; and

- with the SDP key-mgmt attribute according to RFC 4567 [167] and the profile defined in 3GPP TS 33.328 [19C];

and if the UE accepts the MSRP based media, the UE shall:

1) generate the SDP answer with the related MSRP based media:

   a) transported using the MSRP over TLS transport protocol according to RFC 4975 [178], RFC 6714 [214] and the profile defined in 3GPP TS 33.328 [19C]; and

   b) include the SDP key-mgmt attribute according to RFC 4567 [167] and the profile defined in 3GPP TS 33.328 [19C]; and

NOTE 6: SDP fingerprint attribute is not included.

2) indicate the pre-shared key ciphersuites according to RFC 4279 [218] and the profile defined in 3GPP TS 33.328 [19C] in TLS handshake of TLS connection transporting the MSRP based media.

If the terminating UE uses the precondition mechanism (see subclause 5.1.4.1), if the desired QoS resources for one or more media streams have not been reserved at the terminating UE when constructing the SDP offer, the terminating UE shall indicate the related local preconditions for QoS as not met, using the segmented status type, as defined in RFC 3312 [30] and RFC 4032 [64], as well as the strength-tag value "mandatory" for the local segment and the strength-tag value either "optional" or as specified in RFC 3312 [30] and RFC 4032 [64] for the remote segment.

NOTE 7: It is out of scope of this specification which media streams are to be included in the SDP offer.

If the terminating UE uses the precondition mechanism (see subclause 5.1.4.1) and if the desired QoS resources for one or more media streams are available at the terminating UE when the SDP offer is sent, the UE shall indicate the related local preconditions as met, using the segmented status type, as defined in RFC 3312 [30] and RFC 4032 [64], as well as the strength-tag value "mandatory" for the local segment and the strength-tag value either "optional" or as specified in RFC 3312 [30] and RFC 4032 [64] for the remote segment.

If the terminating UE sends an UPDATE request to remove one or more media streams negotiated in the session for which a final response to the INVITE request has not been sent yet, the terminating UE sets the ports of the media streams to be removed from the session to zero in the new SDP offer.

NOTE 8: Upon receiving an initial INVITE request with one or more media streams which the terminating UE supports and one or more media streams which the UE does not support, the UE is not expected to reject the INVITE request just because of the presence of the unsupported media stream.

NOTE 9: Previous versions of this document mandated the use of the SDP inactive attribute in the SDP offer if the desired QoS resources for one or more media streams had not been reserved at the originating UE when constructing the SDP offer unless the originating UE knew that the precondition mechanism was supported by the remote UE. The use can still occur when interoperating with devices based on earlier versions of this document.

6.1.4 Session modification

6.1.4.1 General

This subclause applies after the 2xx response to the initial INVITE request has been sent or received.

6.1.4.2 Generating session modification request

If the precondition mechanism is used for the session modification, the following applies:

a) if the session modification does not increase the QoS requirement of the already established media stream (e.g., all the media streams in a call hold procedure, audio stream in a call upgrade procedure), in the SDP body of the request (re-INVITE, UPDATE, or PRACK), both local and remote QoS of this media shall be indicated as met; and

b) if the session modification increases the QoS requirement of some already established media stream(s) (e.g., request of using a different audio/video codec that requires higher bandwidth), or if the session modification adds a new media stream (e.g., call upgrade), the setting of the current and desired QoS status of the modified or added media stream shall be the same as specified in subclause 6.1.2. If the network fails to modify or reserve the required resources, the UE shall send a CANCEL request to terminate the session modification.

6.1.4.3 Receiving session modification request

If the precondition mechanism is used for the session modification, the settings of the current and desired QoS status shall be the same as specified in subclause 6.1.3. If the network cannot modify or reserve the required resources, the UE shall send a 580 (Precondition-Failure) response towards the UE that initiated the session modification.

6.2 Procedures at the P-CSCF

The P-CSCF shall perform IMS-ALG functionality:

- when the P-CSCF needs to perform procedures for hosted NAT traversal according to Annex F; or
- when the P-CSCF needs to perform procedures for media plane security (see subclause 6.7.2.2);
- when required by the user-related policies provisioned to the P-CSCF (see subclause 5.2.1);
- when the P-CSCF needs to perform ECN procedures (see subclause 6.7.2.3);
- when the P-CSCF needs to perform procedures for OMR (see subclause 6.7.2.4);
- when the P-CSCF needs to perform P-CSCF controlled NA(P)T and NA(P)T-PT (see subclause 6.7.2.5);
- when the P-CSCF needs to perform hosted NAT procedures (see subclause 6.7.2.6);
- when the P-CSCF needs to perform ICE procedures (see subclause 6.7.2.7); or
- when the P-CSCF needs to perform transcoding procedures (see subclause 6.7.2.8).

Upon receiving an initial INVITE request that includes the SDP offer containing only an IPv6 address (in the "c=" parameter) and if the P-CSCF knows that the terminating UE supports only IPv4 addressing and does not perform the IP version interworking as described in subclause 6.7.2.5.1, the P-CSCF may, based on local policy, respond with a 488 (Not Acceptable Here) response including a 301 Warning header field indicating "incompatible network address format".

NOTE 1: How the P-CSCF determines whether the UE supports only IPv4 addressing is implementation specific.

NOTE 2: Upon receiving an initial INVITE request that does not include an SDP offer, the P-CSCF can accept the request and receive an SDP offer in the first reliable response. The SDP offer will reflect the called user's terminal capabilities and user preferences for the session.

When the P-CSCF receives any SIP request containing an SDP offer, the P-CSCF shall examine the media parameters in the received SDP offer.

If the P-CSCF finds any media parameters which are not allowed on the network by local policy or if available by bandwidth authorisation limitation information coming from the IP-CAN (e.g. via PCRF), the P-CSCF shall return a 488 (Not Acceptable Here) response containing an SDP message body. This SDP message body contains either all the media types, codecs and other SDP parameters which are allowed according to the local policy, or, based on configuration by the operator of the P-CSCF, a subset of these allowed parameters. This subset may depend on the content of the received SIP request. For each media line, the P-CSCF shall build the SDP message body in the 488 (Not Acceptable Here) response in the same manner as a UAS builds the SDP message body in a 488 (Not Acceptable Here) response as specifed in RFC 3261 [26]. The P-CSCF shall order the codecs with the most preferred codec listed first. If the SDP offer is encrypted, the P-CSCF may reject the request.

Subject to local policy, if it is not possible to generate a SDP message body (e.g. the available bandwidth is less than the bandwidth of any codec allowed by the local policy), the P-CSCF shall return a 486 (Busy here) response with a 370 Warning header field indicating "insufficient bandwidth".

When the P-CSCF receives a SIP response different from 200 (OK) response containing SDP offer, the P-CSCF shall not examine the media parameters in the received SDP offer, but the P-CSCF shall rather check the succeeding request containing the SDP answer for this offer, and if necessary (i.e. the SDP answer reduced by the UE still breaches local policy, or if available by bandwidth authorisation limitation information coming from the IP-CAN, e.g. via PCRF), the P-CSCF shall return a 488 (Not Acceptable Here) response containing the local policy allowed SDP message body. If the SDP answer is encrypted, the P-CSCF may reject the succeeding request.

When the P-CSCF receives a 200 (OK) response containing SDP offer, the P-CSCF shall examine the media parameters in the received SDP offer. If the P-CSCF finds any media parameters which are not allowed on the network by local policy or if available by bandwidth authorisation limitation information coming from the IP-CAN (e.g. via PCRF), the P-CSCF shall forward the SDP offer and on the receipt of the ACK request containing the SDP answer, the P-CSCF shall immediately terminate the session as described in subclause 5.2.8.1.2. If the SDP offer is encrypted, the P-CSCF shall forward the SDP offer and on the receipt of the ACK request containing the SDP answer, it may immediately terminate the session as described in subclause 5.2.8.1.2.

In case a device performing address and/or port number conversions is provided by a NA(P)T or NA(P)T-PT controlled by the P-CSCF, or by a hosted NAT, the P-CSCF may need to modify the media connection data in SDP message bodies according to the procedures described in annex F or subclause 6.7.2.5.

The P-CSCF shall apply the same SDP policy to the initial request or response containing an SDP message body, and throughout the complete SIP session.

The P-CSCF may inspect, if present, the "b=RS" and "b=RR" lines in order to find out the bandwidth allocation requirements for RTCP.

Subject to local policy, the P-CSCF shall prohibit the negotiation of ECN during SDP offer/answer exchanges associated with multimedia priority service by removing any ECN attribute "a=ecn-capable-rtp" from the SDP offer and shall not invoke ECN for SIP transactions associated with multimedia priority service.

Additional procedures where the P-CSCF acts as an IMS-ALG are given in subclause 6.7.2. The IMS-ALG only applies where there are specific gateway capabilities to be provided.

6.3 Procedures at the S-CSCF

When the S-CSCF receives any SIP request containing an SDP offer, the S-CSCF shall examine the media parameters in the received SDP offer. If the S-CSCF finds any media parameters which are not allowed based on local policy or subscription (i.e. the information in the instances of the Core Network Service Authorization class in the service profile, described in 3GPP TS 29.228 [14]), the S-CSCF shall return a 488 (Not Acceptable Here) response containing an SDP message body. This SDP message body contains either all the media types, codecs and other SDP parameters which are allowed according to the local policy and users subscription or, based on configuration by the operator of the S-CSCF, a subset of these allowed parameters. This subset may depend on the content of the received SIP request. The S-CSCF shall build the SDP message body in the 488 (Not Acceptable Here) response in the same manner as a UAS builds the SDP message body in a 488 (Not Acceptable Here) response as specified in RFC 3261 [26]. If the SDP offer is encrypted, the S-CSCF may reject the request.

When the S-CSCF receives a SIP response different from 200 (OK) response containing SDP offer, the S-CSCF shall not examine the media parameters in the received SDP offer, but the S-CSCF shall rather check the succeeding request containing the SDP answer for this offer, and if necessary (i.e. the SDP answer reduced by the UE still breaches local policy), the S-CSCF shall return a 488 (Not Acceptable Here) response containing the local policy allowed SDP message body. If the SDP answer is encrypted, the S-CSCF may reject the succeeding request.

When the S-CSCF receives a 200 (OK) response containing an SDP offer, the S-CSCF shall examine the media parameters in the received SDP offer. If the S-CSCF finds any media parameters which are not allowed based on local policy or subscription (i.e. the information in the instances of the Core Network Service Authorization class in the service profile, described in 3GPP TS 29.228 [14]), the S-CSCF shall forward the SDP offer and on the receipt of the ACK request containing the SDP answer, the S-CSCF shall immediately terminate the session as described in subclause 5.4.5.1.2. If the SDP offer is encrypted, the S-CSCF shall forward the SDP offer and on the receipt of the ACK request containing the SDP answer, it may immediately terminate the session as described in subclause 5.4.5.1.2.

6.4 Procedures at the MGCF

6.4.1 Calls originating from circuit-switched networks

The usage of SDP by the MGCF is the same as its usage by the UE, as defined in the subclause 6.1 and A.3.2, with the following exceptions:

- in an initial INVITE request generated by a MGCF, the MGCF shall indicate the current status of the local precondition;

- end-to-access edge media security is not applicable to the MGCF; and
- procedures related to the handling of the IP-CAN bearer rejection, modification or release are not applicable to the MGCF.

When sending an SDP message body, the MGCF shall not include the "i=", "u=", "e=", "p=", "r=", and "z=" descriptors in the SDP message body, and the MGCF shall ignore them if received in an SDP message body.

When the MGCF generates and sends an INVITE request for a call originating in a circuit-switched network, the MGCF shall populate the SDP with the codecs supported by the associated MGW.

6.4.2 Calls terminating in circuit-switched networks

The usage of SDP by the MGCF is the same as its usage by the UE, as defined in the subclause 6.1 and A.3.2, with the following exceptions:

a) when the MGCF sends a 183 (Session Progress) response with an SDP message body, the MGCF shall only request confirmation for the result of the resource reservation (as defined in RFC 3312 [30]) at the originating end point if all of the following conditions are true:
   - there are any remaining unfulfilled preconditions at the originating end point;
   - the received initial INVITE request indicates support of SIP preconditions; and
   - local configuration indicates support of SIP preconditions;

b) end-to-access edge media security is not applicable to the MGCF; and

c) procedures related to the handling of the IP-CAN bearer rejection, modification or release are not applicable to the MGCF.

When sending an SDP message body, the MGCF shall not include the "i=", "u=", "e=", "p=", "r=", and "z=" descriptors in the SDP message body, and the MGCF shall ignore them if received in an SDP message body.

6.4.3 Optimal Media Routeing (OMR)

If the MGCF supports OMR it shall also perform the UA procedures described in 3GPP TS 29.079 [11D].

6.4.4 Explicit congestion control support in MGCF

An MGW associated with an MGCF can support Explicit Congestion Notification (ECN) according to RFC 3168 [189], and can act as an ECN endpoint to enable ECN with a local ECN-capable terminal within a local network that properly handles ECN-marked packets.

If the MGCF receives a SDP offer containing ECN attribute "a=ecn-capable-rtp" as specified in RFC 6679 [188], and if the MGCF knows via configuration that the MGW handles ECN-marked packets properly then the MGCF, taking into account the initialisation method the MGW supports, shall return a SDP answer containing the ECN attribute "a=ecn-capable-rtp" according to RFC 6679 [188].

NOTE 1: The "leap" initialisation method is the only initialisation method the MGW supports over the Mn interface in this release.

When creating an SDP offer and if the MGCF knows via configuration that the MGW handles ECN-marked packets properly the MGCF may initiate ECN negotiation in accordance with RFC 6679 [188].

If the MGCF receives the SDP answer also containing ECN attribute "a=ecn-capable-rtp" then the MGCF will instruct the MGW to apply ECN procedures.

6.5 Procedures at the MRFC

Void.

6.6 Procedures at the AS

6.6.1 General

Since an AS may provide a wide range of different services, procedures for the SDP usage for an AS acting as originating UA, terminating UA or third-party call control role are dependent on the service provided to the UA and on the capabilities on the remote UA. There is no special requirements regarding the usage of the SDP, except the requirements for the SDP capabilities described in the following paragraphs and clause A.3:

1) Providing that an INVITE request generated by an AS contains an SDP message body, the AS has the capability of reflecting the originating AS's capabilities, desired QoS and precondition requirements for the session in the SDP message body.

2) When the AS sends a 183 (Session Progress) response with an SDP message body including one or more "m=" media types, it has the capability of requesting confirmation for the result of the resource reservation at the originating endpoint.

When an AS acts as a B2BUA, and it controls media resources using an MRF, it may support OMR. When the AS supports OMR, and it controls media resources using an MRF, it shall also perform the procedures described in 3GPP TS 29.079 [11D].

6.6.2 Transcoding

The AS shall send an SDP offer to the MRFC with the codecs supported by the caller and the codecs to be offered towards the callee, and the IP address and port information received from caller, in seperate media lines. When receiving an SDP answer from the MRFC, the AS shall forward the received selected codecs and IP address and port information in the callee´s media line(s) as an SDP offer towards the callee.

When the callee provides an SDP answer with selected codecs and IP address and port information, the AS shall forward this information within a new SDP offer to the MRFC. When receiving the corresponding SDP answer from the MRFC, the AS shall forward the address and port information within the caller´s media line(s) as an SDP answer towards the caller.

The codecs offered for transcoding are subject to network policy which shall be according to clause T.2.

6.6.3 AS procedures to support WebRTC media optimization procedure

When an AS acts as a B2BUA, and it controls media resources using an MRF, it may support switching to transparent media for WebRTC when those media have been negotiated, as specified in annex U.2.4 of 3GPP TS 23.228 [7]. An AS that supports switching to transparent media for WebRTC shall apply the procedures in the present subclause.

NOTE 1: The AS can in addition apply OMR procedures described in 3GPP TS 29.079 [11D].

If the AS receives an SDP offer that contains any "tra-contact" SDP attribute, and the AS decides to include an MRF in the media path, the AS shall:

1) include the address information as received from the MRF in that contact line and also encapsulate the address information into each received "tra-contact" attribute, replacing previous information; and

2) transparently pass all received "tra-m-line", "tra-att", "tra-SCTP-association", "tra-media-line-number" and "tra-bw" SDP attributes.

NOTE 2: When interacting with the MRF to reserve resources and provide the information needed for media handling the AS will ask for resources suitable for the media described in the SDP offer outside the "tra-m-line", "tra-att" and "tra-bw" SDP attributes.

If an AS receives an SDP answer and the SDP answer includes "tra-m-line" media level SDP attributes, the AS shall:

1) configure the MRF to transparently pass the media described in the received "tra-m-line", "tra-att", "tra-SCTP-association", and "tra-bw" SDP attributes; and

2) transparently pass all received "tra-m-line", "tra-att", "tra-SCTP-association" and "tra-bw" SDP attributes.

NOTE 3: When interacting with MRF the AS will deactivate media plane interworking in the MRF. The AS will use the "tra-SCTP-association" SDP attributes to determine which media streams need to be multiplexed into the same SCTP association.

6.7 Procedures at the IMS-ALG functionality

6.7.1 IMS-ALG in IBCF

6.7.1.1 General

When the IBCF acts as an IMS-ALG, it makes procedures as for an originating UA and terminating UA. The IMS-ALG acts as a B2BUA. The general treatment of the SDP information between originating UA and terminating UA is described in 3GPP TS 29.162 [11A]. For the use of the IMS-ALG for specific capabilities, additional procedures are defined in subsequent subclauses.

Subject to local policy, the IBCF shall prohibit the negotiation of ECN during SDP offer/answer exchanges associated with multimedia priority service by removing any ECN attribute "a=ecn-capable-rtp" from the SDP offer and shall not invoke ECN for SIP transactions associated with multimedia priority service.

NOTE: Disabling ECN in an IBCF does not prevent a P-CSCF (IMS ALG), subject to roaming agreement, from applying ECN over the access network between a UE and the P-CSCF (IMS-ALG).

6.7.1.2 IMS-ALG in IBCF for support of ICE

6.7.1.2.1 General

This subclause describes procedures of an IBCF to support ICE as defined in RFC 8445 [289] and RFC 8839 [290].

If no TrGW is inserted, an IBCF may transparently pass ICE related SDP attibutes to support ICE. The remaining procedures in this subclause are only applicable if the IBCF is inserting a TrGW on the media plane.

When the IBCF with attached TrGW receives SDP candidate information from the SDP offerer the IBCF shall not forward the candidate information towards the SDP answerer. When the IBCF receives SDP candidate information from the SDP answerer the IBCF shall not forward the candidate information towards the SDP offerer. The remaining procedures in this subclause are optional.

NOTE: An IBCF that removes and/or does not provide ICE related SDP attributes (e.g. a=candidate) in the offer/answer exchange will cause the ICE procedures to be aborted and the address and port information in the m and c lines of the SDP offer will be used. If this address and port information contains the relayed candidate address of a STUN Relay server, as recommended by ICE, then an extra media relay server will be used for the session which is not necessary nor desirable.

The IBCF with attached TrGW performs separate ICE procedures towards the SDP offerer and the SDP answerer. The usage of ICE is negotiated separately with the SDP offerer and SDP answerer, and ICE may be applied independently at either side. Furthermore, the IBCF may be configured to apply ICE procedures only towards one network side, e.g. towards the IM CN subsystem it belongs to.

Since the IBCF is not located behind a NAT, it does not request the TrGW to generate keep-alive messages even when acting as a full ICE entity. The IBCF only requests the TrGW to terminate and generate STUN messages used for the candidate selection procedures.

Since the IBCF is not located behind a NAT the IBCF shall only include host candidates in SDP offers and answers generated by the IBCF.

6.7.1.2.2 IBCF full ICE procedures for UDP based streams

6.7.1.2.2.1 General

This subclause describes the IBCF full ICE procedures for UDP based streams.

6.7.1.2.2.2 IBCF receiving SDP offer

When the IBCF receives an SDP offer including ICE candidate information, the IBCF shall send the candidate information for each UDP based stream received in the SDP offer towards the TrGW. The IBCF will request the TrGW to reserve media- and STUN resources towards the SDP offerer, based on the candidate information, in order to allow the TrGW to perform the necessary connectivity checks per the ICE procedures.

If the SDP offerer is acting as an ICE controller entity the IBCF shall act as an ICE controlled entity in the direction towards the SDP offerer. If the SDP offerer is acting as an ICE controlled entity the IBCF shall act as an ICE controller entity in the direction towards the SDP offerer.

6.7.1.2.2.3 IBCF sending SDP offer

Prior to sending an SDP offer, the IBCF may choose to apply related ICE procedues, e.g. if it expects to interact with terminals applying procedures as described in subclause K.5.2, and if both the IBCF and TrGW also support ICE procedures. To invoke these ICE procedures, the IBCF will request the TrGW to reserve media- and STUN resources towards the SDP answerer for each UDP based media stream and include a host candidate attribute for each UDP based stream in the SDP offer, providing the reserved address and port at the TrGW as destination.

The IBCF shall always act as an ICE controller entity towards the SDP answerer.

NOTE: The host candidate address included by the IBCF in the generated SDP offer matches the c- and m line information for the associcated UDP stream in the SDP offer.

6.7.1.2.2.4 IBCF receiving SDP answer

When the IBCF receives an SDP answer including ICE candidate information, the IBCF will send the candidate information for each UDP based stream received in the SDP answer towards the TrGW.

The IBCF will request the TrGW to perform ICE candidate selection procedures towards the SDP answerer. The IBCF will request the TrGW to inform the IBCF, for each UDP stream, which candidate pair has been selected towards the SDP answerer, once the candidate selection procedure towards the SDP answerer has finished.

If the TrGW indicates to the IBCF that, for at least one UDP stream, the selected candidate pair does not match the c- and m- line address information for the associated UDP stream, exchanged between the IBCF and the SDP answerer, and the IBCF acts an ICE controller entity towards the SDP answerer, the IBCF shall send a new offer towards the SDP answerer in order to allign the c- and m- lines address information with the chosen candidate pair for the associated UDP stream.

6.7.1.2.2.5 IBCF sending SDP answer

When the IBCF generates an SDP answer for an offer that included ICE candidate information, the IBCF will request the TrGW to reserve media- and STUN resources towards the SDP offerer for each UDP based media stream and include an SDP host candidate attribute for each UDP based stream in the SDP answer, providing the reserved address and port at the TrGW as destination.

The IBCF shall in the generated SDP answer include host candidate information which matches the c- and m line information for the associated UDP stream in the SDP answer.

The IBCF will request the TrGW to perform ICE candidate selection procedures towards the SDP offerer. The IBCF will request the TrGW to inform the IBCF, for each UDP stream, which candidate pair has been selected towards the SDP offerer, once the candidate selection procedure towards the SDP answerer has finished.

If the TrGW indicates to the IBCF that the selected candidate pair towards the SDP offerer does not match the c- and m-line address information for the associated UDP stream, exchanged between the IBCF and the SDP offerer, and the IBCF acts an ICE controller entity towards the SDP offerer, the IBCF shall send an offer towards the SDP offerer (which will now act as an SDP answerer) in order to allign the c- and m- line address information with the chosen candidate pair for the associated UDP stream.

6.7.1.2.3 IBCF ICE lite procedures for UDP based streams

When the IBCF is using ICE lite procedures for UDP based streams, the IBCF procedures are identical as described in subclause 6.7.1.2.2, with the following exceptions:

- The IBCF always acts as an ICE controlled entity towards the SDP offerer and towards the SDP answerer, and;
- The IBCF requests the TrGW to perform ICE lite candidate selection procedures, as defined in ICE

6.7.1.2.4 ICE procedures for TCP based streams

6.7.1.2.4.1 General

The IBCF shall terminate ICE procedures for TCP based streams. Instead the IBCF will use the mechanism defined in RFC 4145 [83] for establishing TCP based streams, as defined in RFC 6544 [131].

An entity that supports ICE continues the ICE procedures for UDP based streams, even if no candidates are provided for TCP based streams.

NOTE: The IBCF ICE procedures for TCP based streams are identical no matter whether the IBCF uses full ICE- or ICE lite- procedures for UDP based streams.

6.7.1.2.4.2 IBCF receiving SDP offer

When the IBCF receives an SDP offer, the IBCF shall ignore the candidate attributes for TCP based streams. The IBCF shall not send the candidate information for TCP based streams towards the TrGW.

6.7.1.2.4.3 IBCF sending SDP offer

When the IBCF generates an SDP offer the IBCF shall include an "actpass" setup attribute, as defined in RFC 4145 [83], for each TCP based stream, which will cause the SDP answerer to initiate the TCP connections towards the TrGW. The IBCF shall not include any candidate attributes for TCP based streams in the SDP offer.

6.7.1.2.4.4 IBCF receiving SDP answer

Since the IBCF does not include candidates in the SDP offer towards the SDP answerer, there are no ICE specific procedures when the IBCF receives an SDP answer.

NOTE: If the SDP answer contains candidate attributes for TCP based streams, the IBCF simply discards the candidate attributes.

6.7.1.2.4.5 IBCF sending SDP answer

When the IBCF generates an SDP answer the IBCF shall include a "passive" setup attribute, as defined in RFC 4145 [83], for each TCP based stream, which will cause the SDP offerer to initiate the TCP connections towards the TrGW. The IBCF shall not include any candidate attributes for TCP based streams in the SDP answer.

6.7.1.3 IMS-ALG in IBCF for transcoding

Before forwarding the SDP offer to the answerer, the IBCF may add to the selected media one or more codecs to the codec list contained in the SDP offer. The codecs added to the SDP offer are based on local policy and shall be in accordance with the requirements of clause T.2.

NOTE 1: The local policy can be based on supported codecs in the terminating network.

Upon receipt of an SDP answer, the IBCF shall inspect the list of the returned codecs and proceed as follows:

- if the list contains at least one of the codecs belonging to the original SDP offer, the IBCF shall not invoke the transcoding function; and
- if the list contains none of the codecs belonging to the original SDP offer, the IBCF shall select one of the returned codecs introduced in the answer and invoke the transcoding function. In order to perform the transcoding the IBCF shall select one of the codecs originally offered and set to a non-zero port value the related media stream in the answer sent to the offerer.

NOTE 2: The protocol used between IBCF and TrGW to allow the transport plane media trascoding control is out of scope of this specification. The codec selected by the answerer and the one selected by the IBCF and sent to the offerer can be used to instruct the TrGW for the transcoding purposes.

The IBCF shall remove from the SDP the codecs added to the original SDP offer before forwarding the SDP answer to the offerer.

NOTE 3: In accordance with normal SDP procedure the transcoding IBCF informs the answerer of the properties of the chosen codecs (IP-address and ports).

6.7.1.4 IMS-ALG in IBCF for NA(P)T and NA(P)T-PT controlled by the IBCF

6.7.1.4.1 General

This subclause describes the IBCF procedures for supporting the scenario where IP address and/or port conversions occur at the TrGW level in the media path between the UE and the backbone. Two types of address conversions are covered:

- IP version interworking (NA(P)T-PT); and
- IP address/port translation (NA(P)T).

When the IBCF performs procedures for IBCF controlled NA(P)T and NA(P)T-PT, the IBCF shall modify the IP address(es) and port numbers (in case of NA(P)T) in SDP offers and answers, based on the IP address(es) and port number(s) received from the TrGW, as described in subclause 6.7.2.1.

For terminating sessions the IBCF may towards a UE performing the functions of an external attached network indicate in the SDP offer alternate IP address versions (IPv4 and IPv6) by inserting two "altc" attributes as defined in RFC 6947 [228]. The order of setting the two IP addresses in the two "altc" SDP attributes shall be based on local policy. The insertion of the "altc" attributes is independent of their presence in the received SDP offer.

NOTE 1: The insertion of alternate IP versions allows avoiding the rejection of the SDP offer because of incompatible network address formats and when the request terminates in a corporate network enables the corporate network to avoid IP version interworking.

NOTE 2: The handling of alternative IP addresses between the IMS-ALG and the TrGW is defined in 3GPP TS 29.162 [11A].

If the IBCF sends an initial INVITE request that includes only an IPv6 address in the SDP offer, and receives a 488 (Not Acceptable Here) response with 301 Warning header field indicating "incompatible network address format", the IBCF shall send an ACK as per standard SIP procedures. Subsequently, based on operator policy, the IBCF may, by performing the IP version interworking, acquire an IPv4 address or use an existing IPv4 address, and send a new initial INVITE request to the same destination containing only the IPv4 address in the SDP offer.

6.7.1.5 IMS-ALG procedure in IBCF to support WebRTC media optimization procedure

The IMS-ALG in the IBCF may support switching to transparent media for WebRTC when those media have been negotiated, as specified in annex U.2.4 of 3GPP TS 23.228 [7]. An IMS-ALG that supports switching to transparent media for WebRTC shall apply the procedures in the present subclause.

NOTE 1: The IMS-ALG can in addition apply OMR procedures described in 3GPP TS 29.079 [11D].

If the IMS-ALG receives an SDP offer that contains any "tra-contact" SDP attribute, and the IMS-ALG decides to include a TrGw in the media path, the IMS-ALG shall:

1) include the address information as received from the TrGW in that contact line and also encapsulate the address information into each received "tra-contact" attribute, replacing previous information; and
2) transparently pass all received "tra-m-line", "tra-att", "tra-SCTP-association", "tra-media-line-number" and "tra-bw" SDP attributes.

NOTE 2: When interacting with the TrGW to reserve resources and provide the information needed for media handling the IMS-ALG will ask for resources suitable for the media described in the SDP offer outside the "tra-m-line", "tra-att" and "tra-bw" SDP attributes. The details of the interaction between the IMS-ALG and the TrGW are out of scope of this document.

If an IMS-ALG receives an SDP answer and the SDP answer includes "tra-m-line" media level SDP attributes, the IMS-ALG shall:

1) configure the TrGW to transparently pass the media described in the received "tra-m-line", "tra-att", "tra-SCTP-association", and "tra-bw" SDP attributes; and

2) transparently pass all received "tra-m-line", "tra-att", "tra-SCTP-association" and "tra-bw" SDP attributes.

NOTE 3: When interacting with TrGW the IMS-AGW will deactivate media plane interworking in the TrGW. The details of this interaction are out of scope of this document. The IMS-AGW will use the "tra-SCTP-association" SDP attributes to determine which media streams need to be multiplexed into the same SCTP association.

6.7.2 IMS-ALG in P-CSCF

6.7.2.1 General

This subclause specifies the general procedures for the support of SDP in IMS-ALG within the P-CSCF. For the use of the IMS-ALG for specific capabilities, additional procedures are defined in subsequent subclauses.

When the IMS-ALG receives an SDP offer, it shall create a new SDP offer, based the contents of the received SDP offer, modified according to procedures and policies associated with specific capabilities that the IMS-ALG is used for, according to capabilities supported by the IMS-AGW, and to provide the IP address and port information received by the IMS-AGW.

When the IMS-ALG receives an SDP answer, it shall create a new SDP answer, to respond to the originally received SDP offer, modified according to the same procedures and policies that were used to modify the SDP offer.

The P-CSCF may receive multiple provisional responses with an SDP answer due to forking of a request before the first final answer is received. For each SDP answer received in such subsequent provisional responses, the P-CSCF shall apply the procedure in this subclause.

After the session is established, it is possible for both ends of the session to change the media connection data for the session. When the P-CSCF receives a SDP offer/answer where port number(s) or IP address(es) is/are included, there are three different possibilities:

- IP address(es) or/and port number(s) have been added;

- IP address(es) and port number(s) have been reassigned to the end points; or

NOTE 1: If necessary, the P-CSCF will request the IMS-AGW access gateway to release the resources related to the previously assigned IP address(es) and port number(s).

- no change has been made to the IP address(es) and port number(s).

NOTE 2: In the particular case of RTP flows, port conversions also apply to the associated RTCP flows.

6.7.2.2 IMS-ALG in P-CSCF for media plane security

When the P-CSCF acts as an IMS-ALG, it acts as a B2BUA and modifies the SDP as described as described in 3GPP TS 23.334 [7F].

If the P-CSCF indicated support for end-to-access-edge media security using SDES during registration:

1) upon receiving an SDP offer from the served UE containing an end-to-access-edge protected RTP based media, i.e. a RTP media stream:

   - transported using the SRTP transport protocol as defined in RFC 3711 [169];

- with an SDP crypto attribute as defined in RFC 4568 [168]; and

- with the SDP "a=3ge2ae:requested" attribute;

the P-CSCF shall invoke IMS-ALG procedures, will act as defined in 3GPP TS 23.334 [7F] as far as SDP and SRTP is concerned, and shall:

- if the SDP offer contains a Transport Protocol Capability SDP attribute (see RFC 5939 [137]) offering:

  a) "RTP/SAVPF" transport, e.g. "a=tcap:*x* RTP/SAVPF", replace this transport with "RTP/AVPF" within that attribute; and

  b) "RTP/SAVP" transport, e.g. "a=tcap:*x* RTP/SAVP", replace this transport with "RTP/AVP" within that attribute; and

- strip the SDP "a=3ge2ae:requested" attribute and the SDP crypto attribute from the end-to-access-edge protected RTP based media of the received SDP offer; and

2) upon sending an SDP answer to the SDP offer from the served UE, for each end-to-access-edge protected RTP based media of the SDP offer from the served UE which is accepted in the SDP answer, the P-CSCF will act as defined in 3GPP TS 23.334 [7F] as far as SDP and SRTP is concerned and shall:

   - indicate the SRTP transport protocol according to RFC 3711 [169] and the profile defined in 3GPP TS 33.328 [19C]; and

   - include a SDP crypto attribute according to RFC 4568 [168] and the profile defined in 3GPP TS 33.328 [19C].

If the served UE indicated support for end-to-access-edge media security using SDES, during registration, and the P-CSCF indicated support for end-to-access-edge 2ae-media security using SDES during registration:

1) upon receiving an SDP offer from remote user with an RTP based media, for each end-to-access-edge protected RTP based media, i.e. a RTP based media except those for which the result of the SDP offer / answer exchange results in the application of an end-to-end media security mechanism, the P-CSCF shall invoke IMS-ALG procedures, will act as defined in 3GPP TS 23.334 [7F] as far as SDP and RTP is concerned, and shall:

   - remove any SDP crypto attribute and any "a=acap:*x* crypto" SDP attribute (see RFC 5939 [137]);

   - if the SDP offer contains any potential configuration(s) using "RTP/SAVPF" transport or "RTP/SAVP" transport, as offered in corresponding Transport Protocol Capability SDP attribute(s) (see RFC 5939 [137]), (e.g. "a=tcap:*x* RTP/AVPF a=pcfg:*y* t=*x*"), remove those potential configuration(s);

   NOTE: Keeping the related "RTP/SAVPF" transport or "RTP/SAVP" transport within a Transport Protocol Capability SDP attribute that also contains other transports avoids a potential need to renumber other transports and adjust other potential configurations in the SDP offer and the actual configuration in the SDP answer accordingly.

   - if the SDP offer contains a Transport Protocol Capability SDP attribute (see RFC 5939 [137]) offering:

     a) "RTP/AVPF" transport (e.g. "a=tcap:*x* RTP/AVPF"), replace this transport with "RTP/SAVPF" within that attribute; and

     b) "RTP/AVP" transport (e.g. "a=tcap:*x* RTP/AVP"), replace this transport with "RTP/SAVP" within that attribute;

   - if the SDP offer contains any potential configuration(s) with delete-attribute parameter(s) (see RFC 5939 [137]), (e.g. "a=pcfg:1 a=-sm:1"), remove those potential configuration(s);

   - offer SRTP transport protocol according to RFC 3711 [169] and the profile defined in 3GPP TS 33.328 [19C];

   - include a SDP crypto attribute according to RFC 4568 [168] and the profile defined in 3GPP TS 33.328 [19C]; and

   - include a SDP "a=3ge2ae:applied" attribute; and

2) upon receiving an SDP answer to the SDP offer from remote user, for each accepted end-to-access-edge protected RTP based media, the P-CSCF will act as defined in 3GPP TS 23.334 [7F] as far as SDP and RTP is concerned, and shall remove the SDP crypto attribute.

If the P-CSCF indicated support for the end-to-access-edge media security for RTP media using DTLS-SRTP and certificate fingerprints during registration:

1) upon receiving an SDP offer from the served UE containing an end-to-access-edge protected RTP based media, i.e. an RTP based media:

   - transported using "UDP/TLS/RTP/SAVP" or "UDP/TLS/RTP/SAVPF" as the transport protocol according to RFC 5763 [222] and RFC 5764 [223];

   - with the SDP fingerprint attribute as defined in RFC 8122 [241];

   - with the SDP "a=3ge2ae:requested" attribute; and

   - with the SDP tls-id attribute as defined in RFC 8842 [240];

   the P-CSCF shall invoke IMS-ALG procedures, will act as defined in 3GPP TS 23.334 [7F] as far as SDP and "RTP/AVP" or "RTP/AVPF" over UDP is concerned, and shall strip the SDP "a=3ge2ae:requested" attribute, the SDP fingerprint attribute and the SDP tls-id attribute from the RTP based media of the received SDP offer; and

2) upon sending an SDP answer to the SDP offer from the served UE, for each end-to-access-edge protected RTP based media of the SDP offer from the served UE that is accepted in the SDP answer, the P-CSCF will act as defined in 3GPP TS 23.334 [7F] as far as SDP and "RTP/AVP" or "RTP/AVPF" over UDP is concerned and shall:

   - indicate the "UDP/TLS/RTP/SAVP" or "UDP/TLS/RTP/SAVPF" as the transport protocol according to RFC 5763 [222], RFC 5764 [223] and the profile defined in 3GPP TS 33.328 [19C];

   - include the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C]; and

   - include the SDP tls-id attribute as defined in RFC 8842 [240].

If the served UE indicated support for the end-to-access-edge media security for RTP media using DTLS-SRTP and certificate fingerprints during registration, and the P-CSCF indicated support for the end-to-access-edge media security for RTP media using DTLS-SRTP and certificate fingerprints during registration:

1) upon receiving an SDP offer from remote UE with an RTP based media, for each end-to-access-edge protected RTP based media, i.e. an RTP based media except those for which the result of the SDP offer / answer exchange results in the application of an end-to-end security mechanism, the P-CSCF shall invoke IMS-ALG procedures, will act as defined in 3GPP TS 23.334 [7F] as far as SDP and "RTP/AVP" or "RTP/AVPF" over UDP is concerned, and shall:

   - remove any SDP fingerprint attribute;

   - remove any SDP tls-id attribute;

   - offer "UDP/TLS/RTP/SAVP" or "UDP/TLS/RTP/SAVPF" as the transport protocol according to RFC 5763 [222], RFC 5764 [223] and the profile defined in 3GPP TS 33.328 [19C];

   - if the SDP offer contains any potential configuration(s) with delete-attribute parameter(s) (see RFC 5939 [137]), (e.g. "a=pcfg:1 a=-sm:1"), remove those potential configuration(s);

   - include the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C];

   - include the SDP "a=3ge2ae:applied" attribute; and

   - include the SDP tls-id attribute as defined in RFC 8842 [240]; and

2) upon receiving an SDP answer to the SDP offer from remote user, for each accepted end-to-access-edge protected RTP based media, the P-CSCF will act as defined in 3GPP TS 23.334 [7F] as far as SDP and

"RTP/AVP" or "RTP/AVPF" over UDP is concerned, and shall remove the SDP fingerprint attribute and SDP tls-id attribute.

If the P-CSCF indicated support for the end-to-access-edge media security for MSRP using TLS and certificate fingerprints during registration:

1) upon receiving an SDP offer from the served UE containing an end-to-access-edge protected MSRP based media, i.e. an MSRP based media:

   - transported using the MSRP over TLS transport protocol as defined in RFC 4975 [178] and RFC 6714 [214];

   - with the SDP fingerprint attribute as defined in RFC 8122 [241]; and

   - with the SDP "a=3ge2ae:requested" attribute;

   the P-CSCF shall invoke IMS-ALG procedures, will act as defined in 3GPP TS 23.334 [7F] as far as SDP and MSRP is concerned, and shall strip the SDP "a=3ge2ae:requested" attribute and the SDP fingerprint attribute from the end-to-access-edge protected MSRP based media of the received SDP offer; and

2) upon sending an SDP answer to the SDP offer from the served UE, for each end-to-access-edge protected MSRP based media of the SDP offer from the served UE which is accepted in the SDP answer, the P-CSCF will act as defined in 3GPP TS 23.334 [7F] as far as SDP and MSRP is concerned and shall:

   - indicate the MSRP over TLS transport protocol according to RFC 4975 [178], RFC 6714 [214] and the profile defined in 3GPP TS 33.328 [19C]; and

   - include the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C].

If the served UE indicated support for the end-to-access-edge media security for MSRP using TLS and certificate fingerprints during registration, and the P-CSCF indicated support for the end-to-access-edge media security for MSRP using TLS and certificate fingerprints during registration:

1) upon receiving an SDP offer from remote user with an MSRP based media, for each end-to-access-edge protected MSRP based media, i.e. an MSRP based media except those for which the result of the SDP offer / answer exchange results in the application of an end-to-end security mechanism, the P-CSCF shall invoke IMS-ALG procedures, will act as defined in 3GPP TS 23.334 [7F] as far as SDP and MSRP is concerned, and shall:

   - remove any SDP fingerprint attribute;

   - offer MSRP over TLS transport protocol according to RFC 4975 [178], RFC 6714 [214] and the profile defined in 3GPP TS 33.328 [19C];

   - if the SDP offer contains any potential configuration(s) with delete-attribute parameter(s) (see RFC 5939 [137]), (e.g. "a=pcfg:1 a=-sm:1"), remove those potential configuration(s);

   - include the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C]; and

   - include the SDP "a=3ge2ae:applied" attribute; and

2) upon receiving an SDP answer to the SDP offer from remote user, for each accepted end-to-access-edge protected MSRP based media, the P-CSCF will act as defined in 3GPP TS 23.334 [7F] as far as SDP and MSRP is concerned, and shall remove the SDP fingerprint attribute.

If the P-CSCF indicated support for the end-to-access-edge media security for BFCP using TLS and certificate fingerprints during registration:

1) upon receiving an SDP offer from the served UE containing an end-to-access-edge protected BFCP based media, i.e. a BFCP based media:

   - transported using the BFCP over TLS transport protocol as defined in RFC 4583 [108];

   - with the SDP fingerprint attribute as defined in RFC 8122 [241]; and

   - with the SDP "a=3ge2ae:requested" attribute;

the P-CSCF shall invoke IMS-ALG procedures, will act as defined in 3GPP TS 23.334 [7F] as far as SDP and BFCP is concerned, and shall strip the SDP "a=3ge2ae:requested" attribute and the SDP fingerprint attribute from the BFCP based media of the received SDP offer; and

2) upon sending an SDP answer to the SDP offer from the served UE, for each end-to-access-edge protected BFCP based media of the SDP offer from the served UE which is accepted in the SDP answer, the P-CSCF will act as defined in 3GPP TS 23.334 [7F] as far as SDP and BFCP is concerned and shall:

   - indicate the BFCP over TLS transport protocol according to RFC 4583 [108] and the profile defined in 3GPP TS 33.328 [19C]; and

   - include the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C].

If the served UE indicated support for the end-to-access-edge media security for BFCP using TLS and certificate fingerprints during registration, and the P-CSCF indicated support for the end-to-access-edge media security for BFCP using TLS and certificate fingerprints during registration:

1) upon receiving an SDP offer from remote UE with an BFCP based media, for each end-to-access-edge protected BFCP based media, i.e. a BFCP based media except those for which the result of the SDP offer / answer exchange results in the application of an end-to-end security mechanism, the P-CSCF shall invoke IMS-ALG procedures, will act as defined in 3GPP TS 23.334 [7F] as far as SDP and BFCP is concerned, and shall:

   - remove any SDP fingerprint attribute;

   - offer BFCP over TLS transport protocol according to RFC 4583 [108] and the profile defined in 3GPP TS 33.328 [19C];

   - if the SDP offer contains any potential configuration(s) with delete-attribute parameter(s) (see RFC 5939 [137]), (e.g. "a=pcfg:1 a=-sm:1"), remove those potential configuration(s);

   - include the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C]; and

   - include the SDP "a=3ge2ae:applied" attribute; and

2) upon receiving an SDP answer to the SDP offer from remote user, for each accepted end-to-access-edge protected BFCP based media, the P-CSCF will act as defined in 3GPP TS 23.334 [7F] as far as SDP and BFCP is concerned, and shall remove the SDP fingerprint attribute.

If the P-CSCF indicated support for the end-to-access-edge media security for UDPTL over DTLS and certificate fingerprints during registration:

1) upon receiving an SDP offer from the served UE containing an end-to-access-edge protected UDPTL based media, i.e. a UDPTL based media:

   - transported using the UDPTL over DTLS transport protocol as defined in RFC 7345 [217] and RFC 8842 [240];

   - with the SDP fingerprint attribute as defined in RFC 8122 [241];

   - with the SDP "a=3ge2ae:requested" attribute; and

   - with the SDP tls-id attribute as defined in RFC 8842 [240];

   the P-CSCF shall invoke IMS-ALG procedures, will act as defined in 3GPP TS 23.334 [7F] as far as SDP and UDPTL is concerned, and shall strip the SDP "a=3ge2ae:requested" attribute and the SDP fingerprint attribute and the SDP tls-id attribute from the UDPTL based media of the received SDP offer; and

2) upon sending an SDP answer to the SDP offer from the served UE, for each end-to-access-edge protected UDPTL based media of the SDP offer from the served UE which is accepted in the SDP answer, the P-CSCF will act as defined in 3GPP TS 23.334 [7F] as far as SDP and UDPTL is concerned and shall:

   - indicate the UDPTL over DTLS transport protocol according to RFC 7345 [217], RFC 8842 [240] and the profile defined in 3GPP TS 33.328 [19C];

- include the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C]; and

- include the SDP tls-id attribute as defined in RFC 8842 [240].

If the served UE indicated support for the end-to-access-edge media security for UDPTL using DTLS and certificate fingerprints during registration, and the P-CSCF indicated support for the end-to-access-edge media security for UDPTL using DTLS and certificate fingerprints during registration:

1) upon receiving an SDP offer from remote UE with an UDPTL based media, for each end-to-access-edge protected UDPTL based media, i.e. a UDPTL based media except those for which the result of the SDP offer / answer exchange results in the application of an end-to-end security mechanism, the P-CSCF shall invoke IMS-ALG procedures, will act as defined in 3GPP TS 23.334 [7F] as far as SDP and UDPTL is concerned, and shall:

   - remove any SDP fingerprint attribute;

   - remove any SDP tls-id attribute;

   - offer UDPTL over DTLS transport protocol according to RFC 7345 [217], RFC 8842 [240] and the profile defined in 3GPP TS 33.328 [19C];

   - if the SDP offer contains any potential configuration(s) with delete-attribute parameter(s) (see RFC 5939 [137]), (e.g. "a=pcfg:1 a=-sm:1"), remove those potential configuration(s);

   - include the SDP fingerprint attribute according to RFC 8122 [241] and the profile defined in 3GPP TS 33.328 [19C];

   - include the SDP "a=3ge2ae:applied" attribute; and

   - include the SDP tls-id attribute as defined in RFC 8842 [240]; and

2) upon receiving an SDP answer to the SDP offer from remote user, for each accepted end-to-access-edge protected UDPTL based media, the P-CSCF will act as defined in 3GPP TS 23.334 [7F] as far as SDP and UDPTL is concerned, and shall remove the SDP fingerprint attribute and SDP tls-id attribute.

6.7.2.3 IMS-ALG in P-CSCF for explicit congestion control support

6.7.2.3.1 General

An IMS-ALG may support ECN according to RFC 6679 [188].

Subject to local policy, an IMS-ALG shall prohibit the negotiation of ECN during SDP offer/answer exchanges associated with multimedia priority service by removing any ECN attribute "a=ecn-capable-rtp" from the SDP offer and shall not invoke ECN for SIP transactions associated with multimedia priority service.

6.7.2.3.2 Incoming SDP offer with ECN

If the IMS-ALG receives an SDP offer containing the "a=ecn-capable-rtp" attribute as specified in RFC 6679 [188] and:

- the IMS-ALG knows via configuration that the IMS-AGW supports transparently forwarding of ECN bits according to RFC 3168 [189];

- the IMS-ALG knows via configuration that the (IMS) network handles ECN-marked packets properly; and

- the IMS-ALG does not configure the IMS-AGW to transcode,

then the IMS-ALG shall:

- if the "ecn-capable-rtp" attribute includes both the "ice" initialisation method and other initialisation methods, remove the "ice" initialisation method from the "ecn-capable-rtp" attribute and add the attribute with this modification in the outgoing the SDP offer;

- if the "ecn-capable-rtp" attribute only includes the "ice" initialisation method, do not include the "ecn-capable-rtp" attribute it outgoing SDP offer; and

- if the "ecn-capable-rtp" attribute did not includes the "ice" initialisation method include the unmodified "ecn-capable-rtp" attribute within the outgoing SDP offer.

If the IMS-ALG receives an SDP offer containing the ECN attribute "a=ecn-capable-rtp" as specified in RFC 6679 [188] and any of the following conditions apply:

- the IMS-ALG knows by configuration that the IMS-AGW does not support transparent transport of ECN-marked packets;
- the IMS-ALG knows by configuration that the (IMS) network does not properly handle ECN-marked packets; or
- the IMS-ALG does not configure the IMS-AGW to transcode,

then the IMS-ALG shall not include ECN attributes in the outgoing SDP offer, and, if the IMS-ALG knows in addition via configuration that the IMS-AGW supports acting as an ECN endpoint and that the IMS-ALG supports at least some of the initialisation methods offered within the "a=ecn-capable-rtp" attribute, the IMS-ALG shall:

- select an initialisation method supported by the IMS-AGW; and
- return a SDP answer according to the capabilities of the IMS-AGW, containing the "a=ecn-capable-rtp" attribute,

and the IMS-ALG will configure the IMS-AGW to act as an end point for ECN.

If the IMS-ALG receives an SDP answer containing the "a=ecn-capable-rtp" attribute it will instruct the IMS-AGW to transparently forward the ECN bits described in RFC 3168 [189].

6.7.2.3.3 Incoming SDP offer without ECN

If the IMS-ALG receives a SDP offer without the "a=ecn-capable-rtp" attribute and all of the following conditions apply:

- the IMS-ALG knows via configuration that the IMS-AGW supports acting as ECN endpoint; and
- the IMS-ALG knows via configuration that the succeeding network supports ECN,

then the IMS-ALG may include the "a=ecn-capable-rtp" attribute in the offer it forwards towards the succeeding node, indicating the related capabilities of the IMS-AGW.

If the IMS-ALG inserted ECN attributes in the SDP offer and receives an SDP answer containing the "a=ecn-capable-rtp" attribute, the IMS-ALG shall return the SDP answer to the preceding node removing the "a=ecn-capable-rtp" attribute, and will configure the IMS-AGW to act as an ECN endpoint.

6.7.2.4 IMS-ALG in P-CSCF for Optimal Media Routeing (OMR)

Based on operator policy, the P-CSCF shall remove OMR related SDP attributes before it sends an SDP offer or answer towards an UE, as specified in subclause 2.1.9 of 3GPP TS 29.079 [11D].

6.7.2.5 IMS-ALG in P-CSCF for NA(P)T and NA(P)T-PT controlled by the P-CSCF

6.7.2.5.1 General

This subclause describes the P-CSCF procedures for supporting the scenario where IP address and/or port conversions occur at the IMS-AGW level in the media path between the UE and the backbone. Two types of address conversions are covered:

- IP version interworking (NA(P)T-PT); and
- IP address/port translation (NA(P)T).

When the P-CSCF performs procedures for P-CSCF controlled NA(P)T and NA(P)T-PT, it shall modify the IP address(es) and port numbers (in case of NA(P)T) in SDP offers and answers, based on the IP address(es) and port number(s) received from the IMS-AGW, as described in subclause 6.7.2.1.

For terminating sessions the P-CSCF may towards a UE performing the functions of an external attached network indicate in the SDP offer alternate IP address versions (IPv4 and IPv6) by inserting two "altc" attributes as defined in RFC 6947 [228]. The order of setting the two IP addresses in the two "altc" SDP attributes shall be based on local policy. The insertion of the "altc" attributes is independent of their presence in the received SDP offer.

NOTE 1: The insertion of alternate IP versions allows avoiding the rejection of the SDP offer because of incompatible network address formats and when the request terminates in a corporate network enables the corporate network to avoid IP version interworking.

NOTE 2: The handling of alternative IP addresses between the IMS-ALG and the TrGW is defined in 3GPP TS 23.334.

6.7.2.6 IMS-ALG in P-CSCF for support of hosted NAT

6.7.2.6.1 General

When the P-CSCF performs procedures for hosted NAT, it shall modify the IP address(es) and port numbers, based on the IP address(es) and number(s) received from the IMS-AGW, as described in subclause 6.7.2.1.

6.7.2.6.2 Hosted NAT traversal for TCP based streams

When the P-CSCF acts as an IMS-ALG, it acts as a B2BUA and modifies the SDP as described as described in 3GPP TS 23.334 [7F].

6.7.2.7 IMS-ALG in P-CSCF for support of ICE

6.7.2.7.1 General

This subclause describes procedures of a P-CSCF to support ICE, as defined in RFC 8445 [289] and RFC 8839 [290].

NOTE 1: If no IMS-AGW is inserted on the media plane, a P-CSCF might transparently pass ICE related SDP attibutes, in order to support ICE between the UE and remote entities. The remaining procedures in this subclause apply to when the P-CSCF inserts an IMS-ALG on the media plane.

When the P-CSCF with attached IMS-AGW receives SDP candidate information from the offerer, it shall not forward the candidate information towards the answerer. When the P-CSCF receives SDP candidate information from the answerer, it shall not forward the candidate information towards the offerer. The remaining procedures in subclause 6.7.2.7.1 are optional.

NOTE 2: An P-CSCF that removes and/or does not provide ICE related SDP attributes (e.g. a=candidate) in the offer/answer exchange will cause the ICE procedures to be aborted and the address and port information in the m and c lines of the SDP offer will be used. If this address and port information contains the relayed candidate address of a STUN Relay server, as recommended by ICE, then an extra media relay server will be used for the session which is not necessary nor desirable.

The P-CSCF with attached IMS-ALG performs separate ICE procedures towards the offerer and the answerer. The usage of ICE is negotiated separately with the offerer and answerer, and ICE may be applied independently at either side. Furthermore, the P-CSCF may be configured to apply ICE procedures only towards one network side, e.g. towards the IM CN subsystem it belongs to.

NOTE 3: Since the P-CSCF is inserting an IMS-ALG, it can choose to provide the NAT traversal mechanism defined in Annex F towards the UE. In such case the P-CSCF will not provide ICE support towards the UE, but the P-CSCF can still provide ICE support towards the core network in scenarios where ICE is used in the core network, e.g. to support NAT traversal for other access networks with no deploied IMS-ALGs.

Since the P-CSCF is not located behind a NAT, it does not request the IMS-ALG to generate keep-alive messages even when acting as a full ICE entity. The P-CSCF only requests the IMS-ALG to terminate and generate STUN messages used for the candidate selection procedures.

Since the P-CSCF is not located behind a NAT the P-CSCF shall only include host candidates in SDP offers and answers generated by the P-CSCF.

6.7.2.7.2 P-CSCF full ICE procedures for UDP based streams

6.7.2.7.2.1 General

This subclause describes the P-CSCF full ICE procedures for UDP based streams.

6.7.2.7.2.2 P-CSCF receiving SDP offer

When the P-CSCF receives an SDP offer including ICE candidate information, the P-CSCF shall send the candidate information for each UDP based stream received in the SDP offer towards the IMS-ALG. If the SDP offer includes TCP candidate information for a UDP based stream, the P-CSCF may send such candidate information to the IMS-AGW, in addition to the UDP candidate information as defined in RFC 6544 [131]. The P-CSCF shall request the IMS-ALG to reserve media- and STUN resources towards the offerer, based on the candidate information, in order to allow the IMS-ALG to perform the necessary connectivity checks per the ICE procedures.

If the offerer is acting as an ICE controller entity the P-CSCF shall act as an ICE controlled entity in the direction towards the offerer. If the offerer is acting as an ICE controlled entity the P-CSCF shall act as an ICE controller entity in the direction towards the offerer.

6.7.2.7.2.3 P-CSCF sending SDP offer

Prior to sending an SDP offer, the P-CSCF may choose to apply related ICE procedues, e.g. if it expects to interact with terminals applying procedures as described in subclause K.5.2, and if both the P-CSCF and IMS-ALG also support ICE procedures. To invoking these ICE procedures, the P-CSCF shall request the IMS-ALG to reserve media- and STUN resources towards the answerer for each UDP based media stream and include a host candidate attribute for each UDP based stream in the SDP offer, providing the reserved address and port at the IMS-ALG as destination. The P-CSCF may also include host TCP candidate information for UDP based streams in the SDP offer as defined in RFC 6544 [131].

The P-CSCF shall always act as an ICE controller entity towards the answerer.

NOTE: The host candidate address included by the P-CSCF in the generated SDP offer matches the c- and m line information for the associcated UDP stream in the SDP offer.

6.7.2.7.2.4 P-CSCF receiving SDP answer

When the P-CSCF receives an SDP answer including ICE candidate information, the P-CSCF shall send the candidate information for each UDP based stream received in the SDP answer towards the IMS-ALG.

The P-CSCF shall request the IMS-ALG to perform ICE candidate selection procedures towards the answerer. The P-CSCF shall request the IMS-ALG to inform the P-CSCF, for each UDP stream, which candidate pair has been selected towards the answerer, once the candidate selection procedure towards the answerer has finished.

If the IMS-ALG indicates to the P-CSCF that, for at least one UDP stream, the selected candidate pair does not match the c- and m- line address information for the associated UDP stream, exchanged between the P-CSCF and the answerer, and the P-CSCF acts an ICE controller entity towards the answerer, the P-CSCF shall send a new offer towards the answerer in order to allign the c- and m- lines address information with the chosen candidate pair for the associated UDP stream.

6.7.2.7.2.5 P-CSCF sending SDP answer

When the P-CSCF generates an SDP answer for an offer that included ICE candidate information, the P-CSCF shall request the IMS-ALG to reserve media- and STUN resources towards the offerer for each UDP based media stream and include an SDP host candidate attribute for each UDP based stream in the SDP answer, providing the reserved address and port at the IMS-ALG as destination.

The P-CSCF shall in the generated SDP answer include host candidate information which matches the c- and m line information for the associated UDP stream in the SDP answer.

The P-CSCF shall request the IMS-ALG to perform ICE candidate selection procedures towards the offerer. The P-CSCF shall request the IMS-ALG to inform the P-CSCF, for each UDP stream, which candidate pair has been selected towards the offerer, once the candidate selection procedure towards the answerer has finished.

If the IMS-ALG indicates to the P-CSCF that the selected candidate pair towards the offerer does not match the c- and m- line address information for the associated UDP stream, exchanged between the P-CSCF and the offerer, and the P-CSCF acts an ICE controller entity towards the offerer, the P-CSCF shall send an offer towards the offerer (which will now act as an answerer) in order to allign the c- and m- line address information with the chosen candidate pair for the associated UDP stream.

6.7.2.7.3 P-CSCF ICE lite procedures for UDP based streams

When the P-CSCF is using ICE lite procedures for UDP based streams, the P-CSCF procedures are identical as described in subclause 6.7.2.7.2, with the following exceptions:

- The P-CSCF always acts as an ICE controlled entity towards the offerer and towards the answerer; and
- The P-CSCF requests the IMS-ALG to perform ICE lite candidate selection procedures, as defined in RFC 8445 [289].

6.7.2.7.4 ICE procedures for TCP based streams

6.7.2.7.4.1 General

The P-CSCF shall disable ICE procedures for TCP based streams, i.e. streams where TCP is indicated as transport protocol in the m-line. Instead the P-CSCF will use the mechanism defined in RFC 4145 [83] for establishing TCP based streams, as defined in RFC 6544 [131].

NOTE 1: Handling of TCP candidates for UDP based streams is described in subclause 6.7.2.7.2.

NOTE 2: An entity that supports ICE continues the ICE procedures for UDP based streams, even if no candidates are provided for TCP based streams.

6.7.2.7.4.2 P-CSCF receiving SDP offer

When the P-CSCF receives an SDP offer, the P-CSCF shall ignore the candidate attributes for TCP based streams. The P-CSCF shall not send the candidate information for TCP based streams towards the IMS-ALG.

6.7.2.7.4.3 P-CSCF sending SDP offer

When the P-CSCF generates an SDP offer the P-CSCF shall include an "actpass" setup attribute, as defined in RFC 4145 [83], for each TCP based stream, which will cause the answerer to initiate the TCP connections towards the IMS-ALG. The P-CSCF shall not include any candidate attributes for TCP based streams in the SDP offer.

6.7.2.7.4.4 P-CSCF receiving SDP answer

Since the P-CSCF does not include candidates in the SDP offer towards the answerer, there are no ICE specific procedures when the P-CSCF receives an SDP answer.

NOTE: If the SDP answer contains candidate attributes for TCP based streams, the P-CSCF simply discards the candidate attributes.

6.7.2.7.4.5 P-CSCF sending SDP answer

When the P-CSCF generates an SDP answer the P-CSCF shall include a "passive" setup attribute, as defined in RFC 4145 [83], for each TCP based stream, which will cause the offerer to initiate the TCP connections towards the IMS-ALG. The P-CSCF shall not include any candidate attributes for TCP based streams in the SDP answer.

6.7.2.8 IMS-ALG in P-CSCF for transcoding

An IMS-ALG may support procedures to modify SDP for transcoding purposes. The IMS-ALG shall only apply those transcoding procedures if an attached IMS-AGW supports transcoding.

Upon receipt of an SDP offer, based on local policy and SDP signalling inspection, the IMS-ALG may decide to offer transcoding.

To offer transcoding at the IMS-AGW, the IMS-ALG shall add codecs selected by local policy and supported by the IMS-AGW to the SDP offer. The local policy shall be in accordance with the requirements of clause T.2.

Upon receipt of the corresponding SDP answer, the IMS-ALG shall inspect the list of the codecs within the SDP answer and proceed as follows:

- If the list contains at least one of the codecs that was already contained in the previously received SDP offer, no transcoding at the IMS-AGW is required and the IMS-ALG will configure the IMS-AGW accordingly. The IMS-ALG shall remove from the SDP the codecs added to the original offer before forwarding the response to the offerer.

- If only the codecs inserted by the IMS-ALG are contained in the answer, the IMS-ALG will configure the IMS-AGW to transcode. The IMS-ALG shall replace the received codecs in the SDP anwer with the codec it configured the IMS-AGW to use towards the SDP offerer´s direction.

For an IMS-ALG acting as ATCF, the following applies in addition:

- During an originating or terminating session establishment, for media using PS transport towards the UE, the IMS-ALG (ATCF) should pass SDP offers without adding codecs to the SDP offer and pass SDP answers without modification to the contained codecs to avoid the potential need for transcoding in the IMS-AGW before the PS to CS access transfer; and

- during the PS to CS access transfer procedure, the IMS-ALG (ATCF) shall preferentially select from the SDP offer it receives from the MSC server the codec already configured on the corresponding remote leg, if available.

6.7.3 IMS-ALG in ISC gateway function

6.7.3.1 General

When the ISC gateway function acts as an IMS-ALG, it makes procedures as for an originating UA and terminating UA. The IMS-ALG acts as a B2BUA. For the use of the IMS-ALG for specific capabilities, additional procedures are defined in subsequent subclauses.

NOTE: The internal function of the IBCF as an IMS-ALG is defined in 3GPP TS 29.162 [11A], and the capabilities are identical for the ISC gateway function.

6.7.3.2 IMS-ALG in application gateway function for support of ICE

The application gateway function shall act according to the procedures defined for the IBCF in subclause 6.7.1.2.

7 Extensions within the present document

7.1 SIP methods defined within the present document

There are no SIP methods defined within the present document over and above those defined in the referenced IETF specifications.

7.2 SIP header fields defined within the present document

7.2.0 General

This subclause defines additional header fields.

7.2.1 Void

7.2.2 Void

7.2.3 Void

7.2.4 Void

7.2.5 Void

7.2.6 Void

7.2.7 Void

7.2.8 Void

7.2.9 Void

7.2.10 Void

7.2.11 Definition of Restoration-Info header field

7.2.11.1 Introduction

IANA registry: Header Fields registry for the Session Initiation Protocol (SIP)

Header field name: Restoration-Info

Usage: The Restoration-Info header field is used only for informative purposes.

Header field specification reference: 3GPP TS 24.229, http://www.3gpp.org/ftp/Specs/archive/24_series/24.229/

In case of a node failure there are cases where an upstream node can use information about a node failure. The upstream node can use this information for error reporting, or possibly for error recovery.

An upstream node can inform a downstream node about supported error recovery mechanisms. The downstream node can use this information for error recovery.

7.2.11.2 Applicability statement for the Restoration-Info header field

The Restoration-Info header field is applicable within a single private administrative domain or between different administrative domains.

The Restoration-Info header field is applicable when:

1) a node has failed and the SIP node detecting this failure needs to inform a proxy in the administrative domain of the terminating user about the failure; or

2) a proxy located in the private administrative domain of a user wants to send information about the subscriber to a downstream proxy for error recovery.

For case 1) the SIP node detecting the failure can include the Restoration-Info header field set to the value "noresponse" in a 408 (Request Timeout) response to an INVITE request or a 504 (Server Time-out) response to a dialog forming request or standalone transaction,

For case 2) the Restoration-Info header field is included in an initial INVITE request with an "IMSI" header field parameter set to a value identifying the user.

7.2.11.3 Usage of the Restoration-Info header field

A SIP entity that does not receive a response from the next SIP node, may include a Restoration-Info header field in the error response to inform upstream nodes or networks about the downstream node failure. The upstream nodes or networks may use this information to either inform the originating user, to report the failure or to initiate restoration.

A SIP entity in the home network domain may use the Restoration-Info header field to transport an IMSI value to downstream SIP entities. The downstream SIP entity can use this information to initiate restoration for this user.

7.2.11.4 Procedures at the UA

There are no specific procedures specified for a UA. A UAC in a B2BUA may use the information in the Restoration-Info header field for error reporting, or take this information into account when deciding on re-attempting the request. A UAS may include a Retoration-Info header field in an error respons to inform upstream nodes or networks about the downstream node failure.

7.2.11.5 Procedures at the proxy

A SIP proxy that supports this extension and receives a request may insert a Restoration-Info header field prior to forwarding the request. The header field is populated with the IMSI value received in the body of a DIAMETER request as per 3GPP TS 29.228 [14] within the quoted string .

A SIP proxy that supports this extension and receives a request with the Restoration-Info header field, may retrieve the IMSI value from the header field and use it to poulate a DIAMETER request as per 3GPP TS 29.214 [13D] for the purposes of performing PCRF restoration procedures.

A SIP proxy that supports this extension and receives a 408 response with this header field present can use this information for restoration procedures or reporting.

7.2.11.6 Security considerations

The Restoration-Info header field can contain sensitive information. When the Restoration-Info header field contains the IMSI value, it shall be sent only to trusted entities.

A UE is not expected to receive this information.

7.2.11.7 Syntax

The syntax for Restoration-Info header field is specified in table 7.2.11-1.

Table 7.2.11-1: Syntax of Restoration-Info

```
Restoration-Info      = "Restoration-Info" HCOLON pcrf-token / reason / generic-param
pcrf-token            = ("IMSI" / ext-type) EQUAL pcrf-param
pcrf-param            = quoted-string
reason                = "noresponse"
ext-type              = token
```

7.2.11.8 Examples of usage

The Restoration-Info header field can be inserted by the neighbouring upstream SIP node to the SIP node that does not respond. The header field value "noresponse" can be used to inform the upstream SIP entity about the failure. The upstream SIP entity such as a 3GPP S-CSCF can use this information to initiate restoration procedures. The restoration can be in the form of a lower layer message to the terminating UE to indicate that the UE needs to perform a new SIP registration.

The Restoration-Info header field can be used to transport the IMSI value from the S-CSCF to a P-CSCF. The S-CSCF obtains the IMSI value as a string over the 3GPP Cx interface specified in 3GPP TS 29.228 [14]. The downstream node can include the IMSI string received in a diameter request specified in 3GPP TS 29.214 [13D]. The receiver of this diameter request uses the information to find the UE and indicate that the UE needs to perform a new SIP registration.

The indication to the UE that it needs to perform a new SIP registration is sent over a lower layer.

7.2.12 Relayed-Charge header field

7.2.12.1 Introduction

IANA registry: Header Fields registry for the Session Initiation Protocol (SIP)

Header field name: Relayed-Charge

Usage: The Relayed-Charge header field is used only for informative purposes.

Header field specification reference: 3GPP TS 24.229, http://www.3gpp.org/ftp/Specs/archive/24_series/24.229/

The P-Charging-Vector header field is used to carry information relating to charging as it accumulates to various entities within the IM CN subsystem. The information within that header field is applicable to the current dialog or transaction at the point where it is received. Sometimes it is appropriate to carry this accumulated charging information, relating to the same dialog or transaction to other entities within the IM CN subsystem. The Relayed-Charge header field is defined to relay the current contents of the P-Charging-Vector header field as known by one entity to another entity with an indication of the Source entity.

7.2.12.2 Applicability statement for the Relayed-Charge header field

The Relayed-Charge header field is applicable within a single private administrative domain or between different administrative domains where there is a trust relationship between the domains.

The Relayed-Charge header field is not included in a SIP message sent to another network if there is no trust relationship.

The Relayed-Charge header field is applicable whenever the P-Charging-Vector header field would be applicable, as defined by RFC 7315 [52].

7.2.12.3 Usage of the Relayed-Charge header field

A SIP entity that receives a P-Charging-Vector header field may take appropriate fields from the received header field and encode them in the equivalent field within the Relayed-Charge header field, along with a value in the relay-source to indicate the relaying SIP entity.

A SIP UA or SIP proxy that receives a SIP request or response that contains a Relayed-Charge header field can use the values, to produce charging records.

A SIP proxy may remove the Relayed-Charge header field if it is known there is no intended collector of the Relayed-Charge header field subsequent in the path of the request or response.

7.2.12.4 Procedures at the UA

This document does not specify any procedure at a UA located outside the administrative domain of a private network (e.g., PSTN gateway or conference mixer), with regard to the Relayed-Charge header field. UAs need not understand this header field.

However, it might be possible that a UA be located within the administrative domain of a private network (e.g., a PSTN gateway, or conference mixer), and it may interact with the charging entities.

In this case, a UA may insert the Relayed-Charge header field in a SIP request or response when the next hop for the message is a proxy or UA located in the same administrative domain. Similarly, such a UA may use the contents of the Relayed-Charge header field in communicating with the charging entities.

7.2.12.5 Procedures at the proxy

A SIP proxy that supports this extension and receives a request or response without the Relayed-Charge header field MAY insert a Relayed-Charge header field prior to forwarding the message. The header is populated with one or more parameters, as described in the syntax, including but not limited to, a globally unique charging identifier.

If a proxy that supports this extension receives a request or response with the Relayed-Charge header field, it may retrieve the information from the header value to use with application-specific logic, i.e., charging. If the next hop for the message is within the trusted domain, then the proxy should include the Relayed-Charge header field in the outbound message. If the next hop for the message is outside the trusted domain, then the proxy may remove the Relayed-Charge header field.

Per local application-specific logic, the proxy may modify the contents of the Relayed-Charge header field prior to sending the message.

7.2.12.6 Security considerations

It is expected as normal behavior that proxies within a closed network will modify the values of the Relayed-Charge header field and insert it into a SIP request or response. However, these proxies that share this information shall have a trust relationship.

If an untrusted entity were inserted between trusted entities, it could potentially interfere with the charging correlation mechanism. Therefore, an integrity-protection mechanism such as IPsec or other available mechanisms shall be applied in order to prevent such attacks. Since each trusted proxy may need to view or modify the values in the Relayed-Charge header field, the protection should be applied on a hop-by-hop basis.

7.2.12.7 Syntax

The syntax for Relayed-Charge header field is specified in table 7.2.12.1

Table 7.2.12.1: Syntax of Relayed-Charge

```
   relayed-charge      = "Relayed-Charge" HCOLON relayed-charge-list
   relayed-charge-list = relayed-charge-item *(COMMA relayed-charge-item)
   relayed-charge-item = relay-source HCOLON charge-params *(SEMI charge-params)
   relay-source = "PCSCF" / "SCSCF" / "IBCF" / "transitfunction" / "ICSCF" / other-source
   other-source = token

charge-params are as defined for the P-Charging-Vector header field
```

7.2.12.8 Examples of usage

The Relayed-Charge header field is used in situations where there is a need to carry charging information applicable to a dialog or transaction which is not directly pertinent to the next hop. So for example, at the S-CSCF the accumulation of the "transit-ioi" header field parameter for an incoming call is removed and a new accumulation of "transit-ioi" header field parameters started. The received transit-ioi header field parameter accumulation can be passed to the online charging server (acting as an AS in the IM CN subsystem) using the Relayed-Charge header field.

7.2.13 Resource-Share header field

7.2.13.1 Introduction

IANA registry: Header Fields registry for the Session Initiation Protocol (SIP)

Header field name: Resource-Share

Usage: The Resource-Share header field is used only for informative purposes.

Header field specification reference: 3GPP TS 24.229, http://www.3gpp.org/ftp/Specs/archive/24_series/24.229/

The P-CSCF in the 3GPP architecture is responsible for reserving resources in the media plane. The resources reservation procedure includes the possibility to allow resources to be shared between sessions involving the same UE. The possibility to share resources can be dependent on services controlled by application servers.

Since the P-CSCF is service unaware the P-CSCF can benefit from receiving information from application servers regarding potential resource sharing options.

7.2.13.2 Applicability statement for the Resource-Share header field

The Resource-Share header field is applicable within a single private administrative domain or between different administrative domains.

7.2.13.3 Usage of the Resource-Share header field

The P-CSCF can include the Resource-Share header field in the REGISTER request to indicate the support of receiving resource sharing information from application servers in the user's home network or in an subsequent request or response within an existing dialog created by an INVITE request to indicate that resource sharing no longer is possible.

An application server in a user's home network acting as a SIP proxy or a UA may use a Resource-Share header field to transport resource sharing information in any request or response destined for the served user.

The P-CSCF receiving a request or response destined for a served UE containing a Resource-Share header field can use the resource sharing information when reserving resources in the media plane.

7.2.13.4 Procedures at the UA

An application server acting as a UA that supports this extension and receives a request or response destined for the served user containing an SDP offer or answer may insert a Resource-Share header field prior to forwarding the request or response. The value of the header field set to "media-sharing" or "no-media-sharing". When set to "media-sharing" the header field shall further be populated with the "rules" and "timestamp" header field parameters.

7.2.13.5 Procedures at the proxy

When a P-CSCF supporting this extension receives a REGISTER request from a served UE, the P-CSCF may insert a Resource-Share header field prior to forwarding the REGISTER request. The value of the header field is then set to "supported".

When the P-CSCF receives an SDP offer or answer from the served UE in a subsequent request or response within an existing dialog and if the SDP offer or answer contains information conflicting with the applied resource sharing, the P-CSCF may include the Resource-Share header field set to "no-media-sharing" in the request or response sent towards the application server.

When an application server acting as a SIP proxy supporting this extension receives a request or response destined for the served user containing an SDP offer or answer, the SIP proxy may insert a Resource-Share header field prior to forwarding the request or response. The value of the header field set to "media-sharing" or "no-media-sharing". When set to "media-sharing" the header field shall further be populated with the "rules" and "timestamp" header field parameters.

When the P-CSCF supporting this extension receives a request or response destined for the served UE containing the Resource-Share header field with the value "media-sharing", the P-CSCF may extract resource sharing rules from the "rules" header field parameter and use the extracted resource sharing rules to populate a DIAMETER request as per 3GPP TS 29.214 [13D] for the purposes of performing resource sharing procedures.

7.2.13.6 Security considerations

The Resource-Share header field does not contain any information that can disclose user information or the topology of nodes within an operator network.

7.2.13.7 Syntax

The syntax for Resource-Share header field is specified in table 7.2.13.1

Table 7.2.13.1: Syntax of Resource-Share

```
resource-share      = "Resource-Share" HCOLON r-s-param
r-s-param                = r-s-supported / r-s-no-media-sharing / r-s-media-sharing / r-s-other
r-s-supported            = "supported" [SEMI origin] *(SEMI generic-param)
r-s-no-media-sharing     = "no-media-sharing" SEMI origin *(SEMI generic-param)
r-s-media-sharing        = "media-sharing" SEMI origin SEMI resource-sharing-rules SEMI
                           timestamp *(SEMI generic-param)
r-s-other                = other-status *(SEMI generic-param)
other-status             = token
origin                   = "session-initiator" / "session-receiver" / other-origin
other-origin             = token
resource-sharing-rules   = "rules" EQUAL DQUOTE resource-sharing-rule *(COMMA
                           resource-sharing-rule) DQUOTE
resource-sharing-rule    = [ active-resource-sharing-rule ]
active-resource-sharing-rule    = new-sharing-key COLON [ existing-sharing-key-list ]
                           COLON directionality *( COLON generic-rule-param-value )
new-sharing-key          = sharing-key
existing-sharing-key-list = sharing-key *(SLASH sharing-key)
directionality           = "UL" / "DL" / "UL-DL" / other-directionality
other-directionality     = token
sharing-key              = token
generic-rule-param-value  = token
timestamp                = "timestamp" EQUAL 1*DIGIT
```

7.2.13.8 Operation

7.2.13.8.1 General

The values in the "resource-share" header field field are defined as follows:

"supported" indicates that the sender would like to receive information about resource sharing options for sessions involving the UE identified by the "+sip.instance" header field parameter in the Contact header field.

"media-sharing" indicates that an application server has determined that one or more media streams in the session can be subject for resource sharing.

"no-media-sharing" indicates that an application server or the P-CSCF has determined that none of the media streams in the session are subjects for resource sharing.

The Resource-Share header field contains the "origin", "rules" and "timestamp" header field parameters.

7.2.13.8.2 The "origin" header field parameter

The "origin" header field parameter is used to identify the source of the resource sharing information. The values in the "origin" header field field are defined as follows:

"session-initiator" indicates that the application server or the P-CSCF that included the Resource-Share header field is serving the UE sending the initial INVITE request.

"session-receiver" indicates that the application server or the P-CSCF that included the Resource-Share header field is serving the UE receiving the initial INVITE request.

7.2.13.8.3 The "rules" header field parameter

The "rules" header field parameter carries one or more rules for resource sharing. Each rule is included in the same order as the corresponding m-line in the SDP offer/answer and consists of the following parts:

"new-sharing-key" this part is mandatory and identifies a media stream in an exisiting ongoing session or is a new sharing key value when the UE is not already involved in a session subject for resource sharing. The same value of the "new-sharing-key" can only appear in one media stream.

"existing-sharing-key-list" this part is optional and is only included in the INVITE request when the request is forked and if there are UEs (registered via a P-CSCF indicating that receiving resource sharing option information would be useful) already involved in sessions where the media-stream can be shared. Each value in the "existing-sharing-list" identifies a media stream in the ongoing session. In the forking case the "new-sharing-key" includes a new sharing key value to be used by UEs not involved in a session yet. The same value of a sharing key in the "existing-sharing-key-list" can only appear in one media stream.

"directionality" this part indicates in which direction resource sharing applies. "UL" indicates that resource sharing can be applied in the direction from the UE. "DL" indicates that resource sharing can be applied in the direction towards the UE. "UL-DL" indicates that resource sharing can be applied in both directions.

7.2.13.8.4 The "timestamp" header field parameter

The "timestamp" header field parameter indicates when the application server determined the resource sharing rules and is used to determine the most applicable resource sharing option.

NOTE: Since the media streams in several sessions can be shared race conditions can occur due to retransimissions of requests or responses carrying the Resource header field.

The value is a counter unique for each user and is increased and inserted in the header field each time the application server includes a Resource-Share header field in a request or response involving a UE registered by the user.

When the P-CSCF receives a Resorce-Share header field, the P-CSCF extracts and stores the extracted resource sharing rule along with the value of the received "timestamp" header field as follows:

1) if a resource sharing rule identified by the sharing key is not already stored, store the extracted resource sharing rule along with the value of the received "timestamp" header field;

2) if a resource sharing rule identified by the sharing key is already stored with a lower timestamp value than the value of the received "timestamp" header field, replace the stored resource sharing rule with the extracted resource sharing rule along with the value of the received "timestamp" header field; or

3) if a resource sharing rule identified by the sharing key is stored with a higher timestamp value than the value of the received "timestamp" header field, discard the extracted resource sharing rule.

The "timestamp" header field can be reset to "0" when none of the UEs registered by the user is involved in a session any longer.

7.2.13.9 Examples of usage

7.2.13.9.1 Example overview

The following subclauses describe examples on how:

- the P-CSCF indicates in the REGISTER request that P-CSCF supports receiving information about resource sharing;

- the application server sends information about potential resource sharing to the P-CSCF; and

- the P-CSCF extracts resource sharing information for media-streams.

7.2.13.9.2 The P-CSCF indicates in the REGISTER request that P-CSCF supports receiving information about resource sharing

When P-CSCF receives a REGISTER request from a UE served by the P-CSCF, the P-CSCF can include a Resource-Share header field indicating that the P-CSCF supports receiving information about resource sharing.

The example 1 shows the coding when the P-CSCF indicates that the P-CSCF is interested in receiving information about resource sharing in a REGISTER request.

EXAMPLE 1: Resouce-Share: supported

7.2.13.9.3 The application server sends information about potential resource sharing to the P-CSCF

When the application server receives a request or response containing an initial SDP offer/answer with media streams subject for resource sharing, the application server includes the Resource-Share header field with the value "media-sharing" and includes a "origin" header field parameter set to "session-initiator" or "session-receiver" depending on if the application server is serving the user that initiated the session invitation or if the application server is serving the user receiving the session invitation.

The application server includes resource sharing information in a "rules" header field parameter with one resource sharing rule per media stream in the same order the corresponding m-line appears in the SDP. Each resource sharing rule is constructed as follows:

1) if the media stream is subject for resource sharing, the application server:

   - includes a "new-sharing-key" part;

   - if it is the INVITE request and the request will be sent to more than one UE, includes an "existing-resource-sharing-list" part containing one or more sharing keys already in use in other sessions involving UEs that potentially can receive the session invitation due to the forking of the INVITE request; and

   - includes a "directionality" part indicating in which direction resources sharing can apply; or

2) if the media stream can never be shared, includes an empty string.

Finally, the application server includes a "timestamp" header field parameter with a value higher than included in any other Resource-Share header field involving any of the UEs registered by the user.

The example 1 shows the Resource-Share header field when included in the initial SDP answer on the originating side. The SDP answer contains two media streams and both media streams are subject to resource sharing.

EXAMPLE 1: Resouce-Share: media-sharing; session-initiator; rules="k1::UL, k20::UL-DL"; timestamp=55688

The example 2 shows the Resource-Share header field when included in the initial SDP offer on the terminating side. The user has several UEs registered where three UEs are already involved in sessions with media streams subject to resource sharing. The SDP offer contains three media streams where only the first and third media stream is subject to resource sharing identified by K2,K3 and K4 for the first media stream and K21, K22 and k23 for the third media stream in already ongoing sessions. The fact that the second media stream is not subject to resource sharing is indicated as an empty string in second position in the comma delimited list of resource sharing rules in the "rules" header field parameter.

EXAMPLE 2: Resouce-Share: media-sharing; session-receiver; rules="k1:k2/k3/k4:UL,, k20:k21/k22/k23:UL-DL"; timestamp=45678

The example 3 shows the Resource-Share header field when included in a SIP request or SIP response on the originating side when an application server indicates that resources can not be shared due to some service specific reason. This indication can be included already from the beginning of the session or at any point during a session if the SIP proxy or UA determines that resource sharing is not possible any longer.

EXAMPLE 3: Resource-Share: no-media-sharing; session-initiator

7.2.13.9.4 The P-CSCF extracts resource sharing information for media-streams

When the P-CSCF receives an initial SDP answer destined for the served UE in a request or response containing the Resource-Share header field, the P-CSCF extracts the resource sharing rules for each media stream from the "rules" header field parameter in the same order that the corresponding m-line appear in the SDP. The P-CSCF stores and uses the value in the "new-sharing-key" to identify the resource sharing rule for a media stream.

When the P-CSCF receives an initial SDP offer destined for the served UE in a request, the P-CSCF extracts the resource sharing rules for each media stream from the "rules" header field parameter in the same order that the corresponding m-line appear in the SDP. For each extracted resource sharing rule the P-CSCF checks if the UE is involved in any session using a sharing key in the "existing-sharing-key-list" to identify a media-stream, and

- if the UE is involved in a session using a sharing key in the "existing-sharing-key-list" to identify a media-stream, the P-CSCF stores and uses that sharing key value to identify this resource sharing rule for the media stream in this session; or

- if none of the sharing keys in the "existing-sharing-key-list" is used by any session involving the UE or if the "existing-sharing-key-list" is empty, the P-CSCF stores and uses the value in the "new-sharing-key" to identify this resource sharing rule for this media stream in this session.

NOTE: Before storing and using an extracted resource sharing rule the P-CSCF determines the applicability of the rule using the value of the "timestamp" header field parameter as described in subclause 7.2.13.8.4.

If the P-CSCF receives a Resource-Share header field with the value "no-media-sharing" for media streams where resource sharing is already applied due to receipt of a Resource-Share header field with the value "media-sharing" prior to receiving "no-media-sharing" value, the SIP proxy stops media sharing as specified in 3GPP TS 29.214 [13D] annex A.

7.2.14 Definition of Service-Interact-Info header field

7.2.14.1 Introduction

IANA registry: Header Field Parameter Registry for the Session Initiation Protocol (SIP)

Header field name: Service-Interact-Info

Usage: The Service Interact-Infor header field is used only for informative purposes.

Header field specification reference: 3GPP TS 24.229, http://www.3gpp.org/ftp/Specs/archive/24_series/24.229/

One subscriber can subscribe to one or more services provided by different ASs, and one service may be in conflict with one or more other service. Since the conflict can be subject to the status of the service execution, it cannot be avoided during the service provisioning phase.

To avoid such service conflicts, it is needed to have a mechanism to convey information about services executed between the ASes, and an AS can take such information into account to avoid conflicts when executing the local service logic.

7.2.14.2 Applicability statement for the Service-Interact-Info header field

The Service-Interact-Info header field is applicable within a trust domain. The Service-Interact-Info header field can be included in initial SIP requests and responses to initial SIP requests.

AS can include the service identity which has been executed into the Service-Interact-Info header field and also insert service identities which is in conflict with the already executed service.

7.2.14.3 Usage of the Service-Interact-Info header field

Upon receiving a SIP message and executing service logic, the AS should take the information contained in the Service-Interact-Info header field into account. If

1) the executed services indicated in the Service-Interact-Info header field is in conflict with the local service logic; or

2) the local service logic indicated the Service-Interact-Info header field is inconflict with a previously executed service;

the AS should based on local policy decide whether or not to execute the local service.

When certain service logic has been executed, the AS should include the corresponding service identity into the Service-Interact-Info header field. Additionally, the AS can also include identities of any service which may be in conflicit with the executed service.

7.2.14.4 Procedures at the UA

There are no specific procedures specified for a UA. A UAC in a B2BUA can add a Service-Interact-Info header field into the SIP message, or insert a service identity into the Service-Interact-Info header field, or remove the Service-Interact-Info header field when sending a SIP message

7.2.14.5 Procedures at the proxy

A SIP proxy that supports this extension can add a Service-Interact-Info header field into a SIP message, insert a service identity into the Service-Interact-Info header field, or remove the Service-Interact-Info header field when forwarding the SIP message.

7.2.14.6 Security considerations

The Service-Interact-Info header field can contain sensitive information. The Service-Interact-Info header field should be removed when sent outside the trust domain.

A UE is not expected to receive the Service-Interact-Info header field.

7.2.14.7 Syntax

The syntax for Service-Interact-Info header field is specified in table 7.2.14-1.

Table 7.2.14-1: Syntax of Service-Interact-Info

```
Service-Interact-Info    = "Service-Interact-Info" HCOLON executed-service-params*(COMMA executed-
                           service-params)
executed-service-params  = executed-service / avoid-service
executed-service         = "executed-service" EQUAL service-spec
avoid-service            = "avoid-service" EQUAL service-spec
service-spec             = service-id *(SEMI service-param)
service-id               = token/quoted-string
service-param            = generic-param
```

7.2.15 Definition of Cellular-Network-Info header field

7.2.15.1 Introduction

A User Agent (UA) supporting one or more cellular radio access technology (e.g. E-UTRAN) but using a non-cellular IP-CAN to access the IM CN subsystem can use this header field to relay information to its service provider about the radio cell identity of the cellular radio access network the UE most recently camped on. For example, a UE making an emergency call using the Evolved Packet Core (EPC) via Untrusted Wireless Local Access Network (WLAN) as IP-CAN to access the IM CN subsystem uses this header field to convey location information to its service provider.

7.2.15.2 Applicability statement for the Cellular-Network-Info header field

The Cellular-Network-Info field is applicable within a trust domain. The Cellular-Network-Info header field can be included in any SIP requests and responses in which the P-Access-Network-Info header field is present.

7.2.15.3 Usage of the Cellular-Network-Info header field

The Cellular-Network-Info header field is populated with the following contents:

1) the access-type field is set to one of "3GPP-GERAN","3GPP-UTRAN-FDD", "3GPP-UTRAN-TDD", "3GPP-E-UTRAN-FDD", "3GPP-E-UTRAN-TDD", "3GPP-E-UTRAN-ProSe-UNR", "3GPP-NR-FDD", "3GPP-NR-TDD", "3GPP-NR-U-FDD", "3GPP-NR-U-TDD", "3GPP-NR-ProSe-L2UNR", "3GPP-NR-ProSe-L3UNR", "3GPP2-1X", "3GPP2-1X-HRPD", "3GPP2-UMB", "3GPP2-1X-Femto" as appropriate to the additional access technology the information is provided about;

2) if the access-type field is set to "3GPP-GERAN", a cgi-3gpp parameter set to the Cell Global Identity obtained from lower layers of the UE. The Cell Global Identity is a concatenation of MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), LAC (4 hexadeciaml digits) and CI (as described in 3GPP TS 23.003 [3]. The "cgi-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

3) if the access-type field is equal to "3GPP-UTRAN-FDD", or "3GPP-UTRAN-TDD", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), LAC (4 hexadecimal digits) as described in 3GPP TS 23.003 [3] and the UMTS Cell Identity (7 hexadecimal digits) as described in 3GPP TS 25.331 [9A]), obtained from lower layers of the UE. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

4) if the access-type field is equal to "3GPP-E-UTRAN-FDD" or "3GPP-E-UTRAN-TDD", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), Tracking Area Code (4 hexadecimal digits when accessing to EPC and 6 hexadecimal digits when accessing to 5GCN) as described in 3GPP TS 23.003 [3]) and the E-UTRAN Cell Identity (ECI) (7 hexadecimal digits) as described in 3GPP TS 23.003 [3]). The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

EXAMPLE: If MCC is 111, MNC is 22, TAC is 33C4 and ECI is 76B4321, then Cellular-Network-Info header field looks like follows: Cellular-Network-Info: 3GPP-E-UTRAN-FDD;utran-cell-id-3gpp=1112233C476B4321

5) if the access-type field is equal to "3GPP-E-UTRAN-ProSe-UNR", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value) and the E-UTRAN Cell Identity (ECI) (7 hexadecimal digits) as described in 3GPP TS 23.003 [3] obtained from the ProSe-UE-to-network relay that the UE is connected to as specified in 3GPP TS 24.334 [8ZD]. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

EXAMPLE: If MCC is 111, MNC is 22 and ECI is 76B4321, then Cellular-Network-Info header field looks like follows: Cellular-Network-Info: 3GPP-E-UTRAN-ProSe-UNR;utran-cell-id-3gpp=1112276B4321.

6) if the access-type field is set to "3GPP2-1X", a ci-3gpp2 parameter set to the ASCII representation of the hexadecimal value of the string obtained by the concatenation of SID (16 bits), NID (16 bits), PZID (8 bits) and BASE_ID (16 bits) (see 3GPP2 C.S0005-D [85]) in the specified order. The length of the ci-3gpp2 parameter shall be 14 hexadecimal characters. The hexadecimal characters (A through F) shall be coded using the uppercase ASCII characters. If the UE does not know the values for any of the above parameters, the UE shall use the value of 0 for that parameter. For example, if the SID is unknown, the UE shall represent the SID as 0x0000;

NOTE 1: The SID value is represented using 16 bits as opposed to 15 bits as specified in 3GPP2 C.S0005-D [85].

EXAMPLE: If SID = 0x1234, NID = 0x5678, PZID = 0x12, BASE_ID = 0xFFFF, the ci-3gpp2 value is set to the string "1234567812FFFF".

7) if the access-type field is set to "3GPP2-1X-HRPD", a ci-3gpp2 parameter set to the ASCII representation of the hexadecimal value of the string obtained by the concatenation of Sector ID (128 bits) and Subnet length (8 bits) (see 3GPP2 C.S0024-B [86]) and Carrier-ID, if available, (see 3GPP2 X.S0060 [86B]) in the specified order. The length of the ci-3gpp2 parameter shall be 34 or 40 hexadecimal characters depending on whether the Carrier-ID is included. The hexadecimal characters (A through F) shall be coded using the uppercase ASCII characters;

EXAMPLE: If the Sector ID = 0x12341234123412341234123412341234, Subnet length = 0x11, and the Carrier-ID=0x555444, the ci-3gpp2 value is set to the string "12341234123412341234123412341234123411555444".

8) if the access-type field is set to "3GPP2-UMB" 3GPP2 C.S0084-000 [86A], a ci-3gpp2 parameter is set to the ASCII representation of the hexadecimal value of the Sector ID (128 bits) defined in 3GPP2 C.S0084-000 [86A]. The length of the ci-3gpp2 parameter shall be 32 hexadecimal characters. The hexadecimal characters (A through F) shall be coded using the uppercase ASCII characters;

EXAMPLE: If the Sector ID = 0x12341234123412341234123412341234, the ci-3gpp2 value is set to the string "12341234123412341234123412341234".

9) if the access-type field is set to "3GPP2-1X-Femto", a ci-3gpp2-femto parameter set to the ASCII representation of the hexadecimal value of the string obtained by the concatenation of femto MSCID (24 bit), femto CellID (16 bit), FEID (64bit), macro MSCID (24 bits) and macro CellID (16 bits) (3GPP2 X.P0059-200 [86E]) in the specified order. The length of the ci-3gpp2-femto parameter is 36 hexadecimal characters. The hexadecimal characters (A through F) are coded using the uppercase ASCII characters;

10) the cell-info-age parameter indicates the relative time since the information about the cell identity was collected by the UE. The value of the parameter is a number indicating seconds;

NOTE 2: How the UE determines the relative time is up to UE implementation per operator policy or local configuration.

11) if the access-type field is equal to "3GPP-NR-FDD" or "3GPP-NR-TDD", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), Tracking Area Code (6 hexadecimal digits) as described in 3GPP TS 23.003 [3] and the NR Cell Identity (NCI) (9 hexadecimal digits). The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

12) if the access-type field is equal to "3GPP-NR-U-FDD" or "3GPP-NR-U-TDD", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), Tracking Area Code (6 hexadecimal digits) as described in 3GPP TS 23.003 [3] and the NR Cell Identity (NCI) (9 hexadecimal digits). The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212]; and

13) if the access-type field is equal to "3GPP-NR-ProSe-L2UNR" or "3GPP-NR-ProSe-L3UNR", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), Tracking Area Code (6 hexadecimal digits) as described in 3GPP TS 23.003 [3] and the NR Cell Identity (NCI) (9 hexadecimal digits) obtained from the 5G ProSe UE-to-network relay UE that the UE is connected to as specified in 3GPP TS 24.554 [8ZI]. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212].

7.2.15.4 Procedures at the UA

A UA that supports this extension and is willing to disclose the related parameters may insert the Cellular-Network-Info header field in any SIP request or response in which the P-Access-Network-Info header field is allowed to be present.

7.2.15.5 Procedures at the proxy

A SIP proxy shall not modify the value of the Cellular-Network-Info header field.

A SIP proxy shall remove the Cellular-Network-Info header field when the SIP signaling is forwarded to a SIP server located in an untrusted administrative network domain.

A SIP proxy that is providing services to the UA, can act upon the information present in the Cellular-Network-Info header field value, if present, to provide a different service depending on the network or the location through which the UA is accessing the server.A SIP proxy can determine the age of the cell identity information from the cell-info-age parameter. Depending on the recentness of the information the SIP proxy can perform different procedures.

7.2.15.6 Security considerations

The Cellular-Network-Info header field contains sensitive information. The Cellular-Network-Info header field should be removed when sent outside the trust domain.

A UE is not expected to receive the Cellular-Network-Info header field.

7.2.15.7 Syntax

The syntax for Cellular-Network-Info header field is specified in table 7.2.15-1.

Table 7.2.15-1: Syntax of Cellular-Network-Info

```
Cellular-Network-Info   = "Cellular-Network-Info" HCOLON cellular-net-spec
cellular-net-spec       = access-type *(SEMI cellular-access-info)
access-type             = "3GPP-GERAN" / "3GPP-UTRAN-FDD" / "3GPP-UTRAN-TDD" /
                          "3GPP-E-UTRAN-FDD" / "3GPP-E-UTRAN-TDD" / "3GPP2-1X-Femto" /
  "3GPP2-UMB" / "3GPP2-1X-HRPD" / "3GPP2-1X" /
                          "3GPP-E-UTRAN-ProSe-UNR" / "3GPP-NR-FDD" / "3GPP-NR-TDD" /
                          "3GPP-NR-U-FDD" / "3GPP-NR-U-TDD" /
                          "3GPP-NR-ProSe-L2UNR" / "3GPP-NR-ProSe-L3UNR" /
                          token
cellular-access-info              = access-info / cell-info-age
access-info                       = cgi-3gpp / utran-cell-id-3gpp /
                                    ci-3gpp2 / ci-3gpp2-femto / extension-access-info
extension-access-info             = generic-param
cgi-3gpp                          = "cgi-3gpp" EQUAL (token / quoted-string)
utran-cell-id-3gpp                = "utran-cell-id-3gpp" EQUAL (token / quoted-string)
ci-3gpp2                          = "ci-3gpp2" EQUAL (token / quoted-string)
ci-3gpp2-femto                    = "ci-3gpp2-femto" EQUAL (token / quoted-string)
cell-info-age                     = "cell-info-age" EQUAL 1*9DIGIT
```

7.2.16 Priority-Share header field

7.2.16.1 Introduction

IANA registry: Header Field Parameter Registry for the Session Initiation Protocol (SIP)

Header field name: Priority-Share

Usage: The Priority-Share header field is used only for informative purposes.

Header field specification reference: 3GPP TS 24.229, http://www.3gpp.org/ftp/Specs/archive/24_series/24.229/

The Priority-Share header field is used to carry information relating to the possibility to use priority sharing. Priority sharing allows the P-CSCF to instruct the access gateway to use the same bearer for several sessions regardless of the priority of the sessions. When priority sharing is not allowed the P-CSCF will instruct the access gateway to not use priority sharing.

7.2.16.2 Applicability statement for the Priority-Share header field

The Priority-Share header field is applicable within a single private administrative domain or between different administrative domains where there is a trust relationship between the domains.

The Priority-Share header field is not included in a SIP message sent to another network if there is no trust relationship.

The Priority-Share header field is applicable whenever an application/sdp MIME body would be applicable, as defined by RFC 3261 [26].

7.2.16.3 Usage of the Priority-Share header field

A SIP UA or SIP proxy that receives a SIP request or response that contains a Priority-Share header field can use the values as appropriate.

A SIP proxy may remove the Priority-Share header field according to local policy.

7.2.16.4 Procedures at the UA

An application server acting as a UA that supports this extension and receives a request or response without the Priority-Share header field may insert a Priority-Share header field prior to forwarding the message. The header is populated as described in subclause 7.2.16.7.

If an application server acting as a UA that supports this extension receives a request or response with the Priority-Share header field, it may use the information from the header field for application-specific logic, i.e., resource reservation. If information from the header field is used, the header field shall be removed from the request or response.

7.2.16.5 Procedures at the proxy

A SIP proxy that supports this extension and receives a request or response without the Priority-Share header field may insert a Priority-Share header field prior to forwarding the message. The header is populated as described in subclause 7.2.16.7.

If a proxy that supports this extension receives a request or response with the Priority-Share header field, it may use the information from the header field for application-specific logic, i.e., resource reservation. If information from the header field is used, the header field shall be removed from the request or response.

7.2.16.6 Security considerations

The Priority-Share header field does not contain any information that can disclose user information or the topology of nodes within an operator network.

7.2.16.7 Syntax

The syntax for Priority-Share header field is specified in table 7.2.16.1

Table 7.2.16.1: Syntax of Priority-Share

```
priority-share      = "Priority-Share" HCOLON priority-share-options *( SEMI generic-param)
priority-share-options = "allowed" / "not-allowed" / other-options
other-options = token
```

7.2.16.8 Examples of usage

The Priority-Share header field is included by an application server in the home network to inform about the possibility to share resources between session regardless of the priority of a session.

7.2.17 Definition of Response-Source header field

7.2.17.1 Introduction

IANA registry: Header Fields registry for the Session Initiation Protocol (SIP)

Header field name: Response-Source

Usage: the Response-Source header field is used only for informative purposes.

Header field specification reference: 3GPP TS 24.229, http://www.3gpp.org/ftp/Specs/archive/24_series/24.229/

The Response-Source header field is used to carry information related to the originator of an error response. The receiving entities may possibly use this information to decide a more appropriate procedure to invoke in regards with the failure response.

7.2.17.2 Applicability statement for the Response-Source header field

The Response-Source header field is applicable within a single private administrative domain or between different administrative domains where there is a trust relationship between the domains.

7.2.17.3 Usage of the Response-Source header field

A SIP UA or SIP proxy may include the Response-Source header field when responding to a SIP request with an error response to provide the information on who is the sender of the error response using the appropriate URN value as defined in subclause 7.2.17.7.

A SIP UA or SIP proxy that receives a SIP response that contains a Response-Source header field can use the values as appropriate.

7.2.17.4 Procedures at the UA

A UA that supports this extension and rejects a request with an error response may insert a Response-Source header field within the response message. The header is populated as described in subclause 7.2.17.7.

If a UA that supports this extension receives a response with the Response-Source header field, it may take the information from the Response-Source header field into account when handling the response.

NOTE: The Response-Source header field is informational. A UA receiving a response containing a Response-Source header field does not perform any action contrary to the behavior specified in RFC 3261 [26] or other RFCs that specify UA actions upon receiving the specific response code.

7.2.17.5 Procedures at the proxy

A proxy that supports this extension and receives a request for which its internal logic leads to reject the request with an error response may insert a Response-Source header field within the response message. The header is populated as described in subclause 7.2.17.7.

If a proxy that supports this extension receives a response with the Response-Source header field, it may use the information from the header field for its internal logic for error reponses handling.

7.2.17.6 Security considerations

The Response-Source header field will contain a URN identifying the sender that may be considered as sensitive information. The Response-Source header field may be removed when received from outside the trust domain depending on the network policy.

7.2.17.7 Syntax

The ABNF syntax for Response-Source header field is specified in table 7.2.17.7-1.

Table 7.2.17.7-1: Syntax of Response-Source header field

```
Response-Source = "Response-Source" HCOLON source-info
source-info     = source-params *(SEMI source-params)
source-params   = source-urn / token
source-urn      = "fe" EQUAL LAQUOT source-urn-val RAQUOT
source-urn-val  = 1*uric ; defined in RFC 3261
```

The source-urn-val of the source-urn parameter is coded as a URN. The URN identifies the SIP capable functional entity sending a SIP response.

A URN is defined under the "urn:3gpp" label defined in RFC 5279 [253].

The extension of 3gpp-urn is:

urn:3gpp:fe

A formal reference to the publicly available specification:

3GPP TS 24.229

A short phrase describing the function of the extension:

The namespace "fe" is for indicating an IMS functional-entity. See the coding for the namespace extension ns-ext in table 7.2.17.7-2:

Table 7.2.17.7-2: Syntax of urn:3gpp:fe

```
ns-ext      = HCOLON "fe" HCOLON functional-entity
functional-entity   = fe-id *("." fe-param)
fe-id       = "ue" / "p-cscf" / "i-cscf" / "s-cscf" / "e-cscf" / "mgcf" / "bgcf" / "ibcf" / "trf" /
                     "atcf" / "agcf" / "mrfc" / "lrf" / "msc-server" / "as" / token
fe-param        = role / side / token
role        = "tas" / "scc-as" / "ip-sm-gw" / "pf-mcptt-server" / "cf-mcptt-server" / "ncf-mcptt-
                     server" / "cms" / "gms" / "tads" / "iua" / "msc-server-ics" / token
side        = "orig" / "term" / "transit"/ token
```

Contact information for the organization or person making the registration

3GPP Specifications Manager

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The following fe-id values are defined:

- ue: represents the UE;
- p-cscf: represents the P-CSCF;
- i-cscf: represents the I-CSCF;
- s-cscf: represents the S-CSCF;
- e-cscf: represents the E-CSCF;
- mgcf: represents the MGCF;
- bgcf: represents the BGCF;
- ibcf: represents the IBCF;
- trf: represents the TRF;
- atcf: represents the ATCF;
- agcf: represents the AGCF;
- mrfc: represents the MRFC;
- lrf: represents the LRF;
- msc-server: represents the MSC server; and
- as: represents the AS.

The following fe-param values are defined:

- role:
  a. mmtel-as: indicates that the AS is performing the MMTel services role;
  b. scc-as: indicates that the AS is performing the SCC AS role;
  c. ip-sm-gw: indicates that the AS is performing the IP-SM-GW role;
  d. pf-mcptt-server: indicates that the AS is performing the participating MCPTT server role;
  e. cf-mcptt-server: indicates that the AS is performing the controling MCPTT server role;

f. ncf-mcptt-server: indicates that the AS is performing the non-controling MCPTT server role;

g. cms: indicates that the AS is performing the configuration management server role;

h. gms: indicates that the AS is performing the group management server role;

i. tads: indicates that the AS is performing the terminating access domain selection role;

j. iua: indicates that the AS is performing the ICS User Agent role; and

k. msc-server-ics: indicates that the MSC is performing the MSC server enhanced for ICS role.

- side:

a. orig: indicates that this functional entity is in the originating network;

b. term: indicates that this functional entity is in the terminating network;and

c. transit: indicates that this functional entity is in a transit network.

An example of the source-urn header field parameter value is: fe=<urn:3gpp:fe:p-cscf.orig>.

7.2.18 Definition of Attestation-Info header field

7.2.18.1 Introduction

IANA registry: Header Fields registry for the Session Initiation Protocol (SIP)

Header field name: Attestation-Info

Usage: The Attestation-Info header field is used only for informative purposes.

Header field specification reference: 3GPP TS 24.229, http://www.3gpp.org/ftp/Specs/archive/24_series/24.229/

When a node has performed attestation of an identity in an incoming request or has attested the origin of the request, the node can inform a downstream node about what kind of attestation the node has performed. A downstream node such as an application server can use this information to provide the user with more accurate information regarding the attested identity.

7.2.18.2 Applicability statement for the Attestation-Info header field

The Attestation-Info header field is applicable within a single private administrative domain or between different administrative domains.

The Attestation-Info header field is applicable when:

1) a node has performed attestation of an identity in an incoming request; or

2) has performed gateway attestation of the request itself.

Case 1) is when a node has knowledge about the originating identity and can attest this identity based on this knowledge.

Case 2) is when a border node in a network receives a request where the border node has no relation to the originating user and the border node adds a value identifying the source of the request.

7.2.18.3 Usage of the Attestation-Info header field

A node in the originating network attesting the identity of the originating user can add an Attestation-Info header field to inform what relation the network has with the originating user. A node at a border of a network can add an identifier identifying from where the request was received. The Attestation-Info header field informs that this procedure has been performed.

A downstream node can use the Attestation-Info header field when providing analytics functions to inform the terminating user the trust level of the originating identity.

7.2.18.4 Procedures at the UA

There are no specific procedures specified for a UA.

7.2.18.5 Procedures at the proxy

A SIP proxy that supports this extension and receives a request may as part of its procedures insert an Attestation-Info header field prior to forwarding the request. The header field is populated with a value as specified in Table 7.2.18-1.

7.2.18.6 Security considerations

The Attestation-Info header field does not contain any sensitive information.

A UE is not expected to receive this information.

7.2.18.7 Syntax

The syntax for Attestation-Info header field is specified in table 7.2.18-1.

Table 7.2.18-1: Syntax of Attestation-Info

```
Attestation-Info      = "Attestation-Info" HCOLON attestation-level / generic-param
attestation-level     = ("A" / "B" / "C")
```

The meaning of the values "A", "B" and "C" is as defined in RFC 8588 [261] and references therein.

7.2.18.8 Examples of usage

A node in the originating network, such as a 3GPP S-CSCF or an application server, can when attesting the identity of an originating user insert an Attestation-Info header field to provide information on the relation the network has to the originating user. This information can be used when inserting an Identity header field, or can be taken into account when informing the terminating user about the identity of the originating user.

An edge node, such as a 3GPP entry IBCF, receiving a message withouth any Identity header field can use the Attestation-Info header field to inform that the edge node has performed a gateway attestation as specified in RFC 8588 [261].

7.2.19 Definition of Origination-Id header field

7.2.19.1 Introduction

IANA registry: Header Fields registry for the Session Initiation Protocol (SIP)

Header field name: Origination-Id

Usage: The Origination-Id header field is used only for informative purposes.

Header field specification reference: 3GPP TS 24.229, http://www.3gpp.org/ftp/Specs/archive/24_series/24.229/

When a node has performed attestation of an identity in an incoming request the node can add a unique identifier to inform about who attested the identity. When a node has attested from where it received the request, the node can send a unique identifier identifying from where the request was received. A downstream node such as an application server can use this information to provide the user with more accurate information regarding the attested identity.

7.2.19.2 Applicability statement for the Origination-Id header field

The Origination-Id header field is applicable within a single private administrative domain or between different administrative domains.

The Origination-Id header field is applicable when:

1) a node has performed attestation of an identity in an incoming request; or

2) has performed gateway attestation of the request itself.

Case 1) is when a node has knowledge about the originating identity and can attest this identity based on this knowledge.

Case 2) is when a border node in a network receives a request where the border node has no relation to the originating user and the border node adds a value identifying the source of the request.

7.2.19.3 Usage of the Origination-Id header field

A node in the originating network attesting the identity of the originating user can add an Origination-Id header field to identify the node that performed the identity attestation. This value is based on local configuration and regulation. A node at a border of a network can add an Origination-Id header field with a unique identifier identifying from where the request was received.

A downstream node can use the Origination-Id header field when providing analytics functions to inform the terminating user the trust level of the originating identity.

7.2.19.4 Procedures at the UA

There are no specific procedures specified for a UA.

7.2.19.5 Procedures at the proxy

A SIP proxy that supports this extension and receives a request may as part of its procedures insert an Origination-ID header field prior to forwarding the request. The header field is populated with a value as specified in Table 7.2.19-1.

7.2.19.6 Security considerations

The Origination-Id header field can contain a unique value identifying a specific node in the network. A network operator may want to remove this information before transporting to an utrusted entity.

A UE is not expected to receive this information.

7.2.19.7 Syntax

The syntax for Origination-Id header field is specified in table 7.2.19-1.

Table 7.2.19-1: Syntax of Origination-Id

```
Origination-Id        = "Origination-Id" HCOLON originator / token
originator            = UUID
```

The format of the UUID is as defined as in RFC 4122.

7.2.19.8 Examples of usage

A node in the originating network, such as a 3GPP S-CSCF or an application server, can when attesting the identity of an originating user insert an Origination-Id header field to provide information on who attested the identity of the originating user. This information can be used when inserting an Identity header field, or can be taken into account when informing the terminating user about the identity of the originating user.

An edge node, such as a 3GPP entry IBCF, receiving a message without any Identity header field can use the Origination-Id header field to a unique identifier of from where the request is received.

7.2.20 Definition of Additional-Identity header field

7.2.20.1 Introduction

IANA registry: Header Fields registry for the Session Initiation Protocol (SIP)

Header field name: Additional-Identity

Usage: The Additional-Identity header field is used only for informative purposes.

Header field specification reference: 3GPP TS 24.229, http://www.3gpp.org/ftp/Specs/archive/24_series/24.229/

The Additional-Identity header field is used to convey an originating identity on the originating side or a target identity on the terminating side where the served user is not registering this identity but is authorized by the network to use this identity.

On the originating side, when a user has requested such an additional identity to be used for an originating request, the UA can insert this identity in the Additional-Identity header field. When the identity in the Additional-Identity header field has been authorized by the network, the network can remove, ignore or use the Additional-Identity header field. A downstream node such as an application server or UA can use this information to identify the not registered identity on whose behalf the originating user is sending the request.

On the terminating side, when a user is contacted with such an additional identity, and the network decides to inform the terminating user that the user was contacted with this identity, the network can insert this identity in the Additional-Identity header field. A terminating request to the UA can hence contain the Additional-Identity header field with the identity used to reach the terminating user.

7.2.20.2 Applicability statement for the Additional-Identity header field

The Additional-Identity header field is applicable within a single private administrative domain or between different administrative domains.

The Additional-Identity header field is applicable when:

- an originating UA wants to indicate the identity to be used as an originating identity in a multi-identity service;
- a node performs the multi-identity service for an originating UA in an incoming request;
- a node has performed the multi-identity service for a terminating identity in an incoming request; or
- a terminating UA wants to identify the identity used to contact the terminating user.

7.2.20.3 Usage of the Additional-Identity header field

A SIP UA or SIP proxy may include the Additional-Identity header field to indicate:

- in the originating network, the identity to be used for originating requests when the originating user is subscribed to the multi-identity service; and
- in the terminating network, the identity to which the terminating user is contacted when the terminating user is subscribed to the multi-identity service.

7.2.20.4 Procedures at the UA

A SIP UA that supports this extension may as part of its procedures insert the Additional-Identity header field prior to sending the request. The header field is populated with a value as specified in table 7.2.20.7-1.

7.2.20.5 Procedures at the proxy

A SIP proxy that supports this extension and receives a request may as part of its procedures insert an Additional-Identity header field prior to forwarding the request. The header field is populated with a value as specified in table 7.2.20.7-1.

7.2.20.6 Security considerations

Within a 3GPP environment, the Additional-Identity header field is exchanged between a SIP UA and a SIP proxy in the same network. The Additional-Identity header field may also be exchanged between networks when there is a trust relationship for the Additional-Identity header field.

A functional entity at the boundary of the trust domain will remove the Additional-Identity header field when SIP signalling crosses the boundary of the trust domain.

7.2.20.7 Syntax

The syntax for Additional-Identity header field is specified in table 7.2.20.7-1.

Table 7.2.20.7-1: Syntax of the Additional-Identity Header Field

```
Additional-Identity  = "Additional-Identity" HCOLON id-spec / token
id-spec              = name-addr *(SEMI (id-param))
id-param             = generic-param
```

7.2.20.8 Examples of usage

A node in the originating network, such as a UA, can use the Additional-Identity header field to provide to a multi-identity service the information about which identity of the originating user is to be used for this originating request.

A node in the terminating network, such as an application server, when performing the multi-identity service for a terminating user, can insert the Additional-Identity header field to provide information about which identity of the terminating user is to be used as a contacted identity.

7.2.21 Definition of Priority-Verstat header field

7.2.21.1 Introduction

IANA registry: Header Fields registry for the Session Initiation Protocol (SIP)

Header field name: Priority-Verstat

Usage: The Priority-Verstat header field is used only for informative purposes.

Header field specification reference: 3GPP TS 24.229, http://www.3gpp.org/ftp/Specs/archive/24_series/24.229/

When a node has performed verification of a Resource-Priority header field and of a header field value "psap-callback" of a Priority header field (if present) in an incoming request, the node can inform a downstream node whether the Resource-Priority header field and the header field value "psap-callback" of the Priority header field (if present) was populated by an authorized entity and can be trusted. A downstream node can use use this information to determine whether the call should be treated according to the priority level indicated in the Resource-Priority header field and (if the Priority header field was present) whether the call should be treated as emergency call back.

7.2.21.2 Applicability statement for the Priority-Verstat header field

The Priority-Verstat header field is applicable within a single private administrative domain or between different administrative domains.

The Priority-Verstat header field is applicable when a node has performed authentication of a Resource-Priority header field and a header field value "psap-callback" of a Priority header field in an incoming request.

7.2.21.3 Usage of the Priority-Verstat header field

The Priority-Verstat header field is used to indicate the verification status of the Resource-Priority header field and optionally the header field value "psap-callback" of the Priority header field.

7.2.21.4 Procedures at the UA

There are no specific procedures specified for a UA.

7.2.21.5 Procedures at the proxy

A SIP proxy that supports this extension and receives a request may as part of its procedures insert a Priority-Verstat header field prior to forwarding the request. The header field is populated as specified in table 7.2.x-1.

7.2.21.6 Security considerations

A UE is not expected to receive this information.

7.2.21.7 Syntax

The syntax for Priority-Verstat header field is specified in table 7.2.21-1.

Table 7.2.21-1: Syntax of Priority-Verstat

```
Priority-Verstat = "Priority-Verstat" HCOLON verstat-value
verstat-value    = "RPH-Validation-Passed" / "RPH-Validation-Failed" / "No-RPH-Validation" /
                   "ECB-RPH-Validation-Passed" / "ECB-RPH-Validation-Failed" / "No-ECB-RPH-
Validation" / other-value
other-value      = token
```

7.2.21.8 Examples of usage

The Priority-Verstat header field is used in networks which have requirements on authentication of a Resource-Priority header field and a header field value "psap-callback" of a Priority header field to authenticate content of the Resource-Priority header field and the header field value "psap-callback" of the Priority header field.

7.2A Extensions to SIP header fields defined within the present document

7.2A.1 Extension to WWW-Authenticate header field

7.2A.1.1 Introduction

This extension defines a new authentication parameter (auth-param) for the WWW-Authenticate header field used in a 401 (Unauthorized) response to the REGISTER request. For more information, see RFC 7235 [285] subclause 2.1 and clause C.

7.2A.1.2 Syntax

The syntax for for auth-param is specified in table 7.2A.1.

Table 7.2A.1: Syntax of auth-param

```
auth-param  = 1#( integrity-key / cipher-key )
integrity-key   = "ik" EQUAL ik-value
cipher-key  = "ck" EQUAL ck-value
ik-value    = LDQUOT *(HEXDIG) RDQUOT
ck-value    = LDQUOT *(HEXDIG) RDQUOT
```

7.2A.1.3 Operation

This authentication parameter will be used in a 401 (Unauthorized) response in the WWW-Authenticate header field during UE authentication procedure as specified in subclause 5.4.1.

The S-CSCF appends the integrity-key parameter (directive) to the WWW.-Authenticate header field in a 401 (Unauthorized) response. The P-CSCF stores the integrity-key value and removes the integrity-key parameter from the header field prior to forwarding the response to the UE.

The S-CSCF appends the cipher-key parameter (directive) to the WWW-Authenticate header field in a 401 (Unauthorized) response. The P-CSCF removes the cipher-key parameter from the header field prior to forwarding the response to the UE. In the case ciphering is used, the P-CSCF stores the cipher-key value.

7.2A.2 Extension to Authorization header field

7.2A.2.1 Introduction

This extension defines new dig-resp parameters for the Authorization header field used in REGISTER requests. For more information, see RFC 7235 [285] subclause 2.1 and clause C.

7.2A.2.2 Syntax

7.2A.2.2.1 integrity-protected

The syntax of integrity-protected for the Authorization header field is specified in table 7.2A.2.

Table 7.2A.2: Syntax of integrity-protected for Authorization header field

```
dig-resp =/ "integrity-protected" EQUAL ("yes" / "no" / "tls-pending" / "tls-yes" / "ip-assoc-
    pending" / "ip-assoc-yes" / "auth-done" / "tls-connected")
```

7.2A.2.3 Operation

This authentication parameter is inserted in the Authorization header field of all the REGISTER requests. The value of the "integrity-protected" header field parameter in the auth-param parameter is set as specified in subclause 5.2.2. This information is used by S-CSCF to decide whether to challenge the REGISTER request or not, as specified in subclause 5.4.1.

The values in the "integrity-protected" header field field are defined as follows:

"yes": indicates that a REGISTER request received in the P-CSCF is protected using an IPsec security association and IMS AKA is used as authentication scheme.

"no": indicates that a REGISTER request received in the P-CSCF is not protected using an IPsec security association and IMS AKA is used as authentication scheme, i.e. this is an initial REGISTER request with the Authorization header field not containing a challenge response.

"tls-yes": indicates that a REGISTER request is received in the P-CSCF protected over a TLS connection and the Session ID, IP address and port for the TLS connection are already bound to a private user identity. The S-CSCF will decide whether or not to challenge such a REGISTER request based on its policy. This is used in case of SIP digest with TLS.

"tls-pending": indicates that a REGISTER request is received in the P-CSCF protected over a TLS connection and the Session ID, IP address and port for the TLS connection are not yet bound to a private user identity. The S-CSCF shall challenge such a REGISTER request if it does not contain an Authorization header field with a challenge response or if the verification of the challenge response fails. This is used in case of SIP digest with TLS.

"ip-assoc-yes": indicates that a REGISTER request received in the P-CSCF does map to an existing IP association in case SIP digest without TLS is used.

"ip-assoc-pending": indicates that a REGISTER request received in the P-CSCF does not map to an existing IP association, and does contain a challenge response in case SIP digest without TLS is used.

"auth-done": indicates that a REGISTER request is sent from an entity that is trusted and has authenticated the identities used in the REGISTER request. An example for such an entity is the MSC server enhanced for IMS centralized services. The S-CSCF shall skip authentication.

"tls-connected": indicates that a REGISTER request received in the eP-CSCF is issued by a UE over a TLS session established prior to the registration and IMS AKAv2 is used as authentication scheme. This integrity-protected flag value is used for example in case of WebRTC over IMS when the Authentication is IMS-AKA as defined in 3GPP TS 24.371 [8Z].

NOTE 1: In case of SIP digest with TLS is used, but the REGISTER request was not received over TLS, the P-CSCF does not include an "integrity-protected" header field parameter in the auth-param to indicate that an initial REGISTER request was not received over an existing TLS session. The S-CSCF will always challenge such a REGISTER request.

NOTE 2: In case of SIP digest without TLS is used, but the REGISTER request was not received over TLS, the P-CSCF does not include an "integrity-protected" header field parameter in the auth-param to indicate that the REGISTER request does not map to an existing IP association, and does not contain a challenge response. The S-CSCF will always challenge such a REGISTER request.

NOTE 3: The value "yes" is also used when an initial REGISTER request contains an Authorization header field with a challenge response as in this case the IPsec association is already in use, and its use by the UE implicitly authenticates the UE. This is a difference to TLS case where the use of TLS alone does not yet implicitly authenticates the UE. Hence in the TLS case, for an initial REGISTER request containing an Authorization header field with a challenge response the value "tls-pending" and not "tls-yes" is used.

7.2A.3 Tokenized-by header field parameter definition (various header fields)

7.2A.3.1 Introduction

The "tokenized-by" header field parameter is an extension parameter appended to encrypted entries in various SIP header fields as defined in subclause 5.10.4.

7.2A.3.2 Syntax

The syntax for the "tokenized-by" header field parameter is specified in table 7.2A.3:

Table 7.2A.3: Syntax of tokenized-by-param

```
rr-param =  tokenized-by-param / generic-param
via-params =  via-ttl / via-maddr
            / via-received / via-branch
            / tokenized-by-param / via-extension
tokenized-by-param = "tokenized-by" EQUAL hostname
```

The BNF for rr-param and via-params is taken from RFC 3261 [26] and modified accordingly.

7.2A.3.3 Operation

The "tokenized-by" header field parameter is appended by IBCF (THIG) after all encrypted strings within SIP header fields when network configuration hiding is active. The value of the header field parameter is the domain name of the network which encrypts the information.

7.2A.4 P-Access-Network-Info header field

7.2A.4.1 Introduction

The P-Access-Network-Info header field is extended to include specific information relating to particular access technologies.

7.2A.4.2 Syntax

The syntax of the P-Access-Network-Info header field is described in RFC 7315 [52] and RFC 7913 [234]. There are additional coding rules for this header field depending on the type of IP-CAN, according to access technology specific descriptions.

Table 7.2A.4 describes the 3GPP-specific extended syntax of the P-Access-Network-Info header field defined in RFC 7315 [52] and RFC 7913 [234].

Table 7.2A.4: Syntax of extended P-Access-Network-Info header field

```
   daylight-saving-time   = "daylight-saving-time" EQUAL quoted-string
   UE-local-IP-address = "UE-local-IP-address" EQUAL DQUOTE ( IPv4address / IPv6reference ) DQUOTE
   UDP-source-port = "UDP-source-port" EQUAL port
   TCP-source-port = "TCP-source-port" EQUAL port
   ePDG-IP-address = "ePDG-IP-address" EQUAL DQUOTE ( IPv4address / IPv6reference ) DQUOTE
   access-class    =/ "untrusted-non-3GPP-VIRTUAL-EPC" / "VIRTUAL-no-PS" / "WLAN-no-PS" /
                      "3GPP-NR" / "3GPP-NR-U" / "3GPP-NR-SAT" / "3GPP-NR-REDCAP"
   access-type     =/ "3GPP-E-UTRAN-ProSe-UNR" / "xDSL" / "3GPP-NR-FDD" / "3GPP-NR-TDD" /
                      "IEEE-802.11ac" / "3GPP-NR-U-FDD" / "3GPP-NR-U-TDD" / "3GPP-NR-SAT" / "3GPP-
                            NR-ProSe-L2UNR" / "3GPP-NR-ProSe-L3UNR" / "3GPP-NR-REDCAP"
   eps-fb          = "eps-fallback" EQUAL "0" / "1"
```

The daylight-saving-time and the UE-local-IP-address are instances of generic-param from the current extension-access-info component of the P-Access-Network-Info header field defined in RFC 7315 [52] and RFC 7913 [234].

The presence of the "network-provided" header field parameter defined in RFC 7315 [52] indicates a P-Access-Network-Info header field is provided by the P-CSCF, S-CSCF, the AS, the MSC server enhanced for ICS, the MSC server enhanced for SRVCC using SIP interface, the MSC server enhanced for DRVCC using SIP interface or by the MGCF. The content can differ from a P-Access-Network-Info header field without this parameter which is provided by the UE.

The "network-provided" header field parameter can be used with both "access-type" and "access-class" constructs. The "access-class" construct is provided for use where the value is not known to be specific to a particular "access-type" value, e.g. in the case of some values delivered from the PCRF. The "access-class" field can be set only by the P-CSCF, the MSC server enhanced for ICS, the MSC server enhanced for SRVCC using SIP interface, the MSC server enhanced

for DRVCC using SIP interface or by the AS. The "network-provided" header field parameter can be set only by the P-CSCF, S-CSCF, the AS, the MSC server enhanced for ICS, the MSC server enhanced for SRVCC using SIP interface, the MSC server enhanced for DRVCC using SIP interface or by the MGCF. The "local-time-zone" parameter, the "daylight-saving-time" parameter, the "gstn-location" parameter, the "GSTN" value of access-type field and the "untrusted-non-3GPP-VIRTUAL-EPC" value of access-class field shall not be inserted by the UE.

The "local-time-zone" parameter defined in RFC 7315 [52] indicates the time difference between local time and UTC of day. For 3GPP accesses, the "local-time-zone" parameter represents the time zone allocated to the routing area or traffic area which the UE is currently using. As the edge of such areas may overlap, there can be some discrepancy with the actual time zone of the UE where the UE is in the near proximity to a time zone boundary.

The "daylight-saving-time" parameter indicates by how much the local time of the UE has been adjusted due to the use of daylight saving time. Providing the "daylight-saving-time" parameter is optional.

The "UE-local-IP-address" parameter indicates the UE local IP address.

NOTE: The UE local IP address is the source address on the outer header of the IPsec tunnel packets received by the ePDG on the S2b interface.

The "UDP-source-port" parameter indicates that the IKEv2 messages exchanged between the UE and the ePDG are encapsulated in the UDP messages according to IETF RFC 3948 [63A]. The value of the "UDP-source-port" parameter is the UDP source port of the UDP messages:

- received by the ePDG; and
- encapsulating the IKEv2 messages.

The "TCP-source-port" parameter indicates that the IKEv2 messages exchanged between the UE and the ePDG are transported using the firewall traversal tunnel as described in 3GPP TS 24.302 [8U]. The value of the "TCP-source-port" parameter is the TCP source port of the TCP messages:

- received by the ePDG; and
- of the firewall traversal tunnel transporting the IKEv2 messages.

The "ePDG-IP-address" parameter indicates the ePDG IP address used as IKEv2 tunnel endpoint with the UE.

The "eps-fallback" header field parameter is used to indicate that the current access technology is used as a result of EPS fallback. The value "1" indicates that EPS fallback has occurred, the value "0" that EPS fallback has not occurred. The parameter can be set only by the P-CSCF.

7.2A.4.3 Additional coding rules for P-Access-Network-Info header field

The P-Access-Network-Info header field is populated with the following contents:

1) the access-type field set to one of "3GPP-GERAN","3GPP-UTRAN-FDD", "3GPP-UTRAN-TDD", "3GPP-E-UTRAN-FDD", "3GPP-E-UTRAN-TDD", "3GPP-E-UTRAN-ProSe-UNR", "3GPP-NR-FDD", "3GPP-NR-TDD", "3GPP-NR-U-FDD", "3GPP-NR-U-TDD", "3GPP-NR-SAT", "3GPP-NR-ProSe-L2UNR", "3GPP-NR-ProSe-L3UNR", "3GPP-NR-REDCAP", "3GPP2-1X", "3GPP2-1X-HRPD", "3GPP2-UMB", "3GPP2-1X-Femto", "IEEE-802.11", "IEEE-802.11a", "IEEE-802.11b", "IEEE-802.11g", "IEEE-802.11n", "IEEE-802.11ac", "ADSL", "ADSL2", "ADSL2+", "RADSL", "SDSL", "HDSL", "HDSL2", "G.SHDSL", "VDSL", "IDSL", "xDSL", "DOCSIS", "IEEE-802.3", "IEEE-802.3a", "IEEE-802.3e", "IEEE-802.3i", "IEEE-802.3j", "IEEE-802.3u", "IEEE-802.3ab", "IEEE-802.3ae", "IEEE-802.3ah", "IEEE-802.3ak", "IEEE-802.3aq", "IEEE-802.3an", "IEEE-802.3y", "IEEE-802.3z", or "DVB-RCS2" as appropriate to the access technology in use.

1A) the access-class field set to one of "3GPP-GERAN", "3GPP-UTRAN", "3GPP-E-UTRAN", "3GPP-NR", "3GPP-NR-U", "3GPP-NR-SAT", "3GPP-NR-REDCAP", "3GPP-WLAN", "3GPP-GAN", "3GPP-HSPA", "3GPP2", "untrusted-non-3GPP-VIRTUAL-EPC", "VIRTUAL-no-PS", or "WLAN-no-PS" as appropriate to the technology in use. The access-class field set to "untrusted-non-3GPP-VIRTUAL-EPC" indicates the IP-CAN associated with an EPC based untrusted non-3GPP access with unknown radio access technology. The access-class field set to "VIRTUAL-no-PS" indicates an IP-CAN associated with an unknown radio access technology, such that the IP-CAN is not provided by the packet switched domain of the PLMN of the P-CSCF. The access-class field set to "WLAN-no-PS" indicates an IP-CAN associated with WLAN, such that the IP-CAN is not provided by the packet switched domain of the PLMN of the P-CSCF. The access-class field set to "3GPP-NR-

SAT " indicates an IP-CAN associated with satellite NG-RAN. The access-class field set to "3GPP-NR-REDCAP" if the UE has provided the NR RedCap indication to the IP-CAN as described in 3GPP TS 23.501 [257].

2) if the access-type field or the access-class field is set to "3GPP-GERAN", a cgi-3gpp parameter set to the Cell Global Identity obtained from lower layers of the UE. The Cell Global Identity is a concatenation of MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), LAC (4 hexadeciaml digits) and CI (as described in 3GPP TS 23.003 [3]. The "cgi-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

3) if the access-type field is equal to "3GPP-UTRAN-FDD", or "3GPP-UTRAN-TDD", and a UE provides the P-Acces-Network-Info header field, a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), LAC (4 hexadecimal digits) as described in 3GPP TS 23.003 [3] and the UMTS Cell Identity (7 hexadecimal digits) as described in 3GPP TS 25.331 [9A]), obtained from lower layers of the UE. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

3A) if the access-type field is equal to "3GPP-UTRAN-FDD", or "3GPP-UTRAN-TDD", and an entitiy that can use the "network-provided" header field parameter provides the P-Access-Network-Info header field, if available a "utran-sai-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), LAC (4 hexadecimal digits) as described in 3GPP TS 23.003 [3] and SAC (4 hexadecimal digits) as described in 3GPP TS 23.003 [3]. The "utran-sai-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

3B) if the access-class field is equal to "3GPP-UTRAN", or "3GPP-HSPA", if available a "utran-sai-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), LAC (4 hexadecimal digits) as described in 3GPP TS 23.003 [3] and SAC (4 hexadecimal digits) as described in 3GPP TS 23.003 [3]. The "utran-sai-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

4) void

5) if the access-type field is set to "3GPP2-1X", a ci-3gpp2 parameter set to the ASCII representation of the hexadecimal value of the string obtained by the concatenation of SID (16 bits), NID (16 bits), PZID (8 bits) and BASE_ID (16 bits) (see 3GPP2 C.S0005-D [85]) in the specified order. The length of the ci-3gpp2 parameter shall be 14 hexadecimal characters. The hexadecimal characters (A through F) shall be coded using the uppercase ASCII characters. If the UE does not know the values for any of the above parameters, the UE shall use the value of 0 for that parameter. For example, if the SID is unknown, the UE shall represent the SID as 0x0000;

NOTE 1: The SID value is represented using 16 bits as supposed to 15 bits as specified in 3GPP2 C.S0005-D [85].

EXAMPLE: If SID = 0x1234, NID = 0x5678, PZID = 0x12, BASE_ID = 0xFFFF, the ci-3gpp2 value is set to the string "1234567812FFFF".

6) if the access-type field is set to "3GPP2-1X-HRPD", a ci-3gpp2 parameter set to the ASCII representation of the hexadecimal value of the string obtained by the concatenation of Sector ID (128 bits) and Subnet length (8 bits) (see 3GPP2 C.S0024-B [86]) and Carrier-ID, if available, (see 3GPP2 X.S0060 [86B])in the specified order. The length of the ci-3gpp2 parameter shall be 34 or 40 hexadecimal characters depending on whether the Carrier-ID is included. The hexadecimal characters (A through F) shall be coded using the uppercase ASCII characters;

EXAMPLE: If the Sector ID = 0x12341234123412341234123412341234, Subnet length = 0x11, and the Carrier-ID=0x555444, the ci-3gpp2 value is set to the string "12341234123412341234123412341234123411555444".

7) if the access-type field is set to "3GPP2-UMB" 3GPP2 C.S0084-000 [86A], a ci-3gpp2 parameter is set to the ASCII representation of the hexadecimal value of the Sector ID (128 bits) defined in 3GPP2 C.S0084-000 [86A]. The length of the ci-3gpp2 parameter shall be 32 hexadecimal characters. The hexadecimal characters (A through F) shall be coded using the uppercase ASCII characters;

EXAMPLE: If the Sector ID = 0x12341234123412341234123412341234, the ci-3gpp2 value is set to the string "12341234123412341234123412341234".

8) if the access-type field set to one of "IEEE-802.11", "IEEE-802.11a", "IEEE-802.11b", "IEEE-802.11g", "IEEE-802.11n", or "IEEE-802.11ac", an "i-wlan-node-id" parameter is set to the ASCII representation of the hexadecimal value of the AP's MAC address without any delimiting characters;

NOTE 2: The AP's MAC address is provided in the BSSID information element.

EXAMPLE: If the AP's MAC address = 00-0C-F1-12-60-28, then i-wlan-node-id is set to the string "000cf1126028".

NOTE 3: "i-wlan-node-id" parameter is not restricted to I-WLAN. "i-wlan-node-id" parameter can be inserted for a WLAN which is not an I-WLAN.

9) if the access-type field is set to "3GPP2-1X-Femto", a ci-3gpp2-femto parameter set to the ASCII representation of the hexadecimal value of the string obtained by the concatenation of femto MSCID (24 bit), femto CellID (16 bit), FEID (64bit), macro MSCID (24 bits) and macro CellID (16 bits) (3GPP2 X.P0059-200 [86E]) in the specified order. The length of the ci-3gpp2-femto parameter is 36 hexadecimal characters. The hexadecimal characters (A through F) are coded using the uppercase ASCII characters.

10) if the access-type field is set to one of "ADSL", "ADSL2", "ADSL2+", "RADSL", "SDSL", "HDSL", "HDSL2", "G.SHDSL", "VDSL", "IDSL", or "xDSL", the access-info field shall contain a dsl-location parameter obtained from the CLF (see NASS functional architecture);

11) if the access-type field set to "DOCSIS", the access info parameter is not inserted. This release of this specification does not define values for use in this parameter;

12) if the access-type field is equal to "3GPP-E-UTRAN-FDD" or "3GPP-E-UTRAN-TDD", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value) which should be obtained from the E-UTRAN Cell Global Identifier (ECGI), Tracking Area Code (4 hexadecimal digits when accessing to EPC and 6 hexadecimal digits when accessing to 5GCN) as described in 3GPP TS 23.003 [3] and the E-UTRAN Cell Identity (ECI) (7 hexadecimal digits) as described in 3GPP TS 23.003 [3]. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

EXAMPLE: If MCC is 111, MNC is 22, TAC is 33C4 and ECI is 76B4321, then P-Access-Network-Info header field looks like follows: P-Access-Network-Info: 3GPP-E-UTRAN-FDD;utran-cell-id-3gpp=1112233C476B4321;network-provided

NOTE 4: The total length of the "utran-cell-id-3gpp" parameter depends on the various combinations of MNC and TAC possible sizes. The actual length of MNC and TAC parts can be unambiguously deduced from the total length.

NOTE 5: The P-CSCF obtains the ECGI in the 3GPP-User-Location-Info AVP received from the PCRF, while the UE obtains the ECGI from RAN. In roaming scenarios with P-GW in the HPLMN, the MCC-MNC contained in the ECGI retrieved by the P-CSCF can differ from that contained in the ECGI retrieved by the UE. Using MNC and MCC from a different source than ECGI can lead to collision between cell-id values which makes the determination of the UE location not possible or incorrect and disables routing of emergency calls based on location information.

12A) if the access-class field is equal to "3GPP-E-UTRAN", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value) which should be obtained from the E-UTRAN Cell Global Identifier (ECGI), Tracking Area Code (4 hexadecimal digits when accessing to EPC and 6 hexadecimal digits when accessing to 5GCN) as described in 3GPP TS 23.003 [3] and the E-UTRAN Cell Identity (ECI) (7 hexadecimal digits) as described in 3GPP TS 23.003 [3]. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

12B) if the access-type field is equal to "3GPP-E-UTRAN-ProSe-UNR", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value) which should be obtained from the E-UTRAN Cell Global Identifier (ECGI) and the E-UTRAN Cell Identity (ECI) (7 hexadecimal digits) as described in 3GPP TS 23.003 [3] obtained from the ProSe-UE-to-network relay that the UE is connected to as specified in 3GPP TS 24.334 [8ZD]. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in in RFC 20 [212];

EXAMPLE: If MCC is 111, MNC is 22 and ECI is 76B4321, then P-Access-Network-Info header field looks like follows: P-Access-Network-Info: 3GPP-E-UTRAN-ProSe-UNR;utran-cell-id-3gpp=1112276B4321.

12C) if the access-type field is equal to "3GPP-E-UTRAN-FDD" or "3GPP-E-UTRAN-TDD", an "eps-fallback" header field parameter set to an appropriate value;

12D) if the access-class field is equal to "3GPP-E-UTRAN", an "eps-fallback" header field parameter set to an appropriate value;

13) if the access-type field is set to one of "IEEE-802.3", "IEEE-802.3a", "IEEE-802.3e", "IEEE-802.3i", "IEEE-802.3j", "IEEE-802.3u", "IEEE-802.3ab", "IEEE-802.3ae", IEEE-802.3ak", IEEE-802.3aq", IEEE-802.3an", "IEEE-802.3y" or "IEEE-802.3z" and NASS subsystem is used, the access-info field shall contain an eth-location parameter obtained from the CLF (see NASS functional architecture);

14) if the access-type field is set to one of "GPON", "XGPON1" or "IEEE-802.3ah" and NASS is used, the access-info field shall contain an fiber-location parameter obtained from the CLF (see NASS functional architecture);

15) if the access-type field is set to "GSTN", the access-info field may contain a gstn-location parameter if received from the GSTN;

NOTE 6: The "cgi-3gpp", the "utran-cell-id-3gpp", the "ci-3gpp2", the "ci-3gpp2-femto", the "i-wlan-node-id", eth-location, and the "dsl-location" parameters described above among other usage also constitute the location identifiers that are used for emergency services.

16) if the access-type field is set to "DVB-RCS2", the access-info field shall contain a "dvb-rcs2-node-id" parameter which consists of comma-separated list consisting of NCC_ID, satellite_ID, beam_ID, and SVN-MAC as specified in ETSI TS 101 545-2 [194], ETSI TS 101 545-3 [195]; the NCC_ID shall be represented as two digit hexadecimal value, the satellite_ID shall be represented as a two digit hexadecimal value, the beam_ID shall be respresented as a four digit hexadecimal value, and the SVN-MAC shall be represented as six digit hexadecimal value;

EXAMPLE: If the (8 bit) NCC_ID = 0x3A, the (8 bit) satellite_ID = 0xF5, the (16 bit) beam_ID = 0xEA23, and the (24 bit) SVN-MAC = 0xE40AB9, then the "dvb-rcs2-node-id" is set to the string "3A,F5,EA23,E40AB9".

17) the "local-time-zone" parameter in the access-info field is coded as a text string as follows:

UTC±[hh]:[mm]. [hh] is two digits, and [mm] is two digits from four values: "00", "15", "30" or "45", see ISO 8601 [203];

EXAMPLE: "UTC+01:00" indicates that the time difference between local time and UTC of day is one hour.

18) the "daylight-saving-time" parameter in the access-info field is coded as a text string as follows:

[hh]. [hh] is a two digits value from three values "00", "01" or "02" indicating the positive adjustment in hours;

19) void;

20) the operator-specific-GI in the access-info field is coded as a text string and conveys an operator-specifc geographical identifier;

21) if

a) the access-class field is set to "untrusted-non-3GPP-VIRTUAL-EPC"; or

b) the access-class field is set to "3GPP-WLAN" and the WLAN is an untrusted WLAN;

then:

a) if a UE local IP address is available, then a "UE-local-IP-address" parameter set to the UE local IP address;

b) if the IKEv2 messages exchanged between the UE and the ePDG are encapsulated in the UDP messages according to IETF RFC 3948 [63A] and the UDP source port of the UDP messages received by ePDG is available, then a "UDP-source-port" parameter set to the UDP source port of the UDP messages:

- received by the ePDG; and
- encapsulating the IKEv2 messages;

c) if the IKEv2 messages exchanged between the UE and the ePDG are transported using the firewall traversal tunnel as described in 3GPP TS 24.302 [8U] and the TCP source port of the TCP messages of the firewall traversal tunnel received by ePDG is available, then a "TCP-source-port" parameter set to the TCP source port of the TCP messages:

- received by the ePDG; and
- of the firewall traversal tunnel transporting the IKEv2 messages; and

d) if an ePDG IP address used as IKEv2 tunnel endpoint with the UE is available, then an "ePDG-IP-address" parameter set to the ePDG IP address used as IKEv2 tunnel endpoint with the UE;

22) if the access-type field is equal to "3GPP-NR-FDD" or "3GPP-NR-TDD", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), Tracking Area Code (6 hexadecimal digits) as described in 3GPP TS 23.003 [3], the NR Cell Identity (NCI) (9 hexadecimal digits) and optionally, the Network Identifier (NID) (11 hexadecimal digits) as specified in 3GPP TS 23.003 [3]. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212]; and

NOTE 7: NID is included only if a serving network is a Stand-alone Non-Public Network (SNPN) identified by a combination of NID, MCC and MNC. The serving network type can be unambiguously deduced from the total length of the "utran-cell-id-3gpp" parameter.

22A) if the access-class field is equal to "3GPP-NR", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), Tracking Area Code (6 hexadecimal digits) as described in 3GPP TS 23.003 [3], the NR Cell Identity (NCI) (9 hexadecimal digits) and optionally, the NID (11 hexadecimal digits) as specified in 3GPP TS 23.003 [3]. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212].

23) if the access-type field is equal to "3GPP-NR-U-FDD" or "3GPP-NR-U-TDD", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), Tracking Area Code (6 hexadecimal digits) as described in 3GPP TS 23.003 [3], the NR Cell Identity (NCI) (9 hexadecimal digits) and optionally, the NID (11 hexadecimal digits) as specified in 3GPP TS 23.003 [3]. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

23A) if the access-class field is equal to "3GPP-NR-U", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), Tracking Area Code (6 hexadecimal digits) as described in 3GPP TS 23.003 [3], the NR Cell Identity (NCI) (9 hexadecimal digits) and optionally, the NID (11 hexadecimal digits) as specified in 3GPP TS 23.003 [3]. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

24) if the access-type field is equal to "3GPP-NR-SAT", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), Tracking Area Code (6 hexadecimal digits) as described in 3GPP TS 23.003 [3], the NR Cell Identity (NCI) (9 hexadecimal digits) and optionally, the NID (11 hexadecimal digits) as specified in 3GPP TS 23.003 [3]. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

24A) if the access-class field is equal to "3GPP-NR-SAT", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), Tracking Area Code (6 hexadecimal digits) as described in 3GPP TS 23.003 [3], the NR Cell Identity (NCI) (9 hexadecimal digits) and optionally, the NID (11 hexadecimal digits) as specified in 3GPP TS 23.003 [3]. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

25) if the access-type field is equal to "3GPP-NR-ProSe-L2UNR" or "3GPP-NR-ProSe-L3UNR", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), Tracking Area Code (6 hexadecimal digits) as described in 3GPP TS 23.003 [3], and the NR Cell Identity (NCI) (9 hexadecimal digits) obtained from the 5G ProSe UE-to-network relay UE that the UE is connected to as specified in 3GPP TS 24.554 [8ZI]. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212];

26) if the access-class field is equal to "3GPP-NR-REDCAP", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), Tracking Area Code (6 hexadecimal digits) as described in 3GPP TS 23.003 [3], the NR Cell Identity (NCI) (9 hexadecimal digits) and optionally, the NID (11 hexadecimal digits) as specified in 3GPP TS 23.003 [3]. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212]; and

26A) if the access-type field is equal to "3GPP-NR-REDCAP", a "utran-cell-id-3gpp" parameter set to a concatenation of the MCC (3 decimal digits), MNC (2 or 3 decimal digits depending on MCC value), Tracking Area Code (6 hexadecimal digits) as described in 3GPP TS 23.003 [3], the NR Cell Identity (NCI) (9 hexadecimal digits) and optionally, the NID (11 hexadecimal digits) as specified in 3GPP TS 23.003 [3]. The "utran-cell-id-3gpp" parameter is encoded in ASCII as defined in RFC 20 [212].

7.2A.5 P-Charging-Vector header field

7.2A.5.1 Introduction

The P-Charging-Vector header field is extended to include specific charging correlation information needed for IM CN subsystem functional entities.

7.2A.5.2 Syntax

7.2A.5.2.1 General

The syntax of the P-Charging-Vector header field is described in RFC 7315 [52]. There may be additional coding rules for this header field depending on the type of IP-CAN, according to access technology specific descriptions.

Table 7.2A.5 describes 3GPP-specific extensions to the P-Charging-Vector header field defined in RFC 7315 [52].

Table 7.2A.5: Syntax of extensions to P-Charging-Vector header field

```
   access-network-charging-info = (gprs-charging-info / i-wlan-charging-info / xdsl-charging-info /
        packetcable-charging-info / icn-charging-info / eps-charging-info / eth-charging-info/
        loopback-indication / 5gs-charging-info / generic-param)
   gprs-charging-info = ggsn SEMI auth-token [SEMI pdp-info-hierarchy] *(SEMI extension-param)
   ggsn = "ggsn" EQUAL gen-value
   pdp-info-hierarchy = "pdp-info" EQUAL LDQUOT pdp-info *(COMMA pdp-info) RDQUOT
   pdp-info = pdp-item SEMI pdp-sig SEMI gcid [SEMI flow-id]
   pdp-item = "pdp-item" EQUAL DIGIT
   pdp-sig = "pdp-sig" EQUAL ("yes" / "no")
   gcid = "gcid" EQUAL 1*HEXDIG
   auth-token = "auth-token" EQUAL 1*HEXDIG
   flow-id = "flow-id" EQUAL "(" "{" 1*DIGIT COMMA 1*DIGIT "}" *(COMMA "{" 1*DIGIT COMMA 1*DIGIT
        "}")")"
   i-wlan-charging-info = "pdg"
   xdsl-charging-info = bras SEMI auth-token [SEMI xDSL-bearer-info] *(SEMI extension-param)
   bras = "bras" EQUAL gen-value
   xDSL-bearer-info = "dsl-bearer-info" EQUAL LDQUOT dsl-bearer-info *(COMMA dsl-bearer-info) RDQUOT
   dsl-bearer-info = dsl-bearer-item SEMI dsl-bearer-sig SEMI dslcid [SEMI flow-id]
   dsl-bearer-item = "dsl-bearer-item" EQUAL DIGIT
   dsl-bearer-sig = "dsl-bearer-sig" EQUAL ("yes" / "no")
   dslcid = "dslcid" EQUAL 1*HEXDIG
   packetcable-charging-info = packetcable [SEMI bcid]
   packetcable = "packetcable-multimedia"
   bcid = "bcid" EQUAL 1*48(HEXDIG)
   icn-charging-info = icn-bcp *(SEMI itid) [SEMI extension-param]
   icn-bcp = "icn-bcp" EQUAL gen-value
   itid = itc-sig SEMI itc-id SEMI *(flow-id2)
   itc-sig = "itc-sig" EQUAL ("yes" / "no")
   itc-id = "itc-id" EQUAL gen-value
   flow-id2 = "flow-id" EQUAL gen-value
   extension-param = token [EQUAL (token | quoted-string)]
   eps-charging-info = pdngw [SEMI eps-bearer-hierarchy] *(SEMI extension-param)
   pdngw = "pdngw" EQUAL gen-value
   eps-bearer-hierarchy = "eps-info" EQUAL LDQUOT eps-info *(COMMA eps-info) RDQUOT
   eps-info = eps-item SEMI eps-sig SEMI ecid [SEMI flow-id]
   eps-item = "eps-item" EQUAL DIGIT
   eps-sig = "eps-sig" EQUAL ("yes" / "no")
   ecid = "ecid" EQUAL 1*HEXDIG
   eth-charging-info = ip-edge *(SEMI extension-param)
   fiber-charging-info = ip-edge *(SEMI extension-param)
   ip-edge = "ip-edge" EQUAL gen-value
      loopback-indication = "loopback"
   fe-identifier = "fe-identifier" EQUAL fe-id-list   fe-id-list = DQUOTE fe-id-param *(COMMA fe-id-
param) DQUOTE
   fe-id-param = fe-addr/as-addr
   fe-addr = "fe-addr" EQUAL gen-value
   as-addr = "as-addr" EQUAL gen-value "-" ap-id
   ap-id = "ap-id" EQUAL gen-value
   5gs-charging-info = smf [SEMI 5gs-pdu-session-hierarchy] *(SEMI extension-param)
   smf = "smf" EQUAL gen-value
   5gs-pdu-session-hierarchy = "5gs-info" EQUAL LDQUOT 5gs-info *(COMMA 5gs-info) RDQUOT
   5gs-info = 5gs-item SEMI 5gscid [SEMI flow-id]
   5gs-item = "5gs-item" EQUAL DIGIT
   5gscid = "5gscid" EQUAL 1*HEXDIG
```

NOTE: The syntax above is not aligned with the rules for defining new P-Charging-Vector header field parameters as defined in RFC 7315 [52]. Entities that perform syntax check (even if they are not interested in specific header field parameter values) of the header field need to follow the explicit syntax above, as using the rules in RFC 7315 [52] would trigger a parser error.

The access-network-charging-info parameter is an instance of generic-param from the current charge-params component of P-Charging-Vector header field.

The access-network-charging-info parameter includes alternative definitions for different types access networks. The description of these parameters are given in the subsequent subclauses.

The "access-network-charging-info" header field parameter is not included in the P-Charging-Vector for SIP signalling that is not associated with a session.

When the "access-network-charging-info" is included in the P-Charging-Vector and necessary information is not available from the IP-CAN (e.g. via Gx/Rx interface) reference points then null or zero values are included.

For type 1 and type 3 IOIs, the generating SIP entity shall express the "orig-ioi" and "term-ioi" header field parameters in the format of a quoted string as specified in RFC 7315 [52].

If an IOI is a type 1 IOI, the content of the quoted string consists of the "Type 1" string prefix followed by the IOI value. The "Type 1" string prefix is the type-1-prefix value specified in the table 7.2A.5A.

If an IOI is a type 3 IOI, the content of the quoted string consists of the "Type 3" string prefix followed by the IOI value. The "Type 3" string prefix is the type-3-prefix value specified in the table 7.2A.5A.

Table 7.2A.5A: String prefixes

```
type-1-prefix = %x54.79.70.65.20.31 ; "Type 1"
type-3-prefix = %x54.79.70.65.20.33 ; "Type 3"
```

If an IOI is a type 2 IOI, the value of the "orig-ioi" and "term-ioi" header field parameters is set to the IOI value. No string prefix is used.

The receiving SIP entity does not perform syntactic checking of the contents of the IOI parameter (the IOI parameter is passed unmodified to charging entities).

The "loopback" parameter is provided to the charging system of other entities in the signalling path to indicate that loopback has been applied and entities of the IM CN subsystem involved in the loopback, e.g. TRF, can have generated CDRs in their own right.

The "fe-identifier" header field parameter is an instance of generic-param from the current charge-params component of the P-Charging-Vector header field. This header field parameter contains one or more IM CN subsystem functional entity addresses ("fe-addr") and/or AS addresses ("as-addr") and application identifiers ("ap-id") where the IM CN subsystem functional entity does create charging information for the related CDR of this IM CN subsystem functional entity. For AS hosting several applications the AS address can appear several times, each accompanied with a different application identifier based on the application executed by the AS.

7.2A.5.2.2 GPRS as IP-CAN

GPRS is a supported access network (gprs-charging-info parameter). For GPRS there are the following components to track: GGSN address (ggsn parameter), media authorization token (auth token parameter), and a pdp-info parameter that contains the information for one or more PDP contexts. In this release the media authorization token is set to zero. The pdp-info contains one or more pdp-item values followed by a collection of parameters (pdp-sig, gcid, and flow-id). The value of the pdp-item is a unique number that identifies each of the PDP-related charging information within the P-Charging-Vector header field. Each PDP context has an indicator if it is an IM CN subsystem signalling PDP context (pdp-sig parameter), an associated GPRS Charging Identifier (gcid parameter), and a identifier (flow-id parameter). The flow-id parameter contains a sequence of curly bracket delimited flow identifier tuples that identify associated m-lines and relative order of port numbers in an m-line within the SDP from the SIP signalling to which the PDP context charging information applies. For a complete description of the semantics of the flow-id parameter see 3GPP TS 29.214 [13D] Annex B. The gcid, ggsn address and flow-id parameters are transferred from the GGSN to the P-CSCF via the PCRF over the Rx interface (see 3GPP TS 29.214 [13D] and Gx interface (see 3GPP TS 29.212 [13B]).

The gcid value is received in binary format at the P-CSCF (see 3GPP TS 29.214 [13D]). The P-CSCF shall encode it in hexadecimal format before include it into the gcid parameter. On receipt of this header field, a node receiving a gcid shall decode from hexadecimal into binary format.

The "access-network-charging-info" is not included in the P-Charging-Vector for SIP signalling that is not associated with a multimedia session. The access network charging information may be unavailable for sessions that use a general purpose PDP context (for both SIP signalling and media) or that do not require media authorisation.

7.2A.5.2.3 Evolved Packet Core (EPC) via WLAN as IP-CAN

The access-network-charging-info parameter is an instance of generic-param from the current charge-params component of P-Charging-Vector header field.

This version of the specification defines the use of "pdg" for inclusion in the P-Charging-Vector header field. No other extensions are defined for use in I-WLAN in this version of the specification.

Editor's note: WI: TEI12: CR5046: The application of the ABNF element relating to "pdg" to EPS needs to be clarified.

7.2A.5.2.4 xDSL as IP-CAN

The access-network-charging-info parameter is an instance of generic-param from the current charge-params component of P-Charging-Vector header field. The access-network-charging-info parameter includes alternative definitions for different types of access networks. This subclause defines the components of the xDSL instance of the access-network-charging-info.

For xDSL, there are the following components to track: BRAS address (bras parameter), media authorization token (auth-token parameter), and a set of dsl-bearer-info parameters that contains the information for one or more xDSL bearers.

The dsl-bearer-info contains one or more dsl-bearer-item values followed by a collection of parameters (dsl-bearer-sig, dslcid, and flow-id). The value of the dsl-bearer-item is a unique number that identifies each of the dsl-bearer-related charging information within the P-Charging-Vector header field. Each dsl-bearer-info has an indicator if it is an IM CN subsystem signalling dsl-bearer (dsl-bearer-sig parameter), an associated DSL Charging Identifier (dslcid parameter), and a identifier (flow-id parameter). The flow-id parameter contains a sequence of curly bracket delimited flow identifier tuples that identify associated m-lines and relative order of port numbers in an m-line within the SDP from the SIP signalling to which the dsl-bearer charging information applies. For a complete description of the semantics of the flow-id parameter see 3GPP TS 29.214 [13D].

The format of the dslcid parameter is identical to that of ggsn parameter. On receipt of this header field, a node receiving a dslcid shall decode from hexadecimal into binary format.

For a dedicated dsl-bearer for SIP signalling, i.e. no media stream requested for a session, then there is no authorisation activity or information exchange over the Rx and Gx interfaces. Since there are no dslcid, media authorization token or flow identifiers in this case, the dslcid and media authorization token are set to zero and no flow identifier parameters are constructed by the PCRF.

7.2A.5.2.5 DOCSIS as IP-CAN

The access-network-charging-info parameter is an instance of generic-param from the current charge-params component of P-Charging-Vector header field. The access-network-charging-info parameter includes alternative definitions for different types of access networks. This subclause defines the components of the cable instance of the access-network-charging-info. Cable access is based upon the architecture defined by Data Over Cable Service Interface Specification (DOCSIS).

The billing correlation identifier (bcid) uniquely identifies the PacketCable DOCSIS bearer resources associated with the session within the cable operator's network for the purposes of billing correlation. To facilitate the correlation of session and bearer accounting events, a correlation ID that uniquely identifies the resources associated with a session is needed. This is accomplished through the use of the bcid as generated by the PacketCable Multimedia network. This bcid is returned to the P-CSCF within the response to a successful resource request.

The bcid is specified in RFC 3603 [74A]. This identifier is chosen to be globally unique within the system for a window of several months. Consistent with RFC 3603 [74A], the BCID must be encoded as a hexadecimal string of up to 48 characters. Leading zeroes may be suppressed.

If the bcid value is received in binary format by the P-CSCF from the IP-CAN, the P-CSCF shall encode it in hexadecimal format before including it into the bcid parameter. On receipt of this header field, a node using a bcid will normally decode from hexadecimal into binary format.

7.2A.5.2.6 cdma2000® packet data subsystem as IP-CAN

The specific extensions to the P-Charging-Vector header field defined in RFC 7315 [52] when the access network is cdma2000® packet data subsystem are: the icn-charging-info parameter contains one icn-bcp child parameter and one or more child itid parameters. The icn-bcp parameter, identifies the point of attachment where UE has attached itself to the cdma2000® packet data subsystem. The icn-bcp parameter is conveyed to the P-CSCF by the cdma2000® packet data subsystem. Each itid child parameter within icn-charging-info corresponds to one IP-CAN bearer that was established

by the cdma2000® packet data subsystem for the UE. Each itid parameter contains an indicator if it is an IP-CAN subsystem signalling IP-CAN bearer (itc-sig parameter), an associated IP-CAN charging identifier (itc-id parameter), and one or more flow identifiers (flow-id parameter) that identify associated m-lines within the SDP from the SIP signalling. These parameters are transferred from the cdma2000® packet data subsystem to the P-CSCF over the respective interface.

For an IP-CAN bearer that is only used for SIP signalling, i.e. no media stream requested for a session, then there is no authorisation activity or information exchange with the P-CSCF over the respective cdma2000® interfaces. Since there is no itc-id, or flow identifiers in this case, the itc-id is set to zero and no flow identifier parameters are constructed by the P-CSCF.

7.2A.5.2.7 EPS as IP-CAN

For EPS there are the following components to track: P-GW address (pdngw parameter), and a eps-info parameter that contains the information for one or more EPS bearers. The eps-info contains one or more eps-item values followed by a collection of parameters (eps-sig, ecid, and flow-id). The value of the eps-item is a unique number that identifies each of the EPS-bearer-related charging information within the P-Charging-Vector header field. Each EPS bearer context has an associated QCI indicating if it is an IM CN subsystem signalling EPs bearer context (eps-sig parameter), an associated EPS Charging Identifier (ecid parameter), and a identifier (flow-id parameter). The flow-id parameter contains a sequence of curly bracket delimited flow identifier tuples that identify associated m-lines and relative order of port numbers in an m-line within the SDP from the SIP signalling to which the EPS bearer charging information applies. For a complete description of the semantics of the flow-id parameter see 3GPP TS 29.214 [13D] Annex B. The ecid, pdngw address and flow-id parameters are transferred from the P-GW to the P-CSCF via the PCRF over the Rx interface (see 3GPP TS 29.214 [13D] and Gx interface (see 3GPP TS 29.212 [13B]).

The ecid value is received in binary format at the P-CSCF (see 3GPP TS 29.214 [13D]). The P-CSCF shall encode it in hexadecimal format before include it into the ecid parameter. On receipt of this header field, a node receiving a gcid shall decode from hexadecimal into binary format.

The "access-network-charging-info" header field parameter is not included in the P-Charging-Vector for SIP signalling that is not associated with a multimedia session. The access network charging information may be unavailable for sessions that use a general purpose EPS bearer context (for both SIP signalling and media).

7.2A.5.2.8 Ethernet as IP-CAN

The access-network-charging-info parameter is an instance of generic-param from the current charge-params component of P-Charging-Vector header field. For Ethernet accesses, the IP Edge Node address (ip-edge parameter) is tracked. The IP Edge Node is defined in ETSI ES 282 001 [138].

7.2A.5.2.9 Fiber as IP-CAN

The access-network-charging-info parameter is an instance of generic-param from the current charge-params component of P-Charging-Vector header field. For Fiber accesses, the IP Edge Node address (ip-edge parameter) is tracked. The IP Edge Node is defined in ETSI ES 282 001 [138].

7.2A.5.2.10 5GS as IP-CAN

For 5GS there are the following components to track: SMF address (SMF parameter) and a 5gs-info parameter that contains the information for one or more 5GS PDU sessions. The 5gs-info contains one or more 5gs-item values followed by a collection of parameters (5gscid and flow-id). The value of the 5gs-item is a unique number that identifies each of the 5GS PDU session charging information within the P-Charging-Vector header field.

Each 5GS PDU session has an associated 5GS Charging Identifier (5gscid parameter), and an additional information (flow-id parameter). The flow-id parameter contains a sequence of curly bracket delimited parameter tuples that identify associated m-lines and relative order of port numbers in an m-line within the SDP from the SIP signalling to which the 5GS PDU session charging information applies. For a complete description of the semantics of the flow-id parameter see 3GPP TS 29.214 [13D]. The smf address, 5gscid and flow-id parameters are transported to the P-CSCF via the PCRF over the Rx interface (see 3GPP TS 29.214 [13D].

The 5gscid value is received in binary format at the P-CSCF (see 3GPP TS 29.214 [13D]). The P-CSCF shall encode it in hexadecimal format before include it into the 5gscid parameter. On receipt of this header field, a node receiving a 5gscid shall decode from hexadecimal into binary format.

The "access-network-charging-info" header field parameter is not included in the P-Charging-Vector for SIP signalling that is not associated with a multimedia session.

7.2A.5.3 Operation

The operation of this header field is described in subclauses 5.2, 5.3, 5.4, 5.5, 5.6, 5.7 and 5.8.

7.2A.6 Orig parameter definition

7.2A.6.1 Introduction

The "orig" parameter is a uri-parameter intended to:

- tell the S-CSCF that it has to perform the originating services instead of terminating services;
- tell the I-CSCF that it has to perform originating procedures.

7.2A.6.2 Syntax

The syntax for the orig parameter is specified in table 7.2A.6:

Table 7.2A.6: Syntax of orig parameter

```
uri-parameter = transport-param / user-param / method-param / ttl-param / maddr-param / lr-param /
    orig / other-param
orig = "orig"
```

The BNF for uri-parameter is taken from RFC 3261 [26] and modified accordingly.

7.2A.6.3 Operation

The orig parameter is appended to the address of the S-CSCF, I-CSCF or IBCF by the ASs, when those initiate requests on behalf of the user, or to the address of the S-CSCF or I-CSCF by an IBCF acting as entry point, if the network is performing originating service to another network. The S-CSCF will run originating services whenever the orig parameter is present next to its address. The I-CSCF will run originating procedures whenever the orig parameter is present next to its address. The IBCF will preserve the "orig" parameter in the topmost Route header field if received, or it may append the "orig" parameter to the URI in the topmost Route header field (see subclause 5.10.2.3).

7.2A.7 Extension to Security-Client, Security-Server and Security-Verify header fields

7.2A.7.1 Introduction

This extension defines new parameters for the Security-Client, Security-Server and Security-Verify header fields.

This subclause defines the "mediasec" header field parameter that labels any of the Security-Client:, Security-Server, or Security-Verify: header fields as applicable to the media plane and not the signalling plane.

7.2A.7.2 Syntax

7.2A.7.2.1 General

The syntax for the Security-Client, Security-Server and Security-Verify header fields is defined in RFC 3329 [48]. The additional syntax is defined in Annex H of 3GPP TS 33.203 [19].

This specification reuses Security-Client, Security-Server and Security-Verify defined in RFC 3329 [48] and defines the mechanism listed in table 7.2A.7.2.2-2 and the header field parameter "mediasec".

Security mechanisms that apply to the media plane only shall not have the same name as any signalling plane mechanism. If a signalling plane security mechanism name is re-used for the media plane and distinguished only by the "mediasec" parameter, then implementations that do not recognize the "mediasec" parameter may incorrectly use that security mechanism for the signalling plane.

7.2A.7.2.2 "mediasec" header field parameter

The "mediasec" header field parameter may be used in the Security- Client, Security-Server, or Security-Verify header fields defined in RFC 3329 [48] to indicate that a header field applies to the media plane. Any one of the media plane security mechanisms supported by both client and server, if any, may be applied when a media stream is started. Or, a media stream may be set up without security.

Values in the Security-Client, Security-Server, or Security-Verify header fields labelled with the "mediasec" header field parameter are specific to the media plane and specific to the secure media transport protocol used on the media plane.

EXAMPLE: Security-Client: sdes-srtp;mediasec

Usage of the "mediasec" header field parameter in mech-parameters rule of RFC 3329 [48] and the syntax of the "mediasec" header field parameter is shown in table 7.2A.7.2.2-1.

Table 7.2A.7.2.2-1

```
mech-parameters =/ mediasec-param
mediasec-param = "mediasec"
```

The security mechanisms which can be labelled by the "mediasec" header field parameter are listed in the table 7.2A.7.2.2-2, where each line (other than the first line) indicates a token and a media security mechanism for which the token indicates support.

Table 7.2A.7.2.2-2

```
mechanism-name =/ ( sdes-srtp-name / msrp-tls-name / bfcp-tls-name / udptl-dtls-name /
                    dtls-srtp-name / token )
sdes-srtp-name = "sdes-srtp" ; End-to-access-edge media security using SDES.
msrp-tls-name = "msrp-tls" ; End-to-access-edge media security for MSRP using TLS and certificate
fingerprints.
bfcp-tls-name = "bfcp-tls" ; End-to-access-edge media security for BFCP using TLS and certificate
fingerprints.
udptl-dtls-name = "udptl-dtls" ; End-to-access-edge media security for UDPTL using DTLS and
certificate fingerprints.
dtls-srtp-name = "dtls-srtp" ; End-to-access-edge media security for RTP using DTLS-SRTP and
certificate fingerprints.
```

7.2A.7.3 Operation

The operation of the additional parameters for the Security-Client, Security-Server and Security-Verify header fields is defined in Annex H of 3GPP TS 33.203 [19].

Any one of the mechanisms listed in table 7.2A.7.2.2-2 and labelled with the "mediasec" header field parameter can be applied on-the-fly as a media stream is started, unlike mechanisms for signalling one of which is chosen and then applied throughout a session.

Media plane security can be supported independently of any signalling plane security defined in RFC 3329 [4], but in order to protect any cryptographic key carried in SDP signalling plane security as defined in RFC 3329 [4] SHOULD be used. Each media security mechanism can be supported independently.

The message flow is identical to the flow in RFC 3329 [48], but it is not mandatory for the user agent to apply media plane security immediately after it receives the list of supported media plane mechanisms from the server, or any timer after that, nor will the lack of a mutually supported media plane security mechanism prevent SIP session setup.

7.2A.7.4 IANA registration

7.2A.7.4.1 "mediasec" header field parameter

Editor's note: [MEDIASEC_CORE, CR 4156] This subclause forms the basis for IANA registration of the mediasec header field parameter. Registration is intended to be created by an RFC that describes the mediasec header field parameter and creates an IANA registry for its values.

NOTE: This subclause contains information to be provided to IANA for the registration of the media plane security indicator header field parameter.

Contact name, email address, and telephone number:

3GPP Specifications Manager

3gppContact@etsi.org

+33 (0)492944200

Header field in which the parameter can appear:

Security-Client, Security-Server and Security-Verify header fields.

Name of the header field parameter being registered:

mediasec

Whether the parameter only accepts a set of predefined values:

No value is defined for the parameter.

A reference to the RFC where the parameter is defined and to any RFC that defines new values for the parameter:

This parameter is defined in 3GPP TS 24.229.

7.2A.7.4.2 "sdes-srtp" security mechanism

Editor's note: [MEDIASEC_CORE, CR 4156] This subclause forms the basis for IANA registration of the value for the mediasec header field parameter. The registration should be performed by MCC when the registry for mediasec parameter values has been created by IANA.

NOTE: This subclause contains information to be provided to IANA for the registration of the media plane security indicator header field parameter.

Contact name, email address, and telephone number:

3GPP Specifications Manager

3gppContact@etsi.org

+33 (0)492944200

The mechanism-name token:

sdes-srtp

The published RFC describing the details of the corresponding security mechanism:

This mechanism is defined in 3GPP TS 24.229.

7.2A.7.4.3 "msrp-tls" security mechanism

Editor's note: [WI: eMEDIASEC-CT, CR#4624] This subclause forms the basis for IANA registration of the value for the mediasec header field parameter. The registration should be performed by MCC when the registry for mediasec parameter values has been created by IANA.

NOTE: This subclause contains information to be provided to IANA for the registration of the media plane security indicator header field parameter.

Contact name, email address, and telephone number:

3GPP Specifications Manager

3gppContact@etsi.org

+33 (0)492944200

The mechanism-name token:

msrp-tls

The published RFC describing the details of the corresponding security mechanism:

This mechanism is defined in 3GPP TS 24.229.

7.2A.7.4.4 "bfcp-tls" security mechanism

Editor's note: [WI: eMEDIASEC-CT, CR#4624] This subclause forms the basis for IANA registration of the value for the mediasec header field parameter. The registration should be performed by MCC when the registry for mediasec parameter values has been created by IANA.

NOTE: This subclause contains information to be provided to IANA for the registration of the media plane security indicator header field parameter.

Contact name, email address, and telephone number:

3GPP Specifications Manager

3gppContact@etsi.org

+33 (0)492944200

The mechanism-name token:

bfcp-tls

The published RFC describing the details of the corresponding security mechanism:

This mechanism is defined in 3GPP TS 24.229.

7.2A.7.4.5 "udptl-dtls" security mechanism

Editor's note: [WI: eMEDIASEC-CT, CR#4624] This subclause forms the basis for IANA registration of the value for the mediasec header field parameter. The registration should be performed by MCC when the registry for mediasec parameter values has been created by IANA.

NOTE: This subclause contains information to be provided to IANA for the registration of the media plane security indicator header field parameter.

Contact name, email address, and telephone number:

3GPP Specifications Manager

3gppContact@etsi.org

+33 (0)492944200

The mechanism-name token:

udptl-dtls

The published RFC describing the details of the corresponding security mechanism:

This mechanism is defined in 3GPP TS 24.229.

7.2A.7.4.6 " dtls-srtp" security mechanism

Editor's note: [WI: eCryptPr, CR#6554] This subclause forms the basis for IANA registration of the value for the mediasec header field parameter. The registration should be performed by MCC when the registry for mediasec parameter values has been created by IANA.

NOTE: This subclause contains information to be provided to IANA for the registration of the media plane security indicator header field parameter.

Contact name, email address, and telephone number:

3GPP Specifications Manager

3gppContact@etsi.org

+33 (0)492944200

The mechanism-name token:

dtls-srtp

The published RFC describing the details of the corresponding security mechanism:

This mechanism is defined in 3GPP TS 24.229.

7.2A.8 IMS Communication Service Identifier (ICSI)

7.2A.8.1 Introduction

The ICSI is defined to fulfil the requirements as stated in 3GPP TS 23.228 [7]. An ICSI may have specialisations which refine it by adding subclass identifiers separated by dots. Any specialisations of an ICSI shall have an "is a" relationship if the subclasses are removed. For example, a check for ICSI urn:urn-7:3gpp-service.ims.icsi.mmtel will return true when evaluating ICSI urn:urn-7:3gpp-service.ims.icsi.mmtel.hd-video.

7.2A.8.2 Coding of the ICSI

This parameter is coded as a URN. The ICSI URN may be included as:

- a tag-value within the g.3gpp.icsi-ref media feature tag as defined in subclause 7.9.2 and RFC 3840 [62], in which case those characters of the URN that are not part of the tag-value definition in RFC 3840 [62] shall be represented in the percent encoding as defined in RFC 3986 [124];
- a feature cap value within the "g.3gpp.icsi-ref" feature-capability indicator, as defined in subclause 7.9A.1 and RFC 6809 [190], in which case those characters of the URN that are not part of the feature-capability indicator value definition syntax shall be represented in the percent encoding, as defined in RFC 3986 [124]; or
- as a value of the P-Preferred-Service or P-Asserted-Service header fields as defined RFC 6050 [121].

A list of the URNs containing ICSI values registered by 3GPP can be found at http://www.3gpp.org/specifications-groups/34-uniform-resource-name-urn-list

An example of an ICSI for a 3GPP defined IMS communication service is:

```
urn:urn-7:3gpp-service.ims.icsi.mmtel
```

An example of a g.3gpp.icsi-ref media feature tag containing an ICSI for a 3GPP defined IMS communication service is:

```
g.3gpp.icsi-ref="urn%3Aurn-7%3A3gpp-service.ims.icsi.mmtel"
```

An example of a g.3gpp.icsi-ref feature-capability indicator containing an ICSI for a 3GPP defined IMS communication service is:

```
g.3gpp.icsi-ref="urn%3Aurn-7%3A3gpp-service.ims.icsi.mmtel"
```

An example of an ICSI for a 3GPP defined IMS communication service in a P-Preferred-Service header field is

```
P-Preferred-Service: urn:urn-7:3gpp-service.ims.icsi.mmtel
```

An example of an ICSI for a 3GPP defined IMS communication service in a P-Asserted-Service header field is

```
P-Asserted-Service: urn:urn-7:3gpp-service.ims.icsi.mmtel
```

An example of an ICSI for a defined IMS communication service with a specialisation is:

```
P-Asserted-Service: urn:urn-7:3gpp-service.ims.icsi.mmtel.game-v1
```

An example of an ICSI for a 3GPP defined IMS communication service with an organisation-y defined specialisation is:

```
P-Asserted-Service: urn:urn-7:3gpp-service.ims.icsi.mmtel.organisation-y.game-v2
```

7.2A.9 IMS Application Reference Identifier (IARI)

7.2A.9.1 Introduction

The IARI is defined to fulfil the requirements as stated in 3GPP TS 23.228 [7].

7.2A.9.2 Coding of the IARI

This parameter is coded as a URN. The IARI URN may be included as a tag-value within the g.3gpp.iari-ref media feature tag as defined in subclause 7.9.3 and RFC 3840 [62], in which case those characters of the URN that are not part of the tag-value definition in RFC 3840 [62] shall be represented in the percent encoding as defined in RFC 3986 [124].

A list of the URNs containing IARI values registered by 3GPP can be found at http://www.3gpp.org/specifications-groups/34-uniform-resource-name-urn-list

An example of a g.3gpp.iari-ref media feature tag containing an IARI is:

```
g.3gpp.iari-ref="urn%3Aurn-7%3A3gpp-application.ims.iari.game-v1"
```

7.2A.10 "phone-context" tel URI parameter

7.2A.10.1 Introduction

When the request-URI contains a local number, then a phone-context tel URI parameter as described in RFC 3966 [22] shall be present to indicate the related numbering plan.

Procedures for using this parameter are given in subclause 5.1.2A.1.5 and additional coding rules are detailed in subclause 7.2A.10.3.

7.2A.10.2 Syntax

The syntax of the "phone-context" tel URI parameter is described in RFC 3966 [22]. There are additional coding rules for this parameter depending on the type of IP-CAN, according to access technology specific descriptions.

7.2A.10.3 Additional coding rules for "phone-context" tel URI parameter

In case the access network information is available, the entities inserting the "phone-context" tel URI parameter shall populate the "phone-context" tel URI parameter with the following contents:

1) if the IP-CAN is GPRS, then the "phone-context" tel URI parameter is a domain name. It is constructed from the MCC, the MNC and the home network domain name by concatenating the MCC, MNC, and the string "gprs" as domain labels before the home network domain name;

EXAMPLE: If MCC = 216, MNC = 01, then the "phone-context" tel URI parameter is set to '216.01.gprs.home1.net'.

2) if the IP-CAN is Evolved Packet Core (EPC) via WLAN or 5GCN via WLAN, then the "phone-context" tel URI parameter is a domain name.

   a) if all characters of the SSID are allowed by domainlabel syntax definition of clause 3 of RFC 3966 [22], the domain name is constructed from the SSID, AP's MAC address, and the home network domain name by concatenating the SSID, AP's MAC address, and the string "i-wlan" as domain labels before the home network domain name; and

   b) otherwise, the domain name is constructed from AP's MAC address, and the home network domain name by concatenating AP's MAC address, and the string "i-wlan" as domain labels before the home network domain name.

NOTE: The AP's MAC address is provided in the BSSID information element.

EXAMPLE: If SSID = BU-Airport, AP's MAC = 00-0C-F1-12-60-28, and home network domain name is "home1.net", then the "phone-context" tel URI parameter is set to the string "bu-airport.000cf1126028.i-wlan.home1.net".

EXAMPLE: If SSID = <BU Airport>, AP's MAC = 00-0C-F1-12-60-28, and home network domain name is "home1.net", then the "phone-context" tel URI parameter is set to the string "000cf1126028.i-wlan.home1.net".

3) if the IP-CAN is xDSL, then the "phone-context" tel URI parameter is a domain name. It is constructed from the dsl-location (see subclause 7.2A.4) and the home network domain name by concatenating the dsl-location and the string "xdsl" as domain labels before the home network domain name;

4) if the IP-CAN is DOCSIS, then the "phone-context" tel URI parameter is based on data configured locally in the UE;

5) if the IP-CAN is EPS, then the "phone-context" tel URI parameter is a domain name. It is constructed from the MCC, the MNC and the home network domain name by concatenating the MCC, MNC, and the string "eps" as domain labels before the home network domain name;

6) if the IP-CAN is Ethernet, then the "phone-context" parameter is a domain name. It is constructed from the eth-location (see subclause 7.2A.4) and the home network domain name by concatenating the eth-location and the string "ethernet" as domain labels before the home network domain name;

7) if the IP-CAN is Fiber, then the "phone-context" parameter is a domain name. It is constructed from the fiber-location (see subclause 7.2A.4) and the home network domain name by concatenating the fiber-location and the string "fiber" as domain labels before the home network domain name;

8) if the IP-CAN is cdma2000®, then the "phone-context" parameter is a domain name. It is constructed from the subnet id and the home network domain name by concatenating the subnet id as the domain label before the home network domain name;

9) if the IP-CAN is DVB-RCS2, then the "phone-context" tel URI parameter is based on data configured locally in the UE;

10) if the IP-CAN is 5GS via 3GPP access and the serving network is a PLMN, then the "phone-context" tel URI parameter is a domain name. It is constructed from the MCC, the MNC and the home network domain name by concatenating the MCC, MNC, and the string "5gs" as domain labels before the home network domain name; and

11) if the IP-CAN is 5GS via 3GPP access and the serving network is an SNPN, then the "phone-context" tel URI parameter is a domain name. It is constructed from the MCC, the MNC, the NID and the home network domain name by concatenating the MCC, MNC, the SNPN Network Identifier (NID) (11 hexadecimal digits) as specified in 3GPP TS 23.003 [3] and the string "5gs-snpn" as domain labels before the home network domain name.

If the access network information is not available in the UE, then the "phone-context" tel URI parameter is set to the home network domain name preceded by the string "geo-local.".

In case the home domain is indicated in the "phone-context" tel URI parameter, the "phone-context" tel URI parameter is set to the home network domain name (as it is used to address the SIP REGISTER request, see subclause 5.1.1.1A or subclause 5.1.1.1B).

In case the "phone-context" tel URI parameter indicates a network other than the home network or the visited access network, the "phone-context" tel URI parameter is set according to RFC 3966 [22].

7.2A.11 Void

7.2A.11.1 Void

7.2A.11.2 Void

7.2A.11.3 Void

7.2A.12 CPC and OLI tel URI parameter definition

7.2A.12.1 Introduction

The use of the "cpc" and "oli" URI parameters for use in the P-Asserted-Identity in SIP requests is defined.

7.2A.12.2 Syntax

The Calling Party's Category and Originating Line Information are represented as URI parameters for the tel URI scheme and SIP URI representation of telephone numbers. The ABNF syntax is specified in table 7.2A.7 and extends the formal syntax for the tel URI as specified in RFC 3966 [22]:

Table 7.2A.7

```
par =/ cpc / oli
cpc = cpc-tag "=" cpc-value
oli = oli-tag "=" oli-value
cpc-tag = "cpc"
oli-tag = "oli"
cpc-value
= "ordinary" / "test" / "operator" /
"payphone" / "unknown" / "mobile-hplmn" / "mobile-vplmn" / "emergency" /
genvalue
oli-value = 2DIGIT
genvalue = 1*(alphanum / "-" / "." )
```

The Accept-Language header field shall be used to express the language of the operator.

The semantics of these Calling Party's Category values are described below:

ordinary: The caller has been identified, and has no special features.

test: This is a test call that has been originated as part of a maintenance procedure.

operator: The call was generated by an operator position.

payphone: The calling station is a payphone.

unknown: The CPC could not be ascertained.

mobile-hplmn: The call was generated by a mobile device in its home PLMN.

mobile-vplmn: The call was generated by a mobile device in a vistited PLMN.

emergency: The call is an emergency service call.

NOTE 1: The choice of CPC and OLI values and their use are up to the Service Provider. CPC and OLI values can be exchanged across networks if specified in a bilateral agreement between the service providers.

NOTE 2: Additional national/regional CPC values can exist.

The two digit OLI values are decimal codes assigned and administered by North American Numbering Plan Administration.

7.2A.12.3 Operation

The "cpc" and "oli" URI parameters may be supported by IM CN subsystem entities that provide the UA role and by IM CN subsystem entities that provide the proxy role.

The "cpc" and "oli" URI parameters shall not be populated at the originating UE.

In case the "cpc" URI parameter is not included, the call is treated as if the "cpc" URI parameter is set to "ordinary".

Unless otherwise specified in this document, "cpc" and "oli" URI parameters are only passed on by IM CN subsystem entities (subject to trust domain considerations as specified in subclause 4.4.12).

7.2A.13 "sos" SIP URI parameter

7.2A.13.1 Introduction

The "sos" SIP URI parameter is intended to:

- indicate to the S-CSCF that a REGISTER request that includes the "sos" SIP URI parameter is for emergency registration purposes;
- tell the S-CSCF to not apply barring of the public user identity being registered; and
- tell the S-CSCF to not apply initial filter criteria to requests destined for an emergency registered contact.

7.2A.13.2 Syntax

The syntax for the "sos" SIP URI parameter is specified in table 7.2A.8.

Table 7.2A.8: Syntax of sos SIP URI parameter

```
uri-parameter =/ sos-param
sos-param = "sos"
```

The BNF for uri-parameter is taken from RFC 3261 [26] and modified accordingly.

7.2A.13.3 Operation

When a UE includes the "sos" SIP URI parameter in the URI included in the Contact header field of REGISTER request, the REGISTER request is intended for emergency registration.

When a S-CSCF receives a REGISTER request for emergency registration that includes the "sos" SIP URI parameter, the S-CSCF is required to preserve the previously registered contact address. This differs to the registrar operation as defined in RFC 3261 [26] in that the rules for URI comparison for the Contact header field shall not apply and thus, if the URI in the Contact header field matches a previously received URI, then the old contact address shall not be overwritten.

7.2A.14 P-Associated-URI header field

Procedures of RFC 7315 [52] are modified to allow a SIP proxy to remove URIs from the P-Associated-URI header field.

7.2A.15 Void

7.2A.16 Void

7.2A.16.1 Void

7.2A.16.2 Void

7.2A.16.3 Void

7.2A.17 "premium-rate" tel URI parameter definition

7.2A.17.1 Introduction

The use of the "premium-rate" URI parameters for use in the Request-URI in SIP requests is defined.

7.2A.17.2 Syntax

The premium-rate category that a called number belongs to is represented as a URI parameter for the tel URI scheme and SIP URI representation of telephone numbers. The ABNF syntax is as specified in Table 7.2A.17 and extends the formal syntax for the tel URI as specified in RFC 3966 [22]:

Table 7.2A.17

```
par =/ premrate
premrate = premrate-tag "=" premrate-value
premrate-tag = "premium-rate"
premrate-value = "information" / "entertainment"
```

7.2A.17.3 Operation

The "premium-rate" URI parameter may be supported by IM CN subsystem entities that provide the AS role and by IM CN subsystem entities that provide the proxy role.

7.2A.17.4 IANA registration

NOTE: This subclause contains information to be provided to IANA for the registration of the tel-URI parameter "premium-rate".

This parameter needs to be defined in the sub-registry under the tel URI parameters.

Contact name, email address, and telephone number:

3GPP Specifications Manager:

3gppContact@etsi.org

+33 (0)492944200

Name of the parameter:

"premium-rate"

Whether the parameter only accepts a set of predefined values:

"Constrained"

Reference to the RFC or other permanent and readily available public specification defining the parameter and new values:

This parameter and its values are defined in 3GPP TS 24.229.

Description:

This tel URI parameter is used in networks supporting roaming and operator determined barring feature. The tel URI parameter provides a means to identify that a number in a tel URI belongs to a premium rate category in the roaming network. SIP servers in the home network use this information to apply the operator determined barring functionality. An overview of the 3GPP IM CN subsystem can be found in RFC 4083.

7.2A.18 Reason header field

7.2A.18.1 Introduction

The Reason header field is extended to include the additional protocol values.

7.2A.18.2 Syntax

The syntax of the Reason header field is described in RFC 3326 [34A].

Table 7.2A.18 describes 3GPP-specific extension to the Reason header field.

Table 7.2A.18: Syntax of extension to Reason header field

```
protocol          /= "EMM" / "ESM" / "S1AP-RNL" / "S1AP-TL" / "S1AP-NAS" / "S1AP-MISC" /
                     "S1AP-PROT" / "DIAMETER" / "IKEV2" / "RELEASE_CAUSE" / "FAILURE_CAUSE" /
                     "5GMM" / "5GSM" / "NGAP-RNL" / "NGAP-TL" / "NGAP-NAS" / "NGAP-MISC" /
                     "NGAP-PROT"
```

For all the above protocols, the protocol cause is included.

7.2A.18.3 IANA registration of EMM protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: EMM

Protocol cause: Cause value in decimal representation (Note)

Reference: 3GPP TS 24.301 [8J] subclause 9.9.3.9

NOTE: This protocol value can also be used to represent MM cause from 3GPP TS 24.008 [8].

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.18.4 IANA registration of ESM protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: ESM

Protocol cause: Cause value in decimal representation (Note)

Reference: 3GPP TS 24.301 [8J] subclause 9.9.4.4

NOTE: This protocol value can also be used to represent SM cause from 3GPP TS 24.008 [8].

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.18.5 IANA registration of S1AP radio network layer protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: S1AP-RNL

Protocol cause: Radio network layer cause value in decimal representation

Reference: 3GPP TS 36.413

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.18.6 IANA registration of S1AP transport layer protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: S1AP-TL

Protocol cause: Radio network layer cause value in decimal representation

Reference: 3GPP TS 36.413

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.18.7 IANA registration of S1AP non-access stratum protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: S1AP-NAS

Protocol cause: Non-access stratum cause value in decimal representation

Reference: 3GPP TS 36.413

7.2A.18.8 IANA registration of S1AP miscellaneous protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: S1AP-MISC

Protocol cause: Miscellaneous cause value in decimal representation

Reference: 3GPP TS 36.413

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.18.8A IANA registration of S1AP protocol protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: S1AP-PROT

Protocol cause: S1 Protocol cause value in decimal representation

Reference: 3GPP TS 36.413

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.18.9 IANA registration of DIAMETER protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: DIAMETER

Protocol cause: Cause for protocol failure of GTP-C supporting WLAN, as a representation in decimal digits of the received binary value.

Reference: 3GPP TS 29.274 subclause 8.103

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.18.10 IANA registration of IKEV2 protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: IKEV2

Protocol cause: Cause for protocol failure of IKEV2 supporting untrusted WLAN, as a representation in decimal digits of the received binary value.

Reference: 3GPP TS 29.274 subclause 8.103

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.18.10A IANA registration of 5GMM protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: 5GMM

Protocol cause: Cause value in decimal representation

Reference: 3GPP TS 24.501 [258] subclause 9.11.3.2

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.18.10B IANA registration of 5GSM protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: 5GSM

Protocol cause: Cause value in decimal representation

Reference: 3GPP TS 24.501 [258] subclause 9.11.4.2

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.18.10C IANA registration of NGAP radio network layer protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: NGAP-RNL

Protocol cause: Radio network layer cause value in decimal representation

Reference: 3GPP TS 38.413 [295]

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.18.10D IANA registration of NGAP transport layer protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: NGAP-TL

Protocol cause: Radio network layer cause value in decimal representation

Reference: 3GPP TS 38.413 [295]

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.18.10E IANA registration of NGAP non-access stratum protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: NGAP-NAS

Protocol cause: Non-access stratum cause value in decimal representation

Reference: 3GPP TS 38.413 [295]

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.18.10F IANA registration of NGAP miscellaneous protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: NGAP-MISC

Protocol cause: Miscellaneous cause value in decimal representation

Reference: 3GPP TS 38.413 [295]

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.18.10G IANA registration of NGAP protocol protocol value

The following entry is added to the Reason Protocols table within the Session Initiation Protocol (SIP) Parameters.

Protocol value: NGAP-PROT

Protocol cause: S1 Protocol cause value in decimal representation

Reference: 3GPP TS 38.413 [295]

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.18.11 IANA registration of RELEASE_CAUSE protocol value

7.2A.18.11.1 Introduction

This subclause defines an extension to the SIP Reason header field enabling the UE to define release cause events. In a network it is useful for the UE to specify a release cause when sending a BYE request or a CANCEL request. This release cause is for information purpose and can be useful for the remote UE to display to the user. For a network explicit release causes makes it possible to distinguish reasons for releasing a call. The network can then log error cases more accurate.

7.2A.18.11.2 IANA considerations

This document adds to the existing IANA registry for the SIP Reason header field the following protocol value and protocol cause:

Table 7.2A.18.11-1: Addition to the IANA Registry for the SIP Reason header field

Protocol value	Protocol cause	Reference
RELEASE_CAUSE	Cause value in decimal	3GPP TS 24.229

This document adds to the existing IANA registry for SIP Reason header Reason-text strings associated with their respective protocol type and Reason- param cause values:

Table 7.2A.18.11-2: Cause values and Reason-text strings for the RELEASE_CAUSE protocol value

Protocol value	Cause value	Reason-text
RELEASE_CAUSE	1	User ends call
RELEASE_CAUSE	2	RTP/RTCP time-out
RELEASE_CAUSE	3	Media bearer loss
RELEASE_CAUSE	4	SIP timeout - no ACK
RELEASE_CAUSE	5	SIP response time-out
RELEASE_CAUSE	6	Call-setup time-out
RELEASE_CAUSE	7	Redirection failure

7.2A.18.12 IANA registration of FAILURE_CAUSE protocol value

7.2A.18.12.1 Introduction

This subclause defines an extension to the SIP Reason header field to indroduce a new protocol enabling the IMS network entities to define failure cause events. This new indication is intended to be included in SIP error responses with the appropriate cause value and reason text to provide a complementatry indication on the original reason for which this error response has been sent.

7.2A.18.12.2 IANA considerations

This document adds to the existing IANA registry for the SIP Reason header field the following protocol value and protocol cause:

Table 7.2A.18.12-1: Addition to the IANA Registry for the SIP Reason header field

Protocol value	Protocol cause	Reference
FAILURE_CAUSE	Cause value in decimal	3GPP TS 24.229

This document adds to the existing IANA registry for SIP Reason header field the new "FAILURE_CAUSE" protocol parameter value associated with their respective protocol-cause values and reason-text strings:

Table 7.2A.18.12-2: Cause values and Reason-text strings for the FAILURE_CAUSE protocol value

Cause value	Reason-text
1	Media bearer or QoS lost
2	Release of signalling bearer
3	Indication of failed resources allocation

7.2A.19 Thig-path

7.2A.19.1 Introduction

The thig-path header field parameter is defined to enable the P-CSCF which is located in the visited network to subscribe to user's registration-state event package if topology hiding is done on the Path header field.

7.2A.19.2 Coding of the thig-path

The thig-path header field parameter is coded as a URI. The thig-path URI is a SIP URI of the visited network IBCF which applied topology hiding on the Path header field contained in the REGISTER request. The thig-path URI may be included as:

- a fcap-string-value within the "g.3gpp.thig-path" feature-capability indicator, as defined in subclause 7.9A.9 and RFC 6809 [190]; or
- as a value of the P-Asserted-Identity header field.

An example of a g.3gpp.thig-path feature-capability indicator containing thig-path URI is:

```
+g.3gpp.thig-path = "<sip:visit-abc@ibcf-vA1.visited-A.net:5070;lr>"
```

An example of a thig-path URI in a P-Asserted-Identity header field is:

```
P-Asserted-Identity: <sip:visit-abc@ibcf-vA1.visited-A.net:5070;lr>
```

7.2A.20 "verstat" tel URI parameter definition

7.2A.20.1 Introduction

This extension defines the "verstat" tel URI parameter used in the P-Asserted-Identity and the From header fields in a SIP request.

7.2A.20.2 Syntax

The status of the calling number verification performed by the home network is represented as a URI parameter for the tel URI scheme and SIP URI representation of telephone numbers. The ABNF syntax is as specified in Table 7.2A.20.2-1 and extends the formal syntax for the tel URI as specified in RFC 3966 [22]:

Table 7.2A.20.2-1

```
par =/ verstat
verstat = verstat-tag "=" verstat-value
verstat-tag = "verstat"
verstat-value = "TN-Validation-Passed" / "TN-Validation-Failed" / "No-TN-Validation" / other-value
other-value = token
```

7.2A.20.3 Operation

The "verstat" tel URI parameter may be supported by IM CN subsystem entities that provide the AS role and by IM CN subsystem entities that provide the proxy role.

The "verstat" tel URI parameter is inserted by an AS or a proxy in the IM CN subsystem to provide the UE with the calling identity number verification status in an initial INVITE request or when a standalone message is delivered.

Table 7.2A.20.3-1 shows the "verstat" parameter values that are currently defined:

Table 7.2A.20.3-1: Verstat values

Tel URI parameter value	Description
TN-Validation-Passed	**The number passed the validation.**
TN-Validation-Failed	**The number failed the validation.**
No-TN-Validation	**No number validation was performed.**

NOTE: There is no default value for the "verstat" parameter. If new values are defined, specifications need to describe the appropriate procedure if an endpoint receives a parameter value that it does not support.

7.2A.20.4 IANA registration

NOTE: This subclause contains information to be provided to IANA for the registration of the tel URI parameter "verstat".

This parameter needs to be defined in the sub-registry under the tel URI parameters.

Contact name, email address, and telephone number:

3GPP Specifications Manager

3gppContact@etsi.org

+33 (0)492944200

Name of the parameter

"verstat"

Whether the parameter only accepts a set of predefined values

Constrained

Reference to the RFC or other permanent and readily available public specification defining the parameter and new values

This parameter and its values are defined in 3GPP TS 24.229.

Description:

This tel URI parameter is used in networks supporting calling number verification, as described in RFC 8224. The tel URI parameter provides a means to identify that a number in a tel URI or a SIP URI with the user=phone parameter has been verified (verification passed or failed) or to identify that verification was not performed for the number. SIP user agents can use this information to apply functionality based on the verification status. An overview of the 3GPP IM CN subsystem can be found in 3GPP TS 23.228 and 3GPP TS 24.229.

7.2A.21 Extension to "isub-encoding" tel URI parameter

7.2A.21.1 Introduction

This extension defines a new value "user-specified" for the "isub-encoding" tel URI parameter.

7.2A.21.2 Syntax

The syntax for the "isub-encoding" tel URI parameter is defined in IETF RFC 4715 [259].

This specification reuses the "isub-encoding" tel URI parameter and defines the new value "user-specified" as listed in table 7.2A.21.2-1.

Table 7.2A.21.2-1: Syntax of extension of "isub-encoding" tel URI parameter

```
isub-encoding-value =/ "user-specified"
```

The semantics of this "isub-encoding" value are described below:

user-specified: Indication that the "isub" parameter value needs to be encoded using a user-specified encoding type.

7.2A.21.3 IANA registration of "user-specified" tel URI parameter value

7.2A.21.3.1 Introduction

This subclause defines an extension to the SIP "isub-encoding" tel URI parameter to introduce a new value "user-specified" enabling the IMS network entities to identify that the "isub" tel URI parameter has been encoded using a user specified format.

7.2A.21.3.2 IANA considerations

This document adds to the existing IANA registry for the SIP "isub-encoding" tel URI parameter the following value:

Table 7.2A.21.3.2-1: Addition to the IANA Registry for the "isub-encoding" SIP tel URI parameter

tel URI parameter	tel URI parameter value	Reference
isub-encoding	user-specified	3GPP TS 24.229

Contact:

3GPP Specifications Manager
3gppContact@etsi.org
+33 (0)492944200

7.2A.22 scscf-reselection parameter definition

7.2A.22.1 Introduction

The "scscf-reselection" parameter is a SIP URI parameter intended to:

- inform the S-CSCF it has been reselected due to failure of the previously assigned S-CSCF.

7.2A.22.2 Syntax

The syntax for the scscf-reselection parameter is specified in table 7.2A.22.2-1:

Table 7.2A.22.2-1: Syntax of scscf-reselection parameter

```
uri-parameter =/ scscf-reselection
scscf-reselection = "scscf-reselection"
```

The BNF for uri-parameter is taken from RFC 3261 [26] and extended accordingly.

7.2A.22.3 Operation

The "scscf-reselection" parameter is appended to the address of the S-CSCF by the I-CSCF, upon failed communication with the currently assigned S-CSCF. The S-CSCF receiving this parameter includes the S-CSCF reselection indicator set to "true" in the S-CSCF Registration procedure with the HSS, as described in 3GPP TS 29.562 [274], so the change of S-CSCF is accepted by the HSS.

7.3 Option-tags defined within the present document

There are no option-tags defined within the present document over and above those defined in the referenced IETF specifications.

7.4 Status-codes defined within the present document

There are no status-codes defined within the present document over and above those defined in the referenced IETF specifications.

7.5 Session description types defined within the present document

7.5.1 General

This subclause contains definitions for SDP parameters that are specific to SDP usage in the 3GPP IM CN Subsystem and therefore are not described in an RFC.

7.5.2 End-to-access-edge media plane security

7.5.2.1 General

The end-to-access-edge media security-indicator is used to indicate that a UE requests a P-CSCF to apply media plane security or to indicate that a P-CSCF has applied end-to-access-edge media security as defined in 3GPP TS 33.328 [19C].

7.5.2.2 Syntax

3GPP end-to-access-edge media security indicator is a value attribute which is encoded as a media-level SDP attribute with the ABNF syntax defined in table 7.5.1. ABNF is defined in RFC 2234 [20G].

Table 7.5.1: ABNF syntax of 3ge2ae attribute

```
3ge2ae-attribute = "a=3ge2ae:" indicator
indicator = "requested" / "applied" / token
```

"requested": the sender indicates its wish that end-to-access-edge media security is applied.

"applied": the sender indicates that it has applied end-to-access-edge media security.

This version of the specification only defines usage of the "requested" and "applied" attribute values. Other values shall be ignored.

The "3ge2ae" attribute is charset-independent.

7.5.2.3 IANA registration

NOTE: This subclause contains information to be provided to IANA for the registration of the end-to-access-edge security indicator SDP attribute.

Contact name, email address, and telephone number:

3GPP Specifications Manager

3gppContact@etsi.org

+33 (0)492944200

Attribute Name (as it will appear in SDP)

3ge2ae

Long-form Attribute Name in English:

3GPP_e2ae-security-indicator

Type of Attribute

Media level

Is Attribute Value subject to the Charset Attribute?

This Attribute is not dependent on charset.

Purpose of the attribute:

This attribute specifies the end-to-access-edge security-indicator as used for IMS media plane security

Appropriate Attribute Values for this Attribute:

The attribute is a value attribute. The values "requested" and "applied" are defined.

7.5.3 Optimal Media Routeing (OMR) attributes

7.5.3.1 General

The SDP attributes associated with OMR are used to identify and select alternative media plane paths for the purpose of bypassing unneeded media functions in the network, as described in 3GPP TS 29.079 [11D].

7.5.3.2 Semantics

The visited-realm attribute contains an IP realm identifier and transport address for a media function in the media plane that can potentially be used to bypass other allocated media functions.

The secondary-realm attribute contains an IP realm identifier and transport address for an alternate media function in the media plane that can potentially be used to bypass other allocated media functions.

The omr-s-cksum and omr-m-cksum attributes includes checksums for session level information and media level information to identify if the SDP was altered by intermediaries in such a way as to invalidate OMR information present in the SDP.

The "omr-codecs", "omr-m-att" and "omr-s-att" attributes contain codec-related SDP offer information encapsulated by a SIP-ALG in the signalling path that has modified codec related information.

The "omr-m-bw" and "omr-s-bw" attributes contain bandwidth-related SDP offer information encapsulated by a SIP-ALG in the signalling path that has modified codec related information.

Each group of zero or more versions of each of the "omr-codecs", "omr-m-att", "omr-s-att", "omr-m-bw" and "omr-s-bw" attributes for a media line with the same instance number is associated with the visited-realm instance for the modified media line and represents the version of the SDP information for the media line before modifications.

7.5.3.3 Syntax

The syntax specified in table 7.5.2 uses the augmented Backus-Naur Form as described in RFC 2234 [20G].

Table 7.5.2: Syntax OMR attributes

```
visited-realm      = "visited-realm" ":" instance-number SP
realm SP
nettype SP                                      ;from RFC 4566 [39]
addrtype SP                                     ;from RFC 4566 [39]
connection-address SP                           ;from RFC 4566 [39]
port                                            ;from RFC 4566 [39]
[SP rtcp-port [SP rtcp-address]]
*(SP extension-name SP extension-value)

secondary-realm    = "secondary-realm" ":" instance-number SP
realm SP
nettype SP                                      ;from RFC 4566 [39]
addrtype SP                                     ;from RFC 4566 [39]
connection-address SP                           ;from RFC 4566 [39]
port                                            ;from RFC 4566 [39]
[SP rtcp-port [SP rtcp-address]]
*(SP extension-name SP extension-value)

instance-number    = 1*DIGIT

realm              = non-ws-string           ;from RFC 4566 [39]

rtcp-port          = "rtcp-port" SP port

rtcp-address       = "rtcp-address" SP connection-address

extension-name     = token              ;shall be different to existing tokens "previsous-fmt",
                                        "rtcp-port" and "rtcp-address".
extension-value    = non-ws-string

omr-m-cksum      = "omr-m-cksum" ":" 1*HEXDIG

omr-s-cksum      = "omr-s-cksum" ":" 1*HEXDIG

omr-codecs         = "omr-codecs" ":" instance-number SP proto 1*(SP fmt) ;from RFC 4566 [39]

omr-m-att          = "omr-m-att" ":" instance-number SP attribute        ;from RFC 4566 [39]

omr-s-att          = "omr-s-att" ":" instance-number SP attribute        ;from RFC 4566 [39]

omr-m-bw           = "omr-m-bw" ":" instance-number SP bwtype ":" bandwidth ;from RFC 4566 [39]

omr-s-bw           = "omr-s-bw" ":" instance-number SP bwtype ":" bandwidth ;from RFC 4566 [39]
```

This grammar encodes the primary media level information about each visited-realm and secondary-realm instance: the sequence in which the realm was visited, the realm identity, its IP address and port:

<instance-number>: instance-number is a positive decimal integer which identifies the sequence in which this visited-realm was added during the forwarding of an SDP offer. If an IMS-ALG adds second-realm attribute(s), omr-codecs attribute(s), omr-m-att attribute(s), omr-s-att attribute(s), omr-m-bw attribute(s) and/or the omr-s-bw attribute(s) to an SDP offer it will assign the same instance number as assigned to the visited-realm attribute for the forwarded SDP offer. When used in the SDP answer, the instance-number, realm, nettype and addrtype uniquely identify the corresponding visited-realm or secondary-realm instance from the SDP offer.

<realm>: identifies a set of mutually reachable IP endpoints that share a common IP addressing scheme.

Effective application of OMR depends on the scope of each realm being determined by reachability and not by administrative domain. A public IPv4 or IPv6 address reachable from the open internet shall be associated with the special realm "IN". For application to OMR in IPv6 networks, a realm corresponds to an IPv6 autonomous system.

Entity operators must adhere to the following guidelines for creation of an OMR realm string to ensure the integrity of the visited-realm and secondary-realm instance information for their realm(s): 1) Realm strings must be globally unique. It is recommended that a realm string contain a hostname or domain name, following the recommendation in subclause 3.3 of RFC 7616 [286], 2) Realm strings should present a human-readable identifier that can be rendered to a user.

<nettype>, <addrtype> and <connection-address>: are taken from the connection-field (c= line) of RFC 4566 [39]. They describe the IP address associated with the visited-realm or secondary-realm instance, allowing for IPv4 addresses, IPv6 addresses and FQDNs. The connection-address can be either an IP address or an FQDN.

<port>: It is the port associated with the visited-realm or secondary-realm instance as taken from RFC 4566 [39]. Its meaning depends on the network being used for the connection-address, and on the transport protocol selected for the corresponding media line, e.g., UDP or TCP.

<rtcp-port> and <rtcp-address>: taken together are semantically equivalent to the rtcp attribute defined in RFC 3605 [37A]. They optionally encode the RTCP port and address information when the RTCP port number is not exactly one greater than the port for an RTP stream at the same address.

The previous-fmt-list may be supplied within the visited-realm if this attribute is included in an SDP offer and shall not be supplied if this attribute is included in an SDP answer.

The visited-realm and secondary-realm attributes can be extended via <extension-name> and <extension-value>. The grammar allows for new name/value pairs to be added at the end of the attribute.

<omr-m-cksum>: is a hex value calculated on the contents of the media level information per media line.

<omr-s-cksum>: is a hex value calculated on the contents of the session level information.

<omr-codecs> provides the transport format <proto> and list of media formats (e.g., payload type numbers) <fmt> supported by the visited-realm instance immediately preceding the instance identified by <instance-number>. Transport format <proto> and media format <fmt> are defined in RFC 4566 [39] for the SDP m-line.

<omr-m-att> provides a media level SDP attribute <attribute> supported by the visited-realm instance immediately preceding the instance identified by <instance-number>. Attribute <attribute> is defined in RFC 4566 [39] for the SDP a-line.

<omr-s-att> provides a session level SDP attribute <attribute> supported by the visited-realm instance immediately preceding the instance identified by <instance-number>. Attribute <attribute> is defined in RFC 4566 [39] for the SDP a-line.

<omr-m-bw> provides a media level SDP bandwidth described by <bwtype> and <bandwidth> supported by the visited-realm instance immediately preceding the instance identified by <instance-number>. <bwtype> and <bandwidth> are defined in RFC 4566 [39] for the SDP b-line.

<omr-s-bw> provides a session level SDP bandwidth described by <bwtype> and <bandwidth> supported by the visited-realm instance immediately preceding the instance identified by <instance-number>. <bwtype> and <bandwidth> are defined in RFC 4566 [39] for the SDP b-line.

The "visited-realm", "secondary-realm", "omr-m-cksum", "omr-s-cksum", "omr-codecs", "omr-m-att", "omr-s-att" "omr-m-bw" and "omr-s-bw" SDP attributes are media-level attributes.

7.5.3.4 IANA registration

7.5.3.4.1 visited-realm attribute

Contact Name:3GPP Specifications Manager, 3gppContact@etsi.org, +33 (0)492944200

Attribute Name: visited-realm

Long Form: visited-realm

Type of Attribute: media level

Charset Considerations: The attribute is not subject to the charset attribute.

Purpose: This attribute is used in networks employing OMR procedures allowing to bypass border gateways in configurations in which IP realms are re-entered when establishing an end-to-end multimedia session. This attribute is used to identify configurations in which IP realms are re-entered when establishing an end-to-end multimedia session, so that border gateways can be bypassed without compromising their role in securing access to the networks. The

attribute provides a means to identify connection information for visited IP realms to help select the most optimal available path.

Appropriate Values: See table 7.5.2.

7.5.3.4.2 secondary-realm attribute

Contact Name: 3GPP Specifications Manager, 3gppContact@etsi.org, +33 (0)492944200

Attribute Name: secondary-realm

Long Form: secondary-realm

Type of Attribute: media level

Charset Considerations: The attribute is not subject to the charset attribute.

Purpose: This attribute is used in networks employing OMR procedures allowing to bypass border gateways in configurations in which IP realms are re-entered when establishing an end-to-end multimedia session. This attribute is used to identify configurations in which secondary IP realms are available to establish an end-to-end multimedia session, so that border gateways can be bypassed without compromising their role in securing access to the networks. The attribute provides a means to identify connection information for secondary IP realms to help select the most optimal available path.

Appropriate Values: See table 7.5.2.

7.5.3.4.3 omr-s-cksum attribute

Contact Name: 3GPP Specifications Manager, 3gppContact@etsi.org, +33 (0)492944200

Attribute Name: omr-s-cksum

Long Form: omr-s-cksum

Type of Attribute: media level

Charset Considerations: The attribute is not subject to the charset attribute.

Purpose: This attribute is used in networks employing OMR procedures allowing to bypass border gateways in configurations in which IP realms are re-entered when establishing an end-to-end multimedia session. This attribute is used to provide a means to verify that session level SDP information has not been modified by intermediate SIP nodes not supporting the OMR procedures. The attribute provides a checksum calculated value against the session level information. Any OMR information associated with unexpectedly modified media information will be discarded.

Appropriate Values: See table 7.5.2.

7.5.3.4.4 omr-m-cksum attribute

Contact Name: 3GPP Specifications Manager, 3gppContact@etsi.org, +33 (0)492944200

Attribute Name: omr-m-cksum

Long Form: omr-m-cksum

Type of Attribute: media level

Charset Considerations: The attribute is not subject to the charset attribute.

Purpose: This attribute is used in networks employing OMR procedures allowing to bypass border gateways in configurations in which IP realms are re-entered when establishing an end-to-end multimedia session. This attribute is used to provide a means to verify that media level SDP information has not been modified by intermediate SIP nodes not supporting the OMR procedures. The attribute provides a checksum calculated value against the media level information associated with the media stream for which the checksum is provided. Any OMR information associated with unexpectedly modified media information will be discarded.

Appropriate Values: See table 7.5.2.

7.5.3.4.5 omr-codecs attribute

Contact Name: 3GPP Specifications Manager, 3gppContact@etsi.org, +33 (0)492944200

Attribute Name: omr-codecs

Long Form: omr-codecs

Type of Attribute: media level

Charset Considerations: The attribute is not subject to the charset attribute.

Purpose: This attribute is used in networks employing OMR procedures allowing to bypass border gateways in configurations in which IP realms are re-entered when establishing an end-to-end multimedia session. The attribute provides a means to encapsulate codec related SDP information transport format and list of media formats that are applicable if a particular border gateway is bypassed.

Appropriate Values: See table 7.5.2.

7.5.3.4.6 omr-m-att attribute

Contact Name: 3GPP Specifications Manager, 3gppContact@etsi.org, +33 (0)492944200

Attribute Name: omr-m-att

Long Form: omr-m-att

Type of Attribute: media level

Charset Considerations: The attribute is not subject to the charset attribute.

Purpose: This attribute is used in networks employing OMR procedures allowing to bypass border gateways in configurations in which IP realms are re-entered when establishing an end-to-end multimedia session. The attribute provides means to encapsulate a media-level SDP attribute that is applicable if a particular border gateway is bypassed.

Appropriate Values: See table 7.5.2.

7.5.3.4.7 omr-s-att attribute

Contact Name: 3GPP Specifications Manager, 3gppContact@etsi.org, +33 (0)492944200

Attribute Name: omr-s-att

Long Form: omr-s-att

Type of Attribute: media level

Charset Considerations: The attribute is not subject to the charset attribute.

Purpose: This attribute is used in networks employing OMR procedures allowing to bypass border gateways in configurations in which IP realms are re-entered when establishing an end-to-end multimedia session. The attribute provides means to encapsulate a session-level SDP attribute that is applicable if a particular border gateway is bypassed.

Appropriate Values: See table 7.5.2.

7.5.3.4.8 omr-m-bw attribute

Contact Name: 3GPP Specifications Manager, 3gppContact@etsi.org, +33 (0)492944200

Attribute Name: omr-m-bw

Long Form: omr-m-bw

Type of Attribute: media level

Charset Considerations: The attribute is not subject to the charset attribute.

Purpose: This attribute is used in networks employing OMR procedures allowing to bypass border gateways in configurations in which IP realms are re-entered when establishing an end-to-end multimedia session. The attribute provides means to encapsulate a media-level SDP bandwidth that is applicable if a particular border gateway is bypassed.

Appropriate Values: See table 7.5.2.

7.5.3.4.9 omr-s-bw attribute

Contact Name: 3GPP Specifications Manager, 3gppContact@etsi.org, +33 (0)492944200

Attribute Name: omr-s-bw

Long Form: omr-s-bw

Type of Attribute: media level

Charset Considerations: The attribute is not subject to the charset attribute.

Purpose: This attribute is used in networks employing OMR procedures allowing to bypass border gateways in configurations in which IP realms are re-entered when establishing an end-to-end multimedia session. The attribute provides means to encapsulate a session-level SDP bandwidth that is applicable if a particular border gateway is bypassed.

Appropriate Values: See table 7.5.2.

7.5.4 Media plane optimization for WebRTC

7.5.4.1 General

The SDP attributes associated with media plane optimization procedures for WebRTC are used to encapsulate an SDP offer or SDP answer received from a WIC, as described in 3GPP TS 23.228 [7], annex U.2.4.

7.5.4.2 Semantics

The "tra-m-line" and "tra-att" SDP attributes contain media-related SDP information which is applicable if optimized transparent media between WICs are selected. In the SDP offer, the attributes describe the offered transparent media which can be selected. In the SDP answer, the presence of the attributes indicates that the transparent media have been selected and the attributes which have been selected.

The "tra-SCTP-association" SDP attribute indicates for a media line that the related optimized transparent media are transported in the indicated SCTP association. The optimized transparent media related to several media lines can be transported in the same SCTP association.

The "tra-bw" SDP attribute contains bandwidth-related SDP information which is applicable if the optimized transparent media between WICs are selected. In the SDP offer, the attributes describe the bandwidths the offerer wants to receive for transparent media. In the SDP answer, the attributes describe the bandwidths the answerer wants to receive for transparent media.

The "tra-contact" SDP attribute in the SDP offer encapsulate address information which is compared with the address information in contact by the receiving eP-CSCF to detect whether intermediates that do not support switching to transparent media between WICs are in the path.

The "tra-media-line-number" SDP attribute provides the total number of media lines in the SDP, excluding any media lines with port zero, which is compared with the real number of media lines in the SDP, excluding any media lines with port zero, by the receiving eP-CSCF to detect whether intermediates have removed or dissabled media lines.

7.5.4.3 Syntax

The syntax specified in table 7.5.4.3-1 uses the augmented Backus-Naur Form as described in RFC 2234 [20G].

Table 7.5.4.3-1: Syntax of media plane optimization for WebRTC related SDP attributes

```
tra-contact            = "tra-contact" ":" nettype SP addrtype SP connection-address
                             ; from RFC 4566 [39]

tra-m-line             = "tra-m-line" ":" media SP port ["/" integer] proto 1*(SP fmt)
                              ; from RFC 4566 [39]

tra-att                = "tra-att" ":" attribute                ; from RFC 4566 [39]

tra-bw                 = "tra-bw" ":" bwtype ":" bandwidth      ; from RFC 4566 [39]

tra-SCTP-association   = "tra-SCTP-association" ":" SCTP-association-number

tra-media-line-number  = "media-line-number" ":" m-line-number

SCTP-association-number = integer

m-line-number           = integer
```

This grammar encodes the media level information received in an initial SDP offer from a WIC.

<tra-contact>: It is the contact used in the outgoing SDP offer which contains encapsulated media information. It contains nettype, addrtype and connection-address. Nettype, addrtype and connection-address are defined in RFC 4566 [39].

<tra-m-line>: provides the media <media>, port <port>, transport format <proto> and list of media formats (e.g., payload type numbers) <fmt> in the received SDP offer. Media <media>, port <port>, transport format <proto> and media format <fmt> are defined in RFC 4566 [39] for the SDP m-line.

<tra- att> provides an encapsulated SDP attribute <attribute> supported by the sender of the offer. Attribute <attribute> is defined in RFC 4566 [39] for the SDP a-line.

<tra- bw> provides an SDP bandwidth described by <bwtype> and <bandwidth> supported by the sender of the offer. <bwtype> and <bandwidth> are defined in RFC 4566 [39] for the SDP b-line.

<tra-SCTP-association> provides the number <SCTP-association-number> of an SCTP association a media line relates to. If optimized media are selected, the media related to a media line with an "a= tra-SCTP-association" SDP attribute will be transported in that SCTP association, possibly together with media relating to other media lines with a= tra-SCTP-association" SDP attributes with the same <SCTP-association-number>. For a WIC terminating call, the eP-CSCF receiving an offer from the core network containg m-lines with "a= tra-SCTP-association" SDP attributes with the same <SCTP-association-number> will construct a single m-line related to that SCTP association in the offer towards the served WIC.

<tra-media-line-number> provides the total number <m-line-number> of media lines in the SDP, excluding any media lines with port zero.

The "tra-contact", "tra- att", "tra-bw", SDP attributes are session and media-level attributes.

The "tra-m-line" and "tra-SCTPassociation" SDP attributes are media level attributes.

The "tra-media-line-number" SDP attribute is a session level attribute.

7.5.4.4 IANA registration

7.5.4.4.1 tra-contact

Contact Name: 3GPP Specifications Manager, 3gppContact@etsi.org, +33 (0)492944200

Attribute Name: tra-contact

Long Form: tra-contact

Type of Attribute: session and media level

Charset Considerations: The attribute is not subject to the charset attribute.

Purpose: This attribute is used in networks supporting WebRTC-IMS interworking. This attribute is used to encapsulate contact information received from gateways in the SDP offers and SDP answers when setting up a session that supports media plane optimization feature as specified in 3GPP TS 23.228 and 3GPP TS 24.371.

Appropriate Values: See table 7.5.4.3-1.

7.5.4.4.2 tra-m-line

Contact Name: 3GPP Specifications Manager, 3gppContact@etsi.org, +33 (0)492944200

Attribute Name: tra-m-line

Long Form: tra-m-line

Type of Attribute: media level

Charset Considerations: The attribute is not subject to the charset attribute.

Purpose: This attribute is used in networks supporting WebRTC-IMS interworking. This attribute is used to encapsulate an m-line received in an SDP offer or SDP answer into an attribute in an outgoing SDP offer or SDP answer when setting up a session that supports media plane optimization feature as specified in 3GPP TS 23.228 and 3GPP TS 24.371.

Appropriate Values: See table 7.5.4.3-1.

7.5.4.4.3 tra-att

Contact Name: 3GPP Specifications Manager, 3gppContact@etsi.org, +33 (0)492944200

Attribute Name: tra-att

Long Form: tra-att

Type of Attribute: session and media level

Charset Considerations: The attribute is not subject to the charset attribute.

Purpose: This attribute is used in networks supporting WebRTC-IMS interworking. This attribute is used to encapsulate an attribute received in an SDP offer or SDP answer into an attribute in an outgoing SDP offer or SDP answer when setting up a session that supports media plane optimization feature as specified in 3GPP TS 23.228 and 3GPP TS 24.371.

Appropriate Values: See table 7.5.4.3-1.

7.5.4.4.4 tra-bw

Contact Name: 3GPP Specifications Manager, 3gppContact@etsi.org, +33 (0)492944200

Attribute Name: tra-bw

Long Form: tra-bw

Type of Attribute: session and media level

Charset Considerations: The attribute is not subject to the charset attribute.

Purpose: This attribute is used in networks supporting WebRTC-IMS interworking. This attribute is used to encapsulate bandwidth information received in the SDP offers and SDP answers when setting up a session that supports media plane optimization feature as specified in 3GPP TS 23.228 and 3GPP TS 24.371.

Appropriate Values: See table 7.5.4.3-1.

7.5.4.4.5 tra-SCTP-association

Contact Name: 3GPP Specifications Manager, 3gppContact@etsi.org, +33 (0)492944200

Attribute Name: tra-SCTP-association

Long Form: tra-SCTP-association

Type of Attribute: media level

Charset Considerations: The attribute is not subject to the charset attribute.

Purpose: This attribute is used in networks supporting WebRTC-IMS interworking. This attribute is used to indicate that a media line relates to an SCTP association received in the SDP offers and SDP answers when setting up a session that supports media plane optimization feature as specified in 3GPP TS 23.228 and 3GPP TS 24.371.

Appropriate Values: See table 7.5.4.3-1.

7.5.4.4.6 tra- media-line-number

Contact Name: 3GPP Specifications Manager, 3gppContact@etsi.org, +33 (0)492944200

Attribute Name: tra-media-line-number

Long Form: tra-media-line-number

Type of Attribute: session level

Charset Considerations: The attribute is not subject to the charset attribute.

Purpose: This attribute is used in networks supporting WebRTC-IMS interworking. This attribute is used to encapsulate the total number of media lines in the SDP, excluding any media lines with port zero, to detect a removal or dissabling of media lines by intermediate nodes when setting up a session that supports media plane optimization feature as specified in 3GPP TS 23.228 and 3GPP TS 24.371.

Appropriate Values: See table 7.5.4.3-1.

7.5.5 Void

7.5.6 SDP content attribute values

7.5.6.1 General

As defined in RFC 4796 [206], the "a=content" attribute is a media level attribute in SDP.

7.5.6.2 SDP "a=content" attribute "g.3gpp.announcement-no-confirmation" value

7.5.6.2.1 General

The SDP "a=content" attribute "g.3gpp.announcement-no-confirmation" value is used only for informative purposes during an established session, to indicate an SDP media description is for the video announcement service and the operator wishes to play the announcement stream without user confirmation.

The AS may use this value, based on the operator policy, only for the video announcement which is important or needs to be seen by the user immediately.

The UE may accept to play the video announcement without user confirmation when received this value, based on UE's local policy (e.g., local configuration on the UE).

The use of the SDP "a=content" attribute "g.3gpp.announcement-no-confirmation" value shall conform to the procedure specified in 3GPP TS 24.628 [8ZF].

7.5.6.2.2 IANA registration for values of "a=content" attribute

IANA registration table: "content SDP Parameters" table of "Session Description Protocol (SDP) Parameters" registry.

IANA registry: A new value "g.3gpp.announcement-no-confirmation" for the SDP a=content media-level attribute defined in RFC 4796 [206].

Reference: 3GPP TS 24.229, http://www.3gpp.org/ftp/Specs/archive/24_series/24.229/.

7.6 3GPP IM CN subsystem XML body

7.6.1 General

This subclause contains the 3GPP IM CN Subsystem XML body in XML format. The 3GPP IM CN Subsystem XML shall be valid against the 3GPP IM CN Subsystem XML schema defined in table 7.6.1.

Any SIP User Agent or proxy may insert or remove the 3GPP IM CN subsystem XML body or parts of it, as required, in any SIP message. The 3GPP IM CN subsystem XML body shall not be forwarded outside a 3GPP network.

See subclause 7.6.4 and subclause 7.6.5 for the associated MIME type definition.

7.6.2 Document Type Definition

The XML Schema, is defined in table 7.6.1.

Table 7.6.1: IM CN subsystem XML body, XML Schema

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema" elementFormDefault="qualified"
           attributeFormDefault="unqualified" version="1">
   <xs:complexType name="tIMS3GPP">
      <xs:sequence>
         <xs:choice>
            <xs:element name="alternative-service" type="tAlternativeService"/>
            <xs:element name="service-info" type="xs:string"/>
         </xs:choice>
         <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
      </xs:sequence>
      <xs:attribute name="version" type="xs:decimal" use="required"/>
      <xs:anyAttribute/>
   </xs:complexType>
   <xs:complexType name="tAlternativeService">
      <xs:sequence>
         <xs:element ref="type"/>
         <xs:element name="reason" type="xs:string"/>
         <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
      </xs:sequence>
      <xs:anyAttribute/>
   </xs:complexType>

   <!-- root element -->
   <xs:element name="ims-3gpp" type="tIMS3GPP"/>

   <xs:element name="type" type="xs:string"/>

   <!-- action element for //ims-3gpp//alternative-service -->
   <xs:element name="action" type="xs:string"/>

</xs:schema>
```

7.6.3 XML Schema description

This subclause describes the elements of the IM CN subsystem XML Schema as defined in table 7.6.1.

<ims-3gpp>: The <ims-3gpp> element is the root element of the IM CN subsystem XML body. It is always present. XML instance documents of future versions of the XML Schema in table 7.6.1 is valid against the XML Schema in table 7.6.1 in this document. XML instance documents of the XML Schema in table 7.6.1 in the present document have a version attribute value, part of the <ims-3gpp> element, that is equal to the value of the XML Schema version described in the present document.

<service-info>: the transparent element received from the HSS for a particular trigger point are placed within this optional element.

<alternative-service>: in the present document, the alternative service is used as a response for an attempt to establish an emergency session within the IM CN subsystem or as a response to initiate S-CSCF restoration procedures. The element describes an alternative service where the call should success. The alternative service is described by the type of service information. A possible reason cause why an alternative service is suggested may be included.

In the present document, the <alternative-service> element contains a <type> element, a <reason> element, and an optional <action> element.

The <type> element indicates the type of alternative service. The <type> element contains only the values specified in table 7.6.2 in the present document.

Table 7.6.2: ABNF syntax of values of the <type> element

```
emergency-value = %x65.6D.65.72.67.65.6E.63.79 ; "emergency"
restoration-value = %x72.65.73.74.6F.72.61.74.69.6F.6E ; "restoration"
```

The <action> element contains only the values specified in table 7.6.3 in the present document.

Table 7.6.3: ABNF syntax of values of the <action> element

```
emergency-registration-value = %x65.6D.65.72.67.65.6E.63.79.2D.72.65.67.69.73.74.72.61.74.69.6F.6E ;
    "emergency-registration"
initial-registration-value = %x69.6E.69.74.69.61.6C.2D.72.65.67.69.73.74.72.61.74.69.6F.6E ;
    "initial-registration"
anonymous-emergencycall-value =
    %x61.6E.6F.6E.79.6D.6F.75.73.2D.65.6D.65.72.67.65.6E.63.79.63.61.6C.6C ; "anonymous-
    emergencycall"
```

The <reason> element contains an explanatory text with the reason why the session setup has been redirected. A UE may use this information to give an indication to the user.

If included in the IM CN subsystem XML body:

1. the <type> element with the value "emergency" is included as the first child element of the <alternative-service> element;

2. the <type> element with the value "restoration" is included as one of the following:

a) the first child element of the <alternative-service> element; or

b) the third or later child element of the <alternative-service> element;

3. the <action> element with the value "emergency-registration" is includes as the third child element of the <alternative-service> element;

4. the <action> element with value "initial-registration" is included as the third or later child element of the <alternative-service> element; and

5. the <action> element with value "anonymous-emergencycall" is included as the third or later child element of the <alternative-service> element.

NOTE: When included, the <action> and the second occurence of the <type> elements are validated by the <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs="unbounded"/> particle of their parent elements.

7.6.4 MIME type definition

7.6.4.1 Introduction

This subclause defines the MIME type for "application/3gpp-ims+xml". A 3GPP IM CN subsystem XML Document can be identified with this media type.

7.6.4.2 Syntax

The following optional parameters are defined:

- "charset": the parameter has identical semantics to the charset parameter of the "application/xml" media type as specified in RFC 3023 [132].
- "sv" or "schemaversion": the syntax for the "sv" or "schemaversion" parameter is specified in table 7.6.4:

Table 7.6.4: Syntax of the "sv" or "schemaversion" parameter

```
m-parameter    =/ ("sv" / "schemaversion") EQUAL LDQUOT [ sv-value-list ] RDQUOT
sv-value-list  =  sv-value-range *( "," sv-value )
sv-value-range =  sv-value [ "-" sv-value ]
sv-value       =  number / token
number         =  1*DIGIT [ "." 1*DIGIT ]
```

The BNF for m-parameter is taken from RFC 3261 [26] and modified accordingly.

7.6.4.3 Operation

The encoding considerations for "application/3gpp-ims+xml" are identical to those of "application/xml" as described in RFC 3023 [132].

The "sv" or "schemaversion" parameter's value is used to indicate:

- the versions of the 3GPP IM CN Subsystem XML schema that can be used to validate the 3GPP IM CN subsystem XML body (if the MIME type and parameter are present in the Content-Type header field); or
- the accepted versions of the 3GPP IM CN Subsystem XML body (if the MIME type and parameter are present in the Accept header field).

If the "sv" and "schemaversion" parameter are absent, it shall be assumed that version 1 of the XML Schema for the IM CN subsystem XML body is supported.

7.6.5 IANA Registration

NOTE: RFC 4288 [161], subclause 9, states the process that applies in case of changes to the registry of media types. Any future changes to the format or to subclause 7.6.5 would invoke this procedure.

MIME media type name:

application

MIME subtype name:

3gpp-ims+xml

Required parameters:

None

Optional parameters:

"charset" the parameter has identical semantics to the charset parameter of the "application/xml" media type as specified in RFC 3023 [132].

"sv" or "schemaversion" the parameter's value is used to indicate:

- the versions of the 3GPP IP Multimedia (IM) Core Network (CN) subsystem XML schema that can be used to validate the 3GPP IM CN subsystem XML body (if the MIME type and parameter are present in the Content-Type header field); or
- the accepted versions of the 3GPP IM CN Subsystem XML body (if the MIME type and parameter are present in the Accept header field).

If the "sv" and "schemaversion" parameter are absent, it shall be assumed that version 1 of the XML Schema for the IM CN subsystem XML body is supported.

Encoding considerations:

Same as encoding considerations of application/xml as specified in RFC 3023 [132]

Security considerations:

Same as general security considerations for application/xml as specified in subclause 10 of RFC 3023 [132].

In addition, this content type provides a format for exchanging information in SIP, so the security considerations from RFC 3261 [26] apply.

Interoperability considerations:

Same as Interoperability considerations as specified in subclause 3.1 of RFC 3023 [132].

If both "sv" and "schemaversion" are specified, then the value of "schemaversion" is ignored

Published specification:

3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP), stage 3", as published in subclause 7.6.5, version 8.9.0.

Available via <http://www.3gpp.org/specs/numbering.htm>.

Applications which use this media:

Applications that use the 3GPP IM CN Subsystem as defined by 3GPP.

Intended usage:

COMMON

Additional information:

1. Magic number(s): none
2. File extension(s): none
3. Macintosh file type code: none
4. Object Identifiers: none

7.7 SIP timers

The timers T1, T2, T4 A, B, C, D, E, F, G, H and I (defined in RFC 3261 [26]), timers L and M (defined in RFC 6026 [163]), and timer N (defined in RFC 6665 [28]) need modification in some cases to accommodate the delays introduced by the air interface processing and transmission delays. Table 7.7.1 shows recommended values for IM CN subsystem.

Table 7.7.1 lists in the first column, titled "SIP Timer" the timer names as defined in RFC 3261 [26] and RFC 6026 [163].

The second column, titled "value to be applied between IM CN subsystem elements" lists the values recommended for network elements e.g. P-CSCF, S-CSCF, MGCF, when communicating with each other i.e. when no air interface leg is included. These values are identical to those recommended by RFC 3261 [26], RFC 6026 [163], and RFC 6665 [28].

The third column, titled "value to be applied at the UE" lists the values recommended for the UE, when in normal operation the UE generates requests or responses containing a P-Access-Network-Info header field which included a value of "3GPP-GERAN","3GPP-UTRAN-FDD", "3GPP-UTRAN-TDD", "3GPP-E-UTRAN-FDD", "3GPP-E-UTRAN-TDD", "3GPP-E-UTRAN-ProSe-UNR", "3GPP-NR-FDD", "3GPP-NR-TDD", "3GPP-NR-U-FDD", "3GPP-NR-U-TDD", "3GPP-NR-SAT", "3GPP-NR-ProSe-L2UNR", "3GPP-NR-ProSe-L3UNR", "3GPP-NR-REDCAP", "3GPP2-1X", "3GPP2-1X-HRPD", "3GPP2-UMB", "IEEE-802.11", "IEEE-802.11a", "IEEE-802.11b", "IEEE-802.11g", "IEEE-802.11n", "IEEE-802.11ac", or "DVB-RCS2". These are modified when compared to RFC 3261 [26] and RFC 6026 [163] to accommodate the air interface delays. In all other cases, the UE should use the values specified in RFC 3261 [26] or RFC 6026 [163] as indicated in the second column of table 7.7.1.

The fourth column, titled "value to be applied at the P-CSCF toward a UE" lists the values recommended for the P-CSCF when an air interface leg is traversed, and which are used on all SIP transactions on a specific security association where the security association was established using a REGISTER request containing a P-Access-Network-Info header field provided by the UE which included a value of "3GPP-GERAN","3GPP-UTRAN-FDD", "3GPP-UTRAN-TDD", "3GPP-E-UTRAN-FDD", "3GPP-E-UTRAN-TDD", "3GPP-E-UTRAN-ProSe-UNR", "3GPP-NR-FDD", "3GPP-NR-TDD", "3GPP-NR-U-FDD", "3GPP-NR-U-TDD", "3GPP-NR-SAT", "3GPP-NR-ProSe-L2UNR", "3GPP-NR-ProSe-L3UNR", "3GPP-NR-REDCAP", "3GPP2-1X", "3GPP2-1X-HRPD", "3GPP2-UMB", "IEEE-802.11", "IEEE-802.11a", "IEEE-802.11b", "IEEE-802.11g", "IEEE-802.11n", "IEEE-802.11ac", or "DVB-RCS2". These are modified when compared to RFC 3261 [26] and RFC 6026 [163]. In all other cases, the P-CSCF should use the values specified in RFC 3261 [26] and RFC 6026 [163] as indicated in the second column of table 7.7.1.

The final column reflects the timer meaning as defined in RFC 3261 [26], RFC 6026 [163] or RFC 6665 [28].

Table 7.7.1: SIP timers

SIP Timer	Value to be applied between IM CN subsystem elements	Value to be applied at the UE	Value to be applied at the P-CSCF toward a UE	Meaning
T1	500ms default (see NOTE)	2s default	2s default	RTT estimate
T2	4s (see NOTE)	16s	16s	The maximum retransmit interval for non-INVITE requests and INVITE responses
T4	5s (see NOTE)	17s	17s	Maximum duration a message will remain in the network
Timer A	initially T1	initially T1	initially T1	INVITE request retransmit interval, for UDP only
Timer B	64*T1	64*T1	64*T1	INVITE transaction timeout timer
Timer C	> 3min	> 3 min	> 3 min	proxy INVITE transaction timeout
Timer D	> 32s for UDP	>128s	>128s	Wait time for response retransmits
	0s for TCP/SCTP	0s for TCP/SCTP	0s for TCP/SCTP	
Timer E	initially T1	initially T1	initially T1	non-INVITE request retransmit interval, UDP only
Timer F	64*T1	64*T1	64*T1	non-INVITE transaction timeout timer
Timer G	initially T1	initially T1	initially T1	INVITE response retransmit interval
Timer H	64*T1	64*T1	64*T1	Wait time for ACK receipt.
Timer I	T4 for UDP	T4 for UDP	T4 for UDP	Wait time for ACK retransmits
	0s for TCP/SCTP	0s for TCP/SCTP	0s for TCP/SCTP	
Timer J	64*T1 for UDP	64*T1 for UDP	64*T1 for UDP	Wait time for non-INVITE request retransmits
	0s for TCP/SCTP	0s for TCP/SCTP	0s for TCP/SCTP	
Timer K	T4 for UDP	T4 for UDP	T4 for UDP	Wait time for response retransmits
	0s for TCP/SCTP	0s for TCP/SCTP	0s for TCP/SCTP	
Timer L	64*T1	64*T1	64*T1	Wait time for accepted INVITE request retransmits
Timer M	64*T1	64*T1	64*T1	Wait time for retransmission of 2xx to INVITE or additional 2xx from other branches of a forked INVITE
Timer N	64*T1	64*T1	64*T1	Wait time for receipt of a NOTIFY request upon sending SUBSCRIBE
NOTE: As a network option, SIP T1 Timer's value can be extended, along with the necessary modifications of T2 and T4 Timers' values, to take into account the specificities of the supported services when the MRFC and the controlling AS are under the control of the same operator and the controlling AS knows, based on local configuration, that the MRFC implements a longer value of SIP T1 Timer.				

7.8 IM CN subsystem timers

Table 7.8.1 shows recommended values for timers specific to the IM CN subsystem.

Table 7.8.1: IM CN subsystem

Timer	Value to be applied at the UE	Value to be applied at the P-CSCF	Value to be applied at the S-CSCF	Meaning
reg-await-auth	not applicable	not applicable	4 minutes	The timer is used by the S-CSCF during the authentication procedure of the UE for registration. For detailed usage of the timer see subclause 5.4.1.2. The authentication procedure may take in the worst case as long as 2 times Timer F. The IM CN subsystem value for Timer F is 128 seconds.
request-await-auth	not applicable	not applicable	4 minutes	The timer is used by the S-CSCF during the authentication procedure of the UE for requests different than REGISTER. For detailed usage of the timer see subclause 5.4.3.6.1. The authentication procedure may take in the worst case as long as 2 times Timer F. The IM CN subsystem value for Timer F is 128 seconds.
emerg-reg	Configurable value between 8 seconds and 20 seconds	not applicable	not applicable	The timer is used by the UE to supervise the time from deciding that an emergency service is to be established via the IM CN subsystem until completion of the emergency registration procedure, including any required IP-CAN procedures. For detailed usage of the timer see subclause 5.1.6.1.
emerg-request	Configurable value between 5 seconds and 15 seconds	not applicable	not applicable	The timer is used by the UE during initial request for emergency service. For detailed usage of the timer see subclause 5.1.6.8.1.
NoVoPS-dereg	Configurable value between 0 seconds and 65535 seconds	not applicable	not applicable	The timer is used by the UE when the UE receives a VoPS not supported indication from the lower layers and indicates the time the UE needs to wait before the UE deregisters from IMS if the UE is configured with a policy to deregister, see subclause B.3.1.0a, L.3.1.0a, U.3.1.0a and W.3.1.0a
emerg-non3gpp	Configurable value between 5 seconds and 20 seconds	not applicable	not applicable	The timer is used by the UE to supervise the time for searching usable 3GPP access to setup an emergency call before attempting the emergency call via non-3GPP access. For detailed usage of the timer see subclauses R.2.2.6.1 and W.2.2.6.1.

NOTE: The UE and the P-CSCF use the value of the reg-await-auth timer to set the SIP level lifetime of the temporary set of security associations.

7.9 Media feature tags defined within the current document

7.9.1 General

This subclause describes the media feature tag definitions that are applicable for the 3GPP IM CN subsystem.

7.9.2 Definition of media feature tag g.3gpp.icsi-ref

Media feature-tag name: g.3gpp.icsi-ref.

ASN.1 Identifier: 1.3.6.1.8.2.4

Summary of the media feature indicated by this tag: Each value of the Service Reference media feature-tag indicates the software applications supported by the agent. The values for this tag equal the IMS communication Service Identifier (ICSI) values supported by the agent.

The Service Reference media feature tag is defined to fulfil the requirements for forking to an appropriate UE when multiple UEs are registered and dispatch to an appropriate application within the UE based upon the IMS communication Service Identifier (ICSI) values as stated in 3GPP TS 23.228 [7].

Multiple tag-values can be included in the Service Reference media feature-tag.

Values appropriate for use with this feature-tag: Token with an equality relationship.

The feature-tag is intended primarily for use in the following applications, protocols, services, or negotiation mechanisms:

> This feature-tag is most useful in a communications application, for describing the capabilities of a device, such as a phone or PDA.

Examples of typical use: Routeing an IMS Communication Session to a device that supports a particular software application or understands a particular service.

Related standards or documents:

> 3GPP TS 24.229: "IP Multimedia Call Control Protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP), stage 3"

Security Considerations: Security considerations for this media feature-tag are discussed in subclause 11.1 of RFC 3840 [62].

7.9.3 Definition of media feature tag g.3gpp.iari-ref

Media feature-tag name: g.3gpp.iari-ref.

ASN.1 Identifier: 1.3.6.1.8.2.5

Summary of the media feature indicated by this tag: Each value of the Application Reference media feature-tag indicates the software applications supported by the agent. The values for this tag equal IMS Application Reference Identifier (IARI) values supported by the agent

The Application Reference media feature tag is defined to fulfil the requirements for forking to an appropriate UE when multiple UEs are registered and dispatch to an appropriate application within the UE based upon and IMS Application Reference Identifier (IARI) values as stated in 3GPP TS 23.228 [7].

Multiple tag-values can be included in the Application Reference media feature-tag.

Values appropriate for use with this feature-tag: Token with an equality relationship.

The feature-tag is intended primarily for use in the following applications, protocols, services, or negotiation mechanisms:

This feature-tag is most useful in a communications application, for describing the capabilities of a device, such as a phone or PDA.

Examples of typical use: Routeing an IMS Application Session to a device that supports a particular software application or understands a particular application.

Related standards or documents:

3GPP TS 24.229: "IP Multimedia Call Control Protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP), stage 3"

Security Considerations: Security considerations for this media feature-tag are discussed in subclause 11.1 of RFC 3840 [62].

7.9.4 Void

7.9.5 Void

7.9.6 Void

7.9.7 Definition of media feature tag g.3gpp.registration-token

Media feature tag name: g.3gpp.registration-token

ASN.1 Identifier: 1.3.6.1.8.2.27

Summary of the media feature indicated by this media feature tag:

This media feature tag, when included in a third party SIP REGISTER request, indicates the support of using a token to identify the registration used for the request. The mediafeature tag is assigned a value that can be used by the receiving AS to later identify the used registration for initial requests from an originating user or dialog forming responses from a terminating user.

Media feature tag specification reference: 3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

Values appropriate for use with this media feature tag:

String with an equality relationship.

Table 7.9.7-1: ABNF syntax of values of the g.3gpp.registration-token media feature tag

```
g-3gpp-registration-token = "<"qdtext">"
```

The media feature tag is intended primarily for use in the following applications, protocols, services, or negotiation mechanisms: This media feature tag is used to indicate support of using a token to identify the registration used for the current request or response among the set of registrations for the registered URI. As the token is unique per URI, different URIs for different users can have the same value of the token.

Examples of typical use: The S-CSCF includes this media feature tag in a third-party REGISTER request to indicate support of this feature. The value is a unique value identifying this registration among the set of registrations for the registered URI. The S-CSCF includes a token with identical value in subsequent initial requests and responses. An AS supporting this feature can use the value of the token to identify the used registration.

Security Considerations: Security considerations for this media feature-tag are discussed in subclause 11.1 of RFC 3840 [62].

7.9.8 Definition of media feature tag g.3gpp.ps-data-off

Media feature tag name: g.3gpp.ps-data-off.

ASN.1 Identifier: 1.3.6.1.8.2.35

Summary of the feature indicated by this media feature tage: This media feature tag when included in a Contact header field in a REGISTER request indicates the status of the 3GPP PS data off for the registration time.

This media feature tag, when included in the Contact header field in a REGISTER request indicates that the UE supports the 3GPP PS data off. The g.3gpp.ps-data-off media feature tag can take a value that indicates whether the 3GPP PS data off has been activated or deactivated by the user at the UE.

Media feature tag specification reference: 3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

Values appropriate for use with this media feature tag:

String with an equality relationship

Table 7.9.8-1: ABNF syntax of values of the g.3gpp.ps-data-off media feature tag

```
g-3gpp-ps-data-off = "active" / "inactive" / token
```

Examples of typical use: Indicating support and activation status of the 3GPP PS data off function to IMS network entities.

Security Considerations: Security considerations for this media feature tag are discussed in clause 9 of RFC 6809 [190].

7.9.9 Definition of media feature tag g.3gpp.rlos

Media feature-tag name: g.3gpp.rlos

ASN.1 Identifier: 1.3.6.1.8.2.x

Summary of the media feature indicated by this tag: This feature-tag when used in a SIP REGISTER request indicates that the function sending the SIP message supports restricted local operator service.

Values appropriate for use with this feature-tag: Boolean

The feature-tag is intended primarily for use in the following applications, protocols, services, or negotiation mechanisms: This feature-tag is most useful in a communications application, for describing the capabilities of a device, such as a phone or PDA.

Examples of typical use: Indicating that a mobile phone supports the restricted local operator service

Related standards or documents: 3GPP TS 24.229: " IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3"

Security Considerations: Security considerations for this media feature-tag are discussed in subclause 12.1 of IETF RFC 3840 [53].

7.9A Feature-capability indicators defined within the current document

7.9A.1 General

This subclause describes the feature-capability indicators definitions, according to RFC 6809 [190], that are applicable for the 3GPP IM CN subsystem.

7.9A.2 Definition of feature-capability indicator g.3gpp.icsi-ref

Feature-capability indicator name: g.3gpp.icsi-ref.

Summary of the feature indicated by this feature-capability indicator: Each value of the Service Reference feature-capability indicator indicates the software applications supported by the entity. The values for this feature-capability indicator equal the IMS communication Service Identifier (ICSI) values supported by the entity.

Multiple feature-capability indicator values can be included in the Service Reference feature-capability indicators.

When included in the Feature-Caps header field, according to RFC 6809 [190], the value of this feature-capability indicator contains the IMS communication service identifier (ICSI) of the IMS communication service supported for use:

- in the standalone transaction (if included in a request for a standalone transaction or a response associated with it); or
- in the dialog (if included in an initial request for dialog or a response associated with it);

by the entity which included the Feature-Caps header field.

Feature-capability indicator specification reference: 3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

Values appropriate for use with this feature-capability indicator:

When used in a Feature-Caps header field, the g.3gpp-icsi-ref feature-capability indicator is encoded using the feature-cap header field rules specified in clause 6.3 of RFC 6809 [190], where the feature-capability indicator value is an instance of fcap-value-list, listing one or more token values, as specified in RFC 6809 [190].

Examples of typical use: Indicating support of IMS Communication Services to other network entities.

Security Considerations: Security considerations for this feature-capability indicator are discussed in clause 9 of RFC 6809 [190].

7.9A.3 Definition of feature-capability indicators g.3gpp.trf

Feature-capability indicator name: g.3gpp.trf

Summary of the feature indicated by this feature-capability indicator:

This feature-capability indicator, when included in a Feature-Caps header field as specified in RFC 6809 [190] in a SIP INVITE request, indicates that in a roaming scenario, the visited network supports a transit and roaming functionality in order to allow loopback of session requests to the visited network from the home network. When used, it may indicate the URI of the transit and roaming functionality.

Feature-capability indicator specification reference: 3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

Values appropriate for use with this feature-capability indicator:

None or string with an equality relationship. When used in a Feature-Caps header field, the value is string and follows the syntax as described in table 7.9A.1 for g-3gpp-trf. The value of g-3gpp-trf parameter is an instance of fcap-string-value of Feature-Caps header field specified in RFC 6809 [190].

Table 7.9A.1: ABNF syntax of values of the g.3gpp.trf feature-capability indicator

```
g-3gpp-trf = "<" SIP-URI ">"
```

The feature-capability indicator is intended primarily for use in the following applications, protocols, services, or negotiation mechanisms: This feature-capability indicator is used to indicate visited network support of the roaming architecture for voice over IMS with local breakout and to transport the TRF address.

Examples of typical use: A visited network indicating the presence and support of a TRF in a visited network to the home network.

Security Considerations: Security considerations for this feature-capability indicator are discussed in clause 9 of RFC 6809 [190].

7.9A.4 Definition of feature-capability indicator g.3gpp.loopback

Feature-capability indicator name: g.3gpp.loopback

Summary of the feature indicated by this feature-capability indicator:

This feature-capability indicator, when included in a Feature-Caps header field as specified in RFC 6809 [190] in a SIP INVITE request, indicates the support of the roaming architecture for voice over IMS with local breakout.

Feature-capability indicator specification reference: 3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

Values appropriate for use with this feature-capability indicator:

None or a string with an equality relationship.

When used in a Feature-Caps header field, the value is a string identifying the home network and follows the syntax as described in table 7.9A.4-1 for g-3gpp-loopback. The value of g-3gpp-loopback parameter is an instance of fcap-string-value of Feature-Caps header field specified in RFC 6809 [190].

Table 7.9A.4-1: ABNF syntax of values of the g-3gpp-loopback feature-capability indicator

```
g-3gpp-loopback = "<" 1*(qdtext / quoted-pair) ">"
```

The feature-capability indicator is intended primarily for use in the following applications, protocols, services, or negotiation mechanisms: This feature-capability indicator is used to indicate support of the roaming architecture for voice over IMS with local breakout and that the INVITE request is a loopback request.

Examples of typical use: The home network indicating when a loopback INVITE request is sent to a visited network.

Security Considerations: Security considerations for this feature-capability indicator are discussed in clause 9 of RFC 6809 [190].

7.9A.5 Definition of feature-capability indicator g.3gpp.home-visited

Feature-capability indicator name: g.3gpp.home-visited

Summary of the feature indicated by this feature-capability indicator:

This feature-capability indicator, when included in a Feature-Caps header field as specified in RFC 6809 [190] in a SIP INVITE request, indicates that the home network supports loopback to the identified visited network for this session. The loopback is expected to be applied at some subsequent entity to the insertion point. The feature-capability indicator carries a parameter value which indicates the visited network.

Feature-capability indicator specification reference: 3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

Values appropriate for use with this feature-capability indicator:

String with an equality relationship. When used in a Feature-Caps header field, the value follows the syntax as described in table 7.9A.2 for g-3gpp-home-visited. The value of g-3gpp-home-visited parameter is an instance of fcap-string-value of Feature-Caps header field specified in RFC 6809 [190].

Table 7.9A.2: ABNF syntax of values of the g.3gpp.home-visited feature-capability indicator

```
g-3gpp-home-visited = "<" 1*(qdtext / quoted-pair) ">"
```

The value follows that used in the P-Visited-Network-ID header field.

The feature-capability indicator is intended primarily for use in the following applications, protocols, services, or negotiation mechanisms: This feature-capability indicator is used to indicate the home network supports loopback to the identified visited network for this session. The loopback is expected to be applied at some subsequent entity to the insertion point. The feature-capability indicator carries a parameter which indicates the visited network.

Examples of typical use: A home network indicating the home network supports loopback to the identified visited network for this session.

Security Considerations: Security considerations for this feature-capability indicator are discussed in clause 9 of RFC 6809 [190].

7.9A.6 Definition of feature-capability indicator g.3gpp.mrb

Feature-capability indicator name: g.3gpp.mrb

Summary of the feature indicated by this feature-capability indicator:

This feature-capability indicator when included in a Feature-Caps header field as specified in RFC 6809 [190] in a SIP INVITE request indicates that in a roaming scenario, the visited network supports media resource broker functionality for the allocation of multimedia resources in the visited network. When used, it indicates the URI of the visited network MRB.

Feature-capability indicator specification reference:

3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

Values appropriate for use with this feature-capability indicator:

String with an equality relationship. When used in a Feature-Caps header field, the value is string and follows the syntax as described in table 7.9A.3 for g-3gpp-mrb. The value of g-3gpp-mrb parameter is an instance of fcap-string-value of Feature-Caps header field specified in RFC 6809 [190].

Table 7.9A.3: ABNF syntax of values of the g.3gpp.mrb feature-capability indicator

```
g-3gpp-mrb = "<" SIP-URI ">"
```

The feature-capability indicator is intended primarily for use in the following applications, protocols, services, or negotiation mechanisms: This feature-capability indicator is used to indicate the URI of the media resource broker.

Examples of typical use: Indicating the URI of the visited network MRB to the home network.

Security Considerations: Security considerations for this feature-capability indicator are discussed in clause 9 of draft-ietf-sipcore-proxy-feature [190].

7.9A.7 Void

7.9A.8 Definition of feature-capability indicator g.3gpp.registration-token

Feature-capability indicator name: g.3gpp. registration-token

Summary of the feature indicated by this feature-capability indicator:

This feature-capability indicator, when included in a Feature-Caps header field as specified in RFC 6809 [190], indicates the support of using a token to identify the registration used for the request.

This feature-capability indicator can be included in an originating initial request for a dialog or a request for a standalone transaction to identify which registration was used for this request by setting the indicator to the same value as in the g.3gpp.registration-token media feature tag in the Contact header field of the REGISTER request.

This feature-capability indicator can be included in 1xx or 2xx response to a terminating initial request for a dialog or a 2xx response to a request for a standalone transaction to identify which registration was used for the response by setting the indicator to the same value as in the g.3gpp.registration-token media feature tag in the Contact header field of the REGISTER request.

Feature-capability indicator specification reference: 3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

Values appropriate for use with this feature-capability indicator:

String with an equality relationship.

When used in a Feature-Caps header field, the value is a string identifying the used registration and follows the syntax as described in table 7.9A.8-1 for g-3gpp-registration-token. The value of g-3gpp-registration-token parameter is an instance of fcap-string-value of Feature-Caps header field specified in RFC 6809 [190].

Table 7.9A.8-1: ABNF syntax of values of the g.3gpp.registration-token feature-capability indicator

```
g-3gpp-registration-token = "<"qdtext">"
```

The feature-capability indicator is intended primarily for use in the following applications, protocols, services, or negotiation mechanisms: This feature-capability indicator is used to indicate support of using a token to identify the registration used for the current request or response.

Examples of typical use: The S-CSCF includes a media feature tag in a third-party REGISTER request to indicate support of this feature. The value is a unique value identifying this registration among the set of registrations for the registered URI. The S-CSCF includes this token with an identical value as in the previous REGISTER request in subsequent initial requests and responses to indicate its continuous support. An AS supporting this feature can use the value of the token to identify the used registration.

Security Considerations: Security considerations for this feature-capability indicator are discussed in clause 9 of RFC 6809 [190].

7.9A.9 Definition of feature-capability indicator g.3gpp.thig-path

Feature-capability indicator name: g.3gpp.thig-path

Summary of the feature indicated by this feature-capability indicator:

This feature-capability indicator when included in a Feature-Caps header field as specified in RFC 6809 [190] in a 200 (OK) response to the REGISTER request indicates that in a roaming scenario, the visited network IBCF supports topology hiding of a Path header field.

Feature-capability indicator specification reference:

3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

Values appropriate for use with this feature-capability indicator:

String with an equality relationship. When used in a Feature-Caps header field, the value is string and follows the syntax as described in table 7.9A.y for g-3gpp-thig-path. The value of g-3gpp-thig-path parameter is an instance of fcap-string-value of Feature-Caps header field specified in RFC 6809 [190].

Table 7.9A.9-1: ABNF syntax of values of the g.3gpp.thig-path feature-capability indicator

```
g-3gpp-thig-path = "<" SIP-URI ">"
```

The feature-capability indicator is intended primarily for use in the following applications, protocols, services, or negotiation mechanisms: This feature-capability indicator is used to indicate that the visited network IBCF supports topology hiding of a Path header field and to pass to the P-CSCF a SIP URI of the visited network IBCF which applied topology hiding on the Path header field.

Examples of typical use: The visited network IBCF includes the g.3gpp.thig-path feature-capability indicator in a 200 (OK) response to the REGISTER request to pass to the P-CSCF a SIP URI of the visited network IBCF which applied topology hiding on the Path header field contained in the REGISTER request.

Security Considerations: Security considerations for this feature-capability indicator are discussed in clause 9 of RFC 6809 [190].

7.9A.10 Definition of feature-capability indicator g.3gpp.priority-share

Feature-capability indicator name: g.3gpp.priority-share.

Summary of the feature indicated by this feature-capability indicator: When included in a Feature-Caps header field in SIP requests or SIP responses the sender indicates that priority sharing is supported.

Feature-capability indicator specification reference: 3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

Values appropriate for use with this feature-capability indicator:

When used in a Feature-Caps header field, the g.3gpp.priority-share feature-capability indicator is encoded using the feature-cap header field rules specified in subclause 6.3 of RFC 6809 [190], where the feature-capability indicator value is an instance of fcap-value-list, listing one or more token values, as specified in RFC 6809 [190].

Examples of typical use: Indicating support of priority sharing to other network entities.

Security Considerations: Security considerations for this feature-capability indicator are discussed in clause 9 of RFC 6809 [190].

7.9A.11 Definition of feature-capability indicator g.3gpp.verstat

Feature-capability indicator name: g.3gpp.verstat

Summary of the feature indicated by this feature-capability indicator:

This feature-capability indicator, when included in a Feature-Caps header field as specified in RFC 6809 in a 200 (OK) response to a REGISTER request, indicates that the home network supports calling party number verification, as described in RFC 8224.

Feature-capability indicator specification reference:

3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

Values appropriate for use with this feature-capability indicator: Not applicable.

The feature-capability indicator is intended primarily for use in the following applications, protocols, services, or negotiation mechanisms: This feature-capability indicator is used to indicate the support of calling party number verification functionality.

Examples of typical use: Indicating the support of calling number verification in the home network.

Security Considerations: Security considerations for this feature-capability indicator are discussed in clause 9 of RFC 6809.

7.9A.12 Definition of feature-capability indicator g.3gpp.anbr

Feature-capability indicator name: g.3gpp.anbr

Summary of the feature indicated by this feature-capability indicator:

This feature-capability indicator, when included in a Feature-Caps header field as specified in RFC 6809 in a 200 (OK) response to a REGISTER request, indicates that the network supports ANBR as specified in 3GPP TS 26.114 [9B].

Feature-capability indicator specification reference:

3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

Values appropriate for use with this feature-capability indicator: Not applicable.

The feature-capability indicator is intended primarily for use in the following applications, protocols, services, or negotiation mechanisms: This feature-capability indicator is used to indicate the support of ANBR.

Examples of typical use: Indicating the support of ANBR.

Security Considerations: Security considerations for this feature-capability indicator are discussed in clause 9 of RFC 6809 [190].

7.9A.13 Definition of feature-capability indicator g.3gpp.in-call-access-update

Feature-capability indicator name: g.3gpp.in-call-access-update

Summary of the feature indicated by this feature-capability indicator:

This feature-capability indicator, when included in a Feature-Caps header field as specified in RFC 6809 [190] in a SIP INVITE request or a response to a SIP INVITE, indicates that the entity supports in-call access update procedure specified in 3GPP TS 24.229. The value of this feature capability indicator is a SIP URI to where the entity can be reached.

Feature-capability indicator specification reference: 3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

Values appropriate for use with this feature-capability indicator:

String with an equality relationship. When used in a Feature-Caps header field, the value follows the syntax as described in table 7.9A.13-1 for g-3gpp-in-call-access-update. The value of the g-3gpp-in-call-access-update parameter is an instance of fcap-string-value of Feature-Caps header field specified in RFC 6809.

Table 7.9A.13-1: ABNF syntax of values of the g.3gpp.in-call-access-update feature-capability indicator

```
g-3gpp-in-call-access-update = "<" SIP-URI ">"
```

The feature-capability indicator is intended primarily for use in the following applications, protocols, services, or negotiation mechanisms: This feature-capability indicator is used to indicate that an entity reached by the URI in the value supports mid call updates of e.g. location. The mid-call update is performed by a downstream entity when needed.

Examples of typical use: A network entity indicating support for mid-call updates. A downstream network entity performs the update.

Security Considerations: Security considerations for this feature-capability indicator are discussed in clause 9 of RFC 6809.

7.10 Reg-event package extensions defined within the current document

7.10.1 General

This subclause describes the reg-event package extensions that are applicable for the IM CN subsystem.

7.10.2 Reg-Event package extension to transport wildcarded public user identities

7.10.2.1 Structure and data semantics

This subclause defines an extension to the event registration package (RFC 3680 [43]) to transport policy to transport wildcarded public user identities that are encoded using regular expression.

In order to include a wildcared public user identity in the event registration package, the notifier shall

1. if the registration set of the identity whose registration status is notified contains a wildcarded public user identity, add a <wildcardedIdentity> sub-element defined in subclause 7.10.2.2 of this document to the <registration> element of the wildcarded identiy;

2. for the <registration> element containing a <wildcardedIdentity> sub-element:

   a) set the aor attribute to any public user identity that is represented by the wildcarded identity; and

   b) set the <wildcardedIdentity> sub-element inside of the <registration> element to the wildcarded identity as received via the Cx interface.

NOTE: The public user identity that is put into the aor attribute does not have any extra privileges over any other public user identity that is represented by a wildcarded public user identity.

7.10.2.2 XML Schema

Table 7.10.1 in this subclause defines the XML Schema describing the extension to transport wildcarded public user identities which can be included in the reg event package sent from the S-CSCF in NOTIFY requests.

Table 7.10.1: Wildcarded Identity, XML Schema

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema
  targetNamespace="urn:3gpp:ns:extRegExp:1.0"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  elementFormDefault="qualified" attributeFormDefault="unqualified">
    <xs:element name="wildcardedIdentity" type="xs:string"/>
</xs:schema>
```

NOTE: Multiple wildcarded elements can be included in one registration element.

7.10.3 Reg-event package extension for policy transport

7.10.3.1 Scope

This subclause describes coding which extends the registration event package defined in RFC 3680 [43] to transport policy associated with a public user identity.

7.10.3.2 Structure and data semantics

The policy associated with a public user identity shall be encoded as follows:

1. add an <actions> element defined in the RFC 4745 [182] in the <registration> element of the public user identity in the registration information;

NOTE: The <actions> element is validated by the <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"/> particle of the <registration> elements.

2. if the policy to the usage of the communication resource priority (see RFC 4412 [116]) is associated with the public user identity, then for each allowed usage:

   a. include <rph> child element in the <actions> child element of the <registration> element;

   b. set the 'ns' attribute of the <rph> child element of the <actions> child element of the <registration> element to the allowed resource priority namespace as specified in RFC 4412 [116] and as registered in IANA; and

   c. set the 'val' attribute of the <rph> child element of the <actions> child element of the <registration> element to the allowed resource priority value within the allowed resource priority namespace;

3. if the policy to act as privileged sender (the P-CSCF passes identities for all calls) is associated with the public user identity, then include a <privSender> child element in the <actions> child element of the <registration> element;

4. if the policy for special treatment of the P-Private-Network-Indication header field (the P-CSCF allows the UE to make private calls) is associated with the public user identity, then include a <pni> child element in the <actions> child element of the <registration> element, and shall:

   a. if a P-Private-Network-Indication header field shall be forwarded, if received from the attached equipment, set the "insert" attribute of the <pni> element to a "fwd" value;

   b. if a P-Private-Network-Indication header field shall be inserted in all requests received from the attached equipment, insert an "insert" attribute of the <pni> element to a "ins" value; and

   c. if the value of the "insert" attribute is "ins", insert a "domain" attribute with the value of the URI to be set in the P-Private-Network-Indication header field; and

5. if the policy to act as privileged sender for the calls with the P-Private-Network-Indication header field (the P-CSCF allows the UE to make private calls, and the P-CSCF passes identities only for private calls) is associated with the public user identity, then include a <privSenderPNI> child element in the <actions> child element of the <registration> element.

NOTE: If only the <privSender> child element is sent and no <privSenderPNI> child element is sent, then the <privSender> child element applies to both public network traffic and private network traffic (i.e. that with special treatment of the P-Private-Network-Indication header field).

7.10.3.3 XML Schema

Table 7.10.2 in this subclause defines the XML Schema describing the individual policies which can be delivered to the P-CSCF or UE using the reg event package extension for policy transport.

Table 7.10.2: Reg event package extension for policy transport, XML Schema

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema
  targetNamespace="urn:3gpp:ns:extRegInfo:1.0"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  elementFormDefault="qualified" attributeFormDefault="unqualified">
  <xs:element name="rph">
    <xs:complexType>
      <xs:attribute name="ns" type="xs:string"/>
      <xs:attribute name="val" type="xs:string"/>
    </xs:complexType>
  </xs:element>
  <xs:element name="privSender">
    <xs:complexType/>
  </xs:element>
  <xs:element name="pni">
    <xs:complexType>
      <xs:attribute name="insert">
        <xs:simpleType>
          <xs:restriction base="xs:string">
            <xs:enumeration value="fwd"/>
            <xs:enumeration value="ins"/>
          </xs:restriction>
        </xs:simpleType>
      </xs:attribute>
      <xs:attribute name="domain" type="xs:anyURI"/>
    </xs:complexType>
  </xs:element>
  <xs:element name="privSenderPNI">
    <xs:complexType/>
  </xs:element>
</xs:schema>
```

7.11 URNs defined within the present document

7.11.1 Country specific emergency service URN

7.11.1.1 Introduction

The country specific emergency service URN is intended to uniquely identify a type of emergency service for which an emergency service URN, i.e. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69], registered by IANA is not available.

The country specific emergency service URN is intended to be used only when an emergency service URN for a given type of emergency service, with approximately the same caller expectation in terms of services rendered, is not registered by IANA.

The country specific emergency service URN is intended to be used only inside the country where the national regulatory authority defines emergency services only by national numbers. The country specific emergency service URN is not intended to be used by the UE except when indicated by the network.

7.11.1.2 Syntax

The country specific emergency service URN is a service URN with:

1) the top-level service type of "sos" as specified in RFC 5031 [69];

2) the first sub-service of "country-specific";

3) the second sub-service indicating the country where the type of emergency service is deployed. The format of second sub-service is an ISO 3166-1 alpha-2 code as specified in ISO 3166-1 [207]; and

4) the third sub-service uniquely identifying the type of emergency service in the country where the type of emergency service is deployed. The third sub-service is defined by the national regulation of the country where the type of emergency service is deployed. The set of allowable characters for the third sub-service is the same

as that for domain names (see RFC 5031 [69]) and the number of characters for the third sub-service shall be less than 64.

EXAMPLE: urn:service:sos.country-specific.xy.567 can identify a type of emergency service identified by an emergency number 567 in a country identified by "xy" ISO 3166-1 alpha-2 code as specified in ISO 3166-1 [207].

7.11.1.3 Operation

Unless explicitly prohibited, wherever an emergency service URN i.e. a service URN with the top-level service type of "sos" as specified in RFC 5031 [69] can be used, the country specific emergency service URN can also be used.

7.11.1.4 Void

7.11.2 ICSI value for RLOS

7.11.2.1 Introduction

This subclause describes the IMS communications service identifier definitions that is applicable for the usage of restricted local operator service (RLOS).

NOTE: The template has been created using the headers of the table in http://www.3gpp.org/specifications-groups/34-uniform-resource-name-urn-list

7.11.2.2 URN

urn:urn-7:3gpp-service.ims.icsi.rlos

7.11.2.3 Description

This URN indicates that the device has the capabilities to support the restricted local operator service (RLOS).

7.11.2.4 Reference

3GPP TS 24 229: " IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3"

7.11.2.5 Contact

Name: <MCC name>

Email: <MCC email address>

7.11.2.6 Registration of subtype

Yes

7.11.2.7 Remarks

None

7.12 Info package definitions and associated MIME type definitions

7.12.1 DTMF info package and session-info MIME type

7.12.1.1 DTMF info package

7.12.1.1.1 General

This subclause contains the information required for the IANA registration of an info package.

The digit message body and associated MIME type can also be used as defined in other specifications in a legacy manner as defined by RFC 6086 [25].

7.12.1.1.2 Overall description

DTMF tones are normally sent when a user presses a button on the terminal. Each tone, identified by a unique frequency, represents a number (0-9) or a special character. The DTMF info package is used to transport that value. In-order delivery of the DTMF digits is not controlled by the DTMF info package, this would be controlled either by queuing in the sender or by separate interaction with the human user; such mechanisms are out of scope of this registration.

The DTMF info package can be used to transport a single DTMF tone, or a series of tones. If a series of tones is transported in a single SIP INFO request, it is not possible to indicate the duration between each tone in the series.

The DTMF info package is not defined for a specific application. Any application, where sending of DTMF tones using the SIP INFO method is required, can used the DTMF info package.

7.12.1.1.3 Applicability

The info package mechanism for transporting DTMF tones has been chosen because it allows SIP entities that do not have access to the user plane (where DTMF tones can also be transported) to send and receive tones. The mechanism also allows the tones to be sent inside an existing dialog, using the same signalling path as other SIP messages within the dialog, rather than having to establish a separate dialog (DTMF tones can also be transported using subscription event packages).

7.12.1.1.4 Info package name

The name of the DTMF info package is: infoDtmf

7.12.1.1.5 Info package parameters

No parameters are defined for the DTMF info package.

7.12.1.1.6 SIP option tags

No SIP option tags are defined for the DTMF info package.

7.12.1.1.7 INFO message body parts

The DTMF digits are carried in the Overlap digit message body, defined in subclause 7.12.1.2.

The MIME type value for the message body is "application/session-info", defined in subclause 7.12.1.2.

The Content Disposition value for the message body, when associated with the DTMF info package, is "info-package" (see RFC 6086 [25]).

7.12.1.1.8 Info package usage restrictions

No usage restrictions are defined for the DTMF info package.

If SIP entities support multiple mechanisms for sending DTMF tones they need to ensure, using negotiation mechanisms, that each entity is aware of which mechanism is used.

7.12.1.1.9 Rate of INFO requests

No maximum rate or minimum rate is defined for sending INFO requests associated with the DTMF info package.

When DTMF tones are triggered by user interaction, the DTMF tones are normally generated when the user pushes a button. Specific applications can decide upon which rate DTMF tones are generated. However, the DTMF info package does not provide a feedback mechanism to indicate to the sender that the rate of DTMF tones is too slow or fast.

7.12.1.1.10 Info package security considerations

No additional security mechanism is defined for the DTMF info package.

The security of the DTMF info package is based on the generic security mechanism provided for the underlying SIP signalling.

The mechanism should not be used for transferring any data that requires a greater level of security than the underlying SIP signalling.

7.12.1.2 Overlap digit message body

7.12.1.2.1 Scope

This section defines a message body that shall be used for sending additional digits, which have not previously been sent, in SIP INFO messages ("legacy" mode of usage of the INFO method as defined in RFC 6086 [25]) when the in-dialog method is used for overlap dialling.

The same message body can also be used for transporting Dual Tone Multi Frequency (DTMF) tones using SIP INFO requests, using the DTMF info package (see RFC 6086 [25]) defined in subclause 7.12.1.1.

The support of this message body is a network option.

7.12.1.2.2 MIME type

The message body defined in the present annex is registered at IANA as "application/session-info" MIME type.

If the message body is embedded in SIP INFO messages, the Content-Type header shall be set to "application/session-info" and the Content-Disposition header shall be set to "signal" with the handling parameter set to "optional".

7.12.1.2.3 ABNF

session-info = SubsequentDigit

SubsequentDigit = "SubsequentDigit" HCOLON phonedigits

phonedigits = 1*(HEXDIG / "*" / "#")

HEXDIG = DIGIT / "A" / "B" / "C" / "D" / "E" / "F"

7.12.1.2.4 IANA registration template

Within the present subclause, information required for an IANA registration at http://www.iana.org/cgi-bin/mediatypes.pl is provided.

1. Media Type Name

Application

2. Subtype name

"session-info" (Standards Tree)

3. Required parameters

none

4. Optional parameters

none

5. Encoding considerations

binary

6. Security considerations

Modifications of the MIME body by a man-in-the-middle can have severe consequences:

The overlap digits that can be transported with this MIME body influence the routeing of the SIP session that is being setup.

Dual Tone Multi Frequency (DTMF) tones that can also be transported with this MIME body will be interpreted by the application of the end points of the communication for various purposes.

However, this MIME body is used only attached to SIP INFO messages, and modifications of other parts of the SIP signalling will lead to comparable consequences. Protection of the SIP signalling will also protect the present MIME body.

The information transported in this MIME media type does not include active or executable content.

Mechanisms for privacy and intregrity protection of protocol parameters exist. Those mechanisms as well as authentication and further security mechanisms are described in 3GPP TS 24.229.

7. Interoperability considerations

none

8. Published specification

3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP), stage 3".

9. Applications which use this media type

This MIME type is used as a message body within SIP INFO messages.

It is either used for sending additional digits of the callee´s number, which have not previously been sent, in SIP INFO messages ("legacy" mode of usage of the INFO method as defined in IETF RFC 6086) when the in-dialog method is used for overlap dialling.

The same message body can also be used for transporting Dual Tone Multi Frequency (DTMF) tones using SIP INFO requests, using the DTMF info package (see RFC 6086) defined in subclause 7.12.1.1.

10. Additional information

Magic number(s): none

File extension(s): none

Macintosh File Type Code(s): none

Object Identifier(s) or OID(s): none

11. Intended usage

Limited Use

12. Other Information/General Comment

none.

7.12.1.3 Implementation details and examples

Examples of the DTMF info package usage can be found in the following specification:

- 3GPP TS 24.182 [8Q]: "Customized Alerting Tones; Protocol specification".

7.12.2 g.3gpp.current-location-discovery info package and request-for-current-location body

7.12.2.1 g.3gpp.current-location-discovery info package

7.12.2.1.1 General

This subclause contains information required for registration of the g.3gpp.current-location-discovery info package with IANA.

7.12.2.1.2 Overall description

Location of a UA participating in an INVITE-initiated dialog can change during duration of the INVITE-initiated dialog.

The g.3gpp.current-location-discovery info package enables a UA participanting in an INVITE-initiated dialog to indicate a request for location information to the other UA participating in the INVITE-initiated dialog.

7.12.2.1.3 Applicability

A number of solutions for the transportation of the pieces of information identified in the overall description were identified and considered:

1) Use of session related methods for transporting event and state information, e.g. re-INVITE request, UPDATE request.
2) Use of OPTIONS request.
3) Use of SIP MESSAGE method.
4) Use of media plane mechanisms.
5) Use of subscription to the presence event package as described in RFC 3856 [74].
6) Use of SIP INFO method as decribed in RFC 6086 [25], by defining a new info package.

Furthermore, each of the solutions was evaluated.

Use of session related methods was discounted since purpose of the INVITE request and the UPDATE request was to modify the dialog, or the parameters of the session or both and neither the dialog nor the parameters of the session needed to be modified.

Use of the OPTIONS request was discounted since purpose of the OPTIONS request was to query UAS for UAS' capabilites rather than requesting an information available at the UAS.

Use of the MESSAGE request was discounted since the use of the INFO method enabled negotiation of supported event packages in the INVITE transaction while the use of the MESSAGE method did not.

Use of the media plane mechanisms was discounted since the amount of information transferred between the UAs was limited and set up of media stream generated generate extra messages.

Use of the presence event package was discounted since the dialog reuse technique was deprecated accoding to RFC 6665 [28]. Thus, SUBSCRIBE request for the presence event package needed to be sent using a dialog other than any dialog established as result of a INVITE request. However, in some situation - e.g. an emergency session initiated by a UA without a prior registration, there was no way how to ensure delivery of a new initial request for a dialog to the UA. The remote target indicated in Contact header field of:

- the INVITE request; or
- the received response to the INVITE request;

sent by the UA was not necessarily globally routable (e.g. when the UA was behind NAT or when the UA was behind a SIP proxy with a firewall), and the route set indicated in the Record-Route header fields of:

- the INVITE request; or
- the received response to the INVITE request;

sent by the UA might be dedicated to the messages of dialogs established as result of the INVITE request.

Based on the above analyses, the SIP INFO method was chosen.

7.12.2.1.4 Info package name

Info package name is: g.3gpp.current-location-discovery

7.12.2.1.5 Info package parameters

No info package parameters are defined for the g.3gpp.current-location-discovery info package.

7.12.2.1.6 SIP option tags

No SIP option tags are defined for the g.3gpp.current-location-discovery info package.

7.12.2.1.7 INFO message body parts

The MIME type of the body is application/vnd.3gpp.current-location-discovery+xml. The application/vnd.3gpp.current-location-discovery+xml MIME type is defined in 3GPP TS 24.229.

When associated with the g.3gpp.current-location-discovery info package, the Content-Disposition value of the body is "info-package".

7.12.2.1.8 Info package usage restrictions

No usage restrictions are defined for the g.3gpp.current-location-discovery info package.

7.12.2.1.9 Rate of INFO requests

No maximum rate or minimum rate is defined for sending INFO requests associated with the g.3gpp.current-location-discovery info package.

7.12.2.1.10 Info package security considerations

The security of the g.3gpp.current-location-discovery info package is based on the generic security mechanism provided for the underlaying SIP signalling.

As the location information is a sensitive information, unless the location information is requested from a UA who initiated an emergency session, the UA requested to provide the location information needs to authorize the request with the user at the UA.

7.12.2.2 Request-for-current-location body

7.12.2.2.1 General

The request-for-current-location body is of "application/vnd.3gpp.current-location-discovery+xml" MIME type.

The request-for-current-location body is an XML document compliant to the XML schema defined in subclause 7.12.2.2.2, compliant to the additional syntax rules in subclause 7.12.2.2.3, with semantic defined in subclause 7.12.2.2.4.

7.12.2.2.2 XML schema

The XML Schema, is defined in table 7.12.2.2.2.1.

Table 7.12.2.2.2.1: XML schema of application/vnd.3gpp.current-location-discovery+xml MIME type

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema"
    elementFormDefault="qualified"
    attributeFormDefault="unqualified">

  <xs:element name="requestForLocationInformation"
    type="requestForLocationInformationType"/>

  <xs:complexType name="requestForLocationInformationType">
    <xs:sequence>
      <xs:choice>
        <xs:element name="oneShot" type="anyExtType"/>
        <xs:element name="anyExt" type="anyExtType"/>
        <xs:any namespace="##other" processContents="lax"/>
      </xs:choice>
      <xs:element name="anyExt" type="anyExtType" minOccurs="0"/>
      <xs:any namespace="##other" processContents="lax" minOccurs="0"
        maxOccurs="unbounded"/>
    </xs:sequence>
    <xs:anyAttribute namespace="##any" processContents="lax"/>
  </xs:complexType>

  <xs:complexType name="anyExtType">
    <xs:sequence>
      <xs:any namespace="##any" processContents="lax" minOccurs="0"
        maxOccurs="unbounded"/>
    </xs:sequence>
    <xs:anyAttribute namespace="##any" processContents="lax"/>
  </xs:complexType>

</xs:schema>
```

7.12.2.2.3 Additional syntax rules

The <requestForLocationInformation> element is the root element.

The <requestForLocationInformation> root element contains:

1) one of the following elements:
   a) the <oneShot> element;
   b) the <anyExt> element;and
   c) an element from another namespace for the purposes of extensibility;
2) zero or one <anyExt> element;and
3) zero or more elements from other namespaces for the purposes of extensibility; and
4) zero or more attributes from any namespaces for the purpose of extensibility.

The <oneShot> element contains:

1) zero or more elements from any namespaces for the purposes of extensibility; and

2) zero or more attributes from any namespaces for the purpose of extensibility.

The <anyExt> element contains:

1) zero or more elements from any namespaces for the purposes of extensibility; and

2) zero or more attributes from any namespaces for the purpose of extensibility.

7.12.2.2.4 Semantic

The <oneShot> child element of the <requestForLocationInformation> root element indicates that the receiving entity is requested to send the location information once.

The <anyExt> element contains elements defined in future version of this specification.

The receiving entity ignores any unknown XML element and any unknown XML attribute.

7.12.2.2.5 IANA registration

Your name:

<MCC name>

Your email address:

<MCC email>

Media type name:

application

Subtype name:

vnd.3gpp.current-location-discovery+xml

Required parameters:

none.

Optional parameters:

1) "charset" - the parameter has identical semantics to the charset parameter of the "application/xml" media type as specified in section 9.1 of IETF RFC 7303 [247].

Encoding considerations:

binary.

Security considerations:

same as general security considerations for application/xml media type as specified in section 9.1 of RFC 7303 [247]. In addition, this media type provides a format for exchanging information in SIP, so the security considerations from RFC 3261 [26] apply.

The information transported in this MIME media type does not include active or executable content.

Mechanisms for privacy and integrity protection of protocol parameters exist. Those mechanisms as well as authentication and further security mechanisms are described in 3GPP TS 24.229.

This media type includes provisions for directives that institute actions on a recipient's files or other resources. The action is providing the location information. The action is providing the location information of the entity receiving the body. Except when sent a part of an emergency session, the entity receiving the body needs to request the user at the entity to authorize the action.

This media type includes provisions for directives that institute actions that, while not directly harmful to the recipient, may result in disclosure of information that either facilitates a subsequent attack or else violates a recipient's privacy in any way. The action is providing the location information of the entity receiving the body. Except when sent a part of an emergency session, the entity receiving the body needs to request the user at the entity to authorize the action.

This media type does not employ compression.

Interoperability considerations:

Same as general interoperability considerations for application/xml media type as specified in section 9.1 of IETF RFC 7303 [247].

Published specification:

3GPP TS 24.229, (http://www.3gpp.org/ftp/Specs/html-info/24229.htm)

Applications which use this media type:

This MIME type is used for a MIME body within SIP INFO requests.

Fragment identifier considerations:

The handling in section 5 of IETF RFC 7303 [247] applies.

Restrictions on usage:

None

Provisional registration? (standards tree only):

N/A

Additional information:

1) Deprecated alias names for this type: none
2) Magic number(s): none
3) File extension(s): none
4) Macintosh file type code(s): none
5) Object identifier(s) or OID(s): none

Intended usage:

Common.

Person to contact for further information

1) Name: <MCC>
2) Email: <MCC email>
3) Author/change controller:
   i) Author: 3GPP CT1 Working Group/3GPP_TSG_CT_WG1@LIST.ETSI.ORG
   ii) Change controller: <MCC name>/<MCC email address>

7.13 JSON Web Token claims defined within the present document

7.13.1 General

This subclause contains definitions for JSON Web Token claims RFC 7519 [235] usage in the 3GPP IM CN subsystem.

NOTE: The claim names are defined as private claim names, and do not require registration, as defined in RFC 7519 [235].

7.13.2 3GPP-WAF

The 3gpp-waf claim is used to transport the identity of the WAF.

Claim name: 3gpp-waf

Claim value: String

Claim description: WAF identity

7.13.3 3GPP-WWSF

The 3gpp-wwsf claim is used to transport the identity of the WAF.

Claim name: 3gpp-wwsf

Claim value: String

Claim description: WWSF identity

7.13.4 identityHeader

The identityHeader claim is used to transport a SIP Identity header field.

Claim name: identityHeader

Claim value: String

Claim description: Contents of an Identity header field

7.13.5 verstatValue

The verstatValue claim is used to transport the value of a verstat tel URI parameter.

Claim name: verstatValue

Claim value: String

Claim description: The value of a verstat tel URI parameter.

7.13.6 identityHeaders

The identityHeaders claim is used to transport one or more SIP Identity header field(s).

Claim name: identityHeaders

Claim value: Array of strings

Claim description: Array of Identity header fields needed to verify diverting users.

7.13.7 divResult

The divResult claim is used to transport the result for the verified div claims and related identities.

Claim name: divResult

Claim value: Array of one or more [div, verstatValue] tuples

Claim description: The value of a verstat tel URI parameter.

7.13.8 verstatPriority

The verstatPriority claim is used to transport the verification value of the Resource-Priority header field and optionally the header field value "psap-callback" of the Priority header field.

Claim name: verstatPriority

Claim value: String

Claim description: Indicates the result of the verification of the Resource-Priority header field and optionally the header field value "psap-callback" of the Priority header field.

7.14 Dialog event package extensions defined within the present document

7.14.1 General

This subclause describes the dialog event package extensions that are applicable for the IM CN subsystem.

7.14.2 Dialog event package extension to transport UE identity information

7.14.2.1 Structure and data semantics

This subclause defines an extension to the dialog event package (RFC 4235 [171]) to transport UE identity information for UEs belonging to the same subscription.

In order to include UE identity information in the dialog event package, the notifier shall

1. in the <dialog-info> element add one or more <ue-instance> elements defined in subclause 7.14.2.2 and 3GPP TS 24.174 [8ZH], each element containing:

   a) an "identity" attribute set to an identifier of the UE as specified in 3GPP TS 24.174 [8ZH]; and

   b) an "alias" attribute set to a user friendly name of the UE as specified in 3GPP TS 24.174 [8ZH].

7.14.2.2 XML Schema

Table 7.14.1 in this subclause defines the XML Schema describing the extension to include UE identity information which can be included in the dialog event package sent from the TAS in NOTIFY requests.

Table 7.14.1: UE identity information, XML Schema

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema
  targetNamespace="urn:3gpp:extDEN:1.0"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  xmlns:ss="http://uri.etsi.org/ngn/params/xml/simservs/xcap"
  elementFormDefault="qualified" attributeFormDefault="unqualified">
  <xs:include schemaLocation="XCAP.xsd"/>
    <ss:ue-instance/>
</xs:schema>
```

8 SIP compression

8.1 SIP compression procedures at the UE

8.1.1 SIP compression

If in normal operation the UE generates requests or responses containing a P-Access-Network-Info header field which included a value of "3GPP-GERAN","3GPP-UTRAN-FDD", "3GPP-UTRAN-TDD", "3GPP-E-UTRAN-FDD", "3GPP-E-UTRAN-TDD", "3GPP-E-UTRAN-ProSe-UNR", "3GPP-NR-FDD", "3GPP-NR-TDD", "3GPP-NR-U-FDD", "3GPP-NR-U-TDD", "3GPP-NR-SAT", "3GPP-NR-ProSe-L2UNR", "3GPP-NR-ProSe-L3UNR", "3GPP-NR-REDCAP", "3GPP2-1X", "3GPP2-1X-HRPD", "3GPP2-UMB", "IEEE-802.11", "IEEE-802.11a", "IEEE-802.11b", "IEEE-802.11g", "IEEE-802.11n", "IEEE-802.11ac", or "DVB-RCS2", then the UE shall support:

- SigComp as specified in RFC 3320 [32] and as updated by RFC 4896 [118]; and
- the additional requirements specified in RFC 5049 [79], with the exception that the UE shall take a State Memory Size of at least 4096 bytes as a minimum value.

If in normal operation the UE generates requests or responses containing a P-Access-Network-Info header field which included a value of "3GPP-GERAN","3GPP-UTRAN-FDD", "3GPP-UTRAN-TDD", "3GPP-E-UTRAN-FDD", "3GPP-E-UTRAN-TDD", "3GPP-E-UTRAN-ProSe-UNR", "3GPP-NR-FDD", "3GPP-NR-TDD", "3GPP-NR-U-FDD", "3GPP-NR-U-TDD", "3GPP-NR-SAT", "3GPP-NR-ProSe-L2UNR", "3GPP-NR-ProSe-L3UNR", "3GPP-NR-REDCAP", "3GPP2-1X", "3GPP2-1X-HRPD", "3GPP2-UMB", "IEEE-802.11", "IEEE-802.11a", "IEEE-802.11b", "IEEE-802.11g", "IEEE-802.11n", "IEEE-802.11ac", or "DVB-RCS2", then the UE may support:

- the negative acknowledgement mechanism specified in RFC 4077 [65A].

When using SigComp the UE shall send compressed SIP messages in accordance with RFC 3486 [55]. When the UE will create the compartment is implementation specific, but the compartment shall not be created until a set of security associations or a TLS session is set up if signalling security is in use. The UE shall finish the compartment when the UE is deregistered. The UE shall alow state creations and announcements only for messages received in a security association.

NOTE: Exchange of bytecodes during registration will prevent unnecessary delays during session setup.

If the UE supports SigComp:

- the UE shall support the SIP dictionary specified in RFC 3485 [42] and as updated by RFC 4896 [118]. If compression is enabled, the UE shall use the dictionary to compress the first message; and
- if the UE supports the presence user agent or watcher roles as specified in table A.3A/2 and table A.3A/4, the UE may support the presence specific dictionary specified in RFC 5112 [119].

The use of SigComp is not re-negotiated between initial registration and deregistration.

8.1.2 Compression of SIP requests and responses transmitted to the P-CSCF

In normal operation the UE should send the generated requests and responses transmitted to the P-CSCF:

- compressed according to subclause 8.1.1, if the P-Access-Network-Info header field of the initial registration message includes a value of "3GPP-GERAN","3GPP-UTRAN-FDD", "3GPP-UTRAN-TDD", "3GPP2-1X", "3GPP2-1X-HRPD", "3GPP2-UMB", "IEEE-802.11", "IEEE-802.11a", "IEEE-802.11b", IEEE-802.11g", "IEEE-802.11n", "IEEE-802.11ac", or "DVB-RCS2";

- uncompressed, if the P-Access-Network-Info header field of the initial registration message includes a value of "3GPP-E-UTRAN-FDD", "3GPP-E-UTRAN-TDD", "3GPP-E-UTRAN-ProSe-UNR", "3GPP-NR-FDD", "3GPP-NR-TDD","3GPP-NR-U-FDD", "3GPP-NR-U-TDD", "3GPP-NR-SAT", "3GPP-NR-ProSe-L2UNR", "3GPP-NR-ProSe-L3UNR", "3GPP-NR-REDCAP".

In other cases where SigComp is supported, the UE need not compress the requests and responses.

NOTE 1: Compression of SIP messages is an implementation option. However, compression is strongly recommended.

NOTE 2: In an IP-CAN where compression support is mandatory the UE can send even the first message compressed. Sigcomp provides mechanisms to allow the UE to know if state has been created in the P-CSCF or not.

8.1.3 Decompression of SIP requests and responses received from the P-CSCF

If the UE supports SigComp, then the UE shall decompress the compressed requests and responses received from the P-CSCF according to subclause 8.1.1.

NOTE: According to RFC 3486 [55], the UE not supporting SigComp or not indicating willingness to receive compressed messages never receives compressed SIP messages.

If the UE detects a decompression failure at the P-CSCF, the recovery mechanism is implementation specific.

8.2 SIP compression procedures at the P-CSCF

8.2.1 SIP compression

The P-CSCF shall support:

- SigComp as specified in RFC 3320 [32] and as updated by RFC 4896 [118]; and

- the additional requirements specified in RFC 5049 [79], with the exception that the P-CSCF shall take a State Memory Size of at least 4096 bytes as a minimum value.

The P-CSCF may support:

- the negative acknowledgement mechanism specified in RFC 4077 [65A].

When using SigComp the P-CSCF shall send compressed SIP messages in accordance with RFC 3486 [55]. When the P-CSCF will create the compartment is implementation specific, but the compartment shall not be created until a set of security associations are set up. The P-CSCF shall finish the compartment when the UE is deregistered. The P-CSCF shall allow state creations and announcements only for messages received in a security association.

The P-CSCF:

- shall support the SIP dictionary specified in RFC 3485 [42] and as updated by RFC 4896 [118]. If compression is enabled, the P-CSCF shall use the dictionary to compress the first message; and

- may support the presence specific dictionary specified in RFC 5112 [119].

NOTE: Exchange of bytecodes during registration will prevent unnecessary delays during session setup.

8.2.2 Compression of SIP requests and responses transmitted to the UE

For all SIP transactions on a specific security association where the security association was established using a REGISTER request from the UE containing a P-Access-Network-Info header field which included a value of "3GPP-GERAN","3GPP-UTRAN-FDD", "3GPP-UTRAN-TDD", "3GPP-E-UTRAN-ProSe-UNR", "3GPP-NR-FDD", "3GPP-NR-TDD","3GPP-NR-U-FDD", "3GPP-NR-U-TDD", "3GPP-NR-SAT", "3GPP-NR-ProSe-L2UNR", "3GPP-NR-ProSe-L3UNR", "3GPP-E-UTRAN-FDD", "3GPP-E-UTRAN-TDD", "3GPP2-1X", "3GPP2-1X-HRPD", "3GPP2-UMB", "IEEE-802.11", "IEEE-802.11a", "IEEE-802.11b", "IEEE-802.11g", "IEEE-802.11n", "IEEE-802.11ac", or "DVB-RCS2", and the UE has indicated that it supports SigComp and is willing to receive compressed messages in accordance with RFC 3486 [55], then the P-CSCF should compress the requests and responses transmitted to the UE according to subclause 8.2.1. In other cases where SigComp is supported, it need not.

NOTE: Compression of SIP messages is an implementation option. However, compression is strongly recommended.

8.2.3 Decompression of SIP requests and responses received from the UE

The P-CSCF shall decompress the compressed requests and responses received from the UE according to subclause 8.2.1.

If the P-CSCF detects a decompression failure at the UE, the recovery mechanism is implementation specific.

9 IP-Connectivity Access Network aspects when connected to the IM CN subsystem

9.1 Introduction

A UE accessing the IM CN subsystem and the IM CN subsystem itself utilises the services supported by the IP-CAN to provide packet-mode communication between the UE and the IM CN subsystem. General requirements for the UE on the use of these packet-mode services are specified in this clause.

Possible aspects particular to each IP-CAN is described separately for each IP-CAN.

9.2 Procedures at the UE

9.2.1 Connecting to the IP-CAN and P-CSCF discovery

Prior to communication with the IM CN subsystem, the UE shall:

a) establish a connection with the IP-CAN;

b) obtain an IP address using either the standard IETF protocols (e.g., DHCP or IPCP) or a protocol that is particular to the IP-CAN technology that the UE is utilising. The UE shall fix the obtained IP address throughout the period the UE is connected to the IM CN subsystem, i.e. from the initial registration and at least until the last deregistration; and

c) acquire a P-CSCF address(es).

The UE may acquire an IP address via means other than the DHCP. In this case, upon acquiring an IP address, the UE shall request the configuration information (that includes the DNS and P-CSCF addresses) from the DHCP server.

The methods for acquiring a P-CSCF address(es) are:

I. Employ Dynamic Host Configuration Protocol for IPv4 RFC 2131 [40A] or for IPv6 (DHCPv6) RFC 3315 [40]. Employ the DHCP options for SIP servers RFC 3319 [41] or, for IPv6, RFC 3361 [35A]. Employ the DHCP options for Domain Name Servers (DNS) RFC 3646 [56C].

The UE shall either:

- in the DHCP query, request a list of SIP server domain names of P-CSCF(s) and the list of Domain Name Servers (DNS); or
- request a list of SIP server IP addresses of P-CSCF(s).

II. Obtain the P-CSCF address(es) by employing a procedure that the IP-CAN technology supports. (e.g. GPRS).

III. The UE may use pre-configured P-CSCF address(es) (IP address or domain name). For example:

a. The UE selects a P-CSCF from the list stored in ISIM or IMC;

b. The UE selects a P-CSCF from the list in IMS management object.

NOTE 1: Access-specific annexes provide additional guidance on the method to be used by the UE to acquire P-CSCF address(es).

When acquiring a P-CSCF address(es), the UE can freely select either method I or II or III.

NOTE 2: In case a P-CSCF address is provisioned or received as a FQDN, procedures according to RFC 3263 [27A] will provide the resolution of the FQDN.

The UE may also request a DNS Server IP address(es) as specified in RFC 3315 [40] and RFC 3646 [56C] or RFC 2131 [40A].

When:

- the UE obtains a connection with the IP-CAN by performing handover of the connection from another IP-CAN;
- IP address of the UE is not changed during the handover; and
- the UE already communicates with the IM CN subsystem via the connection with the other IP-CAN, e.g. the UE determines that its contact with host portion set to the UE IP address (or FQDN of the UE) associated with the connection with the other IP-CAN has been bound to a public user identity;

the UE shall continue using the P-CSCF address(es) acquired in the other IP-CAN.

9.2.2 Handling of the IP-CAN

The means to ensure that appropriate resources are available for the media flow(s) on the IP-CAN(s) related to a SIP session is dependant on the characteristics for each IP-CAN, and is described separately for each IP-CAN in question.

GPRS is described in annex B. xDSL is described in annex E. DOCSIS is described in annex H. EPS is described in annex L. cdma2000® packet data subsystem is described in annex M. EPC via cdma2000® HRPD is described in annex O. cdma2000® Femtocell network is described in annex Q. Evolved Packet Core (EPC) via WLAN is described in annex R. DVB-RCS2 is described in annex S. 5GS is described in annex U. If a particular handling of the IP-CAN is needed for emergency calls, this is described in the annex for each access technology.

9.2.2A P-CSCF restoration procedure

The UE may support P-CSCF restoration procedures.

An IP-CAN may provide means for detecting a P-CSCF failure.

An UE supporting the P-CSCF restoration procedure should either use the keep-alive procedures described in RFC 6223 [143] or the procedure provided by a IP-CAN for monitoring the P-CSCF status.

NOTE 1: The UE can use other means to monitor the P-CSCF status, e.g. ICMP echo request/response. However, those other means are out of scope of this document.

NOTE 2: A UE registered through the procedures described in RFC 5626 [92] can use the keep-alive mechanism to monitor the status of the P-CSCF.

9.2.3 Special requirements applying to forked responses

Since the UE does not know that forking has occurred until a second provisional response arrives, the UE will request the radio/bearer resources as required by the first provisional response. For each subsequent provisional response that may be received, different alternative actions may be performed depending on the requirements in the SDP answer:

- the UE has sufficient radio/bearer resources to handle the media specified in the SDP of the subsequent provisional response, or
- the UE must request additional radio/bearer resources to accommodate the media specified in the SDP of the subsequent provisional response.

NOTE 1: When several forked responses are received, the resources requested by the UE is the "logical OR" of the resources indicated in the multiple responses to avoid allocation of unnecessary resources. The UE does not request more resources than proposed in the original INVITE request.

NOTE 2: When service-based local policy is applied, the UE receives the same authorization token for all forked requests/responses related to the same SIP session.

When an 199 (Early Dialog Terminated) response for the INVITE request is received for an early dialogue, the UE shall release reserved radio/bearer resources associated with that early dialogue.

When the first final 200 (OK) response for the INVITE request is received for one of the early dialogs, the UE proceeds to set up the SIP session using the radio/bearer resources required for this session. Upon the reception of the first final 200 (OK) response for the INVITE request, the UE shall release all unneeded radio/bearer resources.

10 Media control

10.1 General

The choice of which media control methods below to use is service specific, it depends on the functionality required and physical deployment architectures.

Combinations of the capabilities below are supported by the use of the control channel framework RFC 6230 [146] with associated media control packages.

For security, the principles and protocols described in 3GPP TS 33.210 [19A] shall take precedence over those specified in the referenced specifications in this clause.

For codecs, those described in access specific specifications shall take precedence over those specified in the referenced specifications in this clause.

10.2 Procedures at the AS

10.2.1 General

An AS requesting charging information and authorisation for specific media operations and media usage controlled by the MRFC shall use RFC 6230 [146] together with appropriate packages.

NOTE: This is in addition to the charging related procedures in clause 5 and to the charging information and authorisation requests, defined in 3GPP TS 32.260 [17] which provide charging information and authorisation for SIP session and SDP information.

An AS may support delegation of an XML (such as CCXML or SCXML) script execution to an MRFC. An AS supporting delegation of XML script execution shall use RFC 6230 [146] together with appropriate packages.

The packages, or extensions to existing packages using RFC 6230 [146] framework are not specified in this release.

The AS may support the media server resource consumer interface as defined by RFC 6917 [192]. If supported the AS can support either the in-line mode or the query mode or both.

10.2.2 Tones and announcements

10.2.2.1 General

An AS may support control of the MRFC for tones and announcements. An AS supporting control of the MRFC for tones and announcements shall support one or more of the following methods:

- RFC 4240 [144] announcement service;
- RFC 5552 [145]; or
- RFC 6230 [146] and RFC 6231 [147].

10.2.2.2 Basic network media services with SIP

The AS may support control of the MRFC for basic announcements by the use of RFC 4240 [144] and the announcement service described in RFC 4240 [144] subclause 3.

The media control commands are carried between the AS and the MRFC either directly over the Mr' interface or via the S-CSCF over the ISC and Mr interfaces.

The AS shall provide remote prompts to the MRFC using the AS-MRFC Cr interface.

10.2.2.3 SIP interface to VoiceXML media services

The AS may support control of the MRFC for voice dialogs by the use of RFC 5552 [145].

The media control commands are carried between the AS and the MRFC either directly over the Mr' interface or via the S-CSCF over the ISC and Mr interfaces.

The AS shall provide remote prompts and scripts to the MRFC using the AS-MRFC Cr interface.

Data shall be returned to the AS from the MRFC by either use of the AS-MRFC Cr interface (subclause 4.1 of RFC 5552 [145]), via the ISC interface (subclause 4.2 of RFC 5552 [145]) or via the Mr' interface.

10.2.2.4 Media control channel framework and packages

The AS may support control of the MRFC for interactive voice response by the use of RFC 6231 [147] and RFC 6230 [146].

The AS shall provide remote prompts, media control commands and scripts to the MRFC using the AS-MRFC Cr interface.

The AS shall implement the control client role as described in RFC 6230 [146].

10.2.3 Ad-hoc conferences

10.2.3.1 General

An AS may support control of the MRFC for ad-hoc conferencing. An AS supporting control of the MRFC for ad-hoc conferencing shall support one or more of the following methods:

- RFC 4240 [144] conference service; or
- RFC 6230 [146] and RFC 6505 [148].

10.2.3.2 Basic network media services with SIP

The AS may support control of the MRFC for basic conferencing by the use of RFC 4240 [144] and the conference service described in RFC 4240 [144] subclause 5.

The media control commands are carried between the AS and the MRFC either directly over the Mr' interface or via the S-CSCF over the ISC and Mr interfaces.

10.2.3.3 Media control channel framework and packages

The AS may support control of the MRFC for conference mixing by the use of RFC 6505 [148] and RFC 6230 [146].

An AS may support control of the MRFC for floor controlled conferences (as specified in 3GPP TS 24.147 [8B]), via the use of RFC 6230 [146] together with appropriate packages. The packages, or extensions to existing packages using RFC 6230 [146] framework are not specified in this release.

An AS may support control of the MRFC for session-mode messaging conferences (as specified in 3GPP TS 24.247 [8F]), via the use of RFC 6230 [146] together with appropriate packages. The packages, or extensions to existing packages using RFC 6230 [146] framework are not specified in this release.

The AS shall provide media control commands to the MRFC using the AS-MRFC Cr interface.

The AS shall implement the control client role as described in RFC 6230 [146].

10.2.4 Transcoding

10.2.4.1 General

An AS may support control of the MRFC for transcoding. An AS supporting control of the MRFC for transcoding shall support one or more of the following methods:

- RFC 4240 [144] conference service; or
- RFC 6230 [146] and RFC 6505 [148].

10.2.4.2 Basic network media services with SIP

The AS may support control of the MRFC for transcoding by the use of RFC 4240 [144] and the conference service described in RFC 4240 [144] subclause 5.

The media control commands are carried between the AS and the MRFC either directly over the Mr' interface or via the S-CSCF over the ISC and Mr interfaces.

10.2.4.3 Media control channel framework and packages

The AS may support control of the MRFC for transcoding by the use of RFC 6505 [148] and RFC 6230 [146].

The AS shall provide media control commands to the MRFC using the AS-MRFC Cr interface.

The AS shall implement the control client role as described in RFC 6230 [146].

10.3 Procedures at the MRFC

10.3.1 General

An MRFC required to generate charging information and authorize requests from an AS for specific media operations and media usage shall support RFC 6230 [146] together with appropriate packages.

NOTE: This is in addition to the charging related procedures in clause 5 and to the charging information and authorisation requests, defined in 3GPP TS 32.260 [17] which provide charging information and authorisation for SIP session and SDP information.

An MRFC may support delegated XML (such as CCXML or SCXML) script execution from an AS. An MRFC supporting delegation of XML script execution shall use RFC 6230 [146] together with appropriate packages.

The packages, or extensions to existing packages using RFC 6230 [146] framework above are not specified in this release.

The MRFC may support the media server resource publish interface as defined by RFC 6917 [192].

10.3.2 Tones and announcements

10.3.2.1 General

An MRFC may support control of tones and announcements. An MRFC supporting control of tones and announcements shall support one or more of the following methods:

- RFC 4240 [144] announcement service;
- RFC 5552 [145]; or
- RFC 6230 [146] and RFC 6231 [147].

10.3.2.2 Basic network media services with SIP

The MRFC may support control of basic announcements by the use of RFC 4240 [144] and the announcement service described in RFC 4240 [144] subclause 3.

The media control commands are received from the AS either directly over the Mr' interface or via the S-CSCF over the ISC and Mr interfaces.

The MRFC shall fetch remote prompts from the AS using the AS-MRFC Cr interface.

The MRFC acts as the media server described in RFC 4240 [144].

10.3.2.3 SIP interface to VoiceXML media services

The MRFC may support control of voice dialogs by the use of RFC 5552 [145].

The media control commands are received from the AS either directly over the Mr' interface or via the S-CSCF over the ISC and Mr interfaces.

The MRFC shall fetch via the AS-MRFC Cr interface the remote prompts and scripts from the address included in the "voicexml" URI parameter, if received, in the Request-URI of the SIP INVITE request.

Data shall be returned to the AS from the MRFC by either use of the AS-MRFC Cr interface (subclause 4.1 of RFC 5552 [145]), via the ISC interface (subclause 4.2 of RFC 5552 [145]) or via the Mr' interface.

The MRFC acts as the VoiceXML media server described in RFC 5552 [145].

10.3.2.4 Media control channel framework and packages

The MRFC may support control of interactive voice response by the use of RFC 6231 [147] and RFC 6230 [146].

The MRFC shall fetch remote prompts and scripts from the MRFC using the AS-MRFC Cr interface. The MRFC shall send media control command responses and notifications to the AS using the AS-MRFC Cr interface.

The MRFC shall implement the control server role as described in RFC 6230 [146].

10.3.3 Ad-hoc conferences

10.3.3.1 General

An MRFC may support control of ad-hoc conferencing. An MRFC supporting control of ad-hoc conferencing shall support one or more of the following methods:

- RFC 4240 [144] conference service; or
- RFC 6230 [146] and RFC 6505 [148].

10.3.3.2 Basic network media services with SIP

The MRFC may support control of basic conferencing by the use of RFC 4240 [144] and the conference service described in RFC 4240 [144] subclause 5.

The media control commands are received from the AS either directly over the Mr' interface or via the S-CSCF over the ISC and Mr interfaces.

The MRFC acts as the media server described in RFC 4240 [144].

10.3.3.3 Media control channel framework and packages

The MRFC may support control of conference mixing by the use of RFC 6505 [148] and RFC 6230 [146].

An MRFC may support control of floor controlled conferences (as specified in 3GPP TS 24.147 [8B]), via the use of RFC 6230 [146] together with appropriate packages. The packages, or extensions to existing packages using RFC 6230 [146] framework are not specified in this release. In addition, the MRFC may support the procedures for a multi-stream multiparty multimedia conference using simulcast defined in 3GPP TS 26.114 [9B] annex S.

An MRFC may support control of session-mode messaging conferences (as specified in 3GPP TS 24.247 [8F]), via the use of RFC 6230 [146] together with appropriate packages. The packages, or extensions to existing packages using RFC 6230 [146] framework are not specified in this release.

The MRFC shall send media control command responses and notifications to the AS using the AS-MRFC Cr interface.

The MRFC shall implement the control server role as described in RFC 6230 [146].

10.3.4 Transcoding

10.3.4.1 General

An MRFC may support control of transcoding. An MRFC supporting control of transcoding shall support one or more of the following methods:

- RFC 4240 [144] conference service;
- RFC 6230 [146] and RFC 6505 [148]; or
- RFC 4117 [166]. This is detailed in subclause 5.7.5.6.

10.3.4.2 Basic network media services with SIP

The MRFC may support control of transcoding by the use of RFC 4240 [144] and the conference service described in RFC 4240 [144] subclause 5.

The media control commands are received from the AS either directly over the Mr' interface or via the S-CSCF over the ISC and Mr interfaces.

The MRFC acts as the media server described in RFC 4240 [144].

10.3.4.3 Media control channel framework and packages

The MRFC may support control of transcoding by the use of RFC 6505 [148] and RFC 6230 [146].

The MRFC shall send media control command responses and notifications to the AS using the AS-MRFC Cr interface.

The MRFC shall implement the control server role as described in RFC 6230 [146].

10.4 Procedures at the MRB

The MRB shall support:

a) the in-line unaware MRB interface.

as defined in RFC 5917 [192].

The MRB may support:

a) the media server resource publish interface; and

b) the media server resource consumer interface;

as defined in RFC 6917 [192].

Annex A (normative): Profiles of IETF RFCs for 3GPP usage

A.1 Profiles

A.1.1 Relationship to other specifications

This annex contains a profile to the IETF specifications which are referenced by this specification, and the PICS proformas underlying profiles do not add requirements to the specifications they are proformas for.

This annex provides a profile specification according to both the current IETF specifications for SIP, SDP and other protocols (as indicated by the "RFC status" column in the tables in this annex) which are referenced by this specification and to the 3GPP specifications using SIP (as indicated by the "Profile status" column in the tables in this annex.

In the "RFC status" column the contents of the referenced specification takes precedence over the contents of the entry in the column.

In the "Profile status" column, there are a number of differences from the "RFC status" column. Where these differences occur, these differences take precedence over any requirements of the IETF specifications. Where specification concerning these requirements exists in the main body of the present document, the main body of the present document takes precedence.

Where differences occur in the "Profile status" column, the "Profile status" normally gives more strength to a "RFC status" and is not in contradiction with the "RFC status", e.g. it may change an optional "RFC status" to a mandatory "Profile status". If the "Profile status" weakens the strength of a "RFC status" then additionally this will be indicated by further textual description in the present document.

For all IETF specifications that are not referenced by this document or that are not mentioned within the 3GPP profile of SIP and SDP, the generic rules as defined by RFC 3261 [26] and in addition the rules in clauses 5 and 6 of this specification apply, e.g.

- a proxy which is built in accordance to this specification passes on any unknown method, unknown header field or unknown header field parameter after applying procedures such as filtering, insertion of P-Asserted-Identity header field, etc.;

- an UA which is built in accordance to this specification will

    - handle received unknown methods in accordance to the procedures defined in RFC 3261 [26], e.g. respond with a 501 (Not Implemented) response; and

    - handle unknown header fields and unknown header field parameters in accordance to the procedures defined in RFC 3261 [26], e.g. respond with a 420 (Bad Extension) if an extension identified by an option-tag in the Require header field of the received request is not supported by the UA.

A.1.2 Introduction to methodology within this profile

This subclause does not reflect dynamic conformance requirements but static ones. In particular, a condition for support of a PDU parameter does not reflect requirements about the syntax of the PDU (i.e. the presence of a parameter) but the capability of the implementation to support the parameter.

In the sending direction, the support of a parameter means that the implementation is able to send this parameter (but it does not mean that the implementation always sends it).

In the receiving direction, it means that the implementation supports the whole semantic of the parameter that is described in the main part of this specification.

As a consequence, PDU parameter tables in this subclause are not the same as the tables describing the syntax of a PDU in the reference specification, e.g. RFC 3261 [26] tables 2 and 3. It is not rare to see a parameter which is optional in the syntax but mandatory in subclause below.

The various statii used in this subclause are in accordance with the rules in table A.1.

Table A.1: Key to status codes

Status code	Status name	Meaning
m	mandatory	the capability shall be supported. It is a static view of the fact that the conformance requirements related to the capability in the reference specification are mandatory requirements. This does not mean that a given behaviour shall always be observed (this would be a dynamic view), but that it shall be observed when the implementation is placed in conditions where the conformance requirements from the reference specification compel it to do so. For instance, if the support for a parameter in a sent PDU is mandatory, it does not mean that it shall always be present, but that it shall be present according to the description of the behaviour in the reference specification (dynamic conformance requirement).
o	optional	the capability may or may not be supported. It is an implementation choice.
n/a	not applicable	it is impossible to use the capability. No answer in the support column is required.
x	prohibited (excluded)	It is not allowed to use the capability. This is more common for a profile.
c <integer>	conditional	the requirement on the capability ("m", "o", "n/a" or "x") depends on the support of other optional or conditional items. <integer> is the identifier of the conditional expression.
o.<integer>	qualified optional	for mutually exclusive or selectable options from a set. <integer> is the identifier of the group of options, and the logic of selection of the options.
i	irrelevant	capability outside the scope of the given specification. Normally, this notation should be used in a base specification ICS proforma only for transparent parameters in received PDUs. However, it may be useful in other cases, when the base specification is in fact based on another standard.

In the context of this specification the "i" status code mandates that the implementation does not change the content of the parameter. It is an implementation option if the implementation acts upon the content of the parameter (e.g. by setting filter criteria to known or unknown parts of parameters in order to find out the route a message has to take).

It must be understood, that this 3GPP SIP profile does not list all parameters which an implementation will treat as indicated by the status code "irrelevant". In general an implementation will pass on all unknown messages, header fields and header field parameters, as long as it can perform its normal behaviour.

The following additional comments apply to the interpretation of the tables in this Annex.

NOTE 1: The tables are constructed according to the conventional rules for ICS proformas and profile tables.

NOTE 2: The notation (either directly or as part of a conditional) of "m" for the sending of a parameter and "i" for the receipt of the same parameter, may be taken as indicating that the parameter is passed on transparently, i.e. without modification. Where a conditional applies, this behaviour only applies when the conditional is met.

As an example, the profile for the MGCF is found by first referring to clause 4.1, which states "The MGCF shall provide the UA role". Profiles are divided at the top level into the two roles in table A.2, user agent and proxy. The UA role is defined in subclause A.2.1 and the proxy role is defined in subclause A.2.2. More specific roles are listed in table A.3, table A.3A, table A.3B and table A.3C. The MGCF role is item 6 in table A.3 (the MGCF role is not found in table A.3A or table A.3B). Therefore, all profile entries for the MGCF are found by searching for A.3/6 in subclause A.2.1.

As a further example, to look up support of the Reason header field, table A.4 item 38 lists the Resaon header field as a major capability that is optional for the user agent role. A subsequent search for A.4/38 in subclause A.2.1 shows that the Reason header field is optional for a user agent role to send and receive for ACK, BYE, CANCEL, INVITE, MESSAGE, NOTIFY, OPTIONS, PRACK, PUBLISH, REFER, REGISTER, SUBSCRIBE, and UPDATE requests. Also, table A.162 item 48 lists the Reason header field as a major capability that is optional for the proxy role. A subsequent search for A.162/48 in subclause A.2.2 shows that, if supported, the Reason header field is mandatory to send and irrelevant to receive for ACK, BYE, CANCEL, INVITE, MESSAGE, NOTIFY, OPTIONS, PRACK, PUBLISH, REFER, REGISTER, SUBSCRIBE, and UPDATE requests.

A.1.3 Roles

Table A.2: Roles

Item	Roles	Reference	RFC status	Profile status
1	User agent	[26]	o.1	o.1
2	Proxy	[26]	o.1	o.1
o.1:	It is mandatory to support exactly one of these items.			
NOTE:	For the purposes of the present document it has been chosen to keep the specification simple by the tables specifying only one role at a time. This does not preclude implementations providing two roles, but an entirely separate assessment of the tables shall be made for each role.			

Table A.3: Roles specific to this profile

Item	Roles	Reference	RFC status	Profile status
1	UE	5.1	n/a	o.1
1A	UE containing UICC	5.1	n/a	c5
1B	UE without UICC	5.1	n/a	c5
2	P-CSCF	5.2	n/a	o.1
2A	P-CSCF (IMS-ALG)	[7]	n/a	c6
3	I-CSCF	5.3	n/a	o.1
3A	void			
4	S-CSCF	5.4	n/a	o.1
5	BGCF	5.6	n/a	o.1
6	MGCF	5.5	n/a	o.1
7	AS	5.7	n/a	o.1
7A	AS acting as terminating UA, or redirect server	5.7.2	n/a	c2
7B	AS acting as originating UA	5.7.3	n/a	c2
7C	AS acting as a SIP proxy	5.7.4	n/a	c2
7D	AS performing 3rd party call control	5.7.5	n/a	c2
8	MRFC	5.8	n/a	o.1
8A	MRB	5.8A	n/a	o.1
9	IBCF	5.10	n/a	o.1
9A	IBCF (THIG)	5.10.4	n/a	c4
9B	IBCF (IMS-ALG)	5.10.5, 5.10.7	n/a	c4
9C	IBCF (Screening of SIP signalling)	5.10.6	n/a	c4
9D	IBCF (Privacy protection)	5.10.8	n/a	c4
10	Additional routeing functionality	Annex I	n/a	c3
11	E-CSCF	5.11	n/a	o.1
11A	E-CSCF acting as UA	5.11.1, 5.11.2, 5.11.3	n/a	c7
11B	E-CSCF acting as a SIP Proxy	5.11.1, 5.11.2	n/a	c7
12	LRF	5.12	n/a	o.1
13	ISC gateway function	5.13	n/a	o.1
13A	ISC gateway function (THIG)	5.13.4	n/a	c8
13B	ISC gateway function (IMS-ALG)	5.13.5	n/a	c8
13C	ISC gateway function (Screening of SIP signalling)	5.13.6	n/a	c8
14	Gm based WIC	[8Z]	n/a	o.1
15	Transit function	I.3	n/a	c9

c2: IF A.3/7 THEN o.2 ELSE n/a - - AS.
c3: IF A.3/3 OR A.3/4 OR A.3/5 OR A.3/6 OR A.3/9 THEN o ELSE o.1 - - I-CSCF, S-CSCF, BGCF, MGCF, IBCF.
c4: IF A.3/9 THEN o.3 ELSE n/a - - IBCF.
c5: IF A.3/1 THEN o.4 ELSE n/a - - UE.
c6: IF A.3/2 THEN o ELSE n/a - - P-CSCF.
c7: IF A.3/11 THEN o.5 ELSE n/a - - E-CSCF.
c8: IF A.3/13 THEN o ELSE n/a - - ISC gateway function.
c9 IF A.3/3 OR A.3/4 OR A.3/5 OR A.3/6 OR A.3/9 THEN o ELSE o.1 - - I-CSCF, S-CSCF, BGCF, MGCF, IBCF.
o.1: It is mandatory to support exactly one of these items.
o.2: It is mandatory to support at least one of these items.
o.3: It is mandatory to support at least one of these items.
o.4 It is mandatory to support exactly one of these items.
o.5: It is mandatory to support exactly one of these items.

NOTE: For the purposes of the present document it has been chosen to keep the specification simple by the tables specifying only one role at a time. This does not preclude implementations providing two roles, but an entirely separate assessment of the tables shall be made for each role.

Table A.3A: Roles specific to additional capabilities

Item	Roles	Reference	RFC status	Profile status
1	Presence server	3GPP TS 24.141 [8A]	n/a	c1
2	Presence user agent	3GPP TS 24.141 [8A]	n/a	c2
3	Resource list server	3GPP TS 24.141 [8A]	n/a	c3
4	Watcher	3GPP TS 24.141 [8A]	n/a	c4
11	Conference focus	3GPP TS 24.147 [8B]	n/a	c11
12	Conference participant	3GPP TS 24.147 [8B]	n/a	c6
21	CSI user agent	3GPP TS 24.279 [8E]	n/a	c7
22	CSI application server	3GPP TS 24.279 [8E]	n/a	c8
31	Messaging application server	3GPP TS 24.247 [8F]	n/a	c5
32	Messaging list server	3GPP TS 24.247 [8F]	n/a	c5
33	Messaging participant	3GPP TS 24.247 [8F]	n/a	c2
33A	Page-mode messaging participant	3GPP TS 24.247 [8F]	n/a	c2
33B	Session-mode messaging participant	3GPP TS 24.247 [8F]	n/a	c2
34	Session-mode messaging intermediate node	3GPP TS 24.247 [8F]	n/a	c5
50	Multimedia telephony service participant	3GPP TS 24.173 [8H]	n/a	c2
50A	Multimedia telephony service application server	3GPP TS 24.173 [8H]	n/a	c9
51	Message waiting indication subscriber UA	3GPP TS 24.606 [8I]	n/a	c2
52	Message waiting indication notifier UA	3GPP TS 24.606 [8I]	n/a	c3
53	Advice of charge application server	3GPP TS 24.647 [8N]	n/a	c8
54	Advice of charge UA client	3GPP TS 24.647 [8N]	n/a	c2
55	Ut reference point XCAP server for supplementary services	3GPP TS 24.623 [8P]	n/a	c3
56	Ut reference point XCAP client for supplementary services	3GPP TS 24.623 [8P]	n/a	c2
57	Customized alerting tones application server	3GPP TS 24.182 [8Q]	n/a	c8
58	Customized alerting tones UA client	3GPP TS 24.182 [8Q]	n/a	c2
59	Customized ringing signal application server	3GPP TS 24.183 [8R]	n/a	c8
60	Customized ringing signal UA client	3GPP TS 24.183 [8R]	n/a	c2
61	SM-over-IP sender	3GPP TS 24.341 [8L]	n/a	c2
62	SM-over-IP receiver	3GPP TS 24.341 [8L]	n/a	c2
63	IP-SM-GW	3GPP TS 24.341 [8L]	n/a	c1
71	IP-SM-GW	3GPP TS 29.311 [15A]	n/a	c10
81	MSC Server enhanced for ICS	3GPP TS 24.292 [8O]	n/a	c12
81A	MSC server enhanced for SRVCC using SIP interface	3GPP TS 24.237 [8M]	n/a	c12
81B	MSC server enhanced for DRVCC using SIP interface	3GPP TS 24.237 [8M]	n/a	c12

82	ICS user agent	3GPP TS 24.292 [8O]	n/a	c2
83	SCC application server	3GPP TS 24.292 [8O]	n/a	c9
84	EATF	3GPP TS 24.237 [8M]	n/a	c12
85	In-dialog overlap signalling application server	Annex N.2, Annex N.3.3	n/a	c9
86	In-dialog overlap signalling UA client	Annex N.2, Annex N.3.3	n/a	c2
87	Session continuity controller UE	3GPP TS 24.337 [8ZC]	n/a	c2
88	ATCF (proxy)	3GPP TS 24.237 [8M]	n/a	c13 (note 4)
89	ATCF (UA)	3GPP TS 24.237 [8M]	n/a	c12 (note 4)
91	Malicious communication identification application server	3GPP TS 24.616 [8S]	n/a	c9
92	USSI UE	3GPP TS 24.390 [8W]	n/a	c2
92A	USSI UE supporting user-initiated USSD operations	3GPP TS 24.390 [8W]	n/a	c17
92B	USSI UE supporting network-initiated USSD operations	3GPP TS 24.390 [8W]	n/a	c17
93	USSI AS	3GPP TS 24.390 [8W]	n/a	c3
93A	USSI AS supporting user-initiated USSD operations	3GPP TS 24.390 [8W]	n/a	c18
93B	USSI AS supporting network-initiated USSD operations	3GPP TS 24.390 [8W]	n/a	c18
94	TP UE	3GPP TS 24.103 [7G]	n/a	c14
95	eP-CSCF (P-CSCF enhanced for WebRTC)	3GPP TS 24.371 [8Z]	n/a	c15
101	Business trunking in static mode of operation application server	3GPP TS 24.525 [8ZA]	n/a	c16
102	MCPTT client	3GPP TS 24.379 [8ZE]	n/a	c19
103	MCPTT server	3GPP TS 24.379 [8ZE]	n/a	c20

c1: IF A.3/7A AND A.3/7B THEN o ELSE n/a - - AS acting as terminating UA, or redirect server and AS acting as originating UA.
c2: IF A.3/1 THEN o ELSE n/a - - UE.
c3: IF A.3/7A THEN o ELSE n/a - - AS acting as terminating UA, or redirect server.
c4: IF A.3/1 OR A.3/7B THEN o ELSE n/a - - UE or AS acting as originating UA.
c5: IF A.3/7D AND A.3/8 THEN o ELSE n/a - - AS performing 3rd party call control and MRFC (note 2).
c6: IF A.3/1 OR A.3A/11 THEN o ELSE n/a - - UE or conference focus.
c7: IF A.3/1 THEN o ELSE n/a - - UE.
c8: IF A.3/7D THEN o ELSE n/a - - AS performing 3rd party call control.
c9: IF A.3/7A OR A.3/7B OR A.3/7C OR A.3/7D THEN o ELSE n/a - - AS acting as terminating UA, or redirect server, AS acting as originating UA, AS acting as a SIP proxy, AS performing 3rd party call control.
c10: IF A.3/7A OR A.3/7B OR A.3/7D THEN o ELSE n/a - - AS acting as terminating UA, or redirect server, AS acting as originating UA, AS performing 3rd party call control.
c11: IF A.3/7D THEN o ELSE n/a - - AS performing 3rd party call control.
c12: IF A.2/1 THEN o ELSE n/a - - UA.
c13: IF A.2/2 THEN o ELSE n/a - - proxy.
c14: IF A.3/1 OR A.3A/11 THEN o ELSE n/a - - UE or conference focus.
c15: IF A.3/2A THEN o ELSE n/a - - P-CSCF (IMS-ALG).
c16: IF A.3/7A OR A.3/7B THEN o ELSE n/a - - AS acting as terminating UA, or redirect server, AS acting as originating UA.
c17: IF A.3A/92 THEN o.1 ELSE n/a - - USSI UE.
c18: IF A.3A/93 THEN o.2 ELSE n/a - - USSI AS.
c19: IF A.3/1 THEN o ELSE n/a - - UE.
c20: IF A.3/7 THEN o ELSE n/a - - AS.
o.1: It is mandatory to support at least one of these items.
o.2: It is mandatory to support at least one of these items.

NOTE 1: For the purposes of the present document it has been chosen to keep the specification simple by the tables specifying only one role at a time. This does not preclude implementations providing two roles, but an entirely separate assessment of the tables shall be made for each role.
NOTE 2: The functional split between the MRFC and the AS for page-mode messaging is out of scope of this document and they are assumed to be collocated.
NOTE 3: A.3A/63 is an AS providing the IP-SM-GW role to support the transport level interworking defined in 3GPP TS 24.341 [8L]. A.3A/71 is an AS providing the IP-SM-GW role to support the service level interworking for messaging as defined in 3GPP TS 29.311 [15A].
NOTE 4: An ATCF shall support both the ATCF (proxy) role and the ATCF (UA) role.

Table A.3B: Roles with respect to access technology

Item	Value used in P-Access-Network-Info header field	Reference	RFC status	Profile status
1	3GPP-GERAN	[52] 4.4	o	c1
2	3GPP-UTRAN-FDD	[52] 4.4	o	c1
3	3GPP-UTRAN-TDD	[52] 4.4	o	c1
4	3GPP2-1X	[52] 4.4	o	c1
5	3GPP2-1X-HRPD	[52] 4.4	o	c1
6	3GPP2-UMB	[52] 4.4	o	c1
7	3GPP-E-UTRAN-FDD	[52] 4.4	o	c1
8	3GPP-E-UTRAN-TDD	[52] 4.4	o	c1
8A	3GPP-E-UTRAN-ProSe-UNR	subclause 7.2A.4	n/a	c1
8B	3GPP-NR-FDD	subclause 7.2A.4	n/a	c1
8C	3GPP-NR-TDD	subclause 7.2A.4	n/a	c1
8D	3GPP-NR-U-FDD	subclause 7.2A.4	n/a	c1
8E	3GPP-NR-U-TDD	subclause 7.2A.4	n/a	c1
8F	3GPP-NR-ProSe-L2UNR	subclause 7.2A.4	n/a	c1
8G	3GPP-NR-ProSe-L3UNR	subclause 7.2A.4	n/a	c1
8W	3GPP-NR-SAT	subclause 7.2A.4	n/a	c1
8X	3GPP-NR-REDCAP	subclause 7.2A.4	n/a	c1
9	3GPP2-1X-Femto	[52] 4.4	o	c1
11	IEEE-802.11	[52] 4.4	o	c1
12	IEEE-802.11a	[52] 4.4	o	c1
13	IEEE-802.11b	[52] 4.4	o	c1
14	IEEE-802.11g	[52] 4.4	o	c1
15	IEEE-802.11n	[52] 4.4	o	c1
16	IEEE-802.11ac	[52] 4.4	o	c1
21	ADSL	[52] 4.4	o	c1
22	ADSL2	[52] 4.4	o	c1
23	ADSL2+	[52] 4.4	o	c1
24	RADSL	[52] 4.4	o	c1
25	SDSL	[52] 4.4	o	c1
26	HDSL	[52] 4.4	o	c1
27	HDSL2	[52] 4.4	o	c1
28	G.SHDSL	[52] 4.4	o	c1
29	VDSL	[52] 4.4	o	c1
30	IDSL	[52] 4.4	o	c1
31	xDSL	subclause 7.2A.4	o	c1
41	DOCSIS	[52] 4.4	o	c1
51	DVB-RCS2	[52] 4.4	o	c1
52	3GPP-UTRAN	[52] 4.4	o	c2
53	3GPP-E-UTRAN	[52] 4.4	o	c2
54	3GPP-WLAN	[52] 4.4	o	c2
55	3GPP-GAN	[52] 4.4	o	c2
56	3GPP-HSPA	[52] 4.4	o	c2
57	3GPP2	[52] 4.4	o	c2
58	untrusted-non-3GPP-VIRTUAL-EPC	subclause 7.2A.4	n/a	c2
59	VIRTUAL-no-PS	subclause 7.2A.4	n/a	c2
60	WLAN-no-PS	subclause 7.2A.4	n/a	c2
61	3GPP-NR	Subclause 7.2A.4	n/a	c2
62	3GPP-NR-U	Subclause 7.2A.4	n/a	c2
63	3GPP-NR-SAT	Subclause 7.2A.4	n/a	c2
64	3GPP-NR-REDCAP	Subclause 7.2A.4	n/a	c2

c1: If A.3/1 OR A.3/2 THEN o.1 ELSE n/a - - UE or P-CSCF.
c2: If A.3/2 THEN o.1 ELSE n/a - - P-CSCF.
o.1: It is mandatory to support at least one of these items.

Table A.3C: Modifying roles

Item	Roles	Reference	RFC status	Profile status
1	UE performing the functions of an external attached network	4.1		
2	UE performing the functions of an external attached network operating in static mode	4.1		
NOTE: This table identifies areas where the behaviour is modified from that of the underlying role. Subclause 4.1 indicates which underlying roles are modified for this behaviour.				

Table A.3D: Roles with respect to security mechanism

Item	Security mechanism	Reference	RFC status	Profile status
1	IMS AKA plus IPsec ESP	clause 4.2B.1	n/a	c1
2	SIP digest plus check of IP association	clause 4.2B.1	n/a	c2
3	SIP digest plus Proxy Authentication	clause 4.2B.1	n/a	c2
4	SIP digest with TLS	clause 4.2B.1	n/a	c2
5	NASS-IMS bundled authentication	clause 4.2B.1	n/a	c2
6	GPRS-IMS-Bundled authentication	clause 4.2B.1	n/a	c2
7	Trusted node authentication	clause 4.2B.1	n/a	c3
8	SIP over TLS with client certificate authentication	clause 4.2B.1	n/a	c6
20	End-to-end media security using SDES	clause 4.2B.2	o	c5
20A	End-to-access-edge media security for MSRP using TLS and certificate fingerprints	clause 4.2B.2	n/a	c4
20B	End-to-access-edge media security for BFCP using TLS and certificate fingerprints	clause 4.2B.2	n/a	c4
20C	End-to-access-edge media security for UDPTL using DTLS and certificate fingerprints	clause 4.2B.2	n/a	c4
21	End-to-end media security using KMS	clause 4.2B.2	o	c5
22	End-to-end media security for MSRP using TLS and KMS	clause 4.2B.2	o	c5
30	End-to-access-edge media security using SDES	clause 4.2B.2	n/a	c4
31	End-to-access-edge media security for RTP media using DTLS-SRTP and certificate fingerprints	clause 4.2B.2	n/a	c7

c1: IF (A.3/1A OR A.3/2 OR A.3/3 OR A.3/4) THEN m ELSE IF A.3/1B THEN o ELSE n/a - - UE containing UICC or P-CSCF or I-CSCF or S-CSCF, UE without UICC.
c2: IF (A.3/1 OR A.3/2 OR A.3/3 OR A.3/4) THEN o ELSE n/a - - UE or P-CSCF or I-CSCF or S-CSCF.
c3: IF (A.3/3 OR A.3/4) THEN o ELSE n/a - - I-CSCF or S-CSCF.
c4: IF (A.3/1 OR A.3/2A) THEN o ELSE n/a - - UE or P-CSCF (IMS-ALG).
c5: IF A.3/1 THEN o - - UE.
c6: IF A.3C/2 THEN m ELSE o - - UE performing the functions of an external attached network operating in static mode.
c7: IF (A.3/14 OR A.3A/95) THEN m ELSE IF (A.3/1 OR A.3/2A) THEN o ELSE n/a - - Gm based WIC or eP-CSCF or UE or P-CSCF (IMS-ALG).

A.2 Profile definition for the Session Initiation Protocol as used in the present document

A.2.1 User agent role

A.2.1.1 Introduction

This subclause contains the ICS proforma tables related to the user role. They need to be completed only for UA implementations:

Prerequisite: A.2/1 - - user agent role.

A.2.1.2 Major capabilities

Table A.4: Major capabilities

Item	Does the implementation support	Reference	RFC status	Profile status
	Capabilities within main protocol			
1	client behaviour for registration?	[26] subclause 10.2	o	c3
2	registrar?	[26] subclause 10.3	o	c4
2A	registration of multiple contacts for a single address of record	[26] 10.2.1.2, 16.6	o	o
2B	initiating a session?	[26] subclause 13	o	o
2C	initiating a session which require local and/or remote resource reservation?	[30]	o	c43
3	client behaviour for INVITE requests?	[26] subclause 13.2	c18	c18
4	server behaviour for INVITE requests?	[26] subclause 13.3	c18	c18
5	session release?	[26] subclause 15.1	c18	c18
6	timestamping of requests?	[26] subclause 8.2.6.1	o	o
7	authentication between UA and UA?	[26] subclause 22.2, [287]	c34	c34
8	authentication between UA and registrar?	[26] subclause 22.2, [287]	o	c74
8A	authentication between UA and proxy?	[26] 20.28, 22.3, [287]	o	c75
9	server handling of merged requests due to forking?	[26] 8.2.2.2	m	m
10	client handling of multiple responses due to forking?	[26] 13.2.2.4	m	m
11	insertion of date in requests and responses?	[26] subclause 20.17	o	o
12	downloading of alerting information?	[26] subclause 20.4	o	o
	Extensions			
13	SIP INFO method and package framework?	[25]	o	c100
13A	legacy INFO usage?	[25] 2, 3	o	c90
14	reliability of provisional responses in SIP?	[27]	c19	c44
15	the REFER method?	[36]	o	c33
15A	clarifications for the use of REFER with RFC6665?	[231]	c121	c121
15B	explicit subscriptions for the REFER method?	[232]	o	o
16	integration of resource management and SIP?	[30] [64]	c19	c44
17	the SIP UPDATE method?	[29]	c5	c44
19	SIP extensions for media authorization?	[31]	o	c14
20	SIP specific event notification?	[28]	o	c13
22	acting as the notifier of event information?	[28]	c2	c15
22A	a clarification on the use of GRUUs in the SIP event notification framework?	[233]	c122	c122
23	acting as the subscriber to event information?	[28]	c2	c16
24	session initiation protocol extension header field for registering non-adjacent contacts?	[35]	o	c6
25	private extensions to the Session Initiation Protocol (SIP) for network asserted identity within trusted networks?	[34]	o	m
26	a privacy mechanism for the Session Initiation Protocol (SIP)?	[33]	o	m
26A	request of privacy by the inclusion of a Privacy header indicating any privacy option?	[33]	c9	c11
26B	application of privacy based on the received Privacy header?	[33]	c9	n/a
26C	passing on of the Privacy header transparently?	[33]	c9	c12

26D	application of the privacy option "header" such that those headers which cannot be completely expunged of identifying information without the assistance of intermediaries are obscured?	[33] 5.1	c10	c27
26E	application of the privacy option "session" such that anonymization for the session(s) initiated by this message occurs?	[33] 5.2	c10	c27
26F	application of the privacy option "user" such that user level privacy functions are provided by the network?	[33] 5.3	c10	c27
26G	application of the privacy option "id" such that privacy of the network asserted identity is provided by the network?	[34] 7	c10	n/a
26H	application of the privacy option "history" such that privacy of the History-Info header is provided by the network?	[66] 7.2	c37	c37
27	a messaging mechanism for the Session Initiation Protocol (SIP)?	[50]	o	c7
28	session initiation protocol extension header field for service route discovery during registration?	[38]	o	c17
29	compressing the session initiation protocol?	[55]	o	c8
30	private header extensions to the session initiation protocol for the 3rd-Generation Partnership Project (3GPP)?	[52]	o	m
30A	act as first entity within the trust domain for asserted identity?	[34]	c96	c97
30B	act as entity within trust network that can route outside the trust network?	[34]	c96	c97
30C	act as entity passing on identity transparently independent of trust domain?	[34]	c96	c98
31	the P-Associated-URI header extension?	[52] 4.1, [52A] 4	c21	c22
32	the P-Called-Party-ID header extension?	[52] 4.2, [52A] 4	c21	c23
33	the P-Visited-Network-ID header extension?	[52] 4.3, [52A] 4	c21	c24
34	the P-Access-Network-Info header extension?	[52] 4.4, [52A] 4, [234] 2	c21	c25
35	the P-Charging-Function-Addresses header extension?	[52] 4.5, [52A] 4	c21	c26
36	the P-Charging-Vector header extension?	[52] 4.6, [52A] 4	c21	c26
37	security mechanism agreement for the session initiation protocol?	[48]	o	c20
37A	mediasec header field parameter for marking security mechanisms related to media?	Subclause 7.2A.7	n/a	c101
38	the Reason header field for the session initiation protocol?	[34A]	o	c68
38A	carrying Q.850 codes in Reason header fields in SIP (Session Initiation Protocol) responses?	[130]	o	c82
38B	the location parameter for the SIP Reason header field?	[255]	o	c131
38C	Handling of Identity Header Errors for Secure Telephone Identity Revisited (STIR) (carrying STIR codes in Reason header fields in SIP responses)?	[294]	o	c137

39	an extension to the session initiation protocol for symmetric response routeing?	[56A]	o	c62
40	caller preferences for the session initiation protocol?	[56B]	C29	c29
40A	the proxy-directive within caller-preferences?	[56B] 9.1	o.5	o.5
40B	the cancel-directive within caller-preferences?	[56B] 9.1	o.5	o.5
40C	the fork-directive within caller-preferences?	[56B] 9.1	o.5	o.5
40D	the recurse-directive within caller-preferences?	[56B] 9.1	o.5	o.5
40E	the parallel-directive within caller-preferences?	[56B] 9.1	o.5	o.5
40F	the queue-directive within caller-preferences?	[56B] 9.1	o.5	o.5
41	an event state publication extension to the session initiation protocol?	[70]	o	c30
42	SIP session timer?	[58]	c19	c19
43	the SIP Referred-By mechanism?	[59]	o	c33
44	the Session Inititation Protocol (SIP) "Replaces" header?	[60]	c19	c38 (note 1)
45	the Session Inititation Protocol (SIP) "Join" header?	[61]	c19	c19 (note 1)
46	the callee capabilities?	[62]	o	c35
47	an extension to the session initiation protocol for request history information?	[66]	o	o
47A	application of the "mp" optional header field parameter?	[66]	o	o
47B	application of the "rc" optional header field parameter?	[66]	o	o
47C	application of the "np" optional header field parameter?	[66]	o	o
48	Rejecting anonymous requests in the session initiation protocol?	[67]	o	o
49	session initiation protocol URIs for applications such as voicemail and interactive voice response?	[68]	o	o
49A	Session Initiation Protocol (SIP) cause URI parameter for service number translation?	[230]	c118	c118
50	Session Initiation Protocol's (SIP) non-INVITE transactions?	[84]	m	m
51	the P-User-Database private header extension?	[82] 4	o	c94
52	a uniform resource name for services?	[69]	n/a	c39
53	obtaining and using GRUUs in the Session Initiation Protocol (SIP)?	[93]	o	c40 (note 2)
55	the Stream Control Transmission Protocol (SCTP) as a Transport for the Session Initiation Protocol (SIP)?	[96]	o	c42
56	the SIP P-Profile-Key private header extension?	[97]	n/a	n/a
57	managing client initiated connections in SIP?	[92]	o	c45
58	indicating support for interactive connectivity establishment in SIP?	[102]	o	c46
59	multiple-recipient MESSAGE requests in the session initiation protocol?	[104]	c47	c48
60	SIP location conveyance?	[89]	o	c49
60A	the Location Source parameter for the SIP Geolocation header field?	[xxx]	o	c134
61	referring to multiple resources in the session initiation protocol?	[105]	c50	c50

62	conference establishment using request-contained lists in the session initiation protocol?	[106]	c51	c52
63	subscriptions to request-contained resource lists in the session initiation protocol?	[107]	c53	c53
64	dialstring parameter for the session initiation protocol uniform resource identifier?	[103]	o	c19
65	the P-Answer-State header extension to the session initiation protocol for the open mobile alliance push to talk over cellular?	[111]	o	c60
66	the SIP P-Early-Media private header extension for authorization of early media?	[109] 8	o	c58
67	number portability parameters for the 'tel' URI?	[112]	o	c54
67A	assert or process carrier indication?	[112]	o	c55
67B	local number portability?	[112]	o	c57
69	extending the session initiation protocol Reason header for preemption events	[115]	c69	c69
70	communications resource priority for the session initiation protocol?	[116]	o	c70
70A	inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol?	[116] 4.2	c72	c72
70B	inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol?	[116] 4.2	c72	c72
71	addressing an amplification vulnerability in session initiation protocol forking proxies?	[117]	o	c87
72	the remote application identification of applying signalling compression to SIP	[79] 9.1	o	c8
73	a session initiation protocol media feature tag for MIME application subtypes?	[120]	o	c59
74	SIP extension for the identification of services?	[121]	o	c61
75	a framework for consent-based communications in SIP?	[125]	c76	c76
75A	a relay within the framework for consent-based communications in SIP?	[125]	c77	c78
75B	a recipient within the framework for consent-based communications in SIP?	[125]	c80	c79
76	a mechanism for transporting user-to user-call control information in SIP?	[126]	o	c81
76A	interworking ISDN call control user information with SIP?	[126A]	c109	c109
77	The SIP P-Private-Network-Indication private-header (P-Header)?	[134]	o	o
78	the SIP P-Served-User private header for the 3GPP IM CN subsystem?	[133] 6	o	c93
79	the SIP P-Served-User header extension for Originating CDIV session case?	[239] 4	c126	c127
80	marking SIP messages to be logged?	[140]	o	c85
81	the 199 (Early Dialog Terminated) response code)	[142]	o	c86
82	message body handling in SIP?	[150]	m	m
83	indication of support for keep-alive	[143]	o	c88
84	SIP Interface to VoiceXML Media Services?	[145]	o	c89

85	common presence and instant messaging (CPIM): message format?	[151]	o	c91
86	instant message disposition notification?	[157]	o	c91
87	requesting answering modes for SIP?	[158]	o	c60
89	the early session disposition type for SIP?	[74B]	o	o
91	The Session-ID header?	[162]	o	c102
92	correct transaction handling for 2xx responses to Session Initiation Protocol INVITE requests?	[163]	c18	c18
93	addressing Record-Route issues in the Session Initiation Protocol (SIP)?	[164]	n/a	n/a
94	essential correction for IPv6 ABNF and URI comparison in RFC3261?	[165]	m	m
95	suppression of session initiation protocol REFER method implicit subscription?	[173]	o	c99
96	Alert-Info URNs for the Session Initiation Protocol?	[175]	o	o
97	multiple registrations?	Subclause 3.1	n/a	c103
98	the SIP P-Refused-URI-List private-header?	[183]	o	c104
99	request authorization through dialog Identification in the session initiation protocol?	[184]	o	c105
100	indication of features supported by proxy?	[190]	o	c106
101	registration of bulk number contacts?	[191]	o	c107
102	media control channel framework?	[146]	o	c108
103	S-CSCF restoration procedures?	Subclause 4.14	n/a	c110
104	SIP overload control?	[198]	o	c112
104A	feedback control?	[199]	c113	c113
104B	distribution of load filters?	[201]	c113	c114
105	handling of a 380 (Alternative service) response?	Subclauses 5.1.2A.1.1, 5.1.3.1, 5.1.6.8, and 5.2.10	n/a	c111
106	indication of adjacent network in the Via "received-realm" header field parameter?	[208]	o	c115
107	PSAP callback indicator?	[209]	o	c116
108	SIP URI parameter to indicate traffic leg?	[225]	o	c117
109	PCF or PCRF based P-CSCF restoration?	Subclause 4.14.2	n/a	c119
110	UDM/HSS or HSS based P-CSCF restoration?	Subclause 4.14.2	n/a	c120
111	the Relayed-Charge header field extension?	Subclause 7.2.12	n/a	c123
112	resource sharing?	Subclause 4.15	n/a	c124
113	the Cellular-Network-Info header extension?	Subclause 7.2.15	n/a	c125
114	the Priority-Share header field extension?	Subclause 7.2.16	n/a	c128
115	the Response-Source header field extension?	Subclause 7.2.17	n/a	o
116	authenticated identity management in the Session Initiation Protocol?	[252]	o	c129
117	a SIP response code for unwanted calls extension?	[254]	o	o
118	the 3GPP PS data off extension	Subclause 4.17	n/a	c130
119	Content-ID header field in Session Initiation Protocol (SIP)?	[256]	o	o
120	Next-Generation Pan-European eCall emergency service?	[244]	o	c62
121	the Attestation-Info header field extension?	Subclause 7.2.18	n/a	c132

122	the Origination-Id header field extension?	Subclause 7.2.19	n/a	c132
123	Dynamic services interactions?	Subclause 4.18	n/a	c133
124	the Additional-Identity header field extension?	Subclause 7.2.20	n/a	c135
125	RLOS?	Subclause 4.19	n/a	c136
126	the Priority-Verstat header field extension?	Subclause 7.2.21	n/a	c72

c2:	IF A.4/20 THEN o.1 ELSE n/a - - SIP specific event notification extension.
c3:	IF A.3/1 OR A.3/4 OR A.3A/81 THEN m ELSE n/a - - UE or S-CSCF functional entity or MSC Server enhanced for ICS.
c4:	IF A.3/4 THEN m ELSE IF A.3/7 THEN o ELSE n/a - - S-CSCF or AS functional entity.
c5:	IF A.4/16 THEN m ELSE o - - integration of resource management and SIP extension.
c6:	IF A.3/4 OR A.3/1 OR A.3A/81 THEN m ELSE n/a. - - S-CSCF or UE or MSC Server enhanced for ICS.
c7:	IF A.3/1 OR A.3/4 OR A.3/7A OR A.3/7B OR A.3/7D OR A.3/9B OR A.3/13B OR A.3A/83 OR A.3A/89 THEN m ELSE n/a - - UA or S-CSCF or AS acting as terminating UA or AS acting as originating UA or AS performing 3rd party call control or IBCF (IMS-ALG), ISC gateway function (IMS-ALG), SCC application server, ATCF (UA).
c8:	IF A.3/1 THEN (IF (A.3B/1 OR A.3B/2 OR A.3B/3 OR A.3B/4 OR A.3B/5 OR A.3B/6 OR A.3B/7 OR A.3B/8 OR A.3B/11 OR A.3B/12 OR A.3B/13 OR A.3B/14 OR A.3B/15) THEN m ELSE o) ELSE n/a - - UE behaviour (based on P-Access-Network-Info usage).
c9:	IF A.4/26 THEN o.2 ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c10:	IF A.4/26B THEN o.3 ELSE n/a - - application of privacy based on the received Privacy header.
c11:	IF A.3/1 OR A.3/6 OR A.3A/81 OR A.3A/81A OR A.3A/81B THEN o ELSE IF A.3/9B OR A.3/13B THEN m ELSE n/a - - UE or MGCF, IBCF (IMS-ALG), ISC gateway function (IMS-ALG), MSC Server enhanced for ICS, MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface.
c12:	IF A.3/7D OR A3A/84 OR A.3A/89 THEN m ELSE n/a - - AS performing 3rd-party call control, EATF, ATCF (UA).
c13:	IF A.3/1 OR A.3/2 OR A.3/4 OR A.3/9B OR A.3/11 OR A.3/12 OR A.3/13B OR A.3A/81 THEN m ELSE o - - UE or S-CSCF or IBCF (IMS-ALG) or E-CSCF or LRF or ISC gateway function (IMS-ALG) or MSC Server enhanced for ICS.
c14:	IF A.3/1 AND A4/2B AND (A.3B/1 OR A.3B/2 OR A.3B/3) THEN m ELSE IF A.3/2 THEN o ELSE n/a – UE and initiating sessions and GPRS IP-CAN or P-CSCF.
c15:	IF A.4/20 AND (A.3/4 OR A.3/9B OR A.3/11 OR A.3/13B) THEN m ELSE o – SIP specific event notification extensions and S-CSCF or IBCF (IMS-ALG) or E-CSCF or ISC gateway function (IMS-ALG).
c16:	IF A.4/20 AND (A.3/1 OR A.3/2 OR A.3/9B OR A.3/12 OR A.3/13B OR A.3A/81) THEN m ELSE o - - SIP specific event notification extension and UE or P-CSCF or IBCF (IMS-ALG) or MSC Server enhanced for ICS or LRF or ISC gateway function (IMS-ALG).
c17:	IF A.3/1 OR A.3/4 OR A.3A/81 THEN m ELSE n/a - - UE or S-CSCF or MSC Server enhanced for ICS.
c18:	IF A.4/2B THEN m ELSE n/a - - initiating sessions.
c19:	IF A.4/2B THEN o ELSE n/a - - initiating sessions.
c20:	IF A.3/1 AND (A.3D/1 OR A.3D/4) THEN m ELSE n/a - - UE and (IMS AKA plus IPsec ESP or SIP digest with TLS).
c21:	IF A.4/30 THEN o.4 ELSE n/a - - private header extensions to the session initiation protocol for the 3rd-Generation Partnership Project (3GPP).
c22:	IF A.4/30 AND (A.3/1 OR A.3/4 OR A.3A/81) THEN m ELSE n/a - - private header extensions to the session initiation protocol for the 3rd-Generation Partnership Project (3GPP) and S-CSCF or UE or MSC Server enhanced for ICS.
c23:	IF A.4/30 AND (A.3/1 OR A.3A/81) THEN o ELSE n/a - - private header extensions to the session initiation protocol for the 3rd-Generation Partnership Project (3GPP) and UE or MSC Server enhanced for ICS.
c24:	IF A.4/30 AND (A.3/4 OR A.3A/81 OR A.3A/81A) THEN m ELSE n/a - - private header extensions to the session initiation protocol for the 3rd-Generation Partnership Project (3GPP) and S-CSCF or MSC Server enhanced for ICS or MSC server enhanced for SRVCC using SIP interface.
c25:	IF A.4/30 AND (A.3A/81 OR A.3/4 OR A.3/6 OR A.3/7A OR A.3/7D OR A.3/9B OR A.3/13B OR A3A/84 OR A.3A/81A OR A.3A/81B) THEN m ELSE IF A.4/30 AND A.3/1 AND (A.3B/1OR A.3B/2 OR A.3B/3 OR A.3B/4 OR A.3B/5 OR A.3B/6 OR A.3A/7 OR A.3A/8 OR A.3B/11OR A.3B/12 OR A.3B/13 OR A.3B/14 OR A.3A/15 OR A.3B/41) THEN m ELSE IF A4/30 AND A.3/1 AND (A.3B/21 OR A.3B/22 OR A.3B/23 OR A.3B/24 OR A.3B/25 OR A.3B/26 OR A.3A/27 OR A.3A/28 OR A.3B/29 OR A.3B/30) THEN o ELSE n/a - - private header extensions to the session initiation protocol for the 3rd-Generation Partnership Project (3GPP), MSC Server enhanced for ICS, S-CSCF, MGCF or AS acting as terminating UA or AS acting as third-party call controller or IBCF (IMS-ALG), ISC gateway function (IMS-ALG), UE, EATF, P-Access-Network-Info values or MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface.
c26:	IF A.4/30 AND (A.3A/81 OR (A.3/4 AND A.4/2) OR A.3/6 OR A.3/7A OR A.3/7B or A.3/7D OR A.3/9B OR A.3/13B OR A3A/84 OR A.3A/89 OR A.3A/81A OR A.3A/81B) THEN m ELSE n/a - - private header extensions to the session initiation protocol for the 3rd-Generation Partnership Project (3GPP) MSC Server enhanced for ICS, S-CSCF, registrar, MGCF, AS acting as a terminating UA, or AS acting as an originating UA, or AS acting as third-party call controller, IBCF (IMS-ALG), ISC gateway function (IMS-ALG), EATF, ATCF (UA), MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface.
c27:	IF A.3/7D OR A.3/9D THEN o ELSE x - - AS performing 3rd party call control, IBCF (Privacy).
c29:	IF A.4/40A OR A.4/40B OR A.4/40C OR A.4/40D OR A.4/40E OR A.4/40F THEN m ELSE n/a - - support of any directives within caller preferences for the session initiation protocol.
c30:	IF A.3A/1 OR A.3A/2 THEN m ELSE IF A.3/1 OR A.3/11A OR A.3/2A THEN o ELSE n/a - - presence server, presence user agent, UE, AS, E-CSCF acting as UA, P-CSCF (IMS-ALG).

c33:	IF A.3/9B OR A.3/12 OR A.3/13B OR A.3A/81 OR A.3A/11 OR A.3A/12 OR A.4/44 OR A.3A/81A OR A.3A/81B THEN m ELSE o - - IBCF (IMS-ALG) or LRF or ISC gateway function (IMS-ALG) or MSC Server enhanced for ICS or conference focus or conference participant or the Session Inititation Protocol (SIP) "Replaces" header, MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface.
c34:	IF A.4/44 OR A.4/45 OR A.3/9B OR A.3/13 THEN m ELSE n/a - - the Session Inititation Protocol (SIP) "Replaces" header or the Session Inititation Protocol (SIP) "Join" header or IBCF (IMS-ALG) or ISC gateway function (IMS-ALG).
c35:	IF A.3/4 OR A.3/9B OR A.3/13B OR A.3A/82 OR A.3A/83 OR A.3A/21 OR A.3A/22 OR A3A/84 THEN m ELSE IF (A.3/1 OR A.3/6 OR A.3/7 OR A.3/8 OR A.3A/81 OR A.3A/81A OR A.3A/81B) THEN o ELSE n/a - - S-CSCF or IBCF (IMS-ALG) or ISC gateway function (IMS-ALG) functional entities or ICS user agent or SCC application server or CSI user agent or CSI application server, UE or MGCF or AS or MRFC functional entity or MSC Server enhanced for ICS or EATF or MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface.
c37	IF A.4/47 THEN o.3 ELSE n/a - - an extension to the session initiation protocol for request history information.
c38:	IF A.4/2B AND (A.3A/11 OR A.3A/12 OR A.3/7D) THEN m ELSE IF A.4/2B THEN o ELSE n/a - - initiating sessions, conference focus, conference participant, AS performing 3rd party call control.
c39:	IF A.3/1 THEN m ELSE IF A.3/7B OR A.3/7D OR A.3/9 THEN o ELSE n/a - - UE, AS acting as an originating UA, or AS acting as third-party call controller, IBCF.
c40	IF A.3/4 OR (A.3/1 AND NOT A.3C/1) OR A.3A/81 OR A.4/22 THEN m ELSE IF (A.3/7A OR A.3/7B OR A.3/7D) THEN o ELSE n/a - - S-CSCF, UE, UE performing the functions of an external attached network, MSC Server enhanced for ICS, notifier of event information, AS, AS acting as terminating UA, or redirect server, AS acting as originating UA, AS performing 3rd party call control.
c42:	IF A.3/1 THEN n/a ELSE o - - UE.
c43:	IF A.4/2B THEN o ELSE n/a - - initiating sessions.
c44:	IF A.4/2C THEN m ELSE o - - initiating a session which require local and/or remote resource reservation.
c45:	IF A.4/97 THEN m ELSE n/a - - multiple registrations.
c46	IF A.3/1 OR A.3/4 THEN o ELSE n/a - - UE, S-CSCF.
c47:	IF A.4/27 THEN o ELSE n/a - - a messaging mechanism for the Session Initiation Protocol (SIP).
c48:	IF A.3A/32 AND A.4/27 THEN m ELSE IF A.4/27 THEN o ELSE n/a - - messaging list server, a messaging mechanism for the Session Initiation Protocol (SIP).
c49:	IF A.3/1 OR A.3/9B OR A.3/13B OR A.3A/81 OR A.3/11 OR A.3/12 OR A3A/84 THEN m ELSE o - - UE, IBCF (IMS-ALG), ISC gateway function (IMS-ALG), MSC Server enhanced for ICS, E-CSCF, LRF, EATF.
c50:	IF A.3A/81 OR A.3A/81A OR A.3A/81B THEN n/a ELSE IF A.4/15 THEN o ELSE n/a - - MSC Server enhanced for ICS, MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface, the REFER method.
c51:	IF A.4/2B THEN o ELSE n/a - - initiating a session.
c52:	IF A.3A/11 AND A.4/2B THEN m ELSE IF A.4/2B THEN o ELSE n/a - - conference focus, initiating a session.
c53:	IF A.3A/81 THEN n/a ELSE IF A.4/20 THEN o ELSE n/a - - MSC Server enhanced for ICS, SIP specific event notification.
c54:	IF A.3/1 OR A.3/6 OR A.3/7A OR A.3/7D OR A.3/9 THEN o, ELSE n/a - - UE, MGCF, AS acting as originating UA, AS performing 3rd party call control, IBCF.
c55:	IF A.4/67 THEN m ELSE n/a - - number portability parameters for the 'tel' URI.
c57:	IF A.4/67 THEN m ELSE n/a - - number portability parameters for the 'tel' URI.
c58:	IF A.3/9B OR A.3/13B OR A.3/6 OR A.3A/81 OR A.3A/81A OR A.3A/81B THEN m ELSE o - - IBCF (IMS-ALG), ISC gateway function (IMS-ALG), MGCF, MSC Server enhanced for ICS, MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface.
c59:	IF A.3/4 THEN m ELSE IF (A.3/1 OR A.3/6 OR A.3/7A OR A.3/7B OR A.3/7D OR A.3/8) THEN o ELSE n/a - - S-CSCF, UE, MGCF, AS, AS acting as terminating UA, or redirect server, AS acting as originating UA, AS performing 3rd party call control, or MRFC.
c60:	IF A.3/9B OR A.3/13B THEN m ELSE IF A.3/1 OR A.3/7A OR A.3/7B OR A.3/7D THEN o ELSE n/a - - IBCF (IMS-ALG), ISC gateway function (IMS-ALG), UE, AS acting as terminating UA, AS acting as originating UA, AS performing 3rd party call control.
c61:	IF (A.3/1 OR A.3A/81 OR A.3/6 OR A.3/7A OR A.3/7B OR A.3/7D OR A.3/8 OR A.3/9B OR A.3/13 OR A3A/84 OR A.3A/81A OR A.3A/81B) THEN o ELSE n/a - - UE, MSC Server enhanced for ICS, MGCF, AS, AS acting as terminating UA, or redirect server, AS acting as originating UA, AS performing 3rd party call control, or MRFC or IBCF (IMS-ALG), ISC gateway function (IMS-ALG), EATF, MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface.
c62:	IF A.3/1 THEN o ELSE n/a - - UE.
c68:	IF A.3/2A OR A.3/9 OR A.4/69 OR A.3A/83 THEN m ELSE o - - P-CSCF (IMS-ALG), IBCF, extending the session initiation protocol Reason header for preemption events and Q.850 causes, SCC application server.
c69:	IF A.4/70 THEN o ELSE n/a - - communications resource priority for the session initiation protocol.
c70:	IF A.3/9B OR A.3/13B OR A.3A/102 OR A.3A/103 THEN m ELSE IF A.3/1 OR A.3/6 OR A.3/7 OR A.3/7A OR A.3/7B OR A.3/7D OR A.3A/81 OR A.3A/81A OR A.3A/81B THEN o ELSE n/a - - IBCF (IMS-ALG), ISC gateway function (IMS-ALG), MCPTT client, MCPTT server, UE, MGCF, AS, AS acting as terminating UA,

	or redirect server, AS acting as originating UA, AS performing 3rd party call control, MSC Server enhanced for ICS, MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface.
c72:	IF A.4/70 THEN o ELSE n/a - - communications resource priority for the session initiation protocol

c74:	IF A.3/4 OR A.3/1 THEN o ELSE n/a. - - S-CSCF or UE.
c75:	IF A.3/1 THEN o ELSE n/a. - - UE.
c76:	IF A.4/75A OR A.4/75B THEN m ELSE n/a - - a relay within the framework for consent-based communications in SIP, a recipient within the framework for consent-based communications in SIP.
c77:	IF A.4/59 OR A.4/61 OR A.4/62 OR A.4/63 THEN m ELSE o - - multiple-recipient MESSAGE requests in the session initiation protocol, referring to multiple resources in the session initiation protocol, conference establishment using request-contained lists in the session initiation protocol, subscriptions to request-contained resource lists in the session initiation protocol.
c78:	IF (A.4/59 OR A.4/61 OR A.4/62 OR A.4/63) AND (A.3A/11 OR A.3A/31) THEN m ELSE o - - multiple-recipient MESSAGE requests in the session initiation protocol, referring to multiple resources in the session initiation protocol, conference establishment using request-contained lists in the session initiation protocol, subscriptions to request-contained resource lists in the session initiation protocol, conference focus, messaging application server.
c79:	IF A.3/9B OR A.3/13B OR (A.3/1 AND (A.4/2B OR A.4/15 OR A.4/20 OR A.4/27)) THEN m ELSE IF A.3/6 OR A.3/7A OR A.3/7D THEN o ELSE n/a - - IBCF (IMS-ALG), ISC gateway function (IMS-ALG), UE, initiating a session, the REFER method, SIP specific event notification, a messaging mechanism for the Session Initiation Protocol (SIP), AS acting as terminating UA, or redirect server, AS performing 3rd party call control.
c80:	IF A.4/2B OR A.4/15 OR A.4/20 OR A.4/27 THEN m ELSE n/a - - initiating a session, the REFER method, SIP specific event notification, a messaging mechanism for the Session Initiation Protocol (SIP).
c81:	IF A.3/1 OR A.3/6 OR A.3/7A OR A.3/7B OR A.3/7D THEN o ELSE IF A.3/9B OR A.3/13B THEN m ELSE n/a - - UE, MGCF, AS acting as terminating UA, or redirect server, AS acting as originating UA, AS performing 3rd party call control, IBCF (IMS-ALG), ISC gateway function (IMS-ALG).
c82:	IF A.3/6 OR A.3A/81 OR A.3A/81A OR A.3A/81B THEN m ELSE n/a - - MGCF, MSC server enhanced for ICS, MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface.
c85:	IF A.3/1 OR A.3/6 OR A.3A/81 OR A.3A/81A OR A.3A/81B OR A.3/2 OR A.3/7B THEN m ELSE n/a - - UE, MGCF, MSC Server enhanced for ICS, MSC Server enhanced for SRVCC using SIP interface, MSC Server enhanced for DRVCC using SIP interface, P-CSCF, AS acting as originating UA.
c86:	IF A.4/3 OR A.4/4 THEN m ELSE n/a - - client behaviour for INVITE requests, server behaviour for INVITE requests.
c87:	IF A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C THEN m ELSE o - - IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling).
c88:	IF A.3/1 OR A.3/2 THEN m ELSE o - - UE, P-CSCF.
c89:	IF A.3/7A OR A.3/8 THEN o ELSE n/a - - AS performing 3rd party call control, MRFC.
c90:	IF A.4/13 OR A.3A/53 OR A.3A/54 OR A.3A/91 OR A.3A/85 OR A.3A/86 THEN m ELSE o - - SIP INFO method and package framework, advice of charge application server, advice of charge UA client, malicious communication identification application server, in-dialog overlap signalling application server, in-dialog overlap signalling UA client.
c91:	IF A.3A/61 OR A.3A/62 OR A.3A/63 OR A.3A/71 THEN m ELSE o - - SM-over-IP sender, SM-over-IP receiver, IP-SM-GW, IP-SM-GW.
c93:	IF A.3/7B OR A.3/7D OR A3A/84 THEN o ELSE n/a - - AS acting as originating UA, AS performing 3rd party call control, EATF.
c94:	IF A.3/4 OR A.3/7A OR A.3/7D THEN o ELSE n/a - - S-CSCF and AS acting as terminating UA or redirect server or AS performing 3rd party call control.
c96:	IF A.4/30 THEN o ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c97:	IF (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C) AND A.4/30 THEN m ELSE IF (A.3/7D OR A.3/11 OR A.3C/1) AND A.4/30 THEN o ELSE n/a - - IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling), AS performing 3rd party call control, E-CSCF, UE performing the functions of an external attached network and extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c98:	IF A.3/7D OR A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C OR A.3C/1 OR A3A/84 OR A.3A/89 THEN m ELSE n/a - - AS performing 3rd party call control, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling), UE performing the functions of an external attached network, EATF, ATCF (UA).
c99:	IF A.4/15 AND (A.3/9B OR A.3/9C OR A.13/B OR A.13/C) THEN m ELSE IF A.4/15 THEN o ELSE n/a - - the REFER method, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling).
c100:	IF A.3/6 OR A.3A/57 OR A.3A/58 OR A.3A/59 OR A.3A/60 OR A.3A/81 OR A.3A/81A OR A.3A/81B THEN m ELSE o - - MGCF, customized alerting tones application server, customized alerting tones UA client, customized ringing signal application server, customized ringing signal UA client, MSC server enhanced for ICS, MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface.
c101:	IF A.3D/30 OR A.3D/20A OR A.3D/20B OR A.3D/20C OR A.3D/31 THEN m ELSE n/a - - end-to-access-edge media security using SDES, end-to-access-edge media security for MSRP using TLS and certificate fingerprints, end-to-access-edge media security for BFCP using TLS and certificate fingerprints, end-to-

	access-edge media security for UDPTL using DTLS and certificate fingerprints, end-to-access-edge media security for RTP media using DTLS-SRTP and certificate fingerprints.
c102:	IF A.3A/11 OR A.3A/12 OR A.3/9 THEN m ELSE n/a - - conference focus, conference participant, IBCF.
c103:	IF A.3/1 THEN o ELSE IF A.3/2 OR A.3/4 THEN m ELSE n/a - - UE, P-CSCF, S-CSCF.
c104:	IF A.3/9B OR A.3/13B THEN m ELSE IF A.3/7A OR A.3/7B OR A.3/7D THEN o ELSE n/a - - IBCF (IMS-ALG), ISC gateway function (IMS-ALG), AS acting as terminating UA, AS acting as originating UA, AS performing 3rd party call control.

c105:	IF A.3/9B OR A.3/13B OR A.3A/82 OR A.3A/83 OR A.3A/87 OR A.3A/89 THEN m ELSE o - - IBCF (IMS-ALG), ISC gateway function (IMS-ALG), ICS user agent, SCC application server, Session continuity controller UE, ATCF (UA).
c106:	IF A.3A/50A OR A.3A/83 OR A.3A/89 THEN m ELSE o - - Multimedia telephony application server, SCC application server, ATCF (UA).
c107:	IF A.3C/1 OR A.4/2 THEN o ELSE n/a - - UE performing the functions of an external attached network, registrar.
c108:	IF A.3/7 OR A.3/8 OR A.3/8A THEN o ELSE n/a - - AS, MRFC, MRB.
c109:	IF A.4/76 THEN o ELSE n/a - - a mechanism for transporting user to user call control information in SIP.
c110:	IF A.3/1 THEN m ELSE IF A.3/2 OR A.3/3 OR A.3/4 THEN o ELSE n/a - - UE, P-CSCF, I-CSCF, S-CSCF.
c111:	IF A.3/1 OR A.3/2 THEN m ELSE n/a - - UE, P-CSCF.
c112:	IF NOT (A.3/1 AND NOT A.3C/1) THEN o ELSE n/a - - not UE, UE performing the functions of an external attached network.
c113:	IF A.4/104 THEN o.7 ELSE n/a - - SIP overload control.
c114:	IF A.4/104 THEN IF A.3/4 OR A.3/7 OR A.3/10 THEN o.7 ELSE n/a - - SIP overload control, S-CSCF, AS, additional routeing functionality.
c115:	IF A.3/6 OR A.3/9 OR A.3/7 THEN o ELSE n/a - - MGCF, IBCF, AS
c116:	IF A.3/2A OR A.3/6 OR A.3/7 OR A.3/9 THEN o ELSE IF A.3/1 THEN x ELSE n/a - - P-CSCF (IMS-ALG), MGCF, AS, IBCF, UE.
c117	IF A.3/2 OR A.3/4 OR OR A.3/9 OR A.3A/81 OR A.3A/83 OR A.3A/89 OR A.3A/81A THEN o ELSE n/a - - P-CSCF, S-CSCF, IBCF, MSC server enhanced for ICS, SCC application server, ATCF (UA), MSC server enhanced for SRVCC using SIP interface.
c118:	IF A.4/49 THEN o ELSE n/a - - session initiation protocol URIs for applications such as voicemail and interactive voice response (NOTE 3).
c119:	IF A.3/2A OR A.3/9 THEN o ELSE n/a - - P-CSCF (IMS-ALG), IBCF.
c120:	IF A.3/2A OR A.3/9 THEN o ELSE n/a - - P-CSCF (IMS-ALG), IBCF.
c121:	IF A.4/15 THEN m ELSE n/a - - the REFER method.
c122:	IF A.4/22 THEN m ELSE n/a - - act as a notifier.
c123:	IF A.4/111 AND (A.3/7A OR A.3/7B OR A.3/9A OR A.3/9B OR A.3/13A OR A.3/13B) THEN m ELSE IF A.3/4 OR A.3/7 OR A.3A/102 THEN o ELSE n/a.-.-.the Relayed-Charge header field extension, AS acting as terminating UA, or redirect server, AS acting as originating UA, IBCF (THIG), IBCF (IMS-ALG), ISC gateway function (THIG), ISC gateway function (THIG), S-CSCF, AS, transit function.
c124:	IF A.3/2A OR A.3/9B OR A.3/7 THEN o ELSE n/a - - P-CSCF (IMS-ALG), I-BCF (IMS-ALG), AS.
c125:	IF (A.3/4 OR A.3/6 OR A.3/7A OR A.3/7D OR A.3/9B OR A.3/13B OR A.3A/84 OR A.3A/89 OR A.3/2A OR A.3/8 OR A.3/11A) THEN m ELSE IF A.3/1 AND (A.3B/11 OR A.3B/12 OR A.3B/13 OR A.3B/14 OR A.3B/15) AND (A.3B/1 OR A.3B/2 OR A.3B/3 OR A.3B/4 OR A.3B/5 OR A.3B/6 OR A.3B/7 OR A.3B/8 OR A.3B/9) THEN m ELSE n/a. - - S-CSCF, MGCF, AS acting as terminating UA, AS acting as third-party call controller, IBCF (IMS-ALG), ISC gateway function (IMS-ALG), EATF, ATCF acting as UA, P-CSCF (IMS-ALG), MRFC, E-CSCF acting as UA, UE.
c126:	IF A.4/78 THEN o ELSE n/a - - the SIP P-Served-User private header for the 3GPP IM CN subsystem.
c127:	IF A.4/78 THEN m ELSE n/a - - the SIP P-Served-User private header for the 3GPP IM CN subsystem.
c128:	IF A.3/2A OR A.3/9B OR A.3/7 OR A.3A/103 THEN o ELSE n/a - - P-CSCF (IMS-ALG), IBCF (IMS-ALG), AS, MCPTT server.
c129	IF A.3/6 OR A.3/7 OR A.3/9 OR A.3A/81 OR A.3A/81A OR A.3A/81B THEN o ELSE n/a - - MGCF, AS, IBCF, MSC Server enhanced for ICS, MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface.
c130:	IF A.3/1 OR A.3/7 THEN o ELSE n/a - - UE, AS,
c131:	IF A.3/6 OR A.3A/81 OR A.3A/81A OR A.3A/81B THEN o ELSE n/a - - MGCF, MSC server enhanced for ICS, MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface.
c132:	IF A.3/6 OR A.3/7 OR A.3/9 THEN o ELSE n/a - - MGCF, AS, IBCF.
c133:	IF A.3/2 OR A.3/7 OR A.3/9 THEN o ELSE n/a - - P-CSCF, AS, IBCF.
c134	IF A.4/60 THEN o ELSE n/a - - the Geolocation header field
c135:	IF A.3/1 OR A.3/2 OR A.3/7 OR A.3/9 THEN o ELSE n/a - - UE, P-CSCF, AS, IBCF.
c136:	IF A.3/1 THEN o ELSE n/a - - UE.
c137:	IF A.3/7 OR A.3/9 THEN o ELSE n/a - - AS, IBCF.
o.1:	At least one of these capabilities is supported.
o.2:	At least one of these capabilities is supported.
o.3:	At least one of these capabilities is supported.
o.4:	At least one of these capabilities is supported.
o.5:	At least one of these capabilities is supported.
o.6:	It is mandatory to support at least one of these items.
o.7:	At least one of these capabilities is supported.

NOTE 1: An AS acting as a proxy may be outside the trust domain, and therefore not able to support the capability for that reason; in this case it is perfectly reasonable for the header to be passed on transparently, as specified in the PDU parts of the profile.
NOTE 2: If a UE is unable to become engaged in a service that potentially requires the ability to identify and interact with a specific UE even when multiple UEs share the same single Public User Identity then the UE support can be "o" instead of "m". Examples include telemetry applications, where point-to-point communication is desired between two users.
NOTE 3: AS performing a service number translation (eg. Freephone)

Prerequisite A.4/20 - - SIP specific event notification

Table A.4A: Supported event packages

Item	Does the implementation support	Subscriber			Notifier		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	reg event package?	[43]	c1	c3	[43]	c2	c4
1A	reg event package extension for GRUUs?	[94]	c1	c25	[94]	c2	c4
2	refer package?	[36] 3	c13	c13	[36] 3	c13	c13
3	presence package?	[74] 6	c1	c5	[74] 6	c2	c6
4	eventlist with underlying presence package?	[75], [74] 6	c1	c7	[75], [74] 6	c2	c8
5	presence.winfo template-package?	[72] 4	c1	c9	[72] 4	c2	c10
6	xcap-diff package?	[77] 4	c1	c11	[77] 4	c2	c12
7	conference package?	[78] 3	c1	c21	[78] 3	c1	c22
8	message-summary package?	[65]	c1	c23	[65] 3	c2	c24
9	poc-settings package?	[110]	c1	c26	[110]	c2	c27
11	dialog event package?	[171]	c1	c14	[171]	c2	c15
12	load-control package?	[201]	c29	c30	[201]	c29	c31

c1:	IF A.4/23 THEN o ELSE n/a - - acting as the subscriber to event information.
c2:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.
c3:	IF A.3/1 OR A.3A/81 OR A.3/2 THEN m ELSE IF A.3/7 THEN o ELSE n/a - - UE, MSC Server enhanced for ICS, P-CSCF, AS.
c4:	IF A.3/4 THEN m ELSE IF A.3C/1 THEN o ELSE n/a - - S-CSCF, UE performing the functions of an external attached network.
c5:	IF A.3A/3 OR A.3A/4 THEN m ELSE IF A.4/23 OR A.3/12 THEN o ELSE n/a - - resource list server or watcher, acting as the subscriber to event information, LRF.
c6:	IF A.3A/1 THEN m ELSE IF A.4/22 OR A.3/11A THEN o ELSE n/a - - presence server, acting as the notifier of event information, E-CSCF acting as UA.
c7:	IF A.3A/4 THEN m ELSE IF A.4/23 THEN o ELSE n/a - - watcher, acting as the subscriber to event information.
c8:	IF A.3A/3 THEN m ELSE IF A.4/22 THEN o ELSE n/a - - resource list server, acting as the notifier of event information.
c9:	IF A.3A/2 THEN m ELSE IF A.4/23 THEN o ELSE n/a - - presence user agent, acting as the subscriber to event information.
c10:	IF A.3A/1 THEN m ELSE IF A.4/22 THEN o ELSE n/a - - presence server, acting as the notifier of event information.
c11:	IF A.3A/2 OR A.3A/4 OR A.3A/56 THEN o ELSE IF A.4/23 THEN o ELSE n/a - - presence user agent or watcher or Ut reference point XCAP client for supplementary services, acting as the subscriber to event information.
c12:	IF A.3A/1 OR A.3A/3 OR A.3A/55 THEN m ELSE IF A.4/22 THEN o ELSE n/a - - presence server or resource list server or Ut reference point XCAP server for supplementary services, acting as the notifier of event information.
c13:	IF A.4/15 THEN m ELSE n/a - - the REFER method.
c14:	IF A.3/12 OR A.3A/87 THEN m ELSE IF A.3/1 OR A.3/7B OR A.3/7D THEN o ELSE n/a - - LRF, session continuity controller UE, UE, AS acting as originating UA, AS performing 3rd party call control.
c15:	IF A.3/11 OR A.3A/83 THEN m ELSE IF A.3/1 OR A.3/7A OR A.3/7D THEN o ELSE n/a - - E-CSCF, SCC application server, UE, AS acting as terminating UA, or redirect server, AS performing 3rd party call control.
c21:	IF A.3A/12 THEN m ELSE IF A.4/23 THEN o ELSE n/a - - conference participant or acting as the subscriber to event information.
c22:	IF A.3A/11 THEN m ELSE IF A.4/22 THEN o ELSE n/a - - conference focus or acting as the notifier of event information.
c23:	IF A.3A/52 THEN m ELSE (A.3/1 OR A.3/7A OR A.3/7B) AND A.4/23 THEN o ELSE n/a - - message waiting indication subscriber UA, UE, AS acting as terminating UA, or redirect server, AS acting as originating UA all as subscriber of event information.
c24:	IF A.3A/52 THEN m ELSE (A.3/1 OR A.3/7A OR A.3/7B) AND A.4/22 THEN o ELSE n/a - - message waiting indication notifier UA, UE, AS acting as terminating UA, or redirect server, AS acting as originating UA all as notifier of event information.
c25:	IF A.4A/1 THEN (IF A.3/1 AND A.4/53 THEN m ELSE o) ELSE n/a - - reg event package, UE, reg event package extension for GRUUs.
c26:	IF (A.3/7B OR A.3/1) AND (A.4/23 OR A.4/41) THEN o ELSE n/a - - AS acting as originating UA, UE, acting as the subscriber to event information, an event state publication extension to the session initiation protocol.
c27:	IF (A.4/22 OR A.4/41) AND A.3/1 THEN o ELSE n/a - - UE, acting as the notifier of event information, an event state publication extension to the session initiation protocol.
c28:	IF A.3/1 OR A.3A/81 OR A.3/2 OR A.3/7B THEN m ELSE n/a - - UE, MSC Server enhanced for ICS, P-CSCF, AS acting as originating UA.
c29:	IF A.4/104B THEN m ELSE n/a - - distribution of load filters.
c30:	IF A.4/104B THEN IF A.3/4 OR A.3/7 OR A.3/9 THEN m ELSE n/a - - distribution of load filters. S-CSCF, IBCF, AS.
c31:	IF A.4/104B THEN If A.3/7 THEN m ELSE n/a - - distribution of load filters, AS.

Prerequisite A.4/13 - - SIP INFO method and package framework.

Table A.4B: Supported info packages

Item	Does the implementation support	Sender			Receiver		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	DTMF info package?	7.12.1	n/a	c1	7.12.1	n/a	c1
2	g.3gpp.mid-call?	[8M]	n/a	c2	[8M]	n/a	c3
3	g.3gpp.ussd?	[8W]	n/a	c4	[8W]	n/a	c4
4	g.3gpp.current-location-discovery info package ?	subclause 7.12.2.1	n/a	c5	subclause 7.12.2.1	n/a	c6
5	EmergencyCallData.eCall.MSD Info-Package	[244] 14.9	m	c7	[244] 14.9	m	c7
c1:	IF A.3/6 OR A.3A/57 OR A.3A/58 OR A.3A/59 OR A.3A/60 THEN m ELSE o - - MGCF, customized alerting tones application server, customized alerting tones UA client, customized ringing signal application server, customized ringing signal UA client.						
c2:	IF A.3A/83 THEN o ELSE n/a - - SCC application server.						
c3:	IF A.3A/81 OR A.3A/81B THEN o ELSE n/a - - MSC server enhanced for ICS, MSC server enhanced for DRVCC using SIP interface.						
c4:	IF A.3A/92 OR A.3A/93 THEN m ELSE n/a - - USSI UE, USSI AS.						
c5:	IF A.3/11A OR A.3/2A THEN o ELSE n/a - - E-CSCF acting as UA, P-CSCF (IMS-ALG).						
c6:	IF (A.3/1 AND (A.3B/11 OR A.3B/12 OR A.3B/13 OR A.3B/14 OR A.3B/15)) OR A.3/2A THEN o ELSE n/a - - UE, IEEE-802.11, IEEE-802.11a, IEEE-802.11b, IEEE-802.11g, IEEE-802.11n, IEEE-802.11ac, P-CSCF (IMS-ALG).						
c7:	IF (A.3/1 AND A.4/120) THEN m ELSE n/a - - UE, Next-Generation Pan-European eCall emergency service.						

Table A.4C: Supported media control packages

Item	Does the implementation support	Sender			Receiver		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	msc-ivr/1.0	[147]		c1	[147]		c2
2	msc-mixer/1.0	[148]		c1	[148]		c2
3	mrb-publish/1.0	[192]		c3	[192]		c4
c1:	IF A.3/7D THEN o ELSE n/a - - AS performing 3rd party call control.						
c2:	IF A.3/8 THEN o ELSE n/a - - MRFC.						
c3:	IF A.3/8 THEN o ELSE n/a - - MRFC.						
c4:	IF A.3/8A THEN o ELSE n/a - - MRB.						

A.2.1.3 PDUs

Table A.5: Supported methods

Item	PDU	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	ACK request	[26] 13	c10	c10	[26] 13	c11	c11
2	BYE request	[26] 15.1	c12	c12	[26] 15.1	c12	c12
3	BYE response	[26] 15.1	c12	c12	[26] 15.1	c12	c12
4	CANCEL request	[26] 9	m	m	[26] 9	m	m
5	CANCEL response	[26] 9	m	m	[26] 9	m	m
6	INFO request	[25] 4.2	c21	c21	[25] 4.2	c21	c21
7	INFO response	[25] 4.2	c21	c21	[25] 4.2	c21	c21
8	INVITE request	[26] 13	c10	c10	[26] 13	c11	c11
9	INVITE response	[26] 13	c11	c11	[26] 13	c10	c10
9A	MESSAGE request	[50] 4	c7	c7	[50] 7	c7	c7
9B	MESSAGE response	[50] 4	c7	c7	[50] 7	c7	c7
10	NOTIFY request	[28] 8.1.2	c4	c4	[28] 8.1.2	c3	c3
11	NOTIFY response	[28] 8.1.2	c3	c3	[28] 8.1.2	c4	c4
12	OPTIONS request	[26] 11	m	m	[26] 11	m	m
13	OPTIONS response	[26] 11	m	m	[26] 11	m	m
14	PRACK request	[27] 6	c5	c5	[27] 6	c5	c5
15	PRACK response	[27] 6	c5	c5	[27] 6	c5	c5
15A	PUBLISH request	[70] 11.1.3	c20	c20	[70] 11.1.3	c20	c20
15B	PUBLISH response	[70] 11.1.3	c20	c20	[70] 11.1.3	c20	c20
16	REFER request	[36] 3	c1	c1	[36] 3	c1	c1
17	REFER response	[36] 3	c1	c1	[36] 3	c1	c1
18	REGISTER request	[26] 10	c8	c8	[26] 10	c9	c9
19	REGISTER response	[26] 10	c9	c9	[26] 10	c8	c8
20	SUBSCRIBE request	[28] 8.1.1	c3	c3	[28] 8.1.1	c4	c4
21	SUBSCRIBE response	[28] 8.1.1	c4	c4	[28] 8.1.1	c3	c3
22	UPDATE request	[29] 6.1	c6	c6	[29] 6.2	c6	c6
23	UPDATE response	[29] 6.2	c6	c6	[29] 6.1	c6	c6

c1: IF A.4/15 THEN m ELSE n/a - - the REFER method extension.
c3: IF A.4/23 THEN m ELSE n/a - - recipient for event information.
c4: IF A.4/22 THEN m ELSE n/a - - notifier of event information.
c5: IF A.4/14 THEN m ELSE n/a - - reliability of provisional responses extension.
c6: IF A.4/17 THEN m ELSE n/a - - the SIP update method extension.
c7: IF A.4/27 THEN m ELSE n/a - - the SIP MESSAGE method.
c8: IF A.4/1 THEN m ELSE n/a - - client behaviour for registration.
c9: IF A.4/2 THEN m ELSE n/a - - registrar.
c10: IF A.4/3 THEN m ELSE n/a - - client behaviour for INVITE requests.
c11: IF A.4/4 THEN m ELSE n/a - - server behaviour for INVITE requests.
c12: IF A.4/5 THEN m ELSE n/a - - session release.
c20: IF A.4/41 THEN m ELSE n/a - - event state publication extension.
c21: IF A.4/13 OR A.4/13A THEN m ELSE n/a - - SIP INFO method and package framework, legacy INFO usage.

A.2.1.4 PDU parameters

A.2.1.4.1 Status-codes

Table A.6: Supported status-codes

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	100 (Trying)	[26] 21.1.1	c21	c21	[26] 21.1.1	c11	c11
101	1xx response	[26] 21.1	p21	p21	[26] 21.1	p21	p21
101A	18x response	[26] 21.1	p21	p21	[26] 21.1	p21	p21
2	180 (Ringing)	[26] 21.1.2	c2	c2	[26] 21.1.2	c1	c1
3	181 (Call Is Being Forwarded)	[26] 21.1.3	c2	c2	[26] 21.1.3	c1	c1
4	182 (Queued)	[26] 21.1.4	c2	c2	[26] 21.1.4	c1	c1
5	183 (Session Progress)	[26] 21.1.5	c34	c34	[26] 21.1.5	c1	c1
5A	199 (Early Dialog Terminated)	[142] 11.1	c32	c32	[142] 11.1	c32	c32
102	2xx response	[26] 21.2	p22	p22	[26] 21.1	p22	p22
6	200 (OK)	[26] 21.2.1	m	m	[26] 21.2.1	m	m
7	202 (Accepted)	[28] 8.3.1	c36	c36	[28] 8.3.1	c37	c37
103	3xx response	[26] 21.3	p23	p23	[26] 21.1	p23	p23
8	300 (Multiple Choices)	[26] 21.3.1	m	m	[26] 21.3.1	m	m
9	301 (Moved Permanently)	[26] 21.3.2	m	m	[26] 21.3.2	m	m
10	302 (Moved Temporarily)	[26] 21.3.3	m	m	[26] 21.3.3	m	m
11	305 (Use Proxy)	[26] 21.3.4	m	m	[26] 21.3.4	m	m
12	380 (Alternative Service)	[26] 21.3.5	m	m	[26] 21.3.5	m	m
104	4xx response	[26] 21.4	p24	p24	[26] 21.4	p24	p24
13	400 (Bad Request)	[26] 21.4.1	m	m	[26] 21.4.1	m	m
14	401 (Unauthorized)	[26] 21.4.2	o	c12	[26] 21.4.2	m	m
15	402 (Payment Required)	[26] 21.4.3	n/a	n/a	[26] 21.4.3	n/a	n/a
16	403 (Forbidden)	[26] 21.4.4	m	m	[26] 21.4.4	m	m
17	404 (Not Found)	[26] 21.4.5	m	m	[26] 21.4.5	m	m
18	405 (Method Not Allowed)	[26] 21.4.6	m	m	[26] 21.4.6	m	m
19	406 (Not Acceptable)	[26] 21.4.7	m	m	[26] 21.4.7	m	m
20	407 (Proxy Authentication Required)	[26] 21.4.8	o	o	[26] 21.4.8	m	m
21	408 (Request Timeout)	[26] 21.4.9	c2	c2	[26] 21.4.9	m	m
22	410 (Gone)	[26] 21.4.10	m	m	[26] 21.4.10	m	m
22A	412 (Conditional Request Failed)	[70] 11.2.1	c20	c20	[70] 11.2.1	c20	c20
23	413 (Request Entity Too Large)	[26] 21.4.11	m	m	[26] 21.4.11	m	m
24	414 (Request-URI Too Large)	[26] 21.4.12	m	m	[26] 21.4.12	m	m
25	415 (Unsupported Media Type)	[26] 21.4.13	m	m	[26] 21.4.13	m	m
26	416 (Unsupported URI Scheme)	[26] 21.4.14	m	m	[26] 21.4.14	m	m
26A	417 (Unknown Resource Priority)	[116] 4.6.2	c24	c24	[116] 4.6.2	c24	c24
27	420 (Bad Extension)	[26] 21.4.15	m	c13	[26] 21.4.15	m	m
28	421 (Extension Required)	[26] 21.4.16	o	o	[26] 21.4.16	i	i
28A	422 (Session Interval Too Small)	[58] 6	c7	c7	[58] 6	c7	c7
29	423 (Interval Too Brief)	[26] 21.4.17	c4	c4	[26] 21.4.17	m	m
29A	424 (Bad Location Information)	[89] 4.2	c23	c23	[89] 4.2	c23	c23
29AA	428 Use Identity Header	[252] 6.2.2	c40	c40	[252] 6.2.2	c40	c40
29B	429 (Provide Referrer Identity)	[59] 5	c8	c8	[59] 5	c9	c9
29C	430 (Flow Failed)	[92] 11	n/a	n/a	[92] 11	c22	c22
29D	433 (Anonymity Disallowed)	[67] 4	c14	c14	[67] 4	c14	c14
29DA	436 Bad Identity Info	[252] 6.2.2	c40	c40	[252] 6.2.2	c40	c40
29DB	437 Unsupported Credential	[252] 6.2.2	c40	c40	[252] 6.2.2	c40	c40
29DC	438 Invalid Identity Header	[252] 6.2.2	c40	c40	[252] 6.2.2	c40	c40
29E	439 (First Hop Lacks Outbound Support)	[92] 11	c28	c28	[92] 11	c29	c29

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
29F	440 (Max Breadth Exceeded)	[117] 5	n/a	c30	[117] 5	c31	c31
29G	469 (Bad INFO Package)	[25] 4.2	c33	c33	[25] 4.2	c33	c33
29H	470 (Consent Needed)	[125] 5.9.2	c26	c26	[125] 5.9.2	c27	c27
30	480 (Temporarily Unavailable)	[26] 21.4.18	m	m	[26] 21.4.18	m	m
31	481 (Call/Transaction Does Not Exist)	[26] 21.4.19	m	m	[26] 21.4.19	m	m
32	482 (Loop Detected)	[26] 21.4.20	m	m	[26] 21.4.20	m	m
33	483 (Too Many Hops)	[26] 21.4.21	m	m	[26] 21.4.21	m	m
34	484 (Address Incomplete)	[26] 21.4.22	o	o	[26] 21.4.22	m	m
35	485 (Ambiguous)	[26] 21.4.23	o	o	[26] 21.4.23	m	m
36	486 (Busy Here)	[26] 21.4.24	m	m	[26] 21.4.24	m	m
37	487 (Request Terminated)	[26] 21.4.25	m	m	[26] 21.4.25	m	m
38	488 (Not Acceptable Here)	[26] 21.4.26	m	m	[26] 21.4.26	m	m
39	489 (Bad Event)	[28] 8.3.2	c3	c3	[28] 8.3.2	c3	c3
40	491 (Request Pending)	[26] 21.4.27	m	m	[26] 21.4.27	m	m
41	493 (Undecipherable)	[26] 21.4.28	m	m	[26] 21.4.28	m	m
41A	494 (Security Agreement Required)	[48] 2	c5	c5	[48] 2	c6	c6
105	5xx response	[26] 21.5	p25	p25	[26] 21.5	p25	p25
42	500 (Internal Server Error)	[26] 21.5.1	m	m	[26] 21.5.1	m	m
43	501 (Not Implemented)	[26] 21.5.2	m	m	[26] 21.5.2	m	m
44	502 (Bad Gateway)	[26] 21.5.3	o	o	[26] 21.5.3	m	m
45	503 (Service Unavailable)	[26] 21.5.4	m	m	[26] 21.5.4	m	m
46	504 (Server Time-out)	[26] 21.5.5	m	m	[26] 21.5.5	m	m
47	505 (Version not supported)	[26] 21.5.6	m	m	[26] 21.5.6	m	m
48	513 (Message Too Large)	[26] 21.5.7	m	m	[26] 21.5.7	m	m
49	580 (Precondition Failure)	[30] 8	c35	c35	[30] 8	c35	c35
106	6xx response	[26] 21.6	p26	p26	[26] 21.6	p26	p26
50	600 (Busy Everywhere)	[26] 21.6.1	m	m	[26] 21.6.1	m	m
51	603 (Decline)	[26] 21.6.2	c10	c10	[26] 21.6.2	m	m
52	604 (Does Not Exist Anywhere)	[26] 21.6.3	m	m	[26] 21.6.3	m	m
53	606 (Not Acceptable)	[26] 21.6.4	m	m	[26] 21.6.4	m	m
54	607 (Unwanted)	[254]	o	c38	[254]	o	c39

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
c1:	IF A.5/9 THEN m ELSE n/a - - INVITE response.						
c2:	IF A.5/9 THEN o ELSE n/a - - INVITE response.						
c3:	IF A.4/20 THEN m ELSE n/a - - SIP specific event notification extension.						
c4:	IF A.5/19 OR A.5/21 THEN m ELSE n/a - - REGISTER response or SUBSCRIBE response.						
c5:	IF A.4/37 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						
c6:	IF A.4/37 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						
c7:	IF A.4/42 AND (A.5/9 OR A.5/23) THEN m ELSE n/a - - the SIP session timer AND (INVITE response OR UPDATE response).						
c8:	IF A.4/43 AND A.5/17 THEN o ELSE n/a - - the SIP Referred-By mechanism and REFER response.						
c9:	IF A.4/43 AND A.5/17 THEN m ELSE n/a - - the SIP Referred-By mechanism and REFER response.						
c10:	IF A.4/44 THEN m ELSE o - - the Session Inititation Protocol (SIP) "Replaces" header.						
c11:	IF A.5/3 OR A.5/9 OR A.5/9B OR A.5/11OR A.5/13 OR A.5/15 OR A.5/15B OR A.5/17 OR A.5/19 OR A.5/21 OR A.5/23 THEN m ELSE n/a - - BYE response or INVITE response or MESSAGE response or NOTIFY response or OPTIONS response or PRACK response or PUBLISH response or REFER response or REGISTER response or SUBSCRIBE response or UPDATE response.						
c12:	IF A.3/4 THEN m ELSE o - - S-CSCF.						
c13:	IF A.3/1 OR A.3/2 OR A.3/4 THEN m ELSE o - - UE, P-CSCF, S-CSCF.						
c14:	IF A.4/48 THEN m ELSE n/a - - rejecting anonymous requests in the session initiation protocol.						
c20:	IF A.4/41 THEN m ELSE n/a - - an event state publication extension to the session initiation protocol.						
c21:	IF A.5/3 OR A.5/9 OR A.5/9B OR A.5/11 or A.5/13 OR A.5/15 OR A.5/15B OR A.5/17 OR A.5/19 OR A.5/21 OR A.5/23 THEN o ELSE n/a - - BYE response or INVITE response or MESSAGE response or NOTIFY response or OPTIONS response or PRACK response or PUBLISH response or REFER response or REGISTER response or SUBSCRIBE response or UPDATE response.						
c22:	IF A.4/57 THEN m ELSE n/a - - managing client initiated connections in SIP.						
c23:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.						
c24:	IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						
c26:	IF A.4/75B THEN m ELSE n/a - - a recipient within the framework for consent-based communications in SIP.						
c27:	IF A.4/75A THEN m ELSE n/a - - a relay within the framework for consent-based communications in SIP.						
c28:	IF A.4/2 AND A.4/57 THEN m ELSE n/a - - registrar, managing client initiated connections in SIP.						
c29:	IF A.4/1 AND A.4/57 THEN m ELSE n/a - - client behaviour for registration, managing client initiated connections in SIP.						
c30:	IF A.4/71 AND (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C) THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling).						
c31:	IF A.4/71 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.						
c32:	IF A.5/9 AND A.4/81 THEN m ELSE n/a - - INVITE response and 199 (Early Dialog Terminated) response.						
c33:	IF A.4/13 THEN m ELSE n/a - - SIP INFO method and package framework.						
c34:	IF A.4/16 OR A.3/6 THEN m ELSE IF A.5/9 THEN o ELSE n/a - - initiating a session which require local and/or remote resource reservation, MGCF, INVITE response.						
c35:	IF A.4/16 THEN m ELSE n/a - - integration of resource management and SIP.						
c36:	IF A.5/9B THEN m ELSE n/a - - MESSAGE response.						
c37:	IF A.4/20 OR OR A.5/9B OR A.5/17 THEN m ELSE n/a - - SIP specific event notification extension or MESSAGE response or the REFER response.						
c.38:	IF A.4/117 THEN o ELSE n/a - - a SIP response code for unwanted calls extension.						
c.39:	IF A.4/117 THEN m ELSE n/a - - a SIP response code for unwanted calls extension.						
c.40	IF A.4/116 THEN m ELSE n/a - authenticated identity management in the Session Initiation Protocol						
p21:	A.6/2 OR A.6/3 OR A.6/4 OR A.6/5 OR A.6/5A - - 1xx response.						
p22:	A.6/6 OR A.6/7 - - 2xx response.						
p23:	A.6/8 OR A.6/9 OR A.6/10 OR A.6/11 OR A.6/12 - - 3xx response.						
p24:	A.6/13 OR A.6/14 OR A.6/15 OR A.6/16 OR A.6/17 OR A.6/18 OR A.6/19 OR A.6/20 OR A.6/21 OR A.6/22 OR A.6/22A OR A.6/23 OR A.6/24 OR A.6/25 OR A.6/26 OR A.6/26A OR A.6/27 OR A.6/28 OR A.6/28A OR A.6/29 OR A.6/29A OR A.6/29B OR A.6/29C OR A.6/29D OR A.6/29E OR A.6/29F OR A.6/29G OR A.6/29H OR A.6/30 OR A.6/31 OR A.6/32 OR A.6/33 OR A.6/34 OR A.6/35 OR A.6/36 OR A.6/436 OR A.6/38 OR A.6/39 OR A.6/40 OR A.6/41 OR A.6/41A. - 4xx response.						
p25:	A.6/42 OR A.6/43 OR A.6/44 OR A.6/45 OR A.6/46 OR A.6/47 OR A.6/48 OR A.6/49 - - 5xx response						
p26:	A.6/50 OR A.6/51 OR A.6/52 OR A.6/53 - - 6xx response.						

A.2.1.4.2 ACK method

Prerequisite A.5/1 – ACK request

Table A.7: Supported header fields within the ACK request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Contact	[56B] 9.2	c9	c9	[56B] 9.2	c10	c10
2	Allow-Events	[28] 8.2.2	c1	c1	[28] 8.2.2	c2	c2
3	Authorization	[26] 20.7	c3	c3	[26] 20.7	c3	c3
4	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
5	Cellular-Network-Info	7.2.15	n/a	c26	7.2.15	n/a	c27
6	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
7	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
7A	Content-ID	[256] 3.2	o	c29	[256] 3.2	m	c30
8	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
9	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
10	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
11	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
12	Date	[26] 20.17	c4	c4	[26] 20.17	m	m
13	From	[26] 20.20	m	m	[26] 20.20	m	m
13A	Max-Breadth	[117] 5.8	n/a	c14	[117] 5.8	c15	c15
14	Max-Forwards	[26] 20.22	m	m	[26] 20.22	n/a	c16
15	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
15A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c19	c20	[52] 4.4, [52A] 4, [234] 2	c19	c21
15C	Privacy	[33] 4.2	c6	n/a	[33] 4.2	c6	n/a
15D	P-Charging-Vector	[52] 4.6, [52A] 4	c22	c23	[52] 4.6, [52A] 4	c22	c23
15E	Priority-Share	Subclause 7.2.16	n/a	c28	Subclause 7.2.16	n/a	c28
16	Proxy-Authorization	[26] 20.28	c5	c5	[26] 20.28	n/a	n/a
17	Proxy-Require	[26] 20.29	o	n/a	[26] 20.29	n/a	n/a
17A	Reason	[34A] 2	c8	c8	[34A] 2	c8	c8
17B	Record-Route	[26] 20.30	n/a	c16	[26] 20.30	n/a	c16
17C	Recv-Info	[25] 5.2.3	c17	c17	[25] 5.2.3	c17	c17
17D	Reject-Contact	[56B] 9.2	c9	c9	[56B] 9.2	c10	c10
17E	Relayed-Charge	7.2.12	n/a	c24	7.2.12	n/a	c24
17F	Request-Disposition	[56B] 9.1	c9	c9	[56B] 9.1	c10	c10
18	Require	[26] 20.32	n/a	n/a	[26] 20.32	n/a	n/a
18A	Resource-Priority	[116] 3.1	c11	c11	[116] 3.1	c11	c11
18B	Resource-Share	Subclause 7.2.13	n/a	c25	Subclause 7.2.13	n/a	c25
19	Route	[26] 20.34	m	m	[26] 20.34	n/a	c16
19A	Session-ID	[162]	o	c18	[162]	o	c18
20	Timestamp	[26] 20.38	c7	c7	[26] 20.38	m	m
21	To	[26] 20.39	m	m	[26] 20.39	m	m
22	User-Agent	[26] 20.41	o	o	[26] 20.41	m	m
23	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.
c2:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.
c3:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.
c4:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c5:	IF A.4/8A THEN m ELSE n/a - - authentication between UA and proxy.
c6:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c7:	IF A.4/6 THEN o ELSE n/a - - timestamping of requests.
c8:	IF A.4/38 THEN o ELSE n/a - - the Reason header field for the session initiation protocol.
c9:	IF A.4/40 THEN o ELSE n/a - - caller preferences for the session initiation protocol.
c10:	IF A.4/40 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c11:	IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.
c14:	IF A.4/71 AND (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C) THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling).
c15:	IF A.4/71 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c16:	IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - UE, UE performing the functions of an external attached network.
c17:	IF A.4/13 THEN m ELSE IF A.4/13A THEN m ELSE n/a - - SIP INFO method and package framework, legacy INFO usage.
c18:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c19:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c20:	IF A.4/34 AND A.3/1 OR A.3/2A OR A.3/7 OR A.3A/81 OR A.3A/81A OR A.3A/81B THEN o ELSE n/a - - the P-Access-Network-Info header extension and UE, P-CSCF (IMS-ALG), AS, MSC Server enhanced for ICS, MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface.
c21:	IF A.4/34 AND A.3/1 OR A.3/7 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE, AS.
c22:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c23:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c24:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c25:	IF A.4/112 THEN o ELSE n/a - - resource sharing.
c26:	IF A.4/113 AND A.3/1 OR A.3/2A OR A.3/7 THEN o ELSE n/a - - the Cellular-Network-Info header extension and UE, P-CSCF (IMS-ALG), AS.
c27:	IF A.4/113 AND A.3/7 THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS.
c28:	IF A.4/114 THEN o ELSE n/a - - priority sharing.
c29:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c30:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).

Prerequisite A.5/1 – ACK request

Table A.8: Supported message bodies within the ACK request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.1.4.3 BYE method

Prerequisite A.5/2 - - BYE request

Table A.9: Supported header fields within the BYE request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o	o	[26] 20.1	m	m
1A	Accept-Contact	[56B] 9.2	c18	c18	[56B] 9.2	c22	c22
2	Accept-Encoding	[26] 20.2	o	o	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o	o	[26] 20.3	m	m
3A	Allow	[26] 20.5	o	o	[26] 20.5	m	m
4	Allow-Events	[28] 8.2.2	c1	c1	[28] 8.2.2	c2	c2
5	Authorization	[26] 20.7	c3	c3	[26] 20.7	c3	c3
6	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
6A	Cellular-Network-Info	7.2.15	n/a	c35	7.2.15	n/a	c36
7	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
8	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
8A	Content-ID	[256] 3.2	o	c37	[256] 3.2	m	c38
9	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
10	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
11	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
12	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
13	Date	[26] 20.17	c4	c4	[26] 20.17	m	m
14	From	[26] 20.20	m	m	[26] 20.20	m	m
14A	Geolocation	[89] 4.1	c23	c23	[89] 4.1	c23	c23
14B	Geolocation-Routing	[89] 4.2	c23	c23	[89] 4.2	c23	c23
14C	Max-Breadth	[117] 5.8	n/a	c29	[117] 5.8	c30	c30
15	Max-Forwards	[26] 20.22	m	m	[26] 20.22	n/a	c31
16	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
16A	P-Access-Network-Info	[52] 4.4, [234] 2	c9	c10	[52] 4.4, [234] 2	c9	c11
16B	P-Asserted-Identity	[34] 9.1	n/a	n/a	[34] 9.1	c6	c6
16C	P-Charging-Function-Addresses	[52] 4.5	c13	c14	[52] 4.5	c13	c14
16D	P-Charging-Vector	[52] 4.6	c12	c34	[52] 4.6	c12	c34
16F	P-Preferred-Identity	[34] 9.2	c6	x	[34] 9.2	n/a	n/a
16G	Privacy	[33] 4.2	c7	n/a	[33] 4.2	c7	c7
17	Proxy-Authorization	[26] 20.28	c5	c5	[26] 20.28	n/a	n/a
18	Proxy-Require	[26] 20.29	o	n/a	[26] 20.29	n/a	n/a
18A	Reason	[34A] 2	c17	c21	[34A] 2	c24	c24
19	Record-Route	[26] 20.30	n/a	c31	[26] 20.30	n/a	c31
19A	Referred-By	[59] 3	c19	c19	[59] 3	c20	c20
19B	Reject-Contact	[56B] 9.2	c18	c18	[56B] 9.2	c22	c22
19C	Relayed-Charge	7.2.12	n/a	c33	7.2.12	n/a	c33
19D	Request-Disposition	[56B] 9.1	c18	c18	[56B] 9.1	c22	c22
20	Require	[26] 20.32	m	m	[26] 20.32	m	m
20A	Resource-Priority	[116] 3.1	c25	c25	[116] 3.1	c25	c25
21	Route	[26] 20.34	m	m	[26] 20.34	n/a	c31
21A	Security-Client	[48] 2.3.1	c15	c15	[48] 2.3.1	n/a	n/a
21B	Security-Verify	[48] 2.3.1	c16	c16	[48] 2.3.1	n/a	n/a
21C	Session-ID	[162]	o	c32	[162]	o	c32
22	Supported	[26] 20.37	o	o	[26] 20.37	m	m
23	Timestamp	[26] 20.38	c8	c8	[26] 20.38	m	m
24	To	[26] 20.39	m	m	[26] 20.39	m	m
25	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
25A	User-to-User	[126] 7	c26	c26	[126] 7	c26	c26
26	Via	[26] 20.42	m	m	[20] 20.42	m	m

c1:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.
c2:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.
c3:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.
c4:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c5:	IF A.4/8A THEN m ELSE n/a - - authentication between UA and proxy.
c6:	IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c7:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c8:	IF A.4/6 THEN o ELSE n/a - - timestamping of requests.
c9:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c10:	IF A.4/34 AND (A.3/1 OR A.3/2A OR A.3/7) THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE, P-CSCF (IMS-ALG) or AS.
c11:	IF A.4/34 AND (A.3/2A OR A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the P-Access-Network-Info header extension and P-CSCF (IMS-ALG), AS acting as terminating UA, AS acting as third-party call controller or EATF.
c12:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c13:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c14:	IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c15:	IF A.4/37 OR A.4/37A THEN o ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media (note).
c16:	IF A.4/37 OR A.4/37A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c17:	IF A.4/38 THEN o ELSE n/a - - the Reason header field for the session initiation protocol.
c18:	IF A.4/40 THEN o ELSE n/a - - caller preferences for the session initiation protocol.
c19:	IF A.4/43 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c20:	IF A.4/43 THEN o ELSE n/a - - the SIP Referred-By mechanism.
c21:	IF A.3/2 THEN m ELSE IF A.4/38 THEN o ELSE n/a - - P-CSCF, the Reason header field for the session initiation protocol.
c22:	IF A.4/40 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c23:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c24:	IF A.4/38 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.
c25:	IF A.4/70B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.
c26:	IF A.4/76 THEN o ELSE n/a - - transporting user to user information for call centers using SIP.
c29:	IF A.4/71 AND (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C) THEN m ELSE IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - addressing an amplification vulnerability in session initiation protocol forking proxies, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling), UE, UE performing the functions of an external attached network.
c30:	IF A.4/71 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c31:	IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - UE, UE performing the functions of an external attached network.
c32:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c33:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c34:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c35:	IF A.4/113 AND (A.3/1 OR A.3/2A OR A.3/7) THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE, P-CSCF (IMS-ALG) or AS.
c36:	IF A.4/113 AND (A.3/2A OR A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the Cellular-Network-Info header extension and P-CSCF (IMS-ALG), AS acting as terminating UA, AS acting as third-party call controller or EATF.
c37:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c38:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
NOTE:	Support of this header in this method is dependent on the security mechanism and the security architecture which is implemented. Use of this header in this method is not appropriate to the security mechanism defined by 3GPP TS 33.203 [19].

Prerequisite A.5/2 - - BYE request

Table A.10: Supported message bodies within the BYE request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	XML Schema for PSTN	[11B]	n/a	c1	[11B]	n/a	c1
2	VoiceXML expr / namelist data	[145] 4.2	m	c2	[145] 4.2	m	c2
3	application/vnd.3gpp.ussd	[8W]	n/a	c3	[8W]	n/a	c4
4	application/sdp	[30] 8	o	c5	[30] 8	m	c6
5	application/vnd.etsi.aoc+xml	[8N] 4.7.2	n/a	c7	[8N] 4.7.2	n/a	c8
c1:	IF A.3/6 OR A.3/7A OR A.3/7B OR A.3/7D OR A.3/9B OR A.3/13B THEN o ELSE n/a - - MGCF, AS acting as terminating UA, or redirect server, AS acting as originating UA, AS performing 3rd party call control, IBCF (IMS-ALG), ISC gateway function (IMS-ALG).						
c2:	IF A.4/84 THEN m ELSE n/a - - SIP Interface to VoiceXML Media Services.						
c3:	IF A.3A/93 OR A.3/9 OR A.3/2 OR A.3A/89 THEN m ELSE n/a - - USSI AS, IBCF, P-CSCF, ATCF (UA).						
c4:	IF A.3A/92 OR A.3/9 OR A.3/2 OR A.3A/89 THEN m ELSE n/a - - USSI UE, IBCF, P-CSCF, ATCF (UA).						
c5:	IF A.4/16 THEN o ELSE n/a - - integration of resource management and SIP.						
c6:	IF A.4/16 THEN m ELSE n/a - - integration of resource management and SIP.						
c7	IF A.3A/53 THEN m ELSE n/a - - Advice of charge application server.						
c8	IF A.3A/54 THEN m ELSE n/a - - Advice of charge UA client.						

TableA.11: Void

Prerequisite A.5/3 - - BYE response

Prerequisite: A.6/1 - - Additional for 100 (Trying) response

Table A.11A: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						

Prerequisite A.5/3 - - BYE response for all remaining status-codes

Table A.12: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	c11	c11	[26] 20.5	m	m
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Cellular-Network-Info	7.2.15	n/a	c19	7.2.15	n/a	c20
2	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
3	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
3A	Content-ID	[256] 3.2	o	c21	[256] 3.2	m	c22
4	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
9	From	[26] 20.20	m	m	[26] 20.20	m	m
9A	Geolocation-Error	[89] 4.3	c12	c12	[89] 4.3	c12	c12
10	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
10A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c5	c6	[52] 4.4, [52A] 4, [234] 2	c5	c7
10B	P-Asserted-Identity	[34] 9.1	n/a	n/a	[34] 9.1	c3	c3
10C	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c9	c10	[52] 4.5, [52A] 4	c9	c10
10D	P-Charging-Vector	[52] 4.6, [52A] 4	c8	c18	[52] 4.6, [52A] 4	c8	c18
10F	P-Preferred-Identity	[34] 9.2	c3	x	[34] 9.2	n/a	n/a
10G	Privacy	[33] 4.2	c4	n/a	[33] 4.2	c4	c4
10H	Relayed-Charge	7.2.12	n/a	c17	7.2.12	n/a	c17
10I	Require	[26] 20.32	m	m	[26] 20.32	m	m
10J	Server	[26] 20.35	o	o	[26] 20.35	o	o
10K	Session-ID	[162]	o	c16	[162]	o	c16
11	Timestamp	[26] 20.38	m	m	[26] 20.38	c2	c2
12	To	[26] 20.39	m	m	[26] 20.39	m	m
12A	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
12B	User-to-User	[126] 7	c13	c13	[126] 7	c13	c13
13	Via	[26] 20.42	m	m	[26] 20.42	m	m
14	Warning	[26] 20.43	o (note)	o (note)	[26] 20.43	o	o

c1: IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c2: IF A.4/6 THEN m ELSE n/a - - timestamping of requests.
c3: IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c4: IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c5: IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c6: IF A.4/34 AND (A.3/1 OR A.3/2A ORA.3/7) THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE, P-CSCF (IMS-ALG), or AS.
c7: IF A.4/34 AND (A.3/2A OR A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the P-Access-Network-Info header extension and P-CSCF (IMS-ALG), AS acting as terminating UA, AS acting as third-party call controller or EATF.
c8: IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c9: IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c10: IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c11: IF A.6/18 THEN m ELSE o - - 405 (Method Not Allowed).
c12: IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c13: IF A.4/76 THEN o ELSE n/a - - transporting user to user information for call centers using SIP.
c16: IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c17: IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c18: IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c19: IF A.4/113 AND (A.3/1 OR A.3/2A OR A.3/7) THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE, P-CSCF (IMS-ALG), or AS.
c20: IF A.4/113 AND (A.3/2A OR A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the Cellular-Network-Info header extension and P-CSCF (IMS-ALG), AS acting as terminating UA, AS acting as third-party call controller or EATF.
c21: IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c22: IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).

NOTE: For a 488 (Not Acceptable Here) response, RFC 3261 [26] gives the status of this header as SHOULD rather than OPTIONAL.

Prerequisite A.5/3 - - BYE response

Prerequisite: A.6/102 - - Additional for 2xx response

Table A.13: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Accept-Resource-Priority	[116] 3.2	c5	c5	[116] 3.2	c5	c5
0B	Allow-Events	[28] 8.2.2	c3	c3	[28] 8.2.2	c4	c4
1	Authentication-Info	[26] 20.6	c1	c1	[26] 20.6	c2	c2
4	Supported	[26] 20.37	o	m	[26] 20.37	m	m
c1:	IF A.4/7 THEN o ELSE n/a - - authentication between UA and UA.						
c2:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.						
c3:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.						
c4:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.						
c5:	IF A.4/70B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.						

Prerequisite A.5/3 - - BYE response

Prerequisite: A.6/103 OR A.6/104 OR A.6/105 OR A.6/106 - - Additional for 3xx – 6xx response

Table A.13A: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	o	o	[26] 20.18	o	o
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.4/115 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.5/3 - - BYE response

Prerequisite: A.6/103 OR A.6/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.14: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0B	Contact	[26] 20.10	o (note)	o	[26] 20.10	m	m
NOTE:	RFC 3261 [26] gives the status of this header as SHOULD rather than OPTIONAL.						

Prerequisite A.5/3 - - BYE response

Prerequisite: A.6/14 - - Additional for 401 (Unauthorized) response

Table A.15: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
8	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	m	m
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Prerequisite A.5/3 - - BYE response

Prerequisite: A.6/17 OR A.6/23 OR A.6/30 OR A.6/36 OR A.6/42 OR A.6/45 OR A.6/50 OR A.6/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.16: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Retry-After	[26] 20.33	o	o	[26] 20.33	o	o

Table A.17: Void

Prerequisite A.5/3 - - BYE response

Prerequisite: A.6/19 - - Additional for 407 (Proxy Authentication Required) response

Table A.18: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
6	WWW-Authenticate	[26] 20.44	o	o	[26] 20.44	o	o
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Prerequisite A.5/3 - - BYE response

Prerequisite A.6/25 - - Additional for 415 (Unsupported Media Type) response

Table A.19: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o.1	o.1	[26] 20.1	m	m
2	Accept-Encoding	[26] 20.2	o.1	o.1	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o.1	o.1	[26] 20.3	m	m
o.1	At least one of these capabilities is supported.						

Prerequisite A.5/3 - - BYE response

Prerequisite: A.6/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.19A: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.4/70B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.						

Prerequisite A.5/3 - - BYE response

Prerequisite: A.6/27 - - Additional for 420 (Bad Extension) response

Table A.20: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Unsupported	[26] 20.40	m	m	[26] 20.40	m	m

Prerequisite A.5/3 - - BYE response

Prerequisite: A.6/28 OR A.6/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.20A: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	x	x	[48] 2	c1	c1
c1:	IF A.4/37 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Table A.21: Void

Prerequisite A.5/3 - - BYE response

Prerequisite: A.6/6 - - Additional for 200 (OK) response

Table A.22: Supported message bodies within the BYE response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	VoiceXML expr / namelist data	[145] 4.2	o	c1	[145] 4.2	o	c1
2	application/vnd.etsi.aoc+xml	[8N] 4.7.2	n/a	c2	[8N] 4.7.2	n/a	c3
c1:	IF A.4/84 THEN o ELSE n/a - - SIP Interface to VoiceXML Media Services.						
c2	IF A.3A/53 THEN m ELSE n/a - - Advice of charge application server.						
c3	IF A.3A/54 THEN m ELSE n/a - - Advice of charge UA client.						

A.2.1.4.4 CANCEL method

Prerequisite A.5/4 - - CANCEL request

Table A.23: Supported header fields within the CANCEL request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Contact	[56B] 9.2	c9	c9	[56B] 9.2	c11	c11
5	Authorization	[26] 20.7	c3	c3	[26] 20.7	c3	c3
6	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
8	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
8A	Content-Type	[26] 20.15	c22	c22	[26] 20.15	o	o
9	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
10	Date	[26] 20.17	c4	c4	[26] 20.17	m	m
11	From	[26] 20.20	m	m	[26] 20.20	m	m
11A	Max-Breadth	[117] 5.8	n/a	c16	[117] 5.8	c17	c17
12	Max-Forwards	[26] 20.22	m	m	[26] 20.22	n/a	c18
14	Privacy	[33] 4.2	c6	n/a	[33] 4.2	c6	n/a
15	Reason	[34A] 2	c7	c10	[34A] 2	c12	c12
16	Record-Route	[26] 20.30	n/a	c18	[26] 20.30	n/a	c18
17	Reject-Contact	[56B] 9.2	c9	c9	[56B] 9.2	c11	c11
17A	Relayed-Charge	7.2.12	n/a	c21	7.2.12	n/a	c21
17B	Request-Disposition	[56B] 9.1	c9	c9	[56B] 9.1	c11	c11
17C	Resource-Priority	[116] 3.1	c13	c13	[116] 3.1	c13	c13
18	Route	[26] 20.34	m	m	[26] 20.34	n/a	c18
18A	Session-ID	[162]	o	c19	[162]	o	c19
19	Supported	[26] 20.37	o	o	[26] 20.37	m	m
20	Timestamp	[26] 20.38	c8	c8	[26] 20.38	m	m
21	To	[26] 20.39	m	m	[26] 20.39	m	m
22	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
23	Via	[26] 20.42	m	m	[26] 20.42	m	m

c3: IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.
c4: IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c6: IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c7: IF A.4/38 THEN o ELSE n/a - - the Reason header field for the session initiation protocol.
c8: IF A.4/6 THEN o ELSE n/a - - timestamping of requests.
c9: IF A.4/40 THEN o ELSE n/a - - caller preferences for the session initiation protocol.
c10: IF A.3/2 THEN m ELSE IF A.4/38 THEN o ELSE n/a - - P-CSCF, the Reason header field for the session initiation protocol.
c11: IF A.4/40 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c12: IF A.4/38 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.
c13: IF A.4/70B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.
c16: IF A.4/71 AND (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C) THEN m ELSE IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - addressing an amplification vulnerability in session initiation protocol forking proxies, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling), UE, UE performing the functions of an external attached network..
c17: IF A.4/71 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c18: IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - UE, UE performing the functions of an external attached network.
c19: IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c21: IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c22: IF A.4/16 OR A.24/1 THEN m ELSE o - - integration of resource management and SIP or XML Schema for PSTN.

Prerequisite A.5/4 - - CANCEL request

Table A.24: Supported message bodies within the CANCEL request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	XML Schema for PSTN	[11B]		c1	[11B]		c1
2	application/sdp	[30] 8	o	c2	[30] 8	m	c3
c1:	IF A.3/6 OR A.3/7A OR A.3/7B OR A.3/7D OR A.3/9B OR A.3/13B THEN o ELSE n/a - - MGCF, AS acting as terminating UA, or redirect server, AS acting as originating UA, AS performing 3rd party call control, IBCF (IMS-ALG), ISC gateway function (IMS-ALG).						
c2:	IF A.4/16 THEN o ELSE n/a - - integration of resource management and SIP.						
c3:	IF A.4/16 THEN m ELSE n/a - - integration of resource management and SIP.						

Prerequisite A.5/5 - - CANCEL response for all status-codes

Table A.25: Supported header fields within the CANCEL response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
5	From	[26] 20.20	m	m	[26] 20.20	m	m
5C	Privacy	[33] 4.2	c3	n/a	[33] 4.2	c3	n/a
5D	Relayed-Charge	7.2.12	n/a	c8	7.2.12	n/a	c8
5E	Session-ID	[162]	o	c6	[162]	o	c6
6	Timestamp	[26] 20.38	m	m	[26] 20.38	c2	c2
7	To	[26] 20.39	m	m	[26] 20.39	m	m
7A	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
8	Via	[26] 20.42	m	m	[26] 20.42	m	m
9	Warning	[26] 20.43	o (note)	o	[26] 20.43	o	o
c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						
c2:	IF A.4/6 THEN m ELSE n/a - - timestamping of requests.						
c3:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).						
c6:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.						
c8:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.						
NOTE:	For a 488 (Not Acceptable Here) response, RFC 3261 [26] gives the status of this header as SHOULD rather than OPTIONAL.						

Prerequisite A.5/5 - - CANCEL response

Prerequisite: A.6/102 - - Additional for 2xx response

Table A.26: Supported header fields within the CANCEL response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
4	Supported	[26] 20.37	o	m	[26] 20.37	m	m
c1:	IF A.4/70B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.						

Prerequisite A.5/5 - - CANCEL response

Prerequisite: A.6/103 OR A.6/104 OR A.6/105 OR A.6/106 - - Additional for 3xx – 6xx response

Table A.26A: Supported header fields within the CANCEL response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	o	o	[26] 20.18	o	O
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.4/115 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Table A.27: Void

Prerequisite A.5/5 - - CANCEL response

Prerequisite: A.6/17 OR A.6/23 OR A.6/30 OR A.6/36 OR A.6/42 OR A.6/45 OR A.6/50 OR A.6/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.28: Supported header fields within the CANCEL response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Retry-After	[26] 20.33	o	o	[26] 20.33	o	o

Table A.29: Void

Table A.30: Void

Prerequisite A.5/5 - - CANCEL response

Prerequisite: A.6/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.30A: Supported header fields within the CANCEL response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.4/70B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.						

Prerequisite A.5/5 - - CANCEL response

Table A.31: Supported message bodies within the CANCEL response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.1.4.5 Void

A.2.1.4.6 INFO method

Prerequisite A.5/6 - - INFO request

Table A.32: Supported header fields within the INFO request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o	o	[26] 20.1	m	m
2	Accept-Encoding	[26] 20.2	o	o	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o	o	[26] 20.3	m	m
4	Allow	[26] 20.5	o	o	[26] 20.5	m	m
5	Allow-Events	[28] 8.2.2	c1	c1	[28] 8.2.2	c2	c2
6	Authorization	[26] 20.7	c3	c3	[26] 20.7	c3	c3
7	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
7A	Call-Info	[26] 20.9	o	o	[26] 20.9	o	o
8	Cellular-Network-Info	7.2.15	n/a	c45	7.2.15	n/a	c46
9	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
10	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
10A	Content-ID	[256] 3.2	o	c47	[256] 3.2	m	c48
11	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
12	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
13	Content-Type	[26] 20.15	m	m	[26] 29.15	m	m
14	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
15	Date	[26] 20.17	c4	c4	[26] 20.17	m	m
16	From	[26] 20.20	m	m	[26] 20.20	m	m
17	Geolocation	[89] 4.1	c29	c29	[89] 4.1	c29	c29
17A	Geolocation-Routing	[89] 4.2	c29	c29	[89] 4.2	c29	c29
18	Info-Package	[25] 7.2	c42	c42	[25] 7.2	c42	c42
19	Max-Breadth	[117] 5.8	n/a	c39	[117] 5.8	c40	c40
20	Max-Forwards	[26] 20.22	m	m	[26] 20.22	n/a	c41
21	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
22	P-Access-Network-Info	[52] 4.4, [234] 2	c15	c16	[52] 4.4, [234] 2	c15	c17
23	P-Charging-Function-Addresses	[52] 4.5	c20	c21	[52] 4.5	c20	c21
24	P-Charging-Vector	[52] 4.6	c18	c19	[52] 4.6	c18	c19
26	Privacy	[33] 4.2	c12	c12	[33] 4.2	c12	c12
27	Proxy-Authorization	[26] 20.28	c5	c5	[26] 20.28	n/a	n/a
28	Proxy-Require	[26] 20.29	o	n/a	[26] 20.29	n/a	n/a
29	Reason	[34A] 2	c6	c6	[34A] 2	c6	c6
30	Record-Route	[26] 20.30	n/a	c41	[26] 20.30	n/a	c41
31	Referred-By	[59] 3	c25	c25	[59] 3	c26	c26
32	Relayed-Charge	7.2.12	n/a	c44	7.2.12	n/a	c44
33	Request-Disposition	[56B] 9.1	c24	c24	[56B] 9.1	c28	c28
34	Require	[26] 20.32	m	m	[26] 20.32	m	m
35	Resource-Priority	[116] 3.1	c30	c30	[116] 3.1	c30	c30
36	Route	[26] 20.34	m	m	[26] 20.34	n/a	c41
37	Security-Client	[48] 2.3.1	c22	c22	[48] 2.3.1	n/a	n/a
38	Security-Verify	[48] 2.3.1	c23	c23	[48] 2.3.1	n/a	n/a
38A	Session-ID	[162]	o	c43	[162]	o	c43
39	Subject	[26] 20.35	o	o	[26] 20.36	o	o
40	Supported	[26] 20.37	m	m	[26] 20.37	m	m
41	Timestamp	[26] 20.38	c10	c10	[26] 20.38	m	m
42	To	[26] 20.39	m	m	[26] 20.39	m	m
43	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
44	Via	[26] 20.42	m	m	[26] 20.42	m	m

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
c1:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.						
c2:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.						
c3:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.						
c4:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						
c5:	IF A.4/8A THEN m ELSE n/a - - authentication between UA and proxy.						
c6:	IF A.4/38 THEN o ELSE n/a - - the Reason header field for the session initiation protocol.						
c10:	IF A.4/6 THEN o ELSE n/a - - timestamping of requests.						
c12:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).						
c15:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.						
c16:	IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.						
c17:	IF A.4/34 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.						
c18:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.						
c19:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.						
c20:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.						
c21:	IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.						
c22:	IF A.4/37 OR A.4/37A THEN o ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media? (note 2).						
c23:	IF A.4/37 OR A.4/37A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.						
c24	IF A.4/40 THEN o ELSE n/a - - caller preferences for the session initiation protocol.						
c25:	IF A.4/43 THEN m ELSE n/a - - the SIP Referred-By mechanism.						
c26:	IF A.4/43 THEN o ELSE n/a - - the SIP Referred-By mechanism.						
c28:	IF A.4/40 THEN m ELSE n/a - - caller preferences for the session initiation protocol.						
c29:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.						
c30:	IF A.4/70A THEN m ELSE n/a - - inclusion of INFO, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						
c39:	IF A.4/71 AND (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C) THEN m ELSE IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - addressing an amplification vulnerability in session initiation protocol forking proxies, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling), UE, UE performing the functions of an external attached network.						
c40:	IF A.4/71 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.						
c41:	IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - UE, UE performing the functions of an external attached network.						
c42:	IF A.4/13A THEN n/a ELSE m - - legacy INFO usage.						
c43:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.						
c44:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.						
c45:	IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE.						
c46:	IF A.4/113 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.						
c47:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).						
c48:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).						
NOTE 2:	Support of this header field in this method is dependent on the security mechanism and the security architecture which is implemented. Use of this header field in this method is not appropriate to the security mechanism defined by 3GPP TS 33.203 [19].						

Prerequisite A.5/6 - - INFO request

Table A.33: Supported message bodies within the INFO request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Info-Package	[25]	m	m	[25]	m	m
2	application/vnd.etsi.aoc+xml	[8N] 4.7.2	n/a	c1	[8N] 4.7.2	n/a	c2
3	application/EmergencyCallData.eCall.MSD	[244] 14.3	m	c3	[244] 14.3	m	c4
4	application/EmergencyCallData.Control+xml	[244] 14.4	m	c3	[244] 14.4	m	c3
c1:	IF A.3A/53 THEN m ELSE n/a - - Advice of charge application server.						
c2:	IF A.3A/54 THEN m ELSE n/a - - Advice of charge UA client.						
c3:	IF (A.3/1 AND A.4/120) THEN m ELSE IF ((A.3/2A OR A.3/11A OR A.3A/84) AND A.4/120) THEN i ELSE n/a - - UE, Next-Generation Pan-European eCall emergency service, P-CSCF (IMS-ALG), E-CSCF acting as UA, EATF.						
c4:	IF ((A.3/2A OR A.3/11A OR A.3A/84) AND A.4/120) THEN i ELSE n/a - - P-CSCF (IMS-ALG), E-CSCF acting as UA, EATF, Next-Generation Pan-European eCall emergency service.						

Prerequisite A.5/7 - - INFO response

Prerequisite: A.6/1 - - Additional for 100 (Trying) response

Table A.34: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						

Prerequisite A.5/7 - - INFO response for all remaining status-codes

Table A.35: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	c12	c12	[26] 20.5	m	m
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Call-Info	[26] 20.9	o	o	[26] 20.9	o	o
2A	Cellular-Network-Info	7.2.15	n/a	c19	7.2.15	n/a	c20
3	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
4	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
4A	Content-ID	[256] 3.2	o	c21	[256] 3.2	m	c22
5	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
6	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
7	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
8	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
9	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
10	From	[26] 20.20	m	m	[26] 20.20	m	m
11	Geolocation-Error	[89] 4.3	c14	c14	[89] 4.3	c14	c14
12	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
13	Organization	[26] 20.25	o	o	[26] 20.25	o	o
14	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c5	c6	[52] 4.4, [52A] 4, [234] 2	c5	c7
15	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c10	c11	[52] 4.5, [52A] 4	c10	c11
16	P-Charging-Vector	[52] 4.6, [52A] 4	c8	c9	[52] 4.6, [52A] 4	c8	c9
18	Privacy	[33] 4.2	c4	c4	[33] 4.2	c4	c4
18A	Relayed-Charge	7.2.12	n/a	c18	7.2.12	n/a	c18
19	Require	[26] 20.32	m	m	[26] 20.32	m	m
20	Server	[26] 20.35	o	o	[26] 20.35	o	o
20A	Session-ID	[162]	o	c17	[162]	o	c17
21	Timestamp	[26] 20.38	m	m	[26] 20.38	c2	c2
22	To	[26] 20.39	m	m	[26] 20.39	m	m
23	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
24	Via	[26] 20.42	m	m	[26] 20.42	m	m
25	Warning	[26] 20.43	o	o	[26] 20.43	o	o

c1: IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c2: IF A.4/6 THEN m ELSE n/a - - timestamping of requests.
c4: IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c5: IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c6: IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.
c7: IF A.4/34 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c8: IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c9: IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c10: IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c11: IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c12: IF A.6/18 THEN m ELSE o - - 405 (Method Not Allowed).
c14: IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c17: IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c18: IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c19: IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE.
c20: IF A.4/113 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c21: IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c22: IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).

Prerequisite A.5/7 - - INFO response

Prerequisite: A.6/102 - - Additional for 2xx response

Table A.36: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o	o	[26] 20.1	m	m
2	Accept-Encoding	[26] 20.2	o	o	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o	o	[26] 20.3	m	m
4	Accept-Resource-Priority	[116] 3.2	c5	c5	[116] 3.2	c5	c5
5	Allow-Events	[28] 8.2.2	c3	c3	[28] 8.2.2	c4	c4
6	Authentication-Info	[26] 20.6	c1	c1	[26] 20.6	c2	c2
9	Supported	[26] 20.37	o	o	[26] 20.37	m	m
c1:	IF A.4/7 THEN o ELSE n/a - - authentication between UA and UA.						
c2:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.						
c3:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.						
c4:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.						
c5:	IF A.4/70A THEN m ELSE n/a - - inclusion of INFO, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						

Prerequisite A.5/7 - - INFO response

Prerequisite: A.6/103 OR A.6/104 OR A.6/105 OR A.6/106 - - Additional for 3xx – 6xx response

Table A.37: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	o	o	[26] 20.18	o	o
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.4/115 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.5/7 - - INFO response

Prerequisite: A.6/103 - - Additional for 3xx or 485 (Ambiguous) response

Table A.37A: Void

Prerequisite A.5/7 - - INFO response

Prerequisite: A.6/14 - - Additional for 401 (Unauthorized) response

Table A.38: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
6	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	m	m
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Prerequisite A.5/7 - - INFO response

Prerequisite: A.6/17 OR A.6/23 OR A.6/30 OR A.6/36 OR A.6/42 OR A.6/45 OR A.6/50 OR A.6/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480 (Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.39: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Retry-After	[26] 20.33	o	o	[26] 20.33	o	o

Table A.40: Void

Prerequisite A.5/7 - - INFO response

Prerequisite: A.6/25 - - Additional for 415 (Unsupported Media Type) response

Table A.41: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o.1	o.1	[26] 20.1	m	m
2	Accept-Encoding	[26] 20.2	o.1	o.1	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o.1	o.1	[26] 20.3	m	m
o.1	At least one of these capabilities is supported.						

Prerequisite A.5/7 - - INFO response

Prerequisite: A.6/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.41A: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.4/70A THEN m ELSE n/a - - inclusion of INFO, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						

Prerequisite A.5/7 - - INFO response

Prerequisite: A.6/27 - - Additional for 420 (Bad Extension) response

Table A.42: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Unsupported	[26] 20.40	m	m	[26] 20.40	m	m

Prerequisite A.5/7 - - INFO response

Prerequisite: A.6/28 OR A.6/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.42A: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	x	x	[48] 2	c1	c1
c1:	IF A.4/37 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Table A.43: Void

Table A.44: Void

Prerequisite A.5/7 - - INFO response

Table A.45: Supported message bodies within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.1.4.7 INVITE method

Prerequisite A.5/8 - - INVITE request

Table A.46: Supported header fields within the INVITE request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o	c47	[26] 20.1	m	m
1A	Accept-Contact	[56B] 9.2	c24	c24	[56B] 9.2	c32	c32
2	Accept-Encoding	[26] 20.2	o	o	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o	o	[26] 20.3	m	m
3A	Additional-Identity	7.2.20	n/a	c75	7.2.20	n/a	c76
4	Alert-Info	[26] 20.4	o	o	[26] 20.4	c1	c1
5	Allow	[26] 20.5, [26] 5.1	o (note 1)	o	[26] 20.5, [26] 5.1	m	m
6	Allow-Events	[28] 8.2.2	c2	c2	[28] 8.2.2	c53	c53
6A	Attestation-Info	7.2.18	n/a	c71	7.2.18	n/a	c71
7	Answer-Mode	[158]	c49	c49	[158]	c50	c50
8	Authorization	[26] 20.7	c3	c3	[26] 20.7	c3	c3
9	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
10	Call-Info	[26] 20.9	o	o	[26] 20.9	o	o
10A	Cellular-Network-Info	7.2.15	n/a	c63	7.2.15	n/a	c64
11	Contact	[26] 20.10	m	m	[26] 20.10	m	m
12	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
13	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
13A	Content-ID	[256] 3.2	o	c69	[256] 3.2	m	c70
14	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
15	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
16	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
17	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
18	Date	[26] 20.17	c4	c4	[26] 20.17	m	m
19	Expires	[26] 20.19	o	o	[26] 20.19	o	o
19A	Feature-Caps	[190]	c59	c59	[190]	c58	c58
20	From	[26] 20.20	m	m	[26] 20.20	m	m
20A	Geolocation	[89] 4.1	c33	c33	[89] 4.1	c33	c33
20B	Geolocation-Routing	[89] 4.2	c33	c33	[89] 4.2	c33	c33
20C	History-Info	[66] 4.1	c31	c31	[66] 4.1	c31	c31
20D	Identity	[252] 4	c68	c68	[252] 4	c68	c68
21	In-Reply-To	[26] 20.21	o	o	[26] 20.21	o	o
21A	Join	[61] 7.1	c30	c30	[61] 7.1	c30	c30
21B	Max-Breadth	[117] 5.8	n/a	c45	[117] 5.8	c46	c46
22	Max-Forwards	[26] 20.22	m	m	[26] 20.22	n/a	c52
23	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
23A	Min-SE	[58] 5	c26	c26	[58] 5	c25	c25
24	Organization	[26] 20.25	o	o	[26] 20.25	o	o
24AA	Origination-Id	7.2.19	n/a	c72	7.2.19	n/a	c72
24A	P-Access-Network-Info	[52] 4.4, [234] 2	c15	c16	[52] 4.4, [234] 2	c15	c17
24B	P-Asserted-Identity	[34] 9.1	n/a	c65	[34] 9.1	c7	c7
24C	P-Asserted-Service	[121] 4.1	n/a	c67	[121] 4.1	c38	c38
24D	P-Called-Party-ID	[52] 4.2	x	x	[52] 4.2	c13	c13
24E	P-Charging-Function-Addresses	[52] 4.5	c20	c21	[52] 4.5	c20	c21
24F	P-Charging-Vector	[52] 4.6	c18	c19	[52] 4.6	c18	c19
24H	P-Early-Media	[109] 8	c34	c34	[109] 8	c34	c34
25	P-Media-Authorization	[31] 5.1	n/a	n/a	[31] 5.1	c11	c12
25A	P-Preferred-Identity	[34] 9.2	c7	c5	[34] 9.2	n/a	n/a
25B	P-Preferred-Service	[121] 4.2	c37	c36	[121] 4.2	n/a	n/a
25C	P-Private-Network-Indication	[134]	c42	c42	[134]	c42	c42
25D	P-Profile-Key	[97] 5	n/a	n/a	[97] 5	n/a	n/a
25E	P-Served-User	[133] 6	c51	c51	[133] 6	c51	c51
25F	P-User-Database	[82] 4	n/a	n/a	[82] 4	n/a	n/a
25G	P-Visited-Network-ID	[52] 4.3	x (note 3)	x	[52] 4.3	c14	n/a
26	Priority	[26] 20.26	o	o	[26] 20.26	o	o
26AA	Priority-Share	Subclause 7.2.16	n/a	c66	Subclause 7.2.16	n/a	c66
26AB	Priority-Verstat	Subclause 7.2.21	n/a	c77	Subclause 7.2.21	n/a	c78
26A	Privacy	[33] 4.2	c9	c9	[33] 4.2	c9	c9
26B	Priv-Answer-Mode	[158]	c49	c49	[158]	c50	c50

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
27	Proxy-Authorization	[26] 20.28	c6	c6	[26] 20.28	n/a	n/a
28	Proxy-Require	[26] 20.29	o (note 2)	o (note 2)	[26] 20.29	n/a	n/a
28A	Reason	[34A] 2	c8	c8	[34A] 2	c8	c55
29	Record-Route	[26] 20.30	n/a	c52	[26] 20.30	m	m
29A	Recv-Info	[25] 5.2.3	c48	c48	[25] 5.2.3	c48	c48
30	Referred-By	[59] 3	c27	c27	[59] 3	c28	c28
31	Reject-Contact	[56B] 9.2	c24	c24	[56B] 9.2	c32	c32
31A	Relayed-Charge	7.2.12	n/a	c61	7.2.12	n/a	c61
31B	Replaces	[60] 6.1	c29	c29	[60] 6.1	c29	c29
31C	Reply-To	[26] 20.31	o	o	[26] 20.31	o	o
31D	Request-Disposition	[56B] 9.1	c24	c24	[56B] 9.1	c32	c32
32	Require	[26] 20.32	m	m	[26] 20.32	m	m
32A	Resource-Priority	[116] 3.1	c35	c35	[116] 3.1	c35	c35
32B	Restoration-Info	Subclause 7.2.11	n/a	n/a	Subclause 7.2.11	n/a	c60
32C	Resource-Share	Subclause 7.2.13	n/a	c62	Subclause 7.2.13	n/a	c62
33	Route	[26] 20.34	m	m	[26] 20.34	n/a	c52
33A	Security-Client	[48] 2.3.1	c22	c22	[48] 2.3.1	n/a	n/a
33B	Security-Verify	[48] 2.3.1	c23	c23	[48] 2.3.1	n/a	n/a
33DA	Service-Interact-Info	Subclause 7.2.14	n/a	c73	Subclause 7.2.14	n/a	c74
33D	Session-Expires	[58] 4	c25	c25	[58] 4	c25	c25
33E	Session-ID	[162]	o	c54	[162]	o	c54
34	Subject	[26] 20.36	o	o	[26] 20.36	o	o
35	Supported	[26] 20.37	m	m	[26] 20.37	m	m
35A	Target-Dialog	[184] 7	c56	c56	[184] 7	c57	c57
36	Timestamp	[26] 20.38	c10	c10	[26] 20.38	m	m
37	To	[26] 20.39	m	m	[26] 20.39	m	m
37A	Trigger-Consent	[125] 5.11.2	c39	c39	[125] 5.11.2	c40	c40
38	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
38A	User-to-User	[126] 7	c41	c41	[126] 7	c41	c41
39	Via	[26] 20.42	m	m	[26] 20.42	m	m

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
c1:	IF A.4/12 THEN m ELSE n/a - - downloading of alerting information.						
c2:	IF A.4/22 THEN m ELSE n/a - - acting as the notifier of event information.						
c3:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.						
c4:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						
c5:	IF A.3/1 AND A.4/25 THEN o ELSE n/a - - UE and private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.						
c6:	IF A.4/8A THEN m ELSE n/a - - authentication between UA and proxy.						
c7:	IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.						
c8:	IF A.4/38 THEN o ELSE n/a - - the Reason header field for the session initiation protocol.						
c9:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).						
c10:	IF A.4/6 THEN o ELSE n/a - - timestamping of requests.						
c11:	IF A.4/19 THEN m ELSE n/a - - SIP extensions for media authorization.						
c12:	IF A.3/1 AND A.4/19 THEN m ELSE n/a - - UE, SIP extensions for media authorization.						
c13:	IF A.4/32 THEN o ELSE n/a - - the P-Called-Party-ID extension.						
c14:	IF A.4/33 THEN o ELSE n/a - - the P-Visited-Network-ID extension.						
c15:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.						
c16:	IF A.4/34 AND (A.3/1 OR A.3/2A OR A.3/7 OR A.3A/81 OR A.3A/81A OR A.3A/81B OR A.3/6) THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE, P-CSCF (IMS-ALG), the AS, the MSC server enhanced for ICS, MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface or MGCF.						
c17:	IF A.4/34 AND (A.3/2A OR A.3A/81 OR A.3/7A OR A.3/7D OR A3A/84 OR A.3/6) THEN m ELSE n/a - - the P-Access-Network-Info header extension and P-CSCF (IMS-ALG), the MSC server enhanced for ICS, AS acting as terminating UA, AS acting as third-party call controller, EATF or MGCF.						
c18:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.						
c19:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.						
c20:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.						
c21:	IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.						
c22:	IF A.4/37 OR A.4/37A THEN o ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media (note 4).						
c23:	IF A.4/37 OR A.4/37A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.						
c24:	IF A.4/40 THEN o ELSE n/a - - caller preferences for the session initiation protocol.						
c25:	IF A.4/42 THEN m ELSE n/a - - the SIP session timer.						
c26:	IF A.4/42 THEN o ELSE n/a - - the SIP session timer.						
c27:	IF A.4/43 THEN m ELSE n/a - - the SIP Referred-By mechanism.						
c28:	IF A.4/43 THEN o ELSE n/a - - the SIP Referred-By mechanism.						
c29:	IF A.4/44 THEN m ELSE n/a - - the Session Initiation Protocol (SIP) "Replaces" header.						
c30:	IF A.4/45 THEN m ELSE n/a - - the Session Initiation Protocol (SIP) "Join" header.						
c31:	IF A.4/47 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.						
c32:	IF A.4/40 THEN m ELSE n/a - - caller preferences for the session initiation protocol.						
c33:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.						
c34:	IF A.4/66 THEN m ELSE n/a - - The SIP P-Early-Media private header extension for authorization of early media.						
c35:	IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						
c36:	IF (A.3/1 OR A.3A/81 OR A.3A/81A OR A.3A/81B) AND A.4/74 THEN o ELSE n/a - - UE, MSC Server enhanced for ICS, MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface and SIP extension for the identification of services.						
c37:	IF A.4/74 THEN o ELSE n/a - - SIP extension for the identification of services.						
c38:	IF A.4/74 THEN m ELSE n/a - - SIP extension for the identification of services.						
c39:	IF A.4/75A THEN m ELSE n/a - - a relay within the framework for consent-based communications in SIP.						
c40:	IF A.4/75B THEN m ELSE n/a - - a recipient within the framework for consent-based communications in SIP.						
c41:	IF A.4/76 THEN o ELSE n/a - - transporting user to user information for call centers using SIP.						
c42:	IF A.4/77 THEN m ELSE n/a - - the SIP P-Private-Network-Indication private-header (P-Header).						
c45:	IF A.4/71 AND (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C) THEN m ELSE IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - addressing an amplification vulnerability in session initiation protocol forking proxies, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling), UE, UE performing the functions of an external attached network.						
c46:	IF A.4/71 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.						
c47:	IF A.3/1 AND A.4/2B THEN m ELSE o - - UE and initiating a session.						
c48:	IF A.4/13 THEN m ELSE IF A.4/13A THEN m ELSE n/a - - SIP INFO method and package framework, legacy INFO usage.						

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
c49:	IF A.4/87 THEN o ELSE n/a - - requesting answering modes for SIP.						
c50:	IF A.4/87 THEN m ELSE n/a - - requesting answering modes for SIP.						
c51:	IF A.4/78 THEN m ELSE n/a - - the SIP P-Served-User private header.						
c52:	IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - UE, UE performing the functions of an external attached network.						
c53:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.						
c54:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.						
c55:	IF A.4/38 THEN IF A.3A/83 THEN m ELSE o ELSE n/a - - the Reason header field for the session initiation protocol, SCC application server.						
c56:	IF A.4/99 THEN o ELSE n/a - - request authorization through dialog Identification in the session initiation protocol.						
c57:	IF A.4/99 THEN m ELSE n/a - - request authorization through dialog Identification in the session initiation protocol.						
c58:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.						
c59:	IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.						
c60:	IF A. 4/109 THEN o ELSE n/a - - PCRF based P-CSCF restoration.						
c61:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.						
c62:	IF A.4/112 THEN o ELSE n/a - - resource sharing.						
c63:	IF A.4/113 AND (A.3/1 OR A.3/2A OR A.3/7) THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE, P-CSCF (IMS-ALG) or the AS.						
c64:	IF A.4/113 AND (A.3/2A OR A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE IF A.4/113 AND A.3/6 THEN o ELSE n/a - - the Cellular-Network-Info header extension and P-CSCF (IMS-ALG), AS acting as terminating UA or AS acting as third-party call controller, EATF or MGCF.						
c65:	IF A.4/25 AND (A.3/6 OR A.3/7B OR A.3/8 OR A.3A/81 OR A.3A/81A OR A.3A/81B OR A.3A/83 OR A.3A/89) THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks and MGCF, AS acting as originating UA, MRFC, MSC Server enhanced for ICS, MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface, SCC application server, ATCF (UA).						
c66:	IF A.4/114 THEN o ELSE n/a - - priority sharing.						
c67:	IF A.4/74 AND A.3/7B THEN o ELSE n/a - - SIP extension for the identification of services and AS acting as originating UA.						
c68:	IF A.4/116 AND (A.3/7 OR A.3/9) THEN m ELSE n/a - - authenticated identity management in the Session Initiation Protocol, AS, IBCF.						
c69:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).						
c70:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).						
c71:	IF A.4/121 AND (A.3/6 OR A.3/7 OR A.3/9) THEN m ELSE n/a - - the Attestation-Info header field extension, MGCF, AS, IBCF.						
c72:	IF A.4/122 AND (A.3/6 OR A.3/7 OR A.3/9) THEN m ELSE n/a - - the Origination-Id header field extension, MGCF, AS, IBCF.						
c73:	IF A.4/123 AND (A.3/7 OR A.3/9) THEN m ELSE n/a - - Dynamic services interactions, AS, IBCF.						
c74:	IF A.4/123 AND (A.3/2 OR A.3/7 OR A.3/9) THEN m ELSE n/a - - Dynamic services interactions, P-CSCF, AS, IBCF.						
c75:	IF A.4/124 THEN o ELSE n/a - - the Additional-Identity header field extension.						
c76:	IF A.4/124 THEN m ELSE n/a - - the Additional-Identity header field extension.						
c77:	IF A.4/70 AND A.4/126 AND A.3/9 THEN m ELSE n/a - - communications resource priority for the session initiation protocol, the Priority-Verstat header field extension, IBCF.						
c78:	IF A.4/70 AND A.4/126 THEN o ELSE n/a - - communications resource priority for the session initiation protocol, the Priority-Verstat header field extension.						
o.1:	At least one of these shall be supported.						
NOTE 1:	RFC 3261 [26] gives the status of this header as SHOULD rather than OPTIONAL.						
NOTE 2:	No distinction has been made in these tables between first use of a request on a From/To/Call-ID combination, and the usage in a subsequent one. Therefore the use of "o" etc. above has been included from a viewpoint of first usage.						
NOTE 3:	The strength of this requirement in RFC 7315 [52] is SHOULD NOT, rather than MUST NOT.						
NOTE 4:	Support of this header in this method is dependent on the security mechanism and the security architecture which is implemented. Use of this header in this method is not appropriate to the security mechanism defined by 3GPP TS 33.203 [19].						

Prerequisite A.5/8 - - INVITE request

Table A.47: Supported message bodies within the INVITE request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	XML Schema for PSTN	[11B]		c1	[11B]		c1
2	application/vnd.3gpp.ussd	[8W]		c2	[8W]		c3
3	application/vnd.3gpp.mcptt-info+xml	[8ZE]	n/a	c4	[8ZE]	n/a	c4
4	application/vnd.etsi.aoc+xml	[8N] 4.7.2	n/a	c5	[8N] 4.7.2	n/a	c6
5	application/EmergencyCallData.eCall.MSD	[244] 14.3	m	c7	[244] 14.3	m	c8
c1:	IF A.3/6 OR A.3/7A OR A.3/7B OR A.3/7D OR A.3/9B OR A.3/13B THEN o ELSE n/a - - MGCF, AS acting as terminating UA, or redirect server, AS acting as originating UA, AS performing 3rd party call control, IBCF (IMS-ALG), ISC gateway function (IMS-ALG).						
c2:	IF A.3A/92A OR A.3A/93B OR A.3/9 OR A.3/2 OR A.3A/89 THEN m ELSE n/a - - USSI UE supporting user-initiated USSD operations, USSI AS supporting network-initiated USSD operations, IBCF, P-CSCF, ATCF (UA).						
c3:	IF A.3A/93A OR A.3A/92B OR A.3/9 OR A.3/2 OR A.3A/89 THEN m ELSE n/a - - USSI AS supporting user-initiated USSD operations, USSI UE supporting network-initiated USSD operations, IBCF, P-CSCF, ATCF (UA).						
c4:	IF A.3A/102 OR A.3A/103 THEN m ELSE n/a - - MCPTT client, MCPTT server.						
c5	IF A.3A/53 THEN m ELSE n/a - - Advice of charge application server.						
c6	IF A.3A/54 THEN m ELSE n/a - - Advice of charge UA client.						
c7:	IF (A.3/1 AND A.4/120) THEN m ELSE IF ((A.3/2A OR A.3/11A OR A.3A/84) AND A.4/120) THEN i ELSE n/a - - UE, Next-Generation Pan-European eCall emergency service, P-CSCF (IMS-ALG), E-CSCF acting as UA, EATF.						
c8:	IF ((A.3/2A OR A.3/11A OR A.3A/84 OR A.3/12) AND A.4/120) THEN i ELSE n/a - - P-CSCF (IMS-ALG), E-CSCF acting as UA, EATF, LRF, Next-Generation Pan-European eCall emergency service.						

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/1 - - Additional for 100 (Trying) response

Table A.48: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						

Prerequisite A.5/9 - - INVITE response for all remaining status-codes

Table A.49: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	c12	c12	[26] 20.5	m	m
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Call-Info	[26] 20.9	o	o	[26] 20.9	o	o
1B	Cellular-Network-Info	7.2.15	n/a	c20	7.2.15	n/a	c21
2	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
3	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
3A	Content-ID	[256] 3.2	o	c23	[256] 3.2	m	c24
4	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
8[a]	Expires	[26] 20.19	o	o	[26] 20.19	o	o
9	From	[26] 20.20	m	m	[26] 20.20	m	m
9A	Geolocation-Error	[89] 4.3	c14	c14	[89] 4.3	c14	c14
9B	History-Info	[66] 4.1	c13	c13	[66] 4.1	c13	c13
10	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
11	Organization	[26] 20.25	o	o	[26] 20.25	o	o
11A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c5	c6	[52] 4.4, [52A] 4, [234] 2	c5	c7
11B	P-Asserted-Identity	[34] 9.1	n/a	c22	[34] 9.1	c3	c3
11C	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c10	c11	[52] 4.5, [52A] 4	c11	c11
11D	P-Charging-Vector	[52] 4.6, [52A] 4	c8	c9	[52] 4.6, [52A] 4	c8	c9
11F	P-Preferred-Identity	[34] 9.2	c3	x	[34] 9.2	n/a	n/a
11G	Privacy	[33] 4.2	c4	c4	[33] 4.2	c4	c4
11H	Relayed-Charge	7.2.12	n/a	c19	7.2.12	n/a	c19
11I	Reply-To	[26] 20.31	o	o	[26] 20.31	o	o
11J	Require	[26] 20.32	m	m	[26] 20.32	m	m
11K	Server	[26] 20.35	o	o	[26] 20.35	o	o
11LA	Service-Interact-Info	Subclause 7.2.14	n/a	c25	Subclause 7.2.14	n/a	c26
11L	Session-ID	[162]	o	c18	[162]	o	c18
12	Timestamp	[26] 20.38	m	m	[26] 20.38	c2	c2
13	To	[26] 20.39	m	m	[26] 20.39	m	m
13A	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
13B	User-to-User	[126] 7	c15	c15	[126] 7	c15	c15
14	Via	[26] 20.42	m	m	[26] 20.42	m	m
15	Warning	[26] 20.43	o (note)	o	[26] 20.43	o	o

c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c2:	IF A.4/6 THEN m ELSE n/a - - timestamping of requests.
c3:	IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c4:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c5:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c6:	IF A.4/34 AND (A.3/1 OR A.3/2A OR A.3/7 OR A.3A/81 OR A.3/81A OR A.3A/81B) THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE, P-CSCF (IMS-ALG), AS or MSC server enhanced for ICS or MSC Server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface.
c7:	IF A.4/34 AND (A.3/2A OR A.3A/81 OR A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the P-Access-Network-Info header extension and P-CSCF (IMS-ALG), MSC server enhanced for ICS, AS acting as terminating UA, AS acting as third-party call controller or EATF.
c8:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c9:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c10:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c11:	IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c12:	IF A.6/102 OR A.6/18 THEN m ELSE o - - 2xx response, 405 (Method Not Allowed).
c13:	IF A.4/47 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c14:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c15:	IF A.4/76 THEN o ELSE n/a - - transporting user to user information for call centers using SIP.
c18:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c19:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c20:	IF A.4/113 AND (A.3/1 OR A.3/2A OR A.3/7) THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE, P-CSCF (IMS-ALG) or AS.
c21:	IF A.4/113 AND (A.3/2A OR A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the Cellular-Network-Info header extension and P-CSCF (IMS-ALG), AS acting as terminating UA, AS acting as third-party call controller or EATF.
c22:	IF A.4/25 AND (A.3/6 OR A.3/7B OR A.3/8 OR A.3A/81 OR A.3A/81A OR A.3A/81B OR A.3A/83 OR A.3A/89) THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks and MGCF, AS acting as originating UA, MRFC, MSC Server enhanced for ICS, MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface, SCC application server, ATCF (UA).
c23:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c24:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c25:	IF A.4/123 AND (A.3/7 OR A.3/9) THEN m ELSE n/a - - Dynamic services interactions, AS, IBCF.
c26:	IF A.4/123 AND (A.3/2 OR A.3/7 OR A.3/9) THEN m ELSE n/a - - Dynamic services interactions, P-CSCF, AS, IBCF.
NOTE:	For a 488 (Not Acceptable Here) response, RFC 3261 [26] gives the status of this header as SHOULD rather than OPTIONAL.

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/101A - - Additional for 18x response

Table A.50: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Contact	[26] 20.10	o	m	[26] 20.10	m	m
4A	Feature-Caps	[190]	c17	c17	[190]	c16	c16
5	P-Answer-State	[111]	c13	c13	[111]	c13	c13
5A	P-Early-Media	[109] 8	c14	c14	[109] 8	c14	c14
6	P-Media-Authorization	[31] 5.1	n/a	n/a	[31] 5.1	c11	c12
6AA	Priority-Share	Subclause 7.2.16	n/a	c19	Subclause 7.2.16	n/a	c19
6A	Reason	[130], [294]	o	c15	[130], [294]	o	c15
7	Record-Route	[26] 20.30	o	m	[26] 20.30	m	m
8	Recv-Info	[25] 5.2.3	c4	c4	[25] 5.2.3	c4	c4
8A	Resource-Share	Subclause 7.2.13	n/a	c18	Subclause 7.2.13	n/a	c18
9	RSeq	[27] 7.1	c2	m	[27] 7.1	c3	m
c2:	IF A.4/14 THEN o ELSE n/a - - reliability of provisional responses in SIP.						
c3:	IF A.4/14 THEN m ELSE n/a - - reliability of provisional responses in SIP.						
c4:	IF A.4/13 THEN m ELSE IF A.4/13A THEN m ELSE n/a - - SIP INFO method and package framework, legacy INFO usage.						
c11:	IF A.4/19 THEN m ELSE n/a - - SIP extensions for media authorization.						
c12:	IF A.3/1 AND A.4/19 THEN m ELSE n/a - - UE, SIP extensions for media authorization.						
c13:	IF A.4/65 THEN m ELSE n/a - - the P-Answer-State header extension to the session initiation protocol for the open mobile alliance push to talk over cellular.						
c14:	IF A.4/66 THEN m ELSE n/a - - the SIP P-Early-Media private header extension for authorization of early media.						
c15:	IF A.4/38A OR A.4/38C THEN o ELSE n/a - - use of the Reason header field in Session Initiation Protocol (SIP) responses (carrying Q.850 or STIR codes).						
c16:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.						
c17:	IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.						
c18:	IF A.4/112 THEN o ELSE n/a - - resource sharing.						
c19:	IF A.4/114THEN o ELSE n/a - - priority sharing.						

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/2 - - Additional for 180 (Ringing) response

Table A.50A: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Alert-Info	[26] 20.4	o	c1	[26] 20.4	o	c1
c1:	IF A.4/96 THEN m ELSE o - - Alert-Info URNs for the Session Initiation Protocol.						

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/5A - - Additional for 199 (Early Dialog Terminated) response

Table A.50B: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Contact	[26] 20.10	o	m	[26] 20.10	m	m
5	Reason	[130], [294]	o	c5	[130], [294]	o	c5
7	Record-Route	[26] 20.30	o	m	[26] 20.30	m	m
8	Recv-Info	[25] 5.2.3	c4	c4	[25] 5.2.3	c4	c4
9	RSeq	[27] 7.1	c2	m	[27] 7.1	c3	m
c2:	IF A.4/14 THEN o ELSE n/a - - reliability of provisional responses in SIP.						
c3:	IF A.4/14 THEN m ELSE n/a - - reliability of provisional responses in SIP.						
c4:	IF A.4/13 THEN m ELSE IF A.4/13A THEN m ELSE n/a - - SIP INFO method and package framework, legacy INFO usage.						
c5:	IF A.4/38A OR A.4/38C THEN o ELSE n/a - - use of the Reason header field in Session Initiation Protocol (SIP) responses (carrying Q.850 or STIR codes).						

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/102 - - Additional for 2xx response

Table A.51: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o	o	[26] 20.1	m	m
1A	Accept-Encoding	[26] 20.2	o	o	[26] 20.2	m	m
1B	Accept-Language	[26] 20.3	o	o	[26] 20.3	m	m
1C	Accept-Resource-Priority	[116] 3.2	c15	c15	[116] 3.2	c15	c15
2	Allow-Events	[28] 8.2.2	c3	c3	[28] 8.2.2	c4	c4
3	Answer-Mode	[158]	c6	c6	[158]	c7	c7
4	Authentication-Info	[26] 20.6	c1	c1	[26] 20.6	c2	c2
6	Contact	[26] 20.10	m	m	[26] 20.10	m	m
6A	Feature-Caps	[190]	c18	c18	[190]	c17	c17
7	P-Answer-State	[111]	c14	c14	[111]	c14	c14
8	P-Media-Authorization	[31] 5.1	n/a	n/a	[31] 5.1	c11	c12
8AA	Priority-Share	Subclause 7.2.16	n/a	c20	Subclause 7.2.16	n/a	c20
8A	Priv-Answer-Mode	[158]	c6	c6	[158]	c7	c7
8B	Reason	[294]	o	c21	[294]	o	c21
9	Record-Route	[26] 20.30	m	m	[26] 20.30	m	m
9A	Recv-Info	[25] 5.2.3	c5	c5	[25] 5.2.3	c5	c5
9B	Resource-Share	Subclause 7.2.13	n/a	c19	Subclause 7.2.13	n/a	c19
10	Session-Expires	[58] 4	c13	c13	[58] 4	c13	c13
13	Supported	[26] 20.37	m	m	[26] 20.37	m	m

c1: IF A.4/7 THEN o ELSE n/a - - authentication between UA and UA.
c2: IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.
c3: IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.
c4: IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.
c5: IF A.4/13 THEN m ELSE IF A.4/13A THEN m ELSE n/a - - SIP INFO method and package framework, legacy INFO usage.
c6 IF A.4/87 THEN o ELSE n/a - - requesting answering modes for SIP.
c7 IF A.4/87 THEN m ELSE n/a - - requesting answering modes for SIP.
c11: IF A.4/19 THEN m ELSE n/a - - SIP extensions for media authorization.
c12: IF A.3/1 AND A.4/19 THEN m ELSE n/a - - UE, SIP extensions for media authorization.
c13: IF A.4/42 THEN m ELSE n/a - - the SIP session timer.
c14: IF A.4/65 THEN m ELSE n/a - - the P-Answer-State header extension to the session initiation protocol for the open mobile alliance push to talk over cellular.
c15: IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.
c17: IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.
c18: IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.
c19: IF A.4/112 THEN m ELSE n/a - - resource sharing.
c20: IF A.4/114THEN o ELSE n/a - - priority sharing.
c21: IF A.4/38C THEN o ELSE n/a - - use of the Reason header field in Session Initiation Protocol (SIP) responses (carrying STIR codes).

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/103 OR A.6/104 OR A.6/105 OR A.6/106 - - Additional for 3xx – 6xx response

Table A.51A: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	o	o	[26] 20.18	o	o
2	Reason	[130], [294]	o	c1	[130], [294]	o	c1
3	Response-Source	7.2.17	n/a	c2	7.2.17	n/a	c2
c1:	IF A.4/38A OR A.4/38C THEN o ELSE n/a - - use of the Reason header field in Session Initiation Protocol (SIP) responses (carrying Q.850 or STIR codes).						
c2:	IF A.4/115 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/103 OR A.6/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.52: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Contact	[26] 20.10	o (note 1)	o	[26] 20.10	m	m
NOTE:	The strength of this requirement is RECOMMENDED rather than OPTIONAL.						

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/14 - - Additional for 401 (Unauthorized) response

Table A.53: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
6	Proxy-Authenticate	[26] 20.27	c3	c3	[26] 20.27	c3	c3
13	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	m	m
c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						
c2:	IF A.4/6 THEN m ELSE n/a - - timestamping of requests.						
c3:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/16 - - Additional for 403 (Forbidden) response

Table A.53A: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	P-Refused-URI-List	[183]	c1	c1	[183]	c1	c1
c1:	IF A.4/98 THEN m ELSE n/a -- The SIP P-Refused-URI-List private-header.						

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/17 OR A.6/23 OR A.6/30 OR A.6/36 OR A.6/50 OR A.6/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 600 (Busy Everywhere), 603 (Decline) response

Table A.54: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
8	Retry-After	[26] 20.33	o	o	[26] 20.33	o	o

Table A.55: Void

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.56: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
6	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
11	WWW-Authenticate	[26] 20.44	o	o	[26] 20.44	o	o
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/21 - - Additional for 408 (Request timeout) response

Table A.56A: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Restoration-Info	subclause 7.2.11	n/a	c1	subclause 7.2.11	n/a	n/a
c1:	IF A.4/110 THEN o ELSE n/a - - HSS based P-CSCF restoration.						

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/25 - - Additional for 415 (Unsupported Media Type) response

Table A.57: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o.1	o.1	[26] 20.1	m	m
2	Accept-Encoding	[26] 20.2	o.1	o.1	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o.1	o.1	[26] 20.3	m	m
o.1	At least one of these capabilities is supported.						

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.57A: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/27 - - Additional for 420 (Bad Extension) response

Table A.58: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
10	Unsupported	[26] 20.40	m	m	[26] 20.40	m	m

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/28 OR A.6/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.58A: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	x	x	[48] 2	c1	c1
c1:	IF A.4/37 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/28A - - Additional for 422 (Session Interval Too Small) response

Table A.58B: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Min-SE	[58] 5	c1	c1	[58] 5	c1	c1
c1:	IF A.4/42 THEN o ELSE n/a - - the SIP session timer.						

Table A.59: Void

Table A.60: Void

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/29H - - Additional for 470 (Consent Needed) response

Table A.60A: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Permission-Missing	[125] 5.9.3	m	m	[125] 5.9.3	m	m

Prerequisite A.5/9 - - INVITE response

Prerequisite: A.6/45 - - 503 (Service Unavailable)

Table A.61: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
8	Retry-After	[26] 20.33	o	o	[26] 20.33	o	m

Table A.61A: Void

Prerequisite A.5/9 - - INVITE response

Table A.62: Supported message bodies within the INVITE response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	XML Schema for PSTN	[11B]		c1	[11B]		c1
2	Recipient list	[183]	c2	c2	[183]	c2	c2
3	3GPP IM CN subsystem XML body	subclause 7.6	n/a	c3	subclause 7.6	n/a	c4 (note)
4	application/vnd.3gpp.mcptt-info+xml	[8ZE]	n/a	c5	[8ZE]	n/a	c5
5	application/vnd.etsi.aoc+xml	[8N] 4.7.2	n/a	c6	[8N] 4.7.2	n/a	c7
6	application/EmergencyCallData.Control+xml	[244] 14.4	m	c8	[244] 14.4	m	c9
c1:	IF A.3/6 OR A.3/7A OR A.3/7B OR A.3/7D OR A.3/9B OR A.3/13B THEN o ELSE n/a - - MGCF, AS acting as terminating UA, or redirect server, AS acting as originating UA, AS performing 3rd party call control, IBCF (IMS-ALG), ISC gateway function (IMS-ALG).						
c2:	IF A.3/9B OR A.3/13B THEN m ELSE IF A.3/7A OR A.3/7B OR A.3/7D THEN o ELSE n/a - - IBCF (IMS-ALG), ISC gateway function (IMS-ALG), AS acting as terminating UA, AS acting as originating UA, AS performing 3rd party call control.						

c3:	IF A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C OR (A.4/103 AND A.3/2) OR (A.4/103 AND A.3/4) THEN m ELSE n/a - - IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling), S-CSCF restoration procedures, P-CSCF, S-CSCF.
c4:	IF A.3/1 OR A.3/2 OR A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C THEN m ELSE IF A.3/4 THEN o ELSE n/a - - UE, P-CSCF, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), S-CSCF.
c5:	IF A.3A/102 OR A.3A/103 THEN M ELSE n/a - - MCPTT client, MCPTT server.
c6	IF A.3A/53 THEN m ELSE n/a - - Advice of charge application server.
c7	IF A.3A/54 THEN m ELSE n/a - - Advice of charge UA client.
c8:	IF ((A.3/2A OR A.3/11A OR A.3A/84) AND A.4/120) THEN i ELSE n/a - - P-CSCF (IMS-ALG), E-CSCF acting as UA, EATF, Next-Generation Pan-European eCall emergency service.
c9:	IF (A.3/1 AND A.4/120) THEN m ELSE IF ((A.3/2A OR A.3/11A OR A.3A/84) AND A.4/120) THEN i ELSE n/a - - UE, Next-Generation Pan-European eCall emergency service, P-CSCF (IMS-ALG), E-CSCF acting as UA, EATF.
NOTE:	If a IBCF (IMS-ALG) or a IBCF (Screening of SIP signalling) is unable to receive a 3GPP IM CN subsystem XML body from a S-CSCF in a serving network then the IBCF (IMS-ALG) or the IBCF (Screening of SIP signalling) support can be "o" instead of "m". Examples include an S-CSCF supporting S-CSCF restoration procedures.

A.2.1.4.7A MESSAGE method

Prerequisite A.5/9A - - MESSAGE request

Table A.62A: Supported header fields within the MESSAGE request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Contact	[56B] 9.2	c24	c24	[56B] 9.2	c28	c28
1AA	Additional-Identity	7.2.20	n/a	c58	7.2.20	n/a	c59
1A	Allow	[26] 20.5	o	o	[26] 20.5	m	m
2	Allow-Events	[28] 8.2.2	c1	c1	[28] 8.2.2	c2	c2
2A	Attestation-Info	7.2.18	n/a	c54	7.2.18	n/a	c54
3	Authorization	[26] 20.7	c3	c3	[26] 20.7	c3	c3
4	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
5	Call-Info	[26] 20.9	o	o	[26] 20.9	o	o
5A	Cellular-Network-Info	7.2.15	n/a	c47	7.2.15	n/a	c48
6	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
7	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
7A	Content-ID	[256] 3.2	o	c52	[256] 3.2	m	c53
8	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
9	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
10	Content-Type	[26] 20.15	m	m	[26] 29.15	m	m
11	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
12	Date	[26] 20.17	c4	c4	[26] 20.17	m	m
13	Expires	[26] 20.19	o	o	[26] 20.19	o	o
13A	Feature-Caps	[190]	c45	c45	[190]	c44	c44
14	From	[26] 20.20	m	m	[26] 20.20	m	m
14A	Geolocation	[89] 4.1	c29	c29	[89] 4.1	c29	c29
14B	Geolocation-Routing	[89] 4.2	c29	c29	[89] 4.2	c29	c29
14C	History-Info	[66] 4.1	c27	c27	[66] 4.1	c27	c27
14D	Identity	[252] 4	c51	c51	[252] 4	c51	c51
15	In-Reply-To	[26] 20.21	o	o	[26] 20.21	o	o
15A	Max-Breadth	[117] 5.8	n/a	c39	[117] 5.8	c40	c40
16	Max-Forwards	[26] 20.22	m	m	[26] 20.22	n/a	c42
17	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
18	Organization	[26] 20.25	o	o	[26] 20.25	o	o
18AA	Origination-Id	7.2.19	n/a	c55	7.2.19	n/a	c55
18A	P-Access-Network-Info	[52] 4.4, [234] 2	c15	c16	[52] 4.4, [234] 2	c15	c16
18B	P-Asserted-Identity	[34] 9.1	n/a	c49	[34] 9.1	c11	c11
18C	P-Asserted-Service	[121] 4.1	n/a	c50	[121] 4.1	c33	c33
18D	P-Called-Party-ID	[52] 4.2	x	x	[52] 4.2	c13	c13
18E	P-Charging-Function-Addresses	[52] 4.5	c20	c21	[52] 4.5	c20	c21
18F	P-Charging-Vector	[52] 4.6	c18	c19	[52] 4.6	c18	c19
18H	P-Preferred-Identity	[34] 9.2	c11	c7	[34] 9.2	n/a	n/a
18I	P-Preferred-Service	[121] 4.2	c32	c31	[121] 4.2	n/a	n/a
18J	P-Private-Network-Indication	[134]	c36	c36	[134]	c36	c36
18K	P-Profile-Key	[97] 5	n/a	n/a	[97] 5	n/a	n/a
18L	P-Served-User	[133] 6	c41	c41	[133] 6	c41	c41
18M	P-User-Database	[82] 4	n/a	n/a	[82] 4	n/a	n/a
18N	P-Visited-Network-ID	[52] 4.3	x (note 1)	x	[52] 4.3	c14	n/a
19	Priority	[26] 20.26	o	o	[26] 20.26	o	o
19A	Privacy	[33] 4.2	c12	c12	[33] 4.2	c12	c12
20	Proxy-Authorization	[26] 20.28	c5	c5	[26] 20.28	n/a	n/a
21	Proxy-Require	[26] 20.29	o	n/a	[26] 20.29	n/a	n/a
21A	Reason	[34A] 2	c6	c6	[34A] 2	c6	c6
22A	Referred-By	[59] 3	c25	c25	[59] 3	c26	c26
23	Reject-Contact	[56B] 9.2	c24	c24	[56B] 9.2	c28	c28
23A	Relayed-Charge	7.2.12	n/a	c46	7.2.12	n/a	c46
23B	Reply-To	[26] 20.31	o	o	[26] 20.31	o	o
23C	Request-Disposition	[56B] 9.1	c24	c24	[56B] 9.1	c28	c28
24	Require	[26] 20.32	m	m	[26] 20.32	m	m
24A	Resource-Priority	[116] 3.1	c30	c30	[116] 3.1	c30	c30
25	Route	[26] 20.34	m	m	[26] 20.34	n/a	n/a
25A	Security-Client	[48] 2.3.1	c22	c22	[48] 2.3.1	n/a	n/a
25B	Security-Verify	[48] 2.3.1	c23	c23	[48] 2.3.1	n/a	n/a

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
25CA	Service-Interact-Info	Subclause 7.2.14	n/a	c56	Subclause 7.2.14	n/a	c57
25C	Session-ID	[162]	o	c43	[162]	o	c43
26	Subject	[26] 20.35	o	o	[26] 20.36	o	o
27	Supported	[26] 20.37	c9	m	[26] 20.37	m	m
28	Timestamp	[26] 20.38	c10	c10	[26] 20.38	m	m
29	To	[26] 20.39	m	m	[26] 20.39	m	m
29A	Trigger-Consent	[125] 5.11.2	c34	c34	[125] 5.11.2	c35	c35
30	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
31	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.
c2:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.
c3:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.
c4:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c5:	IF A.4/8A THEN m ELSE n/a - - authentication between UA and proxy.
c6:	IF A.4/38 THEN o ELSE n/a - - the Reason header field for the session initiation protocol.
c7:	IF A.3/1 AND A.4/25 THEN o ELSE n/a - - UE and private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c9:	IF A.4/14 THEN m ELSE o - - support of reliable transport.
c10:	IF A.4/6 THEN o ELSE n/a - - timestamping of requests.
c11:	IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c12:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c13:	IF A.4/32 THEN o ELSE n/a - - the P-Called-Party-ID extension.
c14:	IF A.4/33 THEN o ELSE n/a - - the P-Visited-Network-ID extension.
c15:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c16:	IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.
c17:	IF A.4/34 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c18:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c19:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c20:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c21:	IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c22:	IF A.4/37 OR A.4/37A THEN o ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media (note 2).
c23:	IF A.4/37 OR A.4/37A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c24:	IF A.4/40 THEN o ELSE n/a - - caller preferences for the session initiation protocol.
c25:	IF A.4/43 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c26:	IF A.4/43 THEN o ELSE n/a - - the SIP Referred-By mechanism.
c27:	IF A.4/47 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c28:	IF A.4/40 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c29:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c30:	IF A.4/70A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.
c31:	IF A.3/1 AND A.4/74 THEN o ELSE n/a - - UE and SIP extension for the identification of services.
c32:	IF A.4/74 THEN o ELSE n/a - - SIP extension for the identification of services.
c33:	IF A.4/74 THEN m ELSE n/a - - SIP extension for the identification of services.
c34:	IF A.4/75A THEN m ELSE n/a - - a relay within the framework for consent-based communications in SIP.
c35:	IF A.4/75B THEN m ELSE n/a - - a recipient within the framework for consent-based communications in SIP.
c36:	IF A.4/77 THEN m ELSE n/a - - the SIP P-Private-Network-Indication private-header (P-Header).
c39:	IF A.4/71 AND (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C) THEN m ELSE IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - addressing an amplification vulnerability in session initiation protocol forking proxies, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling), UE, UE performing the functions of an external attached network.
c40:	IF A.4/71 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c41:	IF A.4/78 THEN m ELSE n/a - - the SIP P-Served-User private header.
c42:	IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - UE, UE performing the functions of an external attached network.
c43:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c44:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.
c45:	IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.
c46:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c47:	IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE.
c48:	IF A.4/113 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c49:	IF A.4/25 AND (A.3/7B OR A.3/8) THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks and AS acting as originating UA, MRFC.
c50:	IF A.4/74 AND A.3/7B THEN o ELSE n/a - - SIP extension for the identification of services and AS acting as originating UA.
c51:	IF A.4/116 AND (A.3/7 OR A.3/9) THEN m ELSE n/a - - authenticated identity management in the Session Initiation Protocol, AS, IBCF.
c52:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c53:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
c54:	IF A.4/121 AND (A.3/6 OR A.3/7 OR A.3/9) THEN m ELSE n/a - - the Attestation-Info header field extension, MGCF, AS, IBCF.						
c55:	IF A.4/122 AND (A.3/6 OR A.3/7 OR A.3/9) THEN m ELSE n/a - - the Origination-Id header field extension, MGCF, AS, IBCF.						
c56:	IF A.4/123 AND (A.3/7 OR A.3/9) THEN m ELSE n/a - - Dynamic services interactions, AS, IBCF.						
c57:	IF A.4/123 AND (A.3/2 OR A.3/7 OR A.3/9) THEN m ELSE n/a - - Dynamic services interactions, P-CSCF, AS, IBCF.						
c58:	IF A.4/124 THEN o ELSE n/a - - the Additional-Identity header field extension.						
c59:	IF A.4/124 THEN m ELSE n/a - - the Additional-Identity header field extension.						
NOTE 1:	The strength of this requirement in RFC 7315 [52] is SHOULD NOT, rather than MUST NOT.						
NOTE 2:	Support of this header in this method is dependent on the security mechanism and the security architecture which is implemented. Use of this header in this method is not appropriate to the security mechanism defined by 3GPP TS 33.203 [19].						

Prerequisite A.5/9A - - MESSAGE request

Table A.62B: Supported message bodies within the MESSAGE request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	permission document	[125] 5.4	c1	c1	[125] 5.4	c2	c2
2	application/vnd.3gpp.sms	[4D]	c3	c3	[4D]	c3	c3
3	message/cpim	[151]	c4	c4	[151]	c4	c4
4	message/imdn+xml	[157]	c5	c5	[157]	c5	c5
5	application/vnd.3gpp.mcptt-info+xml	[8ZE]	n/a	c6	[8ZE]	n/a	c6
6	application/vnd.3gpp.mcptt-mbms-usage-info+xml	[8ZE]	n/a	c6	[8ZE]	n/a	c6
7	application/vnd.3gpp.mcptt-location-info+xml	[8ZE]	n/a	c6	[8ZE]	n/a	c6
8	application/vnd.3gpp.mcptt-floor-request+xml	[8ZE]	n/a	c7	[8ZE]	n/a	c7
9	application/vnd.3gpp.mcptt-affiliation-command+xml	[8ZE]	n/a	c6	[8ZE]	n/a	c6
c1:	IF A.4/75A THEN m ELSE n/a - - a relay within the framework for consent-based communications in SIP.						
c2:	IF A.4/75B THEN m ELSE n/a - - a recipient within the framework for consent-based communications in SIP.						
c3:	IF A.3A/61 OR A.3A/62 OR A.3A/63 THEN m ELSE o - - an SM-over-IP sender or an SM-over-IP receiver or an IP-SM-GW for SMS over IP.						
c4:	IF A.3A/71 AND A.4/85 THEN m ELSE n/a - - common presence and instant messaging (CPIM): message format.						
c5:	IF A.3A/71 AND A.4/86 THEN m ELSE n/a - - instant message disposition notification.						
c6:	IF A.3A/102 OR A.3A/103 THEN m ELSE n/a - - MCPTT client, MCPTT server.						
c7:	IF A.3A/103 THEN m ELSE n/a - - MCPTT server.						

Prerequisite A.5/9B - - MESSAGE response

Prerequisite: A.6/1 - - Additional for 100 (Trying) response

Table A.62BA: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						

Prerequisite A.5/9B - - MESSAGE response for all remaining status-codes

Table A.62C: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	c12	c12	[26] 20.5	m	m
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Call-Info	[26] 20.9	o	o	[26] 20.9	o	o
2A	Cellular-Network-Info	7.2.15	n/a	c19	7.2.15	n/a	c20
3	Content-Disposition	[26] 20.11	o (note 1)	o (note 1)	[26] 20.11	m (note 1)	m (note 1)
4	Content-Encoding	[26] 20.12	o (note 1)	o (note 1)	[26] 20.12	m (note 1)	m (note 1)
4A	Content-ID	[256] 3.2	o	c22	[256] 3.2	m	c23
5	Content-Language	[26] 20.13	o (note 1)	o (note 1)	[26] 20.13	m (note 1)	m (note 1)
6	Content-Length	[26] 20.14	m (note 1)	m (note 1)	[26] 20.14	m (note 1)	m (note 1)
7	Content-Type	[26] 20.15	m (note 1)	m (note 1)	[26] 20.15	m (note 1)	m (note 1)
8	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
9	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
9A	Expires	[26] 20.19	o	o	[26] 20.19	o	o
10	From	[26] 20.20	m	m	[26] 20.20	m	m
10A	Geolocation-Error	[89] 4.3	c14	c14	[89] 4.3	c14	c14
10B	History-Info	[66] 4.1	c13	c13	[66] 4.1	c13	c13
11	MIME-Version	[26] 20.24	o (note 1)	o (note 1)	[26] 20.24	m (note 1)	m (note 1)
12	Organization	[26] 20.25	o	o	[26] 20.25	o	o
12A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c5	c6	[52] 4.4, [52A] 4, [234] 2	c5	c7
12B	P-Asserted-Identity	[34] 9.1	n/a	c21	[34] 9.1	c3	c3
12C	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c10	c11	[52] 4.5, [52A] 4	c10	c11
12D	P-Charging-Vector	[52] 4.6, [52A] 4	c8	c9	[52] 4.6, [52A] 4	c8	c9
12F	P-Preferred-Identity	[34] 9.2	c3	x	[34] 9.2	n/a	n/a
12G	Privacy	[33] 4.2	c4	c4	[33] 4.2	c4	c4
12H	Relayed-Charge	7.2.12	n/a	c18	7.2.12	n/a	c18
12I	Reply-To	[26] 20.31	o	o	[26] 20.31	o	o
12J	Require	[26] 20.32	m	m	[26] 20.32	m	m
13	Server	[26] 20.35	o	o	[26] 20.35	o	o
13AA	Service-Interact-Info	Subclause 7.2.14	n/a	c24	Subclause 7.2.14	n/a	c25
13A	Session-ID	[162]	o	c17	[162]	o	c17
14	Timestamp	[26] 20.38	m	m	[26] 20.38	c2	c2
15	To	[26] 20.39	m	m	[26] 20.39	m	m
16	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
17	Via	[26] 20.42	m	m	[26] 20.42	m	m
18	Warning	[26] 20.43	o	o	[26] 20.43	o	o

c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c2:	IF A.4/6 THEN m ELSE n/a - - timestamping of requests.
c3:	IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c4:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c5:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c6:	IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.
c7:	IF A.4/34 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c8:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c9:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c10:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c11:	IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c12:	IF A.6/18 THEN m ELSE o - - 405 (Method Not Allowed).
c13:	IF A.4/47 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c14:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c17:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c18:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c19:	IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE.
c20:	IF A.4/113 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c21:	IF A.4/25 AND (A.3/7B OR A.3/8 OR A.3A/83 OR A.3A/89) THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks and AS acting as originating UA, MRFC, SCC application server, ATCF (UA).
c22:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c23:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c24:	IF A.4/123 AND (A.3/7 OR A.3/9) THEN m ELSE n/a - - Dynamic services interactions, AS, IBCF.
c25:	IF A.4/123 AND (A.3/2 OR A.3/7 OR A.3/9) THEN m ELSE n/a - - Dynamic services interactions, P-CSCF, AS, IBCF.
NOTE 1:	RFC 3428 [50] clause 7 states that all 2xx class responses to a MESSAGE request must not include any body, therefore for 2xx responses to the MESSAGE request the values on Sending side for "RFC status" and "Profile status" are "x", the values for Receiving side for "RFC status" and "Profile Status" are "n/a". RFC 3261 [26] subclause 7.4 states that all responses may contain bodies, therefore for all responses to the MESSAGE request other than 2xx responses, the values on Sending side for "RFC status" and "Profile status" are "o", the values for Receiving side for "RFC status" and "Profile Status" are "m".

Prerequisite A.5/9B - - MESSAGE response

Prerequisite: A.6/102 - - Additional for 2xx response

Table A.62D: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Accept-Resource-Priority	[116] 3.2	c5	c5	[116] 3.2	c5	c5
1	Allow-Events	[28] 8.2.2	c3	c3	[28] 8.2.2	c4	c4
2	Authentication-Info	[26] 20.6	c1	c1	[26] 20.6	c2	c2
3	Feature-Caps	[190]	c8	c8	[190]	c7	c7
3A	Reason	[294]	o	c9	[294]	o	c9
6	Supported	[26] 20.37	o	o	[26] 20.37	m	m
c1:	IF A.4/7 THEN o ELSE n/a - - authentication between UA and UA.						
c2:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.						
c3:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.						
c4:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.						
c5:	IF A.4/70A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						
c7:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.						
c8:	IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.						
c9:	IF A.4/38C THEN o ELSE n/a - - use of the Reason header field in Session Initiation Protocol (SIP) responses (carrying STIR codes).						

Prerequisite A.5/9B - - MESSAGE response

Prerequisite: A.6/103 OR A.6/104 OR A.6/105 OR A.6/106 - - Additional for 3xx – 6xx response

Table A.62DA: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	o	o	[26] 20.18	o	o
1A	Reason	[294]	o	c2	[294]	o	c2
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.4/115 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						
c2:	IF A.4/38C THEN o ELSE n/a - - use of the Reason header field in Session Initiation Protocol (SIP) responses (carrying STIR codes).						

Prerequisite A.5/9B - - MESSAGE response

Prerequisite: A.6/103 - - Additional for 3xx or 485 (Ambiguous) response

Table A.62E: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Contact	[26] 20.10	o (note)	o	[26] 20.10	m	m
NOTE:	The strength of this requirement is RECOMMENDED rather than OPTIONAL.						

Prerequisite A.5/9B - - MESSAGE response

Prerequisite: A.6/14 - - Additional for 401 (Unauthorized) response

Table A.62F: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
6	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	m	m
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Prerequisite A.5/9B - - MESSAGE response

Prerequisite: A.6/17 OR A.6/23 OR A.6/30 OR A.6/36 OR A.6/42 OR A.6/45 OR A.6/50 OR A.6/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.62G: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Retry-After	[26] 20.33	o	o	[26] 20.33	o	o

Table A.62H: Void

Prerequisite A.5/9B - - MESSAGE response

Prerequisite: A.6/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.62I: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
6	WWW-Authenticate	[26] 20.44	o	o	[26] 20.44	o	o
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Table A.62IA: Void

Prerequisite A.5/9B - - MESSAGE response

Prerequisite: A.6/25 - - Additional for 415 (Unsupported Media Type) response

Table A.62J: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o.1	o.1	[26] 20.1	m	m
2	Accept-Encoding	[26] 20.2	o.1	o.1	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o.1	o.1	[26] 20.3	m	m
o.1	At least one of these capabilities is supported.						

Prerequisite A.5/9B - - MESSAGE response

Prerequisite: A.6/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.62JA: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.4/70A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						

Prerequisite A.5/9B - - MESSAGE response

Prerequisite: A.6/27 - - Additional for 420 (Bad Extension) response

Table A.62K: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Unsupported	[26] 20.40	m	m	[26] 20.40	m	m

Prerequisite A.5/9B - - MESSAGE response

Prerequisite: A.6/28 OR A.6/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.62L: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	x	x	[48] 2	c1	c1
c1:	IF A.4/37 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Table A.62M: Void

Prerequisite A.5/9B - - MESSAGE response

Prerequisite: A.6/29H - - Additional for 470 (Consent Needed) response

Table A.62MA: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Permission-Missing	[125] 5.9.3	m	m	[125] 5.9.3	m	m

Prerequisite A.5/9B - - MESSAGE response

Prerequisite: A.6/46 - - Additional for 504 (Server Time-out) response

Table A.62MB: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Restoration-Info	subclause 7.2.11	n/a	c1	subclause 7.2.11	n/a	n/a
c1:	IF A.4/110 THEN o ELSE n/a - - HSS based P-CSCF restoration.						

Prerequisite A.5/9B - - MESSAGE response

Table A.62N: Supported message bodies within the MESSAGE response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.1.4.8 NOTIFY method

Prerequisite A.5/10 - - NOTIFY request

Table A.63: Supported header fields within the NOTIFY request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o	o	[26] 20.1	m	m
1A	Accept-Contact	[56B] 9.2	c19	c19	[56B] 9.2	c23	c23
2	Accept-Encoding	[26] 20.2	o	o	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o	o	[26] 20.3	m	m
3A	Allow	[26] 20.5	o	o	[26] 20.5	m	m
4	Allow-Events	[28] 8.2.2	c1	c1	[28] 8.2.2	c2	c2
5	Authorization	[26] 20.7	c3	c3	[26] 20.7	c3	c3
6	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
6A	Call-Info	[26] 20.9	o	o	[26] 20.9	c25	c25
6B	Cellular-Network-Info	7.2.15	n/a	c38	7.2.15	n/a	c39
6C	Contact	[26] 20.10	m	m	[26] 20.10	m	m
7	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
8	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
8A	Content-ID	[256] 3.2	o	c41	[256] 3.2	m	c42
9	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
10	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
11	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
12	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
13	Date	[26] 20.17	c4	c4	[26] 20.17	m	m
14	Event	[28] 8.2.1	m	m	[28] 8.2.1	m	m
14A	Feature-Caps	[190]	c35	c35	[190]	c34	c34
15	From	[26] 20.20	m	m	[26] 20.20	m	m
15A	Geolocation	[89] 4.1	c24	c24	[89] 4.1	c24	c24
15B	Geolocation-Routing	[89] 4.2	c24	c24	[89] 4.2	c24	c24
15C	History-Info	[66] 4.1	c22	c22	[66] 4.1	c22	c22
15D	Max-Breadth	[117] 5.8	n/a	c26	[117] 5.8	c27	c27
16	Max-Forwards	[26] 20.22	m	m	[26] 20.22	n/a	c32
17	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
17A	P-Access-Network-Info	[52] 4.4, [234] 2	c10	c11	[52] 4.4, [234] 2	c10	c12
17B	P-Asserted-Identity	[34] 9.1	n/a	c40	[34] 9.1	c6	c6
17C	P-Charging-Function-Addresses	[52] 4.5	c14	c15	[52] 4.5	c14	c15
17D	P-Charging-Vector	[52] 4.6	c13	c36	[52] 4.6	c13	c36
17F	P-Preferred-Identity	[34] 9.2	c6	x	[34] 9.2	n/a	n/a
17G	Privacy	[33] 4.2	c7	n/a	[33] 4.2	c7	c7
18	Proxy-Authorization	[26] 20.28	c5	c5	[26] 20.28	n/a	n/a
19	Proxy-Require	[26] 20.29	o	n/a	[26] 20.29	n/a	n/a
19A	Reason	[34A] 2	c18	c18	[34A] 2	c18	c18
20	Record-Route	[26] 20.30	n/a	c32	[26] 20.30	c9	c9
20A	Referred-By	[59] 3	c20	c20	[59] 3	c21	c21
20B	Reject-Contact	[56B] 9.2	c19	c19	[56B] 9.2	c23	c23
20C	Relayed-Charge	7.2.12	n/a	c37	7.2.12	n/a	c37
20D	Request-Disposition	[56B] 9.1	c19	c19	[56B] 9.1	c23	c23
21	Require	[26] 20.32	m	m	[26] 20.32	m	m
22A	Resource-Priority	[116] 3.1	c29	c29	[116] 3.1	c29	c29
22B	Security-Client	[48] 2.3.1	c16	c16	[48] 2.3.1	n/a	n/a
22C	Security-Verify	[48] 2.3.1	c17	c17	[48] 2.3.1	n/a	n/a
22D	Session-ID	[162]	o	c33	[162]	o	c33
22	Route	[26] 20.34	m	m	[26] 20.34	n/a	c32
23	Subscription-State	[28] 8.2.3	m	m	[28] 8.2.3	m	m
24	Supported	[26] 20.37	o	o	[26] 20.37	m	m
25	Timestamp	[26] 20.38	c8	c8	[26] 20.38	m	m
26	To	[26] 20.39	m	m	[26] 20.39	m	m
27	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
28	Via	[26] 20.42	m	m	[26] 20.42	m	m
29	Warning	[26] 20.43	o	o	[26] 20.43	o	o

c1:	IF A.4/20 THEN o ELSE n/a - - SIP specific event notification extension.
c2:	IF A.4/20 THEN m ELSE n/a - - SIP specific event notification extension.
c3:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.
c4:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c5:	IF A.4/8A THEN m ELSE n/a - - authentication between UA and proxy.
c6:	IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c7:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c8:	IF A.4/6 THEN o ELSE n/a - - timestamping of requests.
c9:	IF A.4/15 OR A.4/20 THEN m ELSE n/a - - the REFER method extension or SIP specific event notification extension.
c10:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c11:	IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.
c12:	IF A.4/34 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c13:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c14:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c15:	IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c16:	IF A.4/37 OR A.4/37A THEN o ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media (note).
c17:	IF A.4/37 OR A.4/37A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c18:	IF A.4/38 THEN o ELSE n/a - - the Reason header field for the session initiation protocol.
c19:	IF A.4/40 THEN o ELSE n/a - - caller preferences for the session initiation protocol.
c20:	IF A.4/43 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c21:	IF A.4/43 THEN o ELSE n/a - - the SIP Referred-By mechanism.
c22:	IF A.4/47 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c23:	IF A.4/40 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c24:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c25:	IF A.4/63 THEN m ELSE o - - subscriptions to request-contained resource lists in the session initiation protocol.
c26:	IF A.4/71 AND (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C) THEN m ELSE IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - addressing an amplification vulnerability in session initiation protocol forking proxies, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling), UE, UE performing the functions of an external attached network.
c27:	IF A.4/71 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c29:	IF A.4/70A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.
c32::	IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - UE, UE performing the functions of an external attached network.
c33:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c34:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.
c35:	IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.
c36:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c37:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c38:	IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE.
c39:	IF A.4/113 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c40:	IF A.4/25 AND (A.3/7B OR A.3/8) THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks and AS acting as originating UA, MRFC.
c41:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c42:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
NOTE:	Support of this header in this method is dependent on the security mechanism and the security architecture which is implemented. Use of this header in this method is not appropriate to the security mechanism defined by 3GPP TS 33.203 [19].

Prerequisite A.5/10 - - NOTIFY request

Table A.64: Supported message bodies within the NOTIFY request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	sipfrag	[37] 2	c1	c1	[37]	c1	c1
2	event package (see NOTE)	[28]	m	m	[28]	m	m
c1:	IF A.4/15 THEN m ELSE o - - the REFER method extension						
NOTE:	The appropriate body specified for the supported event package (see table A.4A) is supported.						

Prerequisite A.5/11 - - NOTIFY response

Prerequisite: A.6/1 - - Additional for 100 (Trying) response

Table A.64A: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						

Prerequisite A.5/11 - - NOTIFY response for all remaining status-codes

Table A.65: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	c11	c11	[26] 20.5	m	m
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Cellular-Network-Info	7.2.15	n/a	c18	7.2.15	n/a	c19
2	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
3	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
3A	Content-ID	[256] 3.2	o	c21	[256] 3.2	m	c22
4	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
9	From	[26] 20.20	m	m	[26] 20.20	m	m
9A	Geolocation-Error	[89] 4.3	c12	c12	[89] 4.3	c12	c12
10	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
10A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c5	c6	[52] 4.4, [52A] 4, [234] 2	c5	c7
10B	P-Asserted-Identity	[34] 9.1	n/a	c20	[34] 9.1	c3	c3
10C	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c9	c10	[52] 4.5, [52A] 4	c9	c10
10D	P-Charging-Vector	[52] 4.6, [52A] 4	c8	c16	[52] 4.6, [52A] 4	c8	c16
10F	P-Preferred-Identity	[34] 9.2	c3	x	[34] 9.2	n/a	n/a
10G	Privacy	[33] 4.2	c4	n/a	[33] 4.2	c4	c4
10H	Relayed-Charge	7.2.12	n/a	c17	7.2.12	n/a	c17
10I	Require	[26] 20.32	m	m	[26] 20.32	m	m
10J	Server	[26] 20.35	o	o	[26] 20.35	o	o
10K	Session-ID	[162]	o	c15	[162]	o	c15
11	Timestamp	[26] 20.38	m	m	[26] 20.38	c2	c2
12	To	[26] 20.39	m	m	[26] 20.39	m	m
12A	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
13	Via	[26] 20.42	m	m	[26] 20.42	m	m
14	Warning	[26] 20.43	o (note)	o	[26] 20.43	o	o

c1: IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c2: IF A.4/6 THEN m ELSE n/a - - timestamping of requests.
c3: IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c4: IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c5: IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c6: IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.
c7: IF A.4/34 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c8: IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c9: IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c10: IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c11: IF A.6/18 THEN m ELSE o - - 405 (Method Not Allowed).
c12: IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c15: IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c16: IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c17: IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c18: IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE.
c19: IF A.4/113 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c20: IF A.4/25 AND (A.3/7B OR A.3/8) THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks and AS acting as originating UA, MRFC.
c21: IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c22: IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).

NOTE: RFC 3261 [26] gives the status of this header as SHOULD rather than OPTIONAL.

Prerequisite A.5/11 - - NOTIFY response

Prerequisite: A.6/102 - - Additional for 2xx response

Table A.66: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Accept-Resource-Priority	[116] 3.2	c6	c6	[116] 3.2	c6	c6
0B	Allow-Events	[28] 8.2.2	c4	c4	[28] 8.2.2	c5	c5
1	Authentication-Info	[26] 20.6	c1	c1	[26] 20.6	c2	c2
1A	Contact	[26] 20.10	o	o	[26] 20.10	m	m
1B	Feature-Caps	[190]	c8	c8	[190]	c8	c8
2	Record-Route	[26] 20.30	c3	c3	[26] 20.30	c3	c3
5	Supported	[26] 20.37	m	m	[26] 20.37	m	m
c1:	IF A.4/7 THEN o ELSE n/a - - authentication between UA and UA.						
c2:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.						
c3:	IF A.4/15 OR A.4/20 THEN m ELSE n/a - - the REFER method extension or SIP specific event notification extension.						
c4:	IF A.4/20 THEN o ELSE n/a - - SIP specific event notification extension.						
c5:	IF A.4/20 THEN m ELSE n/a - - SIP specific event notification extension.						
c6:	IF A.4/70A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						
c8:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.						

Prerequisite A.5/11 - - NOTIFY response

Prerequisite: A.6/103 OR A.6/104 OR A.6/105 OR A.6/106 - - Additional for 3xx – 6xx response

Table A.66A: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	o	o	[26] 20.18	o	o
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.4/115 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.5/11 - - NOTIFY response

Prerequisite: A.6/103 - - Additional for 3xx response

Table A.67: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Contact	[26] 20.10	m	m	[26] 20.10	m	m

Prerequisite A.5/11 - - NOTIFY response

Prerequisite: A.6/14 - - Additional for 401 (Unauthorized) response

Table A.68: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
8	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	m	m
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Prerequisite A.5/11 - - NOTIFY response

Prerequisite: A.6/17 OR A.6/23 OR A.6/30 OR A.6/36 OR A.6/42 OR A.6/45 OR A.6/50 OR A.6/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.69: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Retry-After	[26] 20.33	o	o	[26] 20.33	o	o

Table A.70: Void

Table A.71: Void

Prerequisite A.5/11 - - NOTIFY response

Prerequisite: A.6/21 - - Additional for 408 (Request timeout) response

Table A.71A: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Restoration-Info	subclause 7.2.11	n/a	c1	subclause 7.2.11	n/a	n/a
c1:	IF A.4/110 THEN o ELSE n/a - - HSS based P-CSCF restoration.						

Prerequisite A.5/11 - - NOTIFY response

Prerequisite A.6/25 - - Additional for 415 (Unsupported Media Type) response

Table A.72: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o.1	o.1	[26] 20.1	m	m
2	Accept-Encoding	[26] 20.2	o.1	o.1	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o.1	o.1	[26] 20.3	m	m
o.1	At least one of these capabilities is supported.						

Prerequisite A.5/11 - - NOTIFY response

Prerequisite: A.6/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.72A: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.4/70A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						

Prerequisite A.5/11 - - NOTIFY response

Prerequisite: A.6/27 - - Addition for 420 (Bad Extension) response

Table A.73: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Unsupported	[26] 20.40	m	m	[26] 20.40	m	m

Prerequisite A.5/11 - - NOTIFY response

Prerequisite: A.6/28 OR A.6/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.73A: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	x	x	[48] 2	c1	c1
c1:	IF A.4/37 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Table A.74: Void

Prerequisite A.5/11 - - NOTIFY response

Prerequisite: A.6/35 - - Additional for 485 (Ambigious) response

Table A.74A: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Contact	[26] 20.10	o	o	[26] 20.10	m	m

Prerequisite A.5/11 - - NOTIFY response

Prerequisite: A.6/39 - - Additional for 489 (Bad Event) response

Table A.75: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	m	m

Prerequisite A.5/11 - - NOTIFY response

Prerequisite: A.6/46 - - Additional for 504 (Server Time-out) response

Table A.75A: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Restoration-Info	subclause 7.2.11	n/a	c1	subclause 7.2.11	n/a	n/a
c1:	IF A.4/110 THEN o ELSE n/a - - HSS based P-CSCF restoration.						

Prerequisite A.5/11 - - NOTIFY response

Table A.76: Supported message bodies within the NOTIFY response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.1.4.9 OPTIONS method

Prerequisite A.5/12 - - OPTIONS request

Table A.77: Supported header fields within the OPTIONS request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	m	m
1A	Accept-Contact	[56B] 9.2	c21	c21	[56B] 9.2	c26	c26
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	m	m
3A	Allow	[26] 20.5	o	o	[26] 20.5	m	m
4	Allow-Events	[28] 8.2.2	c24	c24	[28] 8.2.2	c1	c1
5	Authorization	[26] 20.7	c2	c2	[26] 20.7	c2	c2
6	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
7	Call-Info	[26] 20.9	o	o	[26] 20.9	o	o
7A	Cellular-Network-Info	7.2.15	n/a	c44	7.2.15	n/a	c45
8	Contact	[26] 20.10	o	o	[26] 20.10	o	o
9	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
10	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
10A	Content-ID	[256] 3.2	o	c48	[256] 3.2	m	c49
11	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
12	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
13	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
14	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
15	Date	[26] 20.17	c3	c3	[26] 20.17	m	m
15A	Feature-Caps	[190]	c42	c42	[190]	c41	c41
16	From	[26] 20.20	m	m	[26] 20.20	m	m
16A	Geolocation	[89] 4.1	c27	c27	[89] 4.1	c27	c27
16B	Geolocation-Routing	[89] 4.2	c27	c27	[89] 4.2	c27	c27
16C	History-Info	[66] 4.1	c25	c25	[66] 4.1	c25	c25
16D	Max-Breadth	[117] 5.8	n/a	c31	[117] 5.8	c32	c32
17	Max-Forwards	[26] 20.22	m	m	[26] 20.22	n/a	c39
18	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
19	Organization	[26] 20.25	o	o	[26] 20.25	o	o
19A	P-Access-Network-Info	[52] 4.4, [234] 2	c11	c12	[52] 4.4, [234] 2	c11	c13
19B	P-Asserted-Identity	[34] 9.1	n/a	c46	[34] 9.1	c6	c6
19C	P-Asserted-Service	[121] 4.1	n/a	c47	[121] 4.1	c30	c30
19D	P-Called-Party-ID	[52] 4.2	x	x	[52] 4.2	c9	c9
19E	P-Charging-Function-Addresses	[52] 4.5	c16	c17	[52] 4.5	c16	c17
19F	P-Charging-Vector	[52] 4.6	c14	c15	[52] 4.6	c14	c15
19H	P-Preferred-Identity	[34] 9.2	c6	c4	[34] 9.2	n/a	n/a
19I	P-Preferred-Service	[121] 4.2	c29	c28	[121] 4.2	n/a	n/a
19J	P-Private-Network-Indication	[134]	c34	c34	[134]	c34	c34
19K	P-Profile-Key	[97] 5	n/a	n/a	[97] 5	n/a	n/a
19L	P-Served-User	[133] 6	c38	c38	[133] 6	c38	c38
19M	P-User-Database	[82] 4	n/a	n/a	[82] 4	n/a	n/a
19N	P-Visited-Network-ID	[52] 4.3	x (note 2)	x	[52] 4.3	c10	n/a
19O	Privacy	[33] 4.2	c8	c8	[33] 4.2	c8	c8
20	Proxy-Authorization	[26] 20.28	c5	c5	[26] 20.28	n/a	n/a
21	Proxy-Require	[26] 20.29	o	o (note 1)	[26] 20.29	n/a	n/a
21A	Reason	[34A] 2	c20	c20	[34A] 2	c20	c20
22	Record-Route	[26] 20.30	n/a	c39	[26] 20.30	n/a	c39
22A	Recv-Info	[25] 5.2.3	c37	c37	[25] 5.2.3	c37	c37
22B	Referred-By	[59] 3	c22	c22	[59] 3	c23	c23
22C	Reject-Contact	[56B] 9.2	c21	c21	[56B] 9.2	c26	c26
22D	Relayed-Charge	7.2.12	n/a	c43	7.2.12	n/a	c43
22E	Request-Disposition	[56B] 9.1	c21	c21	[56B] 9.1	c26	c26
23	Require	[26] 20.32	m	m	[26] 20.32	m	m
23A	Resource-Priority	[116] 3.1	c33	c33	[116] 3.1	c33	c33
24	Route	[26] 20.34	m	m	[26] 20.34	n/a	n/a
24A	Security-Client	[48] 2.3.1	c18	c18	[48] 2.3.1	n/a	n/a
24B	Security-Verify	[48] 2.3.1	c19	c19	[48] 2.3.1	n/a	n/a
24C	Session-ID	[162]	o	c40	[162]	o	c40
25	Supported	[26] 20.37	c6	c6	[26] 20.37	m	m
26	Timestamp	[26] 20.38	c7	c7	[26] 20.38	m	m
27	To	[26] 20.39	m	m	[26] 20.39	m	m

28	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
29	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1: IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.
c2: IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.
c3: IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c4: IF A.3/1 AND A.4/25 THEN o ELSE n/a - - UE and private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c5: IF A.4/8A THEN m ELSE n/a - - authentication between UA and proxy.
c6: IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c7: IF A.4/6 THEN o ELSE n/a - - timestamping of requests.
c8: IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c9: IF A.4/32 THEN o ELSE n/a - - the P-Called-Party-ID extension.
c10: IF A.4/33 THEN o ELSE n/a - - the P-Visited-Network-ID extension.
c11: IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c12: IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.
c13: IF A.4/34 AND (A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA, AS acting as third-party call controller or EATF.
c14: IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c15: IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c16: IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c17: IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c18: IF A.4/37 OR A.4/37A THEN o ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media (note 3).
c19: IF A.4/37 OR A.4/37A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c20: IF A.4/38 THEN o ELSE n/a - - the Reason header field for the session initiation protocol.
c21: IF A.4/40 THEN o ELSE n/a - - caller preferences for the session initiation protocol.
c22: IF A.4/43 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c23: IF A.4/43 THEN o ELSE n/a - - the SIP Referred-By mechanism.
c24: IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.
c25: IF A.4/47 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c26: IF A.4/40 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c27: IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c28: IF (A.3/1 OR A.3A/81) AND A.4/74 THEN o ELSE n/a - - UE, MSC Server enhanced for ICS and SIP extension for the identification of services.
c29: IF A.4/74 THEN o ELSE n/a - - SIP extension for the identification of services.
c30: IF A.4/74 THEN m ELSE n/a - - SIP extension for the identification of services.
c31: IF A.4/71 AND (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13) THEN m ELSE IF A.3/1 AND NOT A.3C/1 - - addressing an amplification vulnerability in session initiation protocol forking proxies, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling), UE, UE performing the functions of an external attached network.
c32: IF A.4/71 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c33: IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.
c34: IF A.4/77 THEN m ELSE n/a - - the SIP P-Private-Network-Indication private-header (P-Header).
c37: IF A.4/13 THEN m ELSE IF A.4/13A THEN m ELSE n/a - - SIP INFO method and package framework, legacy INFO usage.
c38: IF A.4/78 THEN m ELSE n/a - - the SIP P-Served-User private header.
c39: IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - UE, UE performing the functions of an external attached network.
c40: IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c41: IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.
c42: IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.
c43: IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c44: IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE.
c45: IF A.4/113 AND (A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA, AS acting as third-party call controller or EATF.
c46: IF A.4/25 AND (A.3/7B OR A.3/8 OR A.3A/81A) THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks and AS acting as originating UA, MRFC, MSC server enhanced for SRVCC using SIP interface.
c47: IF A.4/74 AND A.3/7B THEN o ELSE n/a - - SIP extension for the identification of services and AS acting as originating UA.
c48: IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c49: IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).

NOTE 1: No distinction has been made in these tables between first use of a request on a From/To/Call-ID combination, and the usage in a subsequent one. Therefore the use of "o" etc. above has been included from a viewpoint of first usage.
NOTE 2: The strength of this requirement in RFC 7315 [52] is SHOULD NOT, rather than MUST NOT.
NOTE 3: Support of this header in this method is dependent on the security mechanism and the security architecture which is implemented. Use of this header in this method is not appropriate to the security mechanism defined by 3GPP TS 33.203 [19].

Prerequisite A.5/12 - - OPTIONS request

Table A.78: Supported message bodies within the OPTIONS request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

Table A.79: Void

Prerequisite A.5/13 - - OPTIONS response

Prerequisite: A.6/1 - - Additional for 100 (Trying) response

Table A.79A: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						

Prerequisite A.5/13 - - OPTIONS response for all remaining status-codes

Table A.80: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	c12	c12	[26] 20.5	m	m
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Call-Info	[26] 20.9	o	o	[26] 20.9	o	o
1B	Cellular-Network-Info	7.2.15	n/a	c20	7.2.15	n/a	c21
2	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
3	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
3A	Content-ID	[256] 3.2	o	c23	[256] 3.2	m	c24
4	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
9	From	[26] 20.20	m	m	[26] 20.20	m	m
9A	Geolocation-Error	[89] 4.3	c14	c14	[89] 4.3	c14	c14
9B	History-Info	[66] 4.1	c13	c13	[66] 4.1	c13	c13
10	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
11	Organization	[26] 20.25	o	o	[26] 20.25	o	o
11A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c5	c6	[52] 4.4, [52A] 4, [234] 2	c5	c7
11B	P-Asserted-Identity	[34] 9.1	n/a	c22	[34] 9.1	c3	c3
11C	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c10	c11	[52] 4.5, [52A] 4	c10	c11
11D	P-Charging-Vector	[52] 4.6, [52A] 4	c8	c9	[52] 4.6, [52A] 4	c8	c9
11F	P-Preferred-Identity	[34] 9.2	c3	x	[34] 9.2	n/a	n/a
11G	Privacy	[33] 4.2	c4	c4	[33] 4.2	c4	c4
11H	Recv-Info	[25] 5.2.3	c17	c17	[25] 5.2.3	c17	c17
11I	Relayed-Charge	7.2.12	n/a	c19	7.2.12	n/a	c19
11J	Require	[26] 20.32	m	m	[26] 20.32	m	m
11K	Server	[26] 20.35	o	o	[26] 20.35	o	o
11L	Session-ID	[162]	o	c18	[162]	o	c18
12	Timestamp	[26] 20.38	m	m	[26] 20.38	c2	c2
13	To	[26] 20.39	m	m	[26] 20.39	m	m
13A	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
14	Via	[26] 20.42	m	m	[26] 20.42	m	m
15	Warning	[26] 20.43	o (note)	o	[26] 20.43	o	o

c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c2:	IF A.4/6 THEN m ELSE n/a - - timestamping of requests.
c3:	IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c4:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c5:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c6:	IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.
c7:	IF A.4/34 AND (A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA, AS acting as third-party call controller, or EATF.
c8:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c9:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c10:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c11:	IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c12:	IF A.6/102 OR A.6/18 THEN m ELSE o - - 2xx response, 405 (Method Not Allowed).
c13:	IF A.4/47 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c14:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c17:	IF A.4/13 THEN m ELSE IF A.4/13A THEN m ELSE n/a - - SIP INFO method and package framework, legacy INFO usage.
c18:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c19:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c20:	IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE.
c21:	IF A.4/113 AND (A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller, or EATF.
c22:	IF A.4/25 AND (A.3/7B OR A.3/8) THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks and AS acting as originating UA, MRFC.
c23:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c24:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
NOTE:	RFC 3261 [26] gives the status of this header as SHOULD rather than OPTIONAL.

Prerequisite A.5/13 - - OPTIONS response

Prerequisite: A.6/102 - - Additional for 2xx response

Table A.81: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	m	m
1A	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	m	m
1B	Accept-Language	[26] 20.3	m	m	[26] 20.3	m	m
1C	Accept-Resource-Priority	[116] 3.2	c14	c14	[116] 3.2	c14	c14
2	Allow-Events	[28] 8.2.2	c3	c3	[28] 8.2.2	c4	c4
3	Authentication-Info	[26] 20.6	c1	c1	[26] 20.6	c2	c2
5	Contact	[26] 20.10	o	o	[26] 20.10	o	o
6	Feature-Caps	[190]	c16	c16	[190]	c15	c15
7	Recv-Info	[25] 5.2.3	c6	c6	[25] 5.2.3	c6	c6
12	Supported	[26] 20.37	m	m	[26] 20.37	m	m

c1:	IF A.4/7 THEN o ELSE n/a - - authentication between UA and UA.
c2:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.
c3:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.
c4:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.
c6:	IF A.4/13 THEN m ELSE n/a - - SIP INFO method and package framework.
c14:	IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.
c15:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.
c16:	IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.

Prerequisite A.5/13 - - OPTIONS response

Prerequisite: A.6/103 OR A.6/104 OR A.6/105 OR A.6/106 - - Additional for 3xx – 6xx response

Table A.81A: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	o	o	[26] 20.18	o	o
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.4/115 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.5/13 - - OPTIONS response

Prerequisite: A.6/103 OR A.6/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.82: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Contact	[26] 20.10	o (note)	o	[26] 20.10	m	m
NOTE:	RFC 3261 [26] gives the status of this header as SHOULD rather than OPTIONAL.						

Prerequisite A.5/13 - - OPTIONS response

Prerequisite: A.6/14 - - Additional for 401 (Unauthorized) response

Table A.83: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
10	WWW-Authenticate	[26] 20.44	o	o	[26] 20.44	o	o
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Prerequisite A.5/13 - - OPTIONS response

Prerequisite: A.6/17 OR A.6/23 OR A.6/30 OR A.6/36 OR A.6/42 OR A.6/45 OR A.6/50 OR A.6/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response.

Table A.84: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Retry-After	[26] 20.33	o	o	[26] 20.33	o	o

Table A.85: Void

Prerequisite A.5/13 - - OPTIONS response

Prerequisite: A.6/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.86: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
8	WWW-Authenticate	[26] 20.44	o	o	[26] 20.44	o	o
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Table A.86A: Void

Prerequisite A.5/13 - - OPTIONS response

Prerequisite: A.6/25 - - Additional for 415 (Unsupported Media Type) response

Table A.87: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o.1	o.1	[26] 20.1	m	m
2	Accept-Encoding	[26] 20.2	o.1	o.1	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o.1	o.1	[26] 20.3	m	m
o.1	At least one of these capabilities is supported.						

Prerequisite A.5/13 - - OPTIONS response

Prerequisite: A.6/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.87A: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						

Prerequisite A.5/13 - - OPTIONS response

Prerequisite: A.6/27 - - Additional for 420 (Bad Extension) response

Table A.88: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
7	Unsupported	[26] 20.40	m	m	[26] 20.40	m	m

Prerequisite A.5/13 - - OPTIONS response

Prerequisite: A.6/28 OR A.6/41A - - Additional 421 (Extension Required), 494 (Security Agreement Required) response

Table A.88A: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	x	x	[48] 2	c1	c1
c1:	IF A.4/37 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Table A.89: Void

Prerequisite A.5/13 - - OPTIONS response

Prerequisite: A.6/46 - - Additional for 504 (Server Time-out) response

Table A.89A: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Restoration-Info	subclause 7.2.11	n/a	c1	subclause 7.2.11	n/a	n/a
c1:	IF A.4/110 THEN o ELSE n/a - - HSS based P-CSCF restoration.						

Prerequisite A.5/13 - - OPTIONS response

Table A.90: Supported message bodies within the OPTIONS response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	application/cccex	[9B]	n/a	c1	[9B]	n/a	c2
c1:	IF A.3A/12 THEN o ELSE n/a - - conference participant.						
c2:	IF A.3A/11 THEN o ELSE n/a - - conference focus.						

A.2.1.4.10 PRACK method

Prerequisite A.5/14 - - PRACK request

Table A.91: Supported header fields within the PRACK request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o	o	[26] 20.1	m	m
1A	Accept-Contact	[56B] 9.2	c15	c15	[56B] 9.2	c18	c18
2	Accept-Encoding	[26] 20.2	o	o	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o	o	[26] 20.3	m	m
3A	Allow	[26] 20.5	o	o	[26] 20.5	m	m
4	Allow-Events	[28] 8.2.2	c1	c1	[28] 8.2.2	c2	c2
5	Authorization	[26] 20.7	c3	c3	[26] 20.7	c3	c3
6	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
6A	Cellular-Network-Info	7.2.15	n/a	c41	7.2.15	n/a	c42
7	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
8	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
8A	Content-ID	[256] 3.2	o	c44	[256] 3.2	m	c45
9	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
10	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
11	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
12	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
13	Date	[26] 20.17	c4	c4	[26] 20.17	m	m
14	From	[26] 20.20	m	m	[26] 20.20	m	m
14A	Max-Breadth	[117] 5.8	n/a	c21	[117] 5.8	c22	c22
15	Max-Forwards	[26] 20.22	m	m	[26] 20.22	n/a	c34
16	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
16A	P-Access-Network-Info	[52] 4.4, [234] 2	c9	c10	[52] 4.4, [234] 2	c9	c11
16B	P-Charging-Function-Addresses	[52] 4.5	c13	c14	[52] 4.5	c13	c14
16C	P-Charging-Vector	[52] 4.6	c12	c40	[52] 4.6	c12	c40
16E	P-Early-Media	[109] 8	c39	c39	[109] 8	c39	c39
16EA	Priority-Share	Subclause 7.2.16	n/a	c43	Subclause 7.2.16	n/a	c43
16F	Privacy	[33] 4.2	c6	n/a	[33] 4.2	c6	n/a
17	Proxy-Authorization	[26] 20.28	c5	c5	[26] 20.28	n/a	n/a
18	Proxy-Require	[26] 20.29	o	n/a	[26] 20.29	n/a	n/a
19	RAck	[27] 7.2	m	m	[27] 7.2	m	m
19A	Reason	[34A] 2	c7	c7	[34A] 2	c7	c7
20	Record-Route	[26] 20.30	n/a	c34	[26] 20.30	n/a	c34
20A	Recv-Info	[25] 5.2.3	c35	c35	[25] 5.2.3	c35	c35
20B	Referred-By	[59] 3	c16	c16	[59] 3	c17	c17
20C	Reject-Contact	[56B] 9.2	c15	c15	[56B] 9.2	c18	c18
20D	Relayed-Charge	7.2.12	n/a	c37	7.2.12	n/a	c37
20E	Request-Disposition	[56B] 9.1	c15	c15	[56B] 9.1	c18	c18
21	Require	[26] 20.32	m	m	[26] 20.32	m	m
21A	Resource-Priority	[116] 3.1	c33	c33	[116] 3.1	c33	c33
21B	Resource-Share	Subclause 7.2.13	n/a	c38	Subclause 7.2.13	n/a	c38
22	Route	[26] 20.34	m	m	[26] 20.34	n/a	c34
22A	Session-ID	[162]	o	c36	[162]	o	c36
23	Supported	[26] 20.37	o	o	[26] 20.37	m	m
24	Timestamp	[26] 20.38	c8	c8	[26] 20.38	m	m
25	To	[26] 20.39	m	m	[26] 20.39	m	m
26	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
27	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.
c2:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.
c3:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.
c4:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c5:	IF A.4/8A THEN m ELSE n/a - - authentication between UA and proxy.
c6:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c7:	IF A.4/38 THEN o ELSE n/a - - the Reason header field for the session initiation protocol.
c8:	IF A.4/6 THEN o ELSE n/a - - timestamping of requests.
c9:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c10:	IF A.4/34 AND (A.3/1 OR A.3/2A OR A.3/7 OR A.3A/81) THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE, P-CSCF (IMS-ALG), AS or MSC server enhanced for ICS.
c11:	IF A.4/34 AND (A.3/2A OR A.3A/81 OR A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the P-Access-Network-Info header extension and P-CSCF (IMS-ALG), MSC server enhanced for ICS, AS acting as terminating UA, AS acting as third-party call controller or EATF.
c12:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c13:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c14:	IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c15:	IF A.4/40 THEN o ELSE n/a - - caller preferences for the session initiation protocol.
c16:	IF A.4/43 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c17:	IF A.4/43 THEN o ELSE n/a - - the SIP Referred-By mechanism.
c18:	IF A.4/40 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c21:	IF A.4/71 AND (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C) THEN m ELSE IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - addressing an amplification vulnerability in session initiation protocol forking proxies, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling), UE, UE performing the functions of an external attached network.
c22:	IF A.4/71 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c33:	IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.
c34:	IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - UE, UE performing the functions of an external attached network.
c35:	IF A.4/13 THEN m ELSE IF A.4/13A THEN m ELSE n/a - - SIP INFO method and package framework, legacy INFO usage.
c36:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c37:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c38	IF A.4/112 THEN o ELSE n/a - - resource sharing.
c39:	IF A.4/66 THEN m ELSE n/a - - The SIP P-Early-Media private header extension for authorization of early media.
c40:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c41:	IF A.4/113 AND (A.3/1 OR A.3/2A OR A.3/7) THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE, P-CSCF (IMS-ALG) or AS.
c42:	IF A.4/113 AND (A.3/2A OR A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the Cellular-Network-Info header extension and P-CSCF (IMS-ALG), AS acting as terminating UA or AS acting as third-party call controller or EATF.
c43:	IF A.4/114 THEN o ELSE n/a - - priority sharing.
c44:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c45:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).

Prerequisite A.5/14 - - PRACK request

Table A.92: Supported message bodies within the PRACK request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

Table A.93: Void

Prerequisite A.5/15 - - PRACK response

Prerequisite: A.6/1 - - Additional for 100 (Trying) response

Table A.93A: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						

Prerequisite A.5/15 - - PRACK response for all remaining status-codes

Table A.94: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	c9	c9	[26] 20.5	m	m
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Cellular-Network-Info	7.2.15	n/a	c17	7.2.15	n/a	c18
2	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
3	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
3A	Content-ID	[256] 3.2	o	c19	[256] 3.2	m	c20
4	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
9	From	[26] 20.20	m	m	[26] 20.20	m	m
10	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
10A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c3	c4	[52] 4.4, [52A] 4, [234] 2	c3	c5
10B	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c7	c8	[52] 4.5, [52A] 4	c7	c8
10C	P-Charging-Vector	[52] 4.6, [52A] 4	c6	c16	[52] 4.6, [52A] 4	c6	c16
10F	Privacy	[33] 4.2	c2	n/a	[33] 4.2	c2	n/a
10G	Recv-Info	[25] 5.2.3	c13	c13	[25] 5.2.3	c13	c13
10H	Relayed-Charge	7.2.12	n/a	c15	7.2.12	n/a	c15
10I	Require	[26] 20.32	m	m	[26] 20.32	m	m
10J	Server	[26] 20.35	o	o	[26] 20.35	o	o
10K	Session-ID	[162]	o	c14	[162]	o	c14
11	Timestamp	[26] 20.38	m	m	[26] 20.38	c2	c2
12	To	[26] 20.39	m	m	[26] 20.39	m	m
12A	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
13	Via	[26] 20.42	m	m	[26] 20.42	m	m
14	Warning	[26] 20.43	o (note)	o	[26] 20.43	o	o
c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						
c2:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).						
c3:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.						
c4:	IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.						
c5:	IF A.4/34 AND (A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA, AS acting as third-party call controller or EATF.						
c6:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.						
c7:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.						
c8:	IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.						
c9:	IF A.6/18 THEN m ELSE o - - 405 (Method Not Allowed)						
c13:	IF A.4/13 THEN m ELSE IF A.4/13A THEN m ELSE n/a - - SIP INFO method and package framework, legacy INFO usage.						
c14:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.						
c15:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.						
c16:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.						
c17:	IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE.						
c18:	IF A.4/113 AND (A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller or EATF.						
c19:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).						
c20:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).						
NOTE:	RFC 3261 [26] gives the status of this header as SHOULD rather than OPTIONAL.						

Prerequisite A.5/15 - - PRACK response

Prerequisite: A.6/102 - - Additional for 2xx response

Table A.95: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Accept-Resource-Priority	[116] 3.2	c14	c14	[116] 3.2	c14	c14
0B	Allow-Events	[28] 8.2.2	c3	c3	[28] 8.2.2	c4	c4
0C	Authentication-Info	[26] 20.6	c1	c1	[26] 20.6	c2	c2
0D	P-Early-Media	[109] 8	c5	c5	[109] 8	c5	c5
1	Priority-Share	Subclause 7.2.16	n/a	c16	Subclause 7.2.16	n/a	c16
2A	Resource-Share	Subclause 7.2.13	n/a	c15	Subclause 7.2.13	n/a	c15
3	Supported	[26] 20.37	m	m	[26] 20.37	m	m
c1:	IF A.4/7 THEN o ELSE n/a - - authentication between UA and UA.						
c2:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.						
c3:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.						
c4:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.						
c5:	IF A.4/66 THEN m ELSE n/a - - the SIP P-Early-Media private header extension for authorization of early media.						
c14:	IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						
c15:	IF A.4/112 THEN o ELSE n/a - - resource sharing.						
c16:	IF A.4/114 THEN o ELSE n/a - - priority sharing.						

Prerequisite A.5/15 - - PRACK response

Prerequisite: A.6/103 OR A.6/104 OR A.6/105 OR A.6/106 - - Additional for 3xx – 6xx response

Table A.95A: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	o	o	[26] 20.18	o	o
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.4/115 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.5/15 - - PRACK response

Prerequisite: A.6/103 OR A.6/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.96: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Contact	[26] 20.10	o (note)	o	[26] 20.10	m	m
NOTE:	RFC 3261 [26] gives the status of this header field as SHOULD rather than OPTIONAL.						

Prerequisite A.5/15 - - PRACK response

Prerequisite: A.6/14 - - Additional for 401 (Unauthorized) response

Table A.97: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
8	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	m	m
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Prerequisite A.5/15 - - PRACK response

Prerequisite: A.6/17 OR A.6/23 OR A.6/30 OR A.6/36 OR A.6/42 OR A.6/45 OR A.6/50 OR A.6/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response.

Table A.98: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Retry-After	[26] 20.33	o	o	[26] 20.33	o	o

Table A.99: Void

Prerequisite A.5/15 - - PRACK response

Prerequisite: A.6/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.100: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
6	WWW-Authenticate	[26] 20.44	o	o	[26] 20.44	o	o
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Prerequisite A.5/15 - - PRACK response

Prerequisite: A.6/25 - - Additional for 415 (Unsupported Media Type) response

Table A.101: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o.1	o.1	[26] 20.1	m	m
2	Accept-Encoding	[26] 20.2	o.1	o.1	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o.1	o.1	[26] 20.3	m	m

Prerequisite A.5/15 - - PRACK response

Prerequisite: A.6/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.101A: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						

Prerequisite A.5/15 - - PRACK response

Prerequisite: A.6/27 - - Additional for 420 (Bad Extension) response

Table A.102: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Unsupported	[26] 20.40	m	m	[26] 20.40	m	m

Prerequisite A.5/15 - - PRACK response

Prerequisite: A.6/28 OR A.6/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.102A: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	x	x	[48] 2	c1	c1
c1:	IF A.4/37 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Table A.103: Void

Prerequisite A.5/15 - - PRACK response

Table A.104: Supported message bodies within the PRACK response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.1.4.10A PUBLISH method

Prerequisite A.5/15A – PUBLISH request

Table A.104A: Supported header fields within the PUBLISH request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Contact	[56B] 9.2	c22	c22	[56B] 9.2	c28	c28
2	Allow	[26] 20.5	o	o	[26] 20.5	m	m
3	Allow-Events	[28] 8.2.2	c1	c1	[28] 8.2.2	c2	c2
4	Authorization	[26] 20.7	c3	c3	[26] 20.7	c3	c3
5	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
6	Call-Info	[26] 20.9	o	o	[26] 20.9	o	o
6A	Cellular-Network-Info	7.2.15	n/a	c43	7.2.15	n/a	c44
6B	Contact	[70] 4	o	o	[70] 6	n/a	n/a
7	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
8	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
8A	Content-ID	[256] 3.2	o	c47	[256] 3.2	m	c48
9	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
10	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
11	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
12	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
13	Date	[26] 20.17	c4	c4	[26] 20.17	m	m
14	Event	[70] 4, 6	m	m	[70] 4, 6	m	m
15	Expires	[26] 20.19, [70] 4, 5, 6	o	o	[26] 20.19, [70] 4, 5, 6	m	m
15A	Feature-Caps	[190]	c41	c41	[190]	c40	c40
16	From	[26] 20.20	m	m	[26] 20.20	m	m
16A	Geolocation	[89] 4.1	c38	c38	[89] 4.1	c38	c38
16B	Geolocation-Routing	[89] 4.2	c38	c38	[89] 4.2	c38	c38
16C	History-Info	[66] 4.1	c27	c27	[66] 4.1	c27	c27
17	In-Reply-To	[26] 20.21	o	o	[26] 20.21	o	o
17A	Max-Breadth	[117] 5.8	n/a	c23	[117] 5.8	c24	c24
18	Max-Forwards	[26] 20.22	m	m	[26] 20.22	n/a	c37
19	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
20	Organization	[26] 20.25	o	o	[26] 20.25	o	o
21	P-Access-Network-Info	[52] 4.4, [234] 2	c15	c16	[52] 4.4, [234] 2	c15	c17
22	P-Asserted-Identity	[34] 9.1	n/a	c45	[34] 9.1	c11	c11
22A	P-Asserted-Service	[121] 4.1	n/a	c46	[121] 4.1	c31	c31
23	P-Called-Party-ID	[52] 4.2	x	x	[52] 4.2	c13	c13
24	P-Charging-Function-Addresses	[52] 4.5	c20	c21	[52] 4.5	c20	c21
25	P-Charging-Vector	[52] 4.6	c18	c19	[52] 4.6	c18	c19
26	P-Preferred-Identity	[34] 9.2	c11	c7	[34] 9.2	n/a	n/a
26A	P-Preferred-Service	[121] 4.2	c31	c30	[121] 4.2	n/a	n/a
26B	P-Private-Network-Indication	[134]	c33	c33	[134]	c33	c33
26C	P-Profile-Key	[97] 5	n/a	n/a	[97] 5	n/a	n/a
26D	P-Served-User	[133] 6	c36	c36	[133] 6	c36	c36
26E	P-User-Database	[82] 4	n/a	n/a	[82] 4	n/a	n/a
27	P-Visited-Network-ID	[52] 4.3	x (note 3)	x	[52] 4.3	c14	n/a
28	Priority	[26] 20.26	o	o	[26] 20.26	o	o
29	Privacy	[33] 4.2	c12	c12	[33] 4.2	c12	c12
30	Proxy-Authorization	[26] 20.28	c5	c5	[26] 20.28	n/a	n/a
31	Proxy-Require	[26] 20.29	o	n/a	[26] 20.29	n/a	n/a
32	Reason	[34A] 2	c8	c8	[34A] 2	c8	c8
33A	Referred-By	[59] 3	c25	c25	[59] 3	c26	c26
34	Reject-Contact	[56B] 9.2	c22	c22	[56B] 9.2	c28	c28
34A	Relayed-Charge	7.2.12	n/a	c42	7.2.12	n/a	c42
34B	Reply-To	[26] 20.31	o	o	[26] 20.31	o	o
35	Request-Disposition	[56B] 9.1	c22	c22	[56B] 9.1	c28	c28
36	Require	[26] 20.32	m	m	[26] 20.32	m	m
36A	Resource-Priority	[116] 3.1	c29	c29	[116] 3.1	c29	c29
37	Route	[26] 20.34	m	m	[26] 20.34	n/a	c37
38	Security-Client	[48] 2.3.1	c9	c9	[48] 2.3.1	n/a	n/a
39	Security-Verify	[48] 2.3.1	c10	c10	[48] 2.3.1	n/a	n/a
39A	Session-ID	[162]	o	c39	[162]	o	c39

40	SIP-If-Match	[70] 11.3.2	o	o	[70] 11.3.2	m	m
41	Subject	[26] 20.36	o	o	[26] 20.36	o	o
42	Supported	[26] 20.37, [26] 7.1	o	o	[26] 20.37, [26] 7.1	m	m
43	Timestamp	[26] 20.38	c6	c6	[26] 20.38	m	m
44	To	[26] 20.39	m	m	[26] 20.39	m	m
45	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
46	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.
c2:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.
c3:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.
c4:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c5:	IF A.4/8A THEN m ELSE n/a - - authentication between UA and proxy.
c6:	IF A.4/6 THEN o ELSE n/a - - timestamping of requests.
c7:	IF A.3/1 AND A.4/25 THEN o ELSE n/a - - UE and private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c8:	IF A.4/38 THEN o ELSE n/a - - the Reason header field for the session initiation protocol.
c9:	IF A.4/37 OR A.4/37A THEN o ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media (note 1).
c10:	IF A.4/37 OR A.4/37A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c11:	IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c12:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c13:	IF A.4/32 THEN o ELSE n/a - - the P-Called-Party-ID extension.
c14:	IF A.4/33 THEN o ELSE n/a - - the P-Visited-Network-ID extension.
c15:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c16:	IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.
c17:	IF A.4/34 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c18:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c19:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c20:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c21:	IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c22:	IF A.4/40 THEN o ELSE n/a - - caller preferences for the session initiation protocol.
c23:	IF A.4/71 AND (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C) THEN m ELSE IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - addressing an amplification vulnerability in session initiation protocol forking proxies, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling), UE, UE performing the functions of an external attached network.
c24:	IF A.4/71 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c25:	IF A.4/43 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c26:	IF A.4/43 THEN o ELSE n/a - - the SIP Referred-By mechanism.
c27:	IF A.4/47 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c28:	IF A.4/40 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c29:	IF A.4/70B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.
c30:	IF (A.3/1 OR A.3A/81) AND A.4/74 THEN o ELSE n/a - - UE, MSC Server enhanced for ICS and SIP extension for the identification of services.
c31:	IF A.4/74 THEN o ELSE n/a - - SIP extension for the identification of services.
c32:	IF A.4/74 THEN m ELSE n/a - - SIP extension for the identification of services.
c33:	IF A.4/77 THEN m ELSE n/a - - the SIP P-Private-Network-Indication private-header (P-Header).
c36:	IF A.4/78 THEN m ELSE n/a - - the SIP P-Served-User private header.
c37:	IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - UE, UE performing the functions of an external attached network.
c38:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c39:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c40:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.
c41:	IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.
c42:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c43:	IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE.
c44:	IF A.4/113 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c45:	IF A.4/25 AND (A.3/7B OR A.3/8) THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks and AS acting as originating UA, MRFC.
c46:	IF A.4/74 AND A.3/7B THEN o ELSE n/a - - SIP extension for the identification of services and AS acting as originating UA.
c47:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c48:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
NOTE 1:	Support of this header in this method is dependent on the security mechanism and the security architecture which is implemented.
NOTE 2:	The strength of this requirement in RFC 7315 [52] is SHOULD NOT, rather than MUST NOT.

Prerequisite A.5/15A - - PUBLISH request

Table A.104B: Supported message bodies within the PUBLISH request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	application/vnd.3gpp.mcptt-info+xml	[8ZE]	n/a	c1	[8ZE]	n/a	c1
2	application/poc-settings+xml	[110]	o	c2	[110]	o	c3
3	application/pidf+xml (NOTE)	[242]	o	c1	[242]	o	c3
c1:	IF A.3A/102 OR A.3A/103 THEN m ELSE n/a - - MCPTT client, MCPTT server.						
c2:	IF A.3A/102 THEN m ELSE n/a - - MCPTT client.						
c3:	IF A.3A/103 THEN m ELSE n/a - - MCPTT server.						
NOTE:	The application/pidf+xml is extended by 3GPP TS 24.379 [8ZE].						

Prerequisite A.5/15B - - PUBLISH response

Prerequisite: A.6/1 - - Additional for 100 (Trying) response

Table A.104BA: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						

Prerequisite A.5/15B - - PUBLISH response for all remaining status-codes

Table A.104C: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	c12	c12	[26] 20.5	m	m
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Call-Info	[26] 24.9	o	o	[26] 24.9	m	m
2A	Cellular-Network-Info	7.2.15	n/a	c19	7.2.15	n/a	c20
3	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
4	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
4A	Content-ID	[256] 3.2	o	c22	[256] 3.2	m	c23
5	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
6	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
7	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
8	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
9	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
9A	Expires	[26] 20.19 [70] 4, 5, 6	o	o	[26] 20.19 [70] 4, 5, 6	o	o
10	From	[26] 20.20	m	m	[26] 20.20	m	m
10A	Geolocation-Error	[89] 4.3	c16	c16	[89] 4.3	c16	c16
10B	History-Info	[66] 4.1	c13	c13	[66] 4.1	c13	c13
11	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
12	Organization	[26] 20.25	o	o	[26] 20.25	o	o
13	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c5	c6	[52] 4.4, [52A] 4, [234] 2	c5	c7
14	P-Asserted-Identity	[34] 9.1	n/a	c21	[34] 9.1	c3	c3
15	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c10	c11	[52] 4.5, [52A] 4	c10	c11
16	P-Charging-Vector	[52] 4.6, [52A] 4	c8	c9	[52] 4.6, [52A] 4	c8	c9
17	P-Preferred-Identity	[34] 9.2	c3	x	[34] 9.2	n/a	n/a
18	Privacy	[33] 4.2	c4	c4	[33] 4.2	c4	c4
18A	Relayed-Charge	7.2.12	n/a	c18	7.2.12	n/a	c18
19	Require	[26] 20.32	m	m	[26] 20.32	m	m
20	Server	[26] 20.35	o	o	[26] 20.35	o	o
20A	Session-ID	[162]	o	c17	[162]	o	c17
21	Timestamp	[26] 20.38	m	m	[26] 20.38	c2	c2
22	To	[26] 20.39	m	m	[26] 20.39	m	m
23	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
24	Via	[26] 20.42	m	m	[26] 20.42	m	m
25	Warning	[26] 20.43	o	o	[26] 20.43	o	o

c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c2:	IF A.4/6 THEN m ELSE n/a - - timestamping of requests.
c3:	IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c4:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c5:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c6:	IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.
c7:	IF A.4/34 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c8:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c9:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c10:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c11:	IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c12:	IF A.6/18 THEN m ELSE o - - 405 (Method Not Allowed).
c13:	IF A.4/47 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c16:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c17:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c18:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c19:	IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE.
c20:	IF A.4/113 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c21:	IF A.4/25 AND (A.3/7B OR A.3/8) THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks and AS acting as originating UA, MRFC.
c22:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c23:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
NOTE:	For a 488 (Not Acceptable Here) response, RFC 3261 [26] gives the status of this header field as SHOULD rather than OPTIONAL.

Prerequisite A.5/15B - - PUBLISH response

Prerequisite: A.6/102 - - Additional for 2xx response

Table A.104D: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Authentication-Info	[26] 20.6	c1	c1	[26] 20.6	c2	c2
3	Expires	[26] 20.19, [70] 4, 5, 6	m	m	[26] 20.19, [70] 4, 5, 6	m	m
3A	Feature-Caps	[190]	c8	c8	[190]	c7	c7
4	SIP-Etag	[70] 11.3.1	m	m	[70] 11.3.1	m	m
5	Supported	[26] 20.37	m	m	[26] 20.37	m	m
c1:	IF A.4/7 THEN o ELSE n/a - - authentication between UA and UA.						
c2:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.						
c7:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.						
c8:	IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.						

Prerequisite A.5/15B - - PUBLISH response

Prerequisite: A.6/6 - - Additional for 200 (OK) response

Table A.104DAA: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.4/70B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.						

Prerequisite A.5/15B - - PUBLISH response

Prerequisite: A.6/103 OR A.6/104 OR A.6/105 OR A.6/106 - - Additional for 3xx – 6xx response

Table A.104DA: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	o	o	[26] 20.18	o	o
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.4/115 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.5/15B - - PUBLISH response

Prerequisite: A.6/103 OR A.6/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.104E: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Contact	[26] 20.10	o	o	[26] 20.10	m	m

Prerequisite A.5/15B - - PUBLISH response

Prerequisite: A.6/8 OR A.6/9 OR A.6/10 OR A.6/11OR A.6/12 – Additional for 401 (Unauthorized) response

Table A.104F: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
5	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	m	m
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Prerequisite A.5/15B - - PUBLISH response

Prerequisite: A.6/17 OR A.6/23 OR A.6/30 OR A.6/36 OR A.6/42 OR A.6/45 OR A.6/50 OR A.6/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.104G: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Retry-After	[26] 20.33	o	o	[26] 20.33	o	o

Table A.104H: Void

Prerequisite A.5/15B - - PUBLISH response

Prerequisite: A.6/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.104I: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
5	WWW-Authenticate	[26] 20.44	o	o	[26] 20.44	o	o
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Table A.104IA: Void

Prerequisite A.5/15B - - PUBLISH response

Prerequisite: A.6/25 - - Additional for 415 (Unsupported Media Type) response

Table A.104J: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o.1	o.1	[26] 20.1	m	m
2	Accept-Encoding	[26] 20.2	o.1	o.1	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o.1	o.1	[26] 20.3	m	m
o.1	At least one of these capabilities is supported.						

Prerequisite A.5/15B - - PUBLISH response

Prerequisite: A.6/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.104JA: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.4/70B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.						

Prerequisite A.5/15B - - PUBLISH response

Prerequisite: A.6/27 - - Additional for 420 (Bad Extension) response

Table A.104K: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Unsupported	[26] 20.40	m	m	[26] 20.40	m	m

Prerequisite A.5/15B - - PUBLISH response

Prerequisite: A.6/28 OR A.6/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.104L: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	x	x	[48] 2	c1	c1
c1:	IF A.4/37 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Prerequisite A.5/15B - - PUBLISH response

Prerequisite: A.6/29 - - Additional for 423 (Interval Too Brief) response

Table A.104M: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Min-Expires	[26] 20.23, [70] 5, 6	m	m	[26] 20.23, [70] 5, 6	m	m

Table A.104N: Void

Prerequisite A.5/15B - - PUBLISH response

Prerequisite: A.6/39 - - Additional for 489 (Bad Event) response

Table A.104O: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	m	m

Prerequisite A.5/15B - - PUBLISH response

Prerequisite: A.6/46 - - Additional for 504 (Server Time-out) response

Table A.104OA: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Restoration-Info	subclause 7.2.11	n/a	c1	subclause 7.2.11	n/a	n/a
c1:	IF A.4/110 THEN o ELSE n/a - - HSS based P-CSCF restoration.						

Prerequisite A.5/15B - - PUBLISH response

Table A.104P: Supported message bodies within the PUBLISH response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.1.4.11 REFER method

Prerequisite A.5/16 - - REFER request

Table A.105: Supported header fields within the REFER request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Accept	[26] 20.1	o	o	[26] 20.1	m	m
0B	Accept-Contact	[56B] 9.2	c22	c22	[56B] 9.2	c25	c25
0C	Accept-Encoding	[26] 20.2	o	o	[26] 20.2	m	m
1	Accept-Language	[26] 20.3	o	o	[26] 20.3	m	m
1AA	Additional-Identity	7.2.20	n/a	c54	7.2.20	n/a	c55
1A	Allow	[26] 20.5	o	o	[26] 20.5	m	m
2	Allow-Events	[28] 8.2.2	c1	c1	[28] 8.2.2	c2	c2
3	Authorization	[26] 20.7	c3	c3	[26] 20.7	c3	c3
4	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
4A	Cellular-Network-Info	7.2.15	n/a	c48	7.2.15	n/a	c49
5	Contact	[26] 20.10	m	m	[26] 20.10	m	m
5A	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
5B	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
5BA	Content-ID	[256] 3.2	o	c52	[256] 3.2	m	c53
5C	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
6	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
7	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
8	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
9	Date	[26] 20.17	c4	c4	[26] 20.17	m	m
10	Expires	[26] 20.19	o	o	[26] 20.19	o	o
10A	Feature-Caps	[190]	c46	c46	[190]	c45	c45
11	From	[26] 20.20	m	m	[26] 20.20	m	m
11A	Geolocation	[89] 4.1	c26	c26	[89] 4.1	c26	c26
11B	Geolocation-Routing	[89] 4.2	c26	c26	[89] 4.2	c26	c26
11C	History-Info	[66] 4.1	c24	c24	[66] 4.1	c24	c24
11D	Max-Breadth	[117] 5.8	n/a	c30	[117] 5.8	c31	c31
12	Max-Forwards	[26] 20.22	m	m	[26] 20.22	n/a	c39
13	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
14	Organization	[26] 20.25	o	o	[26] 20.25	o	o
14A	P-Access-Network-Info	[52] 4.4, [234] 2	c12	c13	[52] 4.4, [234] 2	c12	c14
14B	P-Asserted-Identity	[34] 9.1	n/a	c50	[34] 9.1	c8	c8
14C	P-Asserted-Service	[121] 4.1	n/a	c51	[121] 4.1	c29	c29
14D	P-Called-Party-ID	[52] 4.2, [52A] 4	x	x	[52] 4.2, [52A] 4	c10	c10
14E	P-Charging-Function-Addresses	[52] 4.5	c17	c18	[52] 4.5	c17	c18
14F	P-Charging-Vector	[52] 4.6	c15	c16	[52] 4.6	c15	c16
14H	P-Preferred-Identity	[34] 9.2	c8	c7	[34] 9.2	n/a	n/a
14I	P-Preferred-Service	[121] 4.2	c28	c27	[121] 4.2	n/a	n/a
14J	P-Private-Network-Indication	[134]	c36	c36	[134]	c36	c36
14K	P-Profile-Key	[97] 5	n/a	n/a	[97] 5	n/a	n/a
14L	P-Served-User	[133] 6	c41	c41	[133] 6	c41	c41
14M	P-User-Database	[82] 4	n/a	n/a	[82] 4	n/a	n/a
14N	P-Visited-Network-ID	[52] 4.3	x (note 1)	x	[52] 4.3	c11	n/a
14O	Privacy	[33] 4.2	c9	c9	[33] 4.2	c9	c9
15	Proxy-Authorization	[26] 20.28	c5	c5	[26] 20.28	n/a	n/a
16	Proxy-Require	[26] 20.29	o	n/a	[26] 20.29	n/a	n/a
16A	Reason	[34A] 2	c21	c21	[34A] 2	c21	c21
17	Record-Route	[26] 20.30	n/a	c39	[26] 20.30	m	m
17A	Refer-Sub	[173] 4	c40	c40	[173] 4	c40	c40
18	Refer-To	[36] 3	m	m	[36] 3	m	m
18A	Referred-By	[59] 3	c23	c23	[59] 3	c23	c23
18B	Reject-Contact	[56B] 9.2	c22	c22	[56B] 9.2	c25	c25
18C	Relayed-Charge	7.2.12	n/a	c47	7.2.12	n/a	c47
18D	Request-Disposition	[56B] 9.1	c22	c22	[56B] 9.1	c25	c25
19	Require	[26] 20.32	m	m	[26] 20.32	m	m
19A	Resource-Priority	[116] 3.1	c33	c33	[116] 3.1	c33	c33
20	Route	[26] 20.34	m	m	[26] 20.34	n/a	c39
20A	Security-Client	[48] 2.3.1	c19	c19	[48] 2.3.1	n/a	n/a
20B	Security-Verify	[48] 2.3.1	c20	c20	[48] 2.3.1	n/a	n/a
20C	Session-ID	[162]	o	c42	[162]	o	c42

21	Supported	[26] 20.37, [26] 7.1	o	o	[26] 20.37, [26] 7.1	m	m
21A	Target-Dialog	[184] 7	c43	c43	[184] 7	c44	c44
22	Timestamp	[26] 20.38	c6	c6	[26] 20.38	m	m
23	To	[26] 20.39	m	m	[26] 20.39	m	m
23A	Trigger-Consent	[125] 5.11.2	c34	c34	[125] 5.11.2	c35	c35
24	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
25	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.
c2:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.
c3:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.
c4:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c5:	IF A.4/8A THEN m ELSE n/a - - authentication between UA and proxy.
c6:	IF A.4/6 THEN o ELSE n/a - - timestamping of requests.
c7:	IF A.3/1 AND A.4/25 THEN o ELSE n/a - - UE and private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c8:	IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c9:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c10:	IF A.4/32 THEN o ELSE n/a - - the P-Called-Party-ID extension.
c11:	IF A.4/33 THEN o ELSE n/a - - the P-Visited-Network-ID extension.
c12:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c13:	IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.
c14:	IF A.4/34 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c15:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c16:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c17:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c18:	IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c19:	IF A.4/37 OR A.4/37A THEN o ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media (note 2).
c20:	IF A.4/37 OR A.4/37A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c21:	IF A.4/38 THEN o ELSE n/a - - the Reason header field for the session initiation protocol.
c22:	IF A.4/40 THEN o ELSE n/a - - caller preferences for the session initiation protocol.
c23:	IF A.4/43 THEN m ELSE n/a - - the SIP Referred-By Mechanism.
c24:	IF A.4/47 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c25:	IF A.4/40 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c26:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c27:	IF (A.3/1 OR A.3A/81) AND A.4/74 THEN o ELSE n/a - - UE, MSC Server enhanced for ICS and SIP extension for the identification of services.
c28:	IF A.4/74 THEN o ELSE n/a - - SIP extension for the identification of services.
c29:	IF A.4/74 THEN m ELSE n/a - - SIP extension for the identification of services.
c30:	IF A.4/71 AND (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C) THEN m ELSE IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - addressing an amplification vulnerability in session initiation protocol forking proxies, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling), UE, UE performing the functions of an external attached network.
c31:	IF A.4/71 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c33:	IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.
c34:	IF A.4/75A THEN m ELSE n/a - - a relay within the framework for consent-based communications in SIP.
c35:	IF A.4/75B THEN m ELSE n/a - - a recipient within the framework for consent-based communications in SIP.
c36:	IF A.4/77 THEN m ELSE n/a - - the SIP P-Private-Network-Indication private-header (P-Header).
c39:	IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - UE, UE performing the functions of an external attached network.
c40:	IF A.4/95 THEN m ELSE n/a - - suppression of session initiation protocol REFER method implicit subscription.
c41:	IF A.4/78 THEN m ELSE n/a - - the SIP P-Served-User private header.
c42:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c43:	IF A.4/99 THEN o ELSE n/a - - request authorization through dialog Identification in the session initiation protocol.
c44:	IF A.4/99 THEN m ELSE n/a - - request authorization through dialog Identification in the session initiation protocol.
c45:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.
c46:	IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.
c47:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c48:	IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE.
c49:	IF A.4/113 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c50:	IF A.4/25 AND (A.3/7B OR A.3/8) THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks and AS acting as originating UA, MRFC.
c51:	IF A.4/74 AND A.3/7B THEN o ELSE n/a - - SIP extension for the identification of services and AS acting as originating UA.

c52:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c53:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c54:	IF A.4/124 THEN o ELSE n/a - - the Additional-Identity header field extension.
c55:	IF A.4/124 THEN m ELSE n/a - - the Additional-Identity header field extension.
NOTE 1:	The strength of this requirement in RFC 7315 [52] is SHOULD NOT, rather than MUST NOT.
NOTE 2:	Support of this header field in this method is dependent on the security mechanism and the security architecture which is implemented. Use of this header field in this method is not appropriate to the security mechanism defined by 3GPP TS 33.203 [19].

Prerequisite A.5/16 - - REFER request

Table A.106: Supported message bodies within the REFER request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	application/vnd.3gpp.mid-call+xml	[8M] D	n/a	o	[8M] D	n/a	o
2	application/vnd.3gpp.mcptt-info+xml	[8ZE]	n/a	c1	[8ZE]	n/a	c1
c1:	IF A.3A/102 OR A.3A/103 THEN m ELSE n/a - - MCPTT client, MCPTT server.						

Table A.107: Void

Prerequisite A.5/17 - - REFER response

Prerequisite: A.6/1 - - Additional for 100 (Trying) response

Table A.107A: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						

Prerequisite A.5/17 - - REFER response for all remaining status-codes

Table A.108: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	c12	c12	[26] 20.5	m	m
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Cellular-Network-Info	7.2.15	n/a	c20	7.2.15	n/a	c21
1B	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
2	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
2A	Content-ID	[256] 3.2	o	c23	[256] 3.2	m	c24
3	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
4	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
5	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
6	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
7	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
8	From	[26] 20.20	m	m	[26] 20.20	m	m
8A	Geolocation-Error	[89] 4.3	c15	c15	[89] 4.3	c15	c15
8B	History-Info	[66] 4.1	c14	c14	[66] 4.1	c14	c14
9	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
10	Organization	[26] 20.25	o	o	[26] 20.25	o	o
10A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c5	c6	[52] 4.4, [52A] 4, [234] 2	c5	c7
10B	P-Asserted-Identity	[34] 9.1	n/a	c22	[34] 9.1	c3	c3
10C	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c10	c11	[52] 4.5, [52A] 4	c10	c11
10D	P-Charging-Vector	[52] 4.6, [52A] 4	c8	c9	[52] 4.6, [52A] 4	c8	c9
10F	P-Preferred-Identity	[34] 9.2	c3	x	[34] 9.2	n/a	n/a
10G	Privacy	[33] 4.2	c4	c4	[33] 4.2	c4	c4
10H	Relayed-Charge	7.2.12	n/a	c19	7.2.12	n/a	c19
10I	Require	[26] 20.32	m	m	[26] 20.32	m	m
10J	Server	[26] 20.35	o	o	[26] 20.35	o	o
10K	Session-ID	[162]	o	c18	[162]	o	c18
11	Timestamp	[26] 20.38	m	m	[26] 20.38	c2	c2
12	To	[26] 20.39	m	m	[26] 20.39	m	m
12A	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
13	Via	[26] 20.42	m	m	[26] 20.42	m	m
14	Warning	[26] 20.43	o (note)	o	[26] 20.43	o	o

c1: IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c2: IF A.4/6 THEN m ELSE n/a - - timestamping of requests.
c3: IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c4: IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c5: IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c6: IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.
c7: IF A.4/34 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c8: IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c9: IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c10: IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c11: IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c12: IF A.6/18 THEN m ELSE o - - 405 (Method Not Allowed)
c14: IF A.4/47 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c15: IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c18: IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c19: IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c20: IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info extension and UE.
c21: IF A.4/113 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c22: IF A.4/25 AND (A.3/7B OR A.3/8) THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks and AS acting as originating UA, MRFC.
c23: IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c24: IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).

NOTE: For a 488 (Not Acceptable Here) response, RFC 3261 [26] gives the status of this header field as SHOULD rather than OPTIONAL.

Prerequisite A.5/17 - - REFER response

Prerequisite: A.6/102 - - Additional for 2xx response

Table A.109: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Accept-Resource-Priority	[116] 3.2	c12	c12	[116] 3.2	c12	c12
1	Allow-Events	[28] 8.2.2	c3	c3	[28] 8.2.2	c4	c4
2	Authentication-Info	[26] 20.6	c1	c1	[26] 20.6	c2	c2
2A	Contact	[26] 20.10	m	m	[26] 20.10	m	m
3	Feature-Caps	[190]	c15	c15	[190]	c14	c14
5	Record-Route	[26] 20.30	m	m	[26] 20.30	m	m
6	Refer-Sub	[173] 4	c13	c13	[173] 4	c13	c13
8	Supported	[26] 20.37	m	m	[26] 20.37	m	m
c1:	IF A.4/7 THEN o ELSE n/a - - authentication between UA and UA.						
c2:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.						
c3:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.						
c4:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.						
c12:	IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						
c13:	IF A.4/95 THEN m ELSE n/a - - suppression of session initiation protocol REFER method implicit subscription.						
c14:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.						
c15:	IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.						

Prerequisite A.5/17 - - REFER response

Prerequisite: A.6/103 OR A.6/104 OR A.6/105 OR A.6/106 - - Additional for 3xx – 6xx response

Table A.109A: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0	Contact	[26] 20.10	o	o	[26] 20.10	m	m
1	Error-Info	[26] 20.18	o	o	[26] 20.18	o	o
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.4/115 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Table A.110: Void

Prerequisite A.5/17 - - REFER response

Prerequisite: A.6/14 - - Additional for 401 (Unauthorized) response

Table A.111: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
10	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	m	m
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Prerequisite A.5/17 - - REFER response

Prerequisite: A.6/17 OR A.6/23 OR A.6/30 OR A.6/36 OR A.6/42 OR A.6/45 OR A.6/50 OR A.6/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.112: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
6	Retry-After	[26] 20.33	o	o	[26] 20.33	o	o

Table A.113: Void

Prerequisite A.5/17 - - REFER response

Prerequisite: A.6/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.114: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
8	WWW-Authenticate	[26] 20.44	o	o	[26] 20.44	o	o
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Table A.114A: Void

Prerequisite A.5/17 - - REFER response

Prerequisite: A.6/25 - - Additional for 415 (Unsupported Media Type) response

Table A.115: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o.1	o.1	[26] 20.1	m	m
2	Accept-Encoding	[26] 20.2	o.1	o.1	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o.1	o.1	[26] 20.3	m	m
o.1	At least one of these capabilities is supported.						

Prerequisite A.5/17 - - REFER response

Prerequisite: A.6/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.115A: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						

Prerequisite A.5/17 - - REFER response

Prerequisite: A.6/27 - - Additional for 420 (Bad Extension) response

Table A.116: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
8	Unsupported	[26] 20.40	m	m	[26] 20.40	m	m

Prerequisite A.5/17 - - REFER response

Prerequisite: A.6/28 OR A.6/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.116A: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	x	x	[48] 2	c1	c1
c1:	IF A.4/37 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Table A.117: Void

Prerequisite A.5/17 - - REFER response

Prerequisite: A.6/29H - - Additional for 470 (Consent Needed) response

Table A.117A: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Permission-Missing	[125] 5.9.3	m	m	[125] 5.9.3	m	m

Prerequisite A.5/17 - - REFER response

Prerequisite: A.6/46 - - Additional for 504 (Server Time-out) response

Table A.117B: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Restoration-Info	subclause 7.2.11	n/a	c1	subclause 7.2.11	n/a	n/a
c1:	IF A.4/110 THEN o ELSE n/a - - HSS based P-CSCF restoration.						

Prerequisite A.5/17 - - REFER response

Table A.118: Supported message bodies within the REFER response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.1.4.12 REGISTER method

Prerequisite A.5/18 - - REGISTER request

Table A.119: Supported header fields within the REGISTER request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o	o	[26] 20.1	m	m
2	Accept-Encoding	[26] 20.2	o	o	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o	o	[26] 20.3	m	m
3A	Allow	[26] 20.5	o	o	[26] 20.5	m	m
4	Allow-Events	[28] 8.2.2	c27	c27	[28] 8.2.2	c1	c1
5	Authorization	[26] 20.7, [49]	c2	c29	[26] 20.7, [49]	m	c22
6	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
7	Call-Info	[26] 20.9	o	o	[26] 20.9	o	o
7A	Cellular-Network-Info	7.2.15	n/a	c43	7.2.15	n/a	c44
8	Contact	[26] 20.10	o	m	[26] 20.10	m	m
9	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
10	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
10A	Content-ID	[256] 3.2	o	c45	[256] 3.2	m	c46
11	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
12	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
13	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
14	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
15	Date	[26] 20.17	c3	c3	[26] 20.17	m	m
16	Expires	[26] 20.19	o	o	[26] 20.19	m	m
16A	Feature-Caps	[190]	c40	c40	[190]	c39	c39
17	From	[26] 20.20	m	m	[26] 20.20	m	m
17A	Geolocation	[89] 4.1	c31	c31	[89] 4.1	c31	c31
17B	Geolocation-Routing	[89] 4.2	c31	c31	[89] 4.2	c31	c31
17C	History-Info	[66] 4.1	c28	c28	[66] 4.1	c28	c28
17D	Max-Breadth	[117] 5.8	n/a	c35	[117] 5.8	c36	c36
18	Max-Forwards	[26] 20.22	m	m	[26] 20.22	n/a	n/a
19	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
20	Organization	[26] 20.25	o	o	[26] 20.25	o	o
20A	P-Access-Network-Info	[52] 4.4, [234] 2	c12	c13	[52] 4.4, [234] 2	c12	c14
20B	P-Charging-Function-Addresses	[52] 4.5	c17	c18	[52] 4.5	c17	c18
20C	P-Charging-Vector	[52] 4.6	c15	c16	[52] 4.6	c15	c16
20E	P-User-Database	[82] 4	n/a	n/a	[82] 4	c30	c30
20F	P-Visited-Network-ID	[52] 4.3	x (note 2)	x	[52] 4.3	c10	c11
20G	Path	[35] 4	c4	c5	[35] 4	m	c6
20H	Privacy	[33] 4.2	c9	n/a	[33] 4.2	c9	n/a
21	Proxy-Authorization	[26] 20.28	c8	c8	[26] 20.28	n/a	n/a
22	Proxy-Require	[26] 20.29	o	o (note 1)	[26] 20.29	n/a	n/a
22A	Reason	[34A] 2	c23	c23	[34A] 2	c23	c23
22B	Recv-Info	[25] 5.2.3	c37	c37	[25] 5.2.3	c37	c37
22C	Referred-By	[59] 3	c25	c25	[59] 3	c26	c26
22D	Relayed-Charge	7.2.12	n/a	c41	7.2.12	n/a	c41
22E	Request-Disposition	[56B] 9.1	c24	c24	[56B] 9.1	n/a	n/a
23	Require	[26] 20.32	m	m	[26] 20.32	m	m
23A	Resource-Priority	[116] 3.1	c32	c32	[116] 3.1	c32	c32
23B	Resource-Share	Subclause 7.2.13	n/a	c42	Subclause 7.2.13	n/a	c42
24	Route	[26] 20.34	o	x	[26] 20.34	n/a	n/a
24A	Security-Client	[48] 2.3.1	c19	c20	[48] 2.3.1	n/a	n/a
24B	Security-Verify	[48] 2.3.1	c20	c20	[48] 2.3.1	c21	n/a
24C	Session-ID	[162]	o	c38	[162]	o	c38
25	Supported	[26] 20.37	o	c29	[26] 20.37	m	m
26	Timestamp	[26] 20.38	c7	c7	[26] 20.38	c7	c7
27	To	[26] 20.39	m	m	[26] 20.39	m	m
28	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
29	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.
c2:	IF A.4/8 THEN m ELSE n/a - - authentication between UA and registrar.
c3:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c4:	IF A.4/24 THEN o ELSE n/a - - session initiation protocol extension header field for registering non-adjacent contacts.
c5:	IF A.4/24 THEN x ELSE n/a - - session initiation protocol extension header field for registering non-adjacent contacts.
c6:	IF A.3/4 THEN m ELSE n/a. - - S-CSCF.
c7:	IF A.4/6 THEN m ELSE n/a - - timestamping of requests.
c8:	IF A.4/8A THEN m ELSE n/a - - authentication between UA and proxy.
c9:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c10:	IF A.4/33 THEN o ELSE n/a - - the P-Visited-Network-ID extension.
c11:	IF A.4/33 THEN m ELSE n/a - - the P-Visited-Network-ID extension.
c12:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c13:	IF A.4/34 AND (A.3/1 OR A.3/4) THEN o ELSE n/a - - the P-Access-Network-Info header extension and UE or S-CSCF.
c14:	IF A.4/34 AND (A.3/4 OR A.3/7A) THEN m ELSE n/a - - the P-Access-Network-Info header extension and S-CSCF or AS acting as terminating UA.
c15:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c16:	IF A.4/36 OR A.3/4 THEN m ELSE n/a - - the P-Charging-Vector header extension (including S-CSCF as registrar).
c17:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c18:	IF A.4/35 OR A.3/4 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension (including S-CSCF as registrar).
c19:	IF A.4/37 OR A.4/37A THEN o ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media (note 3).
c20:	IF A.4/37 OR A.4/37A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c21:	IF A.4/37 AND A.4/2 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol and registrar.
c22:	IF A.3/4 THEN m ELSE n/a - - S-CSCF.
c23:	IF A.4/38 THEN o ELSE n/a - - the Reason header field for the session initiation protocol.
c24:	IF A.4/40 THEN o ELSE n/a - - caller preferences for the session initiation protocol.
c25:	IF A.4/43 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c26:	IF A.4/43 THEN o ELSE n/a - - the SIP Referred-By mechanism.
c27:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.
c28:	IF A.4/47 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c29:	IF (A.3/1 OR A.3A/81) THEN m ELSE o - - UE, MSC Server enhanced for ICS.
c30:	IF A.4/48 THEN m ELSE n/a - - the P-User-Database private header extension.
c31:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c32:	IF A.4/70B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.
c35:	IF A.4/71 AND (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C) THEN m ELSE n/a - - IF A.4/71 AND (A.3/9B OR A.3/9C) THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling).
c36:	IF A.4/71 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c37:	IF A.4/13 THEN m ELSE IF A.4/13A THEN m ELSE n/a - - SIP INFO method and package framework, legacy INFO usage.
c38:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c39:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.
c40:	IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.
c41:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c42:	IF A.4/112 AND A.3/2 THEN o ELSE n/a - - resource sharing, AS.
c43:	IF A.4/113 AND (A.3/1 OR A.3/4) THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE or S-CSCF.
c44:	IF A.4/113 AND (A.3/4 OR A.3/7A) THEN m ELSE n/a - - the Cellular-Network-Info header extension and S-CSCF or AS acting as terminating UA.
c45:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c46:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).

NOTE 1: No distinction has been made in these tables between first use of a request on a From/To/Call-ID combination, and the usage in a subsequent one. Therefore the use of "o" etc. above has been included from a viewpoint of first usage.
NOTE 2: The strength of this requirement in RFC 7315 [52] is SHOULD NOT, rather than MUST NOT.
NOTE 3: Support of this header field in this method is dependent on the security mechanism and the security architecture which is implemented.

Prerequisite A.5/18 - - REGISTER request

Table A.120: Supported message bodies within the REGISTER request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	message/sip	[26] 27.5	n/a	c1	[26] 27.5	n/a	c2
2	3GPP IM CN subsystem XML body	subclause 7.6	n/a	c1	subclause 7.6	n/a	c2
3	application/vnd.3gpp.mcptt-info+xml	[8ZE]	n/a	c3	[8ZE]	n/a	c3
c1: IF A.3/4 THEN o ELSE n/a - - S-CSCF. c2: IF A.3/7 THEN o ELSE n/a - - AS. c3: IF A.3A/102 OR A.3A/103 THEN m ELSE n/a - - MCPTT client, MCPTT server.							

Table A.121: Void

Prerequisite A.5/19 - - REGISTER response

Prerequisite: A.6/1 - - Additional for 100 (Trying) response

Table A.121A: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1: IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.							

Prerequisite A.5/19 - - REGISTER response for all remaining status-codes

Table A.122: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	c8	c8	[26] 20.5	m	m
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Call-Info	[26] 20.9	o	o	[26] 20.9	o	o
2	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
3	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
3A	Content-ID	[256] 3.2	o	c15	[256] 3.2	m	c16
4	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
9	From	[26] 20.20	m	m	[26] 20.20	m	m
9A	Geolocation-Error	[89] 4.3	c10	c10	[89] 4.3	c10	c10
9B	History-Info	[66] 4.1	c9	c9	[66] 4.1	c9	c9
10	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
11	Organization	[26] 20.25	o	o	[26] 20.25	o	o
11A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c3	n/a	[52] 4.4, [52A] 4, [234] 2	c3	n/a
11B	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c6	c7	[52] 4.5, [52A] 4	c6	c7
11C	P-Charging-Vector	[52] 4.6, [52A] 4	c4	c5	[52] 4.6, [52A] 4	c4	c5
11E	Privacy	[33] 4.2	c2	n/a	[33] 4.2	c2	n/a
11F	Relayed-Charge	7.2.12	n/a	c14	7.2.12	n/a	c14
11G	Require	[26] 20.32	m	m	[26] 20.32	m	m
11H	Server	[26] 20.35	o	o	[26] 20.35	o	o
11I	Session-ID	[162]	o	c13	[162]	o	c13
12	Timestamp	[26] 20.38	c2	c2	[26] 20.38	m	m
13	To	[26] 20.39	m	m	[26] 20.39	m	m
13A	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
14	Via	[26] 20.42	m	m	[26] 20.42	m	m
15	Warning	[26] 20.43	o (note)	o	[26] 20.43	o	o

c1: IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c2: IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c3: IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c4: IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c5: IF A.4/36 OR A.3/4 THEN m ELSE n/a - - the P-Charging-Vector header extension (including S-CSCF as registrar).
c6: IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c7: IF A.4/35 OR A.3/4 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension (including S-CSCF as registrar).
c8: IF A.6/18 THEN m ELSE o - - 405 (Method Not Allowed).
c9: IF A.4/47 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c10: IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c13: IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c14: IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c15: IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c16: IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).

NOTE: For a 488 (Not Acceptable Here) response, RFC 3261 [26] gives the status of this header field as SHOULD rather than OPTIONAL.

Prerequisite A.5/19 - - REGISTER response

Prerequisite: A.6/102 - - Additional for 2xx response

Table A.123: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o	o	[26] 20.1	o	o
1A	Accept-Encoding	[26] 20.2	o	o	[26] 20.2	m	m
1B	Accept-Language	[26] 20.3	o	o	[26] 20.3	m	m
1C	Accept-Resource-Priority	[116] 3.2	c14	c14	[116] 3.2	c14	c14
2	Allow-Events	[28] 8.2.2	c12	c12	[28] 8.2.2	c13	c13
3	Authentication-Info	[26] 20.6	c6	c6	[26] 20.6	c7	c7
5	Contact	[26] 20.10	o	o	[26] 20.10	m	m
5A	Feature-Caps	[190]	c18	c18	[190]	c17	c17
5B	Flow-Timer	[92] 11	c15	c15	[92] 11	c15	c15
5C	P-Associated-URI	[52] 4.1	c8	c9	[52] 4.1	c10	c11
6	Path	[35] 4	c3	c3	[35] 4	c4	c4
7	Security-Server	Subclause 7.2A.7	n/a	x	Subclause 7.2A.7	n/a	c16
8	Service-Route	[38] 5	c5	c5	[38] 5	c5	c5
9	Supported	[26] 20.37	m	m	[26] 20.37	m	m

c1: IF (A.3/4 AND A.4/2) THEN m ELSE n/a. - - S-CSCF acting as registrar.
c2: IF A.3/4 OR A.3/1 THEN m ELSE n/a. - - S-CSCF or UE.
c3: IF A.4/24 THEN m ELSE n/a - - session initiation protocol extension header field for registering non-adjacent contacts.
c4: IF A.4/24 THEN o ELSE n/a - - session initiation protocol extension header field for registering non-adjacent contacts.
c5: IF A.4/28 THEN m ELSE n/a - - session initiation protocol extension header field for service route discovery during registration.
c6: IF A.4/8 THEN o ELSE n/a - - authentication between UA and registrar.
c7: IF A.4/8 THEN m ELSE n/a - - authentication between UA and registrar.
c8: IF A.4/2 AND A.4/31 THEN m ELSE n/a - - P-Associated-URI header extension and registrar.
c9: IF A.3/1 AND A.4/31 THEN m ELSE n/a - - P-Associated-URI header extension and S-CSCF.
c10: IF A.4/31 THEN o ELSE n/a - - P-Associated-URI header extension.
c11: IF A.4/31 AND A.3/1 THEN m ELSE n/a - - P-Associated-URI header extension and UE.
c12: IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.
c13: IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.
c14: IF A.4/70B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.
c15: IF A.4/57 THEN m ELSE n/a - - managing client initiated connections in SIP.
c16: IF A.4/37A THEN m ELSE n/a - - mediasec header field parameter for marking security mechanisms related to media.
c17: IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.
c18: IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.

Prerequisite A.5/19 - - REGISTER response

Prerequisite: A.6/103 OR A.6/104 OR A.6/105 OR A.6/106 - - Additional for 3xx – 6xx response

Table A.123A: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	o	o	[26] 20.18	o	o
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.4/115 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.5/19 - - REGISTER response

Prerequisite: A.6/103 OR A.6/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.124: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Contact	[26] 20.10	o (note)	o	[26] 20.10	m	m

Prerequisite A.5/19 - - REGISTER response

Prerequisite: A.6/14 - - Additional for 401 (Unauthorized) response

Table A.125: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Proxy-Authenticate	[26] 20.27	c1	x	[26] 20.27	c1	x
6	Security-Server	[48] 2	x	x	[48] 2	n/a	c2
10	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	m	m
c1:	IF A.4/8 THEN m ELSE n/a - - support of authentication between UA and registrar.						
c2:	IF A.4/37 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Prerequisite A.5/19 - - REGISTER response

Prerequisite: A.6/17 OR A.6/23 OR A.6/30 OR A.6/36 OR A.6/42 OR A.6/45 OR A.6/50 OR A.6/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.126: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
6	Retry-After	[26] 20.33	o	o	[26] 20.33	o	o

Table A.127: Void

Prerequisite A.5/19 - - REGISTER response

Prerequisite: A.6/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.128: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Proxy-Authenticate	[26] 20.27	c1	x	[26] 20.27	c1	x
9	WWW-Authenticate	[26] 20.44	o	o	[26] 20.44	o	o
c1:	IF A.4/8 THEN m ELSE n/a - - support of authentication between UA and registrar.						

Prerequisite A.5/19 - - REGISTER response

Prerequisite: A.6/25 - - Additional for 415 (Unsupported Media Type) response

Table A.129: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o.1	o.1	[26] 20.1	m	m
2	Accept-Encoding	[26] 20.2	o.1	o.1	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o.1	o.1	[26] 20.3	m	m
o.1	At least one of these capabilities is supported.						

Prerequisite A.5/19 - - REGISTER response

Prerequisite: A.6/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.129A: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.4/70B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.						

Prerequisite A.5/19 - - REGISTER response

Prerequisite: A.6/27 - - Additional for 420 (Bad Extension) response

Table A.130: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
8	Unsupported	[26] 20.40	m	m	[26] 20.40	m	m

Prerequisite A.5/19 - - REGISTER response

Prerequisite: A.6/28 OR A.6/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.130A: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	c2	c2	[48] 2	c1	c1
c1:	IF A.4/37 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						
c2:	IF A.4/37 AND A.4/2 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol and registrar.						

Prerequisite A.5/19 - - REGISTER response

Prerequisite: A.6/29 - - Additional for 423 (Interval Too Brief) response

Table A.131: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Min-Expires	[26] 20.23	m	m	[26] 20.23	m	m

Table A.132: Void

Prerequisite A.5/19 - - REGISTER response

Table A.133: Supported message bodies within the REGISTER response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.1.4.13 SUBSCRIBE method

Prerequisite A.5/20 - - SUBSCRIBE request

Table A.134: Supported header fields within the SUBSCRIBE request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o	o	[26] 20.1	m	m
1A	Accept-Contact	[56B] 9.2	c22	c22	[56B] 9.2	c26	c26
2	Accept-Encoding	[26] 20.2	o	o	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o	o	[26] 20.3	m	m
3A	Allow	[26] 20.5	o	o	[26] 20.5	m	m
4	Allow-Events	[28] 8.2.2	o	o	[28] 8.2.2	m	m
5	Authorization	[26] 20.7	c3	c3	[26] 20.7	c3	c3
6	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
6A	Call-Info	[26] 20.9	o	o	[26] 20.9	o	o
6B	Cellular-Network-Info	7.2.15	n/a	c48	7.2.15	n/a	c49
6C	Contact	[26] 20.10	m	m	[26] 20.10	m	m
7	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
8	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
8A	Content-ID	[256] 3.2	o	c52	[256] 3.2	m	c53
9	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
10	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
11	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
12	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
13	Date	[26] 20.17	c4	c4	[26] 20.17	m	m
14	Event	[28] 8.2.1	m	m	[28] 8.2.1	m	m
15	Expires	[26] 20.19	o (note 1)	o (note 1)	[26] 20.19	m	m
15A	Feature-Caps	[190]	c46	c46	[190]	c45	c45
16	From	[26] 20.20	m	m	[26] 20.20	m	m
16A	Geolocation	[89] 4.1	c27	c27	[89] 4.1	c27	c27
16B	Geolocation-Routing	[89] 4.2	c27	c27	[89] 4.2	c27	c27
16C	History-Info	[66] 4.1	c25	c25	[66] 4.1	c25	c25
16D	Max-Breadth	[117] 5.8	n/a	c38	[117] 5.8	c39	c39
17	Max-Forwards	[26] 20.22	m	m	[26] 20.22	n/a	c41
18	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
18A	Organization	[26] 20.25	o	o	[26] 20.25	o	o
18B	P-Access-Network-Info	[52] 4.4, [234] 2	c12	c13	[52] 4.4, [234] 2	c12	c14
18C	P-Asserted-Identity	[34] 9.1	n/a	c50	[34] 9.1	c6	c6
18D	P-Asserted-Service	[121] 4.1	n/a	c51	[121] 4.1	c32	c32
18E	P-Called-Party-ID	[52] 4.2	x	x	[52] 4.2	c10	c10
18F	P-Charging-Function-Addresses	[52] 4.5	c17	c18	[52] 4.5	c17	c18
18G	P-Charging-Vector	[52] 4.6	c15	c16	[52] 4.6	c15	c16
18I	P-Preferred-Identity	[34] 9.2	c6	c7	[34] 9.2	n/a	n/a
18J	P-Preferred-Service	[121] 4.2	c31	c30	[121] 4.2	n/a	n/a
18K	P-Private-Network-Indication	[134]	c35	c35	[134]	c35	c35
18L	P-Profile-Key	[97] 5	n/a	n/a	[97] 5	n/a	n/a
18M	P-Served-User	[133] 6	c40	c40	[133] 6	c40	c40
18N	P-User-Database	[82] 4	n/a	n/a	[82] 4	n/a	n/a
18O	P-Visited-Network-ID	[52] 4.3	x (note 2)	x	[52] 4.3	c11	n/a
18P	Priority	[26] 20.26	o	o	[26] 20.26	o	o
18Q	Privacy	[33] 4.2	c9	c9	[33] 4.2	c9	c9
19	Proxy-Authorization	[26] 20.28	c5	c5	[26] 20.28	n/a	n/a
20	Proxy-Require	[26] 20.29	o	n/a	[26] 20.29	n/a	n/a
20A	Reason	[34A] 2	c21	c21	[34A] 2	c21	c21
21	Record-Route	[26] 20.30	n/a	c41	[26] 20.30	m	m
21A	Referred-By	[59] 3	c23	c23	[59] 3	c24	c24
21B	Reject-Contact	[56B] 9.2	c22	c22	[56B] 9.2	c26	c26
21C	Relayed-Charge	7.2.12	n/a	c47	7.2.12	n/a	c47
21D	Request-Disposition	[56B] 9.1	c22	c22	[56B] 9.1	c26	c26
22	Require	[26] 20.32	m	m	[26] 20.32	m	m
22A	Resource-Priority	[116] 3.1	c29	c29	[116] 3.1	c29	c29
23	Route	[26] 20.34	m	m	[26] 20.34	n/a	c41
23A	Security-Client	[48] 2.3.1	c19	c19	[48] 2.3.1	n/a	n/a
23B	Security-Verify	[48] 2.3.1	c20	c20	[48] 2.3.1	n/a	n/a
23C	Session-ID	[162]	o	c42	[162]	o	c42
24	Supported	[26] 20.37	o	o	[26] 20.37	m	m

24A	Target-Dialog	[184] 7	c43	c43	[184] 7	c44	c44
25	Timestamp	[26] 20.38	c8	c8	[26] 20.38	m	m
26	To	[26] 20.39	m	m	[26] 20.39	m	m
26A	Trigger-Consent	[125] 5.11.2	c33	c33	[125] 5.11.2	c34	c34
27	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
28	Via	[26] 20.42	m	m	[26] 20.42	m	m

c3:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.
c4:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c5:	IF A.4/8A THEN m ELSE n/a - - authentication between UA and proxy.
c6:	IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c7:	IF A.3/1 AND A.4/25 THEN o ELSE n/a - - UE and private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c8:	IF A.4/6 THEN o ELSE n/a - - timestamping of requests.
c9:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c10:	IF A.4/32 THEN o ELSE n/a - - the P-Called-Party-ID extension.
c11:	IF A.4/33 THEN o ELSE n/a - - the P-Visited-Network-ID extension.
c12:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c13:	IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.
c14:	IF A.4/34 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c15:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c16:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c17:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c18:	IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c19:	IF A.4/37 OR A.4/37A THEN o ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media (note 3).
c20:	IF A.4/37 OR A.4/37A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c21:	IF A.4/38 THEN o ELSE n/a - - the Reason header field for the session initiation protocol.
c22:	IF A.4/40 THEN o ELSE n/a - - caller preferences for the session initiation protocol.
c23:	IF A.4/43 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c24:	IF A.4/43 THEN o ELSE n/a - - the SIP Referred-By mechanism.
c25:	IF A.4/47 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c26:	IF A.4/40 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c27:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c29:	IF A.4/70A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.
c30:	IF (A.3/1 OR A.3A/81) AND A.4/74 THEN o ELSE n/a - - UE, MSC Server enhanced for ICS and SIP extension for the identification of services.
c31:	IF A.4/74 THEN o ELSE n/a - - SIP extension for the identification of services.
c32:	IF A.4/74 THEN m ELSE n/a - - SIP extension for the identification of services.
c33:	IF A.4/75A THEN m ELSE n/a - - a relay within the framework for consent-based communications in SIP.
c34:	IF A.4/75B THEN m ELSE n/a - - a recipient within the framework for consent-based communications in SIP.
c35:	IF A.4/77 THEN m ELSE n/a - - the SIP P-Private-Network-Indication private-header (P-Header).
c38:	IF A.4/71 AND (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C) THEN m ELSE IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - addressing an amplification vulnerability in session initiation protocol forking proxies, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling), UE, UE performing the functions of an external attached network.
c39:	IF A.4/71 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c40:	IF A.4/78 THEN m ELSE n/a - - the SIP P-Served-User private header.
c41:	IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - UE, UE performing the functions of an external attached network.
c42:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c43:	IF A.4/99 THEN o ELSE n/a - - request authorization through dialog Identification in the session initiation protocol.
c44:	IF A.4/99 THEN m ELSE n/a - - request authorization through dialog Identification in the session initiation protocol.
c45:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.
c46:	IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.
c47:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c48:	IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE.
c49:	IF A.4/113 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c50:	IF A.4/25 AND (A.3/7B OR A.3/8 OR A.3A/81) THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks and AS acting as originating UA, MRFC, MSC Server enhanced for ICS.
c51:	IF A.4/74 AND A.3/7B THEN o ELSE n/a - - SIP extension for the identification of services and AS acting as originating UA.
c52:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).

c53:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
NOTE 1:	The strength of this requirement is RECOMMENDED rather than OPTIONAL.
NOTE 2:	The strength of this requirement in RFC 7315 [52] is SHOULD NOT, rather than MUST NOT.
NOTE 3:	Support of this header field in this method is dependent on the security mechanism and the security architecture which is implemented. Use of this header field in this method is not appropriate to the security mechanism defined by 3GPP TS 33.203 [19].

Prerequisite A.5/20 - - SUBSCRIBE request

Table A.135: Supported message bodies within the SUBSCRIBE request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	application/vnd.3gpp.mcptt-info+xml	[8ZE]	n/a	c1	[8ZE]	n/a	c1
2	application/simple-filter+xml	[243]	o	c1	[243]	n/a	c1
c1	IF A.3A/102 OR A.3A/103 THEN m ELSE n/a - - MCPTT client, MCPTT server.						

Prerequisite A.5/21 - - SUBSCRIBE response

Prerequisite: A.6/1 - - Additional for 100 (Trying) response

Table A.135A: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						

Prerequisite A.5/21 - - SUBSCRIBE response for all remaining status-codes

Table A.136: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	c12	c12	[26] 20.5	m	m
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Call-Info	[26] 20.9	o	o	[26] 20.9	o	o
1B	Cellular-Network-Info	7.2.15	n/a	c19	7.2.15	n/a	c20
2	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
3	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
3A	Content-ID	[256] 3.2	o	c22	[256] 3.2	m	c23
4	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
9	From	[26] 20.20	m	m	[26] 20.20	m	m
9A	Geolocation-Error	[89] 4.3	c14	c14	[89] 4.3	c14	c14
9B	History-Info	[66] 4.1	c13	c13	[66] 4.1	c13	c13
10	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
10A	Organization	[26] 20.25	o	o	[26] 20.25	o	o
10B	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c5	c6	[52] 4.4, [52A] 4, [234] 2	c5	c7
10C	P-Asserted-Identity	[34] 9.1	n/a	c21	[34] 9.1	c3	c3
10D	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c10	c11	[52] 4.5, [52A] 4	c10	c11
10E	P-Charging-Vector	[52] 4.6, [52A] 4	c8	c9	[52] 4.6, [52A] 4	c8	c9
10G	P-Preferred-Identity	[34] 9.2	c3	x	[34] 9.2	n/a	n/a
10H	Privacy	[33] 4.2	c4	c4	[33] 4.2	c4	c4
10I	Relayed-Charge	7.2.12	n/a	c18	7.2.12	n/a	c18
10J	Require	[26] 20.32	m	m	[26] 20.32	m	m
10K	Server	[26] 20.35	o	o	[26] 20.35	o	o
10L	Session-ID	[162]	o	c17	[162]	o	c17
11	Timestamp	[26] 20.38	m	m	[26] 20.38	c2	c2
12	To	[26] 20.39	m	m	[26] 20.39	m	m
12A	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
13	Via	[26] 20.42	m	m	[26] 20.42	m	m
14	Warning	[26] 20.43	o (note)	o	[26] 20.43	o	o

c1: IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c2: IF A.4/6 THEN m ELSE n/a - - timestamping of requests.
c3: IF A.4/25 THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c4: IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c5: IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c6: IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.
c7: IF A.4/34 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c8: IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c9: IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c10: IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c11: IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c12: IF A.6/18 THEN m ELSE o - - 405 (Method Not Allowed).
c13: IF A.4/47 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c14: IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c17: IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c18: IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c19: IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info extension and UE.
c20: IF A.4/113 AND (A.3/7A OR A.3/7D) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller.
c21: IF A.4/25 AND (A.3/7B OR A.3/8) THEN o ELSE n/a - - private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks and AS acting as originating UA, MRFC.
c22: IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c23: IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).

NOTE: For a 488 (Not Acceptable Here) response, RFC 3261 [26] gives the status of this header field as SHOULD rather than OPTIONAL.

Prerequisite A.5/21 - - SUBSCRIBE response

Prerequisite: A.6/102 - - Additional for 2xx response

Table A.137: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Accept-Resource-Priority	[116] 3.2	c5	c5	[116] 3.2	c5	c5
0B	Allow-Events	[28] 8.2.2	o	o	[28] 8.2.2	m	m
1	Authentication-Info	[26] 20.6	c1	c1	[26] 20.6	c2	c2
1A	Contact	[26] 20.10	m	m	[26] 20.10	m	m
2	Expires	[26] 20.19	m	m	[26] 20.19	m	m
2A	Feature-Caps	[190]	c8	c8	[190]	c7	c7
3	Record-Route	[26] 20.30	m	m	[26] 20.30	m	m
6	Supported	[26] 20.37	m	m	[26] 20.37	m	m
c1:	IF A.4/7 THEN o ELSE n/a - - authentication between UA and UA.						
c2:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.						
c5:	IF A.4/70A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						
c7:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.						
c8:	IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.						

Prerequisite A.5/21 - - SUBSCRIBE response

Prerequisite: A.6/103 OR A.6/104 OR A.6/105 OR A.6/106 - - Additional for 3xx – 6xx response

Table A.137A: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	o	o	[26] 20.18	o	o
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.4/115 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.5/21 - - SUBSCRIBE response

Prerequisite: A.6/103 OR A.6/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.138: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Contact	[26] 20.10	m (note)	m	[26] 20.10	m	m
NOTE:	The strength of this requirement is RECOMMENDED rather than MANDATORY for a 485 response.						

Prerequisite A.5/21 - - SUBSCRIBE response

Prerequisite: A.6/14 - - Additional for 401 (Unauthorized) response

Table A.139: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
8	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	m	m
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Prerequisite A.5/21 - - SUBSCRIBE response

Prerequisite: A.6/17 OR A.6/23 OR A.6/30 OR A.6/36 OR A.6/42 OR A.6/45 OR A.6/50 OR A.6/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480 (Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.140: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Retry-After	[26] 20.33	o	o	[26] 20.33	o	o

Table A.141: Void

Prerequisite A.5/21 - - SUBSCRIBE response

Prerequisite: A.6/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.142: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
6	WWW-Authenticate	[26] 20.44	o	o	[26] 20.44	o	o
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Table A.142A: Void

Prerequisite A.5/21 - - SUBSCRIBE response

Prerequisite A.6/25 - - Additional for 415 (Unsupported Media Type) response

Table A.143: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o.1	o.1	[26] 20.1	m	m
2	Accept-Encoding	[26] 20.2	o.1	o.1	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o.1	o.1	[26] 20.3	m	m
o.1	At least one of these capabilities is supported.						

Prerequisite A.5/21 - - SUBSCRIBE response

Prerequisite: A.6/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.143A: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.4/70A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						

Prerequisite A.5/21 - - SUBSCRIBE response

Prerequisite: A.6/27 - - Additional for 420 (Bad Extension) response

Table A.144: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Unsupported	[26] 20.40	m	m	[26] 20.40	m	m

Prerequisite A.5/21 - - SUBSCRIBE response

Prerequisite: A.6/28 OR A.6/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.144A: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	x	x	[48] 2	c1	c1
c1:	IF A.4/37 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Prerequisite A.5/21 - - SUBSCRIBE response

Prerequisite: A.6/29 - - Additional for 423 (Interval Too Brief) response

Table A.145: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Min-Expires	[26] 20.23	m	m	[26] 20.23	m	m

Table A.146: Void

Prerequisite A.5/21 - - SUBSCRIBE response

Prerequisite: A.6/29H - - Additional for 470 (Consent Needed) response

Table A.146A: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Permission-Missing	[125] 5.9.3	m	m	[125] 5.9.3	m	m

Prerequisite A.5/21 - - SUBSCRIBE response

Prerequisite: A.6/39 - - Additional for 489 (Bad Event) response

Table A.147: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	m	m

Table A.148: Void

Prerequisite A.5/21 - - SUBSCRIBE response

Prerequisite: A.6/46 - - Additional for 504 (Server Time-out) response

Table A.148A: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Restoration-Info	subclause 7.2.11	n/a	c1	subclause 7.2.11	n/a	n/a
c1:	IF A.4/110 THEN o ELSE n/a - - HSS based P-CSCF restoration.						

Prerequisite A.5/21 - - SUBSCRIBE response

Table A.149: Supported message bodies within the SUBSCRIBE response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.1.4.14 UPDATE method

Prerequisite A.5/22 - - UPDATE request

Table A.150: Supported header fields within the UPDATE request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o	o	[26] 20.1	m	m
1A	Accept-Contact	[56B] 9.2	c20	c20	[56B] 9.2	c24	c24
2	Accept-Encoding	[26] 20.2	o	o	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o	o	[26] 20.3	m	m
4	Allow	[26] 20.5	o	o	[26] 20.5	m	m
5	Allow-Events	[28] 8.2.2	c2	c2	[28] 8.2.2	c3	c3
6	Authorization	[26] 20.7	c4	c4	[26] 20.7	c4	c4
7	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
8	Call-Info	[26] 20.9	o	o	[26] 20.9	o	o
8A	Cellular-Network-Info	7.2.15	n/a	c40	7.2.15	n/a	c41
9	Contact	[26] 20.10	m	m	[26] 20.10	m	m
10	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
11	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
11A	Content-ID	[256] 3.2	o	c43	[256] 3.2	m	c44
12	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
13	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
14	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
15	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
16	Date	[26] 20.17	c5	c5	[26] 20.17	m	m
16A	Feature-Caps	[190]	c37	c37	[190]	c36	c36
17	From	[26] 20.20	m	m	[26] 20.20	m	m
17A	Geolocation	[89] 4.1	c25	c25	[89] 4.1	c25	c25
17B	Geolocation-Routing	[89] 4.2	c25	c25	[89] 4.2	c25	c25
17C	Max-Breadth	[117] 5.8	n/a	c29	[117] 5.8	c30	c30
18	Max-Forwards	[26] 20.22	m	m	[26] 20.22	n/a	c31
19	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
19A	Min-SE	[58] 5	c21	c21	[58] 5	c21	c21
20	Organization	[26] 20.25	o	o	[26] 20.25	o	o
20A	P-Access-Network-Info	[52] 4.4, [234] 2	c11	c12	[52] 4.4, [234] 2	c11	c13
20B	P-Charging-Function-Addresses	[52] 4.5	c16	c17	[52] 4.5	c16	c17
20C	P-Charging-Vector	[52] 4.6	c14	c15	[52] 4.6	c14	c15
20E	P-Early-Media	[109] 8	c26	c26	[109] 8	c26	c26
20EA	Priority-Share	Subclause 7.2.16	n/a	c42	Subclause 7.2.16	n/a	c42
20F	Privacy	[33] 4.2	c6	n/a	[33] 4.2	c6	n/a
21	Proxy-Authorization	[26] 20.28	c10	c10	[26] 20.28	n/a	n/a
22	Proxy-Require	[26] 20.29	o	n/a	[26] 20.29	n/a	n/a
22A	Reason	[34A] 2	c8	c8	[34A] 2	c8	c8
23	Record-Route	[26] 20.30	n/a	c31	[26] 20.30	n/a	c31
23A	Recv-Info	[25] 5.2.3	c34	c34	[25] 5.2.3	c34	c34
23B	Referred-By	[59] 3	c22	c22	[59] 3	c23	c23
23C	Reject-Contact	[56B] 9.2	c20	c20	[56B] 9.2	c24	c24
23D	Relayed-Charge	7.2.12	n/a	c38	7.2.12	n/a	c38
23E	Request-Disposition	[56B] 9.1	c20	c20	[56B] 9.1	c24	c24
24	Require	[26] 20.32	m	m	[26] 20.32	m	m
24A	Resource-Priority	[116] 3.1	c33	c33	[116] 3.1	c33	c33
24B	Resource-Share	Subclause 7.2.13	n/a	c39	Subclause 7.2.13	n/a	c39
25	Route	[26] 20.34	m	m	[26] 20.34	n/a	c31
25A	Security-Client	[48] 2.3.1	c18	c18	[48] 2.3.1	n/a	n/a
25B	Security-Verify	[48] 2.3.1	c19	c19	[48] 2.3.1	n/a	n/a
25C	Session-Expires	[58] 4	c21	c21	[58] 4	c21	c21
25D	Session-ID	[162]	o	c35	[162]	o	c35
26	Supported	[26] 20.37	o	o	[26] 20.37	m	m
27	Timestamp	[26] 20.38	c9	c9	[26] 20.38	m	m
28	To	[26] 20.39	m	m	[26] 20.39	m	m
29	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
30	Via	[26] 20.42	m	m	[26] 20.42	m	m

c2:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.
c3:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.
c4:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.
c5:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c6:	IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c8:	IF A.4/38 THEN o ELSE n/a - - the Reason header field for the session initiation protocol.
c9:	IF A.4/6 THEN o ELSE n/a - - timestamping of requests.
c10:	IF A.4/8A THEN m ELSE n/a - - authentication between UA and proxy.
c11:	IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c12:	IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.
c13:	IF A.4/34 AND (A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA, AS acting as third-party call controller or EATF.
c14:	IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c15:	IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c16:	IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c17:	IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c18:	IF A.4/37 OR A.4/37A THEN o ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media (note).
c19:	IF A.4/37 OR A.4/37A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c20:	IF A.4/40 THEN o ELSE n/a - - caller preferences for the session initiation protocol.
c21:	IF A.4/42 THEN m ELSE n/a - - the SIP session timer.
c22:	IF A.4/43 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c23:	IF A.4/43 THEN o ELSE n/a - - the SIP Referred-By mechanism.
c24:	IF A.4/40 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c25:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c26:	IF A.4/66 THEN m ELSE n/a - - the SIP P-Early-Media private header extension for authorization of early media.
c29:	IF A.4/71 AND (A.3/9B OR A.3/9C OR A.3/13B OR A.3/13C) THEN m ELSE IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - addressing an amplification vulnerability in session initiation protocol forking proxies, IBCF (IMS-ALG), IBCF (Screening of SIP signalling), ISC gateway function (IMS-ALG), ISC gateway function (Screening of SIP signalling), UE, UE performing the functions of an external attached network.
c30:	IF A.4/71 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c31:	IF A.3/1 AND NOT A.3C/1 THEN n/a ELSE o - - UE, UE performing the functions of an external attached network.
c33:	IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.
c34:	IF A.4/13 THEN m ELSE IF A.4/13A THEN m ELSE n/a - - SIP INFO method and package framework, legacy INFO usage.
c35:	IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c36:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.
c37:	IF A.4/100 AND A.3/1 AND NOT A.3C/1 THEN n/a ELSE IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy, UE, UE performing the functions of an external attached network.
c38:	IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c39:	IF A.4/112 THEN o ELSE n/a - - resource sharing.
c40:	IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE.
c41:	IF A.4/113 AND (A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller or EATF.
c42:	IF A.4/114 THEN o ELSE n/a - - priority sharing.
c43:	IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c44:	IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
NOTE:	Support of this header field in this method is dependent on the security mechanism and the security architecture which is implemented. Use of this header field in this method is not appropriate to the security mechanism defined by 3GPP TS 33.203 [19].

Prerequisite A.5/22 - - UPDATE request

Table A.151: Supported message bodies within the UPDATE request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

Prerequisite A.5/23 - - UPDATE response

Prerequisite: A.6/1 - - Additional for 100 (Trying) response

Table A.151A: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.						

Prerequisite A.5/23 - - UPDATE response for all remaining status-codes

Table A.152: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	c11	c11	[26] 20.5	m	m
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Call-Info	[26] 20.9	o	o	[26] 20.9	o	o
1B	Cellular-Network-Info	7.2.15	n/a	c19	7.2.15	n/a	c20
1C	Contact	[26] 20.10	o	o	[26] 20.10	o	o
2	Content-Disposition	[26] 20.11	o	o	[26] 20.11	m	m
3	Content-Encoding	[26] 20.12	o	o	[26] 20.12	m	m
3A	Content-ID	[256] 3.2	o	c21	[256] 3.2	m	c22
4	Content-Language	[26] 20.13	o	o	[26] 20.13	m	m
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	m	m
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	c1	c1	[26] 20.17	m	m
9	From	[26] 20.20	m	m	[26] 20.20	m	m
9A	Geolocation-Error	[89] 4.3	c13	c13	[89] 4.3	c13	c13
10	MIME-Version	[26] 20.24	o	o	[26] 20.24	m	m
10A	Organization	[26] 20.25	o	o	[26] 20.25	o	o
10B	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c4	c5	[52] 4.4, [52A] 4, [234] 2	c4	c6
10C	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c9	c10	[52] 4.5, [52A] 4	c9	c10
10D	P-Charging-Vector	[52] 4.6, [52A] 4	c7	c8	[52] 4.6, [52A] 4	c7	c8
10F	Privacy	[33] 4.2	c3	n/a	[33] 4.2	c3	n/a
10G	Recv-Info	[25] 5.2.3	c16	c16	[25] 5.2.3	c16	c16
10H	Relayed-Charge	7.2.12	n/a	c18	7.2.12	n/a	c18
10I	Require	[26] 20.31	m	m	[26] 20.31	m	m
10J	Server	[26] 20.35	o	o	[26] 20.35	o	o
10K	Session-ID	[162]	o	c17	[162]	o	c17
11	Timestamp	[26] 20.38	c12	c12	[26] 20.38	c2	c2
12	To	[26] 20.39	m	m	[26] 20.39	m	m
12A	User-Agent	[26] 20.41	o	o	[26] 20.41	o	o
13	Via	[26] 20.42	m	m	[26] 20.42	m	m
14	Warning	[26] 20.43	o (note)	o	[26] 20.43	o	o

c1: IF A.4/11 THEN o ELSE n/a - - insertion of date in requests and responses.
c2: IF A.4/6 THEN m ELSE n/a - - timestamping of requests.
c3: IF A.4/26 THEN o ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c4: IF A.4/34 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c5: IF A.4/34 AND A.3/1 THEN m ELSE n/a - - the P-Access-Network-Info header extension and UE.
c6: IF A.4/34 AND (A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the P-Access-Network-Info header extension and AS acting as terminating UA, AS acting as third-party call controller or EATF.
c7: IF A.4/36 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c8: IF A.4/36 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c9: IF A.4/35 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c10: IF A.4/35 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c11: IF A.6/18 THEN m ELSE o - - 405 (Method Not Allowed)
c12: IF A.4/6 THEN o ELSE n/a - - timestamping of requests.
c13: IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c16: IF A.4/13 THEN m ELSE IF A.4/13A THEN m ELSE n/a - - SIP INFO method and package framework, legacy INFO usage.
c17: IF A.4/91 THEN m ELSE n/a - - the Session-ID header.
c18: IF A.4/111 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c19: IF A.4/113 AND A.3/1 THEN m ELSE n/a - - the Cellular-Network-Info header extension and UE.
c20: IF A.4/113 AND (A.3/7A OR A.3/7D OR A3A/84) THEN m ELSE n/a - - the Cellular-Network-Info header extension and AS acting as terminating UA or AS acting as third-party call controller or EATF.
c21: IF A.4/119 THEN o ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).
c22: IF A.4/119 THEN m ELSE n/a - - Content-ID header field in Session Initiation Protocol (SIP).

NOTE: For a 488 (Not Acceptable Here) response, RFC 3261 [26] gives the status of this header field as SHOULD rather than OPTIONAL.

Prerequisite A.5/23 - - UPDATE response

Prerequisite: A.6/102 - - Additional for 2xx response

Table A.153: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Accept	[26] 20.1	o	o	[26] 20.1	m	m
0B	Accept-Encoding	[26] 20.2	o	o	[26] 20.2	m	m
0C	Accept-Language	[26] 20.3	o	o	[26] 20.3	m	m
0D	Accept-Resource-Priority	[116] 3.2	c14	c14	[116] 3.2	c14	c14
1	Allow-Events	[28] 8.2.2	c4	c4	[28] 8.2.2	c5	c5
2	Authentication-Info	[26] 20.6	c1	c1	[26] 20.6	c2	c2
3	Contact	[26] 20.10	m	m	[26] 20.10	m	m
3A	Feature-Caps	[190]	c16	c16	[190]	c16	c16
3B	P-Early-Media	[109] 8	c6	c6	[109] 8	c6	c6
3C	Priority-Share	Subclause 7.2.16	n/a	c18	Subclause 7.2.16	n/a	c18
3E	Resource-Share	Subclause 7.2.13	n/a	c17	Subclause 7.2.13	n/a	c17
4	Session-Expires	[58]	c3	c3	[58]	c3	c3
6	Supported	[26] 20.37	m	m	[26] 20.37	m	m
c1:	IF A.4/7 THEN o ELSE n/a - - authentication between UA and UA.						
c2:	IF A.4/7 THEN m ELSE n/a - - authentication between UA and UA.						
c3:	IF A.4/42 THEN m ELSE n/a - - the SIP session timer						
c4:	IF A.4/22 THEN o ELSE n/a - - acting as the notifier of event information.						
c5:	IF A.4/23 THEN m ELSE n/a - - acting as the subscriber to event information.						
c6:	IF A.4/66 THEN m ELSE n/a - - the SIP P-Early-Media private header extension for authorization of early media.						
c14:	IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						
c16:	IF A.4/100 THEN m ELSE n/a - - indication of features supported by proxy.						
c17:	IF A.4/112 THEN o ELSE n/a - - resource sharing.						
c18:	IF A.4/114 THEN o ELSE n/a - - priority sharing.						

Prerequisite A.5/23 - - UPDATE response

Prerequisite: A.6/103 OR A.6/104 OR A.6/105 OR A.6/106 - - Additional for 3xx – 6xx response

Table A.153A: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	o	o	[26] 20.18	o	o
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.4/115 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.5/23 - - UPDATE response

Prerequisite: A.6/103 OR A.6/35 - - Additional for 3xx, 485 (Ambiguous) response

Table A.154: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Contact	[26] 20.10	o	o	[26] 20.10	o	o

Prerequisite A.5/23 - - UPDATE response

Prerequisite: A.6/14 - - Additional for 401 (Unauthorized) response

Table A.154A: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
6	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	m	m
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Prerequisite A.5/23 - - UPDATE response

Prerequisite: A.6/17 OR A.6/23 OR A.6/30 OR A.6/36 OR A.6/42 OR A.6/45 OR A.6/50 OR A.6/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.155: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Retry-After	[26] 20.33	o	o	[26] 20.33	o	o

Table A.156: Void

Prerequisite A.5/23 - - UPDATE response

Prerequisite: A.6/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.157: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Proxy-Authenticate	[26] 20.27	c1	c1	[26] 20.27	c1	c1
8	WWW-Authenticate	[26] 20.44	o	o	[26] 20.44	o	o
c1:	IF A.4/7 THEN m ELSE n/a - - support of authentication between UA and UA.						

Prerequisite A.5/23 - - UPDATE response

Prerequisite: A.6/25 - - Additional for 415 (Unsupported Media Type) response

Table A.158: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	o.1	o.1	[26] 20.1	m	m
2	Accept-Encoding	[26] 20.2	o.1	o.1	[26] 20.2	m	m
3	Accept-Language	[26] 20.3	o.1	o.1	[26] 20.3	m	m
o.1	At least one of these capabilities is supported.						

Prerequisite A.5/23 - - UPDATE response

Prerequisite: A.6/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.158A: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.4/70 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						

Prerequisite A.5/23 - - UPDATE response

Prerequisite: A.6/27 - - Additional for 420 (Bad Extension) response

Table A.159: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
7	Unsupported	[26] 20.40	m	m	[26] 20.40	m	m

Prerequisite A.5/23 - - UPDATE response

Prerequisite: A.6/28 OR A.6/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.159A: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	x	x	[48] 2	c1	c1
c1:	IF A.4/37 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Prerequisite A.5/23 - - UPDATE response

Prerequisite: A.6/28A - - Additional for 422 (Session Interval Too Small) response

Table A.159B: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Min-SE	[58] 5	c1	c1	[58] 5	c1	c1
c1:	IF A.4/42 THEN m ELSE n/a - - the SIP session timer.						

Table A.160: Void

Prerequisite A.5/23 - - UPDATE response

Table A.161: Supported message bodies within the UPDATE response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.2 Proxy role

A.2.2.1 Introduction

This subclause contains the ICS proforma tables related to the proxy role. They need to be completed only for proxy implementations.

Prerequisite: A.2/2 - - proxy role

A.2.2.2 Major capabilities

Table A.162: Major capabilities

Item	Does the implementation support	Reference	RFC status	Profile status
	Capabilities within main protocol			
3	initiate session release?	[26] 16	x	c27
4	stateless proxy behaviour?	[26] 16.11	o.1	c29
5	stateful proxy behaviour?	[26] 16.2	o.1	c28
6	forking of initial requests?	[26] 16.1	c1	c31
7	support of indication of TLS connections in the Record-Route header on the upstream side?	[26] 16.7	o	n/a
8	support of indication TLS connections in the Record-Route header on the downstream side?	[26] 16.7	o	n/a
8A	authentication between UA and proxy?	[26] 20.28, 22.3, [287]	o	c85
9	insertion of date in requests and responses?	[26] 20.17	o	o
10	suppression or modification of alerting information data?	[26] 20.4	o	o
11	reading the contents of the Require header before proxying the request or response?	[26] 20.32	o	o
12	adding or modifying the contents of the Require header before proxying the REGISTER request or response	[26] 20.32	o	m
13	adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER?	[26] 20.32	o	o
14	being able to insert itself in the subsequent transactions in a dialog (record-routing)?	[26] 16.6	o	c2
15	the requirement to be able to use separate URIs in the upstream direction and downstream direction when record routeing?	[26] 16.7	c3	c3
16	reading the contents of the Supported header before proxying the response?	[26] 20.37	o	o
17	reading the contents of the Unsupported header before proxying the 420 response to a REGISTER?	[26] 20.40	o	m
18	reading the contents of the Unsupported header before proxying the 420 response to a method other than REGISTER?	[26] 20.40	o	o
19	the inclusion of the Error-Info header in 3xx - 6xx responses?	[26] 20.18	o	o
19A	reading the contents of the Organization header before proxying the request or response?	[26] 20.25	o	o
19B	adding or concatenating the Organization header before proxying the request or response?	[26] 20.25	o	o
19C	reading the contents of the Call-Info header before proxying the request or response?	[26] 20.9	o	o
19D	adding or concatenating the Call-Info header before proxying the request or response?	[26] 20.9	o	o
19E	delete Contact headers from 3xx responses prior to relaying the response?	[26] 20	o	o
19F	proxy reading the contents of a body or including a body in a request or response?	[26]	o	c88
19G	proxy modifying the content of a body	3GPP TS 24.237 [8M]	n/a	c103
	Extensions			

20	SIP INFO method and package framework?	[25]	o	o
20A	legacy INFO usage?	[25] 2, 3	o	o
21	reliability of provisional responses in SIP?	[27]	o	i
22	the REFER method?	[36]	o	o
22A	clarifications for the use of REFER with RFC6665?	[231]	c113	c113
22B	explicit subscriptions for the REFER method?	[232]	o	o
23	integration of resource management and SIP?	[30] [64]	o	i
24	the SIP UPDATE method?	[29]	c4	i
26	SIP extensions for media authorization?	[31]	o	c7
27	SIP specific event notification	[28]	o	i
28	a clarification on the use of GRUUs in the SIP event notification framework?	[232]	n/a	n/a
29	Session Initiation Protocol Extension Header Field for Registering Non-Adjacent Contacts	[35]	o	c6
30	private extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks	[34]	o	m
30A	act as first entity within the trust domain for asserted identity?	[34]	c5	c9
30B	act as entity within trust network that can route outside the trust network?	[34]	c5	c9
30C	act as entity passing on identity transparently independent of trust domain?	[34]	c5	c96
31	a privacy mechanism for the Session Initiation Protocol (SIP)	[33]	o	m
31A	request of privacy by the inclusion of a Privacy header	[33]	n/a	n/a
31B	application of privacy based on the received Privacy header	[33]	c10	c12
31C	passing on of the Privacy header transparently	[33]	c10	c13
31D	application of the privacy option "header" such that those headers which cannot be completely expunged of identifying information without the assistance of intermediaries are obscured?	[33] 5.1	x	x
31E	application of the privacy option "session" such that anonymization for the session(s) initiated by this message occurs?	[33] 5.2	n/a	n/a
31F	application of the privacy option "user" such that user level privacy functions are provided by the network?	[33] 5.3	n/a	n/a
31G	application of the privacy option "id" such that privacy of the network asserted identity is provided by the network?	[34] 7	c11	c12
31H	application of the privacy option "history" such that privacy of the History-Info header is provided by the network?	[66] 7.2	c34	c34
32	Session Initiation Protocol Extension Header Field for Service Route Discovery During Registration	[38]	o	c30
33	a messaging mechanism for the Session Initiation Protocol (SIP)	[50]	o	m
34	Compressing the Session Initiation Protocol	[55]	o	c7

35	private header extensions to the session initiation protocol for the 3rd-Generation Partnership Project (3GPP)?	[52]	o	m
36	the P-Associated-URI header extension?	[52] 4.1, [52A] 4	c14	c15
37	the P-Called-Party-ID header extension?	[52] 4.2, [52A] 4`	c14	c16
38	the P-Visited-Network-ID header extension?	[52] 4.3, [52A] 4, [52B] 3	c14	c17
39	reading, or deleting the P-Visited-Network-ID header before proxying the request or response?	[52] 4.3	c18	n/a
41	the P-Access-Network-Info header extension?	[52] 4.4, [52A] 4, [234] 2	c14	c19
42	act as first entity within the trust domain for access network information?	[52] 4.4	c20	c21
43	act as subsequent entity within trust network for access network information that can route outside the trust network?	[52] 4.4	c20	c22
44	the P-Charging-Function-Addresses header extension?	[52] 4.5, [52A] 4	c14	m
44A	adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response?	[52] 4.6	c25	c26
45	the P-Charging-Vector header extension?	[52] 4.6, [52A] 4	c14	m
46	adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response?	[52] 4.6	c23	c24
47	security mechanism agreement for the session initiation protocol?	[48]	o	c7
47A	mediasec header field parameter for marking security mechanisms related to media?	Subclause 7.2A.7	n/a	c99
48	the Reason header field for the session initiation protocol	[34A]	o	c78
48A	carrying Q.850 codes in reason header fields in SIP (Session Initiation Protocol) responses?	[130]	o	o
48B	the location parameter for the SIP Reason header field?	[255]	o	o
48C	Handling of Identity Header Errors for Secure Telephone Identity Revisited (STIR) (carrying STIR codes in Reason header fields in SIP responses)?	[294]	o	c130
49	an extension to the session initiation protocol for symmetric response routeing	[56A]	o	m
50	caller preferences for the session initiation protocol?	[56B]	c33	c33
50A	the proxy-directive within caller-preferences?	[56B] 9.1	o.4	o.4
50B	the cancel-directive within caller-preferences?	[56B] 9.1	o.4	o.4
50C	the fork-directive within caller-preferences?	[56B] 9.1	o.4	c32
50D	the recurse-directive within caller-preferences?	[56B] 9.1	o.4	o.4
50E	the parallel-directive within caller-preferences?	[56B] 9.1	o.4	c32
50F	the queue-directive within caller-preferences?	[56B] 9.1	o.4	o.4

51	an event state publication extension to the session initiation protocol?	[70]	o	m
52	SIP session timer?	[58]	o	o
53	the SIP Referred-By mechanism?	[59]	o	o
54	the Session Inititation Protocol (SIP) "Replaces" header?	[60]	o	o
55	the Session Inititation Protocol (SIP) "Join" header?	[61]	o	o
56	the callee capabilities?	[62]	o	o
57	an extension to the session initiation protocol for request history information?	[66]	o	o
57A	application of the "mp" optional header field parameter?	[66]	o	o
57B	application of the "rc" optional header field parameter?	[66]	o	o
57C	application of the "np" optional header field parameter?	[66]	o	o
58	Rejecting anonymous requests in the session initiation protocol?	[67]	o	o
59	session initiation protocol URIs for applications such as voicemail and interactive voice response	[68]	o	o
59A	Session Initiation Protocol (SIP) cause URI parameter for service number translation?	[230]	c111	c111
60	the P-User-Database private header extension?	[82]	o	c95
61	Session initiation protocol's non-INVITE transactions?	[84]	m	m
62	a uniform resource name for services	[69]	n/a	c35
63	obtaining and using GRUUs in the Session Initiation Protocol (SIP)	[93]	o	c36
65	the Stream Control Transmission Protocol (SCTP) as a Transport for the Session Initiation Protocol (SIP)?	[96]	o	o (note2)
66	the SIP P-Profile-Key private header extension?	[97]	o	c41
66A	making the first query to the database in order to populate the P-Profile-Key header?	[97]	c38	c39
66B	using the information in the P-Profile-Key header?	[97]	c38	c40
67	managing client initiated connections in SIP?	[92] 11	o	c42
68	indicating support for interactive connectivity establishment in SIP?	[102]	o	o
69	multiple-recipient MESSAGE requests in the session initiation protocol	[104]	n/a	n/a
70	SIP location conveyance?	[89]	o	c94
70A	addition or modification of location in a SIP method?	[89]	c44	c45
70B	passes on locations in SIP method without modification?	[89]	c44	c46
71	referring to multiple resources in the session initiation protocol?	[105]	n/a	n/a
72	conference establishment using request-contained lists in the session initiation protocol?	[106]	n/a	n/a
73	subscriptions to request-contained resource lists in the session initiation protocol?	[107]	n/a	n/a
74	dialstring parameter for the session initiation protocol uniform resource identifier?	[103]	o	n/a

75	the P-Answer-State header extension to the session initiation protocol for the open mobile alliance push to talk over cellular?	[111]	o	c60
76	the SIP P-Early-Media private header extension for authorization of early media?	[109] 8	o	c51
77	number portability parameters for the 'tel' URI?	[112]	o	c47
77A	assert or process carrier indication?	[112]	o	c48
77B	local number portability?	[112]	o	c50
79	extending the session initiation protocol Reason header for preemption events	[115]	c79	c79
80	communications resource priority for the session initiation protocol?	[116]	o	c80
80A	inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol?	[116] 4.2	c82	c82
80B	inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol?	[116] 4.2	c82	c82
81	addressing an amplification vulnerability in session initiation protocol forking proxies?	[117]	c52	c52
82	the remote application identification of applying signalling compression to SIP	[79] 9.1	o	c7
83	a session initiation protocol media feature tag for MIME application subtypes?	[120]	o	c53
84	SIP extension for the identification of services?	[121]	o	c54
84A	act as authentication entity within the trust domain for asserted service?	[121]	c55	c56
85	a framework for consent-based communications in SIP?	[125]	o	m
86	a mechanism for transporting user-to-user call control information in SIP?	[126]	o	c84
87	the SIP P-Private-Network-Indication private-header (P-Header)?	[134]	o	o
88	the SIP P-Served-User private header in the 3GG IM CN subsystem?	[133] 6	o	o
89	the SIP P-Served-User header extension for Originating CDIV session case?	[239] 4	c126	c126
90	marking SIP messages to be logged?	[140]	o	m
91	the 199 (Early Dialog Terminated) response code	[142]	o	c90
92	message body handling in SIP?	[150]	o	c89
93	indication of support for keep-alive?	[143]	o	c51
94	SIP Interface to VoiceXML Media Services?	[145]	o	c91
95	common presence and instant messaging (CPIM): message format?	[151]	o	o
96	instant message disposition notification?	[157]	o	o
97	requesting answering modes for SIP?	[158]	o	o
97A	adding, deleting or reading the Answer-Mode header or Priv-Answer-Mode before proxying the request or response?	[158]	o	c92
99	the early session disposition type for SIP?	[74B]	i	i
101	The Session-ID header?	[162]	o	o

102	correct transaction handling for 2xx responses to Session Initiation Protocol INVITE requests?	[163]	m	m
103	addressing Record-Route issues in the Session Initiation Protocol (SIP)?	[164]	o	o
104	essential correction for IPv6 ABNF and URI comparison in RFC3261?	[165]	m	m
105	suppression of session initiation protocol REFER method implicit subscription?	[173]	o	c100
106	Alert-Info URNs for the Session Initiation Protocol?	[175]	o	o
107	multiple registrations?	Subclause 3.1	n/a	c101
108	the SIP P-Refused-URI-List private-header?	[183]	o	c102
109	request authorization through dialog Identification in the session initiation protocol?	[184]	o	o
110	indication of features supported by proxy?	[190]	o	c104
111	registration of bulk number contacts?	[191]	o	c105
112	media control channel framework?	[146]	n/a	n/a
113	S-CSCF restoration procedures?	Subclause 4.14	n/a	n/a
114	SIP overload control?	[198]	o	o
114A	feedback control?	[199]	c106	c106
114B	distribution of load filters?	[201]	n/a	n/a
115	handling of a 380 (Alternative service) response?	Subclause 5.2.10	n/a	n/a
116	indication of adjacent network in the Via "received-realm" header field parameter?	[208]	o	c107
117	PSAP callback indicator?	[209]	o	c108
118	SIP URI parameter to indicate traffic leg?	[225]	o	c109
119	PCF or PCRF based P-CSCF restoration?	Subclause 4.14.2	n/a	c110
120	UDM/HSS or HSS based P-CSCF restoration?	Subclause 4.14.2	n/a	c112
121	the Relayed-Charge header field extension?	Subclause 7.2.12	n/a	c114
122	resource sharing?	Subclause 4.15	n/a	c115
123	the Cellular-Network-Info header extension?	Subclause 7.2.15	n/a	c116
124	the Priority-Share header field extension?	Subclause 7.2.16	n/a	c127
125	the Response-Source header field extension?	Subclause 7.2.17	n/a	o
126	authenticated identity management in the Session Initiation Protocol?	[252]	o	c128
127	a SIP response code for unwanted calls extension?	[254]	o	o
128	the Attestation-Info header field extension?	Subclause 7.2.18	n/a	o
129	the Origination-Id header field extension?	Subclause 7.2.19	n/a	o
130	Dynamic services interactions?	Subclause 4.18	n/a	c128
131	the Additional-Identity header field extension?	Subclause 7.2.20	n/a	o
132	RLOS?	Subclause 4.19	n/a	c129
133	the Priority-Verstat header field extension?	Subclause 7.2.21	n/a	c82

c1:	IF A.162/5 THEN o ELSE n/a - - stateful proxy behaviour.
c2:	IF A.3/2 OR A.3/9A OR A.3/4 OR A.3/13A OR A.3A/88 THEN m ELSE o - - P-CSCF, IBCF (THIG), S-CSCF, ISC gateway function (THIG), ATCF (proxy).
c3:	IF (A.162/7 AND NOT A.162/8) OR (NOT A.162/7 AND A.162/8) THEN m ELSE IF A.162/14 THEN o ELSE n/a - - TLS interworking with non-TLS else proxy insertion.
c4:	IF A.162/23 THEN m ELSE o - - integration of resource management and SIP.
c5:	IF A.162/30 THEN o ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c6:	IF A.3/2 OR A.3/9A OR A.3A/88 THEN m ELSE n/a - - P-CSCF, IBCF (THIG), ATFC (proxy).
c7:	IF A.3/2 AND (A.3D/1 OR A.3D/4) THEN m ELSE n/a - - P-CSCF and (IMS AKA plus IPsec ESP or SIP digest with TLS).
c9:	IF (A.3/2 OR A.3/4 OR A.3/9A OR A.3/13A) AND A.162/30 THEN m ELSE IF A.3/7C AND A.162/30 THEN o ELSE n/a - - P-CSCF or S-CSCF or IBCF (THIG) or ISC gateway function (THIG) or AS acting as proxy and extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks (NOTE 1).
c10:	IF A.162/31 THEN o.2 ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c11:	IF A.162/31B THEN o ELSE x - - application of privacy based on the received Privacy header.
c12:	IF A.162/31 AND A.3/4 THEN m ELSE IF A.3/11 THEN o ELSE n/a - - S-CSCF, E-CSCF.
c13:	IF A.162/31 AND (A.3/2 OR A.3/3 OR A.3/7C OR A.3/9A OR A.3/13A OR A.3A/88) THEN m ELSE n/a - - P-CSCF, I-CSCF, AS acting as a SIP proxy or IBCF (THIG), ISC gateway function (THIG), ATCF (proxy).
c14:	IF A.162/35 THEN o.3 ELSE n/a - - private header extensions to the session initiation protocol for the 3rd-Generation Partnership Project (3GPP).
c15:	IF A.162/35 AND (A.3/2 OR A.3/3 OR A.3/9A OR A.3/13A) THEN m ELSE n/a - - private header extensions to the session initiation protocol for the 3rd-Generation Partnership Project (3GPP) and P-CSCF or I-CSCF or IBCF (THIG) or ISC gateway function (THIG).
c16:	IF A.162/35 AND (A.3/2 OR A.3/3 OR A.3/4 OR A.3/9A OR A.3/13A) THEN m ELSE n/a - - private header extensions to the session initiation protocol for the 3rd-Generation Partnership Project (3GPP) and P-CSCF or I-CSCF or S-CSCF or IBCF (THIG) or ISC gateway function (THIG).
c17:	IF A.162/35 AND (A.3/2 OR A.3/3 OR A.3/9A OR A.3/13A) THEN m ELSE n/a - - private header extensions to the session initiation protocol for the 3rd-Generation Partnership Project (3GPP) and P-CSCF or I-CSCF or IBCF (THIG) or ISC gateway function (THIG).
c18:	IF A.162/38 THEN o ELSE n/a - - the P-Visited-Network-ID header extension.
c19:	IF A.162/35 AND (A.3/2 OR A.3.3 OR A.3/4 OR A.3/7 THEN m ELSE n/a - - private header extensions to the session initiation protocol for the 3rd-Generation Partnership Project (3GPP) and P-CSCF, I-CSCF, S-CSCF, AS acting as a proxy.
c20:	IF A.162/41 THEN o ELSE n/a - - the P-Access-Network-Info header extension.
c21:	IF A.162/41 AND A.3/2 THEN m ELSE n/a - - the P-Access-Network-Info header extension and P-CSCF.
c22:	IF A.162/41 AND A.3/4 THEN m ELSE n/a - - the P-Access-Network-Info header extension and S-CSCF.
c23:	IF A.162/45 THEN o ELSE n/a - - the P-Charging-Vector header extension.
c24:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c25:	IF A.162/44 THEN o ELSE n/a - - the P-Charging-Function-Addresses header extension.
c26:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function Addresses header extension.
c27:	IF A.3/2 OR A.3/4 THEN m ELSE x - - P-CSCF or S-CSCF.
c28:	IF A.3/2 OR A.3/3 OR A.3/4 THEN m ELSE o.8 - - P-CSCF or I-CSCF or S-CSCF.
c29:	IF A.3/2 OR A.3/4 THEN n/a ELSE IF A.3/3 THEN o ELSE o.8 - - P-CSCF or S-CSCF or I-CSCF.
c30:	IF A.3/2 o ELSE i - - P-CSCF.
c31:	IF A.3/4 THEN m ELSE x - - S-CSCF.
c32:	IF A.3/4 THEN m ELSE o.4 - - S-CSCF.
c33:	IF A.162/50A OR A.162/50B OR A.162/50C OR A.162/50D OR A.162/50E OR A.162/50F THEN m ELSE n/a - - support of any directives within caller preferences for the session initiation protocol.
c34:	IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c35:	IF A.3/2 OR A.3/11 THEN m ELSE IF A.3/7C OR A.3/9 OR A.3/13A THEN o ELSE n/a - - P-CSCF, E-CSCF, AS acting as proxy, IBCF, ISC gateway function (THIG).

c36:	IF A.3/4 THEN m ELSE n/a - - S-CSCF.
c38:	IF A.162/66 THEN o ELSE n/a - - the SIP P-Profile-Key private header.
c39:	IF A.162/66 AND (A.3/3 OR A.3/9A) THEN m ELSE n/a - - the SIP P-Profile-Key private header, I-CSCF or IBCF (THIG).
c40:	IF A.162/66 AND A.3/4 THEN m ELSE n/a - - the SIP P-Profile-Key private header, S-CSCF.
c41:	IF A.3/3 OR A.3/4 OR A.3/9A THEN o ELSE n/a - - I-CSCF or S-CSCF or IBCF (THIG).
c42:	IF A.162/107 THEN m ELSE n/a - - multiple registrations.
c44:	IF A.162/70 THEN o.5 ELSE n/a - - SIP location conveyance.
c45:	IF A.3/11 THEN m ELSE IF A.162/70 AND A.3/7C THEN o.6 ELSE n/a - - E-CSCF, SIP location conveyance, AS acting as a SIP proxy.
c46:	IF A.162/70 AND A.3/2 OR A.3/3 OR A.3/5 OR A.3/10 OR A.3A/88 THEN m ELSE IF A.162/70 AND A.3/7C THEN o.6 ELSE n/a - - SIP location conveyance, P-CSCF, I-CSCF, S-CSCF, BGCF, additional routeing functionality, ATCF (proxy).
c47:	IF A.3/3 OR A.3/4 OR A.3/5 OR A.3/7C THEN o ELSE n/a - - I-CSCF, S-CSCF, BGCF, AS acting as a SIP proxy.
c48:	IF A.162/77 THEN m ELSE n/a - - number portability parameters for the 'tel' URI.
c50:	IF A.162/77 THEN m ELSE n/a - - number portability parameters for the 'tel' URI.
c51:	IF A.3/2 THEN m ELSE o - - P-CSCF.
c52:	IF A.162/6 THEN m ELSE o - - forking of initial requests.
c53:	IF A.3/4 THEN m ELSE n/a - - S-CSCF.
c54:	IF A.3/3 OR A.3/4 OR A.3/7 OR A.3/2 OR A.3/9A OR A.3/13A THEN m ELSE n/a - - I-CSCF, S-CSCF, BGCF, P-CSCF. IBCF (THIG), ISC gateway function (THIG).
c55:	IF A.162/84 THEN o ELSE n/a - - SIP extension for the identification of services.
c56:	IF A.3/4 AND A.162/84 THEN m ELSE n/a - - S-CSCF and SIP extension for the identification of services.
c60:	IF A.3/2 OR A.3/3 OR A.3/4 THEN o ELSE n/a - - P-CSCF, I-CSCF, S-CSCF.
c78:	IF A.3/2 OR A.3/4 OR A.3/9 OR A.162/79 OR A.162/3 THEN m ELSE o - - P-CSCF, S-CSCF, IBCF, extending the session initiation protocol Reason header for preemption events, initiate session release.
c79:	IF A.162/80 THEN o ELSE n/a - - communications resource priority for the session initiation protocol.
c80:	IF A.3/2 OR A.3/3 OR A.3/4 OR A.3/5 OR A.3/7C OR A.3/9A OR A.3/10 OR A.3/13A THEN o ELSE n/a - - P-CSCF, I-CSCF, S-CSCF, BGCF, AS acting as proxy, IBCF (THIG), additional routeing functionality, ISC gateway function (THIG).
c82:	IF A.162/80 THEN o ELSE n/a - - communications resource priority for the session initiation protocol.
c84:	A.3/2 OR A.3/3 OR A.3/4 OR A.3/5 OR A.3/7C OR A.3/9A OR A.3/10 OR A.3/11 OR A.3/13A THEN o ELSE n/a - - P-CSCF, I-CSCF, S-CSCF, BGCF, AS acting as proxy, IBCF (THIG), additional routeing functionality, E-CSCF, ISC gateway function (THIG).
c85:	IF A.3/2 OR A.3/3 OR A.3/4 THEN o ELSE x - - P-CSCF, I-CSCF, S-CSCF.
c88:	IF A.3/2 OR A.3/4 OR A.3/7 OR A.3/7C OR A.3/9C OR A.3/11 OR A.3/13C OR A.3A/88 THEN m ELSE o - - P-CSCF, S-CSCF, AS, AS acting as a SIP proxy, IBCF (Screening of SIP signalling), E-CSCF, ISC gateway function (Screening of SIP signalling), ATCF (proxy).
c89:	IF A.162/19F THEN m ELSE n/a - - proxy reading the contents of a body or including a body in a request or response.
c90:	IF A.3/4 THEN m ELSE i - - S-CSCF.
c91:	IF A.3/4 THEN o ELSE n/a - - S-CSCF.
c92:	IF A.162/92 THEN o ELSE n/a - - requesting answering modes for SIP.
c94:	IF A.3/11 THEN m ELSE o - - E-CSCF.
c95:	IF A.3/3 OR A.3/4 OR A.3/7C THEN o ELSE n/a - - I-CSCF, S-CSCF, AS acting as a SIP proxy.
c96:	IF A.3/2 OR A.3/11 OR A.3A/88 THEN m ELSE n/a - - P-CSCF, E-CSCF, ATCF (proxy).
c99:	IF A.3/2A AND (A.3D/30 OR A.3D/20A OR A.3D/20B OR A.3D/20C OR A.3D/31) THEN m ELSE n/a - - P-CSCF (IMS-ALG) and end-to-access-edge media security using SDES, end-to-access-edge media security for MSRP using TLS and certificate fingerprints, end-to-access-edge media security for BFCP using TLS and certificate fingerprints, end-to-access-edge media security for UDPTL using DTLS and certificate fingerprints, end-to-access-edge media security for RTP media using DTLS-SRTP and certificate fingerprints.
c100:	IF A.4/22 THEN o ELSE n/a - - the REFER method.
c101:	IF A.3/2 OR A.3/4 THEN m ELSE n/a - - P-CSCF, S-CSCF.

c102:	IF A.3/9B THEN m ELSE IF A.3/7A OR A.3/7B OR A.3/7D THEN o ELSE n/a - - IBCF (IMS-ALG), AS acting as terminating UA, AS acting as originating UA, AS performing 3rd party call control.
c103:	IF A.3A/88 THEN m ELSE n/a - - ATCF (proxy).
c104:	IF A.3/2 OR A.3A/50A OR A.3A/83 OR A.3A/88 THEN m ELSE o - - P-CSCF, Multimedia telephony application server, SCC application server, ATCF (proxy).
c105:	IF A.3/2 OR A.3/3 OR A.3/9 THEN o ELSE n/a - - P-CSCF, I-CSCF, IBCF.
c106:	IF A.162/114 THEN o.9 ELSE n/a - - SIP overload control.
c107:	IF A.162/115 THEN o.9 ELSE n/a - - indication of adjacent network in the Via "received-realm" header field parameter.
c108:	IF A.3/2 OR A.3/3 OR A.3/4 OR A.3/7 OR A.3/9 THEN o ELSE n/a - - P-CSCF, I-CSCF, S-CSCF, AS, IBCF.
c109:	IF A.3/2 OR A.3/4 OR A.3/5 OR A.3/9 OR A.3/10 OR A.3A/83 OR A.3A/88 OR A.3/3 THEN o ELSE n/a - - P-CSCF, S-CSCF, BGCF, IBCF, Additional routeing functionality, SCC application server, ATCF (proxy), I-CSCF.
c110:	IF A.3/2 OR A.3/4 OR A.3/9 THEN o ELSE n/a - - P-CSCF, S-CSCF, IBCF.
c111:	IF A.162/59 THEN o ELSE n/a - - session initiation protocol URIs for applications such as voicemail and interactive voice response (NOTE 3).
c112:	IF A.3/2 OR A.3/4 OR A.3/9 THEN o ELSE n/a - - P-CSCF, S-CSCF, IBCF.
c113:	IF A.162/22 THEN m ELSE n/a - - the REFER method.
c114:	IF A.3/4 OR A.3/7 OR A.3A/102 THEN o ELSE n/a.-.-.S-CSCF, AS, transit function.
c115:	IF A.3/2 OR A.3/7C OR A.3/9 THEN o ELSE n/a - - P-CSCF, AS acting as a SIP proxy, IBCF.
c116:	IF A.3/2 OR A.3.3 OR A.3/4 OR A.3/7C OR A.3/9 OR A.3/11B OR A.3A/88 OR A.3/5 THEN m ELSE n/a - - P-CSCF, I-CSCF, S-CSCF, AS acting as a proxy, IBCF, E-CSCF acting as a SIP Proxy, ATCF (proxy), BGCF.
c126:	IF A.162/88 THEN o ELSE n/a - - the SIP P-Served-User private header for the 3GPP IM CN subsystem.
c127:	IF A.3/2 OR A.3/7C OR A.3/9 THEN o ELSE n/a - - P-CSCF, AS acting as a SIP proxy, IBCF.
c128:	IF A.3/7 OR A.3/9 THEN o ELSE n/a - - AS, IBCF.
c129:	IF A.3/2 OR A.3/4 THEN o ELSE n/a - - P-CSCF, S-CSCF.
c130:	IF A.3/4 OR A.3/7C OR A.3/9 THEN o ELSE n/a - - S-CSCF, AS acting as a SIP proxy, IBCF.
o.1:	It is mandatory to support at least one of these items.
o.2:	It is mandatory to support at least one of these items.
o.3:	It is mandatory to support at least one of these items.
o.4	At least one of these capabilities is supported.
o.5:	It is mandatory to support exactly one of these items.
o.6:	It is mandatory to support exactly one of these items.
o.7:	It is mandatory to support at least one of these items.
o.8	It is mandatory to support at least one of these items.
o.9:	At least one of these capabilities is supported.
NOTE 1:	An AS acting as a proxy may be outside the trust domain, and therefore not able to support the capability for that reason; in this case it is perfectly reasonable for the header to be passed on transparently, as specified in the PDU parts of the profile.
NOTE 2:	Not applicable over Gm reference point (UE – P-CSCF).
NOTE 3:	AS performing a service number translation (e.g. Freephone)

A.2.2.3 PDUs

Table A.163: Supported methods

Item	PDU	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	ACK request	[26] 13	m	m	[26] 13	m	m
2	BYE request	[26] 16	m	m	[26] 16	m	m
3	BYE response	[26] 16	m	m	[26] 16	m	m
4	CANCEL request	[26] 16.10	m	m	[26] 16.10	m	m
5	CANCEL response	[26] 16.10	m	m	[26] 16.10	m	m
6	INFO request	[25] 4.2	c2	c2	[25] 4.2	c2	c2
7	INFO response	[25] 4.2	c2	c2	[25] 4.2	c2	c2
8	INVITE request	[26] 16	m	m	[26] 16	m	m
9	INVITE response	[26] 16	m	m	[26] 16	m	m
9A	MESSAGE request	[50] 4	c5	c5	[50] 7	c5	c5
9B	MESSAGE response	[50] 4	c5	c5	[50] 7	c5	c5
10	NOTIFY request	[28] 8.1.2	c3	c3	[28] 8.1.2	c3	c3
11	NOTIFY response	[28] 8.1.2	c3	c3	[28] 8.1.2	c3	c3
12	OPTIONS request	[26] 16	m	m	[26] 16	m	m
13	OPTIONS response	[26] 16	m	m	[26] 16	m	m
14	PRACK request	[27] 6	c6	c6	[27] 6	c6	c6
15	PRACK response	[27] 6	c6	c6	[27] 6	c6	c6
15A	PUBLISH request	[70] 11.1.1	c20	c20	[70] 11.1.1	c20	c20
15B	PUBLISH response	[70] 11.1.1	c20	c20	[70] 11.1.1	c20	c20
16	REFER request	[36] 3	c1	c1	[36] 3	c1	c1
17	REFER response	[36] 3	c1	c1	[36] 3	c1	c1
18	REGISTER request	[26] 16	m	m	[26] 16	m	m
19	REGISTER response	[26] 16	m	m	[26] 16	m	m
20	SUBSCRIBE request	[28] 8.1.1	c3	c3	[28] 8.1.1	c3	c3
21	SUBSCRIBE response	[28] 8.1.1	c3	c3	[28] 8.1.1	c3	c3
22	UPDATE request	[29] 7	c4	c4	[29] 7	c4	c4
23	UPDATE response	[29] 7	c4	c4	[29] 7	c4	c4

c1: IF A.162/22 THEN m ELSE n/a - - the REFER method.
c2: IF A.162/20 OR A.162/20A THEN m ELSE n/a - - SIP INFO method and package framework, legacy INFO usage.
c3: IF A.162/27 THEN m ELSE n/a - - SIP specific event notification.
c4: IF A.162/24 THEN m ELSE n/a - - the SIP UPDATE method.
c5: IF A.162/33 THEN m ELSE n/a - - the SIP MESSAGE method.
c6: IF A.162/21 THEN m ELSE n/a - - reliability of provisional responses.
c20: IF A.4/51 THEN m ELSE n/a

A.2.2.4 PDU parameters

A.2.2.4.1 Status-codes

Table A.164: Supported-status codes

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	100 (Trying)	[26] 21.1.1	c1	c1	[26] 21.1.1	c2	c2
101	1xx response	[26] 21.1	p21	p21	[26] 21.1	p21	p21
101A	18x response	[26] 21.1	p21	p21	[26] 21.1	p21	p21
2	180 (Ringing)	[26] 21.1.2	c3	c3	[26] 21.1.2	c3	c3
3	181 (Call Is Being Forwarded)	[26] 21.1.3	c3	c3	[26] 21.1.3	c3	c3
4	182 (Queued)	[26] 21.1.4	c3	c3	[26] 21.1.4	c3	c3
5	183 (Session Progress)	[26] 21.1.5	c3	c3	[26] 21.1.5	c3	c3
5A	199 (Early Dialog Terminated)	[142] 11.1	c32	c32	[142] 11.1	c32	c32
102	2xx response	[26] 21.2	p22	p22	[26] 21.1	p22	p22
6	200 (OK)	[26] 21.2.1	m	m	[26] 21.2.1	i	m
7	202 (Accepted)	[28] 8.3.1	c34	c34	[28] 8.3.1	c34	c34
103	3xx response	[26] 21.3	p23	p23	[26] 21.1	p23	p23
8	300 (Multiple Choices)	[26] 21.3.1	m	m	[26] 21.3.1	i	i
9	301 (Moved Permanently)	[26] 21.3.2	m	m	[26] 21.3.2	i	i
10	302 (Moved Temporarily)	[26] 21.3.3	m	m	[26] 21.3.3	i	i
11	305 (Use Proxy)	[26] 21.3.4	m	m	[26] 21.3.4	i	i
12	380 (Alternative Service)	[26] 21.3.5	m	m	[26] 21.3.5	i	i
104	4xx response	[26] 21.4	p24	p24	[26] 21.4	p24	p24
13	400 (Bad Request)	[26] 21.4.1	m	m	[26] 21.4.1	i	i
14	401 (Unauthorized)	[26] 21.4.2	m	m	[26] 21.4.2	i	c10
15	402 (Payment Required)	[26] 21.4.3	n/a	n/a	[26] 21.4.3	n/a	n/a
16	403 (Forbidden)	[26] 21.4.4	m	m	[26] 21.4.4	i	i
17	404 (Not Found)	[26] 21.4.5	m	m	[26] 21.4.5	i	i
18	405 (Method Not Allowed)	[26] 21.4.6	m	m	[26] 21.4.6	i	i
19	406 (Not Acceptable)	[26] 21.4.7	m	m	[26] 21.4.7	i	i
20	407 (Proxy Authentication Required)	[26] 21.4.8	m	m	[26] 21.4.8	i	i
21	408 (Request Timeout)	[26] 21.4.9	c3	c3	[26] 21.4.9	i	i
22	410 (Gone)	[26] 21.4.10	m	m	[26] 21.4.10	i	i
22A	412 (Conditional Request Failed)	[70] 11.2.1	c20	c20	[70] 11.2.1	c19	c19
23	413 (Request Entity Too Large)	[26] 21.4.11	m	m	[26] 21.4.11	i	i
24	414 (Request-URI Too Large)	[26] 21.4.12	m	m	[26] 21.4.12	i	i
25	415 (Unsupported Media Type)	[26] 21.4.13	m	m	[26] 21.4.13	i	i
26	416 (Unsupported URI Scheme)	[26] 21.4.14	m	m	[26] 21.4.14	i	i
26A	417 (Unknown Resource Priority)	[116] 4.6.2	c25	c25	[116] 4.6.2	c25	c25
27	420 (Bad Extension)	[26] 21.4.15	m	m	[26] 21.4.15	i	i

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
28	421 (Extension Required)	[26] 21.4.16	m	m	[26] 21.4.16	i	i
28A	422 (Session Interval Too Small)	[58] 6	c8	c8	[58] 6	c8	c8
29	423 (Interval Too Brief)	[26] 21.4.17	c5	c5	[26] 21.4.17	c6	c6
29A	424 (Bad Location Information)	[89] 4.2	c23	c23	[89] 4.2	c24	c24
29AA	428 Use Identity Header	[252] 6.2.2	m	m	[252] 6.2.2	c35	c35
29B	429 (Provide Referrer Identity)	[59] 5	c9	c9	[59] 5	c9	c9
29C	430 (Flow Failed)	[92] 11	o	c21	[92] 11	m	c22
29D	433 (Anonymity Disallowed)	[67] 4	c14	c14	[67] 4	c14	c14
29DA	436 Bad Identity Info	[252] 6.2.2	m	m	[252] 6.2.2	c35	c35
29DB	437 Unsupported Credential	[252] 6.2.2	m	m	[252] 6.2.2	c35	c35
29DC	438 Invalid Identity Header	[252] 6.2.2	m	m	[252] 6.2.2	c35	c35
29E	439 (First Hop Lacks Outbound Support)	[92] 11	c28	c28	[92] 11	c29	c29
29F	440 (Max Breadth Exceeded)	[117] 5	c30	c30	[117] 5	c31	c31
29G	469 (Bad INFO Package)	[25] 4.2	c33	c33	[25] 4.2	c33	c33
29H	470 (Consent Needed)	[125] 5.9.2	c26	c26	[125] 5.9.2	c27	c27
30	480 (Temporarily not available)	[26] 21.4.18	m	m	[26] 21.4.18	i	i
31	481 (Call /Transaction Does Not Exist)	[26] 21.4.19	m	m	[26] 21.4.19	i	i
32	482 (Loop Detected)	[26] 21.4.20	m	m	[26] 21.4.20	i	i
33	483 (Too Many Hops)	[26] 21.4.21	m	m	[26] 21.4.21	i	i
34	484 (Address Incomplete)	[26] 21.4.22	m	m	[26] 21.4.22	i	i
35	485 (Ambiguous)	[26] 21.4.23	m	m	[26] 21.4.23	i	i
36	486 (Busy Here)	[26] 21.4.24	m	m	[26] 21.4.24	i	i
37	487 (Request Terminated)	[26] 21.4.25	m	m	[26] 21.4.25	i	i
38	488 (Not Acceptable Here)	[26] 21.4.26	m	m	[26] 21.4.26	i	i
39	489 (Bad Event)	[28] 8.3.2	c4	c4	[28] 8.3.2	c4	c4
40	491 (Request Pending)	[26] 21.4.27	m	m	[26] 21.4.27	i	i
41	493 (Undecipherable)	[26] 21.4.28	m	m	[26] 21.4.28	i	i
41A	494 (Security Agreement Required)	[48] 2	c7	c7	[48] 2	n/a	n/a
105	5xx response	[26] 21.5	p25	p25	[26] 21.5	p25	p25
42	500 (Internal Server Error)	[26] 21.5.1	m	m	[26] 21.5.1	i	i
43	501 (Not Implemented)	[26] 21.5.2	m	m	[26] 21.5.2	i	i
44	502 (Bad Gateway)	[26] 21.5.3	m	m	[26] 21.5.3	i	i
45	503 (Service Unavailable)	[26] 21.5.4	m	m	[26] 21.5.4	i	i
46	504 (Server Time-out)	[26] 21.5.5	m	m	[26] 21.5.5	i	i
47	505 (Version not supported)	[26] 21.5.6	m	m	[26] 21.5.6	i	i

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
48	513 (Message Too Large)	[26] 21.5.7	m	m	[26] 21.5.7	i	i
49	580 (Precondition Failure)	[30] 8	m	m	[30] 8	i	i
106	6xx response	[26] 21.6	p26	p26	[26] 21.6	p26	p26
50	600 (Busy Everywhere)	[26] 21.6.1	m	m	[26] 21.6.1	i	i
51	603 (Decline)	[26] 21.6.2	m	m	[26] 21.6.2	i	i
52	604 (Does Not Exist Anywhere)	[26] 21.6.3	m	m	[26] 21.6.3	i	i
53	606 (Not Acceptable)	[26] 21.6.4	m	m	[26] 21.6.4	i	i
54	607 (Unwanted)	[254]	m	m	[254]	i	i

c1: IF A.163/3 OR A.163/9 OR A.163/9B OR A.163/11 OR A.163/13 OR A.163/15 OR A.163/15B OR A.163/17 OR A.163/19 OR A.163/21 OR A.163/23 AND A.162/5 THEN m ELSE n/a - - BYE response or INVITE response or MESSAGE response or NOTIFY response or OPTIONS response or PRACK response or PUBLISH response or REFER response or REGISTER response or SUBSCRIBE response or UPDATE response, stateful proxy.

c2: IF A.163/3 OR A.163/9 OR A.163/9B OR A.163/11 OR A.163/13 OR A.163/15 OR A.163/15B OR A.163/17 OR A.163/19 OR A.163/21 OR A.163/23 THEN (IF A.162/5 THEN m ELSE i) ELSE n/a - - BYE response or INVITE response or MESSAGE response or NOTIFY response or OPTIONS response or PRACK response or PUBLISH response or REFER response or REGISTER response or SUBSCRIBE response or UPDATE response, stateful proxy.

c3: IF A.163/9 THEN m ELSE n/a - - INVITE response.

c4: IF A.162/27 THEN m ELSE n/a - - SIP specific event notification.

c5: IF A.163/19 OR A.163/21 THEN m ELSE n/a - - REGISTER response or SUBSCRIBE response.

c6: IF A.163/19 OR A.163/21 THEN i ELSE n/a - - REGISTER response or SUBSCRIBE response.

c7: IF A.162/47 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.

c8: IF A.162/52 THEN m ELSE n/a - - the SIP session timer.

c9: IF A.162/53 AND A.163/17 THEN m ELSE n/a - - the SIP Referred-By mechanism and REFER response.

c10: IF A.3/2 THEN m ELSE i - - P-CSCF.

c14: IF A.162/58 THEN m ELSE n/a - - rejecting anonymous requests in the session initiation protocol.

c19: IF A.162/51 THEN i ELSE n/a - - an event state publication extension to the session initiation protocol.

c20: IF A.162/51 THEN m ELSE n/a - - an event state publication extension to the session initiation protocol.

c21: IF A.4/57 AND A.3/2 THEN o ELSE n/a - - managing client initiated connections in SIP, P-CSCF.

c22: IF A.4/57 AND A.3/4 THEN m ELSE i - - managing client initiated connections in SIP, S-CSCF.

c23: IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.

c24: IF A.162/70 THEN i ELSE n/a - - SIP location conveyance.

c25: IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.

c26: IF A.162/85 THEN m ELSE n/a - - a framework for consent-based communications in SIP.

c27: IF A.162/85 THEN i ELSE n/a - - a framework for consent-based communications in SIP.

c28: IF A.162/57 AND THEN m ELSE n/a - - managing client initiated connections in SIP.

c29: IF A.162/57 AND THEN i ELSE n/a - - managing client initiated connections in SIP.

c30: IF A.162/81 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.

c31: IF A.162/81 THEN i ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.

c32: IF A.162/91 AND A.163/9 THEN m ELSE n/a - - INVITE response and 199 (Early Dialog Terminated) response.

c33 IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.

c34: IF A.162/27 OR A.163/9B OR A.163/17 THEN m ELSE n/a - - SIP specific event notification or MESSAGE response or REFER response.

c35 IF A.162/126 THEN m ELSE i - authenticated identity management in the Session Initiation Protocol

p21: A.164/2 OR A.164/3 OR A.164/4 OR A.164/5 OR A.164/5A - - 1xx response

p22: A.164/6 OR A.164/7 - - 2xx response

p23: A.164/8 OR A.164/9 OR A.164/10 OR A.164/11 OR A.164/12 - - 3xx response

p24: A.164/13 OR A.164/14 OR A.164/15 OR A.164/16 OR A.164/17 OR A.164/18 OR A.164/19 OR A.164/20 OR A.164/21 OR A.164/22 OR A.164/22A OR A.164/23 OR A.164/24 OR A.164/25 OR A.164/26 OR A.164/26A OR A.164/27 OR A.164/28 OR A.164/28A OR A.164/29 OR A.164/29A OR A.164/29B OR A.164/29C OR A.164/29D OR A.164/29E OR A.164/29F OR A.164/29G OR A.164/29H OR A.164/30 OR A.164/31 OR A.164/32 OR A.164/33 OR A.164/34 OR A.164/35 OR A.164/36 OR A.164/436 OR A.164/38 OR A.164/39 OR A.164/40 OR A.164/41 OR A.164/41A. - - 4xx response

p25: A.164/42 OR A.164/43 OR A.164/44 OR A.164/45 OR A.164/46 OR A.164/47 OR A.164/48 OR A.164/49 - - 5xx response

p26: A.164/50 OR A.164/51 OR A.164/52 OR A.164/53 - - 6xx response

A.2.2.4.2 ACK method

Prerequisite A.163/1 - - ACK request

Table A.165: Supported header fields within the ACK request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Contact	[56B] 9.2	c10	c10	[56B] 9.2	c11	c11
2	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
3	Authorization	[26] 20.7	m	m	[26] 20.7	i	i
4	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
5	Cellular-Network-Info	7.2.15	n/a	c27	7.2.15	n/a	c28
6	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
7	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
8	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
9	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
10	Content-Type	[26] 20.15	m	m	[26] 20.15	i	c3
11	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
12	Date	[26] 20.17	m	m	[26] 20.17	c2	c2
13	From	[26] 20.20	m	m	[26] 20.20	m	m
13A	Max-Breadth	[117] 5.8	c15	c15	[117] 5.8	c16	c16
14	Max-Forwards	[26] 20.22	m	m	[26] 20.22	m	m
15	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	c3
15A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c20	c20	[52] 4.4, [52A] 4, [234] 2	c21	c21
15BA	Priority-Share	Subclause 7.2.16	n/a	c29	Subclause 7.2.16	n/a	c29
15C	Privacy	[33] 4.2	c6	c6	[33] 4.2	c7	c7
15D	P-Charging-Vector	[52] 4.6, [52A] 4	c22	c22	[52] 4.6, [52A] 4	c23	c23
16	Proxy-Authorization	[26] 20.28	m	m	[26] 20.28	c4	c4
17	Proxy-Require	[26] 20.29	m	m	[26] 20.29	m	m
17A	Reason	[34A] 2	c8	c8	[34A] 2	c9	c9
17B	Record-Route	[26] 20.30	m	m	[26] 20.30	c26	c26
17C	Recv-Info	[25] 5.2.3	c17	c17	[25] 5.2.3	c18	c18
17D	Reject-Contact	[56B] 9.2	c10	c10	[56B] 9.2	c11	c11
17E	Relayed-Charge	7.2.12	n/a	c24	7.2.12	n/a	c24
17F	Request-Disposition	[56B] 9.1	c10	c10	[56B] 9.1	c11	c11
18	Require	[26] 20.32	m	m	[26] 20.32	c5	c5
18A	Resource-Priority	[116] 3.1	c12	c12	[116] 3.1	c12	c47
18B	Resource-Share	Subclause 4.15	n/a	c25	Subclause 4.15	n/a	c25
19	Route	[26] 20.34	m	m	[26] 20.34	m	m
19A	Session-ID	[162]	c19	c19	[162]	c19	c19
20	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
21	To	[26] 20.39	m	m	[26] 20.39	m	m
22	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
23	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.
c2:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c3:	IF A.3/2 OR A.3/4 THEN m ELSE i - - P-CSCF or S-CSCF.
c4:	IF A.162/8A THEN m ELSE i - - authentication between UA and proxy.
c5:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c6:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c7:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c8:	IF A.162/48 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.
c9:	IF A.162/48 THEN i ELSE n/a - - the Reason header field for the session initiation protocol.
c10:	IF A.162/50 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c11:	IF A.162/50 THEN i ELSE n/a - - caller preferences for the session initiation protocol.
c12:	IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.
c15:	IF A.162/81 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c16:	IF A.162/81 AND A.162/6 THEN m ELSE IF A.162/81 AND NOT A.162/6 THEN i ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, forking of initial requests.
c17:	IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.
c18:	IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.
c19:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c20:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c21:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c22:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c23:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c24:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c25:	IF A.162/122 THEN o ELSE n/a - - resource sharing.
c26:	IF A.162/14 THEN o ELSE i - - the requirement to be able to insert itself in the subsequent transactions in a dialog.
c27:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c28:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c29:	IF A.162/124 THEN o ELSE n/a - - priority sharing.
NOTE:	c1 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.

Prerequisite A.163/1 - - ACK request

Table A.166: Supported message bodies within the ACK request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.2.4.3 BYE method

Prerequisite A.163/2 - - BYE request

Table A.167: Supported header fields within the BYE request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
1A	Accept-Contact	[56B] 9.2	c22	c22	[56B] 9.2	c23	c23
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i
3A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
4	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
5	Authorization	[26] 20.7	m	m	[26] 20.7	i	i
6	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
6A	Cellular-Network-Info	7.2.15	n/a	c37	7.2.15	n/a	c38
7	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	c3
8	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	c3
9	Content-Language	[26] 20.13	m	m	[26] 20.13	i	c3
10	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
11	Content-Type	[26] 20.15	m	m	[26] 20.15	i	c3
12	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
13	Date	[26] 20.17	m	m	[26] 20.17	c2	c2
14	From	[26] 20.20	m	m	[26] 20.20	m	m
14A	Geolocation	[89] 4.1	c26	c26	[89] 4.1	c27	c27
14B	Geolocation-Routing	[89] 4.1	c26	c26	[89] 4.1	c27	c27
14C	Max-Breadth	[117] 5.8	c33	c33	[117] 5.8	c34	c34
15	Max-Forwards	[26] 20.22	m	m	[26] 20.22	m	m
16	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	c3
16A	P-Access-Network-Info	[52] 4.4, [234] 2	c13	c13	[52] 4.4, [234] 2	c14	c14
16B	P-Asserted-Identity	[34] 9.1	c9	c9	[34] 9.1	c10	c10
16C	P-Charging-Function-Addresses	[52] 4.5	c17	c17	[52] 4.5	c18	c18
16D	P-Charging-Vector	[52] 4.6	c15	c15	[52] 4.6	c16	c16
16F	P-Preferred-Identity	[34] 9.2	x	x	[34] 9.2	c8	n/a
16G	Privacy	[33] 4.2	c11	c11	[33] 4.2	c12	c12
17	Proxy-Authorization	[26] 20.28	m	m	[26] 20.28	c4	c4
18	Proxy-Require	[26] 20.29	m	m	[26] 20.29	m	m
18A	Reason	[34A] 2	c20	c20	[34A] 2	c21	c21
19	Record-Route	[26] 20.30	m	m	[26] 20.30	c7	c7
19A	Referred-By	[59] 3	c24	c24	[59] 3	c25	c25
19B	Reject-Contact	[56B] 9.2	c22	c22	[56B] 9.2	c23	c23
19C	Relayed-Charge	7.2.12	n/a	c36	7.2.12	n/a	c36
19D	Request-Disposition	[56B] 9.1	c22	c22	[56B] 9.1	c23	c23
20	Require	[26] 20.32	m	m	[26] 20.32	c5	c5
20A	Resource-Priority	[116] 3.1	c28	c28	[116] 3.1	c28	c28
21	Route	[26] 20.34	m	m	[26] 20.34	m	m
21A	Security-Client	[48] 2.3.1	x	x	[48] 2.3.1	c19	c19
21B	Security-Verify	[48] 2.3.1	x	x	[48] 2.3.1	c19	c19
21C	Session-ID	[162]	c35	c35	[162]	c35	c35
22	Supported	[26] 20.37	m	m	[26] 20.37	c6	c6
23	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
24	To	[26] 20.39	m	m	[26] 20.39	m	m
25	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
25A	User-to-User	[126] 7	c29	c29	[126] 7	c30	c30
26	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.
c2:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c3:	IF A.3/2 OR A.3/4 THEN m ELSE i - - P-CSCF or S-CSCF.
c4:	IF A.162/8A THEN m ELSE i - - authentication between UA and proxy.
c5:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c6:	IF A.162/16 THEN m ELSE i - - reading the contents of the Supported header before proxying the response.
c7:	IF A.162/14 THEN o ELSE i - - the requirement to be able to insert itself in the subsequent transactions in a dialog.
c8:	IF A.162/30A THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity.
c9:	IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c10:	IF A.162/30A OR A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.
c11:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c12:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c13:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c14:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c15:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c16:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c17:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c18:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c19:	IF A.162/47 OR A.162/47A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c20:	IF A.162/48 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.
c21:	IF A.162/48 THEN i ELSE n/a - - the Reason header field for the session initiation protocol.
c22:	IF A.162/50 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c23:	IF A.162/50 THEN i ELSE n/a - - caller preferences for the session initiation protocol.
c24:	IF A.162/53 THEN i ELSE n/a - - the SIP Referred-By mechanism.
c25:	IF A.162/53 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c26:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c27:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c28:	IF A.162/80B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.
c29:	IF A.162/86 THEN m - - transporting user to user information for call centers using SIP.
c30:	IF A.162/86 THEN i - - transporting user to user information for call centers using SIP.
c33:	IF A.162/81 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c34:	IF A.162/81 AND A.162/6 THEN m ELSE IF A.162/81 AND NOT A.162/6 THEN i ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, forking of initial requests.
c35:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c36:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c37:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c38:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
NOTE:	c1 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.

Prerequisite A.163/2 - - BYE request

Table A.168: Supported message bodies within the BYE request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	XML Schema for PSTN	[11B]	n/a	c1	[11B]	n/a	i
2	VoiceXML expr / namelist data	[145] 4.2	m	c2	[145] 4.2	m	c2
3	application/vnd.3gpp.ussd	[8W]	n/a	m	[8W]	n/a	i
c1:	A.3/3 OR A.3/4 OR A.3/5 OR A.3/7C OR A.3/9A OR A.3/10 OR A.3/11 OR A.3/13A THEN m ELSE n/a - - I-CSCF, S-CSCF, BGCF, AS acting as proxy, IBCF (THIG), additional routeing functionality, E-CSCF, ISC gateway function (THIG).						
c2:	IF A.162/94 THEN m ELSE n/a - - SIP Interface to VoiceXML Media Services.						

Table A.169: Void

Prerequisite A.163/3 - - BYE response

Prerequisite: A.164/1 - - Additional for 100 (Trying) response

Table A.169A: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	c2	c2
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF (A.162/9 AND A.162/5) OR A.162/4 THEN m ELSE n/a - - stateful proxy behaviour that inserts date, or stateless proxies.						
c2:	IF A.162/4 THEN i ELSE m - - Stateless proxy passes on.						

Prerequisite A.163/3 - - BYE response

Table A.170: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Cellular-Network-Info	7.2.15	n/a	c23	7.2.15	n/a	c24
2	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	c2
3	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	c2
4	Content-Language	[26] 20.13	m	m	[26] 20.13	i	c2
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	i	c2
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	m	m	[26] 20.17	c1	c1
9	From	[26] 20.20	m	m	[26] 20.20	m	m
9A	Geolocation-Error	[89] 4.3	c15	c15	[89] 4.3	c16	c16
10	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	c2
10A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c12	c12	[52] 4.4, [52A] 4, [234] 2	c13	c13
10B	P-Asserted-Identity	[34] 9.1	c4	c4	[34] 9.1	c5	c5
10C	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c10	c10	[52] 4.5, [52A] 4	c11	c11
10D	P-Charging-Vector	[52] 4.6, [52A] 4	c8	c8	[52] 4.6, [52A] 4	c9	c9
10F	P-Preferred-Identity	[34] 9.2	x	x	[34] 9.2	c3	n/a
10G	Privacy	[33] 4.2	c6	c6	[33] 4.2	c7	c7
10H	Relayed-Charge	7.2.12	n/a	c22	7.2.12	n/a	c22
10I	Require	[26] 20.32	m	m	[26] 20.32	c14	c14
10J	Server	[26] 20.35	m	m	[26] 20.35	i	i
10K	Session-ID	[162]	c21	c21	[162]	c21	c21
11	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
12	To	[26] 20.39	m	m	[26] 20.39	m	m
12A	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
12B	User-to-User	[126] 7	c17	c17	[126] 7	c18	c18
13	Via	[26] 20.42	m	m	[26] 20.42	m	m
14	Warning	[26] 20.43	m	m	[26] 20.43	i	i

c1:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c2:	IF A.3/2 OR A.3/4 THEN m ELSE i - - P-CSCF or S-CSCF.
c3:	IF A.162/30A THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity.
c4:	IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c5:	IF A.162/30A or A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.
c6:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c7:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c8:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c9:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c10:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c11:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c12:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c13:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c14:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c15:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c16:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c17:	IF A.162/86 THEN m - - transporting user to user information for call centers using SIP.
c18:	IF A.162/86 THEN i - - transporting user to user information for call centers using SIP.
c21:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c22:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c23:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c24:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.

Prerequisite A.163/3 - - BYE response

Prerequisite: A.164/102 - - Additional for 2xx response

Table A.171: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Accept-Resource-Priority	[116] 3.2	c4	c4	[116] 3.2	c4	c4
0B	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	i	c1
1	Authentication-Info	[26] 20.6	m	m	[26] 20.6	i	i
2	Record-Route	[26] 20.30	m	m	[26] 20.30	c3	c3
5	Supported	[26] 20.37	m	m	[26] 20.37	i	i
c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.						
c3:	IF A.162/15 THEN o ELSE i - - the requirement to be able to use separate URIs in the upstream direction and downstream direction when record routeing.						
c4:	IF A.162/80B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.						

Prerequisite A.163/3 - - BYE response

Prerequisite: A.164/103 OR A.164/104 OR A.164/105 OR A.164/106 - - Additional for 3xx – 6xx response

Table A.171A: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	m	m	[26] 20.18	i	i
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.162/125 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.163/3 - BYE response

Prerequisite: A.164/103 OR A.164/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.172: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Contact	[26] 20.10	m	m	[26] 20.10	c1	c1
c1:	IF A.162/19E THEN m ELSE i - - deleting Contact headers.						

Prerequisite A.163/3 - - BYE response

Prerequisite: A.164/14 - - Additional for 401 (Unauthorized) response

Table A.173: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
8	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/3 - - BYE response

Prerequisite: A.164/17 OR A.164/23 OR A.164/30 OR A.164/36 OR A.164/42 OR A.164/45 OR A.164/50 OR A.164/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.174: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Retry-After	[26] 20.33	m	m	[26] 20.33	i	i

Table A.175: Void

Prerequisite A.163/3 - - BYE response

Prerequisite: A.164/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.176: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
6	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/3 - - BYE response

Prerequisite: A.164/25 - - Additional for 415 (Unsupported Media Type) response

Table A.177: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i

Prerequisite A.163/3 - - BYE response

Prerequisite: A.164/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.177A: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.162/80B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.						

Prerequisite A.163/3 - - BYE response

Prerequisite: A.164/27 - - Additional for 420 (Bad Extension) response

Table A.178: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Unsupported	[26] 20.40	m	m	[26] 20.40	c3	c3
c3:	IF A.162/18 THEN m ELSE i - - reading the contents of the Unsupported header before proxying the 420 response to a method other than REGISTER.						

Prerequisite A.163/3 - - BYE response

Prerequisite: A.164/28 OR A.164/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.178A: Supported header fields within the BYE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	c1	c1	[48] 2	n/a	n/a
c1:	IF A.162/47 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Table A.179: Void

Prerequisite A163/3 - - BYE response

Prerequisite: A.164/6 - - Additional for 200 (OK) response

Table A.180: Supported message bodies within the BYE response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	VoiceXML expr / namelist data	[145] 4.2	o	c1	[145] 4.2	o	c1
c1:	IF A.162/94 THEN o ELSE n/a - - SIP Interface to VoiceXML Media Services.						

A.2.2.4.4 CANCEL method

Prerequisite A.163/4 - - CANCEL request

Table A.181: Supported header fields within the CANCEL request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Contact	[56B] 9.2	c10	c10	[56B] 9.2	c11	c11
5	Authorization	[26] 20.7	m	m	[26] 20.7	i	i
6	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
8	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
8A	Content-Type	[26] 20.15	c21	c21	[26] 20.15	o	o
9	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
10	Date	[26] 20.17	m	m	[26] 20.17	c2	c2
11	From	[26] 20.20	m	m	[26] 20.20	m	m
11A	Max-Breadth	[117] 5.8	c15	c15	[117] 5.8	c16	c16
12	Max-Forwards	[26] 20.22	m	m	[26] 20.22	m	m
14	Privacy	[33] 4.2	c3	c3	[33] 4.2	c4	c4
15	Reason	[34A] 2	c8	c8	[34A] 2	c9	c9
16	Record-Route	[26] 20.30	m	m	[26] 20.30	c7	c7
17	Reject-Contact	[56B] 9.2	c10	c10	[56B] 9.2	c11	c11
17A	Relayed-Charge	7.2.12	n/a	c20	7.2.12	n/a	c20
17B	Request-Disposition	[56B] 9.1	c10	c10	[56B] 9.1	c11	c11
17C	Resource-Priority	[116] 3.1	c12	c12	[116] 3.1	c12	c12
18	Route	[26] 20.34	m	m	[26] 20.34	m	m
18A	Session-ID	[162]	c17	c17	[162]	c17	c17
19	Supported	[26] 20.37	m	m	[26] 20.37	c6	c6
20	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
21	To	[26] 20.39	m	m	[26] 20.39	m	m
22	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
23	Via	[26] 20.42	m	m	[26] 20.42	m	m

c2: IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c3: IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c4: IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c6: IF A.162/16 THEN m ELSE i - - reading the contents of the Supported header before proxying the response.
c7: IF A.162/14 THEN o ELSE i - - the requirement to be able to insert itself in the subsequent transactions in a dialog.
c8: IF A.162/48 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.
c9: IF A.162/48 THEN i ELSE n/a - - the Reason header field for the session initiation protocol.
c10: IF A.162/50 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c11: IF A.162/50 THEN i ELSE n/a - - caller preferences for the session initiation protocol.
c12: IF A.162/80B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.
c15: IF A.162/81 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c16: IF A.162/81 AND A.162/6 THEN m ELSE IF A.162/81 AND NOT A.162/6 THEN i ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, forking of initial requests.
c17: IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c20: IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c21: IF A.162/23 OR A.182/1 THEN m ELSE o - - integration of resource management and SIP or XML schema for PSTN.

NOTE: c1 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.

Prerequisite A.163/4 - - CANCEL request

Table A.182: Supported message bodies within the CANCEL request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	XML Schema for PSTN	[11B]		c1	[11B]		i
c1:	A.3/3 OR A.3/4 OR A.3/5 OR A.3/7C OR A.3/9A OR A.3/10 OR A.3/11 OR A.3/13A THEN m ELSE n/a - - I-CSCF, S-CSCF, BGCF, AS acting as proxy, IBCF (THIG), additional routeing functionality, E-CSCF, ISC gateway function (THIG).						

Prerequisite A.163/5 - - CANCEL response for all status-codes

Table A.183: Supported header fields within the CANCEL response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	m	m	[26] 20.17	c1	c1
5	From	[26] 20.20	m	m	[26] 20.20	m	m
5C	Privacy	[33] 4.2	c2	c2	[33] 4.2	c3	c3
5D	Relayed-Charge	7.2.12	n/a	c9	7.2.12	n/a	c9
5C	Session-ID	[162]	c6	c6	[162]	c6	c6
6	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
7	To	[26] 20.39	m	m	[26] 20.39	m	m
7A	User-Agent	[26] 20.41	o		[26] 20.41	o	
8	Via	[26] 20.42	m	m	[26] 20.42	m	m
9	Warning	[26] 20.43	m	m	[26] 20.43	i	i
c1:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.						
c2:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).						
c3:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.						
c6:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.						
c9:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.						

Prerequisite A.163/5 - - CANCEL response

Prerequisite: A.164/102 - - Additional for 2xx response

Table A.184: Supported header fields within the CANCEL response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c4	c4	[116] 3.2	c4	c4
2	Record-Route	[26] 20.30	m	m	[26] 20.30	c3	c3
4	Supported	[26] 20.37	m	m	[26] 20.37	i	i
c3:	IF A.162/15 THEN o ELSE i - - the requirement to be able to use separate URIs in the upstream direction and downstream direction when record routeing.						
c4:	IF A.162/80B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.						

Prerequisite A.163/5 - - CANCEL response

Prerequisite: A.164/103 OR A.164/104 OR A.164/105 OR A.164/106 - - Additional for 3xx – 6xx response

Table A.184A: Supported header fields within the CANCEL response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	m	m	[26] 20.18	i	i
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.162/125 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Table A.185: Void

Prerequisite A.163/5 - - CANCEL response

Prerequisite: A.164/17 OR A.164/23 OR A.164/30 OR A.164/36 OR A.164/42 OR A.164/45 OR A.164/50 OR A.164/51 - - Additional 404 (Not Found), 413 (Request for Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.186: Supported header fields within the CANCEL response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Retry-After	[26] 20.33	m	m	[26] 20.33	i	i

Prerequisite A.163/5 - - CANCEL response

Prerequisite: A.164/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.186A: Supported header fields within the CANCEL response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.162/80B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.						

Table A.187: Void

Table A.188: Void

Prerequisite A.163/5 - - CANCEL response

Table A.189: Supported message bodies within the CANCEL response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.2.4.5 Void

A.2.2.4.6 INFO method

Prerequisite A.163/6 - - INFO request

Table A.190: Supported header fields within the INFO request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i
4	Allow	[26] 20.5	m	m	[50] 10	i	i
5	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
6	Authorization	[26] 20.7	m	m	[26] 20.7	i	i
7	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
7A	Call-Info	[26] 20.9	m	m	[26] 20.9	c3	c3
8	Cellular-Network-Info	7.2.15	n/a	c54	7.2.15	n/a	c55
9	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
10	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
11	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
12	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
13	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
14	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
15	Date	[26] 20.17	m	m	[26] 20.17	c2	c2
16	From	[26] 20.20	m	m	[26] 20.20	m	m
17	Geolocation	[89] 4.1	c36	c36	[89] 4.1	c37	c37
17A	Geolocation-Routing	[89] 4.1	c36	c36	[89] 4.1	c37	c37
18	Info-Package	[25] 7.2	c50	c50	[25] 7.2	c51	c51
19	Max-Breadth	[117] 5.8	c48	c48	[117] 5.8	c49	c49
20	Max-Forwards	[26] 20.22	m	m	[26] 20.22	m	m
21	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
22	P-Access-Network-Info	[52] 4.4, [234] 2	c23	c23	[52] 4.4, [234] 2	c24	c24
23	P-Charging-Function-Addresses	[52] 4.5	c21	c21	[52] 4.5	c22	c22
24	P-Charging-Vector	[52] 4.6	c19	c19	[52] 4.6	c20	c20
26	Privacy	[33] 4.2	c12	c12	[33] 4.2	c13	c13
27	Proxy-Authorization	[26] 20.28	m	m	[26] 20.28	c8	c8
28	Proxy-Require	[26] 20.29	m	m	[26] 20.29	m	m
29	Reason	[34A] 2	c26	c26	[34A] 2	c27	c27
30	Record-Route	[26] 20.30	m	m	[26] 20.30	c7	c7
31	Referred-By	[59] 3	c30	c30	[59] 3	c31	c31
32	Relayed-Charge	7.2.12	n/a	c53	7.2.12	n/a	c53
33	Request-Disposition	[56B] 9.1	c28	c28	[56B] 9.1	c28	c28
34	Require	[26] 20.32	m	m	[26] 20.32	c5	c5
35	Resource-Priority	[116] 3.1	c38	c38	[116] 3.1	c38	c38
36	Route	[26] 20.34	m	m	[26] 20.34	m	m
37	Security-Client	[48] 2.3.1	x	x	[48] 2.3.1	c25	c25
38	Security-Verify	[48] 2.3.1	x	x	[48] 2.3.1	c25	c25
38A	Session-ID	[162]	c52	c52	[162]	c52	c52
39	Subject	[26] 20.36	m	m	[26] 20.36	i	i
40	Supported	[26] 20.37	m	m	[26] 20.37	c6	c6
41	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
42	To	[26] 20.39	m	m	[26] 20.39	m	m
43	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
44	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.
c2:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c3:	IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header.
c5:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c6:	IF A.162/16 THEN m ELSE i - - reading the contents of the Supported header before proxying the response.
c7:	IF A.162/14 THEN o ELSE i - - the requirement to be able to insert itself in the subsequent transactions in a dialog.
c8:	IF A.162/8A THEN m ELSE i - - authentication between UA and proxy.
c12:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c13:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c19:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c20:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c21:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c22:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c23:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c24:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c25:	IF A.162/47 OR A.162/47A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c26:	IF A.162/48 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.
c27:	IF A.162/48 THEN i ELSE n/a - - the Reason header field for the session initiation protocol.
c28:	IF A.162/50 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c30:	IF A.162/53 THEN i ELSE n/a - - the SIP Referred-By mechanism.
c31:	IF A.162/53 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c36:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c37:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c38:	IF A.162/80A THEN m ELSE n/a - - inclusion of INFO, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.
c48:	IF A.162/81 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c49:	IF A.162/81 AND A.162/6 THEN m ELSE IF A.162/81 AND NOT A.162/6 THEN i ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, forking of initial requests.
c50:	IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.
c51:	IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.
c52:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c53:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c54:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c55:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
NOTE:	c1 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.

Prerequisite A.163/6 - - INFO request

Table A.191: Supported message bodies within the INFO request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Info-Package	[25]	m	m	[25]	i	i

Prerequisite A.163/7 - - INFO response

Prerequisite: A.164/1 - - Additional for 100 (Trying) response

Table A.192: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	c2	c2
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF (A.162/9 AND A.162/5) OR A.162/4 THEN m ELSE n/a - - stateful proxy behaviour that inserts date, or stateless proxies.						
c2:	IF A.162/4 THEN i ELSE m - - Stateless proxy passes on.						

Prerequisite A.163/7 - - INFO response for all remaining status-codes

Table A.193: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Call-Info	[26] 20.9	m	m	[26] 20.9	c3	c3
2A	Cellular-Network-Info	7.2.15	n/a	c23	7.2.15	n/a	c24
3	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
4	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
5	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
6	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
7	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
8	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
9	Date	[26] 20.17	m	m	[26] 20.17	c1	c1
10	From	[26] 20.20	m	m	[26] 20.20	m	m
11	Geolocation-Error	[89] 4.3	c17	c17	[89] 4.3	c18	c18
12	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
13	Organization	[26] 20.25	m	m	[26] 20.25	c2	c2
14	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c13	c13	[52] 4.4, [52A] 4, [234] 2	c14	c14
15	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c11	c11	[52] 4.5, [52A] 4	c12	c12
16	P-Charging-Vector	[52] 4.6, [52A] 4	c9	c9	[52] 4.6, [52A] 4	c10	c10
18	Privacy	[33] 4.2	c7	c7	[33] 4.2	c8	c8
18A	Relayed-Charge	7.2.12	n/a	c22	7.2.12	n/a	c22
19	Require	[26] 20.32	m	m	[26] 20.32	c15	c15
20	Server	[26] 20.35	m	m	[26] 20.35	i	i
20A	Session-ID	[162]	c21	c21	[162]	c21	c21
21	Timestamp	[26] 20.38	i	i	[26] 20.38	i	i
22	To	[26] 20.39	m	m	[26] 20.39	m	m
23	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
24	Via	[26] 20.42	m	m	[26] 20.42	m	m
25	Warning	[26] 20.43	m	m	[26] 20.43	i	i

c1:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c2:	IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.
c3:	IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header.
c7:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c8:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c9:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c10:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c11:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c12:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c13:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c14:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c15:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c17:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c18:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c21:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c22:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c23:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c24:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.

Prerequisite A.163/7 - - INFO response

Prerequisite: A.164/102 - - Additional for 2xx response

Table A.194: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i
4	Accept-Resource-Priority	[116] 3.2	c4	c4	[116] 3.2	c4	c4
5	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
6	Authentication-Info	[26] 20.6	m	m	[26] 20.6	i	i
8	Record-Route	[26] 20.30	m	m	[26] 20.30	c5	c5
9	Supported	[26] 20.37	m	m	[26] 20.37	i	i
c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.						
c4:	IF A.162/80A THEN m ELSE n/a - - inclusion of INFO, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						
c5:	IF A.162/15 THEN o ELSE i - - the requirement to be able to use separate URIs in the upstream direction and downstream direction when record routeing.						

Prerequisite A.163/7 - - INFO response

Prerequisite: A.164/103 OR A.164/104 OR A.164/105 OR A.164/106 - - Additional for 3xx – 6xx response

Table A.195: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	m	m	[26] 20.18	i	i
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.162/125 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.163/7 - - INFO response

Prerequisite: A.164/103 OR A.164/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.195A: Void

Prerequisite A.163/7 - - INFO response

Prerequisite: A.164/14 - - Additional for 401 (Unauthorized) response

Table A.196: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
6	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/7 - - INFO response

Prerequisite: A.164/17 OR A.164/23 OR A.164/30 OR A.164/36 OR A.164/42 OR A.164/45 OR A.164/50 OR A.164/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.197: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Retry-After	[26] 20.33	m	m	[26] 20.33	i	i

Table A.198: Void

Prerequisite A.163/7 - - INFO response

Prerequisite: A.164/25 - - Additional for 415 (Unsupported Media Type)

Table A.199: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i

Prerequisite A.163/7 - - INFO response

Prerequisite: A.164/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.199A: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.162/80A THEN m ELSE n/a - - inclusion of INFO, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						

Prerequisite A.163/7 - - INFO response

Prerequisite: A.164/27 - - Additional for 420 (Bad Extension) response

Table A.200: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Unsupported	[26] 20.40	m	m	[26] 20.40	c3	c3
c3:	IF A.162/18 THEN m ELSE i - - reading the contents of the Unsupported header before proxying the 420 response to a method other than REGISTER.						

Prerequisite A.163/7 - - INFO response

Prerequisite: A.164/28 OR A.164/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.200A: Supported header fields within the INFO response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	c1	c1	[48] 2	n/a	n/a
c1:	IF A.162/47 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Table A.201: Void

Table A.202: Void

Prerequisite A.163/7 - - INFO response

Table A.203: Supported message bodies within the INFO response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.2.4.7 INVITE method

Prerequisite A.163/8 - - INVITE request

Table A.204: Supported header fields within the INVITE request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
1A	Accept-Contact	[56B] 9.2	c34	c34	[56B] 9.2	c34	c35
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i
3A	Additional-Identity	7.2.20	n/a	c85	7.2.20	n/a	c85
4	Alert-Info	[26] 20.4	c2	c2	[26] 20.4	c3	c3
5	Allow	[26] 20.5	m	m	[26] 20.5	i	i
6	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
7	Answer-Mode	[158]	c67	c67	[158]	c68	c68
7A	Attestation-Info	7.2.18	n/a	c82	7.2.18	n/a	c82
8	Authorization	[26] 20.7	m	m	[26] 20.7	i	i
9	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
10	Call-Info	[26] 20.9	m	m	[26] 20.9	c12	c12
10A	Cellular-Network-Info	7.2.15	n/a	c78	7.2.15	n/a	c79
11	Contact	[26] 20.10	m	m	[26] 20.10	i	i
12	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	c6
13	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	c6
14	Content-Language	[26] 20.13	m	m	[26] 20.13	i	c6
15	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
16	Content-Type	[26] 20.15	m	m	[26] 20.15	i	c6
17	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
18	Date	[26] 20.17	m	m	[26] 20.17	c4	c4
19	Expires	[26] 20.19	m	m	[26] 20.19	i	i
19A	Feature-Caps	[190]	c73	c73	[190]	c73	c73
20	From	[26] 20.20	m	m	[26] 20.20	m	m
20A	Geolocation	[89] 4.1	c47	c47	[89] 4.1	c48	c48
20B	Geolocation-Routing	[89] 4.1	c47	c47	[89] 4.1	c48	c48
20C	History-Info	[66] 4.1	c43	c43	[66] 4.1	c43	c43
20D	Identity	[252] 4	c81	c81	[252] 4	c81	c81
21	In-Reply-To	[26] 20.21	m	m	[26] 20.21	i	i
21A	Join	[61] 7.1	c41	c41	[61] 7.1	c42	c42
21B	Max-Breadth	[117] 5.8	c63	c63	[117] 5.8	c64	c64
22	Max-Forwards	[26] 20.22	m	m	[26] 20.22	m	m
23	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	c6
23A	Min-SE	[58] 5	o	o	[58] 5	o	o
24	Organization	[26] 20.25	m	m	[26] 20.25	c5	c5
24AA	Origination-Id	7.2.19	n/a	c83	7.2.19	n/a	c83
24A	P-Access-Network-Info	[52] 4.4, [234] 2	c28	c28	[52] 4.4, [234] 2	c29	c30
24B	P-Asserted-Identity	[34] 9.1	c15	c15	[34] 9.1	c16	c16
24C	P-Asserted-Service	[121] 4.1	c53	c53	[121] 4.1	c54	c54
24D	P-Called-Party-ID	[52] 4.2	c19	c19	[52] 4.2	c20	c21
24E	P-Charging-Function-Addresses	[52] 4.5	c26	c27	[52] 4.5	c26	c27
24F	P-Charging-Vector	[52] 4.6	c24	c24	[52] 4.6	c25	c25
24H	P-Early-Media	[109] 8	o	c50	[109] 8	o	c50
25	P-Media-Authorization	[31] 5.1	c9	x	[31] 5.1	n/a	n/a
25A	P-Preferred-Identity	[34] 9.2	x	c69	[34] 9.2	c14	c14
25B	P-Preferred-Service	[121] 4.2	x	x	[121] 4.2	c52	c52
25C	P-Private-Network-Indication	[134]	c59	c59	[134]	c59	c59
25D	P-Profile-Key	[97] 5	c45	c45	[97] 5	c46	c46
25E	P-Served-User	[133] 6	c60	c60	[133] 6	c60	c60
25F	P-User-Database	[82] 4	c44	c44	[82] 4	c44	c44
25G	P-Visited-Network-ID	[52] 4.3	c22	o	[52] 4.3	c23	o
26	Priority	[26] 20.26	m	m	[26] 20.26	i	c74
26AA	Priority-Share	Subclause 7.2.16	n/a	c80	Subclause 7.2.16	n/a	c80
26A	Privacy	[33] 4.2	c17	c17	[33] 4.2	c18	c18
26AB	Priority-Verstat	Subclause 7.2.21	n/a	c86	Subclause 7.2.21	n/a	c86
26B	Priv-Answer-Mode	[158]	c67	c67	[158]	c68	c68
27	Proxy-Authorization	[26] 20.28	m	m	[26] 20.28	c13	c13

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
28	Proxy-Require	[26] 20.29, [34] 4	m	m	[26] 20.29, [34] 4	m	m
28A	Reason	[34A] 2	c32	c32	[34A] 2	c33	c33
29	Record-Route	[26] 20.30	m	m	[26] 20.30	c11	c11
29A	Recv-Info	[25] 5.2.3	c65	c65	[25] 5.2.3	c66	c66
30	Referred-By	[59] 3	c37	c37	[59] 3	c38	c38
31	Reject-Contact	[56B] 9.2	c34	c34	[56B] 9.2	c34	c35
31A	Relayed-Charge	7.2.12	n/a	c76	7.2.12	n/a	c76
31B	Replaces	[60] 6.1	c39	c39	[60] 6.1	c40	c40
31C	Reply-To	[26] 20.31	m	m	[26] 20.31	i	i
31D	Request-Disposition	[56B] 9.1	c34	c34	[56B] 9.1	c34	c34
32	Require	[26] 20.32	m	m	[26] 20.32	c7	c7
32A	Resource-Priority	[116] 3.1	c49	c49	[116] 3.1	c49	c49
32B	Restoration-Info	Subclause 7.2.11	n/a	c75	Subclause 7.2.11	n/a	c75
32C	Resource-Share	Subclause 4.15	n/a	c77	Subclause 4.15	n/a	c77
33	Route	[26] 20.34	m	m	[26] 20.34	m	m
33A	Security-Client	[48] 2.3.1	x	x	[48] 2.3.1	c31	c31
33B	Security-Verify	[48] 2.3.1	x	x	[48] 2.3.1	c31	c31
33DA	Service-Interact-Info	Subclause 7.2.14	n/a	c84	Subclause 7.2.14	n/a	c84
33D	Session-Expires	[58] 4	c36	c36	[58] 4	c36	c36
33E	Session-ID	[162]	c70	c70	[162]	c70	c70
34	Subject	[26] 20.36	m	m	[26] 20.36	i	i
35	Supported	[26] 20.37	m	m	[26] 20.37	c8	c8
35A	Target-Dialog	[184] 7	c71	c71	[184] 7	c72	c72
36	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
37	To	[26] 20.39	m	m	[26] 20.39	m	m
37A	Trigger-Consent	[125] 5.11.2	c55	c55	[125] 5.11.2	c56	c56
38	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
38A	User-to-User	[126] 7	c57	c57	[126] 7	c58	c58
39	Via	[26] 20.42	m	m	[26] 20.42	m	m

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.						
c2:	IF A.162/10 THEN n/a ELSE m - - suppression or modification of alerting information data.						
c3:	IF A.162/10 THEN m ELSE i - - suppression or modification of alerting information data.						
c4:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.						
c5:	IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.						
c6:	IF A.3/2 OR A.3/4 THEN m ELSE i - - P-CSCF or S-CSCF.						
c7:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.						
c8:	IF A.162/16 THEN m ELSE i - - reading the contents of the Supported header before proxying the response.						
c9:	IF A.162/26 THEN m ELSE n/a - - SIP extensions for media authorization.						
c11:	IF A.162/14 THEN m ELSE i - - the requirement to be able to insert itself in the subsequent transactions in a dialog.						
c12:	IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header.						
c13:	IF A.162/8A THEN m ELSE i - - authentication between UA and proxy.						
c14:	IF A.162/30A OR A.162/30C THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity, act as entity passing on identity transparently independent of trust domain.						
c15:	IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.						
c16:	IF A.162/30A or A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.						
c17:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).						
c18:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.						
c19:	IF A.162/37 THEN m ELSE n/a - - the P-Called-Party-ID header extension.						
c20:	IF A.162/37 THEN i ELSE n/a - - the P-Called-Party-ID header extension.						
c21:	IF A.162/37 AND A.3/2 THEN m ELSE IF A.162/37 AND (A.3/3 OR A.3/9A) THEN i ELSE n/a - - the P-Called-Party-ID header extension and P-CSCF or (I-CSCF or IBCF (THIG)).						
c22:	IF A.162/38 THEN m ELSE n/a - - the P-Visited-Network-ID header extension.						
c23:	IF A.162/39 THEN m ELSE i - - reading, or deleting the P-Visited-Network-ID header before proxying the request or response.						
c24:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.						
c25:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.						
c26:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.						
c27:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.						
c28:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.						
c29:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.						
c30:	IF A.162/43 OR (A.162/41 AND A.3/2) THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension (with or without P-CSCF).						
c31:	IF A.162/47 OR A.162/47A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.						
c32:	IF A.162/48 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.						
c33:	IF A.162/48 THEN i ELSE n/a - - the Reason header field for the session initiation protocol.						
c34:	IF A.162/50 THEN m ELSE n/a - - caller preferences for the session initiation protocol.						
c35:	IF A.162/50 AND A.4/3 THEN m ELSE IF A.162/50 AND NOT A.4/3 THEN i ELSE n/a - - caller preferences for the session initiation protocol, and S-CSCF.						
c36:	IF A.162/52 THEN m ELSE n/a - - the SIP session timer.						
c37:	IF A.162/53 THEN i ELSE n/a - - the SIP Referred-By mechanism.						
c38:	IF A.162/53 THEN m ELSE n/a - - the SIP Referred-By mechanism.						
c39:	IF A.162/54 THEN m ELSE n/a - - the Session Inititation Protocol (SIP) "Replaces" header.						
c40:	IF A.162/54 THEN i ELSE n/a - - the Session Inititation Protocol (SIP) "Replaces" header.						

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
c41:	IF A.162/55 THEN m ELSE n/a - - the Session Inititation Protocol (SIP) "Join" header.						
c42:	IF A.162/55 THEN i ELSE n/a - - the Session Inititation Protocol (SIP) "Join" header.						
c43:	IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.						
c44:	IF A.162/60 THEN m ELSE n/a - - the P-User-Database private header extension.						
c45:	IF A.162/66A THEN m ELSE n/a - - making the first query to the database in order to populate the P-Profile-Key header.						
c46:	IF A.162/66B THEN m ELSE n/a - - using the information in the P-Profile-Key header.						
c47:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.						
c48:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.						
c49:	IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						
c50:	IF A.162/76 THEN m ELSE n/a - - the SIP P-Early-Media private header extension for authorization of early media.						
c52:	IF A.162/84A THEN m ELSE n/a - - act as authentication entity within the trust domain for asserted service.						
c53:	IF A.162/84 THEN m ELSE n/a - - SIP extension for the identification of services.						
c54:	IF A.162/84 OR A.162/30B THEN m ELSE i - - SIP extension for the identification of services or subsequent entity within trust network that can route outside the trust network.						
c55:	IF A.162/85 THEN m ELSE n/a - - a framework for consent-based communications in SIP.						
c56:	IF A.162/85 THEN i ELSE n/a - - a framework for consent-based communications in SIP.						
c57:	IF A.162/86 THEN m - - transporting user to user information for call centers using SIP.						
c58:	IF A.162/86 THEN i - - transporting user to user information for call centers using SIP.						
c59:	IF A.162/87 THEN m ELSE n/a - - the SIP P-Private-Network-Indication private-header (P-Header).						
c60:	IF A.162/88 THEN m - - the SIP P-Served-User private header.						
c63:	IF A.162/81 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.						
c64:	IF A.162/81 AND A.162/6 THEN m ELSE IF A.162/81 AND NOT A.162/6 THEN i ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, forking of initial requests.						
c65:	IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.						
c66:	IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.						
c67:	IF A.162/97 THEN m ELSE n/a - - requesting answering modes for SIP.						
c68:	IF NOT A.162/97 THEN n/a ELSE IF A.162/97A THEN m ELSE i - - requesting answering modes for SIP, adding, deleting or reading the Answer-Mode header or Priv-Answer-Mode header before proxying the request or response.						
c69:	IF A.162/30C THEN m ELSE x - - act as entity passing on identity transparently independent of trust domain.						
c70:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.						
c71:	IF A.162/109 THEN m ELSE n/a - - request authorization through dialog Identification in the session initiation protocol.						
c72:	IF A.162/109 THEN i ELSE n/a - - request authorization through dialog Identification in the session initiation protocol.						
c73:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.						
c74:	IF A.162/115 THEN m ELSE i - - PSAP callback indicator.						
c75:	IF A.162/119 THEN o ELSE n/a - - PCRF based P-CSCF restoration.						
c76:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.						
c77:	IF A.162/122 THEN o ELSE n/a - - resource sharing.						
c78:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.						
c79:	IF A.162/43 OR (A.162/41A AND A.3/2) THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension (with or without P-CSCF).						
c80:	IF A.162/124 THEN o ELSE n/a - - priority sharing.						
c81:	IF A.162/126 THEN o ELSE n/a - - authenticated identity management in the Session Initiation Protocol.						
c82:	IF A.162/128 THEN o ELSE n/a - - the Attestation-Info header field extension.						
c83:	IF A.162/129 THEN o ELSE n/a - - the Origination-Id header field extension.						
c84:	IF A.162/130 THEN m ELSE n/a - - Dynamic services interactions.						
c85:	IF A.162/131 THEN o ELSE n/a - - the Additional-Identity header field extension.						
c86:	IF A.162/80 AND A.162/133 THEN o ELSE n/a - - communications resource priority for the session initiation protocol, the Priority-Verstat header field extension.						
NOTE:	c1 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.						

Prerequisite A.163/8 - - INVITE request

Table A.205: Supported message bodies within the INVITE request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	XML Schema for PSTN	[11B]	n/a	c1	[11B]	n/a	i
2	application/vnd.3gpp.ussd	[8W]	n/a	m	[8W]	n/a	i
3	application/vnd.3gpp.mcptt-info+xml	[8ZE]	n/a	c1	[8ZE]	n/a	i
c1:	A.3/3 OR A.3/4 OR A.3/5 OR A.3/7C OR A.3/9A OR A.3/10 OR A.3/11 OR A.3/13A THEN m ELSE n/a - - I-CSCF, S-CSCF, BGCF, AS acting as proxy, IBCF (THIG), additional routeing functionality, E-CSCF, ISC gateway function (THIG).						

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/1 - - Additional for 100 (Trying) response

Table A.206: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	c2	c2
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF (A.162/9 AND A.162/5) OR A.162/4 THEN m ELSE n/a - - stateful proxy behaviour that inserts date, or stateless proxies.						
c2:	IF A.162/4 THEN i ELSE m - - Stateless proxy passes on.						

Prerequisite A.163/9 - - INVITE response for all remaining status-codes

Table A.207: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Call-Info	[26] 20.9	m	m	[26] 20.9	c4	c4
1B	Cellular-Network-Info	7.2.15	n/a	c27	7.2.15	n/a	c28
2	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	c3
3	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	c3
4	Content-Language	[26] 20.13	m	m	[26] 20.13	i	c3
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	i	c3
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	m	m	[26] 20.17	c1	c1
8A	Expires	[26] 20.19	m	m	[26] 20.19	i	i
9	From	[26] 20.20	m	m	[26] 20.20	m	m
9A	Geolocation-Error	[89] 4.3	c24	c24	[89] 4.3	c24	c24
9B	History-Info	[66] 4.1	c17	c17	[66] 4.1	c17	c17
10	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	c3
11	Organization	[26] 20.25	m	m	[26] 20.25	c2	c2
11A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c14	c14	[52] 4.4, [52A] 4, [234] 2	c15	c15
11B	P-Asserted-Identity	[34] 9.1	c6	c6	[34] 9.1	c7	c7
11C	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c12	c12	[52] 4.5, [52A] 4	c13	c13
11D	P-Charging-Vector	[52] 4.6, [52A] 4	c10	c10	[52] 4.6, [52A] 4	c11	c11
11F	P-Preferred-Identity	[34] 9.2	x	x	[34] 9.2	c5	n/a
11G	Privacy	[33] 4.2	c8	c8	[33] 4.2	c9	c9
11H	Relayed-Charge	7.2.12	n/a	c26	7.2.12	n/a	c26
11I	Reply-To	[26] 20.31	m	m	[26] 20.31	i	i
11J	Require	[26] 20.32	m	m	[26] 20.32	c16	c16
11K	Server	[26] 20.35	m	m	[26] 20.35	i	i
11LA	Service-Interact-Info	Subclause 7.2.14	n/a	c29	Subclause 7.2.14	n/a	c29
11L	Session-ID	[162]	c25	c25	[162]	c25	c25
12	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
13	To	[26] 20.39	m	m	[26] 20.39	m	m
13A	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
13B	User-to-User	[126] 7	c20	c20	[126] 7	c21	c21
14	Via	[26] 20.42	m	m	[26] 20.42	m	m
15	Warning	[26] 20.43	m	m	[26] 20.43	i	i

c1:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c2:	IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.
c3:	IF A.3/2 OR A.3/4 THEN m ELSE i - - P-CSCF or S-CSCF.
c4:	IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header.
c5:	IF A.162/30A THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity.
c6:	IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c7:	IF A.162/30A or A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.
c8:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c9:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c10:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c11:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c12:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c13:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c14:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c15:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c16:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c17:	IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c18:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c19:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c20:	IF A.162/86 THEN m - - transporting user to user information for call centers using SIP.
c21:	IF A.162/86 THEN i - - transporting user to user information for call centers using SIP.
c24:	IF A.4/60 THEN m ELSE n/a - - SIP location conveyance.
c25:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c26:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c27:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c28:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c29:	IF A.162/130 THEN m ELSE n/a - - Dynamic services interactions.

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/101A - - Additional for 18x response

Table A.208: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Contact	[26] 20.10	m	m	[26] 20.10	i	i
4A	Feature-Caps	[190]	c19	c19	[190]	c19	c19
5	P-Answer-State	[111]	c13	c13	[111]	c14	c14
5A	P-Early-Media	[109] 8	o	c11	[109] 8	o	c11
6	P-Media-Authorization	[31] 5.1	c9	x	[31] 5.1	n/a	n/a
6AA	Priority-Share	Subclause 7.2.16	n/a	c21	Subclause 7.2.16	n/a	c21
6A	Reason	[130], [294]	o	c18	[130], [294]	o	c18
7	Record-Route	[26] 20.30	m	m	[26] 20.30	c15	c15
8	Recv-Info	[25] 5.2.3	c16	c16	[25] 5.2.3	c17	c17
8A	Resource-Share	Subclause 4.15	n/a	c20	Subclause 4.15	n/a	c20
9	RSeq	[27] 7.1	m	m	[27] 7.1	i	i

c9: IF A.162/26 THEN m ELSE n/a - - SIP extensions for media authorization.
c11: IF A.162/76 THEN m ELSE n/a - - the SIP P-Early-Media private header extension for authorization of early media.
c13: IF A.162/75 THEN m ELSE n/a - - the P-Answer-State header extension to the session initiation protocol for the open mobile alliance push to talk over cellular.
c14: IF A.162/75 THEN i ELSE n/a - - the P-Answer-State header extension to the session initiation protocol for the open mobile alliance push to talk over cellular.
c15: IF A.162/14 THEN m ELSE i - - the requirement to be able to insert itself in the subsequent transactions in a dialog.
c16: IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.
c17: IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.
c18: IF A.162/48A OR A.162/48C THEN o ELSE n/a - - use of the Reason header field in Session Initiation Protocol (SIP) responses (carrying Q.850 or STIR codes).
c19: IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.
c20: IF A.162/122 THEN o ELSE n/a - - resource sharing.
c21: IF A.162/124 THEN o ELSE n/a - - priority sharing.

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/2 - - Additional for 180 (Ringing) response

Table A.208A: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Alert-Info	[26] 20.4	m	m	[26] 20.4	i	i

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/5A - - Additional for 199 (Early Dialog Terminated) response

Table A.208B: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Contact	[26] 20.10	m	m	[26] 20.10	i	i
5	Reason	[130], [294]	o	c18	[130], [294]	o	c18
7	Record-Route	[26] 20.30	m	m	[26] 20.30	c15	c15
8	Recv-Info	[25] 5.2.3	c16	c16	[25] 5.2.3	c17	c17
9	RSeq	[27] 7.1	m	m	[27] 7.1	i	i
c15:	IF A.162/14 THEN m ELSE i - - the requirement to be able to insert itself in the subsequent transactions in a dialog.						
c16:	IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.						
c17:	IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.						
c18:	IF A.162/48A OR A.162/48C THEN o ELSE n/a - - use of the Reason header field in Session Initiation Protocol (SIP) responses (carrying Q.850 or STIR codes).						

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/102 - - Additional for 2xx response

Table A.209: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
1A	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
1B	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i
1C	Accept-Resource-Priority	[116] 3.2	c12	c12	[116] 3.2	c12	c12
2	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
3	Answer-Mode	[158]	c19	c19	[158]	c20	c20
4	Authentication-Info	[26] 20.6	m	m	[26] 20.6	i	i
6	Contact	[26] 20.10	m	m	[26] 20.10	i	i
6A	Feature-Caps	[190]	c22	c22	[190]	c22	c22
7	P-Answer-State	[111]	c13	c13	[111]	c14	c14
7A	P-Visited-Network-ID	[52B] 3	o	o	[52B] 3	o	o
8	P-Media-Authorization	[31] 5.1	c9	x	[31] 5.1	n/a	n/a
8AA	Priority-Share	Subclause 7.2.16	n/a	c24	Subclause 7.2.16	n/a	c24
8A	Priv-Answer-Mode	[158]	c19	c19	[158]	c20	c20
8B	Reason	[294]	o	c25	[294]	o	c25
9	Record-Route	[26] 20.30	m	m	[26] 20.30	c3	c3
9A	Recv-Info	[25] 5.2.3	c17	c17	[25] 5.2.3	c18	c18
9B	Resource-Share	Subclause 4.15	n/a	c23	Subclause 4.15	n/a	c23
10	Session-Expires	[58] 4	c11	c11	[58] 4	c11	c11
13	Supported	[26] 20.37	m	m	[26] 20.37	i	i

c1: IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.
c3: IF A.162/14 THEN m ELSE i - - the requirement to be able to insert itself in the subsequent transactions in a dialog.
c9: IF A.162/26 THEN m ELSE n/a - - SIP extensions for media authorization.
c11: IF A.162/52 THEN m ELSE n/a - - the SIP session timer.
c12: IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.
c13: IF A.162/75 THEN m ELSE n/a - - the P-Answer-State header extension to the session initiation protocol for the open mobile alliance push to talk over cellular.
c14: IF A.162/75 THEN i ELSE n/a - - the P-Answer-State header extension to the session initiation protocol for the open mobile alliance push to talk over cellular.
c17: IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.
c18: IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.
c19: IF A.162/97 THEN m ELSE n/a - - requesting answering modes for SIP.
c20: IF NOT A.162/97 THEN n/a ELSE IF A.162/97A THEN m ELSE i - - requesting answering modes for SIP, adding, deleting or reading the Answer-Mode header or Priv-Answer-Mode header before proxying the request or response.
c22: IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.
c23: IF A.162/122 THEN o ELSE n/a - - resource sharing.
c24: IF A.162/124 THEN o ELSE n/a - - priority sharing.
c25: IF A.162/48C THEN o ELSE n/a - - use of the Reason header field in Session Initiation Protocol (SIP) responses (carrying STIR codes).

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/103 OR A.164/104 OR A.164/105 OR A.164/106 - - Additional for 3xx – 6xx response

Table A.209A: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	m	m	[26] 20.18	i	i
2	Reason	[130], [294]	o	c1	[130], [294]	o	c1
3	Response-Source	7.2.17	n/a	c2	7.2.17	n/a	c2
c1:	IF A.162/48A OR A.162/48C THEN o ELSE n/a - - use of the Reason header field in Session Initiation Protocol (SIP) responses (carrying Q.850 or STIR codes).						
c2:	IF A.162/125 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/103 OR A.164/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.210: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Contact	[26] 20.10	m	m	[26] 20.10	c1	c1
c1:	IF A.162/19E THEN m ELSE i - - deleting Contact headers.						

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/14 - - Additional for 401 (Unauthorized) response

Table A.211: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
6	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
15	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.6/16 - - Additional for 403 (Forbidden) response

Table A.211A: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	P-Refused-URI-List	[183]	c1	c1	[183]	c1	c1
c1:	IF A.162/108 THEN m ELSE n/a -- The SIP P-Refused-URI-List private-header.						

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/17 OR A.164/23 OR A.164/30 OR A.164/36 OR A.164/50 OR A.164/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 600 (Busy Everywhere), 603 (Decline) response

Table A.212: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
8	Retry-After	[26] 20.33	m	m	[26] 20.33	i	i

Table A.213: Void

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.214: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
6	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
11	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/21 - - Additional for 408 (Request timeout) response

Table A.214A: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Restoration-Info	subclause 7.2.11	n/a	c1	subclause 7.2.11	n/a	c2
c1:	IF A.162/120 THEN o ELSE n/a - - HSS based P-CSCF restoration.						
c2:	IF A.162/120 THEN m ELSE n/a - - HSS based P-CSCF restoration.						

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/25 - - Additional for 415 (Unsupported Media Type) response

Table A.215: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.215A: Supported header field s within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/27 - - Additional for 420 (Bad Extension) response

Table A.216: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
10	Unsupported	[26] 20.40	m	m	[26] 20.40	c3	c3
c3:	IF A.162/18 THEN m ELSE i - - reading the contents of the Unsupported header before proxying the 420 response to a method other than REGISTER.						

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/28 OR A.164/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.216A: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	c1	c1	[48] 2	n/a	n/a
c1:	IF A.162/47 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Prerequisite A.16/9 - - INVITE response

Prerequisite: A.164/28A - - Additional for 422 (Session Interval Too Small) response

Table A.216B: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Min-SE	[58] 5	c1	c1	[58] 5	c1	c1
c1:	IF A.162/52 THEN m ELSE n/a - - the SIP session timer.						

Table A.217: Void

Table A.217A: Void

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/29H - - Additional for 470 (Consent Needed) response

Table A.217AA: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Permission-Missing	[125] 5.9.3	m	m	[125] 5.9.3	m	m

Prerequisite A.163/9 - - INVITE response

Prerequisite: A.164/45 - - 503 (Service Unavailable)

Table A.217B: Supported header fields within the INVITE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
8	Retry-After	[26] 20.33	m	m	[26] 20.33	i	i

Table A.217C: void

Prerequisite A.163/9 - - INVITE response

Table A.218: Supported message bodies within the INVITE response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	XML Schema for PSTN	[11B]		c1	[11B]		i
2	Recipient List	[183]	c2	c2	[183]	c3	c3
3	3GPP IM CN subsystem XML body	subclause 7.6	n/a	c4	subclause 7.6	n/a	c5
4	application/vnd.3gpp.mcptt-info+xml	[8ZE]	n/a	c1	[8ZE]	n/a	i
c1:	A.3/3 OR A.3/4 OR A.3/5 OR A.3/7C OR A.3/9A OR A.3/10 OR A.3/11 OR A.3/13A THEN m ELSE n/a - - I-CSCF, S-CSCF, BGCF, AS acting as proxy, IBCF (THIG), additional routeing functionality, E-CSCF, ISC gateway function (THIG).						
c2	IF A.3/9B THEN m ELSE IF A.3/7A OR A.3/7B OR A.3/7D THEN m ELSE n/a - - IBCF (IMS-ALG), AS acting as terminating UA, AS acting as originating UA, AS performing 3rd party call control.						
c3	IF A.3/9B THEN m ELSE IF A.3/7A OR A.3/7B OR A.3/7D THEN i ELSE n/a - - IBCF (IMS-ALG), AS acting as terminating UA, AS acting as originating UA, AS performing 3rd party call control.						
c4:	IF A.3/2 OR (A.3/9 AND NOT A.3/9B) OR A.3A/88 THEN m ELSE n/a - - P-CSCF, IBCF, IBCF (IMS-ALG),						

	ATCF (proxy).
c5:	IF A.3/2 OR (A.3/9 AND NOT A.3/9B) OR A.3A/88 THEN i ELSE n/a - - P-CSCF, IBCF, IBCF (IMS-ALG), ATCF (proxy).

A.2.2.4.7A MESSAGE method

Prerequisite A.163/9A - - MESSAGE request

Table A.218A: Supported header fields within the MESSAGE request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Contact	[56B] 9.2	c28	c28	[56B] 9.2	c28	c29
1AA	Additional-Identity	7.2.20	n/a	c79	7.2.20	n/a	c79
1A	Allow	[26] 20.5	m	m	[50] 10	i	i
2	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
2A	Attestation-Info	7.2.18	n/a	c76	7.2.18	n/a	c76
3	Authorization	[26] 20.7	m	m	[26] 20.7	i	i
4	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
5	Call-Info	[26] 20.9	m	m	[26] 20.9	c4	c4
5A	Cellular-Network-Info	7.2.15	n/a	c73	7.2.15	n/a	c74
6	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
7	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
8	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
9	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
10	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
11	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
12	Date	[26] 20.17	m	m	[26] 20.17	c2	c2
13	Expires	[26] 20.19	m	m	[26] 20.19	I	i
13A	Feature-Caps	[190]	c71	c71	[190]	c71	c71
14	From	[26] 20.20	m	m	[26] 20.20	m	m
14A	Geolocation	[89] 4.1	c36	c36	[89] 4.1	c37	c37
14B	History-Info	[66] 4.1	c32	c32	[66] 4.1	c32	c32
14C	Geolocation-Routing	[89] 4.1	c36	c36	[89] 4.1	c37	c37
14D	Identity	[252] 4	c75	c75	[252] 4	c75	c75
15	In-Reply-To	[26] 20.21	m	m	[50] 10	i	i
15A	Max-Breadth	[117] 5.8	c48	c48	[117] 5.8	c49	c49
16	Max-Forwards	[26] 20.22	m	m	[26] 20.22	m	m
17	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
18	Organization	[26] 20.25	m	m	[26] 20.25	c3	c3
18AA	Origination-Id	7.2.19	n/a	c77	7.2.19	n/a	c77
18A	P-Access-Network-Info	[52] 4.4, [234] 2	c23	c23	[52] 4.4, [234] 2	c24	c24
18B	P-Asserted-Identity	[34] 9.1	c10	c10	[34] 9.1	c11	c11
18C	P-Asserted-Service	[121] 4.1	c40	c40	[121] 4.1	c41	c41
18D	P-Called-Party-ID	[52] 4.2	c14	c14	[52] 4.2	c15	c16
18E	P-Charging-Function-Addresses	[52] 4.5	c21	c21	[52] 4.5	c22	c22
18F	P-Charging-Vector	[52] 4.6	c19	c19	[52] 4.6	c20	c20
18H	P-Preferred-Identity	[34] 9.2	x	c69	[34] 9.2	c9	c9
18I	P-Preferred-Service	[121] 4.2	x	x	[121] 4.2	c39	c39
18J	P-Private-Network-Indication	[134]	c44	c44	[134]	c44	c44
18K	P-Profile-Key	[97] 5	c34	c34	[97] 5	c35	c35
18L	P-Served-User	[133] 6	c45	c45	[133] 6	c45	c45
18M	P-User-Database	[82] 4	c33	c33	[82] 4	c33	c33
18N	P-Visited-Network-ID	[52] 4.3	c17	o	[52] 4.3	c18	o
19	Priority	[26] 20.26	m	m	[26] 20.26	i	c50
19A	Privacy	[33] 4.2	c12	c12	[33] 4.2	c13	c13
20	Proxy-Authorization	[26] 20.28	m	m	[26] 20.28	c8	c8
21	Proxy-Require	[26] 20.29	m	m	[26] 20.29	m	m
21A	Reason	[34A] 2	c26	c26	[34A] 2	c27	c27
22A	Referred-By	[59] 3	c30	c30	[59] 3	c31	c31
23	Reject-Contact	[56B] 9.2	c28	c28	[56B] 9.2	c28	c29
23A	Relayed-Charge	7.2.12	n/a	c72	7.2.12	n/a	c72
23B	Reply-To	[26] 20.31	m	m	[26] 20.31	i	i
23C	Request-Disposition	[56B] 9.1	c28	c28	[56B] 9.1	c28	c28
24	Require	[26] 20.32	m	m	[26] 20.32	c5	c5
24A	Resource-Priority	[116] 3.1	c38	c38	[116] 3.1	c38	c38
25	Route	[26] 20.34	m	m	[26] 20.34	m	m
25A	Security-Client	[48] 2.3.1	x	x	[48] 2.3.1	c25	c25
25B	Security-Verify	[48] 2.3.1	x	x	[48] 2.3.1	c25	c25
25D	Service-Interact-Info	Subclause 7.2.14	n/a	c78	Subclause 7.2.14	n/a	c78

26	Subject	[26] 20.36	m	m	[26] 20.36	i	i
25C	Session-ID	[162]	c70	c70	[162]	c70	c70
27	Supported	[26] 20.37	m	m	[26] 20.37	c6	c6
28	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
29	To	[26] 20.39	m	m	[26] 20.39	m	m
29A	Trigger-Consent	[125] 5.11.2	c42	c42	[125] 5.11.2	c43	c43
30	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
31	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.
c2:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c3:	IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.
c4:	IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header.
c5:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c6:	IF A.162/16 THEN m ELSE i - - reading the contents of the Supported header before proxying the response.
c8:	IF A.162/8A THEN m ELSE i - - authentication between UA and proxy.
c9:	IF A.162/30A OR A.162/30C THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity, act as entity passing on identity transparently independent of trust domain.
c10:	IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c11:	IF A.162/30A or A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.
c12:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c13:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c14:	IF A.162/37 THEN m ELSE n/a - - the P-Called-Party-ID header extension.
c15:	IF A.162/37 THEN i ELSE n/a - - the P-Called-Party-ID header extension.
c16:	IF A.162/37 AND A.3/2 THEN m ELSE IF A.162/37 AND (A.3/3 OR A.3/9A) THEN i ELSE n/a - - the P-Called-Party-ID header extension and P-CSCF or (I-CSCF or IBCF (THIG)).
c17:	IF A.162/38 THEN m ELSE n/a - - the P-Visited-Network-ID header extension.
c18:	IF A.162/39 THEN m ELSE i - - reading, or deleting the P-Visited-Network-ID header before proxying the request or response.
c19:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c20:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c21:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c22:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c23:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c24:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c25:	IF A.162/47 OR A.162/47A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c26:	IF A.162/48 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.
c27:	IF A.162/48 THEN i ELSE n/a - - the Reason header field for the session initiation protocol.
c28:	IF A.162/50 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c29:	IF A.162/50 AND A.4/3 THEN m ELSE IF A.162/50 AND NOT A.4/3 THEN i ELSE n/a - - caller preferences for the session initiation protocol, and S-CSCF.
c30:	IF A.162/53 THEN i ELSE n/a - - the SIP Referred-By mechanism.
c31:	IF A.162/53 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c32:	IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c33:	IF A.162/60 THEN m ELSE n/a - - the P-User-Database private header extension.
c34:	IF A.162/66A THEN m ELSE n/a - - making the first query to the database in order to populate the P-Profile-Key header.
c35:	IF A.162/66B THEN m ELSE n/a - - using the information in the P-Profile-Key header.
c36:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c37:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c38:	IF A.162/80A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.
c39:	IF A.162/84A THEN m ELSE n/a - - act as authentication entity within the trust domain for asserted service.

c40:	IF A.162/84 THEN m ELSE n/a - - SIP extension for the identification of services.
c41:	IF A.162/84 OR A.162/30B THEN m ELSE i - - SIP extension for the identification of services or subsequent entity within trust network that can route outside the trust network.
c42:	IF A.162/85 THEN m ELSE n/a - - a framework for consent-based communications in SIP.
c43:	IF A.162/85 THEN i ELSE n/a - - a framework for consent-based communications in SIP.
c44:	IF A.162/87 THEN m ELSE n/a - - the SIP P-Private-Network-Indication private-header (P-Header).
c45:	IF A.162/88 THEN m ELSE n/a - - the SIP P-Served-User private header.
c48:	IF A.162/81 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c49:	IF A.162/81 AND A.162/6 THEN m ELSE IF A.162/81 AND NOT A.162/6 THEN i ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, forking of initial requests.
c50:	IF A.162/115 THEN m ELSE i - - PSAP callback indicator.
c69:	IF A.162/30C THEN m ELSE x - - act as entity passing on identity transparently independent of trust domain.
c70:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c71:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.
c72:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c73:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c74:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c75:	IF A.162/126 THEN o ELSE n/a - - authenticated identity management in the Session Initiation Protocol.
c76:	IF A.162/128 THEN o ELSE n/a - - the Attestation-Info header field extension.
c77:	IF A.162/129 THEN o ELSE n/a - - the Origination-Id header field extension.
c78:	IF A.162/130 THEN m ELSE n/a - - Dynamic services interactions.
c79:	IF A.162/131 THEN o ELSE n/a - - the Additional-Identity header field extension.
NOTE:	c1 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.

Prerequisite A.163/9A - - MESSAGE request

Table A.218B: Supported message bodies within the MESSAGE request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	permission document	[125] 5.4	c1	c1	[125] 5.4	c2	c2
2	application/vnd.3gpp.sms	[4D]	m	m	[4D]	i	i
2	message/cpim	[151]	c3	c3	[151]	c4	c4
3	message/imdn+xml	[157]	c5	c5	[157]	c6	c6
4	application/vnd.3gpp.mcptt-info+xml	[8ZE]	n/a	c7	[8ZE]	n/a	i
5	application/vnd.3gpp.mcptt-mbms-usage-info+xml	[8ZE]	n/a	c7	[8ZE]	n/a	i
6	application/vnd.3gpp.mcptt-location-info+xml	[8ZE]	n/a	c7	[8ZE]	n/a	i
7	application/vnd.3gpp.mcptt-floor-request+xml	[8ZE]	n/a	c7	[8ZE]	n/a	i
8	application/vnd.3gpp.mcptt-affiliation-command+xml	[8ZE]	n/a	c7	[8ZE]	n/a	i
c1:	IF A.162/85 THEN m ELSE n/a - - a framework for consent-based communications in SIP.						
c2:	IF A.162/85 THEN i ELSE n/a - - a framework for consent-based communications in SIP.						
c3:	IF A.162/95 THEN m ELSE n/a - - common presence and instant messaging (CPIM): message format.						
c4:	IF A.162/95 THEN i ELSE n/a - - common presence and instant messaging (CPIM): message format.						
c5:	IF A.162/96 THEN m ELSE n/a - - instant message disposition notification.						
c6:	IF A.162/96 THEN i ELSE n/a - - instant message disposition notification.						
c7:	A.3/3 OR A.3/4 OR A.3/5 OR A.3/7C OR A.3/9A OR A.3/10 OR A.3/11 OR A.3/13A THEN m ELSE n/a - - I-CSCF, S-CSCF, BGCF, AS acting as proxy, IBCF (THIG), additional routeing functionality, E-CSCF, ISC gateway function (THIG).						

Prerequisite A.163/9B - - MESSAGE response

Prerequisite: A.164/1 - - Additional for 100 (Trying) response

Table A.218BA: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	c2	c2
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF (A.162/9 AND A.162/5) OR A.162/4 THEN m ELSE n/a - - stateful proxy behaviour that inserts date, or stateless proxies.						
c2:	IF A.162/4 THEN i ELSE m - - Stateless proxy passes on.						

Prerequisite A.163/9B - - MESSAGE response for all remaining status-codes

Table A.218C: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Call-Info	[26] 20.9	m	m	[26] 20.9	c3	c3
2A	Cellular-Network-Info	7.2.15	n/a	c23	7.2.15	n/a	c24
3	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
4	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
5	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
6	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
7	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
8	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
9	Date	[26] 20.17	m	m	[26] 20.17	c1	c1
9A	Expires	[26] 20.19	m	m	[26] 20.19	i	i
10	From	[26] 20.20	m	m	[26] 20.20	m	m
10A	Geolocation-Error	[89] 4.3	c17	c17	[89] 4.3	c18	c18
10B	History-Info	[66] 4.1	c16	c16	[66] 4.1	c16	c16
11	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
12	Organization	[26] 20.25	m	m	[26] 20.25	c2	c2
12A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c13	c13	[52] 4.4, [52A] 4, [234] 2	c14	c14
12B	P-Asserted-Identity	[34] 9.1	c5	c5	[34] 9.1	c6	c6
12C	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c11	c11	[52] 4.5, [52A] 4	c12	c12
12D	P-Charging-Vector	[52] 4.6, [52A] 4	c9	c9	[52] 4.6, [52A] 4	c10	c10
12F	P-Preferred-Identity	[34] 9.2	x	x	[34] 9.2	c4	n/a
12G	Privacy	[33] 4.2	c7	c7	[33] 4.2	c8	c8
12H	Relayed-Charge	7.2.12	n/a	c22	7.2.12	n/a	c22
12I	Reply-To	[26] 20.31	m	m	[26] 20.31	i	i
12J	Require	[26] 20.32	m	m	[26] 20.32	c15	c15
13	Server	[26] 20.35	m	m	[26] 20.35	i	i
13AA	Service-Interact-Info	Subclause 7.2.14	n/a	c25	Subclause 7.2.14	n/a	c25
13A	Session-ID	[162]	c21	c21	[162]	c21	c21
14	Timestamp	[26] 20.38	i	i	[26] 20.38	i	i
15	To	[26] 20.39	m	m	[26] 20.39	m	m
16	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
17	Via	[26] 20.42	m	m	[26] 20.42	m	m
18	Warning	[26] 20.43	m	m	[26] 20.43	i	i

c1:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c2:	IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.
c3:	IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header.
c4:	IF A.162/30A THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity.
c5:	IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c6:	IF A.162/30A or A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.
c7:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c8:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c9:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c10:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c11:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c12:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c13:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c14:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c15:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c16:	IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c17:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c18:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c21:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c22:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c23:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c24:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c25:	IF A.162/130 THEN m ELSE n/a - - Dynamic services interactions.

Prerequisite A.163/9B - - MESSAGE response

Prerequisite: A.164/102 - - Additional for 2xx response

Table A.218D: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Accept-Resource-Priority	[116] 3.2	c4	c4	[116] 3.2	c4	c4
1	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
2	Authentication-Info	[26] 20.6	m	m	[26] 20.6	i	i
3	Feature-Caps	[190]	c6	c6	[190]	c6	c6
3A	Reason	[294]	o	c7	[294]	o	c7
4	P-Visited-Network-ID	[52B] 3	o	o	[52B] 3	o	o
6	Supported	[26] 20.37	m	m	[26] 20.37	i	i
c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.						
c4:	IF A.162/80A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						
c6:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.						
c7:	IF A.162/48C THEN o ELSE n/a - - use of the Reason header field in Session Initiation Protocol (SIP) responses (carrying STIR codes).						

Prerequisite A.163/9B - - MESSAGE response

Prerequisite: A.164/103 OR A.164/104 OR A.164/105 OR A.164/106 - - Additional for 3xx – 6xx response

Table A.218DA: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	m	m	[26] 20.18	i	i
1A	Reason	[294]	o	c2	[294]	o	c2
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.162/125 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses.						
c2:	IF A.4/38C THEN o ELSE n/a - - use of the Reason header field in Session Initiation Protocol (SIP) responses (carrying STIR codes).						

Prerequisite A.163/9B - - MESSAGE response

Prerequisite: A.164/103 OR A.164/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.218E: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Contact	[26] 20.10	m	m	[26] 20.10	c1	c1
c1:	IF A.162/19E THEN m ELSE i - - deleting Contact headers.						

Prerequisite A.163/9B - - MESSAGE response

Prerequisite: A.164/14 - - Additional for 401 (Unauthorized) response

Table A.218F: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
6	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/9B - - MESSAGE response

Prerequisite: A.164/17 OR A.164/23 OR A.164/30 OR A.164/36 OR A.164/42 OR A.164/45 OR A.164/50 OR A.164/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.218G: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Retry-After	[26] 20.33	m	m	[26] 20.33	i	i

Table A.218H: Void

Prerequisite A.163/9B - - MESSAGE response

Prerequisite: A.164/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.218I: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
6	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Table 218IA: Void

Prerequisite A.163/9B - - MESSAGE response

Prerequisite: A.164/25 - - Additional for 415 (Unsupported Media Type)

Table A.218J: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i

Prerequisite A.163/9B - - MESSAGE response

Prerequisite: A.164/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.218JA: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.162/80A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						

Prerequisite A.163/9B - - MESSAGE response

Prerequisite: A.164/27 - - Additional for 420 (Bad Extension) response

Table A.218K: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Unsupported	[26] 20.40	m	m	[26] 20.40	c3	c3
c3:	IF A.162/18 THEN m ELSE i - - reading the contents of the Unsupported header before proxying the 420 response to a method other than REGISTER.						

Prerequisite A.163/9B - - MESSAGE response

Prerequisite: A.164/28 OR A.164/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.218L: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	c1	c1	[48] 2	n/a	n/a
c1:	IF A.162/47 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Table A.218M: Void

Prerequisite A.163/9B - - MESSAGE response

Prerequisite: A.164/29H - - Additional for 470 (Consent Needed) response

Table A.218MA: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Permission-Missing	[125] 5.9.3	m	m	[125] 5.9.3	m	m

Prerequisite A.163/9B - - MESSAGE response

Prerequisite: A.164/46 - - Additional for 504 (Server Time-out) response

Table A.218MB: Supported header fields within the MESSAGE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Restoration-Info	subclause 7.2.11	n/a	c1	subclause 7.2.11	n/a	n/a
c1:	IF A.4/110 THEN o ELSE n/a - - HSS based P-CSCF restoration.						

Prerequisite A.163/9B - - MESSAGE response

Table A.218N: Supported message bodies within the MESSAGE response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.2.4.8 NOTIFY method

Prerequisite A.163/10 - - NOTIFY request

Table A.219: Supported header fields within the NOTIFY request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
1A	Accept-Contact	[56B] 9.2	c21	c21	[56B] 9.2	c22	c22
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i
3A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
4	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
5	Authorization	[26] 20.7	m	m	[26] 20.7	i	i
6	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
6A	Call-Info	[26] 20.9	m	m	[26] 20.9	c28	c28
6B	Cellular-Network-Info	7.2.15	n/a	c43	7.2.15	n/a	c44
6C	Contact	[26] 20.10	m	m	[26] 20.10	i	i
7	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
8	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
9	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
10	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
11	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
12	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
13	Date	[26] 20.17	m	m	[26] 20.17	c2	c2
14	Event	[28] 8.2.1	m	m	[28] 8.2.1	m	m
14A	Feature-Caps	[190]	c41	c41	[190]	c41	c41
15	From	[26] 20.20	m	m	[26] 20.20	m	m
15A	Geolocation	[89] 4.1	c26	c26	[89] 4.1	c27	c27
15B	Geolocation-Routing	[89] 4.1	c26	c26	[89] 4.1	c27	c27
15C	History-Info	[66] 4.1	c25	c25	[66] 4.1	c25	c25
15D	Max-Breadth	[117] 5.8	c29	c29	[117] 5.8	c30	c30
16	Max-Forwards	[26] 20.22	m	m	[26] 20.22	m	m
17	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
17A	P-Access-Network-Info	[52] 4.4, [234] 2	c16	c16	[52] 4.4. [234] 2	c17	c17
17B	P-Asserted-Identity	[34] 9.1	c8	c8	[34] 9.1	c9	c9
17C	P-Charging-Function-Addresses	[52] 4.5	c14	c14	[52] 4.5	c15	c15
17D	P-Charging-Vector	[52] 4.6	c12	c12	[52] 4.6	c13	c13
17F	P-Preferred-Identity	[34] 9.2	x	x	[34] 9.2	c3	n/a
17G	Privacy	[33] 4.2	c10	c10	[33] 4.2	c11	c11
18	Proxy-Authorization	[26] 20.28	m	m	[26] 20.28	c4	c4
19	Proxy-Require	[26] 20.29	m	m	[26] 20.29	m	m
19A	Reason	[34A] 2	c19	c19	[34A] 2	c20	c20
20	Record-Route	[26] 20.30	m	m	[26] 20.30	c7	c7
20A	Referred-By	[59] 3	c23	c23	[59] 3	c24	c24
20B	Reject-Contact	[56B] 9.2	c21	c21	[56B] 9.2	c22	c22
20C	Relayed-Charge	7.2.12	n/a	c42	7.2.12	n/a	c42
20D	Request-Disposition	[56B] 9.1	c21	c21	[56B] 9.1	c22	c22
21	Require	[26] 20.32	m	m	[26] 20.32	c5	c5
22	Route	[26] 20.34	m	m	[26] 20.34	m	m
22A	Security-Client	[48] 2.3.1	x	x	[48] 2.3.1	c18	c18
22B	Security-Verify	[48] 2.3.1	x	x	[48] 2.3.1	c18	c18
22C	Session-ID	[162]	c40	c40	[162]	c40	c40
23	Subscription-State	[28] 8.2.3	m	m	[28] 8.2.3	i	i
24	Supported	[26] 20.37	m	m	[26] 20.37	c6	c6
24A	Resource-Priority	[116] 3.1	c36	c36	[116] 3.1	c36	c36
25	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
26	To	[26] 20.39	m	m	[26] 20.39	m	m
27	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
28	Via	[26] 20.42	m	m	[26] 20.42	m	m
29	Warning	[26] 20.43	m	m	[26] 20.43	i	i

c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.
c2:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c3:	IF A.162/30A THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity.
c4:	IF A.162/8A THEN m ELSE i - - authentication between UA and proxy.
c5:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c6:	IF A.162/16 THEN m ELSE i - - reading the contents of the Supported header before proxying the response.
c7:	IF A.162/14 THEN (IF A.162/22 OR A.162/27 THEN m ELSE o) ELSE i - - the requirement to be able to insert itself in the subsequent transactions in a dialog or (the REFER method or SIP specific event notification).
c8:	IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c9:	IF A.162/30A or A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.
c10:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c11:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c12:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c13:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c14:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c15:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c16:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c17:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c18:	IF A.162/47 OR A.162/47A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c19:	IF A.162/48 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.
c20:	IF A.162/48 THEN i ELSE n/a - - the Reason header field for the session initiation protocol.
c21:	IF A.162/50 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c22:	IF A.162/50 THEN i ELSE n/a - - caller preferences for the session initiation protocol.
c23:	IF A.162/53 THEN i ELSE n/a - - the SIP Referred-By mechanism.
c24:	IF A.162/53 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c25:	IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c26:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c27:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c28:	IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header.
c29:	IF A.162/81 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c30:	IF A.162/81 AND A.162/6 THEN m ELSE IF A.162/81 AND NOT A.162/6 THEN i ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, forking of initial requests.
c36:	IF A.162/80A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.
c40:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c41:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.
c42:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c43:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c44:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
NOTE:	c1 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.

Prerequisite A.163/10 - - NOTIFY request

Table A.220: Supported message bodies within the NOTIFY request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	sipfrag	[37] 2	m	m	[37] 2	i	i
2	event package	[28]	m	m	[28]	i	i

Prerequisite A.163/11 - - NOTIFY response

Prerequisite: A.164/1 - - Additional for 100 (Trying) response

Table A.220A: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	c2	c2
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF (A.162/9 AND A.162/5) OR A.162/4 THEN m ELSE n/a - - stateful proxy behaviour that inserts date, or stateless proxies.						
c2:	IF A.162/4 THEN i ELSE m - - Stateless proxy passes on.						

Prerequisite A.163/11 - - NOTIFY response for all remaining status-codes

Table A.221: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Cellular-Network-Info	7.2.15	n/a	c20	7.2.15	n/a	c21
2	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
3	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
4	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	m	m	[26] 20.17	c1	c1
9	From	[26] 20.20	m	m	[26] 20.20	m	m
9A	Geolocation-Error	[89] 4.3	c14	c14	[89] 4.3	c15	c15
10	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
10A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c11	c11	[52] 4.4, [52A] 4, [234] 2	c12	c12
10B	P-Asserted-Identity	[34] 9.1	c3	c3	[34] 9.1	c4	c4
10C	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c9	c9	[52] 4.5, [52A] 4	c10	c10
10D	P-Charging-Vector	[52] 4.6, [52A] 4	c7	c7	[52] 4.6, [52A] 4	c8	c8
10F	P-Preferred-Identity	[34] 9.2	x	x	[34] 9.2	c2	n/a
10G	Privacy	[33] 4.2	c5	c5	[33] 4.2	c6	c6
10H	Relayed-Charge	7.2.12	n/a	c19	7.2.12	n/a	c19
10I	Require	[26] 20.32	m	m	[26] 20.32	c13	c13
10J	Server	[26] 20.35	m	m	[26] 20.35	i	i
10K	Session-ID	[162]	c18	c18	[162]	c18	c18
11	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
12	To	[26] 20.39	m	m	[26] 20.39	m	m
12A	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
13	Via	[26] 20.42	m	m	[26] 20.42	m	m
14	Warning	[26] 20.43	m	m	[26] 20.43	i	i

c1:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c2:	IF A.162/30A THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity.
c3:	IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c4:	IF A.162/30A or A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.
c5:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c6:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c7:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c8:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c9:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c10:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c11:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c12:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c13:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c14:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c15:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c18:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c19:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c20:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c21:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.

Prerequisite A.163/11 - - NOTIFY response

Prerequisite: A.164/102 - - Additional for 2xx response

Table A.222: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Accept-Resource-Priority	[116] 3.2	c4	c4	[116] 3.2	c4	c4
0B	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
1	Authentication-Info	[26] 20.6	m	m	[26] 20.6	i	i
1A	Contact	[26] 20.10	m	m	[26] 20.10	i	i
1B	Feature-Caps	[190]	c6	c6	[190]	c6	c6
2	Record-Route	[26] 20.30	m	m	[26] 20.30	c3	c3
5	Supported	[26] 20.37	m	m	[26] 20.37	i	i
c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.						
c3:	IF A.162/15 THEN m ELSE i - - the requirement to be able to use separate URIs in the upstream direction and downstream direction when record routeing.						
c4:	IF A.162/80A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						
c6:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.						

Prerequisite A.163/11 - - NOTIFY response

Prerequisite: A.164/103 OR A.164/104 OR A.164/105 OR A.164/106 - - Additional for 3xx – 6xx response

Table A.222A: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	m	m	[26] 20.18	i	i
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.162/125 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.163/11 - - NOTIFY response

Prerequisite: A.164/103 - - Additional for 3xx response

Table A.223: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Contact	[26] 20.10	m	m	[26] 20.10	c1	c1
c1:	IF A.162/19E THEN m ELSE i - - deleting Contact headers.						

Prerequisite A.163/11 - - NOTIFY response

Prerequisite: A.164/14 - - Additional for 401 (Unauthorized) response

Table A.224: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
8	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/11 - - NOTIFY response

Prerequisite: A.164/17 OR A.164/23 OR A.164/30 OR A.164/36 OR A.164/42 OR A.164/45 OR A.164/50 OR A.164/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.225: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Retry-After	[26] 20.33	m	m	[26] 20.33	i	i

Table A.226: Void

Prerequisite A.163/11 - - NOTIFY response

Prerequisite: A.164/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.227: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
6	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Table A.227A: Void

Prerequisite A.163/11 - - NOTIFY response

Prerequisite: A.164/25 - - Additional for 415 (Unsupported Media Type) response

Table A.228: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i

Prerequisite A.163/11 - - NOTIFY response

Prerequisite: A.164/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.228A: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.162/80A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						

Prerequisite A.163/11 - - NOTIFY response

Prerequisite: A.164/27 - - Additional for 420 (Bad Extension) response

Table A.229: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Unsupported	[26] 20.40	m	m	[26] 20.40	c3	c3
c3:	IF A.162/18 THEN m ELSE i - - reading the contents of the Unsupported header before proxying the 420 response to a method other than REGISTER.						

Prerequisite A.163/11 - - NOTIFY response

Prerequisite: A.164/28 OR A.164/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.229A: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	c1	c1	[48] 2	n/a	n/a
c1:	IF A.162/47 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Table A.230: Void

Prerequisite A.163/11 - - NOTIFY response

Prerequisite: A.164/35 - - Additional for 485 (Ambigious) response

Table A.230A: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Contact	[26] 20.10	m	m	[26] 20.10	i	i

Prerequisite A.163/11 - - NOTIFY response

Prerequisite: A.164/39 - - Additional for 489 (Bad Event) response

Table A.231: Supported header fields within the NOTIFY response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.						
NOTE:	c1 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.						

Prerequisite A.163/11 - - NOTIFY response

Prerequisite: A.164/46 - - Additional for 504 (Server Time-out) response

Table A.231A: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Restoration-Info	subclause 7.2.11	n/a	c1	subclause 7.2.11	n/a	n/a
c1:	IF A.4/110 THEN o ELSE n/a - - HSS based P-CSCF restoration.						

Prerequisite A.163/11 - - NOTIFY response

Table A.232: Supported message bodies within the NOTIFY response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.2.4.9 OPTIONS method

Prerequisite A.163/12 - - OPTIONS request

Table A.233: Supported header fields within the OPTIONS request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
1A	Accept-Contact	[56B] 9.2	c28	c28	[56B] 9.2	c28	c29
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i
3A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
4	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
5	Authorization	[26] 20.7	m	m	[26] 20.7	i	i
6	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
7	Call-Info	[26] 20.9	m	m	[26] 20.9	c4	c4
7A	Cellular-Network-Info	7.2.15	n/a	c59	7.2.15	n/a	c60
8	Contact	[26] 20.10	m	m	[26] 20.10	i	i
9	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
10	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
11	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
12	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
13	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
14	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
15	Date	[26] 20.17	m	m	[26] 20.17	c2	c2
15A	Feature-Caps	[190]	c57	c57	[190]	c57	c57
16	From	[26] 20.20	m	m	[26] 20.20	m	m
16A	Geolocation	[89] 4.1	c36	c36	[89] 4.1	c37	c37
16B	Geolocation-Routing	[89] 4.1	c36	c36	[89] 4.1	c37	c37
16C	History-Info	[66] 4.1	c32	c32	[66] 4.1	c32	c32
16D	Max-Breadth	[117] 5.8	c41	c41	[117] 5.8	c42	c42
17	Max-Forwards	[26] 20.22	m	m	[26] 20.22	m	m
18	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
19	Organization	[26] 20.25	m	m	[26] 20.25	c3	c3
19A	P-Access-Network-Info	[52] 4.4, [234] 2	c23	c23	[52] 4.4, [234] 2	c24	c24
19B	P-Asserted-Identity	[34] 9.1	c10	c10	[34] 9.1	c11	c11
19C	P-Asserted-Service	[121] 4.1	c39	c39	[121] 4.1	c40	c40
19D	P-Called-Party-ID	[52] 4.2	c14	c14	[52] 4.2	c15	c16
19E	P-Charging-Function-Addresses	[52] 4.5	c21	c21	[52] 4.5	c22	c22
19F	P-Charging-Vector	[52] 4.6	c19	c19	[52] 4.6	c20	c20
19H	P-Preferred-Identity	[34] 9.2	x	c54	[34] 9.2	c9	c55
19I	P-Preferred-Service	[121] 4.2	x	x	[121] 4.2	c38	c38
19J	P-Private-Network-Indication	[134]	c48	c48	[134]	c48	c48
19K	P-Profile-Key	[97] 5	c34	c34	[97] 5	c35	c35
19L	P-Served-User	[133] 6	c49	c49	[133] 6	c49	c49
19M	P-User-Database	[82] 4	c33	c33	[82] 4	c33	c33
19N	P-Visited-Network-ID	[52] 4.3	c17	o	[52] 4.3	c18	o
19O	Privacy	[33] 4.2	c12	c12	[33] 4.2	c13	c13
20	Proxy-Authorization	[26] 20.28	m	m	[26] 20.28	c8	c8
21	Proxy-Require	[26] 20.29	m	m	[26] 20.29	m	m
21A	Reason	[34A] 2	c26	c26	[34A] 2	c27	c27
22	Record-Route	[26] 20.30	m	m	[26] 20.30	c7	c7
22A	Recv-Info	[25] 5.2.3	c52	c52	[25] 5.2.3	c53	c53
22B	Referred-By	[59] 3	c30	c30	[59] 3	c31	c31
22C	Reject-Contact	[56B] 9.2	c28	c28	[56B] 9.2	c28	c29
22D	Relayed-Charge	7.2.12	n/a	c58	7.2.12	n/a	c58
22E	Request-Disposition	[56B] 9.1	c28	c28	[56B] 9.1	c28	c28
23	Require	[26] 20.32	m	m	[26] 20.32	c5	c5
23A	Resource-Priority	[116] 3.1	c47	c47	[116] 3.1	c47	c47
24	Route	[26] 20.34	m	m	[26] 20.34	m	m
24A	Security-Client	[48] 2.3.1	x	x	[48] 2.3.1	c25	c25
24B	Security-Verify	[48] 2.3.1	x	x	[48] 2.3.1	c25	c25
24C	Session-ID	[162]	c56	c56	[162]	c56	c56
25	Supported	[26] 20.37	m	m	[26] 20.37	c6	c6
26	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
27	To	[26] 20.39	m	m	[26] 20.39	m	m
28	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i

29	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.						
c2:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.						
c3:	IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.						
c4:	IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header.						
c5:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.						
c6:	IF A.162/16 THEN m ELSE i - - reading the contents of the Supported header before proxying the response.						
c7:	IF A.162/14 THEN o ELSE i - - the requirement to be able to insert itself in the subsequent transactions in a dialog.						
c8:	IF A.162/8A THEN m ELSE i - - authentication between UA and proxy.						
c9:	IF A.162/30A THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity.						
c10:	IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.						
c11:	IF A.162/30A or A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.						
c12:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).						
c13:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.						
c14:	IF A.162/37 THEN m ELSE n/a - - the P-Called-Party-ID header extension.						
c15:	IF A.162/37 THEN i ELSE n/a - - the P-Called-Party-ID header extension.						
c16:	IF A.162/37 AND A.3/2 THEN m ELSE IF A.162/37 AND (A.3/3 OR A.3/9A) THEN i ELSE n/a - - the P-Called-Party-ID header extension and P-CSCF or (I-CSCF or IBCF (THIG)).						
c17:	IF A.162/38 THEN m ELSE n/a - - the P-Visited-Network-ID header extension.						
c18:	IF A.162/39 THEN m ELSE i - - reading, or deleting the P-Visited-Network-ID header before proxying the request or response.						
c19:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.						
c20:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.						
c21:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.						
c22:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.						
c23:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.						
c24:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.						
c25:	IF A.162/47 OR A.162/47A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.						
c26:	IF A.162/48 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.						
c27:	IF A.162/48 THEN i ELSE n/a - - the Reason header field for the session initiation protocol.						
c28:	IF A.162/50 THEN m ELSE n/a - - caller preferences for the session initiation protocol.						
c29:	IF A.162/50 AND A.4/3 THEN m ELSE IF A.162/50 AND NOT A.4/3 THEN i ELSE n/a - - caller preferences for the session initiation protocol, and S-CSCF.						
c30:	IF A.162/53 THEN i ELSE n/a - - the SIP Referred-By mechanism.						
c31:	IF A.162/53 THEN m ELSE n/a - - the SIP Referred-By mechanism.						
c32:	IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.						
c33:	IF A.162/60 THEN m ELSE n/a - - the P-User-Database private header extension.						
c34:	IF A.162/66A THEN m ELSE n/a - - making the first query to the database in order to populate the P-Profile-Key header.						
c35:	IF A.162/66B THEN m ELSE n/a - - using the information in the P-Profile-Key header.						
c36:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.						
c37:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.						
c38:	IF A.162/84A THEN m ELSE n/a - - act as authentication entity within the trust domain for asserted service.						
c39:	IF A.162/84 THEN m ELSE n/a - - SIP extension for the identification of services.						
c40:	IF A.162/84 OR A.162/30B THEN m ELSE i - - SIP extension for the identification of services or subsequent entity within trust network that can route outside the trust network.						
c41:	IF A.162/81 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.						

c42:	IF A.162/81 AND A.162/6 THEN m ELSE IF A.162/81 AND NOT A.162/6 THEN i ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, forking of initial requests.
c47:	IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.
c48:	IF A.162/87 THEN m ELSE n/a - - the SIP P-Private-Network-Indication private-header (P-Header).
c49:	IF A.162/88 THEN m ELSE n/a - - the SIP P-Served-User private header.
c52:	IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.
c53:	IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.
c54:	IF A.162/30C THEN m ELSE x - - act as entity passing on identity transparently independent of trust domain.
c55:	IF A.162/30A OR A.162/30C THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity, act as entity passing on identity transparently independent of trust domain.
c56:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c57:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.
c58:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c59:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c60:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
NOTE:	c1 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.

Prerequisite A.163/12 - - OPTIONS request

Table A.234: Supported message bodies within the OPTIONS request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

Table A.235: Void

Prerequisite A.163/13 - - OPTIONS response

Prerequisite: A.164/1 - - Additional for 100 (Trying) response

Table A.235A: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	c2	c2
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF (A.162/9 AND A.162/5) OR A.162/4 THEN m ELSE n/a - - stateful proxy behaviour that inserts date, or stateless proxies.						
c2:	IF A.162/4 THEN i ELSE m - - Stateless proxy passes on.						

Prerequisite A.163/13 - - OPTIONS response for all remaining status-codes

Table A.236: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Call-Info	[26] 20.9	m	m	[26] 20.9	c3	c3
1B	Cellular-Network-Info	7.2.15	n/a	c25	7.2.15	n/a	c26
2	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
3	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
4	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	m	m	[26] 20.17	c1	c1
9	From	[26] 20.20	m	m	[26] 20.20	m	m
9A	Geolocation-Error	[89] 4.3	c17	c17	[89] 4.3	c18	c18
9B	History-Info	[66] 4.1	c16	c16	[66] 4.1	c16	c16
10	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
11	Organization	[26] 20.25	m	m	[26] 20.25	c2	c2
11A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c13	c13	[52] 4.4, [52A] 4, [234] 2	c14	c14
11B	P-Asserted-Identity	[34] 9.1	c5	c5	[34] 9.1	c6	c6
11C	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c11	c11	[52] 4.5, [52A] 4	c12	c12
11D	P-Charging-Vector	[52] 4.6, [52A] 4	c9	c9	[52] 4.6, [52A] 4	c10	c10
11F	P-Preferred-Identity	[34] 9.2	x	x	[34] 9.2	c4	n/a
11G	Privacy	[33] 4.2	c7	c7	[33] 4.2	c8	c8
11H	Recv-Info	[25] 5.2.3	c21	c21	[25] 5.2.3	c22	c22
11I	Relayed-Charge	7.2.12	n/a	c24	7.2.12	n/a	c24
11J	Require	[26] 20.32	m	m	[26] 20.32	c15	c15
11K	Server	[26] 20.35	m	m	[26] 20.35	i	i
11K	Session-ID	[162]	c23	c23	[162]	c23	c23
12	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
13	To	[26] 20.39	m	m	[26] 20.39	m	m
13A	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
14	Via	[26] 20.42	m	m	[26] 20.42	m	m
15	Warning	[26] 20.43	m	m	[26] 20.43	i	i

c1:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c2:	IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.
c3:	IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header.
c4:	IF A.162/30A THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity.
c5:	IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c6:	IF A.162/30A or A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.
c7:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c8:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c9:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c10:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c11:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c12:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c13:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c14:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c15:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c16:	IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c17:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c18:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c21:	IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.
c22:	IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.
c23:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c24:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c25:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c26:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.

Prerequisite A.163/13 - - OPTIONS response

Prerequisite: A.164/102 - - Additional for 2xx response

Table A.237: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
1A	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
1B	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i
1C	Accept-Resource-Priority	[116] 3.2	c12	c12	[116] 3.2	c12	c12
2	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
3	Authentication-Info	[26] 20.6	m	m	[26] 20.6	i	i
5	Contact	[26] 20.10	m	m	[26] 20.10	i	i
6	Feature-Caps	[190]	c14	c14	[190]	c14	c14
7	Recv-Info	[25] 5.2.3	c7	c7	[25] 5.2.3	c8	c8
9	Record-Route	[26] 20.30	m	m	[26] 20.30	c3	c3
12	Supported	[26] 20.37	m	m	[26] 20.37	i	i
c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.						
c3:	IF A.162/15 THEN o ELSE i - - the requirement to be able to use separate URIs in the upstream direction and downstream direction when record routeing.						
c7:	IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.						
c8:	IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.						
c12:	IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						
c14:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.						

Prerequisite A.163/13 - - OPTIONS response

Prerequisite: A.164/103 OR A.164/104 OR A.164/105 OR A.164/106 - - Additional for 3xx – 6xx response

Table A.237A: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	m	m	[26] 20.18	i	i
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.162/125 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.163/13 - - OPTIONS response

Prerequisite: A.164/103 OR A.164/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.238: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Contact	[26] 20.10	m	m	[26] 20.10	c1	c1
c1:	IF A.162/19E THEN m ELSE i - - deleting Contact headers.						

Prerequisite A.163/13 - - OPTIONS response

Prerequisite: A.164/14 - - Additional for 401 (Unauthorized) response

Table A.239: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
10	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/13 - - OPTIONS response

Prerequisite: A.164/17 OR A.164/23 OR A.164/30 OR A.164/36 OR A.164/42 OR A.164/45 OR A.164/50 OR A.164/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.240: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Retry-After	[26] 20.33	m	m	[26] 20.33	i	i

Table A.241: Void

Prerequisite A.163/13 - - OPTIONS response

Prerequisite: A.164/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.242: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
8	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Table A.242A: Void

Prerequisite A.163/13 - - OPTIONS response

Prerequisite: A.164/25 - - Additional for 415 (Unsupported Media Type) response

Table A.243: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i

Prerequisite A.163/13 - - OPTIONS response

Prerequisite: A.164/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.243A: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						

Prerequisite A.163/13 - - OPTIONS response

Prerequisite: A.164/27 - - Additional for 420 (Bad Extension) response

Table A.244: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
7	Unsupported	[26] 20.40	m	m	[26] 20.40	c3	c3
c3:	IF A.162/18 THEN m ELSE i - - reading the contents of the Unsupported header before proxying the 420 response to a method other than REGISTER.						

Prerequisite A.163/13 - - OPTIONS response

Prerequisite: A.164/28 OR A.164/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.244A: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	c1	c1	[48] 2	n/a	n/a
c1:	IF A.162/47 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Table A.245: Void

Prerequisite A.163/13 - - OPTIONS response

Prerequisite: A.164/46 - - Additional for 504 (Server Time-out) response

Table A.245A: Supported header fields within the OPTIONS response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Restoration-Info	subclause 7.2.11	n/a	c1	subclause 7.2.11	n/a	n/a
c1:	IF A.4/110 THEN o ELSE n/a - - HSS based P-CSCF restoration.						

Prerequisite A.163/13 - - OPTIONS response

Table A.246: Supported message bodies within the OPTIONS response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.2.4.10 PRACK method

Prerequisite A.163/14 - - PRACK request

Table A.247: Supported header fields within the PRACK request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
1A	Accept-Contact	[56B] 9.2	c18	c18	[56B] 9.2	c19	c19
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i
3A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
4	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
5	Authorization	[26] 20.7	m	m	[26] 20.7	i	i
6	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
6A	Cellular-Network-Info	7.2.15	n/a	c51	7.2.15	n/a	c52
7	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	c3
8	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	c3
9	Content-Language	[26] 20.13	m	m	[26] 20.13	i	c3
10	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
11	Content-Type	[26] 20.15	m	m	[26] 20.15	i	c3
12	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
13	Date	[26] 20.17	m	m	[26] 20.17	c2	c2
14	From	[26] 20.20	m	m	[26] 20.20	m	m
14A	Max-Breadth	[117] 5.8	c26	c26	[117] 5.8	c27	c27
15	Max-Forwards	[26] 20.22	m	m	[26] 20.22	m	m
16	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	c3
16A	P-Access-Network-Info	[52] 4.4, [234] 2	c14	c14	[52] 4.4, [234] 2	c15	c15
16B	P-Charging-Function-Addresses	[52] 4.5	c12	c12	[52] 4.5	c13	c13
16C	P-Charging-Vector	[52] 4.6	c10	c10	[52] 4.6	c11	c11
16E	P-Early-Media	[109] 8	o	c22	[109] 8	o	c22
16EA	Priority-Share	Subclause 7.2.16	n/a	c53	Subclause 7.2.16	n/a	c53
16F	Privacy	[33] 4.2	c8	c8	[33] 4.2	c9	c9
17	Proxy-Authorization	[26] 20.28	m	m	[26] 20.28	c4	c4
18	Proxy-Require	[26] 20.29	m	m	[26] 20.29	m	m
19	RAck	[27] 7.2	m	m	[27] 7.2	i	i
19A	Reason	[34A] 2	c16	c16	[34A] 2	c17	c17
20	Record-Route	[26] 20.30	m	m	[26] 20.30	c7	c7
20A	Recv-Info	[25] 5.2.3	c28	c28	[25] 5.2.3	c29	c29
20B	Referred-By	[59] 3	c20	c20	[59] 3	c21	c21
20C	Reject-Contact	[56B] 9.2	c18	c18	[56B] 9.2	c19	c19
20D	Relayed-Charge	7.2.12	n/a	c49	7.2.12	n/a	c49
20E	Request-Disposition	[56B] 9.1	c18	c18	[56B] 9.1	c19	c19
21	Require	[26] 20.32	m	m	[26] 20.32	c5	c5
21A	Resource-Priority	[16] 3.1	c47	c47	[116] 3.1	c47	c47
21B	Resource-Share	Subclause 4.15	n/a	c50	Subclause 4.15	n/a	c50
22	Route	[26] 20.34	m	m	[26] 20.34	m	m
22A	Session-ID	[162]	c48	c48	[162]	c48	c48
23	Supported	[26] 20.37	m	m	[26] 20.37	c6	c6
24	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
25	To	[26] 20.39	m	m	[26] 20.39	m	m
26	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
27	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.
c2:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c3:	IF A.3/2 OR A.3/4 THEN m ELSE i - - P-CSCF or S-CSCF.
c4:	IF A.162/8A THEN m ELSE i - - authentication between UA and proxy.
c5:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c6:	IF A.162/16 THEN m ELSE i - - reading the contents of the Supported header before proxying the response.
c7:	IF A.162/14 THEN o ELSE i - - the requirement to be able to insert itself in the subsequent transactions in a dialog.
c8:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c9:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c10:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c11:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c12:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c13:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c14:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c15:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c16:	IF A.162/48 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.
c17:	IF A.162/48 THEN i ELSE n/a - - the Reason header field for the session initiation protocol.
c18:	IF A.162/50 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c19:	IF A.162/50 THEN i ELSE n/a - - caller preferences for the session initiation protocol.
c20:	IF A.162/53 THEN i ELSE n/a - - the SIP Referred-By mechanism.
c21:	IF A.162/53 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c22:	IF A.162/76 THEN m ELSE n/a - - the SIP P-Early-Media private header extension for authorization of early media.
c26:	IF A.162/81 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c27:	IF A.162/81 AND A.162/6 THEN m ELSE IF A.162/81 AND NOT A.162/6 THEN i ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, forking of initial requests.
c28:	IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.
c29:	IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.
c47:	IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.
c48:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c49:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c50:	IF A.4/112 THEN o ELSE n/a - - resource sharing.
c51:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c52:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c53:	IF A.162/124 THEN o ELSE n/a - - priority sharing.
NOTE:	c1 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.

Prerequisite A.163/14 - - PRACK request

Table A.248: Supported message bodies within the PRACK request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

Table A.249: Void

Prerequisite A.163/15 - - PRACK response

Prerequisite: A.164/1 - - Additional for 100 (Trying) response

Table A.249A: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	c2	c2
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF (A.162/9 AND A.162/5) OR A.162/4 THEN m ELSE n/a - - stateful proxy behaviour that inserts date, or stateless proxies.						
c2:	IF A.162/4 THEN i ELSE m - - Stateless proxy passes on.						

Prerequisite A.163/15 - - PRACK response for all remaining status-codes

Table A.250: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Cellular-Network-Info	7.2.15	n/a	c18	7.2.15	n/a	c19
2	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	c2
3	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	c2
4	Content-Language	[26] 20.13	m	m	[26] 20.13	i	c2
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	i	c2
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	m	m	[26] 20.17	c1	c1
9	From	[26] 20.20	m	m	[26] 20.20	m	m
10	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	c2
10A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c9	c9	[52] 4.4, [52A] 4, [234] 2	c10	c10
10B	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c7	c7	[52] 4.5, [52A] 4	c8	c8
10C	P-Charging-Vector	[52] 4.6, [52A] 4	c5	c5	[52] 4.6, [52A] 4	c6	c6
10F	Privacy	[33] 4.2	c3	c3	[33] 4.2	c4	c4
10G	Recv-Info	[25] 5.2.3	c14	c14	[25] 5.2.3	c15	c15
10H	Relayed-Charge	7.2.12	n/a	c17	7.2.12	n/a	c17
10I	Require	[26] 20.32	m	m	[26] 20.32	c11	c11
10J	Server	[26] 20.35	m	m	[26] 20.35	i	i
10K	Session-ID	[162]	c16	c16	[162]	c16	c16
11	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
12	To	[26] 20.39	m	m	[26] 20.39	m	m
12A	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
13	Via	[26] 20.42	m	m	[26] 20.42	m	m
14	Warning	[26] 20.43	m	m	[26] 20.43	i	i

c1: IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c2: IF A.3/2 OR A.3/4 THEN m ELSE i - - P-CSCF or S-CSCF.
c3: IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c4: IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c5: IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c6: IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c7: IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c8: IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c9: IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c10: IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c11: IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c14: IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.
c15: IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.
c16: IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c17: IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c18: IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c19: IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.

Prerequisite A.163/15 - - PRACK response

Prerequisite: A.164/102 - - Additional for 2xx response

Table A.251: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
0B	Authentication-Info	[26] 20.6	m	m	[26] 20.6	i	i
0C	Accept-Resource-Priority	[116] 3.2	c12	c12	[116] 3.2	c12	c12
0D	P-Early-Media	[109] 8	o	c4	[109] 8	o	c4
0E	Priority-Share	Subclause 7.2.16	n/a	c14	Subclause 7.2.16	n/a	c14
1	Record-Route	[26] 20.30	m	m	[26] 20.30	c3	c3
2	Recv-Info	[25] 5.2.3	c6	c6	[25] 5.2.3	c7	c7
2A	Resource-Share	Subclause 4.15	n/a	c13	Subclause 4.15	n/a	c13
3	Supported	[26] 20.37	m	m	[26] 20.37	i	i
c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.						
c3:	IF A.162/15 THEN o ELSE i - - the requirement to be able to use separate URIs in the upstream direction and downstream direction when record routeing.						
c4:	IF A.162/76 THEN m ELSE n/a - - the SIP P-Early-Media private header extension for authorization of early media.						
c6:	IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.						
c7:	IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.						
c12:	IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						
c13:	IF A.4/112 THEN o ELSE n/a - - resource sharing.						
c14:	IF A.162/124 THEN o ELSE n/a - - priority sharing.						

Prerequisite A.163/3 - - PRACK response

Prerequisite: A.164/103 OR A.164/104 OR A.164/105 OR A.164/106 - - Additional for 3xx – 6xx response

Table A.251A: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	m	m	[26] 20.18	i	i
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.162/125 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.163/15 - - PRACK response

Prerequisite: A.164/103 OR A.164/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.252: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Contact	[26] 20.10	m	m	[26] 20.10	c1	c1
c1:	IF A.162/19E THEN m ELSE i - - deleting Contact headers.						

Prerequisite A.163/15 - - PRACK response

Prerequisite: A.164/14 - - Additional for 401 (Unauthorized) response

Table A.253: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
8	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/15 - - PRACK response

Prerequisite: A.164/17 OR A.164/23 OR A.164/30 OR A.164/36 OR A.164/42 OR A.164/45 OR A.164/50 OR A.164/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.254: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Retry-After	[26] 20.33	m	m	[26] 20.33	i	i

Table A.255: Void

Prerequisite A.163/15 - - PRACK response

Prerequisite: A.164/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.256: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
6	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/15 - - PRACK response

Prerequisite: A.164/25 - - Additional for 415 (Unsupported Media Type) response

Table A.257: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i

Prerequisite A.163/15 - - PRACK response

Prerequisite: A.164/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.257A: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						

Prerequisite A.163/15 - - PRACK response

Prerequisite: A.164/27 - - Addition for 420 (Bad Extension) response

Table A.258: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Unsupported	[26] 20.40	m	m	[26] 20.40	c3	c3
c3:	IF A.162/18 THEN m ELSE i - - reading the contents of the Unsupported header before proxying the 420 response to a method other than REGISTER.						

Prerequisite A.163/15 - - PRACK response

Prerequisite: A.164/28 OR A.164/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.258A: Supported header fields within the PRACK response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	c1	c1	[48] 2	n/a	n/a
c1:	IF A.162/47 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Table A.259: Void

Prerequisite A.163/15 - - PRACK response

Table A.260: Supported message bodies within the PRACK response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.2.4.10A PUBLISH method

Prerequisite A.163/15A - - PUBLISH request

Table A.260A: Supported header fields within the PUBLISH request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Contact	[56B] 9.2	c28	c28	[56B] 9.2	c28	c29
2	Allow	[26] 20.5	m	m	[26] 20.5	i	i
3	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c29	c29
4	Authorization	[26] 20.7	m	m	[26] 20.7	i	i
5	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
6	Call-Info	[26] 24.9	m	m	[26] 24.9	c4	c4
6A	Cellular-Network-Info	7.2.15	n/a	c72	7.2.15	n/a	c73
6B	Contact	[70] 4	o	o	[70] 6	n/a	n/a
7	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
8	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
9	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
10	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
11	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
12	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
13	Date	[26] 20.17	m	m	[26] 20.17	c2	c2
14	Event	[70] 4, 6	m	m	[70] 4, 6	m	m
15	Expires	[26] 20.19, [70] 4, 5, 6	m	m	[26] 20.19, [70] 4, 5, 6	i	i
15A	Feature-Caps	[190]	c70	c70	[190]	c70	c70
16	From	[26] 20.20	m	m	[26] 20.20	m	m
16A	Geolocation	[89] 4.1	c46	c46	[89] 4.1	c47	c47
16B	Geolocation-Routing	[89] 4.1	c46	c46	[89] 4.1	c47	c47
16C	History-Info	[66] 4.1	c32	c32	[66] 4.1	c32	c32
17	In-Reply-To	[26] 20.21	m	m	[26] 20.21	i	i
17A	Max-Breadth	[117] 5.8	c44	c44	[117] 5.8	c45	c45
18	Max-Forwards	[26] 20.22	m	m	[26] 20.22	m	m
19	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
20	Organization	[26] 20.25	m	m	[26] 20.25	c3	c3
21	P-Access-Network-Info	[52] 4.4, [234] 2	c23	c23	[52] 4.4, [234] 2	c24	c24
22	P-Asserted-Identity	[34] 9.1	c10	c10	[34] 9.1	c11	c11
22A	P-Asserted-Service	[121] 4.1	c38	c38	[121] 4.1	c39	c39
23	P-Called-Party-ID	[52] 4.2	c14	c14	[52] 4.2	c15	c16
24	P-Charging-Function-Addresses	[52] 4.5	c21	c21	[52] 4.5	c22	c22
25	P-Charging-Vector	[52] 4.6	c19	c19	[52] 4.6	c20	c20
26	P-Preferred-Identity	[34] 9.2	x	c69	[34] 9.2	c9	c9

26A	P-Preferred-Service	[121] 4.2	x	x	[121] 4.2	c37	c37
26B	P-Private-Network-Indication	[134]	c40	c40	[134]	c40	c40
26C	P-Profile-Key	[97] 5	c34	c34	[97] 5	c35	c35
26D	P-Served-User	[133] 6	c41	c41	[133] 6	c41	c41
26E	P-User-Database	[82] 4	c33	c33	[82] 4	c33	c33
27	P-Visited-Network-ID	[52] 4.3	c17	o	[52] 4.3	c18	o
28	Priority	[26] 20.26	m	m	[26] 20.26	i	c50
29	Privacy	[33] 4.2	c12	c12	[33] 4.2	c13	c13
30	Proxy-Authorization	[26] 20.28	m	m	[26] 20.28	c7	c7
31	Proxy-Require	[26] 20.29	m	m	[26] 20.29	m	m
32	Reason	[34A] 2	c8	c8	[34A] 2	c1	c1
33	Referred-By	[59] 3	c30	c30	[59] 3	c31	c31
34	Reject-Contact	[56B] 9.2	c27	c27	[56B] 9.2	c27	c28
34A	Relayed-Charge	7.2.12	n/a	c71	7.2.12	n/a	c71
34B	Reply-To	[26] 20.31	m	m	[26] 20.31	i	i
35	Request-Disposition	[56B] 9.1	c27	c27	[56B] 9.1	c27	c27
36	Require	[26] 20.32	m	m	[26] 20.32	c5	c5
36A	Resource-Priority	[116] 3.1	c36	c36	[116] 3.1	c36	c36
37	Route	[26] 20.34	m	m	[26] 20.34	m	m
38	Security-Client	[48] 2.3.1	x	x	[48] 2.3.1	c25	c25
39	Security-Verify	[48] 2.3.1	x	x	[48] 2.3.1	c25	c25
39A	Session-ID	[162]	c48	c48	[162]	c48	c48
40	SIP-If-Match	[70] 11.3.2	m	m	[70] 11.3.2	i	i
41	Subject	[26] 20.36	m	m	[26] 20.36	i	i
42	Supported	[26] 20.37	m	m	[26] 20.37	c6	c6
43	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
44	To	[26] 20.39	m	m	[26] 20.39	m	m
45	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
46	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1: IF A.162/48 THEN i ELSE n/a - - the Reason header field for the session initiation protocol.
c2: IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c3: IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.
c4: IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header.
c5: IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c6: IF A.162/16 THEN m ELSE i - - reading the contents of the Supported header before proxying the response.
c7: IF A.162/8A THEN m ELSE i - - authentication between UA and proxy.
c8: IF A.162/48 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.
c9: IF A.162/30A OR A.162/30C THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity, act as entity passing on identity transparently independent of trust domain.
c10: IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c11: IF A.162/30A or A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.
c12: IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c13: IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c14: IF A.162/37 THEN m ELSE n/a - - the P-Called-Party-ID header extension.
c15: IF A.162/37 THEN i ELSE n/a - - the P-Called-Party-ID header extension.
c16: IF A.162/37 AND A.3/2 THEN m ELSE IF A.162/37 AND (A.3/3 OR A.3/9A) THEN i ELSE n/a - - the P-Called-Party-ID header extension and P-CSCF or (I-CSCF or IBCF (THIG).
c17: IF A.162/38 THEN m ELSE n/a - - the P-Visited-Network-ID header extension.
c18: IF A.162/39 THEN m ELSE i - - reading, or deleting the P-Visited-Network-ID header before proxying the request or response.
c19: IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c20: IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c21: IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c22: IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c23: IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c24: IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c25: IF A.162/47 OR A.162/47A THEN o ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media (note 1).
c27: IF A.162/50 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c28: IF A.162/50 AND A.4/3 THEN m ELSE IF A.162/50 AND NOT A.4/3 THEN i ELSE n/a - - caller preferences for the session initiation protocol, and S-CSCF.
c29: IF A.4/20 THEN m ELSE i - - SIP specific event notification extension (note 2).
c30: IF A.162/53 THEN i ELSE n/a - - the SIP Referred-By mechanism.
c31: IF A.162/53 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c32: IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c33: IF A.162/60 THEN m ELSE n/a - - the P-User-Database private header extension.
c34: IF A.162/66A THEN m ELSE n/a - - making the first query to the database in order to populate the P-Profile-Key header.
c35: IF A.162/66B THEN m ELSE n/a - - using the information in the P-Profile-Key header.
c36: IF A.162/80B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.
c37: IF A.162/84A THEN m ELSE n/a - - act as authentication entity within the trust domain for asserted service.
c38: IF A.162/84 THEN m ELSE n/a - - SIP extension for the identification of services.
c39: IF A.162/84 OR A.162/30B THEN m ELSE i - - SIP extension for the identification of services or subsequent entity within trust network that can route outside the trust network.
c40: IF A.162/87 THEN m ELSE n/a - - the SIP P-Private-Network-Indication private-header (P-Header).
c41: IF A.162/88 THEN m ELSE n/a - - the SIP P-Served-User private header.

c44:	IF A.162/81 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c45:	IF A.162/81 AND A.162/6 THEN m ELSE IF A.162/81 AND NOT A.162/6 THEN i ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, forking of initial requests.
c46:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c47:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c48:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c69:	IF A.162/30C THEN m ELSE x - - act as entity passing on identity transparently independent of trust domain.
c50:	IF A.162/115 THEN m ELSE i - - PSAP callback indicator.
c70:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.
c71:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c72:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c73:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
NOTE 1:	Support of this header in this method is dependent on the security mechanism and the security architecture which is implemented.
NOTE 2:	c29 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.

Prerequisite A.163/15A - - PUBLISH request

Table A.260B: Supported message bodies within the PUBLISH request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	application/vnd.3gpp.mcptt-info+xml	[8ZE]	n/a	c1	[8ZE]	n/a	c1
2	application/poc-settings+xml	[110]	o	c1	[110]	o	c1
3	application/pidf+xml	[242]	o	c1	[242]	o	c1
c1:	A.3/3 OR A.3/4 OR A.3/5 OR A.3/7C OR A.3/9A OR A.3/10 OR A.3/11 OR A.3/13A THEN m ELSE n/a - - I-CSCF, S-CSCF, BGCF, AS acting as proxy, IBCF (THIG), additional routeing functionality, E-CSCF, ISC gateway function (THIG).						

Prerequisite A.163/15B - - PUBLISH response

Prerequisite: A.164/1 - - Additional for 100 (Trying) response

Table A.260BA: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	c2	c2
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF (A.162/9 AND A.162/5) OR A.162/4 THEN m ELSE n/a - - stateful proxy behaviour that inserts date, or stateless proxies.						
c2:	IF A.162/4 THEN i ELSE m - - Stateless proxy passes on.						

Prerequisite A.163/15B - - PUBLISH response for all remaining status-codes

Table A.260C: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Call-Info	[26] 24.9	m	m	[26] 24.9	c3	c3
2A	Cellular-Network-Info	7.2.15	n/a	c23	7.2.15	n/a	c24
3	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
4	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
5	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
6	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
7	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
8	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
9	Date	[26] 20.17	m	m	[26] 20.17	c1	c1
9A	Expires	[26] 20.19 [70] 4, 5, 6	m	m	[26] 20.19 [70] 4, 5, 6	i	i
10	From	[26] 20.20	m	m	[26] 20.20	m	m
10A	Geolocation-Error	[89] 4.3	c19	c19	[89] 4.3	c20	c20
10B	History-Info	[66] 4.1	c16	c16	[66] 4.1	c16	c16
11	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
12	Organization	[26] 20.25	m	m	[26] 20.25	c2	c2
13	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c13	c13	[52] 4.4, [52A] 4, [234] 2	c14	c14
14	P-Asserted-Identity	[34] 9.1	c5	c5	[34] 9.1	c6	c6
15	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c11	c11	[52] 4.5, [52A] 4	c12	c12
16	P-Charging-Vector	[52] 4.6, [52A] 4	c9	c9	[52] 4.6, [52A] 4	c10	c10
17	P-Preferred-Identity	[34] 9.2	x	x	[34] 9.2	c4	n/a
18	Privacy	[33] 4.2	c7	c7	[33] 4.2	c8	c8
18A	Relayed-Charge	7.2.12	n/a	c22	7.2.12	n/a	c22
19	Require	[26] 20.32	m	m	[26] 20.32	c15	c15
20	Server	[26] 20.35	m	m	[26] 20.35	i	i
20A	Session-ID	[162]	c21	c21	[162]	c21	c21
21	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
22	To	[26] 20.39	m	m	[26] 20.39	m	m
23	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
24	Via	[26] 20.42	m	m	[26] 20.42	m	m
25	Warning	[26] 20.43	m	m	[26] 20.43	i	i

c1:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c2:	IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.
c3:	IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header.
c4:	IF A.162/30A THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity.
c5:	IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c6:	IF A.162/30A or A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.
c7:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c8:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c9:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c10:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c11:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c12:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c13:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c14:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c15:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c16:	IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c19:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c20:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c21:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c22:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c23:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c24:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.

Prerequisite A.163/15B - - PUBLISH response

Prerequisite: A.164/102 - - Additional for 2xx response

Table A.260D: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Authentication-Info	[26] 20.6	m	m	[26] 20.6	i	i
3	Expires	[26] 20.19, [70] 4, 5, 6	m	m	[26] 20.19, [70] 4, 5, 6	i	i
3A	Feature-Caps	[190]	c6	c6	[190]	c6	c6
4	SIP-Etag	[70] 11.3.1	m	m	[70] 11.3.1	i	i
5	Supported	[26] 20.37	m	m	[26] 20.37	i	i
c6:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.						

Prerequisite A.163/15B - - PUBLISH response

Prerequisite: A.164/6 - - Additional for 200 (OK) response

Table A.260DAA: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.162/80B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.						

Prerequisite A.163/15B - - PUBLISH response

Prerequisite: A.164/103 OR A.164/104 OR A.164/105 OR A.164/106 - - Additional for 3xx – 6xx response

Table A.260DA: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	m	m	[26] 20.18	i	i
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.162/125 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.163/15B - - PUBLISH response

Prerequisite: A.164/103 OR A.164/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.260E: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Contact	[26] 20.10	m	m	[26] 20.10	c1	c1
c1:	IF A.162/19E THEN m ELSE i - - deleting Contact headers.						

Prerequisite A.163/15B - - PUBLISH response

Prerequisite: A.164/8 OR A.164/9 OR A.164/10 OR A.164/11 OR A.164/12 - - Additional for 401 (Unauthorized) response

Table A.260F: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
5	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/15B - - PUBLISH response

Prerequisite: A.164/17 OR A.164/23 OR A.164/30 OR A.164/36 OR A.164/42 OR A.164/45 OR A.164/50 OR A.164/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.260G: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Retry-After	[26] 20.33	m	m	[26] 20.33	i	i

Table A.260H: Void

Prerequisite A.163/15B - - PUBLISH response

Prerequisite: A.164/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.260I: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
5	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Table A.260IA: Void

Prerequisite A.163/15B - - PUBLISH response

Prerequisite: A.164/25 - - Additional for 415 (Unsupported Media Type) response

Table A.260J: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i

Prerequisite A.163/15B - - PUBLISH response

Prerequisite: A.164/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.260JA: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.162/80B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.						

Prerequisite A.163/15B - - PUBLISH response

Prerequisite: A.164/27 - - Additional for 420 (Bad Extension) response

Table A.260K: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Unsupported	[26] 20.40	m	m	[26] 20.40	c3	c3
c3:	IF A.162/18 THEN m ELSE i - - reading the contents of the Unsupported header before proxying the 420 response to a method other than REGISTER.						

Prerequisite A.163/15B - - PUBLISH response

Prerequisite: A.164/28 OR A.164/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.260L: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	c1	c1	[48] 2	n/a	n/a
c1:	IF A.162/47 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Prerequisite A.163/15B - - PUBLISH response

Prerequisite: A.164/29 - - Additional for 423 (Interval Too Brief) response

Table A.260M: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Min-Expires	[26] 20.23, [70] 5, 6	m	m	[26] 20.23, [70] 5, 6	i	i

Table A.260N: Void

Prerequisite A.163/15B - - PUBLISH response

Prerequisite: A.164/39 - - Additional for 489 (Bad Event) response

Table A.260O: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	i	i

Prerequisite A.163/17 - - PUBLISH response

Prerequisite: A.164/46 - - Additional for 504 (Server Time-out) response

Table A.260OA: Supported header fields within the PUBLISH response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Restoration-Info	subclause 7.2.11	n/a	c1	subclause 7.2.11	n/a	n/a
c1:	IF A.4/110 THEN o ELSE n/a - - HSS based P-CSCF restoration.						

Prerequisite A.163/17 - - PUBLISH response

Table A.260P: Supported message bodies within the PUBLISH response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.2.4.11 REFER method

Prerequisite A.163/16 - - REFER request

Table A.261: Supported header fields within the REFER request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Accept	[26] 20.1	m	m	[26] 20.1	i	i
0B	Accept-Contact	[56B] 9.2	c27	c27	[56B] 9.2	c27	c28
0C	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
1	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i
1AA	Additional-Identity	7.2.20	n/a	c77	7.2.20	n/a	c77
1A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
2	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
3	Authorization	[26] 20.7	m	m	[26] 20.7	i	i
4	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
4A	Cellular-Network-Info	7.2.15	n/a	c75	7.2.15	n/a	c76
5	Contact	[26] 20.10	m	m	[26] 20.10	i	i
5A	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
5B	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
5C	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
6	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
7	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
8	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
9	Date	[26] 20.17	m	m	[26] 20.17	c2	c2
10	Expires	[26] 20.19	m	m	[26] 20.19	i	i
10A	Feature-Caps	[190]	cj	cj	[190]	cj	cj
11	From	[26] 20.20	m	m	[26] 20.20	m	m
11A	Geolocation	[89] 4.1	c35	c35	[89] 4.1	c36	c36
11B	Geolocation-Routing	[89] 4.1	c35	c35	[89] 4.1	c36	c36
11C	History-Info	[66] 4.1	c31	c31	[66] 4.1	c31	c31
11D	Max-Breadth	[117] 5.8	c40	c40	[117] 5.8	c41	c41
12	Max-Forwards	[26] 20.22	m	m	[26] 20.22	m	m
13	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
14	Organization	[26] 20.25	m	m	[26] 20.25	c3	c3
14A	P-Access-Network-Info	[52] 4.4, [234] 2	c22	c22	[52] 4.4, [234] 2	c23	c23
14B	P-Asserted-Identity	[34] 9.1	c9	c9	[34] 9.1	c10	c10
14C	P-Asserted-Service	[121] 4.1	c38	c38	[121] 4.1	c39	c39
14D	P-Called-Party-ID	[52] 4.2, [52A] 4	c13	c13	[52] 4.2, [52A] 4	c14	c15
14E	P-Charging-Function-Addresses	[52] 4.5	c20	c20	[52] 4.5	c21	c21
14F	P-Charging-Vector	[52] 4.6	c18	c18	[52] 4.6	c19	c19
14H	P-Preferred-Identity	[34] 9.2	x	c69	[34] 9.2	c8	c8
14I	P-Preferred-Service	[121] 4.2	x	x	[121] 4.2	c37	c37
14J	P-Private-Network-Indication	[134]	c50	c50	[134]	c50	c50
14K	P-Profile-Key	[97] 5	c33	c33	[97] 5	c34	c34
14L	P-Served-User	[133] 6	c53	c53	[133] 6	c53	c53
14M	P-User-Database	[82] 4	c32	c32	[82] 4	c32	c32
14N	P-Visited-Network-ID	[52] 4.3	c16	o	[52] 4.3	c17	o
14O	Privacy	[33] 4.2	c11	c11	[33] 4.2	c12	c12
15	Proxy-Authorization	[26] 20.28	m	m	[26] 20.28	c4	c4
16	Proxy-Require	[26] 20.29	m	m	[26] 20.29	m	m
16A	Reason	[34A] 2	c25	c25	[34A] 2	c26	c26
17	Record-Route	[26] 20.30	m	m	[26] 20.30	c7	c7
17A	Refer-Sub	[173] 4	c54	c54	[173] 4	c55	c55
18	Refer-To	[36] 3	c3	c3	[36] 3	c4	c4
18A	Referred-By	[59] 3	c29	c29	[59] 3	c30	c30
18B	Reject-Contact	[56B] 9.2	c27	c27	[56B] 9.2	c27	c28
18C	Relayed-Charge	7.2.12	n/a	c74	7.2.12	n/a	c74
18D	Request-Disposition	[56B] 9.1	c27	c27	[56B] 9.1	c27	c27
19	Require	[26] 20.32	m	m	[26] 20.32	c5	c5
19A	Resource-Priority	[116] 3.1	c47	c47	[116] 3.1	c47	c47
20	Route	[26] 20.34	m	m	[26] 20.34	m	m
20A	Security-Client	[48] 2.3.1	x	x	[48] 2.3.1	c24	c24
20B	Security-Verify	[48] 2.3.1	x	x	[48] 2.3.1	c24	c24
20C	Session-ID	[162]	c70	c70	[162]	c70	c70
21	Supported	[26] 20.37	m	m	[26] 20.37	c6	c6

21A	Target-Dialog	[184] 7	c71	c71	[184] 7	c72	c72
22	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
23	To	[26] 20.39	m	m	[26] 20.39	m	m
23A	Trigger-Consent	[125] 5.11.2	c48	c48	[125] 5.11.2	c49	c49
24	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
25	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.
c2:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c3:	IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.
c4:	IF A.162/8A THEN m ELSE i - - authentication between UA and proxy.
c5:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c6:	IF A.162/16 THEN m ELSE i - - reading the contents of the Supported header before proxying the response.
c7:	IF A.162/14 THEN m ELSE i - - the requirement to be able to insert itself in the subsequent transactions in a dialog.
c8:	IF A.162/30A OR A.162/30C THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity, act as entity passing on identity transparently independent of trust domain.
c9:	IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c10:	IF A.162/30A or A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.
c11:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c12:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c13:	IF A.162/37 THEN m ELSE n/a - - the P-Called-Party-ID header extension.
c14:	IF A.162/37 THEN i ELSE n/a - - the P-Called-Party-ID header extension.
c15:	IF A.162/37 AND A.3/2 THEN m ELSE IF A.162/37 AND (A.3/3 OR A.3/9A) THEN i ELSE n/a - - the P-Called-Party-ID header extension and P-CSCF or (I-CSCF or IBCF (THIG).
c16:	IF A.162/38 THEN m ELSE n/a - - the P-Visited-Network-ID header extension.
c17:	IF A.162/39 THEN m ELSE i - - reading, or deleting the P-Visited-Network-ID header before proxying the request or response.
c18:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c19:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c20:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c21:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c22:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c23:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c24:	IF A.162/47 OR A.162/47A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c25:	IF A.162/48 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.
c26:	IF A.162/48 THEN i ELSE n/a - - the Reason header field for the session initiation protocol.
c27:	IF A.162/50 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c28:	IF A.162/50 AND A.4/3 THEN m ELSE IF A.162/50 AND NOT A.4/3 THEN i ELSE n/a - - caller preferences for the session initiation protocol, and S-CSCF.
c29:	IF A.162/53 THEN i ELSE n/a - - the SIP Referred-By mechanism.
c30:	IF A.162/53 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c31:	IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c32:	IF A.162/60 THEN m ELSE n/a - - the P-User-Database private header extension.
c33:	IF A.162/66A THEN m ELSE n/a - - making the first query to the database in order to populate the P-Profile-Key header.
c34:	IF A.162/66B THEN m ELSE n/a - - using the information in the P-Profile-Key header.
c35:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c36:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c37:	IF A.162/84A THEN m ELSE n/a - - act as authentication entity within the trust domain for asserted service.
c38:	IF A.162/84 THEN m ELSE n/a - - SIP extension for the identification of services.
c39:	IF A.162/84 OR A.162/30B THEN m ELSE i - - SIP extension for the identification of services or subsequent entity within trust network that can route outside the trust network.

c40:	IF A.162/81 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c41:	IF A.162/81 AND A.162/6 THEN m ELSE IF A.162/81 AND NOT A.162/6 THEN i ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, forking of initial requests.
c47:	IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.
c48:	IF A.162/85 THEN m ELSE n/a - - a framework for consent-based communications in SIP.
c49:	IF A.162/85 THEN i ELSE n/a - - a framework for consent-based communications in SIP.
c50:	IF A.162/87 THEN m ELSE n/a - - the SIP P-Private-Network-Indication private-header (P-Header).
c53:	IF A.162/88 THEN m ELSE n/a - - the SIP P-Served-User private header.
c54:	IF A.162/105 THEN m ELSE n/a - - suppression of session initiation protocol REFER method implicit subscription.
c55:	IF A.162/105 THEN i ELSE n/a - - suppression of session initiation protocol REFER method implicit subscription.
c69:	IF A.162/30C THEN m ELSE x - - act as entity passing on identity transparently independent of trust domain.
c70:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c71:	IF A.162/109 THEN m ELSE n/a - - request authorization through dialog Identification in the session initiation protocol.
c72:	IF A.162/109 THEN i ELSE n/a - - request authorization through dialog Identification in the session initiation protocol.
c73:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.
c74:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c75:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c76:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c77:	IF A.162/131 THEN o ELSE n/a - - the Additional-Identity header field extension.
NOTE:	c1 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.

Prerequisite A.163/16 - - REFER request

Table A.262: Supported message bodies within the REFER request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	application/vnd.3gpp.mid-call+xml	[8M] D	n/a	i	[8M] D	n/a	i
2	application/vnd.3gpp.mcptt-info+xml	[8ZE]	n/a	c1	[8ZE]	n/a	i
c1:	A.3/3 OR A.3/4 OR A.3/5 OR A.3/7C OR A.3/9A OR A.3/10 OR A.3/11 OR A.3/13A THEN m ELSE n/a - - I-CSCF, S-CSCF, BGCF, AS acting as proxy, IBCF (THIG), additional routeing functionality, E-CSCF, ISC gateway function (THIG).						

Table A.263: Void

Prerequisite A.163/17 - - REFER response

Prerequisite: A.164/1 - - Additional for 100 (Trying) response

Table A.263A: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	c2	c2
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF (A.162/9 AND A.162/5) OR A.162/4 THEN m ELSE n/a - - stateful proxy behaviour that inserts date, or stateless proxies.						
c2:	IF A.162/4 THEN i ELSE m - - Stateless proxy passes on.						

Prerequisite A.163/17 - - REFER response for all remaining status-codes

Table A.264: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Cellular-Network-Info	7.2.15	n/a	c22	7.2.15	n/a	c23
1B	Contact	[26] 20.10	m	m	[26] 20.10	i	i
1C	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
2	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
3	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
4	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
5	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
6	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
7	Date	[26] 20.17	m	m	[26] 20.17	c1	c1
8	From	[26] 20.20	m	m	[26] 20.20	m	m
8A	Geolocation-Error	[89] 4.3	c16	c16	[89] 4.3	c17	c17
8B	History-Info	[66] 4.1	c15	c15	[66] 4.1	c15	c15
9	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
10	Organization	[26] 20.25	m	m	[26] 20.25	c2	c2
10A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c12	c12	[52] 4.4, [52A] 4, [234] 2	c13	c13
10B	P-Asserted-Identity	[34] 9.1	c4	c4	[34] 9.1	c5	c5
10C	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c10	c10	[52] 4.5, [52A] 4	c11	c11
10D	P-Charging-Vector	[52] 4.6, [52A] 4	c8	c8	[52] 4.6, [52A] 4	c9	c9
10F	P-Preferred-Identity	[34] 9.2	x	x	[34] 9.2	c3	n/a
10G	Privacy	[33] 4.2	c6	c6	[33] 4.2	c7	c7
10H	Relayed-Charge	7.2.12	n/a	c21	7.2.12	n/a	c21
10I	Require	[26] 20.32	m	m	[26] 20.32	c14	c14
10J	Server	[26] 20.35	m	m	[26] 20.35	i	i
10K	Session-ID	[162]	c20	c20	[162]	c20	c20
11	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
12	To	[26] 20.39	m	m	[26] 20.39	m	m
12A	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
13	Via	[26] 20.42	m	m	[26] 20.42	m	m
14	Warning	[26] 20.43	m	m	[26] 20.43	i	i

c1:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c2:	IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.
c3:	IF A.162/30A THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity.
c4:	IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c5:	IF A.162/30A or A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.
c6:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c7:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c8:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c9:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c10:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c11:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c12:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c13:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c14:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c15:	IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c16:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c17:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c20:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c21:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c22:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c23:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.

Prerequisite A.163/17 - - REFER response

Prerequisite: A.164/102 - - Additional for 2xx response

Table A.265: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Accept-Resource-Priority	[116] 3.2	c12	c12	[116] 3.2	c12	c12
1	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
2	Authentication-Info	[26] 20.6	m	m	[26] 20.6	i	i
3	Feature-Caps	[190]	c14	c14	[190]	c14	c14
5	Record-Route	[26] 20.30	m	m	[26] 20.30	c3	c3
6	Refer-Sub	[173] 4	c4	c4	[173] 4	c5	c5
8	Supported	[26] 20.37	m	m	[26] 20.37	i	i
c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.						
c3:	IF A.162/15 THEN m ELSE i - - the requirement to be able to use separate URIs in the upstream direction and downstream direction when record routeing.						
c4:	IF A.162/105 THEN m ELSE n/a - - suppression of session initiation protocol REFER method implicit subscription.						
c5:	IF A.162/105 THEN I ELSE n/a - - suppression of session initiation protocol REFER method implicit subscription.						
c12:	IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						
c14:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.						

Prerequisite A.163/17 - - REFER response

Prerequisite: A.164/103 OR A.164/104 OR A.164/105 OR A.164/106 - - Additional for 3xx – 6xx response

Table A.265A: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	m	m	[26] 20.18	i	i
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.162/125 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Table A.266: Void

Prerequisite A.163/17 - - REFER response

Prerequisite: A.164/8 OR A.164/9 OR A.164/10 OR A.164/11 OR A.164/12 - - Additional for 401 (Unauthorized) response

Table A.267: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
10	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/17 - - REFER response

Prerequisite: A.164/17 OR A.164/23 OR A.164/30 OR A.164/36 OR A.164/42 OR A.164/45 OR A.164/50 OR A.164/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.268: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
6	Retry-After	[26] 20.33	m	m	[26] 20.33	i	i

Table A.269: Void

Prerequisite A.163/17 - - REFER response

Prerequisite: A.164/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.270: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
8	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Table A.270A: Void

Prerequisite A.163/17 - - REFER response

Prerequisite: A.164/25 - - Additional for 415 (Unsupported Media Type) response

Table A.271: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i

Prerequisite A.163/17 - - REFER response

Prerequisite: A.164/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.271A: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						

Prerequisite A.163/17 - - REFER response

Prerequisite: A.164/27 - - Additional for 420 (Bad Extension) response

Table A.272: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
8	Unsupported	[26] 20.40	m	m	[26] 20.40	c3	c3
c3:	IF A.162/18 THEN m ELSE i - - reading the contents of the Unsupported header before proxying the 420 response to a method other than REGISTER.						

Prerequisite A.163/17 - - REFER response

Prerequisite: A.164/28 OR A.164/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.272A: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	c1	c1	[48] 2	n/a	n/a
c1:	IF A.162/47 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Table A.273: Void

Prerequisite A.163/17 - - REFER response

Prerequisite: A.164/29H - - Additional for 470 (Consent Needed) response

Table A.273A: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Permission-Missing	[125] 5.9.3	m	m	[125] 5.9.3	m	m

Prerequisite A.163/17 - - REFER response

Prerequisite: A.164/46 - - Additional for 504 (Server Time-out) response

Table A.273AA: Supported header fields within the REFER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Restoration-Info	subclause 7.2.11	n/a	c1	subclause 7.2.11	n/a	n/a
c1:	IF A.4/110 THEN o ELSE n/a - - HSS based P-CSCF restoration.						

Prerequisite A.163/17 - - REFER response

Table A.274: Supported message bodies within the REFER response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.2.4.12 REGISTER method

Prerequisite A.163/18 - - REGISTER request

Table A.275: Supported header fields within the REGISTER request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i
3A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
4	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
5	Authorization	[26] 20.7, [49]	m	m	[26] 20.7, [49]	i	i
6	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
7	Call-Info	[26] 20.9	m	m	[26] 20.9	c2	c2
7A	Cellular-Network-Info	7.2.15	n/a	c39	7.2.15	n/a	c40
8	Contact	[26] 20.10	m	m	[26] 20.10	i	i
9	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
10	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
11	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
12	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
13	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
14	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
15	Date	[26] 20.17	m	m	[26] 20.17	m	m
16	Expires	[26] 20.19	m	m	[26] 20.19	i	i
16A	Feature-Caps	[190]	c36	c36	[190]	c36	c36
17	From	[26] 20.20	m	m	[26] 20.20	m	m
17A	Geolocation	[89] 4.1	c26	c26	[89] 4.1	c27	c27
17B	Geolocation-Routing	[89] 4.1	c26	c26	[89] 4.1	c27	c27
17C	History-Info	[66] 4.1	c24	c24	[66] 4,1	c24	c24
17D	Max-Breadth	[117] 5.8	c31	c31	[117] 5.8	c32	c32
18	Max-Forwards	[26] 20.22	m	m	[26] 20.22	m	m
19	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
20	Organization	[26] 20.25	m	m	[26] 20.25	c3	c3
20A	P-Access-Network-Info	[52] 4.4, [234] 2	c16	c16	[52] 4.4, [234] 2	c17	c17
20B	P-Charging-Function-Addresses	[52] 4.5	c14	c14	[52] 4.5	c15	c15
20C	P-Charging-Vector	[52] 4.6	c12	c12	[52] 4.6	c13	c13
20E	P-User-Database	[82] 4	c25	c25	[82] 4	n/a	n/a
20F	P-Visited-Network-ID	[52] 4.3	c10	c10	[52] 4.3	c11	c11
20G	Path	[35] 4.2	c6	c6	[35] 4.2	c6	c6
20H	Privacy	[33] 4.2	c8	c8	[33] 4.2	c9	c9
21	Proxy-Authorization	[26] 20.28	m	m	[26] 20.28	c7	c7
22	Proxy-Require	[26] 20.29	m	m	[26] 20.29	m	m
22A	Reason	[34A] 2	c19	c19	[34A] 2	c20	c20
22B	Recv-Info	[25] 5.2.3	c33	c33	[25] 5.2.3	c34	c34
22C	Referred-By	[59] 3	c22	c22	[59] 3	c23	c23
22D	Relayed-Charge	7.2.12	n/a	c37	7.2.12	n/a	c37
22E	Request-Disposition	[56B] 9.1	c21	c21	[56B] 9.1	c21	c21
23	Require	[26] 20.32	m	m	[26] 20.32	c4	c4
23A	Resource-Priority	[116] 3.1	c28	c28	[116] 3.1	c28	c28
23B	Resource-Share	Subclause 4.15	n/a	c38	Subclause 4.15	n/a	c38
24	Route	[26] 20.34	m	m	[26] 20.34	m	m
24A	Security-Client	[48] 2.3.1	x	x	[48] 2.3.1	c18	c18
24B	Security-Verify	[48] 2.3.1	x	x	[48] 2.3.1	c18	c18
24C	Session-ID	[162]	c35	c35	[162]	c35	c35
25	Supported	[26] 20.37	m	m	[26] 20.37	c5	c5
26	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
27	To	[26] 20.39	m	m	[26] 20.39	m	m
28	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
29	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.
c2:	IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header.
c3:	IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.
c4:	IF A.162/11 OR A.162/12 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c5:	IF A.162/16 THEN m ELSE i - - reading the contents of the Supported header before proxying the response.
c6:	IF A.162/29 THEN m ELSE n/a - - PATH header support.
c7:	IF A.162/8A THEN m ELSE i - - authentication between UA and proxy.
c8:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c9:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c10:	IF A.162/38 THEN m ELSE n/a - - the P-Visited-Network-ID header extension.
c11:	IF A.162/39 THEN m ELSE i - - reading, or deleting the P-Visited-Network-ID header before proxying the request or response.
c12:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c13:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c14:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c15:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c16:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c17:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c18:	IF A.162/47 OR 162/47A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c19:	IF A.162/48 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.
c20:	IF A.162/48 THEN i ELSE n/a - - the Reason header field for the session initiation protocol.
c21:	IF A.162/50 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c22:	IF A.162/53 THEN i ELSE n/a - - the SIP Referred-By mechanism.
c23:	IF A.162/53 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c24:	IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c25:	IF A.162/60 THEN m ELSE n/a - - the P-User-Database private header extension.
c26:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c27:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c28:	IF A.162/80B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.
c31:	IF A.162/81 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c32:	IF A.162/81 AND A.162/6 THEN m ELSE IF A.162/81 AND NOT A.162/6 THEN i ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, forking of initial requests.
c33:	IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.
c34:	IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.
c35:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c36:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.
c37:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c38:	IF A.162/122 THEN m ELSE n/a - - resource sharing.
c39:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c40:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
NOTE:	c1 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.

Prerequisite A.163/18 - - REGISTER request

Table A.276: Supported message bodies within the REGISTER request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	3GPP IM CN subsystem XML body	subclause 7.6	n/a	m	subclause 7.6	n/a	i

Table A.277: Void

Prerequisite A.163/19 - - REGISTER response

Prerequisite: A.164/1 - - Additional for 100 (Trying) response

Table A.277A: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	c2	c2
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF (A.162/9 AND A.162/5) OR A.162/4 THEN m ELSE n/a - - stateful proxy behaviour that inserts date, or stateless proxies.						
c2:	IF A.162/4 THEN i ELSE m - - Stateless proxy passes on.						

Prerequisite A.163/19 - - REGISTER response for all remaining status-codes

Table A.278: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Call-Info	[26] 20.9	m	m	[26] 20.9	c2	c2
2	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
3	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
4	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	m	m	[26] 20.17	m	m
9	From	[26] 20.20	m	m	[26] 20.20	m	m
9A	Geolocation-Error	[89] 4.3	c13	c13	[89] 4.3	c14	c14
9B	History-Info	[66] 4.1	c12	c12	[66] 4.1	c12	c12
10	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
11	Organization	[26] 20.25	m	m	[26] 20.25	c1	c1
11A	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c9	c9	[52] 4.4, [52A] 4, [234] 2	c10	c10
11B	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c7	c7	[52] 4.5, [52A] 4	c8	c8
11C	P-Charging-Vector	[52] 4.6, [52A] 4	c5	c5	[52] 4.6, [52A] 4	c6	c6
11E	Privacy	[33] 4.2	c3	c3	[33] 4.2	c4	c4
11F	Relayed-Charge	7.2.12	n/a	c18	7.2.12	n/a	c18
11G	Require	[26] 20.32	m	m	[26] 20.32	c11	c11
11H	Server	[26] 20.35	m	m	[26] 20.35	i	i
11I	Session-ID	[162]	c17	c17	[162]	c17	c17
12	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
13	To	[26] 20.39	m	m	[26] 20.39	m	m
13A	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
14	Via	[26] 20.42	m	m	[26] 20.42	m	m
15	Warning	[26] 20.43	m	m	[26] 20.43	i	i

c1: IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.
c2: IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header.
c3: IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c4: IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c5: IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c6: IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c7: IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c8: IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c9: IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c10: IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c11: IF A.162/11 OR A.162/12 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c12: IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c13: IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c14: IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c17: IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c18: IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.

Prerequisite A.163/19 - - REGISTER response

Prerequisite: A.164/102 - - Additional for 2xx response

Table A.279: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
1A	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
1B	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i
1C	Accept-Resource-Priority	[116] 3.2	c11	c11	[116] 3.2	c11	c11
2	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
3	Authentication-Info	[26] 20.6	m	m	[26] 20.6	i	i
5	Contact	[26] 20.10	m	m	[26] 20.10	i	i
5A	Feature-Caps	[190]	c16	c16	[190]	c16	c16
5B	Flow-Timer	[92] 11	c12	c12	[92] 11	c13	c14
5C	P-Associated-URI	[52] 4.1	c8	c8	[52] 4.1	c9	c10
6	Path	[35] 4.2	c3	c3	[35] 4.2	c4	c4
7	Security-Server	Subclause 7.2A.7	n/a	c15	Subclause 7.2A.7	n/a	n/a
8	Service-Route	[38] 5	c5	c5	[38] 5	c6	c7
9	Supported	[26] 20.37	m	m	[26] 20.37	i	i

c1: IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.
c3: IF A.162/29 THEN m ELSE n/a - - Path extension support.
c4: IF A.162/29 THEN i ELSE n/a - - Path extension support.
c5: IF A.162/32 THEN m ELSE n/a - - Service-Route extension support.
c6: IF A.162/32 THEN i ELSE n/a - - Service-Route extension support.
c7: IF A.162/32 THEN (IF A.3/2 THEN m ELSE i) ELSE n/a - - Service-Route extension and P-CSCF.
c8: IF A.162/36 THEN m ELSE n/a - - the P-Associated-URI extension.
c9: IF A.162/36 THEN i ELSE n/a - - the P-Associated-URI extension.
c10: IF A.162/36 AND A.3/2 THEN m ELSE IF A.162/36 AND (A.3/3 OR A.3/9A) THEN i ELSE n/a - - the P-Associated-URI extension and P-CSCF or I-CSCF or IBCF (THIG).
c11: IF A.162/80B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.
c12: IF A.162/67 THEN m ELSE n/a - - managing client initiated transactions in SIP.
c13: IF A.162/67 THEN m ELSE n/a - - managing client initiated transactions in SIP, P-CSCF, I-CSCF.
c14: IF A.162/67 AND A.3/2 THEN m ELSE IF A.162/67 AND A.3/3 THEN i ELSE n/a - - managing client initiated transactions in SIP, P-CSCF, I-CSCF.
c15: IF A.162/47A THEN m ELSE n/a - - mediasec header field parameter for marking security mechanisms related to media.
c16: IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.

Prerequisite A.163/19 - - REGISTER response

Prerequisite: A.164/103 OR A.164/104 OR A.164/105 OR A.164/106 - - Additional for 3xx – 6xx response

Table A.279A: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	m	m	[26] 20.18	i	i
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.162/125 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.163/19 - - REGISTER response

Prerequisite: A.164/103 OR A.164/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.280: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Contact	[26] 20.10	m	m	[26] 20.10	c2	c2
c2:	IF A.162/19E THEN m ELSE i - - deleting Contact headers.						

Prerequisite A.163/19 - - REGISTER response

Prerequisite: A.164/14 - - Additional for 401 (Unauthorized) response

Table A.281: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
6	Security-Server	[48] 2	x	c1	[48] 2	n/a	n/a
10	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i
c1:	IF A.162/47 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Prerequisite A.163/19 - - REGISTER response

Prerequisite: A.164/17 OR A.164/23 OR A.164/30 OR A.164/36 OR A.164/42 OR A.164/45 OR A.164/50 OR A.164/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.282: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
6	Retry-After	[26] 20.33	m	m	[26] 20.33	i	i

Table A.283: Void

Prerequisite A.163/19 - - REGISTER response

Prerequisite: A.164/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.284: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
9	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/19 - - REGISTER response

Prerequisite: A.164/25 - - Additional for 415 (Unsupported Media Type) response

Table A.285: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i

Prerequisite A.163/19 - - REGISTER response

Prerequisite: A.164/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.285A: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.162/80B THEN m ELSE n/a - - inclusion of CANCEL, BYE, REGISTER and PUBLISH in communications resource priority for the session initiation protocol.						

Prerequisite A.163/19 - - REGISTER response

Prerequisite: A.164/27 - - Additional for 420 (Bad Extension) response

Table A.286: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
8	Unsupported	[26] 20.40	m	m	[26] 20.40	c3	c3
c3:	IF A.162/17 THEN m ELSE i.						

Prerequisite A.163/19 - - REGISTER response

Prerequisite: A.164/28 OR A.164/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.286A: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	c1	c1	[48] 2	n/a	n/a
c1:	IF A.162/47 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Prerequisite A.163/19 - - REGISTER response

Prerequisite: A.164/29 - - Additional for 423 (Interval Too Brief) response

Table A.287: Supported header fields within the REGISTER response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Min-Expires	[26] 20.23	m	m	[26] 20.23	i	i

Table A.288: Void

Prerequisite A.163/19 - - REGISTER response

Table A.289: Supported message bodies within the REGISTER response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.2.4.13 SUBSCRIBE method

Prerequisite A.163/20 - - SUBSCRIBE request

Table A.290: Supported header fields within the SUBSCRIBE request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
1A	Accept-Contact	[56B] 9.2	c27	c27	[56B] 9.2	c27	c28
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i
3A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
4	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
5	Authorization	[26] 20.7	m	m	[26] 20.7	i	i
6	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
6A	Call-Info	[26] 20.9	m	m	[26] 20.9	c73	c73
6B	Cellular-Network-Info	7.2.15	n/a	c76	7.2.15	n/a	c77
6C	Contact	[26] 20.10	m	m	[26] 20.10	i	i
7	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
8	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
9	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
10	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
11	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
12	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
13	Date	[26] 20.17	m	m	[26] 20.17	c2	c2
14	Event	[28] 8.2.1	m	m	[28] 8.2.1	m	m
15	Expires	[26] 20.19	m	m	[26] 20.19	i	i
15A	Feature-Caps	[190]	c74	c74	[190]	c74	c74
16	From	[26] 20.20	m	m	[26] 20.20	m	m
16A	Geolocation	[89] 4.1	c35	c35	[89] 4.1	c36	c36
16B	Geolocation-Routing	[89] 4.1	c35	c35	[89] 4.1	c36	c36
16C	History-Info	[66] 4.1	c31	c31	[66] 4.1	c31	c31
16D	Max-Breadth	[117] 5.8	c47	c47	[117] 5.8	c48	c48
17	Max-Forwards	[26] 20.22	m	m	[26] 20.22	m	m
18	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
18A	Organization	[26] 20.25	m	m	[26] 20.25	c3	c3
18B	P-Access-Network-Info	[52] 4.4, [234] 2	c22	c22	[52] 4.4, [234] 2	c23	c23
18C	P-Asserted-Identity	[34] 9.1	c9	c9	[34] 9.1	c10	c10
18D	P-Asserted-Service	[121] 4.1	c39	c39	[121] 4.1	c40	c40
18E	P-Called-Party-ID	[52] 4.2	c13	c13	[52] 4.2	c14	c15
18F	P-Charging-Function-Addresses	[52] 4.5	c20	c20	[52] 4.5	c21	c21
18G	P-Charging-Vector	[52] 4.6	c18	c18	[52] 4.6	c19	c19
18I	P-Preferred-Identity	[34] 9.2	x	c69	[34] 9.2	c8	c8
18J	P-Preferred-Service	[121] 4.2	x	x	[121] 4.2	c38	c38
18K	P-Private-Network-Indication	[134]	c43	c43	[134]	c43	c43
18L	P-Profile-Key	[97] 5	c33	c33	[97] 5	c34	c34
18M	P-Served-User	[133] 6	c44	c44	[133] 6	c44	c44
18N	P-User-Database	[82] 4	c32	c32	[82] 4	c32	c32
18O	P-Visited-Network-ID	[52] 4.3	c16	o	[52] 4.3	c17	o
18P	Priority	[26] 20.26	m	m	[26] 20.26	i	c50
18Q	Privacy	[33] 4.2	c11	c11	[33] 4.2	c12	c12
19	Proxy-Authorization	[26] 20.28	m	m	[26] 20.28	c4	c4
20	Proxy-Require	[26] 20.29	m	m	[26] 20.29	m	m
20A	Reason	[34A] 2	c25	c25	[34A] 2	c26	c26
21	Record-Route	[26] 20.30	m	m	[26] 20.30	c7	c7
21A	Referred-By	[59] 3	c29	c29	[59] 3	c30	c30
21B	Reject-Contact	[56B] 9.2	c27	c27	[56B] 9.2	c27	c28
21C	Relayed-Charge	7.2.12	n/a	c75	7.2.12	n/a	c75
21D	Request-Disposition	[56B] 9.1	c27	c27	[56B] 9.1	c27	c27
22	Require	[26] 20.32	m	m	[26] 20.32	c5	c5
22A	Resource-Priority	[116] 3.1	c37	c37	[116] 3.1	c37	c37
23	Route	[26] 20.34	m	m	[26] 20.34	m	m
23A	Security-Client	[48] 2.3.1	x	x	[48] 2.3.1	c24	c24
23B	Security-Verify	[48] 2.3.1	x	x	[48] 2.3.1	c24	c24
23C	Session-ID	[162]	c70	c70	[162]	c70	c70
24	Supported	[26] 20.37	m	m	[26] 20.37	c6	c6
24A	Target-Dialog	[184] 7	c71	c71	[184] 7	c72	c72

25	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
26	To	[26] 20.39	m	m	[26] 20.39	m	m
26A	Trigger-Consent	[125] 5.11.2	c41	c41	[125] 5.11.2	c42	c42
27	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
28	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.
c2:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c3:	IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.
c4:	IF A.162/8A THEN m ELSE i - - authentication between UA and proxy.
c5:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c6:	IF A.162/16 THEN m ELSE i - - reading the contents of the Supported header before proxying the response.
c7:	IF A.162/14 THEN m ELSE i - - the requirement to be able to insert itself in the subsequent transactions in a dialog.
c8:	IF A.162/30A OR A.162/30C THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity, act as entity passing on identity transparently independent of trust domain.
c9:	IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c10:	IF A.162/30A or A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.
c11:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c12:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c13:	IF A.162/37 THEN m ELSE n/a - - the P-Called-Party-ID header extension.
c14:	IF A.162/37 THEN i ELSE n/a - - the P-Called-Party-ID header extension.
c15:	IF A.162/37 AND A.3/2 THEN m ELSE IF A.162/37 AND (A.3/3 OR A.3/9A) THEN i ELSE n/a - - the P-Called-Party-ID header extension and P-CSCF or I-CSCF or IBCF (THIG).
c16:	IF A.162/38 THEN m ELSE n/a - - the P-Visited-Network-ID header extension.
c17:	IF A.162/39 THEN m ELSE i - - reading, or deleting the P-Visited-Network-ID header before proxying the request or response.
c18:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c19:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c20:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c21:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c22:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c23:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c24:	IF A.162/47 OR A.162/47A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c25:	IF A.162/48 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.
c26:	IF A.162/48 THEN i ELSE n/a - - the Reason header field for the session initiation protocol.
c27:	IF A.162/50 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c28:	IF A.162/50 AND A.4/3 THEN m ELSE IF A.162/50 AND NOT A.4/3 THEN i ELSE n/a - - caller preferences for the session initiation protocol, and S-CSCF.
c29:	IF A.162/53 THEN i ELSE n/a - - the SIP Referred-By mechanism.
c30:	IF A.162/53 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c31:	IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c32:	IF A.162/60 THEN m ELSE n/a - - the P-User-Database private header extension.
c33:	IF A.162/66A THEN m ELSE n/a - - making the first query to the database in order to populate the P-Profile-Key header.
c34:	IF A.162/66B THEN m ELSE n/a - - using the information in the P-Profile-Key header.
c35:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c36:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c37:	IF A.162/80A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.
c38:	IF A.162/84A THEN m ELSE n/a - - act as authentication entity within the trust domain for asserted service.
c39:	IF A.162/84 THEN m ELSE n/a - - SIP extension for the identification of services.
c40:	IF A.162/84 OR A.162/30B THEN m ELSE i - - SIP extension for the identification of services or subsequent entity within trust network that can route outside the trust network.

c41:	IF A.162/85 THEN m ELSE n/a - - a framework for consent-based communications in SIP.
c42:	IF A.162/85 THEN i ELSE n/a - - a framework for consent-based communications in SIP.
c43:	IF A.162/87 THEN m ELSE n/a - - the SIP P-Private-Network-Indication private-header (P-Header).
c44	IF A.162/88 THEN m ELSE n/a - - the SIP P-Served-User private header.
c47:	IF A.162/81 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c48:	IF A.162/81 AND A.162/6 THEN m ELSE IF A.162/81 AND NOT A.162/6 THEN i ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, forking of initial requests.
c50:	IF A.162/115 THEN m ELSE i - - PSAP callback indicator.
c69:	IF A.162/30C THEN m ELSE x - - act as entity passing on identity transparently independent of trust domain.
c70:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c71:	IF A.162/109 THEN m ELSE n/a - - request authorization through dialog Identification in the session initiation protocol.
c72:	IF A.162/109 THEN i ELSE n/a - - request authorization through dialog Identification in the session initiation protocol.
c73:	IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header field.
c74:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.
c75:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c76:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c77:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
NOTE:	c1 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.

Prerequisite A.163/20 - - SUBSCRIBE request

Table A.291: Supported message bodies within the SUBSCRIBE request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	application/vnd.3gpp.mcptt-info+xml	[8ZE]	n/a	c1	[8ZE]	n/a	i
2	application/simple-filter+xml	[243]	o	c1	[243]	n/a	i
c1:	A.3/3 OR A.3/4 OR A.3/5 OR A.3/7C OR A.3/9A OR A.3/10 OR A.3/11 OR A.3/13A THEN m ELSE n/a - - I-CSCF, S-CSCF, BGCF, AS acting as proxy, IBCF (THIG), additional routeing functionality, E-CSCF, ISC gateway function (THIG).						

Prerequisite A.163/21 - - SUBSCRIBE response

Prerequisite: A.164/1 - - Additional for 100 (Trying) response

Table A.291A: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	c2	c2
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF (A.162/9 AND A.162/5) OR A.162/4 THEN m ELSE n/a - - stateful proxy behaviour that inserts date, or stateless proxies.						
c2:	IF A.162/4 THEN i ELSE m - - Stateless proxy passes on.						

Prerequisite A.163/21 - - SUBSCRIBE response for all remaining status-codes

Table A.292: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Call-Info	[26] 20.9	m	m	[26] 20.9	c23	c23
1B	Cellular-Network-Info	7.2.15	n/a	c25	7.2.15	n/a	c26
2	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	i
3	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	i
4	Content-Language	[26] 20.13	m	m	[26] 20.13	i	i
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	i	i
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	m	m	[26] 20.17	c1	c1
9	From	[26] 20.20	m	m	[26] 20.20	m	m
9A	Geolocation-Error	[89] 4.3	c20	c20	[89] 4.3	c21	c21
9B	History-Info	[66] 4.1	c15	c15	[66] 4.1	c15	c15
10	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	i
10A	Organization	[26] 20.25	m	m	[26] 20.25	c2	c2
10B	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c12	c12	[52] 4.4, [52A] 4, [234] 2	c13	c13
10C	P-Asserted-Identity	[34] 9.1	c4	c4	[34] 9.1	c5	c5
10D	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c10	c10	[52] 4.5, [52A] 4	c11	c11
10E	P-Charging-Vector	[52] 4.6, [52A] 4	c8	c8	[52] 4.6, [52A] 4	c9	c9
10G	P-Preferred-Identity	[34] 9.2	x	x	[34] 9.2	c3	n/a
10H	Privacy	[33] 4.2	c6	c6	[33] 4.2	c7	c7
10I	Relayed-Charge	7.2.12	n/a	c24	7.2.12	n/a	c24
10J	Require	[26] 20.32	m	m	[26] 20.32	c14	c14
10K	Server	[26] 20.35	m	m	[26] 20.35	i	i
10L	Session-ID	[162]	c22	c22	[162]	c22	c22
11	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
12	To	[26] 20.39	m	m	[26] 20.39	m	m
12A	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
13	Via	[26] 20.42	m	m	[26] 20.42	m	m
14	Warning	[26] 20.43	m	m	[26] 20.43	i	i

c1:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c2:	IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.
c3:	IF A.162/30A THEN m ELSE n/a - - act as first entity within the trust domain for asserted identity.
c4:	IF A.162/30 THEN m ELSE n/a - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks.
c5:	IF A.162/30A or A.162/30B THEN m ELSE i - - extensions to the Session Initiation Protocol (SIP) for asserted identity within trusted networks or subsequent entity within trust network that can route outside the trust network.
c6:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c7:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c8:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c9:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c10:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c11:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c12:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c13:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c14:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c15:	IF A.162/57 THEN m ELSE n/a - - an extension to the session initiation protocol for request history information.
c16:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c17:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c20:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c21:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c22:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c23:	IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header field.
c24:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c25:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c26:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.

Prerequisite A.163/21 - - SUBSCRIBE response

Prerequisite: A.164/102 - - Additional for 2xx response

Table A.293: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Accept-Resource-Priority	[116] 3.2	c4	c4	[116] 3.2	c4	c4
0B	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	i	i
1	Authentication-Info	[26] 20.6	m	m	[26] 20.6	i	i
1A	Contact	[26] 20.10	m	m	[26] 20.10	i	i
2	Expires	[26] 20.19	m	m	[26] 20.19	i	i
2A	Feature-Caps	[190]	c6	c6	[190]	c6	c6
3	Record-Route	[26] 20.30	m	m	[26] 20.30	c3	c3
6	Supported	[26] 20.37	m	m	[26] 20.37	i	i
c3:	IF A.162/15 THEN m ELSE i - - the requirement to be able to use separate URIs in the upstream direction and downstream direction when record routeing.						
c4:	IF A.162/80A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						
c6:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.						

Prerequisite A.163/21 - - SUBSCRIBE response

Prerequisite: A.164/103 OR A.164/104 OR A.164/105 OR A.164/106 - - Additional for 3xx – 6xx response

Table A.293A: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	m	m	[26] 20.18	i	i
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.162/125 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.163/21 - - SUBSCRIBE response

Prerequisite: A.164/103 OR A.164/35 - - Additional for 3xx or 485 (Ambiguous) response

Table A.294: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Contact	[26] 20.10	m	m	[26] 20.10	c1	c1
c1:	IF A.162/19E THEN m ELSE i - - deleting Contact headers.						

Prerequisite A.163/21 - - SUBSCRIBE response

Prerequisite: A.164/14 - - Additional for 401 (Unauthorized) response

Table A.295: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
8	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/21 - - SUBSCRIBE response

Prerequisite: A.164/17 OR A.164/23 OR A.164/30 OR A.164/36 OR A.164/42 OR A.164/45 OR A.164/50 OR A.164/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480 (Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.296: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Retry-After	[26] 20.33	m	m	[26] 20.33	i	i

Table A.297: Void

Prerequisite A.163/21 - - SUBSCRIBE response

Prerequisite: A.164/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.298: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
6	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Table A.298A: Void

Prerequisite A.163/21 - - SUBSCRIBE response

Prerequisite: A.164/25 - - Additional for 415 (Unsupported Media Type) response

Table A.299: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i

Prerequisite A.163/21 - - SUBSCRIBE response

Prerequisite: A.164/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.299A: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.162/80A THEN m ELSE n/a - - inclusion of MESSAGE, SUBSCRIBE, NOTIFY in communications resource priority for the session initiation protocol.						

Prerequisite A.163/21 - - SUBSCRIBE response

Prerequisite: A.164/27 - - Additional for 420 (Bad Extension) response

Table A.300: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Unsupported	[26] 20.40	m	m	[26] 20.40	c3	c3
c3:	IF A.162/18 THEN m ELSE i - - reading the contents of the Unsupported header before proxying the 420 response to a method other than REGISTER.						

Prerequisite A.163/21 - - SUBSCRIBE response

Prerequisite: A.164/28 OR A.164/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.300A: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	c1	c1	[48] 2	n/a	n/a
c1:	IF A.162/47 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Prerequisite A.163/21 - - SUBSCRIBE response

Prerequisite: A.164/29 - - Additional for 423 (Interval Too Brief) response

Table A.301: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Min-Expires	[26] 20.23	m	m	[26] 20.23	i	i

Table A.302: Void

Prerequisite A.163/21 - - SUBSCRIBE response

Prerequisite: A.164/29H - - Additional for 470 (Consent Needed) response

Table A.302A: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Permission-Missing	[125] 5.9.3	m	m	[125] 5.9.3	m	m

Prerequisite A.163/21 - - SUBSCRIBE response

Prerequisite: A.164/39 - - Additional for 489 (Bad Event) response

Table A.303: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.						
NOTE:	c1 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.						

Table A.303A: Void

Prerequisite A.163/21 - - SUBSCRIBE response

Prerequisite: A.164/46 - - Additional for 504 (Server Time-out) response

Table A.303B: Supported header fields within the SUBSCRIBE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Restoration-Info	subclause 7.2.11	n/a	c1	subclause 7.2.11	n/a	n/a
c1:	IF A.4/110 THEN o ELSE n/a - - HSS based P-CSCF restoration.						

Prerequisite A.163/21 - - SUBSCRIBE response

Table A.304: Supported message bodies within the SUBSCRIBE response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.2.2.4.14 UPDATE method

Prerequisite A.163/22 - - UPDATE request

Table A.305: Supported header fields within the UPDATE request

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
1A	Accept-Contact	[56B] 9.2	c21	c21	[56B] 9.2	c22	c22
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i
4	Allow	[26] 20.5	m	m	[26] 20.5	i	i
5	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
6	Authorization	[26] 20.7	m	m	[26] 20.7	i	i
7	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
8	Call-Info	[26] 20.9	m	m	[26] 20.9	c8	c8
8A	Cellular-Network-Info	7.2.15	n/a	c52	7.2.15	n/a	c53
9	Contact	[26] 20.10	m	m	[26] 20.10	i	i
10	Content-Disposition	[26] 20.11	m	m	[26] 20.11	c4	c4
11	Content-Encoding	[26] 20.12	m	m	[26] 20.12	c4	c4
12	Content-Language	[26] 20.13	m	m	[26] 20.13	c4	c4
13	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
14	Content-Type	[26] 20.15	m	m	[26] 20.15	c4	c4
15	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
16	Date	[26] 20.17	m	m	[26] 20.17	c2	c2
16A	Feature-Caps	[190]	c49	c49	[190]	c49	c49
17	From	[26] 20.20	m	m	[26] 20.20	m	m
17A	Geolocation	[89] 4.1	c26	c26	[89] 4.1	c27	c27
17B	Geolocation-Routing	[89] 4.1	c26	c26	[89] 4.1	c27	c27
17C	Max-Breadth	[117] 5.8	c32	c32	[117] 5.8	c33	c33
18	Max-Forwards	[26] 20.22	m	m	[26] 20.22	m	m
19	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	c4
19A	Min-SE	[58] 5	c23	c23	[58] 5	c23	c23
20	Organization	[26] 20.25	m	m	[26] 20.25	c3	c3
20A	P-Access-Network-Info	[52] 4.4, [234] 2	c16	c16	[52] 4.4, [234] 2	c17	c17
20B	P-Charging-Function-Addresses	[52] 4.5	c14	c14	[52] 4.5	c15	c15
20C	P-Charging-Vector	[52] 4.6	c12	c12	[52] 4.6	c13	c13
20E	P-Early-Media	[109] 8	o	c28	[109] 8	o	c28
20EA	Priority-Share	Subclause 7.2.16	n/a	c54	Subclause 7.2.16	n/a	c54
20F	Privacy	[33] 4.2	c10	c10	[33] 4.2	c11	c11
21	Proxy-Authorization	[26] 20.28	m	m	[26] 20.28	c9	c9
22	Proxy-Require	[26] 20.29	m	m	[26] 20.29	m	m
22A	Reason	[34A] 2	c19	c19	[34A] 2	c20	c20
23	Record-Route	[26] 20.30	m	m	[26] 20.30	c7	c7
23A	Recv-Info	[25] 5.2.3	c34	c34	[25] 5.2.3	c35	c35
23B	Referred-By	[59] 3	c24	c24	[59] 3	c25	c25
23C	Reject-Contact	[56B] 9.2	c21	c21	[56B] 9.2	c22	c22
23D	Relayed-Charge	7.2.12	n/a	c50	7.2.12	n/a	c50
23E	Request-Disposition	[56B] 9.1	c21	c21	[56B] 9.1	c22	c22
24	Require	[26] 20.32	m	m	[26] 20.32	c5	c5
24A	Resource-Priority	[116] 3.1	c47	c47	[116] 3.1	c47	c47
24B	Resource-Share	Subclause 4.15	n/a	c51	Subclause 4.15	n/a	c51
25	Route	[26] 20.34	m	m	[26] 20.34	m	m
25A	Security-Client	[48] 2.3.1	x	x	[48] 2.3.1	c18	c18
25B	Security-Verify	[48] 2.3.1	x	x	[48] 2.3.1	c18	c18
25C	Session-Expires	[58] 4	c23	c23	[58] 4	c23	c23
25D	Session-ID	[162]	c48	c48	[162]	c48	c48
26	Supported	[26] 20.37	m	m	[26] 20.37	c6	c6
27	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
28	To	[26] 20.39	m	m	[26] 20.39	m	m
29	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
30	Via	[26] 20.42	m	m	[26] 20.42	m	m

c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.
c2:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c3:	IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.
c4:	IF A.3/2 OR A.3/4 THEN m ELSE i - - P-CSCF or S-CSCF.
c5:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c6:	IF A.162/16 THEN m ELSE i - - reading the contents of the Supported header before proxying the response.
c7:	IF A.162/14 THEN o ELSE i - - the requirement to be able to insert itself in the subsequent transactions in a dialog.
c8:	IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header.
c9:	IF A.162/8A THEN m ELSE i - - authentication between UA and proxy.
c10:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c11:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c12:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c13:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c14:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c15:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c16:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c17:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c18:	IF A.162/47 OR A.162/47A THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol or mediasec header field parameter for marking security mechanisms related to media.
c19:	IF A.162/48 THEN m ELSE n/a - - the Reason header field for the session initiation protocol.
c20:	IF A.162/48 THEN i ELSE n/a - - the Reason header field for the session initiation protocol.
c21:	IF A.162/50 THEN m ELSE n/a - - caller preferences for the session initiation protocol.
c22:	IF A.162/50 THEN i ELSE n/a - - caller preferences for the session initiation protocol.
c23:	IF A.162/52 THEN m ELSE n/a - - the SIP session timer.
c24:	IF A.162/53 THEN i ELSE n/a - - the SIP Referred-By mechanism.
c25:	IF A.162/53 THEN m ELSE n/a - - the SIP Referred-By mechanism.
c26:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c27:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c28:	IF A.162/76 THEN m ELSE n/a - - the SIP P-Early-Media private header extension for authorization of early media.
c32:	IF A.162/81 THEN m ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies.
c33:	IF A.162/81 AND A.162/6 THEN m ELSE IF A.162/81 AND NOT A.162/6 THEN i ELSE n/a - - addressing an amplification vulnerability in session initiation protocol forking proxies, forking of initial requests.
c34:	IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.
c35:	IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.
c47:	IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.
c48:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c49:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.
c50:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c51:	IF A.162/122 THEN o ELSE n/a - - resource sharing.
c52:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c53:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c54:	IF A.162/124 THEN o ELSE n/a - - priority sharing.
NOTE:	c1 refers to the UA role major capability as this is the case of a proxy that also acts as a UA specifically for SUBSCRIBE and NOTIFY.

Prerequisite A.163/22 - - UPDATE request

Table A.306: Supported message bodies within the UPDATE request

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

Prerequisite A.163/23 - - UPDATE response

Prerequisite: A.164/1 - - Additional for 100 (Trying) response

Table A.306A: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
2	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
3	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
4	Date	[26] 20.17	c1	c1	[26] 20.17	c2	c2
5	From	[26] 20.20	m	m	[26] 20.20	m	m
6	To	[26] 20.39	m	m	[26] 20.39	m	m
7	Via	[26] 20.42	m	m	[26] 20.42	m	m
c1:	IF (A.162/9 AND A.162/5) OR A.162/4 THEN m ELSE n/a - - stateful proxy behaviour that inserts date, or stateless proxies.						
c2:	IF A.162/4 THEN i ELSE m - - Stateless proxy passes on.						

Prerequisite A.163/22 - - UPDATE response for all remaining status-codes

Table A.307: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Allow	[26] 20.5	m	m	[26] 20.5	i	i
1	Call-ID	[26] 20.8	m	m	[26] 20.8	m	m
1A	Call-Info	[26] 20.9	m	m	[26] 20.9	c4	c4
1B	Cellular-Network-Info	7.2.15	n/a	c22	7.2.15	n/a	c23
1C	Contact	[26] 20.10	m	m	[26] 20.10	i	i
2	Content-Disposition	[26] 20.11	m	m	[26] 20.11	i	c3
3	Content-Encoding	[26] 20.12	m	m	[26] 20.12	i	c3
4	Content-Language	[26] 20.13	m	m	[26] 20.13	i	c3
5	Content-Length	[26] 20.14	m	m	[26] 20.14	m	m
6	Content-Type	[26] 20.15	m	m	[26] 20.15	i	c3
7	CSeq	[26] 20.16	m	m	[26] 20.16	m	m
8	Date	[26] 20.17	m	m	[26] 20.17	c1	c1
9	From	[26] 20.20	m	m	[26] 20.20	m	m
9A	Geolocation-Error	[89] 4.3	c14	c14	[89] 4.3	c15	c15
10	MIME-Version	[26] 20.24	m	m	[26] 20.24	i	c3
10A	Organization	[26] 20.25	m	m	[26] 20.25	c2	c2
10B	P-Access-Network-Info	[52] 4.4, [52A] 4, [234] 2	c11	c11	[52] 4.4, [52A] 4, [234] 2	c12	c12
10C	P-Charging-Function-Addresses	[52] 4.5, [52A] 4	c9	c9	[52] 4.5, [52A] 4	c10	c10
10D	P-Charging-Vector	[52] 4.6, [52A] 4	c7	c7	[52] 4.6, [52A] 4	c8	c8
10F	Privacy	[33] 4.2	c5	c5	[33] 4.2	c6	c6
10G	Recv-Info	[25] 5.2.3	c18	c18	[25] 5.2.3	c19	c19
10H	Relayed-Charge	7.2.12	n/a	c21	7.2.12	n/a	c21
10I	Require	[26] 20.32	m	m	[26] 20.32	c13	c13
10J	Server	[26] 20.35	m	m	[26] 20.35	i	i
10K	Session-ID	[162]	c20	c20	[162]	c20	c20
11	Timestamp	[26] 20.38	m	m	[26] 20.38	i	i
12	To	[26] 20.39	m	m	[26] 20.39	m	m
12A	User-Agent	[26] 20.41	m	m	[26] 20.41	i	i
13	Via	[26] 20.42	m	m	[26] 20.42	m	m
14	Warning	[26] 20.43	m	m	[26] 20.43	i	i

c1:	IF A.162/9 THEN m ELSE i - - insertion of date in requests and responses.
c2:	IF A.162/19A OR A.162/19B THEN m ELSE i - - reading, adding or concatenating the Organization header.
c3:	IF A.3/2 OR A.3/4 THEN m ELSE i - - P-CSCF or S-CSCF.
c4:	IF A.162/19C OR A.162/19D THEN m ELSE i - - reading, adding or concatenating the Call-Info header.
c5:	IF A.162/31 THEN m ELSE n/a - - a privacy mechanism for the Session Initiation Protocol (SIP).
c6:	IF A.162/31D OR A.162/31G THEN m ELSE IF A.162/31C THEN i ELSE n/a - - application of the privacy option "header" or application of the privacy option "id" or passing on of the Privacy header transparently.
c7:	IF A.162/45 THEN m ELSE n/a - - the P-Charging-Vector header extension.
c8:	IF A.162/46 THEN m ELSE IF A.162/45 THEN i ELSE n/a - - adding, deleting, reading or modifying the P-Charging-Vector header before proxying the request or response or the P-Charging-Vector header extension.
c9:	IF A.162/44 THEN m ELSE n/a - - the P-Charging-Function-Addresses header extension.
c10:	IF A.162/44A THEN m ELSE IF A.162/44 THEN i ELSE n/a - - adding, deleting or reading the P-Charging-Function-Addresses header before proxying the request or response, or the P-Charging-Function-Addresses header extension.
c11:	IF A.162/43 THEN x ELSE IF A.162/41 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c12:	IF A.162/43 THEN m ELSE IF A.162/41 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the P-Access-Network-Info header extension.
c13:	IF A.162/11 OR A.162/13 THEN m ELSE i - - reading the contents of the Require header before proxying the request or response or adding or modifying the contents of the Require header before proxying the request or response for methods other than REGISTER.
c14:	IF A.162/70 THEN m ELSE n/a - - SIP location conveyance.
c15:	IF A.162/70A THEN m ELSE IF A.162/70B THEN i ELSE n/a - - addition or modification of location in a SIP method, passes on locations in SIP method without modification.
c18:	IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.
c19:	IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.
c20:	IF A.162/101 THEN m ELSE n/a - - the Session-ID header.
c21:	IF A.162/121 THEN m ELSE n/a - - the Relayed-Charge header field extension.
c22:	IF A.162/43 THEN x ELSE IF A.162/123 THEN m ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.
c23:	IF A.162/43 THEN m ELSE IF A.162/123 THEN i ELSE n/a - - act as subsequent entity within trust network for access network information that can route outside the trust network, the Cellular-Network-Info header extension.

Prerequisite A.163/23 - - UPDATE response

Prerequisite: A.164/102 - - Additional for 2xx response

Table A.308: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
0A	Accept	[26] 20.1	m	m	[26] 20.1	i	i
0B	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
0C	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i
0D	Accept-Resource-Priority	[116] 3.2	c12	c12	[116] 3.2	c12	c12
1	Allow-Events	[28] 8.2.2	m	m	[28] 8.2.2	c1	c1
2	Authentication-Info	[26] 20.6	m	m	[26] 20.6	i	i
3	Contact	[26] 20.10	m	m	[26] 20.10	i	i
3A	Feature-Caps	[190]	c14	c14	[190]	c14	c14
3B	P-Early-Media	[109] 8	o	c10	[109] 8	o	c11
3BA	Priority-Share	Subclause 7.2.16	n/a	c16	Subclause 7.2.16	n/a	c16
3C	Record-Route	[26] 20.30	m	m	[26] 20.30	c3	c3
3D	Recv-Info	[25] 5.2.3	c5	c5	[25] 5.2.3	c6	c6
3E	Resource-Share	Subclause 4.15	n/a	c15	Subclause 4.15	n/a	c15
4	Session-Expires	[58] 4	c4	c4	[58] 4	c4	c4
6	Supported	[26] 20.37	m	m	[26] 20.37	i	i
c1:	IF A.4/20 THEN m ELSE i - - SIP specific event notification extension.						
c3:	IF A.162/15 THEN o ELSE i - - the requirement to be able to use separate URIs in the upstream direction and downstream direction when record routeing.						
c4:	IF A.162/52 THEN m ELSE n/a - - the SIP session timer.						
c5:	IF A.162/20 THEN m ELSE n/a - - SIP INFO method and package framework.						
c6:	IF A.162/20 THEN i ELSE n/a - - SIP INFO method and package framework.						
c10:	IF A.162/76 THEN m ELSE n/a - - the SIP P-Early-Media private header extension for authorization of early media.						
c11:	IF A.162/76 THEN (IF A.3/2 THEN m ELSE i) ELSE n/a - - P-CSCF, using the information in the P-Early-Media header.						
c12:	IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						
c14:	IF A.162/110 THEN m ELSE n/a - - indication of features supported by proxy.						
c15:	IF A.162/122 THEN o ELSE n/a - - resource sharing.						
c16:	IF A.162/124 THEN o ELSE n/a - - priority sharing.						

Prerequisite A.163/23 - - UPDATE response

Prerequisite: A.164/103 OR A.164/104 OR A.164/105 OR A.164/106 - - Additional for 3xx – 6xx response

Table A.308A: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Error-Info	[26] 20.18	m	m	[26] 20.18	i	i
2	Response-Source	7.2.17	n/a	c1	7.2.17	n/a	c1
c1:	IF A.162/125 THEN o ELSE n/a - - use of the Response-Source header field in SIP error responses?						

Prerequisite A.163/23 - - UPDATE response

Prerequisite: A.164/103 or A.164/35 - - Additional for 3xx, 485 (Ambiguous) response

Table A.309: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
2	Contact	[26] 20.10	m	m	[26] 20.10	c1	c1
c1:	IF A.162/19E THEN m ELSE i - - deleting Contact headers.						

Prerequisite A.163/23 - - UPDATE response

Prerequisite: A.164/14 - - Additional for 401 (Unauthorized) response

Table A.309A: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
6	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/23 - - UPDATE response

Prerequisite: A.164/17 OR A.164/23 OR A.164/30 OR A.164/36 OR A.164/42 OR A.164/45 OR A.164/50 OR A.164/51 - - Additional for 404 (Not Found), 413 (Request Entity Too Large), 480(Temporarily not available), 486 (Busy Here), 500 (Internal Server Error), 503 (Service Unavailable), 600 (Busy Everywhere), 603 (Decline) response

Table A.310: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
5	Retry-After	[26] 20.33	m	m	[26] 20.33	i	i

Table A.311: Void

Prerequisite A.163/23 - - UPDATE response

Prerequisite: A.164/20 - - Additional for 407 (Proxy Authentication Required) response

Table A.312: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
4	Proxy-Authenticate	[26] 20.27	m	m	[26] 20.27	m	m
8	WWW-Authenticate	[26] 20.44	m	m	[26] 20.44	i	i

Prerequisite A.163/23 - - UPDATE response

Prerequisite: A.164/25 - - Additional for 415 (Unsupported Media Type) response

Table A.313: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept	[26] 20.1	m	m	[26] 20.1	i	i
2	Accept-Encoding	[26] 20.2	m	m	[26] 20.2	i	i
3	Accept-Language	[26] 20.3	m	m	[26] 20.3	i	i

Prerequisite A.163/23 - - UPDATE response

Prerequisite: A.164/26A - - Additional for 417 (Unknown Resource-Priority) response

Table A.313A: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Accept-Resource-Priority	[116] 3.2	c1	c1	[116] 3.2	c1	c1
c1:	IF A.162/80 THEN m ELSE n/a - - communications resource priority for the session initiation protocol.						

Prerequisite A.163/23 - - UPDATE response

Prerequisite: A.164/27 - - Additional for 420 (Bad Extension) response

Table A.314: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
7	Unsupported	[26] 20.40	m	m	[26] 20.40	c3	c3
c3:	IF A.162/18 THEN m ELSE i - - reading the contents of the Unsupported header before proxying the 420 response to a method other than REGISTER.						

Prerequisite A.163/23 - - UPDATE response

Prerequisite: A.164/28 OR A.164/41A - - Additional for 421 (Extension Required), 494 (Security Agreement Required) response

Table A.314A: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
3	Security-Server	[48] 2	c1	c1	[48] 2	n/a	n/a
c1:	IF A.162/47 THEN m ELSE n/a - - security mechanism agreement for the session initiation protocol.						

Prerequisite A.163/23 - - UPDATE response

Prerequisite: A.164/28A - - Additional for 422 (Session Interval Too Small) response

Table A.314B: Supported header fields within the UPDATE response

Item	Header field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	Min-SE	[58] 5	c1	c1	[58] 5	c1	c1
c1:	IF A.162/52 THEN m ELSE n/a - - the SIP session timer.						

Table A.315: Void

Prerequisite A.163/23 - - UPDATE response

Table A.316: Supported message bodies within the UPDATE response

Item	Header	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1							

A.3 Profile definition for the Session Description Protocol as used in the present document

A.3.1 Introduction

Void.

A.3.2 User agent role

This subclause contains the ICS proforma tables related to the user agent role. They need to be completed only for UA implementations.

Prerequisite: A.2/1 -- user agent role

A.3.2.1 Major capabilities

Table A.317: Major capabilities

Item	Does the implementation support	Reference	RFC status	Profile status
	Capabilities within main protocol			
	Extensions			
22	integration of resource management and SIP?	[30] [64]	o	c14
23	grouping of media lines?	[53]	c3	c3
24	mapping of media streams to resource reservation flows?	[54]	o	c1
25	SDP bandwidth modifiers for RTCP bandwidth?	[56]	o	o (NOTE 1)
26	TCP-based media transport in the session description protocol?	[83]	o	c2
27	interactive connectivity establishment?	[289], [290]	o	c4
28	session description protocol format for binary floor control protocol streams?	[108]	o	o
29	extended RTP profile for real-time transport control protocol (RTCP)-based feedback (RTP/AVPF)?	[135]	o	c5
30	SDP capability negotiation?	[137]	o	c6
31	Session Description Protocol (SDP) extension for setting up audio media streams over circuit-switched bearers in the Public Switched Telephone Network (PSTN)?	[155]	o	c7
32	miscellaneous capabilities negotiation in the Session Description Protocol (SDP)?	[156]	o	c7
33	transport independent bandwidth modifier for the Session Description Protocol?	[152]	o	c8
34	Secure Real-time Transport Protocol (SRTP)?	[169]	o	c15
35	MIKEY-TICKET?	[170]	o	c10
36	SDES?	[168]	o	c9
37	end-to-access-edge media security using SDES?	7.5.2	n/a	c16
37A	end-to-access-edge media security for MSRP using TLS and certificate fingerprints?	7.5.2	n/a	c22
37B	end-to-access-edge media security for BFCP using TLS and certificate fingerprints?	7.5.2	n/a	c23
37C	end-to-access-edge media security for UDPTL using DTLS and certificate fingerprints?	7.5.2	n/a	c24
37D	end-to-access-edge media security for RTP media using DTLS-SRTP and certificate fingerprints?	7.5.2	n/a	c40
38	SDP media capabilities negotiation?	[172]	o	c12
39	Transcoding Services Invocation in the Session Initiation Protocol (SIP) Using Third Party Call Control (3pcc)?	[166]	o	c13
40	Message Session Relay Protocol?	[178]	o	c17
40A	Connection establishment for media anchoring for the message session relay protocol?	[214]	o	c26
41	a SDP offer/answer mechanism to enable file transfer?	[185]	o	o
42	optimal media routeing	[11D]	n/a	c18
43	ECN for RTP over UDP	[188]	o	c19
44	T.38 FAX?	[202]	n/a	c20
45	support for reduced-size RTCP?	[204]	o	o
46	RTCP extended reports?	[205]	o	o
47	maximum receive SDU size?	[9B]	o	o
48	the SDP content attribute?	[206]	o	c21

49	a general mechanism for RTP header extensions?	[210]	o	o
50	negotiation of generic image attributes in the session description protocol (SDP)?	[211]	o	o
51	connection-oriented media transport over the TLS protocol in the SDP?	[241]	o	c25
52	UDPTL over DTLS?	[217]	o	c27
53	telepresence?	[7G]	o	o
54	SCTP over DTLS?	[219]	o	c28
55	DTLS-SRTP?	[222], [223]	o	c41
56	STUN Usage for Consent Freshness?	[224]	o	c29
57	Alternate Connectivity (ALTC) Attribute?	[228]	o	c30
58	3GPP MTSI RTCP-APP adaptation?	[9B]	n/a	o
59	3GPP MTSI Pre-defined Region-of-Interest (ROI)?	[9B]	n/a	o
60	3GPP MTSI Arbitrary Region-of-Interest (ROI)?	[9B]	n/a	o
61	multiplexing RTP data and control packets on a single port	[237], [237A]	o	o
61A	Exclusive RTP and RTCP multiplexed on one port (a=rtcp-mux-only)?	[246]	o	c34
62	SDP-based data channel negotiation?	[238]	o	c31
63	Media plane optimization for WebRTC?	[8Z]	n/a	c32
64	Enhanced bandwidth negotiation mechanism?	[9B]	n/a	o
65	an SDP offer/answer mechanism to negotiate DTLS protected media?	[240]	o	c33
66	Using simulcast in SDP and RTP sessions?	[249]	o	c35
67	RTP payload format restrictions?	[250]	o	c36
68	Compact Concurrent Codec Negotiation and Capabilities?	[9B]	n/a	c35
69	3GPP MTSI Delay Budget Information (DBI)?	[9B]	n/a	c37
70	Access Network Bitrate Recommendation (ANBR)?	[9B]	n/a	c38
71	Framework for Live Uplink Streaming (FLUS)?	[276]	n/a	c39
72	3GPP MTSI client using data channels?	[9B]	n/a	c39

c1:	IF A.3/1 THEN m ELSE n/a - - UE role.
c2:	IF A.3/9B AND A.3/13B THEN m ELSE IF A.3/1 OR A.3/2A OR A.3/6 OR A.3/7 THEN o ELSE n/a - - IBCF (IMS-ALG), ISC gateway function (IMS-ALG), UE, P-CSCF (IMS-ALG), MGCF, AS.
c3:	IF A.317/24 OR A.317/53 THEN m ELSE o - - mapping of media streams to resource reservation flows, telepresence.
c4:	IF A.3/9B OR A.3/13B THEN m ELSE IF A.3/1 OR A.3/6 THEN o ELSE n/a - - IBCF (IMS-ALG), application gateway function (IMS-ALG), UE, MGCF.
c5:	IF A.3A/50 OR A.3A/50A OR A.3/6 OR A.3/9B OR A.3A/89 OR A.3A/11 OR A.3A/12 THEN m ELSE o - - multimedia telephony service participant, multimedia telephony service application server, MGCF, IBCF (IMS-ALG), ATCF (UA), conference focus, conference participant.
c6:	IF A.3A/50 OR A.3A/50A OR A.3/6 OR A.3/9B OR A.3/13B OR A.3A/89 THEN m ELSE o - - multimedia telephony service participant, multimedia telephony service application server, MGCF, IBCF (IMS-ALG), application gateway function (IMS-ALG), ATCF (UA).
c7:	IF A.3A/82 OR A.3A/83 THEN m ELSE o - - ICS user agent, SCC application server.
c8:	IF A.317/25 AND (A.3/1 OR A.3/6 OR A.3A/89) THEN o ELSE n/a - - SDP bandwidth modifiers for RTCP bandwidth, UE, MGCF, ATCF (UA).
c9:	IF A.3D/30 OR A.3D/20 THEN m ELSE n/a - - end-to-access-edge media security using SDES, end-to-end media security using SDES.
c10:	IF A.3D/21 OR A.3D/22 THEN m ELSE n/a - - end-to-end media security using KMS, end-to-end media security for MSRP using TLS and KMS.
c12:	IF A.3A/82 OR A.3A/83 THEN m ELSE o - - ICS user agent, SCC application server.
c13:	IF IF A.3/7D OR A.3/8 THEN o else n/a - - AS performing 3rd party call control or MRFC.
c14:	IF A.4/2C THEN m ELSE o - - initiating a session which require local and/or remote resource reservation.
c15:	IF A.3D/20 OR A.3D/21 OR A.3D/30 THEN m ELSE n/a - - end-to-end media security using SDES, end-to-end media security using KMS, end-to-access-edge media security using SDES.
c16:	If A.3D/30 THEN m ELSE n/a - - end-to-access-edge media security using SDES.
c17:	IF A.3A/33B OR A.3A/34 THEN m ELSE IF A.3A/8 OR A.3A/9 OR A.3/2A THEN o ELSE n/a - - session-mode messaging participant, session-mode messaging intermediate node, IBCF, MRFC, P-CSCF (IMS-ALG).
c18:	IF A.3/2A OR A.3/6 OR A.3/7 OR A.3/9B OR A.3A/89 OR A.3/13B THEN o ELSE n/a - - P-CSCF (IMS-ALG), MGCF, AS, IBCF (IMS-ALG), ATCF (UA), application gateway function (IMS-ALG).
c19:	IF A.3/2A OR A.3/6 OR A.3/8 OR A.3/9B OR A.3A/81 OR A.3A/89 OR A.3/13B OR A.3A/81A OR A.3A/81B THEN o ELSE n/a - - P-CSCF (IMS-ALG), MGCF, MRFC, IBCF (IMS-ALG), MSC Server enhanced for ICS, ATCF (UA), application gateway function (IMS-ALG), MSC server enhanced for SRVCC using SIP interface, MSC server enhanced for DRVCC using SIP interface.
c20:	IF A.3/1 OR A.3/6 THEN o ELSE n/a - - UE, MGCF.
c21:	IF A.3A/57 OR A.3A/58 OR A.3A/59 OR A.3A/60 OR A.3/2A OR A.3/9B OR A.3A/11 OR A.3A/12 THEN m ELSE o - - Customized alerting tones application server, Customized alerting tones UA client, Customized ringing signal application server, Customized ringing signal UA client, P-CSCF (IMS-ALG), IBCF (IMS-ALG), conference focus, conference participant.
c22:	If A.3D/20A THEN m ELSE n/a - - end-to-access-edge media security for MSRP using TLS and certificate fingerprints.
c23:	If A.3D/20B THEN m ELSE n/a - - end-to-access-edge media security for BFCP using TLS and certificate fingerprints.
c24:	If A.3D/20C THEN m ELSE n/a - - end-to-access-edge media security for UDPTL using DTLS and certificate fingerprints.
c25:	IF (A.317/37A AND A.317/40) OR (A.317/37B AND A.317/28) OR (A.317/37C AND A.317/52) OR (A.317/37D AND A.317/55) THEN m ELSE o - - end-to-access-edge media security for MSRP using TLS and certificate fingerprints, message session relay protocol, end-to-access-edge media security for BFCP using TLS and certificate fingerprints, session description protocol format for binary floor control protocol streams, end-to-access-edge media security for UDPTL using DTLS and certificate fingerprints, UDPTL over DTLS, end-to-access-edge media security for RTP media using DTLS-SRTP and certificate fingerprints, DTLS-SRTP.
c26:	IF A.317/40 THEN m ELSE n/a - - message session relay protocol.
c27:	IF A.317/37C THEN m ELSE o - - end-to-access-edge media security for UDPTL using DTLS and certificate fingerprints.
c28:	IF (A.3/1 AND A.317/53) OR A.3/14 OR A.3A/95 THEN m ELSE o - - UE, telepresence, Gm based WIC, eP-CSCF.
c29:	IF A.3/14 OR A.3A/95 THEN m ELSE o - - Gm based WIC, eP-CSCF.
c30:	IF A.3A/81 OR A.3/9B OR A.3/2A THEN o ELSE n/a - - UE performing the functions of an external attached network, IBCF (IMS-ALG), P-CSCF (IMS-ALG).
c31:	IF A.3/14 OR A.3A/95 THEN o ELSE n/a - - Gm based WIC, eP-CSCF.
c32:	IF A.3A/95 OR A.3/9B THEN o ELSE n/a - - eP-CSCF, IMS-ALG.
c33:	IF A.317/52 OR A.317/54 OR A.317/55 THEN m ELSE n/a - - UDPTL over DTLS, SCTP over DTLS, DTLS-SRTP.
c34:	IF A.3/14 OR A.3A/95 THEN m ELSE n/a - - Gm based WIC, eP-CSCF.
c35:	IF A.3A/11 OR A.3A/12 THEN o ELSE n/a - - conference focus, conference participant.

c36:	IF A.317/66 AND (A.3A/11 OR A.3A/12) THEN o ELSE n/a - - Using simulcast in SDP and RTP sessions, conference focus, conference participant.
c37:	IF A.3/1 OR A.3/2A OR A.3/8 OR A.3/9B THEN o ELSE n/a - - UE, P-CSCF (IMS-ALG), MRFC, IBCF (IMS-ALG).
c38:	IF A.3/1 THEN o ELSE n/a - - UE.
c39:	IF A.3/1 OR A.3/2 THEN o ELSE n/a - - UE, P-CSCF.
c40:	IF A.3D/31 THEN m ELSE n/a - - end-to-access-edge media security for RTP media using DTLS-SRTP and certificate fingerprints.
c41:	IF A.3D/31 THEN m ELSE o - - end-to-access-edge media security for RTP media using DTLS-SRTP and certificate fingerprints.
NOTE 1:	For "video" and "audio" media types that utilise RTP/RTCP, if the RTCP bandwidth level for the session is different than the default RTCP bandwidth as specified in RFC 3556 [56], then, it shall be specified. For other media types, it may be specified.

A.3.2.2 SDP types

Table A.318: SDP types

Item	Type	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
	Session level description						
1	v= (protocol version)	[39] 5.1	m	m	[39] 5.1	m	m
2	o= (owner/creator and session identifier)	[39] 5.2	m	m	[39] 5.2	m	m
3	s= (session name)	[39] 5.3	m	m	[39] 5.3	m	m
4	i= (session information)	[39] 5.4	o	c2	[39] 5.4	m	c3
5	u= (URI of description)	[39] 5.5	o	c4	[39] 5.5	o	n/a
6	e= (email address)	[39] 5.6	o	c4	[39] 5.6	o	n/a
7	p= (phone number)	[39] 5.6	o	c4	[39] 5.6	o	n/a
8	c= (connection information)	[39] 5.7	c5	c5	[39] 5.7	m	m
9	b= (bandwidth information)	[39] 5.8	o	o	[39] 5.8	m	m
	Time description (one or more per description)						
10	t= (time the session is active)	[39] 5.9	m	m	[39] 5.9	m	m
11	r= (zero or more repeat times)	[39] 5.10	o	c4	[39] 5.10	o	n/a
	Session level description (continued)						
12	z= (time zone adjustments)	[39] 5.11	o	n/a	[39] 5.11	o	n/a
13	k= (encryption key)	[39] 5.12	x	x	[39] 5.12	n/a	n/a
14	a= (zero or more session attribute lines)	[39] 5.13	o	o	[39] 5.13	m	m
	Media description (zero or more per description)						
15	m= (media name and transport address)	[39] 5.14	m	m	[39] 5.14	m	m
16	i= (media title)	[39] 5.4	o	c2	[39] 5.4	o	c3
17	c= (connection information)	[39] 5.7	c1	c1	[39] 5.7	m	m
18	b= (bandwidth information)	[39] 5.8	o	o	[39] 5.8	m	m
19	k= (encryption key)	[39] 5.12	x	x	[39] 5.12	n/a	n/a
20	a= (zero or more media attribute lines)	[39] 5.13	o	o	[39] 5.13	m	m
c1:	IF (A.318/15 AND NOT A.318/8) THEN m ELSE IF (A.318/15 AND A.318/8) THEN o ELSE n/a - - "c=" contained in session level description and SDP contains media descriptions.						
c2:	IF A.3/6 THEN x ELSE o - - MGCF.						
c3:	IF A.3/6 THEN n/a ELSE m - - MGCF.						
c4:	IF A.3/6 THEN x ELSE n/a - - MGCF.						
c5:	IF A.318/17 THEN o ELSE m - - "c=" contained in all media description.						

Prerequisite A.318/14 OR A.318/20 - - a= (zero or more session/media attribute lines)

Table A.319: zero or more session / media attribute lines (a=)

Item	Field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	category (a=cat)	[39] 6	c8	c8	[39] 6	c9	c9
2	keywords (a=keywds)	[39] 6	c8	c8	[39] 6	c9	c9
3	name and version of tool (a=tool)	[39] 6	c8	c8	[39] 6	c9	c9
4	packet time (a=ptime)	[39] 6	c10	c10	[39] 6	c11	c11
5	maximum packet time (a=maxptime)	[39] 6 (NOTE 1)	c10	c10	[39] 6 (NOTE 1)	c11	c11
6	receive-only mode (a=recvonly)	[39] 6	o	o	[39] 6	m	m
7	send and receive mode (a=sendrecv)	[39] 6	o	o	[39] 6	m	m
8	send-only mode (a=sendonly)	[39] 6	o	o	[39] 6	m	m
8A	Inactive mode (a=inactive)	[39] 6	o	o	[39] 6	m	m
9	whiteboard orientation (a=orient)	[39] 6	c10	c10	[39] 6	c11	c11
10	conference type (a=type)	[39] 6	c8	c8	[39] 6	c9	c9
11	character set (a=charset)	[39] 6	c8	c8	[39] 6	c9	c9
12	language tag (a=sdplang)	[39] 6	o	o	[39] 6	m	m
13	language tag (a=lang)	[39] 6	o	o	[39] 6	m	m
14	frame rate (a=framerate)	[39] 6	c10	c10	[39] 6	c11	c11
15	quality (a=quality)	[39] 6	c10	c10	[39] 6	c11	c11
16	format specific parameters (a=fmtp)	[39] 6	c10	c10	[39] 6	c11	c11
17	rtpmap attribute (a=rtpmap)	[39] 6	c10	c10	[39] 6	c11	c11
18	current-status attribute (a=curr)	[30] 5	c1	c1	[30] 5	c2	c2
19	desired-status attribute (a=des)	[30] 5	c1	c1	[30] 5	c2	c2
20	confirm-status attribute (a=conf)	[30] 5	c1	c1	[30] 5	c2	c2
21	media stream identification attribute (a=mid)	[53] 3	c3	c3	[53] 3	c4	c4
22	group attribute (a=group)	[53] 4	c5	c5	[53] 3	c6	c6
23	setup attribute (a=setup)	[83] 4	c7	c45	[83] 4	c7	c45
24	connection attribute (a=connection)	[83] 5	c7	c7	[83] 5	c7	c7
24A	DTLS association ID attribute (a=tls-id)	[240] 4	c62	c62	[240] 4	c62	c62
25	IP addresses (a=candidate)	[290]	c12	c12	[290]	c13	c13
26	floor control server determination (a=floorctrl)	[108] 4	c14	c14	[108] 4	c14	c14
27	conference id (a=confid)	[108] 5	c14	c14	[108] 5	c14	c14
28	user id (a=userid)	[108] 5	c14	c14	[108] 5	c14	c14
29	association between streams and floors (a=floorid)	[108] 6	c14	c14	[108] 6	c14	c14
30	RTCP feedback capability attribute (a=rtcp-fb)	[135] 4.2	c15	c15	[135] 4.2	c15	c15
31	extension of the rtcp-fb attribute (a=rtcp-fb)	[136] 7.1, [188] 6.2, [251] 9	c15	c15	[136] 7.1, [251] 9	c15	c15
32	supported capability negotiation extensions (a=csup)	[137] 3.3.1	c16	c16	[137] 3.3.1	c16	c16
33	required capability negotiation extensions (a=creq)	[137] 3.3.2	c16	c16	[137] 3.3.2	c16	c16
34	attribute capability (a=acap)	[137] 3.4.1	c16	c16	[137] 3.4.1	c16	c16
35	transport protocol capability (a=tcap)	[137] 3.4.2	c16	c16	[137] 3.4.2	c16	c16
36	potential configuration (a=pcfg)	[137] 3.5.1 [172] 3.3.6	c16	c16	[137] 3.5.1 [172] 3.3.6	c16	c16

37	actual configuration (a=acfg)	[137] 3.5.2	c16	c16	[137] 3.5.2	c16	c16
38	connection data capability (a=ccap)	[156] 3.1	c17	c17	[156] 3.1	c18	c18
39	maximum packet rate (a=maxprate)	[152] 6.3	c19	c19	[152] 6.3	c19	c19
40	crypto attribute (a=crypto)	[168]	c20	c20	[168]	c20	c20
41	key management attribute (a=key-mgmt)	[167]	c21	c21	[167]	c21	c21
42	3GPP_e2ae-security-indicator (a=3ge2ae)	7.5.2	c22	c22	7.5.2	c22	c22
43	media capability (a=rmcap)	[172] 3.3.1	c23	c23	[172] 3.3.1	c23	c23
43A	media capability (a=omcap)	[172] 3.3.1	c23	c23	[172] 3.3.1	c23	c23
44	media format capability (a=mfcap)	[172] 3.3.2	c23	c23	[172] 3.3.2	c23	c23
45	media-specific capability (a=mscap)	[172] 3.3.3	c23	c23	[172] 3.3.3	c23	c23
46	latent configuration (a=lcfg)	[172] 3.3.5	c44	c44	[172] 3.3.5	c44	c44
47	session capability (a=sescap)	[172] 3.3.8	c24	c24	[172] 3.3.8	c24	c24
48	msrp path (a=path)	[178]	c25	c25	[178]	c25	c25
49	file selector (a=file-selector)	[185] 6	c27	c27	[185] 6	c28	c28
50	file transfer identifier (a= file-transfer-id)	[185] 6	c26	c26	[185] 6	c28	c28
51	file disposition (a=file-disposition)	[185] 6	c26	c26	[185] 6	c28	c28
52	file date (a=file-date)	[185] 6	c26	c26	[185] 6	c28	c28
53	file icon (a=file-icon	[185] 6	c26	c26	[185] 6	c28	c28
54	file range (a=file-range)	[185] 6	c26	c26	[185] 6	c28	c28
55	optimal media routeing visited realm (a=visited-realm)	7.5.3	c29	c29	7.5.3	c29	c29
56	optimal media routeing secondary realm (a=secondary-realm)	7.5.3	c29	c29	7.5.3	c29	c29
57	optimal media routeing media level checksum (a=omr-m-cksum)	7.5.3	c29	c29	7.5.3	c29	c29
58	optimal media routeing session level checksum (a=omr-s-cksum)	7.5.3	c29	c29	7.5.3	c29	c29
59	optimal media routeing codecs (a=omr-codecs)	7.5.3	c29	c29	7.5.3	c29	c29
60	optimal media routeing media attributes (a=omr-m-att)	7.5.3	c29	c29	7.5.3	c29	c29
61	optimal media routeing session attributes (a=omr-s-att)	7.5.3	c29	c29	7.5.3	c29	c29
62	optimal media routeing media bandwidth (a=omr-m-bw)	7.5.3	c29	c29	7.5.3	c29	c29
63	optimal media routeing session bandwidth (a=omr-s-bw)	7.5.3	c29	c29	7.5.3	c29	c29
64	ecn-attribute (a=ecn-capable-rtp)	[188]	c30	c30	[188]	c30	c30
65	T38 FAX Protocol version (a=T38FaxVersion)	[202]	n/a	c31	[202]	n/a	c31
66	T38 FAX Maximum Bit Rate (a=T38MaxBitRate)	[202]	n/a	c31	[202]	n/a	c31
67	T38 FAX Rate Management (a=T38FaxRateManagement)	[202]	n/a	c31	[202]	n/a	c31
68	T38 FAX Maximum Buffer Size (a=T38FaxMaxBuffer)	[202]	n/a	c31	[202]	n/a	c31

69	T38 FAX Maximum Datagram Size (a=T38FaxMaxDatagram)	[202]	n/a	c31	[202]	n/a	c31
70	T38 FAX maximum IFP frame size (a=T38FaxMaxIFP)	[202]	n/a	c32	[202]	n/a	c32
71	T38 FAX UDP Error Correction Scheme (a=T38FaxUdpEC)	[202]	n/a	c32	[202]	n/a	c32
72	T38 FAX UDP Error Correction Depth (a=T38FaxUdpECDepth)	[202]	n/a	c32	[202]	n/a	c32
73	T38 FAX UDP FEC Maximum Span (a=T38FaxUdpFECMaxSpan)	[202]	n/a	c32	[202]	n/a	c32
74	T38 FAX Modem Type (a=T38ModemType)	[202]	n/a	c32	[202]	n/a	c32
75	T38 FAX Vendor Info (a=T38VendorInfo)	[202]	n/a	c32	[202]	n/a	c32
76	reduced-size RTCP (a=rtcp-rsize)	[204]	c33	c33	[204]	c34	c34
77	RTP control protocol extended report parameters (a=rtcp-xr)	[205]	c35	c35	[205]	c36	c36
78	maximum receive SDU size (a=3gpp_MaxRecvSDUSize)	[9B]	c37	c37	[9B]	c38	c38
79	content (a=content)	[206]	c39	c39	[206]	c39	c39
80	generic header extension map definition (a=extmap)	[210]	c40	c40	[210]	c41	c41
81	image attribute (a=imageattr)	[211]	c42	c42	[211]	c43	c43
82	fingerprint (a=fingerprint)	[241]	c46	c46	[241]	c46	c46
83	msrp-cema (a=msrp-cema)	[214]	c47	c47	[214]	c47	c47
84	sctp-port (a=sctp-port)	[219]	c48	c48	[219]	c48	c48
84A	max-message-size (a=max-message-size)	[219]	c68	c68	[219]	c48	c48
85	CS correlation (a=cs-correlation)	[155] 5.2.3.1	c49	c49	[155] 5.2.3.1	c49	c49
86	Alternate Connectivity (ALTC) Attribute (a=altc)	[228]	o	c50	[228]	o	c50
87	3GPP MTSI RTCP-APP adaptation (a=3gpp_mtsi_app_adapt)	[9B]	n/a	c51	[9B]	n/a	c52
88	3GPP MTSI Pre-defined Region-of-Interest (ROI) (a=predefined_ROI)	[9B]	n/a	c53	[9B]	n/a	c54
89	RTP and RTCP multiplexed on one port (a=rtcp-mux)	[237], [237A]	c55	c55	[237], [237A]	c55	c55
90	data channel mapping (a=dcmap)	[238]	c56	c56	[238]	c56	c56
91	data channel subprotocol specific attributes (a=dcsa)	[238]	c55	c56	[238]	c56	c56
92	Media plane optimization for WebRTC Contact (a= tra-contact)	7.5.4	c57	c57	7.5.4	c57	c57
93	Media plane optimization for WebRTC m-line (a= tra-m-line)	7.5.4	c58	c58	7.5.4	c58	c58
94	Media plane optimization for WebRTC attribute (a= tra-att)	7.5.4	c57	c57	7.5.4	c57	c57
95	Media plane optimization for WebRTC bandwidth (a= tra-bw)	7.5.4	c57	c57	7.5.4	c57	c57
96	Media plane optimization for WebRTC SCTP-association (a= tra-SCTP-association)	7.5.4	c58	c58	7.5.4	c58	c58
97	Media plane optimization for WebRTC media line number (a= tra-media-line-number)	7.5.4	c59	c59	7.5.4	c59	c59

98	Enhanced bandwidth negotiation mechanism (a=bw-info)	[9B]	n/a	c60	[9B]	n/a	c61
99	Exclusive RTP and RTCP multiplexed on one port (a=rtcp-mux-only)	[246]	c63	c63	[246]	c63	c63
100	Simulcast stream description (a=simulcast)	[249] 6.1	c64	c64	[249] 6.1	c64	c64
101	Restriction identifier (a=rid)	[250] 10	c65	c65	[250] 10	c66	c66
102	3GPP compact concurrent codec capabilities (a=ccc-list)	[9B]	n/a	c67	[9B]	n/a	c67
103	Delay Budget Information (DBI) RTCP feedback type (a=rtcp-fb:* 3gpp-delay-budget)	[9B] 6.2.8	n/a	c69	[9B] 6.2.8	n/a	c69
104	ANBR Support attribute (a=anbr)	[9B]	n/a	c70	[9B]	n/a	c70
105	Label attribute (a=label)	[277] 4	o	c71	[277] 4	o	c71
106	3GPP QoS hint attribute (a=3gpp-qos-hint)	[9B] 6.2.7.4	n/a	c71	[9B] 6.2.7.4	n/a	c71

c1:	IF A.317/22 AND A.318/20 THEN o ELSE n/a - - integration of resource management and SIP, media level attribute name "a=".
c2:	IF A.317/22 AND A.318/20 THEN m ELSE n/a - - integration of resource management and SIP, media level attribute name "a=".
c3:	IF A.317/23 AND A.318/20 THEN o ELSE n/a - - grouping of media lines, media level attribute name "a=".
c4:	IF A.317/23 AND A.318/20 THEN m ELSE n/a - - grouping of media lines, media level attribute name "a=".
c5:	IF A.317/23 AND A.318/14 THEN o ELSE n/a - - grouping of media lines, session level attribute name "a=".
c6:	IF A.317/23 AND A.318/14 THEN m ELSE n/a - - grouping of media lines, session level attribute name "a=".
c7:	IF A.317/26 AND A.318/20 THEN m ELSE n/a - - TCP-based media transport in the session description protocol, media level attribute name "a=".
c8:	IF A.318/14 THEN o ELSE x - - session level attribute name "a=".
c9:	IF A.318/14 THEN m ELSE n/a - - session level attribute name "a=".
c10:	IF A.318/20 THEN o ELSE x - - media level attribute name "a=".
c11:	IF A.318/20 THEN m ELSE n/a - - media level attribute name "a=".
c12:	IF A.317/27 AND A.318/20 THEN o ELSE n/a - - candidate IP addresses, media level attribute name "a=".
c13:	IF A.317/27 AND A.318/20 THEN m ELSE n/a - - candidate IP addresses, media level attribute name "a=".
c14:	IF A.317/28 AND A.318/20 THEN m ELSE n/a - - session description protocol format for binary floor control protocol streams, media level attribute name "a=".
c15:	IF (A.317/29 AND A.318/20) THEN m ELSE n/a - - extended RTP profile for real-time transport control protocol (RTCP)-based feedback (RTP/AVPF), media level attribute name "a=".
c16:	IF A.317/30 AND A.318/20 THEN m ELSE n/a - - SDP capability negotiation, media level attribute name "a=".
c17:	IF A.317/32 AND A.318/20 THEN o ELSE n/a - - miscellaneous capabilities negotiation in the Session Description Protocol (SDP), media level attribute name "a=".
c18:	IF A.317/32 AND A.318/20 THEN m ELSE n/a - - miscellaneous capabilities negotiation in the Session Description Protocol (SDP), media level attribute name "a=".
c19:	IF A.317/33 AND (A.318/14 OR A.318/20) THEN o ELSE n/a - - bandwidth modifier packet rate parameter, media or session level attribute name "a=".
c20:	IF A.317/34 AND A.317/36 AND A.318/20 THEN m ELSE n/a - - Secure Real-time Transport Protocol, media plane security using SDES, media level attribute name "a=".
c21:	IF ((A.317/34 AND A.3D/21) OR A.3D/22) AND A.317/35 AND A.318/20 THEN m ELSE n/a - - Secure Real-time Transport Protocol, end-to-end media security using KMS, end-to-end media security for MSRP using TLS and KMS, MIKEY-TICKET, media level attribute name "a=".
c22:	IF (A.317/37 OR A.317/37A OR A.317/37B OR A.317/37C OR A.317/37D) AND A.318/20 THEN m ELSE n/a - - end-to-access edge media security using SDES, end-to-access-edge media security for MSRP using TLS and certificate fingerprints, end-to-access-edge media security for BFCP using TLS and certificate fingerprints, end-to-access-edge media security for UDPTL using DTLS and certificate fingerprints, end-to-access-edge media security for RTP media using DTLS-SRTP and certificate fingerprints, media level attribute name "a=".
c23:	IF A.317/38 THEN m ELSE n/a - - SDP media capabilities negotiation.
c24:	IF A.317/38 AND A.318/14 THEN m ELSE n/a - - SDP media capabilities negotiation, session level attribute name "a=".
c25:	IF A.317/40 AND A.318/20 THEN m ELSE n/a - - message session relay protocol, media level attribute name "a=".
c26:	IF A.317/41 AND A.318/20 THEN o ELSE n/a - - a SDP offer/answer mechanism to enable file transfer, media level attribute name "a=".
c27:	IF A.317/41 AND A.318/20 AND (A.3A/31 OR A.3A/33) THEN m ELSE IF A.317/41 AND A.318/20 AND NOT (A.3A/31 OR A.3A/33) THEN o ELSE n/a - - a SDP offer/answer mechanism to enable file transfer, media level attribute name "a=", messaging application server, messaging participant.
c28:	IF A.317/41 AND A.318/20 THEN m ELSE n/a - - a SDP offer/answer mechanism to enable file transfer, media level attribute name "a=".
c29:	IF A.317/42 AND A.318/20 THEN o ELSE n/a - - optimal media routeing, media level attribute name "a=".
c30:	IF A.317/43 THEN m ELSE n/a - - ECN for RTP over UDP, media level attribute name "a=".
c31:	IF A.317/44 AND A.318/20 THEN m ELSE n/a - - T.38 FAX, media level attribute name "a=".
c32:	IF A.317/44 AND A.318/20 THEN o ELSE n/a - - T.38 FAX, media level attribute name "a=".
c33:	IF A.317/45 AND A.318/20 THEN o ELSE n/a - - support for reduced-size RTCP, media level attribute name "a=".
c34:	IF A.317/45 AND A.318/20 THEN m ELSE n/a - - support for reduced-size RTCP, media level attribute name "a=".
c35:	IF A.317/46 AND A.318/20 AND A.318/14 THEN o ELSE n/a - - RTCP extended reports, media level attribute name "a=", session level attribute name "a=".
c36:	IF A.317/46 AND A.318/20 AND A.318/14 THEN m ELSE n/a - - RTCP extended reports, media level attribute name "a=", session level attribute name "a=".
c37:	IF A.317/47 AND A.318/20 AND A.318/14 THEN o ELSE n/a - - maximum receive SDU size, media level attribute name "a=", session level attribute name "a=".
c38:	IF A.317/47 AND A.318/20 AND A.318/14 THEN m ELSE n/a - - maximum receive SDU size, media level attribute name "a=", session level attribute name "a=".

c39:	IF A.317/48 AND A.318/20 THEN m ELSE n/a - - the SDP content attribute, media level attribute name "a=".
c40:	IF A.317/49 AND A.318/20 AND A.318/14 THEN o ELSE n/a - - a general mechanism for RTP header extensions, media level attribute name "a=", session level attribute name "a=".
c41:	IF A.317/49 AND A.318/20 AND A.318/14 THEN m ELSE n/a - - a general mechanism for RTP header extensions, media level attribute name "a=", session level attribute name "a=".
c42:	IF A.317/50 AND A.318/20 THEN o ELSE n/a - - negotiation of generic image attributes in the session description protocol (SDP), media level attribute name "a=".
c43:	IF A.317/50 AND A.318/20 THEN m ELSE n/a - - negotiation of generic image attributes in the session description protocol (SDP), media level attribute name "a=".
c44:	IF A.317/38 AND A.318/20 THEN m ELSE n/a - - SDP media capabilities negotiation, media level attribute name "a=".
c45:	IF (A.317/26 OR A.317/52) AND A.318/20 THEN m ELSE n/a - - TCP-based media transport in the session description protocol, UDPTL over DTLS, media level attribute name "a=".
c46:	IF (A.317/51 OR A.317/55) AND A.318/20 AND A.318/14 THEN m ELSE n/a - - connection-oriented media transport over the TLS protocol in the SDP, DTLS-SRTP, media level attribute name "a=", session level attribute name "a=".
c47:	IF A.317/40A AND A.318/20 THEN m ELSE n/a - - connection establishment for media anchoring for the message session relay protocol, media level attribute name "a=".
c48:	IF A.317/54 AND A.318/20 THEN m ELSE n/a - - SCTP over DTLS, media level attribute name "a=".
c49:	IF A.317/31 AND A.318/20 THEN m ELSE n/a - - Session Description Protocol (SDP) extension for setting up audio media streams over circuit-switched bearers in the Public Switched Telephone Network (PSTN) and SIP, media level attribute name "a=".
c50:	IF A.317/57 AND A.318/20 THEN o ELSE n/a - - Alternate Connectivity (ALTC) Attribute, media level attribute name "a="
c51:	IF A.317/58 AND A.318/20 THEN o ELSE n/a - - 3GPP MTSI RTCP-APP adaptation, media level attribute name "a=".
c52:	IF A.317/58 AND A.318/20 THEN m ELSE n/a - - 3GPP MTSI RTCP-APP adaptation, media level attribute name "a=".
c53:	IF A.317/59 AND A.318/20 THEN o ELSE n/a - - 3GPP MTSI Pre-defined Region-of-Interest (ROI), media level attribute name "a=".
c54:	IF A.317/59 AND A.318/20 THEN m ELSE n/a - - 3GPP MTSI Pre-defined Region-of-Interest (ROI), media level attribute name "a=".
c55:	IF A.317/61 AND A.318/20 THEN m ELSE n/a - - multiplexing RTP data and control packets on a single port, media level attribute name "a=".
c56:	IF A.317/62 AND A.318/20 THEN m ELSE n/a - - SDP-based data channel negotiation, media level attribute name "a=".
c57:	IF A.317/63 AND (A.318/14 OR A.318/20) THEN o ELSE n/a - -, Media plane optimization for WebRTC session or media level attribute name "a=".
c58:	IF A.317/63 AND A.318/20 THEN o ELSE n/a - -, Media plane optimization for WebRTC media level attribute name "a=".
c59:	IF A.317/63 AND A.318/14 THEN o ELSE n/a - -, Media plane optimization for WebRTC session level attribute name "a=".
c60:	IF A.317/64 AND A.318/20 THEN o ELSE n/a - - Enhanced bandwidth negotiation mechanism, media level attribute name "a=".
c61:	IF A.317/64 AND A.318/20 THEN m ELSE n/a - - Enhanced bandwidth negotiation mechanism, media level attribute name "a=".
c62:	IF (A.317/52 OR A.317/54 OR A.317/55) AND A.318/20 THEN m ELSE n/a - - UDPTL over DTLS, SCTP over DTLS, DTLS-SRTP, media level attribute name "a=".
c63:	IF A.317/61A AND A.318/20 THEN m ELSE n/a - - Exclusive RTP and RTCP multiplexed on one port (a=rtcp-mux-only), media level attribute name "a=".
c64:	IF A.317/66 AND A.318/20 THEN m ELSE n/a - - Using simulcast in SDP and RTP sessions, media level attribute name "a=".
c65:	IF A.317/67 AND A.318/20 THEN o ELSE n/a - - RTP payload format restrictions, media level attribute name "a=".
c66:	IF A.317/67 AND A.318/20 THEN m ELSE n/a - - RTP payload format restrictions, media level attribute name "a=".
c67:	IF A.317/68 AND A.318/14 THEN o ELSE n/a - - Compact Concurrent Codec Negotiation and Capabilities, session level attribute name "a=".
c68:	IF A.317/54 AND A.318/20 THEN o ELSE n/a - - SCTP over DTLS, media level attribute name "a=".
c69:	IF A.317/69 AND A.318/20 THEN m ELSE n/a - - Delay Budget Information (DBI), media level attribute name "a=".
c70:	IF A.317/70 AND A.318/20 THEN m ELSE n/a - - Access Network Bitrate Recommendation (ANBR), media level attribute name "a=".
c71:	IF (A.317/71 OR A.317/72) AND A.318/20 THEN o ELSE n/a - - Framework for Live Uplink Streaming (FLUS), 3GPP MTSI client using data channels, media level attribute name "a=".
NOTE 1:	Further specification of the usage of this attribute is defined by specifications relating to individual codecs.

Prerequisite A.319/80 - - a= generic header extension map definition (a=extmap)

Table A.319A: RTP header extensions

Item	Field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	coordination of video orientation (urn:3gpp:video-orientation)	[9B]	n/a	o	[9B]	n/a	o
2	higher granularity coordination of video orientation (urn:3gpp:video-orientation:6)	[9B]	n/a	c1	[9B]	n/a	c1
3	video region-of-interest predefined-roi-sent (urn:3gpp: predefined-roi-sent)	[9B]	n/a	c2	[9B]	n/a	c2
4	video region-of-interest arbitrary-roi-sent (urn:3gpp:roi-sent)	[9B]	n/a	c3	[9B]	n/a	c3
c1:	IF A.319A/1 THEN o ELSE n/a - - coordination of video orientation.						
c2:	IF A.317/59 THEN o ELSE n/a - - 3GPP MTSI Pre-defined Region-of-Interest (ROI).						
c3:	IF A.317/60 THEN o ELSE n/a - - 3GPP MTSI Arbitrary Region-of-Interest (ROI).						

A.3.2.3 Void

Table A.320: Void

Table A.321: Void

Table A.322: Void

Table A.323: Void

Table A.324: Void

Table A.325: Void

Table A.326: Void

Table A.327: Void

A.3.2.4 Void

Table A.327A: Void

A.3.3 Proxy role

This subclause contains the ICS proforma tables related to the user role. They need to be completed only for proxy implementations.

Prerequisite: A.2/2 -- proxy role

A.3.3.1 Major capabilities

Table A.328: Major capabilities

Item	Does the implementation support	Reference	RFC status	Profile status
	Capabilities within main protocol			
0A	application of session policy?	6.2, 6.3	x	c2
	Extensions			
1	integration of resource management and SIP?	[30] [64]	o	n/a
2	grouping of media lines?	[53]	c3	x
3	mapping of media streams to resource reservation flows?	[54]	o	x
4	SDP bandwidth modifiers for RTCP bandwidth?	[56]	o	c1
5	TCP-based media transport in the session description protocol?	[83]	o	c11
6	interactive connectivity establishment?	[289], [290]	o	c4
7	session description protocol format for binary floor control protocol streams?	[108]	o	o
8	extended RTP profile for real-time transport control protocol (RTCP)-based feedback (RTP/AVPF)?	[135]	o	c5
9	SDP capability negotiation?	[137]	o	c9
10	Session Description Protocol (SDP) extension for setting up audio media streams over circuit-switched bearers in the Public Switched Telephone Network (PSTN)?	[155]	o	c6
11	miscellaneous capabilities negotiation in the Session Description Protocol (SDP)?	[156]	o	c6
14	Secure Real-time Transport Protocol (SRTP)?	[169]	o	o
15	MIKEY-TICKET?	[170]	o	o
16	SDES?	[168]	o	o
17	end-to-access edge media security using SDES?	7.5.2	n/a	n/a
17A	end-to-access-edge media security for MSRP using TLS and certificate fingerprints?	7.5.2	n/a	n/a
17B	end-to-access-edge media security for BFCP using TLS and certificate fingerprints?	7.5.2	n/a	n/a
17C	end-to-access-edge media security for UDPTL using DTLS and certificate fingerprints?	7.5.2	n/a	n/a
17D	end-to-access-edge media security for RTP media using DTLS-SRTP and certificate fingerprints?	7.5.2	n/a	n/a
18	SDP media capabilities negotiation?	[172]	o	c8
19	Transcoding Services Invocation in the Session Initiation Protocol (SIP) Using Third Party Call Control (3pcc)?	[166]	m	i
20	Message Session Relay Protocol?	[178]	o	o
20A	Connection establishment for media anchoring for the message session relay protocol?	[214]	o	c12
21	a SDP offer/answer mechanism to enable file transfer?	[185]	o	o
22	optimal media routeing?	[11D]	n/a	o
23	ECN for RTP over UDP?	[188]	o	c10
24	T.38 FAX?	[202]	n/a	o
25	support for reduced-size RTCP?	[204]	o	o
26	RTCP extended reports?	[205]	o	o
27	maximum receive SDU size?	[9B]	o	o
28	the SDP content attribute	[206]	o	o
29	a general mechanism for RTP header extensions?	[210]	o	o

30	negotiation of generic image attributes in the session description protocol (SDP)?	[211]	o	o
31	connection-oriented media transport over the TLS protocol in the SDP?	[241]	o	o
32	UDPTL over DTLS?	[217]	o	o
33	telepresence?	[7G]	o	o
34	SCTP over DTLS?	[219]	o	o
35	DTLS-SRTP?	[222], [223]	o	o
36	STUN Usage for Consent Freshness?	[224]	o	o
38	3GPP MTSI RTCP-APP adaptation?	[9B]	n/a	o
39	3GPP MTSI Pre-defined Region of Interest (ROI)?	[9B]	n/a	o
40	3GPP MTSI Arbitrary Region-of-Interest (ROI)?	[9B]	n/a	o
41	multiplexing RTP data and control packets on a single port	[237], [237A]	o	o
42	Media plane optimization for WebRTC	7.5.4	n/a	o
43	Enhanced bandwidth negotiation mechanism	[9B]	n/a	o
45	an SDP offer/answer mechanism to negotiate DTLS protected media?	[240]	o	c13
46	Using simulcast in SDP and RTP sessions?	[249]	o	o
47	RTP payload format restrictions?	[250]	o	o
48	Compact Concurrent Codec Negotiation and Capabilities?	[9B]	n/a	o
49	3GPP MTSI Delay Budget Information (DBI)?	[9B]	n/a	c14
50	Access Network Bitrate Recommendation (ANBR)?	[9B]	n/a	c15
51	Framework for Live Uplink Streaming (FLUS)?	[276]	n/a	c15
52	3GPP MTSI client using data channels?	[9B]	n/a	c15

c1: IF A.3/2 OR A.3A/88 THEN m ELSE n/a - - P-CSCF, ATCF (proxy).
c2: IF A.3/2 OR A.3/4 THEN o ELSE x – P-CSCF, S-CSCF.
c3: IF A.328/3 THEN m ELSE o - - mapping of media streams to resource reservation flows.
c4: IF A.3/2 OR A.3/4 THEN m ELSE n/a - - P-CSCF, S-CSCF.
c5: IF (A.3A/50A AND A.3/7C) OR A.3/2 OR A.3/4 OR A.3A/88 THEN m ELSE n/a - - multimedia telephony service application server as AS acting as a SIP proxy, P-CSCF, S-CSCF, ATCF (proxy).
c6: IF (A.3A/83 AND A.3/7C) OR A.3/4 THEN m ELSE IF A.3A/88 THEN i ELSE n/a - - SCC application server, AS acting as a SIP proxy, S-CSCF, ATCF (proxy).
c7: IF A.328/18 THEN m ELSE o - - SDP media capabilities negotiation.
c8: IF A.3/2 OR A.3/4 THEN m ELSE IF A.3A/88 THEN i ELSE o - - P-CSCF, S-CSCF, ATCF (proxy).
c9: IF (A.3A/50A AND A.3/7C) OR A.3/2 OR A.3/4 OR A.328/18 OR A.3A/88 THEN m ELSE n/a - - multimedia telephony service application server as AS acting as a SIP proxy, P-CSCF, S-CSCF, SDP media capabilities negotiation, ATCF (proxy).
c10: IF A.3A/88 THEN o ELSE i - - ATCF (proxy).
c11: IF A.3/2 OR A.3/4 OR A.3A/88 THEN m ELSE n/a - - P-CSCF, S-CSCF, ATCF (proxy).
c12: IF A.328/20 THEN m ELSE n/a - - message session relay protocol.
c13: IF A.328/32 OR A.328/34 OR A.328/35 THEN m ELSE n/a - - UDPTL over DTLS, SCTP over DTLS, DTLS-SRTP.
c14: IF A.3/2 OR A.3/4 THEN o ELSE n/a - - P-CSCF, S-CSCF.
c15: If A.3/2 THEN o ELSE n/a - - P-CSCF.

A.3.3.2 SDP types

Table A.329: SDP types

Item	Type	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
	Session level description						
1	v= (protocol version)	[39] 5.1	m	m	[39] 5.1	m	m
2	o= (owner/creator and session identifier).	[39] 5.2	m	m	[39] 5.2	i	i
3	s= (session name)	[39] 5.3	m	m	[39] 5.3	i	i
4	i= (session information)	[39] 5.4	m	m	[39] 5.4	i	i
5	u= (URI of description)	[39] 5.5	m	m	[39] 5.5	i	i
6	e= (email address)	[39] 5.6	m	m	[39] 5.6	i	i
7	p= (phone number)	[39] 5.6	m	m	[39] 5.6	i	i
8	c= (connection information)	[39] 5.7	m	m	[39] 5.7	i	i
9	b= (bandwidth information)	[39] 5.8	m	m	[39] 5.8	i	i
	Time description (one or more per description)						
10	t= (time the session is active)	[39] 5.9	m	m	[39] 5.9	i	i
11	r= (zero or more repeat times)	[39] 5.10	m	m	[39] 5.10	i	i
	Session level description (continued)						
12	z= (time zone adjustments)	[39] 5.11	m	m	[39] 5.11	i	i
13	k= (encryption key)	[39] 5.12	m	m	[39] 5.12	i	i
14	a= (zero or more session attribute lines)	[39] 5.13	m	m	[39] 5.13	i	i
	Media description (zero or more per description)						
15	m= (media name and transport address)	[39] 5.14	m	m	[39] 5.14	m	m
16	i= (media title)	[39] 5.4	m	m	[39] 5.4	i	i
17	c= (connection information)	[39] 5.7	m	m	[39] 5.7	i	i
18	b= (bandwidth information)	[39] 5.8	m	m	[39] 5.8	i	c1
19	k= (encryption key)	[39] 5.12	m	m	[39] 5.12	i	i
20	a= (zero or more media attribute lines)	[39] 5.13	m	m	[39] 5.13	i	c1
c1:	IF A.328/0A THEN m ELSE i - - application of session policy.						

Prerequisite A.329/14 OR A.329/20 - - a= (zero or more session/media attribute lines)

Table A.330: zero or more session / media attribute lines (a=)

Item	Field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	category (a=cat)	[39] 6	m	m	[39] 6	i	i
2	keywords (a=keywds)	[39] 6	m	m	[39] 6	i	i
3	name and version of tool (a=tool)	[39] 6	m	m	[39] 6	i	i
4	packet time (a=ptime)	[39] 6	m	m	[39] 6	i	c9
5	maximum packet time (a=maxptime)	[39] 6 (NOTE 1)	m	m	[39] 6 (NOTE 1)	i	c9
6	receive-only mode (a=recvonly)	[39] 6	m	m	[39] 6	i	c9
7	send and receive mode (a=sendrecv)	[39] 6	m	m	[39] 6	i	c9
8	send-only mode (a=sendonly)	[39] 6	m	m	[39] 6	i	c9
8A	Inactive mode (a=inactive)	[39] 6	m	m	[39] 6	i	c9
9	whiteboard orientation (a=orient)	[39] 6	m	m	[39] 6	i	c9
10	conference type (a=type)	[39] 6	m	m	[39] 6	i	i
11	character set (a=charset)	[39] 6	m	m	[39] 6	i	i
12	language tag (a=sdplang)	[39] 6	m	m	[39] 6	i	c9
13	language tag (a=lang)	[39] 6	m	m	[39] 6	i	c9
14	frame rate (a=framerate)	[39] 6	m	m	[39] 6	i	c9
15	quality (a=quality)	[39] 6	m	m	[39] 6	i	c9
16	format specific parameters (a=fmtp)	[39] 6	m	m	[39] 6	i	c9
17	rtpmap attribute (a=rtpmap)	[39] 6	m	m	[39] 6	i	c9
18	current-status attribute (a=curr)	[30] 5	m	m	[30] 5	c2	c2
19	desired-status attribute (a=des)	[30] 5	m	m	[30] 5	c2	c2
20	confirm-status attribute (a=conf)	[30] 5	m	m	[30] 5	c2	c2
21	media stream identification attribute (a=mid)	[53] 3	c5	x	[53] 3	c6	x
22	group attribute (a=group)	[53] 4	c5	x	[53] 4	c6	x
23	setup attribute (a=setup)	[83] 4	c7	c50	[83] 4	c8	c51
24	connection attribute (a=connection)	[83] 5	c7	c7	[83] 5	c8	c8
24A	DTLS association ID attribute (a=tls-id)	[240] 4	c72	c72	[240] 4	c72	c72
25	candidate IP addresses (a=candidate)	[290]	c9	c9	[290]	c10	c10
26	floor control server determination (a=floorctrl)	[108] 4	c11	c11	[108] 4	c12	c13
27	conference id (a=confid)	[108] 5	c11	c11	[108] 5	c12	c13
28	user id (a=userid)	[108] 5	c11	c11	[108] 5	c12	c13
29	association between streams and floors (a=floorid)	[108] 6	c11	c11	[108] 6	c12	c13
30	RTCP feedback capability attribute (a=rtcp-fb)	[135] 4.2	c14	c14	[135] 4.2	c15	c15
31	extension of the rtcp-fb attribute (a=rtcp-fb)	[136] 7.1 [188] 6.2, [251] 9	c14	c14	[136] 7.1, [251] 9	c15	c15
32	supported capability negotiation extensions (a=csup)	[137] 3.3.1	c16	c16	[137] 3.3.1	c17	c17
33	required capability negotiation extensions (a=creq)	[137] 3.3.2	c16	c16	[137] 3.3.2	c17	c17
34	attribute capability (a=acap)	[137] 3.4.1	c16	c16	[137] 3.4.1	c17	c17
35	transport protocol capability (a=tcap)	[137] 3.4.2	c16	c16	[137] 3.4.2	c17	c17
36	potential configuration (a=pcfg)	[137] 3.5.1 [172] 3.3.6	c16	c16	[137] 3.5.1 [172] 3.3.6	c17	c17

37	actual configuration (a=acfg)	[137] 3.5.2	c16	c16	[137] 3.5.2	c17	c17
38	connection data capability (a=ccap)	[156] 3.1	c18	c18	[156] 3.1	c19	c19
40	crypto attribute (a=crypto)	[168]	c20	c20	[167]	c20	c20
41	key management attribute (a=key-mgmt)	[167]	c21	c21	[168]	c22	c22
42	3GPP_e2ae-security-indicator (a=3ge2ae)	7.5.2	c23	c23	7.5.2	c23	c23
43	media capability (a=rmcap)	[172] 3.3.1	c24	c24	[172] 3.3.1	c26	c26
43A	media capability (a=omcap)	[172] 3.3.1	c24	c24	[172] 3.3.1	c26	c26
44	media format capability (a=mfcap)	[172] 3.3.2	c24	c24	[172] 3.3.2	c26	c26
45	media-specific capability (a=mscap)	[172] 3.3.3	c24	c24	[172] 3.3.3	c26	c26
46	latent configuration (a=lcfg)	[172] 3.3.5	c48	c48	[172] 3.3.5	c49	c49
47	session capability (a=sescap)	[172] 3.3.8	c25	c25	[172] 3.3.8	c27	c27
48	msrp path (a=path)	[178]	c28	c28	[178]	c29	c29
49	file selector (a=file-selector)	[185] 6	c30	c30	[185] 6	c31	c31
50	file transfer identifier (a= file-transfer-id)	[185] 6	c30	c30	[185] 6	c31	c31
51	file disposition (a=file-disposition)	[185] 6	c30	c30	[185] 6	c31	c31
52	file date (a=file-date)	[185] 6	c30	c30	[185] 6	c31	c31
53	file icon (a=file-icon	[185] 6	c30	c30	[185] 6	c31	c31
54	file range (a=file-range)	[185] 6	c30	c30	[185] 6	c31	c31
55	optimal media routeing visited realm (a=visited-realm)	7.5.3	c32	c32	7.5.3	c33	c33
56	optimal media routeing secondary realm (a=secondary-realm)	7.5.3	c32	c32	7.5.3	c33	c33
57	optimal media routeing media level checksum (a=omr-m-cksum)	7.5.3	c32	c32	7.5.3	c33	c33
58	optimal media routeing session level checksum (a=omr-s-cksum)	7.5.3	c32	c32	7.5.3	c33	c33
59	optimal media routeing codecs (a=omr-codecs)	7.5.3	c32	c32	7.5.3	c33	c33
60	optimal media routeing media attributes (a=omr-m-att)	7.5.3	c32	c32	7.5.3	c33	c33
61	optimal media routeing session attributes (a=omr-s-att)	7.5.3	c32	c32	7.5.3	c33	c33
62	optimal media routeing media bandwidth (a=omr-m-bw)	7.5.3	c32	c32	7.5.3	c33	c33
63	optimal media routeing session bandwidth (a=omr-s-bw)	7.5.3	c32	c32	7.5.3	c33	c33
64	ecn-attribute (a=ecn-capable-rtp)	[188]	c34	c34	[188]	c34	c34
65	T38 FAX Protocol version (a=T38FaxVersion)	[202]	n/a	c35	[202]	n/a	c36
66	T38 FAX Maximum Bit Rate (a=T38MaxBitRate)	[202]	n/a	c35	[202]	n/a	c36
67	T38 FAX Rate Management (a=T38FaxRateManagement)	[202]	n/a	c35	[202]	n/a	c36
68	T38 FAX Maximum Buffer Size (a=T38FaxMaxBuffer)	[202]	n/a	c35	[202]	n/a	c36
69	T38 FAX Maximum Datagram Size (a=T38FaxMaxDatagram)	[202]	n/a	c35	[202]	n/a	c36

70	T38 FAX maximum IFP frame size (a=T38FaxMaxIFP)	[202]	n/a	c35	[202]	n/a	c36
71	T38 FAX UDP Error Correction Scheme (a=T38FaxUdpEC)	[202]	n/a	c35	[202]	n/a	c36
72	T38 FAX UDP Error Correction Depth (a=T38FaxUdpECDepth)	[202]	n/a	c35	[202]	n/a	c36
73	T38 FAX UDP FEC Maximum Span (a=T38FaxUdpFECMaxSpan)	[202]	n/a	c35	[202]	n/a	c36
74	T38 FAX Modem Type (a=T38ModemType)	[202]	n/a	c35	[202]	n/a	c36
75	T38 FAX Vendor Info (a=T38VendorInfo)	[202]	n/a	c35	[202]	n/a	c36
76	reduced-size RTCP (a=rtcp-rsize)	[204]	c37	c37	[204]	c38	c38
77	RTP control protocol extended report parameters (a=rtcp-xr)	[205]	c39	c39	[205]	c40	c40
78	maximum receive SDU size (a=3gpp_MaxRecvSDUSize)	[9B]	c41	c41	[9B]	c42	c42
79	content (a=content)	[206]	c43	c43	[206]	c43	c43
80	generic header extension map definition (a=extmap)	[210]	c44	c44	[210]	c45	c45
81	image attribute (a=imageattr)	[211]	c46	c46	[211]	c47	c47
82	fingerprint (a=fingerprint)	[241]	c52	c52	[241]	c53	c53
83	msrp-cema (a=msrp-cema)	[214]	c54	c54	[214]	c55	c55
84	sctp-port (a=sctp-port)	[219]	c56	c56	[219]	c56	c56
84A	max-message-size (a=max-message-size)	[219]	c56	c56	[219]	c56	c56
85	CS correlation (a=cs-correlation)	[155] 5.2.3.1	c57	c57	[155] 5.2.3.1	c57	c57
87	3GPP MTSI RTCP-APP adaptation (a=3gpp_mtsi_app_adapt)	[9B]	n/a	c58	[9B]	n/a	c59
88	3GPP MTSI Pre-defined Region-of-Interest (ROI) (a=predefined_ROI)	[9B]	n/a	c60	[9B]	n/a	c61
89	RTP and RTCP multiplexed on one port (a=rtcp-mux)	[237], [237A]	c62	c62	[237], [237A]	c63	c63
92	Media plane optimization for WebRTC Contact (a= tra-contact)	7.5.4	c64	c64	7.5.4	c65	c65
93	Media plane optimization for WebRTC m-line (a= tra-m-line)	7.5.4	c66	c66	7.5.4	c67	c67
94	Media plane optimization for WebRTC attribute (a= tra-att)	7.5.4	c64	c64	7.5.4	c65	c65
95	Media plane optimization for WebRTC bandwidth (a= tra-bw)	7.5.4	c64	c64	7.5.4	c65	c65
98	Media plane optimization for WebRTC SCTP-association (a= tra-SCTP-association)	7.5.4	c66	c66	7.5.4	c67	c67
97	Media plane optimization for WebRTC media line number (a= tra-media-line-number)	7.5.4	c68	c68	7.5.4	c69	c69
98	Enhanced bandwidth negotiation mechanism (a=bw-info)	[9B]	n/a	c70	[9B]	n/a	c71
99	Exclusive RTP and RTCP multiplexed on one port (a=rtcp-mux-only)	[246]	c62	c62	[246]	c63	c63
100	Simulcast stream description (a=simulcast)	[249] 6.1	c73	c73	[249] 6.1	c74	c74
101	Restriction identifier (a=rid)	[250] 10	c75	c75	[250] 10	c76	c76

102	3GPP compact concurrent codec capabilities (a=ccc-list)	[9B]	n/a	c77	[9B]	n/a	c78
103	Delay Budget Information (DBI) RTCP feedback type (a=rtcp-fb:* 3gpp-delay-budget)	[9B] 6.2.8	n/a	c79	[9B] 6.2.8	n/a	c79
104	ANBR Support attribute (a=anbr)	[9B]	n/a	c80	[9B]	n/a	c80
105	Label attribute (a=label)	[277] 4	o	c81	[277] 4	o	c81
106	3GPP QoS hint attribute (a=3gpp-qos-hint)	[9B] 6.2.7.4	n/a	c81	[9B]	n/a	c81

c2:	IF A.328/1 THEN m ELSE i - - integration of resource management and SIP.
c5:	IF A.328/2 THEN m ELSE n/a - - grouping of media lines.
c6:	IF A.328/3 THEN m ELSE IF A.328/2 THEN i ELSE n/a - - mapping of media streams to resource reservation flows, grouping of media lines.
c7:	IF A.328/5 THEN m ELSE n/a.
c8:	IF A.328/5 THEN i ELSE n/a.
c9:	IF A.329/20 AND A.328/0A THEN m ELSE i - - media level attribute name "a=" and application of session policy.
c9:	IF A.328/6 THEN m ELSE n/a - - interactive connectivity establishment.
c10:	IF A.328/1 AND A.328/6 THEN m ELSE IF A.328/6 THEN i ELSE n/a - - integration of resource management and SIP, interactive connectivity establishment.
c11:	IF A.328/7 THEN m ELSE n/a - - session description protocol format for binary floor control protocol streams.
c12:	IF A.328/7 THEN i ELSE n/a - - session description protocol format for binary floor control protocol streams.
c13:	IF A.328/7 AND A.328/0A AND A.329/20 THEN m ELSE IF A.328/7 AND A.329/20 THEN i ELSE n/a - - session description protocol format for binary floor control protocol streams, media level attribute name "a=" and application of session policy.
c14:	IF (A.328/8 AND A.329/20) THEN m ELSE n/a - - extended RTP profile for real-time transport control protocol (RTCP)-based feedback (RTP/AVPF), media level attribute name "a=".
c15:	IF (A.328/8 AND A.329/20) THEN i ELSE n/a - - extended RTP profile for real-time transport control protocol (RTCP)-based feedback (RTP/AVPF), media level attribute name "a=".
c16:	IF A.328/9 AND A.329/20 THEN m ELSE n/a - - SDP capability negotiation, media level attribute name "a=".
c17:	IF A.328/9 AND A.329/20 THEN i ELSE n/a - - SDP capability negotiation, media level attribute name "a=".
c18:	IF A.328/11 AND A.329/20 THEN o ELSE n/a - - miscellaneous capabilities negotiation in the Session Description Protocol (SDP), media level attribute name "a=".
c19:	IF A.328/11 AND A.329/20 THEN m ELSE n/a - - miscellaneous capabilities negotiation in the Session Description Protocol (SDP), media level attribute name "a=".
c20:	IF A.328/14 AND A.328/16 AND A.329/20 THEN m ELSE n/a - - Secure Real-time Transport Protocol, media plane security using SDES, media level attribute name "a=".
c21:	IF ((A.328/14 AND A.3D/21) OR A.3D/22) AND A.328/15 AND A.329/20 THEN m ELSE n/a - - Secure Real-time Transport Protocol, media plane security using KMS, end-to-end media security for MSRP using TLS and KMS, MIKEY-TICKET, media level attribute name "a=".
c22:	IF ((A.328/14 AND A.3D/21) OR A.3D/22) AND A.328/15 AND A.329/20 THEN i ELSE n/a - - Secure Real-time Transport Protocol, media plane security using KMS, end-to-end media security for MSRP using TLS and KMS, MIKEY-TICKET, media level attribute name "a=".
c23:	IF A.328/17 AND A.329/20 THEN m ELSE n/a - - end to access edge media security, media level attribute name "a=".
c24:	IF A.328/18 THEN m ELSE n/a - - SDP media capabilities negotiation.
c25:	IF A.328/18 AND A.329/14 THEN m ELSE n/a - - SDP media capabilities negotiation, session level attribute name "a=".
c26:	IF A.328/18 AND A.328/0A THEN m ELSE IF A.328/18 THEN i ELSE n/a - - SDP media capabilities negotiation, application of session policy.
c27:	IF A.328/18 AND A.329/14 AND A.328/0A THEN m ELSE IF A.328/18 AND A.329/14 THEN i ELSE n/a - - SDP media capabilities negotiation, session level attribute name "a=", application of session policy.
c28:	IF A.328/20 AND A.329/20 THEN m ELSE n/a - - message session relay protocol, media level attribute name "a=".
c29:	IF A.328/20 AND A.329/20 THEN i ELSE n/a - - message session relay protocol, media level attribute name "a=".
c30:	IF A.328/21 AND A.329/20 THEN m ELSE n/a - - a SDP offer/answer mechanism to enable file transfer, media level attribute name "a=".
c31:	IF A.328/21 AND A.329/20 THEN i ELSE n/a - - a SDP offer/answer mechanism to enable file transfer, media level attribute name "a=".
c32:	IF A.328/22 AND A.329/20 THEN m ELSE n/a - - optimal media routeing, media level attribute name "a=".
c33:	IF A.328/22 AND A.329/20 THEN i ELSE n/a - - optimal media routeing, media level attribute name "a=".
c34:	IF A.328/23 THEN m ELSE i - - ECN for RTP over UDP, media level attribute name "a=".
c35:	IF A.328/24 AND A.329/20 THEN m ELSE n/a - - T.38 FAX, media level attribute name "a=".
c36:	IF A.328/24 AND A.329/20 THEN i ELSE n/a - - T.38 FAX, media level attribute name "a=".
c37:	IF A.328/25 AND A.329/20 THEN m ELSE n/a. - - support for reduced-size RTCP, media level attribute name "a=".
c38:	IF A.328/25 AND A.329/20 THEN i ELSE n/a - - support for reduced-size RTCP, media level attribute name "a=".
c39:	IF A.328/26 AND A.329/20 AND A.329/14 THEN m ELSE n/a -- RTCP extended reports, media level attribute name "a=", session level attribute name "a=".
c40:	IF A.328/26 AND A.329/20 AND A.329/14 THEN i ELSE n/a -- RTCP extended reports, media level attribute name "a=", session level attribute name "a=".
c41:	IF A.328/27 AND A.329/20 AND A.329/14 THEN m ELSE n/a -- maximum receive SDU size, media level attribute name "a=", session level attribute name "a=".

c42: IF A.328/27 AND A.329/20 AND A.329/14 THEN i ELSE n/a -- maximum receive SDU size, media level attribute name "a=", session level attribute name "a=".
c43: IF A.328/28 AND A.329/20 THEN m ELSE n/a - - the SDP content attribute, media level attribute name "a=".
c44: IF A.328/29 AND A.329/20 AND A.329/14 THEN m ELSE n/a - - a general mechanism for RTP header extensions, media level attribute name "a=", session level attribute name "a=".
c45: IF A.328/29 AND A.329/20 AND A.329/14 THEN i ELSE n/a - - a general mechanism for RTP header extensions, media level attribute name "a=", session level attribute name "a=".
c46: IF A.328/30 AND A.329/20 THEN m ELSE n/a - - negotiation of generic image attributes in the session description protocol (SDP), media level attribute name "a=".
c47: IF A.328/30 AND A.329/20 THEN i ELSE n/a - - negotiation of generic image attributes in the session description protocol (SDP), media level attribute name "a=".
c48: IF A.328/18 AND A.329/20 THEN m ELSE n/a - - SDP media capabilities negotiation, media level attribute name "a=".
c49: IF A.328/18 AND A.329/20 AND A.328/0A THEN m ELSE IF A.328/18 AND A.329/20 THEN i ELSE n/a - - SDP media capabilities negotiation, media level attribute name "a=", application of session policy.
c50: IF (A.328/5 OR A.328/32) THEN m ELSE n/a -- TCP-based media transport in the session description protocol, UDPTL over DTLS.
c51: IF (A.328/5 OR A.328/32) THEN i ELSE n/a -- TCP-based media transport in the session description protocol, UDPTL over DTLS.
c52: IF (A.328/82 OR A.328/34) AND A.329/20 AND A.329/14 THEN m ELSE n/a -- connection-oriented media transport over the TLS protocol in the SDP, DTLS-SRTP, media level attribute name "a=", session level attribute name "a=".
c53: IF (A.328/83 OR A.328/34) AND A.329/20 AND A.329/14 THEN i ELSE n/a -- connection-oriented media transport over the TLS protocol in the SDP, DTLS-SRTP, media level attribute name "a=", session level attribute name "a=".
c54: IF A.328/20A AND A.329/20 THEN m ELSE n/a -- connection establishment for media anchoring for the message session relay protocol, media level attribute name "a=".
c55: IF A.328/20A AND A.329/20 THEN i ELSE n/a -- connection establishment for media anchoring for the message session relay protocol, media level attribute name "a=".
c56: IF A.328/33 AND A.329/20 THEN i ELSE n/a -- SCTP on top of DTLS, media level attribute name "a=".
c57: IF A.318/10 AND A.329/20 THEN m ELSE n/a - - Session Description Protocol (SDP) extension for setting up audio media streams over circuit-switched bearers in the Public Switched Telephone Network (PSTN) and SIP, media level attribute name "a=".
c58: IF A.328/38 AND A.329/20 THEN m ELSE n/a -- 3GPP MTSI RTCP-APP adaptation, media level attribute name "a=".
c59: IF A.328/38 AND A.329/20 THEN i ELSE n/a -- 3GPP MTSI RTCP-APP adaptation, media level attribute name "a=".
c60: IF A.328/39 AND A.329/20 THEN o ELSE n/a - - 3GPP MTSI Pre-defined Region-of-Interest (ROI), media level attribute name "a=".
c61: IF A.328/39 AND A.329/20 THEN m ELSE n/a - - 3GPP MTSI Pre-defined Region-of-Interest (ROI), media level attribute name "a=".
c62: IF A.328/41 AND A.329/20 AND A.329/14 THEN m ELSE n/a -- multiplexing RTP data and control packets on a single port, media level attribute name "a=", session level attribute name "a=".
c63: IF A.328/41 AND A.329/20 AND A.329/14 THEN i ELSE n/a -- multiplexing RTP data and control packets on a single port, media level attribute name "a=", session level attribute name "a=".
c64: IF A.328/42 AND (A.329/14 OR A.329/20) THEN m ELSE n/a - -, Media plane optimization for WebRTC session or media level attribute name "a=".
c65: IF A. A.328/42 AND (A.329/14 OR A.329/20) THEN i ELSE n/a - -, Media plane optimization for WebRTC session or media level attribute name "a=".
c66: IF A.328/42 AND A.329/20 THEN m ELSE n/a - -, Media plane optimization for WebRTC media level attribute name "a=".
c67: IF A.328/42 AND A.329/20 THEN i ELSE n/a - -, Media plane optimization for WebRTC media level attribute name "a=".
c68: IF A.328/42 AND A.329/14 THEN m ELSE n/a - -, Media plane optimization for WebRTC session level attribute name "a=".
c69: IF A.328/42 AND A.329/14 THEN i ELSE n/a - -, Media plane optimization for WebRTC session level attribute name "a=".
c70: IF A.328/43 AND A.329/20 THEN o ELSE n/a - - Enhanced bandwidth negotiation mechanism, media level attribute name "a=".
c71: IF A.328/43 AND A.329/20 THEN m ELSE n/a - - Enhanced bandwidth negotiation mechanism, media level attribute name "a=".
c72: IF (A.328/32 OR A.328/34 OR A.328/35) AND A.329/20 THEN m ELSE n/a - - UDPTL over DTLS, SCTP over DTLS, DTLS-SRTP, media level attribute name "a=".
c73: IF A.328/46 AND A.329/20 THEN m ELSE n/a - - Using simulcast in SDP and RTP sessions, media level attribute name "a=".
c74: IF A.328/46 AND A.329/20 THEN i ELSE n/a - - Using simulcast in SDP and RTP sessions, media level attribute name "a=".

c75:	IF A.328/47 AND A.329/20 THEN o ELSE n/a - - RTP payload format restrictions, media level attribute name "a=".
c76:	IF A.328/47 AND A.329/20 THEN i ELSE n/a - - RTP payload format restrictions, media level attribute name "a=".
c77:	IF A.328/48 AND A.329/14 THEN o ELSE n/a - - Compact Concurrent Codec Negotiation and Capabilities, session level attribute name "a=".
c78:	IF A.328/48 AND A.329/14 THEN i ELSE n/a - - Compact Concurrent Codec Negotiation and Capabilities, session level attribute name "a=".
c79:	IF A.328/49 AND A.329/20 m ELSE n/a - - Delay Budget Information (DBI), media level attribute name "a=".
c80:	IF A.328/50 AND A.329/20 THEN m ELSE n/a - - Access Network Bitrate Recommendation (ANBR), media level attribute name "a=".
c81:	IF (A.328/51 OR A.328/52) AND A.329/20 THEN o ELSE n/a - - Framework for Live Uplink Streaming (FLUS), 3GPP MTSI client using data channels, media level attribute name "a=".
NOTE 1:	Further specification of the usage of this attribute is defined by specifications relating to individual codecs.

Prerequisite A.330/80 - - a= generic header extension map definition (a=extmap)

Table A.330A: RTP header extensions

Item	Field	Sending			Receiving		
		Ref.	RFC status	Profile status	Ref.	RFC status	Profile status
1	coordination of video orientation (urn:3gpp:video-orientation)	[9B]	n/a	m	[9B]	n/a	i
2	higher granularity coordination of video orientation (urn:3gpp:video-orientation:6)	[9B]	n/a	m	[9B]	n/a	i
3	video region-of-interest predefined-roi-sent (urn:3gpp:predefined-roi-sent)	[9B]	n/a	m	[9B]	n/a	i
4	video region-of-interest arbitrary-roi-sent (urn:3gpp:roi-sent)	[9B]	n/a	m	[9B]	n/a	i

A.3.3.3 Void

Table A.331: Void

Table A.332: Void

Table A.333: Void

Table A.334: Void

Table A.335: Void

Table A.336: Void

Table A.337: Void

Table A.338: Void

A.3.3.4 Void

Table A.339: Void

A.4 Profile definition for other message bodies as used in the present document

Void.

Annex B (normative): IP-Connectivity Access Network specific concepts when using GPRS to access IM CN subsystem

B.1 Scope

The present annex defines IP-CAN specific requirements for a call control protocol for use in the IP Multimedia (IM) Core Network (CN) subsystem based on the Session Initiation Protocol (SIP), and the associated Session Description Protocol (SDP), where the IP-CAN is General Packet Radio Service (GPRS). The GPRS IP-CAN has a core network:

- with SGSN connected to GGSN providing a UE with a PDP context for the IM CN subsystem-related signalling and media; or
- with SGSN connected to S-GW and S-GW connected to P-GW providing the UE with a PDP context for the IM CN subsystem-related signalling and media.

NOTE: The UE is unaware of how the network provides the UE with a PDP context for the IM CN subsystem-related signalling and media.

The core network can be supported by GERAN and UTRAN radio access networks.

The present annex also defines procedures for invoking CS domain services.

B.2 GPRS aspects when connected to the IM CN subsystem

B.2.1 Introduction

A UE accessing the IM CN subsystem, and the IM CN subsystem itself, utilise the services provided by the core network to provide packet-mode communication between the UE and the IM CN subsystem.

Requirements for the UE on the use of these packet-mode services are specified in this clause. Requirements for the core network in support of this communication are specified in 3GPP TS 29.061 [11], 3GPP TS 29.207 [12] and 3GPP TS 29.212 [13B].

When using the GPRS IP-CAN, each IP-CAN bearer is provided by a PDP context.

B.2.2 Procedures at the UE

B.2.2.1 PDP context activation and P-CSCF discovery

Prior to communication with the IM CN subsystem, the UE shall:

a) if not attached for GPRS services yet, perform a GPRS attach procedure as specified in 3GPP TS 24.008 [8];

b) ensure that a PDP context used for SIP signalling according to the APN and GGSN or P-GW selection criteria described in 3GPP TS 23.060 [4] and 3GPP TS 27.060 [10A] is available. This PDP context shall remain active throughout the period the UE is connected to the IM CN subsystem, i.e. from the initial registration and at least until the deregistration. As a result, the PDP context provides the UE with information that makes the UE able to construct an IPv4 or an IPv6 address;

NOTE 1: During the PDP context activation procedure, the UE and the core network negotiate whether the UE or the GPRS IP-CAN is responsible for the resource reservation applicable to all PDP contexts within the activated PDP address/APN pair, as described in 3GPP TS 24.008 [8].

When the bearer establishment is controlled by the UE, the UE shall choose one of the following options when performing establishment of this PDP context:

I. A dedicated PDP context for SIP signalling:

The UE shall indicate to the core network that this is a PDP context intended to carry IM CN subsystem-related signalling only by setting the IM CN Subsystem Signalling Flag. The UE may also use this PDP context for DNS and DHCP signalling according to the static packet filters as described in 3GPP TS 29.061 [11]. The UE can also set the Signalling Indication attribute within the QoS information element;

II. A general-purpose PDP context:

The UE may decide to use a general-purpose PDP Context to carry IM CN subsystem-related signalling. The UE shall indicate to the core network that this is a general-purpose PDP context by not setting the IM CN Subsystem Signalling Flag. The UE may carry both signalling and media on the general-purpose PDP context. The UE can also set the Signalling Indication attribute within the QoS information element.

NOTE 2: When the bearer establishment is controlled by the GPRS IP-CAN, the core network follows the procedures described in 3GPP TS 29.061 [11] in order to establish a dedicated PDP context for SIP signalling.

If the UE supports the optional configuration parameter "Access_Point_Name_Parameter_Reading_Rule", as defined in 3GPP TS 24.167 [8G] and has been configured with this parameter, then the UE shall use it to retrieve the access point name to use in the PDP context activation procedure.

The UE indicates the IM CN Subsystem Signalling Flag within the Protocol Configuration Options information element of the ACTIVATE PDP CONTEXT REQUEST message or ACTIVATE SECONDARY PDP CONTEXT REQUEST message. Upon successful signalling PDP context establishment the UE receives an indication in the form of IM CN Subsystem Signalling Flag within the Protocol Configuration Options information element. If the flag is not received, the UE shall consider the PDP context as a general-purpose PDP context.

The encoding of the IM CN Subsystem Signalling Flag within the Protocol Configuration Options information element is described in 3GPP TS 24.008 [8].

The UE can indicate a request for prioritised handling over the radio interface by setting the Signalling Indication attribute (see 3GPP TS 23.107 [4A]). The general QoS negotiation mechanism and the encoding of the Signalling Indication attribute within the QoS information element are described in 3GPP TS 24.008 [8]; and

NOTE 3: A general-purpose PDP Context can carry both IM CN subsystem signalling and media, in case the media does not need to be authorized by Policy and Charging control mechanisms as defined in 3GPP TS 29.212 [13B] and Service Based Local Policy mechanisms defined in 3GPP TS 29.207 [12] and the media stream is not mandated by the P-CSCF to be carried in a separate PDP Context.

c) acquire a P-CSCF address(es).

The methods for P-CSCF discovery are:

I. When using IPv4, employ the Dynamic Host Configuration Protocol (DHCP) RFC 2132 [20F], the DHCPv4 options for SIP servers RFC 3361 [35A], and RFC 3263 [27A] as described in subclause 9.2.1. When using IPv6, employ Dynamic Host Configuration Protocol for IPv6 (DHCPv6) RFC 3315 [40], the DHCPv6 options for SIP servers RFC 3319 [41] and DHCPv6 options for Domain Name Servers (DNS) RFC 3646 [56C] as described in subclause 9.2.1.

II. Transfer P-CSCF address(es) within the PDP context activation procedure.

The UE shall indicate the request for a P-CSCF address within the Protocol Configuration Options information element of the ACTIVATE PDP CONTEXT REQUEST message or ACTIVATE SECONDARY PDP CONTEXT REQUEST message.

If the UE is provided with a list of P-CSCF IPv4 or IPv6 addresses in the ACTIVATE PDP CONTEXT ACCEPT message or ACTIVATE SECONDARY PDP CONTEXT ACCEPT message, the UE shall assume that the list is ordered top-down with the first P-CSCF address within the Protocol Configuration Options information element as the P-CSCF address having the highest preference and the last P-CSCF address

within the Protocol Configuration Options information element as the P-CSCF address having the lowest preference.

III. The UE selects a P-CSCF from the list (see 3GPP TS 31.103 [15B]) stored in the ISIM.

IV. The UE selects a P-CSCF from the list in IMS management object.

The UE shall use method IV to select a P-CSCF, if:

- a P-CSCF is to be discovered in the home network;
- the UE is roaming; and
- the IMS management object contains the P-CSCF list.

The UE shall use method III to select the P-CSCF, if:

- a P-CSCF is to be discovered in the home network;
- the UE is roaming;
- either the UE does not contain the IMS management object, or the UE contains the IMS management object, but the IMS management object does not contain the P-CSCF list; and
- the ISIM residing in the UICC supports the P-CSCF list.

The UE can freely select method I or II for P-CSCF discovery, if:

- the UE is in the home network; or
- the UE is roaming and the P-CSCF is to be discovered in the visited network.

The UE can select method IV, if:

- the UE is in the home network; and
- the IMS management object contains the P-CSCF list.

In case method I is selected and several P-CSCF addresses or FQDNs are provided to the UE, the selection of P-CSCF address or FQDN shall be performed as indicated in RFC 3361 [35A] when using IPv4 or RFC 3319 [41] when using IPv6. If sufficient information for P-CSCF address selection is not available, selection of the P-CSCF address by the UE is implementation specific.

NOTE 4: The UE decides whether the P-CSCF is to be discovered in the serving network or in the home network based on local configuration, e.g. whether the application on the UE is permitted to use local breakout.

If the UE is designed to use I above, but receives P-CSCF address(es) according to II, then the UE shall either ignore the received address(es), or use the address(es) in accordance with II, and not proceed with the DHCP request according to I.

If the UE is configured to use Option II above and detects that all P-CSCFs known by the UE have been used when the UE selects a different P-CSCF as a result of:

- receiving 305 (Use Proxy) to the REGISTER request;
- receiving 504 (Server Time-out); or
- expiration of the timer F at the UE,

then the UE should:

- release IP-CAN bearer that is used only for the transport of SIP signalling and that are not used for other non-IMS applications, except emergency IP-CAN bearers;
- perform a new P-CSCF discovery procedure as described in subslause 9.2.1; and
- perform the procedures for initial registration as described in subclause 5.1.1.2.

NOTE 5: The UE may not be able to release the IP-CAN bearer if the IP-CAN bearer is in use by other applications.

When using IPv4, the UE may request a DNS Server IPv4 address(es) via RFC 2132 [20F] or by the Protocol Configuration Options information element when activating a PDP context according to 3GPP TS 27.060 [10A].

When using IPv6, the UE may request a DNS Server IPv6 address(es) via RFC 3315 [40] and RFC 3646 [56C] or by the Protocol Configuration Options information element when activating a PDP context according to 3GPP TS 27.060 [10A].

The encoding of the request and response for IPv4 or IPv6 address(es) for DNS server(s) and list of P-CSCF address(es) within the Protocol Configuration Options information element is described in 3GPP TS 24.008 [8].

When:

- the UE obtains a PDP context used for SIP signalling by performing handover of the connection from another IP-CAN;
- the IP address of the UE is not changed during the handover; and
- the UE already communicates with the IM CN subsystem via the connection with the other IP-CAN, e.g. the UE determines that its contact with host portion set to the UE IP address (or FQDN of the UE) associated with the connection with the other IP-CAN has been bound to a public user identity;

the UE shall continue using the P-CSCF address(es) acquired in the other IP-CAN.

B.2.2.1A Modification of a PDP context used for SIP signalling

The PDP context shall not be modified from a dedicated PDP context for SIP signalling to a general-purpose PDP context or vice versa. The IM CN Subsystem Signalling Flag shall not be set in the Protocol Configuration Options information element of the MODIFY PDP CONTEXT REQUEST message.

The UE shall not indicate the request for a P-CSCF address within the Protocol Configuration Options information element of the MODIFY PDP CONTEXT REQUEST message. The UE shall ignore P-CSCF address(es) if received in the Protocol Configuration Options information element of the MODIFY PDP CONTEXT RESPONSE message.

B.2.2.1B Re-establishment of the PDP context for SIP signalling

If the UE registered a public user identity with an IP address allocated for the APN of the PDP context used for SIP signalling, the PDP context used for SIP signalling is deactivated as result of signalling from the core network (e.g. no longer available as a result of a successful GPRS routeing area update procedure) and:

i) if the signalling from the core network results in requiring the UE to initiate activation of the PDP context used for SIP signalling; or

ii) if the UE needs to continue having a public user identity registered with an IP address allocated for the APN;

and the UE is allowed to send:

- ACTIVATE PDP CONTEXT REQUEST message; or
- ACTIVATE SECONDARY PDP CONTEXT REQUEST message,

to activate the PDP context for SIP signalling, the UE shall:

A) if the non-access stratum is performing the PDP context activation procedure for the APN triggered as result of the signalling from the core network, wait until the PDP context activation procedure for the APN finishes; and

B) perform the procedures in subclause B.2.2.1, bullets a), b) and c).

If:

- none of the bullets i) and ii) of this subclause evaluate to true;
- the UE is not allowed to send the ACTIVATE (SECONDARY) PDP CONTEXT REQUEST message;

- the procedures in bullet B) of this subclause were unable to ensure that the PDP context used for SIP signalling is available; or

- the procedures in bullet B) of this subclause were unable to acquire any P-CSCF address(es);

and if the PDP context used for SIP signalling was a dedicated PDP context for SIP signalling as described in subclause B.2.2.1, the UE shall deactivate all PDP contexts established as a result of SIP signalling according to the 3GPP TS 24.008 [8].

NOTE: 3GPP TS 24.008 [8] specifies conditions that prevent the UE from sending an ACTIVATE (SECONDARY) PDP CONTEXT REQUEST message.

If all PDP contexts for the APN were deactivated at the start of this subclause and the procedures in bullet B) of this subclause ensured that the PDP context used for SIP signalling is available and acquired the P-CSCF address(es), the UE shall perform a new initial registration according to subclause 5.1.1.2.

B.2.2.1C P-CSCF restoration procedure

A UE supporting the P-CSCF restoration procedure performs one of the following procedures:

A. if the UE used method II for P-CSCF discovery and if the UE receives one or more P-CSCF address(es) in the Protocol Configuration Options information element of a MODIFY PDP CONTEXT REQUEST message and the one or more P-CSCF addresse(s) do not include the address of the currently used P-CSCF, then the UE shall acquire a different P-CSCF address from the one or more P-CSCF addresse(s) in the MODIFY PDP CONTEXT REQUEST message. If more than one P-CSCF address with the same container identifier (i.e. "P-CSCF IPv6 Address" or "P-CSCF IPv4 Address") are included, then the UE shall assume that the more than one P-CSCF addresses with the same container identifier are prioritised with the first P-CSCF address with the same container identifier within the Protocol Configuration Options information element as the P-CSCF address with the highest priority.

   If the UE used method II for P-CSCF discovery and if the UE has previously sent the "P-CSCF Re-selection support" PCO indicator at PDP Context Activation and if the UE receives one or more P-CSCF address(es) in the Protocol Configuration Options information element of a MODIFY PDP CONTEXT REQUEST message, then the UE shall acquire a P-CSCF address from the one or more P-CSCF addresse(s) in the MODIFY PDP CONTEXT REQUEST message. If more than one P-CSCF address with the same container identifier (i.e. "P-CSCF IPv6 Address" or "P-CSCF IPv4 Address") are included, then the UE shall assume that the more than one P-CSCF addresses with the same container identifier are prioritised with the first P-CSCF address with the same container identifier within the Protocol Configuration Options information element as the P-CSCF address with the highest priority;

B. if the UE uses RFC 6223 [143] as part of P-CSCF restoration procedures, and if the P-CSCF fails to respond to a keep-alive request, then the UE shall acquire a different P-CSCF address using one of the methods I, III and IV for P-CSCF discovery described in the subclause B.2.2.1.

If the UE has an ongoing session and acquired the new P-CSCF address by using procedure A described above, the UE may wait until the UE has detected that the ongoing session has ended before performing an initial registration as specified in subclause 5.1.

In all other cases, when the UE has acquired the P-CSCF address, the UE not having an ongoing session shall perform an initial registration as specified in subclause 5.1.

NOTE 1: For UEs using procedure A described above, the network ensures that P-CSCF address(es) in the Protocol Configuration Options information element of a MODIFY PDP CONTEXT REQUEST is sent only during P-CSCF restoration procedures as defined in subclause 5 of 3GPP TS 23.380 [7D].

NOTE 2: The P-CSCF can be completely unreachable, so it is up to UE implementation to detect the end of an ongoing session, e.g. using media plane inactivity detection. Services depending on signalling such as CW and MT calls will not work during this time.

B.2.2.2 Session management procedures

The existing procedures for session management as described in 3GPP TS 24.008 [8] shall apply while the UE is connected to the IM CN subsystem.

B.2.2.3 Mobility management procedures

The existing procedures for mobility management as described in 3GPP TS 24.008 [8] shall apply while the UE is connected to the IM CN subsystem.

B.2.2.4 Cell selection and lack of coverage

The existing mechanisms and criteria for cell selection as described in 3GPP TS 25.304 [9] and 3GPP TS 44.018 [20] shall apply while the UE is connected to the IM CN subsystem.

B.2.2.5 PDP contexts for media

B.2.2.5.1 General requirements

The UE can establish media streams that belong to different SIP sessions on the same PDP context.

During establishment of a session, the UE establishes data streams(s) for media related to the session. Such data stream(s) may result in activation of additional PDP context(s). Such additional PDP context(s) shall be established as secondary PDP contexts associated to the PDP context used for signalling. Such secondary PDP contexts for media can be established either by the UE or the core network.

If the bearer establishment is controlled by the UE, the UE starts reserving its local resources whenever it has sufficient information about the media streams, media authorization and used codecs available as specified in 3GPP TS 24.008 [8].

NOTE 1: If the bearer establishment is controlled by the GPRS IP-CAN, the resource reservation requests are initiated by the core network after the P-CSCF has authorised the respective IP flows and provided the QoS requirements over the Rx interface to the PCRF, as described in 3GPP TS 29.214 [13D].

NOTE 2: When the UE has to allocate bandwidth for RTP and RTCP in a PDP context, the UE uses the rules as those outlined in 3GPP TS 29.213 [13C].

B.2.2.5.1A Activation or modification of PDP contexts for media by the UE

If the UE is configured not to initiate resource allocation for media according to 3GPP TS 24.167 [8G] and both UE and the core network are allowed to establish the secondary PDP contents, then the UE shall refrain from establishing the secondary PDP context(s) for media and from modifying existing PDP contexts for media until the UE considers that the network did not initiate resource allocation for the media.

If the UE receives indication within the SDP according to RFC 3524 [54] that media stream(s) belong to group(s), the media stream(s) shall be set up on separate PDP contexts according to the indication of grouping of media streams. The UE may freely group media streams to PDP context(s) in case no indication of grouping of media streams is received from the P-CSCF.

If the capabilities of the originating UE prevents it from establishment of additional PDP contexts according to the media grouping attributes given by the P-CSCF in accordance with RFC 3524 [54], the UE will not establish such grouping of media streams. Instead, the originating UE shall negotiate media parameters for the session according to RFC 3264 [27B].

If the capabilities of the terminating UE prevents it from establishment of additional PDP contexts according to the media grouping attributes given by the P-CSCF in accordance with RFC 3524 [54], the UE will not establish such grouping of media streams. Instead, the terminating UE shall the UE shall handle such SDP offers in accordance with RFC 3388 [53].

The UE can receive a media authorization token in the P-Media-Authorization header field from the P-CSCF according to RFC 3313 [31]. If a media authorization token is received in the P-Media-Authorization header field when a SIP session is initiated, the UE shall:

- either use existing PDP context(s) where another media authorization token is already in use and no indication of grouping of media streams is required; or

- establish separate PDP context(s) for the media; or

- use an existing PDP context where media authorization token is not in use and no indication of grouping of media streams is required.

When a UE modifies a PDP context to indicate a new media authorization token:

- either as a result of establishment of an additional SIP session; or
- modification of media streams for an ongoing SIP session;

the UE shall include all media authorization tokens and all flow identifiers for all ongoing SIP sessions that use this particular PDP context.

If a media authorization token is received in subsequent messages for the same SIP session, the UE shall:

- use the existing PDP context(s) for media;
- modify the existing PDP context(s) for media; or
- establish additional PDP context(s) for media.

If either background or interactive QoS class is needed for the media, then the UE does not need to use the authorization token even if it receives one. In this case the UE may reuse an existing PDP context and it does not need to request PDP context modification unless it needs to modify the QoS.

If existing PDP context(s) where another media authorization token is already in use is re-used for the media, or separate PDP context(s) is established for the media, the UE shall proceed as follows:

- when a SIP session is terminated, the media authorization token is no longer valid and the UE shall not include it in future GPRS session management messages. The UE shall send a MODIFY PDP CONTEXT REQUEST message updating the binding information by deleting the media authorization token and the corresponding flow identifiers that are no longer valid. If a SIP session is terminated and no other SIP sessions are using the PDP context, the UE shall either update the binding information as described above or deactivate the PDP context;
- the UE shall transparently pass the media authorization token received from the P-CSCF in a response to an INVITE request at originating setup or in the INVITE request at terminating setup to the core network. The UE shall signal it by inserting it within the Traffic Flow Template information element in the ACTIVATE SECONDARY PDP CONTEXT REQUEST message or the MODIFY PDP CONTEXT REQUEST message;
- to identify to the core network which flow(s) (identified by m-lines within the SDP) that are transferred within a particular PDP context, the UE shall set the flow identifier(s) within the Traffic Flow Template information element in the ACTIVATE SECONDARY PDP CONTEXT REQUEST message or the MODIFY PDP CONTEXT REQUEST message. Detailed description of how the flow identifiers are constructed is provided in 3GPP TS 29.207 [12];
- if the UE receives several media authorization tokens from the P-CSCF within the same SIP request or response, the first instance of the media authorization token shall be sent to the core network, and subsequent instances are discarded by the UE; and
- the UE shall not include the IM CN Subsystem Signalling Flag when a PDP context for media is established or modified.

The encoding of the media authorization token and the flow identifiers within the Traffic Flow Template information element is described in 3GPP TS 24.008 [8].

B.2.2.5.1B Activation or modification of PDP contexts for media by the core network

If the UE receives an activation request for a PDP context which is associated with the PDP context used for signalling, the UE shall, based on the information contained in the Traffic Flow Template information element, correlate the media PDP context with a currently ongoing SIP session establishment or SIP session modification.

If the UE receives a modification request for a PDP context that is used for one or more media streams in an ongoing SIP session, the UE shall:

1) modify the related PDP context in accordance with the received modification request.

B.2.2.5.1C Deactivation of PDP context for media

When a data stream for media related to a session is released, if the PDP context transporting the data stream is no longer needed and if the PDP context has been activated by the UE, then the UE deactivates the PDP context.

NOTE: The PDP context can be needed e.g. for other data streams of a session or for other applications in the UE.

B.2.2.5.2 Special requirements applying to forked responses

NOTE 1: The procedures in this subclause only apply when the UE requests activation and modification of media bearers. In the case where the core network activates and modifies the media bearers the network takes care of the handling of media bearers in the case of forking.

Since the UE does not know that forking has occurred until a second, provisional response arrives, the UE sets up the PDP context(s) as required by the initial response received. If a subsequent provisional response is received, different alternative actions may be performed depending on the requirements in the SDP answer:

1) the bearer requirements of the subsequent SDP can be accommodated by the existing PDP context(s). The UE performs no activation or modification of PDP contexts.

2) the subsequent SDP introduces different QoS requirements or additional IP flows. The UE modifies the existing PDP context(s), if necessary, according to subclause B.2.2.5.1A.

3) the subsequent SDP introduces one or more additional IP flows. The UE establishes additional PDP context(s) according to subclause B.2.2.5.1A.

NOTE 2: When several forked responses are received, the resources requested by the UE is are the "logical OR" of the resources indicated in the multiple responses to avoid allocation of unnecessary resources. The UE does not request more resources than proposed in the original INVITE request.

NOTE 3: When service-based local policy is applied, the UE receives the same authorization token for all forked requests/responses related to the same SIP session.

When a final answer is received for one of the early dialogs, the UE proceeds to set up the SIP session. The UE shall release all the unneeded radio/bearer resources. Therefore, upon the reception of the first final 200 (OK) response for the INVITE request (in addition to the procedures defined in RFC 3261 [26] subclause 13.2.2.4), the UE shall:

1) in case PDP context(s) were established or modified as a consequence of the INVITE request and forked provisional responses that are not related to the accepted 200 (OK) response, delete the PDP context(s) or modify the delete the PDP context(s) back to their original state.

B.2.2.5.3 Unsuccessful situations

One of the Go, Gq, Rx and Gx interface related error codes can be received by the UE in the ACTIVATE SECONDARY PDP CONTEXT REJECT message or the MODIFY PDP CONTEXT REJECT message. If the UE receives a Go, Gq, Rx and Gx interface related error code, the UE shall either handle the resource reservation failure as described in subclause 6.1.1 or retransmit the message up to three times. The Go, Gq, Rx and Gx interface related error codes are further specified in 3GPP TS 29.207 [12], 3GPP TS 29.209 [13A], 3GPP TS 29.214 [13D] and 3GPP TS 29.212 [13B].

B.2.2.6 Emergency service

B.2.2.6.1 General

Emergency bearers are defined for use in emergency calls in GPRS IP-CAN and core network support of these bearers is indicated to the UE in NAS signalling. Where the UE recognises that a call request is an emergency call and the core network supports emergency bearers, the UE shall use these bearers for both signalling and media on emergency calls made using the IM CN subsystem.

Some jurisdictions allow emergency calls to be made when the UE does not contain an ISIM or USIM, or where the credentials are not accepted. Additionally, where the UE is in state GMM-REGISTERED.LIMITED-SERVICE and GMM-REGISTERED.PLMN-SEARCH, a normal ATTACH has been attempted but it can also be assumed that a

registration in the IM CN subsystem will also fail. In such cases, subject to the lower layers indicating that the network does support emergency bearer services in limited service state (see 3GPP TS 25.331 [9A]), the procedures for emergency calls without registration can be applied, as defined in subclause 5.1.6.8.2. If the GPRS authentication procedure has already succeeded during the latest normal or emergency ATTACH procedure, the UE shall perform an initial emergency registration, as described in subclause 5.1.6.2 before attempting an emergency call as described in subclause 5.1.6.8.3.

NOTE 1: The UE can determine that GPRS authentication procedure has succeeded during the emergency ATTACH procedure when an integrity protection algorithm is received in the RRC signalling SECURITY MODE COMMAND message (see 3GPP TS 25.331 [9A]).

When activating a PDP context to perform emergency registration, the UE shall request a PDP context for emergency bearer services as defined in 3GPP TS 24.008 [8]. The procedures for PDP context activation and P-CSCF discovery, as described in subclause B.2.2.1 of this specification apply accordingly.

In order to find out whether the UE is attached to the home PLMN or to the visited PLMN, the UE shall compare the MCC and MNC values derived from its IMSI with the MCC and MNC of the PLMN the UE is attached to. If the MCC and MNC of the PLMN the UE is attached to donot match with the MCC and MNC derived from the IMSI, then for the purpose of emergency calls in the IM CN subsystem the UE shall consider to be attached to a VPLMN.

NOTE 2: In this respect an equivalent HPLMN, as defined in 3GPP TS 23.122 [4C] will be considered as a visited network.

If the dialled number is equal to an emergency number stored in the ME, in the USIM or in the Local Emergency Number List (as defined in 3GPP TS 24.008 [8]), then the UE shall recognize a number as for an emergency call and performs the procedures in subclause B.2.2.6.1A.

NOTE 3: The Extended Local Emergency Numbers List (see 3GPP TS 24.301 [8J]) does not apply in this IP-CAN.

Upon reception of a 380 (Alternative Service) response to an INVITE request as defined in subclause 5.1.2A.1.1 and subclause 5.1.3.1, if:

- the 380 (Alternate Service) response contains a Contact header field;
- the value of the Contact header field is a service URN; and
- the service URN has a top-level service type of "sos";

then the UE determines that "emergency service information is included" as described 3GPP TS 23.167 [4B].

Upon reception of a 380 (Alternative Service) response to an INVITE request as defined in subclause 5.1.3.1 if the 380 (Alternate Service) response does not contain a Contact header field with service URN that has a top-level service type of "sos", then the UE determines that "no emergency service information is included" as described 3GPP TS 23.167 [4B].

If the "emergency service information is included" as described 3GPP TS 23.167 [4B]:

1) if the URN in the Contact header field matches an emergency service URN in table B.2.2.6.1, then the type of emergency service is the value corresponding to the matching entry in table B.2.2.6.1; and
2) if the URN in the Contact header field does not match any emergency service URN in table B.2.2.6.1, then the type of emergency service is not identified.

NOTE 4: In bullet 2), the URN in the Contact header field either contains "no emergency subservice type" as described in 3GPP TS 23.167 [4B] triggering an emergency call, or contains an "emergency subservice type that does not map into an emergency service category for the CS domain" as described in 3GPP TS 23.167 [4B] triggering a normal call when the dialled number is available or triggering an emergency call when the dialled number is not available. The country specific URN is an example of a "emergency subservice type that does not map into an emergency service category for the CS domain".

When the emergency registration expires, the UE should disconnect the PDP context for emergency bearer services as defined in 3GPP TS 24.008 [8].

Upon receiving a 3xx other than 380 (Alternative service), 4xx, 5xx or 6xx response to an INVITE request for a UE detectable emergency call, the UE shall perform domain selection as specified in 3GPP TS 23.167 [4B] annex H, to re-attempt the emergency call.

B.2.2.6.1A Type of emergency service derived from emergency service category value

The type of emergency service for an emergency number is derived from the settings of the emergency service category value (bits 1 to 5 of the emergency service category value as specified in subclause 10.5.4.33 of 3GPP TS 24.008 [8]). Table B.2.2.6.1 below specifies mappings between a type of emergency service and an emergency service URN. The UE shall use the mapping to match an emergency service URN and a type of emergency service. If a dialled number is an emergency number but does not map to a type of emergency service the service URN shall be "urn:service:sos".

Table B.2.2.6.1: Mapping between type of emergency service and emergency service URN

Type of emergency service	Emergency service URN
Police	urn:service:sos.police
Ambulance	urn:service:sos.ambulance
Fire Brigade	urn:service:sos.fire
Marine Guard	urn:service:sos.marine
Mountain Rescue	urn:service:sos.mountain

NOTE 1: It is not possible for a UE to indicate more than one type of emergency service in an emergency service URN.

If an IP-CAN, capable of providing local emergency numbers, did not provide a local emergency number that matches the dialled number (see subclause 5.1.6.1) and multiple types of emergency service can be derived for a dialled number from the information configured on the USIM then:

- if the UE is in the HPLMN, the UE shall map any one of these types of emergency service to an emergency service URN as specified in table B.2.2.6.1; and

- if the UE is in the VPLMN, the UE shall select "urn:service:sos".

NOTE 2: If the Non-3GPP emergency number indicator within the Non-3GPP NW provided policies IE (see 3GPP TS 24.008 [8]) provided through registration procedures over 3GPP access is set to "use of non-3GPP emergency numbers permitted", the UE also considers WLAN provided local emergency numbers (see 3GPP TS 24.302 [8U], subclause 4.7). If the Non-3GPP NW provided policies IE provided through registration procedures over 3GPP access is set to "use of non-3GPP emergency numbers not permitted", the UE does not consider WLAN provided local emergency numbers. If the Non-3GPP NW provided policies IE is not provided through registration procedures over 3GPP access, the UE does not consider WLAN provided local emergency numbers.

If an IP-CAN, capable of providing local emergency numbers, provided a local emergency number that matches the dialled number (see subclause 5.1.6.1), and:

- if the UE can derive one or more types of emergency service from the information received from the IP-CAN for the dialled number and the UE cannot derive types of emergency service from the information configured on the USIM for the dialled number; or

- if the UE is able to derive identical types of emergency service from both the information received from the IP-CAN for the dialled number and from the information configured on the USIM for the dialled number,

then the UE shall map any one of these emergency service types to an emergency service URN as specified in table B.2.2.6.1.

NOTE 3: How the UE resolves clashes where an emergency number is associated with one or more different types of emergency service configured in the USIM and in information received from the core network, is implementation dependent.

B.2.2.6.1B Type of emergency service derived from extended local emergency number list

Void

B.2.2.6.2 eCall type of emergency service

The UE shall not send an INVITE request with Request-URI set to "urn:service:sos.ecall.manual" or "urn:service:sos.ecall.automatic".

B.2.2.6.3 Current location discovery during an emergency call

Void.

B.2A Usage of SDP

B.2A.0 General

NOTE: When:

- establishing a session which is not an emergency session; or
- modifying a session which is not an emergency session;

and if the IMSVoPS indicator is received in the "Network feature support" Information Element (see 3GPP TS 24.008 [8]), the UE constructs SDP based on the restrictions indicated in the IMSVoPS indicator. Regardless whether the IMSVoPS indicator indicating voice is supported or not, m-lines can be set to "audio" and exclude voice codecs from the SDP answer or SDP offer.

B.2A.1 Impact on SDP offer / answer of activation or modification of PDP contexts for media by the core network

If due to the activation of PDP context from the core network the related SDP media description needs to be changed the UE shall update the related SDP information by sending, within a SIP request, a new SDP offer for each of the existing SIP dialogs,

If the UE receives a modification request from the core network for a PDP context that is used for one or more media streams in an ongoing SIP session, the UE shall:

1) if, due to the modification of the PDP context, the related SDP media description need to be changed, update the related SDP information by sending, with in a SIP request, a new SDP offer for each of the existing SIP dialogs , and respond to the PDP context modification request.

NOTE: The UE can decide to indicate additional media streams as well as additional or different codecs in the SDP offer than those used in the already ongoing session.

B.2A.2 Handling of SDP at the terminating UE when originating UE has resources available and IP-CAN performs network-initiated resource reservation for terminating UE

If the UE receives an SDP offer where the SDP offer includes all media streams for which the originating side indicated its local preconditions as met, if the precondition mechanism is supported by the terminating UE and the IP-CAN performs network-initiated resource reservation for the terminating UE and the available resources are not sufficient for the received offer the terminating UE shall indicate its local preconditions and provide the SDP answer to the originating side without waiting for resource reservation.

NOTE 1: If the resource reservation is controlled by the GPRS IP-CAN, the resource reservation request is initiated by the core network after the P-CSCF has authorised the respective IP flows and provided the QoS requirements over the Rx interface to the PCRF as described in 3GPP TS 29.214 [13D].

NOTE 2: During the PDP context activation procedure the UE and the core network negotiate whether the UE or the GPRS IP-CAN is responsible for the resource reservation applicable to all PDP contexts within the activated PDP address/APN pair as described in 3GPP TS 24.008 [8].

B.2A.3 Emergency service

NOTE: When establishing an emergency session or when modifying an emergency session, the IMSVoPS indicator does not influence handling of SDP offer and SDP answer.

B.3 Application usage of SIP

B.3.1 Procedures at the UE

B.3.1.0 Registration and authentication

The UE shall perform reregistration of a previously registered public user identity bound to any one of its contact addresses when changing to an IP-CAN for which usage is specified in annex R. The reregistration is performed using the new IP-CAN.

NOTE 1: This document does not specify how the UE detects that the used IP-CAN has changed. The information that is forcing the reregistration is also used to generate the content for the P-Access-Network-Info header field.

NOTE 2: The UE will send the reregistration irrespective of whether it has a SIP dialog or not.

If the UE supports the 3GPP PS data off, then the UE shall in all REGISTER requests include the "+g.3gpp.ps-data-off" header field parameter defined in subclause 7.9.8 set to a value indicating the 3GPP PS data off status.

When the UE sends a REGISTER request, if the 3GPP PS data off status is "active", then the UE shall only include media feature tags associated with services that are 3GPP PS data off exempt services in the g.3gpp.icsi-ref media feature tag, as defined in subclause 7.9.2 and RFC 3840 [62], for the IMS communication services it intends to use.

If the UE is registered, and the 3GPP PS data off status is changed or the UE is provided by the network with a new list of 3GPP PS data off exempt services while the 3GPP PS data off status is "active", then the UE shall perform a reregistration of the previously registered public user identity.

B.3.1.0a IMS_Registration_handling policy

The IMS_Registration_handling policy indicates whether the UE deregisters from IMS after a configured amount of time after receiving an indication that the IMS Voice over PS Session is not supported.

The UE may support the IMS_Registration_handling policy.

If the UE supports the IMS_Registration_handling policy, the UE may support being configured with the IMS_Registration_handling policy using one or more of the following methods:

a) the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C];

b) the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

c) the IMS_Registration_Policy node of 3GPP TS 24.167 [8G].

If the UE is configured with both the IMS_Registration_Policy node of 3GPP TS 24.167 [8G] and the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or 3GPP TS 31.103 [15B], then the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file shall take precedence.

NOTE 1: Precedence for files configured on both the USIM and ISIM is defined in 3GPP TS 31.103 [15B].

If the UE is registered with IMS and the IMSVoPS indicator, provided by the lower layers (see 3GPP TS 24.301 [8J]), indicates voice is not supported, the UE shall:

A) if the Stay_Registered_When_VoPS_Not_Supported leaf indicates requirement to stay registered, the UE needs not to deregister and maintains the registration as required for IMS services; or

NOTE 2: The UE will periodically refresh the registration when needed.

B) if the Stay_Registered_When_VoPS_Not_Supported leaf indicates requirement to deregister and the Deregistration_Timer leaf used to configure the NoVoPS-dereg timer defined in table 7.8.1 contains a timer value for the time to wait before deregistrerting from IMS, start a timer with the value indicated in the policy and:

a) if the timer expires before the UE receives an indication from the lower layers that IMS voice is supported:

1) if there is no ongoing IMS session, either performs reregistration as specified in subclause 5.1.1.4 and shall only include feature tags associated with services that are independent of the IMSVoPS indicator or deregister from the IMS following the procedures specified in subclause 5.1.1.6; or

2) if there is ongoing IMS session, and

i) if the UE does not receive indication from the lower layer that the IMS voice is supported before the ongoing IMS session is terminated, either performs reregistration as specified in subclause 5.1.1.4 and shall only include feature tags associated with services that are independent of IMSVoPS indicator or deregister from the IMS following the procedures specified in subclause 5.1.1.6 as soon as the ongoing IMS based service is terminated ; or

ii) if the UE receives indication from the lower layer that the IMS voice is supported before the ongoing IMS session is terminated, cancel the timer; or

NOTE 3: How the UE selects reregistration or deregistration is implementation dependent (e.g., SMS service)

b) if the UE receives an indication from the lower layers that IMS voice is supported before the timer expires, cancel the timer.

If the IMS_Registration_handling policy is not configured, the UE behaviour is implementation specific.

B.3.1.1 P-Access-Network-Info header field

The UE shall always include the P-Access-Network-Info header field where indicated in subclause 5.1.

B.3.1.1A Cellular-Network-Info header field

Not applicable.

B.3.1.2 Availability for calls

The UE indicates to the non-access stratum the status of being available for voice over PS when:

1) the UE is capable of receiving any (but not necessarily all) of the media types which the CS domain supports, such that the media type can also be used when accessing the IM CN subsystem using the current IP-CAN;

2) if the media type of item 1 is an "audio" media type, the UE supports codecs suitable for (conversational) speech, the "audio" media type is not restricted from inclusion in an SDP message according to the media type restriction policy as specified in subclause 6.1.1, and:

a) 3GPP PS data off status is "inactive";

b) 3GPP PS data off status is "active", the UE is in the HPLMN or the EHPLMN, and MMTEL voice is a 3GPP PS data off exempt service; or

c) 3GPP PS data off status is "active", the UE is in the VPLMN, the UE is configured with an indication that MMTEL voice is a 3GPP PS data off exempt service in a VPLMN, and MMTEL voice is a 3GPP PS data off roaming exempt service; and

3) the UE determines a contact has been bound to a public user identity using the IP-CAN, such that this contact is expected to be used for the delivery of incoming requests in the IM CN subsystem relating to such media.

The UE indicates to the non-access stratum the status of being not available for voice over PS when these conditions are no longer met.

NOTE: The status of being not available for voice over PS is used for domain selection for UE originating sessions / calls specified in 3GPP TS 23.221 [6] subclause 7.2a.

B.3.1.2A Availability for SMS

The UE determines that the UE is able to use SMS using IMS if the UE:

I) is capable of using the MIME type "application/vnd.3gpp.sms" (see 3GPP TS 24.341 [8L]), such that the MIME type can also be used when accessing the IM CN subsystem using the current IP-CAN;

II) supports the role of an SM-over-IP sender (see 3GPP TS 24.341 [8L]);

IIA) determines the PDP context used for SIP signalling exists;

III) determines a contact has been bound to a public user identity using the IP-CAN, such that this contact is expected to be used for the delivery of incoming requests in the IM CN subsystem relating to such media;

IV) the UE does not determine that SMS over IP is restricted in 3GPP TS 24.341 [8L] subclause 5.2.1.3; and

V) the 3GPP PS data off status is:

- "inactive";
- "active", the UE is in the HPLMN or the EHPLMN, and SMS over IMS is a 3GPP PS data off exempt service; or
- "active", the UE is in the VPLMN, the UE is configured with an indication that SMS over IMS is a 3GPP PS data off exempt service in a VPLMN, and SMS over IMS is a 3GPP PS data off roaming exempt service.

When above criteria are not matched, the UE determines that SMS using IMS is unavailable.

NOTE: The status that SMS using IMS is unavailable is used for domain selection for UE originating SMS specified in 3GPP TS 23.221 [6] subclause 7.2c.

B.3.1.3 Authorization header field

Void.

B.3.1.4 SIP handling at the terminating UE when precondition is not supported in the received INVITE request, the terminating UE does not have resources available and IP-CAN performs network-initiated resource reservation for the terminating UE

Upon receiving an INVITE request not including the "precondition" option-tag in the Supported header field and not including the "precondition" option-tag in the Require header field, and the IP-CAN performs network-initiated resource reservation for the UE, the UE:

1) if the INVITE request contains an SDP offer and the local resources required at the terminating UE for the received SDP offer are not available:

   a) shall not alert the user; and

   b) shall send 183 (Session Progress) response to the INVITE request without waiting for resource reservation and without alerting the user. If the INVITE request includes a Supported header field indicating support of reliable provisional responses, the UE shall send the 183 (Session Progress) response reliably. In the 183 (Session Progres) response, the UE shall include an SDP answer; and

2) if the INVITE request does not contain an SDP offer and the INVITE request includes a Supported header field indicating support of reliable provisional responses:

   a) shall generate an SDP offer; and

   b) if the local resources required at the terminating UE for the generated SDP offer are not available:

      A) shall not alert the user; and

      B) shall reliably send 183 (Session Progress) response to the INVITE request without waiting for resource reservation and without alerting the user. In the 183 (Session Progres) response, the UE shall include the generated SDP offer.

Upon successful reservation of local resources, if the precondition mechanism is not used by the terminating UE, the UE can send 180 (Ringing) response to the INVITE request and can alert the user.

B.3.1.5 3GPP PS data off

If the 3GPP PS data off status is "active" the UE shall only send initial requests that:

1) are associated with a 3GPP IMS service which enforces 3GPP PS data off;

NOTE: These services are specified in 3GPP TS 22.011 [1C], and enforcement of 3GPP PS data off is described in the respective service specifications.

2) are associated with an emergency service; or

3) are associated with 3GPP PS data off exempt services configured in the UE using one or more of the following methods:

   - the non_3GPP_ICSIs_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in the HPLMN or the EHPLMN, or if the UE is in the VPLMN and the non_3GPP_ICSIs_roaming_exempt node specified in 3GPP TS 24.167 [8G] is not configured;

   - the non_3GPP_ICSIs_roaming_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in the VPLMN;

   - the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], if the UE is in the HPLMN or the EHPLMN, or if the UE is in the VPLMN and the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] is not configured; or

   - the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], if the UE is in the VPLMN.

   If the UE is configured with both the non_3GPP_ICSIs_exempt node of 3GPP TS 24.167 [8G] and the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], then the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] shall take precedence.

   If the UE is configured with both the non_3GPP_ICSIs_roaming_exempt node of 3GPP TS 24.167 [8G] and the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], then the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] shall take precedence.

If the 3GPP PS data off status changes from "inactive" to "active" the UE shall release all dialogs that

1) are not associated with a 3GPP IMS service which enforces 3GPP PS data off;

NOTE: These services are specified in 3GPP TS 22.011 [1C], and enforcement of 3GPP PS data off is described in the respective service specifications.

2) are not associated with an emergency service; and

3) are not associated with 3GPP data off exempt services configured in the UE using one or more of the following methods:

- the non_3GPP_ICSIs_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in the HPLMN or the EHPLMN, or if the UE is in the VPLMN and the non_3GPP_ICSIs_roaming_exempt node specified in 3GPP TS 24.167 [8G] is not configured;

- the non_3GPP_ICSIs_roaming_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in the VPLMN;- the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], if the UE is in the HPLMN or the EHPLMN, or if the UE is in the VPLMN and the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] is not configured; or

- the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], if the UE is in the VPLMN.

If the UE is configured with both the non_3GPP_ICSIs_exempt node of 3GPP TS 24.167 [8G] and the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], then the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] shall take precedence.

If the UE is configured with both the non_3GPP_ICSIs_roaming_exempt node of 3GPP TS 24.167 [8G] and the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], then the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] shall take precedence.

B.3.1.6 Transport mechanisms

No additional requirements are defined.

B.3.1.7 RLOS

Not applicable.

B.3.2 Procedures at the P-CSCF

B.3.2.0 Registration and authentication

Void.

B.3.2.1 Determining network to which the originating user is attached

In order to determine from which network the request was originated the P-CSCF shall check the MCC and MNC fields received in the P-Access-Network-Info header field.

NOTE: The above check can be against more than one MNC code stored in the P-CSCF.

B.3.2.2 Location information handling

Void.

B.3.2.3 Prohibited usage of PDN connection for emergency bearer services

If the P-CSCF detects that a UE uses a PDN connection for emergency bearer services for a non-emergency REGISTER request, the P-CSCF shall reject that request by a 403 (Forbidden) response.

NOTE: By assigning specific IP address ranges for a PDN connection for emergency bearer services and configuring those ranges in P-CSCF, the P-CSCF can detect based on the registered Contact address if UE uses an emergency PDN connection for initial registration.

B.3.2.5 Void

B.3.2.6 Resource sharing

If P-CSCF supports resource sharing, PCC is supported for this access technology and if according to local policy, the P-CSCF shall apply the procedures in subclause L.3.2.6.

B.3.2.7 Priority sharing

If P-CSCF supports priority sharing, PCC is supported for this access technology and if according to operator policy, the P-CSCF shall apply the procedures in subclause L.3.2.7.

B.3.2.8 RLOS

Not applicable.

B.3.3 Procedures at the S-CSCF

B.3.3.1 Notification of AS about registration status

Not applicable

B.3.3.2 RLOS

Not applicable.

B.4 3GPP specific encoding for SIP header field extensions

B.4.1 Void

B.5 Use of circuit-switched domain

When an emergency call is to be set up over the CS domain, the UE shall attempt it according to the procedures described in 3GPP TS 24.008 [8].

NOTE: 3GPP TS 24.301 [8J] specifies additional requirements for the UE in determining the type of setup message the UE sends to the network when an emergency call is to be set up over the CS domain.

Annex C (normative): UICC and USIM Aspects for access to the IM CN subsystem

C.1 Scope

This clause describes the UICC and USIM aspects for access to the IM CN subsystem. Additional requirements related to UICC usage for access to the IM CN subsystem are described in 3GPP TS 33.203 [19].

C.2 Derivation of IMS parameters from USIM

In case the UE is loaded with a UICC that contains a USIM but does not contain an ISIM, the UE shall:

- generate a private user identity;
- generate a temporary public user identity; and
- generate a home network domain name to address the SIP REGISTER request to.

All these three parameters are derived from the IMSI parameter in the USIM, according to the procedures described in 3GPP TS 23.003 [3]. Also in this case, the UE shall derive new values every time the UICC is changed, and shall discard existing values if the UICC is removed.

NOTE: If there is an ISIM and a USIM on a UICC, the ISIM is used for authentication to the IM CN subsystem, as described in 3GPP TS 33.203 [19]. See also subclause 5.1.1.1A.

C.3 ISIM Location in 3GPP Systems

For 3GPP systems, if ISIM is present, it is contained in UICC.

C.3A UICC access to IMS

If the UE supports the UICC access to IMS USAT feature defined in 3GPP TS 31.111 [15D] the following procedures in addition to the UE procedures in this specification apply.

If the $EF_{UICCIARI}$ contains a list of IARIs associated with active applications installed on the UICC in either the USIM or the ISIM, then when performing the user-initiated registration procedure as described in subclause 5.1.1.2 the UE shall include in the REGISTER request the list of IARIs associated with active applications installed on the UICC in addition to any IARI values for active applications installed on the ME in g.3gpp.iari-ref media feature tag(s) in the Contact header field of the REGISTER request as defined in subclause 7.9.3 and RFC 3840 [62].

If the UE receives from the UICC an initial request for a dialog or a standalone transaction then after decapsulating the request as specified in 3GPP TS 31.111 [15D] the UE shall send the request to the IM CN subsystem as specified in subclause 5.1.2A. When the UE receives from the UICC subsequent requests or responses related to dialogs or transactions already established for UICC applications then after decapsulating the request or response as specified in 3GPP TS 31.111 [15D] the UE shall send the request or response to the IM CN subsystem as specified in subclause 5.1.2A.

NOTE: The encapsulated requests and responses transferred between the UICC and UE are complete and valid SIP requests and responses compliant with RFC 3261 [26].

When sending requests or responses received from the UICC to the IM CN subsystem the UE shall modify or include any header fields necessary (such as Route, Via, Contact) in order to conform with the procedures in this specification.

C.4 Update of IMS parameters on the UICC

3GPP TS 31.102 [15C] and 3GPP TS 31.103 [15B] specify the file structure and contents for the preconfigured parameters stored on the USIM and ISIM, respectively, necessary to initiate the registration to the IM CN subsystem. Any of these parameters can be updated via Data Download or a USAT application, as described in 3GPP TS 31.111 [15D]. If one ore more EFs are changed and a REFRESH command is issued by the UICC, then the UE reads the updated parameters from the UICC as specified for the REFRESH command in 3GPP TS 31.111 [15D].

If the UE supports the UICC access to IMS USAT feature defined in 3GPP TS 31.111 [15D] and the $EF_{UICCIARI}$ changes in either the USIM or the ISIM, the UE shall perform the user-initiated reregistration procedure as described in subclause 5.1.1.4 with the new values of the IARI parameter(s) residing on the UICC.

In case of changes to EFs other than the $EF_{UICCIARI}$, the UE is not required to perform deregistration but it shall wait for the network-initated deregistration procedures to occur as described in subclause 5.4.1.5 unless the user initiates deregistration procedures as described in subclause 5.1.1.6. From this point onwards the normal initial registration procedures can occur.

Annex D (normative):
Void

Annex E (normative): IP-Connectivity Access Network specific concepts when using xDSL, Fiber or Ethernet to access IM CN subsystem

E.1 Scope

The present annex defines IP-CAN specific requirements for a call control protocol for use in the IP Multimedia (IM) Core Network (CN) subsystem based on the Session Initiation Protocol (SIP), and the associated Session Description Protocol (SDP), where the IP-CAN is xDSL, Fiber or Ethernet.

NOTE: Fixed-broadband access in this Annex refers to xDSL, Fiber and Ethernet accesses.

E.2 Fixed broadband aspects when connected to the IM CN subsystem

E.2.1 Introduction

A UE accessing the IM CN subsystem, and the IM CN subsystem itself, utilise the services provided by the fixed-broadband access network to provide packet-mode communication between the UE and the IM CN subsystem.

Requirements for the IP Edge node, defined in ETSI ES 282 001 [138]in support of this communication are outside the scope of this document and specified elsewhere.

From the UEs perspective, it is assumed that one or more IP-CAN bearer(s) are provided, in the form of connection(s) managed by the layer 2 (e.g. DSL modem supporting the UE).

In the first instance, it is assumed that the IP-CAN bearer(s) is (are) statically provisioned between the UE and the IP Edge node, defined in ETSI ES 282 001 [138], according to the user's subscription.

It is out of the scope of the current Release to specify whether a single IP-CAN bearer is used to convey both signalling and media flows, or whether several PVC connections are used to isolate various types of IP flows (signalling flows, conversational media, non conversational media…).

The end-to-end characteristics of the fixed-broadband IP-CAN bearer depend on the type of access network, and on network configuration. The description of the network PVC termination (e.g., located in the DSLAM, in the BRAS…) is out of the scope of this annex.

E.2.2 Procedures at the UE

E.2.2.1 Activation and P-CSCF discovery

Fixed-broadband bearer(s) is (are) statically provisioned in the current Release.

Unless a static IP address is allocated to the UE, prior to communication with the IM CN subsystem, the UE shall perform a Network Attachment procedure depending on the used fixed-broadband access type. When using a fixed-broadband access, both IPv4 and IPv6 UEs may access the IM CN subsystem. The UE may request a DNS Server IPv4 address(es) via RFC 2132 [20F] or a DNS Server IPv6 address(es) via RFC 3315 [40].

The methods for P-CSCF discovery are:

I. When using IPv4, employ the Dynamic Host Configuration Protocol (DHCP) RFC 2132 [20F], the DHCPv4 options for SIP servers RFC 3361 [35A], and RFC 3263 [27A] as described in subclause 9.2.1. When using IPv6, employ Dynamic Host Configuration Protocol for IPv6 (DHCPv6) RFC 3315 [40], the DHCPv6 options for SIP servers RFC 3319 [41] and DHCPv6 options for Domain Name Servers (DNS) RFC 3646 [56C] as

described in subclause 9.2.1. In case the DHCP server provides several P-CSCF addresses or FQDNs to the UE, the UE shall select the P-CSCF address or FQDN as indicated in RFC 3319 [41]. If sufficient information for P-CSCF address selection is not available, selection of the P-CSCF address by the UE is implementation specific.

II. The UE selects a P-CSCF from the list in the IMS management object as specified in 3GPP TS 24.167 [8G].

The UE shall use method II to select a P-CSCF if the IMS management object contains the P-CSCF list. Otherwise, the UE shall use method I to select a P-CSCF.

E.2.2.1A Modification of a fixed-broadband connection used for SIP signalling

Not applicable.

E.2.2.1B Re-establishment of a fixed-broadband connection used for SIP signalling

Not applicable.

E.2.2.1C P-CSCF restoration procedure

A UE supporting the P-CSCF restoration procedure uses the keep-alive procedures described in RFC 6223 [143].

If the P-CSCF fails to respond to keep-alive requests the UE shall acquire a different P-CSCF address using any of the methods described in the subclause E.2.2.1 and perform an initial registration as specified in subclause 5.1.

E.2.2.2 Void

E.2.2.3 Void

E.2.2.4 Void

E.2.2.5 Fixed-broadband bearer(s) for media

E.2.2.5.1 General requirements

The UE can establish media streams that belong to different SIP sessions on the same fixed-broadband bearer.

E.2.2.5.1A Activation or modification of fixed-broadband bearers for media by the UE

If the UE receives indication within the SDP according to RFC 3524 [54] that media stream(s) belong to group(s), and if several fixed-broadband bearers are available to the UE for the session, the media stream(s) may be sent on separate fixed-broadband bearers according to the indication of grouping. The UE may freely group media streams to fixed-broadband bearers in case no indication of grouping is received from the P-CSCF.

If the UE receives media grouping attributes in accordance with RFC 3524 [54] that it cannot provide within the available fixed-broadband bearer(s), then the UE shall handle such SDP offers in accordance with RFC 3388 [53].

The UE can receive a media authorization token in the P-Media-Authorization header field from the P-CSCF according to RFC 3313 [31]. If a media authorization token is received in the P-Media-Authorization header field when a SIP session is initiated, the UE shall reuse the existing fixed-broadband bearer(s) and ignore the media authorization token.

E.2.2.5.1B Activation or modification of fixed-broadband bearers for media by the network

Not applicable.

E.2.2.5.1C Deactivation of fixed-broadband bearers for media

Not applicable.

E.2.2.5.2 Special requirements applying to forked responses

Since the UE is unable to perform bearer modification, forked responses place no special requirements on the UE.

E.2.2.5.3 Unsuccessful situations

Not applicable.

E.2.2.6 Emergency service

E.2.2.6.1 General

If attached to network via fixed-broadband access technology, the UE shall always consider being attached to its home operator's network for the purpose of emergency calls.

NOTE: In fixed-broadband the UE is unable to receive any indication from the network, that would allow the UE to determine, whether it is currently attached to its home operator's network or to a different network, so the UE assumes itself always attached to the home operator's network when connected via fixed-broadband access technology.

E.2.2.6.1A Type of emergency service derived from emergency service category value

Not applicable.

E.2.2.6.1B Type of emergency service derived from extended local emergency number list

Not applicable.

E.2.2.6.2 eCall type of emergency service

The UE shall not send an INVITE request with Request-URI set to "urn:service:sos.ecall.manual" or "urn:service:sos.ecall.automatic".

E.2.2.6.3 Current location discovery during an emergency call

Void.

E.2A Usage of SDP

E.2A.0 General

Not applicable.

E.2A.1 Impact on SDP offer / answer of activation or modification of xDSL bearer for media by the network

Not applicable.

E.2A.2 Handling of SDP at the terminating UE when originating UE has resources available and IP-CAN performs network-initiated resource reservation for terminating UE

Not applicable.

E.2A.3 Emergency service

No additional procedures defined.

E.3 Application usage of SIP

E.3.1 Procedures at the UE

E.3.1.0 Registration and authentication

In order to reach IMS in some access networks, the UE may support:

- address and/or port number conversions provided by a NA(P)T or NA(P)T-PT as described in annex F and annex K; and
- UE requested FTT-IMS establishment procedure specified in 3GPP TS 24.322 [8Y].

If a UE supports one or both of these capabilities then a UE may progressively try them to overcome failure to reach the IMS. Use of these capabilities shall have the following priority order:

1) UE uses neither capability because reaching the IMS without an intervening NA(P)T, NA(P)T-PT, or tunnel is preferred.
2) UE may use address and/or port number conversions provided by a NA(P)T or NA(P)T-PT as described in either annex F or annex K.
3) UE may use the UE requested FTT-IMS establishment procedure specified in 3GPP TS 24.322 [8Y]. If the UE uses the UE-requested FTT-IMS establishment procedure specified in 3GPP TS 24.322 [8Y], the UE considers itself to:
   - be configured to send keep-alives;
   - be directly connected to an IP-CAN for which usage of NAT is defined; and
   - be behind a NAT.

Optional procedures apply when the UE is supporting traversal of restrictive non-3GPP access network using STUN/TURN/ICE, as follows:

a) the protection of SIP messages is provided by utilizing TLS as defined in 3GPP TS 33.203 [19];
b) the mechanisms specified in this annex shall only be applicable when the IP traffic to the IMS core does not traverse through the Evolved Packet Core (EPC);
c) the UE shall establish the TLS connection to the P-CSCF on port 443 as defined in 3GPP TS 33.203 [19]. The UE shall use SIP digest with TLS for registration as specified in subclause 5.1. If the TLS connection is established successfully, the UE sends SIP signalling over the TLS connection to the P-CSCF;
d) the UE shall support the keep-alive procedures described in RFC 6223 [143];

NOTE 1: If the UE is configured to use an HTTP proxy, the UE use the HTTP CONNECT method specified in RFC 2817 [220] to request the HTTP proxy to establish the TCP connection with the P-CSCF. Once the UE has received a positive reply from the proxy that the TCP connection has been established, the UE initiates the TLS handshake with the P-CSCF and establishes the TLS connection.

e) the procedures described in subclause K.5.2 apply with the additional procedures described in the present subclause;

f) when using the ICE procedures for traversal of restrictive non-3GPP access network, the UE shall support the ICE TCP as specified in RFC 6544 [131] and TURN TCP as specified in RFC 6062 [221].

g) if the UE is configured to use TURN over TCP on port 80, the UE shall establish the TCP connection to TURN server on port 80. If the UE is configured to use TURN over TLS on port 443, the UE shall establish the TLS connection to the TURN server on port 443 as defined in 3GPP TS 33.203 [19]. If the UE is configured to use both, the UE should prefer to use TURN over TCP on port 80 to avoid TLS overhead;

h) if the connection is established successfully, the UE sends TURN control messages and media packets over the connection as defined in RFC 5766 [101].

NOTE 2: If the UE is configured to use an HTTP proxy, the UE use the HTTP CONNECT method specified in RFC 2817 [220] to request the HTTP proxy to establish the TCP connection with the TURN server. Then, if the UE is configured to use TURN over TLS on port 443 and the UE has received a positive reply from the proxy that the TCP connection has been established, the UE initiates the TLS handshake with the TURN server and establishes the TLS connection.

E.3.1.0a IMS_Registration_handling policy

Not applicable.

E.3.1.1 P-Access-Network-Info header field

The UE may, but need not, include the P-Access-Network-Info header field where indicated in subclause 5.1.

E.3.1.1A Cellular-Network-Info header field

Not applicable.

E.3.1.2 Availability for calls

Not applicable.

E.3.1.2A Availability for SMS

Void.

E.3.1.3 Authorization header field

When using SIP digest or SIP digest without TLS, the UE need not include an Authorization header field on sending a REGISTER request, as defined in subclause 5.1.1.2.1.

NOTE: In case the Authorization header field is absent, the mechanism only supports that one public user identity is associated with only one private user identity. The public user identity is set so that it is possible to derive the private user identity from the public user identity by removing SIP URI scheme and the following parts of the SIP URI if present: port number, URI parameters, and To header field parameters. Therefore, the public user identity used for registration in this case cannot be shared across multiple UEs. Deployment scenarios that require public user identities to be shared across multiple UEs that don't include an private user identity in the initial REGISTER request can be supported as follows:

- Assign each sharing UE a unique public user identity to be used for registration,

- Assign the shared public user identitiess to the implicit registration set of the unique registering public user identities assigned to each sharing UE.

E.3.1.4 SIP handling at the terminating UE when precondition is not supported in the received INVITE request, the terminating UE does not have resources available and IP-CAN performs network-initiated resource reservation for the terminating UE

Not applicable.

E.3.1.5 3GPP PS data off

Not applicable.

E.3.1.6 Transport mechanisms

No additional requirements are defined.

E.3.1.7 RLOS

Not applicable.

E.3.2 Procedures at the P-CSCF

E.3.2.0 Registration and authentication

The P-CSCF may support UEs connected via restrictive non-3GPP access network.

If the P-CSCF supports UEs connected via restrictive non-3GPP access network, when the P-CSCF receives a 200 (OK) response to a REGISTER request, if the contact address of REGISTER request contains an IP address assigned by the EFTF, and the UE's Via header field contains a "keep" header field parameter, then the P-CSCF shall add a value to the "keep" header field parameter of the UE's Via header field of the 200 (OK) response as defined in RFC 6223 [143].

Optional procedures apply when the P-CSCF is supporting traversal of restrictive non-3GPP access network using STUN/TURN/ICE, as follows:

NOTE: In this scenario, the restrictive non-3GPP access network coexists with NA(P)T device located in the customer premises domain:

a) the protection of SIP messages is provided by utilizing TLS as defined in 3GPP TS 33.203 [19];

b) the P-CSCF supporting these additional procedures should use SIP digest with TLS as defined in subclause 5 and the P-CSCF should inserts an IMS-ALG on the media plane;

c) the mechanisms specified in this annex shall only be applicable when the IP traffic to the IMS core does not traverse through the Evolved Packet Core (EPC);

d) the P-CSCF shall support the procedures defined in subclause 5.2, with the exception that the P-CSCF shall use SIP over TLS on port 443 as defined in 3GPP TS 33.203 [19];

e) when the UE has indicated support of the keep-alive mechanism defined in RFC 6223 [143], the P-CSCF shall indicate to the UE that it supports the keep-alive mechanism; and

f) the IMS-ALG in the P-CSCF shall support ICE procedures, as defined in subclause 6.7.2.7.

E.3.2.1 Determining network to which the originating user is attached

In order to determine from which network the request was originated the P-CSCF shall check if the location information received in the network provided and/or UE provided "dsl-location", "eth-location" or "fiber-location" parameter in the P-Access-Network-Info header field(s) belongs to a location in the same country.

NOTE 1: If local policy does not require the insertion of P-Access-Network-Info header field in the P-CSCF even if it is missing in the received initial request, the P-CSCF can assume that the request is initiated by fixed broadband UE in the same country.

NOTE 2: If the location information in the network provided and UE provided "dsl-location", "eth-location" or "fiber-location" parameters (in a request that includes two P-Access-Network-Info header fields) is contradictory, or the two P-Access-Network-Info header fields indicate different access types the P-CSCF ignores either the network provided or the UE provided information according to operator policy.

E.3.2.2 Location information handling

Upon receipt of an initial request for a dialog or standalone transaction or an unknown method, the P-CSCF based on local policy may include a P-Access-Network-Info header field.The value of the "dsl-location", "eth-location" or "fiber-location" parameter shall be the value as received in the Location-Information header in the User-Data Answer command as specified in ETSI ES 283 035 [98].

NOTE: The way the P-CSCF deduce that the request comes from a UE connected through xDSL access is implementation dependent.

E.3.2.3 Void

E.3.2.4 Void

E.3.2.5 Void

E.3.2.6 Resource sharing

Not applicable.

E.3.2.7 Priority sharing

Not applicable.

E.3.2.8 RLOS

Not applicable.

E.3.3 Procedures at the S-CSCF

E.3.3.1 Notification of AS about registration status

Not applicable

E.3.3.2 RLOS

Not applicable.

E.4 3GPP specific encoding for SIP header field extensions

E.4.1 Void

E.5 Use of circuit-switched domain

There is no CS domain in this access technology.

Annex F (normative): Additional procedures in support for hosted NAT

NOTE: This subclause describes the mechanism for support of the hosted NAT scenario. This does not preclude other mechanisms but they are out of the scope of this annex.

F.1 Scope

This annex describes the UE and P-CSCF procedures in support of hosted NAT. In this scenario, both the media flows and the SIP signalling both traverse a NA(P)T device located in the customer premises domain. The term "hosted NAT" is used to address this function.

When receiving an initial SIP REGISTER request without integrity protection, the P-CSCF can, determine whether to perform the hosted NAT procedures for the user identified by the REGISTER request by comparing the address information in the top-most SIP Via header field with the IP level address information from where the request was received. The P-CSCF will use the hosted NAT procedure only when the address information do not match.

NOTE: There is no need for the P-CSCF to resolve a domain name in the Via header field when UDP encapsulated tunnel mode for IPsec is used. The resolution of a domain name in the Via header field is not required by RFC 3261 [26].

In order to provide hosted NAT traversal for SIP REGISTER requests without integrity protection and the associated responses, the P-CSCF makes use of the "received" header field parameter as described in RFC 3261 [26] and, in addition, if UDP is used, makes use of the "rport" header field parameter as described in RFC 3581 [56A]. The hosted NAT traversal for protected SIP messages is provided by applying UDP encapsulation to IPSec packets in accordance with RFC 3948 [63A].

Alternativly to the procedures defined in subclause F.2 which are employed to support the hosted NAT scenario where the security solution is based on UDP encapsulated IPSec as defined in 3GPP TS 33.203 [19], subclause F.4 provides procedures for NAT traversal for security solutions that are not defined in 3GPP TS 33.203 [19]. Use of such security solutions is outside the scope of this document.

F.2 Application usage of SIP

F.2.1 UE usage of SIP

F.2.1.1 General

This subclause describes the UE SIP procedures for supporting hosted NAT scenarios. The description enhances the procedures specified in subclause 5.1.

The UE shall support the symmetric response routeing mechanism according to RFC 3581 [56A].

F.2.1.2 Registration and authentication

F.2.1.2.1 General

The text in subclause 5.1.1.1 applies without changes

F.2.1.2.1A Parameters contained in the ISIM

The text in subclause 5.1.1.1A applies without changes

F.2.1.2.1B Parameters provisioned to a UE without ISIM or USIM

The text in subclause 5.1.1.1B applies without changes.

F.2.1.2.2 Initial registration

The procedures described in subclause 5.1.1.2.1 apply with the additional procedures described in the present subclause.

NOTE 1: In accordance with the definitions given in subclause 3.1 the IP address acquired initially by the UE in a hosted NAT scenario is the UE private IP address.

On sending a REGISTER request, the UE shall populate the header fields as indicated in subclause 5.1.1.2.1 with the exceptions of subitems c) and d) which are modified as follows

The UE shall populate:

c) a Contact header field according to the following rules: if the REGISTER request is sent without integrity protection, the Contact header field shall be set to include SIP URI(s) containing the private IP address of the UE in the hostport parameter or FQDN. If the UE supports GRUU, the UE shall include a "+sip.instance" header field parameter containing the instance ID. If the REGISTER request is integrity protected, the UE shall include the public IP address or FQDNin the hostport parameter. The UE shall only use a FQDN in a protected REGISTER request, if it is ensured that the FQDN resolves to the public IP address of the NAT. If the UE supports GRUU, the UE shall include a "+sip.instance" header field parameter containing the instance ID. The UE shall include all supported ICSI values (coded as specified in subclause 7.2A.8.2) in a g.3gpp.icsi-ref media feature tag as defined in subclause 7.9.2 and RFC 3840 [62] for the IMS communication services it intends to use, and IARI values (coded as specified in subclause 7.2A.9.2), for the IMS applications it intends to use in a g.3gpp.iari-ref media feature tag as defined in subclause 7.9.3 and RFC 3840 [62];

NOTE 2: The UE will learn its public IP address from the "received" header field parameter in the topmost Via header field in the 401 (Unauthorized) response to the unprotected REGISTER request.

d) a Via header field according to the following rules: if the REGISTER request is sent without integrity protection, the Via header field shall be set to include the private IP address or FQDN of the UE in the sent-by field. If the REGISTER request is integrity protected, the UE shall include the public IP address or FQDN in the sent-by field. The UE shall only use a FQDN in a protected REGISTER request, if it is ensured that the FQDN resolves to the public IP address of the NAT. Unless the UE has been configured to not send keep-alives, it shall include a "keep" header field parameter with no value in the Via header field, in order to indicate support of sending keep-alives associated with, the registration, as described in RFC 6223 [143];

NOTE 3: If the UE specifies a FQDN in the host parameter in the Contact header field and in the sent-by field in the Via header field of an unprotected REGISTER request, this FQDN will not be subject to any processing by the P-CSCF or other entities within the IM CN subsystem. The means to ensure that the FQDN resolves to the public IP address of the NAT are outside of the scope of this specification. One option for resolving this is local configuration.

If IMS AKA is used as a security mechanism, on sending a REGISTER request, as defined in subclause 5.1.1.2.1, the UE shall additionally populate the header fields as defined in subclause 5.1.1.2.2, with the exceptions of subitems c), and d) which are modified as follows:

d) the Security-Client header field set to specify the security mechanisms the UE supports, the IPsec layer algorithms the UE supports and the parameters needed for the security association setup. The UE shall support the setup of two pairs of security associations as defined in 3GPP TS 33.203 [19]. The syntax of the parameters needed for the security association setup is specified in Annex H of 3GPP TS 33.203 [19]. The UE shall support the "ipsec-3gpp" security mechanism, as specified in RFC 3329 [48]. The UE shall support the IPSec layer algorithms for integrity protection and for encryption as defined in 3GPP TS 33.203 [19], and shall announce support for them according to the procedures defined in RFC 3329 [48]. In addition to transport mode the UE shall support UDP encapsulated tunnel mode as per RFC 3948 [63A] and shall announce support for both modes as described in TS 33.203 [19];

When a 401 (Unauthorized) response to a REGISTER is received and this response is received without integrity protection, the procedures described in subclause 5.1.1.2.1 apply with the following additions:

The UE shall compare the IP address in the "received" header field parameter with the corresponding value in the sent-by parameter in the topmost Via header field to detect if the UE is behind a NAT. If the comparison indicates that the respective values are the same, the UE concludes that it is not behind a NAT.

- If the UE is not behind a NAT, the UE shall proceed with the procedures described in subclause 5.1 of the main body of this specification;

- If the UE is behind a NAT, the UE shall verify using the Security-Server header field that mode "UDP-enc-tun" is selected. If the verification succeeds the UE shall store the IP address contained in the "received" header field parameter as the UE public IP address. If the verification does not succeed the UE shall abort the registration. When the UE detects that it is behind a NAT, the UE may include a transport=tcp URI parameter in the Contact header when it sends a protected REGISTER.

NOTE 4: The UE includes a transport=tcp parameter to ensure that P-CSCF uses TCP connection when it receives an initial request for a dialog or a request for a standalone transaction destined for the UE.

In addition, when a 401 (Unauthorized) response to a REGISTER is received (with or without integrity protection) the UE shall behave as described in subclause F.2.1.2.5.

When the UE, that is behind a NAT, receives a 400 (Bad Request) response with 301 Warning header field indicating "incompatible network address format" to the unprotected REGISTER request, the UE shall randomly select new values for the protected server port and the protected client port, and perform new initiate registration procedure by sending an unprotected REGISTER request containing the new values in the Security-Client header field.

NOTE 5: This release of the present document does not specify any specific procedures for bulk number registration according to RFC 6140 [191]for hosted NAT scenarios.

F.2.1.2.3 Initial subscription to the registration-state event package

The procedures described in subclause 5.1.1.3 apply with the additional procedures described in subclause F.2.1.4.1.

F.2.1.2.4 User-initiated re-registration

The procedures described in subclause 5.1.1.4.1 apply with the additional procedures described in the present subclause.

On sending a REGISTER request that does not contain a challenge response, the UE shall populate the header fields as indicated in subclause 5.1.1.4.1 with the exception of subitems c) and d) which are modified as follows.

The UE shall populate:

c) a Contact header field set to include SIP URI(s) that contain(s) in the hostport parameter the public IP address of the UE or FQDN, and containing the instance ID of the UE in the "+sip.instance" header field parameter, if the UE supports GRUU. The UE shall only use a FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT. The UE shall include all supported ICSI values (coded as specified in subclause 7.2A.8.2) in a g.3gpp.icsi-ref media feature tag as defined in subclause 7.9.2 and RFC 3840 [62] for the IMS communication services it intends to use, and IARI values (coded as specified in subclause 7.2A.9.2), for the IMS applications it intends to use in a g.3gpp.iari-ref media feature tag as defined in subclause 7.9.3 and RFC 3840 [62]. If the UE has detected it is behind a NAT, the UE may include a transport=tcp URI parameter in the Contact header;

d) a Via header field set to include the public IP address or FQDN of the UE in the sent-by field. The UE shall only use a FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT. For the TCP, the response is received on the TCP connection on which the request was sent. If the UE previously has previously negotiated sending of keep-alives associated with the registration, it shall include a "keep" header field parameter with no value in the Via header field, in order to indicate continous support to send keep-alives, as described in RFC 6223 [143];

NOTE 1: The means to ensure that the FQDN resolves to the public IP address of the NAT are outside of the scope of this specification. One option for resolving this is local configuration.

When the UE, that is behind a NAT, receives a 400 (Bad Request) response with 301 Warning header field indicating "incompatible network address format" to the REGISTER request that does not contain a challenge response, the UE shall randomly select a new value for the protected client port, and send the REGISTER request containing the new values in the Security-Client header field.

NOTE 2: The protected server port stays fixed for a UE until all public user identities of the UE have been de-registered.

NOTE 3: This release of the present document does not specify any specific procedures for bulk number re-registration according to RFC 6140 [191] for hosted NAT scenarios.

F.2.1.2.5 Authentication

F.2.1.2.5.1 IMS AKA - general

The procedures of subclause 5.1.1.5.1 apply with with the additional procedures described in the present subclause.

On receiving a 401 (Unauthorized) response to the REGISTER request and the response is deemed to be valid, the UE shall behave as of subclause 5.1.1.5.1 with the exception of subitem 3) which is modified as follows.

The UE shall:

3) send another REGISTER request using the temporary set of security associations to protect the message. The header fields are populated as defined for the initial request (see subclause F.2.1.2.2), with the addition that the UE shall include an Authorization header field containing the private user identity and the authentication challenge response calculated by the UE using RES and other parameters, as described in RFC 3310 [49] when AKAv1 is used or as described in RFC 4169 [227] when AKAv2 is used. The UE shall also insert the Security-Client header field that is identical to the Security-Client header field that was included in the previous REGISTER request (i.e. the REGISTER request that was challenged with the received 401 (Unauthorized) response). The UE shall also insert the Security-Verify header field into the request, by mirroring in it the content of the Security-Server header field received in the 401 (Unauthorized) response. The UE shall set the Call-ID of the integrity protected REGISTER request which carries the authentication challenge response to the same value as the Call-ID of the 401 (Unauthorized) response which carried the challenge.

Whenever the 200 (OK) response is not received before the temporary SIP level lifetime of the temporary set of security associations expires or a 403 (Forbidden) response is received, the UE shall consider the registration to have failed. The UE shall delete the temporary set of security associations it was trying to establish, and use the old set of security associations. The UE should send an unprotected REGISTER request according to the procedure specified in subclause F.2.1.2.2 if the UE considers the old set of security associations to be no longer active at the P-CSCF.

F.2.1.2.5.2 Void

F.2.1.2.5.3 IMS AKA abnormal cases

The text in subclause 5.1.1.5.3 applies without changes.

F.2.1.2.5.4 SIP digest – general

Not applicable.

F.2.1.2.5.5 SIP digest – abnormal procedures

Not applicable.

F.2.1.2.5.6 SIP digest with TLS – general

Not applicable.

F.2.1.2.5.7 SIP digest with TLS – abnormal procedures

Not applicable.

F.2.1.2.5.8 Abnormal procedures for all security mechanisms

The text in subclause 5.1.1.5.8 applies without changes.

F.2.1.2.5A Network-initiated re-authentication

The text in subclause 5.1.1.5A applies without changes.

F.2.1.2.5B Change of IPv6 address due to privacy

The text in subclause 5.1.1.5B applies without changes.

F.2.1.2.6 User-initiated deregistration

The procedures of subclause 5.1.1.6.1 apply with with the additional procedures described in the present subclause.

On sending a REGISTER request, the UE shall populate the header fields as indicated in subclause 5.1.1.6 with the exception of subitems d) and e) which are modified as follows.

The UE shall populate:

c) a Contact header field set to either the value of "*" or SIP URI(s) that contain(s) in the hostport parameter the IP address of the UE or FQDN; and containing the instance ID of the UE in the "+sip.instance" header field parameter, if the UE supports GRUU. The UE shall only use a FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT;

d) a Via header field set to include the IP address or FQDN of the UE in the sent-by field. The UE shall only use a FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT;

NOTE 1: In case of hosted NAT traversal only the UE public IP addresses are bound to security associations.

NOTE 2: The means to ensure that the FQDN resolves to the public IP address of the NAT are outside of the scope of this specification. One option for resolving this is local configuration.

NOTE 3: This release of the present document does not specify any specific procedures for bulk number re-registration according to RFC 6140 [191] for hosted NAT scenarios.

F.2.1.2.7 Network-initiated deregistration

The procedures of subclause 5.1.1.7 apply with with the additional procedures described in the present subclause.

Upon receipt of a NOTIFY request on the dialog which was generated during subscription to the reg event package as described in subclause 5.1.1.3, including one or more <registration> element(s) which were registered by this UE with:

- the state attribute set to "terminated" and the event attribute set to "rejected" or "deactivated"; or

- the state attribute set to "active" and the state attribute within the <contact> element belonging to this UE set to "terminated", and associated event attribute element to "rejected" or "deactivated";

the UE shall remove all registration details relating to these public user identities. In case of a "deactivated" event attribute, the UE shall start the initial registration procedure as described in subclause F.2.1.2.2. In case of a "rejected" event attribute, the UE shall release all dialogs related to those public user identities.

F.2.1.3 Subscription and notification

The text in subclause 5.1.2 applies without changes.

F.2.1.4 Generic procedures applicable to all methods excluding the REGISTER method

F.2.1.4.1 UE originating case

The procedures described in subclause 5.1.2A.1 apply with the additional procedures described in the present subclause.

When the UE sends any request, the requirements in subclause 5.1.2A.1 are replaced by the following requirements. The UE shall include:

- a Via header field set to include the public IP address of the UE or FQDN and the protected server port in the sent-by field. The UE shall only use a FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT; and if this is a request for a new dialog, and the request includes a Contact header field, then the UE should populate the Contact header field as follows:

  1) if a public GRUU value ("pub-gruu" header field parameter) has been saved associated with the public user identity to be used for this request, and the UE does not indicate privacy of the P-Asserted-Identity, then insert the public GRUU ("pub-gruu" header field parameter) value in the Contact header field as specified in RFC 5627 [93]; or

  2) if a temporary GRUU value ("temp-gruu" header field parameter) has been saved associated with the public user identity to be used for this request, and the UE does indicate privacy of the P-Asserted-Identity, then insert the temporary GRUU ("temp-gruu" header field parameter) value in the Contact header field as specified in RFC 5627 [93].

If this is a request within an existing dialog, and the request includes a Contact header field, and the contact address previously used in the dialog was a GRUU, then the UE should insert the previously used GRUU value in the Contact header field as specified in RFC 5627 [93].

If the UE did not insert a GRUU in the Contact header field, then the UE shall include the public IP address of the UE or FQDN and the protected server port in the hostport parameter in any Contact header field that is otherwise included. The UE shall only use a FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT.

NOTE: The means to ensure that the FQDN resolves to the public IP address of the NAT are outside of the scope of this specification. One option for resolving this is local configuration.

The UE shall discard any SIP response that is not integrity protected and is received from the P-CSCF outside of the registration and authentication procedures. The requirements on the UE within the registration and authentication procedures are defined in subclause F.2.1.2.4.

When a SIP transaction times out, i.e. timer B, timer F or timer H expires at the UE, the UE may behave as if timer F expired, as described in subclause F.2.1.2.3.

F.2.1.4.2 UE terminating case

The procedures described in subclause 5.1.2A.2 apply with the additional procedures described in the present subclause.

When the UE sends any response, the requirements in subclause 5.1.2A.1 are replaced by the following requirement.

If the response includes a Contact header field, and the response is not sent within an existing dialog, then the UE should populate the Contact header field as follows:

1) if a public GRUU value ("pub-gruu" header field parameter) has been saved associated with the public user identity from the P-Called-Party-ID header field, and the UE does not indicate privacy of the P-Asserted-Identity, then insert the public GRUU ("pub-gruu" header field parameter) value in the Contact header field as specified in RFC 5627 [93]; and

2) if a temporary GRUU value ("temp-gruu" header field parameter) has been saved associated with the public user identity from the P-Called-Party-ID header field, and the UE does indicate privacy of the P-Asserted-Identity, then the UE should insert the temporary GRUU ("temp-gruu" header field parameter) value in the Contact header field as specified in RFC 5627 [93].

If the UE did not insert a GRUU in the Contact header field, then the UE shall:

- include the public IP address of the UE or FQDN and the protected server port in the hostport parameter in any Contact header field that is otherwise included. The UE shall only use a FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT.

NOTE: The means to ensure that the FQDN resolves to the public IP address of the NAT are outside of the scope of this specification. One option for resolving this is local configuration.

The UE shall discard any SIP request that is not integrity protected and is received from the P-CSCF outside of the registration and authentication procedures. The requirements on the UE within the registration and authentication procedures are defined in subclause F.2.1.2.

F.2.2 P-CSCF usage of SIP

F.2.2.1 Introduction

This subclause describes the SIP procedures for supporting hosted NAT scenarios.

The description enhances the procedures specified in subclause 5.2.

The P-CSCF shall support the symmetric response routeing mechanism according to RFC 3581 [56A].

NOTE: Symmetric response routeing is used to support hosted NAT and applicable only to initial, unprotected REGISTER requests and corresponding responses.

F.2.2.2 Registration

The procedures described in subclause 5.2.2 apply with the additional procedures described in the present subclause.

When the P-CSCF receives a REGISTER request from the UE, the P-CSCF shall behave as of subclause 5.2.2.1.

If IMS AKA is the security mechanism, when the P-CSCF receives a REGISTER request from the UE, the P-CSCF shall perform the procedures of subclause 5.2.2.2 with the following exception to items 2) and 3):

2) in case the REGISTER request was received without integrity protection, then:

   a) check the existence of the Security-Client header field. If the Security-Client header field is not present, then the P-CSCF shall return a suitable 4xx response. If the Security-Client header field is present the P-CSCF shall:

      - in case the UE indicated support for "UDP-enc-tun" then remove and store it.

      - in case the UE does not indicate support for "UDP-enc-tun" then:

         - if the host portion of the sent-by field in the topmost Via header field contains an IP address that differs from the source address of the IP packet, silently drop the REGISTER;

         - otherwise continue with procedures as of subclause 5.2.2.

NOTE 1: If the UE does not indicate support for "UDP-enc-tun" and the P-CSCF detects that the UE is located behind a NAT device, then the P-CSCF can just drop the REGISTER to avoid unnecessary signalling traffic.

   b) if the host portion of the sent-by field in the topmost Via header field contains a FQDN, or if it contains an IP address that differs from the source address of the IP packet, the P-CSCF shall:

      - add a "received" header field parameter in accordance with the procedure defined in RFC 3581 [56A]. If the "rport" header field parameter is included in the Via header field, the P-CSCF shall also set the value of the "rport" header field parameter to the source port of the requestin accordance with the procedure defined in RFC 3581 [56A]; and

      - check that no any previously registered UE has either the same public IP address (allocated by the NAT and indicated in the Via header field) and the protected server port (specified in the Security-Client header field) or the same public IP address and the protected client port (specified in the Security-Client header field). If there is such UE, the P-CSCF shall return a 400 (Bad Request) response with 301 Warning header field indicating "incompatible network address format" to the unprotected REGISTER request. Otherwise, the P-CSCF shall forward the REGISTER request.

NOTE 2: If two UEs are behind the same NAT, the NAT can assign to them the same public IP address (but different NAT's port). Hence, the two respective UE will have different protected server port numbers, and different protected client port numbers.

3) in case the REGISTER request was received integrity protected, then the P-CSCF shall:

   a) check the security association which protected the request. If the security association is a temporary one, the P-CSCF shall:

- in case the host parameter in the Contact address is in the form of a FQDN, ensure that the given FQDN will resolve (e.g., by reverse DNS lookup) to the IP address bound to the security association;

- in case the P-CSCF has detected earlier that the UE is located behind a NAT, retrieve port_Uenc from the encapsulating UDP header of the packet received and complete configuration of the temporary set of security associations by configuring port_Uenc in each of the temporary security associations;

- check whether the request contains a Security-Verify header field in addition to a Security-Client header field. If there are no such header fields, then the P-CSCF shall return a suitable 4xx response. If there are such header fields, then the P-CSCF shall compare the content of the Security-Verify header field with the content of the Security-Server header field sent earlier and the content of the Security-Client header field with the content of the Security-Client header field received in the challenged REGISTER. If those do not match, then there is a potential man-in-the-middle attack. The request should be rejected by sending a suitable 4xx response. If the contents match, the P-CSCF shall remove the Security-Verify and the Security-Client header field;

b) if the security association the REGISTER request was received on, is an already established one, then the P-CSCF shall:

- remove the Security-Verify header field if it is present;

- check if the Security-Client header field containing new parameter values is present, and:

- if this header field or any required parameter is missing, then the P-CSCF shall return a suitable 4xx response.

- if this header field and the required parameters are present, then the P-CSCF shall check that no any previously registered UE has the same public IP address and the protected client port (specified in the Security-Client header field). If there is such UE, the P-CSCF shall return a 400 (Bad Request) response with 301 Warning header field indicating "incompatible network address format" to the REGISTER request. Otherwise, the P-CSCF shall remove and store the Security-Client header field before forwarding the request to the S-CSCF;

NOTE 3: When sending the protected REGISTER request to the P-CSCF, the UE will not modify the protected server port value, since the protected server port value stays fixed for a UE until all public user identities of the UE have been de-registered.

When the P-CSCF receives a 401 (Unauthorized) response to an unprotected REGISTER request:

1) if this response contains a "received" header field parameter in the Via header field associated with the UE;

2) if the request associated with the response was received:

A) using UDP and this response contains a "rport" header field parameter in the Via header field associated with the UE; or

B) using TCP; and

3) the UE indicated support for "UDP-enc-tun" IPsec mode;

the P-CSCF shall:

1) delete any temporary set of security associations established towards the UE;

2) remove the "ck" and "ik" WWW-Authenticate header field parameters contained in the 401 (Unauthorized) response and bind the values to the proper private user identity and to the temporary set of security associations which will be setup as a result of this challenge. The P-CSCF shall forward the 401 (Unauthorized) response to the UE if and only if the "ck" and "ik" header field parameters have been removed;

3) insert a Security-Server header field in the response, containing the P-CSCF security list and the parameters needed for the security association setup, as specified in Annex H of 3GPP TS 33.203 [19]. The P-CSCF shall support the "ipsec-3gpp" security mechanism, as specified in RFC 3329 [48]. The P-CSCF shall support the IPSec layer algorithms for integrity protection and for encryption as defined in 3GPP TS 33.203 [19]. The P-CSCF shall indicate "UDP-enc-tun" as the only IPsec mode;

4) set up the temporary set of security associations with a temporary SIP level lifetime between the UE and the P-CSCF for the user identified with the private user identity. The P-CSCF shall select UDP encapsulated tunnel mode and shall leave the value for port-Uenc unspecified in each of the temporary security associations. For further details see 3GPP TS 33.203 [19] and RFC 3329 [48]. The P-CSCF shall set the temporary SIP level lifetime for the temporary set of security associations to the value of reg-await-auth timer; and

5) send the 401 (Unauthorized) response unprotected to the UE using the mechanisms described in RFC 3261 [26] and RFC 3581 [56A], i.e. in case UDP is used as transport protocol the P-CSCF shall send the response to the IP address indicated in the "received" header field parameter and to the port indicated in the "rport" header field parameter of the Via header field associated with the UE. In case UDP is used as transport protocol, the P-CSCF shall use the IP address and the port on which the REGISTER request was received as source IP address and the source port when sending the response back to the UE.

When the P-CSCF receives a 401 (Unauthorized) response to a protected REGISTER request and that REGISTER request was protected by an old set of security associations that use UDP encapsulated tunnel mode, the P-CSCF shall:

1) delete any temporary set of security associations established towards the UE;

2) remove the "ck" and "ik" WWW-Authenticate header field parameters contained in the 401 (Unauthorized) response and bind the values to the proper private user identity and to the temporary set of security associations which will be setup as a result of this challenge. The P-CSCF shall forward the 401 (Unauthorized) response to the UE if and only if the "ck" and "ik" header field parameters have been removed;

3) insert a Security-Server header field in the response, containing the P-CSCF security list and the parameters needed for the security association setup, as specified in Annex H of 3GPP TS 33.203 [19]. The P-CSCF shall support the "ipsec-3gpp" security mechanism, as specified in RFC 3329 [48]. The P-CSCF shall support the IPsec layer algorithms for integrity protection and encryption as defined in 3GPP TS 33.203 [19]. The P-CSCF shall indicate "UDP-enc-tun" as the IPsec mode;

4) set up the temporary set of security associations with a temporary SIP level lifetime between the UE and the P-CSCF for the user identified with the private user identity. The P-CSCF shall select UDP encapsulated tunnel mode and shall specify the same port_Uenc that was used in the old set of security associations. The P-CSCF shall set the temporary SIP level lifetime for the temporary set of security associations to the value of reg-await-auth timer; and

5) send the 401 (Unauthorized) response to the UE using the old set of security associations.

Otherwise, when the P-CSCF receives a 401 (Unauthorized) response to an unprotected REGISTER request and:

- this response does not contain a "received" header field parameter in the Via header field associated with the UE;

- this response does not contain "rport" header field parameter in the Via header field associated with the UE and the request associated with the response was received using UDP; or

- when the P-CSCF receives a 401 (Unauthorized) response to a protected REGISTER request and that REGISTER request was protected by an old set of security associations that do not use UDP encapsulated tunnel mode;

the P-CSCF shall proceed as described in subclause 5.2.2.2.

F.2.3 S-CSCF usage of SIP

F.2.3.1 S-CSCF usage of SIP

F.2.3.1.1 Protected REGISTER with IMS AKA as a security mechanism

The procedures at the S-CSCF described in subclause 5.4.1.2.2 apply.

NOTE: When two UEs that are behind the same NAT register their contact addresses, the NAT can assign to them the same public IP address (but different NAT's ports). If these two UEs select the same protected server port number, and register via different P-CSCFs, then they will have the same contact addresses (i.e. same IP address and protected server port). However, any request targeted to either UE will be sent to the respective P-CSCF, hence not causing any ambiguity at the P-CSCF when forwarding the request via NAT.

F.3 Void

F.4 P-CSCF usage of SIP in case UDP encapsulated IPsec is not employed

F.4.1 Introduction

The subclause F.4 describes the SIP procedures for supporting hosted NAT scenarios in case UDP encapsulated IPsec is not employed. In these scenarios the procedures for NAT traversal must take into account that all SIP requests and responses are not protected by an IPsec security association. This subclause also assumes that the UE transmits the SIP messages from the same IP address and port on which the UE expects to receive SIP messages.

F.4.2 Registration

The procedures described in subclause 5.2.2 apply with the additional procedures described in the present clause.

When the P-CSCF receives a REGISTER request from the UE, the P-CSCF shall add the "received" header field parameter to the Via header field set to the source IP address of the packet header in accordance with the procedure defined in RFC 3261 [26] and RFC 3581 [56A]. If the "rport" header field parameter is included in the Via header field, the P-CSCF shall also set the value of the "rport" header field parameter to the source port of the request, in accordance with the procedure defined in RFC 3581 [56A].

When the P-CSCF detects that the UE is behind a NAT:

- if the UE has indicated in the received REGISTER request support of the keep-alive mechanism defined in SIP outbound (RFC 5626 [92]) according to RFC 6223 [143], the P-CSCF shall indicate to the UE that it supports the keep-alive mechanism; and

- if the UE has not indicated in the received REGISTER request support of the keep-alive mechanism defined in SIP outbound (RFC 5626 [92]) according to RFC 6223 [143]:

  a) if a P-CSCF registration timer is running, the P-CSCF should not forward the REGISTER request if received half of the time before expiry of the S-CSCF registration timer, unless the request is intended to update the UE's capabilities according to RFC 3840 [62] or to modify the ICSI values or IARI values that the UE intends to use in the g.ims.app-ref feature tag. If the P-CSCF decides to not forward the REGISTER request, the P-CSCF shall build a 200 (OK) response based on the contents of the 200 (OK) response to the previous REGISTER request and send this response to the UE. If the P-CSCF decides to forward the REGISTER request, the P-CSCF shall set the registration expiration interval to the registration expiration interval value indicated in the received 200 (OK) response to the previous REGISTER request; and

  b) when the P-CSCF receives a 200 (OK) response to the REGISTER request, the P-CSCF shall modify the value of the Expires header field and/or Expires parameter in the Contact header according to the transport protocol. In order to minimize the number of REGISTER requests to the S-CSCF, the P-CSCF may also start a P-CSCF registration timer with a value of 600 seconds if the value received from the S-CSCF was for greater than 1200 seconds, or to half of the time otherwise.

If, upon receiving a REGISTER request from an unregistered user and the P-CSCF discovers that the UE is behind a NAT, the P-CSCF performs the following actions on the Contact header field depending on its content:

- if the Contact header field contains a contact address in the form of an IP address (NOTE), the P-CSCF shall save (for the duration of the registration) this IP address (i.e. the private IP address of the UE) and associated port number (i.e. the private port of the UE) and bind them to the source IP address (i.e. the public IP address of the NAT) and the source port number (i.e. the port number of the NAT) of the IP packet that contained the REGISTER request and forward the REGISTER request;

- if the Contact header field contains more than one contact addresses in the form of an IP address, the P-CSCF shall apply the above procedure to one of those contact addresses by choosing the one with the highest qvalue parameter) and delete any other contact addresses containing an IP address. If the P-CSCF was unable to choose a contact address based on the qvalue, the P-CSCF shall choose one based on local policy and delete any other contact addresses containing an IP address.

NOTE: When the host parameter in the Contact address is in the form of a FQDN, the P-CSCF will resolve the given FQDN (by DNS lookup) to the IP address of the UE. When including the FQDN in the Contact header field the UE insures that the FQDN resolves to the IP address that the UE uses to send the REGISTER request.

When the P-CSCF received a response to the above request, the P-CSCF shall forward the response to the UE using the mechanisms described in RFC 3581 [56A]. In case UDP is used, the P-CSCF shall send the response to the IP address indicated in the "received" header field parameter and to the port indicated in the "rport" header field parameter of the Via header field in the response. If the REGISTER request received from the UE was received over a TCP connection, the P-CSCF shall send the response to the UE over the same TCP connection over which the request was received. The P-CSCF shall transmit the IP packet (containing the response) from the same IP address and port on which the REGISTER request was received.

F.4.3 General treatment for all dialogs and standalone transactions excluding the REGISTER method

F.4.3.1 Introduction

The procedures described in subclause 5.2.6 apply with the additional procedures described in subclause F.4.3.

F.4.3.2 Request initiated by the UE

When the P-CSCF receives, from the UE that is behind a NAT, an initial request for a dialog or a request for a standalone transaction, the P-CSCF shall:

a) if the "rport" header field parameter is included in the Via header field, set the value of the "rport" header field parameter in the Via header field to the source port of the received IP packet that contained the request;

aa) insert the "received" header field parameter in the Via header field containing the source IP address of the received IP packet (that contained the request), as defined in RFC-3581 [56A];

b) if the request is a dialog-forming request that was received over UDP, bind the source IP address (i.e. the public IP address of the NAT) and associated source port number (i.e. the port number of the NAT) of the received IP packet (that contained the initial dialog-forming request) to:

   - the IP address (i.e. the private IP address of the UE) and associated port number (i.e. the private port of the UE) contained in the Contact header field of the received dialog-forming request, if the Contact header field contained an IP address and associated port number, and save the binding; or

   - the saved IP address (i.e. the private IP address of the UE) and associated port number (i.e. the private port of the UE) contained in the Contact header field of the REGISTER request, if the Contact header field of the received dialog-forming request contained a GRUU, and save the binding; and

c) if the dialog-forming request was received over TCP connection, keep this TCP connection up during the entire duration of the dialog;

before forwarding the request, based on the topmost Route header field, in accordance with the procedures of RFC 3261 [26].

When the P-CSCF receives a response to the above request, the P-CSCF shall forward the response to the UE using the mechanisms described in RFC 3581 [56A]. In case UDP is used, the P-CSCF shall send the response to the IP address indicated in the "received" header field parameter and to the port indicated in the "rport" header field parameter of the Via header field of the response. If the dialog-forming request received from the UE was received over the TCP connection, the P-CSCF shall send the response to the UE over the same TCP connection over which the dialog-forming request was received. The P-CSCF shall transmit the IP packet (containing the response) from the same IP address and port on which the initial dialog-forming request was received.

For all subsequent requests belonging to the dialog, received from the UE, the P-CSCF shall:

- insert the "received" header field parameter in the Via header field as described in RFC 3261 [26];
- if the "rport" header field parameter is included in the Via header field, set the value of the "rport" header field parameter in the Via header field as defined in RFC 3581 [56A]; and
- forward the request as described in RFC 3261 [26].

For all subsequent responses belonging to the dialog, destined for the UE, the P-CSCF shall forward the responses using the "received" header field parameter and if UDP is used, set the value of the "rport" header field parameter in the Via header field of the response as defined in RFC 3581 [56A].

For all subsequent requests belonging to the dialog and destined for the UE (that contains the private IP address and associated private port number in the Request-URI), the P-CSCF shall send the requests to the UE either:

- over the TCP connection that was established when the initial INVITE request was received; or
- use UDP. When sending the request using UDP, the P-CSCF shall insert the request in an IP packet, and send the IP packet to the saved IP address (i.e. the public IP address of the NAT) and associated port number (i.e. the port number of the NAT). The P-CSCF shall transmit the IP packet (containing the request) from the same IP address and port on which the REGISTER request was received.

NOTE: When inserting its SIP URI in the Record-Route header field of the dialog-forming request received from the UE, the P-CSCF can include a pointer in the user part of its SIP URI that points to the saved binding used to route the in-dialog requests to the UE. The Route header field of the in-dialog requests will contain the respective pointer in the user part of the P-CSCF's SIP URI.

F.4.3.3 Request terminated by the UE

When the P-CSCF receives an initial request for a dialog or a request for a standalone transaction destined for the UE (it contains the private IP address and associated private port number in the Request-URI), the P-CSCF shall send the requests to the UE either:

- over the TCP connection for the related registration or registration flow (if the multiple registrations mechanism is used), if available (e.g. TCP connection was established during the registration procedure); or
- use UDP. When sending the request using UDP, the P-CSCF shall send the request to the saved IP address (i.e. the public IP address of the NAT) and associated port number (i.e. the port number of the NAT) for the related registration or registration flow (if the multiple registrations mechanism is used) that is bound to the private IP address and associated private port number indicated in the Request-URI and saved during the registration procedure. The P-CSCF shall transmit the IP packet (containing the request) from the same IP address and port on which the REGISTER request was received.

NOTE 1: If the Contact bound to the terminating UE's registration state includes a transport=tcp URI parameter, for the terminating requests, the P-CSCF follows the procedures specified in RFC 3261 [26] subclause 7 and RFC 3263 [27A] subclause 4.1 for selection of the transport, which recommend to use the TCP connection.

For all subsequent requests belonging to the dialog that are received from the UE, the P-CSCF shall:

- insert the "received" header field parameter in the Via header field as defined in RFC 3261 [26];
- if the "rport" header field parameter is included in the Via header field, set the value of the "rport" header field parameter in the Via header field as defined in RFC 3581 [56A]; and
- forward the request as described in RFC 3261 [26].

For all subsequent responses belonging to the dialog, destined or the UE, the P-CSCF shall forward the responses using the "received" header field parameter and if UDP is used, set the value of the "rport" header field parameter in the Via header field of the response as defined in RFC 3581 [56A].

For all subsequent requests belonging to the dialog and destined for the UE (that contains the private IP address and associated private port number in the Request-URI), the P-CSCF shall send the requests to the UE either:

- over the TCP connection, if available; or
- use UDP. When sending the request using UDP, the P-CSCF shall insert the request in an IP packet, and send the IP packet to the saved IP address (i.e. the public IP address of the NAT) and associated port number (i.e. the port number of the NAT). The P-CSCF shall transmit the IP packet (containing the request) from the same IP address and port on which the REGISTER request was received.

NOTE 2: When inserting its SIP URI in the Record-Route header field in a response to the dialog-forming request received from the UE, the P-CSCF can include a pointer in the user part of its SIP URI that points to the saved binding used to route the in-dialog requests to the UE. The Route header field of the in-dialog requests will contain the respective pointer in the user part of the P-CSCF's SIP URI.

F.5 NAT traversal for media flows

To allow the IMS access gateway to perform address latching, for a given UDP-based media stream, the UE shall use the same port number for sending and receiving packets.

To allow early media flows, the UE shall send keepalive messages for each UDP-based media stream as soon as an SDP offer or answer is received in order to allow the IMS access gateway to perform address latching before the call is established.

To keep NAT bindings and firewall pinholes open for the UDP-based media streams, and enable the IMS access gateway to perform address latching, the UE shall send keepalive messages for each media stream as defined in subclause K.5.2.1.

Annex G (informative):
Void

Annex H (normative): IP-Connectivity Access Network specific concepts when using DOCSIS to access IM CN subsystem

H.1 Scope

The present annex defines IP-CAN specific requirements for a call control protocol for use in the IP Multimedia (IM) Core Network (CN) subsystem based on the Session Initiation Protocol (SIP), and the associated Session Description Protocol (SDP), where the IP-CAN is a DOCSIS cable access network.

DOCSIS (Data Over Cable Service Interface Specification) is a term referring to the Recommendation ITU-T J112 [87] Annex B standard for cable modem systems.

H.2 DOCSIS aspects when connected to the IM CN subsystem

H.2.1 Introduction

A UE accessing the IM CN subsystem, and the IM CN subsystem itself, utilise the services provided by the DOCSIS cable access network to provide packet-mode communication between the UE and the IM CN subsystem.

From the perspective of the UE, the necessary IP-CAN bearer for signalling is transparently available to the UE.

The UE is not directly involved in requests for IP-CAN bearer(s) for media flow(s). The IM CN interacts with the PCRF in the DOCSIS IP-CAN to establish IP-CAN bearer(s) for media flow(s), on behalf of the UE.

H.2.2 Procedures at the UE

H.2.2.1 Activation and P-CSCF discovery

Prior to communication with the IM CN subsystem, the UE shall perform a Network Attachment procedure as defined in the CableLabs PacketCable specifications PKT-TR-ARCH-FRM [88]. When using DOCSIS, both IPv4 and IPv6 UEs may access the IM CN subsystem. The procedures for P-CSCF discovery defined in subclause 9.2.1 of this document apply.

H.2.2.1A Modification of IP-CAN used for SIP signalling

Not applicable.

H.2.2.1B Re-establishment of the IP-CAN used for SIP signalling

Not applicable.

H.2.2.1C P-CSCF restoration procedure

A UE supporting the P-CSCF restoration procedure uses the keep-alive procedures described in RFC 6223 [143].

If the P-CSCF fails to respond to the keep-alive request the UE shall acquire a different P-CSCF address using any of the methods described in the subclause H.2.2.1 and perform an initial registration as specified in subclause 5.1.

H.2.2.2 Void

H.2.2.3 Void

H.2.2.4 Void

H.2.2.5 Handling of the IP-CAN for media

H.2.2.5.1 General requirements

The UE does not directly request resources for media flow(s).

H.2.2.5.1A Activation or modification of IP-CAN for media by the UE

Not applicable.

H.2.2.5.1B Activation or modification of IP-CAN for media by the network

Not applicable.

H.2.2.5.1C Deactivation of IP-CAN for media

Not applicable.

H.2.2.5.2 Special requirements applying to forked responses

The UE does not directly request resources for media flow(s). As a result there are no special UE requirements applying to forked responses.

H.2.2.5.3 Unsuccessful situations

Not applicable.

H.2.2.6 Emergency service

H.2.2.6.1 General

If attached to network via DOCSIS access technology, the UE shall always consider being attached to its home operator's network for the purpose of emergency calls.

NOTE: In DOCSIS the UE is unable to receive any indication from the network, that would allow the UE to determine, whether it is currently attached to its home operator's network or to a different network, so the UE assumes itself always attached to the home operator's network when connected via DOCSIS access technology.

H.2.2.6.1A Type of emergency service derived from emergency service category value

Not applicable.

H.2.2.6.1B Type of emergency service derived from extended local emergency number list

Not applicable.

H.2.2.6.2 eCall type of emergency service

The UE shall not send an INVITE request with Request-URI set to "urn:service:sos.ecall.manual" or "urn:service:sos.ecall.automatic".

H.2.2.6.3 Current location discovery during an emergency call

Void.

H.2A Usage of SDP

H.2A.0 General

Not applicable.

H.2A.1 Impact on SDP offer / answer of activation or modification of IP-CAN for media by the network

Not applicable.

H.2A.2 Handling of SDP at the terminating UE when originating UE has resources available and IP-CAN performs network-initiated resource reservation for terminating UE

Not applicable.

H.2A.3 Emergency service

No additional procedures defined.

H.3 Application usage of SIP

H.3.1 Procedures at the UE

H.3.1.0 Void

H.3.1.0a IMS_Registration_handling policy

Not applicable.

H.3.1.1 P-Access-Network-Info header field

If the UE is aware of the access technology, the UE shall include the P-Access-Network-Info header field where indicated in subclause 5.1.

H.3.1.1A Cellular-Network-Info header field

Not applicable.

H.3.1.2 Availability for calls

Not applicable.

H.3.1.2A Availability for SMS

Void.

H.3.1.3 Authorization header field

When using SIP digest or SIP digest without TLS, the UE need not include an Authorization header field on sending a REGISTER request, as defined in subclause 5.1.1.2.1.

NOTE: In case the Authorization header field is absent, the mechanism only supports that one public user identity is associated with only one private user identity. The public user identity is set so that it is possible to derive the private user identity from the public user identity by removing SIP URI scheme and the following parts of the SIP URI if present: port number, URI parameters, and To header field parameters. Therefore, the public user identity used for registration in this case cannot be shared across multiple UEs. Deployment scenarios that require public user identities to be shared across multiple UEs that don't include an private user identity in the initial REGISTER request can be supported as follows:

- Assign each sharing UE a unique public user identity to be used for registration,

- Assign the shared public user identitiess to the implicit registration set of the unique registering public user identities assigned to each sharing UE.

H.3.1.4 SIP handling at the terminating UE when precondition is not supported in the received INVITE request, the terminating UE does not have resources available and IP-CAN performs network-initiated resource reservation for the terminating UE

Not applicable.

H.3.1.5 3GPP PS data off

Not applicable.

H.3.1.6 Transport mechanisms

No additional requirements are defined.

H.3.1.7 RLOS

Not applicable.

H.3.2 Procedures at the P-CSCF

H.3.2.0 Registration and authentication

Void.

H.3.2.1 Determining network to which the originating user is attached

If the access-type field in the P-Access-Network-Info header field indicated DOCSIS access the P-CSCF shall assume that the initial request for a dialog or standalone transaction or an unknown method destined for a PSAP is initiated in the same country.

NOTE 1: If local policy does not require the insertion of P-Access-Network-Info header field in the P-CSCF even if it is missing in the received initial request, the P-CSCF can assume that the request is initiated by fixed broadband UE in the same country.

NOTE 2: If the network provided and UE provided P-Access-Network-Info header fields indicate different access types the P-CSCF ignores the information in either the network provided or the UE provided P-Access-Network-Info header field according to operator policy.

H.3.2.2 Location information handling

Upon receipt of an initial request for a dialog or standalone transaction or an unknown method, the P-CSCF based on local policy may include a P-Access-Network-Info header field.

NOTE: The way the P-CSCF deduces that the request comes from a UE connected through DOCSIS access is implementation dependent.

H.3.2.3 Void

H.3.2.4 Void

H.3.2.5 Void

H.3.2.6 Resource sharing

Not applicable.

H.3.2.7 RLOS

Not applicable.

H.3.3 Procedures at the S-CSCF

H.3.3.1 Notification of AS about registration status

Not applicable.

H.3.3.2 RLOS

Not applicable.

H.4 3GPP specific encoding for SIP header field extensions

H.4.1 Void

H.5 Use of circuit-switched domain

There is no CS domain in this access technology.

Annex I (normative): Additional routeing capabilities in support of traffics in IM CN subsystem

I.1 Scope

Additional routeing functionality is necessary for support of:

- transit traffic as operators may use the IM CN subsystem as a transit network to provide transit functionality for their own CS networks, enterprise networks, or other network operators;
- other traffics such as roaming traffic and incoming traffic destined to CSI UEs (Combining Circuit Switched (CS) and IP Multimedia Subsystem (IMS) services) traffics;
- traffic for the roaming architecture for voice over IMS with local breakout; and
- originating traffic if required by local policy as specified in subclause 5.4.3.2.

Depending on the additional routeing functionalities, the required specific functions may reside in a stand-alone entity or may be collocated with an MGCF, a BGCF, an I-CSCF, an S-CSCF, or an IBCF as appropriate for the specific scenario.

When colocated with an I-CSCF, the additional routeing capabilities may be performed in advance of I-CSCF procedures as specified in subclause 5.3, or after I-CSCF procedures have failed to identify an S-CSCF supporting the user identified by the Request-URI.

When colocated with an MGCF, the generated requests can be routed to an I-CSCF or to possible targets of the routeing procedures defined in subclause I.2.

The BGCF procedures specified in subclause 5.6 are a subset of the more general routeing procedures provided in this annex.

NOTE: Depending on the host entity for the additional routeing functions, the functionality can be accessed as:

a) the last set of functions on the host before forwarding a request (e.g., on an MGCF, an S-CSCF or an IBCF);

b) the first set of functions performed by the host entity when receiving a request at the host entity's entry point (e.g., on a BGCF, I-CSCF or IBCF);

c) a specified point in the logic of the host (e.g., on the I-CSCF at failure to identify an S-CSCF for the Request-URI); or

d) a set of functions associated with a separate entry point (e.g., at a separate entry point associated with a BGCF, I-CSCF, IBCF or stand-alone entity).

I.1A General

For all SIP transactions identified, if priority is supported, as containing an authorised Resource-Priority header field, or, if such an option is supported, relating to a dialog which previously contained an authorised Resource-Priority header field. The additional routeing functionality shall give priority over other transactions or dialogs. This allows special treatment of such transactions or dialogs. If priority is supported, the priority treatment of transactions or dialogs shall be adjusted according to the most recently received authorized Resource-Priority header field or backwards indication value.

NOTE 1: The special treatment can include filtering, higher priority processing, routeing, call gapping. The exact meaning of priority is not defined further in this document, but is left to national regulation and network configuration.

NOTE 2: These SIP transactions are exempt from network management controls.

If logging is in progress for this dialog, check whether a trigger for stopping logging of SIP signalling has occurred, as described in RFC 8497 [140] and configured in the trace management object defined in 3GPP TS 24.323 [8K]. If a stop trigger event has occurred then stop logging of signalling, else determine, by checking its debug configuration, whether to log the response.

With the exception of 305 (Use Proxy) responses, the additional routeing functionality shall not recurse on 3xx responses.

If the additional routeing functionality inserts its own Record-Route header field, then the additional routeing functionality may require the periodic refreshment of the session to avoid hung states. If the network element requires the session to be refreshed, the additional routeing functionality shall apply the procedures described in RFC 4028 [58] clause 8.

NOTE 3: Requesting the session to be refreshed requires support by at least one of the UEs. This functionality cannot automatically be granted, i.e. at least one of the involved UEs needs to support it.

Based on local policy the additional routeing function shall add in requests and in responses in the P-Charging-Vector header field a "transit-ioi" header field parameter with an entry which identifies the operator network which the request or response is transitting or with a void entry.

Based on local policy the additional routeing function shall delete or void in requests and in responses in the P-Charging-Vector header field any received "transit-ioi" header field parameter value.

I.2 Originating, transit and interconnection routeing procedures

The additional routeing functionality, or associated functional entity, performing these additional routeing procedures should analyse the destination address, and determine whether to route to another network, directly, or via the IBCF, or to the BGCF, the MGCF or the I-CSCF in its own network. This analysis may use public (e.g., DNS, ENUM) and/or private database lookups, and/or locally configured data and may, based on operator policy, modify the Request-URI (e.g. to remove number prefixes, to translate local numbers to global numbers, to update the Request-URI with the URI including an obtained ported-to routeing number, etc).

In addition, and based upon local policy, the analysis may include the carrier identified by the "cic" tel-URI parameter of the Request-URI and other signalling information from the incoming request as part of the route determination. Examples of other signalling information are: the content of the P-Access-Network-Info header field, the value of the "cpc" tel URI parameter of the P-Asserted-Identity header field, the value of the "phone-context" Tel URI parameter of the Request-URI, the SDP content, the ICSI values in the Contact header field and the content of the P-Asserted-Service header field.

If the additional routeing functionality decides that the request shall be routed via a specific entity (e.g. IBCF), it shall insert the URI of this entity in the Route header of the request.

When provided as a stand-alone entity, the network element performing these functions need not Record-Route the INVITE request.

When provided as a stand-alone entity, the network element performing these functions shall not apply the procedures of RFC 3323 [33] relating to privacy.

If overlap signalling using the multiple-INVITE method is supported as a network option, several INVITE requests with the same Call ID and same From header field (including "tag" header field parameter) can be received outside of an existing dialog. Such INVITE requests relate to the same call and the additional routeing function shall route such INVITE request received during a certain period of timeto the same next hop.

When colocated with a MGCF, based on local policy for calls originated from circuit-switched networks, if the circuit-switched is a transit network the additional routeing function shall add in requests in the P-Charging-Vector header field a "transit-ioi" header field parameter with an entry which identifies the PSTN network which the request was transitting or with a void entry.

NOTE 1: Only one "transit-ioi" header field parameter entry is added per transit network.

NOTE 2: The local policy can take bilateral agreements between operators into consideration.

The entity implementing the additional routeing functionality shall remove the P-Served-User header field prior to forwarding the request.

If

a) the additional routeing functionality supports indicating the traffic leg as specified in RFC 7549 [225];

b) the Request-URI does not already include an "iotl" SIP URI parameter, and

NOTE 3: If an "iotl" SIP URI parameter is included it contains the value "visitedA-homeB" inserted by the TRF in the roaming architecture for voice over IMS with local breakout scenario.

c) required by local policy;

then the additional routeing functionality shall:

a) if the Request-URI contains a SIP URI, append the "iotl" SIP URI parameter set to "homeA-homeB" to the Request-URI; and

b) if the Request-URI contains a tel URI:

- convert the tel URI in the Request-URI to the form of a SIP URI with user=phone; and

- append an "iotl" SIP URI parameter with a value set to "homeA-homeB" in the Request-URI.

I.3 Providing IMS application services in support of transit & interconnection traffics

I.3.1 Introduction

When the IM CN subsystem provides transit functionality to other operator networks or enterprise networks, it may also provide IMS applications services to the operator network or enterprise network.

The transit service invocation, performed by a transit function, is performed based on local configured transit invocation criteria that are provided for the specific transit scenario.

NOTE: The transit invocation criteria for invocation is intended to be per served network basis, for which transit functionality is provided, and not per subscriber basis.

Similar to the initial filter criteria for a user profile, the transit invocation criteria may have service point triggers, used to generate an ordered list of transit invocation criteria to be applied to a request, based on different information in the request, SIP method, SIP header field, and SIP body. The service invocation procedure shall support suppression/avoidance of conflicting services, multiple invocations of the same service and loopback scenarios.

I.3.2 Procedures

I.3.2.1 Treatment for dialog and standalone transactions

When the transit function receives an initial request for a dialog, or a request for a standalone transaction, and the request is received either from a functional entity within the same trust domain or contains a valid original dialog identifier (see step 3) or the dialog identifier (From, To and Call-ID header fields) relates to an existing request processed by the transit function, then prior to forwarding the request, the transit function shall:

1) check if an original dialog identifier that the transit function previously placed in a Route header field is present in the topmost Route header field of the incoming request.

- If not present, the transit function shall build an ordered list of transit invocation criteria.

- If present, the request has been sent from an AS in response to a previously sent request, an ordered list of transit invocation criteria already exists and the transit function shall not change the ordered list of transit invocation criteria.

2) remove its own SIP URI from the topmost Route header field;

3) check whether the initial request matches any unexecuted transit invocation criteria. If there is a match, then the transit function shall select the first matching unexecuted transit invocation criteria from the ordered list of transit invocation criteria and the transit function shall insert the AS URI to be contacted into the Route header field as the topmost entry followed by its own URI populated;

NOTE: Depending on the result of processing the transit invocation criteria the transit function can contact one or more AS(s) before processing the outgoing Request-URI.

4) if the request is not forwarded to an AS and if local policy requires the application of other additional routeing capabilities, handled by entities other than the transit function, the transit function shall apply the additional routeing capabilities if they are locally available or forward the request to an entity that implements the additional routeing capabilities;

5) determine the destination address (e.g. DNS access) using the URI placed in the topmost Route header field if present, otherwise based on the Request-URI. If the destination requires interconnect functionalities (e.g. the destination address is of an IP address type other than the IP address type used in the IM CN subsystem), the transit function shall forward the request to the destination address via an IBCF in the same network;

6) in case of an initial request for a dialog, based on local policy record-route; and

7) route the request based on SIP routeing procedures.

When the transit function receives a target refresh request, or a a subsequent request other than target refresh request, for a dialog, prior to forwarding the request, the transit function shall:

1) remove its own URI from the topmost Route header field; and

2) forward the request based on the topmost Route header field.

With the exception of 305 (Use Proxy) responses, the transit function shall not recurse on 3xx responses.

I.3.2.1A Handling of header fields related to charging

When the transit function receives a request from a functional entity that is not an AS and if the request is forwarded to an AS the transit function shall:

- store the value of the "orig-ioi" header field parameter received in the P-Charging-Vector header field if present;

NOTE 1: Any received "orig-ioi" header field parameter will be a type 2 IOI. The type 2 IOI identifies the service provider from which the request was sent.

- remove the "orig-ioi" header field parameter from the P-Charging-Vector header field, if present;

- store the value of the "transit-ioi" header field parameter and remove the "transit-ioi" header field parameter, if present; and

- insert in the P-Charging-Vector header field an IOI type 3 value in an "orig-ioi" header field parameter identifying the network sending the request and based on operator option a Relayed-Charge header field with contents set to the value of the received "transit-ioi" header field parameter.

When forwarding the request from an AS to a functional entity that is not an AS the transit function shall:

- store the value of the "orig-ioi" header field parameter received in the P-Charging-Vector header field if present;

NOTE 2: Any received "orig-ioi" header field parameter will be a type 3 IOI. The type 3 IOI identifies the service provider from which the request was sent.

- remove the "orig-ioi" header field parameter from the P-Charging-Vector header field, if present; and

- insert in the P-Charging-Vector header field the "orig-ioi" header field parameter with the IOI type 2 value stored when the initial request for a dialog or the request for a standalone transaction was received from an entity that was not an AS;

- insert the "transit-ioi" header field parameter if previously stored; and

- remove the Relayed-Charge header field, if present.

When the transit function receives a 1xx or 2xx response to a request from a functional entity that is not an AS and if the response is forwarded to an AS the transit function shall:

- store the values of the "orig-ioi", the "term-ioi" and the "transit-ioi" header field parameter received in the P-Charging-Vector header field if present;

NOTE 3: Any received "term-ioi" header field parameter will be a type 2 IOI. The type 2 IOI identifies the service provider from which the response was sent.

- remove the "orig-ioi", the "term-ioi", and the "transit-ioi" header field parameters from the P-Charging-Vector header field, if present; and

- insert in the P-Charging-Vector header field an IOI type 3 value in a "term-ioi" header field parameter identifying the network sending the response, an IOI type 3 value in an "orig-ioi" header field parameter stored when the request was received from an AS, and based on operator option a Relayed-Charge header field with contents set to the value of the received "transit-ioi" header field parameter.

When forwarding any response to a request from an AS to a functional entity that is not an AS the transit function shall remove the Relayed-Charge header field, if present.

When forwarding the 1xx or 2xx response to a request from an AS to a functional entity that is not an AS the transit function shall:

- remove the "orig-ioi" and the "term-ioi" header field parameter from the P-Charging-Vector header field, if present; and

NOTE 4: Any received "term-ioi" and "orig-ioi" header field parameter will be a type 3 IOI. The type 3 IOI identifies the service provider from which the response was sent.

- insert in the P-Charging-Vector header field an "orig-ioi" header field parameter with the IOI type 2 value, an "term-ioi" header field parameter with the IOI type 2 value received in the 1xx or 2xx response to the request from a functional entity that was not an AS, insert the "transit-ioi" header field parameter if previously stored.

I.3.2.2 Original dialog identifier for transit function

The original dialog identifier is an implementation specific token that the transit function encodes into the own transit function URI in a Route header field, prior to forwarding the request to an AS. This is possible because the transit function is the only entity that creates and consumes the value.

The token may identify the original dialog of the request, so in case an AS acting as a B2BUA changes the dialog, the transit function is able to identify the original dialog when the request returns to the transit function. In a case of a standalone transaction, the token indicates that the request has been sent to the transit function from an AS in response to a previously sent request. The token can be encoded in different ways, such as e.g., a character string in the user-part of the transit function URI, a parameter in the transit function URI or port number in the transit function URI.

The transit function shall ensure that the value chosen is unique, in order for the transit function to recognize the value when received in a subsequent message of one or more dialogs and make the proper association between related dialogs that pass through an AS.

An original dialog identifier is sent to each AS invoked due to transit invocation criteria evaluation such that the transit function can associate requests as part of the same sequence that trigger transit invocation criteria evaluation in priority order (and not rely on SIP dialog information that may change due to B2BUA AS).

NOTE: If the same original dialog identifier is included in more than one request from a particular AS (based on service logic in the AS), then the transit function will continue the transit invocation criteria evaluation sequence. If the AS wants transit invocation criteria evaluation to start from the beginning for a request, then AS does not include an original dialog identifier.

I.4 Loopback routeing procedures

I.4.1 Introduction

In order to support traffics for the roaming architecture for voice over IMS with local breakout the additional routeing functionality will perform the procedures described in this subclause. An additional routeing functionality performing the procedures in this subclause is always located in the visited PLMN and is referred to as Transit and Roaming Function (TRF).

The TRF performs local break out and routes the INVITE request via a specific entity e.g. IBCF or BGCF.

Loopback routeing requires support in the visited network and the home network. If the visited network supports loopback routeing then the P-CSCF will, based on local policy, express this support by adding a g.3gpp.trf feature-capability indicator value to the URI of the desired TRF.

The home network decides based on local operator policy if loopback routeing shall be applied. If loopback routeing is applied, the home network routes the INVITE request back to the TRF located in the visited network indicating that loopback routeing is used by including the g.3gpp.loopback feature-capability indicator in a Feature-Caps header field.

In the loopback scenario OMR as specified in 3GPP TS 29.079 [11D] is used to determine the optimal media path between the visited network and the terminating network without passing through the home network.

I.4.2 TRF procedure

When the TRF receives an initial request for a dialog, the TRF shall:

1) retain the "icid-value" header field parameter in the P-Charging-Vector header field;

2) store the value of the "orig-ioi" header field parameter received in the P-Charging-Vector header field, if present, and remove the "orig-ioi" header field parameter from the P-Charging-Vector header field. Insert a type 2 "orig-ioi" header field parameter into the P-Charging-Vector header field. Set the type 2 "orig-ioi" header field parameter to a value that identifies the sending network in which the TRF resides. The TRF shall not include the "term-ioi" header field parameter. Store the value of a "transit-ioi" header field parameter received in the P-Charging-Vector header field, if present, and remove the "transit-ioi" header field parameter from the P-Charging-Vector header field before forwarding the request;

3) if required by local policy, perform number normalization and enum translation in the same way as performed by S-CSCF in subclause 5.4.3.2 step 10);

4) if the P-Access-Network header field is available, determine the entity for local break out (e.g. IBCF or BGCF) using:

   a) the location of the originating user; and

   b) the destination address,

   then include a Route header field set to the URI associated with the determined entity in the forwarded request;

5) create a Record-Route header field containing the TRF own SIP URI;

6) remove the "+g.3gpp.loopback" header field parameter from the Feature-Caps header field of the outgoing request;

6A) if the TRF supports indicating the traffic leg as specified in RFC 7549 [225] and required by local policy:

   a) if the Request-URI in the INVITE request contains a SIP URI, append an "iotl" SIP URI parameter set to "visitedA-homeB" to the Request-URI; and

   b) if the Request-URI in the INVITE request contains a tel URI:

      - convert the tel URI in the Request-URI to the form of a SIP URI with user=phone; and

- append the "iotl" SIP URI parameter set to "visitedA-homeB" in the Request-URI; and

7) route the request based on SIP routeing procedures.

When the TRF receives a 1xx or 2xx response to the INVITE request above, the TRF shall:

- store the value of the "transit-ioi" header field parameter received in the P-Charging-Vector header field and remove the "transit-ioi" header field parameter from the P-Charging-Vector header field, if present;

- remove the "orig-ioi" header field parameter and the "term-ioi" header field parameter from the P-Charging-Vector header field before forwarding the response; and

NOTE: Any received "term-ioi" header field parameter will be a type 2 IOI identifying the sending network of the response.

- insert in the P-Charging-Vector header field the "orig-ioi" header field parameter, if received in the request, and the type 1 "term-ioi" header field parameter in the response. The TRF shall set the type 1 "term-ioi" header field parameter to a value that identifies the network in which the TRF resides and the type 1 "orig-ioi" header field parameter is set to the previously received value of the type 1 "orig-ioi" header field parameter.

When the TRF receives subsequent requests or responses to subsequent requests containing the "+g.3gpp.loopback" header field parameter from the Feature-Caps header field, the TRF shall remove the "+g.3gpp.loopback" header field parameter from the Feature-Caps header field of the outgoing request or the outgoing response.

When the TRF receives responses to initial or subsequent requests from the terminating side, the TRF shall insert in the P-Charging-Vector header field, if present, the "loopback" header field parameter to the outgoing response.

When the TRF receives subsequent requests from the terminating side, the TRF shall insert in the P-Charging-Vector header field, if present, the "loopback" header field parameter to the outgoing request.

When the TRF receives a subsequent request, the TRF shall:

1) retain the "icid-value" header field parameter in the P-Charging-Vector header field;

2) store the value of the "orig-ioi" header field parameter received in the P-Charging-Vector header field, if present, and remove the "orig-ioi" header field parameter from the P-Charging-Vector header field;

3) if the subsequent request is:

   a) received from originating home network and forwarded to terminating home network, insert a type 2 "orig-ioi" header field parameter into the P-Charging-Vector header field, and set the type 2 "orig-ioi" header field parameter to a value that identifies the sending network in which the TRF resides. The TRF shall not include the "term-ioi" header field parameter; or

   b) received from terminating home network and forwarded to originating home network, insert a type 1 "orig-ioi" header field parameter into the P-Charging-Vector header field, and set the type 1 "orig-ioi" header field parameter to a value that identifies the sending network in which the TRF resides. The TRF shall not include the "term-ioi" header field parameter; and

4) store the value of a "transit-ioi" header field parameter received in the P-Charging-Vector header field, if present, and remove the "transit-ioi" header field parameter from the P-Charging-Vector header field before forwarding the request.

When the TRF receives a response to a subsequent request, the TRF shall:

1) store the value of the "transit-ioi" header field parameter received in the P-Charging-Vector header field and remove the "transit-ioi" header field parameter from the P-Charging-Vector header field, if present;

2) remove the "orig-ioi" header field parameter and the "term-ioi" header field parameter from the P-Charging-Vector header field before forwarding the response; and

3) if the response to the subsequent request is:

   a) received from terminating home network and forwarded to originating home network, insert in the P-Charging-Vector header field the "orig-ioi" header field parameter, if received in the request, and the type 1 "term-ioi" header field parameter in the response. The TRF shall set the type 1 "term-ioi" header field

parameter to a value that identifies the network in which the TRF resides and the type 1 "orig-ioi" header field parameter is set to the previously received value of the type 1 "orig-ioi" header field parameter; and

b) received from originating home network and forwarded to terminating home network, insert in the P-Charging-Vector header field the "orig-ioi" header field parameter, if received in the request, and the type 2 "term-ioi" header field parameter in the response. The TRF shall set the type 2 "term-ioi" header field parameter to a value that identifies the network in which the TRF resides and the type 2 "orig-ioi" header field parameter is set to the previously received value of the type 2 "orig-ioi" header field parameter.

I.5 Overload control

I.5.1 Introduction

The additional routeing functionality, or associated functional entity, performing additional routeing procedures described in I.3 may support the event-based overload control mechanism.

I.5.2 Outgoing subscriptions to load-control event

Based on operator policy, the additional routeing functionality may subscribe to the load-control event package with one ore more target SIP entities. The list of target SIP entities is provisioned.

Subscription to the load-control event package is triggered by internal events (e.g. the physical device hosting the SIP entity is power-cycled) or through a management interface.

The AS shall perform subscriptions to the load-control event package to a target entity in accordance with RFC 6665 [28] and with RFC 7200 [201]. When subscribing to the load-control event, additional routeing functionality shall:

1) Send a SUBSCRIBE request in accordance with RFC 6665 [28] and with RFC 7200 [201] to the target entity, with the following elements:

   - an Expires header field set to a network specific value;

2) If the target entity is located in a different network and local policy requires the application of IBCF capabilities, forward the request to an IBCF acting as an exit point.

The additional routeing functionalit shall automatically refresh ongoing subscriptions to the load-control event package either 600 seconds before the expiration time if the initial subscription was for greater than 1200 seconds, or when half of the time has expired if the initial subscription was for 1200 seconds or less.

The additional routeing functionality can terminate a subscription according to RFC 6665 [28].

Annex J (normative):
Void

Annex K (normative): Additional procedures in support of UE managed NAT traversal

K.1 Scope

This annex describes the UE, P-CSCF, and S-CSCF procedures in support of UE managed NAT traversal. For ICE, the IBCF procedures are also described. In this scenario, both the media flows and the SIP signalling both traverse a NA(P)T device located in the customer premises domain. The term "hosted NAT" is used to address this function. This annex does not consider the case where the NAT is behind the P-CSCF as different NAT traversal procedures are necessary for this architectural scenario.

The procedures described in this subclause of this annex rely on the UE to manage the NAT traversal process. As part of the UE management process, the UE can learn whether it is behind a NAT or not, and choose whether the proceedures in this annex are applied or not.

The protection of SIP messages is provided by applying either UDP encapsulation to IPSec packets in accordance with RFC 3948 [63A] and as defined in 3GPP TS 33.203 [19] or by utilizing TLS as defined in 3GPP TS 33.203 [19].

NOTE 1: This annex describes the mechanism for support of UE managed NAT traversal scenario defined in 3GPP TS 23.228 [7]. This does not preclude other mechanisms but they are out of the scope of this annex.

NOTE 2: It is recognized that outbound can be useful for capabilities beyond NAT traversal (e.g. multiple registrations) however this annex does not consider such capabilities at this time. Such capabilites can require additional information elements in the REGISTER request so that the P-CSCF and S-CSCF can distinguish whether to apply procedures as of annex F or annex K.

K.2 Application usage of SIP

K.2.1 Procedures at the UE

K.2.1.1 General

This subclause describes the UE SIP procedures for supporting a UE managed hosted NAT traversal approach. The description enhances the procedures specified in subclause 5.1.

K.2.1.2 Registration and authentication

K.2.1.2.1 General

The text in subclause 5.1.1.1 applies without changes.

K.2.1.2.1A Parameters contained in the ISIM

The text in subclause 5.1.1.1A applies without changes.

K.2.1.2.1B Parameters provisioned to a UE without ISIM or USIM

The text in subclause 5.1.1.1B applies without changes.

K.2.1.2.2 Initial registration

K.2.1.2.2.1 General

The procedures described in subclause 5.1.1.2.1 apply with the additional procedures described in the present subclause.

NOTE 1: In accordance with the definitions given in subclause 3.1 the IP address acquired initially by the UE in a hosted NAT scenario is the UE private IP address.

On sending a REGISTER request, the UE shall populate the header fields as indicated in subitems a) through j) of subclause 5.1.1.2 with the exceptions of subitems c) and d) which are modified as follows.

The UE shall populate:

c) a Contact header field according to the following rules: the Contact header field shall be set to include SIP URI(s) containing the private IP address or FQDN of the UE in the hostport parameter. The UE shall also include an instance ID ("+sip.instance" header field parameter) and "reg-id" header field parameter as described in RFC 5626 [92]. The UE shall include all supported ICSI values (coded as specified in subclause 7.2A.8.2) in a g.3gpp.icsi-ref media feature tag as defined in subclause 7.9.2 and RFC 3840 [62] for the IMS communication services it intends to use, and IARI values (coded as specified in subclause 7.2A.9.2), for theIMS applications it intends to use in a g.3gpp.iari-ref media feature tag as defined in subclause 7.9.3 and RFC 3840 [62];

d) a Via header field set to include the private IP address or FQDN of the UE in the sent-by field. For TCP, the response is received on the TCP connection on which the request was sent. For UDP, the UE shall include the "rport" header field parameter as defined in RFC 3581 [56A].

NOTE 2: The UE will learn its public IP address from the "received" header field parameter in the topmost Via header field in the 401 (Unauthorized) response to the unprotected REGISTER request.

NOTE 3: If the UE specifies a FQDN in the hostport parameter in the Contact header field and in the sent-by field in the Via header field of an unprotected REGISTER request, this FQDN will not be subject to any processing by the P-CSCF or other IMS entities.

When a 401 (Unauthorized) response to a REGISTER request is received with integrity protection the UE shall behave as described in subclause K.2.1.2.5.

When a 401 (Unauthorized) response to a REGISTER request is received and this response is received without integrity protection, the procedures described in subclause 5.1.1.2 apply with the following additions:

The UE shall compare the IP address in the "received" header field parameter with the corresponding value in the sent-by parameter in the topmost Via header field to detect if the UE is behind a NAT. If the comparison indicates that the respective values are the same, the UE concludes that it is not behind a NAT.

- if the UE is not behind a NAT the UE shall proceed with the procedures described in subclause 5.1;

- if the UE is behind a NAT the UE shall verify using the Security-Server header field that either the mechanism-name "tls" or "ipsec-3gpp" and the mode "UDP-enc-tun" is selected. If the verification succeeds the UE shall behave as described in subclause K.2.1.2.5 and store the IP address contained in the "received" header field parameter as the UE's public IP address. If the verification does not succeed the UE shall abort the registration.

On receiving the 200 (OK) response to the REGISTER request, the procedures described in subclause 5.1.1.2 apply with the following additions:

The UE shall determine the P-CSCFs ability to support the keep-alive procedures as described in RFC 5626 [92] by checking whether the "outbound" option-tag is present in the Require header field:

- if no "outbound" option-tag is present, the UE may use some other explicit indication in order to find out whether the P-CSCF supports the outbound edge proxy functionality. Such indication may be acomplished either through UE local configuration means or the UE can examine the 200 (OK) response to its REGISTER request for Path header fields, and if such are present check whether the bottommost Path header field contains the "ob" SIP URI parameter. If the UE determines that the P-CSCF supports the outbound edge proxy functionality, the UE can use the keep-alive techniques defined in subclause K.2.1.5 and RFC 5626 [92] towards the P-CSCF; or

- if an "outbound" option-tag is present, the UE shall initiate keep-alive mechanisms as defined in subclause K.2.1.5 and RFC 5626 [92] towards the P-CSCF.

NOTE 4: Presence of the "outbound" option-tag in the Require header field indicates that both the P-CSCF and S-CSCF fully support the outbound procedures. The number of subsequent outbound registrations for the same private user identity but with a different reg-id value is based on operator policy.

K.2.1.2.2.2 Initial registration using IMS AKA

The procedures described in subclause 5.1.1.2.2 apply with the additional procedures described in the present subclause.

On sending a REGISTER request, the UE shall populate the header fields as indicated in subclause 5.1.1.2.2 with the exceptions of the subitems which are modified as follows:

c) additionally for the Via header field, for UDP, if the REGISTER request is protected by a security association, include the public IP address or FQDN and the protected server port value in the sent-by field. For TCP, if the REGISTER request is protected by a security association, the UE shall include the public IP address or FQDN;

d) the Security-Client header field set to specify the security mechanisms the UE supports, the IPsec layer algorithms the UE supports and the parameters needed for the security association setup. The UE shall support the setup of two pairs of security associations as defined in 3GPP TS 33.203 [19]. The syntax of the parameters needed for the security association setup is specified in annex H of 3GPP TS 33.203 [19]. The UE shall support the "ipsec-3gpp" security mechanism, as specified in RFC 3329 [48]. The UE shall support the IPsec layer algorithms for integrity and confidentiality protection as defined in 3GPP TS 33.203 [19], and shall announce support for them according to the procedures defined in RFC 3329 [48]. In addition to transport mode, the UE shall support UDP encapsulated tunnel mode as per RFC 3948 [73A] and shall announce support for both modes as described in 3GPP TS 33.203 [19];

K.2.1.2.2.3 Initial registration using SIP digest without TLS

The procedures described in subclause 5.1.1.2.3 apply without modification.

K.2.1.2.2.4 Initial registration using SIP digest with TLS

The procedures described in subclause 5.1.1.2.4 apply without modification.

K.2.1.2.2.5 Initial registration using NASS-IMS bundled authentication

The procedures described in subclause 5.1.1.2.5 apply without modification.

K.2.1.2.3 Initial subscription to the registration-state event package

The procedures described in subclause 5.1.1.3 apply with the additional procedures described in subclause K.2.1.4.1.

K.2.1.2.4 User-initiated re-registration

K.2.1.2.4.1 General

The procedures described in subclause 5.1.1.4 apply with the additional procedures described in the present subclause.

On sending a REGISTER request that does not contain a challenge response, the UE shall populate the header fields as indicated in subclause 5.1.1.4.1 with the exception of subitems c) and d) which are modified as follows.

The UE shall populate:

c) a Contact header field set to include SIP URI(s) that contain(s) in the hostport parameter the private IP address of the UE or FQDN, its instance ID ("+sip.instance" header field parameter) along with the same "reg-id" header field parameter used for the initial, successful, registration for the given P-CSCF public identity combination as described in RFC 5626 [92]. The UE shall include all supported ICSI values (coded as specified in subclause 7.2A.8.2) in a g.3gpp.icsi-ref media feature tag as defined in subclause 7.9.2 and RFC 3840 [62] for the IMS communication services it intends to use, and IARI values (coded as specified in subclause 7.2A.9.2), for the IMS applications it intends to use in a g.3gpp.iari-ref media feature tag as defined in subclause 7.9.3 and RFC 3840 [62]; and

d) a Via header field according to the following rules:

- For UDP, the UE shall include the public IP address or FQDN in the sent-by field. The UE shall also include the "rport" header field parameter as defined in RFC 3581 [56A]. The UE shall only use an FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT; or

- For TCP, the UE shall include the public IP address or FQDN of the UE in the sent-by field. The UE shall only use an FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT;

When the timer F expires at the UE, the UE shall:

1) stop processing of all ongoing dialogs and transactions associated with that, if any (i.e. no further SIP signalling will be sent by the UE on behalf of these transactions or dialogs); and

2) after releasing all IP-CAN bearers used for the transport of media according to the procedures in subclause 9.2.2, the UE shall follow the procedures in RFC 5626 [92] to form a new flow to replace the failed one. When registering to create a new flow to replace the failed one, procedures in subclause 5.1.1.2 apply.

NOTE: These actions can also be triggered as a result of the failure of a STUN keep-alive. It is an implementation option whether these actions are also triggered by other means than expiration of timer F, e.g., based on ICMP messages.

If failed registration attempts occur in the process of creating a new flow, the flow recovery procedures defined in RFC 5626 [92] shall apply.

K.2.1.2.4.2 IMS AKA as a security mechanism

The procedures described in subclause 5.1.1.4.2 apply without modification.

K.2.1.2.4.3 SIP Digest without TLS as a security mechanism

The procedures described in subclause 5.1.1.4.3 apply without modification.

K.2.1.2.4.4 SIP Digest with TLS as a security mechanism

The procedures described in subclause 5.1.1.4.4 apply without modification.

K.2.1.2.4.5 NASS-IMS bundled authentication as a security mechanism

The procedures described in subclause 5.1.1.4.5 apply without modification.

K.2.1.2.5 Authentication

K.2.1.2.5.1 IMS AKA – general

The procedures of subclause 5.1.1.5.1 apply with the additional procedures described in the present subclause.

On receiving a 401 (Unauthorized) response to the REGISTER request and the response is deemed to be valid and signalling security is to be used, the UE shall behave as of subclause 5.1.1.5.1 with the exception of subitem 3) which is modified as follows.

The UE shall:

3) send another REGISTER request using the temporary set of security associations to protect the message. The header fields are populated as defined for the initial registration (see subclause K.2.1.2.2), with the addition that the UE shall include an Authorization header field containing the private user identity and if the "algorithm" header field parameter is "AKAv1-MD5" or "AKAv2-SHA-256", the authentication challenge response shall be calculated by the UE using RES and other parameters, as described in RFC 3310 [49] when AKAv1 is used or as described in RFC 4169 [227] when AKAv2 is used. If the "algorithm" header field parameter is "MD5" or "SHA2-256", the UE shall calculate SIP digest-response parameters as indicated in RFC 7616 [286] and RFC 8760 [287] and shall build an Authorization header field based on these parameters. The UE shall also insert the Security-Client header field that is identical to the Security-Client header field that was included in the

previous REGISTER request (i.e. the REGISTER request that was challenged with the received 401 (Unauthorized) response). The UE shall also insert the Security-Verify header field into the request, by mirroring in it the content of the Security-Server header field received in the 401 (Unauthorized) response. The UE shall set the Call-ID of the integrity-protected REGISTER request which carries the authentication challenge response to the same value as the Call-ID of the 401 (Unauthorized) response which carried the challenge.

NOTE: The "AKAv1-MD5" and "MD5" algorithms are only supported for backward compatibility.

For IPsec, if the 200 (OK) response is not received before the temporary SIP level lifetime of the temporary set of security associations expires or a 403 (Forbidden) response is received, the UE shall consider the registration to have failed. The UE shall delete the temporary set of security associations it was trying to establish, and use the old set of security associations. The UE should send an unprotected REGISTER request according to the procedure specified in subclause K.2.1.2.2 if the UE considers the old set of security associations to be no longer active at the P-CSCF.

K.2.1.2.5.2 Void

K.2.1.2.5.3 IMS AKA abnormal cases

The text in subclause 5.1.1.5.3 applies without changes.

K.2.1.2.5.4 SIP digest without TLS – general

The text in subclause 5.1.1.5.4 applies without changes.

K.2.1.2.5.5 SIP digest without TLS – abnormal procedures

The procedures of subclause 5.1.1.5.5 apply with the additional procedures described in the present subclause.

On receiving a 403 (Forbidden) response, the UE shall consider the registration to have failed. If performing SIP digest with TLS, the UE should send an initial REGISTER according to the procedure specified in subclause K.2.1.2.2 if the UE considers the TLS session to be no longer active at the P-CSCF.

K.2.1.2.5.6 SIP digest with TLS – general

The text in subclause 5.1.1.5.6 applies without changes.

K.2.1.2.5.7 SIP digest with TLS – abnormal procedures

The text in subclause 5.1.1.5.7 applies without changes.

K.2.1.2.5.8 NASS-IMS bundled authentication – general

The text in subclause 5.1.1.5.8 applies without changes.

K.2.1.2.5.9 NASS-IMS bundled authentication – abnormal procedures

The text in subclause 5.1.1.5.9 applies without changes.

K.2.1.2.5.10 Abnormal procedures for all security mechanisms

The text in subclause 5.1.1.5.10 applies without changes.

K.2.1.2.5A Network initiated re-authentication

The procedures of subclause 5.1.1.5A apply with the additional procedures described in the present subclause.

On starting the re-authentication procedure sending a REGISTER request that does not contain a challenge response, the UE shall behave as of subclause 5.1.1.5A with the exception of subitem 2) which is modified as follows.

The UE shall:

2) start the re-authentication procedures at the appropriate time (as a result of the S-CSCF procedure described in subclause 5.4.1.6) by initiating a re-registration as described in subclause K.2.1.2.4, if required.

K.2.1.2.5B Change of IPv6 address due to privacy

The text in subclause 5.1.1.5B applies without changes.

K.2.1.2.6 User-initiated deregistration

K.2.1.2.6.1 General

The procedures of subclause 5.1.1.6.1 apply with the additional procedures described in the present subclause.

On sending a REGISTER request, the UE shall populate the header fields as indicated in subclause 5.1.1.6.1 with the exception of subitems c) and d) which are modified as follows.

The UE shall populate:

c) a Contact header field set to either the value of "*" or SIP URI(s) that contain(s) in the hostport parameter the IP address of the UE or FQDN, its instance ID ("+sip.instance" header field parameter) along with the same "reg-id" header field parameter used for the initial, successful, registration for the given P-CSCF public identity combination as described in RFC 5626 [92];. The UE shall only use a FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT;

d) a Via header field according to the following rules:

- For UDP, the UE shall include the public IP address or FQDN. The UE shall also include the "rport" header field parameter as defined in RFC 3581 [56A]. The UE shall only use an FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT; or

- For TCP, the UE shall include the public IP address or FQDN of the UE in the sent-by field. The UE shall only use an FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT;

NOTE: In case of hosted NAT traversal only the UE public IP addresses are bound to security associations or TLS session.

K.2.1.2.6.2 IMS AKA as a security mechanism

The text in subclause 5.1.1.6.2 applies without changes.

K.2.1.2.6.3 SIP digest as a security mechanism

The text in subclause 5.1.1.6.3 applies without changes.

K.2.1.2.6.4 SIP digest with TLS as a security mechanism

The text in subclause 5.1.1.6.4 applies without changes.

K.2.1.2.6.5 Initial registration using NASS-IMS bundled authentication

The text in subclause 5.1.1.6.5 applies without changes.

K.2.1.2.7 Network-initiated deregistration

The procedures of subclause 5.1.1.7 apply with the additional procedures described in the present subclause.

Upon receipt of a NOTIFY request on the dialog which was generated during subscription to the reg event package as described in subclause 5.1.1.3, including one or more <registration> element(s) which were registered by this UE with:

- the state attribute set to "terminated" and the event attribute set to "rejected" or "deactivated"; or

- the state attribute set to "active" and the state attribute within the <contact> element belonging to this UE set to "terminated", and associated event attribute element to "rejected" or "deactivated";

The UE shall remove all registration details relating to these public user identities. In case of a "deactivated" event attribute, the UE shall start the initial registration procedure as described in subclause K.2.1.2.2. In case of a "rejected" event attribute, the UE shall release all dialogs related to those public user identities.

K.2.1.3 Subscription and notification

The text in subclause 5.1.2 applies without changes.

K.2.1.4 Generic procedures applicable to all methods excluding the REGISTER method

K.2.1.4.1 UE-originating case

The procedures described in subclause 5.1.2A.1 apply with the additional procedures described in the present subclause.

When the UE sends any request, the requirements in subclause 5.1.2A.1 are extended by the following requirements. The UE shall include:

- a Via header field according to the following rules:
  - For UDP, the UE shall include the public IP address or FQDN and the protected server port value in the sent-by field. The UE shall also include the "rport" header field parameter as defined in RFC 3581 [56A]. The UE shall only use an FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT; or
  - For TCP, the UE shall include the public IP address or FQDN of the UE in the sent-by field. The UE shall only use an FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT; and
- if the request contains a Contact header field, include a Contact header field according to the following rules:
  - if this is a request for a new or existing dialog, and the UE did insert a GRUU in the Contact header field, then the UE shall also include its instance ID ("+sip.instance" header field parameter), and an "ob" SIP URI parameter as described in RFC 5626 [92]; or
  - if this is a request for a new or existing dialog, and the UE did not insert a GRUU in the Contact header field, then the UE shall include the public IP address of the UE or FQDN and the protected server port value bound to the security association or TLS session in the hostport parameter along with its instance ID ("+sip.instance" header field parameter), and an "ob" SIP URI parameter as described in RFC 5626 [92]. The UE shall only use a FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT.

NOTE: The means to ensure that the FQDN resolves to the public IP address of the NAT are outside of the scope of this specification. One option for resolving this is local configuration.

Where a security association or TLS session exists, the UE shall discard any SIP response that is not protected by the security association or TLS session and is received from the P-CSCF outside of the registration and authentication procedures. The requirements on the UE within the registration and authentication procedures are defined in subclause K.2.1.2.

When a SIP transaction times out, i.e. timer B, timer F or timer H expires at the UE, the UE may behave as if timer F expired, as described in subclause K.2.1.2.4.

K.2.1.4.2 UE-terminating case

The procedures described in subclause 5.1.2A.2 apply with the additional procedures described in the present subclause.

When the UE sends any response, the requirements in subclause 5.1.2A.2 are extended by the following requirement. If the UE did not include a GRUU in the Contact header field, then the UE shall:

- include the public IP address of the UE or FQDN and the protected server port value bound to the security association or TLS session in the hostport parameter in any Contact header field that is otherwise included. The UE shall only use a FQDN, if it is ensured that the FQDN resolves to the public IP address of the NAT.

The UE shall discard any SIP request that is not integrity protected and is received from the P-CSCF outside of the registration and authentication procedures. The requirements on the UE within the registration and authentication procedures are defined in subclause K.2.1.2.

K.2.1.5 Maintaining flows and detecting flow failures

STUN Binding Requests are used by the UE as a keep-alive mechanism to maintain NAT bindings for signalling flows over connectionless transport (for dialogs outside a registration as well as within a registration) as well as to determine whether a flow (as described in RFC 5626 [92]) is still valid (e.g. a NAT reboot could cause the transport parameters to change). As such, the UE acts as a STUN client and shall follow the requirements defined by RFC 8489 [291]. Further, when using UDP encapsulated IPsec, the keep-alive capabilities defined within should not be used.

CRLF as defined in RFC 5626 [92] is used by the UE as a keep-alive mechanism to maintain NAT bindings for signalling flows over connection oriented transports (for dialogs outside a registration as well as within a registration) as well as to determine whether a flow (as described in RFC 5626 [92]) is still valid (e.g. a NAT reboot could cause the transport parameters to change). As such, the UE shall follow the requirements defined by RFC 5626 [92].

If the UE determines that the flow to a given P-CSCF is no longer valid (the UE does not receive a STUN reply (or CRLF) or the reply indicates a new public IP Address) the UE shall consider the flow and any associated security associations invalid and perform the initial registration procedures defined in subclause K.2.1.2.2.

When a NAT is not present, it may not be desirable to send keep-alive requests (i.e. given battery considerations for wireless UEs). As such, if a UE can reliably determine that a NAT is not present (i.e. by comparing the "received" header field parameter in the Via header field in the response to the initial un-protected REGISTER request with the locally assigned IP address) then the UE may not perform the keep-alive procedures.

K.2.1.6 Emergency services

K.2.1.6.1 General

In addition to the procedures in subclause 5.1.6.1, the following additional procedures apply. When receiving and sending requests unprotected, the UE shall transmit and receive all SIP messages using the same IP port.

K.2.1.6.2 Initial emergency registration

When a UE performs an initial emergency registration the UE shall perform the actions as specified in subclause K.2.1.2.2. The remaining procedures described in subclause 5.1.6.2 apply without modification.

K.2.1.6.2A New initial emergency registration

The text in subclause 5.1.6.2A applies without changes.

K.2.1.6.3 Initial subscription to the registration-state event package

The text in subclause 5.1.6.3 applies without changes.

K.2.1.6.4 User-initiated emergency reregistration

The UE shall perform user-initiated emergency reregistration as specified in subclause K.2.1.2.4. The remaining procedures described in subclause 5.1.6.4 apply without modification.

K.2.1.6.5 Authentication

The UE shall perform the authentication procedures as specified in subclause K.2.1.2.5. The remaining procedures described in subclause 5.1.6.5 apply without modification.

K.2.1.6.6 User-initiated emergency deregistration

The text in subclause 5.1.6.6 applies without changes.

K.2.1.6.7 Network-initiated emergency deregistration

The text in subclause 5.1.6.7 applies without changes.

K.2.1.6.8 Emergency session setup

K.2.1.6.8.1 General

The text in subclause 5.1.6.8.1 applies without changes.

K.2.1.6.8.2 Emergency session set-up in case of no registration

The text in subclause 5.1.6.8.2 applies without changes.

K.2.1.6.8.3 Emergency session set-up with an emergency registration

After a successful initial emergency registration, the UE shall apply the procedures as specified in subclause K.2.1.4, subclause 5.1.3, and subclause 5.1.4. The remaining procedures described in subclause 5.1.6.8.3 apply without modification.

K.2.1.6.8.4 Emergency session set-up within a non-emergency registration

The UE shall apply the procedures as specified in subclause K.2.1.4, subclause 5.1.3, and subclause 5.1.4. The remaining procedures described in subclause 5.1.6.8.4 apply without modification.

K.2.1.6.9 Emergency session release

The text in subclause 5.1.6.9 applies without changes.

K.2.2 Procedures at the P-CSCF

K.2.2.1 Introduction

This subclause describes the SIP procedures for supporting hosted NAT scenarios.

The description enhances the procedures specified in subclause 5.2.

K.2.2.2 Registration

K.2.2.2.1 General

The procedures described in subclause 5.2.2.1 apply without changes.

K.2.2.2.2 IMS AKA as a security mechanism

The procedures described in subclause 5.2.2.2 apply with the additional procedures described in the present subclause.

When the P-CSCF receives a REGISTER request from the UE, the P-CSCF shall behave as in subclause 5.2.2.2 with the exception of subitems 2) and 3) which are modified as follows.

2) in case the REGISTER request was received without protection, then:

a) check the existence of the Security-Client header field. If the Security-Client header field is not present and signalling security is used, then the P-CSCF shall return a suitable 4xx response. If the Security-Client header field is present the P-CSCF shall:

- in case the UE indicated support for "UDP-enc-tun" then remove and store it; or

- in case the UE does not indicate support for "UDP-enc-tun" then:

  - if the host portion of the sent-by field in the topmost Via header field contains an IP address that differs from the source address of the IP packet, silently drop the REGISTER request;

- otherwise continue with procedures as of subclause 5.2.2.2;

NOTE 2: If the UE does not indicate support for "UDP-enc-tun" and the P-CSCF detects that the UE is located behind a NAT device, then the P-CSCF can just drop the REGISTER request to avoid unnecessary signalling traffic.

3) in case the REGISTER request was received integrity protected, then the P-CSCF shall:

a) check the security association which protected the request. If IPsec is used and the security association is a temporary one the P-CSCF shall:

- in case the hostport parameter in the Contact address is in the form of a FQDN, ensure that the given FQDN will resolve (e.g., by reverse DNS lookup) to the IP address bound to the security association;

- in case the P-CSCF has detected earlier that the UE is located behind a NAT and IPsec is being used, retrieve port_Uenc from the encapsulating UDP header of the packet received and complete configuration of the temporary set of security associations by configuring port_Uenc in each of the temporary security associations;

- check whether the request contains a Security-Verify header field in addition to a Security-Client header field. If there are no such header fields, then the P-CSCF shall return a suitable 4xx response. If there are such header fields, then the P-CSCF shall compare the content of the Security-Verify header field with the content of the Security-Server header field sent earlier and the content of the Security-Client header field with the content of the Security-Client header field received in the challenged REGISTER request. If those do not match, then there is a potential man-in-the-middle attack. The request should be rejected by sending a suitable 4xx response. If the contents match, the P-CSCF shall remove the Security-Verify and the Security-Client header field;

When the P-CSCF receives a 401 (Unauthorized) response to an unprotected REGISTER request and the P-CSCF previously determined that the UE is behind a NAT and the UE indicated support for "UDP-enc-tun" IPsec mode, the P-CSCF shall:

1) delete any temporary set of security associations established towards the UE;

2) for IPsec, remove the "ck" and "ik" WWW-Authenticate header field parameters contained in the 401 (Unauthorized) response and bind the values to the proper private user identity and to the temporary set of security associations which will be setup as a result of this challenge. The P-CSCF shall forward the 401 (Unauthorized) response to the UE if and only if the "ck" and "ik" header field parameters have been removed;

3) insert a Security-Server header field in the response, containing the P-CSCF security list and the parameters needed.The P-CSCF shall support the setup of two pairs of security associations, as defined in 3GPP TS 33.203 [19]. The syntax of the parameters needed of the IPsec security association setup is specified in annex H of 3GPP TS 33.203 [19]. The P-CSCF shall support the "ipsec-3gpp" security mechanism, as specified in RFC 3329 [48]. The P-CSCF shall support the IPSec layer algorithms for integrity protection and for encryption as defined in 3GPP TS 33.203 [19]. The P-CSCF shall indicate "UDP-enc-tun" as the only IPsec mode.

4) set up the temporary set of security associations with a temporary SIP level lifetime between the UE and the P-CSCF for the user identified with the private user identity. The P-CSCF shall select UDP encapsulated tunnel mode and shall leave the value for port-Uenc unspecified in each of the temporary security associations. For further details see 3GPP TS 33.203 [19] and RFC 3329 [48]. The P-CSCF shall set the temporary SIP level lifetime for the temporary set of security associations to the value of reg-await-auth timer; and

5) send the 401 (Unauthorized) response unprotected to the UE using the mechanisms described in RFC 3261 [26] and RFC 3581 [56A], i.e. the P-CSCF shall send the response to the IP address indicated in the "received" header field parameter and, in case UDP is used, to the port indicated in the "rport" header field parameter (if present) of the Via header field associated with the UE. In case TCP is used as transport protocol, the P-CSCF shall use the port on which the REGISTER request was received as client port for sending the response back to the UE.

When the P-CSCF receives a 401 (Unauthorized) response to a protected REGISTER request and the P-CSCF previously determined that the UE is behind a NAT and that REGISTER request was protected by an old set of security associations that use UDP encapsulated tunnel mode, the P-CSCF shall:

1) delete any temporary set of security associations established towards the UE;

2) remove the "ck" and "ik" WWW-Authenticate header field parameters contained in the 401 (Unauthorized) response and bind the values to the proper private user identity and to the temporary set of security associations which will be setup as a result of this challenge. The P-CSCF shall forward the 401 (Unauthorized) response to the UE if and only if the "ck" and "ik" header field parameters have been removed;

3) insert a Security-Server header field in the response, containing the P-CSCF security list and the parameters needed for the security association setup, as specified in Annex H of 3GPP TS 33.203 [19]. The P-CSCF shall support the "ipsec-3gpp" security mechanism, as specified in RFC 3329 [48]. The P-CSCF shall support the IPsec layer algorithms for integrity protection and encryption as defined in 3GPP TS 33.203 [19]. The P-CSCF shall indicate "UDP-enc-tun" as the IPsec mode;

4) set up the temporary set of security associations with a temporary SIP level lifetime between the UE and the P-CSCF for the user identified with the private user identity. The P-CSCF shall select UDP encapsulated tunnel mode and shall specify the same port_Uenc that was used in the old set of security associations. The P-CSCF shall set the temporary SIP level lifetime for the temporary set of security associations to the value of reg-await-auth timer; and

5) send the 401 (Unauthorized) response to the UE using the old set of security associations and using the rules for sending responses as described in RFC 3261 [26] and RFC 3581 [56A], i.e. the P-CSCF shall send the response to the IP address indicated in the "received" header field parameter and if UDP is used, to the port indicated in the "rport" header field parameter (if present) of the Via header field associated with the UE. Otherwise, when the P-CSCF receives a 401 (Unauthorized) response to an unprotected REGISTER request and:

   - this response does not contain a "received" header field parameter in the Via header field associated with the UE;

   - this response does not contain "rport" header field parameter in the Via header field associated with the UE and the request associated with the response was received using UDP; or

   - when the P-CSCF receives a 401 (Unauthorized) response to a protected REGISTER request and that REGISTER request was protected by an old set of security associations that do not use UDP encapsulated tunnel mode;

   the P-CSCF shall proceed as described in subclause 5.2.2.2 of the main body of this specification.

K.2.2.2.3 SIP digest without TLS as a security mechanism

The text in subclause 5.2.2.3 applies without changes.

K.2.2.2.4 SIP digest with TLS as a security mechanism

The procedures described in subclause 5.2.2.4 apply without changes.

K.2.2.2.5 NASS-IMS bundled authentication as a security mechanism

The text in subclause 5.2.2.5 applies without changes.

K.2.2.3 General treatment for all dialogs and standalone transactions excluding the REGISTER method

K.2.2.3.1 Requests initiated by the UE

K.2.2.3.1.1 General for all requests

The procedures described in subclause 5.2.6.3.1 apply with the additional procedures described in the present subclause.

When the P-CSCF receives from the UE an initial request for a dialog or a request for a standalone transaction, the requirements are extended by the following requirements.

Before forwarding the request, based on the topmost Route header field, in accordance with the procedures of RFC 3261 [26], the P-CSCF shall ensure that all signalling during the lifetime of the dialogue is sent over the same IMS flow set as the dialogue initiating request.

NOTE: The suggested way to ensure all signalling is sent over the same IMS flow set is to form an IMS flow token in the same way that a P-CSCF would form this for the Path header field and insert this IMS flow token in the user portion of the URI used in the record route header field value.

K.2.2.3.1.2 General for all responses

The procedures in subclause 5.2.6.3.2 apply without changes.

K.2.2.3.1.2A Abnormal cases

The text in subclause 5.2.6.3.2A applies without changes.

K.2.2.3.1.3 Initial request for a dialog

The text in subclause 5.2.6.3.3 applies without changes.

K.2.2.3.1.4 Responses to an initial request for a dialog

The text in subclause 5.2.6.3.4 applies without changes.

K.2.2.3.1.5 Target refresh request for a dialog

The text in subclause 5.2.6.3.5 applies without changes.

K.2.2.3.1.6 Responses to a target refresh request for a dialog

The text in subclause 5.2.6.3.6 applies without changes.

K.2.2.3.1.7 Request for a standalone transaction

The text in subclause 5.2.6.3.7 applies without changes.

K.2.2.3.1.8 Responses to a request for a standalone transaction

The text in subclause 5.2.6.3.8 applies without changes.

K.2.2.3.1.9 Subsequent request other than a target refresh request

The text in subclause 5.2.6.3.9 applies without changes.

K.2.2.3.1.10 Responses to a subsequent request other than a target refresh request

Void

K.2.2.3.1.11 Request for an unkown method that does not relate to an existing dialog

The text in subclause 5.2.6.3.11 applies without changes.

K.2.2.3.1.12 Responses to a request for an unkown method that does not relate to an existing dialog

Void

K.2.2.3.2 Requests terminated by the UE

K.2.2.3.2.1 General for all requests

Void

K.2.2.3.2.2 General for all responses

Void

K.2.2.3.2.3 Initial request for a dialog

The procedures described in subclause 5.2.6.4.3 apply with the additional procedures described in the present subclause.

When the P-CSCF receives, destined for the UE, a request, the requirements are extended by the following requirements. The P-CSCF shall:

- forward the request to the terminating UE over the appropriate flow within the denoted IMS flow set.

K.2.2.3.2.4 Responses to an initial request for a dialog

The text in subclause 5.2.6.4.4 applies without changes.

K.2.2.3.2.5 Target refresh request for a dialog

The procedures described in subclause 5.2.6.4.5 apply with the additional procedures described in the present subclause.

When the P-CSCF receives, destined for the UE, a request, the requirements are extended by the following requirements. The P-CSCF shall:

- forward the request to the terminating UE over the appropriate flow within the denoted IMS flow set.

K.2.2.3.2.6 Responses to a target refresh request for a dialog

The text in subclause 5.2.6.4.6 applies without changes.

K.2.2.3.2.7 Request for a standalone transaction

The procedures described in subclause 5.2.6.4.7 apply with the additional procedures described in the present subclause.

When the P-CSCF receives, destined for the UE, a request, the requirements are extended by the following requirements. The P-CSCF shall:

- forward the request to the terminating UE over the appropriate flow within the denoted IMS flow set.

K.2.2.3.2.8 Responses to a request for a standalone transaction

The text in subclause 5.2.6.4.8 applies without changes.

K.2.2.3.2.9 Subsequent request other than a target refresh request

The procedures described in subclause 5.2.6.4.9 apply with the additional procedures described in the present subclause.

When the P-CSCF receives, destined for the UE, a request, the requirements are extended by the following requirements. The P-CSCF shall:

- forward the request to the terminating UE over the appropriate flow within the denoted IMS flow set.

K.2.2.3.2.10 Responses to a subsequent request other than a target refresh request

The text in subclause 5.2.6.4.10 applies without changes.

K.2.2.3.2.11 Request for an unknown method that does not relate to an existing dialog

Void

K.2.2.3.2.12 Responses to a request for an unknown method that does not relate to an existing dialog

Void

K.2.2.4 Void

K.2.2.5 Emergency services

K.2.2.5.1 General

The procedures described in subclause 5.2.10.1 apply without changes.

K.2.2.5.2 General treatment for all dialogs and standalone transactions excluding the REGISTER method – from an unregistered user

The procedures described in subclause 5.2.10.2 apply with the additional procedures described in the present subclause.

When the P-CSCF receives from the UE an initial request for a dialog or a request for a standalone transaction, the requirements are extended by the following requirements.

Before forwarding the request, based on the topmost Route header field, in accordance with the procedures of RFC 3261 [26], the P-CSCF shall ensure that all signalling during the lifetime of the dialogue is sent over the same IMS flow set as the dialogue initiating request.

NOTE: The suggested way to ensure all signalling is sent over the same IMS flow set is to form an IMS flow token in the same way that a P-CSCF would form this for the Path header field and insert this IMS flow token in the user portion of the URI used in the Record-Route header field value.

K.2.2.5.3 General treatment for all dialogs and standalone transactions excluding the REGISTER method after emergency registration

The procedures described in subclause 5.2.10.3 apply with the additional procedures described in the present subclause.

When the P-CSCF receives from the UE an initial request for a dialog, or a standalone transaction, or an unknown method, the following requirements:

1) include in the Request-URI an emergency service URN, i.e. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69], if necessary, and execute the procedure described in step 4, 5, 6, and 7, in subclause 5.2.6.3.3, subclause 5.2.6.3.7, subclause 5.2.6.3.11, subclause 5.2.7.2 and subclause K.2.2.3.1, as appropriate. An additional sub-service type can be added if information on the type of emergency service is known. The entry in the Request-URI that the P-CSCF includes may either be:

   - as received from the UE in the Request-URI in accordance with RFC 5031 [69]; or

   - as deduced from the Request-URI received from the UE.

K.2.2.5.4 General treatment for all dialogs and standalone transactions excluding the REGISTER method – non-emergency registration

The procedures described in subclause 5.2.10.4 apply with the additional procedures described in the present subclause.

When the P-CSCF receives from the UE an initial request for a dialog, or a standalone transaction, or an unknown method, the following requirements are extended:

1) include in the Request-URI an emergency service URN, i.e. a service URN with a top-level service type of "sos" as specified in RFC 5031 [69], if necessary, and execute the procedure described in step 3, 4, 5, 6, and 7, in subclause 5.2.6.3.3, subclause 5.2.6.3.7, subclause 5.2.6.3.11 subclause 5.2.7.2 and subclause K.2.2.3.1, as appropriate. An additional sub-service type can be added if information on the type of emergency service is known. The entry in the Request-URI that the P-CSCF includes may either be:

   - as received from the UE in the Request-URI in accordance with RFC 5031 [69]; or

   - as deduced from the Request-URI received from the UE; and

K.2.2.5.5 Abnormal cases

The text in subclause 5.2.10.5 applies without changes.

K.2.3 Void

K.2.4 Void

K.3 Application usage of SDP

K.3.1 UE usage of SDP

The procedures as of subclause 6.1 apply.

K.3.2 P-CSCF usage of SDP

The procedures as of subclause 6.2 apply.

K.4 Void

K.5 Application usage of ICE

K.5.1 Introduction

The following subclauses describe the usage of the Interactive Connectivity Establishment (ICE) procedures as documented in RFC 8445 [289] and RFC 8839 [290].

K.5.2 UE usage of ICE

K.5.2.1 General

NAT bindings also need to be kept alive for media. RFC 8445 [289] provides requirements for STUN based keepalive mechanisms. UEs that do not implement the ICE procedures as defined in RFC 8445 [289] should implement the keepalive procedures defined in RFC 8445 [289]. In the case where keepalives are required and the other end does not support ICE (such that STUN cannot be used for a keepalive) or the UE can not discover STUN or TURN servers to gather candidates, the UE shall send an empty (no payload) RTP packet with a payload type of 20 as a keepalive as long as the other end has not negotiated the use of this value. If this value has already been negotiated, then some other unused static payload type from table 5 of RFC 3551 [55A] shall be used. When sending an empty RTP packet, the UE shall continue using the sequence number (SSRC) and timestamp as the negotiated RTP steam.

K.5.2.2 Call initiation – UE-origination case

The UE should support the agent requirements for ICE as defined by RFC 8445 [289] when sending the initial INVITE request. RFC 8445 [289] and RFC 8839 [290] provide procedures for:

1) Gathering candidate addresses for RTP and RTCP prior to sending the INVITE;

2) Encoding the candidate addresses in the SDP that is included with the INVITE;

3) Acting as a STUN server to receive binding requests from the remote client when it does connectivity checks;

4) Performing connectivity checks on received candidate addresses for RTP and RTCP;

5) Determining and possibly selecting a better active address based on the requirements in RFC 8445 [289] and RFC 8839 [290];

6) Subsequent offer/answer exchanges; and

7) Sending media.

When supporting the ICE procedures, the UE shall also support the STUN agent requirements as described in RFC 8489 [291] in order to gather STUN addresses, the TURN client requirements as described in RFC 8656 [292] in order to gather TURN Server addresses and the STUN Server requirements defined in RFC 8445 [289] as well as the requirements for STUN Servers defined in RFC 8489 [291] for responding to connectivity checks.

RFC 8445 [289] provides an algorithm for determining the priority of a particular candidate. The following additional requirements are provided to the UE:

1) The type preference assigned for each type of candidate from least to highest should be: Relayed Transport Address, STUN address, local address; and

2) If the UE has a dual IPv4/IPv6 stack, IPv6 addresses may be assigned a higher local preference than IPv4 addresses based on the operator's policy.

RFC 8445 [289] provides guidance on choosing the in-use candidate and recommends that a UE choose relayed candidates as the in-use address. The following additional requirements are provided to the UE:

1) If a TURN server is available, the Relayed Transport Address should be used as the initial active transport address (i.e. as advertised in the m/c lines of the SDP); and

2) If a TURN server is not available, an address obtained via STUN should be used as the initial active transport address.

Regardless of whether the UE supports the above procedures, the UE shall, upon receipt of an SDP answer with candidate addresses, perform connectivity checks on the candidate addresses as described in RFC 8445 [289] and RFC 8839 [290]. In order to perform connectivity checks, the UE shall act as a STUN client as defined in RFC 8489 [291]. Further, the UE shall also follow the procedures in RFC 8445 [289] and RFC 8839 [290] when sending media.

K.5.2.3 Call termination – UE-termination case

The UE should support agent requirements for ICE as defined by RFC 8445 [289] when receiving an initial INVITE request. RFC 8445 [289] and RFC 8839 [290] provide procedures for:

1) Gathering candidate addresses for RTP and RTCP prior to sending the answer as described in RFC 8445 [289];

2) Encoding the candidate addresses in the SDP answer as described in RFC 8839 [290];

3) Acting as a STUN server to receive binding requests from the remote client when it does connectivity checks;

4) Performing connectivity checks on received candidate addresses for RTP and RTCP;

5) Determining and possibly selecting a better active address based on the requirements in RFC 8445 [289] and RFC 8839 [290];

6) Subsequent offer/answer exchanges; and

7) Sending media.

When supporting the ICE procedures, the UE shall also support the STUN agent requirements as described in RFC 8489 [291] in order to gather STUN addresses, the TURN client requirements as described in RFC 8656 [292] in order to gather TURN Server addresses and the STUN Server requirements defined in RFC 8445 [289] as well as the requirements for STUN Servers defined in RFC 8489 [291] for responding to connectivity checks.

RFC 8445 [289] provides an algorithm for determining the priority of a given candidate. The additional requirements for the UE:

1) The priority of candidate addresses from least to highest should be: Relayed Transport Address, STUN address, local address; and

2) If the UE has a dual IPv4/IPv6 stack, IPv6 addresses MAY be placed at a higher priority than IPV4 addresses based on the operator's policy.

RFC 8445 [289] provides guidance on choosing the in-use candidate and recommends that a UE choose relayed candidates as the in-use address. The following additional requirements are provided to the UE:

1) If a TURN server is available, the Relayed Transport Address should be used as the initial active transport address (i.e. as advertised in the m/c lines of the SDP); and

2) If a TURN server is not available, an address obtained via STUN should be used as the initial active transport address.

Regardless of whether the UE supports the above procedures, the UE shall, upon receipt of an SDP offer with candidate addresses, perform connectivity checks on the candidate addresses as described in RFC 8445 [289] and RFC 8839 [290]. In order to perform connectivity checks, the UE shall act as a STUN client as defined in RFC 8489 [291]. Further, the UE shall also follow the procedures in RFC 8445 [289] and RFC 8839 [290] when sending media.

When receiving an SDP offer which does not indicate support for ICE, the UE aborts the ICE procedures and reverts to RFC 3264 [27B] offer/answer procedures; per RFC 8445 [289] and RFC 8839 [290]. However, if the terminating UE is behind a NA(P)T device this may result in the inability to pass media for the session as the terminating UE will respond with its locally assigned IP address which is unreachable. In order to ensure successful media exchange, the terminating UE shall provide either a STUN derived IP address and port or a TURN provided IP address and port in the m/c lines of the SDP answer. If the provided address and port is a TURN address and port, the policy charging and control framework will be unable to establish proper filter criteria as the address is that of the TURN server and not that of the UE or NAT in front of the UE; see RFC 8445 [289] subclause B.3 for further details. To rectify this issue, the terminating UE shall also include a candidate attribute as described in RFC 8445 [289] and RFC 8839 [290] identifying the server reflexive IP address and port (i.e. the IP address and port on the public side of the NAT) used when a TURN provided address and port is provided in the m/c line of the SDP answer.

K.5.3 P-CSCF support of ICE

The P-CSCF procedures to support ICE as specified in RFC 8445 [289] and RFC 8839 [290] are defined in subclause 6.7.2.7.

K.5.4 Void

Annex L (normative): IP-Connectivity Access Network specific concepts when using EPS to access IM CN subsystem

L.1 Scope

The present annex defines IP-CAN specific requirements for a call control protocol for use in the IP Multimedia (IM) Core Network (CN) subsystem based on the Session Initiation Protocol (SIP), and the associated Session Description Protocol (SDP), where the IP-CAN is Evolved Packet System (EPS). The EPS IP-CAN has an EPS core network which can be supported by an E-UTRAN radio access network.

L.2 EPS aspects when connected to the IM CN subsystem via E-UTRAN

L.2.1 Introduction

A UE accessing the IM CN subsystem, and the IM CN subsystem itself, utilise the services provided by EPS to provide packet-mode communication between the UE and the IM CN subsystem.

Requirements for the UE on the use of these packet-mode services are specified in this clause. Requirements for the P-GW in support of this communication are specified in 3GPP TS 29.061 [11], and 3GPP TS 29.212 [13B].

When using the EPS, each IP-CAN bearer is provided by an EPS bearer.

L.2.2 Procedures at the UE

L.2.2.1 EPS bearer context activation and P-CSCF discovery

The policy on the PDN connection established during the EPS attach procedure identifies parameters for composing the ESM messages sent during the EPS attach procedure as specified in 3GPP TS 24.301 [8J], when the UE performs the EPS attach procedure in order to communicate with IM CN subsystem.

The UE may support the policy on the PDN connection established during the EPS attach procedure.

If the UE supports the optional configuration parameter "Access_Point_Name_Parameter_Reading_Rule", as defined in 3GPP TS 24.167 [8G] and has been configured with this parameter, then the UE shall use it to retrieve the access point name to use in the EPS bearer context activation procedure.

If the UE supports the policy on the PDN connection established during the EPS attach procedure, the UE may support being configured with the policy on the PDN connection established during the EPS attach procedure using one or more of the following methods:

a) the EPS_initial_attach_ConRefs node of $EF_{IMSConfigData}$a file described in 3GPP TS 31.102 [15C];

b) the EPS_initial_attach_ConRefs node of $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

c) the EPS_initial_attach_ConRefs node of 3GPP TS 24.167 [8G].

If the UE is configured with both the EPS_initial_attach_ConRefs node of 3GPP TS 24.167 [8G] and the EPS_initial_attach_ConRefs node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or 3GPP TS 31.103 [15B], then the EPS_initial_attach_ConRefs node of the $EF_{IMSConfigData}$ file shall take precedence.

NOTE 1: Precedence for files configured on both the USIM and ISIM is defined in 3GPP TS 31.103 [15B].

Prior to communication with the IM CN subsystem, the UE shall:

a) if not attached for EPS services yet, perform a EPS attach procedure as specified in 3GPP TS 24.301 [8J]. If the UE requests establishment of a PDN connection during the EPS attach procedure, and the UE supports and is configured with the policy on the PDN connection established during the EPS attach procedure, the UE shall compose the ESM messages sent during the EPS attach procedure, according to the policy on the PDN connection established during the EPS attach procedure;

b) ensure that a EPS bearer context used for SIP signalling according to the APN and P-GW selection criteria described in 3GPP TS 23.401 [7B], is available. This EPS bearer context shall remain active throughout the period the UE is connected to the IM CN subsystem, i.e. from the initial registration and at least until the deregistration. As a result, the EPS bearer context provides the UE with information that makes the UE able to construct an IPv4 or an IPv6 address;

NOTE 2: The default EPS bearer context can also be used for SIP signalling as well as any other EPS bearer context.

When the EPS bearer context establishment procedure for the SIP signalling is initiated by the UE:

I. if a default EPS bearer context is not available with the selected P-GW, the UE shall indicate to the network in the PDN CONNECTIVITY REQUEST that the request is for SIP signalling. If the request is authorized, the network establishes a bearer with the appropriate QCI as described in 3GPP TS 24.301 [8J]. The UE may also use this EPS bearer context for DNS and DHCP signalling;

II. if the default EPS bearer context is available with the selected P-GW, and is to be used for SIP signalling no additional steps are needed; and

III. if the default EPS bearer context is available with the selected P-GW and an EPS bearer for SIP signalling with the correct QCI and TFT is to be established, the UE shall indicate to the network, by setting the IM CN Subsystem Signalling Flag in the Protocol Configuration Options information element in the BEARER RESOURCE ALLOCATION REQUEST message, that the request is for SIP signalling. If the request is authorized, the network either establishes a new dedicated bearer or modifies an exisiting bearer with the appropriate QCI and TFT as described in 3GPP TS 24.301 [8J]. The general QoS negotiation mechanism is described in 3GPP TS 24.301 [8J]; and

NOTE 3: An EPS bearer with a QCI value other than the one for signalling can carry both IM CN subsystem signalling and media, in case the media does not need to be authorized by Policy and Charging control mechanisms as defined in 3GPP TS 29.212 [13B] and the media stream is not mandated by the P-CSCF to be carried in a separate EPS bearer.

c) acquire a P-CSCF address(es).

The methods for P-CSCF discovery are:

I. When using IPv4, employ the Dynamic Host Configuration Protocol (DHCP) RFC 2132 [20F], the DHCPv4 options for SIP servers RFC 3361 [35A], and RFC 3263 [27A] as described in subclause 9.2.1. When using IPv6, employ Dynamic Host Configuration Protocol for IPv6 (DHCPv6) RFC 3315 [40], the DHCPv6 options for SIP servers RFC 3319 [41] and DHCPv6 options for Domain Name Servers (DNS) RFC 3646 [56C] as described in subclause 9.2.1.

II. Transfer P-CSCF address(es) within the EPS bearer context activation procedure.

The UE shall indicate the request for a P-CSCF address to the network within the Protocol Configuration Options information element of the PDN CONNECTIVITY REQUEST message or BEARER RESOURCE ALLOCATION REQUEST message.

If the network provides the UE with a list of P-CSCF IPv4 or IPv6 addresses in the ACTIVATE DEFAULT EPS BEARER CONTEXT REQUEST message or ACTIVATE DEDICATED EPS BEARER CONTEXT REQUEST message, the UE shall assume that the list is ordered top-down with the first P-CSCF address within the Protocol Configuration Options information element as the P-CSCF address having the highest preference and the last P-CSCF address within the Protocol Configuration Options information element as the P-CSCF address having the lowest preference.

III. The UE selects a P-CSCF from the list (see 3GPP TS 31.103 [15B]) stored in the ISIM.

IV. The UE selects a P-CSCF from the list in IMS management object.

The UE shall use method IV to select a P-CSCF, if

- a P-CSCF is to be discovered in the home network;
- the UE is roaming; and
- the IMS management object contains the P-CSCF list.

The UE shall use method III to select the P-CSCF, if:

- a P-CSCF is to be discovered in the home network;
- the UE is roaming;
- either the UE does not contain the IMS management object, or the UE contains the IMS management object but the IMS management object does not contain the P-CSCF list; and
- the ISIM residing in the UICC supports the P-CSCF list.

The UE can freely select method I or II for P-CSCF discovery, if:

- the UE is in the home network; or
- the UE is roaming and the P-CSCF is to be discovered in the visited network.

The UE can select method IV, if:

- the UE is in the home network; and
- the IMS management object contains the P-CSCF list.

In case method I is selected and several P-CSCF addresses or FQDNs are provided to the UE, the selection of P-CSCF address or FQDN shall be performed as indicated in RFC 3361 [35A] when using IPv4 or RFC 3319 [41] when using IPv6. If sufficient information for P-CSCF address selection is not available, selection of the P-CSCF address by the UE is implementation specific.

NOTE 4: The UE decides whether the P-CSCF is to be discovered in the serving network or in the home network based on local configuration, e.g. whether the application on the UE is permitted to use local breakout.

If the UE is designed to use I above, but receives P-CSCF address(es) according to II, then the UE shall either ignore the received address(es), or use the address(es) in accordance with II, and not proceed with the DHCP request according to I.

If the UE is configured to use Option II above and detects that all P-CSCFs known by the UE have been used when the UE selects a different P-CSCF as a result of:

- receiving 305 (Use Proxy) to the REGISTER request;
- receiving 504 (Server Time-out); or
- expiration of the timer F at the UE,

then if there are more than one PDN connection that UE is connected to and unless the IP-CAN bearer is in use by other applications, the UE shall:

1) release IP-CAN bearer that is used only for the transport of SIP signalling and that are not used for other non-IMS applications, but shall not release emergency IP-CAN bearers; and
2) unless the UE decides the service is no longer needed,
   a) perform a new P-CSCF discovery procedure as described in subslause 9.2.1; and
   b) perform the procedures for initial registration as described in subclause 5.1.1.2.

When using IPv4, the UE may request a DNS Server IPv4 address(es) via RFC 2132 [20F] or by the Protocol Configuration Options information element when activating a EPS bearer context according to 3GPP TS 24.301 [8J].

When using IPv6, the UE may request a DNS Server IPv6 address(es) via RFC 3315 [40] and RFC 3646 [56C] or by the Protocol Configuration Options information element when activating a EPS bearer context according to 3GPP TS 24.301 [8J].

The encoding of the request and response for IPv4 or IPv6 address(es) for DNS server(s) and list of P-CSCF address(es) within the Protocol Configuration Options information element is described in 3GPP TS 24.301 [8J].

When:

- the UE obtains an EPS bearer context used for SIP signalling by performing handover of the connection from another IP-CAN;
- IP address of the UE is not changed during the handover; and
- the UE already communicates with the IM CN subsystem via the connection with the other IP-CAN, e.g. the UE determines that its contact with host portion set to the UE IP address (or FQDN of the UE) associated with the connection with the other IP-CAN has been bound to a public user identity;

the UE shall continue using the P-CSCF address(es) acquired in the other IP-CAN.

The UE may support the policy on when a UE roaming in a VPLMN is allowed to transfer the PDN connection providing access to IMS between EPC via WLAN and EPS. If the UE roams in the EPS IP-CAN, has a session and the policy indicates "roaming in a VPLMN and having an ongoing session, is not allowed to transfer the PDN connection providing access to IMS between EPC via WLAN and EPS", the UE shall not handover the PDN connection providing access to IMS from EPC via WLAN to EPS.

If the UE roams in the EPS IP-CAN, has a session and the policy indicates "roaming in a VPLMN and having an ongoing session, is allowed to transfer the PDN connection providing access to IMS between EPC via WLAN and EPS", the UE shall, if not prevented by other rules or policies, handover the PDN connection providing access to IMS from EPC via WLAN to EPS.

If the UE roams in the EPS IP-CAN and the policy indicates "roaming in a VPLMN is not allowed to transfer the PDN connection providing access to IMS between EPC via WLAN and EPS, irrespective of if the UE is in a session or not", the UE shall not handover the PDN connection providing access to IMS from EPC via WLAN to EPS. The UE can re-establish a new PDN connection to another IP-CAN type in idle mode, e.g. due to UE domain preference.

If the UE supports the policy on whether a roaming UE when in a session is allowed to transfer the PDN connection providing access to IMS between EPC via WLAN and EPS, the UE may support being configured with the policy on whether a roaming UE when in a session is allowed to transfer the PDN connection providing access to IMS between EPC via WLAN EPS using one or more of the following methods:

a) the Allow_Handover_PDN_connection_WLAN_And_EPS node of $EF_{IMSConfigDat}$a file described in 3GPP TS 31.102 [15C];

b) the Allow_Handover_PDN_connection_WLAN_And_EPS node of $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

c) the Allow_Handover_PDN_connection_WLAN_And_EPS node of 3GPP TS 24.167 [8G].

If the UE is configured with both the Allow_Handover_PDN_connection_WLAN_And_EPS node of 3GPP TS 24.167 [8G] and the Allow_Handover_PDN_connection_WLAN_And_EPS node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or 3GPP TS 31.103 [15B], then the Allow_Handover_PDN_connection_WLAN_And_EPS node of the $EF_{IMSConfigData}$ file shall take precedence.

NOTE 5: Precedence for files configured on both the USIM and ISIM is defined in 3GPP TS 31.103 [15B].

L.2.2.1A Modification of a EPS bearer context used for SIP signalling

The EPS bearer context shall not be modified from being used exclusively for SIP signalling to a general purpose EPS bearer. After the establishment of an EPS bearer context used for SIP signalling, the UE shall not set the IM CN Subsystem Signalling Flag in the Protocol Configuration Options information element of any subsequent BEARER RESOURCE MODIFICATION REQUEST message for that APN. The UE shall ignore the IM CN Subsystem Signalling Flag if received from the network in the Protocol Configuration Options information element.

After the establishment of a EPS bearer context used for SIP signalling, the UE shall not indicate the request for a P-CSCF address to the network within the Protocol Configuration Options information element of any subsequent BEARER RESOURCE MODIFICATION REQUEST message for that APN. The UE shall ignore P-CSCF address(es) if received from the network in the Protocol Configuration Options information element.

L.2.2.1B Re-establishment of the EPS bearer context for SIP signalling

If the UE registered a public user identity with an IP address allocated for the APN of the EPS bearer context used for SIP signalling, the EPS bearer context used for SIP signalling is deactivated as result of signalling from the network and:

i) if the UE is required to perform an initial registration according to subclause L.3.1.2;

ii) if the signalling from the network results in requiring the UE to initiate activation of the PDN connection of the EPS bearer context used for SIP signalling; or

iii) if the UE needs to continue having a public user identity registered with an IP address allocated for the APN;

the UE shall:

A) if the non-access stratum is performing the UE requested PDN connectivity procedure and the EPS bearer context activation procedure(s) for the APN triggered as result of the signalling from the network, wait until the UE requested PDN connectivity procedure and the EPS bearer context activation procedure(s) for the APN finish; and

B) perform the procedures in subclause L.2.2.1, bullets a), b) and c).

If none of the bullets i), ii) and iii) of this subclause evaluate to true, or the procedures in bullet B) of this subclause were unable to ensure that the EPS bearer context used for SIP signalling is available or were unable to acquire any P-CSCF address(es):

1) if the SIP signalling was carried over a dedicated EPS bearer context, the UE shall release all resources established as a result of SIP signalling by sending to the network either:

   a) a BEARER RESOURCE MODIFICATION REQUEST message, if there are EPS bearer contexts to this PDN that are not related SIP sessions; or

   b) a PDN DISCONNECT REQUEST message if all the EPS bearer contexts to this PDN are related to SIP sessions.

NOTE: If the SIP signalling was carried over the default EPS bearer context, all the resources established as a result of SIP signalling are released without any explicit NAS signalling.

If the default EPS bearer context of the PDN connection of the EPS bearer context used for SIP signalling was deactivated at the start of this subclause, and the procedures in bullet B) of this subclause ensured that the EPS bearer context used for SIP signalling is available and acquired the P-CSCF address(es), the UE shall perform a new initial registration according to subclause 5.1.1.2.

L.2.2.1C P-CSCF restoration procedure

A UE supporting the P-CSCF restoration procedure performs one of the following procedures:

A if the UE used method II for P-CSCF discovery and if the UE receives one or more P-CSCF address(es) in the Protocol Configuration Options information element of a Modify EPS Bearer Context Request message the one or more P-CSCF addresse(s) do not include the address of the currently used P-CSCF, then the UE shall acquire a different P-CSCF address from the one or more P-CSCF addresse(s) in the Modify EPS Bearer Context Request message. If more than one P-CSCF address with the same container identifier (i.e. "P-CSCF IPv6 Address" or "P-CSCF IPv4 Address") are included, then the UE shall assume that the more than one P-CSCF addresses with the same container identifier are prioritised with the first P-CSCF address with the same container identifier within the Protocol Configuration Options information element as the P-CSCF address with the highest priority.

   If the UE used method II for P-CSCF discovery and if the UE has previously sent the "P-CSCF Re-selection support" PCO indicator at PDN creation and if the UE receives one or more P-CSCF address(es) in the Protocol

Configuration Options information element of a Modify EPS Bearer Context Request message, then the UE shall acquire a P-CSCF address from the one or more P-CSCF addresse(s) in the Modify EPS Bearer Context Request message. If more than one P-CSCF address with the same container identifier (i.e. "P-CSCF IPv6 Address" or "P-CSCF IPv4 Address") are included, then the UE shall assume that the more than one P-CSCF addresses with the same container identifier are prioritised with the first P-CSCF address with the same container identifier within the Protocol Configuration Options information element as the P-CSCF address with the highest priority;

B if the UE uses RFC 6223 [143] as part of P-CSCF restoration procedures, and if the P-CSCF fails to respond to a keep-alive request, then the UE shall acquire a different P-CSCF address using one of the methods I, III and IV for P-CSCF discovery described in the subclause L.2.2.1.

If the UE has an ongoing session and acquired the new P-CSCF address by using procedure A described above, the UE may wait until the UE has detected that the ongoing session has ended before performing an initial registration as specified in subclause 5.1.

In all other cases, when the UE has acquired the P-CSCF address, the UE not having an ongoing session shall perform an initial registration as specified in subclause 5.1.

NOTE 1: For UEs using procedure A described above, the network ensures that P-CSCF address(es) in the Protocol Configuration Options information element of a Modify EPS Bearer Context Request message is sent only during P-CSCF restoration procedures as defined in subclause 5 of 3GPP TS 23.380 [7D].

NOTE 2: The P-CSCF can be completely unreachable, so it is up to UE implementation to detect the end of an ongoing session, e.g. using media plane inactivity detection. Services depending on signalling such as CW and MT calls will not work during this time.

L.2.2.2 Session management procedures

The existing procedures for session management as described in 3GPP TS 24.301 [8J] shall apply while the UE is connected to the IM CN subsystem.

L.2.2.3 Mobility management procedures

The existing procedures for mobility management as described in 3GPP TS 24.301 [8J] shall apply while the UE is connected to the IM CN subsystem.

L.2.2.4 Cell selection and lack of coverage

The existing mechanisms and criteria for cell selection as described in 3GPP TS 36.304 [19B] shall apply while the UE is connected to the IM CN subsystem.

L.2.2.5 EPS bearer contexts for media

L.2.2.5.1 General requirements

NOTE 1: In EPS, the UE cannot control whether media streams belonging to different SIP sessions are established on the same EPS bearer context or not. During establishment of a session, the UE establishes data streams(s) for media related to the session. Such data stream(s) can result in activation of additional EPS bearer context(s). Either the UE or the network can request for resource allocations for media, but the establishment and modification of the EPS bearer is controlled by the network as described in 3GPP TS 24.301 [8J].

NOTE 2: When the UE wishes to allocate bandwidth for RTP and RTCP, the UE uses the rules as those outlined in 3GPP TS 29.213 [13C].

If the resource allocation is initiated by the UE, the UE starts reserving resources whenever it has sufficient information about the media streams, and used codecs available as specified in 3GPP TS 24.301 [8J].

NOTE 3: If the resource reservation requests are initiated by the EPS IP CAN, then the bearer establishment is initiated by the network after the P-CSCF has authorised the respective IP flows and provided the QoS requirements over the Rx interface to the PCRF as described in 3GPP TS 29.214 [13D].

L.2.2.5.1A Activation or modification of EPS bearer contexts for media by the UE

If the UE is configured not to initiate resource allocation for media according to 3GPP TS 24.167 [8G], then the UE shall refrain from requesting additional EPS bearer context(s) for media until the UE considers that the network did not initiate resource allocation for the media.

L.2.2.5.1B Activation or modification of EPS bearer contexts for media by the network

If the UE receives an activation request from the network for a EPS bearer context which is associated with the EPS bearer context used for signalling, the UE shall, based on the information contained in the Traffic Flow Template information element, correlate the media EPS bearer context with a currently ongoing SIP session establishment or SIP session modification.

If the UE receives a modification request from the network for a EPS bearer context that is used for one or more media streams in an ongoing SIP session, the UE shall:

1) modify the related EPS bearer context in accordance with the request received from the network.

L.2.2.5.1C Deactivation of EPS bearer context for media

When a data stream for media related to a session is released, if the EPS bearer resource transporting the data stream is no longer needed and allocation of the EPS bearer resource was requested by the UE, then the UE releases the EPS bearer resource.

NOTE: The EPS bearer resource can be needed e.g. for other data streams of a session or for other applications in the UE.

L.2.2.5.1D Default EPS bearer context usage restriction policy

The default EPS bearer context usage restriction policy consists of zero or more default EPS bearer context usage restriction policy parts.

The default EPS bearer context usage restriction policy part consists of a mandatory media type condition and an optional ICSI condition.

The default EPS bearer context usage restriction policy does not apply to UE detected emergency calls.

Sending media is restricted according to the default EPS bearer context usage restriction policy, if sending media is restricted according to at least one default EPS bearer context usage restriction policy part of the default EPS bearer context usage restriction policy.

Sending media is restricted according to the default EPS bearer context usage restriction policy part if:

1) the media is to be sent for a media stream negotiated in a session offered or established by SIP signalling;

2) the media stream is of a media type indicated in the media type condition of the EPS bearer context usage restriction policy part;

3) the following is true:

   a) the default EPS bearer context usage restriction policy part does not have the ICSI condition; or

   b) the session is offered or established by SIP signalling related to an IMS communication service identified in the ICSI condition of the default EPS bearer context usage restriction policy part; and

4) the media is to be sent via the default EPS bearer context of the PDN connection for SIP signalling.

The UE may support the default EPS bearer context usage restriction policy.

If the UE supports the default EPS bearer context usage restriction policy:

1) the UE shall not send media restricted according to the default EPS bearer context usage restriction policy; and

2) the UE may support being configured with the default EPS bearer context usage restriction policy using one or more of the following methods:

   a) the Default_EPS_bearer_context_usage_restriction_policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C];

   b) the Default_EPS_bearer_context_usage_restriction_policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

   c) Default_EPS_bearer_context_usage_restriction_policy node of 3GPP TS 24.167 [8G].

   If the UE is configured with both the the Default_EPS_bearer_context_usage_restriction_policy node of Default_EPS_bearer_context_usage_restriction_policy node of 3GPP TS 24.167 [8G] and the Ethe Default_EPS_bearer_context_usage_restriction_policy node of $F_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or 3GPP TS 31.103 [15B], then the $EF_{IMSConfigData}$ file shall take precedence.

NOTE: Precedence for files configured on both the USIM and ISIM is defined in 3GPP TS 31.103 [15B].

L.2.2.5.2 Special requirements applying to forked responses

NOTE 1: The procedures in this subclause only apply when the UE requests activation and modification of media bearers. In the case where the network activates and modifies the media bearers the network takes care of the handling of media bearers in the case of forking.

Since the UE does not know that forking has occurred until a second, provisional response arrives, the UE requests resource allocation as required by the initial response received. If a subsequent provisional response is received, different alternative actions may be performed depending on the requirements in the SDP answer:

1) the bearer requirements of the subsequent SDP can be accommodated by the existing resources requested. The UE performs no further resource requests.

2) the subsequent SDP introduces different QoS requirements or additional IP flows. The UE requests further resource allocation according to subclause L.2.2.5.1.

3) the subsequent SDP introduces one or more additional IP flows. The UE requests further resource allocation according to subclause L.2.2.5.1.

NOTE 2: When several forked responses are received, the resources requested by the UE are the "logical OR" of the resources indicated in the multiple responses to avoid allocation of unnecessary resources. The UE does not request more resources than proposed in the original INVITE request.

When a final answer is received for one of the early dialogs, the UE proceeds to set up the SIP session. The UE shall release all the unneeded IP-CAN resources. Therefore, upon the reception of the first final 200 (OK) response for the INVITE request (in addition to the procedures defined in RFC 3261 [26] subclause 13.2.2.4), the UE shall:

1) in case resources were established or modified as a consequence of the INVITE request and forked provisional responses that are not related to the accepted 200 (OK) response, send release request to release the unneeded resources.

L.2.2.5.3 Unsuccessful situations

One of the Rx and Gx interface related error codes can be received by the UE in either the PDN CONNECTIVITY REJECT message, the BEARER RESOURCE MODIFICATION REJECT message, or the BEARER RESOURCE ALLOCATION REJECT message. If the UE receives a Rx and Gx interface related error code, the UE shall either handle the resource reservation failure as described in subclause 6.1.1 or retransmit the message up to three times. The Rx and Gx interface related error codes are further specified in 3GPP TS 29.214 [13D] and 3GPP TS 29.212 [13B].

L.2.2.6 Emergency service

L.2.2.6.1 General

Emergency bearers are defined for use in emergency calls in EPS and core network support of these bearers is indicated to the UE in NAS signalling. Where the UE recognises that a call request is an emergency call and the core network

supports emergency bearers, the UE shall use these EPS bearer contexts for both signalling and media for emergency calls made using the IM CN subsystem.

Some jurisdictions allow emergency calls to be made when the UE does not contain an ISIM or USIM, or where the credentials are not accepted. Additionally, where the UE is in state EMM-REGISTERED.LIMITED-SERVICE and EMM-REGISTERED.PLMN-SEARCH, a normal ATTACH has been attempted but it can also be assumed that a registration in the IM CN subsystem will also fail. In such cases, subject to the lower layers indicating that the network does support emergency bearer services in limited service state (see 3GPP TS 36.331 [19F]), the procedures for emergency calls without registration can be applied, as defined in subclause 5.1.6.8.2. If the EPS authentication procedure has already succeeded during the latest normal or emergency ATTACH procedure, the UE shall perform an initial emergency registration, as described in subclause 5.1.6.2 before attempting an emergency call as described in subclause 5.1.6.8.3.

NOTE 1: The UE can determine that EPS authentication procedure has succeeded during the emergency ATTACH procedure when a non-null integrity protection algorithm (i.e. other than EIA0 algorithm) is received in the NAS signalling SECURITY MODE COMMAND message.

When activating an EPS bearer context to perform emergency registration, the UE shall request a PDN connection for emergency bearer services as described in 3GPP TS 24.301 [8J]. The procedures for EPS bearer context activation and P-CSCF discovery, as described in subclause L.2.2.1 of this specification apply accordingly.

In order to find out whether the UE is attached to the home PLMN or to the visited PLMN, the UE shall compare the MCC and MNC values derived from its IMSI with the MCC and MNC of the PLMN the UE is attached to. If the MCC and MNC of the PLMN the UE is attached to do not match with the MCC and MNC derived from the IMSI, then for the purpose of emergency calls in the IM CN subsystem the UE shall consider to be attached to a VPLMN.

NOTE 2: In this respect an equivalent HPLMN, as defined in 3GPP TS 23.122 [4C] will be considered as a visited network.

NOTE 3: The UE verifies if a detected emergency number is still present in the Extended Local Emergency Number List after attach to a different PLMN. It is possible for the number to no longer be present in the Extended Local Emergency Number List if:

- the PLMN attached to relies on the Local Emergency Number List for deriving a URN; or
- the previously received Extended Emergency Number List Validity field indicated "Extended Local Emergency Numbers List is valid only in the PLMN from which this IE is received".

If the UE detected an emergency number, the UE subsequently performs an attach procedure or an emergency attach procedure with a different PLMN than the PLMN from which the UE received the last Extended Local Emergency Number List, the dialled number is not stored in the ME, in the USIM and in the Local Emergency Number List; and

- the ATTACH ACCEPT message received from the different PLMN contains the Extended Local Emergency Number List and the emergency number is present in the updated Extended Local Emergency Number List then the UE uses the updated Extended Local Emergency Number List when it performs the procedures in subclause L.2.2.6.1B; and
- the ATTACH ACCEPT message received from the different PLMN contains no Extended Local Emergency Number List or the emergency number is no longer present in the updated Extended Local Emergency Number List then the UE shall attempt UE procedures for SIP that relate to emergency using emergency service URN "urn:service:sos".

If the dialled number is equal to a local emergency number stored in the Extended Local Emergency Number List (as defined in 3GPP TS 24.301 [8J]), then the UE shall recognize such a number as for an emergency call and:

- if the dialled number is equal to an emergency number stored in the ME, or in the USIM, then the UE shall perform either procedures in the subclause L.2.2.6.1B or the procedures in subclause L.2.2.6.1A; and
- if the dialled number in not equal to an emergency number stored in the ME, or in the USIM, then the UE shall perform procedures in the subclause L.2.2.6.1B.

If the dialled number is not equal to a local emergency number stored in the Extended Local Emergency Number List (as defined in 3GPP TS 24.301 [8J]) and:

- if the dialled number is equal to an emergency number stored in the ME, in the USIM or in the Local Emergency Number List (as defined in 3GPP TS 24.008 [8]), then the UE shall recognize such a number as for an emergency call and performs the procedures in subclause L.2.2.6.1A.

Upon reception of a 380 (Alternative Service) response to an INVITE request as defined in subclause 5.1.2A.1.1 and subclause 5.1.3.1, if:

- the 380 (Alternate Service) response contains a Contact header field;
- the value of the Contact header field is a service URN; and
- the service URN has a top-level service type of "sos";

then the UE determines that "emergency service information is included" as described 3GPP TS 23.167 [4B].

Upon reception of a 380 (Alternative Service) response to an INVITE request as defined in subclause 5.1.3.1 if the 380 (Alternate Service) response does not contain a Contact header field with service URN that has a top-level service type of "sos", then the UE determines that "no emergency service information is included" as described 3GPP TS 23.167 [4B].

If the "emergency service information is included" as described 3GPP TS 23.167 [4B]:

1) if the URN in the Contact header field matches an emergency service URN in table L.2.2.6.1, then the type of emergency service is the value corresponding to the matching entry in table L.2.2.6.1; and
2) if the URN in the Contact header field does not match any emergency service URN in table L.2.2.6.1, then the type of emergency service is not identified.

NOTE 4: In bullet 2), the URN in the Contact header field either contains "no emergency subservice type" as described in 3GPP TS 23.167 [4B] triggering an emergency call, or contains an "emergency subservice type that does not map into an emergency service category for the CS domain" as described in 3GPP TS 23.167 [4B] triggering a normal call when the dialled number is available or triggering an emergency call when the dialled number is not available. The country specific URN is an example of a "emergency subservice type that does not map into an emergency service category for the CS domain".

When the emergency registration expires, the UE should disconnect the PDN connection for emergency bearer services as described in 3GPP TS 24.301 [8J].

Upon receiving a 3xx other than 380 (Alternative service), 4xx, 5xx or 6xx response to an INVITE request for a UE detectable emergency call, the UE shall perform domain selection as specified in 3GPP TS 23.167 [4B] annex H, to re-attempt the emergency call.

L.2.2.6.1A Type of emergency service derived from emergency service category value

The type of emergency service for an emergency number is derived from the settings of the emergency service category value (bits 1 to 5 of the emergency service category value as specified in subclause 10.5.4.33 of 3GPP TS 24.008 [8]). Table L.2.2.6.1 below specifies mappings between a type of emergency service and an emergency service URN. The UE shall use the mapping to match an emergency service URN and a type of emergency service. If a dialled number is an emergency number but does not map to a type of emergency service the service URN shall be "urn:service:sos".

Table L.2.2.6.1: Mapping between type of emergency service and emergency service URN

Type of emergency service	Emergency service URN
Police	urn:service:sos.police
Ambulance	urn:service:sos.ambulance
Fire Brigade	urn:service:sos.fire
Marine Guard	urn:service:sos.marine
Mountain Rescue	urn:service:sos.mountain

NOTE 1: It is not possible for a UE to indicate more than one type of emergency service in an emergency service URN.

If an IP-CAN, capable of providing local emergency numbers, did not provide a local emergency number that matches the dialled number (see subclause 5.1.6.1) and multiple types of emergency service can be derived for a dialled number from the information configured on the USIM then:

- if the UE is in the HPLMN, the UE shall map any one of these types of emergency service to an emergency service URN as specified in table L.2.2.6.1; and

- if the UE is in the VPLMN, the UE shall select "urn:service:sos".

NOTE 2: If the Non-3GPP emergency number indicator within the Non-3GPP NW provided policies IE (see 3GPP TS 24.008 [8]) provided through registration procedures over 3GPP access is set to "use of non-3GPP emergency numbers permitted", the UE also considers WLAN provided local emergency numbers (see 3GPP TS 24.302 [8U], subclause 4.7). If the Non-3GPP NW provided policies IE provided through registration procedures over 3GPP access is set to "use of non-3GPP emergency numbers not permitted", the UE does not consider WLAN provided local emergency numbers. If the Non-3GPP NW provided policies IE is not provided through registration procedures over 3GPP access, the UE does not consider WLAN provided local emergency numbers.

If an IP-CAN, capable of providing local emergency numbers, provided a local emergency number that matches the dialled number (see subclause 5.1.6.1), and:

- if the UE can derive one or more types of emergency service from the information received from the IP-CAN for the dialled number and the UE cannot derive types of emergency service from the information configured on the USIM for the dialled number; or

- if the UE is able to derive identical types of emergency service from both the information received from the IP-CAN for the dialled number and from the information configured on the USIM for the dialled number,

then the UE shall map any one of these emergency service types to an emergency service URN as specified in table L.2.2.6.1.

NOTE 3: How the UE resolves clashes where an emergency number is associated with one or more different types of emergency service configured in the USIM and in information received from the access network, is implementation dependent.

L.2.2.6.1B Type of emergency service derived from extended local emergency number list

The Extended Local Emergency Number List (defined in 3GPP TS 24.301 [8J]) can contain sub-services of the associated emergency service URN for the detected emergency number.

If:

- the length of sub-services field is greater than "0", the UE shall construct the emergency service URN using "urn:service:sos" followed by adding a dot followed by the content of the sub-services field; and

- the length of sub-services field is "0", the UE shall use the emergency service URN "urn:service:sos".

EXAMPLE 1: For a detected number, if the sub-service is "gas", then the UE constructs "urn:service:sos.gas" as the associated emergency service URN.

EXAMPLE 2: For a detected number, if no sub-service is provided, then the UE uses "urn:service:sos" as the associated emergency service URN.

L.2.2.6.2 eCall type of emergency service

If the IP-CAN indicates the eCall support indication or the CS domain is not available to the UE, the UE can send an INVITE request with Request-URI set to "urn:service:sos.ecall.manual" or "urn:service:sos.ecall.automatic".

If the IP-CAN does not indicates the eCall support indication and the CS domain is available to the UE, the UE shall not send an INVITE request with Request-URI set to "urn:service:sos.ecall.manual" or "urn:service:sos.ecall.automatic".

L.2.2.6.3 Current location discovery during an emergency call

Void.

L.2A Usage of SDP

L.2A.0 General

NOTE 1: When:

- establishing a session which is not an emergency session; or
- modifying a session which is not an emergency session;

and if the IMSVoPS indicator is received in the EPS network feature support information element (see 3GPP TS 24.301 [8J]) and no persistent EPS bearer context exists at the UE, the UE constructs SDP based on the restrictions indicated in the IMSVoPS indicator. Regardless whether the IMSVoPS indicator indicating voice is supported or not, m-lines can be set to "audio" and exclude voice codecs from the SDP answer or SDP offer. When a persistent EPS bearer context exists, m-lines can be set to "audio" and include voice codecs in the SDP answer or SDP offer.

NOTE 2: When the UE is accessing the IM CN subsystem via E-UTRAN, the appropriate specification for access domain selection is 3GPP TS 23.221 [6].

L.2A.1 Impact on SDP offer / answer of activation or modification of EPS bearer context for media by the network

If, due to the activation of EPS bearer context from the network the related SDP media description needs to be changed, the UE shall update the related SDP information by sending, within a SIP request, a new SDP offer for each of the existing SIP dialogs,

If the UE receives a modification request from the network for a EPS bearer context that is used for one or more media streams in an ongoing SIP session, the UE shall:

1) if, due to the modification of the EPS bearer context, the related SDP media description need to be changed, update the related SDP information by sending, within a SIP request, a new SDP offer for each of the existing SIP dialogs, and respond to the EPS bearer context modification request.

NOTE: The UE can decide to indicate additional media streams as well as additional or different codecs in the SDP offer than those used in the already ongoing session.

L.2A.2 Handling of SDP at the terminating UE when originating UE has resources available and IP-CAN performs network-initiated resource reservation for terminating UE

If the UE receives an SDP offer where the SDP offer includes all media streams for which the originating side indicated its local preconditions as met, if the precondition mechanism is supported by the terminating UE and the IP-CAN performs network-initiated resource reservation for the terminating UE and the available resources are not sufficient for the received offer, the terminating UE shall indicate its local preconditions and provide the SDP answer to the originating side without waiting for resource reservation.

NOTE: If the resource reservation is controlled by the EPS IP-CAN, the resource reservation request is initiated by the network after the P-CSCF has authorised the respective IP flows and provided the QoS requirements over the Rx interface to the PCRF as described in 3GPP TS 29.214 [13D].

L.2A.3 Emergency service

NOTE: When establishing an emergency session or when modifying an emergency session, the IMSVoPS indicator does not influence handling of SDP offer and SDP answer.

L.3 Application usage of SIP

L.3.1 Procedures at the UE

L.3.1.0 Registration and authentication

The UE shall perform reregistration of a previously registered public user identity bound to any one of its contact addresses when changing to an IP-CAN for which usage is specified in annex R or annex W. The reregistration is performed using the new IP-CAN.

NOTE 1: This document does not specify how the UE detects that the used IP-CAN has changed. The information that is forcing the reregistration is also used to generate the content for the P-Access-Network-Info header field.

NOTE 2: The UE will send the reregistration irrespective of whether it has a SIP dialog or not.

If the UE supports the 3GPP PS data off, then the UE shall in all REGISTER requests include the "+g.3gpp.ps-data-off" header field parameter defined in subclause 7.9.8 set to a value indicating the 3GPP PS data off status.

When the UE sends a REGISTER request, if the 3GPP PS data off status is "active", then the UE shall only include media feature tags associated with services that are 3GPP PS data off exempt services in the g.3gpp.icsi-ref media feature tag, as defined in subclause 7.9.2 and RFC 3840 [62], for the IMS communication services it intends to use.

If the UE is registered, and the 3GPP PS data off status is changed or the UE is provided by the network with a new list of 3GPP PS data off exempt services while the 3GPP PS data off status is "active", then the UE shall perform a reregistration of the previously registered public user identity.

A UE supporting ANBR as specified in 3GPP TS 26.114 [9B] shall also support RAN-assisted codec adaptation as specified in 3GPP TS 36.300 [268] and 3GPP TS 36.321 [269].

If the UE supports ANBR, upon receiving a 200 (OK) response to the REGISTER request and if the 200 (OK) response contains a Feature-Caps header field with the g.3gpp.anbr feature-capability indicator the UE shall assume that the network supports RAN-assisted codec adaptation as specified in 3GPP TS 36.300 [268] and 3GPP TS 36.321 [269]. The UE is allowed to include the SDP ‘anbr’ attribute during session establishment as specified in 3GPP TS 26.114 [9B].

L.3.1.0a IMS_Registration_handling policy

The IMS_Registration_handling policy indicates whether the UE deregisters from IMS after a configured amount of time after receiving an indication that the IMS Voice over PS Session is not supported.

The UE may support the IMS_Registration_handling policy.

If the UE supports the IMS_Registration_handling policy, the UE may support being configured with the IMS_Registration_handling policy using one or more of the following methods:

a) the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C];

b) the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

c) the IMS_Registration_Policy node of 3GPP TS 24.167 [8G].

If the UE is configured with both the IMS_Registration_Policy node of 3GPP TS 24.167 [8G] and the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or 3GPP TS 31.103 [15B], then the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file shall take precedence.

NOTE 1: Precedence for files configured on both the USIM and ISIM is defined in 3GPP TS 31.103 [15B].

If the UE is registered with IMS and the IMSVoPS indicator, provided by the lower layers (see 3GPP TS 24.301 [8J]), indicates voice is not supported, the UE shall:

A) if the Stay_Registered_When_VoPS_Not_Supported leaf indicates requirement to stay registered, the UE needs not to deregister and maintains the registration as required for IMS services; or

NOTE 2: The UE will periodically refresh the registration when needed.

B) if the Stay_Registered_When_VoPS_Not_Supported leaf indicates requirement to deregister and the Deregistration_Timer leaf used to configure the NoVoPS-dereg timer defined in table 7.8.1 contains a timer value for the time to wait before deregistrerting from IMS, start a timer with the value indicated in the policy and:

   a) if the timer expires before the UE receives an indication from the lower layers that IMS voice is supported:

      1) if there is no ongoing IMS session, the UE either performs reregistration as specified in subclause 5.1.1.4 and shall only include feature tags associated with services that are independent of IMSVoPS indicator or deregister from the IMS following the procedures specified in subclause 5.1.1.6; or

      2) if there is ongoing IMS session, and

         i) if the UE does not receive indication from the lower layer that the IMS voice is supported before the ongoing IMS session is terminated, the UE either performs reregistration as specified in subclause 5.1.1.4 and shall only include feature tags associated with services that are independent of IMSVoPS indicator or deregister from the IMS following the procedures specified in subclause 5.1.1.6 as soon as the ongoing IMS based service is terminated; or

         ii) if the UE receives indication from the lower layer that the IMS voice is supported before the ongoing IMS session is terminated, cancel the timer; or

NOTE 3: How the UE selects reregistration or deregistration is implementation dependent (e.g., SMS service)

   b) if the UE receives an indication from the lower layers that IMS voice is supported before the timer expires, cancel the timer.

If the IMS_Registration_handling policy is not configured, the UE behaviour is implementation specific.

L.3.1.1 P-Access-Network-Info header field

The UE shall always include the P-Access-Network-Info header field where indicated in subclause 5.1.

NOTE: If the P-Access-Network-Info header field includes radio cell identity, the P-Access-Network-Info header field populated by the UE that supports Multi-RAT Dual Connectivity with the EPC as described in 3GPP TS 37.340 [264] will contain the information about the radio cell identity of the Master RAN node that is serving the UE.

L.3.1.1A Cellular-Network-Info header field

Not applicable.

L.3.1.2 Availability for calls

This subclause documents the minimal requirements for being available for voice communication services when using EPS.

A UE shall perform an initial registration as specified in subclause 5.1.1.2 using an EPS bearer context for SIP signalling (see annex L.2.2.1), if all the following conditions are met:

1) if the UE is operating in one of the following modes of operation (see 3GPP TS 24.301 [8J]):

   a) PS mode 1;

b) CS/PS mode 1 and the UE is attached for EPS-Services only;

2) if the UE is capable of receiving any (but not necessarily all) of the media types which the CS domain supports, such that the media type can also be used when accessing the IM CN subsystem using the current IP-CAN;

3) if:

a) the media type of item 2 is an "audio" media type

b) the UE supports codecs suitable for (conversational) speech; and

c) the "audio" media type is not restricted from inclusion in an SDP message according to the media type restriction policy as specified in subclause 6.1.1

and one of the following is true:

a) 3GPP PS data off status is "inactive";

b) 3GPP PS data off status is "active", the UE is in the HPLMN or the EHPLMN, and MMTEL voice is a 3GPP PS data off exempt service; or

c) 3GPP PS data off status is "active", the UE is in the VPLMN, the UE is configured with an indication that MMTEL voice is a 3GPP PS data off exempt service in a VPLMN, and MMTEL voice is a 3GPP PS data off roaming exempt service.

4) if the UE determines that its contact has not been bound to a public user identity using the IP-CAN, such that the contact is expected to be used for the delivery of incoming requests in the IM CN subsystem relating to the media of item 2 and item 3;

5) if the IMSVoPS indicator, provided by the lower layers (see 3GPP TS 24.301 [8J]), indicates voice is supported;

6) if the procedures to perform the initial registration are enabled (see 3GPP TS 24.305 [8T]); and

7) if the EPS bearer context used for SIP signalling is:

a) available; or

b) not available, and the UE:

i. is allowed to send a PDN CONNECTIVITY REQUEST message to establish an EPS bearer context that is needed for performing the initial registration; or

ii. is allowed to send a BEARER RESOURCE ALLOCATION REQUEST message, wishes to establish an EPS bearer with the correct QCI and TFT for performing the initial registration, and a default EPS bearer context for the APN exists.

NOTE 1: 3GPP TS 24.301 [8J] specifies conditions that prevent the UE from sending a PDN CONNECTIVITY REQUEST message or a BEARER RESOURCE ALLOCATION REQUEST message.

NOTE 2: Regardless of any of the above conditions, a UE might attempt to register with the IM CN subsystem at any time.

EXAMPLE: As an example of the note, a UE configured to preferably attempt to use the EPS to access IM CN subsystem can perform an initial registration as specified in subclause 5.1.1.2, if the conditions in items 2, 3, 4, 5, 6 and 7 in this subclause, evaluate to true.

The UE indicates to the non-access stratum the status of being available for voice over PS when:

I) the UE is capable of receiving any (but not necessarily all) of the media types which the CS domain supports, such that the media type can also be used when accessing the IM CN subsystem using the current IP-CAN;II) if the media type of item I is an "audio" media type, the UE supports codecs suitable for (conversational) speech, the "audio" media type is not restricted from inclusion in an SDP message according to the media type restriction policy as specified in subclause 6.1.1, and:

a) 3GPP PS data off status is "inactive";

b) 3GPP PS data off status is "active", the UE is in the HPLMN or the EHPLMN, and MMTEL voice is a 3GPP PS data off exempt service; or

c) 3GPP PS data off status is "active", the UE is in the VPLMN, the UE is configured with an indication that MMTEL voice is a 3GPP PS data off exempt service in a VPLMN, and MMTEL voice is a 3GPP PS data off roaming exempt service; and

III) the UE determines a contact has been bound to a public user identity using the IP-CAN, such that this contact is expected to be used for the delivery of incoming requests in the IM CN subsystem relating to such media.

The UE indicates to the non-access stratum the status of being not available for voice over PS when:

I) in response to receiving the IMSVoPS indicator indicating voice is supported, the UE:

- initiated an initial registration as specified in subclause 5.1.1.2, received a final response to the REGISTER request sent, but the conditions for indicating the status of being available for voice over PS are not met; or

- did not initiate an initial registration as specified in subclause 5.1.1.2 and, these conditions for indicating the status of being available for voice over PS are not met; or

II) the conditions for indicating the status of being available for voice over PS are no longer met.

NOTE 3: The status of being not available for voice over PS is used for domain selection for UE originating sessions / calls specified in 3GPP TS 23.221 [6] subclause 7.2a.

L.3.1.2A Availability for SMS

The UE determines that the UE is able to use SMS using IMS if the UE:

I) is capable of using the MIME type "application/vnd.3gpp.sms" (see 3GPP TS 24.341 [8L]), such that the MIME type can also be used when accessing the IM CN subsystem using the current IP-CAN;

II) supports the role of an SM-over-IP sender (see 3GPP TS 24.341 [8L]);

IIA) determines the EPS bearer context used for SIP signalling exists;

III) determines a contact has been bound to a public user identity using the IP-CAN, such that this contact is expected to be used for the delivery of incoming requests in the IM CN subsystem relating to such media;

IV) the UE does not determine that SMS over IP is restricted in 3GPP TS 24.341 [8L] subclause 5.2.1.3; and

V) the 3GPP PS data off status is:

- "inactive";

- "active", the UE is in the HPLMN or the EHPLMN, and SMS over IMS is a 3GPP PS data off exempt service; or

- "active", the UE is in the VPLMN, the UE is configured with an indication that SMS over IMS is a 3GPP PS data off exempt service in a VPLMN, and SMS over IMS is a 3GPP PS data off roaming exempt service.

When above criteria are not matched, the UE determines that SMS using IMS is unavailable.

NOTE: The status that SMS using IMS is unavailable is used for domain selection for UE originating SMS specified in 3GPP TS 23.221 [6] subclause 7.2c.

L.3.1.3 Authorization header field

Void.

L.3.1.4 SIP handling at the terminating UE when precondition is not supported in the received INVITE request, the terminating UE does not have resources available and IP-CAN performs network-initiated resource reservation for the terminating UE

Upon receiving an INVITE request not including the "precondition" option-tag in the Supported header field and not including the "precondition" option-tag in the Require header field, and the IP-CAN performs network-initiated resource reservation for the UE, the UE:

1) if the INVITE request contains an SDP offer and the local resources required at the terminating UE for the received SDP offer are not available:

   a) shall not alert the user; and

   b) shall send 183 (Session Progress) response to the INVITE request without waiting for resource reservation and without alerting the user. If the INVITE request includes a Supported header field indicating support of reliable provisional responses, the UE shall send the 183 (Session Progress) response reliably. In the 183 (Session Progres) response, the UE shall include an SDP answer; and

2) if the INVITE request does not contain an SDP offer and the INVITE request includes a Supported header field indicating support of reliable provisional responses:

   a) shall generate an SDP offer;

   b) if the local resources required at the terminating UE for the generated SDP offer are not available:

      A) shall not alert the user; and

      B) shall reliably send 183 (Session Progress) response to the INVITE request without waiting for resource reservation and without alerting the user. In the 183 (Session Progres) response, the UE shall include the generated SDP offer.

Upon successful reservation of local resources, if the precondition mechanism is not used by the terminating UE, the UE can send 180 (Ringing) response to the INVITE request and can alert the user.

L.3.1.5 3GPP PS data off

If the 3GPP PS data off status is "active" the UE shall only send initial requests that:

1) are associated with a 3GPP IMS service which enforces 3GPP PS data off;

NOTE: These services are specified in 3GPP TS 22.011 [1C], and enforcement of 3GPP PS data off is described in the respective service specifications.

2) are associated with an emergency service;

3) are associated with access to RLOS; or

4) are associated with 3GPP PS data off exempt services configured in the UE using one or more of the following methods:

   - the non_3GPP_ICSIs_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in the HPLMN or the EHPLMN, or if the UE is in the VPLMN and the non_3GPP_ICSIs_roaming_exempt node specified in 3GPP TS 24.167 [8G] is not configured;

   - the non_3GPP_ICSIs_roaming_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in the VPLMN;

   - the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], if the UE is in the HPLMN or the EHPLMN, or if the UE is in the VPLMN and the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] is not configured; or

   - the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], if the UE is in the VPLMN.

If the UE is configured with both the non_3GPP_ICSIs_exempt node of 3GPP TS 24.167 [8G] and the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], then the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] shall take precedence.

If the UE is configured with both the non_3GPP_ICSIs_roaming_exempt node of 3GPP TS 24.167 [8G] and the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], then the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] shall take precedence.

If the 3GPP PS data off status changes from "inactive" to "active" the UE shall release all dialogs that

1) are not associated with a 3GPP IMS service which enforces 3GPP PS data off;

NOTE: These services are specified in 3GPP TS 22.011 [1C], and enforcement of 3GPP PS data off is described in the respective service specifications.

2) are not associated with an emergency service; and

3) are not associated with 3GPP data off exempt services configured in the UE using one or more of the following methods:

- the non_3GPP_ICSIs_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in the HPLMN or the EHPLMN, or if the UE is in the VPLMN and the non_3GPP_ICSIs_roaming_exempt node specified in 3GPP TS 24.167 [8G] is not configured;

- the non_3GPP_ICSIs_roaming_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in the VPLMN;

- the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], if the UE is in the HPLMN or the EHPLMN, or if the UE is in the VPLMN and the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] is not configured; or

- the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], if the UE is in the VPLMN.

If the UE is configured with both the non_3GPP_ICSIs_exempt node of 3GPP TS 24.167 [8G] and the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], then the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] shall take precedence.

If the UE is configured with both the non_3GPP_ICSIs_roaming_exempt node of 3GPP TS 24.167 [8G] and the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], then the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] shall take precedence.

L.3.1.6 Transport mechanisms

No additional requirements are defined.

L.3.1.7 RLOS

L.3.1.7.1 General

The support for RLOS as described in this subclause is optional for the UE.

L.3.1.7.2 Registration

A UE containing a UICC on sending an unprotected REGISTER request for RLOS, shall perform the actions as specified in subclause 5.1.1.2 wih the following additions:

a) the UE shall in the REGISTER request include a "+g.3gpp.rlos" Contact header field parameter, as defined in subclause 7.9.9.

NOTE: The UE can choose to use initial registration using IMS AKA or initial registration using GPRS-IMS bundled authentication.

A UE not containing a UICC on sending an uprotected REGISTER request for RLOS shall perform the actions as specified in subclause 5.1.1.2 for registration using GPRS-IMS bundled authentication with the following additions:

a) the UE shall generate a home network domain name according to the rules specified in 3GPP TS 23.003 [3] using the PLMN to which the UE is currently attached and set the Request-URI to the SIP URI of the so generated home network domain name;

b) the UE shall include a To header field set to "sip:unavailable@unknown.invalid" (specified in 3GPP TS 23.003 [3]);

c) the UE shall include a From header field set to "sip:unavailable@unknown.invalid" (specified in 3GPP TS 23.003 [3]); and

d) the UE shall in the REGISTER request include a "+g.3gpp.rlos" Contact header field parameter, as defined in subclause 7.9.9

On reception of a 200 (OK) response to the REGISTER request for RLOS, the UE shall perform the actions as specified in subclause 5.1.1.2 and shall locally store an indication that RLOS session setup is possible. The indication is valid for an implementation specific time.

On receiving a 403 (Forbidden) response containing a Response-Source header field with a "fe" header field parameter set to "<urn:3gpp:fe:s-cscf>" for an initial registration that included a "+g.3gpp.rlos" Contact header field parameter in the REGISTER request, then the UE shall locally store an indication that setup of a RLOS session is possible. The indication is stored for an implementation dependent time.

L.3.1.7.3 Session Setup

L.3.1.7.3.1 Void

L.3.1.7.3.2 RLOS session set-up in case of unsuccessful registration

The UE shall establish a RLOS session as described in this sub-clause only after it has initiated a RLOS registration and has received a 403 (Forbidden) response sent from an S-CSCF.

When establishing a RLOS session in case of an unregistered user, the UE is allowed to receive responses to RLOS requests and requests inside an established RLOS session on the unprotected ports. The UE shall reject or silently discard all other messages. Additionally, the UE shall transmit signalling packets pertaining to the RLOS session from the same IP address and unprotected port on which it expects to receive signalling packets containing the responses to RLOS requests and the requests inside the established RLOS session.

When establishing a RLOS session for an unregistered user, the UE shall use the local IP address and P-CSCF address as used when sending the RLOS registration. The UE shall send only the initial INVITE requests to the port advertised to the UE during the P-CSCF discovery procedure. If the UE does not receive any specific port information during the P-CSCF discovery procedure, the UE shall send the initial INVITE request to the SIP default port values as specified in RFC 3261 [26].

The UE shall apply the procedures as specified in subclause 5.1.2A.1 and subclause 5.1.3 with the following additions:

1) the UE shall set the From header field of the INVITE request to the public user identity as used in the RLOS registration;

2) the UE shall:

   a) if a dial string for a specific RLOS service is available in the UE, use the dial string to build the Request-URI as specified in subclause 5.1.2A.1.3; and

   b) if no dial string for a specific RLOS service is available in the UE use the dummy MSISDN value as defined in 3GPP TS 23.003 [3] to build the Request-URI as specified in subclause 5.1.2A.1.3;

3) the UE shall insert in the INVITE request, a To header field with the same value as in the Request-URI;

4) The UE shall insert a P-Preferred-Service header field according to RFC 6050 [121] set to "urn:urn-7:3gpp-service.ims.icsi.rlos";

5) The UE shall insert an Accept-Contact header field field containing the "+g.3gpp.icsi-ref" header field parameter containing an ICSI value set to "urn:urn-7:3gpp-service.ims.icsi.rlos";

6) the UE shall include in the P-Access-Network-Info header field in any request for a dialog, any subsequent request (except CANCEL requests) or response (except CANCEL responses) within a dialog or any request. Insertion of the P-Access-Network-Info header field into the ACK request is optional. The UE shall populate the P-Access-Network-Info header field with the current point of attachment to the IP-CAN as specified for the access network technology (see subclause 7.2A.4).;

7) The UE shall populate the P-Preferred-Identity header field in the INVITE request with an IMEI based identity in the form of a SIP URI as specified in 3GPP TS 23.003 [3] subclause 13.13;

8) a Contact header field set to include SIP URI that contains in the hostport parameter the IP address of the UE and an unprotected port where the UE will receive incoming requests belonging to this dialog. The UE shall also include a "sip.instance" media feature tag containing Instance ID as described in RFC 5626 [92]. The UE shall also include the "+g.3gpp. icsi-ref" header field parameter containing an ICSI value set to "urn:urn-7:3gpp-service.ims.icsi.rlos" . The UE shall not include either the public or temporary GRUU in the Contact header field;

9) a Via header field set to include the IP address of the UE in the sent-by field and for the UDP the unprotected server port value where the UE will receive responses, while for the TCP, the response is received on the TCP connection on which the RLOS request was sent. For the UDP, the UE shall also include "rport" header field parameter with no value in the top Via header field.;

NOTE 1: The UE inserts the same IP address and port number into the Contact header field and the Via header field, and sends all IP packets to the P-CSCF from this IP address and port number.

10) if the UE has its location information available, or a URI that points to the location information, the UE shall include a Geolocation header field in the INVITE request in the following way:

- if the UE is aware of the URI that points to where the UE's location is stored, include the URI as the Geolocation header field value, as described in RFC 6442 [89]; or

- if the UE is aware of its location information, include the location information in a PIDF location object, in accordance with RFC 4119 [90], include the location object in a message body with the content type application/pidf+xml, and include a Content ID URL, referring to the message body, as the Geolocation header field value, as described RFC 6442 [89], and include a Content-Disposition header field with a disposition type "render" value and a "handling" header field parameter with an "optional" value, as described in RFC 3261 [26];

NOTE 2: If the location information is old or inaccurate, the UE does not consider location information to be available.

11) if the UE includes a Geolocation header field, the UE shall also include a Geolocation-Routing header field with a "yes" header field value, which indicates that the location of the UE can be used by other entities to make routing decisions, as described in RFC 6442 [89]; and

12) if the UE has neither geographical location information available, nor a URI that points to the location information, the UE shall not insert a Geolocation header field in the INVITE request.

The UE shall build a proper preloaded Route header field value for all new dialogs. The UE shall build a Route header field value containing only the P-CSCF URI which was used in the RLOS registration.

When a SIP transaction times out, i.e. timer B, timer F or timer H expires at the UE, the UE may behave as if timer F expired, as described in subclause 5.1.1.4.

If the response for the initial INVITE request indicates that the UE is behind NAT, and the INVITE request was sent over TCP connection, the UE shall keep the TCP connection during the entire duration of the emergency session. In this case the UE will receive all responses to the emergency requests and the requests inside the established emergency session over this TCP connection.

If the Via header field of any provisional response, or of the final 200 (OK) response, for the initial INVITE request contains a "keep" header field parameter with a value, unless the UE detects that it is not behind a NAT, the UE shall start to send keep-alives associated with the session towards the P-CSCF, as described in RFC 6223 [143].

L.3.1.7.3.3 RLOS session set-up in case of successful registration

The UE shall apply the procedures as specified in subclauses 5.1.2A and 5.1.3 with the following additions:

1) the UE shall include in the Request-URI of the initial INVITE request the dummy MSISDN value as defined in 3GPP TS 23.003 [3];

2) the UE shall insert in the INVITE request, a To header field with the same value as in the Request-URI;

3) The UE shall insert a P-Preferred-Service header field according to RFC 6050 [121] set to "urn:urn-7:3gpp-service.ims.icsi.rlos"; and

4) The UE shall insert an Accept-Contact header field field containing the "+g.3gpp.icsi-ref" header field parameter containing an ICSI value set to "urn:urn-7:3gpp-service.ims.icsi.rlos";

L.3.2 Procedures at the P-CSCF

L.3.2.0 Registration and authentication

Void.

L.3.2.1 Determining network to which the originating user is attached

If the P-CSCF is configured to handle emergency requests, in order to determine from which network the request was originated the P-CSCF shall check the MCC and MNC fields received:

- during the registration procedure from the Rx interface as defined in 3GPP°TS°29.214 [13D] (e.g. used for deployments without IMS-level roaming interfaces where the P-CSCF is located in the home network); or

- from the P-Access-Network-Info header field.

NOTE: The above check can be against more than one MNC code stored in the P-CSCF.

L.3.2.2 Location information handling

Void.

L.3.2.3 Prohibited usage of PDN connection for emergency bearer services

If the P-CSCF detects that a UE uses a PDN connection for emergency bearer services for a non-emergency REGISTER request, the P-CSCF shall reject that request by a 403 (Forbidden) response.

NOTE: By assigning specific IP address ranges for a PDN connection for emergency bearer services and configuring those ranges in P-CSCF, the P-CSCF can detect based on the registered Contact address if UE uses an emergency PDN connection for initial registration.

L.3.2.4 Support for paging policy differentiation

The P-CSCF may support paging policy differentiation by marking packet(s) to be sent towards the UE related to that IMS capability. A specific DSCP (IPv4) value and/or a specific Traffic Class (IPv6) value are assigned by local configuration in the P-CSCF.

If local policy requires to provide such marking, the P-CSCF shall identify terminating requests which:

a) contain SDP with an "audio" media line and which are related to a IMS multimedia telephony service session specified in TS 24.173 [8H]; or

b) do not contain an SDP offer but some indication, e.g. a feature capability indicator, indicates that an "audio" media line that would meet network policy for such differentiation, could form part of the subsequent SDP offer.

NOTE 1: Precise details of such indications, if any, are subject to operator policy. Alternatively the operator policy can be to not preferentially page requests without an SDP offer.

For such identified requests:

a) where an unreliable transport mechanism is used as the transport protocol for SIP, the P-CSCF shall mark packets containing an INVITE request; and

b) if a reliable transport mechanism is used as the transport protocol for SIP:

1) if a new reliable transport connection needs to be established, the P-CSCF shall turn on the marking of packets within the reliable transport connection in advance of sending an INVITE request; and

2) if there is an existing reliable transport connection, the P-CSCF may turn on the marking of packets within the reliable transport connection in advance of sending an INVITE request.

In both these cases for a reliable transport connection, the P-CSCF shall turn off the marking of packets within the reliable transport connection at an appropriate time.

L.3.2.5 Void

L.3.2.6 Resource sharing

L.3.2.6.1 Registration

If PCC is supported for this access technology a P-CSCF supporting resource sharing shall insert a Resource-Share header field with the value set to "supported" as described in subclause 7.2.13 and perform the actions in the following subclauses.

L.3.2.6.2 UE-originating case

Upon receiveing an initial INVITE request from a served UE, if the P-CSCF supports resource sharing and local policy requires that resources are reserved already on the SDP offer in the initial INVITE request, the P-CSCF can:

1) for each m-line in the SDP offer, internally generate a set of temporary resource sharing rules where:

a) if the media line is not subject to resouce sharing according to local policies, each resource sharing rule contains a sharing key with a value that is unique and not used by any another media stream in any ongoing session involving the UE;

b) directionality is included according to local policy; and

c) if the media line is subject to resouce sharing according to local policies, each resource sharing rule contains a sharing key with the value as assigned to other ongoing media stream(s) of some ongoing session(s) involving the UE that also use the shared resources; and

2) apply resource-sharing as specified in 3GPP TS 29.214 [13D] using the temporary resource sharing rules.

When the P-CSCF supporting resource receives a 18x response or a 2xx response containing the initial SDP answer, a Resource-Share header field with the value "media-sharing" and the "origin" header field parameter set to "session-initiator", the P-CSCF:

1) shall store resource sharing rules and the value of the corresponding sharing key as described in subclause 7.2.13.5 and use the stored sharing keys to identify resource sharing rules for the media streams in this session; and

2) will apply resource sharing as specified in 3GPP TS 29.214 [13D] using the stored sharing rules.

NOTE 1: If a Resource-Share header field containing the same stored sharing key values and updated resource sharing rules are received in any other session involving the served UE, the updated resource sharing rules overrides the resource sharing rules received in this session.

If P-CSCF supports resource sharing and when the first response to the initial INVITE request (with the exception of the 100 (Trying) response) contains a Resource-Share header field with the value "no-media-sharing" and the "origin" header field parameter set to the value "session-initiator", the P-CSCF shall not share media in this session.

NOTE 2: When resource sharing is not possible, sharing keys identifying the temporary set of resource sharing rules used over the Rx interface if resources were reserved using the SDP offer can still be kept but can never be used in any other session involving this UE as long as this session is ongoing.

If P-CSCF supports resource sharing and when the first response to the initial INVITE request (with the exception of the 100 (Trying) response) containing the initial SDP answer does not contain a Resource-Share header field, the P-CSCF may apply resource sharing using local configuration.

L.3.2.6.3 UE-terminating case

L.3.2.6.3.1 The initial INVITE request contains an initial SDP offer

If P-CSCF supports resource sharing and when P-CSCF receives an initial INVITE request destined to the served user containg the Resource-Share header field parameter with the value "media-sharing" and the "origin" header field parameter set to the value "session-receiver" and if local policy requires resource reservation using the initial SDP offer the P-CSCF shall:

1) store resource sharing rules and corresponding sharing key values as described in subclause 7.2.13.5 and use each stored sharing key to identify resource sharing rules for media streams in this session; and

2) apply resource sharing as specified in 3GPP TS 29.214 [13D] using the stored resource sharing rules.

NOTE: If a Resource-Share header field containing the stored sharing key are received in any other session involving the served UE, the updated resource sharing rules overrides the resource sharing rules received in this session.

Upon receiving a response to the above INVITE request containing an initial SDP answer, if the initial INVITE request containging the initial SDP offer contained the Resource-Share header field set to the value "media-sharing" and the "origin" header field parameter set to "session-receiver" and if local policy requires resource reservation on the initial SDP answer, the P-CSCF shall:

1) store resource sharing rules and corresponding sharing key values as described in subclause 7.2.13.5 and use the stored sharing keys to identify resource sharing rules for media streams in this session; and

2) apply resource sharing as specified in 3GPP TS 29.214 [13D] using the stored resource sharing rules.

If P-CSCF supports resource sharing and when P-CSCF receives an initial INVITE request containg the Resource-Share header field with the value "no-media-sharing" and the "origin" header field parameter set to the value "session-receiver", the P-CSCF shall not share media in this session.

If P-CSCF supports resource sharing and when P-CSCF receives an initial INVITE request not containing a Resource-Share header field, the P-CSCF may apply resource sharing using local configuration.

If the P-CSCF supporting resource sharing receives a request or response from the home network destined to the served user containing a Resource-Share header field with the "origin" header field parameter set to the value "session-initiator" or without the "origin header field parameter, the P-CSCF shall not use the content of the header field and remove the header field from the outgoing response or request.

L.3.2.6.3.2 The 18x response or the 2xx responses contains an initial SDP offer

When the P-CSCF supporting resource sharing receives an 18x response or a 2xx response from a served UE and the response contains an initial SDP offer and local policy requires resource reservation using the initial SDP offer, the P-CSCF shall :

1) for each m-line in the SDP offer, internally generate a set of temporary resource sharing rules where:

- each resource sharing rule contains a sharing key with a value that is unique and not used by any another media stream in any ongoing session involving the UE; and

- directionality is included according to local policy; and

2) apply resource-sharing as specified in 3GPP TS 29.214 [13D] using the temporary resource sharing rules.

Upon receiving a PRACK request or an ACK request containing the initial SDP answer and a Resource-Share header field with the value "media-sharing" and the "origin" header field parameter set to "session-receiver", the P-CSCF shall:

1) store resource sharing rules and the value of the corresponding sharing key as described in subclause 7.2.13.5 and use stored sharing key to identify resource sharing rules for the media streams in this session; and

2) apply resource sharing rules as specified in 3GPP TS 29.214 [13D] using the stored sharing rules.

Upon receiving a PRACK request or an ACK request containing the initial SDP answer and a Resource-Share header field with the value "no-media-sharing", the P-CSCF shall not share resources in this session.

NOTE: If resources can't be shared the temporary set of resource sharing rules used over Rx interface when reserving resources using the initial SDP offer can still be kept but can never be used in any other session involving this UE as long as this dialog is ongoing.

Upon receiving a PRACK request or an ACK request containing the initial SDP answer and the response does not contain a Resource-Share header field, the P-CSCF may apply resource sharing using local configuration.

L.3.2.6.4 Abnormal cases

If P-CSCF receives a request or response from the served UE and there is a conflict with the given instructions over Rx and the UE behaviour, the P-CSCF shall:

- immediately stop resource sharing for all media streams in this session as described in 3GPP TS 29.214 [13D]; and

- if resource sharing options is determined by an AS, include the Resource-Share header field set to the value "no-media-sharing" along with the "origin" header field set to "session-initiator" or "session-receiver" as appropriate in the outgoing request or response.

NOTE 1: How P-CSCF detects that there is a conflict with the given instruction over Rx and the UE behaviour is implementation dependent and not further described.

NOTE 2: A typical example of a conflict between a given instruction over Rx and the UE behaviour is if media sharing is allowed in both uplink and downlink direction and if in the communication waiting use case, the UE sends a 200 (OK) response to the INVITE request without putting the first call on hold. The UE's behaviour is then regarded as unpredictable and resource sharing cannot be used for any session using the stored sharing key value involving that UE.

The P-CSCF shall remove the Resource-Share header field from the request or response received from a served user.

L.3.2.6.5 Resource sharing options updated by AS

At any point in an ongoing session the AS in the home network can include in an subsequent request or response a Resource-Share header field containing updated resource sharing options and when such an update is received the P-CSCF shall store the new sharing rules as described in subclause 7.2.13.8.4 and apply any change as described in 3GPP TS 29.214 [13D].

L.3.2.7 Priority sharing

L.3.2.7.1 General

If P-CSCF supports priority sharing, PCC is supported for this access technology and if according to operator policy, the P-CSCF shall apply the procedures in the following subclauses.

If P-CSCF supports priority sharing, the P-CSCF shall remove the Priority-Share header field from outgoing requests or responses destined to the served UE.

L.3.2.7.2 Registration procedure

The P-CSCF supporting priority sharing and if according to operator policy shall include the g.3gpp.priority-share feature-capability indicator defined in subclause 7.9A.10 in a Feature-Caps header field in the SIP REGISTER request.

L.3.2.7.3 Session establishment procedure

When receiving the Priority-Share header field defined in subclause 7.2.16 in an initial INVITE request or in a 18x or a 2xx response to the initial INVITE request destined to the served UE and if according to operator policy, the P-CSCF shall apply priority sharing according to 3GPP TS 29.214 [13D] based on the value of the Priority-Share header field.

L.3.2.7.4 Subsequent request procedure

When receiving the Priority-Share header field in a re-INVITE request, in a18x or in a 2xx responses to the re-INVITE request destined to the served UE and if according to operator policy, the P-CSCF shall apply priority sharing according to 3GPP TS 29.214 [13D] based on the value of the Priority-Share header field.

When receiving the Priority-Share header field in an subsequent request or in a 2xx responses to the subsequent request destined to the served UE and if according to operator policy, the P-CSCF shall apply priority sharing according to 3GPP TS 29.214 [13D], based on the value of the Priority-Share header field.

L.3.2.8 RLOS

L.3.2.8.1 General

The support for RLOS as described in this subclause is optional for the P-CSCF.

L.3.2.8.2 Registration

A P-CSCF supporting RLOS shall perform the procedures as specified in subclause 5.2.2 and in addition shall perform the procedures specified in this sublclause.

When the P-CSCF receives a REGISTER request from the UE and the REGISTER request contains a "+g.3gpp.rlos" Contact header field parameter, and:

1) if:

   a) the P-Access-Network-Info header field in REGISTER request indicates an IP-CAN for which procedures are defined in an Annex different than Annex L; or

   b) the IP address and APN check, as defined in section Z3.3 of 3GPP TS 23.228 [7] fail;

   then the P-CSCF shall reject the REGISTER request by sending a 403 response including a Response-Source header field with an "fe" header field parameter set to "<urn:3gpp:fe:p-cscf.orig>"; or

2) if the public user identity in the REGISTER request indicates a user for which the operator of the P-CSCF does not have a roaming agreement, then the P-CSCF shall select a preconfigured S-CSCF and insert a Route header field with the S-CSCF URI for originating requests and forward the request to the selected S-CSCF.

When the P-CSCF receives a 403 (Forbidden) response to a REGISTER request that contained a "+g.3gpp.rlos" Contact header field parameter, the P-CSCF shall:

1) create a temporary unauthenticated subscriber record for the registered public user identity;

2) associate the S-CSCF with the registered public user identity and associated contact address;

3) store the registered public user identity as a default public user identity for use with procedures for the P-Asserted-Identity header field for requests received from the UE;

4) store the values received in the P-Charging-Function-Addresses header field;

5) if a "term-ioi" header field parameter is received in the P-Charging-Vector header field, store the value of the received "term-ioi" header field parameter; and

NOTE: Any received "term-ioi" header field parameter will contain a type 1 IOI. The type 1 IOI identifies the home network of the registered user.

6) void.

L.3.2.8.3 Session Setup

L.3.2.8.3.1 General

If the P-CSCF supports RLOS, the P-CSCF shall accept unprotected requests on the IP address and the unprotected port advertised to the UE during the P-CSCF discovery or the SIP default port.

When the P-CSCF sends unprotected responses to the UE, it shall use the same IP address and port where the corresponding request was received.

L.3.2.8.3.2 General treatment for RLOS session setup – requests from an unregistered user

If the P-CSCF receives an initial request for a dialog or standalone transaction, or an unknown method from an unregistered user on the IP address and the unprotected port advertised to the UE during the P-CSCF discovery or the SIP default port,

The P-CSCF shall inspect the Request-URI independent of values of possible entries in the received Route header fields for the presence of the dummy MSISDN or a RLOS service specific dial string. The P-CSCF shall consider the Request URI of the initial request as indicating RLOS, if the Request-URI contains the dummy URI or a RLOS service specific dial string and if the request contains a P-Preferred-Service header field according to RFC 6050 [121] set to "urn:urn-7:3gpp-service.ims.icsi.rlos".

If the P-CSCF detects that the initial request for a dialog or a standalone transaction, or an unknown method indicates RLOS, and the P-CSCF has previously stored a temporary unauthenticated subscriber record for the public user identity contained in the From header field of the request:

1) shall include a topmost Route header field set to the URI of the S-CSCF as stored in the unauthenticated subscriber record;

NOTE 2: How the list of E-CSCF is obtained by the P-CSCF is implementation dependent.

2) shall execute the procedure described in subclause 5.2.6.3.3, subclause 5.2.6.3.7, subclause 5.2.6.3.11 and subclause 5.2.7.2, as appropriate except for:

   - verifying the preloaded route against the received Service-Route header field;

   - routing to IBCF; and

   - inserting a type 1 "orig-ioi" header field parameter in the P-Charging-Vector header field;

3) shall insert a P-Asserted-Identity header field set to the public user identity as stored in the unauthenticated subscriber record; and

4) if the P-CSCF detects that the UE is behind a NAT, and the UE's Via header field contains a "keep" header field parameter, shall add a value to the parameter, to indicate that it is willing to receive keep-alives associated with the dialog from the UE, as defined in RFC 6223 [143].

When the P-CSCF receives any 1xx or 2xx response to the above requests, the P-CSCF shall execute the appropriate procedure for the type of request described in subclause 5.2.6.3.4, subclause 5.2.6.3.8, and subclause 5.2.6.3.12, except that the P-CSCF may rewrite the port number of its own Record-Route entry to an unprotected port where the P-CSCF wants to receive the subsequent incoming requests from the UE belonging to this dialog.

When the P-CSCF receives a target refresh request from the UE for a dialog, the P-CSCF shall execute the procedure described in subclause 5.2.6.3.5, except for inserting a type 1 "orig-ioi" header field parameter in the P-Charging-Vector header field.

When the P-CSCF receives from the UE subsequent requests other than a target refresh request (including requests relating to an existing dialog where the method is unknown), the P-CSCF shall execute the procedure described in subclause 5.2.6.3.9, except for inserting a type 1 "orig-ioi" header field parameter in the P-Charging-Vector header field.

When the P-CSCF receives any 1xx or 2xx response to the above requests, the P-CSCF shall execute the appropriate procedure for the type of request described in subclause 5.2.6.3.5 or subclause 5.2.6.3.9.

L.3.2.8.3.3 General treatment for RLOS session setup – requests from a registered user

The P-CSCF shall follow procedures specified subclause 5.2.6.

L.3.2.9 Support of ANBR and RAN-assisted codec adaptation

If the network supports ANBR as specified in 3GPP TS 26.114 [9B] and RAN-assisted codec adaptation as specified in 3GPP TS 36.300 [268] and 3GPP TS 36.321 [269], then the P-CSCF might be configured to indicate ANBR support.

If the P-CSCF is configured to indicate ANBR support, when the P-CSCF receives the 200 (OK) response to the REGISTER request the P-CSCF shall include the "g.3gpp.anbr" feature-capability indicator in the Feature-Caps header field of the 200 (OK) response to the REGISTER request.

L.3.3 Procedures at the S-CSCF

L.3.3.1 Notification of AS about registration status

Not applicable.

L.3.3.2 RLOS

L.3.3.2.1 General

The support for RLOS as described in this subclause is optional.

L.3.3.2.2 Registration

A S-CSCF supporting RLOS shall perform the procedures as specified in subclause 5.4.1.2 and in addition shall perform the procedures specified in this subclause.

Upon receipt of a REGISTER request that is part of an initial registration as described in subclause 5.4.1.2.1

1) if REGISTER request contains a "+g.3gpp.rlos" Contact header field parameter, the S-CSCF:

   a) if the S-CSCF supports GPRS-IMS-Bundled authentication and the public user identity in the REGISTER request indicates a user for which the operators of the S-CSCF does not have a roaming agreement with the home network operator, shall reject the request by returning a 420 (Bad Extension) response in which the Unsupported header field contains the value "sec-agree"; and

   b) if the S-CSCF does not support GPRS-IMS-Bundled authentication and the public user identity in the REGISTER request indicates a user for which the operators of the S-CSCF does not have a roaming agreement with the home network operator, shall reject the request by returning a 403 (Forbidden) response and include a Response-Source header field with a "fe" header field parameter set to "<urn:3gpp:fe:s-cscf>". The S-CSCF shall create a temporary record for the public user identity which is registered with a default service profile which is valid for an implementation specific time; and

Upon receipt of a REGISTER request without an Authorization header field as described in subclause 5.4.1.2.1E and the REGISTER request contains a "+g.3gpp.rlos" Contact header field paramete, the S-CSCF shall skip the procedures in subclause 5.4.1.2.1E and shall

1) identify the user by the public user identity as received in the To header field of the REGISTER request. The S-CSCF shall derive the private user identity from the public user identity being registered by removing URI

scheme and the following parts of the URI if present: port number, URI parameters, and To header field parameters;

2) check if the P-Visited-Network-ID header field is included in the REGISTER request, and if it is included identify the visited network by the value of this header field;

3) check whether a "received" header field parameter exists in the Via header field provided by the UE. If a "received" header field parameter exists, the S-CSCF shall compare the IP address recorded in the "received" header field parameter against the UE's IP address stored during registration. In case of IPv6 stateless autoconfiguration, the S-CSCF shall compare the prefix of the IP address recorded in the "received" header field parameter against the UE's IP address prefix stored during registration. If no "received" header field parameter exists in the Via header field provided by the UE, then the S-CSCF shall compare IP address recorded in the "sent-by" parameter against the stored UE IP address. In case of IPv6 stateless autoconfiguration, S-CSCF shall compare the prefix of the IP address recorded in the "sent-by" parameter against the UE's IP address prefix stored during registration. In any case, if the stored IP address (or prefix) and the (prefix of the) IP address recorded in the Via header field provided by the UE do not match, the S-CSCF shall query the HSS as described in 3GPP TS 29.228 [14] with the derived private user identity and the public user identity as input and store the received IP address (or prefix) of the UE. If the stored IP address (or prefix) and the (prefix of the) IP address recorded in the Via header field provided by the UE still do not match the S-CSCF shall reject the registration with a 403 (Forbidden) response and skip the following steps;

4) determine the duration of the registration by checking the registration expiration interval value in the received REGISTER request and bind it either to the respective contact address of the UE or to the registration flow and the associated contact address (if the multiple registration mechanism is used). Based on local policy, the S-CSCF may reduce the duration of the registration or send back a 423 (Interval Too Brief) response specifying the minimum allowed time for registration;

5) update registration bindings;

6) create a temporary record for the public user identity being registered with a default service profile;

7) store the "icid-value" header field parameter received in the P-Charging-Vector header field;

8) if an "orig-ioi" header field parameter is received in the P-Charging-Vector header field, store the value of the received "orig-ioi" header field parameter;

9) check whether a Path header field was included in the REGISTER request and construct a list of preloaded Route header fields from the list of entries in the received Path header field. The S-CSCF shall preserve the order of the preloaded Route header fields and bind them either to the contact address of the UE and the contact information that was received in the REGISTER request; and

10) create and send a 200 (OK) response for the REGISTER request as specified in subclause 5.4.1.2.2F;

In case the timer reg-await-auth is running for this user, S-CSCF supports RLOS as specified in TS 23.228 [7] and the REGISTER request contains a "+g.3gpp.rlos" Contact header field parameter, the S-CSCF shall:

1) check if the Call-ID of the request matches with the Call-ID of the 401 (Unauthorized) response which carried the last challenge. The S-CSCF shall only proceed further if the Call-IDs match;

2) stop timer reg-await-auth;

3) check whether an Authorization header field is included, containing:

   a) the private user identity of the user in the "username" header field parameter;

   b) if the "integrity-protected" header field parameter is set to "yes", the "algorithm" header field parameter set to "AKAv1-MD5" or "AKAv2-SHA-256";

NOTE: The "AKAv1-MD5" algorithm is only supported for backward compatibility.

   c) if the "integrity-protected" header field parameter is set to "tls-connected", the "algorithm" header field parameter set to "AKAv2-SHA-256" if the S-CSCF supports the IMS AKA using HTTP Digest AKAv2 without IPSec security association; and

   d) the authentication challenge response needed for the authentication procedure in the "response" header field parameter.

The S-CSCF shall only proceed with the following steps in this paragraph if the authentication challenge response was included;

4) check whether the received authentication challenge response and the expected authentication challenge response (calculated by the S-CSCF using XRES and other parameters as described in RFC 3310 [49] when AKAv1 is used or as described in RFC 4169 [227] when AKAv2 is used) match. The XRES parameter was received from the HSS as part of the Authentication Vector. The S-CSCF shall only proceed with the following steps if the challenge response received from the UE and the expected response calculated by the S-CSCF do not match;

5) if there are public user identities (including the public user identity being registered, if previously registered) that belong to this user that have been previously registered with the same private user identity, and with an old contact address different from the one received in the REGISTER request and if the previous registrations have not expired:

   a) terminate all dialogs, associated with the previously registered public user identities (including the public user identity being registered, if previously registered), with a status code 480 (Temporarily Unavailable) in the Reason header field of the BYE request, as specified in subclause 5.4.5.1.2;

   b) send a NOTIFY request, to the subscribers to the registration event package of the previously registered public user identities, that indicates that all previously registered public user identities (excluding the public user identity being registered) belonging to this user identified with its private user identity, have been deregistered, as described in subclause 5.4.2.1.2. For the public user identity being registered, the NOTIFY request contains the new contact information; and

   c) delete all information associated with the previously registered public user identities;

6) store the following information in the local data:

   a) the public user identities due to the received REGISTER request which is set to the public user identity as received in the REGISTER request. The public user identity is identified as non-barred;

   b) a default service profile that indicates that the public user identity is registered for RLOS;

   c) if S-CSCF restoration procedures are supported, the restoration information if received as specified in 3GPP TS 29.228 [14]; and

   d) if PCRF based P-CSCF restoration procedures are supported, all the user profile(s) corresponding to the public user identities being registered (explicitly or implicitly), including the IMSI, if available;

7) update registration bindings and

   a) bind to each non-barred registered public user identity the registered contact information including all header field parameters contained in the Contact header field and all associated SIP URI parameters, and

   b) if the Contact header field of the REGISTER request contained a "+sip.instance" and a "reg-id" header field parameter, and the SIP URI in the Path header field inserted by the P-CSCF contained an "ob" SIP URI parameter header field, and:

      - if the public user identity has not previously been registered with the same "+sip.instance" and "reg-id" Contact header field parameter values, then create the registration flow in addition to any existing registration flow; or

      - if the public user identity has previously been registered with the same "+sip.instance" and "reg-id" header field parameter values, then determine whether the request refreshes or replaces an existing registration flow. If the request:

         i) refreshes an existing registration flow, then the S-CSCF shall leave the flow intact; or

         ii) replaces the existing registration flow with a new flow, then the S-CSCF shall:

            a) terminate any dialog, as specified in subclause 5.4.5.1.2, with a status code 480 (Temporarily Unavailable) in the Reason header field of the BYE request, associated with the registration flow being replaced; and

b) send a NOTIFY request to the subscribers to the registration event package for the public user identity indicated in the REGISTER request, as described in subclause 5.4.2.1.2;

8) check whether a Path header field was included in the REGISTER request and construct a list of preloaded Route header fields from the list of entries in the received Path header field. The S-CSCF shall preserve the order of the preloaded Route header fields and bind them either to the contact address of the UE or the registration flow and the associated contact address (if the multiple registration mechanism is used) and the contact information that was received in the REGISTER request;

9) determine the duration of the registration by checking the value of the registration expiration interval value in the received REGISTER request and bind it either to the respective contact address of the UE or to the registration flow and the associated contact address (if the multiple registration mechanism is used). Based on local policy, the S-CSCF may reduce the duration of the registration or send back a 423 (Interval Too Brief) response specifying the minimum allowed time for registration.;

10) store the "icid-value" header field parameter received in the P-Charging-Vector header field;

11) if an "orig-ioi" header field parameter is received in the P-Charging-Vector header field, store the value of the received "orig-ioi" header field parameter; and

12) create a 403 (Forbidden) response for the REGISTER request including a Response-Source header field with a "fe" header field parameter set to "<urn:3gpp:fe:s-cscf >" and send the so generated response.

L.3.3.2.3 Session Setup

L.3.3.2.3.1 General

A S-CSCF supporting RLOS shall perform the procedures as specified in subclause 5.4.3.2 and in addition shall perform the procedures specified in this subclause.

When the S-CSCF receives from the served user from an initial request for a dialog or a request for a standalone transaction and performing the procedures in subclause 5.4.3.2 and:

1) if there is no original dialog identifier that the S-CSCF previously placed in a Route header field is present in the topmost Route header field of the incoming request; and

2) if the Request-URI contains the dummy MSISDN value as defined in 3GPP TS 23.003 [3] or a RLOS service specific dial string and a P-Preferred-Service header set to "urn:urn-7:3gpp-service.ims.icsi.rlos" is included in the request;

   the S-CSCF shall build an ordered list of initial filter criteria based on the temporary unauthenticated subscriber record for the public user identiy of the served user instead of building the ordered list of initial filter criteria as described in bullet 3).

L.4 3GPP specific encoding for SIP header field extensions

L.4.1 Void

L.5 Use of circuit-switched domain

There is no CS domain in this access technology.

NOTE 1: If the UE sends an INVITE request including voice codecs which is not successful due to a failure from the lower layers indicating that access is barred for originating calls but not specific to CSFB (see 3GPP TS 24.301 [8J]) and if the CS domain is supported and available, the UE can attempt the voice call via the CS domain.

NOTE 2: If the UE has sent an INVITE request including voice codecs, receives a 500 (Server Internal Error) response to this INVITE request containing no Retry-After header field, including a Reason header field with a protocol value set to "FAILURE_CAUSE" including a cause value header field parameter set to "1" as specified in subclause 7.2A.18.12.2, a Response-Source header field with a "fe" header field parameter set to "<urn:3gpp:fe:p-cscf.orig>", and the UE is attached to both PS and CS domains, the UE can attempt the voice call via the CS domain, e.g. by initiating a service request for CS fallback (see 3GPP TS 24.301 [8J]).

Annex M (normative): IP-Connectivity Access Network specific concepts when using cdma2000® packet data subsystem to access IM CN subsystem

M.1 Scope

This annex defines IP-CAN specific requirements for call control protocol for use in the IM CN subsystem based on the Session Initiation Protocol (SIP), and the associated Session Description Protocol (SDP), where the IP-CAN is the cdma2000® packet data subsystem. It also defines procedures for invoking CS domain services.

M.2 cdma2000® packet data subsystem aspects when connected to the IM CN subsystem

M.2.1 Introduction

A UE accessing the IM CN subsystem, and the IM CN subsystem itself, utilise the services provided by the cdma2000® packet data subsystem to provide packet-mode communication between the UE and the IM CN subsystem.

Requirements for the UE on the use of these packet-mode services are specified in this subclause. Requirements for the IP-CAN bearer control point (i.e. the point where the UE has attached itself to the cdma2000® packet data subsystem) support of this communication are specified in 3GPP2 X.S0011-E [127].

M.2.2 Procedures at the UE

M.2.2.1 Establishment of IP-CAN bearer and P-CSCF discovery

Prior to communication with the IM CN subsystem, the UE shall:

a) establish a connection with the cdma2000® wireless IP network specified in 3GPP2 X.S0011-E [127]. Upon establishment a connection with the cdma2000® wireless IP network, the UE can have an IPv4 address only, an IPv6 address only, or both IPv4 and IPv6 addresses simultaneously;

b) ensure that an IP-CAN bearer used for SIP signalling is available. This IP-CAN bearer shall remain active throughout the period the UE is connected to the IM CN subsystem, i.e. from the initial registration and at least until the last deregistration;

   The UE shall choose one of the following options when performing establishment of this IP-CAN bearer:

   I. A dedicated IP-CAN bearer for SIP signalling:

      The UE shall indicate to the IP-CAN bearer control point that this is an IP-CAN bearer intended to carry IM CN subsystem-related signalling only. The UE may also use this IP-CAN bearer for DNS and DHCP access.

   II. A general-purpose IP-CAN bearer:

      The UE may decide to use a general-purpose IP-CAN bearer to carry IM CN subsystem-related signalling. The UE may carry both signalling and media on the general-purpose IP-CAN bearer; and

c) discover a P-CSCF.

   The methods for P-CSCF discovery are:

I. Use DHCP mechanism

II Retrieve the list of P-CSCF address(es) stored in the IMC

III Obtain the list of P-CSCF address(es) from the IMS management object

The UE can freely select method I, II, or III for P-CSCF discovery. If DHCP is used, the following procedures apply:

Upon establishing an IP-CAN bearer, the UE may use the Dynamic Host Configuration Protocol (DHCP) specified in RFC 2131 [40A] or Dynamic Host Configuration Protocol for IPv6 (DHCPv6) specified in RFC 3315 [40] to discover the P-CSCF.

Prior to accessing the DHCP server, the UE will have obtained an IP address via means other than DHCP and DHCPv6, see 3GPP2 X.S0011-E [127].

If the UE uses DHCP for P-CSCF discovery and the UE is unaware of the address of the DHCP server, the UE shall send the DHCPINFORM using the limited broadcast IP address (i.e., 255.255.255.255) and UDP port 67. If the UE knows the IP address of the DHCP server, the UE shall send the DHCPINFORM to the DHCP server's unicast IP address and UDP port 67. The DHCP server shall send the DHCPACK on the IP address specified in the Client IP Address field of the DHCPINFORM. The DHCP server may include, in the DHCPACK, the SIP Server DHCP Option specified in RFC 3361 [35A], which carries either a list of IPv4 address(es) of the P-CSCF(s) or a list of DNS fully qualified domain name(s) that can be mapped to one or more P-CSCF(s). If the UE uses DHCPv6 for P-CSCF discovery and the UE is unaware of the address of the DHCP Server, the UE shall send an Information Request using the IPv6 multicast address FF02::1:2 and the UDP port 547. If the UE knows the IP address of the DHCPv6 server, the UE shall send the Information Request message to the DHCPv6 server's IP address and UDP port 547. In the Information Request, the UE may request either one or both the SIP Servers Domain Name List option and the SIP Servers IPv6 Address List option specified in RFC 3319 [41]. The DHCP server shall send the Reply to the IP address specified in the Information Request. The DHCP server may include in the Reply either one or both the SIP Servers Domain Name List option and the SIP Servers IPv6 Address List option, as requested by the UE.

In case several P-CSCF's IP addresses or domain names are provided to the UE, the UE shall perform P-CSCF selection according to RFC 3361 [35A] or RFC 3319 [41]. The UE shall perform the procedure for the resolution of domain name according to RFC 3263 [27A].

M.2.2.1AModification of IP-CAN used for SIP signalling

Not applicable.

M.2.2.1BRe-establishment of the IP-CAN used for SIP signalling

Not applicable.

M.2.2.1C P-CSCF restoration procedure

A UE supporting the P-CSCF restoration procedure uses the keep-alive procedures described in RFC 6223 [143].

If the P-CSCF fails to respond to the keep-alive request the UE shall acquire a different P-CSCF address using any of the methods described in the subclause M.2.2.1 and perform an initial registration as specified in subclause 5.1.

M.2.2.2 Void

M.2.2.3 IP-CAN bearer control point support of DHCP based P-CSCF discovery

The IP-CAN bearer control point, or Home Agent in case of Mobile IP with reverse tunneling, may forward the packet to one or more local DHCP servers, or relay the packet to a specific DHCP server. The IP-CAN bearer control point, or Home Agent in case of Mobile IP with reverse tunnelling, shall not forward the DHCPINFORM (or Information-Request) to any UE.

NOTE 1: For forwarding the DHCPINFORM or Information-Request, the IP-CAN bearer control point, or Home Agent in case of Mobile IP with reverse tunnelling, does not change the destination IP address of the packet.

NOTE 2: For relaying the DHCPINFORM or Information-Request, the IP-CAN bearer control point, or Home Agent in case of Mobile IP with reverse tunnelling inserts a DHCP server's IP address in the destination IP address field of the packet.

M.2.2.4 Void

M.2.2.5 Handling of the IP-CAN for media

M.2.2.5.1 General requirements

Not applicable.

M.2.2.5.1A Activation or modification of IP-CAN for media by the UE

Not applicable.

M.2.2.5.1B Activation or modification of IP-CAN for media by the network

Not applicable.

M.2.2.5.1C Deactivation of IP-CAN for media

Not applicable.

M.2.2.5.2 Special requirements applying to forked responses

Not applicable.

M.2.2.5.3 Unsuccessful situations

Not applicable.

M.2.2.6 Emergency service

M.2.2.6.1 General

When establishing an HRPD session to perform emergency registration, the UE shall follow the procedures defined in 3GPP2 X.S0060 [86B].

To determine whether the HRPD UE is attached to the home network or to the visited network, the UE shall compare the Carrier ID values obtained per 3GPP2 X.S0060 [86B]. If the Carrier ID of the network the UE is attached to does not match with the provisioned Carrier ID, then for the purpose of emergency calls in the IM CN subsystem the UE shall consider to be attached to a visited network.

NOTE: For 3GPP2-1X and 3GPP2-UMB, no IP-CAN specific support is provided in the current release. No carrier identification is provided for 3GPP2-1X or 3GPP2-UMB in the P-Access-Network-Info header field, and thus there is no IMS specific procedure for identifying that the UE is in the home network.

M.2.2.6.1A Type of emergency service derived from emergency service category value

Not applicable.

M.2.2.6.1B Type of emergency service derived from extended local emergency number list

Not applicable.

M.2.2.6.2 eCall type of emergency service

The UE shall not send an INVITE request with Request-URI set to "urn:service:sos.ecall.manual" or "urn:service:sos.ecall.automatic".

M.2.2.6.3 Current location discovery during an emergency call

Void.

M.2A Usage of SDP

M.2A.0 General

Not applicable.

M.2A.1 Impact on SDP offer / answer of activation or modification of IP-CAN for media by the network

Not applicable.

M.2A.2 Handling of SDP at the terminating UE when originating UE has resources available and IP-CAN performs network-initiated resource reservation for terminating UE

Not applicable.

M.2A.3 Emergency service

No additional procedures defined.

M.3 Application usage of SIP

M.3.1 Procedures at the UE

M.3.1.0 Void

M.3.1.0a IMS_Registration_handling policy

Not applicable.

M.3.1.1 P-Access-Network-Info header field

The UE shall always include the P-Access-Network-Info header field where indicated in subclause 5.1.

M.3.1.1ACellular-Network-Info header field

Not applicable.

M.3.1.2 Availability for calls

Not applicable.

M.3.1.2AAvailability for SMS

Void.

M.3.1.3 Authorization header field

When using SIP digest or SIP digest without TLS, the UE need not include an Authorization header field on sending a REGISTER request, as defined in subclause 5.1.1.2.1.

NOTE: In case the Authorization header field is absent, the mechanism only supports that one public user identity is associated with only one private user identity. The public user identity is set so that it is possible to derive the private user identity from the public user identity by removing SIP URI scheme and the following parts of the SIP URI if present: port number, URI parameters, and To header field parameters. Therefore, the public user identity used for registration in this case cannot be shared across multiple UEs. Deployment scenarios that require public user identities to be shared across multiple UEs that don't include an private user identity in the initial REGISTER request can be supported as follows:

- Assign each sharing UE a unique public user identity to be used for registration,

- Assign the shared public user identitiess to the implicit registration set of the unique registering public user identities assigned to each sharing UE.

M.3.1.4 SIP handling at the terminating UE when precondition is not supported in the received INVITE request, the terminating UE does not have resources available and IP-CAN performs network-initiated resource reservation for the terminating UE

Not applicable.

M.3.1.5 3GPP PS data off

Not applicable.

M.3.1.6 Transport mechanisms

No additional requirements are defined.

M.3.1.7 RLOS

Not applicable.

M.3.2 Procedures at the P-CSCF

M.3.2.0 Registration and authentication

Void.

M.3.2.1 Determining network to which the originating user is attached

For an HRPD UE, after the initial request for a dialog or standalone transaction or an unknown method is received the P-CSCF shall check the Carrier ID field received in the P-Access-Network-Info header field to determine from which network the request was originated.

NOTE: For 3GPP2-1X and 3GPP2-UMB, no IP-CAN specific support is provided in the current release. No carrier identification is provided for 3GPP2-1X or 3GPP2-UMB in the P-Access-Network-Info header field, and thus there is no IMS specific procedure for identifying that the UE is in the home network.

M.3.2.2 Location information handling

Void

M.3.2.3 Void

M.3.2.4 Void

M.3.2.5 Void

M.3.2.6 Resource sharing

Not applicable.

M.3.2.7 Priority sharing

Not applicable.

M.3.2.8 RLOS

Not applicable.

M.3.3 Procedures at the S-CSCF

M.3.3.1 Notification of AS about registration status

The procedures described in subclause 5.4.1.7 apply with the additional procedures described in the present subclause:

1) in case the received REGISTER request contained a Timestamp header field, the S-CSCF shall insert a Timestamp header field with the value of the Timestamp header field from the received REGISTER request.

M.3.3.2 RLOS

Not applicable.

M.4 3GPP specific encoding for SIP header field extensions

M.4.1 Void

M.5 Use of circuit switched domain

When an emergency call is to be set up over the CS domain, the UE shall attempt it according to the procedures described in 3GPP2 C.S0005-D [85].

Annex N (Normative): Functions to support overlap signalling

N.1 Scope

This annex defines the procedures performed by network entities within the IM CN subsystem to support overlap signalling.

The support of overlap signalling within the IM CN subsystem is optional, and is depended on the network policy.

N.2 Digit collection function

N.2.1 General

The digit collection function is invoked if an entity requires additional digits for a decision where to route a INVITE request.

NOTE 1: The digit collection function is only applicable for the in-dialog method of overlap signalling. Further information about digit collection is provided in subclause 4.9.3.3.

The digit collection function may interact with a routeing database, to reach this decision. The digit collection function shall be performed by an entity acting as a B2BUA. The digit collection function requires the ability to recognise incomplete numbers. The digit collection function may be implemented in different network nodes depending on the operator's deployment strategy (e.g. AS, IBCF).

NOTE 2: An HSS does not support the recognition of incomplete numbers. A routeing database being queried by ENUM also does not support the recognition of incomplete numbers.

NOTE 3: A private routeing database to support the recognition of incomplete numbers e.g. for transit calls or ported numbers can be used.

N.2.2 Collection of digits

N.2.2.1 Initial INVITE request

Upon receiving an initial INVITE request carrying an SDP offer, the digit collection function shall:

1) if the request contains enough digits to forward the request, forward the request towards it's destination;

2) if the digit collection function chooses to collect additional digits in INFO requests, in order to forward the request, and the sender of the INVITE request has indicated support of reliable responses, store the received digits and send a reliable 183 (Session Progress) provisional response in order to establish an early dialog with the sender of the INVITE request. The response shall not contain an SDP answer; or

3) if it is determined that the en-bloc will not be able to forward the request even if additional digits are received,, send a 404 (Not Found) response.

Upon receiving an initial INVITE request without an SDP offer, the digit collection function shall:

1) if the request contains enough digits to forward the request, forward the request towards it's destination; or

2) send a 404 (Not Found) response.

NOTE 1: If the initial INVITE request does not contain an SDP offer, a reliable 18x provisional response generated by the en-bloc conversion function would have to contain an SDP offer. In this case digit collection needs to be performed by the originating SIP entity (e.g. MGCF).

When the digit collection function sends the reliable 183 (Session Progress) provisional response, in order to establish an early dialog with the sender of the INVITE request, the digit collection function shall start a digit collection timer. If the timer expires, the digit collection function shall terminate the call setup by sending a sending a 484 (Address Incomplete) response towards the sender of the INVITE request.

NOTE 2: The digit collection timer is similar to the protection timer used in PSTN/ISDN. The timer value range is between 5 and 15 seconds, and the default value is 10 seconds.

N.2.2.2 Collection of additional digits

Upon receiving an INFO request carrying additional digits, and if an early dialog towards the destination of the initial INVITE request for the call does not exist, the digit collection function shall:

1) send a 200 (OK) response to the INFO request;

2) add the received digits to the previously stored digits for the call;

3) restart the digit collection timer; and

4) check if enough digits have been received in order to forward the initial INVITE request.

When enough digits for the call have been received in order to forward the initial INVITE request, the digit collection function shall add all stored digits to the request URI and forward the request towards its destination and stop the digit collection timer.

Upon receiving an INFO request carrying additional digits, and the initial INVITE request has been forwarded towards its destination, but an early dialog towards the destination of the initial INVITE request for the call does not exist, the digit collection function shall:

1) send a 200 (OK) response to the INFO request; and

2) add the received digits to the previously stored, but yet not forwarded, digits for the call;

When the digit collection function receives a provisional response from the destination of the initial INVITE request, and the digit collection function has received additional digits in INFO requests, the digit collection shall generate and send an INFO request towards the destination of the initial INVITE request. The Request-URI shall contain all digits which have been received and stored since the initial INVITE request was forwarded.

Upon receiving an INFO request carrying additional digits, and if an early dialog towards the destination of the initial INVITE request for the call does exist, the digit collection function shall forward the INFO request on the early dialog towards the destination of the initial SIP INVITE request.

Upon receiving an INFO request carrying additional digits, and if a 180 (Ringing) or a 200 (OK) response to the initial INVITE request for the call has been received, or the digit collection function has received some other indication that enough digits have been forwarded in order for the INVITE request to reach the terminating SIP user, the digit collection function shall, based on operator policy:

1) send a 200 (OK) response to the INFO request and not forward the INFO request; or

2) forward the INFO request on the early dialog towards the destination of the initial SIP INVITE request.

N.2.2.3 Handling of 404 (Not Found) / 484 (Address Incomplete) responses

Upon receiving a 404/484 response to the initial INVITE request, the digit collection function shall acknowledge the response and shall start the digit collection timer. The digit collection function shall not forward the response towards the sender of the INVITE request.

NOTE: If the digit collection function has received a 404/484 response, it will send a new initial INVITE request when it has received additional digits as described above, in order to establish an early dialog towards the destination of the INVITE request. The digit collection timer will re-start if an INFO request with additional digits is received. At timer expiry, the digit collection function will terminate the call as described above

N.2.3 Forwarding of SIP messages by the digit collection function

Apart from 404/484 responses to the initial INVITE request, and INFO requests carrying additional digits received after a 180 (Ringing) or a 200 (OK) response to the initial INVITE request has been received, the digit collection function shall forward all SIP messages. When forwarding SIP messages, the digit collection function shall modify the SIP messages to comply with SIP procedures on both call legs as specified below:

1) The digit collection function will receive a "tag" To header field parameter value from the receiver of the initial INVITE request, which is different from the "tag" To header field parameter value that the digit collection function inserted in the 183 (Session Progress) response that it sent when it received the initial INVITE request. The digit collection function shall modify the "tag" header field parameter value accordingly when forwarding mid-dialog SIP messages.

2) The digit collection function will return a Contact header field in the initial 183 (Session Progress) provisional response to the originating side, which contains a SIP-URI of the digit collection function. The contact information is used in the Request-URI of subsequent mid-dialog SIP requests sent by the originating side, until the digit collection function has received, and forwarded to the originating side, a 183 (Session Progress) provisional response from the destination of the INVITE request. If the Request-URI of the received mid-dialog SIP request contains the SIP-URI of the digit collection function, and the digit collection function forwards the request, the digit collection function shall modify the Request-URI before forwarding the SIP request.

3) The digit collection function will return the Record-Route header fields, which it received in the initial INVITE request, in the initial 183 (Session Progress) provisional response to the originating side. The information is used in the Route header fields of subsequent mid-dialog SIP requests sent by the originating side, until the digit collection function has received, and forwarded to the originating side, a 183 (Session Progress) provisional response from the destination of the INVITE request. If the Route header fields of the received mid-dialog SIP request are based on the Record-Route headers fields which the digit collection function returned in the initial 183 (Session Progress) provisional response, and the digit collection function forwards the request, the digit collection function shall modify the Route header fields before forwarding the SIP request.

N.3 En-bloc conversion function

N.3.1 General

The en-bloc conversion function may be performed in an entity acting as B2BUA. The en-bloc conversion function may be implemented in different network nodes depending on the operator's deployment strategy (e.g. AS, IBCF).

If the initial INVITE request is to be forwarded towards a network, or towards a network entity, that does not support overlap signalling, the en-bloc conversion function shall determine the end of address signalling.

The following methods can be used to determine the end of the address signalling:

1) the maximum number of digits used in a national numbering plan has been received;

2) number analysis, e.g. using a provisioned dial plan, is used to determinte that the complete number of digits has been received; or

3) an inter digit timer expires, and the minimum number of digits required for routeing the call have been received. The timer is started when the initial INVITE request is received, and re-started every time new digit(s) are received.

NOTE: The inter digit timer is similar to the protection timer used in PSTN/ISDN. The timer value range is between 5 and 15 seconds, and the default value is 10 seconds.

The procedures for collecting additional digits are described in subclauses N.3.2 and N.3.3. When end of address signalling has been determined, the en-bloc conversion function shall generate an INVITE request, add all digits to the request and forwards the request towards its destination.

N.3.2 Multiple-INVITE method

Upon reception of an INVITE request, the en-bloc conversion function shall:

1) if an inter digit timer is running for a previously received INVITE request with the same Call-ID and From header, and

   a) if the number of digits within that previous INVITE request is below the number of digits received in the new INVITE request (or as an equivalent test if the CSeqID of the previous INVITE request is below the CSeqID of the new INVITE request), stop the inter-digit timer for that previous INVITE request and send a 484 (Address Incomplete) response for it; and

   b) if the number of digits with the previous INVITE request is above or equal to the number of digits received in the new INVITE request (or as an equivalent test if the CSeqID of the previous INVITE request is below the CSeqID of the new INVITE request), send a 484 (Address Incomplete) response for the new INVITE request; and

2) if the en-bloc conversion function determines that the number received in the INVITE request is complete, forward the INVITE request; and

3) if the en-bloc conversion function determines that it will not be able to forward the request even if additional digits are received, send a 404 (Not Found) response; and

4) if the en-bloc conversion function chooses to collect additional digits, store the INVITE request and start an inter-digit timer to wait for possible INVITE requests with the same Call ID and From header.

When the inter-digit timer expires the en-bloc conversion function shall:

1) if it determines that the number received in the stored INVITE request is incomplete (e. g. by number analysis), terminate the call setup by sending a sending a 484 (Address Incomplete) response towards the sender of the INVITE request; and

2) if it does not determines that the number received in the last INVITE request is incomplete, forward the corresponding stored INVITE request to the next hop.

After forwarding an INVITE request, the en-bloc conversion function shall apply SIP proxy procedures for all subsequent SIP messages within the corresponding dialogue, unless other functionality not related to en-bloc conversion allocated in the same physical norde requires a different behaviour.

N.3.3 In-dialog method

Upon receiving an initial INVITE request carrying an SDP offer, the en-bloc conversion function shall:

1) if it determines that the request contains a complete number, forward the request towards it's destination;

2) if the en-bloc conversion function chooses to collect additional digits in INFO requests before forwarding the request, and the sender of the INVITE request has indicated support of reliable responses, store the received digits, send a reliable 183 (Session Progress) provisional response without an SDP answer in order to establish an early dialog with the sender of the INVITE request, and start an inter digit timer; or

3) if the en-bloc conversion function determines that it will not be able to forward the request even if additional digits are received, send a 404 (Not Found) response.

Upon receiving an initial INVITE request without an SDP offer, the en-bloc conversion function shall:

1) if it is determined that the request contains a complete number, forward the request towards it's destination; or

2) send a 404 (Not Found) response.

NOTE: If the initial INVITE request does not contain an SDP offer, a reliable 18x provisional response generated by the en-bloc conversion function would have to contain an SDP offer. The en-bloc conversion function for the in-dialog method specified here does not support en-bloc conversion for calls with an initial INVITE request that does not contain an SDP offer in the present release. However, en-bloc conversion for an initial INVITE request that does not contain an SDP offer can be performed by the originating SIP entity (e.g. MGCF).

Upon receiving an INFO request carrying additional digits, and an early dialog towards the destination of the initial SIP INVITE request for the call has not been created, the en-bloc conversion function shall:

1) send a 200 (OK) response to the INFO request;

2) add the received digits to the previously stored digits for the call;

3) re-start the inter digit timer; and

4) check if it can determine that a complete number for the call has been received.

When the en-bloc conversion function determines that a complete number for the call has been received, it shall add all stored digits to the initial INVITE request and forward the request towards its destination and stop the inter digit timer.

When the inter-digit timer expires the en-bloc conversion function shall

1) if it determines that the number so far is incomplete (e. g. by number analysis), terminate the call setup by sending a sending a 484 (Address Incomplete) response towards the sender of the INVITE request; or

2) if it does not determines that the number received in the last INVITE request is incomplete, forward the INVITE request to the next hop, including all received digits.

Upon receiving an INFO request carrying additional digits, if the en-bloc conversion function has forwarded the initial INVITE request towards its destination, the en-bloc conversion function shall send a 200 (OK) response to the INFO request and not forward the INFO request.

Apart from INFO requests carrying additional digits received after the initial INVITE request has been forwarded, the en-bloc conversion function shall forward all SIP messages.

The en bloc conversion function will receive a "tag" To header field parameter value from the receiver of the initial INVITE request, which is different from the "tag" To header field parameter value that the digit collection function inserted in the 183 (Session Progress) response that it sent when it received the initial INVITE request. The digit collection function shall modify the "tag" To header field parameter value accordingly when forwarding in-dialog SIP messages.

Annex O (normative): IP-Connectivity Access Network specific concepts when using the EPC via cdma2000® HRPD to access IM CN subsystem

O.1 Scope

The present annex defines IP-CAN specific requirements for a call control protocol for use in the IP Multimedia (IM) Core Network (CN) subsystem based on the Session Initiation Protocol (SIP), and the associated Session Description Protocol (SDP), where the IP-CAN is the Evolved Packet Core (EPC) via a cdma2000® HRPD access network.

O.2 IP-CAN aspects when connected to the IM CN subsystem

O.2.1 Introduction

A UE accessing the IM CN subsystem, and the IM CN subsystem itself, utilise the services provided by the EPC and cdma2000® HRPD access network as specified by 3GPP2 X.P0057 [86C] to provide packet-mode communication between the UE and the IM CN subsystem.

Requirements for the UE on the use of these packet-mode services are specified in this clause. Requirements for the P-GW in support of this communication are specified in 3GPP TS 29.061 [11] and 3GPP TS 29.212 [13B].

Requirements for the use of the EPC via packet cdma2000® HRPD as an IP-CAN are specified in 3GPP2 X.P0057 [86C].

O.2.2 Procedures at the UE

O.2.2.1 IP-CAN bearer context activation and P-CSCF discovery

Prior to communication with the IM CN subsystem, the UE shall:

a) establish a connection with the IP-CAN via the cdma2000® HRPD wireless IP network specified in 3GPP2 X.P0057 [86C]. Upon establishing a connection with the cdma2000® eHRPD wireless IP network, the UE can have an IPv4 address only, an IPv6 address only, or both IPv4 and IPv6 addresses simultaneously;

b) ensure that a IP-CAN bearer context used for SIP signalling is available, according to the APN and P-GW selection criteria described in 3GPP TS 23.402 [7E]. This IP-CAN bearer context shall remain active throughout the period the UE is connected to the IM CN subsystem, i.e. from the initial registration and at least until the deregistration; and

NOTE 1: Any IP-CAN bearer can carry both IM CN subsystem signalling and media if the media does not need to be authorized by Policy and Charging control mechanisms as defined in 3GPP TS 29.212 [13B], and the media stream is not mandated by the P-CSCF to be carried in a separate IP-CAN bearer.

NOTE 2: IP-CAN PDN connection and bearer management procedures are specified in 3GPP2 X.P0057 [86C].

c) acquire a P-CSCF address(es).

The methods for P-CSCF discovery are:

I. Use DHCP mechanism

II Retrieve the list of P-CSCF address(es) stored in the IMC

III Obtain the list of P-CSCFaddress(es) from the IMS management object

IV. Transfer P-CSCF address(es) within the IP-CAN bearer context activation procedure.

The UE shall indicate the request for a P-CSCF address to the network within the Protocol Configuration Options information element of the during PDN connectivity establishment as specified in 3GPP2 X.P0057 [86C].

If the network provides the UE with a list of P-CSCF IPv4 or IPv6 addresses, the UE shall assume that the list is prioritised with the first address within the Protocol Configuration Options information element as the P-CSCF address with the highest priority.

The UE can freely select method I, II, III, or IV for P-CSCF discovery. If DHCP is used, the following procedures apply:

Upon establishing an IP-CAN bearer, the UE may use the Dynamic Host Configuration Protocol (DHCP) specified in RFC 2131 [40A] or Dynamic Host Configuration Protocol for IPv6 (DHCPv6) specified in RFC 3315 [40] to discover the P-CSCF.

Prior to accessing the DHCP server, the UE will have obtained an IP address via means other than DHCP and DHCPv6.

If the UE uses DHCP for P-CSCF discovery and the UE is unaware of the address of the DHCP server, the UE shall send the DHCPINFORM using the limited broadcast IP address (i.e., 255.255.255.255) and UDP port 67. If the UE knows the IP address of the DHCP server, the UE shall send the DHCPINFORM to the DHCP server's unicast IP address and UDP port 67. The DHCP server shall send the DHCPACK on the IP address specified in the Client IP Address field of the DHCPINFORM. The DHCP server may include, in the DHCPACK, the SIP Server DHCP Option specified in RFC 3361 [35A], which carries either a list of IPv4 address(es) of the P-CSCF(s) or a list of DNS fully qualified domain name(s) that can be mapped to one or more P-CSCF(s). If the UE uses DHCPv6 for P-CSCF discovery and the UE is unaware of the address of the DHCP Server, the UE shall send an Information Request using the IPv6 multicast address FF02::1:2 and the UDP port 547. If the UE knows the IP address of the DHCPv6 server, the UE shall send the Information Request message to the DHCPv6 server's IP address and UDP port 547. In the Information Request, the UE may request either one or both the SIP Servers Domain Name List option and the SIP Servers IPv6 Address List option specified in RFC 3319 [41]. The DHCP server shall send the Reply to the IP address specified in the Information Request. The DHCP server may include in the Reply either one or both the SIP Servers Domain Name List option and the SIP Servers IPv6 Address List option, as requested by the UE.

If the UE supports the optional configuration parameter "Access_Point_Name_Parameter_Reading_Rule", as defined in 3GPP TS 24.167 [8G] and has been configured with this parameter, then the UE shall use it to retrieve the access point name to use in the IP CAN bearer context activation procedure.

In case several P-CSCF's IP addresses or domain names are provided to the UE, the UE shall perform P-CSCF selection according to RFC 3361 [35A] or RFC 3319 [41]. The UE shall perform the procedure for the resolution of domain name according to RFC 3263 [27A].

O.2.2.1A Modification of an IP-CAN bearer context used for SIP signalling

The UE shall not modify the IP-CAN bearer from being used exclusively for SIP signalling to a general purpose IP-CAN bearer and vice versa.

After the establishment of a SIP bearer context used for SIP signalling, the UE shall not indicate the request for a P-CSCF address to the network when requesting an IP-CAN bearer modification for that APN. The UE shall ignore P-CSCF address(es) if received from the network as part of the bearer modification procedure.

O.2.2.1B Re-establishment of the IP-CAN bearer context for SIP signalling

If the IP-CAN bearer context for SIP signalling is lost and cannot be re-established:

a. if the SIP signalling was carried over a dedicated IP-CAN bearer, the UE shall release all resources established as a result of SIP signalling by either:

- requesting an IP-CAN bearer modification if there are IP-CAN bearers to this PDN that are not related SIP sessions; or

- initiating disconnection of the PDN connection if all the bearers to this PDN are related to SIP sessions.

NOTE: If the SIP signalling was carried over the default IP-CAN bearer, all the resources established as a result of SIP signalling are released.

O.2.2.1CP-CSCF restoration procedure

A UE supporting the P-CSCF restoration procedure performs one of the following procedures:

A if the UE used method II for P-CSCF discovery and if the UE receives one or more P-CSCF address(es) in an VSNCP Configure-Request message and the one or more P-CSCF addresse(s) do not include the address of the currently used P-CSCF, then the UE shall acquire a different P-CSCF address from the one or more P-CSCF addresse(s) in the VSNCP Configure-Request message. The UE shall assume that the more than one P-CSCF address are prioritised with the first P-CSCF address within the Protocol Configuration Options information element as the P-CSCF address with the highest priority;

B if the UE uses RFC 6223 [143] as part of P-CSCF restoration procedures, and if the P-CSCF fails to respond to a keep-alive request, then the UE shall acquire a different P-CSCF address using one of the methods I, II and III for P-CSCF discovery described in the subclause O.2.2.1.

When a different P-CSCF address is acquired the UE shall perform an initial registration as specified in subclause 5.1.

O.2.2.2 Session management procedures

The existing procedures for session management as described in 3GPP2 X.P0057 [86C] shall apply while the UE is connected to the IM CN subsystem.

O.2.2.3 Mobility management procedures

The existing procedures for mobility management as described in 3GPP2 X.P0057 [86C] shall apply while the UE is connected to the IM CN subsystem.

O.2.2.4 Cell selection and lack of coverage

The existing mechanisms and criteria for cell selection as described in 3GPP2 C.S0014-C [86D] shall apply while the UE is connected to the IM CN subsystem.

O.2.2.5 IP-CAN bearer contexts for media

O.2.2.5.1 General requirements

NOTE 1: The UE cannot control whether media streams belonging to different SIP sessions are established on the same IP-CAN bearer context or not. During establishment of a session, the UE establishes data streams(s) for media related to the session. Such data stream(s) can result in activation of additional IP-CAN bearer context(s). Either the UE or the network can request for resource allocations for media, but the establishment and modification of the IP-CAN bearer is controlled by the network as described in 3GPP2 X.P0057 [86C].

NOTE 2: When the UE wishes to allocate bandwidth for RTP and RTCP, the rules as outlined in 3GPP TS 29.213 [13C] apply. Application of QoS to when using an EPC IP-CAN with cdma2000® HRPD is described in 3GPP2 X.P0057 [86C].

O.2.2.5.1A Activation or modification of IP-CAN bearer contexts for media by the UE

No additional clarifications are needed for the use of the EPC via cdma2000® HRPD as an IP-CAN.

O.2.2.5.1B Activation or modification of IP-CAN bearer contexts for media by the network

If the UE receives an activation request from the network for an IP-CAN bearer context which is associated with the IP-CAN bearer context used for signalling, the UE shall, based on the information contained in the Traffic Flow Template provided by the network, correlate the media IP-CAN bearer context with a currently ongoing SIP session establishment or SIP session modification.

If the UE receives a modification request from the network for an IP-CAN bearer context that is used for one or more media streams in an ongoing SIP session, the UE shall modify the related IP-CAN bearer context in accordance with the request received from the network.

O.2.2.5.1C Deactivation of of IP-CAN bearer contexts for media

Not applicable

O.2.2.5.2 Special requirements applying to forked responses

NOTE 1: The procedures in this subclause only apply when the UE requests activation and modification of media bearers. In the case where the network activates and modifies the media bearers the network takes care of the handling of media bearers in the case of forking.

Since the UE does not know that forking has occurred until a second, provisional response arrives, the UE requests resource allocation as required by the initial response received. If a subsequent provisional response is received, different alternative actions may be performed depending on the requirements in the SDP answer:

1) the bearer requirements of the subsequent SDP can be accommodated by the existing resources requested. The UE performs no further resource requests.

2) the subsequent SDP introduces different QoS requirements or additional IP flows. The UE requests further resource allocation according to subclause O.2.2.5.1.

3) the subsequent SDP introduces one or more additional IP flows. The UE requests further resource allocation according to subclause O.2.2.5.1.

NOTE 2: When several forked responses are received, the resources requested by the UE are the "logical OR" of the resources indicated in the multiple responses to avoid allocation of unnecessary resources. The UE does not request more resources than proposed in the original INVITE request.

When a final answer is received for one of the early dialogs, the UE proceeds to set up the SIP session. The UE shall release all the unneeded IP-CAN resources. Therefore, upon the reception of the first final 200 (OK) response for the INVITE request (in addition to the procedures defined in RFC 3261 [26] subclause 13.2.2.4), the UE shall:

1) in case resources were established or modified as a consequence of the INVITE request and forked provisional responses that are not related to the accepted 200 (OK) response, send release request to release the unneeded resources.

O.2.2.5.3 Unsuccessful situations

Not applicable.

O.2.2.6 Emergency service

O.2.2.6.1 General

Emergency services is not supported when the IP-CAN is the EPC via a cdma2000® HRPD access network.

O.2.2.6.1A Type of emergency service derived from emergency service category value

Not applicable.

O.2.2.6.1B Type of emergency service derived from extended local emergency number list

Not applicable.

O.2.2.6.2 eCall type of emergency service

The UE shall not send an INVITE request with Request-URI set to "urn:service:sos.ecall.manual" or "urn:service:sos.ecall.automatic".

O.2.2.6.3 Current location discovery during an emergency call

Void.

O.2A Usage of SDP

O.2A.0 General

Not applicable.

O.2A.1 Impact on SDP offer / answer of activation or modification of IP-CAN bearer context for media by the network

If, due to the activation of an IP-CAN bearer context from the network the related SDP media description needs to be changed the UE shall update the related SDP information by sending a new SDP offer within a SIP request, which is sent over the existing SIP dialog,

If the UE receives a modification request from the network for an IP-CAN bearer context that is used for one or more media streams in an ongoing SIP session, the UE shall:

1) if, due to the modification of the IP-CAN bearer context, the related SDP media description need to be changed, update the related SDP information by sending a new SDP offer within a SIP request, that is sent over the existing SIP dialog, and respond to the IP-CAN bearer context modification request.

NOTE: The UE can decide to indicate additional media streams as well as additional or different codecs in the SDP offer than those used in the already ongoing session.

O.2A.2 Handling of SDP at the terminating UE when originating UE has resources available and IP-CAN performs network-initiated resource reservation for terminating UE

If the UE receives an SDP offer where the SDP offer includes all media streams for which the originating side indicated its local preconditions as met, if the precondition mechanism is supported by the terminating UE and the IP-CAN performs network-initiated resource reservation for the terminating UE and the available resources are not sufficient for the received offer the terminating UE shall indicate its local preconditions and provide the SDP answer to the originating side without waiting for resource reservation.

NOTE: If the resource reservation is controlled by the network, the resource reservation request is initiated by the network after the P-CSCF has authorised the respective IP flows and provided the QoS requirements over the Rx interface to the PCRF as described in 3GPP TS 29.214 [13D].

O.2A.3 Emergency service

No additional procedures defined.

O.3 Application usage of SIP

O.3.1 Procedures at the UE

O.3.1.0 Void

O.3.1.0a IMS_Registration_handling policy

Not applicable.

O.3.1.1 P-Access-Network-Info header field

The UE shall always include the P-Access-Network-Info header field where indicated in subclause 5.1.

O.3.1.1A Cellular-Network-Info header field

Not applicable.

O.3.1.2 Availability for calls

Not applicable.

O.3.1.2A Availability for SMS

Void.

O.3.1.3 Authorization header field

When using SIP digest or SIP digest without TLS, the UE need not include an Authorization header field on sending a REGISTER request, as defined in subclause 5.1.1.2.1.

NOTE: In case the Authorization header field is absent, the mechanism only supports that one public user identity is associated with only one private user identity. The public user identity is set so that it is possible to derive the private user identity from the public user identity by removing SIP URI scheme and the following parts of the SIP URI if present: port number, URI parameters, and To header field parameters. Therefore, the public user identity used for registration in this case cannot be shared across multiple UEs. Deployment scenarios that require public user identities to be shared across multiple UEs that don't include an private user identity in the initial REGISTER request can be supported as follows:

- Assign each sharing UE a unique public user identity to be used for registration,

- Assign the shared public user identitiess to the implicit registration set of the unique registering public user identities assigned to each sharing UE.

O.3.1.4 SIP handling at the terminating UE when precondition is not supported in the received INVITE request, the terminating UE does not have resources available and IP-CAN performs network-initiated resource reservation for the terminating UE

Not applicable.

O.3.1.5 3GPP PS data off

Not applicable.

O.3.1.6 Transport mechanisms

No additional requirements are defined.

O.3.1.7 RLOS

Not applicable.

O.3.2 Procedures at the P-CSCF

O.3.2.0 Registration and authentication

Void.

O.3.2.1 Determining network to which the originating user is attached

The P-CSCF handling is as defined in subclause M.3.2.

NOTE: Emergency call support for the EPC IP-CAN is not specified in this release. A common P-Access-Network-Info header field value is used for both cdma2000® HRPD based IP-CANs (i.e. HRPD access specified by 3GPP2 X.S0011-E [127], and HRPD access as specified by 3GPP2 X.P0057 [86C]). The result of this is that in both cases the handling in the P-CSCF will be identical. If an operator deploys an IM CN subsystem with both cdma2000® HRPD based IP-CANs, the P-CSCF has no means of distinguishing one from the other. The emergency call handling for the EPC IP-CAN using cdma2000® HRPD access as specified by 3GPP2 X.P0057 [86C] is out of scope for this release of this specification, and therefore all identified emergency calls with a P-Access-Network-Info header field value of "3GPP2-1X-HRPD" will be handled with a 380 (Alternative Service) response when HRPD IP-CAN emergency support is not active.

O.3.2.2 Location information handling

Void.

O.3.2.3 Void

O.3.2.4 Void

O.3.2.5 Void

O.3.2.6 Resource sharing

If the P-CSCF supports resource sharing, PCC is supported for this access technology and if according to local policy, the P-CSCF shall apply the procedures in subclause L.3.2.6.

O.3.2.7 Priority sharing

If the P-CSCF supports priority sharing, PCC is supported for this access technology and if according to operator policy, the P-CSCF shall apply the procedures in subclause L.3.2.7.

O.3.2.8 RLOS

Not applicable.

O.3.3 Procedures at the S-CSCF

O.3.3.1 Notification of AS about registration status

The S-CSCF handling is as defined in subclause M.3.3.1.

O.3.3.2 RLOS

Not applicable.

O.4 3GPP specific encoding for SIP header field extensions

O.4.1 Void

O.5 Use of circuit-switched domain

There is no CS domain in this access technology.

Annex P (informative):
Void

Annex Q (normative): IP-Connectivity Access Network specific concepts when using the cdma2000® 1x Femtocell Network to access IM CN subsystem

Q.1 Scope

The present annex defines IP-CAN specific requirements for a call control protocol for use in the IP Multimedia (IM) Core Network (CN) subsystem based on the Session Initiation Protocol (SIP), and the associated Session Description Protocol (SDP), where the IP-CAN is an IP network as incorporated into the cdma2000® 1x femtocell network subsystem [86E].

Q.2 cdma2000® 1x Femtocell Network aspects when connected to the IM CN subsystem

Q.2.1 Introduction

In the cdma2000® 1x femtocell network subsystem, the cdma2000® 1x Mobile Station (MS) accesses the IM CN subsystem by utilising the services provided by the cdma2000® 1x Femtocell Access Point (FAP) [86E].

NOTE: Protocol between the cdma2000® 1x MS and the cdma2000® 1x FAP is out of scope of this document.

The cdma2000® 1x FAP 3GPP2 X.P0059-200 [86E] acts as a UE toward the IM CN subsystem.

From the perspective of the FAP, it is assumed that one or more IP-CAN bearer(s) are provided, in the form of connection(s) managed by the layer 2.

Q.2.2 Procedures at the UE

Q.2.2.1 Activation and P-CSCF discovery

Unless a static IP address is allocated to the cdma2000® 1x FAP, prior to communication with the IM CN subsystem, the cdma2000® 1x FAP shall perform a Network Attachment procedure depending on the used cdma2000® 1x FAP access type. When using a cdma2000® 1x FAP access, both IPv4 and IPv6 may be used to access the IM CN subsystem. The cdma2000® 1x FAP may request a DNS Server IPv4 address(es) via RFC 2132 [20F] or a DNS Server IPv6 address(es) via RFC 3315 [40].

When using IPv4, the cdma2000® 1x FAP may acquires a P-CSCF address(es) by using the DHCP (see RFC 2132 [20F]), the DHCPv4 options for SIP servers (see RFC 3361 [35A]), and RFC 3263 [27A].

In case the DHCP server provides several P-CSCF addresses or FQDNs to the cdma2000® 1x FAP, the cdma2000® 1x FAP shall select the P-CSCF address or FQDN as indicated in RFC 3361 [35A]. If sufficient information for P-CSCF address selection is not available, selection of the P-CSCF address by the cdma2000® 1x FAP is implementation specific.

When using IPv6, the cdma2000® 1x FAP may acquires a P-CSCF address(es) by using the DHCPv6 (see RFC 3315 [40] and RFC 3646 [56C]), the DHCPv6 options for SIP servers (see RFC 3319 [41]), and RFC 3263 [27H].

In case the DHCP server provides several P-CSCF addresses or FQDNs to the cdma2000® 1x FAP, the cdma2000® 1x FAP shall select the P-CSCF address or FQDN as indicated in RFC 3319 [41]. If sufficient information for P-CSCF address selection is not available, selection of the P-CSCF address by the cdma2000® 1x FAP is implementation specific.

Q.2.2.1A Modification of IP-CAN used for SIP signalling

Not applicable.

Q.2.2.1B Re-establishment of IP-CAN used for SIP signalling

Not applicable.

Q.2.2.2 Void

Q.2.2.3 Void

Q.2.2.4 Void

Q.2.2.5 Handling of the IP-CAN for media

Q.2.2.5.1 General requirements

The cdma2000® 1x FAP uses the bearer used for signalling also for transmission of media.

Q.2.2.5.1A Activation or modification of IP-CAN for media by the UE

Not applicable.

Q.2.2.5.1B Activation or modification of IP-CAN for media by the network

Not applicable.

Q.2.2.5.1C Deactivation of IP-CAN for media

Not applicable

Q.2.2.5.2 Special requirements applying to forked responses

Not applicable.

Q.2.2.5.3 Unsuccessful situations

Not applicable.

Q.2.2.6 Emergency service

Q.2.2.6.1 General

Emergency calls are perceived as regular calls from the perspective of the IM CN subsystem. Entities outside the IM CN subsystem identify and route such calls to PSAP.

Q.2.2.6.1A Type of emergency service derived from emergency service category value

Not applicable.

Q.2.2.6.1B Type of emergency service derived from extended local emergency number list

Not applicable.

Q.2.2.6.2 eCall type of emergency service

The UE shall not send an INVITE request with Request-URI set to "urn:service:sos.ecall.manual" or "urn:service:sos.ecall.automatic".

Q.2.2.6.3 Current location discovery during an emergency call

Void.

Q.2A Usage of SDP

Q.2A.0 General

Not applicable.

Q.2A.1 Impact on SDP offer / answer of activation or modification of IP-CAN for media by the network

Not applicable.

Q.2A.2 Handling of SDP at the terminating UE when originating UE has resources available and IP-CAN performs network-initiated resource reservation for terminating UE

Not applicable.

Q.2A.3 Emergency service

No additional procedures defined.

Q.3 Application usage of SIP

Q.3.1 Procedures at the UE

Q.3.1.0 Void

Q.3.1.0a IMS_Registration_handling policy

Not applicable.

Q.3.1.1 P-Access-Network-Info header field

The cdma2000® 1x FAP shall include the P-Access-Network-Info header field where indicated in subclause 5.1.

Q.3.1.1A Cellular-Network-Info header field

Not applicable.

Q.3.1.2 Availability for calls

Not applicable.

Q.3.1.2A Availability for SMS

Void.

Q.3.1.3 Authorization header field

Void.

Q.3.1.4 SIP handling at the terminating UE when precondition is not supported in the received INVITE request, the terminating UE does not have resources available and IP-CAN performs network-initiated resource reservation for the terminating UE

Not applicable.

Q.3.1.5 3GPP PS data off

Not applicable.

Q.3.1.6 Transport mechanisms

No additional requirements are defined.

Q.3.1.7 RLOS

Not applicable.

Q.3.2 Procedures at the P-CSCF

Q.3.2.0 Registration and authentication

Void.

Q.3.2.1 Determining network to which the originating user is attached

If access-type field in the P-Access-Network-Info header field indicated 3GPP2-1X-Femto access the P-CSCF shall assume that an initial request for a dialog or standalone transaction or an unknown method destined for a PSAP is initiated in the same country.

Q.3.2.2 Location information handling

Not applicable

Q.3.2.3 Void

Q.3.2.4 Void

Q.3.2.5 Void

Q.3.2.6 Resource sharing

Not applicable.

Q.3.2.7 Priority sharing

Not applicable.

Q.3.2.8 RLOS

Not applicable.

Q.3.3 Procedures at the S-CSCF

Q.3.3.1 Notification of AS about registration status

Not applicable

Q.3.3.2 RLOS

Not applicable.

Q.4 3GPP specific encoding for SIP header field extensions

Q.4.1 Void

Q.5 Use of circuit-switched domain

Not applicable

Annex R (normative): IP-Connectivity Access Network specific concepts when using the EPC via WLAN to access IM CN subsystem

R.1 Scope

The present annex defines IP-CAN specific requirements for a call control protocol for use in the IM CN subsystem based on the Session Initiation Protocol (SIP), and the associated Session Description Protocol (SDP), where the IP-CAN is the Evolved Packet Core (EPC) via Wireless Local Access Network (WLAN).

R.2 IP-CAN aspects when connected to the IM CN subsystem

R.2.1 Introduction

A UE accessing the IM CN subsystem, and the IM CN subsystem itself, utilise the services provided by the EPC and the WLAN to provide packet-mode communication between the UE and the IM CN subsystem.

Requirements for the UE on the use of these packet-mode services are specified in this clause.

R.2.2 Procedures at the UE

R.2.2.1 Establishment of IP-CAN bearer and P-CSCF discovery

Prior to communication with the IM CN subsystem:

NOTE 1: The UE performs access network discovery and selection procedures as specified in 3GPP TS 24.302 [8U] and executes access authentication signalling for access to the EPC between the UE and 3GPP AAA server, if applicable, as described in 3GPP TS 24.302 [8U] prior to perform the procedure to obtain a local IP address;

a) the UE establishes an IP-CAN bearer for SIP signalling as follows:

1) if the UE attaches to the EPC via S2b using untrusted WLAN IP access:

A) the UE shall obtain a local IP address using the WLAN IP access;

B) if the UE does not support procedures for access to the EPC via restrictive non-3GPP access network or unless the UE determines that the WLAN used is a restrictive non-3GPP access network, then the UE shall establish an IKEv2 security association and an IPsec ESP security association with ePDG as described in 3GPP TS 24.302 [8U]. If the UE supports the Fixed Access Broadband interworking, the UE shall apply the establishment of tunnel specified in 3GPP TS 24.139 [8X];

NOTE 2: UE can determine that the WLAN used is a restrictive non-3GPP access network if no IKEv2 response is received for the IKEv2 IKE_SA_INIT request, or using means out of scope of this specification.

C) if the UE supports procedures for access to the EPC via restrictive non-3GPP access network, and if the UE determines that the WLAN used is a restrictive non-3GPP access network, then the UE may perform procedures for access to the EPC via restrictive non-3GPP access network as described in 3GPP TS 24.302 [8U], and may establish an IKEv2 security association and an IPsec ESP security association with ePDG via the firewall traversal tunnel;

D) the IKEv2 security association and the IPsec ESP security association (tunnel) shall remain active throughout the period the UE is connected to the IM CN subsystem, i.e. from the initial registration and at least until the deregistration; and

E) the UE may carry both signalling and media on an IPsec ESP security association;

2) if the UE attaches to the EPC via S2c using the WLAN IP access:

A) the UE shall obtain a local IP address;

B) the UE shall establish an IKEv2 security association and an IPsec ESP security association as described in 3GPP TS 24.302 [8U] and 3GPP TS 24.303 [8V]. If the UE supports the Fixed Access Broadband interworking, the UE shall apply the establishment of tunnel specified in 3GPP TS 24.139 [8X];

C) the IKEv2 security association and the IPsec ESP security association (tunnel) shall remain active throughout the period the UE is connected to the IM CN subsystem, i.e. from the initial registration and at least until the deregistration;and

D) the UE may carry both signalling and media on an IPsec ESP security association.

3) if the UE attaches to the EPC via S2a using a trusted WLAN IP access:

A) the IPv4 address and/or IPv6 prefix is allocated as specified in 3GPP TS 24.302 [8U]; and

B) the UE IP address shall remain valid throughout the period the UE is connected to the IM CN subsystem, i.e. from the initial registration and at least until the deregistration;

The UE can determine trust relationship of a non-3GPP IP access network as specified in 3GPP TS 24.302 [8U]; and

b) the UE shall aquire a P-CSCF address(es).

The methods for P-CSCF discovery are:

I. Use DHCP mechanism

II. Use DNS

When using IPv4, the UE may request a DNS Server IPv4 address(es) via RFC 2132 [20F]. When using IPv6, the UE may request a DNS Server IPv6 address(es) via RFC 3315 [40] and RFC 3646 [56C].

III. Obtain the list of P-CSCF address(es) from the IMS management object

IV. Obtain P-CSCF address(es) using signalling for access to the EPC via WLAN.

If the UE attaches to the EPC via S2b using untrusted WLAN IP access, the UE shall request P-CSCF IPv4 address(es), P-CSCF IPv6 address(es) or both using the P_CSCF_IP4_ADDRESS attribute, the P_CSCF_IP6_ADDRESS attribute or both in the CFG_REQUEST configuration payload as described in 3GPP TS 24.302 [8U]. The network can provide the UE with the P-CSCF IPv4 address(es), P-CSCF IPv6 address(es) or both using the P_CSCF_IP4_ADDRESS attribute, the P_CSCF_IP6_ADDRESS attribute or both in the CFG_REPLY configuration payload as described in 3GPP TS 24.302 [8U]. If the UE receives multiple P-CSCF IPv4 or IPv6 addresses, the UE shall assume that the list is ordered top-down with the first P-CSCF address within the CFG_REPLY configuration payload as the P-CSCF address having the highest preference and the last P-CSCF address within the CFG_REPLY configuration payload as the P-CSCF address having the lowest preference.

If the UE attaches to the EPC via S2a using trusted WLAN IP access using single-connection mode, the UE shall indicate request for P-CSCF IPv4 address(es), P-CSCF IPv6 address(es) or both within the PROTOCOL_CONFIGURATION_OPTIONS item of the message with SCM_REQUEST message type as described in 3GPP TS 24.302 [8U]. The network can provide the UE with the P-CSCF IPv4 address(es), P-CSCF IPv6 address(es) or both within the PROTOCOL_CONFIGURATION_OPTIONS item of the message with SCM_RESPONSE message type as described in 3GPP TS 24.302 [8U]. If the UE receives multiple P-CSCF IPv4 or IPv6 addresses, the UE shall assume that the list is ordered top-down with the first P-CSCF address within the PROTOCOL_CONFIGURATION_OPTIONS item as the P-CSCF address having the highest preference and the last P-CSCF address within the PROTOCOL_CONFIGURATION_OPTIONS item as the P-CSCF address having the lowest preference.

If the UE attaches to the EPC via S2a using trusted WLAN IP access using multi-connection mode, the UE shall indicate request for P-CSCF IPv4 address(es), P-CSCF IPv6 address(es) or both within the Protocol Configuration Options information element of the PDN CONNECTIVITY REQUEST message as described in 3GPP TS 24.244 [8ZB]. The network can provide the UE with the P-CSCF IPv4 address(es), P-CSCF IPv6 address(es) or both within the Protocol Configuration Options information element of the PDN CONNECTIVITY ACCEPT message as described in 3GPP TS 24.244 [8ZB]. If the UE receives multiple P-CSCF IPv4 or IPv6 addresses, the UE shall assume that the list is ordered top-down with the first P-CSCF address within the Protocol Configuration Options information element as the P-CSCF address having the highest preference and the last P-CSCF address within the Protocol Configuration Options information element as the P-CSCF address having the lowest preference.

The UE shall use method III to select a P-CSCF, if a P-CSCF is to be discovered in the home network and the WLAN, to which the UE is attached, is connected to a visited network.

The UE can freely select method I, II, III or IV for P-CSCF discovery if:

- the UE is in the home network; or
- the WLAN, to which the UE is attached, is connected to a visited network and the P-CSCF is to be discovered in the visited network.

If DHCP is used, the following procedures apply:

Upon establishing an IP-CAN, the UE may use the Dynamic Host Configuration Protocol (DHCP) specified in RFC 2131 [40A] or Dynamic Host Configuration Protocol for IPv6 (DHCPv6) specified in RFC 3315 [40] to discover the P-CSCF.

Prior to accessing the DHCP server, the UE will have obtained an IP address via means other than DHCP and DHCPv6.

If the UE uses DHCP for P-CSCF discovery and the UE is unaware of the address of the DHCP server, the UE sends the DHCPINFORM using the limited broadcast IP address (i.e., 255.255.255.255) and UDP port 67. If the UE knows the IP address of the DHCP server, the UE shall send the DHCPINFORM to the DHCP server's unicast IP address and UDP port 67. The DHCP server sends the DHCPACK on the IP address specified in the Client IP Address field of the DHCPINFORM. The DHCP server can include, in the DHCPACK, the SIP Server DHCP Option specified in RFC 3361 [35A], which carries either a list of IPv4 address(es) of the P-CSCF(s) or a list of DNS fully qualified domain name(s) that can be mapped to one or more P-CSCF(s). If the UE uses DHCPv6 for P-CSCF discovery and the UE is unaware of the address of the DHCP Server, the UE shall send an Information Request using the IPv6 multicast address FF02::1:2 and the UDP port 547. If the UE knows the IP address of the DHCPv6 server, the UE shall send the Information Request message to the DHCPv6 server's IP address and UDP port 547. In the Information Request, the UE can request either one or both the SIP Servers Domain Name List option and the SIP Servers IPv6 Address List option specified in RFC 3319 [41]. The DHCP server sends the Reply to the IP address specified in the Information Request. The DHCP server can include in the Reply either one or both the SIP Servers Domain Name List option and the SIP Servers IPv6 Address List option, as requested by the UE.

If the UE supports the optional configuration parameter "Access_Point_Name_Parameter_Reading_Rule", as defined in 3GPP TS 24.167 [8G] and has been configured with this parameter, then the UE shall use it to retrieve the access point name to use in the establishment of IP-CAN bearer.

In case several P-CSCF's IP addresses or domain names are provided to the UE, the UE shall perform P-CSCF selection according to RFC 3361 [35A] or RFC 3319 [41]. The UE shall perform the procedure for the resolution of domain name according to RFC 3263 [27A]. If sufficient information for P-CSCF address selection is not available, selection of the P-CSCF address by the UE is implementation specific.

When:

- the UE obtains an IP-CAN bearer for SIP signalling by performing handover of the connection from another IP-CAN;
- IP address of the UE is not changed during the handover; and
- the UE already communicates with the IM CN subsystem via the connection with the other IP-CAN, e.g. the UE determines that its contact with host portion set to the UE IP address (or FQDN of the UE) associated with the connection with the other IP-CAN has been bound to a public user identity;

the UE shall continue using the P-CSCF address(es) acquired in the other IP-CAN.

The UE may support the policy on when a UE roaming in a VPLMN is allowed to transfer the PDN connection providing access to IMS between EPC via WLAN and EPS as specified in subclause L.2.1.1. If the UE roams in the EPS IP-CAN, has a session and the policy indicates "roaming in a VPLMN and having an ongoing session, is not allowed to transfer the PDN connection providing access to IMS between EPC via WLAN and EPS", the UE shall not handover the PDN connection providing access to IMS from EPS to EPC via WLAN.

If the UE roams in the EPS IP-CAN, has a session and the policy indicates "roaming in a VPLMN and having an ongoing session, is allowed to transfer the PDN connection providing access to IMS between EPC via WLAN and EPS", the UE shall, if not prevented by other rules or policies, handover the PDN connection providing access to IMS from EPS to EPC via WLAN.

If the UE roams in the EPS IP-CAN and the policy indicates "roaming in a VPLMN is not allowed to transfer the PDN connection providing access to IMS between EPC via WLAN and EPS, irrespective of if the UE is in a session or not", the UE shall not handover the PDN connection providing access to IMS from EPS to EPC via WLAN. The UE can re-establish a new PDN connection to another IP-CAN type in idle mode, e.g. due to UE domain preference.

The UE may support the policy on when a UE is allowed to transfer a PDN connection providing access to IMS between EPC via WLAN and 5GS as specified in subclause U.2.1.1. If the UE is in a session and the policy indicates "is not allowed to transfer the PDN connection providing access to IMS between EPC via non-3GPP access and 5GCN via NG-RAN" or if the UE is roaming when in a session and the policy indicates "a UE roaming in a VPLMN and having an ongoing IMS session, is not allowed to transfer the PDN connection providing access to IMS between EPC via non-3GPP access and 5GCN via NG-RAN" the UE shall not handover the PDN connection providing access to IMS from 5GS to EPC via WLAN.

If the UE is in a seesion in the EPS IP-CAN and the policy indicates "a UE having an ongoing IMS session, is allowed to transfer the PDN connection providing access to IMS between EPC via non-3GPP access and 5GCN via NG-RAN" or the UE is roaming in an EPS IP-CAN and the policy indicates "a UE roaming in a VPLMN and having an ongoing IMS session, is allowed to transfer the PDN connection providing access to IMS between EPC via non-3GPP access and 5GCN via NG-RAN" the UE shall, if not prevented by other rules or policies, handover the PDN connection providing access to IMS from 5GS to EPC via WLAN.

If the UE is in a 5GS IP-CAN and the policy indicates "a UE is not allowed to transfer a PDN connection providing access to IMS, if any, between EPC via non-3GPP access and 5GCN via NG-RAN, irrespective of if the UE has an ongoing IMS session or not" or the UE is roaming in a 5GS IP-CAN and the policy indicates "a UE roaming in a VPLMN is not allowed to transfer a PDN connection providing access to IMS, if any, between EPC via non-3GPP access and 5GCN via NG-RAN, irrespective of if the UE has an ongoing IMS session or not" the UE shall not handover the PDU session providing access to IMS from 5GS to EPC via WLAN.

R.2.2.1A Modification of an IP-CAN used for SIP signalling

Not applicable.

R.2.2.1B Re-establishment of the IP-CAN used for SIP signalling

If the UE registered a public user identity with an IP address allocated for the APN of the IP-CAN bearer for SIP signalling, the IP-CAN bearer for SIP signalling is deactivated as result of signalling from the network, and:

i) the signalling from the network results in requiring the UE to initiate activation of the IP-CAN bearer for SIP signalling; or

ii) the UE needs to continue having a public user identity registered with an IP address allocated for the APN;

and the UE is allowed to activate the IP-CAN bearer for SIP signalling, the UE shall:

A) if the non-access stratum is performing activation of the IP-CAN bearer for SIP signalling for the APN triggered as result of the signalling from the network, wait until the activation of the IP-CAN bearer for SIP signalling for the APN finishes;

B) if the non-access stratum is not performing activation of the IP-CAN bearer for SIP signalling for the APN, perform the procedures in subclause R.2.2.1, bullet a); and

C) if the IP-CAN bearer for SIP signalling is available:

- perform the procedures in subclause R.2.2.1, bullet b); and
- if a P-CSCF address was acquired, perform a new initial registration according to subclause 5.1.1.2.

R.2.2.1C P-CSCF restoration procedure

A UE supporting the P-CSCF restoration procedure performs one of the following procedures:

A) if the UE used method IV for P-CSCF discovery, the UE has previously sent the "P-CSCF Re-selection support" PCO indicator (in trusted non-3GPP access network) or the P-CSCF_RESELECTION_SUPPORT IKEv2 attribute (in untrusted non-3GPP access network) during the PDN connection establishment, and the UE receives one or more P-CSCF address(es) during the TWAG initiated PDN connectivity modification procedure (in trusted non-3GPP access network) or the ePDG initiated modification (in untrusted non-3GPP access network), then the UE shall acquire a P-CSCF address from the one or more P-CSCF addresse(s). If more than one P-CSCF addresses of the same IP address type are included, then the UE shall assume that the more than one P-CSCF addresses of the same IP address type are prioritised with the first P-CSCF address within the Protocol Configuration Options information element or within the IKEv2 configuration payload as the P-CSCF address having the highest priority; and

B) if the UE uses RFC 6223 [143] as part of P-CSCF restoration procedures, and the P-CSCF fails to respond to keep-alive requests, then the UE shall acquire a different P-CSCF address using any of the methods described in the subclause R.2.2.1.

When the UE has acquired the P-CSCF address, the UE shall perform an initial registration as specified in subclause 5.1.

NOTE: For UEs using procedure A described above, the network ensures that P-CSCF address(es) received in the Protocol Configuration Options information element during TWAG initiated PDN connectivity modification procedure (in trusted non-3GPP access network), and P-CSCF address(es) received in the P-CSCF_IP6_ADDRESS attribute, the P-CSCF_IP4_ADDRESS attribute or both during ePDG initiated modification (in untrusted non-3GPP access network) are sent only during P-CSCF restoration procedures as defined in subclause 5 of 3GPP TS 23.380 [7D].

R.2.2.2 Void

R.2.2.3 IP-CAN support of DHCP based P-CSCF discovery

When using WLAN IP access via S2c to access the EPC, the Home Agent (HA) in case of Mobile IP with reverse tunneling, can forward the packet to one or more local DHCP servers, or relay the packet to a specific DHCP server. The HA in case of Mobile IP with reverse tunnelling, does not forward the DHCPINFORM (or Information-Request) to any UE.

NOTE 1: For forwarding the DHCPINFORM or Information-Request, the HA in case of Mobile IP with reverse tunnelling, does not change the destination IP address of the packet.

NOTE 2: For relaying the DHCPINFORM or Information-Request, the HA in case of Mobile IP with reverse tunnelling inserts a DHCP server's IP address in the destination IP address field of the packet.

R.2.2.4 Void

R.2.2.5 Tunnel procedures for media

R.2.2.5.1 General requirements

The UE can establish media streams that belong to different SIP sessions on the same tunnel when accessing the EPC via untrusted WLAN.

During establishment of a session, the UE establishes data streams(s) for media related to the session. When using untrusted WLAN IP access via S2c to access the EPC, such data stream(s) may result in activation of additional IPsec ESP security associations (tunnels).

If the capabilities of the originating UE, or operator policy at the ePDG prevents the originating UE from establishment of additional IPsec ESP security associations (tunnels) according to the media grouping attributes given by the P-CSCF in accordance with RFC 3524 [54], the UE will not establish such grouping of media streams. Instead, the originating UE shall negotiate media parameters for the session according to RFC 3264 [27B].

If the capabilities of the terminating UE or operator policy at the ePDG prevents the originating UE from establishment of additional IPsec ESP security associations (tunnels) according to the media grouping attributes given by the P-CSCF in accordance with RFC 3524 [54], the UE will not establish such grouping of media streams. Instead, the terminating UE shall handle such SDP offers in accordance with RFC 3388 [53].

The UE can receive a media authorization token in the P-Media-Authorization header field from the P-CSCF according to RFC 3313 [31]. If a media authorization token is received in the P-Media-Authorization header field when a SIP session is initiated, the UE shall reuse the existing tunnel and ignore the media authorization token.

R.2.2.5.1A Modification of tunnel for media by the UE

The tunnel modification procedure for the UE shall follow the procedures specified in 3GPP TS 24.302 [8U]. If the UE supports the Fixed Access Broadband interworking, the modification of tunnel specified in 3GPP TS 24.139 [8X] shall be applied.

R.2.2.5.1B Modification of tunnel for media by the network

The UE shall follow the procedure specified in 3GPP TS 24.302 [8U] for the modification of tunnel. If the UE supports the Fixed Access Broadband interworking, the modification of tunnel specified in 3GPP TS 24.139 [8X] shall be applied.

If the UE attaches to the EPC via S2b using untrusted WLAN IP access, and IKEv2 multiple bearer PDN connectivity is used in the PDN connection according to 3GPP TS 24.302 [8U], then:

1) if:

   A) an additional IPSec ESP tunnel is established according to 3GPP TS 24.302 [8U]; or

   B) an IPSec ESP tunnel is modified according to 3GPP TS 24.302 [8U];

   the UE shall, based on the information contained in the TFT Notify payload, correlate the media IPSec ESP tunnel with SDP media descriptions of a currently ongoing SIP session establishment or SIP session modification.

R.2.2.5.1C Deactivation of tunnel for media

Not applicable.

R.2.2.5.2 Special requirements applying to forked responses

Since the UE is unable to perform bearer modification, forked responses place no special requirements on the UE.

R.2.2.5.3 Unsuccessful situations

Not applicable.

R.2.2.6 Emergency service

R.2.2.6.1 General

In this release of the specification, a WLAN, conforming to the requirements in this annex, defines emergency bearers. Emergency session is supported over the WLAN access if the UE has failed or has not been able to use 3GPP access to set up an emergency session as described in 3GPP TS 23.167 [4B] annex J.

IMS emergency session is also supported for UEs with unavailable IMSI (i.e. a UE without USIM) or unauthenticated IMSI. Some jurisdictions allow emergency calls to be made when the UE does not contain an ISIM or USIM, or where the credentials are not accepted.

When the UE is attached over a WLAN access and detects an emergency call attempt, if the UE supports the emerg-non3gpp timer defined in table 7.8.1, the UE shall start the emerg-non3gpp timer when starting a domain selection searching for a 3GPP access usable to establish an emergency call. The UE shall stop the timer when a 3GPP access supporting emergency call is found. When the emerg-non3gpp timer expires, the UE shall consider that it has failed to use 3GPP access to setup the emergency call and shall attempt to setup the emergency call over the available WLAN access.

The UE may support being configured for the emerg-non3gpp timer using one or more of the following methods:

a) the Timer_Emerg_non3gpp leaf of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C];

b) the Timer_Emerg_non3gpp leaf of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

c) the Timer_Emerg_non3gpp leaf of 3GPP TS 24.167 [8G].

EPC procedures for emergency session using WLAN are defined for both trusted WLAN access via S2a, depending on the TWAN usage mode, and untrusted WLAN access via S2b to access EPC.

When the IM CN subsystem is selected as the domain for the emergency call attempt, and the UE uses:

- untrusted WLAN access via S2b, the UE determines that the EPC supports emergency bearer services by selecting or using an ePDG that has indicated its capability of support for emergency services, as specified in subclause 7.2.1A of 3GPP TS 24.302 [8U]; or

- trusted WLAN access via S2a, the UE determines that the EPC, accessed in usage modes multi-connection mode or single-connection mode, supports emergency bearer services if the CONNECTION_MODE_CAPABILITY item in the EAP-Request/AKA'-Challenge message indicates support of emergency services, as specified in 3GPP TS 24.302 [8U].

When the IM CN subsystem is selected as the domain for the emergency call attempt, and the UE uses untrusted WLAN access via S2b, the UE determines whether it is currently attached to its home operator's network (e.g. HPLMN) or not (e.g. VPLMN) after it has determined that the core network supports emergency bearer services.

When establishing an IMS emergency session using trusted WLAN access via S2a, the UE shall establish an IMS emergency session over trusted WLAN access depending on the usage mode used to access EPC. When using the usage mode single-connection mode, subclause 6.4.2.6.2A of 3GPP TS 24.302 [8U] applies. When using the usage mode multi-connection mode, subclause 6.4.2.6.3A of 3GPP TS 24.302 [8U] applies. The procedures for attaching to the EPC via S2a using a trusted WLAN IP access, as described in subclause R.2.2.1 of this specification apply accordingly.

When establishing an IMS emergency session using untrusted WLAN access via S2b, the UE shall establish an IMS emergency session over untrusted non-3GPP access as specified in 3GPP TS 24.302 [8U]. The procedures for attaching to the EPC via S2b using untrusted WLAN IP access, as described in subclause R.2.2.1 of this specification apply accordingly.

If the ME is equipped with a UICC, in order to find out whether the UE is attached to the home PLMN or to the visited PLMN, the UE shall compare the MCC and MNC values derived from its IMSI with the MCC and MNC of the PLMN the UE is attached to. If the MCC and MNC of the PLMN the UE is attached to do not match with the MCC and MNC derived from the IMSI, then for the purposes of emergency calls in the IM CN subsystem the UE shall consider to be attached to a VPLMN. If the ME is not equipped with a UICC, the procedure to find d out whether the UE is attached to the home PLMN or to the visited PLMN for the purpose of emergency calls in the IM CN subsystem, is implementation specific.

NOTE 1: The UE verifies if a detected emergency number is still present in the Extended Local Emergency Number List applicable to the PLMN it is currently using. It is possible for the number to no longer be present in the Extended Local Emergency Number List if:

- the PLMN attached to relies on the Local Emergency Number List for deriving a URN; or
- the previously received Extended Emergency Number List Validity field indicated "Extended Local Emergency Numbers List is valid only in the PLMN from which this IE is received".

If the UE detected an emergency number, the UE subsequently uses a different PLMN than the PLMN from which the UE received the last Extended Local Emergency Number List:

NOTE 2: The UE has either attached to or authenticated (e.g. due to EAP-3GPP-LimitedService based access authentication, see 3GPP TS 24.302 [8U]) with a PLMN via WLAN prior to detecting the emergency number or because of detecting the emergency number.

- the dialled number is not stored in the ME, in the USIM and in the Local Emergency Number List;

then:

a) if the UE supports provision and handling of local emergency numbers as defined in 3GPP TS 24.302 [8U], the UE has received the local emergency numbers using any of the methods defined in subclause 4.7 in 3GPP TS 24.302 [8U], the dialled number matches a received local emergency number, then the UE derives a URN as defined in 3GPP TS 24.302 [8U] or using the procedures in subclause R.2.2.6.1A, depending on the method used to provision the local emergency number;

b) otherwise, the UE shall attempt UE procedures for SIP that relate to emergency using emergency service URN "urn:service:sos".

If the dialled number is equal to a local emergency number stored in the Extended Local Emergency Number List (as defined in 3GPP TS 24.301 [8J]), then the UE shall recognize such a number as for an emergency call and:

- if the dialled number is equal to an emergency number stored in the ME, or in the USIM, then the UE shall perform either procedures in the subclause R.2.2.6.1B or the procedures in subclause R.2.2.6.1A; and
- if the dialled number in not equal to an emergency number stored in the ME, or in the USIM, then the UE shall perform procedures in the subclause R.2.2.6.1B.

If the dialled number is not equal to a local emergency number stored in the Extended Local Emergency Number List (as defined in 3GPP TS 24.301 [8J]) and:

- if the dialled number is equal to an emergency number stored in the ME, in the USIM or in the Local Emergency Number List (as defined in 3GPP TS 24.008 [8]), then the UE shall recognize such a number as for an emergency call and performs the procedures in subclause R.2.2.6.1A.

Once IPsec tunnel setup is completed, the UE shall follow the procedures described in subclause R.2.2.1 of this specification for establishment of IP-CAN bearer and P-CSCF discovery accordingly.

Upon reception of a 380 (Alternative Service) response to an INVITE request as defined in subclause 5.1.2A.1.1 and subclause 5.1.3.1, if:

- the 380 (Alternate Service) response contains a Contact header field;
- the value of the Contact header field is a service URN; and
- the service URN has a top-level service type of "sos";

then the UE determines that "emergency service information is included" as described 3GPP TS 23.167 [4B].

Upon reception of a 380 (Alternative Service) response to an INVITE request as defined in subclause 5.1.3.1, if the 380 (Alternate Service) response does not contain a Contact header field with service URN that has a top-level service type of "sos", then the UE determines that "no emergency service information is included" as described 3GPP TS 23.167 [4B].

Upon reception of a 380 (Alternative Service) response to an INVITE request as defined in subclause 5.1.2A.1.1 and subclause 5.1.3.1, the UE shall proceed as follows:

1) if a 3GPP access network is available and the UE has not already attempted to use a 3GPP access network to set up an emergency session as described in 3GPP TS 23.167 [4B] annex J, when the UE selects a domain in accordance with the conventions and rules specified in 3GPP TS 22.101 [1A] and 3GPP TS 23.167 [4B], the UE shall attempt to select a domain of the 3GPP access network, and:

   - if the CS domain is selected, the UE behaviour is defined in subclause 7.1.2 of 3GPP TS 23.167 [4B] and in annex B, annex L or annex U; and

   - if the IM CN subsystem is selected, the UE shall apply the procedures in subclause 5.1.6 with the exception of selecting a domain for the emergency call attempt;

   In addition, when the UE determines that "it has not been able to use 3GPP access to set up an emergency session" in accordance with subclause J.1 of 3GPP TS 23.167 [4B], the UE shall apply the procedures in subclause 5.1.6 using WLAN, with the exception of selecting a domain for the emergency call attempt; and

2) if a 3GPP access network is not available, then the UE shall apply the procedures in subclause 5.1.6 using WLAN, with the exception of selecting a domain for the emergency call attempt.

When the emergency session ends, the UE:

1) shall release the tunnel as described in 3GPP TS 24.302 [8U]; and

2) if EPC via WLAN is the preferred IP-CAN to access IM CN subsystem or if no 3GPP access is available:

   a) if the UE did not select the currently selected ePDG using procedures for selection of ePDG for non-emergency services, shall select an ePDG for non-emergency services as described in 3GPP TS 24.302 [8U];

   b) if the UE does not have an IP-CAN bearer for non-emergency SIP signalling, shall follow the procedures described in subclause R.2.2.1 for establishment of an IP-CAN bearer for SIP signalling and P-CSCF discovery; and

   c) if the UE determines that its contact associated with the IP-CAN bearer for non-emergency SIP signalling is not bound to a public user identity, shall perform an initial registration as specified in subclause 5.1.1.2 using the IP-CAN bearer for SIP signalling.

R.2.2.6.1A Type of emergency service derived from emergency service category value

The type of emergency service for an emergency number is derived from the settings of the emergency service category value (bits 1 to 5 of the emergency service category value as specified in subclause 10.5.4.33 of 3GPP TS 24.008 [8]). Table R.2.2.6.1 below specifies mappings between a type of emergency service and an emergency service URN. The UE shall use the mapping to match an emergency service URN and a type of emergency service. If a dialled number is an emergency number but does not map to a type of emergency service the service URN shall be "urn:service:sos".

Table R.2.2.6.1: Mapping between type of emergency service and emergency service URN

Type of emergency service	Emergency service URN
Police	urn:service:sos.police
Ambulance	urn:service:sos.ambulance
Fire Brigade	urn:service:sos.fire
Marine Guard	urn:service:sos.marine
Mountain Rescue	urn:service:sos.mountain

NOTE 1: It is not possible for a UE to indicate more than one type of emergency service in an emergency service URN.

If:

- the UE considers itself in the country of the HPLMN;

NOTE 2: It is out of scope of the present annex to define how the UE determines whether it considered itself in the country of the HPLMN. When the UE is in coverage of a 3GPP RAT, it can, for example, use the information derived from the available PLMN(s). In this case, the UE can match the MCC broadcasted on the BCCH of the 3GPP access against the UE's IMSI to determine if they belong to the same country, as defined in 3GPP TS 23.122 [4C]. If the UE is not in coverage of a 3GPP RAT, the UE can use other techniques, including user-provided location, for determining whether it is located in its home country or not.

- multiple types of emergency services can be derived for a dialled number from the information configured on the USIM; and

- no IP-CAN provided a local emergency number that matches the dialled number (see subclause 5.1.6.1);

NOTE 3: If the Non-3GPP emergency number indicator within the Non-3GPP NW provided policies IE (see 3GPP TS 24.008 [8]) provided through registration procedures over 3GPP access is set to "use of non-3GPP emergency numbers permitted", the UE also considers WLAN provided local emergency numbers (see 3GPP TS 24.302 [8U], subclause 4.7). If the Non-3GPP NW provided policies IE provided through registration procedures over 3GPP access is set to "use of non-3GPP emergency numbers not permitted", the UE does not consider WLAN provided local emergency numbers. If the Non-3GPP NW provided policies IE is not provided through registration procedures over 3GPP access, the UE does not consider WLAN provided local emergency numbers.

NOTE 4: A UE, only connected to a PLMN through non-3GPP access, considers the WLAN provided local emergency numbers if the applicable conditions in subclause 4.7 of 3GPP TS 24.302 [8U], are met.

then the UE shall map any one of these types of emergency service to an emergency service URN as specified in table R.2.2.6.1.

If the UE considers itself in the country of the HPLMN and an IP-CAN provided a local emergency number that matches the dialled number (see subclause 5.1.6.1), and if the UE:

- can derive one or more types of emergency service from the information received from the IP-CAN for the dialled number and the UE cannot derive types of emergency service from the information configured on the USIM for the dialled number; or

- derives identical types of emergency service from both the information received from the IP-CAN for the dialled number and from the information configured on the USIM for the dialled number;

then the UE shall map any one of these emergency service types to an emergency service URN as specified in table R.2.2.6.1.

NOTE 5: How the UE resolves clashes where an emergency number is associated with one or more different types of emergency service configured in the USIM and in information received from an IP-CAN, is implementation dependent.

R.2.2.6.1B Type of emergency service derived from extended local emergency number list

The Extended Local Emergency Number List (defined in 3GPP TS 24.301 [8J]) can contain sub-services of the associated emergency service URN for the detected emergency number.

If:

- the length of sub-services field is greater than "0", the UE shall construct the emergency service URN using "urn:service:sos" followed by adding a dot followed by the content of the sub-services field; and

- the length of sub-services field is "0", the UE shall use the emergency service URN "urn:service:sos".

EXAMPLE 1: For a detected number, if the sub-service is "gas", then the UE constructs "urn:service:sos.gas" as the associated emergency service URN.

EXAMPLE 2: For a detected number, if no sub-service is provided, then the UE uses "urn:service:sos" as the associated emergency service URN.

R.2.2.6.2 eCall type of emergency service

The UE shall not send an INVITE request with Request-URI set to "urn:service:sos.ecall.manual" or "urn:service:sos.ecall.automatic".

R.2.2.6.3 Current location discovery during an emergency call

The UE may support the current location discovery during an emergency call specified in subclause 5.1.6.8.2, subclause 5.1.6.8.3, subclause 5.1.6.8.4, and subclause 5.1.6.12.

R.2A Usage of SDP

R.2A.0 General

Not applicable.

R.2A.1 Impact on SDP offer / answer of activation or modification of IP-CAN for media by the network

If the UE is attached to the EPC via S2b using untrusted WLAN IP access, and IKEv2 multiple bearer PDN connectivity is used in the PDN connection according to 3GPP TS 24.302 [8U], then:

1) if an additional IPSec ESP tunnel is established according to 3GPP TS 24.302 [8U] and the related SDP media description needs to be changed, the UE shall update the related SDP information by sending a new SDP offer within a SIP request, which is sent over the existing SIP dialog; and

2) if an IPSec ESP tunnel is modified according to 3GPP TS 24.302 [8U] and the related SDP media description need to be changed, the UE shall update the related SDP information by sending a new SDP offer within a SIP request, that is sent over the existing SIP dialog.

NOTE: The UE can decide to indicate additional media streams as well as additional or different codecs in the SDP offer than those used in the already ongoing session.

R.2A.2 Handling of SDP at the terminating UE when originating UE has resources available and IP-CAN performs network-initiated resource reservation for terminating UE

Not applicable.

R.2A.3 Emergency service

No additional procedures defined.

R.3 Application usage of SIP

R.3.1 Procedures at the UE

R.3.1.0 Registration and authentication

Editor's note: [WID: TURAN-CT, CR#4993]: Best usage of this FTT-IMS establishment is when the UE does not establish the IKEv2 security association which is an option not specified in subclause R.2.2.1. This requires further study.

In order to reach the IM CN subsystem in some untrusted access networks, the UE may support:

- address and/or port number conversions provided by a NA(P)T or NA(P)T-PT as described in annex F and annex K; or
- the IP UE requested FTT-IMS establishment procedure specified in 3GPP TS 24.322 [8Y], which is applicable to direct access to an external IP network and not applicable to access through a PLMN.

If a UE supports one or both of these capabilities then a UE may progressively try them to overcome failure to reach the IPM CN subsystem. Use of these capabilities shall have the following priority order:

1) UE does not use capability because reaching the IMS without an intervening NA(P)T, NA(P)T-PT, or tunnel is preferred.
2) UE may use address and/or port number conversions provided by a NA(P)T or NA(P)T-PT as described in either annex F or annex K.
3) UE may use the UE requested FTT-IMS establishment procedure specified in 3GPP TS 24.322 [8Y]. If the UE uses the UE-requested FTT-IMS establishment procedure specified in 3GPP TS 24.322 [8Y], the UE considers itself to:
   - be configured to send keep-alives;
   - be directly connected to an IP-CAN for which usage of NAT is defined; and
   - be behind a NAT.

Optional procedures apply when the UE is supporting traversal of restrictive non-3GPP access network using STUN/TURN/ICE, as follows:

a) the protection of SIP messages is provided by utilizing TLS as defined in 3GPP TS 33.203 [19];

b) the mechanisms specified in this annex shall only be applicable when the IP traffic to the IMS core does not traverse through the Evolved Packet Core (EPC);

c) the UE shall establish the TLS connection to the P-CSCF on port 443 as defined in 3GPP TS 33.203 [19]. The UE shall use SIP digest with TLS for registration as specified in subclause 5.1. If the TLS connection is established successfully, the UE sends SIP signalling over the TLS connection to the P-CSCF;

d) the UE shall support the keep-alive procedures described in RFC 6223 [143];

NOTE 1: If the UE is configured to use an HTTP proxy, the UE use the HTTP CONNECT method specified in RFC 2817 [220] to request the HTTP proxy to establish the TCP connection with the P-CSCF. Once the UE has received a positive reply from the proxy that the TCP connection has been established, the UE initiates the TLS handshake with the P-CSCF and establishes the TLS connection.

e) the procedures described in subclause K.5.2 apply with the additional procedures described in the present subclause;

f) when using the ICE procedures for traversal of restrictive non-3GPP access network, the UE shall support the ICE TCP as specified in RFC 6544 [131] and TURN TCP as specified in RFC 6062 [221].

g) if the UE is configured to use TURN over TCP on port 80, the UE shall establish the TCP connection to TURN server on port 80. If the UE is configured to use TURN over TLS on port 443, the UE shall establish the TLS connection to the TURN server on port 443 as defined in 3GPP TS 33.203 [19]. If the UE is configured to use both, the UE should prefer to use TURN over TCP on port 80 to avoid TLS overhead;

h) if the connection is established successfully, the UE sends TURN control messages and media packets over the connection as defined in RFC 5766 [101].

NOTE 2: If the UE is configured to use an HTTP proxy, the UE use the HTTP CONNECT method specified in RFC 2817 [220] to request the HTTP proxy to establish the TCP connection with the TURN server. Then, if the UE is configured to use TURN over TLS on port 443 and the UE has received a positive reply from the proxy that the TCP connection has been established, the UE initiates the TLS handshake with the TURN server and establishes the TLS connection.

The UE shall perform reregistration of a previously registered public user identity bound to any one of its contact addresses when changing to an IP-CAN for which usage is specified in annex B, annex L or annex U. The reregistration is performed using the new IP-CAN.

NOTE 3: This document does not specify how the UE detects that the used IP-CAN has changed. The information that is forcing the reregistration is also used to generate the content for the P-Access-Network-Info header field.

NOTE 4: The UE will send the reregistration irrespective of whether it has a SIP dialog or not.

R.3.1.0a IMS_Registration_handling policy

Not applicable.

R.3.1.1 P-Access-Network-Info header field

The UE shall always include the P-Access-Network-Info header field where indicated in subclause 5.1.

R.3.1.1A Cellular-Network-Info header field

The UE:

1) using the Evolved Packet Core (EPC) via Untrusted Wireless Local Access Network (WLAN) as IP-CAN to access the IM CN subsystem; and

2) supporting one or more cellular radio access technology (e.g. E-UTRAN);

shall always include the Cellular-Network-Info header field specified in subclause 7.2.15, if the information is available, in every request or response in which the P-Access-Network-Info header field is present.

NOTE: If the Cellular-Network-Info header field includes radio cell identity, then the Cellular-Network-Info header field populated by the UE that supports Multi-RAT Dual Connectivity with the EPC as described in 3GPP TS 37.340 [264] will contain the information about the radio cell identity of the Master RAN node that is serving the UE.

R.3.1.2 Availability for calls

Not applicable.

R.3.1.2A Availability for SMS

Void.

R.3.1.3 Authorization header field

When using SIP digest or SIP digest without TLS, the UE need not include an Authorization header field on sending a REGISTER request, as defined in subclause 5.1.1.2.1.

NOTE: In case the Authorization header field is absent, the mechanism only supports that one public user identity is associated with only one private user identity. The public user identity is set so that it is possible to derive the private user identity from the public user identity by removing SIP URI scheme and the following parts of the SIP URI if present: port number, URI parameters, and To header field parameters. Therefore, the public user identity used for registration in this case cannot be shared across multiple UEs. Deployment scenarios that require public user identities to be shared across multiple UEs that don't include an private user identity in the initial REGISTER request can be supported as follows:

- Assign each sharing UE a unique public user identity to be used for registration,
- Assign the shared public user identitiess to the implicit registration set of the unique registering public user identities assigned to each sharing UE.

R.3.1.4 SIP handling at the terminating UE when precondition is not supported in the received INVITE request, the terminating UE does not have resources available and IP-CAN performs network-initiated resource reservation for the terminating UE

Not applicable.

R.3.1.5 3GPP PS data off

Not applicable.

R.3.1.6 Transport mechanisms

The transport mechansims as defined in subclause 4.2A are used with an additional requirement:

a) If the UE has attached to the EPC via untrusted non-3gpp access and uses IPSec tunnel mode, in order to reduce the risk of UDP fragmentation, the UE shall decrement the IPSec tunnel overhead from the path MTU between the UE and the ePDG prior applying the transport selection for SIP requests as defined in RFC 3261 [26] subclause 18.1.1.

NOTE: The method for discovering the maximum transmission unit for non-3GPP is implementation dependent and out of scope of 3GPP.

R.3.1.7 RLOS

Not applicable.

R.3.2 Procedures at the P-CSCF

R.3.2.0 Registration and authentication

The P-CSCF may support UEs connected via restrictive non-3GPP access network.

If the P-CSCF supports UEs connected via restrictive non-3GPP access network, when the P-CSCF receives a 200 (OK) response to a REGISTER request, if the contact address of REGISTER request contains an IP address assigned by the EFTF, and the UE's Via header field contains a "keep" header field parameter, then the P-CSCF shall add a value to the "keep" header field parameter of the UE's Via header field of the 200 (OK) response as defined in RFC 6223 [143].

Optional procedures apply when the P-CSCF is supporting traversal of restrictive non-3GPP access network using STUN/TURN/ICE, as follows:

NOTE: In this scenario, the restrictive non-3GPP access network coexists with NA(P)T device located in the customer premises domain:

a) the protection of SIP messages is provided by utilizing TLS as defined in 3GPP TS 33.203 [19];

b) the P-CSCF supporting these additional procedures should use SIP digest with TLS as defined in subclause 5 and the P-CSCF should inserts an IMS-ALG on the media plane;

c) the mechanisms specified in this annex shall only be applicable when the IP traffic to the IMS core does not traverse through the Evolved Packet Core (EPC);

d) the P-CSCF shall support the procedures defined in subclause 5.2, with the exception that the P-CSCF shall use SIP over TLS on port 443 as defined in 3GPP TS 33.203 [19];

e) when the UE has indicated support of the keep-alive mechanism defined in RFC 6223 [143], the P-CSCF shall indicate to the UE that it supports the keep-alive mechanism; and

f) the IMS-ALG in the P-CSCF shall support ICE procedures, as defined in subclause 6.7.2.7.

R.3.2.1 Determining network to which the originating user is attached

If the P-CSCF is configured to handle emergency requests, in order to determine from which network the request was originated the P-CSCF shall,

- if PCRF is used for this UE and 3GPP-User-Location-Info as specified in 3GPP TS 29.214 [13D] is available (see subclause 5.2.1), check the MCC and MNC received in 3GPP-User-Location-Info; and

- if PCRF is not used for this UE or 3GPP-User-Location-Info is not available, check the MCC and MNC fields received in the Cellular-Network-Info header field in the emergency request.

NOTE 1: The Cellular-Network-Info header field includes the MCC and the MNC of the cellular radio access network on which the UE most recently camped. The UE is not necessarily attached to a network via that cell or still camped on that cell.

NOTE 2: The above check can be against more than one MNC code stored in the P-CSCF.

R.3.2.2 Location information handling

Void.

R.3.2.3 Prohibited usage of PDN connection for emergency bearer services

If the P-CSCF detects that a UE uses a PDN connection for emergency bearer services for a non-emergency REGISTER request, the P-CSCF shall reject that request by a 403 (Forbidden) response.

NOTE: By assigning specific IP address ranges for a PDN connection for emergency bearer services and configuring those ranges in P-CSCF, the P-CSCF can detect based on the registered Contact address if UE uses an emergency PDN connection for initial registration.

R.3.2.4 Void

R.3.2.5 Void

R.3.2.6 Resource sharing

If PCC is supported for this access technology a P-CSCF supporting resource sharing shall apply the procedures in subclause L.3.2.6.

NOTE: Resource sharing has in this version of the specification no meaning in the WLAN IP CAN. However, since transfer to EPS IP CAN is seamless from P-CSCF point of view the resource sharing options used over the Rx interface when the UE is attached to the WLAN IP CAN could be used when the UE moves to EPS IP CAN and since Rx requires the resource sharing options to be included in the initial AAR the resource sharing options need to be included also when the UE attached to the WLAN IP CAN initiates or receives a INVITE request.

R.3.2.7 Priority sharing

If PCC is supported for this access technology a P-CSCF supporting priority sharing and if according to operator policy shall apply the procedures in subclause L.3.2.7.

R.3.2.8 RLOS

Not applicable.

R.3.3 Procedures at the S-CSCF

R.3.3.1 Notification of AS about registration status

Not applicable.

R.3.3.2 RLOS

Not applicable.

R.4 3GPP specific encoding for SIP header field extensions

R.4.1 Void

R.5 Use of circuit-switched domain

Void.

Annex S (normative): IP-Connectivity Access Network specific concepts when using DVB-RCS2 to access IM CN subsystem

S.1 Scope

The present annex defines IP-CAN specific requirements for a call control protocol for use in the IP Multimedia (IM) Core Network (CN) subsystem based on the Session Initiation Protocol (SIP), and the associated Session Description Protocol (SDP), where the IP-CAN is DVB-RCS2 satellite access network.

DVB-RCS2 (Second Generation DVB Interactive Satellite System) is a term referring to the ETSI standard ETSI TS 101 454-1 [193], ETSI EN 301 545-2 [194], ETSI TS 101 545-3 [195] for 2nd generation DVB satellite based access network technology with a return channel for two-way communication.

S.2 DVB-RCS2 aspects when connected to the IM CN subsystem

S.2.1 Introduction

A UE accessing the IM CN subsystem, and the IM CN subsystem itself, utilise the services provided by the DVB-RCS2 satellite access network to provide packet-mode communication between the UE and the IM CN subsystem.

From the perspective of the UE, the necessary IP-CAN bearer for signalling is transparently available to the UE.

The UE is not directly involved in requests for IP-CAN bearer(s) for media flow(s). The IM CN interacts with the PCRF in the DVB-RCS2 IP-CAN to establish IP-CAN bearer(s) for media flow(s), on behalf of the UE.

S.2.2 Procedures at the UE

S.2.2.1 Activation and P-CSCF discovery

Prior to communication with the IM CN subsystem, the UE shall:

a) establish a connection to a DVB-RCS2 RCST depending on local configuration; the RCST shall have completed the RCST commissioning and initialization procedures as described in ETSI TS 101 545-3 [195], and thus have achieved IP connectivity to the NCC of an SVN;

b) obtain an IP address using the standard IETF protocols (e.g., DHCP or IPCP). The UE shall fix the obtained IP address throughout the period the UE is connected to the IM CN subsystem, i.e. from the initial registration and at least until the last deregistration; and

c) acquire a P-CSCF address(es), according to one of the methods described in subclause 9.2.1; the method available to the UE is determined by the SVN to which the RCST is connected.

S.2.2.1A Modification of IP-CAN bearer used for SIP signalling

Not applicable.

S.2.2.1B Re-establishment of IP-CAN bearer used for SIP signalling

Not applicable.

S.2.2.1C P-CSCF restoration procedure

A UE supporting the P-CSCF restoration procedure uses the keep-alive procedures described in RFC 6223 [143].

If the P-CSCF fails to respond to keep-alive requests the UE shall acquire a different P-CSCF address using any of the methods described in the subclause S.2.2.1 and perform an initial registration as specified in subclause 5.1.

S.2.2.2 Void

S.2.2.3 Void

S.2.2.4 Void

S.2.2.5 Handling of the IP-CAN for media

S.2.2.5.1 General requirements

The UE does not directly request resources for media flow(s).

S.2.2.5.1A Activation or modification of IP-CAN for media by the UE

Not applicable.

S.2.2.5.1B Activation or modification of IP-CAN for media by the network

Not applicable.

S.2.2.5.1C Deactivation of IP-CAN for media

Not applicable.

S.2.2.5.2 Special requirements applying to forked responses

The UE does not directly request resources for media flow(s). As a result there are no special UE requirements applying to forked responses.

S.2.2.5.3 Unsuccessful situations

Not applicable.

S.2.2.6 Emergency service

S.2.2.6.1 General

Emergency service is not supported when the IP-CAN is a DVB-RCS2 satellite access network.

S.2.2.6.1A Type of emergency service derived from emergency service category value

Not applicable.

S.2.2.6.1B Type of emergency service derived from extended local emergency number list

Not applicable.

S.2.2.6.2 eCall type of emergency service

The UE shall not send an INVITE request with Request-URI set to "urn:service:sos.ecall.manual" or "urn:service:sos.ecall.automatic".

S.2.2.6.3 Current location discovery during an emergency call

Void.

S.2A Usage of SDP

S.2A.0 General

Not applicable.

S.2A.1 Impact on SDP offer / answer of activation or modification of satellite bearer for media by the network

Not applicable.

S.2A.2 Handling of SDP at the terminating UE when originating UE has resources available and IP-CAN performs network-initiated resource reservation for terminating UE

Not applicable.

S.2A.3 Emergency service

No additional procedures defined.

S.3 Application usage of SIP

S.3.1 Procedures at the UE

S.3.1.0 Void

S.3.1.0a IMS_Registration_handling policy

Not applicable.

S.3.1.1 P-Access-Network-Info header field

The UE may, but need not, include the P-Access-Network-Info header field where indicated in subclause 5.1.

S.3.1.1A Cellular-Network-Info header field

Not applicable.

S.3.1.2 Availability for calls

Not applicable.

S.3.1.2A Availability for SMS

Void.

S.3.1.3 Authorization header field

Void

S.3.1.4 SIP handling at the terminating UE when precondition is not supported in the received INVITE request, the terminating UE does not have resources available and IP-CAN performs network-initiated resource reservation for the terminating UE

Not applicable.

S.3.1.5 3GPP PS data off

Not applicable.

S.3.1.6 Transport mechanisms

No additional requirements are defined.

S.3.1.7 RLOS

Not applicable.

S.3.2 Procedures at the P-CSCF

S.3.2.0 Registration and authentication

Void.

S.3.2.1 Determining network to which the originating user is attached

In order to determine from which network the request was originated the P-CSCF shall check if the location information received in the network provided and/or UE provided "dvb-rcs2-node-id" parameter in the P-Access-Network-Info header field(s) indicates that the UE is connected to the same network as the P-CSCF or not.

NOTE 1: If local policy does not require the insertion of P-Access-Network-Info header field in the P-CSCF even if it is missing in the received initial request, the P-CSCF can assume that the request is initiated by a UE in the same network as the P-CSCF.

NOTE 2: If the location information in the network provided and UE provided "dvb-rcs2-node-id" parameters (in a request that includes two P-Access-Network-Info header fields) is contradictory, or the two P-Access-Network-Info header fields indicate different access types the P-CSCF ignores either the network provided or the UE provided information according to operator policy.

S.3.2.2 Location information handling

Upon receipt of an initial request for a dialog or standalone transaction or an unknown method, the P-CSCF based on local policy may include a P-Access-Network-Info header field. The value of the "dvb-rcs2-node-id" parameter shall be provided by the IP-CAN.

S.3.2.3 Void

S.3.2.4 Void

S.3.2.5 Void

S.3.2.6 Resource sharing

Not applicable.

S.3.2.7 Priority sharing

Not applicable.

S.3.2.8 RLOS

Not applicable.

S.3.3 Procedures at the S-CSCF

S.3.3.1 Notification of AS about registration status

Not applicable

S.3.3.2 RLOS

Not applicable.

S.4 3GPP specific encoding for SIP header field extensions

S.4.1 Void

S.5 Use of circuit-switched domain

There is no CS domain in this access technology.

Annex T (Normative): Network policy requirements for the IM CN subsystem

T.1 Scope

This annex details areas where network policy is subject to additional requirements to those specified in the main part of this document.

T.2 Application of network policy for the support of transcoding

When providing transcoding functions at the P-CSCF, at the IBCF and at an AS, the set of codecs to be forwarded as the SDP offer to the remote user is subject to network policy. In order to give support to the codecs and media quality originally requested by the offerer, the network policy shall meet the following requirements.

NOTE 1: RFC 3264 [27B] recommends to list codecs in priority order, so by adding network inserted codecs to the end of the codec list will give higher priority to previous codecs that might have been inserted by the originating UE.

A) An intermediate entity should attempt to support the original request for codecs from the UE.

B) An intermediate entity should only remove a codec from the codec list to meet policy requirements of the local access of the user.

C) A modification (i.e. any combination of reordering, removal or addition) to the codec list should only be made, such that the resultant SDP offer / answer exchange results in media of equal or better end-to-end quality than if the modification had not been made, subject to policy restrictions of the access of the local user.

NOTE 2: Transcoding between codecs of higher quality can provide better end-to-end quality than using a common codec of lower quality.

D) A modification (i.e. any combination of reordering, removal or addition) to the codec list should only be made, such that the resultant SDP offer / answer exchange prefers solutions that do not use a transcoder rather than ones that do use transcoder, subject to meeting the policy restrictions in B) and meeting the best end-to-end media quality in C) above.

E) Additions to the codec list that are provided by the network entity shall be supported by transcoding between at least one of the offered codecs contained in the SDP offer and the added codecs.

F) An intermediate entity shall not insert a codec to the codec list if end-to-end media security mechanism is required for the related media.

G) If an intermediate entity performs transcoding during an ongoing session and receives a SIP message containing a subsequent SDP offer including the codec that is currently in use on the incoming call leg, the entity should include the codec that is currently in use on the outgoing call leg when forwarding the SIP message.

H) If an intermediate entity performs transcoding during an ongoing session and receives a SIP message containing a subsequent SDP offer not including the codec that is currently in use on the incoming call leg, the entity should include the codec that is currently in use on the outgoing call leg when forwarding the SIP message, subject to meeting the policy restrictions in E).

I) If an intermediate entity does not perform transcoding during an ongoing session and receives a SIP message containing a subsequent SDP offer including the codec that is currently in use, the entity should not add any codecs in the subsequent SDP offer.

Annex U (normative): IP-Connectivity Access Network specific concepts when using 5GS to access IM CN subsystem

U.1 Scope

The present annex defines IP-CAN specific requirements for a call control protocol for use in the IP Multimedia (IM) Core Network (CN) subsystem based on the Session Initiation Protocol (SIP), and the associated Session Description Protocol (SDP), where the IP-CAN is 5G System (5GS). The 5GS IP-CAN has a 5GS core network which can be supported by a NG-RAN.

U.2 IP-CAN aspects when connected to the IM CN subsystem

U.2.1 Introduction

A UE accessing the IM CN subsystem, and the IM CN subsystem itself, utilise the services provided by 5GS to provide packet-mode communication between the UE and the IM CN subsystem.

Requirements for the UE on the use of these packet-mode services are specified in this clause.

When using the 5GS, each IP-CAN bearer is provided by a 5GS QoS flow.

U.2.2 Procedures at the UE

U.2.2.1 Establishment of IP-CAN bearer and P-CSCF discovery

Prior to communication with the IM CN subsystem, the UE shall:

a) if not registered for 5GS services, perform a registration procedure in 5GS as specified in 3GPP TS 24.501 [258];

b) ensure that a 5GS PDU session and a QoS flow used for SIP signalling of that PDU session is available. This 5GS QoS flow shall remain active throughout the period the UE is connected to the IM CN subsystem, i.e. from the initial registration and at least until the deregistration. As a result, the 5GS PDU session provides the UE with information that makes the UE able to construct an IPv4 or an IPv6 address;

   when establishing a 5GS PDU session, if a service which requires service continuity provided by SSC mode 1 as specified in 3GPP TS 23.501 [257], (e.g. IMS Multimedia Telephony Service as specified in 3GPP TS 24.173 [8H]), is to be used within that PDU session, the UE shall set SSC mode 1 for that PDU session;

   when establishing a 5GS PDU session with a QoS flow used for SIP signaling, the UE shall indicate to the network, by setting the IM CN Subsystem Signalling Flag in the extended Protocol Configuration Options information element specified in 3GPP TS 24.501 [258] in the PDU SESSION ESTABLISHMENT REQUEST message, that the request is for SIP signalling;

c) use the QoS flow associated with the default QoS rule for SIP signalling. The UE may also use this 5GS QoS flow for Domain Name Server (DNS) and Dynamic Host Configuration Protocol (DHCP) signalling; and

d) acquire a P-CSCF address(es).

   The methods for P-CSCF discovery are:

I. When using IPv4, employ the DHCP RFC 2132 [20F], the DHCPv4 options for SIP servers RFC 3361 [35A], and RFC 3263 [27A] as described in subclause 9.2.1. When using IPv6, employ Dynamic Host Configuration Protocol for IPv6 (DHCPv6) RFC 3315 [40], the DHCPv6 options for SIP servers RFC 3319 [41] and DHCPv6 options for DNS RFC 3646 [56C] as described in subclause 9.2.1.

II. Transfer P-CSCF address(es) within the 5GS PDU session establishement procedure.

The UE shall indicate the request for a P-CSCF address to the network within the extended Protocol Configuration Options information element of the PDU SESSION ESTABLISHMENT REQUEST message.

If the network provides the UE with a list of P-CSCF IPv4 or IPv6 addresses in the PDU SESSION ESTABLISHMENT ACCEPT message, the UE shall assume that the list is ordered top-down with the first P-CSCF address within the extended Protocol Configuration Options information element as the P-CSCF address having the highest preference and the last P-CSCF address within the extended Protocol Configuration Options information element as the P-CSCF address having the lowest preference.

III. The UE selects a P-CSCF from the list (see 3GPP TS 31.103 [15B]) stored in the ISIM.

IV. The UE selects a P-CSCF from the list in IMS management object.

The UE shall use method IV to select a P-CSCF, if

- a P-CSCF is to be discovered in the home network;
- the UE is roaming; and
- the IMS management object contains the P-CSCF list.

The UE shall use method III to select the P-CSCF, if:

- a P-CSCF is to be discovered in the home network;
- the UE is roaming;
- either the UE does not contain the IMS management object, or the UE contains the IMS management object but the IMS management object does not contain the P-CSCF list; and
- the ISIM residing in the UICC supports the P-CSCF list.

The UE can freely select method I or II for P-CSCF discovery, if:

- the UE is in the home network; or
- the UE is roaming and the P-CSCF is to be discovered in the visited network.

The UE can select method IV, if:

- the UE is in the home network; and
- the IMS management object contains the P-CSCF list.

In case method I is selected and several P-CSCF addresses or FQDNs are provided to the UE, the selection of P-CSCF address or FQDN shall be performed as indicated in RFC 3361 [35A] when using IPv4 or RFC 3319 [41] when using IPv6. If sufficient information for P-CSCF address selection is not available, selection of the P-CSCF address by the UE is implementation specific.

NOTE: The UE decides whether the P-CSCF is to be discovered in the serving network or in the home network based on local configuration, e.g. whether the application on the UE is permitted to use local breakout.

If the UE is designed to use I above, but receives P-CSCF address(es) according to II, then the UE shall either ignore the received address(es), or use the address(es) in accordance with II, and not proceed with the DHCP request according to I.

If the UE is configured to use Option II above and detects that all P-CSCFs known by the UE have been used when the UE selects a different P-CSCF as a result of:

- receiving 305 (Use Proxy) to the REGISTER request;

- receiving 504 (Server Time-out); or
- expiration of the timer F at the UE,

then unless the PDU session is in use by other applications, the UE shall:

1) release the PDU session of the 5GS QoS flow that is used only for the transport of SIP signalling and that are not used for other non-IMS applications, but shall not release emergency PDU session; and
2) unless the UE decides the service is no longer needed,
   a) perform a new P-CSCF discovery procedure as described in subslause 9.2.1; and
   b) perform the procedures for initial registration as described in subclause 5.1.1.2.

When using IPv4, the UE may request a DNS Server IPv4 address(es) via RFC 2132 [20F] or by the extended Protocol Configuration Options information element when establishing the PDU session according to 3GPP TS 24.501 [258].

When using IPv6, the UE may request a DNS Server IPv6 address(es) via RFC 3315 [40] and RFC 3646 [56C] or by the extended Protocol Configuration Options information element when establishing the PDU session according to 3GPP TS 24.501 [258].

When:

- the UE obtains a 5GS QoS flow used for SIP signalling by performing handover of the connection from another IP-CAN;
- IP address of the UE is not changed during the handover; and
- the UE already communicates with the IM CN subsystem via the connection with the other IP-CAN, e.g. the UE determines that its contact with host portion set to the UE IP address (or FQDN of the UE) associated with the connection with the other IP-CAN has been bound to a public user identity;

the UE shall continue using the P-CSCF address(es) acquired in the other IP-CAN.

The UE may support the policy on whether a roaming UE when in a session is allowed to handover the PDN connection providing access to IMS between EPC via WLAN and 5GS, or is allowed to handover a PDU session providing access to IMS via non 3GPP access, the UE may support being configured with the policy on whether UE when in a session is allowed to transfer the PDN connection or PDU session providing access to IMS between EPC via WLAN using one or more of the following methods:

a) the Allow_Handover_PDU_session_non-3GPP_and_NG-RAN node of $EF_{IMSConfigDat}$a file described in 3GPP TS 31.102 [15C];
b) the Allow_Handover_PDU_session_non-3GPP_and_NG-RANnode of $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B];
c) the Allow_Handover_PDU_session_non-3GPP_and_NG-RAN node of 3GPP TS 24.167 [8G];
d) the Allow_Handover_PDN_connection_non-3GPP_and_NG-RAN node of $EF_{IMSConfigDat}$a file described in 3GPP TS 31.102 [15C];
e) the Allow_Handover_PDN_connection_non-3GPP_and_NG-RAN node of $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and
f) the Allow_Handover_PDN_connection_non-3GPP_and_NG-RAN node of 3GPP TS 24.167 [8G].

If the UE is configured with both the Allow_Handover_PDU_session_non-3GPP_and_NG-RAN node of 3GPP TS 24.167 [8G] and the Allow_Handover_PDU_session_non-3GPP_and_NG-RAN node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or 3GPP TS 31.103 [15B], then the Allow_Handover_PDU_session_non-3GPP_and_NG-RAN node of the $EF_{IMSConfigData}$ file shall take precedence.

If the UE is configured with both the Allow_Handover_PDN_connection_non-3GPP_and_NG-RAN node of 3GPP TS 24.167 [8G] and the Allow_Handover_PDN_connection_non-3GPP_and_NG-RAN node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or 3GPP TS 31.103 [15B], then the Allow_Handover_PDN_connection_non-3GPP_and_NG-RAN node of the $EF_{IMSConfigData}$ file shall take precedence.

NOTE 5: Precedence for files configured on both the USIM and ISIM is defined in 3GPP TS 31.103 [15B].

U.2.2.1A Modification of the PDU session of the 5GS QoS flow used for SIP signalling

The UE shall not modify the 5GS QoS flow from being used exclusively for SIP signalling.

When a 5GS QoS flow used for SIP signalling is established, the UE shall not set the IM CN Subsystem Signalling Flag in the extended Protocol Configuration Options information element specified in 3GPP TS 24.501 [258] of any subsequent PDU SESSION MODIFICATION REQUEST message for that DNN.

The UE shall ignore the IM CN Subsystem Signalling Flag if received from the network in the extended Protocol Configuration Options information element.

After the establishment of a 5GS QoS flow used for SIP signalling, the UE shall not indicate the request for a P-CSCF address to the network within the extended Protocol Configuration Options information element of any subsequent PDU SESSION MODIFICATION REQUEST message for that DNN.

The UE shall ignore P-CSCF address(es) if received from the network in the extended Protocol Configuration Options information element of a PDU SESSION MODIFICATION COMMAND message which is triggered by a PDU SESSION MODIFICATION REQUEST message.

U.2.2.1B Re-establishment of the PDU session with the 5GS QoS flow used for SIP signalling

If the UE registered a public user identity with an IP address allocated for the DNN of the PDU session with the 5GS QoS flow used for SIP signalling, the PDU session with the 5GS QoS flow used for SIP signalling is deactivated as result of signalling from the network and:

i) if the UE is required to perform an initial registration according to subclause U.3.1.2;

ii) if the signalling from the network results in requiring the UE to initiate activation of the PDU session of the 5GS QoS flow used for SIP signalling; or

iii) if the UE needs to continue having a public user identity registered with an IP address allocated for the DNN;

the UE shall:

A) if the non-access stratum is performing the UE requested PDU session establishment procedure for the DNN triggered as result of the signalling from the network, wait until the UE requested PDU session establishment procedure for the DNN finish; and

B) perform the procedures in subclause U.2.2.1, bullets a), b) and c).

If none of the bullets i), ii) and iii) of this subclause evaluate to true, or the procedures in bullet B) of this subclause were unable to ensure that the 5GS QoS flow used for SIP signalling is available or were unable to acquire any P-CSCF address(es):

1) if the SIP signalling was carried over a 5GS QoS flow not associated with the default QoS rule, the UE shall release all resources established as a result of SIP signalling by sending to the network either:

   a) a PDU SESSION MODIFICATION REQUEST message, if there are 5GS QoS flows of this PDU session that are not related SIP sessions; or

   b) a PDU SESSION RELEASE REQUEST message if all the 5GS QoS flows of this PDU session are related to SIP sessions.

If the 5GS QoS flow associated with default QoS rule used for SIP signalling was deactivated as described at the start of this subclause, and the procedures in bullet B) of this subclause ensured that the 5GS QoS flow used for SIP signalling is available and acquired the P-CSCF address(es), the UE shall perform a new initial registration according to subclause 5.1.1.2.

U.2.2.1C P-CSCF restoration procedure

A UE supporting the P-CSCF restoration procedure performs one of the following procedures:

A if the UE used method II for P-CSCF discovery and if the UE receives one or more P-CSCF address(es) in the extended Protocol Configuration Options information element of a PDU SESSION MODIFICATION COMMAND message and the one or more P-CSCF addresse(s) do not include the address of the currently used P-CSCF, then the UE shall acquire a different P-CSCF address from the one or more P-CSCF addresse(s) in the PDU SESSION MODIFICATION COMMAND message. If more than one P-CSCF address with the same container identifier (i.e. "P-CSCF IPv6 Address" or "P-CSCF IPv4 Address") are included, then the UE shall assume that the more than one P-CSCF addresses with the same container identifier are prioritised with the first P-CSCF address with the same container identifier within the Protocol Configuration Options information element as the P-CSCF address with the highest priority.

If the UE used method II for P-CSCF discovery and if the UE has previously sent the "P-CSCF Re-selection support" PCO indicator at PDU session creation and if the UE receives one or more P-CSCF address(es) in the extended Protocol Configuration Options information element of a PDU SESSION MODIFICATION COMMAND message, then the UE shall acquire a P-CSCF address from the one or more P-CSCF addresse(s) in the PDU SESSION MODIFICATION COMMAND message. If more than one P-CSCF address with the same container identifier (i.e. "P-CSCF IPv6 Address" or "P-CSCF IPv4 Address") are included, then the UE shall assume that the more than one P-CSCF addresses with the same container identifier are prioritised with the first P-CSCF address with the same container identifier within the Protocol Configuration Options information element as the P-CSCF address with the highest priority;

B if the UE uses RFC 6223 [143] as part of P-CSCF restoration procedures, and if the P-CSCF fails to respond to a keep-alive request, then the UE shall acquire a different P-CSCF address using one of the methods I, III and IV for P-CSCF discovery described in the subclause U.2.2.1.

If the UE has an ongoing session and acquired the new P-CSCF address by using procedure A described above, the UE may wait until the UE has detected that the ongoing session has ended before performing an initial registration as specified in subclause 5.1.

In all other cases, when the UE has acquired the P-CSCF address, the UE not having an ongoing session shall perform an initial registration as specified in subclause 5.1.

NOTE 1: For UEs using procedure A described above, the network ensures that P-CSCF address(es) in the extended Protocol Configuration Options information element of a PDU SESSION MODIFICATION COMMAND message is sent only during P-CSCF restoration procedures as defined in subclause 5 of 3GPP TS 23.380 [7D].

NOTE 2: The P-CSCF can be completely unreachable, so it is up to UE implementation to detect the end of an ongoing session, e.g. using media plane inactivity detection. Services depending on signalling such as CW and MT calls will not work during this time.

U.2.2.2 Session management procedures

The procedures for session management as described in 3GPP TS 24.501 [258] shall apply while the UE is connected to the IM CN subsystem.

U.2.2.3 Mobility management procedures

The procedures for mobility management as described in 3GPP TS 24.501 [258] shall apply while the UE is connected to the IM CN subsystem.

U.2.2.4 Cell selection and lack of coverage

The existing mechanisms and criteria for cell selection as described in 3GPP TS 38.304 [260] or 3GPP TS 36.304 [19B] shall apply while the UE is connected to the IM CN Subsystem.

U.2.2.5 5GS QoS flow for media

U.2.2.5.1 General requirements

NOTE 1: During establishment of a session, the UE establishes data streams(s) for media related to the session. Either the UE or the network can request for resource allocations for media, but the establishment and modification of the 5GS QoS flow is controlled by the network as described in 3GPP TS 24.501 [258].

If the resource allocation is initiated by the UE, the UE starts reserving resources whenever it has sufficient information about the media streams, and used codecs available as specified in 3GPP TS 24.501 [258].

NOTE 2: If the resource reservation requests are initiated by the network, then the establishment of 5GS QoS flow for media is initiated by the network after the P-CSCF has authorised the respective 5GS QoS flows and provided the QoS requirements to the PCF.

U.2.2.5.1A Activation or modification of QoS flows for media by the UE

If the UE is configured not to initiate resource allocation for media according to 3GPP TS 24.167 [8G], then the UE shall refrain from requesting additional 5GS QoS flow(s) for media until the UE considers that the network did not initiate resource allocation for the media.

U.2.2.5.1B Activation or modification of QoS flows for media by the network

If the UE receives an activation request from the network for a 5GS QoS flow for media which is associated with the 5GS QoS flow used for signalling, the UE shall correlate the media 5GS QoS flow with a currently ongoing SIP session establishment or SIP session modification.

If the UE receives a modification request from the network for a 5GS QoS flow that is used for one or more media streams in an ongoing SIP session, the UE shall:

1) modify the related PDU session context in accordance with the request received from the network.

U.2.2.5.1C Deactivation of a QoS flow for media

When a data stream for media related to a session is released, if the 5GS QoS flow transporting the data stream is no longer needed and allocation of the 5GS QoS flow was requested by the UE, then the UE releases the 5GS QoS flow.

NOTE: The 5GS QoS flow can be needed e.g. for other data streams of a session or for other applications in the UE.

U.2.2.5.1D Default QoS flow usage restriction policy

The default QoS flow usage restriction policy consists of zero or more default QoS flow usage restriction policy parts.

The default QoS flow usage restriction policy part consists of a mandatory media type condition and an optional ICSI condition.

The default QoS flow usage restriction policy does not apply to UE detected emergency calls.

Sending media is restricted according to the default QoS flow usage restriction policy, if sending media is restricted according to at least one default QoS flow usage restriction policy part of the default QoS flow usage restriction policy.

Sending media is restricted according to the default QoS flow usage restriction policy part if:

1) the media is to be sent for a media stream negotiated in a session offered or established by SIP signalling;

2) the media stream is of a media type indicated in the media type condition of the QoS flow usage restriction policy part;

3) the following is true:

   a) the default QoS flow usage restriction policy part does not have the ICSI condition; or

b) the session is offered or established by SIP signalling related to an IMS communication service identified in the ICSI condition of the default QoS flow usage restriction policy part; and

4) the media is to be sent via the default QoS flow of the PDU session for SIP signalling.

The UE may support the default QoS flow usage restriction policy.

If the UE supports the default QoS flow usage restriction policy:

1) the UE shall not send media restricted according to the default QoS flow usage restriction policy; and

2) the UE may support being configured with the default QoS flow usage restriction policy using one or more of the following methods:

a) the Default_QoS_Flow_usage_restriction_policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C];

b) the Default_QoS_Flow_usage_restriction_policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

c) the Default_QoS_Flow_usage_restriction_policy node of 3GPP TS 24.167 [8G].

If the UE is configured with both the Default_QoS_Flow_usage_restriction_policy node of 3GPP TS 24.167 [8G] and the Default_QoS_Flow_usage_restriction_policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or 3GPP TS 31.103 [15B], then the Default_QoS_Flow_usage_restriction_policy node of the $EF_{IMSConfigData}$ file shall take precedence.

NOTE: Precedence for files configured on both the USIM and ISIM is defined in 3GPP TS 31.103 [15B].

U.2.2.5.2 Special requirements applying to forked responses

NOTE 1: The procedures in this subclause only apply when the UE requests activation and modification of 5GS QoS flows for media . In the case where the network activates and modifies the 5GS QoS flows for media the network takes care of the handling of 5GS QoS flows in the case of forking.

Since the UE does not know that forking has occurred until a second, provisional response arrives, the UE requests resource allocation as required by the initial response received. If a subsequent provisional response is received, different alternative actions may be performed depending on the requirements in the SDP answer:

1) The resource requirements of the subsequent SDP can be accommodated by the existing resources requested. The UE performs no further resource requests.

2) The subsequent SDP introduces different QoS requirements or additional IP flows. The UE requests further resource allocation according to subclause U.2.2.5.1.

3) The subsequent SDP introduces one or more additional IP flows. The UE requests further resource allocation according to subclause U.2.2.5.1.

NOTE 2: When several forked responses are received, the resources requested by the UE are the "logical OR" of the resources indicated in the multiple responses to avoid allocation of unnecessary resources. The UE does not request more resources than proposed in the original INVITE request.

When a final answer is received for one of the early dialogs, the UE proceeds to set up the SIP session. The UE shall release all the unneeded IP-CAN resources. Therefore, upon the reception of the first final 200 (OK) response for the INVITE request (in addition to the procedures defined in RFC 3261 [26] subclause 13.2.2.4), the UE shall:

1) in case resources were established or modified as a consequence of the INVITE request and forked provisional responses that are not related to the accepted 200 (OK) response, send release request to release the unneeded resources.

U.2.2.5.3 Unsuccessful situations

The UE can receive resource reservation related error codes in a PDU SESSION MODIFICATION REJECT as described in 3GPP TS 24.501 [258]. If the UE receives a resource reservation related error code, the UE shall either handle the resource reservation failure as described in subclause 6.1.1 or retransmit the message up to three times.

U.2.2.6 Emergency service

U.2.2.6.1 General

For the purposes of this document, an emergency PDU session is the equivalent of emergency bearers; i.e. the 5GS defines emergency bearers for the support of emergency calls. Emergency PDU session is defined for use in emergency calls in 5GS and core network support of emergency PDU session is indicated to the UE in NAS signalling. Where the UE recognises that a call request is an emergency call and the core network supports emergency PDU session, the UE shall use emergency PDU session for both signalling and media for emergency calls made using the IM CN subsystem.

Some jurisdictions allow emergency calls to be made when the UE does not contain an UICC, or where the credentials are not accepted. Additionally, where the UE is in state 5GMM-REGISTERED.LIMITED-SERVICE or 5GMM-REGISTERED.PLMN-SEARCH, a normal registration in 5GS has been attempted but it can also be assumed that a registration in the IM CN subsystem will also fail. In such cases, subject to the lower layers indicating that the network does support emergency bearer services in limited service state (see 3GPP TS 36.331 [19F] or 3GPP TS 38.331 [19G]), the procedures for emergency calls without registration can be applied, as defined in subclause 5.1.6.8.2. If the 5GS primary authentication procedure has already succeeded during the latest normal or emergency registration procedure in 5GS, the UE shall perform an initial emergency registration, as described in subclause 5.1.6.2 before attempting an emergency call as described in subclause 5.1.6.8.3.

NOTE 1: The UE can determine that 5GS primary authentication procedure has succeeded during the emergency registration procedure in 5GS when a non-null integrity protection algorithm (i.e. other than 5G-IA0 algorithm) is received in the NAS signalling SECURITY MODE COMMAND message.

To perform emergency registration, the UE shall request to establish an emergency PDU session as described in 3GPP TS 24.501 [258]. The procedures for PDU session establishment and P-CSCF discovery, as described in subclause U.2.2.1 of this specification apply accordingly.

In the present document, "EMS is Y" as described in 3GPP TS 23.167 [4B] refers to one of the following conditions:

a) if the UE is in an NR cell connected to 5GCN, the network indicates in the REGISTRATION ACCEPT message that EMC is set to either "Emergency services supported in NR connected to 5GCN only" or "Emergency services supported in NR connected to 5GCN and E-UTRA connected to 5GCN" as described in 3GPP TS 24.501 [258]; or

b) if the UE is in an E-UTRA cell connected to 5GCN, the network indicates in the REGISTRATION ACCEPT message that EMC is set to either "Emergency services supported in E-UTRA connected to 5GCN only" or "Emergency services supported in NR connected to 5GCN and E-UTRA connected to 5GCN" as described in 3GPP TS 24.501 [258].

In the present document, "EMS is N" as described in 3GPP TS 23.167 [4B] refers to one of the following conditions:

a) if the UE is in an NR cell connected to 5GCN, the network indicates in the REGISTRATION ACCEPT message that EMC is set to either "Emergency services not supported" or "Emergency services supported in E-UTRA connected to 5GCN only" as described in 3GPP TS 24.501 [258]; or

b) if the UE is in an E-UTRA cell connected to 5GCN, the network indicates in the REGISTRATION ACCEPT message that EMC is set to either "Emergency services not supported" or "Emergency services supported in NR connected to 5GCN only" as described in 3GPP TS 24.501 [258].

In the present document, "ESFB is Y" as described in 3GPP TS 23.167 [4B] refers to one of the following conditions:

a) if the UE is in an NR cell connected to 5GCN, the network indicates in the REGISTRATION ACCEPT message that EMF is set to either "Emergency service fallback supported in NR connected to 5GCN only" or "Emergency service fallback supported in NR connected to 5GCN and E-UTRA connected to 5GCN" as described in 3GPP TS 24.501 [258]; or

b) if the UE is in an E-UTRA cell connected to 5GCN, the network indicates in the REGISTRATION ACCEPT message that EMF is set to either "Emergency service fallback supported in E-UTRA connected to 5GCN only" or "Emergency service fallback supported in NR connected to 5GCN and E-UTRA connected to 5GCN" as described in 3GPP TS 24.501 [258].

In the present document, "ESFB is N" as described in 3GPP TS 23.167 [4B] refers to one of the following conditions:

a) if the UE is in an NR cell connected to 5GCN, the network indicates in the REGISTRATION ACCEPT message that EMF is set to either "Emergency service fallback not supported" or "Emergency service fallback supported in E-UTRA connected to 5GCN only" as described in 3GPP TS 24.501 [258]; or

b) if the UE is in an E-UTRA cell connected to 5GCN, the network indicates in the REGISTRATION ACCEPT message that EMF is set to either "Emergency service fallback not supported" or "Emergency service fallback supported in NR connected to 5GCN only" as described in 3GPP TS 24.501 [258].

Emergency services fallback is defined to direct or redirect the UE towards either E-UTRA connected to 5GCN or EPS and support of emergency service fallback is indicated to the UE in NAS signalling.

In order to find out whether the UE is attached to the home PLMN or to a visited PLMN, the UE shall compare the MCC and MNC values derived from its IMSI with the MCC and MNC of the PLMN the UE is attached to. If the MCC and MNC of the PLMN the UE is attached to do not match with the MCC and MNC derived from the IMSI, then for the purpose of emergency calls in the IM CN subsystem the UE shall consider to be attached to a VPLMN.

NOTE 2: In this respect an equivalent HPLMN, as defined in 3GPP TS 23.122 [4C] will be considered as a visited network.

If the UE selected an entry of the "list of subscriber data" in the SNPN access mode as specified in 3GPP TS 23.122 [4C], in order to find out whether the UE is attached to a subscribed SNPN or to a non-subscribed SNPN , the UE shall compare the SNPN identity of the selected entry of the "list of subscriber data" with the SNPN identity of the SNPN the UE is attached to. If the SNPN identity of the SNPN the UE is attached to does not match the SNPN identity of the selected entry of the "list of subscriber data", then for the purpose of emergency calls in the IM CN subsystem the UE shall consider to be attached to a non-subscribed SNPN.

If the UE selected the PLMN subscription in the SNPN access mode as specified in 3GPP TS 23.122 [4C], then for the purpose of emergency calls in the IM CN subsystem the UE shall consider to be attached to a non-subscribed SNPN.

If the dialled number is equal to a local emergency number stored in the Extended Local Emergency Number List (as defined in 3GPP TS 24.301 [8J]), then the UE shall recognize such a number as for an emergency call and:

- if the dialled number is equal to an emergency number stored in the ME, or in the USIM, then the UE shall perform either procedures in the subclause U.2.2.6.1B or the procedures in subclause U.2.2.6.1A; and

- if the dialled number in not equal to an emergency number stored in the ME, or in the USIM, then the UE shall perform procedures in the subclause U.2.2.6.1B.

If the dialled number is not equal to a local emergency number stored in the Extended Local Emergency Number List (as defined in 3GPP TS 24.301 [8J]) and:

- if the dialled number is equal to an emergency number stored in the ME, in the USIM or in the Local Emergency Number List (as defined in 3GPP TS24.008 [8]), then the UE shall recognize such a number as for an emergency call and performs the procedures in subclause U.2.2.6.1A.

NOTE 3: The UE verifies if a detected emergency number is still present in the Extended Local Emergency Number List after registering to a different PLMN. It is possible for the number to no longer be present in the Extended Local Emergency Number List if:

- the PLMN attached to relies on the Local Emergency Number List for deriving a URN; or

- the previously received Extended Emergency Number List Validity field indicated "Extended Local Emergency Numbers List is valid only in the PLMN from which this IE is received".

If the UE detected an emergency number, the UE subsequently performs a registration procedure or an emergency registration procedure with a different PLMN than the PLMN from which the UE received the last Extended Local Emergency Number List, the dialled number is not stored in the ME, in the USIM and in the Local Emergency Number List, and:

- the REGISTRATION ACCEPT message received from the different PLMN contains the Extended Local Emergency Number List and the emergency number is present in the updated Extended Local Emergency Number List then the UE uses the updated Extended Local Emergency Number List when it performs the procedures in subclause U.2.2.6.1B; and

- the REGISTRATION ACCEPT message received from the different PLMN contains no Extended Local Emergency Number List or the emergency number is no longer present in the updated Extended Local Emergency Number List then the UE shall attempt UE procedures for SIP that relate to emergency using emergency service URN "urn:service:sos".

Upon reception of a 380 (Alternative Service) response to an INVITE request as defined in subclause 5.1.2A.1.1 and subclause 5.1.3.1, if:

- the 380 (Alternate Service) response contains a Contact header field;
- the value of the Contact header field is a service URN; and
- the service URN has a top-level service type of "sos";

then the UE determines that "emergency service information is included" as described 3GPP TS 23.167 [4B].

Upon reception of a 380 (Alternative Service) response to an INVITE request as defined in subclause 5.1.3.1 if the 380 (Alternate Service) response does not contain a Contact header field with service URN that has a top-level service type of "sos", then the UE determines that "no emergency service information is included" as described 3GPP TS 23.167 [4B].

If the "emergency service information is included" as described 3GPP TS 23.167 [4B]:

1) if the URN in the Contact header field matches an emergency service URN in table U.2.2.6.1, then the type of emergency service is the value corresponding to the matching entry in table U.2.2.6.1; and
2) if the URN in the Contact header field does not match any emergency service URN in table U.2.2.6.1, then the type of emergency service is not identified.

NOTE 4: In bullet 2), the URN in the Contact header field either contains "no emergency subservice type" as described in 3GPP TS 23.167 [4B] triggering an emergency call, or contains an "emergency subservice type that does not map into an emergency service category for the CS domain" as described in 3GPP TS 23.167 [4B] triggering a normal call when the dialled number is available or triggering an emergency call when the dialled number is not available. The country specific URN is an example of a "emergency subservice type that does not map into an emergency service category for the CS domain".

When the emergency registration expires, the UE should disconnect the emergency PDU session.

Upon receiving a 3xx other than 380 (Alternative service), 4xx, 5xx or 6xx response to an INVITE request for a UE detectable emergency call, the UE shall perform domain selection as specified in 3GPP TS 23.167 [4B] annex H, to re-attempt the emergency call.

U.2.2.6.1A Type of emergency service derived from emergency service category value

The type of emergency service for an emergency number is derived from the settings of the emergency service category value (bits 1 to 5 of the emergency service category value as specified in subclause 10.5.4.33 of 3GPP TS 24.008 [8]). Table U.2.2.6.1 below specifies mappings between a type of emergency service and an emergency service URN. The UE shall use the mapping to match an emergency service URN and a type of emergency service. If a dialled number is an emergency number but does not map to a type of emergency service the service URN shall be "urn:service:sos".

Table U.2.2.6.1: Mapping between type of emergency service and emergency service URN

Type of emergency service	Emergency service URN
Police	urn:service:sos.police
Ambulance	urn:service:sos.ambulance
Fire Brigade	urn:service:sos.fire
Marine Guard	urn:service:sos.marine
Mountain Rescue	urn:service:sos.mountain

NOTE 1: It is not possible for a UE to indicate more than one type of emergency service in an emergency service URN.

If an IP-CAN, capable of providing local emergency numbers, did not provide a local emergency number that matches the dialled number (see subclause 5.1.6.1) and multiple types of emergency service can be derived for a dialled number from the information configured on the UICC then:

- if the UE is in the HPLMN or a subscribed SNPN, the UE shall map any one of these types of emergency service to an emergency service URN as specified in table U.2.2.6.1; and

- if the UE is in a VPLMN or a non-subscribed SNPN, the UE shall select "urn:service:sos".

If an IP-CAN, capable of providing local emergency numbers, provided a local emergency number that matches the dialled number (see subclause 5.1.6.1), and:

- if the UE can derive one or more types of emergency service from the information received from the IP-CAN for the dialled number and the UE cannot derive types of emergency service from the information configured on the UICC for the dialled number; or

- if the UE is able to derive identical types of emergency service from both the information received from the IP-CAN for the dialled number and from the information configured on the UICC for the dialled number,

then the UE shall map any one of these emergency service types to an emergency service URN as specified in table U.2.2.6.1.

NOTE 2: How the UE resolves clashes where an emergency number is associated with one or more different types of emergency service configured in the USIM and in information received from the core network, is implementation dependent.

U.2.2.6.1B Type of emergency service derived from extended local emergency number list

The Extended Local Emergency Number List (defined in 3GPP TS 24.301 [8J]) can contain sub-services of the associated emergency service URN for the detected emergency number.

If:

- the length of sub-services field is greater than "0", the UE shall construct the emergency service URN using "urn:service:sos" followed by adding a dot followed by the content of the sub-services field; and

- the length of sub-services field is "0", the UE shall use the emergency service URN "urn:service:sos".

EXAMPLE 1: For a detected number, if the sub-service is "gas", then the UE constructs "urn:service:sos.gas" as the associated emergency service URN.

EXAMPLE 2: For a detected number, if no sub-service is provided, then the UE uses "urn:service:sos" as the associated emergency service URN.

U.2.2.6.2 eCall type of emergency service

If the IP-CAN indicates the eCall support indication or the CS domain is not available to the UE, the UE can send an INVITE request with Request-URI set to "urn:service:sos.ecall.manual" or "urn:service:sos.ecall.automatic".

If the IP-CAN does not indicate the eCall support indication and the CS domain is available to the UE, the UE shall not send an INVITE request with Request-URI set to "urn:service:sos.ecall.manual" or "urn:service:sos.ecall.automatic".

U.2.2.6.3 Current location discovery during an emergency call

Void.

U.2.2.6.4 Emergency services in single-registration mode

NOTE: This subclause covers only the case where the UE selects the IM CN subsystem in accordance with the conventions and rules specified in 3GPP TS 23.167 [4B] and describes the IP-CAN specific procedure. It does not preclude the use of CS domain. When a CS system based on 3GPP TS 24.008 [8] is to be used, clause B.5 applies.

When the UE operates in single-registration mode as described in 3GPP TS 24.501 [258] and the UE recognises that a call request is an emergency call, if:

1) the IM CN subsystem is selected in accordance with the conventions and rules specified in 3GPP TS 23.167 [4B]; and

2) the UE is currently registered to the 5GS services while the UE is in an NR cell connected to 5GCN;

then the following treatment is applied:

1) if the EMC indicates "Emergency services not supported":

   a) if the UE supports emergency services fallback as specified in 3GPP TS 23.501 [257] and the emergency services fallback is available (i.e., "ESFB is Y" as described in 3GPP TS 23.167 [4B]), the UE shall attempt emergency services fallback as specified in 3GPP TS 24.501 [258]. If the UE receives from the lower layers an indication that the emergency services fallback attempt failed, the UE may behave as described in bullet b) below assuming that the emergency services fallback is not available;

   b) if the UE supports emergency services fallback as specified in 3GPP TS 23.501 [257] and the emergency services fallback is not available (i.e., "ESFB is N" as described in 3GPP TS 23.167 [4B]) and if:

      i) the EMF is set to "Emergency services fallback supported in E-UTRA connected to 5GCN only" and the UE is capable of accessing 5GCN via E-UTRA, the UE shall either:

         A) attempt to select an E-UTRA cell connected to 5GCN. If the UE finds a suitable E-UTRA cell connected to 5GCN, the UE shall attempt emergency services fallback as specified in 3GPP TS 24.501 [258] via E-UTRA connected to 5GCN. If the UE does not find a suitable E-UTRA cell connected to 5GCN or the UE receives from the lower layers an indication that the emergency services fallback attempt failed, the UE may attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; or

         B) attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; or

      ii) the EMF is set to "Emergency services fallback not supported" or the UE is not capable of accessing 5GCN via E-UTRA, the UE shall request the lower layers to disable the N1 mode capability for 3GPP access as specified in 3GPP TS 24.501 [257] and attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; and

   c) if the UE does not support emergency services fallback as specified in 3GPP TS 23.501 [257], the UE shall request the lower layers to disable the N1 mode capability for 3GPP access as specified in 3GPP TS 24.501 [257] and attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration;

2) if the EMC indicates "Emergency services supported in E-UTRA connected to 5GCN only":

   a) if the UE supports emergency services fallback as specified in 3GPP TS 23.501 [257] and the emergency services fallback is available (i.e., "ESFB is Y" as described in 3GPP TS 23.167 [4B]), the UE shall attempt emergency services fallback as specified in 3GPP TS 24.501 [258]. If the UE receives from the lower layers an indication that the emergency services fallback attempt failed, the UE may behave as described in bullet b) below assuming that the emergency services fallback is not available; and

   b) if the UE does not support emergency services fallback as specified in 3GPP TS 23.501 [257] or the emergency services fallback is not available (i.e., "ESFB is N" as described in 3GPP TS 23.167 [4B]) and if:

i) the UE is capable of accessing 5GCN via E-UTRA, the UE shall attempt to select an E-UTRA cell connected to 5GCN. If the UE finds a suitable E-UTRA cell connected to 5GCN, the UE shall trigger establishment of an emergency PDU session as specified in 3GPP TS 24.501 [258] via E-UTRA connected to 5GCN. If the UE does not find a suitable E-UTRA cell connected to 5GCN, the UE may attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; or

ii) the UE is not capable of accessing 5GCN via E-UTRA, the UE shall request the lower layers to disable the N1 mode capability for 3GPP access as specified in 3GPP TS 24.501 [257] and attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; and

3) if the EMC indicates "Emergency services supported in NR connected to 5GCN only" or "Emergency services supported in NR connected to 5GCN and E-UTRA connected to 5GCN", the UE shall trigger establishment of an emergency PDU session as specified in 3GPP TS 24.501 [258].

When the UE operates in single-registration mode as described in 3GPP TS 24.501 [258] and the UE recognises that a call request is an emergency call, if:

1) the IM CN subsystem is selected in accordance with the conventions and rules specified in 3GPP TS 23.167 [4B]; and

2) the UE is currently registered to the 5GS services while the UE is in an E-UTRA cell connected to 5GCN;

then the following treatment is applied:

1) if the EMC indicates "Emergency services not supported":

a) if the UE supports emergency services fallback as specified in 3GPP TS 23.501 [257] and the emergency services fallback is available (i.e., "ESFB is Y" as described in 3GPP TS 23.167 [4B]), the UE shall attempt emergency services fallback as specified in 3GPP TS 24.501 [258]. If the UE receives from the lower layers an indication that the emergency services fallback attempt failed, the UE may behave as described in bullet b) below assuming that the emergency services fallback is not available;

b) if the UE supports emergency services fallback as specified in 3GPP TS 23.501 [257] and the emergency services fallback is not available (i.e., "ESFB is N" as described in 3GPP TS 23.167 [4B]) and if:

i) the EMF is set to "Emergency services fallback supported in NR connected to 5GCN only" and the UE is capable of accessing 5GCN via NR, the UE shall either:

A) attempt to select an NR cell connected to 5GCN. If the UE finds a suitable NR cell connected to 5GCN, the UE shall attempt emergency services fallback as specified in 3GPP TS 24.501 [258] via NR connected to 5GCN. If the UE does not find a suitable NR cell connected to 5GCN or the UE receives from the lower layers an indication that the emergency services fallback attempt failed, the UE may attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; or

B) attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; or

ii) the EMF is set to "Emergency services fallback not supported" or the UE is not capable of accessing 5GCN via NR, the UE shall request the lower layers to disable the N1 mode capability for 3GPP access as specified in 3GPP TS 24.501 [257] and attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to

"emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; and

c) if the UE does not support emergency services fallback as specified in 3GPP TS 23.501 [257], the UE shall request the lower layers to disable the N1 mode capability for 3GPP access as specified in 3GPP TS 24.501 [257] and attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration;

2) if the EMC indicates "Emergency services supported in NR connected to 5GCN only":

a) if the UE supports emergency services fallback as specified in 3GPP TS 23.501 [257] and the emergency services fallback is available (i.e., "ESFB is Y" as described in 3GPP TS 23.167 [4B]), the UE shall attempt emergency services fallback as specified in 3GPP TS 24.501 [258]. If the UE receives from the lower layers an indication that the emergency services fallback attempt failed, the UE may behave as described in bullet b) below assuming that the emergency services fallback is not available; and

b) if the UE does not support emergency services fallback as specified in 3GPP TS 23.501 [257] or the emergency services fallback is not available (i.e., "ESFB is N" as described in 3GPP TS 23.167 [4B]) and if:

i) the UE is capable of accessing 5GCN via NR, the UE shall attempt to select an NR cell connected to 5GCN. If the UE finds a suitable NR cell connected to 5GCN, the UE shall trigger establishment of an emergency PDU session as specified in 3GPP TS 24.501 [258] via NR connected to 5GCN. If the UE does not find a suitable NR cell connected to 5GCN, the UE may attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; or

ii) the UE is not capable of accessing 5GCN via NR, the UE shall request the lower layers to disable the N1 mode capability for 3GPP access as specified in 3GPP TS 24.501 [257] and attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; and

3) if the EMC indicates "Emergency services supported in E-UTRA connected to 5GCN only" or "Emergency services supported in NR connected to 5GCN and E-UTRA connected to 5GCN", the UE shall trigger establishment of an emergency PDU session as specified in 3GPP TS 24.501 [258].

U.2.2.6.5 Emergency services in dual registration mode

NOTE 1: This subclause covers only the case where the UE selects the IM CN subsystem in accordance with the conventions and rules specified in 3GPP TS 23.167 [4B] and describes the IP-CAN specific procedure. It does not preclude the use of CS domain. When a CS system based on 3GPP TS 24.008 [8] is to be used, clause B.5 applies.

When the UE operates in dual-registration mode as described in 3GPP TS 24.501 [258] and the UE recognises that a call request is an emergency call, if:

1) the IM CN subsystem is selected in accordance with the conventions and rules specified in 3GPP TS 23.167 [4B]; and

2) the UE currently camps on an NR cell connected to 5GCN;

then the following treatment is applied:

1) if the EMC indicates "Emergency services not supported" and:

a) if the UE is attached for EPS services and the network provided the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the

procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; or

b) if the UE is not attached for EPS services, and:

i) if emergency services fallback is available (i.e. "ESFB is Y" as described in 3GPP TS 23.167 [4B]) and the UE supports emergency services fallback as specified in 3GPP TS 23.501 [257], the UE shall attempt emergency services fallback as specified in 3GPP TS 24.501 [258]. If the UE receives from the lower layers an indication that the emergency services fallback attempt failed, the UE may behave as described in bullet ii) below assuming that the emergency services fallback is not available; or

ii) if emergency services fallback is not available (i.e. "ESFB is N" as described in 3GPP TS 23.167 [4B]) or the UE does not support emergency services fallback as specified in 3GPP TS 23.501 [257], the UE shall attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration;

2) if the EMC indicates "Emergency services supported in E-UTRA connected to 5GCN only" and:

a) if the UE is attached for EPS services and:

i) if the network provided the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; or

ii) if the network provided the UE with the EMC BS set to "emergency bearer services in S1 mode not supported" as described in 3GPP TS 24.301 [8J] and the UE is capable of accessing 5GCN via E-UTRA, the UE shall be locally detached for EPS services and the UE shall attempt to select an E-UTRA cell connected to 5GCN. If the UE finds a suitable E-UTRA cell connected to 5GCN, the UE shall trigger establishment of an emergency PDU session as specified in 3GPP TS 24.501 [258] via E-UTRA connected to 5GCN; or

b) if the UE is not attached for EPS services and:

i) if emergency services fallback is available (i.e., "ESFB is Y" as described in 3GPP TS 23.167 [4B]) and the UE supports emergency services fallback as specified in 3GPP TS 23.501 [257], the UE shall attempt emergency services fallback as specified in 3GPP TS 24.501 [258]. If the UE receives from the lower layers an indication that the emergency services fallback attempt failed, the UE may behave as described in bullet ii) below assuming that the emergency services fallback is not available; or

ii) if emergency services fallback is not available (i.e., "ESFB is N" as described in 3GPP TS 23.167 [4B]) or the UE does not support emergency services fallback as specified in 3GPP TS 23.501 [257], and:

A) if the UE is capable of accessing 5GCN via E-UTRA, the UE shall attempt to select an E-UTRA cell connected to 5GCN. If the UE finds a suitable E-UTRA cell connected to 5GCN, the UE shall trigger establishment of an emergency PDU session as specified in 3GPP TS 24.501 [258] via E-UTRA connected to 5GCN. If the UE does not find a suitable E-UTRA cell connected to 5GCN, the UE may attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; or

B) if the UE is not capable of accessing 5GCN via E-UTRA, the UE shall attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; and

3) if the EMC indicates "Emergency services supported in NR connected to 5GCN only" or "Emergency services supported in NR connected to 5GCN and E-UTRA connected to 5GCN", and:

a) if the UE is not attached for EPS services or the UE is attached for EPS services but the network provided the UE with the EMC BS set to "emergency bearer services in S1 mode not supported" as described in

3GPP TS 24.301 [8J], the UE shall trigger establishment of an emergency PDU session as specified in 3GPP TS 24.501 [258]; or

b) if the UE is attached for EPS services and the network provided the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall either:

i) trigger establishment of an emergency PDU session as specified in 3GPP TS 24.501 [258]; or

ii) perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration.

NOTE 2: In the above case, the UE chooses between items i) and ii) based on UE implementation.

When the UE operates in dual-registration mode as described in 3GPP TS 24.501 [258] and the UE recognises that a call request is an emergency call, if:

1) the IM CN subsystem is selected in accordance with the conventions and rules specified in 3GPP TS 23.167 [4B]; and

2) the UE currently camps on an E-UTRA cell connected to 5GCN and the UE is not attached for EPS services;

then the following treatment is applied:

1) if the EMC indicates "Emergency services not supported" and:

a) if emergency services fallback is available (i.e. "ESFB is Y" as described in 3GPP TS 23.167 [4B]) and the UE supports emergency services fallback as specified in 3GPP TS 23.501 [257], the UE shall attempt emergency services fallback as specified in 3GPP TS 24.501 [258]. If the UE receives from the lower layers an indication that the emergency services fallback attempt failed, the UE may behave as described in bullet b) below assuming that the emergency services fallback is not available; or

b) if emergency services fallback is not available (i.e. "ESFB is N" as described in 3GPP TS 23.167 [4B]) or the UE does not support emergency services fallback as specified in 3GPP TS 23.501 [257], the UE shall attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC, the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; and

2) if the EMC indicates "Emergency services supported in NR connected to 5GCN only" and:

a) if emergency services fallback is available (i.e., "ESFB is Y" as described in 3GPP TS 23.167 [4B]) and the UE supports emergency services fallback as specified in 3GPP TS 23.501 [257], the UE shall attempt emergency services fallback as specified in 3GPP TS 24.501 [258]. If the UE receives from the lower layers an indication that the emergency services fallback attempt failed, the UE may behave as described in bullet b) below assuming that the emergency services fallback is not available; or

b) if emergency services fallback is not available (i.e., "ESFB is N" as described in 3GPP TS 23.167 [4B]) or the UE does not support emergency services fallback as specified in 3GPP TS 23.501 [257] and:

i) if the UE is capable of accessing 5GCN via NR, the UE shall attempt to select an NR cell connected to 5GCN. If the UE finds a suitable NR cell connected to 5GCN, the UE shall trigger establishment of an emergency PDU session as specified in 3GPP TS 24.501 [258] via NR connected to 5GCN. If the UE cannot find a suitable NR cell connected to 5GCN, the UE may attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; and

ii) if the UE is not capable of accessing 5GCN via NR, the UE shall attempt to select an E-UTRA cell connected to EPC. If the UE finds a suitable E-UTRA cell connected to EPC and the network provides the UE with the EMC BS set to "emergency bearer services in S1 mode supported" as described in 3GPP TS 24.301 [8J], the UE shall perform the procedures as described in subclause L.2.2.6 to activate an EPS bearer context to perform emergency registration; and

3) if the EMC indicates "Emergency services supported in E-UTRA connected to 5GCN only" or "Emergency services supported in NR connected to 5GCN and E-UTRA connected to 5GCN", the UE shall trigger establishment of an emergency PDU session as specified in 3GPP TS 24.501 [258].

If the UE operating in dual-registration mode and camping on an E-UTRA cell connected to 5GCN, is also attached for EPS service, the domain selection is out of the scope of the specification.

U.2A Usage of SDP

U.2A.0 General

No additional procedures defined.

U.2A.1 Impact on SDP offer / answer of activation or modification of IP-CAN for media by the network

If, due to the activation of 5GS QoS flow from the network the related SDP media description needs to be changed, the UE shall update the related SDP information by sending a new SDP offer within a SIP request, which is sent over the existing SIP dialog.

If the UE receives a modification request from the network for a 5GS QoS flow that is used for one or more media streams in an ongoing SIP session, the UE shall:

1) if, due to the modification of the 5GS QoS flow, the related SDP media description need to be changed, update the related SDP information by sending a new SDP offer within a SIP request, that is sent over the existing SIP dialog, and respond to the 5GS QoS flow modification request.

NOTE: The UE can decide to indicate additional media streams as well as additional or different codecs in the SDP offer than those used in the already ongoing session.

U.2A.2 Handling of SDP at the terminating UE when originating UE has resources available and IP-CAN performs network-initiated resource reservation for terminating UE

If the UE receives an SDP offer where the SDP offer includes all media streams for which the originating side indicated its local preconditions as met, if the precondition mechanism is supported by the terminating UE and the IP-CAN performs network-initiated resource reservation for the terminating UE and the available resources are not sufficient for the received offer, the terminating UE shall indicate its local preconditions and provide the SDP answer to the originating side without waiting for resource reservation.

U.2A.3 Emergency service

NOTE: When establishing an emergency session or when modifying an emergency session, the IMSVoPS indicator does not influence handling of SDP offer and SDP answer.

U.3 Application usage of SIP

U.3.1 Procedures at the UE

U.3.1.0 Registration and authentication

The UE shall perform reregistration of a previously registered public user identity bound to any one of its contact addresses when changing to an IP-CAN for which usage is specified in annex R or annex W. The reregistration is performed using the new IP-CAN.

NOTE 1: This document does not specify how the UE detects that the used IP-CAN has changed. The information that is forcing the reregistration is also used to generate the content for the P-Access-Network-Info header field.

NOTE 2: The UE will send the reregistration irrespective of whether it has a SIP dialog or not.

If the UE supports the 3GPP PS data off, then the UE shall in all REGISTER requests include the "+g.3gpp.ps-data-off" header field parameter defined in subclause 7.9.8 set to a value indicating the 3GPP PS data off status.

When the UE sends a REGISTER request, if the 3GPP PS data off status is "active", then the UE shall only include media feature tags associated with services that are 3GPP PS data off exempt services in the g.3gpp.icsi-ref media feature tag, as defined in subclause 7.9.2 and RFC 3840 [62], for the IMS communication services it intends to use.

If the UE is registered, and the 3GPP PS data off status is changed, then the UE shall perform a reregistration of the previously registered public user identity.

A UE supporting ANBR as specified in 3GPP TS 26.114 [9B] shall also support RAN-assisted codec adaptation as specified in 3GPP TS 38.300 [270] and 3GPP TS 38.321 [271].

If the UE supports ANBR, upon receiving a 200 (OK) response to the REGISTER request and if the 200 (OK) response contains a Feature-Caps header field with the g.3gpp.anbr feature-capability indicator the UE shall assume that the Network supports RAN-assisted codec adaptation as specified in 3GPP TS 38.300 [270] and 3GPP TS 38.321 [271]. The UE is allowed to include the SDP ‘anbr’ attribute during session establishment as specified in 3GPP TS 26.114 [9B].

U.3.1.0A IMS_Registration_handling policy

The IMS_Registration_handling policy indicates whether the UE deregisters from IMS after a configured amount of time after receiving an indication that the IMS Voice over PS Session is not supported.

The UE may support the IMS_Registration_handling policy.

If the UE supports the IMS_Registration_handling policy, the UE may support being configured with the IMS_Registration_handling policy using one or more of the following methods:

a) the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C];

b) the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

c) the IMS_Registration_Policy node of 3GPP TS 24.167 [8G].

If the UE is configured with both the IMS_Registration_Policy node of 3GPP TS 24.167 [8G] and the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or 3GPP TS 31.103 [15B], then the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file shall take precedence.

NOTE 1: Precedence for files configured on both the USIM and ISIM is defined in 3GPP TS 31.103 [15B].

If the UE is registered with IMS and the IMSVoPS indicator, provided by the lower layers (see 3GPP TS 24.501 [258]), indicates voice is not supported, the UE shall:

A) if the Stay_Registered_When_VoPS_Not_Supported leaf indicates requirement to stay registered, the UE needs not to deregister and maintains the registration as required for IMS services; or

NOTE 2: The UE will periodically refresh the registration when needed.

B) if the Stay_Registered_When_VoPS_Not_Supported leaf indicates requirement to deregister and the Deregistration_Timer leaf used to configure the NoVoPS-dereg timer defined in table 7.8.1 contains a timer value for the time to wait before deregistrerting from IMS, start a timer with the value indicated in the policy and:

a) if the timer expires before the UE receives an indication from the lower layers that IMS voice is supported:

1) if there is no ongoing IMS session, the UE either performs reregistration as specified in subclause 5.1.1.4 and shall only include feature tags associated with services that are independent of IMSVoPS indicator or deregister from the IMS following the procedures specified in subclause 5.1.1.6; or

2) if there is ongoing IMS session, and

i) if the UE does not receive indication from the lower layer that the IMS voice is supported before the ongoing IMS session is terminated, the UE either performs reregistration as specified in subclause 5.1.1.4 and shall only include feature tags associated with services that are independent of IMSVoPS indicator or deregister from the IMS following the procedures specified in subclause 5.1.1.6 as soon as the ongoing IMS based service is terminated; or

ii) if the UE receives indication from the lower layer that the IMS voice is supported before the ongoing IMS session is terminated, cancel the timer; or

NOTE 3: How the UE selects reregistration or deregistration is implementation dependent (e.g., SMS service)

b) if the UE receives an indication from the lower layers that IMS voice is supported before the timer expires, cancel the timer.

If the IMS_Registration_handling policy is not configured, the UE behaviour is implementation specific.

U.3.1.1 P-Access-Network-Info header field

The UE shall always include the P-Access-Network-Info header field where indicated in subclause 5.1.

NOTE: If the P-Access-Network-Info header field includes radio cell identity, the P-Access-Network-Info header field populated by the UE that supports Multi-RAT Dual Connectivity with the 5GCN as described in 3GPP TS 37.340 [264] will contain the information about the radio cell identity of the Master RAN node that is serving the UE.

U.3.1.1A Cellular-Network-Info header field

Not applicable.

U.3.1.2 Availability for calls

This subclause documents the minimal requirements for being available for voice communication services when using 5GS.

A UE shall perform an initial registration as specified in subclause 5.1.1.2 using a QoS flow for SIP signalling (see annex U.2.2.1), if all the following conditions are met:

1) if the UE is operating in the "voice centric" way;

2) if the UE is capable of receiving any (but not necessarily all) of the media types which the CS domain supports, such that the media type can also be used when accessing the IM CN subsystem using:

a) the 5GS IP-CAN via NR;

b) the 5GS IP-CAN via E-UTRA; or

NOTE 1: The use of 5GS IP-CAN via E-UTRA can also be the result of an inter-RAT fallback during setup of the IMS voice call. This can occur, for example, when a UE not supporting the media type in 5GS IP-CAN via NR initiates an IMS voice call in 5GS IP-CAN via NR.

c) the EPS IP-CAN;

NOTE 2: EPS can be used as IP-CAN as the result of an EPS fallback during setup of the IMS voice call. This can occur, for example, when a UE not supporting the media type in 5GS IP-CAN via NR initiates an IMS voice call in 5GS IP-CAN via NR, or when a UE not supporting the media type in 5GS IP-CAN via E-UTRA initiates an IMS voice call in 5GS IP-CAN via E-UTRA.

3) if:

a) the media type of item 2 is an "audio" media type;

b) the UE supports codecs suitable for (conversational) speech; and

c) the "audio" media type is not restricted from inclusion in an SDP message according to the media type restriction policy as specified in subclause 6.1.1;

and one of the following is true:

a) 3GPP PS data off status is "inactive";

b) 3GPP PS data off status is "active", the UE is in the HPLMN or EHPLMN or a subscribed SNPN, and MMTEL voice is a 3GPP PS data off exempt service; or

c) 3GPP PS data off status is "active", the UE is in a VPLMN or a non-subscribed SNPN, the UE is configured with an indication that MMTEL voice is a 3GPP PS data off roaming exempt service in the VPLMN or MMTEL voice is a 3GPP data off non-subscribed exempt service in the non-subscribed SNPN;

4) if the UE determines that its contact has not been bound to a public user identity using the IP-CAN, such that the contact is expected to be used for the delivery of incoming requests in the IM CN subsystem relating to the media of item 2 and item 3;

5) if the IMSVoPS indicator, provided by the lower layers indicates voice is supported;

6) if the procedures to perform the initial registration are enabled (see 3GPP TS 24.305 [8T]); and

7) if the PDU session used for IMS is:

a) available; or

b) not available, and the UE is allowed to send a PDU SESSION ESTABLISHMENT REQUEST message to establish a PDU session with 5GS QoS flow that is needed for performing the initial registration as described in U.2.2.1.

NOTE 3: Regardless of any of the above conditions, a UE might attempt to register with the IM CN subsystem at any time.

EXAMPLE: As an example of the note, a UE configured to preferably attempt to use the 5GS to access IM CN subsystem can perform an initial registration as specified in subclause 5.1.1.2, if the conditions in items 2, 3, 4, 5, 6 and 7 in this subclause, evaluate to true.

The UE indicates to the non-access stratum the status of being available for voice over PS when:

I) the UE is capable of receiving any (but not necessarily all) of the media types which the CS domain supports, such that the media type can also be used when accessing the IM CN subsystem using:

a) the 5GS IP-CAN via NR;b) the 5GS IP-CAN via E-UTRA; or

NOTE 4: The use of 5GS IP-CAN via E-UTRA can also be the result of an inter-RAT fallback during setup of the IMS voice call. This can occur, for example, when a UE not supporting the media type in 5GS IP-CAN via NR initiates an IMS voice call in 5GS IP-CAN via NR.

c) the EPS IP-CAN;

NOTE 5: EPS can be used as IP-CAN as the result of an EPS fallback during setup of the IMS voice call. This can occur, for example, when a UE not supporting the media type in 5GS IP-CAN via NR initiates an IMS voice call in 5GS IP-CAN via NR, or when a UE not supporting the media type in 5GS IP-CAN via E-UTRA initiates an IMS voice call in 5GS IP-CAN via E-UTRA.

II) if the media type of item I is an "audio" media type, the UE supports codecs suitable for (conversational) speech, the "audio" media type is not restricted from inclusion in an SDP message according to the media type restriction policy as specified in subclause 6.1.1; and:

a) 3GPP PS data off status is "inactive";

b) 3GPP PS data off status is "active", the UE is in the HPLMN, the EHPLMN or a subscribed SNPN, and MMTEL voice is a 3GPP PS data off exempt service; or

c) 3GPP PS data off status is "active", the UE is in a VPLMN or a non-subscribed SNPN, the UE is configured with an indication that MMTEL voice is a 3GPP PS data off roaming exempt service in the VPLMN or MMTEL voice is a 3GPP PS data off non-subscribed exempt service in the non-subscribed SNPN; and

III) the UE determines a contact has been bound to a public user identity using the IP-CAN, such that this contact is expected to be used for the delivery of incoming requests in the IM CN subsystem relating to such media.

The UE indicates to the non-access stratum the status of being not available for voice over PS when:

I) in response to receiving the IMSVoPS indicator indicating voice is supported, the UE:

- initiated an initial registration as specified in subclause 5.1.1.2, received a final response to the REGISTER request sent, but the conditions for indicating the status of being available for voice over PS are not met; or

- did not initiate an initial registration as specified in subclause 5.1.1.2 and, these conditions for indicating the status of being available for voice over PS are not met; or

II) the conditions for indicating the status of being available for voice over PS are no longer met.

NOTE 6: The status of being not available for voice over PS is used for domain selection for UE originating sessions / calls specified in 3GPP TS 23.501 [257] subclause 5.16.3.5.

U.3.1.2A Availability for SMS

The UE determines that the UE is able to use SMS using IMS if the UE:

I) is capable of using the MIME type "application/vnd.3gpp.sms" (see 3GPP TS 24.341 [8L]), such that the MIME type can also be used when accessing the IM CN subsystem using the current IP-CAN;

II) supports the role of an SM-over-IP sender (see 3GPP TS 24.341 [8L]);

IIA) determines the PDU session used for IMS exists;

III) determines a contact has been bound to a public user identity using the IP-CAN, such that this contact is expected to be used for the delivery of incoming requests in the IM CN subsystem relating to such media; and

IV) the UE does not determine that SMS over IP is restricted in 3GPP TS 24.341 [8L] subclause 5.2.1.3; and

V) the 3GPP PS data off status is:

- "inactive";

- "active", the UE is in the HPLMN, the EHPLMN or a subscribed SNPN, and SMS over IMS is a 3GPP PS data off exempt service; or

- "active", the UE is in a VPLMN or a non-subscribed SNPN, the UE is configured with an indication that SMS over IMS is a 3GPP PS data off roaming exempt service in the VPLMN or SMS over IMS is a 3GPP PS data off non-subscribed exempt service in the non-subscribed SNPN.

When above criteria are not met, the UE determines that SMS using IMS is unavailable.

NOTE: The status that SMS using IMS is unavailable is used for domain selection for UE originating SMS specified in 3GPP TS 23.501 [257] subclause 5.16.3.8.

U.3.1.3 Authorization header field

Void.

U.3.1.4 SIP handling at the terminating UE when precondition is not supported in the received INVITE request, the terminating UE does not have resources available and IP-CAN performs network-initiated resource reservation for the terminating UE

Upon receiving an INVITE request not including the "precondition" option-tag in the Supported header field and not including the "precondition" option-tag in the Require header field, and the IP-CAN performs network-initiated resource reservation for the UE, the UE:

1) if the INVITE request contains an SDP offer and the local resources required at the terminating UE for the received SDP offer are not available:

   a) shall not alert the user; and

   b) shall send 183 (Session Progress) response to the INVITE request without waiting for resource reservation and without alerting the user. If the INVITE request includes a Supported header field indicating support of reliable provisional responses, the UE shall send the 183 (Session Progress) response reliably. In the 183 (Session Progres) response, the UE shall include an SDP answer; and

2) if the INVITE request does not contain an SDP offer and the INVITE request includes a Supported header field indicating support of reliable provisional responses:

   a) shall generate an SDP offer;

   b) if the local resources required at the terminating UE for the generated SDP offer are not available:

      A) shall not alert the user; and

      B) shall reliably send 183 (Session Progress) response to the INVITE request without waiting for resource reservation and without alerting the user. In the 183 (Session Progres) response, the UE shall include the generated SDP offer.

Upon successful reservation of local resources, if the precondition mechanism is not used by the terminating UE, the UE can send 180 (Ringing) response to the INVITE request and can alert the user.

U.3.1.5 3GPP PS data off

If the 3GPP PS data off status is "active" the UE shall only send initial requests that:

1) are associated with a 3GPP IMS service which enforces 3GPP PS data off;

NOTE 1: These services are specified in 3GPP TS 22.011 [1C], and enforcement of 3GPP PS data off is described in the respective service specifications.

2) are associated with an emergency service; or

3) are associated with 3GPP PS data off exempt services configured in the UE using one or more of the following methods:

   - the non_3GPP_ICSIs_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in the HPLMN or the EHPLMN, or if the UE is in the VPLMN and the non_3GPP_ICSIs_roaming_exempt node specified in 3GPP TS 24.167 [8G] is not configured;

   - the non_3GPP_ICSIs_roaming_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in the VPLMN;

   - the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], if the UE is in the HPLMN or the EHPLMN, or if the UE is in the VPLMN and the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] is not configured;

- the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], if the UE is in the VPLMN;

- the /<X>/SNPN_Configuration/<X>/3GPP_PS_data_off/non_3GPP_ICSIs_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in the subscribed SNPN, or if the UE is in a non-subscribed SNPN and the non_3GPP_ICSIs_non-subscribed_exempt node specified in 3GPP TS 24.167 [8G] is not configured; or

- the non_3GPP_ICSIs_non-subscribed_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in a non-subscribed SNPN.

If the UE is configured with both the non_3GPP_ICSIs_exempt node of 3GPP TS 24.167 [8G] and the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], then the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] shall take precedence.

If the UE is configured with both the non_3GPP_ICSIs_roaming_exempt node of 3GPP TS 24.167 [8G] and the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], then the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] shall take precedence.

If the 3GPP PS data off status changes from "inactive" to "active" the UE shall release all dialogs that

1) are not associated with a 3GPP IMS service which enforces 3GPP PS data off;

NOTE 2: These services are specified in 3GPP TS 22.011 [1C], and enforcement of 3GPP PS data off is described in the respective service specifications.

2) are not associated with an emergency service; and

3) are not associated with 3GPP data off exempt services configured in the UE using one or more of the following methods:

- the non_3GPP_ICSIs_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in the HPLMN or the EHPLMN, or if the UE is in the VPLMN and the non_3GPP_ICSIs_roaming_exempt node specified in 3GPP TS 24.167 [8G] is not configured;

- the non_3GPP_ICSIs_roaming_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in a VPLMN;

- the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], if the UE is in the VPLMN;

- the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], if the UE is in the HPLMN or the EHPLMN, or if the UE is in the VPLMN and the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] is not configured; or

- the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], if the UE is in the VPLMN.

- the /<X>/SNPN_Configuration/<X>/3GPP_PS_data_off/non_3GPP_ICSIs_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in the subscribed SNPN, or if the UE is in a non-subscribed SNPN and the non_3GPP_ICSIs_non-subscribed_exempt node specified in 3GPP TS 24.167 [8G] is not configured;

- the non_3GPP_ICSIs_non-subscribed_exempt node specified in 3GPP TS 24.167 [8G], if the UE is in a non-subscribed SNPN.

If the UE is configured with both the non_3GPP_ICSIs_exempt node of 3GPP TS 24.167 [8G] and the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], then the non_3GPP_ICSIs_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] shall take precedence.

If the UE is configured with both the non_3GPP_ICSIs_roaming_exempt node of 3GPP TS 24.167 [8G] and the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C], then the non_3GPP_ICSIs_roaming_exempt node in the $EF_{3GPPPSDATAOFFservicelist}$ file described in 3GPP TS 31.102 [15C] shall take precedence.

U.3.1.6 RLOS

Not applicable.

U.3.1.7 SIP handling at the originating UE when redirecting the UE from NG-RAN to E-UTRAN fails

When a failure occurs in the process of redirecting the UE from NG-RAN to E-UTRAN and the UE is aware of the failure, the UE shall send out a CANCEL request to cancel the INVITE request that includes a Reason header field with a protocol value set to "RELEASE_CAUSE" and a "cause" header field parameter with the value of "7" as specified in subclause 7.2A.18.11.2.

U.3.1.8 Unified Access Control

The following information is provided to the non-access stratum:

- MO--IMS-registration-related-signalling-started; and
- MO-IMS-registration-related-signalling-ended;

Prior to sending a REGISTER request which is not for emergency registration, the UE sends the MO-IMS-registration-related-signalling-started to the non-access stratum and

a) if the barring result is "not-barred", continues with registration procedure as described in subclause 5.1.1; and

b) if the barring result is "barred", aborts the registration procedure;

When the UE needs to send SUBSCRIBE request for the reg event package, then the UE sends the MO-IMS-registration-related-signalling-started to the non-access stratum before sending SUBSCRIBE request and

a) if the barring result is "not-barred", continues with subscribe procedure as described in subclause 5.1.1.3; and

b) if the barring result is "barred", aborts the subscribe procedure;

When a procedure for MO IMS registration related signalling ends, i.e.

- a final response to the REGISTER request which is not for emergency registration is received;
- a final response to the SUBSCRIBE request for the reg event package is received;
- timer F expires at the UE; or
- a fatal transport error for sending the REGISTER request or the SUBSCRIBE request is reported by the transport layer, as desribed in IETF RFC 3261 [26];

the UE sends the MO-IMS-registration-related-signalling-ended to the non-access stratum.

U.3.1.9 Abnormal cases

Upon sending MO-IMS-registration-related-signalling-started indication to the non-access stratum as described in subclause U.3.1.8, if:

a) the UE receives, from the lower layers, a notification that the service request

   1) was not accepted due to congestion; or

   2) resulted in starting timer T3525 (see 3GPP TS 24.501 [258]); and

b) a procedure for MO IMS registration related signalling has not ended;

then:

a) the UE shall abort the procedure for MO-IMS-registration-related-signalling and send MO-IMS-registration-related-signalling-ended indication to the non-access stratum; and

b) if an alternative radio access network is available, the UE may attempt the procedure for MO-IMS-registration-related signalling on the alternative radio access network.

U.3.2 Procedures at the P-CSCF

U.3.2.0 Registration and authentication

Void.

U.3.2.1 Determining network to which the originating user is attached

If the P-CSCF is configured to handle emergency requests, in order to determine from which network the request was originated the P-CSCF shall check the MCC and MNC fields received:

- during the registration procedure from the Rx interface as defined in 3GPP TS 29.214 [13D] (e.g. used for deployments without IMS-level roaming interfaces where the P-CSCF is located in the home network); or
- from the P-Access-Network-Info header field.

NOTE: The above check can be against more than one MNC code stored in the P-CSCF.

U.3.2.2 Location information handling

Void.

U.3.2.3 Prohibited usage of PDU session for emergency services

If the P-CSCF detects that a UE uses a PDU session for emergency services for a non-emergency REGISTER request, the P-CSCF shall reject that request by a 403 (Forbidden) response.

NOTE: By assigning specific IP address ranges for a PDU session for emergency bearer services and configuring those ranges in P-CSCF, the P-CSCF can detect based on the registered Contact address if UE uses an emergency PDU session for initial registration.

U.3.2.4 Support for paging policy differentiation

The P-CSCF may support paging policy differentiation by marking packet(s) to be sent towards the UE related to that IMS capability. A specific DSCP (IPv4) value and/or a specific Traffic Class (IPv6) value are assigned by local configuration in the P-CSCF.

If local policy requires to provide such marking, the P-CSCF shall identify terminating requests which:

a) contain SDP with an "audio" media line and which are related to a IMS multimedia telephony service session specified in 3GPP TS 24.173 [8H]; or

b) do not contain an SDP offer but some indication, e.g. a feature capability indicator, indicates that an "audio" media line that would meet network policy for such differentiation, could form part of the subsequent SDP offer.

NOTE 1: Precise details of such indications, if any, are subject to operator policy. Alternatively the operator policy can be to not preferentially page requests without an SDP offer.

For such identified requests:

a) where an unreliable transport mechanism is used as the transport protocol for SIP, the P-CSCF shall mark packets containing an INVITE request; and

b) if a reliable transport mechanism is used as the transport protocol for SIP:

   1) if a new reliable transport connection needs to be established, the P-CSCF shall turn on the marking of packets within the reliable transport connection in advance of sending an INVITE request; and

2) if there is an existing reliable transport connection, the P-CSCF may turn on the marking of packets within the reliable transport connection in advance of sending an INVITE request.

In both these cases for a reliable transport connection, the P-CSCF shall turn off the marking of packets within the reliable transport connection at an appropriate time.

U.3.2.5 Void

U.3.2.6 Resource sharing

This feature is not supported in this release.

U.3.2.7 Priority sharing

This feature is not supported in this release.

U.3.2.8 RLOS

Not applicable.

U.3.2.9 Support of ANBR and RAN-assisted codec adaptation

If the network supports ANBR as specified in 3GPP TS 26.114 [9B] and RAN-assisted codec adaptation as specified in 3GPP TS 38.300 [270] and 3GPP TS 38.321 [271], then the P-CSCF might be configured to indicate ANBR support.

If the P-CSCF is configured to indicate ANBR support, when the P-CSCF receives the 200 (OK) response to the REGISTER request the P-CSCF shall include the "g.3gpp.anbr" feature-capability indicator in the Feature-Caps header field of the 200 (OK) response to the REGISTER request.

U.3.3 Procedures at the S-CSCF

U.3.3.1 Notification of AS about registration status

Not applicable.

U.3.3.2 RLOS

Not applicable.

U.4 3GPP specific encoding for SIP header field extensions

U.4.1 Void

U.5 Use of circuit-switched domain

There is no CS domain in this access technology.

Annex V (normative): HTTP Profiling

V.1 Scope

The present annex defines the HTTP messages and data types sent over reference points specified in the present document.

V.2 Ms reference point

V.2.1 General

The Ms reference point is used to request signing of an Identity header field or request verification of a signed assertion in an Identity header field.

HTTP POST method is used for the verification request.

HTTP 200 (OK) is used when the AS for verification has successfully processed the verification request.

HTTP POST method is used for the signing request.

HTTP 200 (OK) is used when the AS for signing has successfully processed the signing request.

HTTP POST method is used for the diversion signing request.

HTTP 200 (OK) is used when the AS for signing has successfully processed the diversion signing request.

HTTP POST method is used for the Resource-Priority header field signing request.

HTTP 200 (OK) is used when the AS for signing has successfully processed the Resource-Priority header field signing request.

HTTP POST method is used for the Resource-Priority and Priority header fields signing request.

HTTP 200 (OK) is used when the AS for signing has successfully processed the Resource-Priority and Priority header field signing request.

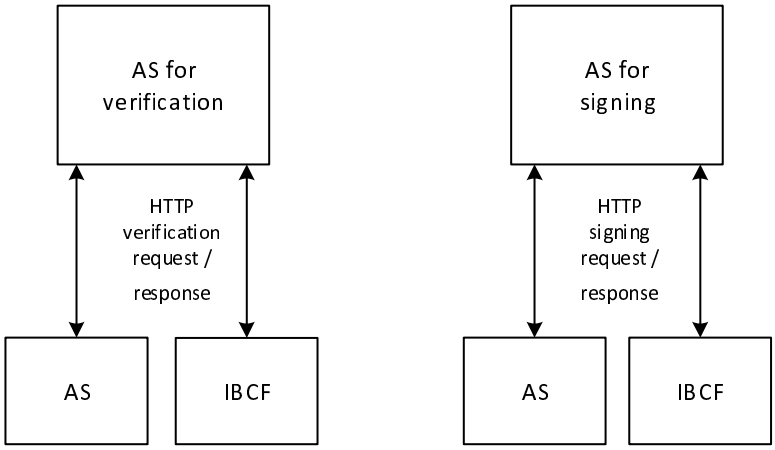


Figure V.2.1-1: Usage of the Ms reference point

V.2.2 Resource structure

API resources are defined with respect to a "server root". The server root is a URI:

- {hostname}:{port}/{RoutingPath},

The resource URI structure is:

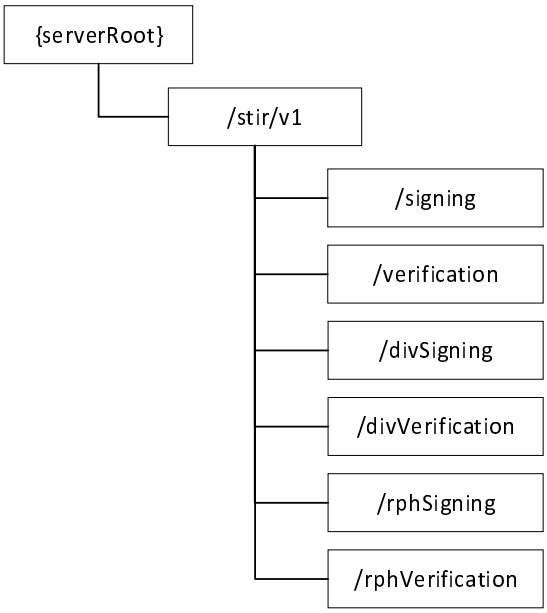


Figure V.2.2-1: Resource structure for the resource exposed over the Ms reference point

NOTE: v1 is the version number of the API.

Table V.2.2-1: Variables for the server root

Variable	Description	Presence
hostname	Host name used to reach the resource.	M
port	Port where the resource is reached	M
RoutingPath	Path identifying the resource	M

V.2.3 Request requirements

V.2.3.1 General

V.2.3.2 Request header requirements

Table V.2.3.2-1 lists request header field requirements.

Table V.2.3.2-1: Header fields included in the requests

Header field name	Description	Presence
Content-Type	Describes the format of the request body. Shall be set to "application/json"	M
Accept	Describes the supported format of the response body. Shall be set to "application/json" if present	O

V.2.4 Response requirements

V.2.4.1 General

V.2.4.2 Response header requirements

Table V.2.4.2-1: Header fields included in the responses

Header field name	Description	Presence
Content-Type	Describes the format of the response body. Shall be set to "application/json"	M

V.2.4.3 Error response requirements

V.2.4.3.1 General

If the server cannot process the request, the server provides an HTTP error response. The error response contains a JSON object specifying the error type.

The server provides a service error when the server is unable to process the request.

The server provides a policy error when the server is able to process the request, but not able to complete the service execution due to a policy restriction.

V.2.4.3.2 Service errors

Table V.2.4.3.2-1: Service failure descriptions

Exception ID	Exception text	HTTP status code	Exception variables	Description
-	Error: Missing request body.	400	-	The request could not be processed due to missing request body.
-	Error: Missing mandatory parameter.	400	-	The request could not be processed due to missing parameters.
-	Error: Requested response body type is not supported.	406	-	The request could not be processed due to a not supported message body format.
-	Error: Requested resource not found.	404	-	The request could not be processed due to no resource available related to the Request-URI
-	Error: Unsupported request body type.	415	-	The request could not be processed due to not supported message body.
-	Error: Invalid parameter value.	400	-	The request could not be processed due to invalid parameter value.
-	Error: Failed to parse message body.	400	-	The request could not be processed due to failure to parse the message body.
-	Error: Missing mandatory Content-Length headers	411	-	The request could not be processed due to a missing Content-Length header.

V.2.4.3.3 Policy errors

Table V.2.4.3.3-1: Policy failure descriptions

Exception ID	Exception text	HTTP status code	Exception variables	Description
-	Method not allowed	405	-	The resource was invoked with unsupported operation
-	Internal server error.	500	-	The request failed due to internal error

V.2.5 signing

V.2.5.1 General

To construct a PASSporT SHAKEN object specified in RFC 8588 [261], a PASSporT rph object specified in RFC 8443 [279] and RFC 9027 [278], or a PASSporT div object, specified in RFC 8946 [265], the client sends an HTTP POST request to the AS for signing containing the SIP request information plus any additional claim information to be signed by the target PASSporT type. If the request was successfully processed, then the received signingResponse contains an Identity header field value populated with a signed PASSporT of the requested type along with the appropriate Identity header field parameters.

Unsuccessful requests are responded with an HTTP 4xx or 5xx response.

V.2.5.2 Data types

Table V.2.5.2-1 specifies the data types included in the signing request. The signing request contains the claims included in:

- a PASSporT SHAKEN JSON Web Token, specified in RFC 8588 [261];
- a PASSporT div JSON Web Token specified in RFC 8946 [265]; or
- a PASSporT rph JSON Web Token specified in RFC 8443 [279] and RFC 9027 [278].

Table V.2.5.2-1: Data types for the signingRequest

Parameter	Type; Value	Presence	Description
ppt	string; set to the value of the "ppt" parameter in the protected header of the PASSporT to be created	O	This parameter is included to inform the AS for signing the type of PASSporT that is to be constructed and signed.
attest	string; "A", "B" or "C"	O	Identifying the relation between the service provider attesting the identity and the subscriber. Specified in RFC 8588 [261].
dest	array of identity claim JSON objects representing destination identities; tn or uri	M	Identifying the called user taken from the To header field for a "shaken" or "rph" PASSporT as specified in RFC 8224 [252], and from the Request-URI after retargeting for a "div" PASSporT as specified in RFC 8946 [265].
div	identity claim JSON object, tn or uri. A hi element should be included.	O	Identifying the diverting user; i.e., the user identified in the Request-URI before retargeting, taken from the History-Info header field as pecified in RFC 8946 [265].
iat	integer; time and date of issuance of the PASSporT token	M	Time since 1 January 1970 in Numeric Date format as specified in RFC 7519 [235].
orig	identity claim JSON object; tn or uri	M	Identifying the calling user. Specified in RFC 8225 [262].
origid	String; UUID	O	Specified in RFC 8588 [261]
rph	array of strings that correspond to the r-values indicated in the SIP Resource-Priority header field	O	Contains assertion of the priority level of the user to be used for a given communication session as specified in RFC 8443 [279] and RFC 9027 [278].
sph	string "psap-callback"	O	Contains header field value "psap-callback" of the SIP Priority header field as specified in RFC 9027 [278].

The signing request provides the following two options for identifying the PASSporT type to be constructed:

- the PASSporT type is indicated by invoking the signing resource associated with that PASSporT type, as defined in subclause V.2.2; or
- the PASSporT type is identified by including a ppt parameter when invoking the /signing resource defined in subclause V.2.2.

Table V.2.5.2-1a indicates representative signingRequest parameter inclusion rules per PASSporT type.

Table V.2.5.2-1a: signingRequest parameters per PASSporT Type

PASSporT Type	Parameter							
	attest	dest	div	iat	orig	origid	rph	sph
shaken	M	M	X	M	M	M	X	X
div	X	M	M	M	M	X	X	X
rph	X	M	X	M	M	X	M	O
NOTE: "M" means "mandatory", "O" means "optional", and "X" means "not applicable".								

Table V.2.5.2-2 further specifies the data types contained in the signing request parameters.

Table V.2.5.2-2: Data types for the signingRequest parameters

Parameter	Type; Value	Presence	Description
hi	string. An "index" header field parameter as specified in RFC 7044 [66]	O	The "index" header field parameter is included in the entry identifying the diverting user in the History-Info header field.
tn	string; allowed characters as for local-number-digits and global-number-digits defined in RFC 3966 [22]	M	The number needs to be in canonical form according to RFC 8224 [252] section 8.3 (see NOTE).
uri	string; A SIP URI as specified in RFC 3261 [26] following the generic guidelines in RFC [3986].	O	Used if the "orig" or "dest" is given in a SIP URI.
NOTE: The TAS is expected to provide a valid globally routable number.			

Table V.2.5.2-3 specifies the data types included in the signing response.

Table V.2.5.2-3: Data types for the signingResponse

Parameter	Type; Value	Presence	Description
identityHeader	string; Identity header field value as specified in RFC 8224 [252]	M	This string cannot be NULL

V.2.6 verification

V.2.6.1 General

To verify one or more received PASSporTs, the client sends a verification request in the form of an HTTP POST request to the AS for verification containing the Identity header field(s) populated with the PASSporT object(s) to be verified. The verification request also contains the following information:

- the originating identity and optionally all the diverting identities; and/or
- the Resource-Priority header field value and optionally the header field value "psap-callback" of the Priority header field.

The verificationResponse contains the outcome of the verification in a verstat claim with values as specified for the verstat tel URI parameter in subclause 7.2A.20 and in a verstatPriority claim with values as specified for the Priority-Verstat header field in subclause 7.2.21. The verificationResponse can optionally contain the claims of PASSporT(s) that passed verification.

The verificationResponse can also contain verification results information for each verified PASSporT, as follows:

- if verification passed, then verificationResponse contains the validated claims of the PASSporT;
- if verification failed, then verificationResponse contains information describing the reason for the failure.

Unsuccessful requests are responded with an HTTP 4xx or 5xx response.

V.2.6.2 Data types

Table V.2.6.2-1 specifies the data types included in the verification request.

Table V.2.6.2-1: Data types for the verificationRequest

Parameter	Type; Value	Presence	Description
identityHeader	string; Identity header field value for the originating identity as specified in RFC 8224 [252] and RFC 8588 [261].	M	This parameter contains the "shaken" Identity header field value to be verified (see NOTE 1).
identityHeaders	array of string; Identity header field values as specified in RFC 8224 [252] , RFC 8443 [279], RFC 8946 [265], RFC 9027 [278]. One identityHeader claim per received Identity header field is sent.	O	This array contains the "div" and "rph" Identity header field values to be verified (see NOTE 2).
to	string; identity claim JSON object; tn or uri	M	The destination identity taken from the To header field. Used when no div claim is included.
dest	string; identity claim JSON object; tn or uri	O	The destination identity taken from the Request-URI in the incoming request.
time	integer; Numeric date format defined in RFC 7519 [235]	M	Time based on the Date header field in the incoming request.
from	string; identity claim JSON object; tn or uri	M	The asserted identity, taken from the P-Asserted-Identity or the From header field of the incoming request
protectedHeaders	array of string; header field(s)	O	Contains the SIP header field(s) protected by claims in the PASSporT(s) of the identityHeaders array.
NOTE 1) For "shaken" PASSporT verification, an identityHeader parameter containing a "shaken" Identity header field shall be included in the verification request. NOTE 2) For "rph" PASSporT verification, an identityHeaders parameter containing an "rph" Identity header field shall be included in the verification request. If the identityHeader parameter contains a "shaken" Identity header field, and/or the identityHeaders parameter contains an "rph" Identity header field, then all "div" Identity header field(s) received in an INVITE request shall be included in the identityHeaders parameter of the verification request. If "rph" PASSporT verification is not to be performed, and if the INVITE request does not contain any "div" Identity header fields, then the identityHeaders parameter shall not be included in the verification request.			

Invocation of the verification request results in the verification of the Identity header fields in the identityHeader and identityHeaders parameters. In addition, a verification request invocation can verify the integrity of SIP header fields protected by the "div" and "rph" PASSporTs. When verification of SIP header field integrity is required, the protected SIP header field(s) information shall be conveyed in the protectedHeaders parameter.

The dest parameter is included when the verification request includes a div claim, and when the verification service is required to distinguish between a legitimately retargeted request and a maliciously replayed request.

Table V.2.6.2-1a indicates representative inclusion rules for the verification request identityHeader and identityHeaders parameters for the different combinations of PASSporT types to be verified.

Table V.2.6.2-1a: Verification request identityHeader and identityHeaders parameter inclusion rules

PASSporT type(s) to be verified	Parameter	
	identityHeader	identityHeaders
shaken	M	X
shaken, div	M	M
rph	X	M
rph, div	X	M
shaken, rph	M	M
shaken, rph, div	M	M
NOTE: "M" means "mandatory" and "X" means "not applicable".		

Table V.2.6.2-2 specifies the data types included in the verification response.

Table V.2.6.2-2: Data types for the verificationResponse

Parameter	Type; Value	Presence	Description
divResult	array of one or more [div, verstatValue] tuples	O	Parameter informing of the result of the verification of diverting identities. For each verified identity the verstat parameter is added to the verified identity.
verstatValue	string; set to a value defined in table 7.2A.20.3-1	O	Parameter informing of the result of the verification of originating identity. To be used in the verstat parameter added to the verified identity. The parameter is mandatory if the request contained a PASSporT SHAKEN JSON Web Token.
verstatPriority	string; set to a value defined in table 7.2.21-1	O	Parameter informing of the result of the verification of the Resource-Priority header field and optionally the header field value "psap-callback" of the Priority header field.
verifyResults	array of one or more verifyResult, as defined in table V.2.6.2-3	O	Each array entry contains the verification results of a PASSporT contained in the request.

Table V.2.6.2-3 specifies the mandatory data types of the verifyResult parameter included in the verification response.

Table V.2.6.2-3: Data types for mandatory verifyResult parameters

Parameter	Type; Value	Presence	Description
verifyResult	structured data type containing [ppt, status, validClaims, reasonCode, reasonText, passport] tuple	M	Contains the verification results of a single Identity header field contained in the identityHeader parameter or an entry of the identityHeaders array of the verification request.The ppt and status parameters are always present. The inclusion of the other parameters in the tuple depends on the value of the status parameter.
ppt	string, set to the value of the ppt parameter in the protected header of the PASSporT.	M	Identifies the type of PASSporT associated with this array entry.
status	String, set to the value pass, fail or none.	M	Identifies the verification result of the PASSporT associated with this array entry.

Each verifyResult entry of the verifyResults array conveys the verification results of an Identity header field contained in the identityHeader parameter or contained in an entry of the identityHeaders array of the verification request.

Additional verifyResult parameters are returned based on the value of the status parameter as follows:

- a status parameter with a value of "fail" returns the parameters shown in table V.2.6.2-4;
- a status parameter with a value of "pass" returns the parameter shown in table V.2.6.2-5; and
- a status parameter with a value of "none" returns no additional parameters.

A status parameter with a value of "none" indicates that the PASSporT type is not supported.

Table V.2.6.2-4 specifies the optional data types of the verifyResult parameter included in the verification response.

Table V.2.6.2-4: Data types of optional verifyResult parameters

Parameter	Type; Value	Presence	Description
reasonCode	integer	O	Identifies the 4xx failure reason code of the failing PASSporT, as defined in RFC 8224 [252].
reasonText	string	O	Identifies the failure text associated with the 4xx failure reason code, as specified in RFC 8224 [252].
passport	string	O	Contains the failing PASSporT
NOTE: These parameters are optional since they are included only when the verifyResult "status" parameter has a value of "fail".			

Table V.2.6.2-5 specifies the additional data types included in the verification response when the status parameter contains a value of "pass".

Table V.2.6.2-5: Data types of additional verifyResults parameter for status of "pass"

Parameter	Type; Value	Presence	Description
validClaims	JSON object	O	The validClaims parameter contains the payload of the verified PASSporT.

The validClaims parameter can be used:

- to verify the integrity of SIP header field information associated with the validated claims, where a mismatch results in a verification failure; or
- to ensure that SIP header field contents contain the information authorized by the validated claims, where a mismatch is resolved by updating the SIP header field to match the validated claims.

Annex W (normative): IP-Connectivity Access Network specific concepts when using the 5GCN via WLAN to access IM CN subsystem

W.1 Scope

The present annex defines IP-CAN specific requirements for a call control protocol for use in the IM CN subsystem based on the Session Initiation Protocol (SIP), and the associated Session Description Protocol (SDP), where the IP-CAN is the 5GCN via Wireless Local Access Network (WLAN).

W.2 IP-CAN aspects when connected to the IM CN subsystem

W.2.1 Introduction

A UE accessing the IM CN subsystem, and the IM CN subsystem itself, utilise the services provided by the 5GCN and the WLAN to provide packet-mode communication between the UE and the IM CN subsystem.

Requirements for the UE on the use of these packet-mode services are specified in this clause.

W.2.2 Procedures at the UE

W.2.2.1 Establishment of IP-CAN bearer and P-CSCF discovery

NOTE: The UE performs access network discovery and selection procedures as specified in 3GPP TS 24.502 [263] and executes access authentication signalling as described in 3GPP TS 24.502 [263] prior to perform the procedure to obtain a local IP address;

The UE handles an IP-CAN bearer for SIP signalling as follows:

1) the UE shall obtain a local IP address;

2) the UE shall establish an IKEv2 security association and an IPsec ESP security association as described in 3GPP TS 24.502 [263]; and

3) the IKEv2 security association and the IPsec ESP security association (tunnel) shall remain active throughout the period the UE is connected to the IM CN subsystem, i.e. from the initial registration and at least until the deregistration; and

In addition the procedures specified in Annex U.2.2.1 apply

The UE may support the policy on when a UE is allowed to transfer a PDU session providing access to IMS between 5GC via non-3GPP access and 5GS as specified in subclause U.2.1.1. If the UE is in a session and the policy indicates "a UE having an ongoing IMS session, is not allowed to transfer the PDU session providing access to IMS between 5GCN via non-3GPP access and 5GCN via NG-RAN" or if the UE is roaming when in a session and the policy indicates "a UE roaming in a VPLMN and having an ongoing IMS session, is not allowed to transfer the PDU session providing access to IMS between 5GCN via non-3GPP access and 5GCN via NG-RAN" the UE shall not handover the PDU session providing access to IMS from 5GC via non-3GPP access to 5GS.

If the UE is in a seesion in the EPS IP-CAN and the policy indicates "a UE is not allowed to transfer a PDU session providing access to IMS, if any, between 5GCN via non-3GPP access and 5GCN via NG-RAN" or the UE is roaming in an EPS IP-CAN and the policy indicates "a UE roaming in a VPLMN and having an ongoing IMS session, is allowed to transfer the PDU session providing access to IMS between 5GCN via non-3GPP access and 5GCN via NG-RAN" the

UE shall, if not prevented by other rules or policies, handover the PDU session providing access to IMS from 5GC via non-3GPP access to 5GS.

If the UE is in a 5GS IP-CAN and the policy indicates "a UE is not allowed to transfer a PDU session providing access to IMS, if any, between 5GCN via non-3GPP access and 5GCN via NG-RAN, irrespective of if the UE has an ongoing IMS session or not" or the UE is roaming in a 5GS IP-CAN and the policy indicates "a UE roaming in a VPLMN is not allowed to transfer a PDU session providing access to IMS, if any, between 5GCN via non-3GPP access and 5GCN via NG-RAN, irrespective of if the UE has an ongoing IMS session or not" the UE shall not handover the PDU session providing access to IMS from 5GC via non-3GPP access to 5GS.

W.2.2.1A Modification of an IP-CAN used for SIP signalling

The procedures specified in Annex U.2.2.1A apply.

W.2.2.1B Re-establishment of the IP-CAN used for SIP signalling

The procedures specified in Annex U.2.2.1B apply.

W.2.2.1C P-CSCF restoration procedure

The procedures specified in Annex U.2.2.1C apply.

W.2.2.2 Session management procedures

The procedures specified in Annex U.2.2.2 apply.

W.2.2.3 Mobility management procedures

The procedures specified in Annex U.2.2.3 apply.

W.2.2.4 Cell selection and lack of coverage

Not applicable.

W.2.2.5 5GS QoS flow for media

W.2.2.5.1 General requirements

NOTE 1: During establishment of a session, the UE establishes data streams(s) for media related to the session. Either the UE or the network can request for resource allocations for media, but the establishment and modification of the 5GS QoS flow is controlled by the network as described in 3GPP TS 24.501 [258].

If the resource allocation is initiated by the UE, the UE starts reserving resources whenever it has sufficient information about the media streams, and used codecs available as specified in 3GPP TS 24.501 [258] and 3GPP TS 24.502 [263].

NOTE 2: If the resource reservation requests are initiated by the network, then the establishment of 5GS QoS flow for media is initiated by the network after the P-CSCF has authorised the respective 5GS QoS flows and provided the QoS requirements to the PCF.

W.2.2.5.1A Activation or modification of QoS flows for media by the UE

If the UE is configured not to initiate resource allocation for media according to 3GPP TS 24.167 [8G], then the UE shall refrain from requesting additional 5GS QoS flow(s) for media until the UE considers that the network did not initiate resource allocation for the media.

W.2.2.5.1B Activation or modification of QoS flows for media by the network

If the UE receives an activation request from the network for a 5GS QoS flow for media which is associated with the 5GS QoS flow used for signalling, the UE shall correlate the media 5GS QoS flow with a currently ongoing SIP session establishment or SIP session modification.

If the UE receives a modification request from the network for a 5GS QoS flow that is used for one or more media streams in an ongoing SIP session, the UE shall:

1) modify the related PDU session context in accordance with the request received from the network.

W.2.2.5.1C Deactivation of a QoS flow for media

When a data stream for media related to a session is released, if the 5GS QoS flow transporting the data stream is no longer needed and allocation of the 5GS QoS flow was requested by the UE, then the UE releases the 5GS QoS flow.

NOTE: The 5GS QoS flow can be needed e.g. for other data streams of a session or for other applications in the UE.

W.2.2.5.2 Special requirements applying to forked responses

The procedures specified in Annex U.2.2.5.2 apply.

W.2.2.5.3 Unsuccessful situations

Not applicable.

W.2.2.6 Emergency service

W.2.2.6.1 General

Emergency session is supported over the WLAN access if the UE has failed or has not been able to use 3GPP access to set up an emergency session as described in 3GPP TS 23.167 [4B] annex L. IMS emergency session is also supported for UEs with unavailable IMSI (i.e. a UE without USIM) or unauthenticated IMSI.

Some jurisdictions allow emergency calls to be made when the UE does not contain an ISIM or USIM, or where the credentials are not accepted.

The UE determines that the 5GCN supports emergency services via WLAN when the Emergency service support for non-3GPP (EMCN3) access indicator in the REGISTRATION ACCEPT message indicates emergency services are supported over non-3GPP access as defined in subclause 9.11.3.5 of 3GPP TS 24.501 [258].

When the UE is registered over a WLAN access and detects an emergency call attempt, if the UE supports the emerg-non3gpp timer defined in table 7.8.1 and has determined that 5GCN supports emergency services via WLAN the UE shall start the emerg-non3gpp timer when starting a domain selection searching for a 3GPP access usable to establish an emergency call. The UE shall stop the timer when a 3GPP access supporting emergency call is found. When the emerg-non3gpp timer expires the UE shall consider that it has failed to use 3GPP access to setup the emergency call and shall attempt to setup the emergency call over the available WLAN access.

The UE may support being configured for the emerg-non3gpp timer using one or more of the following methods:

a) the Timer_Emerg_non3gpp leaf of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C];

b) the Timer_Emerg_non3gpp leaf of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

c) the Timer_Emerg_non3gpp leaf of 3GPP TS 24.167 [8G].

When the IM CN subsystem is selected as the domain for the emergency call attempt, the UE determines whether it is currently attached to its home operator's network (e.g. HPLMN) or not (e.g. VPLMN) after it has determined that the 5GCN supports emergency services via WLAN.

To perform emergency registration, the UE shall request to establish an emergency PDU session as described in 3GPP TS 24.501 [258]. The procedures for PDU session establishment and P-CSCF discovery, as described in subclause W.2.2.1 of this specification apply accordingly.

If the ME is equipped with a UICC, in order to find out whether the UE is attached to the home PLMN or to the visited PLMN, the UE shall compare the MCC and MNC values derived from its IMSI with the MCC and MNC of the PLMN the UE is attached to. If the MCC and MNC of the PLMN the UE is attached to do not match with the MCC and MNC derived from the IMSI, then for the purposes of emergency calls in the IM CN subsystem the UE shall consider to be attached to a VPLMN. If the ME is not equipped with a UICC, the procedure to find d out whether the UE is attached to the home PLMN or to the visited PLMN for the purpose of emergency calls in the IM CN subsystem, is implementation specific.

NOTE: The UE verifies if a detected emergency number is still present in the Extended Local Emergency Number List applicable to the PLMN it is currently using. It is possible for the number to no longer be present in the Extended Local Emergency Number List if:

- the PLMN attached to relies on the Local Emergency Number List for deriving a URN; or
- the previously received Extended Emergency Number List Validity field indicated "Extended Local Emergency Numbers List is valid only in the PLMN from which this IE is received".

If the UE detected an emergency number, the UE subsequently uses a different PLMN than the PLMN from which the UE received the last Extended Local Emergency Number List, the dialled number is not stored in the ME, in the USIM and in the Local Emergency Number List, and:

- the REGISTRATION ACCEPT message received from the different PLMN contains the Extended Local Emergency Number List and the emergency number is present in the updated Extended Local Emergency Number List then the UE uses the updated Extended Local Emergency Number List when it performs the procedures in subclause W.2.2.6.1B; and
- the REGISTRATION ACCEPT message received from the different PLMN contains no Extended Local Emergency Number List or the emergency number is no longer present in the updated Extended Local Emergency Number List then the UE shall attempt UE procedures for SIP that relate to emergency using emergency service URN "urn:service:sos".

If the dialled number is equal to a local emergency number stored in the Extended Local Emergency Number List (as defined in 3GPP TS 24.301 [8J]), then the UE shall recognize such a number as for an emergency call and:

- if the dialled number is equal to an emergency number stored in the ME, or in the USIM, then the UE shall perform either procedures in the subclause W.2.2.6.1B or the procedures in subclause W.2.2.6.1A; and
- if the dialled number in not equal to an emergency number stored in the ME, or in the USIM, then the UE shall perform procedures in the subclause W.2.2.6.1B.

If the dialled number is not equal to a local emergency number stored in the Extended Local Emergency Number List (as defined in 3GPP TS 24.301 [8J]) and:

- if the dialled number is equal to an emergency number stored in the ME, in the USIM or in the Local Emergency Number List (as defined in 3GPP TS 24.008 [8]), then the UE shall recognize such a number as for an emergency call and performs the procedures in subclause W.2.2.6.1A.

Once IPsec tunnel setup is completed, the UE shall follow the procedures described in subclause W.2.2.1 of this specification for establishment of IP-CAN bearer and P-CSCF discovery accordingly.

Upon reception of a 380 (Alternative Service) response to an INVITE request as defined in subclause 5.1.2A.1.1 and subclause 5.1.3.1, if:

- the 380 (Alternate Service) response contains a Contact header field;
- the value of the Contact header field is a service URN; and
- the service URN has a top-level service type of "sos";

then the UE determines that "emergency service information is included" as described 3GPP TS 23.167 [4B].

Upon reception of a 380 (Alternative Service) response to an INVITE request as defined in subclause 5.1.3.1, if the 380 (Alternate Service) response does not contain a Contact header field with service URN that has a top-level service type of "sos", then the UE determines that "no emergency service information is included" as described 3GPP TS 23.167 [4B].

Upon reception of a 380 (Alternative Service) response to an INVITE request as defined in subclause 5.1.2A.1.1 and subclause 5.1.3.1, the UE shall proceed as follows:

1) if a 3GPP access network is available and the UE has not already attempted to use a 3GPP access network to set up an emergency session as described in 3GPP TS 23.167 [4B] annex L, when the UE selects a domain in accordance with the conventions and rules specified in 3GPP TS 22.101 [1A] and 3GPP TS 23.167 [4B], the UE shall attempt to select a domain of the 3GPP access network, and:

   - if the CS domain is selected, the UE behaviour is defined in subclause 7.1.2 of 3GPP TS 23.167 [4B] and in annex B; and

   - if the IM CN subsystem is selected, the UE shall apply the procedures in subclause 5.1.6 with the exception of selecting a domain for the emergency call attempt;

   In addition, when the UE determines that "it has not been able to use 3GPP access to set up an emergency session" in accordance with subclause L.1 of 3GPP TS 23.167 [4B], the UE shall apply the procedures in subclause 5.1.6 using WLAN, with the exception of selecting a domain for the emergency call attempt; and

2) if a 3GPP access network is not available, then the UE shall apply the procedures in subclause 5.1.6 using WLAN, with the exception of selecting a domain for the emergency call attempt.

When the emergency registration expires, the UE should disconnect the emergency PDU session.

W.2.2.6.1A Type of emergency service derived from emergency service category value

Annex U.2.2.6.1A applies.

W.2.2.6.1B Type of emergency service derived from extended local emergency number list

Annex U.2.2.6.1B applies.

W.2.2.6.2 eCall type of emergency service

The UE shall not send an INVITE request with Request-URI set to "urn:service:sos.ecall.manual" or "urn:service:sos.ecall.automatic".

W.2.2.6.3 Current location discovery during an emergency call

The UE may support the current location discovery during an emergency call specified in subclause 5.1.6.8.2, subclause 5.1.6.8.3, subclause 5.1.6.8.4, and subclause 5.1.6.12.

W.2A Usage of SDP

W.2A.0 General

Not applicable.

W.2A.1 Impact on SDP offer / answer of activation or modification of IP-CAN for media by the network

The procedures specified in Annex U.2A.1 apply.

W.2A.2 Handling of SDP at the terminating UE when originating UE has resources available and IP-CAN performs network-initiated resource reservation for terminating UE

Not applicable.

W.2A.3 Emergency service

No additional procedures defined.

W.3 Application usage of SIP

W.3.1 Procedures at the UE

W.3.1.0 Registration and authentication

The procedures specified in Annex U.3.1.0 apply with the following clarification:

- the UE shall perform reregistration of a previously registered public user identity bound to any one of its contact addresses when changing to an IP-CAN for which usage is specified in annex U or annex L.

W.3.1.0aIMS_Registration_handling policy

The IMS_Registration_handling policy indicates whether the UE deregisters from IMS after a configured amount of time after receiving an indication that the IMS Voice over PS Session is not supported.

The UE may support the IMS_Registration_handling policy.

If the UE supports the IMS_Registration_handling policy, the UE may support being configured with the IMS_Registration_handling policy using one or more of the following methods:

a) the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C];

b) the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.103 [15B]; and

c) the IMS_Registration_Policy node of 3GPP TS 24.167 [8G].

If the UE is configured with both the IMS_Registration_Policy node of 3GPP TS 24.167 [8G] and the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file described in 3GPP TS 31.102 [15C] or 3GPP TS 31.103 [15B], then the IMS_Registration_Policy node of the $EF_{IMSConfigData}$ file shall take precedence.

NOTE 1: Precedence for files configured on both the USIM and ISIM is defined in 3GPP TS 31.103 [15B].

If the UE is registered with IMS and the IMSVoPS indicator, provided by the lower layers (see 3GPP TS 24.501 [258]), indicates voice is not supported, the UE shall:

A) if the Stay_Registered_When_VoPS_Not_Supported leaf indicates requirement to stay registered, the UE needs not to deregister and maintains the registration as required for IMS services; or

NOTE 2: The UE will periodically refresh the registration when needed.

B) if the Stay_Registered_When_VoPS_Not_Supported leaf indicates requirement to deregister and the Deregistration_Timer leaf used to configure the NoVoPS-dereg timer defined in table 7.8.1 contains a timer value for the time to wait before deregistrerting from IMS, start a timer with the value indicated in the policy and:

a) if the timer expires before the UE receives an indication from the lower layers that IMS voice is supported:

1) if there is no ongoing IMS session, the UE either performs reregistration as specified in subclause 5.1.1.4 and shall only include feature tags associated with services that are independent of IMSVoPS indicator or deregister from the IMS following the procedures specified in subclause 5.1.1.6; or

2) if there is ongoing IMS session, and

   i) if the UE does not receive indication from the lower layer that the IMS voice is supported before the ongoing IMS session is terminated, the UE either performs reregistration as specified in subclause 5.1.1.4 and shall only include feature tags associated with services that are independent of IMSVoPS indicator or deregister from the IMS following the procedures specified in subclause 5.1.1.6 as soon as the ongoing IMS based service is terminated; or

   ii) if the UE receives indication from the lower layer that the IMS voice is supported before the ongoing IMS session is terminated, cancel the timer; or

NOTE 3: How the UE selects reregistration or deregistration is implementation dependent (e.g., SMS service)

b) if the UE receives an indication from the lower layers that IMS voice is supported before the timer expires, cancel the timer.

If the IMS_Registration_handling policy is not configured, the UE behaviour is implementation specific.

W.3.1.1 P-Access-Network-Info header field

The UE shall always include the P-Access-Network-Info header field where indicated in subclause 5.1.

W.3.1.1A Cellular-Network-Info header field

The UE:

1) using the 5GCN via Wireless Local Access Network (WLAN) as IP-CAN to access the IM CN subsystem; and

2) supporting one or more cellular radio access technology (e.g. NR);

shall always include the Cellular-Network-Info header field specified in subclause 7.2.15, if the information is available, in every request or response in which the P-Access-Network-Info header field is present.

W.3.1.2 Availability for calls

Not applicable.

W.3.1.2A Availability for SMS

Void.

W.3.1.3 Authorization header field

Void.

W.3.1.4 SIP handling at the terminating UE when precondition is not supported in the received INVITE request, the terminating UE does not have resources available and IP-CAN performs network-initiated resource reservation for the terminating UE

Not applicable.

W.3.1.5 3GPP PS data off

Not applicable.

W.3.1.6 Transport mechanisms

Void.

W.3.1.7 RLOS

Not applicable.

W.3.2 Procedures at the P-CSCF

W.3.2.0 Registration and authentication

Void.

W.3.2.1 Determining network to which the originating user is attached

The procedures specified in Annex U.3.2.1 apply.

W.3.2.2 Location information handling

Void.

W.3.2.3 Prohibited usage of PDU session for emergency bearer services

The procedures specified in Annex U.3.2.3 apply.

W.3.2.4 Support for paging policy differentiation

Void.

W.3.2.5 Void

W.3.2.6 Resource sharing

The feature is not supported in this release of the specification.

W.3.2.7 Priority sharing

The feature is not supported in this release of the specificaiton

W.3.2.8 RLOS

Not applicable.

W.3.3 Procedures at the S-CSCF

W.3.3.1 Notification of AS about registration status

Not applicable.

W.3.3.2 RLOS

Not applicable.

W.4 3GPP specific encoding for SIP header field extensions

W.4.1 Void

W.5 Use of circuit-switched domain

Void.

Annex X (informative): Support of SBA in IMS

X.1 Scope

This annex describes support for SBA for IMS nodes.

IMS nodes can use the SBA interfaces described in the present Annex as an alternative to the Diameter Rx and Cx and Sh reference points based on configuration. To support co-existence of IMS nodes supporting SBA services and IMS nodes not supporting SBA services SBI, enabled IMS nodes can support both SBI and non-SBI interfaces.

While the main body of the present document only describes usage of Diameter Rx and Cx and Sh reference points, the usage of the equivalent SBA services is a valid option.

NOTE 1: This version of the specification does not specify the details for usage of SBIs by P-CSCF, S-CSCF, I-CSCF and AS in the main body of the specification.

NOTE 2: This annex is intended to be used in conjunction with 5GC, see Annex U and Annex W.

X.2 Reference points to support SBA in IMS

The following IMS related reference points are realized by service-based interfaces:

- **N5**: Reference point between the PCF and an AF.

NOTE 1: The P-CSCF acts as an AF from the PCF point of view. The N5 Reference point is defined in 3GPP TS 23.501 [257] and the related protocol specification is in 3GPP TS 29.514 [273]. It provides equivalent functionality to the Diameter-based Rx reference point.

- **N70**: Reference point between an SBI capable I/S-CSCF and an SBI capable HSS.

NOTE 2: The N70 Reference point is defined in 3GPP TS 23.501 [257] and the related protocol specification is in 3GPP TS 29.562 [274]. It provides equivalent functionality to the Diameter-based Cx reference point.

- **N71**: Reference point between an SBI capable IMS AS and an SBI capable HSS.

NOTE 3: The N71 Reference point is defined in 3GPP TS 23.501 [257] and the related protocol specification is in 3GPP TS 29.562 [274]. It provides equivalent functionality to the Diameter-based Sh reference point.

X.3 Services to support SBA in IMS

If a P-CSCF uses the Npcf_PolicyAuthorization service, it will apply Npcf_PolicyAuthorization service operations (defined in 3GPP TS 29.514[273]) instead of Rx procedures (defined in 3GPP TS 29.214[13D]) and will interact with the PCF instead of the PCRF.

- **Npcf_PolicyAuthorization**: This service is provided by the PCF. This service is to authorise an AF request and to create policies as requested by the authorized AF for the PDU Session to which the AF session is bound. This service also allows the NF service consumer to subscribe/unsubscribe the notification of events.

NOTE 1: The P-CSCF acts as an AF from the PCF point of view. The Npcf_PolicyAuthorization service is defined in 3GPP TS 23.502 [275] and the related protocol specification is in 3GPP TS 29.514 [273]. It provides equivalent functionality to the Diameter-based Rx reference point.

If an I-CSCF or an S-CSCF uses the Nhss_ims services, it will apply Nhss_ims service operations instead of Cx procedures mentioned throughout the present document and will interact with an SBI capable HSS.

- **Nhss_imsUEContextManagement**: This service is provided by an SBI capable HSS. It enables service operations related to the management of a UE context.

- **Nhss_imsSubscriberDataManagement**: This service is provided by an SBI capable HSS. It enables service operations related to subscriber data management.

- **Nhss_imsUEAuthentication**: This service is provided by an SBI capable HSS. It enables a service operation related to the authentication between the end user and the home IMS network.

NOTE 2: The Nhss_imsUEContextManagement, Nhss_imsSubscriberDataManagement, and Nhss_imsUEAuthentication services are defined in annex AA of 3GPP TS 23.228 [7] and the related protocol specification is in 3GPP TS 29.562 [274]. They provide equivalent functionality to the Diameter-based Cx and Sh reference point.

NOTE 3: The Nhss_imsUEAuthentication service is not consumed by I-CSCF.

NOTE 4: The Nhss_imsUEAuthentication and Nhss_imsUEContextManagement services are not consumed by AS.

If an I-CSCF or an S-CSCF uses the Nnrf_NFDiscovery service, it will apply Nnrf_NFDiscovery service operations instead of Cx procedures mentioned throughout the present document and will interact with an SBI capable NRF.

NOTE 5: The Nnrf_NFDiscovery service is defined in 3GPP TS 23.502 [275] and the related protocol specification is in 3GPP TS 29.510 [288].

Annex Y (informative): Change history

Change history								
Date	**TSG #**	**TSG Doc.**	**CR**	**Rev**	**Subject/Comment**	**Old**	**New**	**WG doc**
					Version 0.0.0 Editor's internal draft			
					Version 0.0.1 Editor's internal draft			
					Version 0.0.2 Editor's internal draft			
		N1-001060			Version 0.0.3 Submitted to CN1 SIP adhoc #1			
19/10/00		N1-001109			Version 0.0.4 Reflecting results of initial CN1 discussion			
19/10/00		N1-001115			Version 0.0.5 Reflecting output of CN1 SIP adhoc#1 discussion			
09/11/00					Version 0.0.6 Revision to include latest template and styles			
		N1-010092			Version 0.0.7 Reflecting updates of some IETF drafts			
14/02/01		N1-010269			Version 0.0.8 Revision to include temporary annex B incorporating valuable source material			
18/03/01		N1-010378 rev			Version 0.1.0 incorporating results of CN1 discussion at CN1 #16			
12/04/01		N1-010737			Version 0.2.0 incorporating results of CN1 discussions at SIP adhoc #4			
11/06/01		N1-010935			Version 0.3.0 incorporating results of CN1 discussions at CN1 #16			
23/07/01		N1-011103			Version 0.4.0 incorporating results of CN1 discussions at CN1 #18 (agreed documents N1-011028, N1-011050, N1-011055, N1-011056)			
12/09/01		N1-011385			Version 0.5.0 incorporating results of CN1 discussions at CN1 #19 (agreed documents N1-011109, N1-011152, N1-011195, N1-011312, N1-011319, N1-011343)			
04/10/01		N1-011470			Version 0.6.0 incorporating results of CN1 discussions at CN1 #19bis (agreed documents N1-011346, N1-011373, N1-011389, N1-011390, N1-011392, N1-011393, N1-011394, N1-011408, N1-011410, N1-011426)			
19/10/01		N1-011643			Version 0.7.0 incorporating results of CN1 discussions at CN1 #20 (agreed documents N1-011477, N1-011479, N1-011498, N1-011523, N1-011548, N1-011585, N1-011586, N1-011592, N1-011611, N1-011629)			
16/11/01		N1-011821			Version 0.8.0 incorporating results of CN1 discussions at CN1 #20bis (agreed documents N1-011685, N1-011690, N1-011741, N1-011743, N1-011759, N1-011760, N1-011761, N1-011765c, N1-011767, N1-011769, N1-011770, N1-011771, N1-011774, N1-011777, N1-011779, N1-011780) N1-011712 was agreed but determined to have no impact on the specification at this time.			
30/11/01		N1-020010			Version 1.0.0 incorporating results of CN1 discussions at CN1 #21 (agreed documents N1-011828, N1-011829, N1-011836, N1-011899 [revision marks not used on moved text - additional change from chairman's report incorporated], implementation of subclause 3.1 editor's note based on discussion of N1-011900 [chairman's report], N1-011905, N1-011984, N1-011985, N1-011986, N1-011988, N1-011989, N1-012012 [excluding points 2 and 16], N1-012013, N1-012014 [excluding point 1], N1-012015, N1-012021, N1-012022, N1-012025, N1-012031, N1-012045, N1-012056, N1-012057) CN1 agreed for presentation for information to CN plenary.			
18/01/02		N1-020189			Version 1.1.0 incorporating results of CN1 discussions at CN1 SIP ad-hoc (agreed documents N1-020015, N1-020053, N1-020064, N1-020101, N1-020123, N1-020124, N1-020142, N1-020146, N1-020147, N1-020148, N1-020151, N1-020157, N1-020159, N1-020165). Also N1-012000 (agreed at previous meeting) required, subclause 5.2.6 to be deleted and this change has been enacted			

Change history								
Date	**TSG #**	**TSG Doc.**	**CR**	**Rev**	**Subject/Comment**	**Old**	**New**	**WG doc**
01/02/02		N1-020459			Version 1.2.0 incorporating results of CN1 discussions at CN1 #22 (agreed documents N1-020198, N1-020396, N1-020398, N1-020399, N1-020408, N1-020417, N1-020418, N1-020419, N1-020421, N1-020422, N1-020436, N1-020437, N1-020449)			
01/02/02		N1-020569			Version 1.2.1 issues to correct cut and paste error in incorporation of Annex B into main document. Affected subclause 5.1.1.3. Change to clause 7 title that was incorrectly applied to subclause 7.2 also corrected.			
22/02/02					Advanced to version 2.0.0 based on agreement of N1-020515. Version 2.0.0 incorporating results of CN1 discussions at CN1 #22bis (agreed documents N1-020466, N1-020468, N1-020469, N1-020472, N1-020473, N1-020500, N1-020504, N1-020507, N1-020511, N1-020512, N1-020521, N1-020583, N1-020584, N1-020602, N1-020603, N1-020604, N1-020611, N1-020612, N1-020613, N1-020614, N1-020615, N1-020617, N1-020623, N1-020624, N1-020625, N1-020626, N1-020627, N1-020642, N1-020643, N1-020646, N1-020649, N1-020656, N1-020659, N1-020668, N1-020669, N1-020670, N1-020671). In addition N1-020409, agreed at CN1#22 but missed from the previous version, was also implemented. References have been resequenced.			
02/03/02					Editorial clean-up by ETSI/MCC.	2.0.0	2.0.1	
11/03/02	TSG CN#15	NP-020049			The draft was approved, and 3GPP TS 24.229 was then to be issued in Rel-5 under formal change control.	2.0.1	5.0.0	
2002-06	NP-16	NP-020230	004	1	S-CSCF Actions on Authentication Failure	5.0.0	5.1.0	N1-020903
2002-06	NP-16	NP-020230	005	2	Disallow Parallel Registrations	5.0.0	5.1.0	N1-020959
2002-06	NP-16	NP-020230	007	1	Hiding	5.0.0	5.1.0	N1-020910
2002-06	NP-16	NP-020312	008	8	Support for services for unregistered users	5.0.0	5.1.0	
2002-06			009	1	Not implemented nor implementable. In the meeting report CN1#24 under doc N1-021513 it is shown that CR095r2 supercedes 009r1 if 095r2 was to be approved in CN#16 (but unfortunately 009r1 was also approved in the the CN#16 draft minutes).			N1-020921
2002-06	NP-16	NP-020231	019		MGCF procedure clarification	5.0.0	5.1.0	N1-020788
2002-06	NP-16	NP-020231	020	2	MGCF procedure error cases	5.0.0	5.1.0	N1-020960
2002-06	NP-16	NP-020231	022	1	Abbreviations clean up	5.0.0	5.1.0	N1-020949
2002-06	NP-16	NP-020231	023		Clarification of SIP usage outside IM CN subsystem	5.0.0	5.1.0	N1-020792
2002-06	NP-16	NP-020314	024	3	Replacement of COMET by UPDATE	5.0.0	5.1.0	
2002-06	NP-16	NP-020231	025	3	Incorporation of current RFC numbers	5.0.0	5.1.0	N1-021091
2002-06	NP-16	NP-020231	026	1	Clarification of B2BUA usage in roles	5.0.0	5.1.0	N1-020941
2002-06	NP-16	NP-020231	028	4	Determination of MO / MT requests in I-CSCF(THIG)	5.0.0	5.1.0	N1-021248
2002-06	NP-16	NP-020231	030	2	P-CSCF release of an existing session	5.0.0	5.1.0	N1-021006
2002-06	NP-16	NP-020232	031	1	S-CSCF release of an existing session	5.0.0	5.1.0	N1-020939
2002-06	NP-16	NP-020232	033	3	SDP procedure at the UE	5.0.0	5.1.0	N1-020971
2002-06	NP-16	NP-020232	035	1	AS Procedures corrections	5.0.0	5.1.0	N1-020934

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Date	**TSG #**	**TSG Doc.**	**CR**	**Rev**	**Subject/Comment**	**Old**	**New**	**WG doc**
2002-06	NP-16	NP-020232	036	8	Corrections to SIP Compression	5.0.0	5.1.0	N1-021499
2002-06	NP-16	NP-020232	037	1	Enhancement of S-CSCF and I-CSCF Routing Procedures for interworking with external networks	5.0.0	5.1.0	N1-020928
2002-06	NP-16	NP-020232	041	2	Delivery of IMS security parameters from S-CSCF to the P-CSCF by using proprietary auth-param	5.0.0	5.1.0	N1-021003
2002-06	NP-16	NP-020232	045		Cleanup of request / response terminology - clause 5	5.0.0	5.1.0	N1-020835
2002-06	NP-16	NP-020232	046		Cleanup of request / response terminology - clause 6	5.0.0	5.1.0	N1-020836
2002-06	NP-16	NP-020232	047	2	Simplification of profile tables	5.0.0	5.1.0	N1-021059
2002-06	NP-16	NP-020232	049		Forking options	5.0.0	5.1.0	N1-020839
2002-06	NP-16	NP-020315	050	1	Media-Authorization header corrections	5.0.0	5.1.0	
2002-06	NP-16	NP-020233	051	1	Clause 5.4 editorials (S-CSCF)	5.0.0	5.1.0	N1-020950
2002-06	NP-16	NP-020233	053	2	Integrity protection signalling from the P-CSCF to the S-CSCF	5.0.0	5.1.0	N1-021007
2002-06	NP-16	NP-020233	054		Representing IM CN subsystem functional entities in profile table roles	5.0.0	5.1.0	N1-020847
2002-06	NP-16	NP-020233	055		Clause 4 editorials	5.0.0	5.1.0	N1-020848
2002-06	NP-16	NP-020233	056		Clause 5.8 editorials (MRFC)	5.0.0	5.1.0	N1-020849
2002-06	NP-16	NP-020233	057	1	Annex A editorials, including precondition additions	5.0.0	5.1.0	N1-021001
2002-06	NP-16	NP-020233	058	2	Representing the registrar as a UA	5.0.0	5.1.0	N1-021054
2002-06	NP-16	NP-020233	059		Additional definitions	5.0.0	5.1.0	N1-020852
2002-06	NP-16	NP-020312	060	11	Restructuring of S-CSCF Registration Sections	5.0.0	5.1.0	
2002-06	NP-16	NP-020234	061	2	Determination of MOC / MTC at P-CSCF and S-CSCF	5.0.0	5.1.0	N1-021060
2002-06	NP-16	NP-020234	062		Correction to the terminating procedures	5.0.0	5.1.0	N1-020927
2002-06	NP-16	NP-020234	063		Loose Routing for Network Initiated Call Release Procedures	5.0.0	5.1.0	N1-020940
2002-06	NP-16	NP-020234	064		Incorporation of previously agreed corrections to clause 5.2.5.2 (N1-020416)	5.0.0	5.1.0	N1-021004
2002-06	NP-16	NP-020234	065		Clause 7.2 editorial corrections	5.0.0	5.1.0	N1-021005
2002-06	NP-16	NP-020234	067	2	S-CSCF routing of MO calls	5.0.0	5.1.0	N1-021097
2002-06	NP-16	NP-020234	068	1	I-CSCF routeing of dialog requests	5.0.0	5.1.0	N1-021078
2002-06	NP-16	NP-020234	069	2	Definition of the Tokanised-by parameter	5.0.0	5.1.0	N1-021096
2002-06	NP-16	NP-020235	070	3	SDP procedures at UE	5.0.0	5.1.0	N1-021453
2002-06	NP-16	NP-020235	073	2	Updates to the procedures involving the iFCs, following the Oulu iFC changes	5.0.0	5.1.0	N1-021440
2002-06	NP-16	NP-020235	074	1	Addition of DHCPv6 references to 24.229	5.0.0	5.1.0	N1-021086
2002-06	NP-16	NP-020235	075	1	Clarification to URL and address assignments	5.0.0	5.1.0	N1-021083
2002-06	NP-16	NP-020235	079	3	Downloading the implicitely registered public user identities from the S-CSCF to P-CSCF	5.0.0	5.1.0	N1-021510
2002-06	NP-16	NP-020235	080	3	Clarification of GPRS aspects	5.0.0	5.1.0	N1-021486
2002-06	NP-16	NP-020235	081	2	Introduction of Subscription Locator Function Interrogation at I-CSCF in 24.229	5.0.0	5.1.0	N1-021469
2002-06	NP-16	NP-020235	082	1	Introduction of Visited_Network_ID p-header	5.0.0	5.1.0	N1-021433
2002-06	NP-16	NP-020236	084	1	MRFC register addresses	5.0.0	5.1.0	N1-021434
2002-06	NP-16	NP-020236	085	1	MRFC INVITE interface editor's notes	5.0.0	5.1.0	N1-021470

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Date	TSG #	TSG Doc.	CR	Rev	Subject/Comment	Old	New	WG doc
2002-06	NP-16	NP-020236	086	1	MRFC OPTIONS interface editor's notes	5.0.0	5.1.0	N1-021471
2002-06	NP-16	NP-020236	087		MRFC PRACK & INFO editor's notes	5.0.0	5.1.0	N1-021159
2002-06	NP-16	NP-020236	088	1	MGCF OPTIONS interface editor's notes	5.0.0	5.1.0	N1-021472
2002-06	NP-16	NP-020236	089		MGCF reINVITE editor's notes	5.0.0	5.1.0	N1-021161
2002-06	NP-16	NP-020237	090		3PCC AS editor's notes	5.0.0	5.1.0	N1-021162
2002-06	NP-16	NP-020237	091		AS acting as terminating UA editor's notes	5.0.0	5.1.0	N1-021163
2002-06	NP-16	NP-020237	092	1	AS acting as originating UA editor's notes	5.0.0	5.1.0	N1-021466
2002-06	NP-16	NP-020237	093	2	Charging overview clause	5.0.0	5.1.0	N1-021512
2002-06	NP-16	NP-020237	094	1	Procedures for original-dialog-id P-header	5.0.0	5.1.0	N1-021456
2002-06	NP-16	NP-020237	095	2	Procedures for charging-vector P-header	5.0.0	5.1.0	N1-021513
2002-06	NP-16	NP-020237	096	1	Procedures for charging-function-addresses P-header	5.0.0	5.1.0	N1-021458
2002-06	NP-16	NP-020237	097	1	SDP types	5.0.0	5.1.0	N1-021467
2002-06	NP-16	NP-020237	100		Removal of State from profile tables	5.0.0	5.1.0	N1-021173
2002-06	NP-16	NP-020238	101		Editor's note cleanup - clause 3	5.0.0	5.1.0	N1-021174
2002-06	NP-16	NP-020238	102		Editor's note cleanup - clause 4	5.0.0	5.1.0	N1-021175
2002-06	NP-16	NP-020238	103		Editor's note cleanup - clause 5.1 and deletion of void subclauses	5.0.0	5.1.0	N1-021176
2002-06	NP-16	NP-020238	104	1	Editor's note cleanup - clause 5.2 and deletion of void subclauses	5.0.0	5.1.0	N1-021487
2002-06	NP-16	NP-020238	105		Editor's note cleanup - clause 5.3	5.0.0	5.1.0	N1-021178
2002-06	NP-16	NP-020238	106		Editor's note cleanup - clause 5.4 and deletion of void subclauses	5.0.0	5.1.0	N1-021179
2002-06	NP-16	NP-020238	107		Editor's note cleanup - clause 5.5 and deletion of void subclauses	5.0.0	5.1.0	N1-021180
2002-06	NP-16	NP-020238	110		Editor's note cleanup - clause 6	5.0.0	5.1.0	N1-021183
2002-06	NP-16	NP-020238	111		Editor's note cleanup - clause 9	5.0.0	5.1.0	N1-021184
2002-06	NP-16	NP-020239	113	1	SIP Default Timers	5.0.0	5.1.0	N1-021465
2002-06	NP-16	NP-020239	114	1	Correction of the subscription to the registration event package	5.0.0	5.1.0	N1-021436
2002-06	NP-16	NP-020239	115	1	Support for ISIMless UICC	5.0.0	5.1.0	N1-021441
2002-06	NP-16	NP-020239	119	1	SIP procedures at UE	5.0.0	5.1.0	N1-021452
2002-06	NP-16	NP-020239	121	2	New requirements in the P-CSCF	5.0.0	5.1.0	N1-021509
2002-06	NP-16	NP-020239	122		SDP procedures at MGCF	5.0.0	5.1.0	N1-021264
2002-06	NP-16	NP-020239	124	1	S-CSCF allocation	5.0.0	5.1.0	N1-021443
2002-06	NP-16	NP-020240	129	1	Introduction of P-Access-Network-Info header	5.0.0	5.1.0	N1-021498
2002-06	NP-16	NP-020240	130	2	Usage of Path and P-Service Route	5.0.0	5.1.0	N1-021508
2002-06	NP-16	NP-020240	133		Removal of Referred-By header from specification	5.0.0	5.1.0	N1-021354
2002-06	NP-16	NP-020240	134		Handling of Record-Route header in profile tables	5.0.0	5.1.0	N1-021357
2002-06	NP-16	NP-020312	135	1	Asserted identities and privacy	5.0.0	5.1.0	
2002-06	NP-16	NP-020240	136		Removal of caller preferences from specification	5.0.0	5.1.0	N1-021359
2002-06	NP-16	NP-020240	137		Substitution of REFER references	5.0.0	5.1.0	N1-021360

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Date	**TSG #**	**TSG Doc.**	**CR**	**Rev**	**Subject/Comment**	**Old**	**New**	**WG doc**
2002-06	NP-16	NP-020240	138		Removal of session timer from specification	5.0.0	5.1.0	N1-021361
2002-09	NP-17	NP-020489	141	2	Adding MESSAGE to 24.229	5.1.0	5.2.0	
2002-09	NP-17	NP-020375	142		Public user identity to use for third party register	5.1.0	5.2.0	N1-021563
2002-09	NP-17	NP-020375	143	1	Replace P-Original-Dialog-ID header with unique data in Route header	5.1.0	5.2.0	N1-021797
2002-09	NP-17	NP-020375	145		Synchronize text with latest I-D for P-headers for charging	5.1.0	5.2.0	N1-021569
2002-09	NP-17	NP-020488	146	2	Service profiles and implicitly registered public user identities	5.1.0	5.2.0	
2002-09	NP-17	NP-020376	147		S-CSCF decides when to include	5.1.0	5.2.0	N1-021571
2002-09	NP-17	NP-020376	148		Clean up XML in clause 7.6	5.1.0	5.2.0	N1-021572
2002-09	NP-17	NP-020376	149		Fix clause 5.2.7.4 header	5.1.0	5.2.0	N1-021573
2002-09	NP-17	NP-020376	150		Removal of forward reference to non P-CSCF procedures	5.1.0	5.2.0	N1-021589
2002-09	NP-17	NP-020376	151		Deregistration of public user identities	5.1.0	5.2.0	N1-021590
2002-09	NP-17	NP-020376	152		Reauthentication trigger via other means	5.1.0	5.2.0	N1-021591
2002-09	NP-17	NP-020487	153	3	Registration with integrity protection	5.1.0	5.2.0	
2002-09	NP-17	NP-020485	154	2	Explicit listing of need to route response messages	5.1.0	5.2.0	
2002-09	NP-17	NP-020377	157	1	Include IP address in ICID	5.1.0	5.2.0	N1-021816
2002-09	NP-17	NP-020377	158		Reference updates	5.1.0	5.2.0	N1-021604
2002-09	NP-17	NP-020377	159		Abbreviation updates	5.1.0	5.2.0	N1-021605
2002-09	NP-17	NP-020377	163	1	Clarifications of allocation of IP address	5.1.0	5.2.0	N1-021817
2002-09	NP-17	NP-020377	171	1	Verifications at the P-CSCF for subsequent request	5.1.0	5.2.0	N1-021802
2002-09	NP-17	NP-020377	174	1	Clarification of IMS signalling flag	5.1.0	5.2.0	N1-021781
2002-09	NP-17	NP-020377	176	1	Definition of a general-purpose PDP context for IMS	5.1.0	5.2.0	N1-021783
2002-09	NP-17	NP-020372	177	2	Request for DNS IPv6 server address	5.1.0	5.2.0	N1-021833
2002-09	NP-17	NP-020378	178		Error cases for PDP context modification	5.1.0	5.2.0	N1-021679
2002-09	NP-17	NP-020378	183	1	Incorporation of draft-ietf-sip-sec-agree-04.txt	5.1.0	5.2.0	N1-021791
2002-09	NP-17	NP-020378	185	1	User Initiated De-registration	5.1.0	5.2.0	N1-021787
2002-09	NP-17	NP-020378	186	1	Mobile initiated de-registration	5.1.0	5.2.0	N1-021788
2002-09	NP-17	NP-020378	187	1	CallID of REGISTER requests	5.1.0	5.2.0	N1-021786
2002-09	NP-17	NP-020378	188	1	Correction to the I-CSCF routing procedures	5.1.0	5.2.0	N1-021803
2002-09	NP-17	NP-020378	189	1	Registration procedures at P-CSCF	5.1.0	5.2.0	N1-021793
2002-09	NP-17	NP-020378	192	1	Corrections related to the P-Access-Network-Info header	5.1.0	5.2.0	N1-021827
2002-09	NP-17	NP-020378	194	1	Chapter to decribe the registration event	5.1.0	5.2.0	N1-021794
2002-09	NP-17	NP-020484	196		Definition of abbreviation IMS	5.1.0	5.2.0	
2002-12	NP-18	NP-020558	140	4	Support of non-IMS forking	5.2.0	5.3.0	N1-022446
2002-12	NP-18	NP-020565	144	2	Identification of supported IETF drafts within this release	5.2.0	5.3.0	N1-022114
2002-12	NP-18	NP-020558	161	3	Clarifications and editorials to SIP profile	5.2.0	5.3.0	N1-022412

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2002-12	NP-18	NP-020558	175	5	Clarifications of the binding and media grouping	5.2.0	5.3.0	N1-022494
2002-12	NP-18	NP-020558	179	2	Support of originating requests from Application Servers	5.2.0	5.3.0	N1-022106
2002-12	NP-18	NP-020558	197		Wrong references in 4.1	5.2.0	5.3.0	N1-021902
2002-12	NP-18	NP-020558	198		Alignment of the MGCF procedures to RFC 3312	5.2.0	5.3.0	N1-021903
2002-12	NP-18	NP-020558	199	1	Service Route Header and Path Header interactions	5.2.0	5.3.0	N1-022080
2002-12	NP-18	NP-020558	202		Addition of clause 6 though clause 9 references to conformance clause	5.2.0	5.3.0	N1-021919
2002-12	NP-18	NP-020558	203	1	URL and address assignments	5.2.0	5.3.0	N1-022115
2002-12	NP-18	NP-020559	204	3	Fix gprs-charging-info definition and descriptions	5.2.0	5.3.0	N1-022426
2002-12	NP-18	NP-020559	206		Alignment of the SDP attributes related to QoS integration with IETF	5.2.0	5.3.0	N1-021930
2002-12	NP-18	NP-020559	207	1	Update of the 3GPP-generated SIP P- headers document references	5.2.0	5.3.0	N1-022116
2002-12	NP-18	NP-020559	208	1	Handling of INVITE requests that do not contain SDP	5.2.0	5.3.0	N1-022098
2002-12	NP-18	NP-020559	209	2	UE Registration	5.2.0	5.3.0	N1-022471
2002-12	NP-18	NP-020559	211	1	Usage of private user identity during registration	5.2.0	5.3.0	N1-022083
2002-12	NP-18	NP-020559	212	1	P-CSCF subscription to the users registration-state event	5.2.0	5.3.0	N1-022084
2002-12	NP-18	NP-020559	213	2	Handling of MT call by the P-CSCF	5.2.0	5.3.0	N1-022154
2002-12	NP-18	NP-020559	215		P-CSCF acting as a UA	5.2.0	5.3.0	N1-021939
2002-12	NP-18	NP-020559	216	1	S-CSCF handling of protected registrations	5.2.0	5.3.0	N1-022085
2002-12	NP-18	NP-020560	217	1	S-CSCF handling of subscription to the users registration-state event	5.2.0	5.3.0	N1-022086
2002-12	NP-18	NP-020560	218	1	Determination of MO or MT in I-CSCF	5.2.0	5.3.0	N1-022102
2002-12	NP-18	NP-020560	220		Definition of the NAI and RTCP abbreviations	5.2.0	5.3.0	N1-021944
2002-12	NP-18	NP-020560	222	4	Go related error codes in the UE	5.2.0	5.3.0	N1-022495
2002-12	NP-18	NP-020560	223	1	Clarifications on CCF/ECF addresses	5.2.0	5.3.0	N1-022120
2002-12	NP-18	NP-020560	225	2	Clarifications on dedicated PDP Context for IMS signalling	5.2.0	5.3.0	N1-022156
2002-12	NP-18	NP-020560	228	3	Clarifications on the use of charging correlation information	5.2.0	5.3.0	N1-022425
2002-12	NP-18	NP-020560	232	1	Expires information in REGISTER response	5.2.0	5.3.0	N1-022095
2002-12	NP-18	NP-020560	235	2	Indication of successful establishment of Dedicated Signalling PDP context to the UE	5.2.0	5.3.0	N1-022129
2002-12	NP-18	NP-020560	237		P-CSCF sending 100 (Trying) Response for reINVITE	5.2.0	5.3.0	N1-021998
2002-12	NP-18	NP-020561	239	1	Correction on P-Asserted-Id, P-Preferred-Id, Remote-Party-ID	5.2.0	5.3.0	N1-022100
2002-12	NP-18	NP-020561	240	1	Clarifications to subclause 9.2.5	5.2.0	5.3.0	N1-022137

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2002-12	NP-18	NP-020561	242		ENUM translation	5.2.0	5.3.0	N1-022020
2002-12	NP-18	NP-020561	243	1	AS routing	5.2.0	5.3.0	N1-022107
2002-12	NP-18	NP-020561	245	1	Warning header	5.2.0	5.3.0	N1-022108
2002-12	NP-18	NP-020561	246	3	S-CSCF procedure tidyup	5.2.0	5.3.0	N1-022497
2002-12	NP-18	NP-020561	247	1	P-CSCF procedure tidyup	5.2.0	5.3.0	N1-022125
2002-12	NP-18	NP-020561	248	2	UE procedure tidyup	5.2.0	5.3.0	N1-022472
2002-12	NP-18	NP-020561	249	3	MESSAGE corrections part 1	5.2.0	5.3.0	N1-022455
2002-12	NP-18	NP-020561	250	2	MESSAGE corrections part 2	5.2.0	5.3.0	N1-022456
2002-12	NP-18	NP-020562	251	2	Security association clarifications	5.2.0	5.3.0	N1-022440
2002-12	NP-18	NP-020562	252	1	The use of security association by the UE	5.2.0	5.3.0	N1-022433
2002-12	NP-18	NP-020562	253	1	UE integrity protected re-registration	5.2.0	5.3.0	N1-022434
2002-12	NP-18	NP-020562	255	3	Handling of default public user identities by the P-CSCF	5.2.0	5.3.0	N1-022496
2002-12	NP-18	NP-020562	263		Fixing ioi descriptions	5.2.0	5.3.0	N1-022266
2002-12	NP-18	NP-020562	264	1	Fix descriptions for ECF/CCF addresses	5.2.0	5.3.0	N1-022447
2002-12	NP-18	NP-020562	266	2	Alignment with draft-ietf-sipping-reg-event-00 and clarification on network initiated deregistration	5.2.0	5.3.0	N1-022493
2002-12	NP-18	NP-020563	267	1	Correction to network initiated re-authentication procedure	5.2.0	5.3.0	N1-022449
2002-12	NP-18	NP-020563	268	1	Registration Expires Timer Default Setting	5.2.0	5.3.0	N1-022439
2002-12	NP-18	NP-020563	269	1	Clarification on Sh interface for charging purposes	5.2.0	5.3.0	N1-022465
2002-12	NP-18	NP-020563	270	2	Clarifications on the scope	5.2.0	5.3.0	N1-022500
2002-12	NP-18	NP-020563	273	1	Add charging info for SUBSCRIBE	5.2.0	5.3.0	N1-022467
2002-12	NP-18	NP-020563	274	1	Profile revisions for RFC 3261 headers	5.2.0	5.3.0	N1-022413
2002-12	NP-18	NP-020563	275		Consistency changes for SDP procedures at MGCF	5.2.0	5.3.0	N1-022345
2002-12	NP-18	NP-020563	276		Proxy support of PRACK	5.2.0	5.3.0	N1-022350
2002-12	NP-18	NP-020563	277		Clarification of transparent handling of parameters in profile	5.2.0	5.3.0	N1-022351
2002-12	NP-18	NP-020564	279	1	Meaning of refresh request	5.2.0	5.3.0	N1-022444
2002-12	NP-18	NP-020564	280		Removal of Caller Preferences dependency	5.2.0	5.3.0	N1-022362
2002-12	NP-18	NP-020564	281	1	P-Access-Network-Info clarifications	5.2.0	5.3.0	N1-022445
2002-12	NP-18	NP-020564	282		Clarification on use of the From header by the UE	5.2.0	5.3.0	N1-022370
2002-12	NP-18	NP-020634	283	2	Support of comp=sigcomp parameter	5.2.0	5.3.0	
2002-12	NP-18	NP-020668	284	4	SDP media policy rejection	5.2.0	5.3.0	
2002-12	NP-18	NP-020567	285	1	Fallback for compression failure	5.2.0	5.3.0	N1-022481
2002-12	NP-18	NP-020564	287	1	SA related procedures	5.2.0	5.3.0	N1-022459

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2002-12	NP-18	NP-020568	290	1	Emergency Service correction	5.2.0	5.3.0	N1-022461
2002-12	NP-18	NP-020663	278	4	P-CSCF does not strip away headers	5.2.0	5.3.0	N1-022499
2002-12	NP-18	NP-020557	289		PCF to PDF	5.2.0	5.3.0	N1-022387
2003-03	NP-19	NP-030049	291		Minor correction and consistency changes to general part of profile	5.3.0	5.4.0	N1-030012
2003-03	NP-19	NP-030049	292		SIP profile minor correction and consistency changes	5.3.0	5.4.0	N1-030013
2003-03	NP-19	NP-030049	293	1	Network asserted identity procedure corrections for the UE	5.3.0	5.4.0	N1-030261
2003-03	NP-19	NP-030049	294	1	Asserted identity inclusion in SIP profile	5.3.0	5.4.0	N1-030300
2003-03	NP-19	NP-030049	296		Profile references relating to registration	5.3.0	5.4.0	N1-030023
2003-03	NP-19	NP-030049	297	2	Reference corrections	5.3.0	5.4.0	N1-030301
2003-03	NP-19	NP-030050	300	1	488 message with a subset of allowed media parameters	5.3.0	5.4.0	N1-030245
2003-03	NP-19	NP-030050	301	1	Handling of Emergency Numbers in P-CSCF	5.3.0	5.4.0	N1-030239
2003-03	NP-19	NP-030050	302	2	Correction of the registration state event package	5.3.0	5.4.0	N1-030268
2003-03	NP-19	NP-030050	305	2	User initiated de-registration at P-CSCF	5.3.0	5.4.0	N1-030295
2003-03	NP-19	NP-030050	306	2	Network-initiated deregistration at UE, P-CSCF, and S-CSCF	5.3.0	5.4.0	N1-030296
2003-03	NP-19	NP-030050	307	2	UE deregistration during established dialogs	5.3.0	5.4.0	N1-030297
2003-03	NP-19	NP-030050	308	2	S-CSCF handling of deregistration during established dialogs	5.3.0	5.4.0	N1-030298
2003-03	NP-19	NP-030050	309	1	S-CSCF handling of established dialogs upon deregistration	5.3.0	5.4.0	N1-030233
2003-03	NP-19	NP-030050	310	2	S-CSCF handling of established dialogs upon registration-lifetime expiration	5.3.0	5.4.0	N1-030299
2003-03	NP-19	NP-030051	311	1	P-CSCF handling of established dialogs upon registration-lifetime expiration	5.3.0	5.4.0	N1-030235
2003-03	NP-19	NP-030051	312	1	Correction of Authentication procedure	5.3.0	5.4.0	N1-030240
2003-03	NP-19	NP-030051	313		Mixed Path header and Service-Route operation	5.3.0	5.4.0	N1-030127
2003-03	NP-19	NP-030051	315	2	Clarifications on updating the authorization token	5.3.0	5.4.0	N1-030255
2003-03	NP-19	NP-030051	318	2	Consideration of P-CSCF/PDF	5.3.0	5.4.0	N1-030307
2003-03	NP-19	NP-030051	319	2	Clarification on GPRS charging information	5.3.0	5.4.0	N1-030308
2003-03	NP-19	NP-030051	323	1	P-Access-Network-Info procedure corrections for the UE	5.3.0	5.4.0	N1-030250
2003-03	NP-19	NP-030051	324	1	P-Access-Network-Info procedure corrections for the S-CSCF	5.3.0	5.4.0	N1-030251
2003-03	NP-19	NP-030051	326	1	Updating user agent related profile tables	5.3.0	5.4.0	N1-030260
2003-03	NP-19	NP-030052	327	2	Cleanup and clarification to the registration and authentication procedure	5.3.0	5.4.0	N1-030282
2003-03	NP-19	NP-030052	328	1	Corrections to the reg event package	5.3.0	5.4.0	N1-030230

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2003-03	NP-19	NP-030052	330	2	Clarifications for setting up separate PDP contexts in case of SBLP	5.3.0	5.4.0	N1-030288
2003-03	NP-19	NP-030052	331	2	Handling of the P-Media-Autohorization header	5.3.0	5.4.0	N1-030289
2003-03	NP-19	NP-030052	333	3	Removal of P-Asserted-Identity from clause 7 of 24.229	5.3.0	5.4.0	N1-030310
2003-03	NP-19	NP-030052	334		P-CSCF general procedure corrections	5.3.0	5.4.0	N1-030182
2003-03	NP-19	NP-030052	335	2	Usage of Contact in UE's registration procedure	5.3.0	5.4.0	N1-030281
2003-03	NP-19	NP-030052	337		Usage of P-Asserted-Identity for responses	5.3.0	5.4.0	N1-030193
2003-03	NP-19	NP-030052	339	2	Authorization for registration event package	5.3.0	5.4.0	N1-030285
2003-03	NP-19	NP-030052	341	1	P-CSCF subscription to reg event	5.3.0	5.4.0	N1-030284
2003-06	NP-20	NP-030275	295	4	Security agreement inclusion in SIP profile	5.4.0	5.5.0	N1-030939
2003-06	NP-20	NP-030275	322	5	3GPP P-header inclusion in SIP profile	5.4.0	5.5.0	N1-030938
2003-06	NP-20	NP-030275	332	5	Change of IP address for the UE	5.4.0	5.5.0	N1-030923
2003-06	NP-20	NP-030275	342		Removal of the requirement for UE re-authentication initiated by HSS	5.4.0	5.5.0	N1-030349
2003-06	NP-20	NP-030275	343	2	UE behaviour on reception of 420 (Bad Extension) message	5.4.0	5.5.0	N1-030552
2003-06	NP-20	NP-030275	347	2	Handling of DTMF	5.4.0	5.5.0	N1-030551
2003-06	NP-20	NP-030276	348	1	Format of Tel URL in P-Asserted-Id	5.4.0	5.5.0	N1-030510
2003-06	NP-20	NP-030276	349		Delete Note on header stripping/SDP manipulation	5.4.0	5.5.0	N1-030387
2003-06	NP-20	NP-030276	354	1	Clarifications on using DNS procedures	5.4.0	5.5.0	N1-030520
2003-06	NP-20	NP-030276	356	4	Addition of procedures at the AS for SDP	5.4.0	5.5.0	N1-030942
2003-06	NP-20	NP-030276	357	1	Usage of P-Associated-URI	5.4.0	5.5.0	N1-030499
2003-06	NP-20	NP-030276	359	1	Network-initiated deregistration at UE and P-CSCF	5.4.0	5.5.0	N1-030501
2003-06	NP-20	NP-030276	360	2	Barred identities	5.4.0	5.5.0	N1-030550
2003-06	NP-20	NP-030276	365	1	PDP contex subject to SBLP cannot be reused by other IMS sessions	5.4.0	5.5.0	N1-030513
2003-06	NP-20	NP-030276	368	1	User authentication failure cleanups	5.4.0	5.5.0	N1-030506
2003-06	NP-20	NP-030277	369	3	S-CSCF behavior correction to enable call forwarding	5.4.0	5.5.0	N1-030931
2003-06	NP-20	NP-030277	370	1	SUBSCRIBE request information stored at the P-CSCF and S-CSCF	5.4.0	5.5.0	N1-030521
2003-06	NP-20	NP-030277	371	1	Profile Tables - Transparency	5.4.0	5.5.0	N1-030858
2003-06	NP-20	NP-030277	375	1	Profile Tables - Major Capability Corrections	5.4.0	5.5.0	N1-030860
2003-06	NP-20	NP-030277	376	2	Profile Tables - Deletion of Elements not used in 24.229	5.4.0	5.5.0	N1-030921
2003-06	NP-20	NP-030277	377	1	Use of the QoS parameter 'signalling information' for a signalling PDP context	5.4.0	5.5.0	N1-030840
2003-06	NP-20	NP-030277	378	2	Deregistration of a PUID (not the last one)	5.4.0	5.5.0	N1-030919

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2003-06	NP-20	NP-030277	379	2	'Last registered public user identity' terminology change	5.4.0	5.5.0	N1-030920
2003-06	NP-20	NP-030277	380	1	Check Integrity Protection for P-Access-Network-Info header	5.4.0	5.5.0	N1-030881
2003-06	NP-20	NP-030278	381	1	PCSCF setting of Integrity protection indicator and checking of Security Verify header	5.4.0	5.5.0	N1-030882
2003-06	NP-20	NP-030278	383	1	Consistent treatment of register and de-register	5.4.0	5.5.0	N1-030884
2003-06	NP-20	NP-030278	384	1	Optionality of sending CK is removed	5.4.0	5.5.0	N1-030885
2003-06	NP-20	NP-030278	385	1	Addition of note and Correction of References regarding security associations and registration	5.4.0	5.5.0	N1-030886
2003-06	NP-20	NP-030278	387	1	Subscription/Registration refresh time	5.4.0	5.5.0	N1-030887
2003-06	NP-20	NP-030278	388	1	Corrections to use of IK	5.4.0	5.5.0	N1-030863
2003-06	NP-20	NP-030278	390		Mobile-originating case at UE	5.4.0	5.5.0	N1-030647
2003-06	NP-20	NP-030278	394	2	Re-authentication procedure.	5.4.0	5.5.0	N1-030917
2003-06	NP-20	NP-030278	395		Replacement of SIP URL with SIP URI	5.4.0	5.5.0	N1-030652
2003-06	NP-20	NP-030279	397	2	Notification about registration state	5.4.0	5.5.0	N1-030926
2003-06	NP-20	NP-030279	402	1	Handling of P-Asserted ID in MGCF	5.4.0	5.5.0	N1-030848
2003-06	NP-20	NP-030279	404	1	S-CSCF initiated release of calls to circiut switched network	5.4.0	5.5.0	N1-030873
2003-06	NP-20	NP-030279	405	2	Supported Integrity algorithms	5.4.0	5.5.0	N1-030927
2003-06	NP-20	NP-030279	407	1	RFC 3524, Single Reservation Flows	5.4.0	5.5.0	N1-030851
2003-06	NP-20	NP-030279	410	1	Clarification of the S-CSCF's handling of the P-access-network-info header	5.4.0	5.5.0	N1-030868
2003-06	NP-20	NP-030279	411	2	Port numbers in the RR header entries	5.4.0	5.5.0	N1-030941
2003-06	NP-20	NP-030279	412	2	Registration abnormal cases	5.4.0	5.5.0	N1-030928
2003-06	NP-20	NP-030280	415		Minor correction to section 5.4.5.1.2	5.4.0	5.5.0	N1-030720
2003-06	NP-20	NP-030280	417	1	Introduction of RTCP bandwidth	5.4.0	5.5.0	N1-030872
2003-06	NP-20	NP-030280	418	1	Registratin Event - Shortend	5.4.0	5.5.0	N1-030844
2003-06	NP-20	NP-030280	419	1	HSS / S-CSCF text relating to user deregistration	5.4.0	5.5.0	N1-030845
2003-06	NP-20	NP-030280	421		Handling of unknown methods at the P-CSCF	5.4.0	5.5.0	N1-030743
2003-06	NP-20	NP-030280	422	1	Definitions and abbreviations update	5.4.0	5.5.0	N1-030870
2003-06	NP-20	NP-030280	423		Removal of hanging paragraph	5.4.0	5.5.0	N1-030752
2003-06	NP-20	NP-030280	424		Access network charging information	5.4.0	5.5.0	N1-030753
2003-06	NP-20	NP-030280	425	1	UE procedure tidyup	5.4.0	5.5.0	N1-030871
2003-06	NP-20	NP-030281	426		P-CSCF procedure tidyup	5.4.0	5.5.0	N1-030755
2003-06	NP-20	NP-030281	427		I-CSCF procedure tidyup	5.4.0	5.5.0	N1-030756
2003-06	NP-20	NP-030281	428		S-CSCF procedure tidyup	5.4.0	5.5.0	N1-030757
2003-06	NP-20	NP-030281	429		BGCF procedure tidyup	5.4.0	5.5.0	N1-030758
2003-06	NP-20	NP-030281	430		AS procedure tidyup	5.4.0	5.5.0	N1-030759

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2003-06	NP-20	NP-030281	431		MRFC procedure tidyup	5.4.0	5.5.0	N1-030760
2003-06	NP-20	NP-030281	434	1	SDP procedure tidyup	5.4.0	5.5.0	N1-030852
2003-06	NP-20	NP-030281	438	2	Profile Tables – Further Corrections	5.4.0	5.5.0	N1-030935
2003-06	NP-20	NP-030281	439	3	AS's subscription for the registration state event package	5.4.0	5.5.0	N1-030940
2003-06	NP-20	NP-030281	440		Temporary Public User Identity in re- and de-REGISTER requests	5.4.0	5.5.0	N1-030792
2003-09	NP-21	NP-030412	444	2	All non-REGISTER requests must be integrity protected	5.5.0	5.6.0	N1-031328
2003-09	NP-21	NP-030412	445		Download of all service profiles linked to PUID being registered and implicitly registered	5.5.0	5.6.0	N1-031010
2003-09	NP-21	NP-030412	448	3	Authentication at UE	5.5.0	5.6.0	N1-031326
2003-09	NP-21	NP-030412	449	1	Nework authentication failure at the UE	5.5.0	5.6.0	N1-031242
2003-09	NP-21	NP-030412	451	3	Handling of security association	5.5.0	5.6.0	N1-031327
2003-09	NP-21	NP-030412	452	1	Re-authentication timer at S-CSCF	5.5.0	5.6.0	N1-031274
2003-09	NP-21	NP-030412	455	2	Authentication failure at S-CSCF	5.5.0	5.6.0	N1-031285
2003-09	NP-21	NP-030413	456	2	Subscription termination sent by the S-CSCF	5.5.0	5.6.0	N1-031276
2003-09	NP-21	NP-030413	457		Subscription termination at the P-CSCF	5.5.0	5.6.0	N1-031032
2003-09	NP-21	NP-030413	458		Network -initiated deregistration at P-CSCF	5.5.0	5.6.0	N1-031033
2003-09	NP-21	NP-030349	459	2	Notification about registration status at AS	5.5.0	5.6.0	
2003-09	NP-21	NP-030413	461	1	Service profile	5.5.0	5.6.0	N1-031233
2003-09	NP-21	NP-030413	466	1	Requirements on Preconditions	5.5.0	5.6.0	N1-031246
2003-09	NP-21	NP-030413	467	1	Call forwarding cleanup	5.5.0	5.6.0	N1-031238
2003-09	NP-21	NP-030413	468		Update of references	5.5.0	5.6.0	N1-031094
2003-09	NP-21	NP-030414	470	1	Adding P-Asserted-Identity headers to NE initiated subscriptions	5.5.0	5.6.0	N1-031314
2003-09	NP-21	NP-030414	479	1	Replace USIM by ISIM for user identity storage	5.5.0	5.6.0	N1-031247
2003-09	NP-21	NP-030414	481	1	24.229 R5 CR: Corrections to Profile Tables	5.5.0	5.6.0	N1-031248
2003-09	NP-21	NP-030414	482		24.229 R5 CR: Setting of SUBSCRIBE exipiration time	5.5.0	5.6.0	N1-031140
2003-09	NP-21	NP-030414	483	3	24.229 R5 CR: Alignment of IMS Compression with RFC 3486	5.5.0	5.6.0	N1-031335
2003-09	NP-21	NP-030418	465	1	Alignment with TS for policy control over Gq interface	5.6.0	6.0.0	N1-031267
2003-09	NP-21	NP-030418	472	1	I-CSCF procedures for openness	5.6.0	6.0.0	N1-031304
2003-09	NP-21	NP-030433	473	3	Registration from multiple terminals and forking	5.6.0	6.0.0	
2003-09	NP-21	NP-030419	480	3	Access Independent IMS	5.6.0	6.0.0	N1-031333
2003-12	NP-22	NP-030482	487	1	Registration amendments in profile	6.0.0	6.1.0	N1-031627
2003-12	NP-22	NP-030482	489		Privacy considerations for the UE	6.0.0	6.1.0	N1-031351
2003-12	NP-22	NP-030476	493		INVITE dialog amendments in profile	6.0.0	6.1.0	N1-031359

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2003-12	NP-22	NP-030482	494		Correction of I-CSCF handling of multiple private user identities with same public user identity	6.0.0	6.1.0	N1-031375
2003-12	NP-22	NP-030476	496	1	P-Asserted-Identity in SUBSCRIBE requests	6.0.0	6.1.0	N1-031632
2003-12	NP-22	NP-030482	497		Addition of reference to Gq interface	6.0.0	6.1.0	N1-031378
2003-12	NP-22	NP-030476	503	2	Update of HSS information at deregistration	6.0.0	6.1.0	N1-031720
2003-12	NP-22	NP-030482	507		Unavailable definitions	6.0.0	6.1.0	N1-031392
2003-12	NP-22	NP-030476	509		Reference corrections	6.0.0	6.1.0	N1-031394
2003-12	NP-22	NP-030484	510	1	UICC related changes for IMS commonality and interoperability	6.0.0	6.1.0	N1-031682
2003-12	NP-22	NP-030484	511		Interoperability and commonality; definition of scope	6.0.0	6.1.0	N1-031427
2003-12	NP-22	NP-030484	512		Interoperability and commonality; addition of terminology	6.0.0	6.1.0	N1-031428
2003-12	NP-22	NP-030484	513		Interoperability and commonality; media grouping	6.0.0	6.1.0	N1-031429
2003-12	NP-22	NP-030484	515		Interoperability and commonality; charging information	6.0.0	6.1.0	N1-031431
2003-12	NP-22	NP-030482	518	1	Profile support of RFC 3326: The Reason Header Field for the Session Initiation Protocol	6.0.0	6.1.0	N1-031681
2003-12	NP-22	NP-030482	519		Profile support of RFC 3581: An Extension to the Session Initiation Protocol (SIP) for Symmetric Response Routing	6.0.0	6.1.0	N1-031439
2003-12	NP-22	NP-030484	522	1	Clause 9 restructuring	6.0.0	6.1.0	N1-031684
2003-12	NP-22	NP-030477	524	2	Correct use of RAND during re-synchronisation failures	6.0.0	6.1.0	N1-031712
2003-12	NP-22	NP-030478	526	1	Correction to description or RES/XRES usage	6.0.0	6.1.0	N1-031617
2003-12	NP-22	NP-030483	529		Corrections on charging specification number	6.0.0	6.1.0	N1-031469
2003-12	NP-22	NP-030581	531	3	Corrections on ICID for REGISTER	6.0.0	6.1.0	
2003-12	NP-22	NP-030478	543	1	Correction of user initiated re-registration	6.0.0	6.1.0	N1-031619
2003-12	NP-22	NP-030483	551	1	IMS trust domain in Rel 6	6.0.0	6.1.0	N1-031622
2003-12	NP-22	NP-030478	556	1	P-CSCF and UE handling of Security Associations	6.0.0	6.1.0	N1-031624
2003-12	NP-22	NP-030483	560	2	SDP offer handling in SIP responses in S-CSCF and P-CSCF	6.0.0	6.1.0	N1-031727
2003-12	NP-22	NP-030483	564	1	SIP compression	6.0.0	6.1.0	N1-031705
2003-12	NP-22	NP-030478	566		Sending challenge	6.0.0	6.1.0	N1-031580
2003-12	NP-22	NP-030480	568	2	Reg-await-auth timer value	6.0.0	6.1.0	N1-031716
2003-12	NP-22	NP-030480	571	1	Network initiated deregistration	6.0.0	6.1.0	N1-031707
2003-12	NP-22	NP-030483	572		Text harmonisation with 3GPP2	6.0.0	6.1.0	N1-031589
2003-12	NP-22	NP-030483	573	1	Procedures in the absence of UICC	6.0.0	6.1.0	N1-031680
2003-12	NP-22	NP-030483	575	1	P-Access-Network-Info changes	6.0.0	6.1.0	N1-031683
2004-03	NP-23	NP-040027	488	3	Completion of major capabilities table in respect of privacy	6.1.0	6.2.0	N1-040406

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2004-03	NP-23	NP-040027	499	5	P-CSCF integrity protection	6.1.0	6.2.0	N1-040500
2004-03	NP-23	NP-040032	578	1	UE requesting no-fork	6.1.0	6.2.0	N1-040184
2004-03	NP-23	NP-040032	579	1	Inclusion of caller preferences into profile	6.1.0	6.2.0	N1-040284
2004-03	NP-23	NP-040027	586	1	Network-initiated re-authentication	6.1.0	6.2.0	N1-040391
2004-03	NP-23	NP-040032	588	1	Re-authentication - Abnormal cases	6.1.0	6.2.0	N1-040393
2004-03	NP-23	NP-040027	592	1	Integrity protected correction	6.1.0	6.2.0	N1-040398
2004-03	NP-23	NP-040032	596	1	Sec-agree parameter in "Proxy-Require" header	6.1.0	6.2.0	N1-040400
2004-03	NP-23	NP-040027	600	2	Handling of record-route in target refresh and subsequent request	6.1.0	6.2.0	N1-040481
2004-03	NP-23	NP-040035	603		Cleanup for IP-CAN and GPRS	6.1.0	6.2.0	N1-040304
2004-03	NP-23	NP-040032	604		Forking in S-CSCF	6.1.0	6.2.0	N1-040325
2004-03	NP-23	NP-040108	605	3	Determination of S-CSCF role	6.1.0	6.2.0	
2004-03	NP-23	NP-040134	608	3	Unprotected deregistration	6.1.0	6.2.0	
2004-03	NP-23	NP-040029	610		Sending authentication challenge	6.1.0	6.2.0	N1-040331
2004-03	NP-23	NP-040033	613		Reference to PDF operation	6.1.0	6.2.0	N1-040334
2004-03	NP-23	NP-040029	615	1	Support of MESSAGE (Profile Tables)	6.1.0	6.2.0	N1-040466
2004-03	NP-23	NP-040033	616	2	Introduction of PSI Routing to 24.229	6.1.0	6.2.0	N1-040487
2004-03	NP-23	NP-040033	617	1	P-CSCF Re-selection	6.1.0	6.2.0	N1-040463
2004-03	NP-23	NP-040033	618		I-CSCF does not re-select S-CSCF during re-registration	6.1.0	6.2.0	N1-040344
2004-03	NP-23	NP-040033	620	1	Handling of media authorization token due to messaging	6.1.0	6.2.0	N1-040430
2004-06	NP-24	NP-040191	621	2	Forking requests terminating at the served user	6.2.0	6.3.0	N1-040739
2004-06	NP-24	NP-040191	624	1	Abbreviations	6.2.0	6.3.0	N1-040691
2004-06	NP-24	NP-040191	625	5	Removal of restriction for multiple SIP sessions on a single PDP context	6.2.0	6.3.0	N1-041053
2004-06	NP-24	NP-040191	626	3	Record route in S-CSCF	6.2.0	6.3.0	N1-041061
2004-06	NP-24	NP-040189	627	3	Correction of reception of media authorization token	6.2.0	6.3.0	N1-040994
2004-06	NP-24	NP-040191	628	3	Introduction of PSI Routing to 24.229	6.2.0	6.3.0	N1-041059
2004-06	NP-24	NP-040198	629	2	Addition of PRESNC material	6.2.0	6.3.0	N1-040996

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2004-06	NP-24	NP-040189	631	1	Missing statements regarding P-Charging-Function-Addresses header	6.2.0	6.3.0	N1-040987
2004-06	NP-24	NP-040191	634	1	Multiple registrations	6.2.0	6.3.0	N1-041054
2004-06	NP-24	NP-040192	635	1	Network-initiated deregistration	6.2.0	6.3.0	N1-041055
2004-06	NP-24	NP-040192	636		Network-initiated re-authentication	6.2.0	6.3.0	N1-040778
2004-06	NP-24	NP-040192	637	1	Mobile-initiated deregistration	6.2.0	6.3.0	N1-041056
2004-06	NP-24	NP-040192	638	1	Notification about registration state	6.2.0	6.3.0	N1-041057
2004-06	NP-24	NP-040189	642	3	Syntax of the extension to the P-Charging-Vector header field	6.2.0	6.3.0	N1-041100
2004-06	NP-24	NP-040192	643	2	Session Timer	6.2.0	6.3.0	N1-041095
2004-06	NP-24	NP-040193	644	3	Session initiation without preconditions	6.2.0	6.3.0	N1-041096
2004-06	NP-24	NP-040192	645	1	IMS Conferencing: Inclusion of Profile Tables to TS 24.229	6.2.0	6.3.0	N1-041015
2004-06	NP-24	NP-040189	649	1	Revisions due to published version of draft-ietf-sipping-reg-event	6.2.0	6.3.0	N1-040992
2004-06	NP-24	NP-040198	652		Creation of separate event package table for UA role	6.2.0	6.3.0	N1-041066
2004-09	NP-25	NP-040380	658		Correction of User identity verification at the AS	6.3.0	6.4.0	N1-041344
2004-09	NP-25	NP-040381	666	1	NOTIFY requests	6.3.0	6.4.0	N1-041586
2004-09	NP-25	NP-040381	654	4	Callee capabilities and Registration	6.3.0	6.4.0	N1-041315
2004-09	NP-25	NP-040381	668	2	Network deregistration	6.3.0	6.4.0	N1-041614
2004-09	NP-25	NP-040381	682	1	SDP parameters received by the S-CSCF and the P-CSCF in the 200 OK message	6.3.0	6.4.0	N1-041592
2004-09	NP-25	NP-040381	661	1	Call Release	6.3.0	6.4.0	N1-041589
2004-09	NP-25	NP-040381	659		Multiple public ID registration	6.3.0	6.4.0	N1-041350
2004-09	NP-25	NP-040381	660		Standalone transactions	6.3.0	6.4.0	N1-041351
2004-09	NP-25	NP-040381	663		Unprotected REGISTER	6.3.0	6.4.0	N1-041354
2004-09	NP-25	NP-040381	662	1	Session timer	6.3.0	6.4.0	N1-041590
2004-09	NP-25	NP-040381	665		Contact in SUBSCRIBE request	6.3.0	6.4.0	N1-041372
2004-09	NP-25	NP-040381	650	2	Support of draft-ietf-sip-replaces	6.3.0	6.4.0	N1-041391
2004-09	NP-25	NP-040381	657	1	Support of draft-ietf-sip-join	6.3.0	6.4.0	N1-041393
2004-09	NP-25	NP-040381	656	1	Support of draft-ietf-sip-referredby	6.3.0	6.4.0	N1-041263
2004-09	NP-25	NP-040381	678		Support of TLS	6.3.0	6.4.0	N1-041462
2004-09	NP-25	NP-040381	688	2	Filtering of the P-Access-Network-Info header by the S-CSCF and privacy rules	6.3.0	6.4.0	N1-041641
2004-09	NP-25	NP-040382	692	2	Ipv6 IPv4 interworking	6.3.0	6.4.0	N1-041630
2004-09	NP-25	NP-040383	689	2	Addition of session set-up not requiring preconditions and reliable transport of provisional responses.	6.3.0	6.4.0	N1-041632
2004-09	NP-25	NP-040385	697		Missing value for the event attribute within the <contact> element of NOTIFY body	6.3.0	6.4.0	N1-041540

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2004-09	NP-25	NP-040385	698		HSS initiated deregistration	6.3.0	6.4.0	N1-041549
2004-09	NP-25	NP-040385	673		Syntax correction for the P-Charging-Vector header	6.3.0	6.4.0	N1-041434
2004-09	NP-25	NP-040385	699	1	Network initiated deregistration upon UE roaming and registration to a new network	6.3.0	6.4.0	N1-041629
2004-12	NP-26	NP-040506	651	4	Downloading the user profile based on User-Data-Request-Type	6.4.0	6.5.0	N1-042031
2004-12	NP-26	NP-040506	703	2	SDP Encryption	6.4.0	6.5.0	N1-042095
2004-12	NP-26	NP-040506	704	1	RTCP streams	6.4.0	6.5.0	N1-042019
2004-12	NP-26	NP-040506	709		Contact in 200(OK) response	6.4.0	6.5.0	N1-041725
2004-12	NP-26	NP-040506	710	1	P-Access-Network-Info header	6.4.0	6.5.0	N1-042020
2004-12	NP-26	NP-040506	711	1	P-Called-Party-ID header	6.4.0	6.5.0	N1-041954
2004-12	NP-26	NP-040506	713	1	IMS-ALG routing	6.4.0	6.5.0	N1-042021
2004-12	NP-26	NP-040506	714	1	Public User Identity	6.4.0	6.5.0	N1-042022
2004-12	NP-26	NP-040506	715	1	"Pres" and "im" URIs	6.4.0	6.5.0	N1-042023
2004-12	NP-26	NP-040502	723	1	Correction Term IOI handling	6.4.0	6.5.0	N1-041956
2004-12	NP-26	NP-040502	725	1	Request handling in S-CSCF originating case	6.4.0	6.5.0	N1-041958
2004-12	NP-26	NP-040502	727	1	Request handling in S-CSCF - terminating case	6.4.0	6.5.0	N1-041960
2004-12	NP-26	NP-040506	728		SBLP and non-realtime PDP contexts	6.4.0	6.5.0	N1-041797
2004-12	NP-26	NP-040590	730	2	Reference updates	6.4.0	6.5.0	N1-042085
2004-12	NP-26	NP-040590	733	3	Support for extended SigComp	6.4.0	6.5.0	N1-042117
2004-12	NP-26	NP-040590	734	2	Correction to subclause 5.1.3 of TS 24,229	6.4.0	6.5.0	N1-042120
2004-12	NP-26	NP-040590	735	1	Correction to subclause 5.1.4.1.2.3 of TS 24,.229	6.4.0	6.5.0	N1-042084
2004-12	NP-26	NP-040502	738	1	Population of Via header when using REGISTER method	6.4.0	6.5.0	N1-041962
2004-12	NP-26	NP-040590	739		Tel-URI related reference updates	6.4.0	6.5.0	N1-041869
2004-12	NP-26	NP-040590	741	1	Throttling	6.4.0	6.5.0	N1-042086
2004-12	NP-26	NP-040590	742		Editorial correction resulting from CR665	6.4.0	6.5.0	N1-041881
2004-12	NP-26	NP-040590	743		Unprotected REGISTER corrections	6.4.0	6.5.0	N1-041882
2004-12	NP-26	NP-040590	744	1	Corrections to receiving SDP offer in 200 (OK) response	6.4.0	6.5.0	N1-042087
2004-12	NP-26	NP-040590	745	1	Privacy corrections	6.4.0	6.5.0	N1-042085
2004-12	NP-26	NP-040590	747	2	Syntax of the P-Charging-Vector	6.4.0	6.5.0	N1-042105
2004-12	NP-26	NP-040590	752	2	Unavailability of the access-network-charging-info when the session is established without SBLP	6.4.0	6.5.0	N1-042106
2004-12	NP-26	NP-040590	753	1	SIP messages carrying the access-network-charging-info for sessions without preconditions	6.4.0	6.5.0	N1-042089
2004-12	NP-26	NP-040590	755	1	Network-initiated deregistration for multiple UEs sharing the same user public identity and for the	6.4.0	6.5.0	N1-042090

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					old contact information of a roaming UE registered in a new network			
2004-12	NP-26	NP-040502	765	1	Interaction between S-CSCF and HSS in Network initiated deregistration procedure	6.4.0	6.5.0	N1-041966
2004-12	NP-26	NP-040502	768	1	Downloading of user profile	6.4.0	6.5.0	N1-042103
2005-01					Fix Word problem	6.5.0	6.5.1	
2005-03	NP-27	NP-050069	839		Filter criteria matching and generation of third-party REGISTER request for network-initiated deregistration	5.11.1	5.12.0	N1-050220
2005-03	NP-27	NP-050069	785		Deregistration effect on active sessions	6.5.1	6.6.0	N1-050052
2005-03	NP-27	NP-050069	784		Deregistration effect on active sessions	5.11.1	5.12.0	N1-050051
2005-03	NP-27	NP-050069	809	1	IOI storage at MGCF	5.11.1	5.12.0	N1-050295
2005-03	NP-27	NP-050069	840		Filter criteria matching and generation of third-party REGISTER request for network-initiated deregistration	6.5.1	6.6.0	N1-050221
2005-03	NP-27	NP-050069	806	1	Use of original dialog identifier at AS	6.5.1	6.6.0	N1-050292
2005-03	NP-27	NP-050069	807	2	Checking Request-URI for terminating requests at the S-CSCF	5.11.1	5.12.0	N1-050401
2005-03	NP-27	NP-050069	805	1	Use of original dialog identifier at AS	5.11.1	5.12.0	N1-050291
2005-03	NP-27	NP-050069	808	2	Checking Request-URI for terminating requests at the S-CSCF	6.5.1	6.6.0	N1-050402
2005-03	NP-27	NP-050069	810	1	IOI storage at MGCF	6.5.1	6.6.0	N1-050296
2005-03	NP-27	NP-050073	794		RFC 3966	6.5.1	6.6.0	N1-050080
2005-03	NP-27	NP-050073	848	1	Removal of I-CSCF normative requirement on Cx interface	6.5.1	6.6.0	N1-050299
2005-03	NP-27	NP-050073	841		Filtering of the P-Access-Network-Info header by the S-CSCF and privacy rules	6.5.1	6.6.0	N1-050225
2005-03	NP-27	NP-050073	817		Editorial corrections	6.5.1	6.6.0	N1-050129
2005-03	NP-27	NP-050073	786	1	Cleanups resulting from CR changes for last version	6.5.1	6.6.0	N1-050324
2005-03	NP-27	NP-050073	821	1	Handling topmost Route header at the P-CSCF	6.5.1	6.6.0	N1-050297
2005-03	NP-27	NP-050073	790		Registration - Abnormal Case	6.5.1	6.6.0	N1-050076
2005-03	NP-27	NP-050074	832	1	Corrections to the tables for 'PUBLISH'	6.5.1	6.6.0	N1-050341
2005-03	NP-27	NP-050074	822	1	Corrections to the UE tables for 'major capabilities'	6.5.1	6.6.0	N1-050332
2005-03	NP-27	NP-050074	825	1	Corrections to the UE tables for 'ACK'	6.5.1	6.6.0	N1-050334
2005-03	NP-27	NP-050074	826	1	Corrections to the tables for 'CANCEL'	6.5.1	6.6.0	N1-050335
2005-03	NP-27	NP-050074	827	1	Corrections to the tables for 'INVITE'	6.5.1	6.6.0	N1-050336
2005-03	NP-27	NP-050074	828	1	Corrections to the tables for 'MESSAGE'	6.5.1	6.6.0	N1-050337
2005-03	NP-27	NP-050074	829	1	Corrections to the tables for 'NOTIFY'	6.5.1	6.6.0	N1-050338
2005-03	NP-27	NP-050074	830	1	Corrections to the tables for 'OPTIONS'	6.5.1	6.6.0	N1-050339
2005-03	NP-27	NP-050074	834	1	Corrections to the tables for 'REGISTER'	6.5.1	6.6.0	N1-050343

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2005-03	NP-27	NP-050074	831	1	Corrections to the tables for 'PRACK'	6.5.1	6.6.0	N1-050340
2005-03	NP-27	NP-050074	833	1	Corrections to the tables for 'REFER'	6.5.1	6.6.0	N1-050342
2005-03	NP-27	NP-050074	835	1	Corrections to the tables for 'SUBSCRIBE'	6.5.1	6.6.0	N1-050344
2005-03	NP-27	NP-050074	836	1	Corrections to the tables for 'UPDATE'	6.5.1	6.6.0	N1-050345
2005-03	NP-27	NP-050074	837	1	Corrections to the tables for SDP	6.5.1	6.6.0	N1-050346
2005-03	NP-27	NP-050074	824	1	Removal of the UE table for 'status codes'	6.5.1	6.6.0	N1-050351
2005-03	NP-27	NP-050074	823	1	Corrections to the tables for 'BYE'	6.5.1	6.6.0	N1-050333
2005-03	NP-27	NP-050075	846	2	Correction to the Registration procedure	6.5.1	6.6.0	N1-050413
2005-03	NP-27	NP-050075	850	1	Addition of IMS-ALF to profile tables	6.5.1	6.6.0	N1-050348
2005-03	NP-27	NP-050075	851	2	Press and im URIs in incoming requests	6.5.1	6.6.0	N1-050395
2005-03	NP-27	NP-050075	788	1	MO - Calls to IPv4 SIP terminals	6.5.1	6.6.0	N1-050387
2005-03	NP-27	NP-050075	818	3	Corrections to subclause 5.5 in TS 24.229	6.5.1	6.6.0	N1-050414
2005-03	NP-27	NP-050075	801	3	Default handling associated with the trigger at the S-CSCF	6.5.1	6.6.0	N1-050418
2005-03	NP-27	NP-050075	803	4	Default handling associated with the trigger for third party registration	6.5.1	6.6.0	N1-050421
2005-03	NP-27	NP-050078	795	1	Sip-profile package in major capabilities	6.5.1	6.6.0	N1-050306
2005-03	NP-27	NP-050127	849	2	Corrections to addition of session set-up not requiring preconditions and reliable transport of provisional responses	6.5.1	6.6.0	
2005-06	CP-28	CP-050059	879		Correction Reg-Await-Auth Timer	6.6.0	6.7.0	C1-050522
2005-06	CP-28	CP-050059	881		Security Association in P-CSCF	6.6.0	6.7.0	C1-050524
2005-06	CP-28	CP-050059	871	1	Port 5060	6.6.0	6.7.0	C1-050674
2005-06	CP-28	CP-050059	891	2	SIP headers storage for P-CSCF initiated session release	6.6.0	6.7.0	C1-050777
2005-06	CP-28	CP-050059	921	1	Correction of error in the specification of the extension to Authorization header	6.6.0	6.7.0	C1-050689
2005-06	CP-28	CP-050059	886	2	Handling of P-Associated URI header	6.6.0	6.7.0	C1-050783
2005-06	CP-28	CP-050059	907	2	Clarification to the procedures at the I-CSCF	6.6.0	6.7.0	C1-050785
2005-06	CP-28	CP-050061	894	1	Re-registration failure	6.6.0	6.7.0	C1-050709
2005-06	CP-28	CP-050061	892		Completion of status-code tables in SIP profile	6.6.0	6.7.0	C1-050571
2005-06	CP-28	CP-050061	865	1	Unsubscribe by P-CSCF	6.6.0	6.7.0	C1-050671
2005-06	CP-28	CP-050061	866	1	Protected initial registration	6.6.0	6.7.0	C1-050708
2005-06	CP-28	CP-050061	916	1	Clarify that S-CSCF shall support Supported and Require headers	6.6.0	6.7.0	C1-050684
2005-06	CP-28	CP-050061	862		Shared public user identities	6.6.0	6.7.0	C1-050599
2005-06	CP-28	CP-050061	860	1	P-CSCF - routing of REGISTER requests	6.6.0	6.7.0	C1-050701
2005-06	CP-28	CP-050061	870	1	Correction of table A.104A	6.6.0	6.7.0	C1-050711
2005-06	CP-28	CP-050061	887	1	Contact address in REGISTER response	6.6.0	6.7.0	C1-050716

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2005-06	CP-28	CP-050061	890	1	P-CSCF Record-Route processing for target refresh requests/responses	6.6.0	6.7.0	C1-050717
2005-06	CP-28	CP-050061	893	1	AS originated requests on behalf of PSI	6.6.0	6.7.0	C1-050719
2005-06	CP-28	CP-050061	896	1	Routing PSI at terminating side	6.6.0	6.7.0	C1-050720
2005-06	CP-28	CP-050061	856	2	Notification about registration state	6.6.0	6.7.0	C1-050789
2005-06	CP-28	CP-050061	861	3	Registration failure at UE	6.6.0	6.7.0	C1-050790
2005-06	CP-28	CP-050061	899	2	Correction of the references for the integration of resource management procedures	6.6.0	6.7.0	C1-050791
2005-06	CP-28	CP-050061	902	2	Clarification on P-CSCF-intiated call release	6.6.0	6.7.0	C1-050792
2005-06	CP-28	CP-050061	863	3	Error handling in UE in case of RFC 3524	6.6.0	6.7.0	C1-050793
2005-06	CP-28	CP-050061	895	3	UE registration failure because the selected S-CSCF is unreachable	6.6.0	6.7.0	C1-050802
2005-06	CP-28	CP-050061	787	6	MT- SDP offer with IPv4 address.	6.6.0	6.7.0	C1-050794
2005-06	CP-28	CP-050061	858	1	S-CSCF redirecting	6.6.0	6.7.0	C1-050700
2005-06	CP-28	CP-050064	872	2	I-WLAN information for IMS	6.6.0	6.7.0	C1-050729
2005-06	CP-28	CP-050074	901		MWI RFC3842	6.6.0	7.0.0	C1-050600
2005-06	CP-28	CP-050075	905	1	3xx response and non-SDP bodies handking by proxies	6.6.0	7.0.0	C1-050775
2005-09	CP-29	CP-050346	986		Modifications to 24.229 to allow multiple IPsec security association per IKE_Security association	7.0.0	7.1.0	
2005-09	CP-29	CP-050355	930	1	Correction Profile Table A.119	7.0.0	7.1.0	C1-051061
2005-09	CP-29	CP-050355	946		Public User identity in 3rd party REG	7.0.0	7.1.0	C1-050906
2005-09	CP-29	CP-050355	957	1	Removal of Access Network Charging Information by the S-CSCF	7.0.0	7.1.0	C1-051081
2005-09	CP-29	CP-050355	965		Optional ccf	7.0.0	7.1.0	C1-050986
2005-09	CP-29	CP-050355	969	1	Contact header in REGISTER requests	7.0.0	7.1.0	C1-051177
2005-09	CP-29	CP-050359	932		SigComp-Corrections	7.0.0	7.1.0	C1-050877
2005-09	CP-29	CP-050359	962	1	IETF reference corrections	7.0.0	7.1.0	C1-051074
2005-09	CP-29	CP-050359	968	1	AS procedure correction	7.0.0	7.1.0	C1-051085
2005-09	CP-29	CP-050367	924		Incorporation of draft-ietf-sip-history	7.0.0	7.1.0	C1-050838
2005-09	CP-29	CP-050367	938		Contact header	7.0.0	7.1.0	C1-050887
2005-09	CP-29	CP-050367	939	1	Reason header - loss of radio coverage	7.0.0	7.1.0	C1-051158
2005-09	CP-29	CP-050367	947	3	Changes to TS 24.229 to ease interworking with non precondition terminals	7.0.0	7.1.0	C1-051213
2005-09	CP-29	CP-050367	958	2	Contents of P-Associated-URI header in 200 (OK) response to REGISTER	7.0.0	7.1.0	C1-051206
2005-09	CP-29	CP-050367	960	3	Consideration on 3rd Party Service Provider in Trust Domain	7.0.0	7.1.0	C1-051208
2005-09	CP-29	CP-050367	971	1	Correction of requirement to insert P-Asserted-Identity header	7.0.0	7.1.0	C1-051166
2005-09	CP-29	CP-050368	950	3	privacy and trust rules for History header	7.0.0	7.1.0	C1-051199

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2005-10					missing word in subclause 5.4.1.2.2, bullet 10b) is added by MCC	7.1.0	7.1.1	
2005-12	CP-30	CP-050538	1049		Replace "originated" with "terminated"	7.1.1	7.2.0	C1-051479
2005-12	CP-30	CP-050538	1046	2	Mobile originating call related requests	7.1.1	7.2.0	C1-051668
2005-12	CP-30	CP-050538	1012	1	Correction to section 5.4.3.2 t of TS 24.229	7.1.1	7.2.0	C1-051563
2005-12	CP-30	CP-050538	1026		Handling of P-Charging-Function-Adress	7.1.1	7.2.0	C1-051424
2005-12	CP-30	CP-050538	1071		Correction Syntax P-Charging Vector	7.1.1	7.2.0	C1-051508
2005-12	CP-30	CP-050541	1002	1	Modification to the definition of Security Association	7.1.1	7.2.0	C1-051576
2005-12	CP-30	CP-050542	0982	3	Access Type of P-Access-Network-Info header	7.1.1	7.2.0	C1-051675
2005-12	CP-30	CP-050542	1059		Replace "served" by "Originating" UE	7.1.1	7.2.0	C1-051489
2005-12	CP-30	CP-050542	1017		Correction to subclause 5.7.5.1. of TS 24229	7.1.1	7.2.0	C1-051382
2005-12	CP-30	CP-050542	1073	2	Short Session Setup in IMS	7.1.1	7.2.0	C1-051656
2005-12	CP-30	CP-050542	1054		Adjusting section reference in section 6.3	7.1.1	7.2.0	C1-051484
2005-12	CP-30	CP-050542	1029	1	B2B UA AS handling	7.1.1	7.2.0	C1-041597
2005-12	CP-30	CP-050542	1062	2	Correction to 3rd party registration procedures for SESSION_TERMINATED default handling	7.1.1	7.2.0	C1-051672
2005-12	CP-30	CP-050542	0994		cdma2000	7.1.1	7.2.0	C1-051336
2005-12	CP-30	CP-050542	1043		Correction of a reference in some tables in Appendix A	7.1.1	7.2.0	C1-051473
2005-12	CP-30	CP-050542	1005	2	Refreshes of SUBSCRIBE to reg-event (Fix for Rel 7)	7.1.1	7.2.0	C1-051670
2005-12	CP-30	CP-050542	1065	1	Charging terms correction	7.1.1	7.2.0	C1-051618
2005-12	CP-30	CP-050548	1081		Change of originating and terminating terminal terminology	7.1.1	7.2.0	C1-051535
2005-12	CP-30	CP-050548	1069	2	IBCF	7.1.1	7.2.0	C1-051587
2005-12	CP-30	CP-050550	1055		Editorial Changes	7.1.1	7.2.0	C1-051485
2005-12	CP-30	CP-050550	0996	1	UE initiated deregistration	7.1.1	7.2.0	C1-051649
2005-12	CP-30	CP-050550	1027	1	Mobile originated Request for unregistered user	7.1.1	7.2.0	C1-051653
2005-12	CP-30	CP-050550	0990	1	Authentication related Clarification	7.1.1	7.2.0	C1-051560
2005-12	CP-30	CP-050550	1019	2	Receipt of SIP URI with user equal phone at I-CSCF	7.1.1	7.2.0	C1-051671
2005-12	CP-30	CP-050550	0995	2	Default public user ID	7.1.1	7.2.0	C1-051691
2005-12	CP-30	CP-050550	0997	1	P-Preferred-Identity header	7.1.1	7.2.0	C1-051650
2005-12	CP-30	CP-050550	1082	1	P-CSCF discovery	7.1.1	7.2.0	C1-051681
2005-12	CP-30	CP-050677	1085	2	Incorporating of TR 24.819 fixed broadband access impacts into TS 24.229	7.1.1	7.2.0	
2006-03	CP-31	CP-060106	1187	-	Removal of Warning header non-compliance with RFC 3261	7.2.0	7.3.0	C1-060328
2006-03	CP-31	CP-060106	1117	1	IMS AKA - SQN resync clarifications	7.2.0	7.3.0	C1-060453

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2006-03	CP-31	CP-060106	1114	1	IMS AKA - content of initial authentication header	7.2.0	7.3.0	C1-060450
2006-03	CP-31	CP-060106	1204	-	Syntax and operation for Security-Client, Security-Server and Security-Verify headers	7.2.0	7.3.0	C1-060387
2006-03	CP-31	CP-060107	1148	1	UE processing 305 (Use Proxy)	7.2.0	7.3.0	C1-060507
2006-03	CP-31	CP-060107	1164	1	Clarifications on P-CSCF discovery	7.2.0	7.3.0	C1-060459
2006-03	CP-31	CP-060107	1161	1	DHCPv6 options for Domain Name Servers	7.2.0	7.3.0	C1-060456
2006-03	CP-31	CP-060110	1136	1	SDP answer	7.2.0	7.3.0	C1-060472
2006-03	CP-31	CP-060110	1206	-	Inclusion of Ma reference point	7.2.0	7.3.0	C1-060392
2006-03	CP-31	CP-060110	1134	-	Preconditions required	7.2.0	7.3.0	C1-060192
2006-03	CP-31	CP-060110	1156	1	Tables Change in Appendix A	7.2.0	7.3.0	C1-060478
2006-03	CP-31	CP-060110	1132	1	P-Asserted-Identity	7.2.0	7.3.0	C1-060476
2006-03	CP-31	CP-060111	1219	-	Reference Update of TS24.229, Rel7	7.2.0	7.3.0	C1-060483
2006-03	CP-31	CP-060111	1119	2	IMS Short Session Setup - Clarifications	7.2.0	7.3.0	C1-060595
2006-03	CP-31	CP-060111	1189	3	Definition of principles for IOI exchange and storage	7.2.0	7.3.0	C1-060610
2006-03	CP-31	CP-060111	1129	2	Tel URI	7.2.0	7.3.0	C1-060593
2006-03	CP-31	CP-060117	1210	1	Coding of P-Access-Network-Info header for 3GPP2 IMS	7.2.0	7.3.0	C1-060494
2006-03	CP-31	CP-060118	1103	1	Editor's Note on xDSL bearer	7.2.0	7.3.0	C1-060119
2006-03	CP-31	CP-060118	1095	1	Reference to new annexes on NAT	7.2.0	7.3.0	C1-060116
2006-03	CP-31	CP-060118	1101	-	Replaces header in Profile Tables	7.2.0	7.3.0	C1-060051
2006-03	CP-31	CP-060118	1093	2	P-Access-Network-Info header absence for emergency call detection	7.2.0	7.3.0	C1-060339
2006-03	CP-31	CP-060118	1196	1	correction for the procedure of changing media data	7.2.0	7.3.0	C1-060518
2006-03	CP-31	CP-060118	1197	1	Editorial Changes	7.2.0	7.3.0	C1-060519
2006-03	CP-31	CP-060118	1092	3	Optionality of P-Access-Network-Info header	7.2.0	7.3.0	C1-060338
2006-03	CP-31	CP-060118	1086	1	Addition of TISPAN supported internet-drafts	7.2.0	7.3.0	C1-060337
2006-03	CP-31	CP-060118	1089	1	IBCF corrections	7.2.0	7.3.0	C1-060110
2006-03	CP-31	CP-060118	1106	4	Completion of IBCF routing procedures	7.2.0	7.3.0	C1-060498
2006-03	CP-31	CP-060118	1088	4	IBCF enhancements	7.2.0	7.3.0	C1-060603
2006-03	CP-31	CP-060119	1177	1	PacketCable Extensions to P-Charging-Vector header	7.2.0	7.3.0	C1-060512
2006-03	CP-31	CP-060120	1098	4	Emergency service S-CSCF impact	7.2.0	7.3.0	C1-060601
2006-03	CP-31	CP-060120	1097	5	Emergency service - P-CSCF impact	7.2.0	7.3.0	C1-060600
2006-03	CP-31	CP-060120	1099	5	Emergency service - E-CSCF impact	7.2.0	7.3.0	C1-060599
2006-03	CP-31	CP-060120	1096	5	Emergency service - UE impact	7.2.0	7.3.0	C1-060602
2006-03	CP-31	CP-060121	1183	-	Transfer of Text from the Combinational Services TR 24.879 to TS 24.229	7.2.0	7.3.0	C1-060311

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2006-03	CP-31	CP-060124	1138	2	Session termination by P-CSCF	7.2.0	7.3.0	C1-060605
2006-03	CP-31	CP-060124	1157	3	Support for RFC 4145	7.2.0	7.3.0	C1-060621
2006-03	CP-31	CP-060124	1184	3	Registration of multiple PUIs - CR	7.2.0	7.3.0	C1-060608
2006-03	CP-31	CP-060124	1137	1	Session termination by S-CSCF	7.2.0	7.3.0	C1-060533
2006-03	CP-31	CP-060124	1152	1	Editorial Changes	7.2.0	7.3.0	C1-060539
2006-03	CP-31	CP-060124	1107	1	Reference Update of TS24.229	7.2.0	7.3.0	C1-060123
2006-03	CP-31	CP-060124	1125	-	Pre-loaded Route header	7.2.0	7.3.0	C1-060183
2006-03	CP-31	CP-060142	1226	1	Transport of HSS address from I-CSCF to S-CSCF	7.2.0	7.3.0	-
2006-03	CP-31	CP-060153	1222	2	Mandationof RFC 4320 fixes for issues found with the Session Initiation Protocol's (SIP) Non-INVITE Transactions	7.2.0	7.3.0	-
2006-03	CP-31	CP-060176	1225	2	Support of call forwarding at the S-CSCF	7.2.0	7.3.0	-
2006-06	CP-32	CP-060232	1290	2	Realm Parameter Handling	7.3.0	7.4.0	
2006-06	CP-32	CP-060249	1242	3	SDP answer	7.3.0	7.4.0	
2006-06	CP-32	CP-060262	1309	2	Hiding correction	7.3.0	7.4.0	C1-061115
2006-06	CP-32	CP-060262	1306	2	3rd-party registration	7.3.0	7.4.0	C1-061098
2006-06	CP-32	CP-060262	1303	1	One private identity one contact	7.3.0	7.4.0	C1-061095
2006-06	CP-32	CP-060264	1274	2	Re-authentication during deregistration	7.3.0	7.4.0	C1-061113
2006-06	CP-32	CP-060265	1312		I-CSCF registration procedure correction	7.3.0	7.4.0	C1-060829
2006-06	CP-32	CP-060266	1265	1	IOI overview	7.3.0	7.4.0	C1-060997
2006-06	CP-32	CP-060266	1271	1	Introduction of signalling encryption	7.3.0	7.4.0	C1-060999
2006-06	CP-32	CP-060266	1348		UE behavior after timer F expiry	7.3.0	7.4.0	C1-060897
2006-06	CP-32	CP-060266	1236	2	P-Asserted-ID	7.3.0	7.4.0	C1-061119
2006-06	CP-32	CP-060266	1238	1	Via header in the initial registration	7.3.0	7.4.0	C1-060975
2006-06	CP-32	CP-060266	1327	1	Incorrect requirement on I-CSCF	7.3.0	7.4.0	C1-061079
2006-06	CP-32	CP-060270	1247	1	Emergency PUID	7.3.0	7.4.0	C1-061054
2006-06	CP-32	CP-060270	1266	1	Inclusion of draft-ietf-ecrit-service-urn	7.3.0	7.4.0	C1-061009
2006-06	CP-32	CP-060270	1229		Emergency service S-CSCF impact	7.3.0	7.4.0	C1-060642
2006-06	CP-32	CP-060270	1360		Inclusion of E-CSCF in subclause 3.1 and subclause 4.1	7.3.0	7.4.0	C1-060923
2006-06	CP-32	CP-060270	1249	2	Emergency call release	7.3.0	7.4.0	C1-061121
2006-06	CP-32	CP-060270	1338	1	Adding RDF in E-CSCF procedure	7.3.0	7.4.0	C1-061060
2006-06	CP-32	CP-060270	1358	1	Priority handling for emergency calls at the E-CSCF	7.3.0	7.4.0	C1-061017
2006-06	CP-32	CP-060270	1357	1	Priority handling for emergency calls at the S-CSCF	7.3.0	7.4.0	C1-061015
2006-06	CP-32	CP-060270	1356	1	Priority handling for emergency calls at the P-CSCF	7.3.0	7.4.0	C1-061013

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2006-06	CP-32	CP-060270	1354		Inclusion of session timer procedures at the E-CSCF	7.3.0	7.4.0	C1-060917
2006-06	CP-32	CP-060270	1340	2	TEL URI associated with emergency IMPU	7.3.0	7.4.0	C1-061120
2006-06	CP-32	CP-060270	1337	1	Getting local emergency numbers	7.3.0	7.4.0	C1-061010
2006-06	CP-32	CP-060270	1336	1	Some corrections in IMS emergency calls	7.3.0	7.4.0	C1-061059
2006-06	CP-32	CP-060271	1258	1	UDP encapsulation of IPSec	7.3.0	7.4.0	C1-061019
2006-06	CP-32	CP-060271	1318	1	Extensions to P-Access-Network-Info header for DOCSIS Access	7.3.0	7.4.0	C1-061025
2006-06	CP-32	CP-060271	1317	2	PRACK	7.3.0	7.4.0	C1-061026
2006-06	CP-32	CP-060271	1267	1	IBCF corrections	7.3.0	7.4.0	C1-061022
2006-06	CP-32	CP-060271	1259	1	IBCF initiated call release	7.3.0	7.4.0	C1-061021
2006-06	CP-32	CP-060271	1345	1	Correction of the reference document	7.3.0	7.4.0	C1-061082
2006-06	CP-32	CP-060274	1234	1	Final NOTIFY	7.3.0	7.4.0	C1-060989
2006-06	CP-32	CP-060274	1255		Full notification	7.3.0	7.4.0	C1-060686
2006-06	CP-32	CP-060274	1260		Reg event package parameters in notification	7.3.0	7.4.0	C1-060704
2006-06	CP-32	CP-060274	1261		Subscription refreshing	7.3.0	7.4.0	C1-060705
2006-06	CP-32	CP-060274	1217	2	Definition of B2BUA	7.3.0	7.4.0	C1-061074
2006-06	CP-32	CP-060274	1277	1	Usage of associated public user identities	7.3.0	7.4.0	C1-060964
2006-06	CP-32	CP-060274	1321		Verification by I-CSCF of trust domain origin for incoming requests	7.3.0	7.4.0	C1-060844
2006-06	CP-32	CP-060274	1322		Miscellaneous Correction	7.3.0	7.4.0	C1-060845
2006-06	CP-32	CP-060274	1328	1	Resilience to registration and authentication errors	7.3.0	7.4.0	C1-061080
2006-06	CP-32	CP-060274	1335	1	The Correction on the description for the information of registration status	7.3.0	7.4.0	C1-060986
2006-06	CP-32	CP-060274	1361		Reference updates	7.3.0	7.4.0	C1-060924
2006-06	CP-32	CP-060283	1366		Emergency service – UE impact	7.3.0	7.4.0	
2006-06	CP-32	CP-060284	1367		Emergency service- E-CSCF impact	7.3.0	7.4.0	
2006-06	CP-32	CP-060335	1232	3	Handling of P-Charging-Addresses	7.3.0	7.4.0	
2006-06	CP-32	CP-060345	1365	1	Registration of several unrelated public user identities	7.3.0	7.4.0	
2006-06	CP-32	CP-060352	1228	4	Emergency service P-CSCF-impact	7.3.0	7.4.0	C1-061134
2006-09	CP-33	CP-060452	1461	1	Correction of Realm Parameter Handling for S-CSCF procedures	7.4.0	7.5.0	C1-061732
2006-09	CP-33	CP-060452	1467		SDP reference revision	7.4.0	7.5.0	C1-061657
2006-09	CP-33	CP-060452	1475	2	"Response" value in unprotected Register requests	7.4.0	7.5.0	C1-061845
2006-09	CP-33	CP-060463	1351	3	Treatment of emergency requests other than INVITE requests at the P-CSCF	7.4.0	7.5.0	C1-061357
2006-09	CP-33	CP-060463	1352	3	Treatment of emergency requests other than INVITE requests at the E-CSCF	7.4.0	7.5.0	C1-061358

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2006-09	CP-33	CP-060463	1369	1	UE emergency deregistration	7.4.0	7.5.0	C1-061304
2006-09	CP-33	CP-060463	1370	1	Emergency subscription	7.4.0	7.5.0	C1-061305
2006-09	CP-33	CP-060463	1371	1	P-CSCF emergency subscription	7.4.0	7.5.0	C1-061306
2006-09	CP-33	CP-060463	1373	2	S-CSCF emergency registration	7.4.0	7.5.0	C1-061350
2006-09	CP-33	CP-060463	1374	2	Handling of Emergency registration in S-CSCF	7.4.0	7.5.0	C1-061349
2006-09	CP-33	CP-060463	1375	2	Handling of emergency registration at the UE	7.4.0	7.5.0	C1-061351
2006-09	CP-33	CP-060463	1379	4	Location handling E-CSCF	7.4.0	7.5.0	C1-061913
2006-09	CP-33	CP-060463	1380	1	Clarification of Emergency Session Setup without prior IMS Registration	7.4.0	7.5.0	C1-061311
2006-09	CP-33	CP-060463	1381	1	Clarifications to subclause 5.1.6.1	7.4.0	7.5.0	C1-061313
2006-09	CP-33	CP-060463	1383	1	Non-INVITE requests	7.4.0	7.5.0	C1-061314
2006-09	CP-33	CP-060463	1384	2	IP-CAN for emergency calls	7.4.0	7.5.0	C1-061355
2006-09	CP-33	CP-060463	1390	1	Adoption of terminology from draft-ietf-ecrit-requirements	7.4.0	7.5.0	C1-061315
2006-09	CP-33	CP-060463	1391	3	Minor corrections to EMC1 text from previous CRs	7.4.0	7.5.0	C1-061367
2006-09	CP-33	CP-060463	1414	2	Handling of loacation information at E-CSCF	7.4.0	7.5.0	C1-061860
2006-09	CP-33	CP-060463	1440	2	P-Asserted-Identity in P-CSCF handling	7.4.0	7.5.0	C1-061861
2006-09	CP-33	CP-060463	1443	4	Handling of PSAP address mapping result at E-CSCF	7.4.0	7.5.0	C1-061919
2006-09	CP-33	CP-060465	1413	1	Miscellaneous Corrections in Annex F	7.4.0	7.5.0	C1-061826
2006-09	CP-33	CP-060465	1420	1	Transit IMS	7.4.0	7.5.0	C1-061827
2006-09	CP-33	CP-060465	1425	1	P-CSCF procedures for session release when QoS resources are unavailable	7.4.0	7.5.0	C1-061830
2006-09	CP-33	CP-060465	1427	1	Make SDP bandwidth modifiers optional for standard RTCP usage	7.4.0	7.5.0	C1-061832
2006-09	CP-33	CP-060465	1430	3	Addition of the cpc parameter to TS24.229	7.4.0	7.5.0	C1-061882
2006-09	CP-33	CP-060466	1385	4	Introduction of GRUU in 24.229	7.4.0	7.5.0	C1-061858
2006-09	CP-33	CP-060466	1386	5	S-SCSF procedures to support GRUU	7.4.0	7.5.0	C1-061915
2006-09	CP-33	CP-060468	1405		Original dialog identifier	7.4.0	7.5.0	C1-061408
2006-09	CP-33	CP-060468	1406		No-fork	7.4.0	7.5.0	C1-061409
2006-09	CP-33	CP-060468	1409		Connection addres - zero	7.4.0	7.5.0	C1-061412
2006-09	CP-33	CP-060468	1415		Reference for populating the "Anonymous" From header	7.4.0	7.5.0	C1-061439
2006-09	CP-33	CP-060468	1439	1	Usage of P-Associated-URI	7.4.0	7.5.0	C1-061759
2006-09	CP-33	CP-060468	1450		Clarification of network initiated deregistration to match reginfo format	7.4.0	7.5.0	C1-061585
2006-09	CP-33	CP-060468	1456	2	Authentication between UA and UA	7.4.0	7.5.0	C1-061851
2006-09	CP-33	CP-060468	1457	2	Treatment by S-CSCF of profile changes for registered PUIs	7.4.0	7.5.0	C1-061853
2006-09	CP-33	CP-060468	1458	1	Completionof RFC 4320 fixes for 100 Trying responses Non-INVITE Transactions RFC 4320	7.4.0	7.5.0	C1-061765

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					fixes for 100 Trying responses Non-INVITE Transactions tration			
2006-09	CP-33	CP-060468	1463		Correction to S-CSCF procedures for UE-originated requests	7.4.0	7.5.0	C1-061646
2006-09	CP-33	CP-060468	1464	1	SCTP transport	7.4.0	7.5.0	C1-061766
2006-09	CP-33	CP-060504	1257	4	SDP usage at MGCF	7.4.0	7.5.0	C1-061847
2006-09	CP-33	CP-060504	1417	1	Type 3 orig-ioi in I-CSCF	7.4.0	7.5.0	C1-061744
2006-09	CP-33	CP-060504	1469		SDP corrections	7.4.0	7.5.0	C1-061659
2006-09	CP-33	CP-060504	1471		SDP completion	7.4.0	7.5.0	C1-061661
2006-09	CP-33	CP-060504	1478	1	Updates to Profile Tables UE Major Capabilities	7.4.0	7.5.0	C1-061754
2006-09	CP-33	CP-060504	1481		Removal of Editor's notes in 24.229, rel-6	7.4.0	7.5.0	C1-061745
2006-09	CP-33	CP-060504	1483		Final codec selection	7.4.0	7.5.0	C1-061850
2006-09	CP-33	CP-060526	1418	3	Originating requests on behalf of an unregistered user	7.4.0	7.5.0	C1-061758
2006-09					Version 7.5.1 created by MCC to correct styles	7.5.0	7.5.1	
2006-12	CP-34	CP-060655	1502	-	RFC reference update	7.5.1	7.6.0	C1-061977
2006-12	CP-34	CP-060655	1506	-	SDP group attribute correction	7.5.1	7.6.0	C1-061981
2006-12	CP-34	CP-060655	1504	1	Addressing editor's notes relating to trust domains	7.5.1	7.6.0	C1-062304
2006-12	CP-34	CP-060655	1546	-	Join header correction	7.5.1	7.6.0	C1-062205
2006-12	CP-34	CP-060655	1508	2	Processing the successful response at S-CSCF	7.5.1	7.6.0	C1-062434
2006-12	CP-34	CP-060655	1449	2	Correction of S-CSCF construction and UE interpretation of registration event notification	7.5.1	7.6.0	C1-062317
2006-12	CP-34	CP-060655	1514	1	Removal of more Editor's notes in 24.229, rel-6	7.5.1	7.6.0	C1-062310
2006-12	CP-34	CP-060659	1491	2	Location handling for emergency	7.5.1	7.6.0	C1-062437
2006-12	CP-34	CP-060659	1521	1	Location information for IMS emergency	7.5.1	7.6.0	C1-062293
2006-12	CP-34	CP-060659	1529	2	Emergency re-registration due to mobility	7.5.1	7.6.0	C1-062436
2006-12	CP-34	CP-060659	1515	1	Removal of Editor's notes on emergency call in clause 4	7.5.1	7.6.0	C1-062292
2006-12	CP-34	CP-060659	1484	1	Corrections to emergency call procedures for P-Asserted-Identity header	7.5.1	7.6.0	C1-062289
2006-12	CP-34	CP-060659	1543	-	Next hop is the BGCF	7.5.1	7.6.0	C1-062181
2006-12	CP-34	CP-060659	1536	-	Editorial corrections to emergency call text	7.5.1	7.6.0	C1-062142
2006-12	CP-34	CP-060659	1542	1	minor correction to EMC of UE and PCSCF	7.5.1	7.6.0	C1-062299
2006-12	CP-34	CP-060659	1490	2	Emergency call on existing registration	7.5.1	7.6.0	C1-062435
2006-12	CP-34	CP-060660	1486	2	Introduction of communication service concept in TS 24229	7.5.1	7.6.0	C1-062451
2006-12	CP-34	CP-060662	1494	1	Tel URI translation	7.5.1	7.6.0	C1-062325
2006-12	CP-34	CP-060662	1523	1	I-CSCF procedure	7.5.1	7.6.0	C1-062333
2006-12	CP-34	CP-060662	1544	-	Clarification of UEs initial SDP offer	7.5.1	7.6.0	C1-062189

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2006-12	CP-34	CP-060662	1493	1	Alias URI	7.5.1	7.6.0	C1-062324
2006-12	CP-34	CP-060662	1525	1	Clarification of iFC execution for UE-terminated requests at S-CSCF	7.5.1	7.6.0	C1-062334
2006-12	CP-34	CP-060662	1533	1	SIP response code to unknown method	7.5.1	7.6.0	C1-062336
2006-12	CP-34	CP-060662	1537	-	Originating requests on behalf of an unregistered user	7.5.1	7.6.0	C1-062143
2006-12	CP-34	CP-060662	1538	-	Treatment by S-CSCF of profile changes for registered PUIs	7.5.1	7.6.0	C1-062144
2006-12	CP-34	CP-060662	1547	-	Corrections to Profile table for RFC 4320 compliance	7.5.1	7.6.0	C1-062210
2006-12	CP-34	CP-060662	1539	-	Miscellaneous editorial corrections	7.5.1	7.6.0	C1-062145
2006-12	CP-34	CP-060662	1509	1	No-forking at AS	7.5.1	7.6.0	C1-062329
2006-12	CP-34	CP-060662	1528	2	P-Visited-Network-ID on ISC interface	7.5.1	7.6.0	C1-062442
2006-12	CP-34	CP-060662	1487	1	Introduction of P-Profile Key in TS 24.229	7.5.1	7.6.0	C1-062322
2006-12	CP-34	CP-060662	1522	1	Local numbering	7.5.1	7.6.0	C1-062338
2006-12	CP-34	CP-060662	1495	2	BGCF procedures	7.5.1	7.6.0	C1-062440
2006-12	CP-34	CP-060662	1498	2	AS acting as PSI	7.5.1	7.6.0	C1-062441
2006-12	CP-34	CP-060662	1524	-	Clarification of the URI in UE-terminating requests at the P-CSCF	7.5.1	7.6.0	C1-062061
2006-12	CP-34	CP-060662	1549	1	Core Network Service Authorizatrion	7.5.1	7.6.0	C1-062339
2006-12	CP-34	CP-060663	1527	3	Align with GRUU IETF draft 11	7.5.1	7.6.0	C1-062512
2006-12	CP-34	CP-060663	1496	1	I-CSCF processing GRUU	7.5.1	7.6.0	C1-062340
2006-12	CP-34	CP-060663	1497	1	S-CSCF processing GRUU	7.5.1	7.6.0	C1-062341
2006-12	CP-34	CP-060663	1422	3	GRUU processing by non-UE User Agents	7.5.1	7.6.0	C1-062343
2006-12	CP-34	CP-060667	1426	3	Allowing an asserted display name to be conveyed with a Public Identity	7.5.1	7.6.0	C1-062427
2006-12	CP-34	CP-060667	1429	4	Update to NAT Traversal procedures in support of Outbound and ICE	7.5.1	7.6.0	C1-062515
2006-12	CP-34	CP-060667	1540	2	Annex I (Transit IMS) improvements	7.5.1	7.6.0	C1-062516
2007-03	CP-35	CP-070130	1566	-	Session Establishment Interworking with Rel-5 UEs	7.6.0	7.7.0	C1-070052
2007-03	CP-35	CP-070130	1638	-	Inclusion of draft-ietf-sip-uri-list-message in SIP profile	7.6.0	7.7.0	C1-070266
2007-03	CP-35	CP-070130	1619	-	Clarifications on resource reservation	7.6.0	7.7.0	C1-070180
2007-03	CP-35	CP-070130	1621	1	Routeing B2BUA handling of Replaces header	7.6.0	7.7.0	C1-070439
2007-03	CP-35	CP-070132	1609	-	Establishing an emergency session	7.6.0	7.7.0	C1-070147
2007-03	CP-35	CP-070132	1575	-	Deletion of editors note in subclause 5.1.6.5	7.6.0	7.7.0	C1-070068
2007-03	CP-35	CP-070132	1639	-	Identification of emergency calls	7.6.0	7.7.0	C1-070276
2007-03	CP-35	CP-070132	1593	1	Limitation on Emergency Registration	7.6.0	7.7.0	C1-070424
2007-03	CP-35	CP-070132	1586	1	Tidyup UE clause	7.6.0	7.7.0	C1-070418
2007-03	CP-35	CP-070132	1654	1	Double reference removal	7.6.0	7.7.0	C1-070381

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2007-03	CP-35	CP-070132	1605	1	Emergency PUID	7.6.0	7.7.0	C1-070419
2007-03	CP-35	CP-070132	1569	1	Handling of parallel emergency registration	7.6.0	7.7.0	C1-070413
2007-03	CP-35	CP-070132	1574	1	Deletion of editors note in subclause 5.1.6.2	7.6.0	7.7.0	C1-070414
2007-03	CP-35	CP-070132	1568	1	Connecting to an Emergency APN	7.6.0	7.7.0	C1-070409
2007-03	CP-35	CP-070132	1581	1	Deletion of Editor' s notes in 5.2.10	7.6.0	7.7.0	C1-070416
2007-03	CP-35	CP-070132	1641	-	Correction of service-urn	7.6.0	7.7.0	C1-070278
2007-03	CP-35	CP-070132	1589	-	Correction of CR#1484r1 implementation error (subclause 5.1.6.8.3)	7.6.0	7.7.0	C1-070111
2007-03	CP-35	CP-070132	1610	-	Emergency session-no registration	7.6.0	7.7.0	C1-070148
2007-03	CP-35	CP-070134	1612	2	Emergency treatment at P-CSCF	7.6.0	7.7.0	C1-070563
2007-03	CP-35	CP-070134	1635	1	Remove the term ESRP	7.6.0	7.7.0	C1-070430
2007-03	CP-35	CP-070134	1607	2	Emergency call at P-CSCF	7.6.0	7.7.0	C1-070443
2007-03	CP-35	CP-070134	1632	1	Backward compatibility for using 380 response	7.6.0	7.7.0	C1-070429
2007-03	CP-35	CP-070134	1653	3	Location for emergency	7.6.0	7.7.0	C1-070618
2007-03	CP-35	CP-070134	1626	1	Handling of re-registration when user redial emergency number	7.6.0	7.7.0	C1-070426
2007-03	CP-35	CP-070134	1582	2	Deletion of editors notes in 5.11 and 5.4.8	7.6.0	7.7.0	C1-070615
2007-03	CP-35	CP-070134	1567	3	Home Network Indication for Emergency Calls	7.6.0	7.7.0	C1-070640
2007-03	CP-35	CP-070134	1631	2	Correction to emergency call procedure with non-emergency registration for P-Asserted-Identity header	7.6.0	7.7.0	C1-070617
2007-03	CP-35	CP-070137	1634	1	Profile definition for CSI application server	7.6.0	7.7.0	C1-070469
2007-03	CP-35	CP-070138	1660	1	Format of dsl-location	7.6.0	7.7.0	C1-070552
2007-03	CP-35	CP-070138	1595	1	Deletion of EN's in clause 5.10	7.6.0	7.7.0	C1-070547
2007-03	CP-35	CP-070138	1594	-	Deletion of EN's in Annex G	7.6.0	7.7.0	C1-070132
2007-03	CP-35	CP-070139	1613	2	Annex K NAT Traversal Procedural and References Updates	7.6.0	7.7.0	C1-070626
2007-03	CP-35	CP-070139	1617	1	Routing of SIP URI "user=phone" when domain doesn't own target user	7.6.0	7.7.0	C1-070551
2007-03	CP-35	CP-070139	1614	1	Annex A updates for Annex K NAT Traversal Procedurals	7.6.0	7.7.0	C1-070550
2007-03	CP-35	CP-070140	1598	1	Forked MESSAGE request	7.6.0	7.7.0	C1-070451
2007-03	CP-35	CP-070140	1558	1	Removal of notes for screening functionality	7.6.0	7.7.0	C1-070441
2007-03	CP-35	CP-070140	1556	1	Handling of special characters in the local service number	7.6.0	7.7.0	C1-070458
2007-03	CP-35	CP-070140	1655	2	Forwarding a request by transit functions in the S-CSCF	7.6.0	7.7.0	C1-070586
2007-03	CP-35	CP-070140	1587	1	Terminating case in S-CSCF	7.6.0	7.7.0	C1-070449
2007-03	CP-35	CP-070140	1559	-	Completion of SIP timers functionality	7.6.0	7.7.0	C1-070039
2007-03	CP-35	CP-070140	1588	1	P-User-Database	7.6.0	7.7.0	C1-070450
2007-03	CP-35	CP-070140	1560	1	Removal of notes for SIGCOMP functionality	7.6.0	7.7.0	C1-070442

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2007-03	CP-35	CP-070140	1557	-	Removal of normative statements in NOTEs	7.6.0	7.7.0	C1-070037
2007-03	CP-35	CP-070140	1604	1	Forwarding P-Charging-Vector outside the home network	7.6.0	7.7.0	C1-070453
2007-03	CP-35	CP-070140	1555	1	Removal of Editor's notes for message bodies	7.6.0	7.7.0	C1-070440
2007-03	CP-35	CP-070140	1652	-	Correction for local numbers	7.6.0	7.7.0	C1-070341
2007-03	CP-35	CP-070140	1601	-	Tel URI translation	7.6.0	7.7.0	C1-070139
2007-03	CP-35	CP-070140	1646	1	Align definition of Alias URI with the description in 23.228	7.6.0	7.7.0	C1-070455
2007-03	CP-35	CP-070140	1600	2	Dual IP addresses	7.6.0	7.7.0	C1-070584
2007-03	CP-35	CP-070142	1642	-	SIP extensions covering URI-lists	7.6.0	7.7.0	C1-070279
2007-03	CP-35	CP-070148	1564	1	Network Initiated / Modified Media PDP Contexts	7.6.0	7.7.0	C1-070447
2007-03	CP-35	CP-070149	1643	-	SDP usage in association with BFCP (additions to SDP profile)	7.6.0	7.7.0	C1-070282
2007-03	CP-35	CP-070151	1648	2	S-CSCF inserts P-Called-Party-ID before forwarding request towards served user	7.6.0	7.7.0	C1-070588
2007-03	CP-35	CP-070151	1597	1	Instance ID	7.6.0	7.7.0	C1-070461
2007-03	CP-35	CP-070151	1615	1	Signalling Public User Identity to AS when request URI is Temp-GRUU	7.6.0	7.7.0	C1-070463
2007-03	CP-35	CP-070214	1640	3	Location conveyance revisions	7.6.0	7.7.0	
2007-03	CP-35	CP-070242	1576	3	Deletion of editors notes in subclauses 5.1.6.8.2, 5.1.6.8.3, 5.1.6.8.4	7.6.0	7.7.0	
2007-03	CP-35	CP-070252	1658	4	Profile for IBCF	7.6.0	7.7.0	
2007-03	CP-35	CP-070254	1580	3	PCC introduction to TS 24.229	7.6.0	7.7.0	
2007-03	CP-35	CP-070255	1630	3	Corrections for the handling of target refresh requests at the S-CSCF	7.6.0	7.7.0	
2007-03	CP-35	CP-070271	1623	5	Further alignment with phonebcp draft	7.6.0	7.7.0	
2007-06	CP-36	CP-070370	1749	1	Correction of coding rules of P-Access-Network-Info header	7.7.0	7.8.0	C1-071435
2007-06	CP-36	CP-070370	1689	2	Inclusion of "addressing an amplification vulnerability in session initiation protocol forking proxies" (draft-ietf-sip-fork-loop-fix) in the SIP profile	7.7.0	7.8.0	C1-071409
2007-06	CP-36	CP-070373	1666	2	Protocol between E-CSCF and LRF	7.7.0	7.8.0	C1-071040
2007-06	CP-36	CP-070373	1690	-	Further alignment with phonebcp draft	7.7.0	7.8.0	C1-070779
2007-06	CP-36	CP-070373	1763	1	Emergency registration clarification	7.7.0	7.8.0	C1-071441
2007-06	CP-36	CP-070373	1665	1	Definition of identities used for emergency call	7.7.0	7.8.0	C1-070957
2007-06	CP-36	CP-070374	1714	1	Alignment of layout of access technology specific annexes	7.7.0	7.8.0	C1-071032
2007-06	CP-36	CP-070374	1715	1	GPRS IP-CAN change of normative requirement out of scope to informative	7.7.0	7.8.0	C1-071033
2007-06	CP-36	CP-070374	1732	2	Clarification on iFC execution	7.7.0	7.8.0	C1-071460
2007-06	CP-36	CP-070374	1721	1	UE un-subscribing to reg-event	7.7.0	7.8.0	C1-071419

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2007-06	CP-36	CP-070374	1722	-	MO Record-Route at P-CSCF	7.7.0	7.8.0	C1-071051
2007-06	CP-36	CP-070374	1723	1	MT Record-Route at P-CSCF	7.7.0	7.8.0	C1-071420
2007-06	CP-36	CP-070374	1727	1	Double registration	7.7.0	7.8.0	C1-071422
2007-06	CP-36	CP-070374	1730	1	Inclusion of new mandatory elements of SigComp	7.7.0	7.8.0	C1-071423
2007-06	CP-36	CP-070374	1731	1	Use of a presence specific dictionary in SigComp	7.7.0	7.8.0	C1-071424
2007-06	CP-36	CP-070374	1720	1	Registration and deregistration	7.7.0	7.8.0	C1-071418
2007-06	CP-36	CP-070374	1746	1	Correction to P-CSCF procedures for cancellation of a session currently being established	7.7.0	7.8.0	C1-071431
2007-06	CP-36	CP-070374	1762	1	Originating a terminating request in an AS	7.7.0	7.8.0	C1-071433
2007-06	CP-36	CP-070374	1769	2	Clarification to Original Dialog Identifier	7.7.0	7.8.0	C1-071463
2007-06	CP-36	CP-070374	1761	-	Local numbering clarification	7.7.0	7.8.0	C1-071196
2007-06	CP-36	CP-070374	1760	1	PANI related corrections	7.7.0	7.8.0	C1-071437
2007-06	CP-36	CP-070374	1743	1	The precondition mechanism may be required in subsequent SDP offer/answer exchanges	7.7.0	7.8.0	C1-071430
2007-06	CP-36	CP-070374	1772	-	Minor miscellaneous clean-up	7.7.0	7.8.0	C1-071231
2007-06	CP-36	CP-070374	1739	1	P-CSCF processing of P-Early-Media	7.7.0	7.8.0	C1-071428
2007-06	CP-36	CP-070374	1738	3	Originating UE sending of P-Early-Media	7.7.0	7.8.0	C1-071462
2007-06	CP-36	CP-070374	1737	2	Originating UE processing of P-Early-Media	7.7.0	7.8.0	C1-071461
2007-06	CP-36	CP-070375	1692	-	Profile support for a session initiation protocol event package and data format for various settings in support for the push-to-talk over cellular service (RFC4354)	7.7.0	7.8.0	C1-070781
2007-06	CP-36	CP-070375	1562	4	Completion of Phone-context parameter in rel-7	7.7.0	7.8.0	C1-071009
2007-06	CP-36	CP-070375	1700	-	Translation of non-international format numbers	7.7.0	7.8.0	C1-070810
2007-06	CP-36	CP-070375	1680	-	Outgoing Request URI=pres or IM URI processing clarification and misc clean-up	7.7.0	7.8.0	C1-070705
2007-06	CP-36	CP-070375	1691	1	Profile support for the P-Answer-State header extension to the session initiation protocol for the open mobile alliance push to talk over cellular (draft-allen-sipping-poc-p-answer-state-header)	7.7.0	7.8.0	C1-070987
2007-06	CP-36	CP-070375	1678	1	Qvalue	7.7.0	7.8.0	C1-070984
2007-06	CP-36	CP-070375	1704	-	Minor miscellaneous clean-up	7.7.0	7.8.0	C1-070824
2007-06	CP-36	CP-070375	1703	-	Filter criteria evaluation when the AS changes the P-Asserted-Identity	7.7.0	7.8.0	C1-070823
2007-06	CP-36	CP-070378	1718	1	Addition to network initiated PDP context	7.7.0	7.8.0	C1-071346
2007-06	CP-36	CP-070380	1679	-	Cleanup of Signalling Public GRUU to AS	7.7.0	7.8.0	C1-070704
2007-06	CP-36	CP-070380	1663	-	Provide GRUU functionality in case of hosted NAT	7.7.0	7.8.0	C1-070663
2007-06	CP-36	CP-070380	1756	1	GRUU Alignment with stage 2	7.7.0	7.8.0	C1-071456

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2007-06	CP-36	CP-070380	1686	2	Alternate GRUU for AS acting on behalf of Public User Identity	7.7.0	7.8.0	C1-071010
2007-06	CP-36	CP-070380	1713	2	Cleanup of GRUU	7.7.0	7.8.0	C1-071238
2007-06	CP-36	CP-070380	1766	1	Management of GRUU	7.7.0	7.8.0	C1-071457
2007-06	CP-36	CP-070380	1711	2	Use of GRUU for Emergency Sessions	7.7.0	7.8.0	C1-071458
2007-06	CP-36	CP-070383	1773	-	IMS Communication Service ID registration	7.7.0	7.8.0	C1-071234
2007-06	CP-36	CP-070383	1645	6	IMS Communication Service ID 24.229	7.7.0	7.8.0	C1-071475
2007-06	CP-36	CP-070388	1735	2	Correction on the handlng of CPC parameter regarding trust domain	7.7.0	7.8.0	C1-071464
2007-06	CP-36	CP-070388	1662	-	Tidyup open issues from FBI work item	7.7.0	7.8.0	C1-070662
2007-06	CP-36	CP-070388	1596	5	Update to NAT Traversal procedures in support of Outbound and ICE	7.7.0	7.8.0	C1-071400
2007-06	CP-36	CP-070388	1740	1	IBCF processing of P-Early-Media	7.7.0	7.8.0	C1-071404
2007-06	CP-36	CP-070388	1742	1	IBCF Path header	7.7.0	7.8.0	C1-071405
2007-06	CP-36	CP-070436	1696	3	Endorsement of P-Early-Media header draft	7.7.0	7.8.0	
2007-06	CP-36	CP-070447	1698	3	Report of new transit scenario documented in stage 2	7.7.0	7.8.0	-
2007-06	CP-36	CP-070450	1771	3	THIG processing correction to ensure conformity to RFC 3261	7.7.0	7.8.0	-
2007-06	CP-36	CP-070496	1717	4	PCC impact	7.7.0	7.8.0	-
2007-06	CP-36	CP-070393	1751	1	Resource-Priority header and trust domains	7.7.0	8.0.0	C1-071446
2007-06	CP-36	CP-070393	1695	2	Inclusion policy for Resource-Priority header in support of multimedia priority service	7.7.0	8.0.0	C1-071443
2007-06	CP-36	CP-070393	1694	2	Inclusion of "communications resource priority for the session initiation protocol" (RFC4412) in the SIP profile	7.7.0	8.0.0	C1-071444
2007-06	CP-36	CP-070393	1693	1	Inclusion of "extending the session initiation protocol Reason header for preemption events" (RFC4411) in the SIP profile	7.7.0	8.0.0	C1-070918
2007-06	CP-36	CP-070396	1682	2	IMS Enhancements to Support Number Portability (NP) for Cable Networks	7.7.0	8.0.0	C1-070994
2007-06	CP-36	CP-070396	1681	4	Enhancements to Support Preferred Circuit Carrier Access and Dial-Around for Cable Networks	7.7.0	8.0.0	C1-071294
2007-09	CP-37	CP-070578	1945		Correction of the Authorization Header in the Profile Table	8.0.0	8.1.0	C1-072085
2007-09	CP-37	CP-070578	1811		Integrity param in De- and ReREGISTER	8.0.0	8.1.0	C1-071573
2007-09	CP-37	CP-070579	1905	2	Clarification of DTD	8.0.0	8.1.0	C1-072150
2007-09	CP-37	CP-070580	1795	2	Unprotected registration at UE	8.0.0	8.1.0	C1-072153
2007-09	CP-37	CP-070580	1876		IETF reference updates	8.0.0	8.1.0	C1-071772
2007-09	CP-37	CP-070580	1924	1	P-Access-Network-Info header clarification	8.0.0	8.1.0	C1-072042
2007-09	CP-37	CP-070580	1922	1	Optional rport parameter in UE	8.0.0	8.1.0	C1-072039
2007-09	CP-37	CP-070580	1797	1	Unprotected registration at S-CSCF	8.0.0	8.1.0	C1-072052

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2007-09	CP-37	CP-070584	1866		Emergency Registration without eAPN	8.0.0	8.1.0	C1-071728
2007-09	CP-37	CP-070585	1878		IETF reference updates relating to emergency call feature	8.0.0	8.1.0	C1-071776
2007-09	CP-37	CP-070585	1892	1	Correction of emergency procedures unregistered user case	8.0.0	8.1.0	C1-072018
2007-09	CP-37	CP-070585	1894		Emergency registration timer in visited network	8.0.0	8.1.0	C1-071808
2007-09	CP-37	CP-070585	1927		Contents of From header when initiating an emergency session within a emergency registration	8.0.0	8.1.0	C1-071874
2007-09	CP-37	CP-070586	1861	1	Correction for the URNs of IMS Communication Service Identifier and IMS Application Reference Identifier	8.0.0	8.1.0	C1-071956
2007-09	CP-37	CP-070586	1909	2	Completing UE ICSI/IARI procedures	8.0.0	8.1.0	C1-072162
2007-09	CP-37	CP-070586	1842	1	S-CSCF option to add P-Asserted-Service in UE-originated case	8.0.0	8.1.0	C1-071952
2007-09	CP-37	CP-070586	1911	2	Completing S-CSCF ICSI/IARI procedures	8.0.0	8.1.0	C1-072164
2007-09	CP-37	CP-070586	1826	1	Cleanup of text related to contact header dealing with ICSI	8.0.0	8.1.0	C1-071942
2007-09	CP-37	CP-070586	1838	2	Description of the ICSI as an assigned identifier	8.0.0	8.1.0	C1-072159
2007-09	CP-37	CP-070586	1929	1	ICSI Alignments with reqs 2, 3 and 11	8.0.0	8.1.0	C1-071947
2007-09	CP-37	CP-070586	1942	1	UE usage of ServideID received from the network	8.0.0	8.1.0	C1-072181
2007-09	CP-37	CP-070586	1840		Correction of application server handling of ICSI and IARI values	8.0.0	8.1.0	C1-071676
2007-09	CP-37	CP-070590	1807	3	Trust Domain in IMS	8.0.0	8.1.0	C1-072185
2007-09	CP-37	CP-070590	1799	1	Unprotected registration at P-CSCF	8.0.0	8.1.0	C1-072054
2007-09	CP-37	CP-070590	1793	1	Protected registration	8.0.0	8.1.0	C1-072046
2007-09	CP-37	CP-070590	1804	1	No multiple simultanous Registration	8.0.0	8.1.0	C1-072056
2007-09	CP-37	CP-070590	1864	1	Corrections of tables in Annex A	8.0.0	8.1.0	C1-072065
2007-09	CP-37	CP-070590	1879	1	Essential corrections to P-Early-Media header procedures	8.0.0	8.1.0	C1-072062
2007-09	CP-37	CP-070590	1881		IETF SigComp reference updates	8.0.0	8.1.0	C1-071779
2007-09	CP-37	CP-070590	1934		SIP related reference update	8.0.0	8.1.0	C1-071888
2007-09	CP-37	CP-070590	1913	1	Removal of IBCF Route Headers Editors Note	8.0.0	8.1.0	C1-072073
2007-09	CP-37	CP-070590	1854	1	Clarification on P-Profile-Key	8.0.0	8.1.0	C1-072063
2007-09	CP-37	CP-070592	1817		Resolve FFS for AS-GRUU	8.0.0	8.1.0	C1-071581
2007-09	CP-37	CP-070596	1885	2	Update Emergency NAT Traversal Procedures Annex K	8.0.0	8.1.0	C1-072078
2007-09	CP-37	CP-070596	1883	1	Update GRUU NAT Traversal Procedures Annex-K	8.0.0	8.1.0	C1-071926
2007-09	CP-37	CP-070600	1750	3	Resource-Priority and priority	7.8.0	8.1.0	C1-072132
2007-09	CP-37	CP-070600	1919	2	Addition of MGCF for optional support of Resource-Priority	8.0.0	8.1.0	C1-072184

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2007-09	CP-37	CP-070601	1815	2	Updates to Annex K in support of SIP Digest and TLS procedures	8.0.0	8.1.0	C1-072137
2007-09	CP-37	CP-070601	1812	4	UE Digest and TLS Procedures	8.0.0	8.1.0	C1-072172
2007-09	CP-37	CP-070601	1814	4	S-CSCF Digest and TLS Procedures	8.0.0	8.1.0	C1-072174
2007-09	CP-37	CP-070601	1813	4	P-CSCF Digest and TLS Procedures	8.0.0	8.1.0	C1-072173
2007-09	CP-37	CP-070603	1847	1	Cleanup of SigComp dictionary support	8.0.0	8.1.0	C1-072144
2007-09	CP-37	CP-070603	1896	1	S-CSCF procedure corrections	8.0.0	8.1.0	C1-072089
2007-09	CP-37	CP-070603	1935		Restructuring of subclause 5.2.6 (General treatment for all dialogs and standalone transactions excluding the REGISTER method) of the P-CSCF	8.0.0	8.1.0	C1-071891
2007-09	CP-37	CP-070603	1788	2	Request-URI in registration	8.0.0	8.1.0	C1-072154
2007-09	CP-37	CP-070670	1907	3	Definition of feature tag for IARI/ICSI	8.0.0	8.1.0	C1-072006
2007-09	CP-37	CP-070674	1791	2	Emergency registration	8.0.0	8.1.0	C1-072016
2007-09	CP-37	CP-070676	1851	4	P-CSCF behaviour upon loss of SIP signalling transport	8.0.0	8.1.0	C1-072178
2007-09	CP-37	CP-070691	1926	5	UE setting of IARI	8.0.0	8.1.0	C1-072166
2007-12	CP-38	CP-070735	2077	1	Update P-Early-Media Reference	8.1.0	8.2.0	C1-072750
2007-12	CP-38	CP-070785	2065		Authenticating with AKAv1-MD5	8.1.0	8.2.0	C1-072533
2007-12	CP-38	CP-070785	2115		Proxy profile corrections	8.1.0	8.2.0	C1-072922
2007-12	CP-38	CP-070785	2111		Corrections to RFC 3329 entries in profile	8.1.0	8.2.0	C1-072918
2007-12	CP-38	CP-070785	2041	1	Corrections for re-authenticating user	8.1.0	8.2.0	C1-072553
2007-12	CP-38	CP-070785	2049	3	Introduction of versioning and conventions	8.1.0	8.2.0	C1-072989
2007-12	CP-38	CP-070788	2028	1	Coverage of access technology specific text	8.1.0	8.2.0	C1-072746
2007-12	CP-38	CP-070788	2017	2	Action on missing "integrity-protected" parameter	8.1.0	8.2.0	C1-073179
2007-12	CP-38	CP-070788	2035	1	MGCF does not act as a proxy	8.1.0	8.2.0	C1-072565
2007-12	CP-38	CP-070788	2070	1	Correction to subclause 7.2A.5.2.2	8.1.0	8.2.0	C1-073052
2007-12	CP-38	CP-070791	1999	1	380 at normal call setup	8.1.0	8.2.0	C1-072670
2007-12	CP-38	CP-070791	2062	2	Miscellaneous EMC1 corrections	8.1.0	8.2.0	C1-072748
2007-12	CP-38	CP-070791	2120		Introductory text for emergency service	8.1.0	8.2.0	C1-072930
2007-12	CP-38	CP-070794	1990		Correct sub-section references in Annex-K	8.1.0	8.2.0	C1-072295
2007-12	CP-38	CP-070794	2023		Correction of outbound and ice option tag support in profile tables	8.1.0	8.2.0	C1-072383
2007-12	CP-38	CP-070795	1986	1	Align with draft-gruu-reg-ev-09	8.1.0	8.2.0	C1-072752
2007-12	CP-38	CP-070795	2043	1	Addition of GRUU to emergency set-up when registration exists	8.1.0	8.2.0	C1-072599
2007-12	CP-38	CP-070799	2067	1	P-CSCF Releases/Rejects session due to PCRF responses	8.1.0	8.2.0	C1-073067
2007-12	CP-38	CP-070805	2053	2	Terminating UE ICSI procedures	8.1.0	8.2.0	C1-072708

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2007-12	CP-38	CP-070805	2021	1	Correction to digest and TLS Procedures for Annex K	8.1.0	8.2.0	C1-072508
2007-12	CP-38	CP-070805	1951	1	Correction to the examples for ICSI and IARI values	8.1.0	8.2.0	C1-072490
2007-12	CP-38	CP-070805	2014	2	Encoding of ICSI and IARI within the g.ims.app_ref feature tag	8.1.0	8.2.0	C1-072704
2007-12	CP-38	CP-070805	2051	1	Multiple IARI/ICSI values in g.ims.app_ref feature tag	8.1.0	8.2.0	C1-072512
2007-12	CP-38	CP-070805	1969	1	One ICSI value per P-Preferred-Service header	8.0.0	8.2.0	C1-072496
2007-12	CP-38	CP-070805	1963	1	Change of name for feature tag g.ims.app_ref	8.0.0	8.2.0	C1-072492
2007-12	CP-38	CP-070806	2008	2	Handling of invalid and unauthorized media based on Communication Service Identifiers	8.1.0	8.2.0	C1-072702
2007-12	CP-38	CP-070806	2092	2	S-CSCF Processing of P-Preferred-Service and P-Asserted-Service	8.1.0	8.2.0	C1-073204
2007-12	CP-38	CP-070806	2107	2	The received list of ICSIs from the Network	8.1.0	8.2.0	C1-073206
2007-12	CP-38	CP-070806	2088		ICSI in Annex F	8.1.0	8.2.0	C1-072841
2007-12	CP-38	CP-070806	2019	2	Miscellaneous service identifier corrections	8.1.0	8.2.0	C1-073106
2007-12	CP-38	CP-070806	1965	3	Minor corrections to P-Preferred and P-Asserted Service headers	8.1.0	8.2.0	C1-073102
2007-12	CP-38	CP-070806	1976	2	Correction to S-CSCF handling of IMS communication service	8.1.0	8.2.0	C1-072700
2007-12	CP-38	CP-070807	2005	1	No SIPS	8.1.0	8.2.0	C1-072593
2007-12	CP-38	CP-070807	1961	1	Route header verification at P-CSCF	8.1.0	8.2.0	C1-072587
2007-12	CP-38	CP-070807	1955	1	Update of the reference for P-Profile-Key Private Header (P-Header)	8.1.0	8.2.0	C1-072487
2007-12	CP-38	CP-070807	2012		Reference alignment	8.1.0	8.2.0	C1-072364
2007-12	CP-38	CP-070807	2037	1	AS does not subscribe to reg-event package when user is unregistered	8.1.0	8.2.0	C1-072597
2007-12	CP-38	CP-070807	2045	2	Correction of mutually exclusive ICSI and GRUU	8.1.0	8.2.0	C1-072706
2007-12	CP-38	CP-070807	2055		Update of P-Answer-State header draft Reference	8.1.0	8.2.0	C1-072446
2007-12	CP-38	CP-070808	2057	2	Clarification of UE handling of the P-Early-Media header.	8.1.0	8.2.0	C1-072723
2007-12	CP-38	CP-070808	2100	1	Access Network Info for I-WLAN	8.1.0	8.2.0	C1-073075
2007-12	CP-38	CP-070808	2003	2	Service Profile Change	8.1.0	8.2.0	C1-072718
2007-12	CP-38	CP-070808	1957	4	Correction to the IBCF subsection in relation with trusted domain	8.1.0	8.2.0	C1-072687
2007-12	CP-38	CP-070808	2072	2	Correction to procedure when registration timer times out	8.1.0	8.2.0	C1-073173
2007-12	CP-38	CP-070808	2103	1	Access Network Info for 3GPP2/UMB	8.1.0	8.2.0	C1-073057
2007-12	CP-38	CP-070810	2081	3	Correction of multiple Contact headers in abnormal case	8.1.0	8.2.0	C1-073226
2007-12	CP-38	CP-070810	2117	1	Miscellaneous editorial corrections (part 3)	8.1.0	8.2.0	C1-073165
2007-12	CP-38	CP-070810	1932	4	Incorporation of draft-ietf-consent-framework	8.1.0	8.2.0	C1-073166

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2007-12	CP-38	CP-070810	2098	1	Superfluous requirements for removing charging information at terminating P-CSCF	8.1.0	8.2.0	C1-073164
2007-12	CP-38	CP-070810	1974	1	Synchronization When Service Profile Being Modified	8.1.0	8.2.0	C1-072661
2007-12	CP-38	CP-070810	2029	3	Miscellaneous editorial corrections	8.1.0	8.2.0	C1-072764
2007-12	CP-38	CP-070810	2059	3	Miscellaneous editorial corrections (part 2)	8.1.0	8.2.0	C1-073162
2007-12	CP-38	CP-070811	2078	1	Clarification on interconnect functionalities	8.1.0	8.2.0	C1-073163
2007-12	CP-38	CP-070812	2086	1	Semantics for values in "integrity-protected" field	8.1.0	8.2.0	C1-073112
2007-12	CP-38	CP-070812	2060	3	Public user identity and private user identity derivation in UEs without UICC	8.1.0	8.2.0	C1-073201
2007-12	CP-38	CP-070812	2006	1	Digest Support in Profile Tables	8.1.0	8.2.0	C1-072623
2007-12	CP-38	CP-070812	2026	1	Security-related references and definitions	8.1.0	8.2.0	C1-072761
2007-12	CP-38	CP-070812	2025	3	Introduction to security mechanisms	8.1.0	8.2.0	C1-073175
2007-12	CP-38	CP-070812	1982	6	Updates to integrity protection for digest and TLS	8.1.0	8.2.0	C1-073202
2007-12	CP-38	CP-070814	2085	4	Addition of SIP header to support UUS1	8.1.0	8.2.0	C1-073208
2007-12	CP-38	CP-070816	2024	5	Integration of text for digest and TLS plus digest into the main body of the specification	8.1.0	8.2.0	C1-073200
2007-12	CP-38	CP-070864	1953	5	Clarifications on NW-init and resource reservation	8.1.0	8.2.0	C1-073069
2007-12	CP-38	CP-070875	1997	4	Corrections for emergency procedures	8.1.0	8.2.0	C1-072991
2008-03	CP-39	CP-080120	2174		Reference correction for RFC 4244	8.2.0	8.3.0	C1-080147
2008-03	CP-39	CP-080120	2149		Handling of the reason header in requests at the MGCF	8.2.0	8.3.0	C1-080045
2008-03	CP-39	CP-080120	2162	1	Correction on handling of P-Charging-Vector at IBCF	8.2.0	8.3.0	C1-080515
2008-03	CP-39	CP-080120	2181	1	Correction of Alias	8.2.0	8.3.0	C1-080517
2008-03	CP-39	CP-080120	2176		SDP with precondition	8.2.0	8.3.0	C1-080149
2008-03	CP-39	CP-080126	2201	2	Handling of Service ID in interworking cases	8.2.0	8.3.0	C1-080630
2008-03	CP-39	CP-080126	2155	2	Clarification on the use of IARI in the contact header	8.2.0	8.3.0	C1-080635
2008-03	CP-39	CP-080126	2183	2	UE behaviour when no ICSI is contained in the Accept-Contact header	8.2.0	8.3.0	C1-080531
2008-03	CP-39	CP-080130	2143	1	Procedure at S-CSCF	8.2.0	8.3.0	C1-080600
2008-03	CP-39	CP-080130	2144		Empty RES	8.2.0	8.3.0	C1-080009
2008-03	CP-39	CP-080130	2145	1	Alias URI	8.2.0	8.3.0	C1-080601
2008-03	CP-39	CP-080130	2146	2	Notification at S-CSCF	8.2.0	8.3.0	C1-080631
2008-03	CP-39	CP-080130	2156	1	Correction of example of IARI coding	8.2.0	8.3.0	C1-080526
2008-03	CP-39	CP-080130	2160	1	Correction on the value used for P-Preferred-Identity header at UE	8.2.0	8.3.0	C1-080513
2008-03	CP-39	CP-080130	2170	1	Correction to user initiated mergency re-registration	8.2.0	8.3.0	C1-080405

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2008-03	CP-39	CP-080130	2187	1	IPv4 and IPv6 support	8.2.0	8.3.0	C1-080609
2008-03	CP-39	CP-080130	2188	4	P-CSCF awareness for 3GPP accesses	8.2.0	8.3.0	C1-080658
2008-03	CP-39	CP-080130	2196	2	Annex K: ICE procedures for the IBCF	8.2.0	8.3.0	C1-080643
2008-03	CP-39	CP-080131	2192	1	Completion of CIC and DAI requirements for MGCF	8.2.0	8.3.0	C1-080472
2008-03	CP-39	CP-080132	2163	1	Miscelanous Corrections on SIP Digest	8.2.0	8.3.0	C1-080473
2008-03	CP-39	CP-080132	2189	1	Enhancements to security introduction text	8.2.0	8.3.0	C1-080474
2008-03	CP-39	CP-080134	2190	1	Inclusion of NASS bundled authentication	8.2.0	8.3.0	C1-080518
2008-03	CP-39	CP-080139	2164	1	SIP XML addition for support of transit specific content	8.2.0	8.3.0	C1-080533
2008-03	CP-39	CP-080140	2138	2	IP-CAN procedure for cdma2000	8.2.0	8.3.0	C1-080411
2008-03	CP-39	CP-080140	2141	2	P-CSCF interface to IP-CAN	8.2.0	8.3.0	C1-080413
2008-03	CP-39	CP-080140	2140	2	Access-network-charging-info for cdma2000 access	8.2.0	8.3.0	C1-080412
2008-03	CP-39	CP-080141	2197	1	Wildcarded Public User Identity: P-CSCF impact	8.2.0	8.3.0	C1-080612
2008-03	CP-39	CP-080141	2198	2	Wildcarded Public User Identity: S-CSCF impact	8.2.0	8.3.0	C1-080644
2008-03	CP-39	CP-080199	2147	4	NAT traversal	8.2.0	8.3.0	
2008-03	CP-39	CP-080201	2151	5	Handling of the reason header in responses	8.2.0	8.3.0	
2008-06	CP-40	CP-080338	2288	1	Correction to de-registration procedure when registration expirated	8.3.0	8.4.0	C1-081936
2008-06	CP-40	CP-080340	2215	-	Revision of references to documents from IETF ECRIT working group	8.3.0	8.4.0	C1-080854
2008-06	CP-40	CP-080341	2243	1	Correction to P-CSCF session release procedures	8.3.0	8.4.0	C1-081336
2008-06	CP-40	CP-080341	2275	2	Addition of AVPF support	8.3.0	8.4.0	C1-082022
2008-06	CP-40	CP-080341	2258	1	Correction on identifiers distinguishing the dialog	8.3.0	8.4.0	C1-081338
2008-06	CP-40	CP-080341	2238	1	Removal of reason header annex	8.3.0	8.4.0	C1-081334
2008-06	CP-40	CP-080341	2217	-	Revision of references to documents from IETF	8.3.0	8.4.0	C1-080858
2008-06	CP-40	CP-080341	2277	1	Addition of the SDP Capability Negotiaion mechanism	8.3.0	8.4.0	C1-081932
2008-06	CP-40	CP-080343	2158	6	Handling of SDP at the terminating UE	8.3.0	8.4.0	C1-082050
2008-06	CP-40	CP-080344	2290	-	Correction of GRUU references	8.3.0	8.4.0	C1-081799
2008-06	CP-40	CP-080349	2236	-	Revision of references to documents from IETF SIP working group	8.3.0	8.4.0	C1-080860
2008-06	CP-40	CP-080353	2203	1	Emergency calls - NAT traversal at UE	8.3.0	8.4.0	C1-081228
2008-06	CP-40	CP-080353	2204	1	NAT traversal for emergency calls at P-CSCF	8.3.0	8.4.0	C1-081229
2008-06	CP-40	CP-080353	2220	1	PANI header text revision	8.3.0	8.4.0	C1-081346
2008-06	CP-40	CP-080353	2225	1	Addition of 802.11n to P-Access-Network-Info header	8.3.0	8.4.0	C1-081348
2008-06	CP-40	CP-080353	2205	3	"im" URI	8.3.0	8.4.0	C1-081411
2008-06	CP-40	CP-080353	2254	2	Annex K: Moving of IBCF ICE procedures	8.3.0	8.4.0	C1-081469

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2008-06	CP-40	CP-080353	2168	9	Correction of 3GPP IM CN subsystem XML handling	8.3.0	8.4.0	C1-081481
2008-06	CP-40	CP-080353	2221	1	Media transcoding control functionality in IBCF	8.3.0	8.4.0	C1-081347
2008-06	CP-40	CP-080353	2219	1	PANI header coding	8.3.0	8.4.0	C1-081345
2008-06	CP-40	CP-080353	2209	1	Alias URI	8.3.0	8.4.0	C1-081343
2008-06	CP-40	CP-080353	2136	7	3GPP IM CN subsystem XML Schema version	8.3.0	8.4.0	C1-081480
2008-06	CP-40	CP-080353	2255	3	Annex K: ICE procedures for the P-CSCF	8.3.0	8.4.0	C1-081470
2008-06	CP-40	CP-080354	2284	2	SDP Enhancements to support resource allocation	8.3.0	8.4.0	C1-082045
2008-06	CP-40	CP-080354	2218	2	B2BUA AS influence of filter criteria evaluation	8.3.0	8.4.0	C1-082033
2008-06	CP-40	CP-080354	2263	1	Multiple contact addresses	8.3.0	8.4.0	C1-082041
2008-06	CP-40	CP-080354	2280	-	Annex A : SIP Record-Route header table correction	8.3.0	8.4.0	C1-081605
2008-06	CP-40	CP-080354	2206	2	"rport" and "received" parameters at P-CSCF	8.3.0	8.4.0	C1-081871
2008-06	CP-40	CP-080354	2282	1	Display Name in Reg Event	8.3.0	8.4.0	C1-082027
2008-06	CP-40	CP-080354	2285	-	Update IETF draft reference	8.3.0	8.4.0	C1-081701
2008-06	CP-40	CP-080354	2207	2	UE handling the "rport" parameter	8.3.0	8.4.0	C1-081872
2008-06	CP-40	CP-080355	2234	4	Annex K alignment with main body and cleanup	8.3.0	8.4.0	C1-082043
2008-06	CP-40	CP-080355	2291	1	Determining when to invoke SIP Digest procedures in S-CSCF	8.3.0	8.4.0	C1-081944
2008-06	CP-40	CP-080355	2269	1	Cleanup of SIP Digest/TLS procedures	8.3.0	8.4.0	C1-081942
2008-06	CP-40	CP-080359	2260	2	P-CSCF: Aligning P-Profile-Key behaviour for Wildcarded public user identities with Wildcarded PSI	8.3.0	8.4.0	C1-081476
2008-06	CP-40	CP-080359	2212	4	Dial string handling	8.3.0	8.4.0	C1-082110
2008-06	CP-40	CP-080359	2261	2	Trustdomain: Adding P-Profile-Key header to the trustdomain	8.3.0	8.4.0	C1-081477
2008-06	CP-40	CP-080359	2239	1	Trust domain changes for identity headers for business communication	8.3.0	8.4.0	C1-081206
2008-06	CP-40	CP-080359	2259	2	I-CSCF: Aligning P-Profile-Key behaviour for Wildcarded public user identities with Wildcarded PSI procedures	8.3.0	8.4.0	C1-081475
2008-06	CP-40	CP-080359	2232	2	Delivering Request-URI to UE managing several terminals	8.3.0	8.4.0	C1-081474
2008-06	CP-40	CP-080359	2262	-	Private network indication annex A changes	8.3.0	8.4.0	C1-081210
2008-06	CP-40	CP-080359	2240	3	Handling of private network indication	8.3.0	8.4.0	C1-081953
2008-06	CP-40	CP-080360	2273	1	Event package usage for Message Waiting Indication (MWI) service	8.3.0	8.4.0	C1-081901
2008-06	CP-40	CP-080360	2226	3	XML-support of transit specific content Tables	8.3.0	8.4.0	C1-081905
2008-06	CP-40	CP-080364	2222	3	Depth of IMS service level trace	8.3.0	8.4.0	C1-081955
2008-06	CP-40	CP-080366	2252	1	Emergency CS call set up procedures for non-3GPP systems	8.3.0	8.4.0	C1-081465
2008-06	CP-40	CP-080366	2268	1	Different IP addresses	8.3.0	8.4.0	C1-081945

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2008-06	CP-40	CP-080366	2251	1	Remove specific codec requirement	8.3.0	8.4.0	C1-081464
2008-06	CP-40	CP-080400	2208	2	"rport" parameter	8.3.0	8.4.0	-
2008-06	CP-40	CP-080402	2296	-	IARI and ICSI in different feature tags	8.3.0	8.4.0	-
2008-06	CP-40	CP-080417	2211	5	Call forwarding in IMS	8.3.0	8.4.0	-
2008-06					Editorial change done by MCC	8.4.0	8.4.1	
2008-09	CP-41	CP-080643	2177	7	Allow Multiple Registrations in Rel 8 by using Outbound	8.4.1	8.5.0	
2008-09	CP-41	CP-080539	2178	6	Add Timestamp in Register Request	8.4.1	8.5.0	C1-082810
2008-09	CP-41	CP-080527	2297	1	Cleanup of P-CSCF procedures for inclusion of "tls-yes" and "tls-pending"	8.4.1	8.5.0	C1-082623
2008-09	CP-41	CP-080538	2298	1	Introduction of GIBA (Early IMS) procedures	8.4.1	8.5.0	C1-082657
2008-09	CP-41	CP-080527	2299	1	Add reference to draft-dotson-sip-mutual-auth	8.4.1	8.5.0	C1-082621
2008-09	CP-41	CP-080523	2301	1	Correction of DHCP reference	8.4.1	8.5.0	C1-082620
2008-09	CP-41	CP-080523	2302		Reference correction	8.4.1	8.5.0	C1-082142
2008-09	CP-41	CP-080515	2306	1	Annex A: Correction of SDP connection information	8.4.1	8.5.0	C1-082611
2008-09	CP-41	CP-080523	2308	1	Backward compability issue with P-Access-Network-Info ABNF extension	8.4.1	8.5.0	C1-082625
2008-09	CP-41	CP-080517	2314		Addition of AVPF support and SDP capability negotiation mechanism	8.4.1	8.5.0	C1-082268
2008-09	CP-41	CP-080520	2316		Profile corrections for outbound	8.4.1	8.5.0	C1-082270
2008-09	CP-41	CP-080531	2319		Support of Direct Ethernet access as IP-CAN	8.4.1	8.5.0	C1-082324
2008-09	CP-41	CP-080520	2323	1	Update Outbound Reference	8.4.1	8.5.0	C1-082626
2008-09	CP-41	CP-080523	2325	2	Error Response for Different S-CSCF Assignment	8.4.1	8.5.0	C1-082770
2008-09	CP-41	CP-080527	2328	1	Annex K Technical Corrections	8.4.1	8.5.0	C1-082622
2008-09	CP-41	CP-080528	2329	1	Adding P-Debug-ID to SIP Profile Tables	8.4.1	8.5.0	C1-082752
2008-09	CP-41	CP-080528	2330	2	Subscribing to the debug event package	8.4.1	8.5.0	C1-082781
2008-09	CP-41	CP-080522	2331	4	EPS as IP-CAN	8.4.1	8.5.0	C1-083637
2008-09	CP-41	CP-080523	2333	2	Alignment of IP-CAN specific annexes	8.4.1	8.5.0	C1-082778
2008-09	CP-41	CP-080516	2336		Emergency PUID	8.4.1	8.5.0	C1-082864
2008-09	CP-41	CP-080667	2340	3	Initial emergency registration	8.4.1	8.5.0	
2008-09	CP-41	CP-080516	2342	2	Emergency session set-up	8.4.1	8.5.0	C1-083532
2008-09	CP-41	CP-080516	2344	1	P-CSCF handling of emergency sessions	8.4.1	8.5.0	C1-083391
2008-09	CP-41	CP-080516	2346	3	S-CSCF handling of emergency registration	8.4.1	8.5.0	C1-083534
2008-09	CP-41	CP-080523	2347	1	Informative Explanation and Corrections of Profile Tables	8.4.1	8.5.0	C1-083353
2008-09	CP-41	CP-080523	2350	1	More than one contact address per UE	8.4.1	8.5.0	C1-083351
2008-09	CP-41	CP-080528	2351	1	IMS Trace for entities not on the path of the register request	8.4.1	8.5.0	C1-083383

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2008-09	CP-41	CP-080528	2352	1	Start and stop procedures for IMS trace	8.4.1	8.5.0	C1-083384
2008-09	CP-41	CP-080636	2353	1	Align emergency session handling outside a security association or TLS session	8.4.1	8.5.0	
2008-09	CP-41	CP-080637	2354	3	Addressing privacy requirement	8.4.1	8.5.0	
2008-09	CP-41	CP-080523	2359	2	SDP Offer	8.4.1	8.5.0	C1-083398
2008-09	CP-41	CP-080515	2362		SDP referencing error for IBCF (IMS-ALG)	8.4.1	8.5.0	C1-082927
2008-09	CP-41	CP-080523	2363	2	Addition of draft-ietf-sip-199-00	8.4.1	8.5.0	C1-083399
2008-09	CP-41	CP-080523	2365	1	Usage of draft-holmberg-sip-keep-01 for emergency session	8.4.1	8.5.0	C1-083395
2008-09	CP-41	CP-080537	2366	1	Mediactrl and netann specifications	8.4.1	8.5.0	C1-083363
2008-09	CP-41	CP-080536	2369	1	S-CSCF and AS procedures with Enhanced Filter Criteria	8.4.1	8.5.0	C1-083501
2008-09	CP-41	CP-080617	2371	2	Correct handling for <reason> element	8.4.1	8.5.0	
2008-09	CP-41	CP-080539	2375		Modification of ci-3gpp2 parameter	8.4.1	8.5.0	C1-083200
2008-09	CP-41	CP-080668	2377	3	Alignment of usage of terms ISIM and ISIM Application	8.4.1	8.5.0	
2008-09	CP-41	CP-080524	2378	1	Introduction additional methods of P-CSCF discovery to support IMS Local Breakout	8.4.1	8.5.0	C1-083400
2008-09	CP-41	CP-080515	2381		Alignment with current version of draft-ietf-sip-fork-loop-fix	8.4.1	8.5.0	C1-083246
2008-09	CP-41	CP-080522	2386	1	Relationship to IP-CAN	8.4.1	8.5.0	C1-083424
2008-09					Editorial change done by MCC	8.5.0	8.5.1	
2008-12	CP-42	CP-080942	2324	9	Introduction of IMC in support of common IMS	8.5.1	8.6.0	-
2008-12	CP-42	CP-080847	2327	5	SDP Enhancements to support resource allocation	8.5.1	8.6.0	C1-084937
2008-12	CP-42	CP-080840	2332	3	Additional changes for private network indication	8.5.1	8.6.0	C1-084441
2008-12	CP-42	CP-080847	2358	7	Prevent DDOS attack on PSAP	8.5.1	8.6.0	C1-085454
2008-12	CP-42	CP-080840	2383	1	Modifications to private network indication in profile	8.5.1	8.6.0	C1-084080
2008-12	CP-42	CP-080847	2388	3	Annex A fixes regarding draft-ietf-sip-199	8.5.1	8.6.0	C1-085202
2008-12	CP-42	CP-080847	2389	1	Annex A fixes regarding draft-holmberg-sip-keep	8.5.1	8.6.0	C1-084278
2008-12	CP-42	CP-080847	2394	-	Correction on setting P-Served-User	8.5.1	8.6.0	C1-083694
2008-12	CP-42	CP-080847	2396	1	Clarification on ICSI and IARI	8.5.1	8.6.0	C1-084203
2008-12	CP-42	CP-080847	2402	2	Interface identifier	8.5.1	8.6.0	C1-085204
2008-12	CP-42	CP-080844	2403	2	UE subscription to reg-evet	8.5.1	8.6.0	C1-084420
2008-12	CP-42	CP-080844	2405	3	UE - multiple contacts registration	8.5.1	8.6.0	C1-085205
2008-12	CP-42	CP-080844	2406	1	UE - multiple contacts authentication and deregistration	8.5.1	8.6.0	C1-084282
2008-12	CP-42	CP-080844	2407	1	UE using multiple contacts	8.5.1	8.6.0	C1-084283
2008-12	CP-42	CP-080845	2408	4	Introduction of additional methods of P-CSCF discovery for EPS to support IMS Local Breakout	8.5.1	8.6.0	C1-085206

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2008-12	CP-42	CP-080956	2409	5	UE procedures when multiple P-CSCF discovery procedures are supported	8.5.1	8.6.0	-
2008-12	CP-42	CP-080854	2411	1	Cr addition to section 4	8.5.1	8.6.0	C1-084230
2008-12	CP-42	CP-080854	2412	2	Netann, mediactrl text improvements	8.5.1	8.6.0	C1-084434
2008-12	CP-42	CP-080854	2413	2	Media control for charging, delegation	8.5.1	8.6.0	C1-085256
2008-12	CP-42	CP-080847	2421	-	Trademark CDMA terminology	8.5.1	8.6.0	C1-083983
2008-12	CP-42	CP-080843	2423	2	Aligning initial INVITE request usage of Accept header field and profile tables	8.5.1	8.6.0	C1-084438
2008-12	CP-42	CP-080858	2424	1	Clarification of security-verify for TLS	8.5.1	8.6.0	C1-084234
2008-12	CP-42	CP-080840	2425	2	Setting of the Phone-contex paramater when IP-CAN is Ethernet	8.5.1	8.6.0	C1-085201
2008-12	CP-42	CP-080847	2427	-	P-CSCF call release upon reception of indication that no ressource is available.	8.5.1	8.6.0	C1-084024
2008-12	CP-42	CP-080847	2428	2	Removing of the cpc parameter by the terminating S-CSCF removes CPC	8.5.1	8.6.0	C1-084435
2008-12	CP-42	CP-080844	2430	2	Clarification of abnormal case for deregistration	8.5.1	8.6.0	C1-085158
2008-12	CP-42	CP-080847	2431	-	P-CSCF handling of "integrity-protected"	8.5.1	8.6.0	C1-084048
2008-12	CP-42	CP-080839	2432	2	Registration Procedure for ICS	8.5.1	8.6.0	C1-085200
2008-12	CP-42	CP-080870	2434	1	SMSIP related changes for the profile tables	8.5.1	8.6.0	C1-084202
2008-12	CP-42	CP-080853	2435	1	Adding roles defined for service level interworking for messaging to the profile table	8.5.1	8.6.0	C1-084270
2008-12	CP-42	CP-080840	2436	-	Downloading of information to the P-CSCF	8.5.1	8.6.0	C1-084082
2008-12	CP-42	CP-080835	2440	2	Adding reference to Internet Draft on sos URI parameter for emergency calls	8.5.1	8.6.0	C1-085260
2008-12	CP-42	CP-080857	2441	-	Update reference for DAI Parameter for the "tel" URI	8.5.1	8.6.0	C1-084120
2008-12	CP-42	CP-080847	2442	3	Inclusion of draft-ietf-sip-body-handling in the profile tables	8.5.1	8.6.0	C1-085209
2008-12	CP-42	CP-080856	2443	3	Deterministic Routeing for overlap signalling	8.5.1	8.6.0	C1-085239
2008-12	CP-42	CP-080840	2444	1	Allowing P-Asserted Identity from an UE	8.5.1	8.6.0	C1-085254
2008-12	CP-42	CP-080835	2446	-	Emergency call	8.5.1	8.6.0	C1-084649
2008-12	CP-42	CP-080843	2448	1	Deregistration in 200 (OK)	8.5.1	8.6.0	C1-085435
2008-12	CP-42	CP-080939	2449	2	Revision of 24.229-2449r1 (C1-085416)	8.5.1	8.6.0	-
2008-12	CP-42	CP-080844	2450	2	Usage of outbound in call setup	8.5.1	8.6.0	C1-085450
2008-12	CP-42	CP-080844	2451	-	Multiple registrations at P-CSCF	8.5.1	8.6.0	C1-084655
2008-12	CP-42	CP-080940	2452	2	Revision of 24.229-2452r1 (C1-085418)	8.5.1	8.6.0	-
2008-12	CP-42	CP-080844	2454	1	Multiple registrations at S-CSCF	8.5.1	8.6.0	C1-085419
2008-12	CP-42	CP-080869	2456	-	Correction of ICSI and IARI feature tag name	8.5.1	8.6.0	C1-084689
2008-12	CP-42	CP-080862	2457	2	Inclusion and Modification of Resource-Priority header at P-CSCF	8.5.1	8.6.0	C1-085451
2008-12	CP-42	CP-080854	2458	1	Media control related profile table updates	8.5.1	8.6.0	C1-085255

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2008-12	CP-42	CP-080854	2459	1	Mediactrl reference updates	8.5.1	8.6.0	C1-085257
2008-12	CP-42	CP-080839	2460	2	Instance ID definition	8.5.1	8.6.0	C1-085459
2008-12	CP-42	CP-080844	2462	2	GRUU and Multilple registration	8.5.1	8.6.0	C1-085468
2008-12	CP-42	CP-080959	2464	4	Overlap signalling procedures	8.5.1	8.6.0	-
2008-12	CP-42	CP-080841	2469	-	Reference updates (release 6 ietf dependencies)	8.5.1	8.6.0	C1-084898
2008-12	CP-42	CP-080843	2471	-	Reference updates (release 7 ietf dependencies)	8.5.1	8.6.0	C1-084903
2008-12	CP-42	CP-080858	2472	1	No domain field for SIP digest	8.5.1	8.6.0	C1-085261
2008-12	CP-42	CP-080858	2473	1	Digest Authentication of Non-Register requests	8.5.1	8.6.0	C1-085262
2008-12	CP-42	CP-080855	2477	1	Minor corrections to configuration of entities for trace	8.5.1	8.6.0	C1-085128
2008-12	CP-42	CP-080843	2479	-	Inclusion of missing RFC 3351 reference	8.5.1	8.6.0	C1-085011
2008-12	CP-42	CP-080847	2480	2	Documentation of INFO within the IM CN subsystem	8.5.1	8.6.0	C1-085424
2008-12	CP-42	CP-080847	2481	-	Removal of TrGw normative requirements from IBCF	8.5.1	8.6.0	C1-085015
2008-12	CP-42	CP-080847	2482	-	Editorial consistency and best practice	8.5.1	8.6.0	C1-085016
2008-12	CP-42	CP-080965	2483	3	Updates to profile tables to include ICS additions	8.5.1	8.6.0	-
2008-12	CP-42	CP-080849	2484	-	Cleanup of various GIBA Editor's notes	8.5.1	8.6.0	C1-085025
2008-12	CP-42	CP-080853	2485	1	Addition of cpim/message and message/imdn+xml	8.5.1	8.6.0	C1-085291
2008-12	CP-42	CP-080847	2494	3	Documenting RFC 5373	8.5.1	8.6.0	C1-085483
2008-12	CP-42	CP-080873	2495	1	S-CSCF and AS procedures with Enhanced Filter Criteria	8.5.1	8.6.0	C1-085292
2008-12	CP-42	CP-080847	2498	2	Call release by the P-CSCF upon resource reservation faillure	8.5.1	8.6.0	C1-085467
2008-12	CP-42	CP-080847	2499	1	Hosted NAT traversal for media flows	8.5.1	8.6.0	C1-085430
2008-12	CP-42	CP-080846	2501	1	Reference updates (release 8 ietf dependencies)	8.5.1	8.6.0	C1-085426
2008-12	CP-42	CP-080847	2502	-	Corrections to security overview	8.5.1	8.6.0	C1-085093
2008-12	CP-42	CP-080847	2505	-	Identification of public user identity in absence of Authorization header	8.5.1	8.6.0	C1-085131
2008-12	CP-42				Editorial cleanup by ETSI EditHelp! and MCC	8.5.1	8.6.0	
2009-03	CP-43	CP-090134	2438	7	Correction of non UE detectable emergency call procedures	8.6.0	8.7.0	C1-091088
2009-03	CP-43	CP-090121	2507		Correction of URN-value for Service Identifiers	8.6.0	8.7.0	C1-090012
2009-03	CP-43	CP-090134	2508	1	Re-selection of S-CSCF during Terminating and Originating Procedures	8.6.0	8.7.0	C1-090991
2009-03	CP-43	CP-090146	2509	2	Re-selection of S-CSCF during Terminating and Originating Procedures when restoration is supported.	8.6.0	8.7.0	C1-091066
2009-03	CP-43	CP-090245	2510	4	Returning an error to trigger a new registration when IMS restoration is supported	8.6.0	8.7.0	-

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2009-03	CP-43	CP-090225	2511	4	Re-selection of S-CSCF during Re-registration when IMS restoration is supported	8.6.0	8.7.0	-
2009-03	CP-43	CP-090134	2514	1	Outbound with IMS AKA	8.6.0	8.7.0	C1-090992
2009-03	CP-43	CP-090134	2515	2	Registration procedure at the S-CSCF	8.6.0	8.7.0	C1-091041
2009-03	CP-43	CP-090134	2516	3	P-CSCFprocessing 200 (OK)	8.6.0	8.7.0	C1-091085
2009-03	CP-43	CP-090134	2517	4	Multiple de-registrations	8.6.0	8.7.0	C1-091111
2009-03	CP-43	CP-090134	2519	1	Instance-ID in INVITE	8.6.0	8.7.0	C1-090997
2009-03	CP-43	CP-090134	2520		Multiple contact addresses	8.6.0	8.7.0	C1-090042
2009-03	CP-43	CP-090130	2524	3	Support for eHRPD	8.6.0	8.7.0	C1-091381
2009-03	CP-43	CP-090155	2525	1	Adding the role of The Early Session Disposition Type	8.6.0	8.7.0	C1-090950
2009-03	CP-43	CP-090134	2527		Cleanup inclusion of draft-ietf-sip-body-handling in the profile tables	8.6.0	8.7.0	C1-090201
2009-03	CP-43	CP-090116	2529	2	Aligning with draft-ietf-sip-location-conveyance-12	8.6.0	8.7.0	C1-091040
2009-03	CP-43	CP-090134	2530	1	Addressing privacy requirement for emergency calls	8.6.0	8.7.0	C1-090999
2009-03	CP-43	CP-090116	2532	1	Correcting condition for using indicating use of emergency registration	8.6.0	8.7.0	C1-090959
2009-03	CP-43	CP-090224	2534	3	Overlap signalling en-bloc conversion procedures	8.6.0	8.7.0	-
2009-03	CP-43	CP-090209	2535	3	Overlap signalling digit collection procedures	8.6.0	8.7.0	-
2009-03	CP-43	CP-090134	2537	1	Correction of registration duration value	8.6.0	8.7.0	C1-091024
2009-03	CP-43	CP-090127	2540	1	Corrections to E-UTRAN specific aspects	8.6.0	8.7.0	C1-090850
2009-03	CP-43	CP-090134	2541		Miscellaneous corrections to annex B	8.6.0	8.7.0	C1-090377
2009-03	CP-43	CP-090142	2543	1	Miscellaneous corrections to Annex M	8.6.0	8.7.0	C1-090985
2009-03	CP-43	CP-090142	2544	1	Phone-context parameter value for cdma2000®	8.6.0	8.7.0	C1-090986
2009-03	CP-43	CP-090142	2545	1	Common IMS for MGW and MRF	8.6.0	8.7.0	C1-090987
2009-03	CP-43	CP-090134	2546	4	Deterministic behaviour for Call Forwarding	8.6.0	8.7.0	C1-091122
2009-03	CP-43	CP-090136	2547	1	Overlap Corrections	8.6.0	8.7.0	C1-090962
2009-03	CP-43	CP-090116	2550	1	Alignment of emergency indication with draft-patel-ecrit-sos-parameter-03	8.6.0	8.7.0	C1-090968
2009-03	CP-43	CP-090272	2553	3	Use of multiple access technologies in IMS	8.6.0	8.7.0	-
2009-03	CP-43	CP-090134	2555		Alignment of authentication parameter terminology	8.6.0	8.7.0	C1-090534
2009-03	CP-43	CP-090134	2556		Use of access-class and access-type constructs in the P-Access-Network-Info header field	8.6.0	8.7.0	C1-090535
2009-03	CP-43	CP-090134	2558		P-Served-User header field corrections (profile)	8.6.0	8.7.0	C1-090537
2009-03	CP-43	CP-090134	2560		Editorial consistency and best practice	8.6.0	8.7.0	C1-090539
2009-03	CP-43	CP-090141	2561	1	Removal of redundant NASS bundled authentication text for S-CSCF	8.6.0	8.7.0	C1-090969
2009-03	CP-43	CP-090150	2564	1	Emergency call handling for CS media	8.6.0	8.7.0	C1-090908

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2009-03	CP-43	CP-090118	2574	2	Correction to Annex A / SIP extensions for media authorization	8.6.0	8.7.0	C1-091120
2009-03	CP-43	CP-090275	2578	4	Correction to Annex A /P-Access-Network-Info	8.6.0	8.7.0	-
2009-03	CP-43	CP-090134	2579	2	Correction to Annex A /P-User-Database header	8.6.0	8.7.0	C1-091084
2009-03	CP-43	CP-090134	2582	2	Routeing B2BUA transparency	8.6.0	8.7.0	C1-091078
2009-03	CP-43	CP-090134	2583	1	Call release by P-CSCF- Editorial correction	8.6.0	8.7.0	C1-091013
2009-03	CP-43	CP-090118	2584	1	References correction	8.6.0	8.7.0	C1-091014
2009-03	CP-43	CP-090142	2595	1	Corrections for cdma2000® HRPD Emergency Services	8.6.0	8.7.0	C1-090988
2009-03	CP-43	CP-090127	2596		Corrections to EPS as IMS access technology Annex	8.6.0	8.7.0	C1-090685
2009-03	CP-43	CP-090135	2597	1	Update of references to SIP debug internet drafts	8.6.0	8.7.0	C1-090970
2009-03	CP-43	CP-090159	2598	1	Handling of provisioned mode of the resource allocation used for IMS media	8.6.0	8.7.0	C1-091069
2009-03	CP-43	CP-090237	2601	2	Reference correction	8.6.0	8.7.0	C1-091115
2009-06	CP-44	CP-090428	2518	5	Flow- token in the Record-Route	8.7.0	8.8.0	C1-091475
2009-06	CP-44	CP-090398	2539	8	Mechanism for UE to identify a SIP URI that has an associated tel URI	8.7.0	8.8.0	C1-092241
2009-06	CP-44	CP-090428	2557	3	Application server usage of P-Served-User header field	8.7.0	8.8.0	C1-092077
2009-06	CP-44	CP-090399	2605	2	P-CSCF releasing a dialog	8.7.0	8.8.0	C1-092084
2009-06	CP-44	CP-090399	2607	2	S-CSCF releasing a dialog	8.7.0	8.8.0	C1-092086
2009-06	CP-44	CP-090428	2608	2	GRUU translation	8.7.0	8.8.0	C1-092087
2009-06	CP-44	CP-090428	2610	1	Correct backwards emergency notification procedure	8.7.0	8.8.0	C1-092072
2009-06	CP-44	CP-090428	2611		Correction of implementation error of CR2537r1	8.7.0	8.8.0	C1-091494
2009-06	CP-44	CP-090428	2612	1	BGCF routing	8.7.0	8.8.0	C1-092074
2009-06	CP-44	CP-090403	2614		Correction of 3GPP URN link	8.7.0	8.8.0	C1-091504
2009-06	CP-44	CP-090428	2616	2	RFC 2833 substituted by RFC 4733	8.7.0	8.8.0	C1-092050
2009-06	CP-44	CP-090428	2617		Call Forwarding Leftover	8.7.0	8.8.0	C1-091510
2009-06	CP-44	CP-090415	2618	1	Correction Identity handling for NGCN	8.7.0	8.8.0	C1-091974
2009-06	CP-44	CP-090419	2619		Reference Update draft-ietf-mmusic-sdp-cs	8.7.0	8.8.0	C1-091513
2009-06	CP-44	CP-090428	2620	1	RFC reference fix	8.7.0	8.8.0	C1-092075
2009-06	CP-44	CP-090428	2625	1	Deterministic XML schema	8.7.0	8.8.0	C1-092204
2009-06	CP-44	CP-090398	2634		Emergency call treatment of P-Preferred-Identity header field in profile	8.7.0	8.8.0	C1-091649
2009-06	CP-44	CP-090405	2635	1	Subdivision of digit collection text	8.7.0	8.8.0	C1-091967
2009-06	CP-44	CP-090428	2636		Editorial changes	8.7.0	8.8.0	C1-091655
2009-06	CP-44	CP-090398	2639	1	Correcting emergency registration support and access type	8.7.0	8.8.0	C1-092003

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2009-06	CP-44	CP-090397	2645	1	Correction to Annex A /Caller preferences directives	8.7.0	8.8.0	C1-092079
2009-06	CP-44	CP-090428	2657	2	Alignment of Cx reference point procedures withTS 29.228 procedures	8.7.0	8.8.0	C1-092211
2009-06	CP-44	CP-090415	2658	2	Correction to GRUU procedures to ensure that sessions using UE assigned Public GRUUs don't fail	8.7.0	8.8.0	C1-092219
2009-06	CP-44	CP-090428	2659		Removing obsolete Editor's Note	8.7.0	8.8.0	C1-091854
2009-06	CP-44	CP-090428	2660	1	Correction of instance ID related Editor's Note and text	8.7.0	8.8.0	C1-092076
2009-06	CP-44	CP-090398	2662		Version update for "sos" URI parameter Internet Draft	8.7.0	8.8.0	C1-091857
2009-06	CP-44	CP-090428	2663		Contact Header in PUBLISH method	8.7.0	8.8.0	C1-091879
2009-06	CP-44	CP-090428	2666		Removing non-essential and incorrect statement regarding ordering of codec formats in the SDP offer	8.7.0	8.8.0	C1-092114
2009-06	CP-44	CP-090400	2667	1	Correction to Annex A /P-User-Database	8.7.0	8.8.0	C1-092209
2009-06	CP-44	CP-090430	2644	2	Addition of capability for delivering the original Request-URI	8.8.0	9.0.0	C1-092227
2009-09	CP-45	CP-090696	2671	2	Service-Route/Path header handling for fetching bindings	9.0.0	9.1.0	C1-093049
2009-09	CP-45	CP-090644	2674	2	Inconsistency between text and XML schema	9.0.0	9.1.0	C1-093709
2009-09	CP-45	CP-090650	2675		Confusing text in L.2.2.5.1A	9.0.0	9.1.0	C1-092401
2009-09	CP-45	CP-090658	2676	3	Emergency call handling in P-CSCF and UE	9.0.0	9.1.0	C1-093070
2009-09	CP-45	CP-090649	2679	1	TISPAN IBCF review comment fixes	9.0.0	9.1.0	C1-092903
2009-09	CP-45	CP-090696	2680		TISPAN review comments - minor fixes	9.0.0	9.1.0	C1-092407
2009-09	CP-45	CP-090657	2682	1	Contact port in non REGISTER request with AKA	9.0.0	9.1.0	C1-092409
2009-09	CP-45	CP-090696	2684	1	reg/debug event package subscription headers	9.0.0	9.1.0	C1-092987
2009-09	CP-45	CP-090664	2686	2	Connection of complex UEs to IMS	9.0.0	9.1.0	C1-093739
2009-09	CP-45	CP-090737	2689	2	Calling party category (cpc)	9.0.0	9.1.0	-
2009-09	CP-45	CP-090696	2691	1	UE procedure on registration failure	9.0.0	9.1.0	C1-093015
2009-09	CP-45	CP-090658	2693	1	Correction of BGCF procedures	9.0.0	9.1.0	C1-092989
2009-09	CP-45	CP-090696	2694	2	Topology hiding on Path header	9.0.0	9.1.0	C1-093016
2009-09	CP-45	CP-090682	2695	1	Create XML source files	9.0.0	9.1.0	C1-093029
2009-09	CP-45	CP-090667	2697	1	Correcting preventing of DDOS attack on registrar	9.0.0	9.1.0	C1-092952
2009-09	CP-45	CP-090657	2700		Correcting mismatch in conditions for non-UE detectable emergency call	9.0.0	9.1.0	C1-092494
2009-09	CP-45	CP-090659	2702	1	The "comp" parameter	9.0.0	9.1.0	C1-093702
2009-09	CP-45	CP-090659	2704		Routing procedure	9.0.0	9.1.0	C1-092501
2009-09	CP-45	CP-090664	2706		UE as an externally attached network	9.0.0	9.1.0	C1-092503
2009-09	CP-45	CP-090725	2710	2	Require with the option-tag "outbound"	9.0.0	9.1.0	-

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2009-09	CP-45	CP-090658	2712	1	Outbound support	9.0.0	9.1.0	C1-092994
2009-09	CP-45	CP-090657	2718	2	Contact header in registration	9.0.0	9.1.0	C1-093704
2009-09	CP-45	CP-090659	2720	1	S-CSCF not supporting Outbound registration	9.0.0	9.1.0	C1-093002
2009-09	CP-45	CP-090648	2722	2	NAT traversal without outbound	9.0.0	9.1.0	C1-093041
2009-09	CP-45	CP-090651	2724		Duplicate subclauses in Annex O	9.0.0	9.1.0	C1-092530
2009-09	CP-45	CP-090664	2727	2	P-CSCF handling alignments for priviledged senders	9.0.0	9.1.0	C1-093486
2009-09	CP-45	CP-090664	2729	1	P-CSCF handling for NCGN as regular UE	9.0.0	9.1.0	C1-092932
2009-09	CP-45	CP-090664	2731	5	S-CSCF handling alignments for NGCN	9.0.0	9.1.0	C1-093910
2009-09	CP-45	CP-090664	2741	2	Use of GRUU by UEs that perform the functions of an external attached network	9.0.0	9.1.0	C1-093905
2009-09	CP-45	CP-090658	2743		Correction of alignment of Cx reference point procedures withTS 29.228 procedures	9.0.0	9.1.0	C1-092658
2009-09	CP-45	CP-090659	2745		Reference update for draft-montemurro-gsma-imei-urn	9.0.0	9.1.0	C1-092660
2009-09	CP-45	CP-090696	2746	1	Annex K: P-CSCF alignment	9.0.0	9.1.0	C1-093017
2009-09	CP-45	CP-090696	2747	1	Annex K: S-CSCF alignment	9.0.0	9.1.0	C1-093018
2009-09	CP-45	CP-090696	2748		Annex K: Removal of IBCF modifications	9.0.0	9.1.0	C1-092664
2009-09	CP-45	CP-090658	2752	2	Keep-alives for emergency calls	9.0.0	9.1.0	C1-093043
2009-09	CP-45	CP-090649	2755	1	P-CSCF forwarding request towards entry point	9.0.0	9.1.0	C1-092910
2009-09	CP-45	CP-090659	2759	1	Re-INVITE for precondition status indication	9.0.0	9.1.0	C1-093011
2009-09	CP-45	CP-090658	2761	1	Digest URI verification fix	9.0.0	9.1.0	C1-093034
2009-09	CP-45	CP-090696	2762		SDP in session modification messages	9.0.0	9.1.0	C1-092678
2009-09	CP-45	CP-090658	2764		Correction of table condition: AoC roles	9.0.0	9.1.0	C1-092680
2009-09	CP-45	CP-090732	2766	5	Aligning IANA registration of MIME type "application/3gpp-ims+xml"	9.0.0	9.1.0	-
2009-09	CP-45	CP-090690	2767	4	Emergency call introduction	9.0.0	9.1.0	C1-093946
2009-09	CP-45	CP-090690	2768	1	Emergency call changes to Annex B (GPRS)	9.0.0	9.1.0	C1-092825
2009-09	CP-45	CP-090690	2769	1	Emergency call changes to Annex L (EPS)	9.0.0	9.1.0	C1-092826
2009-09	CP-45	CP-090667	2778	1	How the P-CSCF forwards the request to the next hop excluding the REGISTER method.	9.0.0	9.1.0	C1-093006
2009-09	CP-45	CP-090696	2779	1	Clarification of a target refresh request.	9.0.0	9.1.0	C1-093007
2009-09	CP-45	CP-090660	2780	1	No Proxy-Authentication-Info header	9.0.0	9.1.0	C1-093721
2009-09	CP-45	CP-090664	2781	2	No P-P-I from NGCN	9.0.0	9.1.0	C1-093790
2009-09	CP-45	CP-090696	2784	1	Trust domain clarification	9.0.0	9.1.0	C1-093753
2009-09	CP-45	CP-090696	2785	1	Clarification of Handling of geo-local numbers	9.0.0	9.1.0	C1-093754
2009-09	CP-45	CP-090645	2789		IOI Handling	9.0.0	9.1.0	C1-093266
2009-09	CP-45	CP-090671	2791	1	Invalid Registration	9.0.0	9.1.0	C1-093745
2009-09	CP-45	CP-090665	2793	1	IBCF and P-Asserted-Identity	9.0.0	9.1.0	C1-093783

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2009-09	CP-45	CP-090657	2797	1	Correct the preconditions for NBA mechanism	9.0.0	9.1.0	C1-093760
2009-09	CP-45	CP-090682	2800	4	Correction of dialog correlation	9.0.0	9.1.0	C1-093985
2009-09	CP-45	CP-090696	2801		Corrections to SDP profile table entries	9.0.0	9.1.0	C1-093449
2009-09	CP-45	CP-090657	2803	1	Adding RFC 3890 and maximum packet rate to SDP profile tables	9.0.0	9.1.0	C1-093762
2009-09	CP-45	CP-090679	2806	2	Correcting duplicate mentioning of 802.3y	9.0.0	9.1.0	C1-093913
2009-09	CP-45	CP-090647	2813		Update of reference to I-D for sos URI parameter and miscellaneous reference corrections	9.0.0	9.1.0	C1-093574
2009-09	CP-45	CP-090659	2815	2	Use of ports for SIP between UE and P-CSCF	9.0.0	9.1.0	C1-093908
2009-09	CP-45	CP-090659	2817	1	Profile table correction on the support of security mechanism	9.0.0	9.1.0	C1-093578
2009-09	CP-45	CP-090696	2819	1	Correction on the summary of security mechanism	9.0.0	9.1.0	C1-093767
2009-09	CP-45	CP-090657	2827	1	Clarification on identity usage for NBA	9.0.0	9.1.0	C1-093769
2009-09	CP-45	CP-090664	2829		Describe the right behaviour of the IBCF	9.0.0	9.1.0	C1-093609
2009-12	CP-46	CP-090923	2834	3	Correction to introduce support for IMSVoPS	9.1.0	9.2.0	C1-095602
2009-12	CP-46	CP-090923	2835	2	Transcoding Control at MRF using RFC 4117	9.1.0	9.2.0	C1-094737
2009-12	CP-46	CP-090890	2839		Inclusion of draft-ietf-sipcore-invfix	9.1.0	9.2.0	C1-094120
2009-12	CP-46	CP-090890	2843	1	Inclusion of draft-ietf-sip-ipv6-abnf-fix	9.1.0	9.2.0	C1-094531
2009-12	CP-46	CP-090891	2847		Change of ua-profile package to xcap-diff package	9.1.0	9.2.0	C1-094131
2009-12	CP-46	CP-090892	2850		Release 7 IETF reference updates for emergency call	9.1.0	9.2.0	C1-094134
2009-12	CP-46	CP-090940	2854		Inclusion of draft-ietf-sip-record-route-fix	9.1.0	9.2.0	C1-094152
2009-12	CP-46	CP-090940	2855	1	Correction of support of trust domain boundaries for identity	9.1.0	9.2.0	C1-094566
2009-12	CP-46	CP-090923	2856	1	Inclusion of roles for XCAP client / server at the Ut reference point for supplementary services	9.1.0	9.2.0	C1-094538
2009-12	CP-46	CP-090920	2858		Update of draft-ietf-sip-body-handling reference to RFC 5621	9.1.0	9.2.0	C1-094215
2009-12	CP-46	CP-090940	2860		xsd file alignment with main document	9.1.0	9.2.0	C1-094316
2009-12	CP-46	CP-090940	2861	1	Textual layout errors in Annex A	9.1.0	9.2.0	C1-094568
2009-12	CP-46	CP-090936	2863	2	Media plane security	9.1.0	9.2.0	C1-094729
2009-12	CP-46	CP-090940	2866	1	3rd party registration failure	9.1.0	9.2.0	C1-094336
2009-12	CP-46	CP-090923	2689	4	Detecting requests destined for a PSAP	9.1.0	9.2.0	C1-095704
2009-12	CP-46	CP-091016	2875	5	Alignment of 24.229 with draft-ietf-sipcore-info-events	9.1.0	9.2.0	-
2009-12	CP-46	CP-090940	2877	1	Correction of indication to the user that an emergency call was made	9.1.0	9.2.0	C1-094582
2009-12	CP-46	CP-090940	2881	2	Annex A /183 (Session Progress) response	9.1.0	9.2.0	C1-094733
2009-12	CP-46	CP-090890	2885		Annex A / c and m paramters in media description in SDP	9.1.0	9.2.0	C1-094382

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2009-12	CP-46	CP-090890	2889		Annex A / User-Agent in PUBLISH responses	9.1.0	9.2.0	C1-094387
2009-12	CP-46	CP-091049	2891	3	Annex A / Allow events	9.1.0	9.2.0	-
2009-12	CP-46	CP-090940	2892	1	Annex A /MIME-Version header	9.1.0	9.2.0	C1-094571
2009-12	CP-46	CP-090940	2893	2	Annex A / Require header	9.1.0	9.2.0	C1-094734
2009-12	CP-46	CP-090940	2894	1	Application of trust domains to the P-Early-media header field	9.1.0	9.2.0	C1-094573
2009-12	CP-46	CP-090923	2895	2	Allowing direct routing between AS and MRFC	9.1.0	9.2.0	C1-094736
2009-12	CP-46	CP-090936	2900	3	Registration of IMS media plane security capabilities	9.1.0	9.2.0	C1-094730
2009-12	CP-46	CP-090893	2905		Updating of outbound and related references	9.1.0	9.2.0	C1-094826
2009-12	CP-46	CP-090894	2908		Updating of GRUU references	9.1.0	9.2.0	C1-094832
2009-12	CP-46	CP-090940	2909		Miscellaneous editorial corrections	9.1.0	9.2.0	C1-094850
2009-12	CP-46	CP-090892	2912	1	Removal of outstanding Editor's notes for EMC1	9.1.0	9.2.0	C1-095486
2009-12	CP-46	CP-090896	2914		Removal of outstanding Editor's note for ServID	9.1.0	9.2.0	C1-094855
2009-12	CP-46	CP-090903	2916		Removal of outstanding Editor's note for Overlap	9.1.0	9.2.0	C1-094857
2009-12	CP-46	CP-090940	2924	2	Definition of globally Globally Routeable SIP URI.	9.1.0	9.2.0	C1-095676
2009-12	CP-46	CP-090940	2925	1	Handling of Request-URI with tel URI and sip URI containing user=phone by the BGCF	9.1.0	9.2.0	C1-095438
2009-12	CP-46	CP-090940	2926	2	Additional routeing capabilities	9.1.0	9.2.0	C1-095677
2009-12	CP-46	CP-090902	2932	1	Handling of Route by the I-CSCF	9.1.0	9.2.0	C1-095607
2009-12	CP-46	CP-090902	2934	1	Annex A/ P-Charging-Vector	9.1.0	9.2.0	C1-095606
2009-12	CP-46	CP-090902	2936	2	REGISTERs for Keeping NAT binding /Annex F	9.1.0	9.2.0	C1-095703
2009-12	CP-46	CP-090938	2940	1	Ml reference point additions – general aspects	9.1.0	9.2.0	C1-095467
2009-12	CP-46	CP-090938	2941	1	Ml reference point additions – location determination summary	9.1.0	9.2.0	C1-095468
2009-12	CP-46	CP-090938	2942	3	Ml reference point additions – E-CSCF changes	9.1.0	9.2.0	C1-095726
2009-12	CP-46	CP-090938	2943	3	Ml reference point additions – new LRF functionality	9.1.0	9.2.0	C1-095727
2009-12	CP-46	CP-090938	2944		Ml reference point additions – profile changes	9.1.0	9.2.0	C1-094995
2009-12	CP-46	CP-090902	2946		Correction of profile table on the role for UE	9.1.0	9.2.0	C1-094997
2009-12	CP-46	CP-090936	2951	2	Indicating End-to-Access Edge Media Plane Security in session set-up	9.1.0	9.2.0	C1-095700
2009-12	CP-46	CP-090895	2954	2	Correct Phone-Context parameter coding	9.1.0	9.2.0	C1-095688
2009-12	CP-46	CP-090940	2955	2	Human readable UE name	9.1.0	9.2.0	C1-095648
2009-12	CP-46	CP-090927	2959	2	E-CSCF invoking EATF	9.1.0	9.2.0	C1-095718
2009-12	CP-46	CP-090930	2960	2	IMEI in unauthenticated emergency call in EPS and GPRS	9.1.0	9.2.0	C1-095714
2009-12	CP-46	CP-090930	2961	1	Emergency bearer activation in EPS and GPRS	9.1.0	9.2.0	C1-095309
2009-12	CP-46	CP-090892	2964		Alignment of 24.229 with draft-patel-ecrit-sos-parameter-07	9.1.0	9.2.0	C1-095069

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2009-12	CP-46	CP-090892	2967	1	Removal of editor's note in 5.4.8.2 – use of "sos" in GRUU	9.1.0	9.2.0	C1-095489
2009-12	CP-46	CP-090923	2971	1	Reason header in provisional responses	9.1.0	9.2.0	C1-095472
2009-12	CP-46	CP-090940	2976		Correcting SIP interface to VoiceXML media services	9.1.0	9.2.0	C1-095187
2009-12	CP-46	CP-090940	2980	1	Annex A: Support of INFO for CAT and CRS	9.1.0	9.2.0	C1-095445
2009-12	CP-46	CP-090940	2981	2	Removal of editor's note on 199 provisional response	9.1.0	9.2.0	C1-095649
2009-12	CP-46	CP-090983	2970	2	Update to annex J based on draft-patel-dispatch-cpc-oli-parameter	9.1.0	9.2.0	-
2010-03	CP-47	CP-100131	2810	3	Correcting handling of emergency session requests made by unregistered users	9.2.0	9.3.0	C1-101129
2010-03	CP-47	CP-100110	2930	4	Handling of Request-URI with tel URI and sip URI containing user=phone by the S-CSCF	9.2.0	9.3.0	C1-100993
2010-03	CP-47	CP-100104	2958	4	Emergency session with P-CSCF in visited network	9.2.0	9.3.0	C1-100720
2010-03	CP-47	CP-100110	2990	1	IETF reference updates (IMSProtoc2 related)	9.2.0	9.3.0	C1-100210
2010-03	CP-47	CP-100124	2992	3	Support of draft-ietf-mmusic-sdp-media-capabilities	9.2.0	9.3.0	C1-101151
2010-03	CP-47	CP-100153	2994	5	Adding 1XRTT Femto support for the 3GPP2-1X access type	9.2.0	9.3.0	C1-101180
2010-03	CP-47	CP-100149	2996	1	Correction for e2ae syntax	9.2.0	9.3.0	C1-100200
2010-03	CP-47	CP-100153	2997	2	Implications of resource reservation failure	9.2.0	9.3.0	C1-100704
2010-03	CP-47	CP-100143	2998	1	RFC 4488 in Annex A	9.2.0	9.3.0	C1-100176
2010-03	CP-47	CP-100153	3000	1	Removing an Editor's note in the reference section	9.2.0	9.3.0	C1-100135
2010-03	CP-47	CP-100153	3001	4	Handling of Subscription context information by intermediary entities	9.2.0	9.3.0	C1-101116
2010-03	CP-47	CP-100151	3002	1	Editorial update: adding missing defenitions, correcting typos and inconsistencies	9.2.0	9.3.0	C1-100198
2010-03	CP-47	CP-100151	3003	3	Correcting providing of additional location information to LRF	9.2.0	9.3.0	C1-101117
2010-03	CP-47	CP-100149	3004	1	Editorial amendments for end to access edge media security	9.2.0	9.3.0	C1-100233
2010-03	CP-47	CP-100149	3005	2	Improvements to end to access edge security text	9.2.0	9.3.0	C1-100780
2010-03	CP-47	CP-100149	3006	1	MGCF is not involved in e2ae security	9.2.0	9.3.0	C1-100234
2010-03	CP-47	CP-100149	3007	1	UE requirements in the absence of P-CSCF support of end to access edge security	9.2.0	9.3.0	C1-100202
2010-03	CP-47	CP-100149	3008	1	Profile additions for end to access edge security	9.2.0	9.3.0	C1-100203
2010-03	CP-47	CP-100149	3009	1	Coverage of media security in the security introduction	9.2.0	9.3.0	C1-100204
2010-03	CP-47	CP-100151	3010	1	Making the E-CSCF responsible for the domain of incoming Request-URI	9.2.0	9.3.0	C1-100230
2010-03	CP-47	CP-100151	3011	1	Usage of P-Charging-Vector header within the emergency call architecture	9.2.0	9.3.0	C1-100199

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2010-03	CP-47	CP-100151	3013	1	Delivery of location by the E-CSCF	9.2.0	9.3.0	C1-100159
2010-03	CP-47	CP-100151	3014	2	Structure of reference identifier	9.2.0	9.3.0	C1-100941
2010-03	CP-47	CP-100151	3015	1	Handling of editor's note on subscribing to all dialogs	9.2.0	9.3.0	C1-100160
2010-03	CP-47	CP-100109	3017		Resolution of editor's notes related to PRIOR	9.2.0	9.3.0	C1-100208
2010-03	CP-47	CP-100230	3019	1	Removal of editor's notes relating to learning of trust domain boundaries and information saved during registration	9.2.0	9.3.0	-
2010-03	CP-47	CP-100135	3020	1	Correcting IP-CAN documentation	9.2.0	9.3.0	C1-100944
2010-03	CP-47	CP-100153	3024		P-CSCF Note correction	9.2.0	9.3.0	C1-100339
2010-03	CP-47	CP-100153	3025		Authentication-Info header field	9.2.0	9.3.0	C1-100340
2010-03	CP-47	CP-100153	3026	4	DTMF Info Package definition	9.2.0	9.3.0	C1-101119
2010-03	CP-47	CP-100110	3028		Removal of editor's note: 199 (Early Dialog Terminated) option-tag	9.2.0	9.3.0	C1-100366
2010-03	CP-47	CP-100111	3031		Removal of editor's note: Annex K NAT traversal	9.2.0	9.3.0	C1-100369
2010-03	CP-47	CP-100107	3035		Closure of SAES related editor's notes	9.2.0	9.3.0	C1-100419
2010-03	CP-47	CP-100117	3037		Addressing editor's notes relating to NASS bundled authentication	9.2.0	9.3.0	C1-100421
2010-03	CP-47	CP-100110	3039		Removal of editor's notes relating to emergency call	9.2.0	9.3.0	C1-100423
2010-03	CP-47	CP-100110	3043		Removal of outstanding Editor's note on IOI	9.2.0	9.3.0	C1-100436
2010-03	CP-47	CP-100107	3045		Incorrect NAS message in Annex L	9.2.0	9.3.0	C1-100454
2010-03	CP-47	CP-100135	3048	2	Delete EN pertaining to RFC 4117	9.2.0	9.3.0	C1-101156
2010-03	CP-47	CP-100122	3053		Incorrect trigger in I-CSCF for restoration procedures	9.2.0	9.3.0	C1-100462
2010-03	CP-47	CP-100112	3054	1	Clean up editor's notes on subscription to debug event package	9.2.0	9.3.0	C1-100983
2010-03	CP-47	CP-100149	3055	1	Exchanging media plane security capabilities at registration	9.2.0	9.3.0	C1-100971
2010-03	CP-47	CP-100218	3056	2	Profile table changes for exchanging media plane security capabilities at registration	9.2.0	9.3.0	-
2010-03	CP-47	CP-100153	3057	1	Corrections to profile table entries related to security agreement	9.2.0	9.3.0	C1-100973
2010-03	CP-47	CP-100110	3059	1	Inclusion of draft alert-urns for INVITE Responses	9.2.0	9.3.0	C1-100954
2010-03	CP-47	CP-100119	3063		Reference update of draft-ietf-mediactrl-vxml	9.2.0	9.3.0	C1-100518
2010-03	CP-47	CP-100118	3065	1	Address the UUS related Editor's Note	9.2.0	9.3.0	C1-100986
2010-03	CP-47	CP-100110	3069	1	Correcting missing reference	9.2.0	9.3.0	C1-100991
2010-03	CP-47	CP-100153	3072	4	Session ID profile table alignment	9.2.0	9.3.0	C1-101176
2010-03	CP-47	CP-100105	3075	1	Annex A/ Fixing of missing status support in Tables	9.2.0	9.3.0	C1-100982
2010-03	CP-47	CP-100105	3078		Annex A/ P-Media-Authorization support	9.2.0	9.3.0	C1-100666

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2010-03	CP-47	CP-100105	3081		Annex A / integration of resource management and SIP	9.2.0	9.3.0	C1-100670
2010-03	CP-47	CP-100247	3082	2	Additional routeing capabilities	9.2.0	9.3.0	-
2010-03	CP-47	CP-100138	3083	3	P-CSCF Restoration Procedures	9.2.0	9.3.0	C1-101262
2010-03	CP-47	CP-100110	3086		New version of IETF draft-yu-tel-dai	9.2.0	9.3.0	C1-100684
2010-03	CP-47	CP-100110	3092		Abnormal Digest procedures fix	9.2.0	9.3.0	C1-100692
2010-03	CP-47	CP-100128	3094		IMDN reference update	9.2.0	9.3.0	C1-100694
2010-03	CP-47	CP-100140	3095	1	I4 applicability and EATF functionality	9.2.0	9.3.0	C1-100940
2010-03	CP-47	CP-100153	3096		Failure of GPRS and EPS resource reservation	9.2.0	9.3.0	C1-100703
2010-03	CP-47	CP-100142	3097	3	Addition of Dialog Event package to profile tables in support of Inter-UE transfer	9.2.0	9.3.0	C1-101162
2010-03	CP-47	CP-100151	3098		Correction of reference to RFC 4235	9.2.0	9.3.0	C1-100966
2010-03	CP-47	CP-100144	3099		Emergency call clarifications in the absence of registration	9.2.0	9.3.0	C1-100774
2010-03	CP-47	CP-100110	3101		Correct authentication and registration referencing for emergency registration	9.2.0	9.3.0	C1-100805
2010-03	CP-47	CP-100107	3103		P-Access-Network-Info correction for LTE	9.2.0	9.3.0	C1-100808
2010-03	CP-47	CP-100104	3106		Update reference for draft-patel-ecrit-sos-parameter	9.2.0	9.3.0	C1-100811
2010-03	CP-47	CP-100216	3033	2	Updating of SAES related references	9.2.0	9.3.0	-
2010-03	CP-47				Editorial correction	9.3.0	9.3.1	-
2010-06	CP-48	CP-100364	3012	3	Completion of dialog event package usage	9.3.1	9.4.0	C1-101860
2010-06	CP-48	CP-100363	3118	1	Profile table changes for SDES media plane security role	9.3.1	9.4.0	C1-101889
2010-06	CP-48	CP-100363	3119		Using SDES cryptro attribute	9.3.1	9.4.0	C1-101395
2010-06	CP-48	CP-100346	3121		Wrong requirements for ICS MSC in profile tables	9.3.1	9.4.0	C1-101399
2010-06	CP-48	CP-100337	3129		Reference updates	9.3.1	9.4.0	C1-101472
2010-06	CP-48	CP-100359	3130	1	norefersub corrections	9.3.1	9.4.0	C1-101859
2010-06	CP-48	CP-100364	3131		Charging tidyup	9.3.1	9.4.0	C1-101487
2010-06	CP-48	CP-100359	3136	1	MSC Server assisted mid-call feature - conferencing	9.3.1	9.4.0	C1-102032
2010-06	CP-48	CP-100340	3142	1	RFC4694 for IBCF	9.3.1	9.4.0	C1-101814
2010-06	CP-48	CP-100364	3148		3xx response replaced by response	9.3.1	9.4.0	C1-101584
2010-06	CP-48	CP-100340	3151	1	Use of P-Served-User header field in user location procedure	9.3.1	9.4.0	C1-101812
2010-06	CP-48	CP-100340	3155	2	IBCF and Content-Disposition	9.3.1	9.4.0	C1-102031
2010-06	CP-48	CP-100351	3158	1	Addition of MSRP SDP a=path attribute	9.3.1	9.4.0	C1-101820
2010-06	CP-48	CP-100363	3161	1	Roles relating to media plane security	9.3.1	9.4.0	C1-101890
2010-06	CP-48	CP-100354	3162	2	IMS available	9.3.1	9.4.0	C1-102103
2010-06	CP-48	CP-100367	3040	1	Identifying an emergency call at the P-CSCF	9.4.0	10.0.0	C1-101504

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2010-06	CP-48	CP-100367	3110		Handling of Privacy header	9.4.0	10.0.0	C1-101838
2010-06	CP-48	CP-100367	3113	2	S-CSCF triggering of Additional Routeing capability	9.4.0	10.0.0	C1-102042
2010-06	CP-48	CP-100367	3114	2	xPON access type values in P-Access-Network-Info	9.4.0	10.0.0	C1-102043
2010-06	CP-48	CP-100367	3116	1	Digit manipulation	9.4.0	10.0.0	C1-101843
2010-06	CP-48	CP-100371	3124	1	Digest authentication without Authorization header	9.4.0	10.0.0	C1-102012
2010-06	CP-48	CP-100367	3126	1	Corrections for NASS-Bundled authentication	9.4.0	10.0.0	C1-101844
2010-06	CP-48	CP-100367	3134		Miscellaneous editorial issues	9.4.0	10.0.0	C1-101503
2010-06	CP-48	CP-100371	3137		Usage of "trusted node authentication"	9.4.0	10.0.0	C1-101509
2010-06	CP-48	CP-100367	3146	1	Annex A, Table A.4, item 2C, reference update	9.4.0	10.0.0	C1-101845
2010-09	CP-49	CP-100510	3168	3	Outbound reregistration at P-CSCF	10.0.0	10.1.0	C1-102822
2010-09	CP-49	CP-100500	3171	3	Initial registration for GPRS-IMS at S-CSCF	10.0.0	10.1.0	C1-102848
2010-09	CP-49	CP-100511	3172	5	Privacy protection in IBCF	10.0.0	10.1.0	C1-103526
2010-09	CP-49	CP-100639	3176	3	Alignment with RFC 5552	10.0.0	10.1.0	-
2010-09	CP-49	CP-100511	3178	6	User-related policy data enforcement by the P-CSCF	10.0.0	10.1.0	C1-103517
2010-09	CP-49	CP-100640	3180	3	Handling of aliases URIs	10.0.0	10.1.0	-
2010-09	CP-49	CP-100641	3181	3	Structure of the Request URI sent by a UE	10.0.0	10.1.0	-
2010-09	CP-49	CP-100481	3188	2	Home network check for (E)UTRAN access	10.0.0	10.1.0	C1-103041
2010-09	CP-49	CP-100482	3196	1	Updates to references pertaining to Internet Drafts for tel URI parameters	10.0.0	10.1.0	C1-102676
2010-09	CP-49	CP-100519	3197	1	Usage of alternative P-CSCF address during registration	10.0.0	10.1.0	C1-102631
2010-09	CP-49	CP-100496	3198	8	Mandate registeration with IMS in order to receive audio/voice services	10.0.0	10.1.0	C1-103536
2010-09	CP-49	CP-100510	3200		Annex A, Reason header	10.0.0	10.1.0	C1-102448
2010-09	CP-49	CP-100652	3205	3	Emergency registration in HPLMN	10.0.0	10.1.0	-
2010-09	CP-49	CP-100486	3209	1	Keep-alive corrections	10.0.0	10.1.0	C1-102624
2010-09	CP-49	CP-100511	3211	4	Passing policy with subscription information to UE and P-CSCF	10.0.0	10.1.0	C1-103504
2010-09	CP-49	CP-100486	3214	1	Wildcarded identity AVP correction	10.0.0	10.1.0	C1-102685
2010-09	CP-49	CP-100486	3217		Subclause reference correction	10.0.0	10.1.0	C1-102492
2010-09	CP-49	CP-100483	3221		Update of draft-rosenberg-sip-app-media-tag reference	10.0.0	10.1.0	C1-102532
2010-09	CP-49	CP-100511	3222	3	Location number	10.0.0	10.1.0	C1-103543
2010-09	CP-49	CP-100487	3226		Updates to references pertaining to Internet Drafts for tel URI parameters	10.0.0	10.1.0	C1-102679
2010-09	CP-49	CP-100511	3236	2	Insertion of IMS access gateway by P-CSCF	10.0.0	10.1.0	C1-103518
2010-09	CP-49	CP-100511	3237	4	Enforcement of P-Early-Media indication by P-CSCF	10.0.0	10.1.0	C1-103544

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2010-09	CP-49	CP-100508	3239		EN pertaining to Media Plane Securtiy	10.0.0	10.1.0	C1-103039
2010-09	CP-49	CP-100481	3243	2	Detecting valid emergency identifiers	10.0.0	10.1.0	C1-103542
2010-09	CP-49	CP-100501	3245	2	Emergency PDN connection usage control in P-CSCF	10.0.0	10.1.0	C1-103513
2010-09	CP-49	CP-100510	3249	1	IBCF procedures for SIP message	10.0.0	10.1.0	C1-103382
2010-09	CP-49	CP-100519	3250	2	Indicating wildcarded IMPU in reg-event	10.0.0	10.1.0	C1-103528
2010-09	CP-49	CP-100501	3252	1	Wildcarded Identities handling	10.0.0	10.1.0	C1-103354
2010-09	CP-49	CP-100481	3256	2	Correction of Stage 3 misalignment with Stage 1 and Stage 2 on use of SIP 380 response.	10.0.0	10.1.0	C1-103389
2010-09	CP-49	CP-100519	3257	3	SigComp disabling	10.0.0	10.1.0	C1-103530
2010-09	CP-49	CP-100486	3258	2	Ensuring PSAP receives correctly formatted request	10.0.0	10.1.0	C1-103568
2010-09	CP-49	CP-100486	3261	1	Mandate registeration with IMS in order to receive audio/voice services	10.0.0	10.1.0	C1-103508
2010-12	CP-50	CP-100843	3305	2	SRVCC enhancements - ATCF invocation	10.1.0	10.2.0	C1-104362
2010-12	CP-50	CP-100728	3267	1	Protected AKA registration at S-CSCF	10.1.0	10.2.0	C1-104197
2010-12	CP-50	CP-100728	3270	1	Protected Digest registration at S-CSCF	10.1.0	10.2.0	C1-104300
2010-12	CP-50	CP-100728	3273	2	Unprotected registration at S-CSCF	10.1.0	10.2.0	C1-104370
2010-12	CP-50	CP-100750	3278		Supported header field corrected	10.1.0	10.2.0	C1-103619
2010-12	CP-50	CP-100728	3281	1	Update reference	10.1.0	10.2.0	C1-104310
2010-12	CP-50	CP-100725	3285		Correcting mixed references in IBCF	10.1.0	10.2.0	C1-103761
2010-12	CP-50	CP-100728	3288	3	Conference and IBCF IMS_ALG and removal of an Editor's note.	10.1.0	10.2.0	C1-105071
2010-12	CP-50	CP-100735	3291		Correcting errors in S-CSCF restoration procedures	10.1.0	10.2.0	C1-103773
2010-12	CP-50	CP-100728	3301		Incorrect sequence of steps in P-CSCF	10.1.0	10.2.0	C1-104316
2010-12	CP-50	CP-100723	3304		Emergency registration and normal registration	10.1.0	10.2.0	C1-104183
2010-12	CP-50	CP-100738	3314	1	Updating IMEI URN draft reference	10.1.0	10.2.0	C1-104328
2010-12	CP-50	CP-100721	3319		IETF reference updates	10.1.0	10.2.0	C1-103921
2010-12	CP-50	CP-100722	3324		IETF reference updates	10.1.0	10.2.0	C1-103926
2010-12	CP-50	CP-100726	3328		IETF reference updates	10.1.0	10.2.0	C1-103936
2010-12	CP-50	CP-100728	3331		IETF reference updates	10.1.0	10.2.0	C1-104337
2010-12	CP-50	CP-100728	3334		EN removal: Retry-After Header field value in 503 response	10.1.0	10.2.0	C1-103955
2010-12	CP-50	CP-100728	3340		EN removal: Network inserted codecs	10.1.0	10.2.0	C1-103961
2010-12	CP-50	CP-100723	3344	1	Further modifications required to SIP 380 response to remove new requirements.	10.1.0	10.2.0	C1-104187
2010-12	CP-50	CP-100864	3345	4	Inclusion of IMEI in the sip.instance of the initial SIP-Register request	10.1.0	10.2.0	C1-105086
2010-12	CP-50	CP-100733	3348		Handling of editor's note relating to private network traffic breakout and breakin	10.1.0	10.2.0	C1-103984

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2010-12	CP-50	CP-100726	3354	2	Inclusion of file transfer attributes	10.1.0	10.2.0	C1-104986
2010-12	CP-50	CP-100752	3355	2	IBCF and 3xx responses	10.1.0	10.2.0	C1-104595
2010-12	CP-50	CP-100752	3356		Non E.164 Tel URI conversion failure	10.1.0	10.2.0	C1-104464
2010-12	CP-50	CP-100750	3357	2	max-time and base-time parameters provision	10.1.0	10.2.0	C1-105207
2010-12	CP-50	CP-100752	3358		reference correction	10.1.0	10.2.0	C1-104466
2010-12	CP-50	CP-100728	3361	1	AKA registration at S-CSCF	10.1.0	10.2.0	C1-104991
2010-12	CP-50	CP-100728	3364	2	Autentication already performed	10.1.0	10.2.0	C1-105203
2010-12	CP-50	CP-100728	3367	1	Digest registration at S-CSCF	10.1.0	10.2.0	C1-104997
2010-12	CP-50	CP-100728	3370	1	Bundle registration	10.1.0	10.2.0	C1-105000
2010-12	CP-50	CP-100720	3377	1	Codec and DTMF correction	10.1.0	10.2.0	C1-104980
2010-12	CP-50	CP-100728	3380		Definition: multiple registrations	10.1.0	10.2.0	C1-104535
2010-12	CP-50	CP-100871	3383	1	Reference update: draft-ietf-sipcore-199	10.1.0	10.2.0	-
2010-12	CP-50	CP-100724	3387		Reference update: draft-ietf-sipcore-keep	10.1.0	10.2.0	C1-104547
2010-12	CP-50	CP-100864	3388	2	Modifications to priority handling in support of MPS	10.1.0	10.2.0	C1-105095
2010-12	CP-50	CP-100885	3389	3	Updating the restoration procedure definition	10.1.0	10.2.0	-
2010-12	CP-50	CP-100752	3390		Adding RFC 5318 to major capabilities tables	10.1.0	10.2.0	C1-105226
2010-12	CP-50	CP-100728	3393	1	Handling of the isfocus media feature tag in P-CSCF	10.1.0	10.2.0	C1-105003
2010-12	CP-50	CP-100752	3394		Annex A, Table A.4, item 29+72 and Table A.4A, prerequisite	10.1.0	10.2.0	C1-104618
2010-12	CP-50	CP-100728	3397		"ob" parameter in case of no registration	10.1.0	10.2.0	C1-105006
2010-12	CP-50	CP-100728	3401	2	Addition of Target-Dialog header and capability in Annex A	10.1.0	10.2.0	C1-105074
2010-12	CP-50	CP-100766	3405	2	Alternative emergency session handling in non-roaming cases (P-CSCF)	10.1.0	10.2.0	C1-105052
2010-12	CP-50	CP-100766	3406	2	Alternative emergency session handling in non-roaming cases (S-CSCF)	10.1.0	10.2.0	C1-105053
2010-12	CP-50	CP-100766	3407		Alternative emergency session handling in non-roaming cases (E-CSCF)	10.1.0	10.2.0	C1-104682
2010-12	CP-50	CP-100749	3409		Removal of erroneous passing on of IOI value to PSAP	10.1.0	10.2.0	C1-104718
2010-12	CP-50	CP-100766	3411		Additions to E-CSCF functionality for IESE	10.1.0	10.2.0	C1-104721
2010-12	CP-50	CP-100766	3412		IBCF detection and routeing of emergency call	10.1.0	10.2.0	C1-104722
2010-12	CP-50	CP-100864	3413	3	Introduction to priority schemes in the IM CN subsystem	10.1.0	10.2.0	C1-105221
2010-12	CP-50	CP-100766	3414	1	Addition to introductory clauses in support of IESE	10.1.0	10.2.0	C1-104974
2010-12	CP-50	CP-100725	3415	1	Correction of the usage for type 3 IOI	10.1.0	10.2.0	C1-105051
2010-12	CP-50	CP-100864	3425	2	P-CSCF behaviour for insufficient bandwidth	10.1.0	10.2.0	C1-105058
2010-12	CP-50	CP-100752	3416	1	Text corrections	10.1.0	10.2.0	C1-104969

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2010-12	CP-50	CP-100727	3420		Update of IETF reference	10.1.0	10.2.0	C1-104842
2011-03	CP-51	CP-110181	3371	4	Sending of location information from LRF to E-CSCF	10.2.0	10.3.0	C1-110671
2011-03	CP-51	CP-110181	3429	2	Response code in Reason header field	10.2.0	10.3.0	C1-110659
2011-03	CP-51	CP-110164	3432	1	UE initiated deregistration	10.2.0	10.3.0	C1-110581
2011-03	CP-51	CP-110181	3433		Other databases	10.2.0	10.3.0	C1-110010
2011-03	CP-51	CP-110181	3434	1	Clarification of possible triggers for network-initiated reauthentication	10.2.0	10.3.0	C1-110560
2011-03	CP-51	CP-110161	3435	6	Update to IMS registration procedures due to USAT initiated Refresh for ISIM/USIM EFs	10.2.0	10.3.0	C1-111511
2011-03	CP-51	CP-110184	3436	1	Optimal Media Routeing – SDP attribute syntax definition	10.2.0	10.3.0	C1-110558
2011-03	CP-51	CP-110184	3437	1	Update SDP profile table for Optimal Media Routeing	10.2.0	10.3.0	C1-110559
2011-03	CP-51	CP-110196	3439	1	Modifications to S-CSCF procedures in support of MPS	10.2.0	10.3.0	C1-110562
2011-03	CP-51	CP-110196	3440	1	Modifications to P-CSCF and IBCF procedures in support of MPS	10.2.0	10.3.0	C1-110563
2011-03	CP-51	CP-110201	3441	1	Select E-CSCF upon S-SCSF failure	10.2.0	10.3.0	C1-110557
2011-03	CP-51	CP-110158	3445	1	Correct P-CSCF handling of requests for emergency services with Route header fields	10.2.0	10.3.0	C1-110567
2011-03	CP-51	CP-110196	3450		Clarification on P-CSCF behaviour in case of insufficient bandwidth	10.2.0	10.3.0	C1-110180
2011-03	CP-51	CP-110166	3453	2	New Reference for Alert-URN	10.2.0	10.3.0	C1-111349
2011-03	CP-51	CP-110187	3457	5	Explicit Congestion Notification (ECN) for RTP over UDP	10.2.0	10.3.0	C1-111360
2011-03	CP-51	CP-110196	3458	4	Clarify the P-CSCF restoration procedure	10.2.0	10.3.0	C1-111271
2011-03	CP-51	CP-110164	3461	1	Reference update: draft-ietf-mmusic-ice-tcp	10.2.0	10.3.0	C1-110578
2011-03	CP-51	CP-110164	3464	1	Reference update: RFC 6086	10.2.0	10.3.0	C1-110589
2011-03	CP-51	CP-110159	3468		Reference update: draft-ietf-sipcore-keep	10.2.0	10.3.0	C1-110267
2011-03	CP-51	CP-110164	3471	3	P-CSCF Path SIP URI and IMS flow token correction	10.2.0	10.3.0	C1-111283
2011-03	CP-51	CP-110196	3474	1	Encoding of PANI for E-UTRAN	10.2.0	10.3.0	C1-110442
2011-03	CP-51	CP-110196	3475	3	Insertion of "orig" parameter by IBCF	10.2.0	10.3.0	C1-110665
2011-03	CP-51	CP-110196	3479	5	Removal of reference CPC and OLI Internet Draft	10.2.0	10.3.0	C1-111329
2011-03	CP-51	CP-110158	3483	2	Specifying "sos" URI parameter in 24.229	10.2.0	10.3.0	C1-111087
2011-03	CP-51	CP-110196	3484	7	Example of IMS Registration conditions, taking into account the network operator's preference for selection of the voice domain	10.2.0	10.3.0	C1-111270
2011-03	CP-51	CP-110181	3486	2	Removal of Sigcomp disabling	10.2.0	10.3.0	C1-111266
2011-03	CP-51	CP-110164	3489		New registration	10.2.0	10.3.0	C1-110842
2011-03	CP-51	CP-110309	3490	6	Inclusion of MEID in the sip.instance of the SIP-Register request	10.2.0	10.3.0	-

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2011-03	CP-51	CP-110181	3491	1	Disabling SigComp by default in E-UTRAN	10.2.0	10.3.0	C1-111221
2011-03	CP-51	CP-110164	3495	1	S-CSCF Service-Route SIP URI	10.2.0	10.3.0	C1-111274
2011-03	CP-51	CP-110184	3496	4	Introduction of OMR procedures in AS, MGCF and P-CSCF	10.2.0	10.3.0	C1-111359
2011-03	CP-51	CP-110181	3497		Removal of editor's note: different sets of policies for a user	10.2.0	10.3.0	C1-111235
2011-03	CP-51	CP-110181	3498		Removal of editor's note: additional policy elements	10.2.0	10.3.0	C1-110939
2011-03	CP-51	CP-110164	3501		Reference update and procedure correction: 199	10.2.0	10.3.0	C1-111277
2011-03	CP-51	CP-110162	3502	1	Contact header clarification	10.2.0	10.3.0	C1-111240
2011-03	CP-51	CP-110160	3507	1	MGCF procedure corrections related to SIP preconditions	10.2.0	10.3.0	C1-111259
2011-03	CP-51	CP-110164	3510		Erroneous row reference in Table A.50A	10.2.0	10.3.0	C1-111000
2011-03	CP-51	CP-110164	3514	1	Correction reference	10.2.0	10.3.0	C1-111280
2011-03	CP-51	CP-110176	3517	2	Correction to the header field indicating where the request comes from in E-CSCF procedures	10.2.0	10.3.0	C1-111325
2011-03	CP-51	CP-110181	3519	1	Editorial corrections to S-CSCF registration subclauses	10.2.0	10.3.0	C1-111241
2011-03	CP-51	CP-110181	3520	3	Clarification of authentication of 380 and 504 responses with multiple registration	10.2.0	10.3.0	C1-111337
2011-03	CP-51	CP-110181	3521	1	Provision of phone-context parameter value via MO	10.2.0	10.3.0	C1-111245
2011-03	CP-51	CP-110010	3522	3	P-CSCF graceful shutdown	10.2.0	10.3.0	-
2011-06	CP-52	CP-110450	3532	1	Reference update: 199	10.3.0	10.4.0	C1-112024
2011-06	CP-52	CP-110445	3536	1	Reference update: RFC 6223	10.3.0	10.4.0	C1-112013
2011-06	CP-52	CP-110450	3539		Annex A: RFC 6086 reference corrections	10.3.0	10.4.0	C1-111556
2011-06	CP-52	CP-110468	3540	1	Removal of Annex F.3	10.3.0	10.4.0	C1-112015
2011-06	CP-52	CP-110468	3541	1	Moving of P-CSCF ICE procedures (Annex K.3.2 and K.5.3)	10.3.0	10.4.0	C1-112016
2011-06	CP-52	CP-110468	3542	1	Removal of Annex G	10.3.0	10.4.0	C1-112014
2011-06	CP-52	CP-110468	3545	1	S-CSCF-initiated session release	10.3.0	10.4.0	C1-112025
2011-06	CP-52	CP-110450	3548	1	Service-Route at the UE	10.3.0	10.4.0	C1-112040
2011-06	CP-52	CP-110450	3551	1	Service-Route at the P-CSCF	10.3.0	10.4.0	C1-112043
2011-06	CP-52	CP-110450	3554	2	Service-Route at the S-CSCF	10.3.0	10.4.0	C1-112227
2011-06	CP-52	CP-110450	3557	1	Path header field at the S-CSCF	10.3.0	10.4.0	C1-112049
2011-06	CP-52	CP-110450	3560	1	S-CSCF releasing the dialogs	10.3.0	10.4.0	C1-112052
2011-06	CP-52	CP-110450	3563	1	NOTIFY request	10.3.0	10.4.0	C1-112028
2011-06	CP-52	CP-110450	3566	1	Network Initiated deregistration at S-CSCF	10.3.0	10.4.0	C1-112031
2011-06	CP-52	CP-110450	3569	2	Network Initiated deregistration at P-CSCF	10.3.0	10.4.0	C1-112223
2011-06	CP-52	CP-110450	3572	1	Network Initiated deregistration at UE	10.3.0	10.4.0	C1-112037
2011-06	CP-52	CP-110468	3573		UE initiated deregistration	10.3.0	10.4.0	C1-111590

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2011-06	CP-52	CP-110448	3578	1	P-Access-Network-Info : ABNF correction	10.3.0	10.4.0	C1-112004
2011-06	CP-52	CP-110468	3579	1	Moving of IBCF ICE procedures (Annex K.5.4)	10.3.0	10.4.0	C1-112017
2011-06	CP-52	CP-110531	3583	1	SRVCC enhancements in Annex A	10.3.0	10.4.0	-
2011-06	CP-52	CP-110469	3584		ENs on P-CSCF invoking ATCF	10.3.0	10.4.0	C1-111614
2011-06	CP-52	CP-110465	3585	1	Inclusion of MEID in the sip.instance of the SIP-Register request	10.3.0	10.4.0	C1-112201
2011-06	CP-52	CP-110465	3586	1	Clarification of scope of section 5.1.6 on Emergency Call	10.3.0	10.4.0	C1-112089
2011-06	CP-52	CP-110447	3591	1	Fraud prevention for deregistration for ICS	10.3.0	10.4.0	C1-112061
2011-06	CP-52	CP-110474	3596	2	UICC Access to IMS	10.3.0	10.4.0	C1-112249
2011-06	CP-52	CP-110447	3599	1	Updating IMEI URN draft reference	10.3.0	10.4.0	C1-112058
2011-06	CP-52	CP-110468	3602	2	Insertion of "gated" parameter by the P-CSCF	10.3.0	10.4.0	C1-112231
2011-06	CP-52	CP-110451	3605	1	Removal of dial around indicator	10.3.0	10.4.0	C1-112235
2011-06	CP-52	CP-110477	3606	1	OMR designation as media level attributes in profile	10.3.0	10.4.0	C1-112096
2011-06	CP-52	CP-110472	3612	2	Application server detection and routeing of emergency call	10.3.0	10.4.0	C1-112232
2011-06	CP-52	CP-110468	3613	1	Removal of duplicate material in P-CSCF emergency call handling	10.3.0	10.4.0	C1-112094
2011-06	CP-52	CP-110468	3622		Miscellaneous 24.229 corrections	10.3.0	10.4.0	C1-111949
2011-06	CP-52	CP-110521	3611	3	Removal of repetition of IOI header field parameters	10.3.0	10.4.0	-
2011-06	CP-52	CP-110535	3518	4	Reference Location for Emergency Service	10.4.0	11.0.0	-
2011-09	CP-53	CP-110686	3624	1	Reference update	11.0.0	11.1.0	C1-112731
2011-09	CP-53	CP-110654	3633	3	Correcting errors in S-CSCF restoration procedure	11.0.0	11.1.0	C1-113584
2011-09	CP-53	CP-110656	3641	2	P-Profile-Key header field corrections in I-CSCF	11.0.0	11.1.0	C1-112915
2011-09	CP-53	CP-110693	3648	3	Emergency session when IMS voice over PS is not supported	11.0.0	11.1.0	C1-113170
2011-09	CP-53	CP-110666	3651		EATF in Annex A	11.0.0	11.1.0	C1-112512
2011-09	CP-53	CP-110695	3657	2	Correction on call initiation procedure at the MGCF	11.0.0	11.1.0	C1-112945
2011-09	CP-53	CP-110686	3658		Editorial corrections on SIP header field name	11.0.0	11.1.0	C1-112519
2011-09	CP-53	CP-110689	3659	1	Address the Editor's Note in RLI	11.0.0	11.1.0	C1-112725
2011-09	CP-53	CP-110704	3665	3	Additional IOI correction for SIP responses	11.0.0	11.1.0	-
2011-09	CP-53	CP-110653	3669	1	Replacement of draft-garcia-mmusic-sdp-misc-cap with draft-garcia-mmusic-sdp-miscellaneous-caps	11.0.0	11.1.0	C1-113294
2011-09	CP-53	CP-110686	3670	4	Filtering of P-Associated-URI at P-CSCF	11.0.0	11.1.0	C1-113440
2011-09	CP-53	CP-110674	3676	2	Modification on roles of ATCF	11.0.0	11.1.0	C1-112928
2011-09	CP-53	CP-110651	3683	1	Emergency call – correction of requests covered at the P-CSCF	11.0.0	11.1.0	C1-112832

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2011-09	CP-53	CP-110658	3687		IETF reference update	11.0.0	11.1.0	C1-112647
2011-09	CP-53	CP-110681	3689	1	Removal of "select an E-CSCF"	11.0.0	11.1.0	C1-112754
2011-09	CP-53	CP-110686	3691	2	Release of Media Bearers	11.0.0	11.1.0	C1-112937
2011-09	CP-53	CP-110686	3693	2	Network identified by IOI header field parameter	11.0.0	11.1.0	C1-112960
2011-09	CP-53	CP-110648	3700		"P-Visited-Network-ID" correction	11.0.0	11.1.0	C1-113004
2011-09	CP-53	C1-110715	3701	2	Emergency session handling correction	11.0.0	11.1.0	-
2011-09	CP-53	CP-110681	3703		Deletion of Editor's Note Concerning P-CSCF Dialstring Recognition	11.0.0	11.1.0	C1-113087
2011-09	CP-53	CP-110693	3707	1	Emergency Session Setup – Incorrect Reference	11.0.0	11.1.0	C1-113445
2011-09	CP-53	CP-110677	3713	2	Policy passing when different policies are related to different IMPIs sharing an IMPU	11.0.0	11.1.0	C1-113698
2011-09	CP-53	CP-110681	3715		ENs on XML namespace registration	11.0.0	11.1.0	C1-113176
2011-09	CP-53	CP-110656	3719	1	Adding Call-Info to SUBSCRIBE in annex A	11.0.0	11.1.0	C1-113529
2011-09	CP-53	CP-110653	3730		Updating IMEI URN draft reference	11.0.0	11.1.0	C1-113287
2011-09	CP-53	CP-110653	3732	2	Including draft-holmberg-sipcore-proxy-feature	11.0.0	11.1.0	C1-113594
2011-09	CP-53	CP-110687	3740		Transit IOI principles	11.0.0	11.1.0	C1-113595
2011-09	CP-53	CP-110681	3744		Deletion of Editor's Note in 24.229 on authentication mechanism (Rel-10)	11.0.0	11.1.0	C1-113374
2011-09	CP-53	CP-110681	3746	1	Deletion of Editor's Note in 24.229 on aor attribute (Rel-10)	11.0.0	11.1.0	C1-113476
2011-09	CP-53	CP-110661	3758		Deletion of Editor's Note in 24.229 on NASS error message (Rel-8)	11.0.0	11.1.0	C1-113388
2011-09	CP-53	CP-110686	3759		Inter-operator identifier corrections	11.0.0	11.1.0	C1-113392
2011-09	CP-53	CP-110736	3762	2	Correction on EMC handling of S-CSCF	11.0.0	11.1.0	-
2011-09	CP-53	CP-110690	3763		3GPP2 reference corrections	11.0.0	11.1.0	C1-113396
2011-12	CP-54	CP-110887	3673	9	"Default handling" triggering correction	11.1.0	11.2.0	C1-115232
2011-12	CP-54	CP-110887	3766	3	AS determination of the served user identity	11.1.0	11.2.0	C1-114942
2011-12	CP-54	CP-110887	3771		Editorial correction of the P-CSCF behavior for TCP connection	11.1.0	11.2.0	C1-113821
2011-12	CP-54	CP-110873	3773	1	Update draft-atarius-dispatch-meid-urn	11.1.0	11.2.0	C1-114376
2011-12	CP-54	CP-110887	3786	1	Correction on conditional expression of Major Capabilities	11.1.0	11.2.0	C1-114384
2011-12	CP-54	CP-110852	3790	4	P-CSCF behaviour for emergency calls when failure occurs	11.1.0	11.2.0	C1-115362
2011-12	CP-54	CP-110887	3794	3	Adding missing handling in NOTIFY body for a registration event	11.1.0	11.2.0	C1-114462
2011-12	CP-54	CP-110887	3803	1	P-Profile-Key header field corrections in AS	11.1.0	11.2.0	C1-114214
2011-12	CP-54	CP-110887	3807	1	P-Profile-Key header field corrections in S-CSCF	11.1.0	11.2.0	C1-114215
2011-12	CP-54	CP-110887	3809		S-CSCF flow selection correction	11.1.0	11.2.0	C1-114106

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2011-12	CP-54	CP-110887	3810		S-CSCF determing supported IP version by UE for media	11.1.0	11.2.0	C1-114107
2011-12	CP-54	CP-110881	3812	3	ICSI to visited network	11.1.0	11.2.0	C1-115170
2011-12	CP-54	CP-110868	3819	1	Removal of editor's notes relating to insertion of P-Access-Network-Info header field by a proxy	11.1.0	11.2.0	C1-114206
2011-12	CP-54	CP-110887	3820		Editorial corrections on SIP header field name	11.1.0	11.2.0	C1-114534
2011-12	CP-54	CP-110887	3821	1	Addition of IEEE802.3ah to P-Access-Network-Info header	11.1.0	11.2.0	C1-115154
2011-12	CP-54	CP-110887	3822		Editorial correction on de-registration of emergency service	11.1.0	11.2.0	C1-114536
2011-12	CP-54	CP-110856	3827	1	Incorrect reference to RFC 5261	11.1.0	11.2.0	C1-115009
2011-12	CP-54	CP-110873	3834	2	proxy-feature I-D reference update	11.1.0	11.2.0	C1-115288
2011-12	CP-54	CP-110861	3840		Inclusion of media feature tag ASN.1 identifiers	11.1.0	11.2.0	C1-114594
2011-12	CP-54	CP-110887	3845		Record-Route reference correction	11.1.0	11.2.0	C1-114600
2011-12	CP-54	CP-110887	3846	1	Number of emergency registrations	11.1.0	11.2.0	C1-115167
2011-12	CP-54	CP-110850	3850	2	Reference update: Reason header in SIP responses	11.1.0	11.2.0	C1-115274
2011-12	CP-54	CP-110887	3855	4	Additional granularity for IMS Communication Service Identifier	11.1.0	11.2.0	C1-115348
2011-12	CP-54	CP-110869	3857		Correction UE handling compression	11.1.0	11.2.0	C1-114687
2011-12	CP-54	CP-110880	3859	1	Routing of emergency requests via S-CSCF	11.1.0	11.2.0	C1-115178
2011-12	CP-54	CP-110887	3860	1	S-CSCF terminating procedures	11.1.0	11.2.0	C1-115155
2011-12	CP-54	CP-110873	3862	2	Transcoding Control at the IMS-ALG in the P-CSCF and related ECN corrections.	11.1.0	11.2.0	C1-115340
2011-12	CP-54	CP-110887	3863	1	T1 Timer value for MRFC	11.1.0	11.2.0	C1-115158
2011-12	CP-54	CP-110881	3868	1	Adding availability for SMS over IMS determination	11.1.0	11.2.0	C1-114971
2011-12	CP-54	CP-110881	3869	1	ICSI included by AS in Feature-Caps header field in terminating requests	11.1.0	11.2.0	C1-115171
2011-12	CP-54	CP-110881	3870		Indicating Multimedia Telephony Application Server in Feature-Caps header field	11.1.0	11.2.0	C1-114779
2011-12	CP-54	CP-110865	3874	1	3GPP2 reference corrections	11.1.0	11.2.0	C1-115192
2011-12	CP-54	CP-110887	3879	2	Correction on MRFC handling when receiving an INVITE message	11.1.0	11.2.0	C1-115235
2011-12	CP-54	CP-110885	3891	2	Additional routeing function behaviour for transit ioi	11.1.0	11.2.0	C1-115231
2011-12	CP-54	CP-110887	3892	2	Clarification on I-CSCF routeing procedure for incoming call with Request-URI in SIP URI format	11.1.0	11.2.0	C1-115252
2012-01					Correction of formatting in tables of annex A	11.2.0	11.2.1	
2012-03	CP-55	CP-120118	3835	4	IPXS: Application invocation procedures	11.2.1	11.3.0	C1-120849
2012-03	CP-55	CP-120165	3844	3	Updating of UUS references	11.2.1	11.3.0	-
2012-03	CP-55	CP-120096	3900		Corrections on the conditions of MSRP SDP a=path attribute	11.2.1	11.3.0	C1-120114

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2012-03	CP-55	CP-120117	3901		Addition of procedures in case of Fiber access network	11.2.1	11.3.0	C1-120148
2012-03	CP-55	CP-120117	3902		Removal of Editor's Note about access-info of P-Access-Network-Info header	11.2.1	11.3.0	C1-120149
2012-03	CP-55	CP-120124	3903	1	ICSI to visited network - ENs	11.2.1	11.3.0	C1-120778
2012-03	CP-55	CP-120117	3905	2	S-CSCF behavior when the number of simultaneous registrations for the same UE is reached.	11.2.1	11.3.0	C1-120880
2012-03	CP-55	CP-120117	3906	3	P-CSCF address provided by OMA DM for fixed access (Annex E).	11.2.1	11.3.0	C1-120906
2012-03	CP-55	CP-120124	3909	3	Use of Contact Parameters in a 3XX Response from an LRF	11.2.1	11.3.0	C1-120898
2012-03	CP-55	CP-120093	3916	1	Geo-Redundancy Registration	11.2.1	11.3.0	C1-120556
2012-03	CP-55	CP-120107	3918	2	P-CSCF forwarding REGISTER when ATCF is used	11.2.1	11.3.0	C1-120869
2012-03	CP-55	CP-120112	3920		IMS-ALG in the P-CSCF is invoked for Transcoding Control	11.2.1	11.3.0	C1-120212
2012-03	CP-55	CP-120117	3921	2	P-Served-User to BGCF	11.2.1	11.3.0	C1-120854
2012-03	CP-55	CP-120090	3928	1	Location Conveyance: Reference update	11.2.1	11.3.0	C1-120562
2012-03	CP-55	CP-120117	3933	1	Location Conveyance: Location Forwarding to MGCF and PSAP	11.2.1	11.3.0	C1-120563
2012-03	CP-55	CP-120112	3940	1	Reference update: draft-holmberg-sipcore-proxy-feature	11.2.1	11.3.0	C1-120619
2012-03	CP-55	CP-120112	3942	1	UE usage of Feature-Caps	11.2.1	11.3.0	C1-120617
2012-03	CP-55	CP-120124	3945	1	GRUU: UE self-assigned GRUU	11.2.1	11.3.0	C1-120766
2012-03	CP-55	CP-120116	3947	2	IMS_IOI_CH input on IBCF behaviour	11.2.1	11.3.0	C1-120848
2012-03	CP-55	CP-120115	3948	1	GINI input on profile tables	11.2.1	11.3.0	C1-120696
2012-03	CP-55	CP-120092	3957	1	Updating references to IMEI URN and XML body handling drafts	11.2.1	11.3.0	C1-120583
2012-03	CP-55	CP-120112	3960	1	Removing contradictory statement from User initiated deregstration procedure	11.2.1	11.3.0	C1-120621
2012-03	CP-55	CP-120117	3961		Editorial corrections	11.2.1	11.3.0	C1-120321
2012-03	CP-55	CP-120124	3962	1	Clarification on forking related issues	11.2.1	11.3.0	C1-120768
2012-03	CP-55	CP-120115	3967	2	Add general support for RFC 6140	11.2.1	11.3.0	C1-120850
2012-03	CP-55	CP-120115	3968	2	Add complex UE support for RFC 6140 mainline GIN registration functionality	11.2.1	11.3.0	C1-120851
2012-03	CP-55	CP-120115	3969	2	Add S-CSCF support for RFC 6140 mainline GIN registration functionality	11.2.1	11.3.0	C1-120852
2012-03	CP-55	CP-120119	3970	3	Introduction of MRB functional entity	11.2.1	11.3.0	C1-120896
2012-06	CP-56	CP-120299	3896	2	Reference update for MIKEY_TICKET RFC	11.3.0	11.4.0	C1-121104
2012-06	CP-56	CP-120314	3904	6	P-Served-User and session case	11.3.0	11.4.0	C1-122360
2012-06	CP-56	CP-120307	3946	2	P-CSCF releasing the session when resource is lost	11.3.0	11.4.0	C1-121545

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2012-06	CP-56	CP-120307	3952	8	Correcting procedure for propagating service profile update to the UE	11.3.0	11.4.0	C1-122450
2012-06	CP-56	CP-120324	3971	6	Addition of the transit and roaming function	11.3.0	11.4.0	C1-122413
2012-06	CP-56	CP-120323	3975	4	PANI header support of network provided location information	11.3.0	11.4.0	C1-122508
2012-06	CP-56	CP-120323	3976	4	Distribution of location information- AS procedures	11.3.0	11.4.0	C1-122509
2012-06	CP-56	CP-120289	3983		Correction on SDP Profile Status	11.3.0	11.4.0	C1-121042
2012-06	CP-56	CP-120314	3984	1	Editorial correction on SDP Profile Status	11.3.0	11.4.0	C1-121540
2012-06	CP-56	CP-120286	3989		GRUU: S-CSCF URI matching	11.3.0	11.4.0	C1-121054
2012-06	CP-56	CP-120284	3993	1	Reference update: draft-salud-alert-info-urns	11.3.0	11.4.0	C1-121416
2012-06	CP-56	CP-120307	3997	2	Restoration procedures missing in entry IBCF	11.3.0	11.4.0	C1-122250
2012-06	CP-56	CP-120306	3998	2	Missing emergency call procedure in S-CSCF	11.3.0	11.4.0	C1-121658
2012-06	CP-56	CP-120324	4000	6	Loopback routeing	11.3.0	11.4.0	C1-122412
2012-06	CP-56	CP-120322	4005	1	Removal of EN regarding PUI format	11.3.0	11.4.0	C1-121529
2012-06	CP-56	CP-120303	4011		Correcting implementation error, dai parameter	11.3.0	11.4.0	C1-121178
2012-06	CP-56	CP-120307	4012		E-CSCF handling of PAI in responses	11.3.0	11.4.0	C1-121179
2012-06	CP-56	CP-120307	4013		Editorial corrections to 24.229	11.3.0	11.4.0	C1-121190
2012-06	CP-56	CP-120314	4018	1	Correcting incorrect references in P-CSCF procedures when emergency call failure occurs	11.3.0	11.4.0	C1-121406
2012-06	CP-56	CP-120314	4019	7	Correcting IBCF and profile tables for use of 3GPP IM CN subsystem XML body in restoration procedures	11.3.0	11.4.0	C1-122415
2012-06	CP-56	CP-120322	4020	7	Addition of GRUU procedures for RFC6140 procedures	11.3.0	11.4.0	C1-122480
2012-06	CP-56	CP-120286	4025	1	Correcting contradictory statements regarding GRUU handling by IBCF	11.3.0	11.4.0	C1-121411
2012-06	CP-56	CP-120314	4026	2	Transparent passing of contact feature tags by B2BUA AS	11.3.0	11.4.0	C1-121719
2012-06	CP-56	CP-120307	4027	1	Use of Contact Parameters in a 3XX Response from an LRF	11.3.0	11.4.0	C1-121546
2012-06	CP-56	CP-120301	4030	1	Reference update: draft-ietf-avtcore-ecn-for-rtp	11.3.0	11.4.0	C1-122285
2012-06	CP-56	CP-120427	4031	2	Update to reference titles in TS 24.229	11.3.0	11.4.0	-
2012-06	CP-56	CP-120314	4032	1	Transparency to GRUU of B2BUA AS	11.3.0	11.4.0	C1-122369
2012-06	CP-56	CP-120314	4033	2	Addressing potential abuse of 3xx responses	11.3.0	11.4.0	C1-122414
2012-06	CP-56	CP-120303	4038	2	Correcting ambiguity in restoration procedures definitions	11.3.0	11.4.0	C1-122445
2012-06	CP-56	CP-120307	4041	1	Adding the related-icid in charging overview	11.3.0	11.4.0	C1-122365
2012-06	CP-56	CP-120295	4045		Updating of UUS references	11.3.0	11.4.0	C1-121944
2012-06	CP-56	CP-120292	4049		IETF reference update (mixer-control)	11.3.0	11.4.0	C1-121948
2012-06	CP-56	CP-120290	4058	1	Correction on profile of REFER request	11.3.0	11.4.0	C1-122268
2012-06	CP-56	CP-120314	4059		Contact header field parameter values	11.3.0	11.4.0	C1-121970

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2012-06	CP-56	CP-120314	4064	1	Adding 3gpp body xml schema to archive	11.3.0	11.4.0	C1-122372
2012-06	CP-56	CP-120314	4069	3	New technology annex when using the EPC via WLAN to access IM CN subsystem	11.3.0	11.4.0	C1-122513
2012-06	CP-56	CP-120290	4074		Handling of EN relating to granularity of access class	11.3.0	11.4.0	C1-122089
2012-06	CP-56	CP-120314	4075	2	Provision of access-type values in the P-CSCF, and Support of network location reporting for IMS functionality over GxGxx interfaces	11.3.0	11.4.0	C1-122491
2012-06	CP-56	CP-120314	4076	2	Correction to the technology annex when using I- WLAN to access IM CN subsystem	11.3.0	11.4.0	C1-122486
2012-06	CP-56	CP-120314	4077		P-CSCF handling UE port along with IP address during registration	11.3.0	11.4.0	C1-122116
2012-06	CP-56	CP-120307	4079	2	Use of Contact Parameters in a 3XX Response from an LRF	11.3.0	11.4.0	C1-122499
2012-09	CP-57	CP-120583	4039	5	SMS domain selection	11.4.0	11.5.0	C1-123296
2012-09	CP-57	CP-120566	4068	5	Emergency sub-service type handling	11.4.0	11.5.0	C1-123416
2012-09	CP-57	CP-120597	4078	1	Support of MRB Query mode in 3GPP TS 24.229	11.4.0	11.5.0	C1-122938
2012-09	CP-57	CP-120603	4082	2	Application servers and RAVEL	11.4.0	11.5.0	C1-123288
2012-09	CP-57	CP-120586	4083	1	Annex A updates for USSI	11.4.0	11.5.0	C1-123264
2012-09	CP-57	CP-120588	4084	1	Correction of correction to profile tables for use of 3GPP IM CN subsystem XML body in restoration procedures	11.4.0	11.5.0	C1-123172
2012-09	CP-57	CP-120588	4085	1	Correcting profile tables for use of 3GPP IM CN subsystem XML body in response to request for emergency services	11.4.0	11.5.0	C1-123168
2012-09	CP-57	CP-120582	4088	2	Reference update and technical changes: draft-ietf-sipcore-proxy-feature	11.4.0	11.5.0	C1-123348
2012-09	CP-57	CP-120601	4089	1	Annex A: P-Access-Network-Info in ACK	11.4.0	11.5.0	C1-123256
2012-09	CP-57	CP-120569	4094	1	Correction of SDP Profile about RFC 4145	11.4.0	11.5.0	C1-123104
2012-09	CP-57	CP-120582	4097	1	Feature-Caps header field part of trust domain	11.4.0	11.5.0	C1-123158
2012-09	CP-57	CP-120599	4098		Removing an EN regarding missing charging related headers	11.4.0	11.5.0	C1-122680
2012-09	CP-57	CP-120603	4099		Removing an EN regarding preservation of parameters in AS	11.4.0	11.5.0	C1-122686
2012-09	CP-57	CP-120603	4100		Removing EN regarding number normalization and enum translation	11.4.0	11.5.0	C1-122687
2012-09	CP-57	CP-120603	4107	1	IOI usage between TRF and terminating side	11.4.0	11.5.0	C1-123284
2012-09	CP-57	CP-120603	4108	1	Co-location of TRF	11.4.0	11.5.0	C1-123285
2012-09	CP-57	CP-120569	4114	1	Correct handling of PPR in S-CSCF	11.4.0	11.5.0	C1-123109
2012-09	CP-57	CP-120583	4115	1	Correction ue initiated deregistration	11.4.0	11.5.0	C1-123295
2012-09	CP-57	CP-120577	4118	1	Condition for usage of Session-ID header filed within MESSAGE response	11.4.0	11.5.0	C1-123134
2012-09	CP-57	CP-120597	4119		Reference update: draft-ietf-mediactrl-mrb	11.4.0	11.5.0	C1-122765
2012-09	CP-57	CP-120597	4120	3	Visited network MRB information	11.4.0	11.5.0	C1-123396

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2012-09	CP-57	CP-120583	4121	3	PCSCF discovery clarification	11.4.0	11.5.0	C1-123440
2012-09	CP-57	CP-120600	4122	2	Clarifications of used identities for registration procedures	11.4.0	11.5.0	C1-123355
2012-09	CP-57	CP-120588	4123	3	DVB-RCS2 satellite access network as IP-CAN for IMS	11.4.0	11.5.0	C1-123428
2012-09	CP-57	CP-120600	4124	3	Add reg-event changes for RFC 6140	11.4.0	11.5.0	C1-123378
2012-09	CP-57	CP-120583	4126	1	P-CSCF registration context lost	11.4.0	11.5.0	C1-123297
2012-09	CP-57	CP-120588	4131	1	Correction DHCP mechanism for P-CSCF discovery in Annex M	11.4.0	11.5.0	C1-123293
2012-09	CP-57	CP-120588	4132	1	Correction to DHCP mechanism for P-CSCF discovery in Annex O	11.4.0	11.5.0	C1-123294
2012-09	CP-57	CP-120588	4133		Correction to Annex 9.2	11.4.0	11.5.0	C1-122849
2012-09	CP-57	CP-120570	4137	1	Reference update: draft-ietf-mmusic-ice-tcp	11.4.0	11.5.0	C1-123130
2012-09	CP-57	CP-120601	4140	1	Network provided location information inserted by the MSC server enhanced for ICS	11.4.0	11.5.0	C1-123259
2012-09	CP-57	CP-120591	4146		Specification of ISC gateway function – general clauses	11.4.0	11.5.0	C1-122928
2012-09	CP-57	CP-120591	4147	1	Specification of ISC gateway function – SIP procedures	11.4.0	11.5.0	C1-123271
2012-09	CP-57	CP-120591	4148	1	Specification of application gateway function – SDP procedures	11.4.0	11.5.0	C1-123272
2012-09	CP-57	CP-120588	4151		Reversal of terminology change in annex D	11.4.0	11.5.0	C1-122939
2012-09	CP-57	CP-120588	4152	1	Emergency priority using the Resource-Priority header field	11.4.0	11.5.0	C1-123173
2012-09	CP-57	CP-120641	4153	3	Description of overload control	11.4.0	11.5.0	-
2012-09	CP-57	CP-120642	4154	3	Support of overload control	11.4.0	11.5.0	-
2012-09	CP-57	CP-120664	4156	5	Media plane security	11.4.0	11.5.0	-
2012-09	CP-57	CP-120576	4159	1	mediasec ref deletions	11.4.0	11.5.0	C1-123141
2012-09	CP-57	CP-120603	4162		Updates to charging introduction for RAVEL	11.4.0	11.5.0	C1-122968
2012-09	CP-57	CP-120588	4163	1	Condition for restoration procedures causing UE reregistration	11.4.0	11.5.0	C1-123174
2012-09	CP-57	CP-120588	4164	1	Missing procedure for NASS-IMS bundled authentication at S-CSCF	11.4.0	11.5.0	C1-123171
2012-09	CP-57	CP-120574	4180	2	Emergency and normal registration independence	11.4.0	11.5.0	C1-123374
2012-09	CP-57	CP-120606	4185	2	Support of T.38 related SDP attributes	11.4.0	11.5.0	C1-123400
2012-09	CP-57	CP-120578	4188		Correcting incorrect references in P-CSCF procedures when emergency call failure occurs	11.4.0	11.5.0	C1-123164
2012-09	CP-57	CP-120656	4189	1	Reference list correction to align with the corrected TS 29.212 title	11.4.0	11.5.0	-
2012-12	CP-58	CP-120802	4106	5	Additional guidance on use of 3xx responses	11.5.0	11.6.0	C1-124983
2012-12	CP-58	CP-120802	4127	2	Update the general requirements for tunnel procedures	11.5.0	11.6.0	C1-123886
2012-12	CP-58	CP-120802	4128	4	IP address obtained on S2a interface	11.5.0	11.6.0	C1-124274

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2012-12	CP-58	CP-120802	4129	4	Tunnel modification by the UE	11.5.0	11.6.0	C1-124275
2012-12	CP-58	CP-120802	4130	5	Tunnel modification by the network	11.5.0	11.6.0	C1-124276
2012-12	CP-58	CP-120804	4149	3	Specification of ISC gateway function – SIP profile	11.5.0	11.6.0	C1-124261
2012-12	CP-58	CP-120804	4150	2	Specification of application gateway function – SDP profile	11.5.0	11.6.0	C1-124100
2012-12	CP-58	CP-120787	4187	1	IANA registration of OMR parameters	11.5.0	11.6.0	C1-123575
2012-12	CP-58	CP-120783	4196		Delete IETF mediasec draft reference	11.5.0	11.6.0	C1-123520
2012-12	CP-58	CP-120783	4197		IMS media security profile table cleanup	11.5.0	11.6.0	C1-123521
2012-12	CP-58	CP-120793	4201	4	Contents of From and To header fields in SUBSCRIBE message	11.5.0	11.6.0	C1-124950
2012-12	CP-58	CP-120802	4202	4	Correction on handling of rn parameter and npdi parameter at S-CSCF.	11.5.0	11.6.0	C1-125015
2012-12	CP-58	CP-120821	4203	2	Support of T.38 SDP attributes in IMS	11.5.0	11.6.0	C1-124158
2012-12	CP-58	CP-120801	4204	1	Transit-ioi is removed from fowarded message to visited network	11.5.0	11.6.0	C1-124092
2012-12	CP-58	CP-120815	4205	5	Overload control clarifications	11.5.0	11.6.0	C1-125010
2012-12	CP-58	CP-120812	4206	3	Removing ENs about IBCF and OMR	11.5.0	11.6.0	C1-124262
2012-12	CP-58	CP-120812	4207		Removing an EN regarding PSI	11.5.0	11.6.0	C1-123620
2012-12	CP-58	CP-120812	4212		Correcting the UE-originating case indication	11.5.0	11.6.0	C1-123630
2012-12	CP-58	CP-120793	4214	6	Correcting procedures for re-establishment a context for SIP signalling	11.5.0	11.6.0	C1-124952
2012-12	CP-58	CP-120773	4219	4	Correction of emergency sub-service type handling	11.5.0	11.6.0	C1-124284
2012-12	CP-58	CP-120793	4220	2	Remaining corrections to emergency sub-service type handling	11.5.0	11.6.0	C1-124181
2012-12	CP-58	CP-120782	4223	2	Corrections to E-CSCF and LRF handling for emergency calls	11.5.0	11.6.0	C1-124764
2012-12	CP-58	CP-120812	4224		Application servers and RAVEL	11.5.0	11.6.0	C1-123577
2012-12	CP-58	CP-120777	4228	1	Correction of 3GPP IM CN subsystem XML handling	11.5.0	11.6.0	C1-123972
2012-12	CP-58	CP-120793	4229	3	PCSCF discovery Annex L editorial	11.5.0	11.6.0	C1-124989
2012-12	CP-58	CP-120776	4234		Table A.162, item 61 referencing incorrect document	11.5.0	11.6.0	C1-123669
2012-12	CP-58	CP-120802	4235	1	SDP impacts due to IP-CAN bearer release	11.5.0	11.6.0	C1-124006
2012-12	CP-58	CP-120802	4236	1	Precondition and INVITE without SDP offer	11.5.0	11.6.0	C1-124007
2012-12	CP-58	CP-120802	4237	1	User rejecting media stream during set up of multimedia session	11.5.0	11.6.0	C1-124008
2012-12	CP-58	CP-120812	4239	1	Decission on loop back routing in S-CSCF	11.5.0	11.6.0	C1-124102
2012-12	CP-58	CP-120793	4241	1	Correct Defintion of Temporarily Authorized Resource-Priority	11.5.0	11.6.0	C1-124003
2012-12	CP-58	CP-120791	4243	2	Reference update: draft-ietf-sipcore-proxy-feature	11.5.0	11.6.0	C1-124766

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2012-12	CP-58	CP-120791	4244	1	Feature-Caps header field in target refresh requests and responses	11.5.0	11.6.0	C1-124122
2012-12	CP-58	CP-120804	4247	1	Specification of application gateway function – SDP procedures	11.5.0	11.6.0	C1-124098
2012-12	CP-58	CP-120810	4248	1	Profiles change for P-Access-Network-Info header	11.5.0	11.6.0	C1-124111
2012-12	CP-58	CP-120810	4249	3	Correction to the coding of UE-time-zone	11.5.0	11.6.0	C1-124273
2012-12	CP-58	CP-120810	4250	2	Removal of Editor's Note on NPLI inserted by both P-CSCF and AS	11.5.0	11.6.0	C1-124223
2012-12	CP-58	CP-120793	4251	1	Correct emergency call description when roaming	11.5.0	11.6.0	C1-124116
2012-12	CP-58	CP-120782	4254		Dialog state notification clarification	11.5.0	11.6.0	C1-123752
2012-12	CP-58	CP-120785	4257	1	Emergency and normal registration independence	11.5.0	11.6.0	C1-123994
2012-12	CP-58	CP-120815	4258	5	Overload control -Inconstancies correction	11.5.0	11.6.0	C1-125009
2012-12	CP-58	CP-120815	4262	3	Event-based overload control procedures	11.5.0	11.6.0	C1-124856
2012-12	CP-58	CP-120780	4263	2	Correction to integrity-protected usage in S-CSCF	11.5.0	11.6.0	C1-124150
2012-12	CP-58	CP-120809	4273		Mz Reference Point – ISC alternative	11.5.0	11.6.0	C1-124301
2012-12	CP-58	CP-120812	4275	1	Removing the g.3gpp.loopback in TRF	11.5.0	11.6.0	C1-124850
2012-12	CP-58	CP-120788	4281		Reference update: RFC 6679	11.5.0	11.6.0	C1-124369
2012-12	CP-58	CP-120775	4285	3	Updating IMEI URN draft reference	11.5.0	11.6.0	C1-125006
2012-12	CP-58	CP-120793	4286	2	Default ICSI value selected by S-CSCF	11.5.0	11.6.0	C1-124951
2012-12	CP-58	CP-120810	4287		Correction of "UE-time-zone" to "local-time-zone" in TS 24.229	11.5.0	11.6.0	C1-124433
2012-12	CP-58	CP-120801	4288		Transit IOI general description	11.5.0	11.6.0	C1-124439
2012-12	CP-58	CP-120801	4289	1	Including transit-IOI in SIP responses	11.5.0	11.6.0	C1-124843
2012-12	CP-58	CP-120802	4294	1	Removal of internal references from IBCF procedures	11.5.0	11.6.0	C1-124772
2012-12	CP-58	CP-120815	4296	2	Closure of open issues in IOC work item	11.5.0	11.6.0	C1-125008
2012-12	CP-58	CP-120793	4299	1	P-CSCF registration context lost – text correction	11.5.0	11.6.0	C1-124896
2012-12	CP-58	CP-120793	4300		NAT detection by the UE- text correction	11.5.0	11.6.0	C1-124521
2012-12	CP-58	CP-120780	4305		Correction on integrity-protected handling in S-CSCF	11.5.0	11.6.0	C1-124528
2012-12	CP-58	CP-120793	4311	1	Correction to challenge response examination in P-CSCF	11.5.0	11.6.0	C1-124903
2013-03	CP-59	CP-130093	4139	6	IBCF don't change the dialog-ID	11.6.0	11.7.0	C1-130555
2013-03	CP-59	CP-130117	4238	3	Network provided cell identity for UTRAN	11.6.0	11.7.0	C1-130501
2013-03	CP-59	CP-130110	4246	4	Specification of ISC gateway function – SIP procedures	11.6.0	11.7.0	C1-130336
2013-03	CP-59	CP-130117	4301	5	Network provided Geographical Identifier	11.6.0	11.7.0	C1-130915

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2013-03	CP-59	CP-130099	4314	4	Correct determination of the status of being available for voice over PS	11.6.0	11.7.0	C1-130892
2013-03	CP-59	CP-130116	4315	3	Correcting availability for calls when dedicated bearer suitable for SIP signalling is used	11.6.0	11.7.0	C1-130897
2013-03	CP-59	CP-130117	4324	1	P-CSCF handling of PANI in case PCRF is used	11.6.0	11.7.0	C1-130502
2013-03	CP-59	CP-130098	4327	1	Security Client header in case of Media Sec	11.6.0	11.7.0	C1-130536
2013-03	CP-59	CP-130107	4331	1	Support for P-Charging-Vector in SIP ACK	11.6.0	11.7.0	C1-130512
2013-03	CP-59	CP-130107	4332	1	Replacement of RFC3455 by draft-drage-sipping-rfc3455bis-06	11.6.0	11.7.0	C1-130564
2013-03	CP-59	CP-130107	4333	1	Clarification of handling IOI parameter at S-CSCF	11.6.0	11.7.0	C1-130565
2013-03	CP-59	CP-130107	4334	3	Clarification of handling IOI parameters at transit function	11.6.0	11.7.0	C1-130895
2013-03	CP-59	CP-130102	4337		Reference update: RFC 6809	11.6.0	11.7.0	C1-130143
2013-03	CP-59	CP-130116	4440		Correcting reference to PSU extension	11.6.0	11.7.0	C1-130164
2013-03	CP-59	CP-130116	4441	1	ICSIs in Contact and Accept-Contact header fields	11.6.0	11.7.0	C1-130746
2013-03	CP-59	CP-130101	4447	2	Annex A updates for RFC5506	11.6.0	11.7.0	C1-130797
2013-03	CP-59	CP-130101	4450	2	Annex A updates for RFC3611	11.6.0	11.7.0	C1-130814
2013-03	CP-59	CP-130116	4451	2	Annex A updates for 3gpp_MaxRecvSDUSize SDP attributes	11.6.0	11.7.0	C1-130827
2013-03	CP-59	CP-130094	4462	1	Correction of subclause reference in dialstring related procedures	11.6.0	11.7.0	C1-130739
2013-03	CP-59	CP-130098	4469		Delete IETF mediasec draft reference	11.6.0	11.7.0	C1-130328
2013-03	CP-59	CP-130101	4471		Correction of references to annex G and annex F	11.6.0	11.7.0	C1-130334
2013-03	CP-59	CP-130121	4472		Interaction between overload control mechanisms	11.6.0	11.7.0	C1-130337
2013-03	CP-59	CP-130098	4480	1	IANA registration issues for a=3ge2ae	11.6.0	11.7.0	C1-130539
2013-03	CP-59	CP-130096	4483	1	Profile changes relating to CRS	11.6.0	11.7.0	C1-130542
2013-03	CP-59	CP-130135	4456	2	Improvements to the UE Keepalive Procedure for UE managed NAT Traversal	11.7.0	12.0.0	C1-130901
2013-03	CP-59	CP-130135	4461		Correction of header styles	11.7.0	12.0.0	C1-130269
2013-03	CP-59	CP-130188	4477	3	Emergency service URN for country specific types of emergency service	11.7.0	12.0.0	-
2013-06	CP-60	CP-130261	4448	2	Annex A updates for RFC5285	12.0.0	12.1.0	C1-132481
2013-06	CP-60	CP-130262	4449	3	Annex A updates for RFC6236	12.0.0	12.1.0	C1-132315
2013-06	CP-60	CP-130268	4476	4	PSAP Callback Indicator	12.0.0	12.1.0	C1-131784
2013-06	CP-60	CP-130222	4499	1	Updating MMUSIC draft references for ICS User Agent and SCC Application Server	12.0.0	12.1.0	C1-132292
2013-06	CP-60	CP-130240	4502	1	Reference update of draft-atarius-dispatch-meid-urn	12.0.0	12.1.0	C1-131543
2013-06	CP-60	CP-130257	4503		Missplaced release procedure	12.0.0	12.1.0	C1-131038

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2013-06	CP-60	CP-130257	4504	4	P-Visited-Network-ID clarifications	12.0.0	12.1.0	C1-131775
2013-06	CP-60	CP-130227	4505	1	Not reachable for SIP signalling due to unclear requirement on re-establishment of the SIP signalling bearer	12.0.0	12.1.0	C1-131410
2013-06	CP-60	CP-130231	4506	3	UE indicating type of emergency service when 380 response is received	12.0.0	12.1.0	C1-131924
2013-06	CP-60	CP-130257	4507	5	EN correction on IANA registration	12.0.0	12.1.0	C1-131851
2013-06	CP-60	CP-130387	4508	2	Adjacent Network Indicator	12.0.0	12.1.0	-
2013-06	CP-60	CP-130257	4509	2	IANA registered URN for emergency service offered in one country only	12.0.0	12.1.0	C1-131741
2013-06	CP-60	CP-130252	4516		Syntax for "local-time-zone" incorrect	12.0.0	12.1.0	C1-131084
2013-06	CP-60	CP-130257	4517	2	Adding the procedure of the "icid-value" header field parameter in SIP response	12.0.0	12.1.0	C1-131714
2013-06	CP-60	CP-130242	4529	1	"transit-ioi" header field parameter correction	12.0.0	12.1.0	C1-131633
2013-06	CP-60	CP-130253	4531		Removing RAVEL EN related to charging indicators	12.0.0	12.1.0	C1-131151
2013-06	CP-60	CP-130253	4533	1	IOI exchange when loopback routeing occurs	12.0.0	12.1.0	C1-131621
2013-06	CP-60	CP-130253	4535	1	Handling of transit IOI when loopback routeing occurs	12.0.0	12.1.0	C1-131623
2013-06	CP-60	CP-130265	4537	1	Type 1 IOI for originating leg P-CSCF to S-CSCF	12.0.0	12.1.0	C1-131614
2013-06	CP-60	CP-130265	4539	1	Type 1 IOI for terminating leg P-CSCF to S-CSCF	12.0.0	12.1.0	C1-131615
2013-06	CP-60	CP-130265	4541	3	Type 1 IOI P-CSCF to S-CSCF for SUBSCRIBE	12.0.0	12.1.0	C1-132480
2013-06	CP-60	CP-130265	4547	5	Resolving IMS dangling session issue	12.0.0	12.1.0	C1-132658
2013-06	CP-60	CP-130230	4548	1	Emergency bearer support and IMS Emergency correction	12.0.0	12.1.0	C1-131414
2013-06	CP-60	CP-130233	4557	3	Removal of dialled digits option from emergency call generation	12.0.0	12.1.0	C1-132564
2013-06	CP-60	CP-130251	4569	2	Removal of Editor's Note on maximum number of registration flows	12.0.0	12.1.0	C1-131635
2013-06	CP-60	CP-130257	4576	2	Phone-context : text clarification	12.0.0	12.1.0	C1-132668
2013-06	CP-60	CP-130220	4585	3	Prevent receipt of normal call at the UE while it is attached for emergency services only	12.0.0	12.1.0	C1-132627
2013-06	CP-60	CP-130257	4586	1	S-CSCF procedures for inserting the P-Acces-Network-Info header	12.0.0	12.1.0	C1-132322
2013-06	CP-60	CP-130265	4587		Correcting the list of header fields parameters in 4.5.1	12.0.0	12.1.0	C1-131908
2013-06	CP-60	CP-130265	4588	1	Reason header field in CANCEL	12.0.0	12.1.0	C1-132478
2013-06	CP-60	CP-130388	4589	4	PSAP Callback Indicator: Annex A	12.0.0	12.1.0	-
2013-06	CP-60	CP-130257	4590		Correction on title of IETF RFC 5031	12.0.0	12.1.0	C1-131945
2013-06	CP-60	CP-130257	4591	1	Encoding for "integrity-protected" parameter	12.0.0	12.1.0	C1-132323
2013-06	CP-60	CP-130265	4604	1	Editorial CR against UE procedures for emergency services	12.0.0	12.1.0	C1-132483

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2013-06	CP-60	CP-130225	4609	2	Consolidation of DTMF info package definition in 3GPP TS 24.229	12.0.0	12.1.0	C1-132580
2013-06	CP-60	CP-130265	4611		Rport for UDP only	12.0.0	12.1.0	C1-132126
2013-09	CP-61	CP-130504	4574	1	Indication of network supported services to the UE in Feature-Caps header	12.1.0	12.2.0	C1-133633
2013-09	CP-61	CP-130504	4575	6	Nonce caching handling to optimize authentication procedures	12.1.0	12.2.0	C1-133661
2013-09	CP-61	CP-130507	4602	3	Ignore a PSAP callback indicator defined in draft-ietf-ecrit-psap-callback at the UE	12.1.0	12.2.0	C1-133522
2013-09	CP-61	CP-130481	4616	1	Updating IMEI URN draft reference	12.1.0	12.2.0	C1-133252
2013-09	CP-61	CP-130513	4617	1	Updating TS 24.229 to RFC 6665	12.1.0	12.2.0	C1-133236
2013-09	CP-61	CP-130511	4620		Clarification of Service-Route header usage	12.1.0	12.2.0	C1-132718
2013-09	CP-61	CP-130504	4621	3	Clarifying that ISO 3166-1 alpha-2 codes are not neccesarily country codes	12.1.0	12.2.0	C1-133631
2013-09	CP-61	CP-130512	4622	2	Access via WLAN connected to EPC when WLAN is a restrictive non-3GPP access network	12.1.0	12.2.0	C1-133575
2013-09	CP-61	CP-130504	4623	2	Fixing issues in RTP media security	12.1.0	12.2.0	C1-133524
2013-09	CP-61	CP-130504	4626	2	Correct "otherwise" condition, causing emergency call failure or delay	12.1.0	12.2.0	C1-133525
2013-09	CP-61	CP-130493	4629		Via header field handling on Privacy protection at IBCF	12.1.0	12.2.0	C1-132756
2013-09	CP-61	CP-130483	4634	1	Reference update of draft-vanelburg-dispatch-private-network-ind	12.1.0	12.2.0	C1-133265
2013-09	CP-61	CP-130511	4635	2	P-CSCF discovery	12.1.0	12.2.0	C1-133579
2013-09	CP-61	CP-130511	4638	2	S-CSCF matching between ICSIs and SIP request content	12.1.0	12.2.0	C1-133636
2013-09	CP-61	CP-130504	4639	1	Overload control application	12.1.0	12.2.0	C1-133476
2013-09	CP-61	CP-130504	4640		Correction on the supported SIP status codes profile	12.1.0	12.2.0	C1-132824
2013-09	CP-61	CP-130511	4642	1	Priority Consideration for SIP Overload Control	12.1.0	12.2.0	C1-133310
2013-09	CP-61	CP-130511	4643	3	Switching to CS domain when the PS voice call initiation is failed due to access class barring (24.229) alternative2	12.1.0	12.2.0	C1-133638
2013-09	CP-61	CP-130501	4648	3	Identifying the home network to TRF	12.1.0	12.2.0	C1-133611
2013-09	CP-61	CP-130511	4652	3	Clarification on UE reaction to IP-CAN bearer changes during a VoLTE/VT call	12.1.0	12.2.0	C1-133639
2013-09	CP-61	CP-130512	4653	2	Tunnelling over restrictive access networks IMS case	12.1.0	12.2.0	C1-133580
2013-09	CP-61	CP-130504	4654		draft-ietf-ecrit-psap-callback reference update	12.1.0	12.2.0	C1-133024
2013-09	CP-61	CP-130507	4655		draft-ietf-ecrit-psap-callback reference update	12.1.0	12.2.0	C1-133025
2013-09	CP-61	CP-130507	4656		Addition of AS to profile tables	12.1.0	12.2.0	C1-133031
2013-09	CP-61	CP-130488	4666	1	Fixing essential issues in RTP media security	12.1.0	12.2.0	C1-133505
2013-12	CP-62	CP-130749	4624	3	End-to-access-edge media security for MSRP, BFCP and UDPTL	12.2.0	12.3.0	C1-134612

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2013-12	CP-62	CP-130753	4660	5	Include DST adjustment in local time zone	12.2.0	12.3.0	C1-135174
2013-12	CP-62	CP-130753	4667	3	Adding country-specific URN to configurable lists	12.2.0	12.3.0	C1-134510
2013-12	CP-62	CP-130758	4669	2	Improve requirement to ignore the PSAP callback indicator	12.2.0	12.3.0	C1-134330
2013-12	CP-62	CP-130722	4677	1	Reference update of draft-vanelburg-dispatch-private-network-ind	12.2.0	12.3.0	C1-134200
2013-12	CP-62	CP-130728	4681		Reference update: draft-kaplan-insipid-session-id	12.2.0	12.3.0	C1-133695
2013-12	CP-62	CP-130748	4683	1	Reference update: draft-ietf-soc-overload-control, draft-ietf-soc-overload-rate-control and draft-ietf-soc-load-control-event-package	12.2.0	12.3.0	C1-134439
2013-12	CP-62	CP-130743	4685	1	Reference update: RFC 6917 (draft-ietf-mediactrl-mrb)	12.2.0	12.3.0	C1-134189
2013-12	CP-62	CP-130764	4686	1	Editor's note in access via WLAN connected to EPC when WLAN is a restrictive non-3GPP access network	12.2.0	12.3.0	C1-134102
2013-12	CP-62	CP-130804	4687	2	Solution for tunnelling of IMS services over restrictive non-3GPP access networks - keep-alive using RFC 6223	12.2.0	12.3.0	-
2013-12	CP-62	CP-130721	4692	1	Reference update: RFC 7006 (draft-ietf-mmusic-sdp-miscellaneous-caps)	12.2.0	12.3.0	C1-134210
2013-12	CP-62	CP-130766	4694	2	Addition of Definitions for Business Trunking	12.2.0	12.3.0	C1-134447
2013-12	CP-62	CP-130766	4695	2	Addition of Business Trunking Features for Signalling Security	12.2.0	12.3.0	C1-134448
2013-12	CP-62	CP-130766	4696	3	Addition of Business Trunking Features at UE	12.2.0	12.3.0	C1-134478
2013-12	CP-62	CP-130767	4697	3	Addition of Business Trunking Features at P-CSCF	12.2.0	12.3.0	C1-134975
2013-12	CP-62	CP-130753	4701	3	Nonce caching and digest authentication-corrections	12.2.0	12.3.0	C1-134525
2013-12	CP-62	CP-130805	4712	4	Tunnelling over restrictive access networks IMS case	12.2.0	12.3.0	-
2013-12	CP-62	CP-130733	4716	3	Use of ECN in Multimedia Priority Service	12.2.0	12.3.0	C1-134374
2013-12	CP-62	CP-130814	4717	5	Updating the depiction of the creation of subscription dialog	12.2.0	12.3.0	-
2013-12	CP-62	CP-130729	4722		Fixing remaining errors in media security	12.2.0	12.3.0	C1-133921
2013-12	CP-62	CP-130753	4723	3	Correction to formatting of text in Abnormal cases related to mobile originated sessions	12.2.0	12.3.0	C1-135173
2013-12	CP-62	CP-130753	4724	1	Correcting table number for IM CN Subsystem XML body schema	12.2.0	12.3.0	C1-135172
2013-12	CP-62	CP-130731	4730	4	Correction when receiving a 380 (Alternative Service) response indicating "emergency" to UE non-detectable IMS emergency call	12.2.0	12.3.0	C1-134906
2013-12	CP-62	CP-130753	4731	3	Time zone in Pacific/Kiritimati	12.2.0	12.3.0	C1-135175
2013-12	CP-62	CP-130753	4736	1	Reversal of change relating to digest authentication	12.2.0	12.3.0	C1-134233
2013-12	CP-62	CP-130720	4741		Reference Update: draft-ietf-salud-alert-info-urns	12.2.0	12.3.0	C1-134537

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2013-12	CP-62	CP-130753	4745	1	Number of retry attempts when receiving invalid challenges in IMS AKA procedure	12.2.0	12.3.0	C1-135109
2013-12	CP-62	CP-130736	4746	1	Incorrect preconditions for inclusion of P-Refused-URI-List and Retry-After header fields in INVITE	12.2.0	12.3.0	C1-135119
2013-12	CP-62	CP-130753	4747	1	Inclusion of b=RR and b=RS at session-level in SDP	12.2.0	12.3.0	C1-135111
2013-12	CP-62	CP-130744	4751		Correct Annex A due to 3xx response containing a Contact header field	12.2.0	12.3.0	C1-134571
2013-12	CP-62	CP-130753	4753	1	P-ANI encoding correction	12.2.0	12.3.0	C1-135009
2013-12	CP-62	CP-130753	4754	1	Proxy-Authentication-Info header field not defined for SIP	12.2.0	12.3.0	C1-135106
2013-12	CP-62	CP-130746	4756		utran-sai-id-3gpp and utran-sai-3gpp	12.2.0	12.3.0	C1-134610
2013-12	CP-62	CP-130749	4757		End-to-end media security for MSRP using TLS and KMS	12.2.0	12.3.0	C1-134613
2013-12	CP-62	CP-130758	4760	2	EN Removal: Support of the PSAP callback indicator in non-INVITE requests	12.2.0	12.3.0	C1-135149
2013-12	CP-62	CP-130747	4762	1	Home network bypassing of E.164 translation	12.2.0	12.3.0	C1-135008
2013-12	CP-62	CP-130721	4767	1	Updating IMEI URN draft reference	12.2.0	12.3.0	C1-134988
2013-12	CP-62	CP-130753	4769	2	Nodes that set the access-class parameter	12.2.0	12.3.0	C1-135180
2013-12	CP-62	CP-130753	4770	1	Correction term-ioi handling	12.2.0	12.3.0	C1-135176
2013-12	CP-62	CP-130770	4776	2	Mapping of Target-Dialog identifiers	12.2.0	12.3.0	C1-135151
2013-12	CP-62	CP-130758	4713	2	Reference update: draft-ietf-ecrit-psap-callback	12.2.0	12.3.0	C1-134650
2014-03	CP-63	CP-140153	4748	5	Usage of b=AS	12.3.0	12.4.0	C1-140636
2014-03	CP-63	CP-140126	4785	2	Clarify that SDP for sessions with voice media can be updated in a TA indicating voice over PS is not supported	12.3.0	12.4.0	C1-140401
2014-03	CP-63	CP-140150	4786	1	Reference update of draft-ietf-mmusic-udptl-dtls	12.3.0	12.4.0	C1-140593
2014-03	CP-63	CP-140146	4787		Solution for tunnelling of IMS services over restrictive non-3GPP access networks - keep-alive using RFC 6223 in Annex D	12.3.0	12.4.0	C1-140018
2014-03	CP-63	CP-140133	4789		Re-adding 3gpp body xml schema to archive	12.3.0	12.4.0	C1-140032
2014-03	CP-63	CP-140128	4791	2	Correction on syntax of values of feature capability indicators	12.3.0	12.4.0	C1-140661
2014-03	CP-63	CP-140115	4796		Reference update of draft-vanelburg-dispatch-private-network-ind	12.3.0	12.4.0	C1-140052
2014-03	CP-63	CP-140153	4797	1	Collection of transit-ioi example	12.3.0	12.4.0	C1-140611
2014-03	CP-63	CP-140115	4802	1	Correcting P-Private-Network-Indication descriptions in response	12.3.0	12.4.0	C1-140471
2014-03	CP-63	CP-140153	4804	1	Alignment between Annex I and the core specification	12.3.0	12.4.0	C1-140614
2014-03	CP-63	CP-140149	4805		Correction of P-CSCF procedures for Business Trunking	12.3.0	12.4.0	C1-140138
2014-03	CP-63	CP-140121	4809	3	Non UE detected emergency call correction for retry in CS domain	12.3.0	12.4.0	C1-140778

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2014-03	CP-63	CP-140149	4816	1	Business trunking in IBCF	12.3.0	12.4.0	C1-140587
2014-03	CP-63	CP-140128	4818		Correcting g.3gpp.home visited feature-capability indicator name	12.3.0	12.4.0	C1-140190
2014-03	CP-63	CP-140114	4823	1	Updating IMEI URN draft reference	12.3.0	12.4.0	C1-140500
2014-03	CP-63	CP-140113	4830	2	Allocating a GRUU when the explictly registered IMPU is barred	12.3.0	12.4.0	C1-140693
2014-03	CP-63	CP-140153	4842	2	daylight-saving-time definition clarifications	12.3.0	12.4.0	C1-140675
2014-03	CP-63	CP-140153	4843	1	Use of "daylight-saving-time" parameter	12.3.0	12.4.0	C1-140624
2014-03	CP-63	CP-140113	4845	2	Generation of Public GRUU	12.3.0	12.4.0	C1-140699
2014-03	CP-63	CP-140153	4846	1	P-CSCF and GRUU	12.3.0	12.4.0	C1-140610
2014-06	CP-64	CP-140332	4771	6	Firewall traversal for IMS services based on ICE	12.4.0	12.5.0	C1-141589
2014-06	CP-64	CP-140322	4855	3	Non UE detected emergency call correction: additional cases for retry attempt using EMERGENCY SETUP in CS domain	12.4.0	12.5.0	C1-142524
2014-06	CP-64	CP-140320	4856	1	IMS based telepresence: 24.229 SDP impacts	12.4.0	12.5.0	C1-141464
2014-06	CP-64	CP-140296	4858	3	Clarification of payload type usage for telephony-events	12.4.0	12.5.0	C1-141629
2014-06	CP-64	CP-140317	4860	2	Reference update: RFC 7090 (draft-ietf-ecrit-psap-callback)	12.4.0	12.5.0	C1-142349
2014-06	CP-64	CP-140296	4865	3	Corrections for voice centric UE to continue being reachable for IMS voice when P-CSCF serving the UE stops being available	12.4.0	12.5.0	C1-141555
2014-06	CP-64	CP-140318	4866	1	Reference update of draft-ietf-mmusic-udptl-dtls	12.4.0	12.5.0	C1-141481
2014-06	CP-64	CP-140322	4867		SIP timer table update due to RFC 6026	12.4.0	12.5.0	C1-140907
2014-06	CP-64	CP-140330	4868	1	Routeing to MSC server in S-CSCF	12.4.0	12.5.0	C1-141497
2014-06	CP-64	CP-140302	4871	1	IOI exchange between the SCC AS and ATCF	12.4.0	12.5.0	C1-141500
2014-06	CP-64	CP-140322	4888	5	Proactive media transcoding in IMS	12.4.0	12.5.0	C1-142533
2014-06	CP-64	CP-140324	4889	2	Indicating traffic leg in dialog creating and stand-alone requests	12.4.0	12.5.0	C1-142410
2014-06	CP-64	CP-140295	4894	1	Reference update of draft-vanelburg-dispatch-private-network-ind	12.4.0	12.5.0	C1-141806
2014-06	CP-64	CP-140322	4902	2	Support of RFC 3263 for P-CSCF discovery procedure	12.4.0	12.5.0	C1-141592
2014-06	CP-64	CP-140291	4908		Syntax for OLI is incorrect	12.4.0	12.5.0	C1-141035
2014-06	CP-64	CP-140294	4913	1	GRUU validation	12.4.0	12.5.0	C1-141367
2014-06	CP-64	CP-140322	4914	1	Issue with "Tokenized-by"	12.4.0	12.5.0	C1-141482
2014-06	CP-64	CP-140300	4922		Profile table changes for SDES media plane security role	12.4.0	12.5.0	C1-141057
2014-06	CP-64	CP-140318	4923		Profile tables changes for e2e media security for MSRP using TLS and KMS	12.4.0	12.5.0	C1-141058
2014-06	CP-64	CP-140318	4924		Profile tables changes for mediasec header field	12.4.0	12.5.0	C1-141059

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2014-06	CP-64	CP-140322	4930	3	Remote strength tag value in subsequent SDP offers	12.4.0	12.5.0	C1-141658
2014-06	CP-64	CP-140322	4934		Correction of I-CSCF IOI Note	12.4.0	12.5.0	C1-141141
2014-06	CP-64	CP-140322	4935		IOI in NOTIFY response	12.4.0	12.5.0	C1-141142
2014-06	CP-64	CP-140316	4936	4	Barring premium rate number	12.4.0	12.5.0	C1-142407
2014-06	CP-64	CP-140322	4938	2	Fail-safe emergency PDN connection release	12.4.0	12.5.0	C1-142504
2014-06	CP-64	CP-140307	4940	1	EN removal: Feature-capability indicator IANA registration status	12.4.0	12.5.0	C1-142256
2014-06	CP-64	CP-140311	4942	1	Correction of g.3gpp.icsi-ref feature-capability indicator definition	12.4.0	12.5.0	C1-142258
2014-06	CP-64	CP-140321	4943	1	Web Real Time Communication (WebRTC) Access to IMS: Annex A impacts	12.4.0	12.5.0	C1-142336
2014-06	CP-64	CP-140322	4944	3	Applicability of i-wlan-node-id parameter in P-Access-Network-Info	12.4.0	12.5.0	C1-142529
2014-06	CP-64	CP-140299	4961	2	Solution for editor's note related to draft-ietf-sipcore-rfc4244bis-00	12.4.0	12.5.0	C1-142451
2014-06	CP-64	CP-140292	4967	2	URI matching for terminating requests	12.4.0	12.5.0	C1-142441
2014-06	CP-64	CP-140324	4968	2	Traffic leg URI parameter in SUBSCRIBE from P-CSCF	12.4.0	12.5.0	C1-142413
2014-06	CP-64	CP-140324	4969	1	Trust domain for traffic leg URI parameter	12.4.0	12.5.0	C1-142362
2014-06	CP-64	CP-140324	4970	2	Traffic leg URI parameter – Annex A	12.4.0	12.5.0	C1-142414
2014-06	CP-64	CP-140294	4975	1	Updating CS-SDP draft reference to RFC 7195	12.4.0	12.5.0	C1-142241
2014-06	CP-64	CP-140322	4979	2	Session refresh request retry	12.4.0	12.5.0	C1-142525
2014-06	CP-64	CP-140304	4982	1	Update draft-atarius-dispatch-meid-urn	12.4.0	12.5.0	C1-142254
2014-06	CP-64	CP-140322	4984	2	Reference mismatch between Rel-11 and Rel-12	12.4.0	12.5.0	C1-142505
2014-06	CP-64	CP-140322	4985	1	Race case for subscription to reg-event	12.4.0	12.5.0	C1-142367
2014-06	CP-64	CP-140321	4988	3	WebRTC access to IMS general part	12.4.0	12.5.0	C1-142514
2014-06	CP-64	CP-140305	4992	1	Reference update for overload control	12.4.0	12.5.0	C1-142263
2014-06	CP-64	CP-140332	4993	1	Removal of TURAN material from non-supported access technology I-WLAN	12.4.0	12.5.0	C1-142140
2014-06	CP-64	CP-140315	4994	2	Reconstruction of identity text	12.4.0	12.5.0	C1-142512
2014-06	CP-64	CP-140321	4995	1	Support of webrtc functional entities	12.4.0	12.5.0	C1-142333
2014-06	CP-64	CP-140322	4996	2	Clarification on description of Loose-Route indication	12.4.0	12.5.0	C1-142460
2014-06	CP-64	CP-140324	4997	2	Indicating II-NNI traversal scenario - REGISTER	12.4.0	12.5.0	C1-142412
2014-06	CP-64	CP-140322	4998	1	Correction on the value of transit-ioi parameter.	12.4.0	12.5.0	C1-142337
2014-06	CP-64	CP-140324	4999	1	Indicating the use case towards TRF	12.4.0	12.5.0	C1-142358
2014-06	CP-64	CP-140306	5002		EN removal: Feature-capability indicator IANA registration status	12.4.0	12.5.0	C1-142476
2014-09	CP-65	CP-140649	4844	2	SDP renegotiation not impacting TLS for BFCP	12.5.0	12.6.0	C1-143235
2014-09	CP-65	CP-140656	4895	6	Capability indication by P-CSCF	12.5.0	12.6.0	C1-143287

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2014-09	CP-65	CP-140634	4949	3	Resolving interoperability problems caused by insufficiently documented UE precondition handling - alternative A	12.5.0	12.6.0	C1-143296
2014-09	CP-65	CP-140656	4956	3	Annex F : selection of the transport protocol by the P-CSCF	12.5.0	12.6.0	C1-143120
2014-09	CP-65	CP-140633	5012	1	Updating IMEI URN draft reference to RFC 7254	12.5.0	12.6.0	C1-143085
2014-09	CP-65	CP-140649	5013		Reference update: draft-ietf-mmusic-udptl-dtls	12.5.0	12.6.0	C1-142589
2014-09	CP-65	CP-140644	5015		EN removal: g.3gpp.icsi-ref feature-capability indicator IANA registration status	12.5.0	12.6.0	C1-142591
2014-09	CP-65	CP-140652	5016	2	History-Info anonymized in case of privacy	12.5.0	12.6.0	C1-143284
2014-09	CP-65	CP-140652	5017	1	History-Info optional header field parameters	12.5.0	12.6.0	C1-143232
2014-09	CP-65	CP-140652	5018	1	Updating references from RFC 4244 to RFC 7044	12.5.0	12.6.0	C1-143285
2014-09	CP-65	CP-140655	5019		WebRTC support of SCTP over DTLS	12.5.0	12.6.0	C1-142638
2014-09	CP-65	CP-140665	5020	3	Correction of P-CSCF 380 response message to triggering CS emergency calls	12.5.0	12.6.0	C1-143406
2014-09	CP-65	CP-140656	5021	1	Sync Failure during initial IMS registration	12.5.0	12.6.0	C1-143116
2014-09	CP-65	CP-140656	5022	1	Blocked Subsriber in S-CSCF	12.5.0	12.6.0	C1-143117
2014-09	CP-65	CP-140641	5025	1	ICE in IBCF	12.5.0	12.6.0	C1-143102
2014-09	CP-65	CP-140648	5026	1	IANA registration for permium rate tel URI parameter	12.5.0	12.6.0	C1-143204
2014-09	CP-65	CP-140658	5027		Reference update: draft-holmberg-dispatch-iotl	12.5.0	12.6.0	C1-142705
2014-09	CP-65	CP-140660	5028	4	Including IMSI in terminating INVITE for P-CSCF restoration	12.5.0	12.6.0	C1-143393
2014-09	CP-65	CP-140656	5029	2	Replacing tel URI after ENUM lookup clarification	12.5.0	12.6.0	C1-143288
2014-09	CP-65	CP-140658	5030	2	Identifying roaming architecture for voice over IMS with local breakout	12.5.0	12.6.0	C1-143282
2014-09	CP-65	CP-140658	5031	2	II-NNI traversal scenario - General	12.5.0	12.6.0	C1-143283
2014-09	CP-65	CP-140658	5032	1	Adding II-NNI traversal scenario by BGCF	12.5.0	12.6.0	C1-143227
2014-09	CP-65	CP-140658	5033	1	Indicatiing II-NNI traversal scenario in transit cases	12.5.0	12.6.0	C1-143228
2014-09	CP-65	CP-140658	5034	1	Using the "iotl" parameter in IBCF	12.5.0	12.6.0	C1-143229
2014-09	CP-65	CP-140655	5036	1	Reference update: draft-ietf-mmusic-sctp-sdp	12.5.0	12.6.0	C1-143215
2014-09	CP-65	CP-140646	5037		P-Preferred Identity reference correction	12.5.0	12.6.0	C1-142822
2014-09	CP-65	CP-140648	5038	2	Identifying when to translate a premium number in IBCF	12.5.0	12.6.0	C1-143278
2014-09	CP-65	CP-140660	5039	3	HSS based P-CSCF restoration procedures	12.5.0	12.6.0	C1-143391
2014-09	CP-65	CP-140660	5040	2	IBCF detecting a non-working P-CSCF for HSS based solution	12.5.0	12.6.0	C1-143381
2014-09	CP-65	CP-140656	5041	1	Correction of C1-142367, SUBSCRIBE race case	12.5.0	12.6.0	C1-143124
2014-09	CP-65	CP-140656	5042	4	Transit IOI handling towards an AS	12.5.0	12.6.0	C1-143394

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2014-09	CP-65	CP-140658	5043		Trust domain for iotl parameter	12.5.0	12.6.0	C1-142871
2014-09	CP-65	CP-140665	5044	2	Support of additional Reason header field containing information about bearer loss	12.5.0	12.6.0	C1-143335
2014-09	CP-65	CP-140665	5045	1	Creation of new protocol values within the Reason header field	12.5.0	12.6.0	C1-143017
2014-09	CP-65	CP-140665	5046	2	Adjustment of I-WLAN support from release 12	12.5.0	12.6.0	C1-143401
2014-09	CP-65	CP-140656	5047		Correction of transit-ioi handling at S-CSCF	12.5.0	12.6.0	C1-142881
2014-09	CP-65	CP-140654	5050		Telepresence annex A corrections	12.5.0	12.6.0	C1-142944
2014-09	CP-65	CP-140656	5052	1	Annex A – Correction of 380 response major capability	12.5.0	12.6.0	C1-143126
2014-09	CP-65	CP-140681	4932	7	AS determination of used registration	12.5.0	12.6.0	-
2014-12	CP-66	CP-140839	4698	7	Major Capabilities for Business Trunking	12.6.0	12.7.0	C1-144967
2014-12	CP-66	CP-140831	4955	12	Adding an option enabling P-CSCF to send 380 response triggering CS EMERGENCY SETUP to default PSAP	12.6.0	12.7.0	C1-145053
2014-12	CP-66	CP-140831	4983	5	Inclusion of "precondition" option-tag in messages sent by MT UE	12.6.0	12.7.0	C1-144006
2014-12	CP-66	CP-140829	5055		Reference update: RFC 7345 (draft-ietf-mmusic-udptl-dtls)	12.6.0	12.7.0	C1-143413
2014-12	CP-66	CP-140823	5057	1	Reference Update: RFC7315	12.6.0	12.7.0	C1-143983
2014-12	CP-66	CP-140831	5061	3	Replacing draft-kaplan-insipid-session-id with draft-ietf-insipid-session-id	12.6.0	12.7.0	C1-144331
2014-12	CP-66	CP-140831	5063	1	Correcting for termination of received-transit-ioi	12.6.0	12.7.0	C1-143995
2014-12	CP-66	CP-140831	5065	2	Including Transit IOI exchange cases for visited network	12.6.0	12.7.0	C1-144244
2014-12	CP-66	CP-140831	5066	4	Corrections of handling P-Charging-Vector for NOTIFY method	12.6.0	12.7.0	C1-144973
2014-12	CP-66	CP-140831	5068	1	Clarification of handling icid-value of P-Charging-Vector header	12.6.0	12.7.0	C1-143999
2014-12	CP-66	CP-140831	5069	1	Correction about handling icid-value at TRF	12.6.0	12.7.0	C1-144000
2014-12	CP-66	CP-140816	5074	1	Reference update from draft-vanelburg-dispatch-private-network-ind to RFC 7316	12.6.0	12.7.0	C1-143965
2014-12	CP-66	CP-140849	5075	5	Support of altc	12.6.0	12.7.0	C1-144373
2014-12	CP-66	CP-140826	5078	2	Editor's note on ICSI usage	12.6.0	12.7.0	C1-144772
2014-12	CP-66	CP-140841	5080	3	Alt2 Including IMSI in terminating INVITE for P-CSCF restoration	12.6.0	12.7.0	C1-144238
2014-12	CP-66	CP-140831	5082	1	Response of terminating UE to SDP offer including incapable media streams	12.6.0	12.7.0	C1-144005
2014-12	CP-66	CP-140865	5088	1	ICE for TCP in P-CSCF	12.6.0	12.7.0	C1-144096
2014-12	CP-66	CP-140839	5089		Clarifying how to identify a private network	12.6.0	12.7.0	C1-143663
2014-12	CP-66	CP-140852	5090	5	The "iotl" SIP URI parameter included in the destination used for charging	12.6.0	12.7.0	C1-144995
2014-12	CP-66	CP-140852	5091	2	Aligning the list of conditions	12.6.0	12.7.0	C1-144194
2014-12	CP-66	CP-140852	5093	2	Removing an editor's note in BGCF	12.6.0	12.7.0	C1-144241

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2014-12	CP-66	CP-140852	5094		Identifying further II-NNI traversal scenarios	12.6.0	12.7.0	C1-143681
2014-12	CP-66	CP-140837	5095	5	S-CSCF action when timer reg-await-auth is still running	12.6.0	12.7.0	C1-145001
2014-12	CP-66	CP-140831	5100	2	Correcting match conditions in 5.4.3.3	12.6.0	12.7.0	C1-144757
2014-12	CP-66	CP-140826	5104	1	EN: combination of emergency call types	12.6.0	12.7.0	C1-143993
2014-12	CP-66	CP-140837	5105	1	P-Charging-Function-Addresses only to P-CSCF in own network	12.6.0	12.7.0	C1-144141
2014-12	CP-66	CP-140865	5106	5	procedures using AKAv2 for WebRTC	12.6.0	12.7.0	C1-145047
2014-12	CP-66	CP-140841	5109	1	PCO list handling in P-CSCF restoration	12.6.0	12.7.0	C1-144121
2014-12	CP-66	CP-140845	5110	1	Correction of definitions	12.6.0	12.7.0	C1-144086
2014-12	CP-66	CP-140831	5112		Reference to tables from text	12.6.0	12.7.0	C1-143822
2014-12	CP-66	CP-140831	5113	2	Description of relaying charging parameters in subclause 4.5	12.6.0	12.7.0	C1-144282
2014-12	CP-66	CP-140831	5114	5	Definition of relayed charging parameters	12.6.0	12.7.0	C1-145056
2014-12	CP-66	CP-140841	5115	8	HSS-based P-CSCF restoration: S-CSCF procedures	12.6.0	12.7.0	C1-144997
2014-12	CP-66	CP-140841	5116	4	P-CSCF restoration: Annex A additions	12.6.0	12.7.0	C1-144375
2014-12	CP-66	CP-140841	5117	2	HSS based P-CSCF restoration: IBCF detecting non-working P-CSCF	12.6.0	12.7.0	C1-144199
2014-12	CP-66	CP-140841	5118	1	HSS based P-CSCF restoration; P-CSCF has restarted	12.6.0	12.7.0	C1-144130
2014-12	CP-66	CP-140831	5119	1	Clarifications for AS determines used registration	12.6.0	12.7.0	C1-144142
2014-12	CP-66	CP-140865	5120	5	The definition of WAF id and the related procedures	12.6.0	12.7.0	C1-144888
2014-12	CP-66	CP-140848	5121	3	NAT traversal for ETSI E2NA	12.6.0	12.7.0	C1-144878
2014-12	CP-66	CP-140852	5122		Reference update: draft-holmberg-dispatch-iotl	12.6.0	12.7.0	C1-144296
2014-12	CP-66	CP-140828	5128		Reference update: SIP overload control	12.6.0	12.7.0	C1-144304
2014-12	CP-66	CP-140831	5129	1	Minor corrections for phone-context	12.6.0	12.7.0	C1-144776
2014-12	CP-66	CP-140831	5130		RFC7044 reference correction	12.6.0	12.7.0	C1-144318
2014-12	CP-66	CP-140831	5132	1	Definition of type of emergency service	12.6.0	12.7.0	C1-144777
2014-12	CP-66	CP-140852	5133	1	Correction of confusing note	12.6.0	12.7.0	C1-144969
2014-12	CP-66	CP-140841	5134	3	Adding Restoration text to clause 4.	12.6.0	12.7.0	C1-144996
2014-12	CP-66	CP-140837	5135		Clarification of text on determination of originating or terminating case.	12.6.0	12.7.0	C1-144376
2014-12	CP-66	CP-140831	5138	1	Correcting E-CSCF P-Charging-Vector handling for SUBSCRIBE from LRF	12.6.0	12.7.0	C1-144794
2014-12	CP-66	CP-140831	5139	1	Clarification on IOI between a P-CSCF and an E-CSCF	12.6.0	12.7.0	C1-144795
2014-12	CP-66	CP-140852	5140	1	Updated proxy capability to include also I-CSCF	12.6.0	12.7.0	C1-144882
2014-12	CP-66	CP-140837	5141	1	Clarification of the use of transit-ioi	12.6.0	12.7.0	C1-144784

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2014-12	CP-66	CP-140838	5152	2	Movement of material in annex U to an access specific annex	12.6.0	12.7.0	C1-145048
2014-12	CP-66	CP-140856	5153		Support of network-initiated USSD operations in profile tables	12.6.0	12.7.0	C1-144554
2014-12	CP-66	CP-140837	5154		Removal of I-WLAN as IP-CAN	12.6.0	12.7.0	C1-144557
2014-12	CP-66	CP-140866	5157	1	3gpp_mtsi_app_adapt SDP attribute	12.6.0	12.7.0	C1-144899
2014-12	CP-66	CP-140837	5162		Make the "utran-sai-3gpp" parameter optional	12.6.0	12.7.0	C1-144951
2014-12	CP-66	CP-140860	5062	2	Reducing likelihood of ghost calls when precondition is not used, the terminating UE does not have resources available and IP-CAN performs network-initiated resource reservation for the terminating UE	12.7.0	13.0.0	C1-144898
2014-12	CP-66	CP-140857	5076		P-CSCF discovery using signalling for access to EPC via WLAN connected using S2a and S2b	12.7.0	13.0.0	C1-143589
2014-12	CP-66	CP-140861	5131	1	new cause-param value for service number translation	12.7.0	13.0.0	C1-144896
2014-12	CP-66	CP-140860	5136	1	No bandwidth information when port is zero	12.7.0	13.0.0	C1-144793
2014-12	CP-66	CP-140859	5148	1	Paging policy differentiation	12.7.0	13.0.0	C1-144897
2015-03	CP-67	CP-150082	4125	3	Hosted NAT traversal for MSRP media flows	13.0.0	13.1.0	C1-150791
2015-03	CP-67	CP-150079	4466	12	Determination of host part of SIP URI used in Request-URI when representing telephone numbers as SIP URIs	13.0.0	13.1.0	C1-150584
2015-03	CP-67	CP-150079	5035	6	Insertion of "operator-specific-GI" in PANI by the AS	13.0.0	13.1.0	C1-150785
2015-03	CP-67	CP-150082	5108	6	Reference to FQDN	13.0.0	13.1.0	C1-150342
2015-03	CP-67	CP-150079	5164	3	phone-context setting	13.0.0	13.1.0	C1-150897
2015-03	CP-67	CP-150176	5165	3	P-Early-Media & Preconditions interaction clarification	13.0.0	13.1.0	-
2015-03	CP-67	CP-150079	5167	2	Mapping between cell-id and Geolocation Identifier	13.0.0	13.1.0	C1-150789
2015-03	CP-67	CP-150082	5168	3	Not acquiring P-CSCF addresses when UE communicates with IM CN subsystem and handover between IP-CANs occurs	13.0.0	13.1.0	C1-150793
2015-03	CP-67	CP-150060	5178	3	Correction to attribute "a=inactive" in initial INVITE	13.0.0	13.1.0	C1-150740
2015-03	CP-67	CP-150050	5184		Reference update for UUSIW	13.0.0	13.1.0	C1-150110
2015-03	CP-67	CP-150058	5186		Abbreviations for eMEDIASEC-CT requirements	13.0.0	13.1.0	C1-150112
2015-03	CP-67	CP-150079	5187		Correction of reference identifier for RFC 6026	13.0.0	13.1.0	C1-150113
2015-03	CP-67	CP-150082	5188	1	TWAN Release Cause	13.0.0	13.1.0	C1-150591
2015-03	CP-67	CP-150074	5190		Reference update: draft-holmberg-dispatch-iotl	13.0.0	13.1.0	C1-150119
2015-03	CP-67	CP-150060	5192		Reference update: draft-holmberg-sipcore-received-realm	13.0.0	13.1.0	C1-150123
2015-03	CP-67	CP-150053	5195		Reference update: RFC 7135 (was draft-polk-local-emergency-rph-namespace	13.0.0	13.1.0	C1-150126
2015-03	CP-67	CP-150060	5197	1	Reference update: draft-ietf-insipid-session-id	13.0.0	13.1.0	C1-150554

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2015-03	CP-67	CP-150072	5199		IMS WebRTC reference updates	13.0.0	13.1.0	C1-150133
2015-03	CP-67	CP-150074	5201		Correction of syntax and description with regards to the iotl SIP URI parameter	13.0.0	13.1.0	C1-150171
2015-03	CP-67	CP-150060	5203	1	Correcting P-Charging-Vector for target refresh request on P-CSCF	13.0.0	13.1.0	C1-150556
2015-03	CP-67	CP-150175	5215	3	Correction of P-Charging-Vector access-network-charging-info syntax – Alt 2	13.0.0	13.1.0	-
2015-03	CP-67	CP-150044	5221	1	Handling of P-Charging-Function-Addresses	13.0.0	13.1.0	C1-150540
2015-03	CP-67	CP-150060	5223	1	"utran-sai-id-3gpp" vs ""utran-sai- 3gpp"	13.0.0	13.1.0	C1-150550
2015-03	CP-67	CP-150063	5225	2	PS_TO_CS_HANDOVER AVP in P-CSCF	13.0.0	13.1.0	C1-150778
2015-03	CP-67	CP-150084	5226		Resource sharing – Definitions	13.0.0	13.1.0	C1-150234
2015-03	CP-67	CP-150084	5227	2	Resource sharing – General part	13.0.0	13.1.0	C1-150725
2015-03	CP-67	CP-150084	5228	2	Resource sharing – Resource-Share header field	13.0.0	13.1.0	C1-150782
2015-03	CP-67	CP-150084	5229	1	Resource sharing – P-CSCF procedures	13.0.0	13.1.0	C1-150688
2015-03	CP-67	CP-150084	5231	1	Resource sharing – AS procedure	13.0.0	13.1.0	C1-150689
2015-03	CP-67	CP-150067	5233	3	Update REFER to reflect RFC 6665	13.0.0	13.1.0	C1-150875
2015-03	CP-67	CP-150066	5241	2	Clause 4 improvements: restoration procedures.	13.0.0	13.1.0	C1-150824
2015-03	CP-67	CP-150066	5243	2	Applicability statement improvement for the Restoration-Info	13.0.0	13.1.0	C1-150774
2015-03	CP-67	CP-150066	5245	2	P-CSCF restoration: Restoration-Info Annex A corrections	13.0.0	13.1.0	C1-150776
2015-03	CP-67	CP-150060	5247	1	Relayed-Charge header field in Annex A	13.0.0	13.1.0	C1-150558
2015-03	CP-67	CP-150060	5249	1	Addition of transit function to profile tables	13.0.0	13.1.0	C1-150560
2015-03	CP-67	CP-150082	5250	1	P-CSCF restoration name alignment	13.0.0	13.1.0	C1-150596
2015-03	CP-67	CP-150060	5252	2	Revisions to definition of Relayed-Charge header field	13.0.0	13.1.0	C1-150708
2015-03	CP-67	CP-150079	5255		Chargiong related definitions	13.0.0	13.1.0	C1-150316
2015-03	CP-67	CP-150082	5256	1	Addition of missing word SIP parameters in general description of trust domain	13.0.0	13.1.0	C1-150599
2015-03	CP-67	CP-150079	5258	1	Clarification of IOI between a P-CSCF and an E-CSCF on request from a user with/without emergency registration	13.0.0	13.1.0	C1-150574
2015-03	CP-67	CP-150080	5260		Reference update: draft-mohali-dispatch-cause-for-service-number	13.0.0	13.1.0	C1-150393
2015-06	CP-68	CP-150324	5163	6	Closure of TCP connections	13.1.0	13.2.0	C1-151649
2015-06	CP-68	CP-150328	5166	7	Determination of the registration duration by the S-CSCF	13.1.0	13.2.0	C1-152280
2015-06	CP-68	CP-150324	5169	3	Precondition and swap of 200 for UPDATE and 180/200 for INVITE	13.1.0	13.2.0	C1-151445
2015-06	CP-68	CP-150324	5175	5	Correcting undesired consequences of 503 response in P-CSCF terminating call handling	13.1.0	13.2.0	C1-151595
2015-06	CP-68	CP-150324	5176	8	Correcting IBCF hiding of the Service Route procedure	13.1.0	13.2.0	C1-151795

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2015-06	CP-68	CP-150324	5264	4	New header field for dynamic service interaction	13.1.0	13.2.0	C1-151651
2015-06	CP-68	CP-150287	5270	1	Reference Update: RFC7462 (alert-info urns)	13.1.0	13.2.0	C1-151400
2015-06	CP-68	CP-150297	5276	1	Clarifying condition for not routeing via S-CSCF for an emergency call	13.1.0	13.2.0	C1-151439
2015-06	CP-68	CP-150322	5277	2	Moving P-CSCF procedures to annex L	13.1.0	13.2.0	C1-151583
2015-06	CP-68	CP-150322	5278	1	Including resource sharing in annex A	13.1.0	13.2.0	C1-152150
2015-06	CP-68	CP-150322	5279		Removing S-CSCF impact	13.1.0	13.2.0	C1-150999
2015-06	CP-68	CP-150322	5280	2	Solving editor's note about INVITE without SDP offer	13.1.0	13.2.0	C1-151584
2015-06	CP-68	CP-150322	5281	1	Resource sharing per media stream in the AS	13.1.0	13.2.0	C1-151520
2015-06	CP-68	CP-150321	5282	4	P-CSCF priority order	13.1.0	13.2.0	C1-152149
2015-06	CP-68	CP-150322	5283	5	Resource sharing procedures and updated ABNF	13.1.0	13.2.0	C1-152269
2015-06	CP-68	CP-150324	5284		Removing MSC server enhanced for ICS from proxy major capabilities	13.1.0	13.2.0	C1-151008
2015-06	CP-68	CP-150324	5285		Defining an MSC server enhanced for SRVCC role in annex A	13.1.0	13.2.0	C1-151009
2015-06	CP-68	CP-150293	5290	1	Corrections to media plane security	13.1.0	13.2.0	C1-151428
2015-06	CP-68	CP-150324	5291	2	Correcting P-Charging-Vector for response to subsequent request on P-CSCF	13.1.0	13.2.0	C1-151590
2015-06	CP-68	CP-150324	5294	5	Correcting undesired consequences of 503 response in P-CSCF originating call handling	13.1.0	13.2.0	C1-152272
2015-06	CP-68	CP-150311	5302		Correction to Annex A table for p-cscf restoration	13.1.0	13.2.0	C1-151054
2015-06	CP-68	CP-150306	5305	1	IETF Update on IMS Telepresence	13.1.0	13.2.0	C1-151509
2015-06	CP-68	CP-150324	5313		Applying THIG on Path header field, using new Feature-capability indicator	13.1.0	13.2.0	C1-151165
2015-06	CP-68	CP-150294	5321	3	Update 3GPP2 referrence to reflect the correct published version	13.1.0	13.2.0	C1-151603
2015-06	CP-68	CP-150324	5322	1	Editorial change to eliminate confusion on UE supporting multiple registrations	13.1.0	13.2.0	C1-151449
2015-06	CP-68	CP-150324	5323	6	Expires in 3rd party REGISTER	13.1.0	13.2.0	C1-152496
2015-06	CP-68	CP-150324	5324	1	Correction Restoration-Info in Annex A	13.1.0	13.2.0	C1-151452
2015-06	CP-68	CP-150307	5326		Moving misplaced subclause 5.7.1.3B	13.1.0	13.2.0	C1-151230
2015-06	CP-68	CP-150300	5329	4	Translation of geo-local numbers	13.1.0	13.2.0	C1-152211
2015-06	CP-68	CP-150324	5330	2	Anonymous User Identity in the From header field	13.1.0	13.2.0	C1-151592
2015-06	CP-68	CP-150300	5334	3	Loopback indication in responses	13.1.0	13.2.0	C1-152263
2015-06	CP-68	CP-150321	5335		P-CSCF priority order text alignment	13.1.0	13.2.0	C1-151669
2015-06	CP-68	CP-150324	5336		Removal of redundant Via header from INVITE response table.	13.1.0	13.2.0	C1-151765
2015-06	CP-68	CP-150324	5338	1	Removal of redundant Require header from SUBSCRIBE 2xx response table.	13.1.0	13.2.0	C1-152225

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2015-06	CP-68	CP-150292	5344	1	Modification of missing Allow-Events header field status in annex A	13.1.0	13.2.0	C1-152172
2015-06	CP-68	CP-150324	5345	1	Support of Expires header field in responses to PUBLISH request in annex A	13.1.0	13.2.0	C1-152226
2015-06	CP-68	CP-150328	5347	1	Aligning TLS profiles used by CT1 specifications with SA3 agreed TLS profile	13.1.0	13.2.0	C1-152230
2015-06	CP-68	CP-150324	5349	1	Correction of profile status of Contact header field in annex A	13.1.0	13.2.0	C1-152227
2015-06	CP-68	CP-150324	5350		Correcting profile status of WWW-Authenticate and Proxy-Authenticate header field in annex A	13.1.0	13.2.0	C1-151790
2015-06	CP-68	CP-150324	5351	1	Modification of prerequisites for CANCEL response in annex A	13.1.0	13.2.0	C1-152228
2015-06	CP-68	CP-150324	5352	2	Correction of status code for the PUBLISH response in annex A	13.1.0	13.2.0	C1-152285
2015-06	CP-68	CP-150328	5353	1	UE accessing IM CN subsystem using PDP context provided by SGSN connected to S-GW and P-GW	13.1.0	13.2.0	C1-152060
2015-06	CP-68	CP-150328	5354		Not acquiring P-CSCF addresses when UE communicates with IM CN subsystem and handover to GPRS IP-CANs occurs	13.1.0	13.2.0	C1-151806
2015-06	CP-68	CP-150324	5355	2	Correcting undesired consequences of 503 response in MGCF handling	13.1.0	13.2.0	C1-152273
2015-06	CP-68	CP-150324	5364	2	Annex A status correction of Record-Route	13.1.0	13.2.0	C1-152274
2015-06	CP-68	CP-150324	5365		Annex A status correction of Supported header field	13.1.0	13.2.0	C1-151824
2015-06	CP-68	CP-150324	5366	1	Annex A status correction of Server header field	13.1.0	13.2.0	C1-152235
2015-06	CP-68	CP-150307	5368		Correction to P-Access-Network-Info	13.1.0	13.2.0	C1-151872
2015-06	CP-68	CP-150328	5369		Improvement of reference to S-CSCF restoration procedures	13.1.0	13.2.0	C1-151877
2015-06	CP-68	CP-150324	5370	1	Clarification on BSSID usage	13.1.0	13.2.0	C1-152240
2015-06	CP-68	CP-150318	5372		Reference update: RFC 7549 (draft-holmberg-dispatch-iotl)	13.1.0	13.2.0	C1-151906
2015-06	CP-68	CP-150324	5373	1	P-CSCF public user identity matching	13.1.0	13.2.0	C1-152246
2015-06	CP-68	CP-150289	5379	1	Content-Disposition for pidf+xml message bodies	13.1.0	13.2.0	C1-152161
2015-06	CP-68	CP-150324	5380	3	P-Early Media; Annex A corrections	13.1.0	13.2.0	C1-152295
2015-06	CP-68	CP-150307	5382	1	Corrections Relayed-Charge Annex A	13.1.0	13.2.0	C1-152221
2015-06	CP-68	CP-150307	5384	3	Relayed-charge clarifications	13.1.0	13.2.0	C1-152289
2015-06	CP-68	CP-150324	5385	1	Geographical Identifier insertion when TWAN is used	13.1.0	13.2.0	C1-152241
2015-06	CP-68	CP-150325	5386	3	Reference update: draft-mohali-dispatch-cause-for-service-number	13.1.0	13.2.0	C1-152497
2015-06	CP-68	CP-150325	5387		Service access number translation by an AS	13.1.0	13.2.0	C1-152006
2015-06	Post CT-68				Deletion of superfluous empty rows in tables of annex A	13.2.0	13.2.1	
2015-09	CP-69	CP-150520	5303	5	S-CSCF stores AS IP address	13.2.1	13.3.0	C1-153039

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2015-09	CP-69	CP-150520	5363	3	Annex A status correction of Record-Route for MESSAGE method	13.2.1	13.3.0	C1-152680
2015-09	CP-69	CP-150521	5388		Updated Reference: draft-mohali-dispatch-cause-for-service-number	13.2.1	13.3.0	C1-152519
2015-09	CP-69	CP-150521	5389	3	Service Access Number at terminating UE	13.2.1	13.3.0	C1-153271
2015-09	CP-69	CP-150530	5392		Incorrect reference to S-CSCF restoration procedures	13.2.1	13.3.0	C1-152569
2015-09	CP-69	CP-150535	5393		Resource sharing in CDMA2000 access	13.2.1	13.3.0	C1-152570
2015-09	CP-69	CP-150535	5394		Resolving the editor's notes about UE involvement	13.2.1	13.3.0	C1-152571
2015-09	CP-69	CP-150535	5395		Aligning the general description with the actual procedures	13.2.1	13.3.0	C1-152573
2015-09	CP-69	CP-150535	5396		Minor improvement of the P-CSCF resource sharing procedures	13.2.1	13.3.0	C1-152574
2015-09	CP-69	CP-150535	5397		Stop resource sharing when receiving a conflicting SDP offer	13.2.1	13.3.0	C1-152575
2015-09	CP-69	CP-150535	5398		P-CSCF indicating that resource sharing is no longer possible	13.2.1	13.3.0	C1-152576
2015-09	CP-69	CP-150535	5399		Updating resource sharing options	13.2.1	13.3.0	C1-152577
2015-09	CP-69	CP-150530	5400	1	Remove sending of 403 due to MAC address error	13.2.1	13.3.0	C1-153045
2015-09	CP-69	CP-150514	5402	1	SIP timer table update due to RFC 6665 – missing timer N added	13.2.1	13.3.0	C1-153024
2015-09	CP-69	CP-150533	5403	2	Support of Emergency services over WLAN access to EPC	13.2.1	13.3.0	C1-153217
2015-09	CP-69	CP-150530	5405	1	Domain selection for UE originating voice and SMS	13.2.1	13.3.0	C1-153046
2015-09	CP-69	CP-150530	5406		Correction for Route Header in REGISTER request	13.2.1	13.3.0	C1-152619
2015-09	CP-69	CP-150530	5410	1	Annex A: support of PANI in INVITE request by MGCF	13.2.1	13.3.0	C1-153035
2015-09	CP-69	CP-150504	5413		Reference update: RFC 7415	13.2.1	13.3.0	C1-152660
2015-09	CP-69	CP-150510	5415	1	Defining an "MSC server enhanced for DRVCC" role in annex A	13.2.1	13.3.0	C1-152735
2015-09	CP-69	CP-150530	5416	1	General principle of ICID and IOI	13.2.1	13.3.0	C1-153048
2015-09	CP-69	CP-150530	5417	1	Apply general principle of ICID and IOI to P-CSCF	13.2.1	13.3.0	C1-153049
2015-09	CP-69	CP-150530	5418	2	Apply general principle of ICID and IOI to S-CSCF	13.2.1	13.3.0	C1-153065
2015-09	CP-69	CP-150530	5419	2	Apply general principle of ICID and IOI to AS	13.2.1	13.3.0	C1-153066
2015-09	CP-69	CP-150530	5420		Apply general principle of ICID and IOI to I-CSCF.	13.2.1	13.3.0	C1-152670
2015-09	CP-69	CP-150530	5421	1	Apply general principle of ICID and IOI to MGCF	13.2.1	13.3.0	C1-153052
2015-09	CP-69	CP-150530	5422		Apply general principle of ICID and IOI to BGCF	13.2.1	13.3.0	C1-152672
2015-09	CP-69	CP-150530	5423	1	Apply general principle of ICID and IOI to IBCF	13.2.1	13.3.0	C1-153053

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2015-09	CP-69	CP-150530	5424		Apply general principle of ICID and IOI to E-CSCF	13.2.1	13.3.0	C1-152674
2015-09	CP-69	CP-150530	5425	2	Apply general principle of ICID and IOI to ISC gateway function	13.2.1	13.3.0	C1-153067
2015-09	CP-69	CP-150530	5426	4	Apply general principle of ICID and IOI to TF and TRF	13.2.1	13.3.0	C1-153079
2015-09	CP-69	CP-150520	5427		Correction of profile status on Allow header field	13.2.1	13.3.0	C1-152677
2015-09	CP-69	CP-150520	5428		Profile status modification for P-Charging-Vector and Relayed-Charge in Annex A	13.2.1	13.3.0	C1-152679
2015-09	CP-69	CP-150586	5448	2	Support for Video Region-of-Interest (ROI) Signaling	13.2.1	13.3.0	-
2015-09	CP-69	CP-150513	5431		No-response value for Restoration-Info header field	13.2.1	13.3.0	C1-152716
2015-09	CP-69	CP-150520	5432	1	UE to perform reregistration on change of IPCAN	13.2.1	13.3.0	C1-153041
2015-09	CP-69	CP-150512	5434		Non-dialable callback number in PAI before LRF is invoked	13.2.1	13.3.0	C1-152720
2015-09	CP-69	CP-150520	5436		Correcting the spelling of "privilege"	13.2.1	13.3.0	C1-152795
2015-09	CP-69	CP-150520	5437		Correcting the Contact header field entry of PUBLISH in profile tables	13.2.1	13.3.0	C1-152796
2015-09	CP-69	CP-150520	5438	1	Corrections related to reg event XML	13.2.1	13.3.0	C1-153042
2015-09	CP-69	CP-150520	5439	1	Delete duplicated SHALL	13.2.1	13.3.0	C1-153043
2015-09	CP-69	CP-150520	5440	3	Missing Content-Type in Supported header field of CANCEL method	13.2.1	13.3.0	C1-153081
2015-09	CP-69	CP-150520	5441		EPC via WLAN - not reachable for SIP signalling upon loss of IP-CAN bearer for SIP signalling	13.2.1	13.3.0	C1-152847
2015-09	CP-69	CP-150530	5442	2	Supported Media Types	13.2.1	13.3.0	C1-153069
2015-09	CP-69	CP-150513	5444		IANA registration for Restoration-Info header field	13.2.1	13.3.0	C1-152877
2015-09	CP-69	CP-150508	5446		IANA registration for Relayed-Charge header field	13.2.1	13.3.0	C1-152879
2015-09	CP-69	CP-150520	5447		AS handling of expires=0	13.2.1	13.3.0	C1-152880
2015-09	CP-69				Correction of table numbering	13.3.0	13.3.1	
2015-12	CP-70	CP-150699	5449	1	Updated Reference: draft-mohali-dispatch-cause-for-service-number	13.3.1	13.4.0	C1-153702
2015-12	CP-70	CP-150699	5450	1	service access number terminology definition	13.3.1	13.4.0	C1-153703
2015-12	CP-70	CP-150705	5452	1	Support for Video Region-of-Interest (ROI) Signaling	13.3.1	13.4.0	C1-153718
2015-12	CP-70	CP-150709	5453	1	Various editorial corrections	13.3.1	13.4.0	C1-153720
2015-12	CP-70	CP-150701	5454	1	P-CSCF restoration in Annex R	13.3.1	13.4.0	C1-154591
2015-12	CP-70	CP-150698	5457		Annex A - incorrect linkage to INFO request and INFO response	13.3.1	13.4.0	C1-153339
2015-12	CP-70	CP-150720	5458		Resource sharing in CDMA2000 access using EPC	13.3.1	13.4.0	C1-153366
2015-12	CP-70	CP-150709	5459		Cleanup of duplicate words	13.3.1	13.4.0	C1-153371

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2015-12	CP-70	CP-150709	5460		Cleanup of table numbers in clause 4	13.3.1	13.4.0	C1-153372
2015-12	CP-70	CP-150872	5461	1	draft-gundavelli-ipsecme-3gpp-ims-options became RFC7651	13.3.1	13.4.0	-
2015-12	CP-70	CP-150698	5463	3	Missing Content-Type in Supported header field of CANCEL method in Table A.181	13.3.1	13.4.0	C1-154505
2015-12	CP-70	CP-150714	5464	2	Remove non-access technology specific requirement from access technology annex	13.3.1	13.4.0	C1-153979
2015-12	CP-70	CP-150714	5466	1	WLAN Location Information in support of emergency session	13.3.1	13.4.0	C1-153695
2015-12	CP-70	CP-150698	5468		MSC server roles	13.3.1	13.4.0	C1-153579
2015-12	CP-70	CP-150698	5470	3	Protocol values for SIP reason header field	13.3.1	13.4.0	C1-154502
2015-12	CP-70	CP-150685	5472	1	Protocol value for S1 protocol errors missing	13.3.1	13.4.0	C1-153726
2015-12	CP-70	CP-150698	5473	3	Updates on UE to perform reregistration on change of IPCAN	13.3.1	13.4.0	C1-153767
2015-12	CP-70	CP-150712	5474	3	Media Optimization for WebRTC – SDP syntax	13.3.1	13.4.0	C1-154516
2015-12	CP-70	CP-150689	5476		Removal of the IANA specific subclauses for headers	13.3.1	13.4.0	C1-153613
2015-12	CP-70	CP-150698	5477		Correct name of "network-provided"	13.3.1	13.4.0	C1-153615
2015-12	CP-70	CP-150689	5479	2	P-CSCF restoration corrections	13.3.1	13.4.0	C1-154534
2015-12	CP-70	CP-150698	5480		Removing confusing inactive statement	13.3.1	13.4.0	C1-153620
2015-12	CP-70	CP-150690	5482	1	Update RFC 6665 related references to RFC 7614, RFC 7621 and RFC 7647	13.3.1	13.4.0	C1-153747
2015-12	CP-70	CP-150680	5485	2	P-Access-Network-Info ABNF Update	13.3.1	13.4.0	C1-154597
2015-12	CP-70	CP-150693	5487	1	JSON Web Token Claims for transport of WAF and WWSF identities	13.3.1	13.4.0	C1-154530
2015-12	CP-70	CP-150880	5493	2	Call modification: removing mandatory usage of the preconditions mechanism	13.3.1	13.4.0	-
2015-12	CP-70	CP-150693	5495		Reference update: RFC 7675 (draft-ietf-rtcweb-stun-consent-freshness)	13.3.1	13.4.0	C1-154030
2015-12	CP-70	CP-150684	5497		Reference update: draft-ietf-mmusic-sctp-sdp	13.3.1	13.4.0	C1-154048
2015-12	CP-70	CP-150715	5498		MCPTT: SDP Considerations	13.3.1	13.4.0	C1-154051
2015-12	CP-70	CP-150718	5499	1	Support of RTP / RTCP transport multiplexing in IMS call control signalling	13.3.1	13.4.0	C1-154581
2015-12	CP-70	CP-150720	5500	1	Correct AS and S-CSCF for resource sharing	13.3.1	13.4.0	C1-154500
2015-12	CP-70	CP-150717	5501		Paging Policy Differentiation, complete TCP cases	13.3.1	13.4.0	C1-154127
2015-12	CP-70	CP-150712	5502	1	Negotiation of contents of data channels (MSRP) – Annex A	13.3.1	13.4.0	C1-154515
2015-12	CP-70	CP-150698	5503	1	Determination of used registration – standalone request	13.3.1	13.4.0	C1-154501
2015-12	CP-70	CP-150681	5506	1	Adding the loopback-indication in subsequent requests	13.3.1	13.4.0	C1-154570
2015-12	CP-70	CP-150681	5511	1	Loopback indication in Responses	13.3.1	13.4.0	C1-154567

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2015-12	CP-70	CP-150712	5512	1	Media Optimization for WebRTC – Optional Procedures for intermediate nodes	13.3.1	13.4.0	C1-154518
2015-12	CP-70	CP-150709	5513		Cleanup of duplicate words-2	13.3.1	13.4.0	C1-154168
2015-12	CP-70	CP-150709	5514		Reference correction in Table A.3	13.3.1	13.4.0	C1-154172
2015-12	CP-70	CP-150709	5515	1	Reference correction for IBCF performing media transcoding control	13.3.1	13.4.0	C1-154519
2015-12	CP-70	CP-150709	5516		Normative word in normal notes	13.3.1	13.4.0	C1-154174
2015-12	CP-70	CP-150677	5522		Call modification: UE shall not indicate the requirement for the precondition mechanism by using the Require header field mechanism.	13.3.1	13.4.0	C1-154234
2015-12	CP-70	CP-150708	5524	2	New access type value for ProSe UE-to-Network Relay	13.3.1	13.4.0	C1-154457
2015-12	CP-70	CP-150699	5526	1	Service access number scope	13.3.1	13.4.0	C1-154506
2015-12	CP-70	CP-150709	5527	3	Handling of failure responses by the UE during call setup	13.3.1	13.4.0	C1-154881
2015-12	CP-70	CP-150698	5528	2	Correcting ambiguity when a list of UE actions are specified	13.3.1	13.4.0	C1-154720
2015-12	CP-70	CP-150698	5529	1	Update condition c22 of Content-Type in Supported header field of CANCEL method in Table A.23	13.3.1	13.4.0	C1-154504
2015-12	CP-70	CP-150698	5530		Correct description of P-CSCF restoration	13.3.1	13.4.0	C1-154395
2016-03	CP-71	CP-160136	5361	10	Coding of Type 1 IOIs and of Type 3 IOIs	13.4.0	13.5.0	C1-161541
2016-03	CP-71	CP-160083	5467	6	Addition of UE Provided Location Information for Untrusted WLAN access	13.4.0	13.5.0	C1-160941
2016-03	CP-71	CP-160082	5523	5	Align terms with definitions in TS 24.302 and reword untestable conditions when accessing IM CN subsystem via WLAN IP access	13.4.0	13.5.0	C1-161539
2016-03	CP-71	CP-160136	5532	2	Usage of the contact address previously registered	13.4.0	13.5.0	C1-160494
2016-03	CP-71	CP-160136	5536	2	JSON Web Token Claims for transport of WAF and WWSF identities: missing note	13.4.0	13.5.0	C1-160495
2016-03	CP-71	CP-160136	5537	1	P-Access-Network-Info ABNF Update	13.4.0	13.5.0	C1-160444
2016-03	CP-71	CP-160136	5538	4	MGCF call modification: removing mandatory usage of the preconditions mechanism	13.4.0	13.5.0	C1-161354
2016-03	CP-71	CP-160076	5539		MCPTT: Removal of SDP considerations	13.4.0	13.5.0	C1-160078
2016-03	CP-71	CP-160066	5542		Correction of the incorrect referenced subclause number.	13.4.0	13.5.0	C1-160155
2016-03	CP-71	CP-160064	5548	2	MGCF call initiation: removing posibility for using require header field in preconditions	13.4.0	13.5.0	C1-160487
2016-03	CP-71	CP-160136	5549	1	UE session modification clarification	13.4.0	13.5.0	C1-160447
2016-03	CP-71	CP-160074	5550	1	Profile table for media plane optimization webrtc attributes	13.4.0	13.5.0	C1-160450
2016-03	CP-71	CP-160136	5551	1	Reg-token in case of no registration	13.4.0	13.5.0	C1-160448
2016-03	CP-71	CP-160136	5552		IANA registration part for Service Interact-Info	13.4.0	13.5.0	C1-160224
2016-03	CP-71	CP-160074	5553		Update reference for draft-ietf-mmusic-data-channel-sdpneg	13.4.0	13.5.0	C1-160229

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2016-03	CP-71	CP-160136	5562	4	Clarification of emergency registration handling	13.4.0	13.5.0	C1-160618
2016-03	CP-71	CP-160081	5563		Support for Video Region-of-Interest (ROI) Signaling	13.4.0	13.5.0	C1-160336
2016-03	CP-71	CP-160080	5564	1	Support of enhanced bandwidth negotiation mechanism for MTSI sessions	13.4.0	13.5.0	C1-161333
2016-03	CP-71	CP-160080	5565		Support of enhanced bandwidth negotiation mechanism by MTSI UE	13.4.0	13.5.0	C1-160819
2016-03	CP-71	CP-160083	5568		Network-provided PANI when a UE accesses IMS from general Internet using S2b	13.4.0	13.5.0	C1-160855
2016-03	CP-71	CP-160138	5569	2	P-Access-Network-Info header field in retransmitted SIP messages	13.4.0	13.5.0	-
2016-03	CP-71	CP-160136	5570	1	Precondition option tag in responses within a dialog other than INVITE responses	13.4.0	13.5.0	C1-161383
2016-03	CP-71	CP-160136	5571	1	Clarification of base-time description	13.4.0	13.5.0	C1-161502
2016-03	CP-71	CP-160136	5577	2	Restriction on requesting early media authorization from UE	13.4.0	13.5.0	C1-161506
2016-03	CP-71				Editorial corrections	13.5.0	13.5.1	
2016-06	CP-72	CP-160317	5561	8	Handling of 380 in Annex R	13.5.1	13.6.0	C1-161953
2016-06	CP-72	CP-160317	5580	2	Corrections for PSAP callback after emergency session via untrusted WLAN	13.5.1	13.6.0	C1-162307
2016-06	CP-72	CP-160317	5581	3	Corrections for the "VIRTUAL" access type	13.5.1	13.6.0	C1-162982
2016-06	CP-72	CP-160298	5584	1	orig-cdiv session case definition from draft-mohali-dispatch-originating-cdiv-parameter-01	13.5.1	13.6.0	C1-162093
2016-06	CP-72	CP-160317	5591	2	Corrections to Cellular-Network-Info handling	13.5.1	13.6.0	C1-162507
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2016-06	CP-72	CP-160320	5617	2	SDP offer/answer for TLS and DTLS protected media	13.5.1	13.6.0	C1-163122
2016-06	CP-72	CP-160322	5623	1	Adding draft-holmberg-dispatch-mcptt-rp-namespace to annex A	13.5.1	13.6.0	C1-163008
2016-06	CP-72	CP-160298	5626	1	Reference update: draft-holmberg-dispatch-pani-abnf	13.5.1	13.6.0	C1-162864
2016-06	CP-72	CP-160311	5636		Corrections of errors in PANI and CNI	13.5.1	13.6.0	C1-162584
2016-06	CP-72	CP-160321	5585		Service access number draft update	13.5.1	13.6.0	C1-161619
2016-06	CP-72	CP-160328	5566	4	Coding of the user part in the contact address	13.6.0	14.0.0	C1-162075
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2016-06	CP-72	CP-160328	5599	2	P-Access-Network-Info header in ACK request	13.6.0	14.0.0	C1-162281
2016-06	CP-72	CP-160328	5600	2	P-Charging-Vector header in ACK request	13.6.0	14.0.0	C1-162282

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2016-06	CP-72	CP-160328	5607	5	Clarifications of call setup and session modifications using preconditions	13.6.0	14.0.0	C1-163038
2016-06	CP-72	CP-160328	5608	1	Resource-Priority description minor correction	13.6.0	14.0.0	C1-162074
2016-06	CP-72	CP-160326	5609	6	Addition of eCall URNs	13.6.0	14.0.0	C1-163032
2016-06	CP-72	CP-160328	5612	2	Indication of release cause	13.6.0	14.0.0	C1-163036
2016-06	CP-72	CP-160332	5615	1	Mandatory support of RTP/RTCP multiplexing	13.6.0	14.0.0	C1-162876
2016-06	CP-72	CP-160328	5630	1	Request forwarding in IBCF	13.6.0	14.0.0	C1-162881
2016-06	CP-72	CP-160326	5631	4	MSD transfer for eCall over IMS	13.6.0	14.0.0	C1-163060
2016-06	CP-72	CP-160328	5632	3	Handling of 488 response with 301 warn-code in network	13.6.0	14.0.0	C1-163150
2016-06	CP-72	CP-160328	5639	1	Clarifications of QoS attributes during session modifications	13.6.0	14.0.0	C1-162886
2016-06	CP-72	CP-160329	5634	1	Media type restriction policy enforcement	13.6.0	14.0.0	C1-162941
2016-06					Editorial fixes	14.0.0	14.0.1	

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2016-09	CT#73	CP-160495	5641	4	A	Introducing priority sharing	14.1.0
2016-09	CT#73	CP-160515	5642	4	B	Default EPS bearer context usage restriction policy	14.1.0
2016-09	CT#73	CP-160509	5644		A	Missing parameters for lawful interception of calls established over untrusted non-3GPP access connected to EPC	14.1.0
2016-09	CT#73	CP-160514	5645		F	Handling re-INVITE request collisions in application servers	14.1.0
2016-09	CT#73	CP-160518	5648	1	F	Correction of reference to Ix interface	14.1.0
2016-09	CT#73	CP-160518	5649		F	Correction of referenced specification for Ix interface	14.1.0
2016-09	CT#73	CP-160508	5652		A	Updated ref to draft-mohali-dispatch-cause-for-service-number-07	14.1.0
2016-09	CT#73	CP-160483	5662	1	A	P-CSCF includes access-network-charging-info in reliable response	14.1.0
2016-09	CT#73	CP-160517	5663	3	B	Phase 2 support of Emergency services over WLAN	14.1.0
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2016-09	CT#73	CP-160490	5667	1	A	Registration-token in 3rd party REGISTER	14.1.0
2016-09	CT#73	CP-160515	5668	2	C	Support of Media_type_restirction_policy only conditional	14.1.0
2016-09	CT#73	CP-160514	5669	1	F	Error in Table A.4 for Resource-Priority	14.1.0
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2016-09	CT#73	CP-160512	5672	1	C	Update of eCall over IMS procedures	14.1.0
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2016-09	CT#73	CP-160514	5674	1	F	Corrections to resource sharing	14.1.0
2016-09	CT#73	CP-160514	5678		F	Correct reference for codec insertion	14.1.0
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2016-09	CT#73	CP-160484	5696		A	Reference update: RFC 7913	14.1.0
2016-09	CT#73	CP-160509	5698	1	A	Correction on support of Cellular-Network-Info header	14.1.0
2016-09	CT#73	CP-160514	5699	1	F	Correcting reference for DTLS-SRTP in annex A	14.1.0
2016-09	CT#73	CP-160518	5701	1	F	Reference update draft-ietf-mmusic-mux-exclusive 24.229	14.1.0
2016-09	CT#73	CP-160521	5702	1	B	Enable usage P-Visited-Network-ID header for S8HR	14.1.0
2016-12	CT#74	CP-160752	5650	3	B	Clarification on SDP offer answer for transcoding	14.2.0
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2016-12	CT#74	CP-160752	5716	1	F	Modification of the duplicated SIP-ISUP interworking procedure at MGCF	14.2.0
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